

A Detailed Notebook on Ratner's Theorems

Volumes I–XIV

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Foundations, rigidity, effective dynamics, and arithmetic applications

Publication scope and external input

The fourteen volumes form one proof-owned route from Lie groups and ergodic theory through Ratner rigidity to effective homogeneous dynamics and quantitative arithmetic applications.

The sole deliberate project-specific external research input is the Dani–Margulis structural interface stated in Volume III: countability of the homogeneous support class and closed/discrete exterior-power orbits for a fixed support subgroup. All other project-specific research-scale statements used in the active dependency graph are proved in the series. Standard graduate prerequisites are declared separately in the volume front matter; literature citations otherwise serve attribution, comparison, and theorem concordance rather than silent proof ownership.

For physical publication the series is also assembled as Book A (Volumes I–IV), Book B (Volumes V–X), and Book C (Volumes XI–XIV). The present PDF is the complete electronic edition.

Volumes in this edition

- **Volume I.** Foundations of Homogeneous Dynamics (206 pages)
- **Volume II.** Ergodic Theory and Unipotent Dynamics (201 pages)
- **Volume III.** Ratner Rigidity: Measure Classification, Orbit Closures, and Equidistribution (263 pages)
- **Volume IV.** Arithmetic Applications, Structure Theory, and Further Directions (67 pages)
- **Volume V.** Algebraic and Arithmetic Infrastructure for Post–Ratner Dynamics (165 pages)
- **Volume VI.** Mautner, Howe–Moore, and Mixing on Homogeneous Spaces (81 pages)
- **Volume VII.** Limits of Homogeneous Measures and the Mozes–Shah Principle (103 pages)
- **Volume VIII.** Polynomial and Tame Trajectories on Homogeneous Spaces (97 pages)
- **Volume IX.** Expanding Translates, Horospheres, Curves, and Submanifolds (77 pages)
- **Volume X.** Counting on Homogeneous Varieties (89 pages)
- **Volume XI.** Spectral Gap, Sobolev Theory, and Quantitative Mixing (77 pages)
- **Volume XII.** Effective Expanding Translates and Effective Counting (69 pages)
- **Volume XIII.** Effective Unipotent Dynamics and Effective Closing (121 pages)

- **Volume XIV.** Quantitative Oppenheim and Effective Arithmetic Applications (145 pages)

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Foundations of Homogeneous Dynamics

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Dependency and proof-ownership contract

Purpose. Topological and measure-theoretic foundations.

Prerequisites. elementary real analysis, metric/topological preliminaries, and finite-dimensional linear algebra. Appendix A proves the inverse-function, regular-value, submersion, smooth-cutoff, Riesz–Markov, dominated-convergence, and Stone–Weierstrass tools used later; Volume II proves the remaining conditional-expectation, disintegration, tightness, and Lusin machinery.

New objects. Lie groups, Haar measure, quotients, Radon measures. Every project-specific technical term is defined before its first load-bearing use or recalled by an exact backward reference.

Main outputs. foundational toolkit for all later volumes.

External-input rule. Every nontrivial imported result is either part of the declared graduate-level prerequisites, an explicitly named external research input in the prerequisite map, or is proved in this series before use (possibly under an explicit deferred-proof contract). A citation by itself never constitutes a proof.

Reader map and proof route

Question of the volume. How does one put Lie groups, Haar measure, lattices, quotient spaces, recurrence, and weak convergence on a foundation strong enough that every later dynamical argument has a precise measure-theoretic meaning?

Prerequisite checkpoint. The reader should know graduate real analysis, basic topology, linear algebra, and the elementary theory of smooth manifolds. The foundational analysis used by the series is now proof-owned inside this volume. Appendix A contains the inverse-function theorem, regular-value and submersion consequences, the compact bump/partition construction, Riesz–Markov representation and uniqueness, dominated convergence, and Stone–Weierstrass. Conditional expectation, disintegration, tightness compactness, and Lusin are constructed in Volume II. These results play an infrastructural role: none contains a homogeneous-dynamics classification theorem.

Load-bearing route. Read the chapters on Lie groups and homogeneous spaces first, then Haar measure and quotient measures, then lattices/recurrence, and finally weak convergence and compactness. The later laboratory and appendices are for calculation and technical reinforcement; they are not substitutes for the main proofs.

What to keep in view. There are three recurring distinctions: left versus right translation, finite invariant measure versus mere local finiteness, and a statement on G versus the descended statement on G/Γ . Most normalization errors later in the series can be traced to confusing one of these pairs.

Sanity check. For the affine group $x \mapsto ax + b$ with $a > 0$, compute the Jacobian of left and right multiplication before guessing a Haar density. The mismatch is a small model for the modular-function bookkeeping that reappears whenever a subgroup is not unimodular.

CHAPTER 1

Topological Groups, Lie Groups, and One-Parameter Subgroups

1. Why groups enter dynamics

A large part of homogeneous dynamics begins with a deceptively simple construction. Take a Lie group G and a subgroup $\Gamma < G$. In the applications that motivate these notes, Γ is usually closed and often discrete; this matters because quotients by nonclosed subgroups can have very poor topology. Form the left-coset space

$$X = G/\Gamma.$$

When a topology is needed, X carries the quotient topology. A subgroup $U < G$ then acts on X by left translation,

$$u \cdot (g\Gamma) = (ug)\Gamma.$$

This is well defined even when Γ is not normal: if $g_1\Gamma = g_2\Gamma$, then $g_1 = g_2\gamma$ for some $\gamma \in \Gamma$, and hence

$$ug_1\Gamma = ug_2\gamma\Gamma = ug_2\Gamma.$$

The dynamical problem is to understand the orbit

$$Ug\Gamma = \{ug\Gamma : u \in U\},$$

its closure, and the measures that are invariant under the U -action.

The abstract notation hides a great deal of geometry. For example, when

$$G = \mathrm{SL}_d(\mathbb{R}), \quad \Gamma = \mathrm{SL}_d(\mathbb{Z}),$$

a coset $g\Gamma$ can be identified with the unimodular lattice $g\mathbb{Z}^d$ in $\mathbb{R}^d$. Thus a matrix flow on G/Γ moves lattices. A diagonal matrix can make some lattice directions longer and others shorter; a unipotent matrix can shear the lattice. Here a matrix u is *unipotent* if $u - I$ is nilpotent, equivalently if all complex eigenvalues of u are equal to 1. Statements about recurrence or orbit closure can therefore become statements about how lattices degenerate, how short vectors appear, or how arithmetic data are distributed.

This is already a useful lesson: one should not think of G/Γ as a formal quotient only. The quotient often parametrizes concrete geometric or

arithmetic objects, and the group action is a geometric deformation of those objects.

The most basic regularity hierarchy is

$$\boxed{\text{abstract group} \longrightarrow \text{topological group} \longrightarrow \text{Lie group}.}$$

An abstract group remembers multiplication and inversion. A topological group also remembers convergence and continuity. A Lie group has a smooth manifold structure, so infinitesimal methods become available: tangent spaces, vector fields, differential equations, and the exponential map.

The purpose of this chapter is not merely to collect definitions. The goal is to understand how the algebraic and topological structures constrain one another, and why those constraints are powerful enough to support dynamics. Non-obvious constructions are accompanied by an explanation of the mechanism before the formal proof.

Worked Example 1.1.1 (The space of unimodular lattices). Let $d \geq 2$ and

$$X_d = \mathrm{SL}_d(\mathbb{R}) / \mathrm{SL}_d(\mathbb{Z}).$$

A full-rank lattice in $\mathbb{R}^d$ is called *unimodular* if its covolume is 1; equivalently, if $\Lambda = g\mathbb{Z}^d$ for some g with $|\det g| = 1$. Associate to $g \in \mathrm{SL}_d(\mathbb{Z})$ the lattice

$$\Lambda_g = g\mathbb{Z}^d.$$

This is well defined. Indeed, if $g_1 \mathrm{SL}_d(\mathbb{Z}) = g_2 \mathrm{SL}_d(\mathbb{Z})$, then $g_2^{-1}g_1 \in \mathrm{SL}_d(\mathbb{Z})$, and hence

$$g_1\mathbb{Z}^d = g_2(g_2^{-1}g_1)\mathbb{Z}^d = g_2\mathbb{Z}^d,$$

because an integer unimodular matrix preserves $\mathbb{Z}^d$.

Conversely, if $g_1\mathbb{Z}^d = g_2\mathbb{Z}^d$, then $g_2^{-1}g_1$ preserves $\mathbb{Z}^d$. Its columns are integer vectors, and the same is true of its inverse. Therefore $g_2^{-1}g_1 \in \mathrm{GL}_d(\mathbb{Z})$. Since both g_1 and g_2 have determinant 1, the determinant is 1, so actually

$$g_2^{-1}g_1 \in \mathrm{SL}_d(\mathbb{Z}).$$

Thus X_d parametrizes unimodular lattices exactly. Conversely, a covolume-one lattice admits a basis matrix of determinant ± 1 ; changing the sign of one basis vector if necessary gives determinant 1, so it is represented by an element of $\mathrm{SL}_d(\mathbb{R})$.

Now let

$$a_t = \mathrm{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_d t}), \quad \lambda_1 + \dots + \lambda_d = 0.$$

Then

$$a_t(g\mathbb{Z}^d) = (a_t g)\mathbb{Z}^d.$$

So the diagonal flow stretches some ambient coordinate directions and contracts others while preserving covolume. This concrete model is one of the reasons quotients of Lie groups are so effective in number theory.

2. Topological groups

Definition 1.2.1 (Topological group). A *topological group* is a group G equipped with a topology such that

$$m : G \times G \rightarrow G, \quad m(g, h) = gh,$$

and

$$\iota : G \rightarrow G, \quad \iota(g) = g^{-1},$$

are continuous. The product $G \times G$ carries the product topology.

The point of the definition is that algebraic operations must respect small perturbations. If $g_n \rightarrow g$ and $h_n \rightarrow h$, then

$$g_n h_n \rightarrow gh,$$

and if $g_n \rightarrow g$, then

$$g_n^{-1} \rightarrow g^{-1}.$$

In dynamics this compatibility is indispensable: an orbit obtained by multiplying by nearby group elements must move nearby points to nearby points.

Example 1.2.2 (Additive vector groups). $(\mathbb{R}^n, +)$ is a topological group. Addition and $x \mapsto -x$ are continuous. This example is abelian and linear, but many local arguments in an arbitrary Lie group reduce to this model after choosing coordinates near the identity.

Example 1.2.3 (The circle and torus). The unit circle

$$S^1 = \{z \in \mathbb{C} : |z| = 1\}$$

with multiplication is a compact topological group. The homomorphism

$$\mathbb{R} \rightarrow S^1, \quad t \mapsto e^{2\pi i t},$$

has kernel $\mathbb{Z}$, so $S^1 \simeq \mathbb{R}/\mathbb{Z}$. More generally,

$$\mathbb{T}^d = \mathbb{R}^d / \mathbb{Z}^d$$

is a compact abelian topological group.

Example 1.2.4 (Matrix groups). Identify $M_n(\mathbb{R})$ with $\mathbb{R}^{n^2}$. The general linear group

$$\mathrm{GL}_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det A \neq 0\}$$

is a topological group. Matrix multiplication is polynomial in the entries, and

$$A^{-1} = \frac{1}{\det A} \operatorname{adj}(A)$$

is continuous where $\det A \neq 0$. Two basic subgroups are

$$\operatorname{SL}_n(\mathbb{R}) = \{A \in \operatorname{GL}_n(\mathbb{R}) : \det A = 1\}$$

and

$$\operatorname{O}(n) = \left\{ A \in \operatorname{GL}_n(\mathbb{R}) : A^T A = I \right\}.$$

With the subspace topology, each is again a topological group.

Worked Example 1.2.5 (The real Heisenberg group). A less elementary example, important in homogeneous dynamics, nilpotent geometry, and additive combinatorics, is

$$H_3(\mathbb{R}) = \left\{ \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, z \in \mathbb{R} \right\}.$$

Writing an element as (x, y, z) , matrix multiplication gives

$$(x, y, z)(x', y', z') = (x + x', y + y', z + z' + xy').$$

The extra term xy' is the first sign that the group is nonabelian. Using the commutator convention $[g, h] = ghg^{-1}h^{-1}$, a direct calculation gives

$$[(x, y, z), (x', y', z')] = (0, 0, xy' - x'y).$$

Thus the failure of commutativity is entirely in the central z -coordinate. The subgroup $\{(0, 0, z) : z \in \mathbb{R}\}$ is central, so every commutator is central and every commutator of commutators is trivial. In this sense $H_3(\mathbb{R})$ is a two-step nilpotent group.

Both multiplication and inversion are polynomial in (x, y, z) ; in particular,

$$(x, y, z)^{-1} = (-x, -y, -z + xy).$$

Thus $H_3(\mathbb{R})$ is a topological group. Its integer subgroup $H_3(\mathbb{Z})$ leads to a compact nilmanifold and is a basic model for dynamics beyond tori.

3. Translations and the homogeneous local structure

For $g \in G$, define left and right translations by

$$L_g(x) = gx, \quad R_g(x) = xg.$$

Proposition 1.3.1. *For every $g \in G$, both L_g and R_g are homeomorphisms.*

PROOF. The map L_g is continuous because it is obtained from the continuous multiplication map by fixing the first variable. Algebraically,

$$L_{g^{-1}} \circ L_g = L_e = \text{id}_G,$$

and multiplication in the opposite order gives $L_g \circ L_{g^{-1}} = L_e = \text{id}_G$. Since $L_{g^{-1}}$ is also continuous, it is a continuous inverse for L_g . Hence L_g is a homeomorphism. The proof for R_g is identical. $\square$

This elementary fact deserves more attention than its proof length suggests. A general manifold may look different at different points if additional structure is present. In a topological group, every point is obtained from the identity by a homeomorphism that is part of the group structure itself. Consequently, any local topological statement proved near e can be transported to any $g \in G$.

Proposition 1.3.2 (Equivalent continuity criterion). *A group G with a topology is a topological group if and only if the map*

$$F : G \times G \rightarrow G, \quad F(x, y) = xy^{-1},$$

is continuous.

PROOF. If multiplication and inversion are continuous, then F is the composition

$$(x, y) \mapsto (x, y^{-1}) \mapsto xy^{-1},$$

so it is continuous.

Conversely, assume F is continuous. Fixing $x = e$ gives

$$y \mapsto F(e, y) = y^{-1},$$

so inversion is continuous. Since inversion is now known to be continuous,

$$xy = F(x, y^{-1})$$

is a composition of continuous maps. Hence multiplication is continuous as well. $\square$

3.1. Shrinking neighborhoods. Topological group arguments constantly use the ability to take a desired neighborhood and replace it by a smaller one whose products remain controlled.

Lemma 1.3.3 (Symmetric neighborhood lemma). *Let U be a neighborhood of e . There is a neighborhood V of e such that*

$$V^{-1} = V, \quad VV \subset U.$$

Guiding Idea 1.3.4. The group operation is continuous at (e, e) , so if the target set U contains $ee = e$, then sufficiently small perturbations of both inputs still multiply into U . The only extra step is to make the neighborhood symmetric so that taking inverses does not leave it.

PROOF. Continuity of multiplication at (e, e) means that there are neighborhoods W_1 and W_2 of e for which

$$W_1 W_2 \subset U.$$

Put $W = W_1 \cap W_2$. Then W is a neighborhood of e and

$$WW \subset W_1 W_2 \subset U.$$

Because inversion is continuous and fixes e , the set W^{-1} is also a neighborhood of e . Define

$$V = W \cap W^{-1}.$$

Then

$$V^{-1} = W^{-1} \cap W = V,$$

and $V \subset W$ gives

$$VV \subset WW \subset U.$$

□

Corollary 1.3.5. *For every neighborhood U of e and every integer $m \geq 1$, there is a symmetric neighborhood V of e such that*

$$V^m \subset U.$$

PROOF. Choose r so that $2^r \geq m$. Starting with $U_0 = U$, recursively use theorem 1.3.3 to find symmetric neighborhoods U_{j+1} with

$$U_{j+1}^2 \subset U_j$$

for $0 \leq j < r$. Then

$$U_r^{2^r} \subset U_0 = U.$$

Since $e \in U_r$, every product of $m \leq 2^r$ elements of U_r can be padded with identities to a product of 2^r elements, so $U_r^m \subset U$. □

4. Subgroups, closures, and connected components

Proposition 1.4.1. *If $H < G$ is a subgroup of a topological group, then its closure $\overline{H}$ is a subgroup.*

PROOF. A useful subgroup test is that a nonempty subset $K \subset G$ is a subgroup if $xy^{-1} \in K$ whenever $x, y \in K$. Let $x, y \in \overline{H}$, and let O be any open neighborhood of xy^{-1} . The map

$$F : G \times G \rightarrow G, \quad F(a, b) = ab^{-1},$$

is continuous at (x, y) . Hence there are neighborhoods U of x and V of y such that

$$UV^{-1} \subset O.$$

Because $x, y \in \overline{H}$, choose $h \in U \cap H$ and $k \in V \cap H$. Then

$$hk^{-1} \in H \cap O.$$

Thus every neighborhood O of xy^{-1} meets H , so $xy^{-1} \in \overline{H}$. The subgroup test now gives that $\overline{H}$ is a subgroup.

This argument avoids any countability or metrizable assumption; it works in an arbitrary topological group. $\square$

Worked Example 1.4.2 (A dense cyclic subgroup of the circle). Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ and

$$H = \{e^{2\pi i n \alpha} : n \in \mathbb{Z}\} \subset S^1.$$

Then H is proper because it is countable, but H is dense in S^1 .

Here is an elementary proof that also explains the mechanism. For $x \in \mathbb{R}$, write

$$\|x\|_{\mathbb{R}/\mathbb{Z}} = \min_{m \in \mathbb{Z}} |x - m|$$

for the distance of x to the nearest integer. Fix $N \geq 1$ and divide $\mathbb{R}/\mathbb{Z}$ into N intervals of length $1/N$. Among the $N + 1$ points

$$0, \alpha, 2\alpha, \dots, N\alpha \pmod{1},$$

two lie in the same interval. Their difference gives an integer $1 \leq q \leq N$ such that

$$0 < \|q\alpha\|_{\mathbb{R}/\mathbb{Z}} < \frac{1}{N}.$$

The strict positivity uses the irrationality of α . Replacing q by $-q$ if necessary, let $\beta \in (0, 1/N)$ be the representative modulo 1 of $q\alpha$ or $-q\alpha$. Then every point $k\beta \pmod{1}$ belongs to H .

Now put $M = \lfloor 1/\beta \rfloor$. On the circle, the points

$$0, \beta, 2\beta, \dots, M\beta$$

have consecutive gaps equal to β , except for the final gap from $M\beta$ back to 1, which is

$$1 - M\beta < \beta.$$

Hence these points form a β -net, and therefore a $1/N$ -net, in $\mathbb{R}/\mathbb{Z}$. Since N is arbitrary, every nonempty open arc meets H , proving that H is dense.

The example shows why closure is a natural operation in dynamics. An orbit may be algebraically simple and yet topologically fill a much larger group or homogeneous space.

4.1. The identity component. Let G° denote the connected component of the identity.

Proposition 1.4.3. *G° is a closed normal subgroup of G .*

Guiding Idea 1.4.4. There are two separate points. First, connectedness is preserved by continuous maps, so multiplying one connected set by the inverse of another produces another connected set. Second, conjugation is a homeomorphism fixing the identity, so it must preserve the largest connected set containing the identity.

PROOF. Connected components are closed in any topological space, so G° is closed.

To prove the subgroup property, consider the continuous map

$$F : G^\circ \times G^\circ \rightarrow G, \quad F(x, y) = xy^{-1}.$$

The product $G^\circ \times G^\circ$ is connected, hence its image

$$G^\circ(G^\circ)^{-1}$$

is connected. This image contains $e = ee^{-1}$. Since G° is the maximal connected subset containing e , we have

$$G^\circ(G^\circ)^{-1} \subset G^\circ.$$

The subgroup test now shows that G° is a subgroup.

For normality, fix $g \in G$. Conjugation

$$C_g(x) = gxg^{-1}$$

is a homeomorphism and $C_g(e) = e$. Hence $C_g(G^\circ)$ is connected and contains e , so

$$gG^\circ g^{-1} \subset G^\circ.$$

Apply the same argument with g^{-1} to obtain

$$g^{-1}G^\circ g \subset G^\circ.$$

Conjugating this last inclusion by g gives $G^\circ \subset gG^\circ g^{-1}$. Therefore equality holds. $\square$

Proposition 1.4.5. *The connected component containing $g \in G$ is*

$$gG^\circ = G^\circ g.$$

PROOF. Left translation L_g is a homeomorphism, so gG° is connected and contains g . Hence it lies in the connected component C_g of g .

Conversely, $g^{-1}C_g$ is connected because $L_{g^{-1}}$ is a homeomorphism, and it contains e . Therefore $g^{-1}C_g \subset G^\circ$, or $C_g \subset gG^\circ$. Thus $C_g = gG^\circ$. Normality of G° gives $gG^\circ = G^\circ g$. $\square$

5. Open subgroups and generation by small neighborhoods

Proposition 1.5.1. *Every open subgroup $H < G$ is closed.*

PROOF. Every left coset gH is open because L_g is a homeomorphism. Distinct left cosets partition G , and

$$G \setminus H = \bigcup_{g \notin H} gH.$$

The right-hand side is open, so H is closed. $\square$

Corollary 1.5.2. *A connected topological group has no proper open subgroup.*

PROOF. By Proposition 1.5.1, every open subgroup is also closed. A nonempty subset that is both open and closed in a connected space is the whole space. $\square$

Proposition 1.5.3. *If G is connected and U is any neighborhood of e , then*

$$\langle U \rangle = G.$$

PROOF. Let $H = \langle U \rangle$. We may replace U by an open neighborhood contained in it, so suppose U is open. Since H contains U ,

$$H = \bigcup_{h \in H} hU.$$

Each hU is open, hence H is open. By theorem 1.5.1, H is also closed. Connectedness of G forces the nonempty clopen subset H to be all of G . $\square$

This proposition says something conceptually strong: in a connected group, arbitrarily small motions generate every global motion. In a Lie group, once the exponential map supplies a neighborhood of the identity from tangent vectors, this becomes a bridge from infinitesimal information to the whole connected group.

6. Locally compact groups

Definition 1.6.1. A Hausdorff space X is *locally compact* if every point has a neighborhood whose closure is compact. A locally compact group is a Hausdorff topological group whose underlying space is locally compact.

Proposition 1.6.2. *Every finite-dimensional Lie group is locally compact.*

PROOF. Let G be an n -dimensional Lie group and $g \in G$. Choose a coordinate chart

$$\phi : U \rightarrow V \subset \mathbb{R}^n,$$

with $g \in U$. Since V is open, choose $r > 0$ such that the closed Euclidean ball

$$\overline{B(\phi(g), r)}$$

is contained in V . Set

$$W = \phi^{-1}(B(\phi(g), r)).$$

Then W is a neighborhood of g . Let

$$K = \phi^{-1}(\overline{B(\phi(g), r)}).$$

The closed ball is compact by Heine–Borel, so K is compact. Since a manifold is Hausdorff, compact subsets are closed; hence K is closed in G . Because $W \subset K$, the closure of W in G satisfies

$$\overline{W}^G \subset K.$$

The closure $\overline{W}^G$ is a closed subset of the compact set K , and is therefore compact. Thus g has a neighborhood with compact closure in G . $\square$

Local compactness is the natural topological setting for invariant integration on groups. Even before introducing any measure, it is useful to know that one can trap local arguments inside compact sets, extract convergent subsequences, and use finite coverings.

7. Quotients by subgroups

Let $H < G$ be a subgroup. The set of left cosets is

$$G/H = \{gH : g \in G\}.$$

The quotient map is

$$\pi : G \rightarrow G/H, \quad \pi(g) = gH.$$

The quotient topology is defined by declaring $A \subset G/H$ open exactly when $\pi^{-1}(A)$ is open in G .

Proposition 1.7.1. *The quotient map $\pi : G \rightarrow G/H$ is open.*

PROOF. Let $U \subset G$ be open. We compute the full inverse image of its image:

$$\pi^{-1}(\pi(U)) = UH = \bigcup_{h \in H} Uh.$$

Every Uh is open because right translation by h is a homeomorphism. Therefore UH is open. By definition of the quotient topology, this means $\pi(U)$ is open. $\square$

THEOREM 1.7.2. *G/H is Hausdorff if and only if H is closed.*

Guiding Idea 1.7.3. The direction “Hausdorff quotient $\Rightarrow$ closed subgroup” is immediate because a point in a Hausdorff space is closed. The converse is more interesting. Distinct cosets xH and yH mean exactly that $x^{-1}y \notin H$. If H is closed, then $x^{-1}y$ has a neighborhood avoiding H . Continuity of $(a, b) \mapsto a^{-1}b$ lets us pull that separation back to neighborhoods of x and y .

PROOF. Assume first that G/H is Hausdorff. The singleton $\{H\}$ is closed, so

$$H = \pi^{-1}(\{H\})$$

is closed in G .

Conversely, assume H is closed, and take distinct cosets $xH \neq yH$. Then $x^{-1}y \notin H$. Since $G \setminus H$ is open and the map

$$F : G \times G \rightarrow G, \quad F(a, b) = a^{-1}b,$$

is continuous, there are neighborhoods U of x and V of y such that

$$U^{-1}V \subset G \setminus H.$$

We claim $\pi(U)$ and $\pi(V)$ are disjoint. If not, there would exist $u \in U$ and $v \in V$ with $uH = vH$, hence $u^{-1}v \in H$. But $u^{-1}v \in U^{-1}V \subset G \setminus H$, a contradiction.

Finally, theorem 1.7.1 shows that $\pi(U)$ and $\pi(V)$ are open. Thus the two cosets have disjoint open neighborhoods, and G/H is Hausdorff. $\square$

Worked Example 1.7.4 (Why nonclosed subgroups produce pathological quotients). Take the dense cyclic subgroup $H < S^1$ from theorem 1.4.2. Since H is not closed, S^1/H is not Hausdorff. In fact the pathology is stronger: every nonempty open subset of S^1/H is the whole quotient.

To see this, let $O \subset S^1/H$ be nonempty and open. Then $\pi^{-1}(O)$ is a nonempty open subset of S^1 that is saturated under multiplication by H . Pick an open arc $I \subset \pi^{-1}(O)$. Because H is dense, the translates hI , $h \in H$, cover S^1 . Since $\pi^{-1}(O)$ contains all these translates, it equals S^1 . Hence $O = S^1/H$.

So closedness is not a cosmetic technical assumption. By theorem 1.7.2, it is precisely what guarantees that the quotient is Hausdorff; for a dense proper subgroup, as in this example, the failure can be much more extreme.

8. Smooth manifolds and Lie groups

Definition 1.8.1 (Lie group). A *Lie group* is a smooth finite-dimensional manifold G equipped with a group structure such that multiplication

$$G \times G \rightarrow G, \quad (g, h) \mapsto gh,$$

and inversion

$$G \rightarrow G, \quad g \mapsto g^{-1},$$

are smooth.

Every Lie group is therefore a topological group. The added smooth structure allows us to differentiate group operations. This is the decisive step: a nonlinear group can be studied infinitesimally through the vector space

$$\mathfrak{g} = T_e G.$$

Example 1.8.2. $(\mathbb{R}^n, +)$ is an n -dimensional Lie group. Its tangent space at the identity 0 is canonically $\mathbb{R}^n$ itself.

Example 1.8.3. The torus $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$ is an n -dimensional compact Lie group. Locally it looks exactly like $\mathbb{R}^n$; the lattice quotient changes the global topology but not the local differential structure.

8.1. The general linear group.

Proposition 1.8.4. $\mathrm{GL}_n(\mathbb{R})$ is an open subset of $M_n(\mathbb{R})$ and hence an n^2 -dimensional smooth manifold.

PROOF. The determinant is continuous and

$$\mathrm{GL}_n(\mathbb{R}) = \det^{-1}(\mathbb{R} \setminus \{0\}).$$

Since $\mathbb{R} \setminus \{0\}$ is open, $\mathrm{GL}_n(\mathbb{R})$ is open in $M_n(\mathbb{R}) \simeq \mathbb{R}^{n^2}$. Any open subset of $\mathbb{R}^{n^2}$ inherits a smooth manifold structure of dimension n^2 . $\square$

Matrix multiplication is polynomial in the entries. Inversion is smooth because

$$A^{-1} = \frac{1}{\det A} \mathrm{adj}(A)$$

and the denominator never vanishes on $\mathrm{GL}_n(\mathbb{R})$. Hence $\mathrm{GL}_n(\mathbb{R})$ is a Lie group.

8.2. The special linear group.

Proposition 1.8.5 (Differential of the determinant). *For $A \in \mathrm{GL}_n(\mathbb{R})$ and $X \in M_n(\mathbb{R})$,*

$$D(\det)_A(X) = \det(A) \mathrm{tr}(A^{-1}X).$$

Guiding Idea 1.8.6. The derivative at a general invertible matrix is reduced to the derivative at the identity by factoring

$$A + tX = A(I + tA^{-1}X).$$

At the identity, the linear part of the determinant is just the trace. This is the infinitesimal form of the relation between determinant and volume.

PROOF. Write

$$\det(A + tX) = \det(A) \det(I + tA^{-1}X).$$

So it is enough to understand $\det(I + tY)$ to first order. Expand the determinant using permutations:

$$\det(I + tY) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^n (\delta_{i,\sigma(i)} + tY_{i,\sigma(i)}).$$

The constant term is 1, coming from the identity permutation. A term linear in t can only arise by choosing exactly one tY_{ii} and identity entries in every other row. Thus only the identity permutation contributes linearly, and the linear coefficient is

$$\sum_{i=1}^n Y_{ii} = \operatorname{tr} Y.$$

Hence

$$\det(I + tY) = 1 + t \operatorname{tr} Y + O(t^2).$$

Taking $Y = A^{-1}X$ gives

$$\det(A + tX) = \det(A) \left(1 + t \operatorname{tr}(A^{-1}X) + O(t^2) \right),$$

which yields the derivative formula. □

THEOREM 1.8.7. $\operatorname{SL}_n(\mathbb{R})$ is a Lie group of dimension $n^2 - 1$.

PROOF. We have

$$\operatorname{SL}_n(\mathbb{R}) = \det^{-1}(1).$$

For $A \in \operatorname{SL}_n(\mathbb{R})$, apply theorem 1.8.5 to the tangent direction $X = A$:

$$D(\det)_A(A) = \det(A) \operatorname{tr}(A^{-1}A) = n \neq 0.$$

Thus $D(\det)_A$ is a nonzero linear functional at every point of $\det^{-1}(1)$. Therefore 1 is a regular value of the determinant. The regular value theorem implies that $\operatorname{SL}_n(\mathbb{R})$ is an embedded submanifold of codimension 1 in $M_n(\mathbb{R})$, hence of dimension $n^2 - 1$.

Multiplication and inversion preserve determinant 1 and are restrictions of smooth maps on $\operatorname{GL}_n(\mathbb{R})$, so they are smooth on $\operatorname{SL}_n(\mathbb{R})$. □

Worked Example 1.8.8 ($\operatorname{SL}_2(\mathbb{R})$ as a three-dimensional manifold). Writing

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the defining equation is $ad - bc = 1$. Thus $\mathrm{SL}_2(\mathbb{R})$ is a smooth hypersurface in $\mathbb{R}^4$. Three particularly important curves through the identity are

$$a_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix}, \quad u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \quad v_s = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix}.$$

They correspond to diagonal stretching and two opposite shearing directions.

9. Smooth translations and invariant vector fields

In a Lie group, theorem 1.3.1 improves from a topological statement to a smooth one.

Proposition 1.9.1. *For every $g \in G$, L_g and R_g are diffeomorphisms.*

PROOF. Both are smooth because multiplication is smooth. Their inverses are $L_{g^{-1}}$ and $R_{g^{-1}}$, which are also smooth. $\square$

Hence

$$(dL_g)_e : T_e G \rightarrow T_g G$$

is a linear isomorphism. Every tangent space can therefore be identified with the tangent space at the identity by a canonical map supplied by the group structure.

Let $X \in T_e G$. Define a vector field $\tilde{X}$ by

$$\tilde{X}(g) = (dL_g)_e X.$$

Proposition 1.9.2. *$\tilde{X}$ is smooth and left invariant:*

$$(dL_h)_g \tilde{X}(g) = \tilde{X}(hg).$$

PROOF. The smoothness follows from differentiating the smooth multiplication map with respect to its second variable: as g varies, $(dL_g)_e$ varies smoothly.

For invariance, use the identity

$$L_h \circ L_g = L_{hg}.$$

Differentiate at e :

$$(dL_h)_g \circ (dL_g)_e = (dL_{hg})_e.$$

Applying both sides to X gives the claimed formula. $\square$

Every left-invariant vector field is determined by its value at the identity. Thus

$$T_e G \longleftrightarrow \{\text{left-invariant smooth vector fields on } G\}$$

is a linear bijection.

10. One-parameter subgroups

Definition 1.10.1. A *one-parameter subgroup* of a Lie group G is a smooth homomorphism

$$\gamma : (\mathbb{R}, +) \rightarrow G.$$

Thus

$$\gamma(s + t) = \gamma(s)\gamma(t).$$

Automatically $\gamma(0) = e$ and $\gamma(-t) = \gamma(t)^{-1}$.

Proposition 1.10.2. *If γ is a one-parameter subgroup and $X = \gamma'(0)$, then*

$$\gamma'(t) = (dL_{\gamma(t)})_e X.$$

PROOF. For fixed t ,

$$\gamma(t + s) = \gamma(t)\gamma(s) = L_{\gamma(t)}(\gamma(s)).$$

Differentiate with respect to s at $s = 0$:

$$\gamma'(t) = (dL_{\gamma(t)})_e \gamma'(0).$$

□

The proposition says that a one-parameter subgroup is the integral curve through e of the left-invariant vector field determined by its initial velocity.

THEOREM 1.10.3 (Existence and uniqueness). *For every $X \in T_e G$, there exists a unique one-parameter subgroup*

$$\gamma_X : \mathbb{R} \rightarrow G$$

with $\gamma'_X(0) = X$.

Guiding Idea 1.10.4. Local ODE theory gives an integral curve only for small time. The nontrivial point is why it exists for all $t \in \mathbb{R}$. The reason is special to invariant vector fields: once a short piece of trajectory is known near the identity, left translation copies that short piece starting from any point already reached. The group law therefore prevents finite-time blow-up.

PROOF. Let $\tilde{X}$ be the left-invariant vector field associated with X . By the local existence and uniqueness theorem for ODEs on manifolds, there are $\varepsilon > 0$ and a unique smooth curve

$$\gamma : (-\varepsilon, \varepsilon) \rightarrow G$$

satisfying

$$\gamma(0) = e, \quad \gamma'(t) = \tilde{X}(\gamma(t)).$$

We first derive a local group law. Fix s with $|s| < \varepsilon$. Whenever all terms are defined, consider

$$\eta(t) = \gamma(s)\gamma(t).$$

By left invariance,

$$\eta'(t) = (dL_{\gamma(s)})_{\gamma(t)} \tilde{X}(\gamma(t)) = \tilde{X}(\gamma(s)\gamma(t)) = \tilde{X}(\eta(t)).$$

Also $\eta(0) = \gamma(s)$. The shifted curve $t \mapsto \gamma(s+t)$ solves the same ODE and has the same value at $t = 0$. Uniqueness gives

$$\gamma(s+t) = \gamma(s)\gamma(t)$$

whenever $s, t, s+t$ lie in the local interval.

Now choose $0 < \delta < \varepsilon/2$. For arbitrary $t \in \mathbb{R}$, choose an integer n so large that $|t/n| < \delta$. Define

$$\Gamma(t) = \gamma(t/n)^n.$$

We must check that this does not depend on the chosen subdivision. If m is another sufficiently large integer, take a common multiple N of m and n . The local group law implies

$$\gamma(t/n) = \gamma(t/N)^{N/n}.$$

This iteration is legitimate because every intermediate sum $k(t/N)$ occurring before t/n has absolute value at most $|t/n| < \delta < \varepsilon$. Hence

$$\gamma(t/n)^n = \gamma(t/N)^N = \gamma(t/m)^m.$$

Thus $\Gamma(t)$ is well defined.

We now verify the homomorphism law explicitly. Given arbitrary $s, t \in \mathbb{R}$, choose N so large that

$$|s/N|, |t/N|, |(s+t)/N| < \delta.$$

The local group law gives

$$\gamma((s+t)/N) = \gamma(s/N)\gamma(t/N).$$

It also gives

$$\gamma(s/N)\gamma(t/N) = \gamma(t/N)\gamma(s/N),$$

because both products equal $\gamma((s+t)/N)$. Hence the two factors commute, and by the subdivision-independence already proved,

$$\begin{aligned} \Gamma(s+t) &= \gamma((s+t)/N)^N \\ &= (\gamma(s/N)\gamma(t/N))^N \\ &= \gamma(s/N)^N \gamma(t/N)^N \\ &= \Gamma(s)\Gamma(t). \end{aligned}$$

For $|t| < \delta$ we may take the subdivision $n = 1$, so $\Gamma(t) = \gamma(t)$. Hence Γ agrees with the original local solution near 0 and is smooth there. The homomorphism identity then expresses Γ near any t_0 as

$$\Gamma(t_0 + h) = \Gamma(t_0)\Gamma(h),$$

showing smoothness everywhere. Its initial derivative is X .

For uniqueness, let Γ_1 and Γ_2 be one-parameter subgroups with the same initial derivative. By theorem 1.10.2, both solve the same ODE through e , so they agree on a neighborhood of 0. For arbitrary t , divide t into sufficiently many small pieces and use the homomorphism law to conclude $\Gamma_1(t) = \Gamma_2(t)$. $\square$

Remark 1.10.5. The proof explains why left-invariant vector fields are complete. This is not true for arbitrary vector fields. On $\mathbb{R}$, the vector field $x^2 d/dx$ has the solution $x(t) = 1/(1-t)$ through $x(0) = 1$, which blows up at $t = 1$. Translation invariance is exactly the extra structure that rules out this behavior.

11. The exponential map

Let $\mathfrak{g} = T_e G$. For $X \in \mathfrak{g}$, denote the unique one-parameter subgroup from theorem 1.10.3 by $t \mapsto \exp(tX)$.

Definition 1.11.1 (Exponential map). The *exponential map* is

$$\exp : \mathfrak{g} \rightarrow G, \quad \exp(X) = \gamma_X(1).$$

Equivalently, $\gamma_X(t) = \exp(tX)$.

For a fixed X ,

$$\exp((s+t)X) = \exp(sX) \exp(tX).$$

The qualification “for a fixed X ” is essential. In a nonabelian group, generally

$$\exp(X + Y) \neq \exp(X) \exp(Y).$$

Before differentiating the exponential map, one point must be justified: $\exp$ is a smooth map, not merely a pointwise-defined collection of one-parameter subgroups.

Proposition 1.11.2 (Smooth dependence of the exponential). *The map*

$$\mathbb{R} \times \mathfrak{g} \rightarrow G, \quad (t, X) \mapsto \exp(tX),$$

is smooth. In particular, $\exp : \mathfrak{g} \rightarrow G$ is smooth.

PROOF. Consider the parameter-dependent differential equation

$$g'(t) = (dL_{g(t)})_e X, \quad g(0) = e,$$

where $X \in \mathfrak{g}$ is regarded as a parameter. The right-hand side

$$(g, X) \mapsto (dL_g)_e X$$

is smooth in both variables: it is the differential, in the second variable at e , of the smooth multiplication map $(g, h) \mapsto gh$.

The smooth-dependence theorem for ordinary differential equations says that the local solution depends smoothly on time, the initial condition, and smooth parameters. Applied here, it shows that $(t, X) \mapsto \exp(tX)$ is smooth wherever the solution is defined. By theorem 1.10.3, these solutions are defined for every $t \in \mathbb{R}$ and every $X \in \mathfrak{g}$. Hence the map is smooth on all of $\mathbb{R} \times \mathfrak{g}$. Setting $t = 1$ gives the smoothness of $X \mapsto \exp X$. $\square$

Proposition 1.11.3. *Under the natural identifications $T_0\mathfrak{g} \simeq \mathfrak{g}$ and $T_eG = \mathfrak{g}$,*

$$(d\exp)_0 = \text{id}_{\mathfrak{g}}.$$

PROOF. Let $X \in \mathfrak{g}$ and consider the line $c(t) = tX$ in $\mathfrak{g}$. Then

$$(d\exp)_0 X = \left. \frac{d}{dt} \right|_{t=0} \exp(tX).$$

By the definition of the one-parameter subgroup generated by X , the derivative on the right is exactly X . $\square$

THEOREM 1.11.4 (Exponential coordinates). *There are neighborhoods U of 0 in $\mathfrak{g}$ and V of e in G such that*

$$\exp : U \rightarrow V$$

is a diffeomorphism.

PROOF. By theorem 1.11.3, the derivative of $\exp$ at 0 is an invertible linear map. The inverse function theorem therefore provides neighborhoods U and V on which $\exp$ is a diffeomorphism. $\square$

The theorem is local, not global. It says that every sufficiently small group element has a unique logarithm X near 0. It does not say that every element of G is an exponential, or that an element far from the identity has a unique logarithm.

Worked Example 1.11.5 (A local logarithm for matrices). If $A = I + Y$ with $\|Y\| < 1$ in a submultiplicative matrix norm, then the series

$$\log(I + Y) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} Y^k$$

converges absolutely. As in the scalar power series, one has

$$\exp(\log(I + Y)) = I + Y.$$

This gives a concrete model of exponential coordinates near the identity in a matrix group.

12. Matrix exponentials and concrete flows

For $X \in M_n(\mathbb{R})$, define

$$e^X = \sum_{k=0}^{\infty} \frac{X^k}{k!}.$$

The series converges absolutely in any matrix norm.

Proposition 1.12.1. *For fixed $X \in M_n(\mathbb{R})$,*

$$\gamma_X(t) = e^{tX}$$

is a one-parameter subgroup of $\mathrm{GL}_n(\mathbb{R})$ and

$$\frac{d}{dt}e^{tX} = Xe^{tX} = e^{tX}X.$$

PROOF. Termwise differentiation is justified by uniform convergence of the power series on compact t -intervals:

$$\frac{d}{dt}e^{tX} = \sum_{k=1}^{\infty} \frac{kt^{k-1}X^k}{k!} = X \sum_{j=0}^{\infty} \frac{t^j X^j}{j!} = Xe^{tX}.$$

Since X commutes with every polynomial in X , this also equals $e^{tX}X$.

To prove the homomorphism law, note that sX and tX commute. Multiplying the two absolutely convergent series and collecting total degree m gives

$$e^{sX}e^{tX} = \sum_{m=0}^{\infty} \left(\sum_{k=0}^m \frac{s^k t^{m-k}}{k!(m-k)!} \right) X^m = \sum_{m=0}^{\infty} \frac{(s+t)^m}{m!} X^m = e^{(s+t)X}.$$

Taking $s = -t$ yields $e^{tX}e^{-tX} = I$, so every e^{tX} is invertible. $\square$

Worked Example 1.12.2 (Nilpotent shears in higher dimension). For $i \neq j$, let E_{ij} be the matrix with a single 1 in position (i, j) . Since $E_{ij}^2 = 0$,

$$e^{sE_{ij}} = I + sE_{ij}.$$

Thus

$$u_{ij}(s) = I + sE_{ij}$$

is a one-parameter unipotent subgroup of $\mathrm{SL}_d(\mathbb{R})$. Acting on a vector, it adds s times the j th coordinate to the i th coordinate. Acting on a lattice, it performs a continuous shear.

Worked Example 1.12.3 (A higher-dimensional diagonal flow). Choose real numbers $\lambda_1, \dots, \lambda_d$ with sum zero and let

$$H = \mathrm{diag}(\lambda_1, \dots, \lambda_d).$$

Then

$$e^{tH} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_d t}) \in \text{SL}_d(\mathbb{R}).$$

The condition $\sum_i \lambda_i = 0$ is exactly what preserves determinant 1.

Worked Example 1.12.4 (Unipotent blocks beyond rank one). Let

$$N = \begin{pmatrix} 0 & I_m \\ 0 & 0 \end{pmatrix}$$

in block form. Then $N^2 = 0$ and

$$e^{tN} = \begin{pmatrix} I_m & tI_m \\ 0 & I_m \end{pmatrix}.$$

Such block shears occur naturally when studying homogeneous spaces associated with systems of linear forms rather than a single one-dimensional horocycle.

Worked Example 1.12.5 (The Heisenberg exponential). For

$$X = \begin{pmatrix} 0 & x & z \\ 0 & 0 & y \\ 0 & 0 & 0 \end{pmatrix},$$

we have $X^3 = 0$ and

$$e^X = I + X + \frac{1}{2}X^2 = \begin{pmatrix} 1 & x & z + \frac{1}{2}xy \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix}.$$

The quadratic correction $xy/2$ is a concrete trace of noncommutativity. Even though the exponential series terminates, it is not simply $I + X$ unless $X^2 = 0$.

13. One-parameter subgroups of matrix groups

THEOREM 1.13.1. *Every smooth one-parameter subgroup*

$$\gamma : \mathbb{R} \rightarrow \text{GL}_n(\mathbb{R})$$

has the form

$$\gamma(t) = e^{tX}$$

for a unique $X \in M_n(\mathbb{R})$.

PROOF. Set $X = \gamma'(0)$. From

$$\gamma(t+s) = \gamma(t)\gamma(s)$$

and differentiation with respect to s at $s = 0$, we obtain

$$\gamma'(t) = \gamma(t)X.$$

The matrix-valued function $Y(t) = e^{tX}$ satisfies

$$Y'(t) = Y(t)X, \quad Y(0) = I.$$

The same initial-value problem is satisfied by γ . Regard a matrix as a vector in $\mathbb{R}^{n^2}$. The equation $Y' = YX$ is a linear ODE with globally Lipschitz right-hand side, so its solution with prescribed initial value is unique. Therefore $\gamma(t) = e^{tX}$ for all t .

The generator is unique because differentiating at 0 recovers it:

$$X = \gamma'(0).$$

□

Proposition 1.13.2. *For every real matrix X ,*

$$\det(e^X) = e^{\operatorname{tr} X}.$$

PROOF. A useful proof avoids choosing eigenvectors. Define

$$f(t) = \det(e^{tX}).$$

By theorem 1.8.5 and theorem 1.12.1,

$$\begin{aligned} f'(t) &= \det(e^{tX}) \operatorname{tr} \left(e^{-tX} \frac{d}{dt} e^{tX} \right) \\ &= f(t) \operatorname{tr} \left(e^{-tX} e^{tX} X \right) \\ &= f(t) \operatorname{tr} X. \end{aligned}$$

Thus f solves the scalar ODE

$$f'(t) = (\operatorname{tr} X)f(t), \quad f(0) = 1.$$

Hence

$$f(t) = e^{t \operatorname{tr} X}.$$

Setting $t = 1$ gives the claim. □

Consequently,

$$e^{tX} \in \operatorname{SL}_n(\mathbb{R}) \quad \text{for all } t$$

if and only if $\operatorname{tr} X = 0$. The tangent space itself can also be read directly from the regular level-set description: by theorem 1.8.5,

$$T_I \operatorname{SL}_n(\mathbb{R}) = \ker D(\det)_I = \{X \in M_n(\mathbb{R}) : \operatorname{tr} X = 0\}.$$

We denote this vector space by

$$\mathfrak{sl}_n(\mathbb{R}).$$

The exponential calculation above shows, in addition, that every such tangent vector generates a one-parameter subgroup lying entirely in $\mathrm{SL}_n(\mathbb{R})$. For $n = 2$, every element has the form

$$\begin{pmatrix} a & b \\ c & -a \end{pmatrix}.$$

14. Lie group homomorphisms and functoriality of the exponential

Definition 1.14.1. A Lie group homomorphism $\phi : G \rightarrow H$ is a smooth group homomorphism.

Proposition 1.14.2. For a Lie group homomorphism $\phi : G \rightarrow H$ and $X \in T_e G$,

$$\phi(\exp_G(tX)) = \exp_H(t d\phi_e X).$$

In particular,

$$\phi(\exp_G X) = \exp_H(d\phi_e X).$$

Guiding Idea 1.14.3. The exponential is characterized by one-parameter subgroups. A homomorphism carries each such subgroup to another one. Differentiation gives its new initial velocity, and uniqueness forces the formula.

PROOF. The curve

$$\eta(t) = \phi(\exp_G(tX))$$

is a homomorphism $\mathbb{R} \rightarrow H$ because both $t \mapsto \exp_G(tX)$ and ϕ are homomorphisms. Its derivative at 0 is

$$\eta'(0) = d\phi_e(X).$$

By uniqueness of the one-parameter subgroup generated by a tangent vector,

$$\eta(t) = \exp_H(t d\phi_e X).$$

□

Worked Example 1.14.4 (The determinant as an infinitesimal trace). The determinant

$$\det : \mathrm{GL}_n(\mathbb{R}) \rightarrow \mathbb{R}^\times$$

is a Lie group homomorphism. Its derivative at I is

$$d(\det)_I(X) = \mathrm{tr} X.$$

Applying theorem 1.14.2 gives again

$$\det(e^X) = e^{\mathrm{tr} X},$$

where the exponential on the right is the ordinary exponential in the multiplicative group $\mathbb{R}_{>0}$.

15. The Lie product formula

The closed subgroup theorem requires one important approximation principle: adding infinitesimal directions corresponds, to first order, to multiplying small group elements.

THEOREM 1.15.1 (Lie product formula). *For $X, Y \in \mathfrak{g}$ and fixed $t \in \mathbb{R}$,*

$$\exp(t(X + Y)) = \lim_{n \rightarrow \infty} \left(\exp\left(\frac{tX}{n}\right) \exp\left(\frac{tY}{n}\right) \right)^n.$$

The convergence is uniform for t in compact intervals.

Guiding Idea 1.15.2. For a small step size s , the product

$$\exp(sX) \exp(sY)$$

has logarithm

$$s(X + Y) + O(s^2).$$

The useful observation is that the same small group element is repeated n times. Once we take its local logarithm, there is no need to estimate an error after each multiplication separately: if

$$\exp(sX) \exp(sY) = \exp Z_s,$$

then its n th power is exactly $\exp(nZ_s)$. The second-order error in Z_s is therefore multiplied only by n , giving an error of size $O(1/n)$ when $s = t/n$.

PROOF. Choose exponential coordinates near e : there are neighborhoods $U \subset \mathfrak{g}$ and $V \subset G$ for which $\exp : U \rightarrow V$ is a diffeomorphism. Let $\log : V \rightarrow U$ be the inverse.

For s sufficiently close to 0, define

$$F(s) = \log(\exp(sX) \exp(sY)).$$

Then $F(0) = 0$. We compute its derivative. The curve

$$c(s) = \exp(sX) \exp(sY)$$

has derivative $X + Y$ at $s = 0$: the differential of multiplication at (e, e) sends (X, Y) to $X + Y$. Moreover,

$$(d\log)_e = ((d\exp)_0)^{-1} = I.$$

Hence

$$F'(0) = X + Y.$$

Fix any norm $\|\cdot\|$ on the finite-dimensional vector space $\mathfrak{g}$. Because F is smooth, Taylor's theorem gives

$$F(s) = s(X + Y) + R(s), \quad \|R(s)\| \leq Cs^2$$

for small s . If t is restricted to a fixed compact interval, the same constant can be chosen for all $s = t/n$ once n is sufficiently large.

Set $s = t/n$ and write

$$Z_{n,t} = F(t/n).$$

By definition of F ,

$$\exp(tX/n) \exp(tY/n) = \exp Z_{n,t}.$$

Therefore, using the one-parameter subgroup law for the single vector $Z_{n,t}$,

$$(\exp(tX/n) \exp(tY/n))^n = (\exp Z_{n,t})^n = \exp(nZ_{n,t}).$$

But

$$nZ_{n,t} = t(X + Y) + nR(t/n),$$

and

$$\|nR(t/n)\| \leq C \frac{t^2}{n} \longrightarrow 0.$$

Thus

$$nZ_{n,t} \longrightarrow t(X + Y)$$

uniformly for t in compact intervals. The vectors $t(X + Y)$, with t in a fixed compact interval, form a compact subset of $\mathfrak{g}$. For large n , the vectors $nZ_{n,t}$ remain in a fixed compact neighborhood of that set. The exponential map is uniformly continuous on this compact neighborhood. Therefore

$$(\exp(tX/n) \exp(tY/n))^n \longrightarrow \exp(t(X + Y))$$

uniformly for t in the chosen compact interval. □

Remark 1.15.3. The proof is cleaner than a step-by-step error estimate because the repeated factor is identical at every step. Taking a logarithm converts the n fold product into multiplication of one tangent vector by n .

16. Closed subgroups of Lie groups

A subgroup of a Lie group can be topologically very irregular when it is not closed. The dense cyclic subgroup of S^1 is the simplest example. The remarkable fact is that closedness alone rules out such irregularity.

THEOREM 1.16.1 (Closed subgroup theorem). *Let G be a Lie group and $H < G$ a closed subgroup. Then H has a unique smooth manifold structure for which it is an embedded Lie subgroup of G .*

16.1. Proof architecture. Before proving the theorem, it is useful to understand the strategy.

A closed subgroup H may not initially come with any manifold structure. So we cannot start by taking its tangent space in the ordinary manifold sense. Instead we *define* the would-be tangent directions by

$$\mathfrak{h} = \{X \in \mathfrak{g} : \exp(tX) \in H \text{ for every } t \in \mathbb{R}\}.$$

The proof has three conceptual steps.

- (1) Show that $\mathfrak{h}$ is a vector subspace. Scalar multiplication is immediate. Addition is nontrivial and uses the Lie product formula together with closedness of H .
- (2) Show that any sequence of small elements of H determines, after rescaling, a direction in $\mathfrak{h}$. This is the “direction lemma.” Closedness is used again to pass from approximate small motions to an entire one-parameter subgroup.
- (3) Choose a complement $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$. Exponential coordinates then look locally like $(X, Y) \mapsto \exp X \exp Y$. The direction lemma shows that H cannot accumulate in a nonzero transverse $\mathfrak{m}$ -direction. Therefore locally H is exactly the slice $Y = 0$, hence a submanifold.

The important idea is that *closedness converts infinitesimal accumulation into actual one-parameter symmetry*.

Lemma 1.16.2. $\mathfrak{h}$ is a vector subspace of $\mathfrak{g}$.

PROOF. If $X \in \mathfrak{h}$ and $a \in \mathbb{R}$, then for every t ,

$$\exp(t(aX)) = \exp((ta)X) \in H,$$

so $aX \in \mathfrak{h}$.

Now let $X, Y \in \mathfrak{h}$. By the Lie product formula,

$$\exp(t(X + Y)) = \lim_{n \rightarrow \infty} \left(\exp\left(\frac{tX}{n}\right) \exp\left(\frac{tY}{n}\right) \right)^n.$$

Every factor in every approximating product belongs to H , hence the entire approximating product lies in H . Since H is closed, the limit lies in H . Thus

$$\exp(t(X + Y)) \in H$$

for every t , proving $X + Y \in \mathfrak{h}$. □

Lemma 1.16.3 (Direction lemma). *Fix any norm on the finite-dimensional vector space $\mathfrak{g}$. Suppose $Z_j \in \mathfrak{g} \setminus \{0\}$ satisfy*

$$Z_j \rightarrow 0, \quad \exp Z_j \in H,$$

and

$$\frac{Z_j}{\|Z_j\|} \rightarrow Z.$$

Then $Z \in \mathfrak{h}$.

Guiding Idea 1.16.4. The element $\exp Z_j$ is a very small step in H . If the normalized direction tends to Z , then taking roughly $t/\|Z_j\|$ repeated steps should approximate motion by distance t in direction Z . Because every repeated step remains in H and H is closed, the limiting motion $\exp(tZ)$ also lies in H .

PROOF. Fix $t \geq 0$. Choose integers

$$m_j = \left\lfloor \frac{t}{\|Z_j\|} \right\rfloor.$$

Then

$$0 \leq t - m_j \|Z_j\| < \|Z_j\|,$$

so

$$m_j \|Z_j\| \rightarrow t.$$

Therefore

$$m_j Z_j = m_j \|Z_j\| \frac{Z_j}{\|Z_j\|} \rightarrow tZ.$$

Since $\exp Z_j \in H$ and H is a subgroup,

$$(\exp Z_j)^{m_j} \in H.$$

All powers lie on the same one-parameter subgroup, so

$$(\exp Z_j)^{m_j} = \exp(m_j Z_j).$$

Continuity of $\exp$ gives

$$\exp(m_j Z_j) \rightarrow \exp(tZ).$$

Closedness of H implies $\exp(tZ) \in H$.

For $t < 0$, apply the already proved case to $-t$ and use inversion:

$$\exp(tZ) = \exp((-t)Z)^{-1} \in H.$$

Thus $\exp(tZ) \in H$ for all real t , so $Z \in \mathfrak{h}$. □

Choose a vector-space complement

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}.$$

Define, near $(0, 0)$,

$$\Phi : \mathfrak{h} \times \mathfrak{m} \rightarrow G, \quad \Phi(X, Y) = \exp X \exp Y.$$

Lemma 1.16.5. *The differential at $(0, 0)$ is*

$$d\Phi_{(0,0)}(X, Y) = X + Y.$$

Hence Φ is a diffeomorphism from a neighborhood of $(0, 0)$ onto a neighborhood of e .

PROOF. Consider a tangent vector (X, Y) and the curve $t \mapsto (tX, tY)$. Then

$$\Phi(tX, tY) = \exp(tX) \exp(tY).$$

At $t = 0$, the derivative of the first factor is X and of the second is Y . The differential of multiplication at (e, e) adds these tangent vectors, so

$$d\Phi_{(0,0)}(X, Y) = X + Y.$$

Because $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$, the map $(X, Y) \mapsto X + Y$ is a linear isomorphism. The inverse function theorem now gives the local diffeomorphism statement. $\square$

Lemma 1.16.6 (No transverse accumulation). *After shrinking the local product chart, there are neighborhoods $U_{\mathfrak{h}} \subset \mathfrak{h}$, $U_{\mathfrak{m}} \subset \mathfrak{m}$, and $W \subset G$ such that*

$$\Phi : U_{\mathfrak{h}} \times U_{\mathfrak{m}} \rightarrow W$$

is a diffeomorphism and

$$H \cap W = \{\exp X : X \in U_{\mathfrak{h}}\}.$$

PROOF. If $X \in U_{\mathfrak{h}}$, then $X \in \mathfrak{h}$, so the one-parameter subgroup $t \mapsto \exp(tX)$ lies in H and in particular $\exp X \in H$. The chart was chosen with $\exp(U_{\mathfrak{h}}) \subset W$. Therefore

$$\{\exp X : X \in U_{\mathfrak{h}}\} \subset H \cap W.$$

Suppose the reverse inclusion fails in every sufficiently small such chart. Shrink the product neighborhoods to balls of radius $1/j$ around $(0, 0)$ and choose, for each j , an element of H in the corresponding image whose transverse coordinate is nonzero. In this way we obtain elements $g_j \in H$ with $g_j \rightarrow e$ and unique decompositions

$$g_j = \exp X_j \exp Y_j,$$

where $X_j \in \mathfrak{h}$, $Y_j \in \mathfrak{m}$, and $Y_j \neq 0$.

Because $\exp X_j \in H$ and $g_j \in H$,

$$\exp Y_j = (\exp X_j)^{-1} g_j \in H.$$

The local coordinates tend to $(0, 0)$ as $g_j \rightarrow e$, so $Y_j \rightarrow 0$. Normalize:

$$\hat{Y}_j = \frac{Y_j}{\|Y_j\|}.$$

The unit sphere in the finite-dimensional vector space $\mathfrak{m}$ is compact. Passing to a subsequence,

$$\widehat{Y}_j \rightarrow Y$$

for some $Y \in \mathfrak{m}$ with $\|Y\| = 1$.

Now apply the direction lemma to Y_j . Since $\exp Y_j \in H$, it gives $Y \in \mathfrak{h}$. Hence

$$Y \in \mathfrak{h} \cap \mathfrak{m} = \{0\},$$

contradicting $\|Y\| = 1$. Therefore no nonzero transverse coordinate can occur sufficiently near e . $\square$

PROOF OF THEOREM 1.16.1. By theorem 1.16.6, near the identity the subgroup H is exactly

$$\exp(U_{\mathfrak{h}}),$$

which is an embedded submanifold of dimension $\dim \mathfrak{h}$.

For any $h \in H$, left translation carries this local piece to

$$h(H \cap W) = H \cap hW.$$

Since L_h is a diffeomorphism of G , each $H \cap hW$ is an embedded submanifold. These sets cover H , so they define a smooth manifold structure on H compatible with its inclusion in G .

Multiplication and inversion on H are restrictions of the corresponding smooth maps on G . To be explicit about the codomain: in the local product coordinates from theorem 1.16.5, the embedded submanifold H is the slice whose $\mathfrak{m}$ -coordinate is zero. A smooth G -valued map whose image lies in that slice has zero transverse coordinate and is therefore smooth as an H -valued map. Hence the restricted multiplication and inversion are smooth. Thus H is a Lie group and the inclusion $H \hookrightarrow G$ is a smooth embedding.

It remains to justify uniqueness. Suppose the same subset $H \subset G$ is given another Lie-group structure for which the inclusion

$$i : H \hookrightarrow G$$

is a smooth embedding. Let $\mathfrak{h}'$ be its tangent space at the identity, viewed inside $\mathfrak{g}$ through di_e . If $Y \in \mathfrak{h}'$, the one-parameter subgroup $\exp_H(tY)$ maps under i to a one-parameter subgroup of G , so theorem 1.14.2 gives

$$i(\exp_H(tY)) = \exp_G(t di_e Y).$$

Hence $di_e Y \in \mathfrak{h}$.

Conversely, if $X \in \mathfrak{h}$, then the curve $t \mapsto \exp_G(tX)$ has image in the subset H . Because i is an embedding, a map into H is smooth precisely when its composition with i is smooth in local embedded-submanifold coordinates. Thus this curve is smooth as an H -valued curve and has tangent vector whose image under di_e is X . Therefore $di_e(\mathfrak{h}') = \mathfrak{h}$.

The exponential maps of the two Lie-group structures consequently agree, through the inclusion, on a neighborhood of the identity. Both give the same local slice $\exp_G(U_{\mathfrak{h}}) \subset H$. Left translations by elements of H then transport this identical local smooth structure to every point of H . Hence the embedded Lie-group structure is unique. $\square$

Remark 1.16.7 (Where closedness is used). Closedness enters twice, and both uses are essential. First, it lets the limit in the Lie product formula remain in H , making $\mathfrak{h}$ closed under addition. Second, it lets repeated small elements converge to a full one-parameter subgroup in the direction lemma. Without closedness, a dense cyclic subgroup of a torus is the counterexample to both geometric conclusions.

17. Research-facing examples of closed subgroups

The closed subgroup theorem is powerful because many groups arising from equations are visibly closed before one knows anything about their manifold structure.

Worked Example 1.17.1 (Stabilizer of a quadratic form). Let $Q(x) = x^T A x$ be a nondegenerate quadratic form on $\mathbb{R}^d$, with $A = A^T$ invertible. Its special orthogonal group is

$$\mathrm{SO}(Q) = \{g \in \mathrm{SL}_d(\mathbb{R}) : g^T A g = A\}.$$

First note that this set is a subgroup. If $g^T A g = A$ and $h^T A h = A$, then

$$(gh)^T A (gh) = h^T (g^T A g) h = h^T A h = A.$$

Also $g^T A g = A$ is equivalent, after multiplying by g^{-T} on the left and g^{-1} on the right, to

$$(g^{-1})^T A g^{-1} = A.$$

Thus products and inverses stay in the set. The map

$$g \mapsto g^T A g$$

is continuous, so $\mathrm{SO}(Q)$ is closed in $\mathrm{SL}_d(\mathbb{R})$. Therefore the closed subgroup theorem makes it a Lie subgroup.

To compute its tangent space at the identity, take a smooth curve $g(t) \in \mathrm{SO}(Q)$ with $g(0) = I$ and $g'(0) = X$. Differentiate

$$g(t)^T A g(t) = A.$$

At $t = 0$ this gives

$$X^T A + A X = 0.$$

This proves that every tangent vector satisfies the displayed linear equation. For the converse, suppose X satisfies $X^T A + AX = 0$. Since $(e^{tX})^T = e^{tX^T}$, we have

$$\frac{d}{dt} (e^{tX^T} A e^{tX}) = e^{tX^T} (X^T A + AX) e^{tX} = 0.$$

The expression equals A at $t = 0$, so

$$e^{tX^T} A e^{tX} = A$$

for all t . Hence e^{tX} preserves Q . Taking traces after multiplying $X^T A + AX = 0$ on the left by A^{-1} gives

$$2 \operatorname{tr} X = 0,$$

so $\det(e^{tX}) = e^{t \operatorname{tr} X} = 1$. Therefore $e^{tX} \in \operatorname{SO}(Q)$, and X is indeed a tangent direction. We have proved the equality

$$T_I \operatorname{SO}(Q) = \{X \in \mathfrak{sl}_d(\mathbb{R}) : X^T A + AX = 0\}.$$

This is a typical research pattern: define a stabilizer by an algebraic equation, invoke closedness to obtain a Lie subgroup, then differentiate the equation and verify that every solution of the linearized equation exponentiates back into the stabilizer.

Worked Example 1.17.2 (The affine special linear group). The group of volume-preserving affine transformations of $\mathbb{R}^d$ is

$$\operatorname{ASL}_d(\mathbb{R}) = \operatorname{SL}_d(\mathbb{R}) \ltimes \mathbb{R}^d,$$

with multiplication

$$(A, x)(B, y) = (AB, x + Ay).$$

It can be embedded as a closed matrix subgroup of $\operatorname{SL}_{d+1}(\mathbb{R})$ via

$$(A, x) \mapsto \begin{pmatrix} A & x \\ 0 & 1 \end{pmatrix}.$$

Indeed, the image is characterized by the closed conditions that the last row is $(0, \dots, 0, 1)$.

Define

$$\operatorname{ASL}_d(\mathbb{Z}) = \operatorname{SL}_d(\mathbb{Z}) \ltimes \mathbb{Z}^d.$$

The quotient

$$\operatorname{ASL}_d(\mathbb{R}) / \operatorname{ASL}_d(\mathbb{Z})$$

parametrizes unimodular affine lattices $g\mathbb{Z}^d + x$ as subsets of $\mathbb{R}^d$: changing g by $g\gamma$ with $\gamma \in \operatorname{SL}_d(\mathbb{Z})$ leaves the underlying lattice unchanged, and changing x by an element of $g\mathbb{Z}^d$ leaves the translated lattice unchanged. Conversely, if

$$g\mathbb{Z}^d + x = g'\mathbb{Z}^d + x'$$

as subsets, taking pairwise differences of points on the two sides gives $g\mathbb{Z}^d = g'\mathbb{Z}^d$. Hence $g' = g\gamma$ for some $\gamma \in \mathrm{SL}_d(\mathbb{Z})$, and equality of the translates then gives $x' - x \in g\mathbb{Z}^d$. Thus the two affine transformations differ by right multiplication by an element of $\mathrm{ASL}_d(\mathbb{Z})$. Such spaces occur when an arithmetic problem involves a lattice together with a marked translation, rather than a lattice alone.

Worked Example 1.17.3 (Upper unitriangular groups). Let

$$\mathrm{UT}_d(\mathbb{R}) = \{g \in \mathrm{GL}_d(\mathbb{R}) : g_{ii} = 1, g_{ij} = 0 \text{ for } i > j\}.$$

The defining coordinate equations are closed, and matrix multiplication preserves them. Inverses do as well: if $g = I + N$ with N strictly upper triangular, then $N^d = 0$ and

$$g^{-1} = I - N + N^2 - \cdots + (-1)^{d-1}N^{d-1},$$

which is again upper unitriangular. Thus $\mathrm{UT}_d(\mathbb{R})$ is a closed subgroup and hence, by the closed subgroup theorem, a Lie subgroup. Every $g \in \mathrm{UT}_d(\mathbb{R})$ is unipotent because $g - I = N$ is nilpotent.

For $d = 3$ this is precisely the Heisenberg group in matrix form. For larger d it provides a hierarchy of nilpotent groups used throughout modern homogeneous dynamics.

18. Discrete subgroups

Definition 1.18.1. A subgroup $\Gamma < G$ is *discrete* if the subspace topology on Γ is discrete. Equivalently, there is a neighborhood U of e with

$$U \cap \Gamma = \{e\}.$$

Indeed, if such a U exists, then $\gamma U \cap \Gamma = \{\gamma\}$ for every $\gamma \in \Gamma$; conversely, if the subspace topology is discrete, the singleton $\{e\}$ is open in Γ and therefore equals $U \cap \Gamma$ for some open $U \subset G$.

Proposition 1.18.2. *Every discrete subgroup of a Lie group is closed.*

PROOF. Choose a neighborhood U of e with $U \cap \Gamma = \{e\}$. By theorem 1.3.3, choose a symmetric neighborhood V such that

$$V^2 \subset U.$$

Then

$$V^2 \cap \Gamma = \{e\}.$$

Suppose $\gamma_n \in \Gamma$ converges to some $g \in G$. Since $\gamma_n \rightarrow g$, for sufficiently large m, n both $g^{-1}\gamma_m$ and $g^{-1}\gamma_n$ lie in V . Hence

$$\gamma_m^{-1}\gamma_n = (g^{-1}\gamma_m)^{-1}(g^{-1}\gamma_n) \in V^{-1}V = V^2.$$

But $\gamma_m^{-1}\gamma_n \in \Gamma$, so it must equal e . Thus γ_n is eventually constant. Its limit g is therefore an element of Γ .

Lie groups are metrizable, so sequential closedness implies closedness. $\square$

Worked Example 1.18.3 ($\mathrm{SL}_d(\mathbb{Z})$). $\mathrm{SL}_d(\mathbb{Z})$ is discrete in $\mathrm{SL}_d(\mathbb{R})$. If two integer matrices are within distance less than 1 in the maximum-entry norm, then each pair of corresponding integer entries is equal. Thus every matrix in $\mathrm{SL}_d(\mathbb{Z})$ has a neighborhood containing no other integer matrix.

Worked Example 1.18.4 (Integer Heisenberg subgroup). Inside $H_3(\mathbb{R})$, let

$$H_3(\mathbb{Z}) = \{(x, y, z) : x, y, z \in \mathbb{Z}\}.$$

The multiplication law

$$(x, y, z)(x', y', z') = (x + x', y + y', z + z' + xy')$$

preserves integrality, so this is a subgroup. It is discrete because integer coordinates are separated in $\mathbb{R}^3$. The quotient $H_3(\mathbb{R})/H_3(\mathbb{Z})$ is compact. To see this directly, start from $(x, y, z) \in H_3(\mathbb{R})$. Right multiplication by an integer element (m, n, p) gives

$$(x, y, z)(m, n, p) = (x + m, y + n, z + p + xn).$$

Choose integers m, n so that $x + m$ and $y + n$ lie in $[0, 1)$, and then choose p so that $z + p + xn$ lies in $[0, 1)$. Thus every right coset has a representative in the compact cube $[0, 1]^3$. The quotient map sends this cube onto the whole quotient, so the quotient is compact. Such a compact quotient of a nilpotent Lie group by a discrete subgroup is called a *nilmanifold*. Here the underlying group law is genuinely nonabelian, in contrast with a torus.

19. Connected Lie groups and exponentials

Proposition 1.19.1. *If G is connected, then*

$$G = \langle \exp(\mathfrak{g}) \rangle.$$

Equivalently, every element of G is a finite product of exponentials.

PROOF. By theorem 1.11.4, there is a neighborhood U of 0 such that $V = \exp(U)$ is a neighborhood of e . By theorem 1.5.3, V generates the connected group G . Since $V \subset \exp(\mathfrak{g})$,

$$G = \langle V \rangle \subset \langle \exp(\mathfrak{g}) \rangle \subset G.$$

$\square$

Warning 1.19.2. The conclusion does not say $G = \exp(\mathfrak{g})$. A finite product of exponentials need not collapse to one exponential in a nonabelian group.

19.1. Connected components of $\mathrm{GL}_n(\mathbb{R})$. The determinant shows that $\mathrm{GL}_n(\mathbb{R})$ has at least two connected components. We now justify that it has exactly two.

Proposition 1.19.3. *The sets*

$$\mathrm{GL}_n^+(\mathbb{R}) = \{A : \det A > 0\}, \quad \mathrm{GL}_n^-(\mathbb{R}) = \{A : \det A < 0\}$$

are the two connected components of $\mathrm{GL}_n(\mathbb{R})$.

Guiding Idea 1.19.4. The sign of the determinant cannot change along a continuous path without passing through determinant zero. The only real work is to show that every positive-determinant matrix can be connected to I . A useful decomposition separates “shape” from “rotation”: $A = QP$, where P is positive definite and Q is orthogonal. The positive definite part can be contracted linearly to I , and an orientation-preserving orthogonal matrix is a product of planar rotations, each of which can be rotated continuously to I .

PROOF. Continuity of the determinant implies the two sets are disjoint open subsets whose union is $\mathrm{GL}_n(\mathbb{R})$, so no connected set can meet both.

Take $A \in \mathrm{GL}_n^+(\mathbb{R})$. Define

$$P = (A^T A)^{1/2}, \quad Q = AP^{-1}.$$

The matrix $A^T A$ is symmetric positive definite, so it has an orthonormal eigenbasis with positive eigenvalues. Taking the positive square roots of those eigenvalues defines the unique positive definite square root P . Then

$$Q^T Q = P^{-1} A^T A P^{-1} = I,$$

so $Q \in O(n)$. Moreover

$$\det Q = \frac{\det A}{\det P} > 0,$$

thus $Q \in SO(n)$.

The path

$$P_s = (1 - s)P + sI, \quad 0 \leq s \leq 1,$$

remains positive definite because for every nonzero v ,

$$v^T P_s v = (1 - s)v^T P v + s\|v\|^2 > 0.$$

Thus P is connected to I through invertible positive-determinant matrices.

Because an orthogonal matrix is normal, $\mathbb{R}^n$ admits an orthogonal decomposition into Q -invariant subspaces of dimension one or two. On a two-dimensional invariant subspace, Q acts as a planar rotation. On a one-dimensional invariant subspace it acts by $+1$ or -1 . Since $\det Q = 1$, the number of -1 blocks is even; pair them, regarding each pair as a two-dimensional rotation through angle π . Thus, in a suitable orthonormal basis,

Q is block diagonal with planar rotation blocks and 1×1 blocks equal to 1. Replacing every rotation angle θ by $(1-s)\theta$ produces a continuous path in $SO(n)$ from Q to I . Combining the path from $A = QP$ to Q through QP_s with the path from Q to I shows that $\mathrm{GL}_n^+(\mathbb{R})$ is path connected.

Finally, multiplication by the fixed reflection

$$R = \mathrm{diag}(-1, 1, \dots, 1)$$

is a homeomorphism from $\mathrm{GL}_n^+(\mathbb{R})$ onto $\mathrm{GL}_n^-(\mathbb{R})$. Hence the negative component is also path connected. Therefore these are exactly the two connected components. $\square$

20. Linear flows on tori

Let $v \in \mathbb{R}^d$. The map

$$\gamma_v(t) = tv + \mathbb{Z}^d$$

is a one-parameter subgroup of $\mathbb{T}^d$.

The orbit is periodic exactly when there exists $T > 0$ such that

$$Tv \in \mathbb{Z}^d.$$

Indeed, this condition is precisely the assertion that $\gamma_v(T) = \gamma_v(0)$. Equivalently, v lies on a one-dimensional real subspace containing a nonzero integer vector. Nonperiodic behavior can be much larger: in dimension two, an irrational slope already gives a dense orbit, as the next example shows.

Worked Example 1.20.1 (A dense line in $\mathbb{T}^2$). Let $v = (1, \alpha)$ with $\alpha \notin \mathbb{Q}$. We claim

$$\left\{ (t, \alpha t) \bmod \mathbb{Z}^2 : t \in \mathbb{R} \right\}$$

is dense in $\mathbb{T}^2$.

Fix a target point of $\mathbb{T}^2$ and choose real representatives $x_0, y_0 \in [0, 1)$ for its two coordinates. Let $\varepsilon > 0$. It is enough to find t whose first coordinate is x_0 modulo 1 and whose second coordinate has circle distance less than ε from y_0 . Put

$$t = x_0 + n,$$

where $n \in \mathbb{Z}$. Then the first coordinate is exactly x_0 modulo 1, and the second is

$$\alpha x_0 + n\alpha \pmod{1}.$$

By density of $\{n\alpha \pmod{1} : n \in \mathbb{Z}\}$ in $\mathbb{T}$, choose n so that

$$\|\alpha x_0 + n\alpha - y_0\|_{\mathbb{R}/\mathbb{Z}} < \varepsilon.$$

Hence every neighborhood of every point meets the orbit.

Worked Example 1.20.2 (A proper dense connected subgroup). The preceding orbit is the image of the continuous homomorphism

$$\mathbb{R} \rightarrow \mathbb{T}^2, \quad t \mapsto (t, \alpha t).$$

Its kernel is trivial: if both t and αt are integers, irrationality of α forces $t = 0$. Its derivative is the nonzero vector $(1, \alpha)$, so the homomorphism is an injective immersion. Its image is connected because $\mathbb{R}$ is connected, and it is dense by the previous example. It is also proper: a point of the image whose first coordinate is 0 must come from an integer time $t = n$, so the intersection of the image with the vertical circle $\{0\} \times \mathbb{T}$ is only the countable set $\{(0, n\alpha) : n \in \mathbb{Z}\}$. It therefore cannot contain the whole vertical circle, and hence cannot equal $\mathbb{T}^2$. Hence the image of an injective Lie-group immersion can be connected and dense without being an embedded closed subgroup.

This is precisely why the adjective “closed” is indispensable when one wants a quotient or an embedded homogeneous subspace with good topology.

21. Compactness, noncompactness, and geometric behavior

Proposition 1.21.1. $O(n)$ is compact.

PROOF. If $A \in O(n)$, its columns are unit vectors, so every entry satisfies $|A_{ij}| \leq 1$. Hence $O(n)$ is bounded in $M_n(\mathbb{R}) \simeq \mathbb{R}^{n^2}$. It is closed because it is the inverse image of I under the continuous map $A \mapsto A^T A$. By Heine–Borel, $O(n)$ is compact. $\square$

Proposition 1.21.2. $\mathrm{SL}_n(\mathbb{R})$ is noncompact for every $n \geq 2$.

PROOF. The matrices

$$b_t = \mathrm{diag}(e^t, e^{-t}, 1, \dots, 1)$$

all lie in $\mathrm{SL}_n(\mathbb{R})$, but their operator norms tend to infinity as $t \rightarrow \infty$. Thus $\mathrm{SL}_n(\mathbb{R})$ is unbounded and hence noncompact. $\square$

The difference between compact and noncompact ambient groups has major dynamical consequences. In a compact manifold (hence a compact metrizable space), every sequence has a convergent subsequence. In a noncompact homogeneous space, a trajectory may drift toward regions represented by matrices with increasingly extreme singular values. For spaces of lattices, this corresponds to lattices developing very short vectors in some directions while compensating with long directions in others.

22. Products and semidirect products

If G and H are Lie groups, then $G \times H$ is a Lie group under componentwise multiplication. Its tangent space at the identity is

$$T_{(e,e)}(G \times H) \simeq T_e G \oplus T_e H.$$

A more interesting construction is the semidirect product.

Definition 1.22.1 (Semidirect product). Suppose H acts smoothly on a Lie group N by automorphisms,

$$\alpha : H \rightarrow \text{Aut}(N),$$

meaning that each $\alpha(h)$ is a group automorphism of N and the evaluation map

$$H \times N \rightarrow N, \quad (h, n) \mapsto \alpha(h)(n),$$

is smooth. Then $N \rtimes_\alpha H$ is the manifold $N \times H$ with multiplication

$$(n_1, h_1)(n_2, h_2) = (n_1 \alpha(h_1)(n_2), h_1 h_2).$$

Its inverse is

$$(n, h)^{-1} = (\alpha(h^{-1})(n^{-1}), h^{-1}).$$

Because the group operations in N and H and the action α are smooth, these formulas make $N \rtimes_\alpha H$ a Lie group.

Worked Example 1.22.2 (The affine group of the line). The transformations $x \mapsto ax + b$ with $a > 0$ form a group represented by

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}.$$

Multiplication is

$$(b, a)(b', a') = (b + ab', aa'),$$

so the group is $\mathbb{R} \rtimes \mathbb{R}_{>0}$, where a acts on translations by $b' \mapsto ab'$.

Conjugation reveals the interaction:

$$\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a^{-1} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & ab \\ 0 & 1 \end{pmatrix}.$$

Thus one subgroup expands or contracts another. This simple mechanism is a prototype for many diagonal–unipotent interactions.

23. Centers, centralizers, and normalizers

Definition 1.23.1. The center of G is

$$Z(G) = \{z \in G : zg = gz \text{ for every } g \in G\}.$$

The centralizer of a subset $S \subset G$ is

$$C_G(S) = \{g \in G : gs = sg \text{ for every } s \in S\}.$$

The normalizer of a subgroup $H < G$ is

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

Proposition 1.23.2. *Let G be a Lie group. Then $Z(G)$ and $C_G(S)$ are closed for every subset $S \subset G$. If $H < G$ is closed, then $N_G(H)$ is closed.*

PROOF. For fixed $s \in G$, the map

$$f_s(g) = gsg^{-1}s^{-1}$$

is continuous. Hence

$$C_G(s) = f_s^{-1}(\{e\})$$

is closed. Then

$$C_G(S) = \bigcap_{s \in S} C_G(s)$$

is closed, and $Z(G) = C_G(G)$ is closed.

For the normalizer, first consider

$$A = \{g \in G : gHg^{-1} \subset H\}.$$

For a fixed $h \in H$, the condition $ghg^{-1} \in H$ is closed in g , because the map $g \mapsto ghg^{-1}$ is continuous and H is closed. Therefore

$$A = \bigcap_{h \in H} \{g \in G : ghg^{-1} \in H\}$$

is closed. For the reverse conjugation condition,

$$B = \{g \in G : g^{-1}Hg \subset H\} = \bigcap_{h \in H} \{g \in G : g^{-1}hg \in H\}.$$

For each fixed h , the map $g \mapsto g^{-1}hg$ is continuous, so the corresponding inverse image of the closed set H is closed. Thus B is closed. But

$$N_G(H) = A \cap B,$$

because the two inclusions are equivalent to $gHg^{-1} = H$. Thus $N_G(H)$ is closed. $\square$

Worked Example 1.23.3 (Center of $\mathrm{SL}_2(\mathbb{R})$). Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

commute with every element of $\mathrm{SL}_2(\mathbb{R})$. Commuting with

$$\begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}$$

for some $t \neq 0$ forces $b = c = 0$. Thus A is diagonal. Commuting with

$$\begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$$

forces its diagonal entries to agree, so $A = aI$. Since $\det A = a^2 = 1$,

$$Z(\mathrm{SL}_2(\mathbb{R})) = \{\pm I\}.$$

Warning 1.23.4. There is no general theorem that $N_G(H)/H$ is finite. For example, if $H = \{e\}$ then $N_G(H) = G$. Any finiteness statement about a normalizer quotient needs additional structural hypotheses.

24. Conjugation and the adjoint action

For $g \in G$, conjugation is the Lie group automorphism

$$C_g(h) = ghg^{-1}.$$

Since $C_g(e) = e$, differentiation gives a linear automorphism

$$\mathrm{Ad}(g) = (dC_g)_e : \mathfrak{g} \rightarrow \mathfrak{g}.$$

Proposition 1.24.1. *For $g \in G$ and $X \in \mathfrak{g}$,*

$$g \exp(X) g^{-1} = \exp(\mathrm{Ad}(g)X).$$

PROOF. Apply theorem 1.14.2 to the Lie group automorphism C_g . Its differential at the identity is $\mathrm{Ad}(g)$, so

$$C_g(\exp X) = \exp(dC_{ge}X) = \exp(\mathrm{Ad}(g)X).$$

□

The notation “adjoint action” is justified by the following identity.

Proposition 1.24.2. *The map*

$$\mathrm{Ad} : G \rightarrow \mathrm{GL}(\mathfrak{g}), \quad g \mapsto \mathrm{Ad}(g),$$

is a group homomorphism. Thus

$$\mathrm{Ad}(gh) = \mathrm{Ad}(g) \mathrm{Ad}(h), \quad \mathrm{Ad}(g^{-1}) = \mathrm{Ad}(g)^{-1}.$$

PROOF. Conjugations compose according to

$$C_{gh} = C_g \circ C_h.$$

Differentiating at the identity and using the chain rule gives

$$\text{Ad}(gh) = (dC_g)_e \circ (dC_h)_e = \text{Ad}(g) \text{Ad}(h).$$

Taking $h = g^{-1}$ gives the inverse formula. The dependence of $(dC_g)_e$ on g is smooth because conjugation depends smoothly on (g, h) , so Ad is in fact a Lie group homomorphism into $\text{GL}(\mathfrak{g})$. $\square$

For a matrix group,

$$\text{Ad}(g)X = gXg^{-1}.$$

Indeed,

$$ge^{tX}g^{-1} = e^{t(gXg^{-1})}.$$

25. Expansion and contraction under diagonal conjugation

The following calculation is one of the most useful concrete identities in homogeneous dynamics.

Let

$$a_t = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_d t}), \quad \sum_i \lambda_i = 0,$$

and let

$$u_{ij}(s) = I + sE_{ij} \quad (i \neq j).$$

Then

$$a_t E_{ij} a_t^{-1} = e^{(\lambda_i - \lambda_j)t} E_{ij},$$

and therefore

$$(25.1) \quad \boxed{a_t u_{ij}(s) a_t^{-1} = u_{ij} \left(e^{(\lambda_i - \lambda_j)t} s \right)}.$$

PROOF. Left multiplication by a_t multiplies row i by $e^{\lambda_i t}$, while right multiplication by a_t^{-1} multiplies column j by $e^{-\lambda_j t}$. Since E_{ij} has only one nonzero entry, the combined factor is $e^{(\lambda_i - \lambda_j)t}$. $\square$

The formula sorts the shear directions into three types:

- if $\lambda_i > \lambda_j$, the parameter expands exponentially as $t \rightarrow +\infty$;
- if $\lambda_i < \lambda_j$, it contracts exponentially;
- if $\lambda_i = \lambda_j$, it is unchanged.

This is a research-level example of how linear algebra in the ambient group directly organizes dynamical behavior.

Worked Example 1.25.1 (The $\mathrm{SL}_2(\mathbb{R})$ case). Take

$$a_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix}, \quad u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}.$$

Here the diagonal exponents are $\lambda_1 = 1/2$ and $\lambda_2 = -1/2$, so (25.1) gives

$$a_t u_s a_{-t} = u_{e^t s}.$$

For the opposite shear

$$v_s = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix},$$

the exponent is -1 , giving

$$a_t v_s a_{-t} = v_{e^{-t} s}.$$

Thus the same diagonal flow expands one unipotent direction and contracts the opposite one.

26. The basic subgroups of $\mathrm{SL}_2(\mathbb{R})$

Four explicit subgroups are worth keeping in view:

$$A = \left\{ \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix} : t \in \mathbb{R} \right\},$$

$$N = \left\{ \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} : s \in \mathbb{R} \right\}, \quad N^- = \left\{ \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix} : s \in \mathbb{R} \right\},$$

and

$$K = \mathrm{SO}(2) = \left\{ \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} : \theta \in \mathbb{R} \right\}.$$

Their infinitesimal generators are

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Thus

$$a_t = e^{tH/2}, \quad u_s = e^{sX}, \quad v_s = e^{sY}, \quad k_\theta = e^{\theta J}.$$

The value of these examples is not that 2×2 matrices are intrinsically special. Rather, they isolate in the smallest possible dimension three behaviors that persist in larger groups: diagonal expansion, unipotent shearing, and compact rotation.

27. More homogeneous spaces arising in research

The language developed in this chapter already allows several important parameter spaces to be described cleanly.

Worked Example 1.27.1 (Lattices with a marked point). The affine quotient from theorem 1.17.2,

$$\mathrm{ASL}_d(\mathbb{R}) / \mathrm{ASL}_d(\mathbb{Z}),$$

can be viewed as the space of pairs $(\Lambda, [v])$ where Λ is a unimodular lattice and $[v] \in \mathbb{R}^d / \Lambda$. Indeed, right multiplication by $(\gamma, m) \in \mathrm{SL}_d(\mathbb{Z}) \ltimes \mathbb{Z}^d$ changes

$$(g, v) \quad \text{to} \quad (g\gamma, v + gm).$$

The lattice $g\gamma\mathbb{Z}^d$ is still $g\mathbb{Z}^d = \Lambda$, while $v + gm$ represents the same class as v modulo Λ . Conversely, two representatives of the same pair differ in exactly this way.

This space is a natural setting for inhomogeneous Diophantine approximation, where one studies how a lattice approximates a target that is not the origin.

Worked Example 1.27.2 (Quadratic forms as an orbit). Let Q_0 be a fixed nondegenerate quadratic form with matrix A_0 . $\mathrm{SL}_d(\mathbb{R})$ acts on quadratic forms by change of variables:

$$(g \cdot Q_0)(x) = Q_0(g^{-1}x).$$

The stabilizer of Q_0 is $\mathrm{SO}(Q_0)$, so the orbit of Q_0 is in canonical set-theoretic bijection with

$$\mathrm{SL}_d(\mathbb{R}) / \mathrm{SO}(Q_0).$$

The matrix of $g \cdot Q_0$ is $g^{-T}A_0g^{-1}$. Hence its determinant is

$$\det(g)^{-2} \det(A_0) = \det(A_0),$$

and Sylvester's law of inertia shows that its signature is also unchanged. Thus every form in this orbit has the same determinant and signature as Q_0 . At this point we are asserting the orbit-coset bijection only; identifying quotient and orbit smooth structures requires an additional quotient-manifold argument, and no such stronger claim is needed here. This conversion from an algebraic object to a group quotient is one of the standard gateways by which number-theoretic problems enter homogeneous dynamics.

Worked Example 1.27.3 (Flags and parabolic stabilizers). A line $\ell \subset \mathbb{R}^d$ has a stabilizer

$$P_\ell = \{g \in \mathrm{SL}_d(\mathbb{R}) : g\ell = \ell\}.$$

It is closed. To see this without assuming any topology on projective space, choose a nonzero vector v spanning ℓ and linear functionals $\varphi_2, \dots, \varphi_d$ whose common kernel is ℓ . Then

$$g\ell = \ell \iff \varphi_j(gv) = 0 \text{ for } j = 2, \dots, d.$$

Each map $g \mapsto \varphi_j(gv)$ is continuous, so P_ℓ is an intersection of closed zero sets. Hence P_ℓ is a Lie subgroup. The coset space $\mathrm{SL}_d(\mathbb{R})/P_\ell$ is in canonical bijection with projective space, the set of lines through the origin. Stabilizers of more elaborate flags are commonly called parabolic subgroups; the present observation uses only the elementary stabilizer construction.

28. Stabilizers and homogeneous parametrizations

A recurring way to produce a homogeneous space is to start with a group action and fix one object. The subgroup preserving that object is its stabilizer.

Definition 1.28.1. If a group G acts on a set Y and $y \in Y$, the stabilizer of y is

$$\mathrm{Stab}_G(y) = \{g \in G : g \cdot y = y\}.$$

Proposition 1.28.2 (Orbit–stabilizer parametrization). *Let G act on a set Y , fix $y \in Y$, and set $H = \mathrm{Stab}_G(y)$. Then, at the level of sets,*

$$G/H \longrightarrow G \cdot y, \quad gH \longmapsto g \cdot y,$$

is a well-defined bijection.

PROOF. The main point is to understand why passing from g to the coset gH loses exactly the information that is irrelevant to $g \cdot y$.

Suppose first that $g_1H = g_2H$. Then $g_2^{-1}g_1 \in H$, and hence

$$(g_2^{-1}g_1) \cdot y = y.$$

Applying g_2 gives

$$g_1 \cdot y = g_2 \cdot y,$$

so the map is well defined.

It is surjective by the definition of the orbit $G \cdot y$. For injectivity, suppose

$$g_1 \cdot y = g_2 \cdot y.$$

Applying g_2^{-1} gives

$$(g_2^{-1}g_1) \cdot y = y,$$

so $g_2^{-1}g_1 \in H$. Hence $g_1H = g_2H$. □

The proposition is elementary but conceptually important. A quotient G/H often means: “take all possible images of one reference object under G , and identify two group elements when they produce the same object.” This interpretation is usually more informative than thinking only in terms of cosets.

Worked Example 1.28.3 (Projective space as a homogeneous space). Assume $d \geq 2$. Let $G = \mathrm{SL}_d(\mathbb{R})$ act on the set of lines in $\mathbb{R}^d$, and let

$$\ell_0 = \mathbb{R}e_1.$$

Its stabilizer is

$$P = \{g \in \mathrm{SL}_d(\mathbb{R}) : g\ell_0 = \ell_0\}.$$

The action is transitive. Indeed, given a line $\ell = \mathbb{R}v$ with $v \neq 0$, extend v to a basis $v, v_2, \dots, v_d$. Rescale one of $v_2, \dots, v_d$ so that the resulting basis matrix has determinant 1. That matrix belongs to $\mathrm{SL}_d(\mathbb{R})$ and sends ℓ_0 to ℓ . Therefore theorem 1.28.2 gives a canonical bijection

$$\mathrm{SL}_d(\mathbb{R})/P \longleftrightarrow \mathbb{P}^{d-1}(\mathbb{R}).$$

The quotient notation therefore encodes the familiar projective space of directions. No topological or smooth identification is being used in this statement.

29. The symplectic group: a modern matrix example

Another closed subgroup that appears throughout dynamics, representation theory, and Hamiltonian geometry is the real symplectic group. It is useful because the same pattern used for orthogonal groups and quadratic forms works with a skew-symmetric bilinear form.

Let

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

Define

$$\mathrm{Sp}_{2n}(\mathbb{R}) = \{g \in \mathrm{GL}_{2n}(\mathbb{R}) : g^T J g = J\}.$$

Proposition 1.29.1. $\mathrm{Sp}_{2n}(\mathbb{R})$ is a closed Lie subgroup of $\mathrm{GL}_{2n}(\mathbb{R})$. Its tangent space at the identity is

$$\mathfrak{sp}_{2n}(\mathbb{R}) = \{X : X^T J + JX = 0\}.$$

PROOF. We first verify the subgroup property. If $g^T J g = J$ and $h^T J h = J$, then

$$(gh)^T J (gh) = h^T (g^T J g) h = h^T J h = J.$$

Moreover, $g^T J g = J$ is equivalent to

$$(g^{-1})^T J g^{-1} = J,$$

so inverses also preserve J .

Now consider the continuous map

$$F : \mathrm{GL}_{2n}(\mathbb{R}) \rightarrow M_{2n}(\mathbb{R}), \quad F(g) = g^T J g.$$

Then

$$\mathrm{Sp}_{2n}(\mathbb{R}) = F^{-1}(\{J\})$$

is closed. The closed subgroup theorem therefore makes it a Lie subgroup.

To determine its tangent space, let $g(t)$ be a smooth curve in the symplectic group with

$$g(0) = I, \quad g'(0) = X.$$

Differentiate

$$g(t)^T J g(t) = J.$$

Using the product rule at $t = 0$ gives

$$X^T J + J X = 0.$$

Conversely, if X satisfies this equation, then $(e^{tX})^T = e^{tX^T}$ and

$$\frac{d}{dt}(e^{tX^T} J e^{tX}) = e^{tX^T} (X^T J + J X) e^{tX} = 0.$$

Since the expression equals J at $t = 0$, it equals J for every t . Thus e^{tX} lies in the symplectic group, confirming that every such X is a genuine tangent direction. $\square$

The defining equation can be made explicit. Write

$$X = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

with $n \times n$ blocks. The relation $X^T J + J X = 0$ becomes

$$D = -A^T, \quad B = B^T, \quad C = C^T.$$

Hence

$$\mathfrak{sp}_{2n}(\mathbb{R}) = \left\{ \begin{pmatrix} A & B \\ C & -A^T \end{pmatrix} : B = B^T, C = C^T \right\}.$$

There are n^2 free parameters in A and $n(n+1)/2$ in each of the symmetric matrices B and C , so

$$\dim \mathrm{Sp}_{2n}(\mathbb{R}) = n(2n+1).$$

This calculation illustrates a general method worth internalizing:

$$\begin{aligned} \text{preserved tensor} &\implies \text{closed stabilizer} \\ &\implies \text{differentiate the preservation law} \end{aligned}$$

It works for symmetric forms, skew forms, volume forms, flags, and many other structures.

30. Polynomial one-parameter motions from nilpotence

Unipotent flows are special partly because their matrix entries vary polynomially in time. This elementary fact is already visible at the level of the exponential map.

Proposition 1.30.1. *If $N \in M_d(\mathbb{R})$ is nilpotent and $N^r = 0$, then*

$$e^{tN} = I + tN + \frac{t^2}{2!}N^2 + \cdots + \frac{t^{r-1}}{(r-1)!}N^{r-1}.$$

In particular, every matrix entry of e^{tN} is a polynomial in t of degree at most $r - 1$.

PROOF. The exponential series is

$$e^{tN} = \sum_{k=0}^{\infty} \frac{t^k N^k}{k!}.$$

Since $N^k = 0$ for all $k \geq r$, every term from $k = r$ onward vanishes. The infinite series therefore truncates to the displayed polynomial. $\square$

Worked Example 1.30.2 (A three-step shear). Take

$$N = E_{12} + E_{23} + E_{34} \in M_4(\mathbb{R}).$$

Then

$$N^2 = E_{13} + E_{24}, \quad N^3 = E_{14}, \quad N^4 = 0.$$

Hence

$$e^{tN} = I + t(E_{12} + E_{23} + E_{34}) + \frac{t^2}{2}(E_{13} + E_{24}) + \frac{t^3}{6}E_{14}.$$

A displacement initially inserted into the nearest-off-diagonal entries propagates into more distant coordinates with quadratic and cubic time dependence. This polynomial propagation is a concrete reason that unipotent dynamics behaves very differently from diagonal dynamics, where the corresponding factors are exponential in time.

Worked Example 1.30.3 (Conjugating a nearby matrix by a unipotent flow). Let $u_t = e^{tN}$ with $N^r = 0$, and fix a matrix A . Then

$$u_t A u_t^{-1} = e^{tN} A e^{-tN}.$$

Both exponentials are matrix polynomials in t , so every entry of the conjugate is a polynomial in t . Thus, in these matrix coordinates, the size of the conjugated perturbation grows at most polynomially in $|t|$; depending on A , some polynomial coefficients may of course vanish. This contrasts with diagonal conjugation, where exponential factors occur.

This observation is elementary, but it gives useful intuition for rigidity arguments: polynomial divergence can leave long time intervals on which nearby trajectories remain meaningfully comparable.

31. One-parameter subgroups in affine groups

The exponential can also be computed explicitly in semidirect products. This example is useful whenever a problem contains both a linear deformation and a translation.

Embed $\text{ASL}_d(\mathbb{R})$ into $\text{SL}_{d+1}(\mathbb{R})$ by

$$(A, v) \longleftrightarrow \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix}.$$

An infinitesimal element has the form

$$Z = \begin{pmatrix} B & w \\ 0 & 0 \end{pmatrix}, \quad \text{tr } B = 0.$$

Proposition 1.31.1. *For such Z ,*

$$e^{tZ} = \begin{pmatrix} e^{tB} & \int_0^t e^{sB} w \, ds \\ 0 & 1 \end{pmatrix}.$$

Guiding Idea 1.31.2. The linear part evolves by the usual matrix exponential. The translation does not simply become tw unless $B = 0$: while translation is being accumulated, the linear flow continuously moves the translation direction. The integral records that accumulated transport.

PROOF. Define

$$A(t) = e^{tB}, \quad v(t) = \int_0^t e^{sB} w \, ds,$$

and set

$$F(t) = \begin{pmatrix} A(t) & v(t) \\ 0 & 1 \end{pmatrix}.$$

Then

$$A'(t) = A(t)B,$$

and, by the fundamental theorem of calculus,

$$v'(t) = e^{tB}w = A(t)w.$$

Therefore

$$F'(t) = \begin{pmatrix} A(t)B & A(t)w \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} A(t) & v(t) \\ 0 & 1 \end{pmatrix} \begin{pmatrix} B & w \\ 0 & 0 \end{pmatrix} = F(t)Z.$$

Also $F(0) = I$. By uniqueness of the matrix ODE $Y' = YZ$, $Y(0) = I$, we have $F(t) = e^{tZ}$. $\square$

Two special cases clarify the formula. If $B = 0$, then

$$e^{tZ} = \begin{pmatrix} I & tw \\ 0 & 1 \end{pmatrix},$$

a pure translation. If $w = 0$, one gets the pure linear flow

$$\begin{pmatrix} e^{tB} & 0 \\ 0 & 1 \end{pmatrix}.$$

In general the two effects are coupled, and the integral is exactly the coupling term.

32. Conceptual summary

The central ideas of the chapter can be compressed into a few principles.

1. A topological group is locally homogeneous. Translations are homeomorphisms. Therefore the identity is not merely a distinguished algebraic point; it is a universal local model for the whole group.

2. Closedness is the bridge from algebra to geometry. A nonclosed subgroup may be dense and topologically pathological. A closed subgroup of a Lie group is automatically an embedded Lie subgroup. The heart of the closed subgroup theorem is the direction lemma: sufficiently small elements accumulating in a direction force the entire one-parameter subgroup in that direction to lie inside the closed subgroup.

3. The exponential map linearizes the group near the identity. Every tangent vector $X \in \mathfrak{g}$ generates a unique global one-parameter subgroup $\exp(tX)$, and $\exp$ is a local diffeomorphism near 0. Thus small nonlinear group motions can be described by ordinary vectors.

4. Matrix exponentials turn structure into explicit calculations.

Nilpotent matrices produce polynomial one-parameter groups, diagonal matrices produce exponential stretching, and conjugation compares the two. The formula

$$a_t u_{ij}(s) a_t^{-1} = u_{ij}(e^{(\lambda_i - \lambda_j)t} s)$$

is a particularly transparent example of group structure becoming dynamical information.

5. Quotients are parameter spaces, not merely formal constructions.

Examples include unimodular lattices,

$$\mathrm{SL}_d(\mathbb{R}) / \mathrm{SL}_d(\mathbb{Z}),$$

affine lattices,

$$\mathrm{ASL}_d(\mathbb{R}) / \mathrm{ASL}_d(\mathbb{Z}),$$

and families of quadratic forms,

$$\mathrm{SL}_d(\mathbb{R}) / \mathrm{SO}(Q).$$

Thinking of these quotients as moduli spaces of concrete objects is often the most efficient way to build intuition.

6. Small motions can generate large global behavior. In a connected Lie group, every neighborhood of the identity generates the whole group. On a torus, a one-dimensional subgroup can even be dense. These facts are elementary precursors of a broad rigidity-versus-density theme in homogeneous dynamics.

Notation and concept map

Symbol	Meaning
G	a topological or Lie group
e	identity element of G
G°	identity component
L_g, R_g	left and right translation
G/H	left-coset quotient
$\mathfrak{g} = T_e G$	tangent space at the identity
$\exp : \mathfrak{g} \rightarrow G$	exponential map
$\text{Ad}(g)$	differential at e of conjugation by g
$Z(G)$	center of G
$C_G(S)$	centralizer of S
$N_G(H)$	normalizer of H
$\text{SL}_d(\mathbb{R})/\text{SL}_d(\mathbb{Z})$	space of unimodular lattices
$\text{ASL}_d(\mathbb{R})$	$\text{SL}_d(\mathbb{R}) \ltimes \mathbb{R}^d$, the special affine group
$\text{SO}(Q)$	stabilizer in $\text{SL}_d(\mathbb{R})$ of a quadratic form Q

CHAPTER 2

Lie Algebras and Infinitesimal Structure

1. Why an infinitesimal algebra is needed

A Lie group is nonlinear: multiplication need not commute, subgroups can curve through the ambient manifold, and conjugation can expand one direction while contracting another. Near the identity, however, the group is modeled by the vector space

$$\mathfrak{g} = T_e G.$$

The tangent space alone remembers only first-order linear information. To recover the first nontrivial trace of noncommutativity one needs one additional operation, the *Lie bracket*

$$[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}.$$

The central principle of this chapter is that the bracket is not an arbitrary decoration on the tangent space. It is forced by the group law. If two infinitesimal motions commute, their bracket is zero; if they fail to commute, the bracket records the leading second-order displacement created by going around a small commutator loop.

Guiding Idea 2.1.1 (From group multiplication to a bilinear operation). Take two small motions $\exp(sX)$ and $\exp(tY)$. The group commutator

$$\exp(sX) \exp(tY) \exp(-sX) \exp(-tY)$$

is equal to the identity to first order in s and t . The first potentially nonzero term is of size st , and its infinitesimal direction is $[X, Y]$. Thus the bracket is naturally a second-order object even though it lives in the first-order tangent space.

2. Vector fields and the intrinsic definition of the bracket

A smooth vector field X on a manifold M acts on smooth functions as a derivation,

$$X : C^\infty(M) \rightarrow C^\infty(M), \quad f \mapsto Xf.$$

For vector fields X, Y , define their commutator on functions by

$$[X, Y]f = X(Yf) - Y(Xf).$$

At first sight this is merely an operator on functions. The following proposition is the key point.

Proposition 2.2.1. *For smooth vector fields X, Y on a smooth manifold M , the operator*

$$f \mapsto X(Yf) - Y(Xf)$$

is again a smooth vector field.

PROOF. The issue is locality and the Leibniz rule. For $f, g \in C^\infty(M)$,

$$\begin{aligned} [X, Y](fg) &= X((Yf)g + f(Yg)) - Y((Xf)g + f(Xg)) \\ &= X(Yf)g + (Yf)(Xg) + (Xf)(Yg) + fX(Yg) \\ &\quad - Y(Xf)g - (Xf)(Yg) - (Yf)(Xg) - fY(Xg) \\ &= [X, Y](f)g + f[X, Y](g). \end{aligned}$$

Thus $[X, Y]$ is a derivation. In local coordinates $x_1, \dots, x_n$, write

$$X = \sum_i a_i \frac{\partial}{\partial x_i}, \quad Y = \sum_j b_j \frac{\partial}{\partial x_j}.$$

A direct computation gives

$$[X, Y] = \sum_k \left(\sum_i a_i \frac{\partial b_k}{\partial x_i} - \sum_i b_i \frac{\partial a_k}{\partial x_i} \right) \frac{\partial}{\partial x_k},$$

whose coefficients are smooth. Hence the derivation is represented by a smooth vector field. $\square$

The vector-field bracket is bilinear and antisymmetric. Its least obvious algebraic property is the Jacobi identity.

Proposition 2.2.2 (Jacobi identity). *For smooth vector fields X, Y, Z ,*

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0.$$

PROOF. View vector fields as linear operators on $C^\infty(M)$. The bracket is the operator commutator $[A, B] = AB - BA$. Expanding,

$$\begin{aligned} [A, [B, C]] &= ABC - ACB - BCA + CBA, \\ [B, [C, A]] &= BCA - BAC - CAB + ACB, \\ [C, [A, B]] &= CAB - CBA - ABC + BAC. \end{aligned}$$

Every term cancels exactly once with its negative. $\square$

Definition 2.2.3 (Lie algebra). A real *Lie algebra* is a real vector space $\mathfrak{g}$ together with a bilinear map

$$[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$$

such that

$$[X, X] = 0$$

for every X , and the Jacobi identity holds. Over a field of characteristic different from 2, $[X, X] = 0$ is equivalent to antisymmetry,

$$[X, Y] = -[Y, X].$$

Let G be a Lie group. Each $X \in T_e G$ determines a unique left-invariant vector field

$$\tilde{X}(g) = (dL_g)_e X.$$

Lemma 2.2.4. *If $\tilde{X}$ and $\tilde{Y}$ are left-invariant, then $[\tilde{X}, \tilde{Y}]$ is left-invariant.*

PROOF. For a diffeomorphism $F : M \rightarrow M$, push-forward preserves vector-field brackets:

$$F_*[X, Y] = [F_*X, F_*Y].$$

This follows directly from the chain rule applied to the derivation definition. Taking $F = L_g$ and using both identities $(L_g)_*\tilde{X} = \tilde{X}$ and $(L_g)_*\tilde{Y} = \tilde{Y}$ gives

$$(L_g)_*[\tilde{X}, \tilde{Y}] = [\tilde{X}, \tilde{Y}].$$

□

Definition 2.2.5 (Lie algebra of a Lie group). The Lie algebra of G is the vector space

$$\mathfrak{g} = T_e G$$

with bracket defined by

$$[X, Y] = [\tilde{X}, \tilde{Y}](e).$$

The left-invariant extension identifies $\mathfrak{g}$ with the Lie algebra of all left-invariant vector fields. Consequently the Jacobi identity on vector fields proves the Jacobi identity in $\mathfrak{g}$.

3. Matrix Lie algebras

For matrix groups the bracket has the familiar form $XY - YX$. It is worth proving this rather than adopting it as a convention, because the proof fixes the sign convention used throughout the volume.

THEOREM 2.3.1 (Bracket for matrix groups). *Let $G < \mathrm{GL}_n(\mathbb{R})$ be a matrix Lie group and $\mathfrak{g} = T_e G$. Then for $X, Y \in \mathfrak{g}$,*

$$[X, Y] = XY - YX.$$

PROOF. For $X \in \mathfrak{g}$, the left-invariant field at $A \in G$ is

$$\tilde{X}(A) = AX,$$

because $L_A(B) = AB$ and therefore $(dL_A)_I(X) = AX$.

Regard $\tilde{X}$ and $\tilde{Y}$ as restrictions of linear vector fields on the ambient vector space $M_n(\mathbb{R})$:

$$F_X(A) = AX, \quad F_Y(A) = AY.$$

Their derivatives are

$$(DF_X)_A(H) = HX, \quad (DF_Y)_A(H) = HY.$$

Hence

$$\begin{aligned} [F_X, F_Y](A) &= (DF_Y)_A(F_X(A)) - (DF_X)_A(F_Y(A)) \\ &= AXY - AYX \\ &= A(XY - YX). \end{aligned}$$

Evaluating at $A = I$ gives $XY - YX$. □

Worked Example 2.3.2 (The Lie algebra of $\mathrm{SL}_n(\mathbb{R})$). Differentiating $\det(e^{tX}) = e^{t \operatorname{tr} X}$ at $t = 0$ shows directly that

$$\mathfrak{sl}_n(\mathbb{R}) = \{X \in M_n(\mathbb{R}) : \operatorname{tr} X = 0\}.$$

The commutator preserves trace because

$$\operatorname{tr}(XY - YX) = \operatorname{tr}(XY) - \operatorname{tr}(YX) = 0.$$

Thus $\mathfrak{sl}_n(\mathbb{R})$ is closed under the bracket.

Worked Example 2.3.3 (Orthogonal and pseudo-orthogonal algebras). Let J be a nonsingular symmetric matrix and

$$G = \{g \in \mathrm{GL}_n(\mathbb{R}) : g^T J g = J\}.$$

Differentiating $g(t)^T J g(t) = J$ at $t = 0$ gives

$$X^T J + JX = 0.$$

Thus

$$\mathfrak{g} = \{X : X^T J + JX = 0\}.$$

If $J = I$ this is the space of skew-symmetric matrices $\mathfrak{so}(n)$; if J has signature (p, q) it is $\mathfrak{so}(p, q)$.

Worked Example 2.3.4 (The symplectic algebra). Let

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$$

and

$$\mathrm{Sp}_{2n}(\mathbb{R}) = \{g : g^T J g = J\}.$$

Then

$$\mathfrak{sp}_{2n}(\mathbb{R}) = \{X : X^T J + JX = 0\}.$$

Writing

$$X = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

with $n \times n$ blocks gives

$$D = -A^T, \quad B = B^T, \quad C = C^T.$$

Hence

$$\dim \mathfrak{sp}_{2n}(\mathbb{R}) = n^2 + \frac{n(n+1)}{2} + \frac{n(n+1)}{2} = n(2n+1).$$

4. Differentials of homomorphisms

A smooth homomorphism of Lie groups has a linear shadow at the identity. The fact that this shadow respects brackets is fundamental.

THEOREM 2.4.1. *Let $\phi : G \rightarrow H$ be a Lie group homomorphism. Then*

$$d\phi_e : \mathfrak{g} \rightarrow \mathfrak{h}$$

is a Lie algebra homomorphism:

$$d\phi_e([X, Y]) = [d\phi_e X, d\phi_e Y].$$

PROOF. A homomorphism intertwines left translations:

$$\phi \circ L_g = L_{\phi(g)} \circ \phi.$$

Differentiating shows that ϕ_* sends the left-invariant field generated by X to the left-invariant field generated by $d\phi_e X$. Push-forward by a smooth map that is locally a diffeomorphism is often used to discuss brackets, but ϕ need not be locally invertible. To avoid any hidden hypothesis, compute on functions after choosing local coordinates on the image, or equivalently use flows. The flow of $\tilde{X}$ is right multiplication by $\exp(tX)$:

$$\Phi_t(g) = g \exp(tX).$$

Because $\phi(\exp(tX)) = \exp(t d\phi_e X)$,

$$\phi(\Phi_t(g)) = \phi(g) \exp(t d\phi_e X),$$

so the fields are ϕ -related. Brackets of related vector fields are related. Evaluating at the identity gives the claimed equality. $\square$

Corollary 2.4.2. *The kernel of a Lie algebra homomorphism is an ideal, and the image is a Lie subalgebra.*

PROOF. If $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$ is a Lie algebra homomorphism, then $\phi([X, Y]) = [\phi(X), \phi(Y)]$. Thus $X \in \ker \phi$ implies $[X, Y] \in \ker \phi$ for every Y , and brackets of elements in $\text{im } \phi$ again lie in $\text{im } \phi$. $\square$

5. The adjoint representation

Conjugation is the internal symmetry by which a group acts on itself. Differentiating at the identity gives a linear representation on its Lie algebra.

Definition 2.5.1. For $g \in G$, let

$$C_g(h) = ghg^{-1}.$$

The *adjoint representation* is

$$\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g}), \quad \text{Ad}(g) = (dC_g)_e.$$

Proposition 2.5.2. *Ad is a Lie group homomorphism:*

$$\text{Ad}(gh) = \text{Ad}(g) \text{Ad}(h).$$

PROOF. Since $C_{gh} = C_g \circ C_h$, the chain rule gives

$$(dC_{gh})_e = (dC_g)_e (dC_h)_e.$$

$\square$

For matrix groups,

$$\text{Ad}(g)X = gXg^{-1}.$$

Indeed,

$$g \exp(tX) g^{-1} = \exp(tgXg^{-1}),$$

and differentiation at $t = 0$ gives the formula.

Differentiating Ad at the identity produces

$$\text{ad} = d(\text{Ad})_e : \mathfrak{g} \rightarrow \text{End}(\mathfrak{g}).$$

THEOREM 2.5.3. *For $X, Y \in \mathfrak{g}$,*

$$\text{ad}(X)Y = [X, Y].$$

Consequently,

$$\text{Ad}(\exp X) = e^{\text{ad } X}.$$

PROOF. Because Ad is a Lie group homomorphism, compatibility with exponentials gives

$$\text{Ad}(\exp(tX)) = \exp(t d \text{Ad}_e X).$$

It remains to identify $d \text{Ad}_e X$. For a matrix group, differentiating the conjugation formula entrywise gives

$$\begin{aligned} \left. \frac{d}{dt} \right|_0 \text{Ad}(e^{tX})Y &= \left. \frac{d}{dt} \right|_0 e^{tX} Y e^{-tX} \\ &= XY - YX = [X, Y]. \end{aligned}$$

For a general Lie group, it is worth checking the sign rather than appealing to analogy with matrices. Let $\tilde{X}$ and $\tilde{Y}$ be the left-invariant vector fields determined by X and Y . The flow of $\tilde{X}$ is

$$\Phi_t(g) = g \exp(tX) = R_{\exp(tX)}(g).$$

For vector fields, the bracket can be recovered from the flow by

$$[\tilde{X}, \tilde{Y}] = \left. \frac{d}{dt} \right|_0 (\Phi_{-t})_* \tilde{Y}.$$

Evaluate this identity at the identity element. To compute $((\Phi_{-t})_* \tilde{Y})_e$, note that the source point is $\exp(tX)$, and therefore

$$\begin{aligned} ((\Phi_{-t})_* \tilde{Y})_e &= (dR_{\exp(-tX)})_{\exp(tX)}(dL_{\exp(tX)})_e Y \\ &= \left. \frac{d}{ds} \right|_0 \exp(tX) \exp(sY) \exp(-tX) \\ &= \text{Ad}(\exp(tX))Y. \end{aligned}$$

Consequently

$$[X, Y] = \left. \frac{d}{dt} \right|_0 \text{Ad}(\exp(tX))Y = (d \text{Ad}_e X)Y.$$

This proves $d \text{Ad}_e X = \text{ad } X$ intrinsically, with the sign fixed by the chosen convention for left-invariant vector fields. The exponential formula then follows from the homomorphism property of Ad . $\square$

Remark 2.5.4. The sign in $\text{ad}(X)Y = [X, Y]$ is convention-sensitive. It agrees with our choice of left-invariant vector fields and with the matrix commutator $XY - YX$.

Worked Example 2.5.5 (Conjugating a unipotent direction). In $\mathfrak{sl}_2(\mathbb{R})$ let

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then $[H, X] = 2X$. If

$$a_t = \exp(tH/2) = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix},$$

then

$$\text{Ad}(a_t)X = e^{t \text{ad}(H/2)}X = e^t X.$$

Exponentiating once more gives

$$a_t \exp(sX) a_t^{-1} = \exp(e^t sX).$$

This calculation explains why conjugation by a diagonal subgroup produces exponential scaling of unipotent parameters.

6. Commutator geometry

The bracket can be recovered from a small commutator loop. The precise statement below is often more useful than the slogan that the bracket measures noncommutativity.

Proposition 2.6.1 (Infinitesimal commutator formula). *For $X, Y \in \mathfrak{g}$,*

$$\exp(sX) \exp(tY) \exp(-sX) \exp(-tY) = \exp(st[X, Y] + o(st))$$

as $(s, t) \rightarrow (0, 0)$ in any regime with $st \neq 0$ and with the error interpreted in local logarithmic coordinates. In particular,

$$[X, Y] = \left. \frac{\partial^2}{\partial s \partial t} \right|_{(0,0)} \log(\exp(sX) \exp(tY) \exp(-sX) \exp(-tY)).$$

PROOF. Use

$$\exp(sX) \exp(tY) \exp(-sX) = \exp(t \text{Ad}(\exp(sX))Y).$$

Since

$$\text{Ad}(\exp(sX))Y = e^{s \text{ad} X}Y = Y + s[X, Y] + O(s^2),$$

the first three factors equal

$$\exp(tY + st[X, Y] + O(s^2t)).$$

Multiplying by $\exp(-tY)$ cancels the first-order tY term. The differential of the local multiplication map at $(0, 0)$ is addition, so in logarithmic coordinates the remaining leading term is $st[X, Y]$; all other terms are $o(st)$ when both variables tend to zero. $\square$

Worked Example 2.6.2 (A nontrivial commutator in the Heisenberg group).

Let

$$X = E_{12}, \quad Y = E_{23} \in \mathfrak{ut}_3(\mathbb{R}).$$

Then

$$[X, Y] = E_{13},$$

and E_{13} commutes with both X and Y . Because $X^2 = Y^2 = 0$,

$$e^{sX} = I + sX, \quad e^{tY} = I + tY.$$

A direct multiplication gives the exact identity

$$e^{sX} e^{tY} e^{-sX} e^{-tY} = I + stE_{13} = e^{stE_{13}}.$$

Here the infinitesimal commutator formula has no higher-order error because the relevant Lie algebra is two-step nilpotent.

7. Subalgebras, ideals, and normalizers

Definition 2.7.1. A vector subspace $\mathfrak{h} \subset \mathfrak{g}$ is a *Lie subalgebra* if

$$[\mathfrak{h}, \mathfrak{h}] \subset \mathfrak{h}.$$

It is an *ideal* if

$$[\mathfrak{g}, \mathfrak{h}] \subset \mathfrak{h}.$$

Ideals are the infinitesimal analogues of normal subgroups.

Proposition 2.7.2. *If $H \triangleleft G$ is a closed normal Lie subgroup, then $\mathfrak{h} = \text{Lie}(H)$ is an ideal in $\mathfrak{g}$.*

PROOF. Normality gives $gHg^{-1} = H$, hence

$$\text{Ad}(g)\mathfrak{h} = \mathfrak{h}$$

for all $g \in G$. Fix $X \in \mathfrak{g}$ and $Y \in \mathfrak{h}$. The curve

$$t \mapsto \text{Ad}(\exp(tX))Y$$

lies in the vector space $\mathfrak{h}$. Differentiating at 0 gives

$$[X, Y] \in \mathfrak{h}.$$

□

The infinitesimal normalizer is equally important.

Definition 2.7.3. For a Lie subalgebra $\mathfrak{h} \subset \mathfrak{g}$, define

$$\mathfrak{n}_{\mathfrak{g}}(\mathfrak{h}) = \{X \in \mathfrak{g} : [X, \mathfrak{h}] \subset \mathfrak{h}\}.$$

Proposition 2.7.4. $\mathfrak{n}_{\mathfrak{g}}(\mathfrak{h})$ is a Lie subalgebra. If $H < G$ is a closed Lie subgroup, then

$$\mathrm{Lie}(N_G(H)) \subset \mathfrak{n}_{\mathfrak{g}}(\mathfrak{h}).$$

If H is connected, equality holds.

PROOF. If X, Y normalize $\mathfrak{h}$ and $Z \in \mathfrak{h}$, Jacobi gives

$$[[X, Y], Z] = [X, [Y, Z]] - [Y, [X, Z]] \in \mathfrak{h},$$

so the infinitesimal normalizer is a subalgebra.

If $X \in \mathrm{Lie}(N_G(H))$, then $\exp(tX)$ normalizes H for all t . Hence

$$\mathrm{Ad}(\exp(tX))\mathfrak{h} = \mathfrak{h}.$$

Differentiating at $t = 0$ gives $[X, \mathfrak{h}] \subset \mathfrak{h}$.

Conversely, suppose H is connected and $[X, \mathfrak{h}] \subset \mathfrak{h}$. Then $e^{t \mathrm{ad} X}$ preserves $\mathfrak{h}$, so

$$\mathrm{Ad}(\exp(tX))\mathfrak{h} = \mathfrak{h}.$$

The connected Lie subgroups H and $\exp(tX)H\exp(-tX)$ have the same Lie algebra. A connected Lie subgroup is generated by exponentials of its Lie algebra, so they coincide. Thus $\exp(tX) \in N_G(H)$ for every t , and $X \in \mathrm{Lie}(N_G(H))$. $\square$

Warning 2.7.5. Connectedness is genuinely used in the reverse implication. A Lie algebra does not detect discrete components. Two closed subgroups can have the same Lie algebra while differing by additional discrete elements.

8. Derived and lower central series

Two descending constructions measure different kinds of noncommutativity.

Definition 2.8.1. The *derived series* of $\mathfrak{g}$ is

$$\mathfrak{g}^{(0)} = \mathfrak{g}, \quad \mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}].$$

The algebra is *solvable* if $\mathfrak{g}^{(m)} = 0$ for some m .

Definition 2.8.2. The *lower central series* is

$$\mathfrak{g}_1 = \mathfrak{g}, \quad \mathfrak{g}_{k+1} = [\mathfrak{g}, \mathfrak{g}_k].$$

The algebra is *nilpotent* if $\mathfrak{g}_m = 0$ for some m .

Proposition 2.8.3. Every nilpotent Lie algebra is solvable.

PROOF. We claim by induction that

$$\mathfrak{g}^{(k)} \subset \mathfrak{g}_{2^k}.$$

For $k = 0$ this is equality. If it holds at k , then

$$\mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}] \subset [\mathfrak{g}_{2^k}, \mathfrak{g}_{2^k}] \subset \mathfrak{g}_{2^{k+1}}.$$

The last inclusion follows from repeated use of Jacobi, or from the general relation $[\mathfrak{g}_i, \mathfrak{g}_j] \subset \mathfrak{g}_{i+j}$. If the lower central series reaches 0, so does the derived series. $\square$

Worked Example 2.8.4 (Upper unitriangular matrices). Let $\mathfrak{u}_n$ be the strictly upper triangular matrices. A basis is E_{ij} with $i < j$, and

$$[E_{ij}, E_{kl}] = \delta_{jk}E_{il} - \delta_{li}E_{kj}.$$

Define the *height* of E_{ij} to be $j - i$. Bracketing adds heights whenever the result is nonzero. Hence the r th term of the lower central series is spanned by E_{ij} with $j - i \geq r$. Therefore $\mathfrak{u}_n$ is nilpotent of class $n - 1$.

The corresponding group U_n of upper unitriangular matrices has polynomial multiplication and inversion. If $X \in \mathfrak{u}_n$, then $X^n = 0$, so

$$e^X = I + X + \frac{X^2}{2!} + \cdots + \frac{X^{n-1}}{(n-1)!}.$$

Thus exponentials in nilpotent matrix groups are polynomial rather than infinite series.

Warning 2.8.5 (Two uses of the word nilpotent). A Lie algebra can be nilpotent in the lower-central-series sense. An individual endomorphism T can also be nilpotent when $T^m = 0$. For a matrix Lie algebra, an element X may be nilpotent as a matrix even when the ambient Lie algebra is not nilpotent. For example, $E_{12} \in \mathfrak{sl}_2(\mathbb{R})$ is nilpotent as a matrix, whereas $\mathfrak{sl}_2(\mathbb{R})$ is not a nilpotent Lie algebra.

9. The algebra $\mathfrak{sl}_2(\mathbb{R})$

Set

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Every traceless 2×2 matrix is a unique linear combination of H, X, Y .

Proposition 2.9.1. *The basic bracket relations are*

$$[H, X] = 2X, \quad [H, Y] = -2Y, \quad [X, Y] = H.$$

PROOF. Direct multiplication gives

$$HX = X, \quad XH = -X,$$

so $[H, X] = 2X$. Direct multiplication also gives $HY = -Y$ and $YH = Y$, hence $[H, Y] = -2Y$. Finally

$$XY = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad YX = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

so $[X, Y] = H$. □

The action of $\text{ad } H$ diagonalizes:

$$\mathfrak{sl}_2(\mathbb{R}) = \mathbb{R}Y \oplus \mathbb{R}H \oplus \mathbb{R}X,$$

with eigenvalues $-2, 0, 2$. This elementary decomposition is a model for root-space decompositions in larger semisimple Lie algebras.

Proposition 2.9.2 (Adjoint action of the basic one-parameter subgroups).
Let

$$a_t = e^{tH/2}, \quad u_s = e^{sX}, \quad v_s = e^{sY}.$$

Then

$$\text{Ad}(a_t)X = e^t X, \quad \text{Ad}(a_t)Y = e^{-t} Y, \quad \text{Ad}(a_t)H = H,$$

and

$$\begin{aligned} \text{Ad}(u_s)X &= X, \\ \text{Ad}(u_s)H &= H - 2sX, \\ \text{Ad}(u_s)Y &= Y + sH - s^2X. \end{aligned}$$

PROOF. The a_t formulas follow from the eigenvalues of $\text{ad}(H/2)$. For u_s use

$$\text{Ad}(u_s) = e^{s \text{ad } X}.$$

Now

$$\text{ad } X(H) = -2X, \quad (\text{ad } X)^2(H) = 0,$$

so $e^{s \text{ad } X}H = H - 2sX$. Also

$$\text{ad } X(Y) = H, \quad (\text{ad } X)^2(Y) = [X, H] = -2X, \quad (\text{ad } X)^3(Y) = 0.$$

Hence

$$e^{s \text{ad } X}Y = Y + sH + \frac{s^2}{2}(-2X) = Y + sH - s^2X.$$

□

Guiding Idea 2.9.3 (Why unipotent conjugation is polynomial). If X is such that $\text{ad } X$ is nilpotent, then

$$\text{Ad}(e^{tX}) = e^{t \text{ad } X}$$

is a finite polynomial in t . Thus conjugating a nearby group element by a unipotent flow creates polynomial, rather than exponential, divergence in Lie-algebra coordinates. This is a structural property, not an accident of 2×2 matrices.

Worked Example 2.9.4 (A cubic shear in $\text{SL}_3(\mathbb{R})$). Let

$$X = E_{12} + E_{23} \in \mathfrak{sl}_3(\mathbb{R}).$$

Then $X^3 = 0$ and

$$u_t = e^{tX} = I + t(E_{12} + E_{23}) + \frac{t^2}{2}E_{13}.$$

For any $Z \in \mathfrak{sl}_3(\mathbb{R})$,

$$\text{Ad}(u_t)Z = e^{t \text{ad } X}Z.$$

Since $\text{ad } X$ is nilpotent on the finite-dimensional algebra, this expression is polynomial in t . The degree depends on Z ; different transverse directions can therefore separate at different polynomial rates under the same unipotent motion.

10. The Killing form in the $\mathfrak{sl}_2$ model

The bilinear form

$$B(X, Y) = \text{tr}(\text{ad } X \text{ ad } Y)$$

is called the Killing form. We need only its elementary properties here.

Proposition 2.10.1. *For any finite-dimensional Lie algebra, the Killing form is symmetric and invariant:*

$$B([Z, X], Y) + B(X, [Z, Y]) = 0.$$

PROOF. Symmetry follows from $\text{tr}(AB) = \text{tr}(BA)$. For invariance, Jacobi says

$$\text{ad}[Z, X] = [\text{ad } Z, \text{ad } X].$$

Therefore

$$\begin{aligned} B([Z, X], Y) &= \text{tr}([\text{ad } Z, \text{ad } X] \text{ ad } Y) \\ &= \text{tr}(\text{ad } Z \text{ ad } X \text{ ad } Y - \text{ad } X \text{ ad } Z \text{ ad } Y), \end{aligned}$$

while

$$\begin{aligned} B(X, [Z, Y]) &= \operatorname{tr}(\operatorname{ad} X [\operatorname{ad} Z, \operatorname{ad} Y]) \\ &= \operatorname{tr}(\operatorname{ad} X \operatorname{ad} Z \operatorname{ad} Y - \operatorname{ad} X \operatorname{ad} Y \operatorname{ad} Z). \end{aligned}$$

The middle terms cancel, and cyclicity of trace gives

$$\operatorname{tr}(\operatorname{ad} Z \operatorname{ad} X \operatorname{ad} Y) = \operatorname{tr}(\operatorname{ad} X \operatorname{ad} Y \operatorname{ad} Z).$$

□

Worked Example 2.10.2 (Killing form on $\mathfrak{sl}_2$). With basis H, X, Y one finds

$$B(H, H) = 8, \quad B(X, Y) = B(Y, X) = 4,$$

and

$$B(H, X) = B(H, Y) = B(X, X) = B(Y, Y) = 0.$$

Thus the Gram matrix is

$$\begin{pmatrix} 8 & 0 & 0 \\ 0 & 0 & 4 \\ 0 & 4 & 0 \end{pmatrix},$$

which is nondegenerate and has signature $(2, 1)$ after a change of basis. This is the infinitesimal reason that $\operatorname{PSL}_2(\mathbb{R})$ is closely related to the identity component of an orthogonal group of signature $(2, 1)$.

11. Semidirect products at the Lie algebra level

Suppose a Lie group H acts smoothly by automorphisms $\rho(h)$ on a Lie group N . The semidirect product $G = N \rtimes H$ has multiplication

$$(n, h)(n', h') = (n\rho(h)n', hh').$$

Differentiating the action gives a Lie algebra representation

$$d\rho : \mathfrak{h} \rightarrow \operatorname{Der}(\mathfrak{n}),$$

where a derivation D satisfies

$$D[U, V] = [DU, V] + [U, DV].$$

Proposition 2.11.1. *The Lie algebra of $N \rtimes H$ is the vector space $\mathfrak{n} \oplus \mathfrak{h}$ with bracket*

$$[(U, X), (V, Y)] = ([U, V] + d\rho(X)V - d\rho(Y)U, [X, Y]).$$

PROOF. The formula is obtained by differentiating the group commutator. A clean verification is to realize (U, X) as a left-invariant vector field on $N \times H$. The H component brackets as usual. In the N component, the bracket of the two N directions contributes $[U, V]$, while differentiating the dependence of the transported N direction on the H coordinate contributes

$d\rho(X)V - d\rho(Y)U$. Bilinearity and Jacobi follow either from this derivation or directly because it is the bracket of a Lie group. $\square$

Worked Example 2.11.2 (Special affine group). For

$$\mathrm{ASL}_d(\mathbb{R}) = \mathbb{R}^d \rtimes \mathrm{SL}_d(\mathbb{R}),$$

the action is the standard linear action. Writing elements of the Lie algebra as (v, X) with $v \in \mathbb{R}^d$ and $X \in \mathfrak{sl}_d(\mathbb{R})$,

$$[(v, X), (w, Y)] = (Xw - Yv, [X, Y]).$$

This formula is useful whenever linear and translational dynamics are coupled. The translation part is an abelian ideal, but it is usually not central because Xw need not vanish.

12. Stabilizers of tensors and forms

Many subgroups in homogeneous dynamics arise as stabilizers of algebraic data. The tangent condition is obtained simply by differentiating the invariance equation.

Proposition 2.12.1 (Infinitesimal stabilizer). *Let $\rho : G \rightarrow \mathrm{GL}(V)$ be a finite-dimensional representation and let $v \in V$. Put*

$$H = \{g \in G : \rho(g)v = v\}.$$

Then H is closed, and

$$\mathrm{Lie}(H) = \{X \in \mathfrak{g} : d\rho(X)v = 0\}.$$

PROOF. Closedness follows because H is the inverse image of the closed singleton $\{v\}$ under the continuous map $g \mapsto \rho(g)v$. If $X \in \mathrm{Lie}(H)$, then $\rho(e^{tX})v = v$ for every t , so differentiating at 0 gives $d\rho(X)v = 0$.

Conversely, suppose $d\rho(X)v = 0$. Since

$$\rho(e^{tX}) = e^{t d\rho(X)},$$

we have

$$\rho(e^{tX})v = e^{t d\rho(X)}v = v$$

for all t . Hence $e^{tX} \in H$ for all t , so $X \in \mathrm{Lie}(H)$. $\square$

Worked Example 2.12.2 (Quadratic forms). Let $Q(x) = x^T A x$ with $A = A^T$ nonsingular. The action of $g \in \mathrm{SL}_d(\mathbb{R})$ on quadratic forms may be written

$$(g \cdot Q)(x) = Q(g^{-1}x).$$

The stabilizer condition is equivalent to

$$g^T A g = A.$$

Differentiating $e^{tX^T} A e^{tX} = A$ at 0 gives

$$X^T A + A X = 0.$$

Thus

$$\text{Lie}(\text{SO}(Q)) = \left\{ X \in \mathfrak{sl}_d(\mathbb{R}) : X^T A + A X = 0 \right\}.$$

When Q has signature (p, q) this algebra is isomorphic, after a linear change of coordinates, to $\mathfrak{so}(p, q)$.

13. Exterior powers and subspaces

Exterior powers package subspaces and determinants into linear representations. This simple device repeatedly converts a nonlinear geometric condition into a statement about a vector in a representation.

Let V be a d -dimensional vector space. For $1 \leq k \leq d$, the space $\bigwedge^k V$ is generated by symbols

$$v_1 \wedge \cdots \wedge v_k$$

subject to multilinearity and alternation. A linear map $g : V \rightarrow V$ acts by

$$\bigwedge^k g(v_1 \wedge \cdots \wedge v_k) = g v_1 \wedge \cdots \wedge g v_k.$$

Proposition 2.13.1. *Let $W \subset V$ be a k -dimensional subspace with basis $w_1, \dots, w_k$, and set*

$$\omega_W = w_1 \wedge \cdots \wedge w_k \in \bigwedge^k V.$$

Then a linear isomorphism g preserves W if and only if

$$\left(\bigwedge^k g \right) \omega_W \in \mathbb{R}^\times \omega_W.$$

PROOF. If $gW = W$, then $g w_1, \dots, g w_k$ is another basis of W , so its wedge differs from ω_W by the nonzero determinant of the change-of-basis matrix.

Conversely, suppose

$$g w_1 \wedge \cdots \wedge g w_k = c w_1 \wedge \cdots \wedge w_k, \quad c \neq 0.$$

A decomposable nonzero k -vector determines its underlying k -plane as

$$W = \{ v \in V : v \wedge \omega_W = 0 \}.$$

Thus the two decomposable vectors determine the same plane, giving $gW = W$. $\square$

Worked Example 2.13.2 (Primitive-sublattice covolume). Let $v_1, \dots, v_k \in \mathbb{Z}^d$ be linearly independent and let

$$\Delta = \mathbb{Z}v_1 + \dots + \mathbb{Z}v_k.$$

The Euclidean norm of

$$v_1 \wedge \dots \wedge v_k$$

is the k -dimensional volume of the parallelepiped spanned by the v_i . Under $g \in \mathrm{SL}_d(\mathbb{R})$ this becomes

$$\|gv_1 \wedge \dots \wedge gv_k\|.$$

Thus the sizes of sublattices of $g\mathbb{Z}^d$ can be monitored by linear representations $\wedge^k g$. This is a prototypical example of *linearization*: encode geometric degeneration by the smallness of a vector in a finite-dimensional representation.

14. A controlled Baker–Campbell–Hausdorff expansion

The full Baker–Campbell–Hausdorff series is not needed for the basic geometry developed here, but its first terms explain why brackets arise when multiplying exponentials.

THEOREM 2.14.1 (BCH to second order). *For X, Y sufficiently small in a finite-dimensional Lie algebra of a Lie group,*

$$\log(\exp X \exp Y) = X + Y + \frac{1}{2}[X, Y] + R_3(X, Y),$$

where, in any fixed norm and sufficiently small neighborhood,

$$\|R_3(X, Y)\| \leq C(\|X\| + \|Y\|)^3$$

for some constant C .

Guiding Idea 2.14.2. The linear term must be $X + Y$ because the differential of multiplication at (e, e) is addition. There is only one universal antisymmetric bilinear correction compatible with the group law, namely a scalar multiple of $[X, Y]$. Comparing the small commutator determines the coefficient $1/2$.

PROOF. Work in exponential coordinates around e and define

$$F(X, Y) = \log(\exp X \exp Y).$$

This is smooth near $(0, 0)$. Since $F(X, 0) = X$ and $F(0, Y) = Y$, its Taylor expansion has the form

$$F(X, Y) = X + Y + B(X, Y) + R_3(X, Y),$$

where B is bilinear and the remainder is cubic in the stated norm sense.

Associativity of group multiplication imposes identities on B , but a direct comparison with the commutator is shorter. Replacing (X, Y) by (sX, tY) and using the expansion four times gives

$$\log(e^{sX}e^{tY}e^{-sX}e^{-tY}) = st(B(X, Y) - B(Y, X)) + o(st).$$

The infinitesimal commutator formula says that the left side is

$$st[X, Y] + o(st).$$

Hence

$$B(X, Y) - B(Y, X) = [X, Y].$$

On the other hand, setting $Y = X$ and using $e^Xe^X = e^{2X}$ gives $B(X, X) = 0$; polarization implies that B is antisymmetric. Therefore

$$2B(X, Y) = [X, Y],$$

so $B(X, Y) = \frac{1}{2}[X, Y]$. Because F is C^3 , choose a closed ball K about the origin on which $\|D^3F\| \leq M$. Taylor's formula with integral remainder gives, for X, Y and the segment joining $(0, 0)$ to (X, Y) inside K , a remainder bounded by $CM(\|X\| + \|Y\|)^3$ for an absolute combinatorial constant C . This is the stated cubic bound after shrinking the neighborhood if necessary. $\square$

Remark 2.14.3. For a nilpotent Lie algebra the BCH series terminates after finitely many brackets, because sufficiently long iterated brackets vanish. This is one reason nilpotent groups admit unusually explicit coordinates.

15. Adjoint orbits and infinitesimal centralizers

The adjoint representation turns conjugacy into an ordinary linear group action. For $X \in \mathfrak{g}$, its adjoint orbit is

$$\mathcal{O}_X = \{\text{Ad}(g)X : g \in G\}.$$

Its stabilizer is the group centralizer

$$C_G(X) = \{g \in G : \text{Ad}(g)X = X\}.$$

At the Lie algebra level define

$$\mathfrak{c}_{\mathfrak{g}}(X) = \{Y \in \mathfrak{g} : [Y, X] = 0\}.$$

Proposition 2.15.1. $C_G(X)$ is a closed Lie subgroup with Lie algebra $\mathfrak{c}_{\mathfrak{g}}(X)$. The tangent space to the adjoint orbit at X is

$$T_X\mathcal{O}_X = [\mathfrak{g}, X] = \{[Y, X] : Y \in \mathfrak{g}\}.$$

PROOF. The map $g \mapsto \text{Ad}(g)X$ is continuous, so the stabilizer of X is closed. If $Y \in \text{Lie}(C_G(X))$, then

$$\text{Ad}(e^{tY})X = X$$

for all t , and differentiation gives $[Y, X] = 0$. Conversely, if $[Y, X] = 0$, then

$$\text{Ad}(e^{tY})X = e^{t \text{ad } Y}X = X,$$

so $e^{tY} \in C_G(X)$ for all t . Thus the Lie algebra is the stated centralizer.

The orbit map $g \mapsto \text{Ad}(g)X$ has differential

$$Y \mapsto \left. \frac{d}{dt} \right|_0 \text{Ad}(e^{tY})X = [Y, X].$$

Its image is therefore $[\mathfrak{g}, X]$, which is the tangent space to the immersed orbit. $\square$

Worked Example 2.15.2 (Adjoint orbits in $\mathfrak{sl}_2(\mathbb{R})$). For

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

the centralizer in $\mathfrak{sl}_2$ is $\mathbb{R}H$: if $Y = aH + bX + cY_-$ commutes with H , the root relations force $b = c = 0$. Hence the adjoint orbit of H has dimension 2.

For the nilpotent element $X = E_{12}$, a direct calculation gives

$$\mathfrak{c}_{\mathfrak{sl}_2}(X) = \mathbb{R}X.$$

Its nonzero nilpotent orbit is also two-dimensional. These are not merely dimension counts: the stabilizers explain which perturbations of a semisimple or nilpotent element are genuine conjugacy changes and which are internal symmetries.

16. Polynomial degree under unipotent conjugation

The statement that unipotent conjugation is polynomial can be quantified in every finite-dimensional representation.

Proposition 2.16.1. *Let $\rho : G \rightarrow \text{GL}(V)$ be a finite-dimensional representation and suppose $u_t = \exp(tX)$ with $d\rho(X)$ nilpotent of index m , meaning*

$$d\rho(X)^m = 0.$$

Then for every $v \in V$,

$$\rho(u_t)v = \sum_{j=0}^{m-1} \frac{t^j}{j!} d\rho(X)^j v.$$

In particular, every coordinate of $\rho(u_t)v$ in a fixed basis is a polynomial of degree at most $m - 1$.

PROOF. Compatibility of representations with exponentials gives

$$\rho(u_t) = e^{t d\rho(X)}.$$

The exponential series terminates after the $(m-1)$ st term because all higher powers vanish. $\square$

Worked Example 2.16.2 (Exterior powers of a shear). Let $u_t = I + tE_{12} \in \mathrm{SL}_d(\mathbb{R})$. On $\mathbb{R}^d$ the action is linear in t . On $\bigwedge^k \mathbb{R}^d$ one still has

$$\bigwedge^k u_t = e^{t d(\bigwedge^k)(E_{12})},$$

and the induced infinitesimal operator is nilpotent. For this rank-one shear it actually squares to zero, so every wedge coordinate is also affine in t . For a longer Jordan block, higher polynomial degrees occur. Thus exterior-power linearization preserves the polynomial nature of unipotent motion while making rational subspaces visible.

17. Derivations and infinitesimal automorphisms

A linear map $D : \mathfrak{g} \rightarrow \mathfrak{g}$ is a *derivation* if

$$D[X, Y] = [DX, Y] + [X, DY].$$

The Jacobi identity shows that every $\mathrm{ad} Z$ is a derivation; such derivations are called inner.

Proposition 2.17.1. *If Φ_t is a one-parameter group of Lie-group automorphisms of G , then*

$$A_t = d(\Phi_t)_e$$

is a one-parameter group of Lie-algebra automorphisms. If

$$D = \left. \frac{d}{dt} \right|_0 A_t,$$

then D is a derivation.

PROOF. Every A_t preserves brackets because it is the differential of a group homomorphism. Thus

$$A_t[X, Y] = [A_t X, A_t Y].$$

Differentiate at $t = 0$ and use $A_0 = I$:

$$D[X, Y] = [DX, Y] + [X, DY].$$

$\square$

Conjugation by $\exp(tZ)$ is a one-parameter automorphism group, and its derivative is $e^{t\text{ad}Z}$; differentiating at zero recovers the inner derivation $\text{ad}Z$. Semidirect products arise when an external Lie algebra acts by derivations in exactly the same way.

18. Two instructive subalgebras of $\mathfrak{sl}_2$

The subspaces

$$\mathfrak{n} = \mathbb{R}X, \quad \mathfrak{a} = \mathbb{R}H,$$

are one-dimensional abelian subalgebras. The two-dimensional space

$$\mathfrak{b} = \mathbb{R}H \oplus \mathbb{R}X$$

is also a subalgebra because $[H, X] = 2X$. It is solvable but not nilpotent:

$$[\mathfrak{b}, \mathfrak{b}] = \mathbb{R}X, \quad [\mathfrak{b}, \mathbb{R}X] = \mathbb{R}X.$$

Thus the derived series reaches zero after one more commutator,

$$\mathfrak{b}^{(2)} = 0,$$

while the lower central series stabilizes at $\mathbb{R}X$ and never reaches zero.

This tiny example separates two notions that are easily conflated. Solvability allows the algebra to become abelian after repeated commutators of the current algebra with itself. Nilpotence demands that repeated commutation with the whole original algebra eventually kills every direction. The diagonal element H keeps rescaling X , preventing nilpotence of $\mathfrak{b}$.

19. Conceptual summary

The Lie algebra retains exactly the local information most useful for dynamics. One-parameter subgroups become vectors; conjugation becomes the linear action Ad ; infinitesimal conjugation becomes ad ; normal subgroups become ideals; and polynomial shearing by unipotent elements becomes the finite expansion of $e^{t\text{ad}X}$ when $\text{ad}X$ is nilpotent.

The most important identities to keep available are

$$[X, Y] = XY - YX \quad \text{for matrix groups,}$$

$$\text{Ad}(\exp X) = e^{\text{ad}X},$$

$$\text{ad}(X)Y = [X, Y],$$

and, in the $\mathfrak{sl}_2$ basis,

$$[H, X] = 2X, \quad [H, Y] = -2Y, \quad [X, Y] = H.$$

These formulas are not isolated computations. They are the local algebraic mechanism behind expansion, contraction, and polynomial divergence in homogeneous dynamics.

CHAPTER 3

The Structure of $\mathrm{SL}_2(\mathbb{R})$

1. Why $\mathrm{SL}_2(\mathbb{R})$ is the basic model

Among noncompact Lie groups, $\mathrm{SL}_2(\mathbb{R})$ is small enough that almost every structural statement can be proved by hand, but rich enough to display the main phenomena of homogeneous dynamics. It contains compact rotations, diagonal expansion and contraction, and unipotent shears. Its action on the projective line is explicit, and its quotient by the center is the full orientation-preserving isometry group of the hyperbolic plane.

Throughout this chapter we use the normalization

$$a_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix}.$$

With this choice,

$$a_t u_s a_{-t} = u_{e^t s}, \quad u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}.$$

The factor $1/2$ in a_t is not cosmetic: it makes the parameter t agree with hyperbolic arclength when a_t is interpreted geometrically.

We write

$$G = \mathrm{SL}_2(\mathbb{R}), \quad K = \mathrm{SO}(2), \quad A = \{a_t : t \in \mathbb{R}\},$$

$$N = \{u_s : s \in \mathbb{R}\}, \quad N^- = \{v_s : s \in \mathbb{R}\}, \quad v_s = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix}.$$

We also use the upper triangular subgroup

$$B = AN \cup (-I)AN = \left\{ \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} : a \in \mathbb{R}^\times, b \in \mathbb{R} \right\}.$$

2. The projective action

The real projective line $\mathbb{P}^1(\mathbb{R})$ is the set of one-dimensional subspaces of $\mathbb{R}^2$. Writing a nonzero vector as $(x, y)^T$, its projective class is denoted

$[x : y]$. On the affine chart $y \neq 0$ we may use the coordinate $z = x/y \in \mathbb{R}$; the missing point $[1 : 0]$ is denoted ∞ . Thus as a set and topological space,

$$\mathbb{P}^1(\mathbb{R}) \simeq \mathbb{R} \cup \{\infty\},$$

where neighborhoods of ∞ correspond to complements of compact subsets of $\mathbb{R}$.

Every $g \in \mathrm{GL}_2(\mathbb{R})$ maps lines to lines, so it acts on $\mathbb{P}^1(\mathbb{R})$. Scalar matrices act trivially; consequently the projective action of $\mathrm{SL}_2(\mathbb{R})$ has kernel $\{\pm I\}$.

Proposition 3.2.1 (Möbius formula). *Let*

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{R}).$$

On the affine coordinate $z = x/y$, the action is

$$g \cdot z = \frac{az + b}{cz + d},$$

with the usual conventions when the denominator vanishes or $z = \infty$.

PROOF. The line represented by $(z, 1)^T$ is sent to

$$g \begin{pmatrix} z \\ 1 \end{pmatrix} = \begin{pmatrix} az + b \\ cz + d \end{pmatrix}.$$

If $cz + d \neq 0$, its affine coordinate is the displayed quotient. If $cz + d = 0$, the image vector has second coordinate zero and hence represents ∞ . The formula at ∞ follows by applying g to $(1, 0)^T$. $\square$

Worked Example 3.2.2 (Three basic actions). For the standard subgroups,

$$u_s \cdot z = z + s,$$

$$a_t \cdot z = e^t z,$$

and

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \cdot z = -\frac{1}{z}.$$

Translations, dilations, and inversion already generate a large portion of the projective geometry of G .

3. Stabilizers and homogeneous descriptions

Let $e_1 = (1, 0)^T$ and $e_2 = (0, 1)^T$.

Proposition 3.3.1. *The stabilizer in G of the point $[1 : 0] = \infty$ is B . Hence the map*

$$G/B \rightarrow \mathbb{P}^1(\mathbb{R}), \quad gB \mapsto g \cdot \infty,$$

is a well-defined bijection and, with the natural quotient topology, a homeomorphism.

PROOF. A matrix $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ fixes the line $\mathbb{R}e_1$ exactly when $ge_1 = (a, c)^T$ lies on that line, equivalently $c = 0$. The determinant-one condition then forces $d = a^{-1}$, giving precisely B .

The orbit map $G \rightarrow \mathbb{P}^1(\mathbb{R})$, $g \mapsto g \cdot \infty$, is continuous and surjective because G acts transitively: given a nonzero vector (x, y) , choose a second column (r, s) so that $xs - yr = 1$. The fibers are left cosets of B , giving the bijection.

To verify the topology without appealing to a quotient-manifold theorem, construct local sections. On the affine chart $z \in \mathbb{R}$, the matrix

$$s_0(z) = \begin{pmatrix} z & -1 \\ 1 & 0 \end{pmatrix}$$

has determinant 1 and sends ∞ to z . Near ∞ , use the coordinate $w = y/x$ on lines $[1 : w]$ and the section

$$s_\infty(w) = \begin{pmatrix} 1 & 0 \\ w & 1 \end{pmatrix}.$$

Both sections are continuous. If $q : G \rightarrow G/B$ is the quotient map and $\Phi : G/B \rightarrow \mathbb{P}^1(\mathbb{R})$ the induced bijection, then on either chart

$$\Phi^{-1}(z) = q(s_0(z)), \quad \Phi^{-1}([1 : w]) = q(s_\infty(w)),$$

so Φ^{-1} is continuous locally and therefore globally. Hence Φ is a homeomorphism. $\square$

A different quotient records nonzero vectors rather than lines.

Proposition 3.3.2. *The map*

$$G/N \rightarrow \mathbb{R}^2 \setminus \{0\}, \quad gN \mapsto ge_1,$$

is a G -equivariant diffeomorphism.

PROOF. The stabilizer of e_1 is exactly N : if

$$g \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

then the first column of g is $(1, 0)^T$, and determinant one forces

$$g = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}.$$

The action is transitive on nonzero vectors. Given $v = (x, y)^T \neq 0$, choose $w = (r, s)^T$ so that $xs - yr = 1$; then the matrix with columns v, w lies in G and sends e_1 to v . Thus the orbit-stabilizer map is bijective.

To see smoothness explicitly rather than only invoke the homogeneous-space theorem, use two charts. If $x \neq 0$, one may choose

$$g(v) = \begin{pmatrix} x & 0 \\ y & x^{-1} \end{pmatrix};$$

if $y \neq 0$, one may choose

$$g(v) = \begin{pmatrix} x & -y^{-1} \\ y & 0 \end{pmatrix}.$$

These give smooth local sections of the orbit map. Hence the induced bijection has smooth local inverse and is a diffeomorphism. $\square$

Remark 3.3.3. The distinction between G/N and G/B is geometrically important. Passing from N to B forgets the scale of a nonzero vector and remembers only its direction.

4. Iwasawa decomposition: $G = KAN$

The Iwasawa decomposition is best understood as Gram–Schmidt applied to the columns of a matrix.

THEOREM 3.4.1 (Iwasawa decomposition). *Every $g \in \mathrm{SL}_2(\mathbb{R})$ can be written uniquely as*

$$g = ka_t u_s, \quad k \in K, \quad t, s \in \mathbb{R}.$$

Thus multiplication defines a diffeomorphism

$$K \times A \times N \rightarrow G.$$

Guiding Idea 3.4.2. Write the columns of g as v_1, v_2 . First rotate the normalized first column to e_1 . Its length becomes the positive diagonal factor. After that, determinant one forces the second diagonal entry, while the remaining off-diagonal component is exactly the shear parameter.

PROOF. Let

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

and let

$$r = \sqrt{a^2 + c^2} > 0.$$

The first column $v_1 = (a, c)^T$ has length r . Define

$$k = \frac{1}{r} \begin{pmatrix} a & -c \\ c & a \end{pmatrix} \in \mathrm{SO}(2).$$

Then $k^{-1}v_1 = (r, 0)^T$. Hence

$$k^{-1}g = \begin{pmatrix} r & \beta \\ 0 & \delta \end{pmatrix}$$

for some β, δ . The determinant is one, so $r\delta = 1$ and $\delta = r^{-1}$. Choose t by $e^{t/2} = r$. Then

$$a_{-t}k^{-1}g = \begin{pmatrix} 1 & r^{-1}\beta \\ 0 & 1 \end{pmatrix} = u_s,$$

where $s = r^{-1}\beta$. Thus $g = ka_tu_s$.

For uniqueness, suppose

$$ka_tu_s = k'a_{t'}u_{s'}.$$

Then

$$k^{-1}k' = a_tu_su_{-s'}a_{-t'} \in K \cap AN.$$

An element of AN is upper triangular with positive diagonal entries. If it is also orthogonal, its columns are orthonormal; the lower-left entry is zero, forcing the upper-right entry to be zero, and positivity forces both diagonal entries to be 1. Hence $K \cap AN = \{I\}$. Therefore $k = k'$ and

$$a_tu_s = a_{t'}u_{s'}.$$

Comparing diagonal entries gives $t = t'$, then $s = s'$. Smoothness of the parameters follows from the explicit formulas above. $\square$

Worked Example 3.4.3 (Explicit Iwasawa coordinates). For

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

set $r = \sqrt{a^2 + c^2}$. Then

$$e^{t/2} = r,$$

and a short computation gives

$$s = \frac{ab + cd}{a^2 + c^2}.$$

Indeed, the second column of $k^{-1}g$ has first coordinate $(ab+cd)/r$, so dividing by r gives s . These formulas are useful for converting between matrix entries and KAN coordinates.

5. The NAK coordinates and the upper half-plane

There is another global factorization,

$$G = NAK,$$

which should not be confused with KAN . The parameters are different.

Proposition 3.5.1 (NAK decomposition). *Every $g \in G$ can be written uniquely as*

$$g = u_x a_t k, \quad x, t \in \mathbb{R}, k \in K.$$

PROOF. Apply Iwasawa decomposition to g^{-1} :

$$g^{-1} = k' a_{t'} u_{s'}.$$

Taking inverses gives

$$g = u_{-s'} a_{-t'} k'^{-1}.$$

Existence follows. Uniqueness is proved exactly as for KAN , using $NA \cap K = \{I\}$. $\square$

The NAK coordinates are adapted to the action on the upper half-plane

$$\mathbb{H} = \{z = x + iy \in \mathbb{C} : y > 0\}.$$

The Möbius formula extends to $\mathbb{H}$.

Proposition 3.5.2. *For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G$ and $z \in \mathbb{H}$,*

$$\Im(gz) = \frac{\Im z}{|cz + d|^2}.$$

In particular G preserves $\mathbb{H}$.

PROOF. Since a, b, c, d are real,

$$\begin{aligned} gz - \overline{gz} &= \frac{az + b}{cz + d} - \frac{a\bar{z} + b}{c\bar{z} + d} \\ &= \frac{(ad - bc)(z - \bar{z})}{|cz + d|^2} = \frac{z - \bar{z}}{|cz + d|^2}. \end{aligned}$$

Divide by $2i$. $\square$

Now

$$u_x a_t \cdot i = x + ie^t.$$

Since K fixes i , the NAK decomposition says precisely that every g is determined by the point $g \cdot i \in \mathbb{H}$ together with a rotation in the stabilizer of i .

6. Cartan decomposition

The Cartan decomposition separates a matrix into two rotations and one symmetric stretching.

THEOREM 3.6.1 (Cartan decomposition). *Every $g \in G$ can be written*

$$g = k_1 a_t k_2, \quad k_1, k_2 \in K, t \geq 0.$$

The number t is uniquely determined by g . The two K factors are not always unique.

PROOF. The positive-definite symmetric matrix $g^T g$ has determinant 1. By the spectral theorem there exists $k_2 \in K$ such that

$$k_2(g^T g)k_2^{-1} = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix}$$

with $\lambda \geq 1$ after possibly exchanging the eigenspaces by a rotation through $\pi/2$. Write $\lambda = e^t$ with $t \geq 0$. Then

$$a_t^2 = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}.$$

Set

$$k_1 = gk_2^{-1}a_{-t}.$$

We compute

$$\begin{aligned} k_1^T k_1 &= a_{-t} k_2^{-T} g^T g k_2^{-1} a_{-t} \\ &= a_{-t} a_t^2 a_{-t} = I. \end{aligned}$$

Also $\det k_1 = 1$, so $k_1 \in K$, and $g = k_1 a_t k_2$.

The eigenvalues of $g^T g$ are intrinsic, so $e^{\pm t}$ and hence $t \geq 0$ are uniquely determined. $\square$

Remark 3.6.2. The operator norm of g for the Euclidean norm is $e^{t/2}$ in the Cartan decomposition. Thus the Cartan parameter measures how far g is from the compact subgroup K in a very concrete linear-algebraic sense.

7. Bruhat decomposition

Let

$$w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The Bruhat decomposition separates G according to whether the lower-left entry is zero.

THEOREM 3.7.1 (Bruhat decomposition).

$$G = B \sqcup BwB.$$

The two pieces are disjoint. The second is open and dense.

PROOF. If $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has $c = 0$, then $g \in B$.

Assume $c \neq 0$. Then the following factorization is exact:

$$g = \begin{pmatrix} 1 & a/c \\ 0 & 1 \end{pmatrix} w \begin{pmatrix} c & d \\ 0 & c^{-1} \end{pmatrix}.$$

The first and third factors lie in B . To verify the identity rather than infer it from dimension counting, multiply:

$$w \begin{pmatrix} c & d \\ 0 & c^{-1} \end{pmatrix} = \begin{pmatrix} 0 & -c^{-1} \\ c & d \end{pmatrix},$$

and hence

$$\begin{pmatrix} 1 & a/c \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -c^{-1} \\ c & d \end{pmatrix} = \begin{pmatrix} a & (ad-1)/c \\ c & d \end{pmatrix}.$$

Because $ad - bc = 1$, the upper-right entry is $(ad - 1)/c = b$. Thus every matrix with $c \neq 0$ belongs to BwB .

The pieces are disjoint because elements of B fix ∞ , whereas w sends ∞ to 0 and BwB consists of elements sending ∞ to a finite point. The condition $c \neq 0$ is open and dense in the hypersurface $ad - bc = 1$, so BwB is open dense. $\square$

It is often useful to have an explicit Gauss decomposition on the big cell.

Proposition 3.7.2 (Gauss decomposition). *If*

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G$$

and $a \neq 0$, then

$$g = v_{c/a} \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} u_{b/a}.$$

PROOF. Multiplying the right side gives

$$\begin{pmatrix} 1 & 0 \\ c/a & 1 \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} \begin{pmatrix} 1 & b/a \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & b \\ c & cb/a + a^{-1} \end{pmatrix}.$$

Since $ad - bc = 1$,

$$d = \frac{1 + bc}{a} = \frac{bc}{a} + a^{-1}.$$

$\square$

8. Conjugacy types from the trace

For $g \in G$, the characteristic polynomial is

$$\lambda^2 - (\operatorname{tr} g)\lambda + 1.$$

Thus the discriminant $(\operatorname{tr} g)^2 - 4$ determines the qualitative eigenvalue picture.

Definition 3.8.1. A noncentral $g \in \operatorname{SL}_2(\mathbb{R})$ is called

- *elliptic* if $|\operatorname{tr} g| < 2$;
- *parabolic* if $|\operatorname{tr} g| = 2$;
- *hyperbolic* if $|\operatorname{tr} g| > 2$.

Proposition 3.8.2 (Normal forms). *Let $g \in G$ be noncentral.*

(1) *If $|\operatorname{tr} g| < 2$, then g is conjugate in G to a rotation*

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

for some $\theta \notin \pi\mathbb{Z}$.

(2) *If $|\operatorname{tr} g| = 2$, then g is conjugate to $\pm u_s$ for some $s \neq 0$. Rescaling by a diagonal conjugation shows that, within each allowed sign/orientation class, s may be normalized to ± 1 .*

(3) *If $|\operatorname{tr} g| > 2$, then g is conjugate to $\pm a_t$ for a unique $t > 0$ after accounting for the sign of the eigenvalues.*

PROOF. If $|\operatorname{tr} g| > 2$, the characteristic polynomial has two distinct real reciprocal eigenvalues. Choosing eigenvectors gives a real diagonalization. By rescaling the eigenbasis one can arrange determinant one for the conjugating matrix. The diagonal entries are λ, λ^{-1} with $|\lambda| > 1$, so g has the stated form.

If $|\operatorname{tr} g| < 2$, the eigenvalues are complex conjugates $e^{\pm i\theta}$. Choose a complex eigenvector $v = p + iq$. The real plane basis p, q identifies the action with rotation through θ (possibly after changing the sign of one basis vector to make the conjugating matrix orientation-preserving). Rescaling makes the conjugating matrix determinant one.

If $|\operatorname{tr} g| = 2$ and $g \neq \pm I$, then the characteristic polynomial has a repeated eigenvalue ± 1 but g is not diagonalizable. Its Jordan form is $\pm \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$ with $s \neq 0$. Diagonal conjugation satisfies

$$a_t u_s a_{-t} = u_{e^t s},$$

so the magnitude of s can be normalized. The sign of s is not changed by positive diagonal scaling; whether it is changed by a conjugation inside the chosen group depends on the precise central/sign convention, so it is safest to retain the indicated ± 1 distinction. $\square$

Remark 3.8.3. In $\operatorname{PSL}_2(\mathbb{R})$ the central signs disappear. Hyperbolic elements have two fixed points on $\mathbb{P}^1(\mathbb{R})$, parabolic elements one, and elliptic elements none on the real boundary. This projective description is often more geometric than the matrix normal form.

9. Centralizers and normalizers of the basic subgroups

The following computations are elementary but useful because they prevent several common confusions.

Proposition 3.9.1. *In $G = \mathrm{SL}_2(\mathbb{R})$:*

$$C_G(A) = A \cup (-I)A,$$

$$N_G(A) = (A \cup (-I)A) \sqcup w(A \cup (-I)A),$$

$$C_G(N) = \{\pm u_s : s \in \mathbb{R}\},$$

$$N_G(N) = B.$$

PROOF. If g commutes with every a_t , then it preserves the two eigenspaces of a_t for $t \neq 0$, so g is diagonal; determinant one gives $C_G(A)$. If g normalizes A , conjugation permutes the two one-dimensional weight spaces, so g is either diagonal or anti-diagonal. The anti-diagonal elements form $wC_G(A)$.

For N , let $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The equation $gu_s = u_sg$ for all s gives $c = 0$ and $a = d$; determinant one yields $a = d = \pm 1$, giving $C_G(N)$.

If g normalizes N , then it preserves the unique common fixed point of N on $\mathbb{P}^1(\mathbb{R})$, namely ∞ . Therefore $g \in B$. Conversely, a direct computation shows that

$$\begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix} u_s \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix}^{-1} = u_{a^2 s},$$

so every element of B normalizes N . □

Warning 3.9.2. The normalizer of A is a finite extension of A in this specific example, but such a conclusion is not true for arbitrary subgroups of arbitrary Lie groups. It follows here from the two weight lines and the fact that the only possible permutation is their transposition.

10. Generation by unipotent subgroups

THEOREM 3.10.1. *$\mathrm{SL}_2(\mathbb{R})$ is generated by N and N^- . In particular it is generated by unipotent one-parameter subgroups.*

PROOF. The key is to produce the Weyl element and the diagonal subgroup. First,

$$u_1 v_{-1} u_1 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = -w.$$

Thus w lies in the group generated by N, N^- up to the central element $-I$; but $(-w)^2 = -I$, so $-I$ lies there as well, hence so does w .

For $r \neq 0$, define

$$h(r) = u_r v_{-1/r} u_r.$$

A direct multiplication gives

$$h(r) = \begin{pmatrix} 0 & r \\ -r^{-1} & 0 \end{pmatrix}.$$

Then

$$h(r)h(1)^{-1} = \begin{pmatrix} r & 0 \\ 0 & r^{-1} \end{pmatrix}.$$

For $r > 0$ this is $a_{2 \log r}$, so A is generated by N, N^- . We have therefore obtained A, N, w , and the central element $-I$. Every element of the upper triangular subgroup has the form

$$(\pm I)a_t u_s,$$

so B is generated by A, N , and $-I$. The Bruhat decomposition $G = B \sqcup BwB$ now shows that every element of G lies in the subgroup generated by N, N^- . $\square$

11. The adjoint representation and $\mathrm{SO}(2, 1)$

We now make precise the relation between $\mathrm{PSL}_2(\mathbb{R})$ and an orthogonal group of signature $(2, 1)$. The cleanest construction uses the Killing form on $\mathfrak{sl}_2(\mathbb{R})$.

Let

$$B(X, Y) = \mathrm{tr}(\mathrm{ad} X \mathrm{ad} Y).$$

The invariance of the Killing form implies

$$B(\mathrm{Ad}(g)X, \mathrm{Ad}(g)Y) = B(X, Y).$$

Hence $\mathrm{Ad}(G)$ lies in the orthogonal group $O(B)$.

Proposition 3.11.1. *The kernel of*

$$\mathrm{Ad} : \mathrm{SL}_2(\mathbb{R}) \rightarrow \mathrm{GL}(\mathfrak{sl}_2(\mathbb{R}))$$

is $\{\pm I\}$.

PROOF. If $\mathrm{Ad}(g) = I$, then g commutes with $\exp(tX)$ for every $X \in \mathfrak{sl}_2(\mathbb{R})$ and $t \in \mathbb{R}$. Since G is connected and generated by exponentials, g commutes with all of G . The center of G is $\{\pm I\}$. Conversely central elements act trivially by conjugation. $\square$

THEOREM 3.11.2. *After identifying the Killing form with a quadratic form of signature $(2, 1)$,*

$$\mathrm{Ad}(\mathrm{SL}_2(\mathbb{R})) = \mathrm{SO}(2, 1)^\circ.$$

Consequently,

$$\mathrm{PSL}_2(\mathbb{R}) \cong \mathrm{SO}(2, 1)^\circ.$$

PROOF. Using

$$[H, X] = 2X, \quad [H, Y] = -2Y, \quad [X, Y] = H,$$

a direct matrix calculation of the operators $\mathrm{ad} H, \mathrm{ad} X, \mathrm{ad} Y$ gives

$$B(H, H) = 8, \quad B(X, Y) = B(Y, X) = 4,$$

while

$$B(H, X) = B(H, Y) = B(X, X) = B(Y, Y) = 0.$$

Consequently, in the basis

$$H, \quad X + Y, \quad X - Y,$$

the Gram matrix of B is exactly

$$\mathrm{diag}(8, 8, -8).$$

Thus the Killing form has signature $(2, 1)$ and $\mathrm{Ad}(G) \subset O(2, 1)$. Since G is connected, the image lies in the identity component $\mathrm{SO}(2, 1)^\circ$; the determinant of $\mathrm{Ad}(g)$ is continuous and equals 1 at the identity.

The differential of Ad is

$$\mathrm{ad} : \mathfrak{sl}_2(\mathbb{R}) \rightarrow \mathfrak{so}(B).$$

Its kernel is the center of the Lie algebra. The relations $[H, X] = 2X$, $[H, Y] = -2Y$, $[X, Y] = H$ show directly that the center is zero. Thus ad is injective. Both domain and target have dimension 3, so ad is an isomorphism.

Therefore $\mathrm{Ad}(G)$ is a connected Lie subgroup of $\mathrm{SO}(2, 1)^\circ$ with the same Lie algebra. A connected Lie subgroup with full Lie algebra is open; an open subgroup of a connected group is the whole group. Hence the image is $\mathrm{SO}(2, 1)^\circ$. The kernel computation then gives the quotient isomorphism. $\square$

Worked Example 3.11.3 (Binary quadratic forms). There is an explicit intertwiner between the adjoint representation and the action on binary quadratic forms; displaying it removes any ambiguity in the phrase “the same representation.” Put

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

For

$$X = \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \in \mathfrak{sl}_2(\mathbb{R}),$$

define

$$q_X(v) = v^T J X v.$$

Since

$$JX = \begin{pmatrix} c & -a \\ -a & -b \end{pmatrix}$$

is symmetric,

$$q_X(x, y) = cx^2 - 2axy - by^2.$$

The map $X \mapsto q_X$ is a linear isomorphism from $\mathfrak{sl}_2(\mathbb{R})$ to $\text{Sym}^2((\mathbb{R}^2)^*)$, the space of binary quadratic forms.

For $g \in \text{SL}_2(\mathbb{R})$ we have $g^T J g = J$, equivalently $g^{-T} J = J g$. Therefore

$$\begin{aligned} q_{\text{Ad}(g)X}(v) &= v^T J (g X g^{-1}) v \\ &= v^T g^{-T} J X g^{-1} v \\ &= q_X(g^{-1} v). \end{aligned}$$

Thus $X \mapsto q_X$ intertwines the adjoint action with the change-of-variables action

$$(g \cdot q)(v) = q(g^{-1} v).$$

Moreover,

$$\text{disc}(q_X) = (-2a)^2 - 4c(-b) = 4(a^2 + bc) = -4 \det X.$$

This discriminant is a quadratic form of signature $(2, 1)$, and it is preserved by the action. The central element $-I$ acts trivially, so the representation factors through $\text{PSL}_2(\mathbb{R})$. The displayed intertwiner shows directly—not merely by a dimension count—that this quadratic-form model is equivalent to the adjoint model above.

12. A concrete catalogue of one-parameter subgroups

Every nonzero $X \in \mathfrak{sl}_2(\mathbb{R})$ satisfies a simple quadratic identity. Write

$$X = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}.$$

Then

$$X^2 = (a^2 + bc)I = -\det(X)I.$$

Hence the sign of $a^2 + bc$ controls the exponential.

Proposition 3.12.1. *Let $\kappa = a^2 + bc = -\det X$.*

(1) *If $\kappa > 0$, then*

$$e^{tX} = \cosh(\sqrt{\kappa}t)I + \frac{\sinh(\sqrt{\kappa}t)}{\sqrt{\kappa}}X.$$

(2) *If $\kappa = 0$, then $X^2 = 0$ and*

$$e^{tX} = I + tX.$$

(3) *If $\kappa < 0$, writing $\omega = \sqrt{-\kappa}$,*

$$e^{tX} = \cos(\omega t)I + \frac{\sin(\omega t)}{\omega}X.$$

PROOF. Separate the exponential series into even and odd powers and use $X^{2m} = \kappa^m I$ and $X^{2m+1} = \kappa^m X$. The three cases are exactly the hyperbolic, nilpotent, and trigonometric power series. $\square$

Thus one-parameter subgroups of $\mathrm{SL}_2(\mathbb{R})$ naturally fall into hyperbolic, unipotent, and elliptic types. The classification is already visible infinitesimally.

13. Cross-ratios and projective configurations

The projective action becomes especially transparent on configurations of boundary points. For four distinct points $x_1, x_2, x_3, x_4 \in \mathbb{P}^1(\mathbb{R})$, define the cross-ratio in an affine chart by

$$[x_1, x_2; x_3, x_4] = \frac{(x_1 - x_3)(x_2 - x_4)}{(x_1 - x_4)(x_2 - x_3)}.$$

The formula extends continuously when one of the points is ∞ by canceling the corresponding leading factors.

Proposition 3.13.1. *The cross-ratio is invariant under the projective action of $\mathrm{PSL}_2(\mathbb{R})$.*

PROOF. For $T(z) = (az + b)/(cz + d)$,

$$T(x) - T(y) = \frac{x - y}{(cx + d)(cy + d)}.$$

Substituting this expression into the four factors of the cross-ratio, every denominator $(cx_i + d)$ occurs once in the numerator and once in the denominator, so all cancel. $\square$

Worked Example 3.13.2 (Three-transitivity with an orientation constraint). Given three distinct boundary points (x_1, x_2, x_3) , there is a unique projective transformation in $\mathrm{PGL}_2(\mathbb{R})$ sending them to $(\infty, 0, 1)$. Requiring the transformation to lie in $\mathrm{PSL}_2(\mathbb{R})$ imposes preservation of the cyclic orientation of the boundary circle. Once three points are normalized, the cross-ratio of a fourth point is exactly its remaining projective coordinate. This is why cross-ratios are the natural scalar invariants of four boundary points.

14. Polar decomposition

The Cartan decomposition can be recast in a form useful for matrix estimates.

THEOREM 3.14.1 (Polar decomposition). *Every $g \in \mathrm{SL}_2(\mathbb{R})$ can be written uniquely as*

$$g = kp,$$

where $k \in \mathrm{SO}(2)$ and p is symmetric positive definite with determinant 1.

PROOF. Set

$$p = (g^T g)^{1/2},$$

the unique positive-definite square root supplied by the spectral theorem. Then p is symmetric positive definite and

$$\det p = \sqrt{\det(g^T g)} = 1.$$

Define $k = gp^{-1}$. Then

$$k^T k = p^{-1} g^T g p^{-1} = I,$$

and $\det k = 1$, so $k \in K$.

If $g = k_1 p_1 = k_2 p_2$ are two such decompositions, then

$$p_2 p_1^{-1} = k_2^{-1} k_1$$

is both orthogonal and conjugate to a positive-definite matrix. More directly,

$$g^T g = p_1^2 = p_2^2,$$

and uniqueness of the positive square root gives $p_1 = p_2$, hence $k_1 = k_2$. $\square$

Diagonalizing p by K gives the Cartan decomposition. Thus polar decomposition separates a matrix into a pure rotation and a pure symmetric deformation, while Cartan decomposition diagonalizes the symmetric deformation as well.

15. Topology and the central double cover

The Iwasawa coordinates make the basic topology of the group explicit.

Proposition 3.15.1. *$\mathrm{SL}_2(\mathbb{R})$ is connected and has fundamental group isomorphic to $\mathbb{Z}$; in particular it is not simply connected. The quotient map*

$$\mathrm{SL}_2(\mathbb{R}) \longrightarrow \mathrm{PSL}_2(\mathbb{R})$$

is a two-sheeted covering with deck group $\{\pm I\}$.

PROOF. The Iwasawa decomposition is a diffeomorphism

$$K \times A \times N \longrightarrow G.$$

Here $K \cong S^1$ and $A, N \cong \mathbb{R}$, so G is connected. Contracting the A and N coordinates to the identity defines a deformation retraction of G onto K . Hence

$$\pi_1(G) \cong \pi_1(K) \cong \pi_1(S^1) \cong \mathbb{Z}.$$

The subgroup $\{\pm I\}$ is discrete and central. Choose an identity neighborhood U with $U \cap (-U) = \emptyset$. Its image in the quotient has preimage $U \sqcup (-U)$,

and the quotient map restricts to a homeomorphism (indeed a diffeomorphism) on each sheet. Translating this local picture gives a two-sheeted covering everywhere, with the nontrivial deck transformation $g \mapsto -g$. $\square$

The central quotient does not change the local Lie algebra or local differential geometry, but it changes stabilizers globally. For instance, in $\mathrm{SL}_2(\mathbb{R})$ both I and $-I$ act trivially on $\mathbb{H}$, so the stabilizer of i is $\mathrm{SO}(2)$ and the action is not effective: its central elements $\{\pm I\}$ induce the same isometry. Passing to $\mathrm{PSL}_2(\mathbb{R})$ removes precisely this redundancy.

Warning 3.15.2. Whenever a homogeneous identification is claimed to be one-to-one, the center must be checked. Statements such as “ $T^1\mathbb{H} = \mathrm{SL}_2(\mathbb{R})$ ” are false literally: the correct effective group is $\mathrm{PSL}_2(\mathbb{R})$, while $\mathrm{SL}_2(\mathbb{R})$ gives a twofold cover.

16. Root subgroups and an explicit commutator identity

The upper and lower unipotent groups interact nontrivially. The following identity makes the interaction completely explicit.

Proposition 3.16.1 (Opposite-unipotent factorization). *For $s, t \in \mathbb{R}$ with $1 + st \neq 0$,*

$$u_s v_t = v_{t/(1+st)} \begin{pmatrix} 1+st & 0 \\ 0 & (1+st)^{-1} \end{pmatrix} u_{s/(1+st)}.$$

PROOF. Multiply the right side. The first two factors give

$$\begin{pmatrix} 1+st & 0 \\ t & (1+st)^{-1} \end{pmatrix}.$$

Multiplying by $u_{s/(1+st)}$ yields

$$\begin{pmatrix} 1+st & s \\ t & 1 \end{pmatrix} = u_s v_t.$$

$\square$

This identity shows concretely how multiplying opposite unipotent directions creates a diagonal component. It is a rank-one version of the root-subgroup identities that organize much larger semisimple groups.

17. Conceptual summary

The structure of $\mathrm{SL}_2(\mathbb{R})$ is controlled by four elementary subgroups:

$$K = \mathrm{SO}(2), \quad A, \quad N, \quad N^-.$$

The KAN decomposition is Gram–Schmidt; the NAK decomposition is adapted to the upper-half-plane action; the Cartan decomposition is singular-value decomposition; and the Bruhat decomposition separates the closed triangular cell from one open dense cell.

The projective quotients

$$G/N \simeq \mathbb{R}^2 \setminus \{0\}, \quad G/B \simeq \mathbb{P}^1(\mathbb{R})$$

show how stabilizers turn group quotients into concrete geometric spaces. The conjugation identity

$$a_t u_s a_{-t} = u_{e^t s}$$

exhibits expansion of a unipotent direction, while the adjoint formulas

$$\mathrm{Ad}(u_s)Y = Y + sH - s^2X$$

exhibit polynomial shearing. Finally,

$$\mathrm{PSL}_2(\mathbb{R}) \cong \mathrm{SO}(2, 1)^\circ$$

links this matrix group simultaneously to hyperbolic geometry, quadratic forms, and three-dimensional Lorentzian geometry.

CHAPTER 4

Homogeneous Spaces and Quotient Geometry

1. Actions, orbits, and stabilizers

A *left action* of a group G on a set X is a map $(g, x) \mapsto gx$ satisfying

$$ex = x, \quad g(hx) = (gh)x.$$

If G is a Lie group and X a smooth manifold, the action is called smooth when the map

$$G \times X \rightarrow X$$

is smooth.

For $x \in X$, define the orbit and stabilizer

$$Gx = \{gx : g \in G\}, \quad G_x = \{g \in G : gx = x\}.$$

The orbit is the part of X accessible from x by the action; the stabilizer measures the redundancy in the parametrization $g \mapsto gx$.

Proposition 4.1.1. *For a continuous action of a topological group on a Hausdorff space, G_x is closed.*

PROOF. The orbit map

$$\pi_x : G \rightarrow X, \quad g \mapsto gx$$

is continuous, and

$$G_x = \pi_x^{-1}(\{x\}).$$

Since X is Hausdorff, $\{x\}$ is closed. □

Thus for a smooth action on a manifold, stabilizers are closed Lie subgroups by the closed subgroup theorem.

Proposition 4.1.2 (Set-theoretic orbit–stabilizer identification). *The map*

$$G/G_x \rightarrow Gx, \quad gG_x \mapsto gx,$$

is a well-defined bijection.

PROOF. If $gG_x = hG_x$, then $h^{-1}g \in G_x$, so $gx = h(h^{-1}g)x = hx$. Thus the map is well defined. It is surjective by definition of the orbit. If $gx = hx$, then $h^{-1}g \in G_x$, so $gG_x = hG_x$. □

The main geometric problem is to determine when this bijection is a homeomorphism or diffeomorphism. The quotient-manifold theorem supplies the correct smooth structure on G/H whenever H is closed.

2. The quotient-manifold theorem

Let $H < G$ be a closed subgroup with Lie algebras $\mathfrak{h} \subset \mathfrak{g}$. Choose a vector-space complement

$$\mathfrak{g} = \mathfrak{m} \oplus \mathfrak{h}.$$

No invariance of $\mathfrak{m}$ is assumed.

Guiding Idea 4.2.1 (A local slice transverse to the subgroup). Near the identity, the closed subgroup theorem identifies H with the exponential of $\mathfrak{h}$. Directions in a complementary space $\mathfrak{m}$ move transversely to the cosets of H . The map

$$(X, h) \mapsto \exp(X)h$$

should therefore give local coordinates on G in which the H -cosets are obtained by varying only the second coordinate. Forgetting that coordinate gives local coordinates on G/H .

THEOREM 4.2.2 (Quotient-manifold theorem). *Let G be a Lie group and $H < G$ a closed subgroup. Then G/H admits a unique smooth manifold structure such that the quotient map*

$$\pi : G \rightarrow G/H$$

is a smooth submersion. Its dimension is

$$\dim(G/H) = \dim G - \dim H.$$

The natural left action of G on G/H is smooth.

PROOF. We construct charts explicitly. Let $\mathfrak{g} = \mathfrak{m} \oplus \mathfrak{h}$. Consider

$$F : \mathfrak{m} \times H \rightarrow G, \quad F(X, h) = \exp(X)h.$$

Near $(0, e)$, its differential is

$$dF_{(0,e)}(X, Y) = X + Y,$$

under the identifications $T_e H = \mathfrak{h}$ and $T_e G = \mathfrak{g}$. This is an isomorphism because $\mathfrak{g} = \mathfrak{m} \oplus \mathfrak{h}$. The inverse function theorem therefore gives neighborhoods $U \subset \mathfrak{m}$, $V \subset H$, and $W \subset G$ such that

$$F : U \times V \rightarrow W$$

is a diffeomorphism.

We need a slice that controls full right H -cosets, not only points with the H coordinate in V . Shrink W if needed. If $\exp(X)H = \exp(Y)H$ with $X, Y \in U$, then

$$\exp(X) = \exp(Y)h$$

for some $h \in H$. When U is sufficiently small, the element

$$h = \exp(-Y) \exp(X)$$

lies in the H -neighborhood V occurring in the local product chart. We then have two representations of the same element of W ,

$$F(X, e) = \exp(X) = F(Y, h).$$

Injectivity of $F : U \times V \rightarrow W$ forces $X = Y$ and $h = e$. Thus the map

$$\sigma : U \rightarrow G/H, \quad X \mapsto \exp(X)H$$

is injective.

Its image is open. Indeed,

$$\pi^{-1}(\sigma(U)) = \exp(U)H,$$

which is open near every one of its points because right translation by H transports the open local product neighborhoods. Since π is an open map, $\sigma(U)$ is open in G/H .

Declare $\sigma : U \rightarrow \sigma(U)$ to be a coordinate chart. Left translating this chart by elements of G covers G/H . Transition maps are smooth because they are obtained from the smooth group multiplication, inversion, and the local inverse of F . This defines a smooth structure.

In the chart at eH , the quotient map has local form

$$U \times V \rightarrow U, \quad (X, h) \mapsto X,$$

so it is a submersion and the dimension formula follows.

The left action is smooth in these charts because it is induced by smooth multiplication on G . Finally, uniqueness follows because any smooth structure making π a submersion admits local smooth sections by the submersion theorem; those sections force exactly the same quotient charts. $\square$

Remark 4.2.3. The topology underlying this smooth structure is the ordinary quotient topology. Closedness of H is essential for Hausdorffness. If H is not closed, G/H can fail even to be a reasonable topological space, so no compatible manifold structure should be expected.

3. Tangent space of a homogeneous space

Let $o = eH \in G/H$ be the base point. The differential of the quotient map

$$d\pi_e : \mathfrak{g} \rightarrow T_o(G/H)$$

is surjective because π is a submersion.

THEOREM 4.3.1. *The kernel of $d\pi_e$ is $\mathfrak{h}$. Therefore*

$$T_{eH}(G/H) \cong \mathfrak{g}/\mathfrak{h}.$$

At a general point gH there are two useful, but different, quotient descriptions:

(1) translating the base-point tangent space by g gives

$$\mathfrak{g}/\mathfrak{h} \longrightarrow T_{gH}(G/H), \quad X + \mathfrak{h} \longmapsto \left. \frac{d}{dt} \right|_0 g \exp(tX)H;$$

(2) differentiating the global left G -action at the point gH gives

$$\mathfrak{g}/\text{Ad}(g)\mathfrak{h} \longrightarrow T_{gH}(G/H), \quad X + \text{Ad}(g)\mathfrak{h} \longmapsto \left. \frac{d}{dt} \right|_0 \exp(tX)gH.$$

Both maps are vector-space isomorphisms.

PROOF. In the local product coordinates used above, $d\pi_e$ is projection

$$\mathfrak{m} \oplus \mathfrak{h} \rightarrow \mathfrak{m},$$

so its kernel is $\mathfrak{h}$. This gives the base-point quotient.

For the first general-point formula, left multiplication by g induces a diffeomorphism of G/H carrying eH to gH . Its differential transports the already established isomorphism at the base point and gives the displayed curve.

For the second formula, consider the orbit map

$$\rho_{gH} : G \rightarrow G/H, \quad a \mapsto agH.$$

Its stabilizer is gHg^{-1} , whose Lie algebra is $\text{Ad}(g)\mathfrak{h}$. Hence the differential $d\rho_{gH}|_e$ is surjective with precisely that kernel, and the first isomorphism theorem gives the second quotient. The two descriptions are related by $\text{Ad}(g)$; neither should be substituted for the other without specifying the convention. $\square$

Warning 4.3.2 (Keep the tangent convention visible). There are several natural identifications of tangent spaces with $\mathfrak{g}$, differing by left or right translation and sometimes by $\text{Ad}(g)$. A formula such as $T_{gH}(G/H) \cong \mathfrak{g}/\mathfrak{h}$ is true only after specifying how the tangent space at g is transported back to the identity. Accordingly, every formula below specifies the convention whenever the distinction matters.

4. The isotropy representation

The stabilizer H fixes the base point $o = eH$. Therefore each $h \in H$ acts linearly on $T_o(G/H)$ by the differential of left translation.

THEOREM 4.4.1 (Isotropy action). *Under the canonical identification*

$$T_o(G/H) \cong \mathfrak{g}/\mathfrak{h},$$

the differential of the action of $h \in H$ is

$$X + \mathfrak{h} \longmapsto \text{Ad}(h)X + \mathfrak{h}.$$

In other words, the isotropy representation is the quotient of $\text{Ad}|_H$ on $\mathfrak{g}/\mathfrak{h}$.

PROOF. Take $X \in \mathfrak{g}$. The corresponding tangent vector at o is represented by the curve

$$c(t) = \exp(tX)H.$$

Acting by h gives

$$hc(t) = h \exp(tX)H.$$

Since $h \in H$, $hH = H$, and

$$h \exp(tX)H = h \exp(tX)h^{-1}H = \exp(t \text{Ad}(h)X)H.$$

Differentiating at 0 gives the stated action. If X is replaced by $X + Y$ with $Y \in \mathfrak{h}$, then $\text{Ad}(h)Y \in \mathfrak{h}$, so the formula is well defined on the quotient. $\square$

Worked Example 4.4.2 (The tangent representation of the sphere). Let $G = \text{SO}(n)$ and let $H = \text{SO}(n-1)$ be embedded as the stabilizer of e_n . Then

$$G/H \cong S^{n-1}.$$

At e_n , the tangent space is $e_n^\perp \cong \mathbb{R}^{n-1}$. The isotropy action of H is exactly its standard action on $\mathbb{R}^{n-1}$. The abstract quotient representation $\mathfrak{so}(n)/\mathfrak{so}(n-1)$ therefore recovers the ordinary tangent representation of rotations fixing the north pole.

5. Fundamental vector fields

Let G act smoothly on a manifold M . Each $X \in \mathfrak{g}$ defines a vector field by differentiating the one-parameter action.

Definition 4.5.1. For a left action, define the *fundamental vector field*

$$X_M(x) = \left. \frac{d}{dt} \right|_0 \exp(tX)x.$$

There is a sign subtlety in the bracket.

Proposition 4.5.2. *With the above convention for a left action,*

$$[X_M, Y_M] = -[X, Y]_M.$$

If instead one defines $X_M(x) = \frac{d}{dt}\big|_0 \exp(-tX)x$, the map $X \mapsto X_M$ is a Lie algebra homomorphism.

PROOF. The flow of X_M is $\Phi_t(x) = \exp(tX)x$. Push-forward by this flow gives

$$(\Phi_t)_*Y_M = (\text{Ad}(\exp(tX))Y)_M$$

with the point moved by the action. The Lie derivative convention for vector fields is

$$[X_M, Y_M] = \frac{d}{dt}\bigg|_0 (\Phi_{-t})_*Y_M.$$

Thus

$$[X_M, Y_M] = \frac{d}{dt}\bigg|_0 (\text{Ad}(\exp(-tX))Y)_M = -[X, Y]_M.$$

Replacing X by $-X$ in the definition removes the sign. $\square$

Remark 4.5.3. This sign is one of the most common sources of silent errors when switching between left group actions, right group actions, and left-invariant vector fields on the group itself. The safest practice is to derive the sign from the flow convention rather than remember it mechanically.

6. Orbits as immersed submanifolds

For a smooth action, the orbit map

$$\pi_x : G \rightarrow M, \quad g \mapsto gx$$

has differential at the identity

$$d(\pi_x)_e(X) = X_M(x).$$

Its kernel is $\mathfrak{g}_x = \text{Lie}(G_x)$.

THEOREM 4.6.1 (Orbit manifold). *For every $x \in M$, the quotient G/G_x is a smooth manifold and the natural map*

$$\iota_x : G/G_x \rightarrow M, \quad gG_x \mapsto gx,$$

is an injective immersion onto the orbit Gx . Thus every orbit carries a canonical immersed-manifold structure of dimension

$$\dim G - \dim G_x.$$

PROOF. The stabilizer G_x is closed, so G/G_x is a smooth manifold. The orbit map factors as

$$G \xrightarrow{\pi} G/G_x \xrightarrow{\iota_x} M.$$

At the identity, $d\pi_e$ has kernel $\mathfrak{g}_x$. We now check carefully that $d(\pi_x)_e$ has the same kernel. If $X \in \text{Lie}(G_x)$, then $\exp(tX)x = x$ for all t , so certainly $d(\pi_x)_e X = 0$. Conversely, suppose $d(\pi_x)_e X = 0$. The curve

$$c(t) = \exp(tX)x$$

is the integral curve through x of the fundamental vector field X_M . Since $X_M(x) = 0$, the constant curve $t \mapsto x$ is another integral curve with the same initial value. Uniqueness for ordinary differential equations gives $c(t) = x$ for all t for which the flow is defined, and the group law then gives this for every $t \in \mathbb{R}$. Hence $\exp(tX) \subset G_x$ and $X \in \text{Lie}(G_x)$.

Thus the two differentials have the same kernel. The induced differential of ι_x is injective at eG_x . Equivariance transports this to every point, so ι_x is an immersion. It is injective by the set-theoretic orbit-stabilizer argument. $\square$

Warning 4.6.2 (Immersed does not mean embedded). The image orbit can fail to have the subspace topology of its canonical manifold structure. The irrational line

$$t \mapsto (t, \alpha t) \bmod \mathbb{Z}^2$$

in $\mathbb{T}^2$ is an injectively immersed copy of $\mathbb{R}$ whose image is dense. As a subspace of the torus it is not homeomorphic to $\mathbb{R}$. Thus orbit geometry and orbit closure are genuinely different questions.

7. Proper actions and embedded orbits

Definition 4.7.1. A continuous action of a locally compact group G on a Hausdorff space X is *proper* if the map

$$G \times X \rightarrow X \times X, \quad (g, x) \mapsto (gx, x),$$

is proper, meaning inverse images of compact sets are compact.

Proposition 4.7.2. *If a Lie group acts properly on a manifold, every stabilizer is compact and every orbit is closed and embedded.*

PROOF. Fix x . The preimage of the compact singleton $\{(x, x)\}$ under the properness map is

$$G_x \times \{x\},$$

so G_x is compact.

If $g_n x \rightarrow y$, choose a relatively compact neighborhood C containing x and all sufficiently large $g_n x$. Properness implies that the set of g carrying x into C has compact closure modulo the fixed point; more directly, (g_n, x)

lies in the inverse image of a compact subset of $X \times X$, so a subsequence g_{n_j} converges to some g . Then $y = gx$, proving the orbit is closed.

The canonical injective immersion $G/G_x \rightarrow X$ is proper when restricted to compact sets by the same argument. A proper injective immersion is an embedding. $\square$

For discrete subgroups acting by translations on a Lie group, proper discontinuity is the corresponding elementary notion. If $\Gamma < G$ is discrete, right multiplication by Γ on G is free and proper, so G/Γ is a smooth manifold and $G \rightarrow G/\Gamma$ is a covering map.

8. Invariant geometric structures

Homogeneous spaces allow geometric structures to be described entirely at one point.

Proposition 4.8.1 (Invariant tensors). *Let G/H be a homogeneous space and $o = eH$. A G -invariant tensor field on G/H is uniquely determined by its value at o . A tensor at o extends to a G -invariant tensor field if and only if it is fixed by the isotropy representation of H .*

PROOF. If T is invariant, then

$$T_{gH} = g_*T_o,$$

so T_o determines the field. Conversely, start with a tensor T_o fixed by H . Define

$$T_{gH} = g_*T_o.$$

If $gH = g'h$, more precisely $g' = gh$ with $h \in H$, then

$$g'_*T_o = g_*h_*T_o = g_*T_o,$$

so the definition is independent of the representative. Smoothness follows from local smooth sections of $G \rightarrow G/H$. $\square$

Corollary 4.8.2 (Invariant Riemannian metrics). *G/H admits a G -invariant Riemannian metric if and only if the H -representation on $\mathfrak{g}/\mathfrak{h}$ preserves some positive-definite inner product.*

PROOF. Apply the preceding invariant-tensor correspondence to positive-definite symmetric bilinear forms on $T_{eH}(G/H) \cong \mathfrak{g}/\mathfrak{h}$. Positive definiteness is preserved under transport by the G -action. $\square$

Worked Example 4.8.3 (Compact isotropy gives an invariant metric). If H is compact, choose any inner product $\langle \cdot, \cdot \rangle_0$ on $\mathfrak{g}/\mathfrak{h}$ and average it over H using normalized Haar probability:

$$\langle v, w \rangle = \int_H \langle hv, hw \rangle_0 dh.$$

The result is positive definite and H -invariant. Hence every homogeneous space G/H with compact H admits a G -invariant Riemannian metric.

Warning 4.8.4. If H is noncompact, an invariant positive-definite inner product may fail to exist. For instance, the isotropy action of the diagonal subgroup on suitable quotient directions can contain nontrivial expansion, which no positive-definite inner product can preserve.

9. Associated bundles

Let $\rho : H \rightarrow \mathrm{GL}(V)$ be a finite-dimensional representation. Define an equivalence relation on $G \times V$ by

$$(g, v) \sim (gh, \rho(h)^{-1}v).$$

The quotient

$$G \times_H V = (G \times V) / \sim$$

is the *associated vector bundle* over G/H , with projection

$$[g, v] \mapsto gH.$$

Proposition 4.9.1. $G \times_H V \rightarrow G/H$ is a smooth vector bundle with fiber V .

PROOF. Take a local smooth section $s : U \rightarrow G$ of $G \rightarrow G/H$. Every g over U can be written uniquely as

$$g = s(x)h$$

with $h \in H$ locally. Then

$$[g, v] = [s(x), \rho(h)v],$$

which identifies the bundle over U with $U \times V$. On overlaps, the transition function is given by the H -valued change of local section followed by ρ , hence is smooth and linear on fibers. $\square$

The tangent bundle itself has this form.

Proposition 4.9.2. There is a natural G -equivariant vector-bundle isomorphism

$$T(G/H) \cong G \times_H (\mathfrak{g}/\mathfrak{h}),$$

where H acts on $\mathfrak{g}/\mathfrak{h}$ by the isotropy representation.

PROOF. Send

$$[g, X + \mathfrak{h}]$$

to the derivative at $t = 0$ of

$$g \exp(tX)H.$$

If g is replaced by gh and $X + \mathfrak{h}$ by $\text{Ad}(h)^{-1}X + \mathfrak{h}$, then

$$gh \exp(t \text{Ad}(h)^{-1}X)H = g \exp(tX)hH = g \exp(tX)H,$$

so the map is well defined. It is fiberwise linear, bijective, smooth in local quotient charts, and G -equivariant. $\square$

10. Classical homogeneous spaces

Worked Example 4.10.1 (Spheres). $\text{SO}(n)$ acts transitively on S^{n-1} . The stabilizer of e_n is the block subgroup isomorphic to $\text{SO}(n-1)$. Hence

$$S^{n-1} \cong \text{SO}(n)/\text{SO}(n-1).$$

The dimension formula gives

$$\frac{n(n-1)}{2} - \frac{(n-1)(n-2)}{2} = n-1.$$

Worked Example 4.10.2 (Real projective space). $\text{GL}_n(\mathbb{R})$ acts transitively on the set of lines in $\mathbb{R}^n$. The stabilizer of the line $\mathbb{R}e_1$ is a parabolic block-upper-triangular subgroup. Thus $\mathbb{P}^{n-1}(\mathbb{R})$ is homogeneous. If one uses $\text{SL}_n(\mathbb{R})$ instead, the action remains transitive and the stabilizer is the intersection with that parabolic subgroup.

Worked Example 4.10.3 (Grassmannians). Let $\text{Gr}(k, n)$ be the set of k -dimensional subspaces of $\mathbb{R}^n$. The action of $\text{GL}_n(\mathbb{R})$ is transitive. The stabilizer of

$$E = \text{span}(e_1, \dots, e_k)$$

consists of block matrices

$$\begin{pmatrix} A & B \\ 0 & D \end{pmatrix}$$

with $A \in \text{GL}_k(\mathbb{R})$ and $D \in \text{GL}_{n-k}(\mathbb{R})$. Hence

$$\text{Gr}(k, n) \cong \text{GL}_n(\mathbb{R})/P_{k, n-k}.$$

The tangent space at E can be identified with

$$\text{Hom}(E, \mathbb{R}^n/E),$$

of dimension $k(n-k)$. To see this from the quotient algebra, the off-diagonal lower-left block gives exactly a linear map $E \rightarrow \mathbb{R}^n/E$.

11. Spaces of lattices as homogeneous spaces

Let

$$X_d = \mathrm{SL}_d(\mathbb{R}) / \mathrm{SL}_d(\mathbb{Z}).$$

The map

$$g \mathrm{SL}_d(\mathbb{Z}) \longmapsto g\mathbb{Z}^d$$

is a bijection from X_d onto the set of unimodular lattices in $\mathbb{R}^d$: two matrices give the same lattice exactly when they differ on the right by an element of $\mathrm{SL}_d(\mathbb{Z})$.

Because $\mathrm{SL}_d(\mathbb{Z})$ is discrete, its Lie algebra is zero. Therefore

$$T_{g \mathrm{SL}_d(\mathbb{Z})} X_d$$

has dimension $d^2 - 1$ and is locally modeled on $\mathfrak{sl}_d(\mathbb{R})$. The quotient map

$$\mathrm{SL}_d(\mathbb{R}) \rightarrow X_d$$

is a local diffeomorphism: around each g , choose a neighborhood U small enough that

$$U^{-1}U \cap \mathrm{SL}_d(\mathbb{Z}) = \{I\}.$$

Then U projects injectively to X_d .

Worked Example 4.11.1 (Moving a lattice infinitesimally). Let $X \in \mathfrak{sl}_d(\mathbb{R})$ and $\Lambda = g\mathbb{Z}^d$. The curve

$$t \mapsto e^{tX} g\mathbb{Z}^d$$

represents the tangent direction X at the lattice Λ after transporting the tangent space by left translation. If X is diagonal, the deformation stretches coordinate directions with total logarithmic volume change $\mathrm{tr} X = 0$. If X is nilpotent, the deformation is a shear.

The affine analogue is

$$Y_d = \mathrm{ASL}_d(\mathbb{R}) / \mathrm{ASL}_d(\mathbb{Z}),$$

where

$$\mathrm{ASL}_d(\mathbb{Z}) = \mathbb{Z}^d \rtimes \mathrm{SL}_d(\mathbb{Z})$$

for the convention $\mathrm{ASL}_d(\mathbb{R}) = \mathbb{R}^d \rtimes \mathrm{SL}_d(\mathbb{R})$. A point represents an affine unimodular lattice

$$v + g\mathbb{Z}^d.$$

This space remembers both the linear shape of the lattice and a translation parameter.

12. Orbits of quadratic forms

Let $V = \text{Sym}_d(\mathbb{R})$ be the vector space of real symmetric $d \times d$ matrices, identified with quadratic forms $Q_A(x) = x^T A x$. Let $G = \text{SL}_d(\mathbb{R})$ act by

$$g \cdot A = (g^{-1})^T A g^{-1},$$

so that

$$(g \cdot Q_A)(x) = Q_A(g^{-1}x).$$

Fix a nonsingular form A . Its stabilizer is

$$H_A = \{g \in \text{SL}_d(\mathbb{R}) : g^T A g = A\}.$$

The orbit is therefore an immersed homogeneous space G/H_A .

Proposition 4.12.1. *The tangent space to the G -orbit of A at A is*

$$\{-X^T A - A X : X \in \mathfrak{sl}_d(\mathbb{R})\} \subset \text{Sym}_d(\mathbb{R}).$$

Its kernel as a map of X is $\mathfrak{h}_A = \{X : X^T A + A X = 0\}$.

PROOF. Differentiate

$$e^{tX} \cdot A = e^{-tX^T} A e^{-tX}$$

at $t = 0$:

$$\left. \frac{d}{dt} \right|_0 e^{-tX^T} A e^{-tX} = -X^T A - A X.$$

The derivative vanishes exactly when X belongs to the infinitesimal stabilizer. $\square$

Remark 4.12.2. This calculation is a useful prototype: an arithmetic object such as a quadratic form is placed inside a linear representation, and the orbit becomes a homogeneous space. Stabilizer calculations then translate geometric questions about the form into subgroup geometry.

13. Left quotients and right quotients

Both

$$G/H = \{gH\}$$

and

$$H \backslash G = \{Hg\}$$

occur naturally. They are diffeomorphic via inversion:

$$G/H \rightarrow H \backslash G, \quad gH \mapsto Hg^{-1}.$$

PROOF. If $gH = g'h$, equivalently $g' = gh$ for some $h \in H$, then

$$Hg'^{-1} = Hh^{-1}g^{-1} = Hg^{-1}.$$

The inverse map is the same construction. Smoothness follows from smooth inversion on G and the quotient-manifold structures. $\square$

The distinction is nevertheless important for dynamics. On G/H , left multiplication by G is canonical. On $H \backslash G$, right multiplication by G is canonical. Inversion converts one convention into the other but also reverses the order of multiplication. Whenever a flow is written as right translation on $\Gamma \backslash G$ or left translation on G/Γ , the quotient convention should therefore be visible.

14. Equivariant maps between homogeneous spaces

Homogeneous spaces admit a useful classification of equivariant maps. It turns questions about maps into subgroup inclusions.

Proposition 4.14.1. *Let $H, K < G$ be closed subgroups. A G -equivariant map*

$$F : G/H \rightarrow G/K$$

is determined by the image of eH . If

$$F(eH) = g_0K,$$

then such a map exists if and only if

$$H \subset g_0Kg_0^{-1}.$$

When this holds,

$$F(gH) = gg_0K.$$

PROOF. Equivariance forces

$$F(gH) = gF(eH) = gg_0K,$$

so any equivariant map with the prescribed value at eH has exactly this value at every coset; uniqueness follows pointwise. This formula is well defined precisely when every $h \in H$ fixes g_0K :

$$hg_0K = g_0K \iff g_0^{-1}hg_0 \in K.$$

This is equivalent to $H \subset g_0Kg_0^{-1}$. Smoothness follows from the quotient-manifold theorem because the formula is induced by smooth multiplication on G . $\square$

Worked Example 4.14.2 (Forgetting scale). For $G = \mathrm{SL}_2(\mathbb{R})$ we have $N < B$. Therefore there is a canonical G -equivariant map

$$G/N \rightarrow G/B.$$

Under

$$G/N \cong \mathbb{R}^2 \setminus \{0\}, \quad G/B \cong \mathbb{P}^1(\mathbb{R}),$$

this map is exactly

$$v \mapsto \mathbb{R}v.$$

The subgroup inclusion $N < B$ is therefore the algebraic expression of “forget the length and remember only the direction.”

15. Double cosets and relative position

Suppose H acts by left multiplication on G/K . Two points gK and $g'K$ lie in the same H -orbit exactly when

$$g' \in HgK.$$

Thus the set of H -orbits on G/K is the double-coset space

$$H \backslash G/K.$$

The stabilizer in H of gK is

$$H \cap gKg^{-1}.$$

Hence the orbit has canonical immersed structure

$$H/(H \cap gKg^{-1}).$$

Worked Example 4.15.1 (Relative position of two boundary points). Let $G = \mathrm{PSL}_2(\mathbb{R})$ and B the stabilizer of ∞ . The double cosets

$$B \backslash G/B$$

correspond to B -orbits on the boundary G/B . There are exactly two: the fixed point ∞ and its complement $\mathbb{R}$. Algebraically this is the two-cell Bruhat decomposition

$$G = B \sqcup BwB.$$

Thus a double-coset decomposition is often a precise form of the geometric statement that two objects can occupy only finitely many relative positions.

16. Subgroup orbits inside G/Γ

Let $\Gamma < G$ be discrete and let $H < G$ be a closed subgroup. Fix

$$x = g\Gamma \in G/\Gamma.$$

The H -orbit is

$$Hx = Hg\Gamma/\Gamma.$$

Its stabilizer in H is

$$H_x = H \cap g\Gamma g^{-1}.$$

Because $g\Gamma g^{-1}$ is discrete, H_x is discrete in H .

Proposition 4.16.1. *The orbit map induces an injective immersion*

$$\Phi : H/(H \cap g\Gamma g^{-1}) \longrightarrow G/\Gamma, \quad h(H \cap g\Gamma g^{-1}) \mapsto hg\Gamma.$$

If the orbit Hx is closed, then Φ is a homeomorphism onto Hx and hence an embedding.

Guiding Idea 4.16.2 (Why closedness is enough here). A closed injectively immersed submanifold need not be embedded in complete generality: a remote part of an immersed curve can accumulate back on a local branch. Homogeneous orbits have extra structure. The source H/H_x is σ -compact, while a closed orbit in the manifold G/Γ is locally compact and therefore a Baire space. Baire category forces one compact piece of the source to cover an open piece of the orbit. Equivariance moves that control to any chosen point, and a local immersion chart then excludes all remote branches.

PROOF. Write

$$\Lambda = H \cap g\Gamma g^{-1}, \quad Y = Hx.$$

The stabilizer computation and the orbit theorem give the continuous bijective immersion

$$\Phi : H/\Lambda \rightarrow Y.$$

It remains only to prove continuity of the inverse when Y is closed.

Because H/Λ is a second-countable locally compact manifold, it is σ -compact. Choose compact sets $K_1 \subset K_2 \subset \cdots$ whose union is H/Λ . Each $\Phi(K_n)$ is compact, hence closed in the Hausdorff space Y . Since Y is closed in the manifold G/Γ , it is locally compact Hausdorff and therefore a Baire space. From

$$Y = \bigcup_{n \geq 1} \Phi(K_n)$$

we conclude that some $\Phi(K_n)$ has nonempty interior in Y . By H -equivariance and transitivity of the action on Y , after translating both source and target

we obtain a compact set $K \subset H/\Lambda$ and an open neighborhood O of x in Y such that

$$O \subset \Phi(K).$$

Consequently

$$\Phi^{-1}(O) \subset K.$$

Now let V be any neighborhood of the base point $e\Lambda$ in H/Λ . Since Φ is an immersion, after shrinking inside V we can choose a neighborhood V_0 of $e\Lambda$ on which Φ is an embedding onto a local immersed patch. The compact set

$$K \setminus V_0$$

has compact image under Φ , and this image does not contain x because Φ is injective. Therefore

$$W = O \setminus \Phi(K \setminus V_0)$$

is an open neighborhood of x in Y . If $z \in \Phi^{-1}(W)$, then $z \in K$ because $W \subset O$, while $z \notin K \setminus V_0$ by the definition of W . Hence $z \in V_0 \subset V$. We have proved

$$\Phi^{-1}(W) \subset V,$$

which is continuity of Φ^{-1} at the base point. Equivariance gives continuity at every point. Thus Φ is a homeomorphism onto Y ; being also an injective immersion, it is an embedding. $\square$

The phrase *closed homogeneous orbit* therefore has a precise meaning: it is an embedded copy of a quotient of a subgroup by a discrete stabilizer.

THEOREM 4.16.3 (Invariant measure on a closed subgroup orbit). *Assume G is a Lie group, $\Gamma < G$ is discrete, $H < G$ is closed, and Hx is closed for $x = g\Gamma$. Put*

$$\Lambda = H \cap g\Gamma g^{-1}.$$

Then:

- (1) *Hx carries an H -invariant Radon measure if and only if*

$$\Delta_H(\lambda) = 1$$

for every $\lambda \in \Lambda$;

- (2) *if H is unimodular, such a measure always exists;*

- (3) *this invariant measure is finite if and only if Λ is a lattice in H .*

PROOF. The orbit is identified with H/Λ . We prove the required criterion directly in this discrete-fiber case.

Let dh be left Haar measure on H . Right translation by $\lambda \in \Lambda$ satisfies

$$d(h\lambda) = \Delta_H(\lambda)^{-1} dh$$

in the sense of measures on sets. If $\Delta_H(\lambda) = 1$ for every $\lambda \in \Lambda$, then dh is right Λ -invariant. Because $H \rightarrow H/\Lambda$ is a covering map, choose a small open set U whose right translates $U\lambda$ are disjoint. Define the quotient measure on the image of U by transporting $dh|_U$. Right Λ -invariance makes this definition independent of the chosen lift, and the local definitions patch to an H -invariant Radon measure on H/Λ .

Conversely, suppose H/Λ carries a nonzero H -invariant Radon measure $d\mu$. On compactly supported functions define

$$I(f) = \int_{H/\Lambda} \sum_{\lambda \in \Lambda} f(h\lambda) d\mu(h\Lambda).$$

The sum is locally finite, and I is a nonzero left Haar measure on H . Replacing $f(h)$ by $f(h\lambda_0)$ merely permutes the summation index, so $I(f(\cdot\lambda_0)) = I(f)$. On the other hand, the defining modular relation for the left Haar measure I gives

$$I(f(\cdot\lambda_0)) = \Delta_H(\lambda_0)I(f).$$

Choosing $f \geq 0$ with $I(f) > 0$ yields $\Delta_H(\lambda_0) = 1$. This proves the first assertion without using any more general quotient-measure theorem.

If H is unimodular, the condition is automatic. Finally, the invariant measure has finite total mass exactly when the discrete subgroup Λ is a lattice in H , which is the definition of a lattice. $\square$

Worked Example 4.16.4 (A periodic unipotent orbit). Let

$$G = \mathrm{SL}_2(\mathbb{R}), \quad \Gamma = \mathrm{SL}_2(\mathbb{Z}), \quad H = N.$$

At the standard point $x = e\Gamma$,

$$N \cap \Gamma = \{u_n : n \in \mathbb{Z}\} \cong \mathbb{Z}.$$

Thus

$$Nx \cong N/(N \cap \Gamma) \cong \mathbb{R}/\mathbb{Z}.$$

It is a closed circle carrying normalized Lebesgue measure. The calculation illustrates every piece of the general theorem: the orbit stabilizer is an arithmetic intersection, N is unimodular, and the stabilizer is a lattice in N .

17. Closedness versus finite volume

Closedness and finite invariant volume are logically distinct. A subgroup orbit can be closed but have infinite volume.

Worked Example 4.17.1 (Closed but infinite-volume orbit). Take

$$G = \mathbb{R}^2, \quad \Gamma = \mathbb{Z}^2,$$

and let $H = \mathbb{R} \times \{0\}$. The H -orbit through the origin in the torus is a closed circle, and its stabilizer is $\mathbb{Z} \times \{0\}$, a lattice in H , so the orbit has finite H -volume.

Now instead take $G = \mathbb{R}^3$, $\Gamma = \mathbb{Z}^2 \times \{0\}$ and $H = \mathbb{R}^3$ itself. The quotient $G/\Gamma \cong \mathbb{T}^2 \times \mathbb{R}$ is closed as the full orbit but has infinite invariant volume. The stabilizer Γ is discrete but not a lattice in $\mathbb{R}^3$ because it has rank $2 < 3$.

Conversely, in the standard locally compact setting a finite invariant measure supported on one immersed homogeneous orbit strongly constrains the orbit topology: the stabilizer must be a lattice and the quotient has finite volume, but embedded closedness is an additional geometric assertion unless it is built into the term “homogeneous subspace.” Keeping these notions separate avoids circular arguments.

18. Homogeneous maps and stabilizer inclusions as a proof tool

A useful proof strategy is to replace a geometric construction by a stabilizer calculation. Suppose one suspects that a parameter space Y is G/H . Choose a base object $y_0 \in Y$, compute its stabilizer H , and prove transitivity. The set-theoretic bijection follows immediately. To obtain a diffeomorphism, produce local sections or apply the orbit-manifold theorem. To obtain an invariant measure, test the modular functions. In this way topology, smooth structure, and measure are verified in separate steps rather than bundled into one informal identification.

This separation is particularly valuable in research arguments, where a sentence such as “the orbit is H/Λ ” may conceal three different claims: a bijection, an embedded topology, and compatibility of invariant measures. Each has its own hypothesis.

19. Conceptual summary

A closed subgroup $H < G$ determines a smooth homogeneous space G/H whose geometry is controlled at the base point by the quotient representation

$$T_{eH}(G/H) \cong \mathfrak{g}/\mathfrak{h}.$$

The stabilizer acts on this tangent space by

$$X + \mathfrak{h} \mapsto \text{Ad}(h)X + \mathfrak{h}.$$

This single representation governs invariant metrics, tensor fields, and associated bundles.

For a general smooth action, every orbit is canonically an immersed homogeneous space G/G_x . It need not be embedded or closed: dense irrational torus lines are the basic warning. Spaces of lattices, affine lattices, Grassmannians, projective spaces, spheres, and orbits of quadratic forms all fit the

same framework. The value of the framework is precisely that geometric and arithmetic parameter spaces can be treated by Lie-group methods without losing their concrete interpretation.

CHAPTER 5

Haar Measure and the Modular Function

1. Why an invariant measure is structural

A homogeneous dynamical system is not only a topological space with a group action. Its most rigid statements are measure-theoretic: recurrence, ergodicity, invariant probability measures, and equidistribution all require a canonical way to measure sets. On a Lie group the natural measure is Haar measure.

There are two possible invariance requirements. A measure m on G is *left invariant* if

$$m(gE) = m(E)$$

for all Borel sets E and all $g \in G$, and *right invariant* if

$$m(Eg) = m(E).$$

For abelian or compact groups the distinction disappears. For a general nonabelian group it does not. The discrepancy is measured by the modular function.

Throughout, a *Radon measure* on a finite-dimensional manifold means a Borel measure finite on compact sets and inner regular on open sets. Such measures integrate compactly supported continuous functions and interact well with smooth coordinate changes.

2. Invariant densities on a Lie group

A density on an n -dimensional vector space V is a function on ordered bases that transforms by absolute determinant. Unlike a top-degree differential form, a positive density does not require an orientation. Every smooth manifold admits smooth positive densities.

Let G be an n -dimensional Lie group. Choose a positive density ω_e on $\mathfrak{g} = T_e G$. Transport it by left translation:

$$\omega_g = (L_{g^{-1}})^* \omega_e.$$

Equivalently, ω_g is the unique density satisfying

$$(L_g)^* \omega_g = \omega_e.$$

Proposition 5.2.1. *The resulting smooth density ω is left invariant:*

$$L_h^* \omega = \omega$$

for every $h \in G$.

PROOF. At $g \in G$,

$$(L_h^* \omega)_g = (dL_h)_g^* \omega_{hg}.$$

By definition,

$$\omega_{hg} = ((dL_{(hg)^{-1}})_{hg})^* \omega_e.$$

Since

$$L_{(hg)^{-1}} \circ L_h = L_{g^{-1}},$$

the chain rule gives

$$(dL_h)_g^* \omega_{hg} = ((dL_{g^{-1}})_g)^* \omega_e = \omega_g.$$

□

Integrating ω gives a Radon measure m_G . Smooth positive densities are locally comparable with Lebesgue measure in charts, so they are finite on compact sets.

THEOREM 5.2.2 (Existence of Haar measure for Lie groups). *Every Lie group admits a nonzero left-invariant Radon measure and a nonzero right-invariant Radon measure.*

PROOF. The left-invariant density above gives a left-invariant Radon measure. Applying inversion $\iota(g) = g^{-1}$ to a left Haar measure gives a right-invariant measure:

$$m_R(E) = m_L(E^{-1}).$$

Indeed,

$$m_R(Eg) = m_L((Eg)^{-1}) = m_L(g^{-1}E^{-1}) = m_L(E^{-1}) = m_R(E).$$

□

3. Uniqueness

THEOREM 5.3.1 (Uniqueness of Haar measure for Lie groups). *Any two nonzero left-invariant Radon measures on a finite-dimensional Lie group are positive scalar multiples of one another. The same is true for right-invariant Radon measures.*

This is exactly the scope used throughout the series. A complete approximate-identity proof is given in Appendix A. The more general locally compact-group theorem is standard background but is neither needed nor claimed as an internally proved result here.

Guiding Idea 5.3.2 (Why there can be only one). Left translations act transitively on G . If two locally finite measures are both unchanged by every left translation, then whatever ratio they have on one sufficiently small neighborhood must be reproduced on every translate. Regularity allows arbitrary compact sets to be assembled from such small pieces. The only freedom that survives is one global multiplicative constant.

Because of uniqueness, we may speak of *a* left Haar measure, understanding that its normalization is arbitrary unless additional structure fixes it. On a compact group it is customary to normalize total mass to 1.

Worked Example 5.3.3 (Euclidean space). On $(\mathbb{R}^d, +)$, Lebesgue measure dx is invariant under translations. Uniqueness says that every translation-invariant Radon measure is $c dx$ for some $c \geq 0$.

Worked Example 5.3.4 (Discrete groups). A discrete group carries counting measure

$$m(E) = \#E$$

for finite sets and the natural extension to all subsets. It is both left and right invariant. Any Haar measure is a positive multiple of counting measure.

4. The modular function: convention and definition

Fix a left Haar measure m . For $g \in G$, the measure

$$E \mapsto m(Eg)$$

is again left invariant, because

$$m(hEg) = m(Eg).$$

By uniqueness, there exists a positive scalar depending only on g .

Definition 5.4.1 (Modular function). Our convention is

$$m(Eg) = \Delta_G(g)^{-1}m(E).$$

The function

$$\Delta_G : G \rightarrow \mathbb{R}_{>0}$$

is called the *modular function*.

Warning 5.4.2 (Conventions differ). Some authors define the reciprocal modular function. Every formula involving Δ_G should therefore be checked against its defining identity. In these notes the defining identity is always

$$m(Eg) = \Delta_G(g)^{-1}m(E).$$

Proposition 5.4.3. Δ_G is a continuous group homomorphism:

$$\Delta_G(gh) = \Delta_G(g)\Delta_G(h).$$

PROOF. For Borel E of finite positive measure,

$$\begin{aligned} m(Egh) &= \Delta_G(h)^{-1} m(Eg) \\ &= \Delta_G(h)^{-1} \Delta_G(g)^{-1} m(E). \end{aligned}$$

Comparing with the definition for gh gives the homomorphism law.

For continuity, choose $f \in C_c(G)$, $f \geq 0$, with $\int f dm > 0$. Right translation is continuous in the C_c topology, and

$$\int_G f(xg^{-1}) dm(x) = \Delta_G(g)^{-1} \int_G f(x) dm(x)$$

with the chosen convention after a change of variables. The left side depends continuously on g by uniform continuity on a common compact support neighborhood and dominated convergence. Hence Δ_G is continuous. $\square$

Definition 5.4.4. A locally compact group is *unimodular* if

$$\Delta_G \equiv 1.$$

Equivalently, a left Haar measure is also right invariant.

5. The Lie-group formula $\Delta_G(g) = |\det \text{Ad}(g)|$

For Lie groups the modular function is explicitly computable from the adjoint representation.

THEOREM 5.5.1. *Let G be a Lie group. With the convention above,*

$$\Delta_G(g) = |\det \text{Ad}(g)|.$$

PROOF. Let ω be a left-invariant positive density. The pullback $R_g^* \omega$ is again left invariant, so it must be a positive scalar multiple of ω . Compute that scalar at the identity.

The differential

$$(dR_g)_e : T_e G \rightarrow T_g G$$

followed by left translation back to the identity gives

$$(dL_{g^{-1}})_g (dR_g)_e X.$$

For the curve $\exp(tX)$,

$$g^{-1}(\exp(tX)g) = g^{-1} \exp(tX)g = \exp(t \text{Ad}(g^{-1})X).$$

Therefore the composite linear map is $\text{Ad}(g^{-1})$. Since densities transform by absolute determinant,

$$(R_g^* \omega)_e = \left| \det \text{Ad}(g^{-1}) \right| \omega_e = |\det \text{Ad}(g)|^{-1} \omega_e.$$

Thus

$$R_g^* \omega = |\det \text{Ad}(g)|^{-1} \omega.$$

Integrating over E gives

$$m(Eg) = |\det \operatorname{Ad}(g)|^{-1} m(E).$$

Comparing with the definition yields the formula. $\square$

Corollary 5.5.2 (Infinitesimal modular character). *For $X \in \mathfrak{g}$,*

$$\Delta_G(\exp X) = e^{\operatorname{tr}(\operatorname{ad} X)}.$$

PROOF. Since

$$\operatorname{Ad}(\exp X) = e^{\operatorname{ad} X},$$

we have

$$\det \operatorname{Ad}(\exp X) = \det(e^{\operatorname{ad} X}) = e^{\operatorname{tr}(\operatorname{ad} X)}.$$

The exponential is positive, so absolute values are unnecessary. $\square$

Corollary 5.5.3. *A connected Lie group is unimodular if and only if*

$$\operatorname{tr}(\operatorname{ad} X) = 0$$

for every $X \in \mathfrak{g}$.

PROOF. If the traces vanish, Δ_G equals 1 on every exponential. A connected Lie group is generated by exponentials, so $\Delta_G \equiv 1$. The converse follows by differentiating $\Delta_G(\exp(tX)) = 1$. $\square$

6. Important unimodular classes

Proposition 5.6.1. *Every abelian Lie group is unimodular.*

PROOF. Conjugation is trivial, so $\operatorname{Ad}(g) = I$. $\square$

Proposition 5.6.2. *Every compact Lie group is unimodular.*

PROOF. The image $\Delta_G(G)$ is a compact subgroup of the multiplicative group $\mathbb{R}_{>0}$. Applying logarithm gives a compact subgroup of $(\mathbb{R}, +)$, which must be $\{0\}$. Hence $\Delta_G(G) = \{1\}$. $\square$

Proposition 5.6.3. *Every connected nilpotent Lie group is unimodular.*

PROOF. If $\mathfrak{g}$ is nilpotent, then for each X the operator $\operatorname{ad} X$ is nilpotent. One way to see this is that

$$(\operatorname{ad} X)^k Y = [X, [X, \dots, [X, Y] \dots]]$$

lies in the $(k+1)$ st term of the lower central series, which eventually vanishes. A nilpotent linear operator has trace zero. Thus $\operatorname{tr}(\operatorname{ad} X) = 0$ for all X , and connectedness gives unimodularity. $\square$

Proposition 5.6.4. $\operatorname{SL}_n(\mathbb{R})$ *is unimodular.*

PROOF. For $g \in \mathrm{SL}_n(\mathbb{R})$, conjugation on the full matrix space $M_n(\mathbb{R})$ is

$$T_g : X \mapsto gXg^{-1}.$$

Left multiplication $X \mapsto gX$ has determinant $(\det g)^n = 1$, because it acts independently on the n columns. Right multiplication by g^{-1} has determinant $(\det g^{-1})^n = 1$, because it acts independently on the rows. Hence $\det T_g = 1$ on $M_n(\mathbb{R})$.

The decomposition

$$M_n(\mathbb{R}) = \mathbb{R}I \oplus \mathfrak{sl}_n(\mathbb{R})$$

is invariant under conjugation, and T_g acts as the identity on $\mathbb{R}I$. Therefore its determinant on $\mathfrak{sl}_n(\mathbb{R})$ is also 1. Hence $\Delta_{\mathrm{SL}_n}(g) = 1$. $\square$

7. A nonunimodular model: the affine group

Consider

$$G = \{(a, b) : a > 0, b \in \mathbb{R}\}$$

with multiplication corresponding to $x \mapsto ax + b$:

$$(a, b)(a', b') = (aa', b + ab').$$

Proposition 5.7.1. *A left Haar measure is*

$$dm_L(a, b) = \frac{da db}{a^2},$$

while a right Haar measure is

$$dm_R(a, b) = \frac{da db}{a}.$$

The modular function is

$$\Delta_G(a, b) = a.$$

PROOF. Left translation by (a_0, b_0) sends

$$(a, b) \mapsto (a_0a, b_0 + a_0b).$$

The Euclidean Jacobian is a_0^2 . Meanwhile the first coordinate changes from a to a_0a , so

$$\frac{d(a_0a) d(b_0 + a_0b)}{(a_0a)^2} = \frac{a_0^2 da db}{a_0^2 a^2} = \frac{da db}{a^2}.$$

Thus $a^{-2}da db$ is left invariant.

For right translation,

$$(a, b)(a_0, b_0) = (aa_0, b + ab_0).$$

The Jacobian is a_0 . Hence

$$m_L(E(a_0, b_0)) = a_0^{-1}m_L(E).$$

By our convention, $\Delta_G(a_0, b_0) = a_0$.

For the right Haar measure, a direct Jacobian computation gives invariance of $a^{-1}da db$. $\square$

Worked Example 5.7.2 (Adjoint verification). Use Lie algebra basis

$$H = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

in the matrix realization

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}.$$

Then $[H, X] = X$. The element (a, b) acts on the translation direction by a factor a , while the induced action on the quotient by translations is trivial. Hence

$$\det \text{Ad}(a, b) = a,$$

in agreement with the modular function.

8. Haar measure on closed subgroups

Every closed subgroup $H < G$ is itself a Lie group and therefore has its own left Haar measure m_H and modular function Δ_H . There is no reason for m_H to be the restriction of m_G . If H has smaller dimension, the ambient Haar measure of H is usually zero.

Worked Example 5.8.1 (Diagonal and unipotent subgroups of SL_2). For

$$A = \{a_t : t \in \mathbb{R}\}, \quad N = \{u_s : s \in \mathbb{R}\},$$

the maps $t \mapsto a_t$ and $s \mapsto u_s$ are group isomorphisms from $(\mathbb{R}, +)$. Thus Haar measures are constant multiples of dt and ds . Both groups are abelian, hence unimodular.

The upper triangular subgroup AN is not unimodular.

Proposition 5.8.2. *For $P = AN$ with elements $a_t u_s$, the modular function is*

$$\Delta_P(a_t u_s) = e^t.$$

PROOF. Its Lie algebra is spanned by $H/2$ and X with

$$[H/2, X] = X.$$

Thus $\text{ad}(H/2)$ has trace 1 on $\mathfrak{p}$, while $\text{ad } X$ has trace 0. Therefore

$$\Delta_P(a_t) = e^t.$$

The subgroup N lies in the kernel because $\text{ad } X$ is nilpotent. Since every element is $a_t u_s$ and Δ is a homomorphism, the formula follows. $\square$

9. Integration formulas and change of variables

Let m be left Haar measure. The defining relation implies the following useful formula.

Proposition 5.9.1. *For $f \in C_c(G)$ and $g \in G$,*

$$\int_G f(xg) dm(x) = \Delta_G(g) \int_G f(x) dm(x).$$

Equivalently,

$$\int_G f(xg^{-1}) dm(x) = \Delta_G(g)^{-1} \int_G f(x) dm(x).$$

Also

$$\int_G f(gx) dm(x) = \int_G f(x) dm(x).$$

PROOF. The last identity is left invariance. For the first, test initially on $f = \mathbf{1}_E$:

$$\int_G \mathbf{1}_E(xg) dm(x) = m(Eg^{-1}) = \Delta_G(g^{-1})^{-1} m(E) = \Delta_G(g) m(E).$$

By linearity the identity holds for nonnegative simple functions. For $f \in C_c(G)$ with $f \geq 0$, choose simple functions $s_n \uparrow f$ and use monotone convergence on both sides; then decompose an arbitrary real or complex f into positive and negative real and imaginary parts. Thus the identity holds for every $f \in C_c(G)$. Replacing g by g^{-1} gives the equivalent second formula. This indicator-function check is the safest way to prevent reciprocal errors in the modular convention. $\square$

In a unimodular group inversion also preserves Haar measure.

Proposition 5.9.2. *If G is unimodular and m is Haar measure, then*

$$\int_G f(x^{-1}) dm(x) = \int_G f(x) dm(x).$$

PROOF. The push-forward of a left Haar measure by inversion is right Haar. In a unimodular group the original measure is also right Haar. By uniqueness the two differ by a scalar. Applying inversion twice forces the scalar to have square 1, and positivity makes it 1 after matching the normalization at one compact set; equivalently, choose the right Haar measure to be the same normalization. $\square$

10. Haar measure in $\mathrm{SL}_2(\mathbb{R})$ coordinates

Normalize Haar probability dk on $K = \mathrm{SO}(2)$, so that $\int_K dk = 1$. Haar measure on G is determined only up to an overall scalar, so coordinate formulas are stated with a constant $C > 0$.

THEOREM 5.10.1 (*NAK coordinate formula*). *In the coordinates*

$$g = u_x a_t k_\theta,$$

a left Haar measure on G has the form

$$dg = C e^{-t} dx dt d\theta,$$

or, after normalizing angular measure $dk = d\theta/(2\pi)$,

$$dg = C' e^{-t} dx dt dk.$$

Equivalently, with $y = e^t$,

$$dg = C' \frac{dx dy}{y^2} dk.$$

PROOF. A direct way to avoid any hidden invariance assumption is to compute the left Maurer–Cartan form

$$\Theta = g^{-1} dg,$$

whose components in a fixed basis of $\mathfrak{sl}_2(\mathbb{R})$ are left-invariant one-forms. Their wedge is therefore a left-invariant volume form.

Write

$$g = u_x a_t k_\theta, \quad X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Then

$$u_x^{-1} du_x = X dx, \quad a_t^{-1} da_t = \frac{H}{2} dt, \quad k_\theta^{-1} dk_\theta = J d\theta.$$

Using the product rule for $g^{-1} dg$,

$$\Theta = \mathrm{Ad}(k_\theta^{-1}) \left(\mathrm{Ad}(a_t^{-1})(X) dx + \frac{H}{2} dt \right) + J d\theta.$$

Since

$$\mathrm{Ad}(a_t^{-1})X = e^{-t}X,$$

the coefficient matrix of the three one-forms relative to the basis $(X, H/2, J)$ has determinant e^{-t} times the determinant contributed by $\mathrm{Ad}(k_\theta^{-1})$. The latter has absolute determinant 1 because K is compact (equivalently because G is unimodular). Hence the absolute value of the wedge is

$$C e^{-t} dx dt d\theta.$$

This is a left-invariant density, so it is Haar measure. Substituting $y = e^t$ gives $e^{-t} dt = dy/y^2$. $\square$

THEOREM 5.10.2 (*KAN coordinate formula*). *In the Iwasawa coordinates*

$$g = ka_t u_s,$$

Haar measure has the form

$$dg = Ce^t dk dt ds.$$

PROOF. G is unimodular, so inversion preserves Haar measure. Inverting $u_x a_t k$ gives

$$(u_x a_t k)^{-1} = k^{-1} a_{-t} u_{-x}.$$

Thus the NAK variables (x, t, k) become KAN variables

$$(k', t', s) = (k^{-1}, -t, -x).$$

Haar probability on K is invariant under inversion, and the Euclidean Jacobian in (x, t) is 1. The factor e^{-t} therefore becomes $e^{t'}$. $\square$

Warning 5.10.3. The opposite factors e^{-t} and e^t are not contradictory. They correspond to different orders of coordinates. Confusing NAK with KAN is a common source of sign errors in hyperbolic and homogeneous calculations.

11. Compact quotients and normalization

If $\Gamma < G$ is discrete and G/Γ carries a finite G -invariant measure, it is natural to normalize the total mass to 1. This produces an invariant probability space. The existence of such a finite quotient measure is precisely what distinguishes a lattice from an arbitrary discrete subgroup.

Even before quotient measures are constructed explicitly, Haar measure explains why the notion is geometric: a fundamental region for Γ should have finite Haar volume. When G is unimodular, left and right conventions are compatible; in a nonunimodular group, finite-volume quotients are much more constrained.

12. Modular functions of vector semidirect products

The affine example has a useful generalization. Let H act linearly on $V = \mathbb{R}^d$ through a smooth representation

$$\rho : H \rightarrow \mathrm{GL}(V),$$

and form

$$G = V \rtimes_{\rho} H, \quad (v, h)(w, k) = (v + \rho(h)w, hk).$$

Proposition 5.12.1. *Let dh be left Haar measure on H . A left Haar measure on G is*

$$dg = \frac{dv dh}{|\det \rho(h)|},$$

where dv is Lebesgue measure on V . Its modular function is

$$\Delta_G(v, h) = \Delta_H(h) |\det \rho(h)|.$$

PROOF. Left multiplication by (v_0, h_0) sends

$$(v, h) \mapsto (v_0 + \rho(h_0)v, h_0h).$$

Lebesgue measure in v acquires the Jacobian $|\det \rho(h_0)|$, while left Haar measure dh is unchanged. The denominator at the new H coordinate is

$$|\det \rho(h_0h)| = |\det \rho(h_0)| |\det \rho(h)|,$$

so the factors cancel. Thus the displayed measure is left invariant.

To compute right translation by (v_0, h_0) , observe that

$$(v, h)(v_0, h_0) = (v + \rho(h)v_0, hh_0).$$

For fixed h , translation in the v variable has Jacobian 1. Right translation $h \mapsto hh_0$ changes left Haar measure on H by the factor $\Delta_H(h_0)^{-1}$ on sets, while the denominator changes by

$$|\det \rho(hh_0)| = |\det \rho(h)| |\det \rho(h_0)|.$$

Hence the measure of a right-translated set is multiplied by

$$(\Delta_H(h_0) |\det \rho(h_0)|)^{-1}.$$

Comparison with the defining modular convention proves the formula. $\square$

Worked Example 5.12.2 (The special affine group). For

$$\mathrm{ASL}_d(\mathbb{R}) = \mathbb{R}^d \rtimes \mathrm{SL}_d(\mathbb{R}),$$

$H = \mathrm{SL}_d(\mathbb{R})$ is unimodular and $|\det \rho(h)| = |\det h| = 1$. Therefore $\mathrm{ASL}_d(\mathbb{R})$ is unimodular. A left Haar measure is simply the product of Lebesgue measure on translations and Haar measure on $\mathrm{SL}_d(\mathbb{R})$.

Worked Example 5.12.3 (Scalar expansion in dimension d). Let $H = \mathbb{R}$ act on $\mathbb{R}^d$ by

$$\rho(t) = e^t I.$$

Then $|\det \rho(t)| = e^{dt}$, so the semidirect product has

$$\Delta_G(v, t) = e^{dt}.$$

The exponent d counts the number of expanding translation directions. The one-dimensional $ax + b$ group is the case $d = 1$.

13. Divergence and the modular character

There is an infinitesimal interpretation of $\text{tr}(\text{ad } X)$. Let ω be a left Haar density and let X^R be the right-invariant vector field generated by X ,

$$X^R(g) = (dR_g)_e X.$$

Its flow is left multiplication by e^{tX} and therefore preserves left Haar measure. By contrast, the left-invariant vector field $X^L(g) = (dL_g)_e X$ has flow $g \mapsto ge^{tX}$, a right translation, and its flow changes Haar measure by the modular factor.

Proposition 5.13.1. *With respect to a left Haar density,*

$$\text{div}(X^L) = -\text{tr}(\text{ad } X),$$

while

$$\text{div}(X^R) = 0.$$

PROOF. The flow of X^L is $R_{e^{tX}}$. Since

$$R_{e^{tX}}^* \omega = \Delta_G(e^{tX})^{-1} \omega = e^{-t \text{tr}(\text{ad } X)} \omega,$$

differentiating at $t = 0$ gives the Lie derivative

$$\mathcal{L}_{X^L} \omega = -\text{tr}(\text{ad } X) \omega.$$

By definition the coefficient is the divergence. The flow of X^R is left translation and preserves ω , so its divergence is zero. $\square$

This result explains geometrically why the trace of $\text{ad } X$ is the infinitesimal modular character: it is the rate at which the flow of the left-invariant vector field changes left Haar volume.

14. Haar measure on $GL_n(\mathbb{R})$

Lebesgue measure dA on the vector space $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$ is not Haar measure on $GL_n(\mathbb{R})$ because left multiplication changes Euclidean volume.

Proposition 5.14.1. *A left and right Haar measure on $GL_n(\mathbb{R})$ is*

$$dg = |\det g|^{-n} dA.$$

Thus $GL_n(\mathbb{R})$ is unimodular.

PROOF. Under left multiplication $A \mapsto g_0 A$, each of the n columns is multiplied by g_0 , so the Euclidean Jacobian is

$$|\det g_0|^n.$$

The determinant factor transforms as

$$|\det(g_0 A)|^{-n} = |\det g_0|^{-n} |\det A|^{-n},$$

which cancels the Jacobian. Hence the measure is left invariant.

Under right multiplication $A \mapsto Ag_0$, each of the n rows is transformed by g_0^T , again with total Jacobian $|\det g_0|^n$, and the same determinant factor cancels it. Therefore the measure is also right invariant. $\square$

Remark 5.14.2. The formula is singular at the determinant-zero hypersurface, but this causes no problem: Haar measure lives only on the open set $\mathrm{GL}_n(\mathbb{R})$. On compact subsets bounded away from singular matrices, the density is smooth and locally comparable to ordinary Lebesgue measure.

15. Why compact subgroups never create modular obstructions

Let $K < G$ be compact. Since $\Delta_G|_K$ is a continuous homomorphism from a compact group to $\mathbb{R}_{>0}$, it is trivial. Thus right translation by elements of K preserves every left Haar measure on G , even when G itself is nonunimodular.

This small observation has two useful consequences. First, quotienting by a compact subgroup never fails the invariant-measure criterion. Second, compact angular variables in decompositions such as NAK can be normalized independently by Haar probability on K without introducing a modular correction.

16. Conceptual summary

A Lie group carries a canonical measure class obtained by transporting a density from the identity. The normalization is arbitrary, but the measure class and its invariance are intrinsic. The failure of left Haar measure to be right invariant is measured by

$$\Delta_G(g) = |\det \mathrm{Ad}(g)|,$$

with our convention

$$m(Eg) = \Delta_G(g)^{-1}m(E).$$

At the Lie algebra level,

$$\Delta_G(\exp X) = e^{\mathrm{tr}(\mathrm{ad} X)}.$$

Abelian, compact, connected nilpotent, and $\mathrm{SL}_n(\mathbb{R})$ groups are unimodular. The affine group is the basic counterexample. In $\mathrm{SL}_2(\mathbb{R})$, keeping the normalization

$$a_t = \mathrm{diag}(e^{t/2}, e^{-t/2})$$

fixed gives the coordinate formulas

$$dg = Ce^{-t} dx dt dk$$

in NAK order and

$$dg = Ce^t dk dt ds$$

in *KAN* order. These formulas are concrete tests for every modular convention used in the volume.

CHAPTER 6

Invariant Measures on Homogeneous Spaces

1. The basic question

Let G be a Lie group and $H < G$ a closed subgroup. The homogeneous space G/H always carries a smooth transitive G -action, but it need not carry a nonzero G -invariant measure. The obstruction is exactly the mismatch between the modular functions of G and H .

The result is simple to state:

$$\boxed{G/H \text{ carries a nonzero } G\text{-invariant Radon measure} \iff \Delta_G|_H = \Delta_H.}$$

The purpose of this chapter is to prove the statement in a way that makes the mechanism visible. For smooth invariant densities, the obstruction is read directly from the isotropy action on the tangent space. To prove necessity for an arbitrary invariant Radon measure, we lift the measure along the H -fibers and compare modular factors. A quotient integration formula then relates Haar integration on G to integration along the fibers and over the base G/H .

2. Densities on the tangent quotient

At the base point $o = eH$,

$$T_o(G/H) \cong \mathfrak{g}/\mathfrak{h}.$$

The isotropy action of $h \in H$ is induced by $\text{Ad}(h)$ on this quotient.

A positive density on a finite-dimensional vector space V is preserved by a linear isomorphism T exactly when

$$|\det T| = 1.$$

Therefore a G -invariant density on G/H can exist only if the isotropy action of every $h \in H$ has determinant of absolute value 1.

Lemma 6.2.1 (Determinant on the tangent quotient). *For $h \in H$,*

$$\left| \det(\text{Ad}(h)|_{\mathfrak{g}/\mathfrak{h}}) \right| = \frac{\Delta_G(h)}{\Delta_H(h)}.$$

PROOF. Because H is a subgroup, $\text{Ad}(h)$ preserves $\mathfrak{h}$. Choose any vector-space complement $\mathfrak{m}$ so that

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}.$$

With respect to this decomposition, $\text{Ad}(h)$ has block upper-triangular form

$$\begin{pmatrix} A & * \\ 0 & C \end{pmatrix},$$

where A is the restriction to $\mathfrak{h}$ and C represents the induced map on $\mathfrak{g}/\mathfrak{h}$. Hence

$$|\det \text{Ad}_G(h)| = |\det A| |\det C|.$$

The restriction A is exactly $\text{Ad}_H(h)$ on $\mathfrak{h}$. Using the Lie-group modular formula,

$$\Delta_G(h) = |\det \text{Ad}_G(h)|, \quad \Delta_H(h) = |\det \text{Ad}_H(h)| = |\det A|.$$

Solving for $|\det C|$ gives the claim. $\square$

3. The invariant-measure criterion

THEOREM 6.3.1 (Invariant measure criterion). *Let G be a Lie group and $H < G$ a closed subgroup. The following are equivalent.*

- (1) G/H admits a nonzero G -invariant Radon measure.
- (2) G/H admits a smooth positive G -invariant density.
- (3)

$$\Delta_G(h) = \Delta_H(h)$$

for every $h \in H$.

When such a measure exists, it is unique up to multiplication by a positive scalar.

Guiding Idea 6.3.2 (The whole obstruction sits at one point). A transitive group action moves the base point to every other point. Thus to build an invariant density, choose a density at eH and translate it. This is well defined exactly when different group elements representing the same point produce the same density. The ambiguity is the stabilizer H , so the only condition is that H preserve the chosen density on $\mathfrak{g}/\mathfrak{h}$.

PROOF. Assume first that the modular functions agree on H . By the determinant lemma, the isotropy representation of H on $\mathfrak{g}/\mathfrak{h}$ has absolute determinant 1. Choose any positive density ω_o on $T_o(G/H)$. Since a linear map acts on the one-dimensional density line by the factor $|\det|$, every isotropy element fixes ω_o .

For $x = gH$, define

$$\omega_x = g_*\omega_o.$$

If $g' = gh$ with $h \in H$, then

$$g'_*\omega_o = g_*h_*\omega_o = g_*\omega_o,$$

so the definition is independent of the representative. Local smooth sections of $G \rightarrow G/H$ show that ω is smooth. By construction it is G -invariant, and integrating it gives a nonzero invariant Radon measure. Thus (3) implies (2) and (1).

Now assume only that a nonzero G -invariant Radon measure μ exists on G/H . Fix left Haar measure dh on H and define, for $f \in C_c(G)$,

$$I(f) = \int_{G/H} \int_H f(gh) dh d\mu(gH).$$

The inner integral is independent of the representative gH , and local quotient charts plus compact support make I a nonzero Radon measure on G . Invariance of μ shows that I is left invariant, hence by Haar uniqueness

$$I = c dg$$

for some left Haar measure dg on G and $c > 0$.

Fix $h_0 \in H$ and replace $f(g)$ by $f(gh_0)$. Along the H fiber,

$$\int_H f(ghh_0) dh = \Delta_H(h_0) \int_H f(gh) dh,$$

because our modular convention gives $\int \phi(hh_0)dh = \Delta_H(h_0) \int \phi(h)dh$. Therefore

$$I(f(\cdot h_0)) = \Delta_H(h_0)I(f).$$

On the other hand, since $I = c dg$,

$$I(f(\cdot h_0)) = \Delta_G(h_0)I(f).$$

Choose $f \geq 0$ with $I(f) > 0$. Comparing the two identities gives

$$\Delta_H(h_0) = \Delta_G(h_0).$$

Since h_0 was arbitrary, (1) implies (3).

For uniqueness, let μ_1, μ_2 be two nonzero invariant Radon measures and form the corresponding lifted left Haar measures

$$I_j(f) = \int_{G/H} \int_H f(gh) dh d\mu_j(gH).$$

Haar uniqueness gives $I_1 = cI_2$ for some $c > 0$. To recover the base measures locally, choose a quotient chart $U \subset G/H$ with smooth section $s : U \rightarrow G$,

choose $\psi \in C_c(H)$ with $\int_H \psi dh = 1$, and let $\varphi \in C_c(U)$. After shrinking to a relatively compact chart if necessary, define

$$f(s(x)h) = \varphi(x)\psi(h)$$

on the corresponding trivialization and extend it by zero. Then

$$I_j(f) = \int_U \varphi d\mu_j.$$

Hence $\mu_1 = c\mu_2$ on every quotient chart, and therefore globally. This proves uniqueness without any additional quotient formula. $\square$

4. A local fiber-normalization lemma

To integrate first along H and then over G/H , we need enough compactly supported functions on G whose integral along each relevant H -fiber is prescribed.

For $f \in C_c(G)$ define

$$P_H f(gH) = \int_H f(gh) dh,$$

where dh is a fixed left Haar measure on H .

Lemma 6.4.1. *$P_H f$ is well defined and continuous on G/H .*

PROOF. If $g' = gh_0$ with $h_0 \in H$, then

$$\int_H f(g'h) dh = \int_H f(gh_0h) dh.$$

The substitution $k = h_0h$ preserves left Haar measure, so this equals $\int_H f(gk) dk$.

For continuity, work in a local trivialization of the quotient map, so elements over a neighborhood $U \subset G/H$ are written $s(x)h$. Then

$$P_H f(x) = \int_H f(s(x)h) dh.$$

Fix a relatively compact sub-neighborhood $U_0 \Subset U$. Compactness of $\text{supp } f$ implies that there is one compact set $C \subset H$ such that $f(s(x)h) = 0$ for all $x \in U_0$ and $h \notin C$. On $U_0 \times C$ the integrand is uniformly continuous, and it is bounded by an integrable constant multiple of $\mathbf{1}_C$. Dominated convergence therefore gives continuity of $P_H f$. $\square$

Lemma 6.4.2 (Fiber normalization). *Let $K \subset G/H$ be compact. There exists $\beta \in C_c(G)$, $\beta \geq 0$, such that*

$$\int_H \beta(gh) dh = 1$$

for every $gH \in K$.

PROOF. The point of the construction is to normalize the integral separately in local product coordinates and then glue the local functions on the base. Choose finitely many quotient neighborhoods $U_i \subset G/H$ with smooth sections

$$s_i : U_i \rightarrow G$$

whose union contains K . After shrinking the neighborhoods, choose functions

$$\eta_i \in C_c(U_i), \quad \eta_i \geq 0,$$

such that

$$\sum_i \eta_i(x) = 1$$

on a neighborhood of K . Choose also $\psi \in C_c(H)$, $\psi \geq 0$, with

$$\int_H \psi(h) dh = 1.$$

Every point of $\pi^{-1}(U_i)$ has a unique expression $s_i(x)h$. Define on this set

$$\beta_i(s_i(x)h) = \eta_i(x)\psi(h).$$

Because $\text{supp } \eta_i$ is compactly contained in U_i , this function extends by zero to a continuous compactly supported function on G ; in fact its support is contained in the compact image of $\text{supp } \eta_i \times \text{supp } \psi$ under $(x, h) \mapsto s_i(x)h$.

Now let $x \in U_i$ and let $h_0 \in H$. Using left invariance of Haar measure on H ,

$$\begin{aligned} \int_H \beta_i(s_i(x)h_0h) dh &= \eta_i(x) \int_H \psi(h_0h) dh \\ &= \eta_i(x) \int_H \psi(h) dh \\ &= \eta_i(x). \end{aligned}$$

Thus the value of the fiber integral depends only on the coset x , not on the representative chosen in that fiber. Finally set

$$\beta = \sum_i \beta_i.$$

The sum is finite, hence $\beta \in C_c(G)$, and for $x \in K$ we have

$$\int_H \beta(gh) dh = \sum_i \eta_i(x) = 1.$$

□

Corollary 6.4.3. *For every $\varphi \in C_c(G/H)$ there exists $f \in C_c(G)$ such that*

$$P_H f = \varphi.$$

PROOF. Apply the lemma to $K = \text{supp } \varphi$ and set

$$f(g) = \varphi(gH)\beta(g).$$

Then

$$P_H f(gH) = \varphi(gH)P_H \beta(gH) = \varphi(gH)$$

on the support, and both sides vanish off it after the construction is localized. $\square$

5. Weil's quotient integration formula

Assume now that

$$\Delta_G|_H = \Delta_H.$$

Fix left Haar measures dg on G and dh on H .

THEOREM 6.5.1 (Quotient integration formula). *There exists a unique normalization of the invariant measure $d\mu$ on G/H such that for every $f \in C_c(G)$,*

$$\int_G f(g) dg = \int_{G/H} \int_H f(gh) dh d\mu(gH).$$

Changing the normalization of dh rescales $d\mu$ inversely.

PROOF. Let μ_0 be any nonzero G -invariant Radon measure on G/H , whose existence has already been established by the invariant-density construction. Define a positive linear functional on $C_c(G)$ by

$$I(f) = \int_{G/H} \int_H f(gh) dh d\mu_0(gH).$$

Local trivializations and compact support show that the integral is finite and defines a Radon measure on G .

We claim I is left invariant. For $a \in G$, let $L_a f(g) = f(a^{-1}g)$. Then

$$I(L_a f) = \int_{G/H} \int_H f(a^{-1}gh) dh d\mu_0(gH).$$

Set $x = a^{-1}gH$. Since μ_0 is G -invariant, the outer measure is unchanged, giving $I(L_a f) = I(f)$. Hence I is a left Haar measure on G . By uniqueness,

$$I = c dg$$

for some $c > 0$. Replace μ_0 by $c^{-1}\mu_0$ to obtain the formula.

For uniqueness, suppose μ_1, μ_2 both satisfy the formula with the same dg, dh . Given $\varphi \in C_c(G/H)$, choose $f \in C_c(G)$ with $P_H f = \varphi$. Then

$$\int_{G/H} \varphi d\mu_i = \int_G f dg$$

for $i = 1, 2$. Therefore the measures agree on $C_c(G/H)$ and hence are equal. $\square$

Remark 6.5.2. The equality of modular functions is hidden in the claim that the fiber construction is compatible with a G -invariant base measure. If the modular functions do not match, right translation along a fiber changes ambient Haar volume and fiber Haar volume by different factors, so no invariant transverse measure can compensate consistently.

6. Compact stabilizers

Corollary 6.6.1. *If H is compact, then G/H admits a G -invariant measure.*

PROOF. $\Delta_H \equiv 1$ because H is compact. The restriction $\Delta_G|_H$ is a continuous homomorphism from a compact group to $\mathbb{R}_{>0}$, so it is also identically 1. The criterion applies. $\square$

Worked Example 6.6.2 (Hyperbolic plane as G/K). For $G = \mathrm{PSL}_2(\mathbb{R})$ and $K \cong \mathrm{SO}(2)$, the quotient G/K carries a G -invariant measure. In upper-half-plane coordinates $z = x + iy$, it is a scalar multiple of

$$\frac{dx \, dy}{y^2}.$$

Indeed, the NAK Haar formula

$$dg = C \frac{dx \, dy}{y^2} dk$$

and normalized Haar probability dk on K give the quotient formula directly.

7. The quotients G/N , G/A , and G/B in SL_2

Let $G = \mathrm{SL}_2(\mathbb{R})$. Since G is unimodular, $\Delta_G \equiv 1$.

Worked Example 6.7.1 (G/N). N is abelian and hence unimodular. Therefore G/N carries a G -invariant measure. Under the diffeomorphism

$$G/N \cong \mathbb{R}^2 \setminus \{0\}, \quad gN \mapsto ge_1,$$

this measure is a scalar multiple of ordinary Lebesgue measure $dx \, dy$. The reason is transparent: $\mathrm{SL}_2(\mathbb{R})$ acts linearly with determinant one and hence preserves Lebesgue measure.

Worked Example 6.7.2 (G/A). $A \cong (\mathbb{R}, +)$ is abelian, so $\Delta_A = 1$. Thus G/A admits a G -invariant measure. In the projective interpretation, after passing to $\mathrm{PSL}_2(\mathbb{R})$ to remove the central sign, G/A parametrizes oriented

geodesics, equivalently ordered pairs of distinct boundary points. An explicit invariant density in affine boundary coordinates is proportional to

$$\frac{du dv}{(u - v)^2}.$$

To check invariance, for a Möbius map $T(z) = (az + b)/(cz + d)$,

$$T'(z) = \frac{1}{(cz + d)^2}$$

and

$$T(u) - T(v) = \frac{u - v}{(cu + d)(cv + d)}.$$

Therefore

$$\frac{dT(u) dT(v)}{(T(u) - T(v))^2} = \frac{du dv}{(u - v)^2}.$$

Worked Example 6.7.3 (G/B and the projective line). The subgroup B is not unimodular. Its connected positive-diagonal component is AN . Since conjugation by

$$a_t = \text{diag}(e^{t/2}, e^{-t/2})$$

sends u_s to $u_{e^t s}$, the semidirect-product modular formula gives

$$\Delta_{AN}(a_t u_s) = e^t.$$

Thus Δ_B is nontrivial, whereas $\Delta_G \equiv 1$. The criterion implies that

$$G/B \cong \mathbb{P}^1(\mathbb{R})$$

does *not* carry a nonzero G -invariant Radon measure.

This is a useful warning because $\mathbb{P}^1(\mathbb{R})$ is compact. Compactness alone does not produce a measure invariant under a noncompact transitive group. By contrast, the compact subgroup K does preserve the usual angular probability measure on the circle, but that measure is not invariant under all of G .

8. Projective space in higher dimension

Let $G = \text{SL}_d(\mathbb{R})$ act on $\mathbb{P}^{d-1}(\mathbb{R})$. The stabilizer P of the line $\mathbb{R}e_1$ consists of block upper triangular matrices

$$\begin{pmatrix} a & * \\ 0 & B \end{pmatrix}, \quad a \det B = 1.$$

It is a parabolic subgroup and is nonunimodular for $d \geq 2$.

Proposition 6.8.1. *There is no nonzero $\text{SL}_d(\mathbb{R})$ -invariant Radon measure on $\mathbb{P}^{d-1}(\mathbb{R})$.*

PROOF. G is unimodular. It suffices to show P is not. Consider the one-parameter subgroup

$$b_t = \text{diag}(e^{(d-1)t}, e^{-t}, \dots, e^{-t}),$$

which lies in P and has determinant one. The unipotent directions E_{1j} , $j \geq 2$, lie in $\mathfrak{p}$ and satisfy

$$\text{Ad}(b_t)E_{1j} = e^{dt}E_{1j}.$$

The remaining diagonal/block directions contribute determinant one to the modular character, while these $d-1$ directions contribute a nontrivial factor $e^{d(d-1)t}$. Thus Δ_P is nontrivial. The invariant-measure criterion rules out a G -invariant measure. $\square$

9. Discrete subgroups

If $\Gamma < G$ is discrete, then Γ has counting Haar measure and

$$\Delta_\Gamma \equiv 1.$$

Therefore:

Corollary 6.9.1. *G/Γ admits a G -invariant Radon measure if and only if*

$$\Delta_G(\gamma) = 1$$

for every $\gamma \in \Gamma$.

PROOF. For a discrete subgroup, counting measure is Haar measure and $\Delta_\Gamma \equiv 1$. The quotient invariant-measure criterion proved above therefore reduces exactly to the displayed condition. $\square$

In particular, if G is unimodular, every quotient by a discrete subgroup carries an invariant measure. The measure can be infinite; finiteness is the additional lattice condition.

With counting measure on Γ , the quotient integration formula becomes

$$\int_G f(g) dg = \int_{G/\Gamma} \sum_{\gamma \in \Gamma} f(g\gamma) d\mu(g\Gamma).$$

For $f \in C_c(G)$ the sum is locally finite because the action of a discrete subgroup by right translations is proper.

10. Fundamental domains

Suppose $\Gamma < G$ is discrete. A measurable set $F \subset G$ is a *measurable fundamental domain* if almost every right coset $g\Gamma$ meets F in exactly one point. When such an F is chosen and boundary ambiguities have Haar measure zero, the quotient measure satisfies

$$\mu(G/\Gamma) = m_G(F).$$

More generally,

$$\int_{G/\Gamma} \varphi(g\Gamma) d\mu = \int_F \varphi(g\Gamma) dg$$

for suitable functions φ .

The formula should be interpreted measure-theoretically rather than as a demand for a geometrically pretty region. Fundamental domains can have complicated boundaries, and different domains differ on pieces rearranged by Γ , but Haar invariance ensures the same total covolume.

11. Invariant probability and finite volume

If the invariant measure on G/H has finite total mass, normalize it by

$$\mu(G/H) = 1.$$

Then $(G/H, \mu)$ becomes a probability space on which every element of G acts measure preservingly.

Finite volume is much stronger than existence of an invariant measure. For example,

$$\mathrm{SL}_2(\mathbb{R})/N \cong \mathbb{R}^2 \setminus \{0\}$$

has invariant Lebesgue measure but infinite total mass. On the other hand,

$$\mathrm{SL}_2(\mathbb{R})/\mathrm{SL}_2(\mathbb{Z})$$

has finite invariant volume even though it is noncompact. The latter fact reflects the geometry of cusps rather than compactness.

12. When invariant measure fails: quasi-invariant measures

Failure of the modular criterion does not mean that G/H has no useful measure class. Smooth homogeneous spaces always carry smooth positive densities; what fails is invariance under the full group. The correct replacement is quasi-invariance.

Definition 6.12.1. A Radon measure μ on a G -space X is *quasi-invariant* if for every $g \in G$, the translated measure $g_*\mu$ has the same null sets as μ . Equivalently, $g_*\mu$ and μ are mutually absolutely continuous.

If μ is quasi-invariant, fix the convention

$$(g_*\mu)(E) = \mu(g^{-1}E)$$

and define

$$J(g, x) = \frac{d(g_*\mu)}{d\mu}(x).$$

Then the chain rule for Radon–Nikodym derivatives gives, for almost every x ,

$$J(gh, x) = J(g, x)J(h, g^{-1}x).$$

Indeed, $(gh)_*\mu = g_*(h_*\mu)$; under push-forward by g , the density $J(h, \cdot)$ is evaluated at $g^{-1}x$, and the remaining change of measure is $J(g, x)$.

Proposition 6.12.2. *Every smooth positive density on a homogeneous space G/H defines a quasi-invariant Radon measure.*

PROOF. For fixed $g \in G$, pullback of a smooth positive density by the diffeomorphism $x \mapsto gx$ is another smooth positive density. The ratio of two positive smooth densities is a positive smooth function. Hence the translated measure has a positive smooth Radon–Nikodym derivative relative to the original measure, and the two measures have identical null sets. $\square$

Worked Example 6.12.3 (Lebesgue measure class on the projective line). Although $\mathrm{PSL}_2(\mathbb{R})$ has no invariant Radon measure on $\mathbb{P}^1(\mathbb{R})$, ordinary Lebesgue measure dx on the affine chart is quasi-invariant. For

$$T(x) = \frac{ax + b}{cx + d},$$

$$T'(x) = \frac{1}{(cx + d)^2}.$$

Thus on intervals avoiding the pole, if $y = T(x)$ then

$$d(T_*dx)(y) = \left| (T^{-1})'(y) \right| dy = \frac{dy}{(-cy + a)^2}.$$

The density is strictly positive wherever the affine chart is valid. The single point ∞ has measure zero, so these chart calculations define a preserved smooth measure class on the whole projective line. This is the natural setting when a compact boundary carries no G -invariant probability but still has a canonical smooth measure class.

13. A density-line interpretation of the modular obstruction

The space of densities on the tangent space $\mathfrak{g}/\mathfrak{h}$ is one-dimensional. The isotropy group H therefore acts on that line by a positive character

$$\chi_{G/H}(h) = \left| \det(\text{Ad}(h)|_{\mathfrak{g}/\mathfrak{h}}) \right| = \frac{\Delta_G(h)}{\Delta_H(h)}.$$

The invariant-measure criterion can be restated as

$$G/H \text{ has invariant density} \iff \chi_{G/H} \equiv 1.$$

When $\chi_{G/H}$ is nontrivial, any choice of base-point density is rescaled by isotropy. This makes it impossible to translate the density consistently around the quotient. Thus the modular obstruction is literally a one-dimensional representation of the stabilizer.

Worked Example 6.13.1 (Upper triangular stabilizer in SL_2). For $G/B \cong \mathbb{P}^1(\mathbb{R})$, the tangent space at ∞ is one-dimensional. Conjugation by

$$a_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix}$$

acts on a local coordinate near ∞ by a factor e^{-t} (use $w = 1/z$ as the local coordinate). Hence it rescales one-dimensional density by e^{-t} rather than preserving it. This local calculation is the same obstruction detected globally by $\Delta_B(a_t) = e^t$.

14. Invariant measures on subgroup orbits

Let G be unimodular, $\Gamma < G$ discrete, and $H < G$ closed. For

$$x = g\Gamma \in G/\Gamma,$$

the stabilizer of the H -action is

$$\Lambda_x = H \cap g\Gamma g^{-1}.$$

If the orbit is embedded and closed, it is naturally

$$H/\Lambda_x.$$

Proposition 6.14.1. *Assume H is unimodular and Hx is closed. Then Hx carries an H -invariant Radon measure. It carries an H -invariant probability measure if and only if Λ_x is a lattice in H .*

PROOF. Λ_x is discrete, hence unimodular. Since H is unimodular, the quotient criterion gives an invariant measure on H/Λ_x . Its total mass is finite exactly when Λ_x is a lattice, by definition. Transport the measure through the orbit embedding. $\square$

Worked Example 6.14.2 (A diagonal orbit in the space of lattices). Take $G = \mathrm{SL}_2(\mathbb{R})$, $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, $H = A$, and the standard lattice point $x = e\Gamma$. Since

$$A \cap \Gamma = \{I\}$$

for the positive diagonal subgroup, the orbit is immersed as a copy of $\mathbb{R}$. It is not periodic and its invariant A -measure is infinite. In fact $a_t\mathbb{Z}^2$ develops a vector of length $e^{-|t|/2}$ in one time direction or the other and leaves compact sets. This contrasts sharply with the periodic N -orbit, whose stabilizer is a copy of $\mathbb{Z}$.

15. Changing Haar normalizations

The quotient formula

$$\int_G f(g) dg = \int_{G/H} \int_H f(gh) dh d\mu(gH)$$

contains three measures, but only two independent normalizations. If

$$dg' = a dg, \quad dh' = b dh,$$

then the compatible quotient measure is

$$d\mu' = \frac{a}{b} d\mu.$$

This elementary bookkeeping matters when comparing formulas from different sources. A numerical covolume such as $\pi/3$ has meaning only after the ambient Haar normalization is specified. Qualitative statements—finite versus infinite volume, probability after normalization, invariance, and ergodicity—do not depend on this scalar choice.

16. Push-forward measures under equivariant quotient maps

Suppose $H < K < G$ are closed and both G/H and G/K admit invariant measures. The canonical equivariant map

$$p : G/H \rightarrow G/K, \quad gH \mapsto gK,$$

can be understood as integration along the fiber K/H .

If K/H has finite K -invariant volume, normalize it to probability. Then the push-forward of a suitably normalized G -invariant measure on G/H is the G -invariant measure on G/K . Locally this is Fubini's theorem in a product chart; globally it is a repeated application of Weil integration. The statement is especially transparent when K is compact: averaging over the compact fiber loses angular or framing information while preserving invariant volume on the base.

Worked Example 6.16.1 (From unit tangent vectors to base points). For $G = \mathrm{PSL}_2(\mathbb{R})$ and compact $K = \mathrm{PSO}(2)$, Haar measure on

$$G \cong T^1\mathbb{H}$$

pushes forward under

$$G \rightarrow G/K \cong \mathbb{H}$$

to hyperbolic area after the K -fiber is normalized to total mass 1. In NAK coordinates this is exactly

$$C \frac{dx \, dy}{y^2} \, dk \mapsto C \frac{dx \, dy}{y^2}.$$

17. Conceptual summary

The measure theory of a homogeneous space is determined by a single local compatibility condition. The isotropy action of H on

$$T_{eH}(G/H) \cong \mathfrak{g}/\mathfrak{h}$$

preserves a density exactly when

$$\frac{\Delta_G(h)}{\Delta_H(h)} = 1.$$

Thus

$$G/H \text{ has invariant measure } \iff \Delta_G|_H = \Delta_H.$$

Once this condition holds, Haar integration decomposes into fiber and base integration:

$$\int_G f(g) \, dg = \int_{G/H} \int_H f(gh) \, dh \, d\mu(gH).$$

The examples G/N , G/A , and G/B show why the criterion is substantive: two quotients of the same unimodular group carry invariant measures, while the compact projective quotient G/B does not. The obstruction is not topology but the modular character of the stabilizer.

CHAPTER 7

Discrete Subgroups, Lattices, and Cusp Geometry

1. Discrete subgroups versus lattices

A discrete subgroup gives a smooth quotient, but discreteness alone says nothing about the size of that quotient. The additional finite-volume condition is the definition of a lattice.

Definition 7.1.1 (Lattice). Let G be a locally compact group. A discrete subgroup $\Gamma < G$ is a *lattice* if G/Γ carries a finite nonzero G -invariant Radon measure. Equivalently, after normalization, G/Γ carries a G -invariant probability measure.

Definition 7.1.2. A lattice is *uniform* if G/Γ is compact and *nonuniform* if G/Γ is noncompact.

The word “uniform” in this definition includes the lattice property; we do not silently assume that every cocompact discrete subgroup in an arbitrary locally compact group has already been shown to satisfy the modular compatibility condition.

Worked Example 7.1.3 (The torus). $\mathbb{Z}^d$ is a uniform lattice in $\mathbb{R}^d$. The quotient

$$\mathbb{R}^d/\mathbb{Z}^d = \mathbb{T}^d$$

is compact, and the invariant probability measure is Lebesgue measure on $[0, 1)^d$ transported to the quotient.

Worked Example 7.1.4 (A nonlattice discrete subgroup). The subgroup $\mathbb{Z} < \mathbb{R}^2$ embedded as $\mathbb{Z} \times \{0\}$ is discrete, but

$$\mathbb{R}^2/(\mathbb{Z} \times \{0\}) \cong S^1 \times \mathbb{R}$$

has infinite invariant area. Thus it is not a lattice.

2. A group containing a lattice is unimodular

This fact is structurally important: finite invariant volume rules out ambient modular expansion.

THEOREM 7.2.1. *If a Lie group G contains a lattice Γ , then G is unimodular.*

PROOF. Since G/Γ has an invariant measure, the quotient-measure criterion gives

$$\Delta_G(\gamma) = 1$$

for every $\gamma \in \Gamma$, because a discrete group is unimodular. Hence the modular function descends to a positive continuous function

$$F : G/\Gamma \rightarrow \mathbb{R}_{>0}, \quad F(g\Gamma) = \Delta_G(g).$$

It satisfies

$$F(hx) = \Delta_G(h)F(x).$$

Suppose for contradiction that some $h \in G$ has $c = \Delta_G(h) > 1$. Let

$$A = \{x \in G/\Gamma : 1 \leq F(x) < c\}.$$

Then the sets

$$h^n A = \{x : c^n \leq F(x) < c^{n+1}\}, \quad n \in \mathbb{Z},$$

are pairwise disjoint and their union is all of G/Γ : every positive number lies in exactly one interval $[c^n, c^{n+1})$.

The invariant finite measure μ gives

$$\mu(h^n A) = \mu(A)$$

for every n . If $\mu(A) = 0$, the countable union has measure zero, contradicting $\mu(G/\Gamma) > 0$. Thus $\mu(A) > 0$, but then the union of infinitely many disjoint translates has infinite measure, contradicting finiteness. Therefore no h has $\Delta_G(h) > 1$. Applying the same conclusion to h^{-1} rules out $\Delta_G(h) < 1$. Hence $\Delta_G \equiv 1$. $\square$

Corollary 7.2.2. *A nonunimodular Lie group cannot contain a lattice.*

PROOF. The preceding theorem shows that the existence of a finite invariant measure on G/Γ forces $\Delta_G \equiv 1$, which is precisely unimodularity. $\square$

Thus the affine $ax + b$ group provides a basic example of a Lie group with no lattices.

3. Covolume and fundamental domains

Let G be unimodular and $\Gamma < G$ discrete. Fix Haar measure dg and the induced invariant quotient measure $d\mu$.

Definition 7.3.1. If $\mu(G/\Gamma) < \infty$, the number

$$\text{covol}(\Gamma) = \mu(G/\Gamma)$$

is the *covolume* of Γ relative to the chosen normalization of Haar measure.

If $F \subset G$ is a measurable fundamental domain for the right action of Γ , then

$$\text{covol}(\Gamma) = m_G(F)$$

provided boundaries and multiple representatives are discarded on null sets. The quotient integration formula makes this precise.

Proposition 7.3.2 (Unfolding). *For $f \in C_c(G)$,*

$$\int_G f(g) dg = \int_{G/\Gamma} \sum_{\gamma \in \Gamma} f(g\gamma) d\mu(g\Gamma).$$

If F is a measurable fundamental domain and Φ is a nonnegative or integrable function on G/Γ , then

$$\int_{G/\Gamma} \Phi(g\Gamma) d\mu = \int_F \Phi(g\Gamma) dg.$$

PROOF. Both identities are the quotient integration formula of Volume I, Chapter 6, specialized to counting Haar measure on the discrete subgroup Γ . For the second identity, integrate the Γ -periodic lift of Φ over one measurable fundamental domain. $\square$

The first formula is often called *unfolding*: a sum over lattice translates on the quotient is unfolded into a single integral over the ambient group.

4. Euclidean lattices and covolume

A full-rank lattice in $\mathbb{R}^d$ has the form

$$\Lambda = g\mathbb{Z}^d, \quad g \in \text{GL}_d(\mathbb{R}).$$

Its covolume is

$$\text{covol}(\Lambda) = |\det g|.$$

This is independent of the chosen lattice basis because changing the basis multiplies g on the right by an element of $\text{GL}_d(\mathbb{Z})$, whose determinant has absolute value 1.

A lattice is unimodular when its covolume is 1. The quotient

$$X_d = \text{SL}_d(\mathbb{R}) / \text{SL}_d(\mathbb{Z})$$

parametrizes unimodular lattices via $g \mathrm{SL}_d(\mathbb{Z}) \mapsto g\mathbb{Z}^d$: changing g on the right by an integer unimodular matrix changes only the chosen lattice basis, and every covolume-one lattice admits an oriented basis matrix of determinant 1.

Define the first minimum

$$\lambda_1(\Lambda) = \min \{ \|v\| : v \in \Lambda \setminus \{0\} \}.$$

The minimum exists because a lattice is discrete and a closed Euclidean ball is compact.

Proposition 7.4.1. *The function*

$$\lambda_1 : X_d \rightarrow \mathbb{R}_{>0}$$

is continuous.

PROOF. Fix $\Lambda = g\mathbb{Z}^d$. Because $\mathrm{SL}_d(\mathbb{Z})$ is discrete, there is a neighborhood V of the identity such that

$$V^{-1}V \cap \mathrm{SL}_d(\mathbb{Z}) = \{I\}.$$

After replacing V by a smaller neighborhood if necessary, the quotient map is injective on gV and hence identifies gV homeomorphically with a neighborhood of Λ in X_d . Thus nearby lattices may be represented uniquely by matrices g' close to g .

Choose $R > \lambda_1(\Lambda)$. If g' is sufficiently close to g , then

$$\|g'g^{-1} - I\|_{\mathrm{op}} < \frac{1}{2}.$$

Consequently, for every $n \in \mathbb{Z}^d$,

$$\|g'n\| = \|(g'g^{-1})(gn)\| \geq \frac{1}{2} \|gn\|.$$

Hence any vector with $\|gn\| > 2R$ has $\|g'n\| > R$ and cannot realize the shortest-vector length of $g'\mathbb{Z}^d$, because a fixed shortest vector of Λ still has length $< R$ after a sufficiently small perturbation. Only the finitely many $n \in \mathbb{Z}^d$ satisfying $\|gn\| \leq 2R$ therefore need to be considered. On this finite set the functions $g' \mapsto \|g'n\|$ are continuous, so their minimum is continuous. Thus λ_1 is continuous at Λ . $\square$

5. Minkowski's convex-body theorem

The geometry of numbers supplies the compactness input needed for spaces of lattices.

THEOREM 7.5.1 (Minkowski's first theorem). *Let $\Lambda \subset \mathbb{R}^d$ be a lattice of covolume D , and let $K \subset \mathbb{R}^d$ be a convex, centrally symmetric measurable set. If*

$$\text{vol}(K) > 2^d D,$$

then K contains a nonzero point of Λ .

PROOF. The argument is a volume pigeonhole principle modulo the lattice. Put $L = \frac{1}{2}K$. Then

$$\text{vol}(L) = 2^{-d} \text{vol}(K) > D.$$

If L is unbounded or has infinite volume, choose first a bounded measurable subset $A \subset L$ with $\text{vol}(A) > D$; this is possible by intersecting L with increasingly large Euclidean balls. If L already has finite volume, simply take $A = L$. Thus in either case we have a measurable $A \subset L$ with

$$D < \text{vol}(A) < \infty.$$

Let F be a measurable fundamental parallelepiped for Λ , with $\text{vol}(F) = D$. Decompose A according to the translates $F + \lambda$:

$$\text{vol}(A) = \sum_{\lambda \in \Lambda} \text{vol}(A \cap (F + \lambda)) = \int_F \sum_{\lambda \in \Lambda} \mathbf{1}_A(x + \lambda) dx.$$

Since the integral is strictly larger than $\text{vol}(F)$, the integer-valued sum in the last expression is at least 2 on a set of positive measure. Hence there are distinct points $x, y \in A$ with

$$x - y \in \Lambda \setminus \{0\}.$$

Because $A \subset \frac{1}{2}K$ and K is centrally symmetric and convex,

$$x - y \in \frac{1}{2}K - \frac{1}{2}K = \frac{1}{2}K + \frac{1}{2}K = K.$$

Thus K contains a nonzero lattice point. □

Corollary 7.5.2. *For every unimodular lattice $\Lambda \subset \mathbb{R}^d$,*

$$\lambda_1(\Lambda) \leq C_d$$

for a constant depending only on d .

PROOF. Apply Minkowski's theorem to a Euclidean ball of radius R whose volume exceeds 2^d . □

For Mahler compactness we need a standard strengthening involving successive minima. We state it in the form required here.

Definition 7.5.3. For $1 \leq j \leq d$, the j th successive minimum $\lambda_j(\Lambda)$ is the infimum of $r > 0$ such that the ball $B(0, r)$ contains j linearly independent vectors of Λ .

THEOREM 7.5.4 (Minkowski's second theorem, Euclidean form). *There are constants $0 < c_d < C_d < \infty$ depending only on d such that every full-rank lattice Λ satisfies*

$$c_d \operatorname{covol}(\Lambda) \leq \lambda_1(\Lambda) \cdots \lambda_d(\Lambda) \leq C_d \operatorname{covol}(\Lambda).$$

PROOF. This is the Euclidean-ball specialization of Minkowski's second theorem, which is proved in full in Appendix 1. Applying Theorem B.1.1 to the Euclidean unit ball B gives

$$\frac{2^d}{d! \operatorname{vol}(B)} \operatorname{covol}(\Lambda) \leq \lambda_1 \cdots \lambda_d \leq \frac{2^d}{\operatorname{vol}(B)} \operatorname{covol}(\Lambda).$$

□

Remark 7.5.5 (What this theorem contributes). For covolume-one lattices, a lower bound on λ_1 forces an upper bound on λ_d . The complete proof in the appendix makes explicit the nontrivial packing-and-slicing mechanism behind this implication; it is not being used here as an unexplained geometry-of-numbers black box.

Proposition 7.5.6 (Bounded basis lemma). *For each d and $\varepsilon > 0$ there is $M = M(d, \varepsilon)$ such that every unimodular lattice Λ with*

$$\lambda_1(\Lambda) \geq \varepsilon$$

has a basis $v_1, \dots, v_d$ satisfying

$$\|v_i\| \leq M$$

for all i .

PROOF. By Minkowski's second theorem,

$$\lambda_d(\Lambda) \leq C_d \varepsilon^{-(d-1)} =: R.$$

Choose linearly independent lattice vectors $w_1, \dots, w_d$ with $\|w_i\| \leq 2R$. They generate a sublattice $\Lambda' \subset \Lambda$. Its index is

$$[\Lambda : \Lambda'] = \frac{\operatorname{covol}(\Lambda')}{\operatorname{covol}(\Lambda)} = |\det(w_1, \dots, w_d)|.$$

Hadamard's inequality gives

$$[\Lambda : \Lambda'] \leq \prod_i \|w_i\| \leq (2R)^d =: Q.$$

Thus the index is an integer bounded solely in terms of d, ε .

Choose a basis $b_1, \dots, b_d$ of Λ and put the basis vectors into the matrix $B = (b_1 \ \cdots \ b_d)$. Since each w_j lies in Λ , there is an integer matrix A with

$$W := (w_1 \ \cdots \ w_d) = BA.$$

Its determinant satisfies

$$1 \leq |\det A| = [\Lambda : \Lambda'] \leq Q.$$

Use integer row operations to put A into row Hermite normal form: there is $U \in \mathrm{GL}_d(\mathbb{Z})$ such that

$$H_0 = UA$$

is triangular, has positive diagonal entries, and each reduced off-diagonal entry is between 0 and the relevant diagonal entry. Since the product of the diagonal entries is $|\det A| \leq Q$, every entry of H_0 is bounded by Q . Set

$$B' = BU^{-1}.$$

The columns of B' are again a basis of Λ , while

$$W = BA = B'UA = B'H_0, \quad \text{so} \quad B' = WH_0^{-1}.$$

Now

$$H_0^{-1} = \frac{\mathrm{adj}(H_0)}{\det H_0}.$$

The denominator is a positive integer and hence at least 1, while every entry of the adjugate is a sum of at most $(d-1)!$ products of $d-1$ entries of H_0 . Consequently every entry of H_0^{-1} is bounded by a constant $C(d, Q)$. Each column v_i of B' is therefore a linear combination of the w_j with coefficients bounded by $C(d, Q)$, and

$$\|v_i\| \leq dC(d, Q) \max_j \|w_j\| \leq 2dC(d, Q)R.$$

This proves the uniform bounded-basis assertion. $\square$

Remark 7.5.7. The exact value of $M(d, \varepsilon)$ is irrelevant for compactness. What matters is uniformity: once very short vectors are excluded, one can choose an entire lattice basis inside one bounded Euclidean ball.

6. Mahler's compactness criterion

THEOREM 7.6.1 (Mahler compactness). *Let $\mathcal{K} \subset X_d$ be a set of unimodular lattices. Then $\mathcal{K}$ has compact closure in X_d if and only if there exists $\varepsilon > 0$ such that*

$$\lambda_1(\Lambda) \geq \varepsilon$$

for every $\Lambda \in \mathcal{K}$.

Guiding Idea 7.6.2. A unimodular lattice can leave every compact subset only by developing a short nonzero vector. Volume is fixed at 1, so a direction cannot become extremely thin without some compensating expansion elsewhere. Mahler's theorem says that this intuitive picture is exact.

PROOF. If $\overline{\mathcal{K}}$ is compact, continuity and positivity of λ_1 imply that λ_1 attains a positive minimum on the compact closure. This gives the necessary lower bound.

Conversely, suppose $\lambda_1(\Lambda) \geq \varepsilon$ on $\mathcal{K}$. By the bounded basis lemma, every $\Lambda \in \mathcal{K}$ has a basis $v_1, \dots, v_d$ with $\|v_i\| \leq M$. Form the basis matrix

$$g = (v_1 \ \cdots \ v_d).$$

Since the lattice is unimodular, $|\det g| = 1$. By changing the sign of one basis vector if needed, arrange $\det g = 1$. Hence $g \in \mathrm{SL}_d(\mathbb{R})$ represents Λ .

All entries of g are bounded by M , and $\det g = 1$. The set

$$C = \{g \in \mathrm{SL}_d(\mathbb{R}) : \text{each column has norm at most } M\}$$

is closed and bounded in $M_d(\mathbb{R})$, hence compact. Every lattice in $\mathcal{K}$ has a representative in C . The quotient map $\mathrm{SL}_d(\mathbb{R}) \rightarrow X_d$ is continuous, so its image $\pi(C)$ is compact and contains $\mathcal{K}$. Thus $\mathcal{K}$ has compact closure. $\square$

Corollary 7.6.3 (Escape of mass geometrically). *A sequence $\Lambda_n \in X_d$ leaves every compact subset if and only if*

$$\lambda_1(\Lambda_n) \rightarrow 0$$

along a subsequence.

PROOF. This is the sequential contrapositive of Mahler's compactness criterion proved above: a family is relatively compact exactly when its first minimum is bounded away from zero. $\square$

7. Diagonal and unipotent examples in X_2

Let

$$a_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix}.$$

Then

$$a_t \mathbb{Z}^2$$

contains the vector $a_t e_2 = e^{-t/2} e_2$, so

$$\lambda_1(a_t \mathbb{Z}^2) \leq e^{-t/2} \rightarrow 0.$$

Mahler's criterion therefore shows that $a_t \mathbb{Z}^2$ diverges as $t \rightarrow +\infty$.

By contrast,

$$u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$$

fixes e_1 . The orbit $u_s\mathbb{Z}^2$ need not diverge simply because $|s| \rightarrow \infty$, since integer shears satisfy

$$u_n \in \mathrm{SL}_2(\mathbb{Z})$$

and hence

$$u_{s+n}\mathbb{Z}^2 = u_s\mathbb{Z}^2.$$

Thus the u_s -orbit of the standard lattice is periodic modulo $\mathrm{SL}_2(\mathbb{Z})$, with period 1.

Worked Example 7.7.1 (A closed unipotent orbit). The map

$$\mathbb{R}/\mathbb{Z} \rightarrow X_2, \quad s + \mathbb{Z} \mapsto u_s \mathrm{SL}_2(\mathbb{Z})$$

is continuous and injective, hence embeds a circle. Its image is exactly the N -orbit of the standard lattice. This is a basic example of a closed finite-volume orbit of a unipotent subgroup.

8. The modular group and a finite-volume fundamental domain

Let

$$\Gamma = \mathrm{PSL}_2(\mathbb{Z})$$

act on the upper half-plane by Möbius transformations. It is generated by

$$T : z \mapsto z + 1, \quad S : z \mapsto -1/z.$$

Define

$$\mathcal{F} = \left\{ z = x + iy \in \mathbb{H} : |x| \leq \frac{1}{2}, |z| \geq 1 \right\}.$$

Lemma 7.8.1. *For every $z \in \mathbb{H}$, there exists $\gamma \in \mathrm{PSL}_2(\mathbb{Z})$ such that $\gamma z \in \mathcal{F}$.*

PROOF. For

$$\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

$$\Im(\gamma z) = \frac{y}{|cz + d|^2}.$$

Among coprime integer pairs (c, d) , the positive quantities $|cz + d|$ have a positive minimum: as $|c| + |d| \rightarrow \infty$, the quantity tends to infinity outside finitely many possibilities because $y > 0$ gives $|cz + d| \geq |c|y$. Choose coprime (c, d) minimizing it and complete (c, d) to a matrix $\gamma \in \mathrm{PSL}_2(\mathbb{Z})$. Then $\Im(\gamma z)$ is maximal in the Γ -orbit.

Translate by a power of T so that the real part lies in $[-1/2, 1/2]$; translation does not change the imaginary part. Call the resulting point w . If $|w| < 1$, then

$$\Im(Sw) = \frac{\Im w}{|w|^2} > \Im w,$$

contradicting maximality. Hence $|w| \geq 1$, so $w \in \mathcal{F}$. $\square$

Lemma 7.8.2. *If z lies in the interior of $\mathcal{F}$ and γz also lies in the interior for $\gamma \in \mathrm{PSL}_2(\mathbb{Z})$, then γ is the identity.*

PROOF. First observe that for $z \in \mathcal{F}$ and coprime integers c, d ,

$$|cz + d| \geq 1.$$

If $c = 0$, then $|d| = 1$. If $|c| \geq 2$, then

$$|cz + d|^2 \geq c^2 y^2 \geq 4 \cdot \frac{3}{4} > 1,$$

because $\mathcal{F}$ has $y \geq \sqrt{3}/2$. If $|c| = 1$ and $d = 0$, the inequality is $|z| \geq 1$. If $|c| = 1$ and $d \neq 0$, then $|cx + d| \geq 1/2$, so

$$|cz + d|^2 = (cx + d)^2 + c^2 y^2 \geq \frac{1}{4} + \frac{3}{4} = 1.$$

Therefore for $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$\Im(\gamma z) = \frac{\Im z}{|cz + d|^2} \leq \Im z.$$

Applying the same statement to $\gamma z \in \mathcal{F}$ and γ^{-1} gives the reverse inequality. Hence equality holds. For interior points the equality cases above force $c = 0$, hence $d = \pm 1$ and γ is a translation. Two interior points with real parts strictly between $-1/2$ and $1/2$ can differ by an integer only if that integer is zero. Thus γ is the identity in $\mathrm{PSL}_2(\mathbb{Z})$. $\square$

Boundary points have the familiar identifications, but boundaries have hyperbolic area zero. Thus $\mathcal{F}$ is a measurable fundamental domain.

THEOREM 7.8.3. *The hyperbolic area of $\mathcal{F}$ is*

$$\int_{\mathcal{F}} \frac{dx dy}{y^2} = \frac{\pi}{3}.$$

In particular $\mathrm{PSL}_2(\mathbb{Z})$ is a nonuniform lattice in $\mathrm{PSL}_2(\mathbb{R})$.

PROOF. For $|x| \leq 1/2$, the condition $|z| \geq 1$ is

$$y \geq \sqrt{1 - x^2}.$$

Therefore

$$\begin{aligned} \text{area}(\mathcal{F}) &= \int_{-1/2}^{1/2} \int_{\sqrt{1-x^2}}^{\infty} \frac{dy \, dx}{y^2} \\ &= \int_{-1/2}^{1/2} \frac{dx}{\sqrt{1-x^2}} \\ &= 2 \arcsin(1/2) = \frac{\pi}{3}. \end{aligned}$$

The hyperbolic quotient is naturally the left quotient

$$\Gamma \backslash \mathbb{H} = \Gamma \backslash G / K.$$

Lifting the fundamental region $\mathcal{F}$ through the compact K -fiber and using

$$dg = C \frac{dx \, dy}{y^2} dk$$

shows that $\Gamma \backslash G$ has finite Haar volume. Because G is unimodular, inversion preserves Haar measure and gives a diffeomorphism

$$\Gamma \backslash G \longrightarrow G / \Gamma, \quad \Gamma g \mapsto g^{-1} \Gamma.$$

Hence G / Γ also has finite invariant volume, which is exactly the lattice condition.

To verify noncompactness rather than infer it from the unbounded shape of a fundamental region, consider $y = e^t \rightarrow \infty$. Under inversion the class of a_t in $\Gamma \backslash G$ corresponds to $a_{-t} \Gamma$ in G / Γ . The associated lattice $a_{-t} \mathbb{Z}^2$ contains

$$a_{-t} e_1 = e^{-t/2} e_1,$$

whose length tends to zero. Mahler compactness therefore shows that these classes leave every compact subset. Thus the quotient is genuinely noncompact. $\square$

9. The cusp as a short-vector phenomenon

For $z = x + iy \in \mathbb{H}$, set

$$g_z = u_x a_{\log y} = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \sqrt{y} & 0 \\ 0 & 1/\sqrt{y} \end{pmatrix}.$$

Then $g_z \cdot i = z$. Associate the unimodular lattice

$$\Lambda_z = g_z \mathbb{Z}^2.$$

The vector e_2 maps to

$$g_z e_2 = \begin{pmatrix} x/\sqrt{y} \\ 1/\sqrt{y} \end{pmatrix},$$

so for $|x| \leq 1/2$,

$$\|g_z e_2\| \leq \frac{\sqrt{5}/2}{\sqrt{y}}.$$

Thus as $y \rightarrow \infty$ in the standard cusp,

$$\lambda_1(\Lambda_z) \rightarrow 0.$$

Mahler compactness therefore identifies the hyperbolic cusp with degeneration of lattices through short vectors.

Conversely, on the standard fundamental domain a very short vector forces large height, and in this two-dimensional model one can see the lower bound directly rather than appeal only to compactness. Suppose

$$|x| \leq \frac{1}{2}, \quad |z| \geq 1, \quad y \leq Y.$$

The condition $|z| \geq 1$ implies

$$y \geq \sqrt{1-x^2} \geq \frac{\sqrt{3}}{2}.$$

For a nonzero integer vector $(m, n)^T$,

$$g_z \begin{pmatrix} m \\ n \end{pmatrix} = \begin{pmatrix} \sqrt{y} m + xn/\sqrt{y} \\ n/\sqrt{y} \end{pmatrix}.$$

If $n \neq 0$, its norm is at least

$$\frac{|n|}{\sqrt{y}} \geq \frac{1}{\sqrt{Y}}.$$

If $n = 0$, then $m \neq 0$ and its norm is at least

$$|m|\sqrt{y} \geq \sqrt{\frac{\sqrt{3}}{2}}.$$

Consequently

$$\lambda_1(\Lambda_z) \geq \min \left\{ Y^{-1/2}, \sqrt{\frac{\sqrt{3}}{2}} \right\}$$

throughout the part of the standard fundamental domain with $y \leq Y$. Thus $\lambda_1(\Lambda_z) \rightarrow 0$ can occur only when $y \rightarrow \infty$. In this precise sense the geometric notion of entering the cusp and the lattice-theoretic notion of developing a short vector are equivalent.

10. Injectivity radius and local quotient geometry

Let G carry any left-invariant Riemannian metric and let $X = G/\Gamma$ with Γ discrete. At $x = g\Gamma$, define the group-theoretic injectivity scale

$$\text{inj}(x) = \frac{1}{2} \inf_{\gamma \in \Gamma \setminus \{e\}} d_G(e, g\gamma g^{-1}).$$

The factor $1/2$ ensures that the ball of radius $\text{inj}(x)$ around g projects injectively to X .

Proposition 7.10.1. *For every $x \in X$, $\text{inj}(x) > 0$. On compact subsets of X , the injectivity scale is bounded below uniformly.*

PROOF. The conjugate $g\Gamma g^{-1}$ is discrete, so the identity is isolated and the infimum over nontrivial elements is positive. For uniformity on a compact set, cover it by finitely many quotient neighborhoods on which the covering map admits injective lifts of a fixed positive radius after possibly shrinking each neighborhood. The minimum of the finitely many radii is positive. $\square$

In spaces of lattices, degeneration of injectivity radius is closely tied to short algebraic data. The precise quantitative relation depends on the chosen metric, but the conceptual picture is robust: near the cusp, nontrivial lattice symmetries become visible at smaller and smaller ambient scales.

11. An elementary two-dimensional reduction argument

In dimension two, the compactness mechanism can be seen without the full machinery of successive minima. Let $\Lambda \subset \mathbb{R}^2$ be unimodular and choose a shortest nonzero vector v_1 . Choose $v_2 \in \Lambda$ so that v_1, v_2 form a basis and replace v_2 by

$$v_2 - nv_1$$

with $n \in \mathbb{Z}$ chosen to make the component of v_2 parallel to v_1 have absolute value at most $\|v_1\|/2$.

Because the parallelogram has area 1, the perpendicular component of v_2 has magnitude

$$\frac{1}{\|v_1\|}.$$

Therefore

$$\|v_2\|^2 \leq \frac{\|v_1\|^2}{4} + \frac{1}{\|v_1\|^2}.$$

Minkowski's first theorem gives a dimension-dependent upper bound on $\|v_1\|$, while a hypothesis $\|v_1\| \geq \varepsilon$ bounds the reciprocal term. Hence both basis vectors are uniformly bounded. This proves the difficult direction of Mahler compactness for X_2 directly.

The argument is a miniature version of lattice reduction: subtract integer multiples of previously chosen basis vectors to control parallel components, while fixed covolume controls the remaining transverse components.

12. Siegel-set coordinates for the modular cusp

The NAK decomposition writes

$$g = u_x a_t k.$$

For the modular group, integer translations change x by integers, so one reduces to

$$|x| \leq \frac{1}{2}.$$

The condition $|z| \geq 1$ imposes a lower bound on $y = e^t$, namely

$$y \geq \sqrt{1 - x^2} \geq \frac{\sqrt{3}}{2}.$$

Thus the standard fundamental region sits inside a set of the form

$$\left\{ u_x a_t k : |x| \leq \frac{1}{2}, t \geq t_0, k \in K \right\}$$

together with a compact correction near its lower boundary. Such a region is the simplest example of a *Siegel set*: the unipotent coordinate is kept in a bounded interval, the compact coordinate is harmless, and noncompactness is concentrated in a positive diagonal cone.

The Haar density

$$e^{-t} dx dt dk$$

shows immediately why the cusp can be noncompact but finite volume:

$$\int_{t_0}^{\infty} e^{-t} dt < \infty.$$

This calculation isolates an important general mechanism. A cusp extends infinitely far in a diagonal direction, while the Haar Jacobian decays exponentially fast enough to make the total volume finite.

13. Diophantine approximation as a short-vector calculation

The space of lattices turns rational approximation into a geometric question with almost no abstraction. Fix $\alpha \in \mathbb{R}$ and let

$$u_\alpha = \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}, \quad \Lambda_\alpha = u_\alpha \mathbb{Z}^2.$$

For integers p, q ,

$$u_\alpha \begin{pmatrix} -p \\ q \end{pmatrix} = \begin{pmatrix} q\alpha - p \\ q \end{pmatrix}.$$

Apply the diagonal flow

$$a_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix}.$$

The squared norm becomes

$$e^t |q\alpha - p|^2 + e^{-t} q^2.$$

Proposition 7.13.1 (Balancing identity). *For $q \neq 0$ and $q\alpha - p \neq 0$,*

$$\min_{t \in \mathbb{R}} \left(e^t |q\alpha - p|^2 + e^{-t} q^2 \right) = 2|q| |q\alpha - p|.$$

The minimizing time satisfies

$$e^t = \frac{|q|}{|q\alpha - p|}.$$

PROOF. Put

$$A = |q\alpha - p|^2, \quad B = q^2.$$

The arithmetic–geometric mean inequality gives

$$Ae^t + Be^{-t} \geq 2\sqrt{AB},$$

with equality exactly when $Ae^t = Be^{-t}$, or $e^t = \sqrt{B/A}$. Substituting A, B gives the formulas. $\square$

Thus a good rational approximation

$$|q\alpha - p| \ll \frac{1}{|q|}$$

produces a short vector somewhere along the diagonal orbit $a_t \Lambda_\alpha$. Conversely, a short vector in this orbit gives integers p, q with a correspondingly small product

$$|q| |q\alpha - p|.$$

Worked Example 7.13.2 (Badly approximable numbers and bounded diagonal orbits). Suppose there is $c > 0$ such that

$$|q\alpha - p| \geq \frac{c}{|q|}$$

for every $p \in \mathbb{Z}$ and nonzero $q \in \mathbb{Z}$. The relevant dynamical statement concerns the *forward* orbit $\{a_t \Lambda_\alpha : t \geq 0\}$. We check the two kinds of integer vectors separately. If $q = 0$ and $p \neq 0$, then

$$a_t u_\alpha(-p, 0)^T = (-e^{t/2} p, 0)^T,$$

so such vectors expand, rather than contract, for $t \geq 0$. If $q \neq 0$, then

$$e^t |q\alpha - p|^2 + e^{-t} q^2 \geq 2|q| |q\alpha - p| \geq 2c.$$

Hence every nonzero vector in $a_t\Lambda_\alpha$ has norm at least $\min(1, \sqrt{2c})$ for every $t \geq 0$. Mahler compactness therefore implies that the forward diagonal orbit is relatively compact.

Conversely, if the forward orbit is relatively compact, Mahler gives a uniform lower bound on the norms of all its nonzero vectors. First note that $q\alpha - p$ cannot vanish for nonzero q : if it did, then $u_\alpha(-p, q)^T = (0, q)^T$, whose image under a_t has norm $e^{-t/2}|q| \rightarrow 0$, contradicting the Mahler lower bound. Now, for each $q \neq 0$, choose an integer p nearest to $q\alpha$. Then $0 < |q\alpha - p| \leq 1/2$, and the balancing time $e^t = |q|/|q\alpha - p|$ satisfies $t \geq 0$. Evaluating at this time gives a uniform positive lower bound for $|q| |q\alpha - p|$. Thus boundedness of the forward orbit is equivalent to bad approximability of α . This is the simplest exact instance of the dictionary between Diophantine inequalities and trajectories in a homogeneous space.

14. The same dictionary in higher dimension

The preceding calculation is not a peculiarity of continued fractions. It already contains the higher-dimensional mechanism used throughout homogeneous Diophantine approximation.

Let $m \geq 1$ and let

$$\alpha = (\alpha_1, \dots, \alpha_m) \in \mathbb{R}^m.$$

Embed simultaneous rational approximation into the space of unimodular lattices in dimension $m + 1$ by

$$u_\alpha = \begin{pmatrix} I_m & \alpha \\ 0 & 1 \end{pmatrix} \in \mathrm{SL}_{m+1}(\mathbb{R}), \quad \Lambda_\alpha = u_\alpha \mathbb{Z}^{m+1},$$

where α is viewed as a column vector. For $\mathbf{p} \in \mathbb{Z}^m$ and $q \in \mathbb{Z}$,

$$u_\alpha \begin{pmatrix} -\mathbf{p} \\ q \end{pmatrix} = \begin{pmatrix} q\alpha - \mathbf{p} \\ q \end{pmatrix}.$$

Use the determinant-one diagonal flow

$$a_t = \mathrm{diag}(\underbrace{e^{t/m}, \dots, e^{t/m}}_{m \text{ times}}, e^{-t}).$$

Then

$$(7.12) \quad \left\| a_t u_\alpha \begin{pmatrix} -\mathbf{p} \\ q \end{pmatrix} \right\|^2 = e^{2t/m} \|q\alpha - \mathbf{p}\|^2 + e^{-2t} q^2.$$

The two terms have a transparent meaning. Forward time expands the approximation error in the first m coordinates and contracts the denominator coordinate. A very good rational approximation is precisely what allows the two effects to balance at a small common scale.

Proposition 7.14.1 (Higher-dimensional balancing). *Let $q \neq 0$ and put*

$$E = \|q\alpha - \mathbf{p}\| > 0.$$

The minimum over $t \in \mathbb{R}$ of the right-hand side of (7.12) is

$$(7.13) \quad C_m |q|^{2/(m+1)} E^{2m/(m+1)}, \quad C_m = m^{1/(m+1)} + m^{-m/(m+1)}.$$

In particular, this minimum is bounded below uniformly in $\mathbf{p}, q$ if and only if the products

$$|q|^{1/m} \|q\alpha - \mathbf{p}\|$$

are bounded below uniformly away from zero.

PROOF. Write

$$F(t) = E^2 e^{2t/m} + q^2 e^{-2t}.$$

Differentiation gives

$$F'(t) = \frac{2}{m} E^2 e^{2t/m} - 2q^2 e^{-2t}.$$

The unique critical point satisfies

$$\frac{1}{m} E^2 e^{2t/m} = q^2 e^{-2t},$$

or equivalently

$$e^{2t(1+1/m)} = \frac{mq^2}{E^2}.$$

Since $F(t) \rightarrow \infty$ as $t \rightarrow \pm\infty$, this critical point is the global minimum. Substituting the balancing relation gives

$$F_{\min} = C_m |q|^{2/(m+1)} E^{2m/(m+1)}.$$

Taking the power $(m+1)/(2m)$ shows that a positive lower bound for $F_{\min}$ is equivalent, up to a constant depending only on m , to a positive lower bound for $|q|^{1/m} E$. $\square$

Worked Example 7.14.2 (Simultaneously badly approximable vectors).

Call $\alpha \in \mathbb{R}^m$ *simultaneously badly approximable* if there is $c > 0$ such that

$$(7.14) \quad \|q\alpha - \mathbf{p}\| \geq c|q|^{-1/m} \quad (\mathbf{p} \in \mathbb{Z}^m, q \in \mathbb{Z} \setminus \{0\}).$$

Then the forward orbit

$$\{a_t \Lambda_\alpha : t \geq 0\}$$

is relatively compact in $\mathrm{SL}_{m+1}(\mathbb{R})/\mathrm{SL}_{m+1}(\mathbb{Z})$. Indeed, vectors with $q = 0$ expand in forward time, while for $q \neq 0$ formula (7.13) and (7.14) give a uniform positive lower bound for the length of every nonzero lattice vector. Mahler compactness then applies.

Conversely, if the forward orbit is relatively compact, Mahler gives a uniform positive lower bound on all nonzero vectors along it. First, $q\alpha - \mathbf{p}$

cannot vanish for any $q \neq 0$: otherwise $u_{\alpha}(-\mathbf{p}, q)^T = (0, q)^T$, and its image under a_t would have norm $e^{-t}|q| \rightarrow 0$. For each $q \neq 0$, choose $\mathbf{p}$ by rounding the coordinates of $q\alpha$, so that $0 < E \leq \sqrt{m}/2$. For all sufficiently large $|q|$ the balancing time in the proof above is nonnegative; evaluating at that time yields (7.14). The finitely many remaining denominators are absorbed by decreasing the constant c .

Thus bounded forward diagonal orbits encode bad simultaneous rational approximation in every dimension. The one-dimensional calculation is the case $m = 1$, not a separate phenomenon.

The point of this example is conceptual as well as arithmetic. A Diophantine inequality becomes a compactness statement because the unipotent matrix stores the approximation error and denominator in one lattice vector, while the diagonal flow chooses the scale at which those two quantities should be compared.

15. Primitive sublattices and higher-rank degeneration

A rank- k primitive sublattice $\Delta < \mathbb{Z}^d$ with basis $v_1, \dots, v_k$ determines the wedge vector

$$v_{\Delta} = v_1 \wedge \cdots \wedge v_k \in \bigwedge^k \mathbb{Z}^d,$$

well defined up to sign. For $g \in \mathrm{SL}_d(\mathbb{R})$,

$$\left\| \left(\bigwedge^k g \right) v_{\Delta} \right\|$$

is the k -dimensional covolume of $g\Delta$ inside its real span.

Let

$$a_t = \mathrm{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_d t}), \quad \sum_i \lambda_i = 0.$$

On a coordinate wedge

$$e_{i_1} \wedge \cdots \wedge e_{i_k},$$

the action is multiplication by

$$e^{(\lambda_{i_1} + \cdots + \lambda_{i_k})t}.$$

Thus higher exterior powers record not only individual short vectors but collapsing rational subspaces of every dimension.

Mahler compactness for unimodular lattices can be stated using only rank-one vectors, because fixed total covolume and geometry of numbers force all other degenerations to create a short vector somewhere. In quantitative arguments, however, exterior-power covolumes are often more informative:

they identify which rational subspace is responsible for the approach to a cusp.

Worked Example 7.15.1 (A rank-two collapse in SL_3). Take

$$a_t = \mathrm{diag}(e^t, e^t, e^{-2t}).$$

The coordinate plane $\mathrm{span}(e_1, e_2)$ expands in area by e^{2t} , while planes containing e_3 can contract. For example,

$$\bigwedge^2 a_t(e_1 \wedge e_3) = e^{-t}(e_1 \wedge e_3).$$

Thus the covolume of the rank-two sublattice generated by e_1, e_3 shrinks like e^{-t} . At the same time e_3 itself shrinks like e^{-2t} , so Mahler detects the escape already at rank one. In more complicated sheared lattices, the exterior-power viewpoint identifies the responsible rational subspace before a convenient short basis vector is obvious.

16. Conceptual summary

A lattice is a discrete subgroup whose homogeneous quotient has finite invariant volume. Finite volume forces the ambient group to be unimodular. For Euclidean lattices, the geometry of the quotient X_d is controlled by the shortest vector:

$$\mathcal{K} \subset X_d \text{ precompact} \iff \inf_{\Lambda \in \mathcal{K}} \lambda_1(\Lambda) > 0.$$

This is Mahler's compactness criterion.

For the modular surface, the standard region

$$|x| \leq \frac{1}{2}, \quad |z| \geq 1$$

has finite hyperbolic area $\pi/3$ but extends to infinite height. The high cusp corresponds exactly to a short vector in the associated unimodular lattice. Thus the geometric phrase “escape to infinity” acquires an algebraic detector: some nonzero lattice vector becomes small.

CHAPTER 8

Hyperbolic Geometry, Geodesic Flow, and Horocycles

1. The upper half-plane metric

The upper half-plane is

$$\mathbb{H} = \{z = x + iy \in \mathbb{C} : y > 0\}.$$

Equip it with the Riemannian metric

$$ds^2 = \frac{dx^2 + dy^2}{y^2}.$$

The associated area element is

$$dA = \frac{dx \, dy}{y^2}.$$

The metric has constant curvature -1 . The full curvature calculation is not required for the group-theoretic constructions here, but one elementary feature should be checked immediately: vertical scaling is an isometry because both numerator and denominator scale by the same factor.

2. Möbius transformations are isometries

Let

$$G = \mathrm{PSL}_2(\mathbb{R}) = \mathrm{SL}_2(\mathbb{R}) / \{\pm I\}.$$

It acts on $\mathbb{H}$ by

$$gz = \frac{az + b}{cz + d}.$$

THEOREM 8.2.1. *The action of G on $\mathbb{H}$ preserves the hyperbolic metric and the orientation.*

PROOF. For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$g'(z) = \frac{1}{(cz + d)^2}$$

because $ad - bc = 1$. Also

$$\Im(gz) = \frac{\Im z}{|cz + d|^2}.$$

If $w = gz$, then

$$|dw|^2 = |g'(z)|^2 |dz|^2 = \frac{|dz|^2}{|cz + d|^4}$$

and

$$(\Im w)^2 = \frac{y^2}{|cz + d|^4}.$$

Therefore

$$\frac{|dw|^2}{(\Im w)^2} = \frac{|dz|^2}{y^2}.$$

The map is holomorphic, hence orientation preserving. $\square$

Proposition 8.2.2. *The action of G on $\mathbb{H}$ is transitive, and the stabilizer of i is*

$$K = \text{PSO}(2).$$

Consequently,

$$\mathbb{H} \cong G/K.$$

PROOF. For $z = x + iy$,

$$u_x a_{\log y} \cdot i = x + iy,$$

so the action is transitive. If $g \cdot i = i$, solving

$$\frac{ai + b}{ci + d} = i$$

gives $a = d$ and $b = -c$. Together with $ad - bc = 1$, this means $a^2 + c^2 = 1$, so g is represented by a rotation matrix. Modulo $\pm I$ this is K . $\square$

The invariant measure on G/K obtained from Haar measure is therefore the hyperbolic area measure, up to scalar. In the coordinates

$$g = u_x a_t k, \quad y = e^t,$$

the left Haar density is $Ce^{-t} dx dt dk$. Since $dt = dy/y$ and $e^{-t} = 1/y$, this becomes

$$dg = C \frac{dx dy}{y^2} dk.$$

3. Hyperbolic distance

Along a vertical line $x = x_0$, the hyperbolic length from iy_1 to iy_2 is

$$\left| \int_{y_1}^{y_2} \frac{dy}{y} \right| = |\log(y_2/y_1)|.$$

In particular,

$$d_{\mathbb{H}}(i, e^t i) = |t|.$$

A useful explicit formula for arbitrary points is

$$\cosh d(z, w) = 1 + \frac{|z - w|^2}{2\Im z \Im w}.$$

We give a proof because it makes the role of G transparent.

Proposition 8.3.1 (Distance formula). *For $z, w \in \mathbb{H}$,*

$$\cosh d(z, w) = 1 + \frac{|z - w|^2}{2\Im z \Im w}.$$

PROOF. Both sides are invariant under G . Invariance of the left side is the isometry theorem. For the right side, one checks using

$$gz - gw = \frac{z - w}{(cz + d)(cw + d)}$$

and the formula for imaginary parts that the quotient

$$\frac{|z - w|^2}{\Im z \Im w}$$

is invariant.

Thus move z to i . It remains to justify explicitly that an element of K can move the second point onto the imaginary axis. The Cayley map

$$C(\zeta) = \frac{\zeta - i}{\zeta + i}$$

sends $\mathbb{H}$ biholomorphically to the unit disk and sends i to 0. Under this coordinate, the stabilizer K acts by ordinary rotations of the disk (a direct substitution with a rotation matrix verifies this). Hence some $k \in K$ sends $C(w)$ to the real interval $[0, 1)$; applying C^{-1} sends that interval to the positive imaginary axis. We may therefore assume $w = ie^t$ with $t \geq 0$. Then $d(i, w) = t$, while

$$\begin{aligned} 1 + \frac{|i - ie^t|^2}{2e^t} &= 1 + \frac{(e^t - 1)^2}{2e^t} \\ &= \frac{e^t + e^{-t}}{2} = \cosh t. \end{aligned}$$

This proves the formula. □

4. Geodesics and the boundary

The boundary at infinity of $\mathbb{H}$ is naturally

$$\partial\mathbb{H} = \mathbb{R} \cup \{\infty\} = \mathbb{P}^1(\mathbb{R}).$$

The G -action extends continuously to this boundary by the same Möbius formula.

THEOREM 8.4.1. *The unoriented hyperbolic geodesics in $\mathbb{H}$ are exactly*

- *vertical Euclidean lines, and*
- *Euclidean semicircles orthogonal to the real axis.*

Every complete geodesic is uniquely determined by its two distinct boundary endpoints.

PROOF. The vertical line through 0 is the orbit

$$t \mapsto a_t i = i e^t.$$

Its speed in the hyperbolic metric is

$$\frac{|d(e^t)/dt|}{e^t} = 1.$$

A direct Christoffel-symbol calculation shows it is a geodesic, or one may use the fact that it locally minimizes length between points on the line by

$$\int \frac{\sqrt{x'(t)^2 + y'(t)^2}}{y(t)} dt \geq \left| \int \frac{y'(t)}{y(t)} dt \right|.$$

Every pair of distinct boundary points can be sent by a Möbius transformation in G to 0 and ∞ . Since isometries take geodesics to geodesics, the inverse image of the vertical line is a geodesic. Möbius maps take generalized circles (circles or lines) to generalized circles and preserve angles, so these inverse images are precisely circles or lines orthogonal to the boundary. Uniqueness follows by transporting back to the vertical model. $\square$

An *oriented* geodesic has an ordered pair of endpoints

$$(\xi^-, \xi^+) \in (\partial\mathbb{H} \times \partial\mathbb{H}) \setminus \Delta,$$

where ξ^- is the backward endpoint and ξ^+ the forward endpoint.

5. The space of oriented geodesics

The standard upward vertical geodesic has ordered endpoints $(0, \infty)$. Its stabilizer in $G = \mathrm{PSL}_2(\mathbb{R})$ is A .

Proposition 8.5.1. *There is a G -equivariant diffeomorphism*

$$G/A \cong (\partial\mathbb{H} \times \partial\mathbb{H}) \setminus \Delta,$$

$$gA \mapsto (g \cdot 0, g \cdot \infty).$$

PROOF. If g fixes both 0 and ∞ individually, then a matrix representative is diagonal, hence lies in A modulo the center. Thus A is the stabilizer of the ordered pair. The action is transitive on ordered pairs of distinct boundary points: given (ξ^-, ξ^+) , choose a Möbius map sending $(0, \infty)$ to that pair. The orbit-stabilizer theorem gives the homogeneous identification. Smoothness follows in boundary charts away from the diagonal or from the quotient-manifold theorem. $\square$

The invariant measure on this two-dimensional space has a simple boundary formula.

Proposition 8.5.2. *In affine coordinates $u \neq v$ on $\mathbb{R} \subset \partial\mathbb{H}$, the measure*

$$d\nu(u, v) = \frac{du dv}{(u - v)^2}$$

is invariant under G .

PROOF. For $T(z) = (az + b)/(cz + d)$,

$$T'(z) = \frac{1}{(cz + d)^2}$$

and

$$T(u) - T(v) = \frac{u - v}{(cu + d)(cv + d)}.$$

Therefore

$$\frac{dT(u) dT(v)}{(T(u) - T(v))^2} = \frac{du dv}{(u - v)^2}.$$

The formula extends across charts containing ∞ . $\square$

6. The unit tangent bundle as G

Let $T^1\mathbb{H}$ be the unit tangent bundle. Let v_0 be the upward-pointing unit tangent vector at i . Define

$$\Phi : G \rightarrow T^1\mathbb{H}, \quad \Phi(g) = dg_i(v_0),$$

where the notation means the tangent vector based at gi obtained by differentiating the isometry g .

THEOREM 8.6.1. *Φ is a G -equivariant diffeomorphism. Thus*

$$T^1\mathbb{H} \cong \mathrm{PSL}_2(\mathbb{R}).$$

PROOF. G acts transitively on unit tangent vectors. Indeed, first move the base point to i ; the stabilizer K acts transitively on the unit circle in $T_i\mathbb{H}$, so a rotation aligns the vector with v_0 . The stabilizer of v_0 is trivial in $\mathrm{PSL}_2(\mathbb{R})$: an isometry fixing i lies in K , and the only rotation fixing a specified unit tangent vector is the identity. Hence the orbit map is bijective.

Its differential is an isomorphism because both manifolds have dimension 3 and the stabilizer Lie algebra is zero. Therefore it is a local diffeomorphism. A bijective local diffeomorphism is a diffeomorphism. For $h \in G$, the point represented by hg is sent to h applied to the point represented by g , so the map intertwines the left G -actions; this is the required equivariance. $\square$

Remark 8.6.2. For $\mathrm{SL}_2(\mathbb{R})$ rather than $\mathrm{PSL}_2(\mathbb{R})$, the map has kernel $\{\pm I\}$. Keeping track of the center avoids a common factor-of-two ambiguity in quotient identifications.

7. The geodesic flow as right translation

Let $\varphi_t : T^1\mathbb{H} \rightarrow T^1\mathbb{H}$ be the geodesic flow: follow the unit-speed geodesic determined by a unit tangent vector for time t .

THEOREM 8.7.1. *Under the identification $T^1\mathbb{H} \cong G$, the geodesic flow is right multiplication by a_t :*

$$\varphi_t(g) = ga_t, \quad a_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix}.$$

PROOF. At the identity vector v_0 , the curve

$$t \mapsto a_t i = i e^t$$

is the upward vertical geodesic, and the hyperbolic distance from i is $|t|$, so the parameter is unit speed. The tangent direction is also transported correctly by da_t .

For a general vector represented by g , the corresponding geodesic is the image under the isometry g of the vertical geodesic through i with upward unit tangent. Therefore at time t its unit tangent vector is represented by ga_t . $\square$

This is an important convention: on $T^1\mathbb{H} \cong G$, geometry produces a *right* action.

8. Horocycles

A horocycle can be defined geometrically as a level set of a Busemann function, but in the upper half-plane model the picture is elementary. Horocycles centered at ∞ are horizontal lines $y = \text{constant}$; those centered at a

finite boundary point are Euclidean circles tangent to the real axis at that point.

At height 1,

$$u_s i = s + i.$$

Thus the N -orbit of i is the horizontal horocycle through i centered at ∞ .

Definition 8.8.1. Under $T^1\mathbb{H} \cong G$, define the stable horocycle flow by

$$h_s^+(g) = gu_s$$

and the unstable horocycle flow by

$$h_s^-(g) = gv_s.$$

The terminology is justified by conjugation.

Proposition 8.8.2 (Contraction and expansion).

$$a_{-t}u_s a_t = u_{e^{-t}s}, \quad a_{-t}v_s a_t = v_{e^t s}.$$

Consequently,

$$h_s^+(g)a_t = ga_t u_{e^{-t}s},$$

so points on the same N -horocycle approach one another under forward geodesic flow in local group coordinates. The N^- direction expands forward and contracts backward.

PROOF. The matrix identities follow from direct multiplication, or from the Lie algebra relations

$$[H/2, X] = X, \quad [H/2, Y] = -Y.$$

For the right-action relation,

$$gu_s a_t = ga_t(a_{-t}u_s a_t) = ga_t u_{e^{-t}s}.$$

□

Guiding Idea 8.8.3 (Stable does not mean stationary). The horocycle parameter itself is not preserved by geodesic flow. It is rescaled. Two vectors differing by u_s at time 0 differ by $u_{e^{-t}s}$ after time t when compared in the moving frame. Stability is therefore a statement about conjugation, not about the base points lying on the same fixed horizontal line for all time.

9. Busemann functions and horocycles

For a geodesic ray $\gamma(t)$ tending to a boundary point ξ , define the Busemann function

$$\beta_\xi(z, w) = \lim_{t \rightarrow \infty} (d(z, \gamma(t)) - d(w, \gamma(t))).$$

In the hyperbolic plane the limit can be computed explicitly, which also proves its existence and independence of the ray. For $\xi = \infty$, take $\gamma(t) = ie^t$. The distance formula gives

$$\cosh d(z, ie^t) = 1 + \frac{|z - ie^t|^2}{2(\Im z)e^t} = \frac{e^t}{2\Im z} + O(1)$$

as $t \rightarrow \infty$. Since $\operatorname{arcosh} R = \log(2R) + o(1)$,

$$d(z, ie^t) = t - \log(\Im z) + o(1).$$

Subtracting the corresponding formula for w yields

$$\beta_\infty(z, w) = \log \frac{\Im w}{\Im z}.$$

For a finite endpoint $\xi \in \mathbb{R}$, the Möbius transformation

$$T_\xi(z) = -\frac{1}{z - \xi}$$

belongs to $\operatorname{PSL}_2(\mathbb{R})$ and sends ξ to ∞ . Isometry invariance therefore gives

$$\begin{aligned} \beta_\xi(z, w) &= \beta_\infty(T_\xi z, T_\xi w) \\ &= \log \left(\frac{(\Im w)|z - \xi|^2}{(\Im z)|w - \xi|^2} \right). \end{aligned}$$

This explicit formula proves the asserted limit for every boundary point. Its level sets are horizontal lines when $\xi = \infty$ and, after a Möbius transformation, Euclidean circles tangent to the real axis when ξ is finite.

The group N fixes ∞ and preserves each horizontal horocycle. The diagonal group A also fixes ∞ but moves one horocycle to another by scaling height.

10. Boundary coordinates for a unit tangent vector

A unit tangent vector determines more than its oriented geodesic: it also specifies a position along that geodesic. Thus one expects coordinates

$$(\xi^-, \xi^+, r),$$

where $\xi^- \neq \xi^+$ are the two endpoints and $r \in \mathbb{R}$ is a signed position parameter.

One convenient normalization uses a Busemann coordinate. For $v \in T^1\mathbb{H}$, let $v^-, v^+ \in \partial\mathbb{H}$ be the backward and forward endpoints of the geodesic determined by v , and let $\pi(v)$ be its base point. Fix $o = i$ and set

$$r(v) = \beta_{v^+}(o, \pi(v)).$$

Then

$$v \longmapsto (v^-, v^+, r(v))$$

is a smooth coordinate system on the open set of endpoint charts, with the expected changes of coordinate at ∞ .

With the definition above, geodesic flow satisfies exactly

$$(v^-, v^+, r) \mapsto (v^-, v^+, r + t).$$

Indeed, moving time t toward v^+ increases the chosen Busemann coordinate by t .

These coordinates separate the three pieces of information in a frame: two boundary endpoints determine the oriented geodesic, and one real coordinate determines the point along it.

11. Quotient surfaces and right-flow conventions

Let $\Gamma < G = \mathrm{PSL}_2(\mathbb{R})$ be discrete. The quotient

$$\Gamma \backslash \mathbb{H}$$

is a hyperbolic orbifold in general. If Γ is torsion-free, it is a hyperbolic surface.

Because

$$T^1\mathbb{H} \cong G,$$

the unit tangent bundle quotient is naturally

$$\Gamma \backslash G.$$

If the action is free, this is exactly $T^1(\Gamma \backslash \mathbb{H})$. If elliptic stabilizers occur, the statement is interpreted in the orbifold sense.

The geodesic and horocycle flows descend because they are right multiplications:

$$\Gamma g \mapsto \Gamma g a_t,$$

$$\Gamma g \mapsto \Gamma g u_s.$$

Left multiplication by Γ commutes with right multiplication by G .

Warning 8.11.1 (Relation with G/Γ). Many homogeneous-dynamics texts use the left-coset space G/Γ and left multiplication flows, whereas hyperbolic geometry naturally produces the right-coset convention $\Gamma \backslash G$ with right multiplication. Inversion gives a diffeomorphism

$$\Gamma \backslash G \rightarrow G/\Gamma, \quad \Gamma g \mapsto g^{-1}\Gamma.$$

Under inversion, right multiplication by a_t becomes left multiplication by a_{-t} :

$$(ga_t)^{-1}\Gamma = a_{-t}g^{-1}\Gamma.$$

Thus the two conventions describe the same dynamics with time reversed for the diagonal flow. Writing this conversion explicitly is safer than switching conventions silently.

12. Closed geodesics and hyperbolic elements

Assume Γ is torsion-free for simplicity. A geodesic-flow orbit through Γg is periodic of period $T > 0$ exactly when

$$\Gamma ga_T = \Gamma g,$$

that is, when

$$ga_Tg^{-1} \in \Gamma.$$

Proposition 8.12.1. *Primitive periodic geodesic-flow orbits on $\Gamma \backslash G$, with their flow orientation, correspond to conjugacy classes of primitive hyperbolic elements of Γ . More generally, if*

$$\gamma = ga_Tg^{-1}, \quad T > 0,$$

then T is the translation length of γ along its axis; if $\gamma = \delta^m$ with δ primitive, the underlying periodic orbit has minimal period T/m .

PROOF. If a flow orbit has minimal period $T_0 > 0$, then $\Gamma ga_{T_0} = \Gamma g$, so

$$\delta = ga_{T_0}g^{-1} \in \Gamma.$$

It is hyperbolic. The return-time set

$$L_g = \{t \in \mathbb{R} : ga_tg^{-1} \in \Gamma\}$$

is a closed subgroup of $\mathbb{R}$. Discreteness of Γ implies that it is discrete, so for a periodic orbit it has the form $T_0\mathbb{Z}$ with $T_0 > 0$. The element $\delta = ga_{T_0}g^{-1}$ is therefore the primitive generator of $\Gamma \cap gAg^{-1}$; in particular it cannot be a proper positive power inside that intersection.

Conversely, a hyperbolic element γ has two ordered fixed boundary points. Choosing g that sends $(0, \infty)$ to these points diagonalizes it as ga_Tg^{-1} for a unique $T > 0$ after fixing the forward orientation. Then Γg lies on a periodic geodesic-flow orbit. The subgroup $\Gamma \cap gAg^{-1}$ is discrete in $gAg^{-1} \cong \mathbb{R}$, hence

cyclic; its primitive generator corresponds to the minimal period. Replacing g by ηg with $\eta \in \Gamma$ conjugates the return subgroup and does not change the orbit, so conjugacy is the correct equivalence relation. If $\gamma = \delta^m$, its translation length is m times that of the primitive generator, explaining the final statement. $\square$

In terms of the trace of a lift to $\mathrm{SL}_2(\mathbb{R})$,

$$|\mathrm{tr} \gamma| = 2 \cosh(T/2),$$

so

$$T = 2 \operatorname{arccosh}(|\mathrm{tr} \gamma|/2).$$

13. Periodic horocycles and cusps

A right N -orbit through Γg is periodic exactly when

$$gNg^{-1} \cap \Gamma$$

is a lattice in the one-dimensional group $gNg^{-1} \cong \mathbb{R}$. Equivalently, there is some nonzero s with $gu_s g^{-1} \in \Gamma$. In that case the set of return parameters

$$L_g = \left\{ s \in \mathbb{R} : gu_s g^{-1} \in \Gamma \right\}$$

is a nonzero discrete subgroup of $\mathbb{R}$, hence $L_g = s_* \mathbb{Z}$ for a unique $s_* > 0$. The minimal period of the horocycle orbit is precisely s_* .

Worked Example 8.13.1 (The standard modular horocycle). For $\Gamma = \mathrm{PSL}_2(\mathbb{Z})$,

$$u_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in \Gamma.$$

Hence the N -orbit of Γe is periodic with period 1. Geometrically it projects to a closed horocycle around the cusp at ∞ .

At height y , the horizontal segment $x \in [0, 1]$ has hyperbolic length

$$\int_0^1 \frac{dx}{y} = \frac{1}{y}.$$

Thus closed horocycles become shorter as they move deeper into the cusp.

Worked Example 8.13.2 (How geodesic flow rescales a periodic horocycle). Suppose the right N -orbit through Γg has minimal period $s_* > 0$. After geodesic time t the base point in $\Gamma \backslash G$ is Γga_t . Its return parameters are those s for which

$$(ga_t)u_s(ga_t)^{-1} \in \Gamma.$$

Using

$$a_t u_s a_{-t} = u_{e^t s},$$

this condition becomes

$$gu_{e^t s} g^{-1} \in \Gamma,$$

that is,

$$e^t s \in s_* \mathbb{Z}.$$

Hence the new minimal period is

$$s_*(t) = e^{-t} s_*.$$

This is an exact group-theoretic version of the geometric statement that closed cusp horocycles shrink exponentially under forward geodesic flow. For the standard modular horocycle, $s_* = 1$, so the horocycle through Γa_t has N -period e^{-t} .

There are two parameters in the picture and it is useful not to confuse them. The number $s_*(t)$ is the period in the right- N flow parameter; the projected hyperbolic length of the corresponding closed horocycle is the same exponential scale in the standard normalization, but that agreement is a consequence of the chosen unit-speed geometric model, not the definition of the group period.

14. Haar measure and Liouville measure

Haar measure on $G \cong T^1\mathbb{H}$ is the natural Liouville measure for the geodesic flow. In NAK coordinates,

$$dg = C \frac{dx dy}{y^2} dk.$$

The first factor is hyperbolic area on the base and dk is angular measure on unit tangent directions.

Since G is unimodular, right multiplication by a_t and u_s preserves Haar measure. Therefore geodesic and horocycle flows preserve Liouville measure on $T^1\mathbb{H}$ and on every finite-volume quotient $\Gamma \backslash G$.

Proposition 8.14.1. *If $\Gamma < G$ is a lattice, normalized Haar measure on $\Gamma \backslash G$ is invariant under each of the flows*

$$g \mapsto ga_t, \quad g \mapsto gu_s, \quad g \mapsto gv_s.$$

PROOF. Haar measure on the unimodular group G is both left and right invariant. It descends to the quotient by the left action of Γ , and right translations commute with that left action. Hence the descended measure is right invariant. $\square$

15. Stable and unstable geometry in matrix form

The stable/unstable decomposition can be read directly from the adjoint action. In

$$\mathfrak{sl}_2(\mathbb{R}) = \mathbb{R}Y \oplus \mathbb{R}H \oplus \mathbb{R}X,$$

$$\mathrm{Ad}(a_t)Y = e^{-t}Y, \quad \mathrm{Ad}(a_t)H = H, \quad \mathrm{Ad}(a_t)X = e^tX.$$

For right geodesic flow, nearby points are compared by conjugating the displacement in the opposite order. If $g' = g \exp(Z)$, then

$$g'a_t = ga_t(a_{-t} \exp(Z)a_t) = ga_t \exp(\mathrm{Ad}(a_{-t})Z).$$

Thus an X displacement contracts like e^{-t} , while a Y displacement expands like e^t . This matches the right- N stable and right- N^- unstable conventions above.

Worked Example 8.15.1 (Why a lower-triangular perturbation shears). Take

$$v_\varepsilon = \begin{pmatrix} 1 & 0 \\ \varepsilon & 1 \end{pmatrix}.$$

Compare $u_t v_\varepsilon$ with u_t . In the moving frame,

$$u_t v_\varepsilon u_{-t}$$

can be computed explicitly:

$$\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \varepsilon & 1 \end{pmatrix} \begin{pmatrix} 1 & -t \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 + t\varepsilon & -t^2\varepsilon \\ \varepsilon & 1 - t\varepsilon \end{pmatrix}.$$

The displacement develops a term of size $t^2\varepsilon$ in the upper-right entry. This is the matrix manifestation of polynomial shearing under a unipotent flow. The same phenomenon was encoded infinitesimally by

$$\mathrm{Ad}(u_t)Y = Y + tH - t^2X.$$

16. A geometric dictionary

The following identifications summarize the rank-one model. Here $G = \mathrm{PSL}_2(\mathbb{R})$.

Homogeneous object	Geometric meaning
G/K	points of $\mathbb{H}$
G/A	oriented complete geodesics
G	unit tangent vectors $T^1\mathbb{H}$
right A action	geodesic flow
right N action	stable horocycle flow
right N^- action	unstable horocycle flow
G/B	boundary points $\partial\mathbb{H}$
$\Gamma\backslash G$	unit tangent bundle/orbifold quotient

The table is useful only if the stabilizers and conventions behind it are remembered. For example, G/A records an *oriented* geodesic because A fixes the ordered pair $(0, \infty)$; the Weyl element swaps the endpoints and therefore reverses orientation rather than belonging to A .

17. Hopf coordinates and Haar measure

Boundary endpoints together with one Busemann coordinate give particularly efficient coordinates on $T^1\mathbb{H}$. Write

$$\mathcal{B}^{(2)} = (\partial\mathbb{H} \times \partial\mathbb{H}) \setminus \Delta.$$

For a unit tangent vector v , let (v^-, v^+) be its endpoints and set

$$r = \beta_{v^+}(i, \pi(v)).$$

Then, away from the usual boundary-chart changes at ∞ ,

$$v \longleftrightarrow (v^-, v^+, r)$$

identifies $T^1\mathbb{H}$ with $\mathcal{B}^{(2)} \times \mathbb{R}$.

Proposition 8.17.1 (Geodesic flow in Hopf coordinates). *With the above sign convention,*

$$\varphi_t(v^-, v^+, r) = (v^-, v^+, r + t).$$

PROOF. Geodesic flow does not change the complete oriented geodesic, so the two endpoints are fixed. Along a unit-speed geodesic moving toward v^+ , the Busemann coordinate relative to v^+ changes at unit rate. In the normalized model $v^- = 0$, $v^+ = \infty$, the base point is ie^r and geodesic flow sends it to ie^{r+t} . Since $\beta_\infty(i, ie^s) = s$, the Busemann coordinate changes from r to $r + t$, which is exactly the asserted formula. Equivariance under G gives the general case. $\square$

The G -invariant measure on oriented geodesics was

$$\frac{du dv}{(u - v)^2}.$$

Since dr is translation invariant under geodesic flow, one expects Haar measure to factor as their product.

THEOREM 8.17.2 (Haar measure in Hopf coordinates). *Up to an overall positive constant, Haar measure on*

$$G \cong T^1\mathbb{H}$$

is

$$dg = \frac{du \, dv \, dr}{(u - v)^2}$$

in affine endpoint coordinates $u = v^-$, $v = v^+$.

PROOF. The measure

$$d\nu(u, v) = \frac{du \, dv}{(u - v)^2}$$

is G -invariant on oriented geodesics. The Busemann coordinate changes under $g \in G$ by addition of a function of the endpoints, because Busemann cocycles satisfy

$$\beta_{g\xi}(gz, gw) = \beta_\xi(z, w).$$

Thus the transformation in the r coordinate is a translation, whose Jacobian is 1. Consequently $d\nu \, dr$ is G -invariant on $T^1\mathbb{H}$. It is a nonzero smooth Radon measure, hence by Haar uniqueness it is a scalar multiple of Haar measure. $\square$

This formula is often the cleanest way to see simultaneously that geodesic flow preserves Haar measure and that the endpoint pair is a first integral of the flow.

18. Stable leaves characterized geometrically

The group calculation says that right N -orbits contract under forward geodesic flow. There is a geometric characterization in terms of endpoints and Busemann coordinates.

Proposition 8.18.1. *Two unit tangent vectors $v, w \in T^1\mathbb{H}$ lie on the same stable horocycle if and only if*

$$v^+ = w^+$$

and

$$\beta_{v^+}(i, \pi(v)) = \beta_{w^+}(i, \pi(w)).$$

Equivalently, they have the same forward endpoint and their base points lie on the same horocycle centered at that endpoint.

PROOF. For the normalized forward endpoint ∞ , equal Busemann coordinate means equal height because

$$\beta_\infty(i, x + iy) = \log y.$$

Thus the two base points lie on one horizontal line. The group N acts simply transitively along that horizontal horocycle with the compatible tangent directions. Applying an element of G that sends a general endpoint to ∞ gives the statement for every endpoint. $\square$

Corollary 8.18.2. *If v, w lie on the same stable horocycle, then*

$$d_{T^1\mathbb{H}}(\varphi_t v, \varphi_t w) \rightarrow 0$$

as $t \rightarrow +\infty$ for any left-invariant Riemannian metric on $G \cong T^1\mathbb{H}$, and more generally for any metric uniformly equivalent to such a metric.

PROOF. Write $w = vu_s$ in group coordinates. Then

$$\varphi_t(w) = vu_s a_t = va_t u_{e^{-t}s}.$$

Since $u_{e^{-t}s} \rightarrow e$, the two points converge in the moving local chart. $\square$

19. Exact commutation relations of the flows

Let g_t denote geodesic flow and h_s^+ stable horocycle flow, both acting on the right:

$$g_t(x) = xa_t, \quad h_s^+(x) = xu_s.$$

The group identity $u_s a_t = a_t u_{e^{-t}s}$ gives

$$g_t \circ h_s^+ = h_{e^{-t}s}^+ \circ g_t.$$

Indeed, for $x = g$,

$$(g_t \circ h_s^+)(g) = (gu_s)a_t = ga_t u_{e^{-t}s} = (h_{e^{-t}s}^+ \circ g_t)(g).$$

Equivalently, replacing s by $e^t s$ gives

$$h_s^+ \circ g_t = g_t \circ h_{e^t s}^+.$$

Writing the relation in one fixed composition convention prevents the common mistake of reversing the exponent.

Differentiating the group relation at $s = 0$ recovers the Lie algebra relation

$$\text{Ad}(a_t)X = e^t X.$$

Thus the dynamical commutation law and the root-space eigenvalue are the same statement at group and Lie-algebra levels.

20. Hyperbolic elements and their axes

Let $\gamma \in \mathrm{PSL}_2(\mathbb{R})$ be hyperbolic. It has two distinct fixed boundary points γ^-, γ^+ . The geodesic joining them is called the *axis* of γ .

Proposition 8.20.1. *γ preserves its axis and acts on it by translation through length*

$$\ell(\gamma) = 2 \operatorname{arcosh}(|\operatorname{tr} \tilde{\gamma}|/2),$$

where $\tilde{\gamma}$ is either lift to $\mathrm{SL}_2(\mathbb{R})$.

PROOF. Choose g sending $(0, \infty)$ to (γ^-, γ^+) . Then

$$g^{-1}\gamma g = a_T$$

for some $T \neq 0$, after orienting the axis so $T > 0$. On the vertical geodesic, a_T translates hyperbolic arclength by T , so the same is true on the conjugate axis. A lift of a_T has trace

$$e^{T/2} + e^{-T/2} = 2 \cosh(T/2).$$

Absolute trace removes the choice of central lift, giving the formula. □

This proof shows why traces of hyperbolic matrices encode geometric lengths: diagonalization converts a matrix eigenvalue into translation distance along an invariant geodesic.

21. Cusp excursions in geometric coordinates

On the modular surface, the cusp at ∞ is represented by large y . A geodesic with forward endpoint ∞ has $y(t)$ growing exponentially, while a stable horocycle keeps the Busemann height fixed at a given time. The relation

$$a_t u_s a_{-t} = u_{e^t s}$$

means that when the same horocyclic displacement is viewed before applying the forward geodesic flow, its Euclidean parameter must be magnified exponentially.

This explains a visual feature of the cusp: a horizontal interval of Euclidean length 1 has hyperbolic length $1/y$. As geodesic flow raises height from y to $e^t y$, the same quotient horocycle becomes shorter by the factor e^{-t} . The algebraic contraction of the stable N parameter and the geometric shortening of cusp horocycles are exactly the same scaling law.

22. Conceptual summary

Hyperbolic geometry packages the algebra of $\mathrm{PSL}_2(\mathbb{R})$ into a concrete dynamical model. The homogeneous identifications

$$\mathbb{H} \cong G/K, \quad T^1\mathbb{H} \cong G, \quad \{\text{oriented geodesics}\} \cong G/A$$

turn geometry into group multiplication.

With

$$a_t = \mathrm{diag}(e^{t/2}, e^{-t/2}),$$

the geodesic flow is right multiplication by a_t and the parameter t is actual hyperbolic time. The right- N and right- N^- actions are the stable and unstable horocycle flows because

$$a_{-t}u_s a_t = u_{e^{-t}s}, \quad a_{-t}v_s a_t = v_{e^t s}.$$

On a quotient $\Gamma \backslash G$, hyperbolic elements of Γ give closed geodesics and parabolic/unipotent elements give periodic horocycles around cusps. Haar measure becomes Liouville measure, tying the group-theoretic and geometric pictures together without changing the underlying dynamics.

CHAPTER 9

Worked Geometry Laboratory: Haar Measure and the Modular Cusp

1. Why this laboratory is here

The abstract statements about Haar measure and quotient geometry become much easier to remember once one has computed a non-unimodular group and a cusp by hand. The two calculations below also explain why modular functions and height functions later enter nondivergence estimates.

Worked Example 9.1.1 (Left and right Haar measure on the affine group). Let $G = \{(a, b) : a > 0, b \in \mathbb{R}\}$ with $(a, b)(a', b') = (aa', b + ab')$. We claim that

$$d\mu_L(a, b) = a^{-2} da db, \quad d\mu_R(a, b) = a^{-1} da db.$$

For left translation by (a_0, b_0) the change of variables is $(a, b) \mapsto (a_0 a, b_0 + a_0 b)$, with Jacobian a_0^2 . Hence $(a_0 a)^{-2} a_0^2 = a^{-2}$, proving left invariance. For right translation, $(a, b) \mapsto (aa_0, b + ab_0)$ has Jacobian a_0 , and $(aa_0)^{-1} a_0 = a^{-1}$, proving right invariance. Applying right translation to μ_L multiplies it by a_0^{-1} ; this is the modular correction. Thus the affine group is not unimodular.

Worked Example 9.1.2 (Haar measure in SL_2 Iwasawa coordinates). Write $g = n_x a_y k_\theta$ with $z = x + iy \in \mathbb{H}$, $y > 0$. The quotient $G/K \simeq \mathbb{H}$ carries the G -invariant metric $ds^2 = (dx^2 + dy^2)/y^2$, whose area element is $dx dy/y^2$. Normalizing Haar measure on K by $d\theta/(2\pi)$ therefore gives

$$dg = c \frac{dx dy}{y^2} d\theta.$$

For $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, the standard domain $|x| \leq 1/2$, $|z| \geq 1$ has a cusp $y \rightarrow \infty$ but finite area since $\int_Y^\infty y^{-2} dy = Y^{-1}$. The calculation is the basic model for “noncompact but finite volume.”

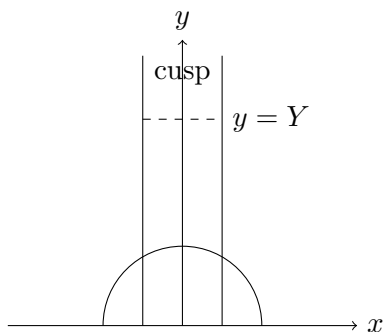


FIGURE 1. The modular fundamental domain. The horizontal cross-section has Euclidean width 1 but hyperbolic length $1/y$, so the cusp has finite area.

APPENDIX A

Analytic and Differential-Geometric Background

This appendix collects a small number of standard tools used repeatedly in the volume. The purpose is not to replace a full course in manifold theory or measure theory, but to make the hypotheses and logical dependencies explicit.

1. Inverse and implicit function theorems

Lemma A.1.1 (Contraction principle in finite dimensions). *Let $B \subset \mathbb{R}^n$ be a nonempty closed subset and let $T : B \rightarrow B$ satisfy*

$$\|T(u) - T(v)\| \leq q\|u - v\|, \quad 0 \leq q < 1.$$

Then T has a unique fixed point in B .

PROOF. Starting from $u_0 \in B$, put $u_{j+1} = T(u_j)$. Inductively,

$$\|u_{j+1} - u_j\| \leq q^j \|u_1 - u_0\|,$$

so for $k > j$,

$$\|u_k - u_j\| \leq \frac{q^j}{1 - q} \|u_1 - u_0\|.$$

Thus (u_j) is Cauchy. Closed subsets of $\mathbb{R}^n$ are complete, so $u_j \rightarrow u \in B$. Continuity of T gives $T(u) = u$. If $T(v) = v$, then $\|u - v\| \leq q\|u - v\|$, hence $u = v$. $\square$

THEOREM A.1.2 (Inverse function theorem). *Let $U \subset \mathbb{R}^n$ be open and $F : U \rightarrow \mathbb{R}^n$ continuously differentiable. If DF_x is invertible at $x \in U$, then there are neighborhoods U_0 of x and V_0 of $F(x)$ such that*

$$F : U_0 \rightarrow V_0$$

is a C^1 diffeomorphism. If F is C^k (respectively smooth), its inverse is C^k (respectively smooth).

PROOF. Translations and the linear change of target coordinates given by DF_x^{-1} reduce us to the case

$$x = 0, \quad F(0) = 0, \quad DF_0 = I.$$

Choose $r > 0$ with $\overline{B(0, r)} \subset U$ and

$$\|DF_z - I\| \leq \frac{1}{2} \quad (\|z\| \leq r).$$

The mean-value formula along the segment from u to v gives

$$(1.1) \quad \|(F(u) - u) - (F(v) - v)\| \leq \frac{1}{2}\|u - v\|.$$

In particular,

$$(1.2) \quad \frac{1}{2}\|u - v\| \leq \|F(u) - F(v)\| \leq \frac{3}{2}\|u - v\|.$$

so F is injective on $B(0, r)$.

For $y \in B(0, r/2)$ define

$$T_y(z) = z + y - F(z).$$

By (1.1), T_y is a contraction with constant $1/2$. Moreover

$$\|T_y(z)\| \leq \|y\| + \|z - F(z)\| \leq \frac{r}{2} + \frac{1}{2}\|z\| \leq r$$

for $\|z\| \leq r$. Lemma A.1.1 therefore gives a unique $z \in \overline{B(0, r)}$ with $T_y(z) = z$, equivalently $F(z) = y$. Thus

$$B(0, r/2) \subset F(B(0, r)).$$

Let $V_0 = B(0, r/2)$ and let $U_0 = F^{-1}(V_0) \cap B(0, r)$. The inverse map $G: V_0 \rightarrow U_0$ is Lipschitz by (1.2):

$$\|G(y) - G(y')\| \leq 2\|y - y'\|.$$

It remains to differentiate G . Fix $y = F(z)$ and write $y + h = F(z + k)$. Since F is differentiable,

$$h = DF_z k + o(\|k\|).$$

The Lipschitz estimate gives $\|k\| = O(\|h\|)$, hence

$$k = (DF_z)^{-1}h + o(\|h\|).$$

Therefore

$$DG_y = (DF_{G(y)})^{-1}.$$

The right-hand side is continuous because matrix inversion is continuous on GL_n , so G is C^1 . If F is C^k , the displayed formula and induction show that G is C^k ; the smooth case follows for every k . $\square$

Corollary A.1.3 (Quantitative inverse-function estimate). *Let $F : B(x_0, r) \rightarrow \mathbb{R}^n$ be C^1 , put $A = DF_{x_0}$, and assume*

$$\|A^{-1}\| \leq M, \quad \|A^{-1}(DF_x - A)\| \leq \frac{1}{2} \quad (x \in B(x_0, r)).$$

Then F is injective on $B(x_0, r)$,

$$B\left(F(x_0), \frac{r}{2M}\right) \subset F(B(x_0, r)),$$

and the inverse on that ball is $2M$ -Lipschitz. In particular, if $\|D^2F\| \leq L$ on the ball, the hypothesis holds whenever $r \leq (2ML)^{-1}$.

PROOF. Apply the normalized proof above to

$$H(u) = A^{-1}(F(x_0 + u) - F(x_0)).$$

The last assertion follows from the mean-value estimate $\|DF_x - A\| \leq L\|x - x_0\|$. $\square$

The manifold version follows by inserting coordinate charts. We now derive the two consequences used throughout, so the regular-value and submersion theorems are no longer independent inputs.

THEOREM A.1.4 (Local form of a submersion). *If $F : M^m \rightarrow N^n$ has surjective differential at x , then there are local coordinates near x and $F(x)$ in which*

$$F(x_1, \dots, x_m) = (x_1, \dots, x_n).$$

In particular, a smooth submersion admits local smooth sections.

PROOF. Choose coordinate charts sending x and $F(x)$ to the origins, so it is enough to consider a smooth map $F : U \subset \mathbb{R}^m \rightarrow \mathbb{R}^n$ with $F(0) = 0$ and DF_0 surjective. After a linear change of coordinates in the source, the restriction of DF_0 to the first n coordinate directions is invertible. Write a source point as $(u, v) \in \mathbb{R}^n \times \mathbb{R}^{m-n}$ and define

$$\Phi(u, v) = (F(u, v), v).$$

Its derivative at the origin is the block matrix

$$D\Phi_0 = \begin{pmatrix} D_u F(0) & D_v F(0) \\ 0 & I_{m-n} \end{pmatrix},$$

which is invertible because $D_u F(0)$ is. The inverse function theorem gives neighborhoods on which Φ is a diffeomorphism. Using $(y, v) = \Phi(u, v)$ as source coordinates and y as target coordinates, F is exactly the projection $(y, v) \mapsto y$.

For the section statement, fix $v = 0$ in these coordinates: $y \mapsto \Phi^{-1}(y, 0)$ is smooth and is sent to y by F . $\square$

THEOREM A.1.5 (Regular-value theorem). *Let $F : M^m \rightarrow N^n$ be smooth and let $y \in N$. If dF_x is surjective for every $x \in F^{-1}(y)$, then $F^{-1}(y)$ is an embedded submanifold of dimension $m - n$, with*

$$T_x F^{-1}(y) = \ker dF_x.$$

PROOF. Fix $x \in F^{-1}(y)$. By Theorem A.1.4, choose local coordinates $(z_1, \dots, z_m)$ at x and $(w_1, \dots, w_n)$ at y in which

$$F(z_1, \dots, z_m) = (z_1, \dots, z_n).$$

In this chart the level set $F^{-1}(y)$ is given by $z_1 = \dots = z_n = 0$. It is therefore locally an embedded copy of an open subset of $\mathbb{R}^{m-n}$, and these local descriptions are compatible because they are restrictions of the ambient smooth charts. Its tangent space is cut out by $dz_1 = \dots = dz_n = 0$, which is precisely the kernel of dF_x . $\square$

2. Smooth bump functions and compact partitions of unity

The manuscript only needs bump functions near compact sets and finite subordinate partitions there. Both can be constructed directly from Euclidean bumps, so no global paracompactness theorem is needed for these uses.

Lemma A.2.1 (Euclidean bump with prescribed compact core). *If $K \subset U \subset \mathbb{R}^d$, with K compact and U open, there is $\rho \in C_c^\infty(U)$ satisfying $0 \leq \rho \leq 1$ and $\rho = 1$ on a neighborhood of K .*

PROOF. For $a \in K$, choose radii $0 < r_a < R_a$ with $\overline{B(a, R_a)} \subset U$. Compactness gives finitely many balls $B(a_j, r_j)$ covering K . It therefore suffices to construct a smooth function that is one on $\overline{B(a_j, r_j)}$ and zero outside $B(a_j, R_j)$.

Set

$$\theta(t) = \begin{cases} e^{-1/t}, & t > 0, \\ 0, & t \leq 0, \end{cases} \quad \chi(t) = \frac{\theta(1-t)}{\theta(1-t) + \theta(t)}.$$

Then χ is smooth, equals 1 for $t \leq 0$, and equals 0 for $t \geq 1$. Hence

$$\rho_j(x) = \chi\left(\frac{\|x - a_j\|^2 - r_j^2}{R_j^2 - r_j^2}\right)$$

is smooth, equals one on $\overline{B(a_j, r_j)}$, and is supported in $B(a_j, R_j)$. Finally

$$\rho = 1 - \prod_j (1 - \rho_j)$$

has the desired properties. $\square$

Lemma A.2.2 (Manifold bump near a compact set). *Let M be a finite-dimensional smooth manifold, $K \subset M$ compact, and U an open neighborhood of K . Then there is $\rho \in C_c^\infty(U)$ with $0 \leq \rho \leq 1$ and $\rho = 1$ on a neighborhood of K .*

PROOF. Choose finitely many coordinate neighborhoods W_j with compact closures in U and smaller open sets V_j whose closures lie in W_j and cover K . Apply Lemma A.2.1 in each chart to obtain $\rho_j \in C_c^\infty(W_j)$ equal to one near $\overline{V_j}$; extension by zero is smooth because the support is compactly contained in W_j . The function $1 - \prod_j (1 - \rho_j)$ is then supported in U and equals one near K . $\square$

Proposition A.2.3 (Compact smooth partition of unity). *Let K be a compact subset of a smooth manifold and $U_1, \dots, U_m$ open sets covering K . There exist smooth functions $\eta_i \geq 0$ with compact support in U_i such that*

$$\sum_{i=1}^m \eta_i = 1$$

on a neighborhood of K .

PROOF. Choose compact sets $K_i \subset U_i$ whose interiors still cover K ; this follows by taking finitely many relatively compact coordinate balls subordinate to the given finite cover and grouping them by U_i . By Lemma A.2.2, choose $\rho_i \in C_c^\infty(U_i)$ equal to one on a neighborhood of K_i . Then $\rho = \sum_i \rho_i$ is positive on an open neighborhood W of K . Choose, again by Lemma A.2.2, $\chi \in C_c^\infty(W)$ equal to one near K . Define on W

$$\eta_i = \chi \frac{\rho_i}{\rho}$$

and extend by zero. Because χ is compactly supported in W , this extension is smooth. Near K the sum is $\chi = 1$, and each η_i is supported in U_i . $\square$

3. Stone–Weierstrass from an elementary lattice argument

THEOREM A.3.1 (Stone–Weierstrass, real and self-adjoint complex forms). *Let K be compact Hausdorff.*

- (1) *If $A \subset C(K, \mathbb{R})$ is a unital subalgebra that separates points, then A is uniformly dense in $C(K, \mathbb{R})$.*
- (2) *If $A \subset C(K, \mathbb{C})$ is a unital subalgebra, separates points, and is closed under complex conjugation, then A is uniformly dense in $C(K, \mathbb{C})$.*

PROOF. We first prove the real statement. Write $\overline{A}$ for the uniform closure. It is again a unital algebra. We claim that it is also closed under absolute values, hence under pointwise maxima and minima.

Fix $f \in \overline{A}$ and put $M = \|f\|_\infty$. If $M = 0$ there is nothing to prove. On $[-1, 1]$ the function $|t|$ is the uniform limit of polynomials in t with real coefficients. For completeness, one may approximate $\sqrt{s}$ on $[0, 1]$ by truncating the absolutely and uniformly convergent binomial series

$$\sqrt{s} = \sqrt{1 - (1 - s)} = 1 - \sum_{k \geq 1} c_k (1 - s)^k, \quad c_k > 0,$$

whose coefficients satisfy $c_k = O(k^{-3/2})$; composing with $s = t^2$ gives polynomial approximation to $|t|$. Rescaling gives polynomials p_j with $p_j(f) \rightarrow |f|$ uniformly. Since $\overline{A}$ is an algebra, $|f| \in \overline{A}$. Therefore

$$\max(f, g) = \frac{f + g + |f - g|}{2}, \quad \min(f, g) = \frac{f + g - |f - g|}{2}$$

belong to $\overline{A}$ whenever f, g do.

Now fix $h \in C(K, \mathbb{R})$ and $\varepsilon > 0$. For each pair $x, y \in K$ with $x \neq y$, separation and the presence of constants let us choose $a_{x,y} \in A$ satisfying $a_{x,y}(x) = h(x)$ and $a_{x,y}(y) = h(y)$: if $b(x) \neq b(y)$, an affine function of b interpolates the two prescribed values. For $y = x$, simply use the constant function $h(x)$.

Fix x . The open sets

$$U_{x,y} = \{z : a_{x,y}(z) > h(z) - \varepsilon\}$$

cover K as y varies (with the harmless choice $a_{x,x} = h(x)$ replaced by any interpolant equal to $h(x)$ at x). By compactness choose $y_1, \dots, y_r$ whose sets cover K , and put

$$g_x = \max_{1 \leq j \leq r} a_{x,y_j} \in \overline{A}.$$

Then $g_x > h - \varepsilon$ on all of K , while $g_x(x) = h(x)$. Hence $V_x = \{z : g_x(z) < h(z) + \varepsilon\}$ is an open neighborhood of x . Choose finitely many $x_1, \dots, x_s$ with $K = \bigcup_i V_{x_i}$ and put

$$g = \min_{1 \leq i \leq s} g_{x_i} \in \overline{A}.$$

Every $g_{x_i} > h - \varepsilon$, so $g > h - \varepsilon$ everywhere; and at each point some $g_{x_i} < h + \varepsilon$, so $g < h + \varepsilon$. Thus $\|g - h\|_\infty < \varepsilon$, proving the real theorem.

For the complex statement, let $A_{\mathbb{R}} = \{\Re f : f \in A\}$. Self-adjointness gives $\Re f = (f + \bar{f})/2 \in A$, so $A_{\mathbb{R}}$ is a real unital algebra. If $x \neq y$, choose $f \in A$ with $f(x) \neq f(y)$; either its real or imaginary part separates x, y , and $\Im f = \Re(-if) \in A_{\mathbb{R}}$. The real theorem therefore makes $A_{\mathbb{R}}$ dense in $C(K, \mathbb{R})$. Approximate the real and imaginary parts of an arbitrary complex continuous function separately and combine them. $\square$

4. Positive functionals, Radon measures, and dominated convergence

The measure-theoretic arguments later in the series repeatedly turn a positive linear functional on test functions into a measure and then identify measures by their test integrals. We prove that mechanism here in the locally compact second-countable setting used by homogeneous spaces. We also derive dominated convergence from monotone convergence, which is part of the ordinary measure-theory prerequisite.

Lemma A.4.1 (Cutoff between a compact set and an open set). *Let X be a locally compact second-countable Hausdorff space. If $K \subset U$, with K compact and U open, then there is $\chi \in C_c(X)$ such that*

$$0 \leq \chi \leq 1, \quad \chi = 1 \text{ on } K, \quad \text{supp } \chi \subset U.$$

PROOF. A locally compact second-countable Hausdorff space is metrizable and admits a countable relatively compact open basis. Compactness lets us choose an open V with $K \subset V \subset \bar{V} \subset U$ and $\bar{V}$ compact. In a compatible metric d , the functions

$$a(x) = d(x, K), \quad b(x) = d(x, X \setminus V)$$

are continuous, and $a + b > 0$. The function

$$\chi_0(x) = \frac{b(x)}{a(x) + b(x)}$$

is continuous, equals one on K , and vanishes outside V . Since its support is contained in the compact set $\bar{V}$, it belongs to $C_c(X)$. $\square$

THEOREM A.4.2 (Riesz–Markov representation in the form used here). *Let X be locally compact, second-countable and Hausdorff, and let*

$$L : C_c(X, \mathbb{R}) \longrightarrow \mathbb{R}$$

be a positive linear functional. Then there is a unique Radon measure μ such that

$$L(f) = \int_X f \, d\mu \quad (f \in C_c(X)).$$

If L is bounded on $C_0(X)$, the same statement holds for $L : C_0(X) \rightarrow \mathbb{R}$ and $\mu(X) = \|L\|$ when $L \neq 0$ is positive.

PROOF. We separate construction, regularity, and identification of the integral.

Step 1: local boundedness. If $K \subset X$ is compact, choose $\chi \in C_c(X)$ with $\chi = 1$ on K by Lemma A.4.1. For every $f \in C_c(X)$ supported in K ,

$$-\|f\|_\infty \chi \leq f \leq \|f\|_\infty \chi.$$

Positivity therefore gives

$$(RM1) \quad |L(f)| \leq L(\chi)\|f\|_\infty.$$

Thus L is uniformly bounded on each fixed compact support.

Step 2: define the size of open sets. For open $U \subset X$ set

$$\mu_0(U) = \sup\{L(f) : f \in C_c(X), 0 \leq f \leq 1, \text{ supp } f \subset U\}.$$

This is monotone, and it is finitely additive on disjoint open sets. Indeed, the inequality $\mu_0(U) + \mu_0(V) \leq \mu_0(U \cup V)$ for disjoint U, V follows by adding admissible functions. Conversely, if f is admissible for $U \cup V$, then the compact set $\text{supp } f$ is the disjoint union of its intersections with U and V . Lemma A.4.1, applied inside U and V , gives continuous cutoffs that split f into two nonnegative functions supported in the two open sets; hence $L(f) \leq \mu_0(U) + \mu_0(V)$.

For arbitrary $A \subset X$ define

$$\mu^*(A) = \inf \left\{ \sum_{j=1}^{\infty} \mu_0(U_j) : A \subset \bigcup_{j=1}^{\infty} U_j, U_j \text{ open} \right\}.$$

The empty cover convention gives $\mu^*(\emptyset) = 0$. If $A \subset B$, every open cover of B also covers A , so taking infima gives $\mu^*(A) \leq \mu^*(B)$. Concatenating covers of countably many sets proves countable subadditivity. Hence μ^* is an outer measure.

We need one point that is sometimes hidden inside a reference to the Riesz theorem: for every open U ,

$$(RM2) \quad \mu^*(U) = \mu_0(U).$$

The inequality $\mu^*(U) \leq \mu_0(U)$ follows from the one-set cover. For the reverse inequality, let $U \subset \bigcup_j U_j$ be an open cover and let f be admissible for U . The compact set $K = \text{supp } f$ has a finite subcover $U_{j_1}, \dots, U_{j_m}$. Applying Lemma A.4.1 successively to the compact pieces of K gives a finite continuous partition $\eta_1 + \dots + \eta_m = 1$ on K with $0 \leq \eta_i \leq 1$ and $\text{supp } \eta_i \subset U_{j_i}$. Then

$$L(f) = \sum_i L(f\eta_i) \leq \sum_i \mu_0(U_{j_i}) \leq \sum_j \mu_0(U_j).$$

Taking the supremum over f and then the infimum over covers proves (RM2).

The proof of (RM2) also shows countable subadditivity of μ_0 on open sets: if $U \subset \bigcup_j U_j$ and f is admissible for U , compactness of $\text{supp } f$ reduces to finitely many U_j , and the same finite partition gives $L(f) \leq \sum_j \mu_0(U_j)$. Consequently

$$(RM2') \quad \mu^*(A) = \inf\{\mu_0(U) : A \subset U, U \text{ open}\}.$$

Indeed a one-set cover gives one inequality, and replacing an arbitrary open cover by its union gives the other.

Because X is metrizable, μ^* is a metric outer measure. If $d(A, B) = \delta > 0$ and U is any open set containing $A \cup B$, the open sets

$$U_A = U \cap \{d(x, A) < \delta/3\}, \quad U_B = U \cap \{d(x, B) < \delta/3\}$$

are disjoint and contain A and B . Finite additivity of μ_0 on disjoint open sets gives

$$\mu_0(U) \geq \mu_0(U_A) + \mu_0(U_B) \geq \mu^*(A) + \mu^*(B).$$

Taking the infimum in (RM2') and combining with subadditivity proves $\mu^*(A \cup B) = \mu^*(A) + \mu^*(B)$.

We also give the Carathéodory step. Let F be closed and $E \subset X$. If $\mu^*(E) = \infty$ the desired inequality is automatic, so suppose it is finite. Put

$$E_j = E \cap \{2^{-j-1} \leq d(x, F) < 2^{-j}\} \quad (j \geq 0).$$

The union of the even-indexed E_j is a union of mutually separated layers, and the same is true of the odd-indexed layers. Repeated metric additivity therefore gives

$$(RM2'') \quad \sum_j \mu^*(E_j) \leq 2\mu^*(E) < \infty.$$

For $r = 2^{-N}$, the sets $E \cap F$ and $E \cap \{d(x, F) \geq r\}$ are separated, hence

$$\mu^*(E) \geq \mu^*(E \cap F) + \mu^*(E \cap \{d(x, F) \geq r\}).$$

The part of $E \setminus F$ omitted by the second term is contained in $\bigcup_{j \geq N} E_j$, whose outer measure tends to zero by (RM2''). Subadditivity now lets $N \rightarrow \infty$ and yields

$$\mu^*(E) \geq \mu^*(E \cap F) + \mu^*(E \setminus F).$$

Thus every closed set is Carathéodory measurable; the measurable sets form a σ -algebra, so every Borel set is measurable. Restricting μ^* to the Borel σ -algebra gives a Borel measure μ satisfying

$$(RM3) \quad \mu(U) = \mu_0(U) \quad (U \text{ open}).$$

Step 3: Radon regularity. If K is compact, choose a relatively compact open $U \supset K$. Every admissible function for U is supported in the fixed compact set $\bar{U}$, so (RM1) gives $\mu(U) = \mu_0(U) < \infty$; hence μ is finite on compact sets. Outer regularity is exactly (RM2').

For inner regularity on an open set U , let $0 \leq f \leq 1$ be admissible for U and put $K = \text{supp } f$. If $K \subset \bigcup_j U_j$ is any open cover, the finite-partition

argument used above gives $L(f) \leq \sum_j \mu_0(U_j)$. Taking the infimum over covers shows $L(f) \leq \mu(K)$. Therefore, using (RM3),

$$\mu(U) = \mu_0(U) \leq \sup_{K \in U} \mu(K) \leq \mu(U),$$

so equality holds. For completeness, let E be Borel with $\mu(E) < \infty$ and fix $\varepsilon > 0$. Outer regularity gives an open $U \supset E$ with $\mu(U) \leq \mu(E) + \varepsilon < \infty$, and hence $\mu(U \setminus E) < \varepsilon$. Inner regularity of the open set U gives compact $K_0 \subset U$ with $\mu(U \setminus K_0) < \varepsilon$. Outer regularity applied to the Borel set $U \setminus E$ gives an open $V \supset U \setminus E$ with $\mu(V) \leq \mu(U \setminus E) + \varepsilon < 2\varepsilon$. Then $K = K_0 \setminus V$ is compact and contained in E , while

$$\mu(E \setminus K) \leq \mu(U \setminus K_0) + \mu(V) < 3\varepsilon.$$

Thus every finite-measure Borel set is inner regular. For an arbitrary Borel set, intersect with an increasing compact exhaustion and apply the finite-measure statement on each relatively compact piece. Hence μ is Radon.

Step 4: identify L with integration. Fix $0 \leq f \in C_c(X)$ and choose $\chi \in C_c(X)$ equal to one on $\text{supp } f$. We compare both quantities by horizontal slices. Let $M > \|f\|_\infty$ and divide $[0, M]$ into intervals of width ε . For $i \geq 1$, put

$$K_i = \{x : f(x) \geq i\varepsilon\}.$$

Choose $\phi_i \in C_c(X)$ with $0 \leq \phi_i \leq 1$, $\phi_i = 1$ on K_i , and

$$\text{supp } \phi_i \subset \{f > (i - \frac{1}{2})\varepsilon\}$$

whenever $K_i \neq \emptyset$. From (RM3) and the definition of μ_0 one obtains the two elementary comparison rules

$$(RM4) \quad \mu(K) \leq L(\phi) \quad \text{if } \phi \geq 1 \text{ on a neighborhood of } K,$$

and

$$(RM5) \quad L(h) \leq \mu(U) \quad \text{if } 0 \leq h \leq 1, \text{ supp } h \subset U.$$

The first follows by testing open neighborhoods V of K contained in a set where $\phi > 1 - \delta$ and letting $\delta \downarrow 0$; the second is the definition of $\mu_0(U)$.

Now

$$\varepsilon \sum_{i \geq 1} \mu(K_i) \leq L\left(\varepsilon \sum_{i \geq 1} \phi_i\right).$$

At each point, at most $f(x)/\varepsilon + 1$ of the ϕ_i can be nonzero, so

$$0 \leq \varepsilon \sum_i \phi_i \leq f + \varepsilon \chi.$$

Therefore

$$(RM6) \quad \varepsilon \sum_i \mu(K_i) \leq L(f) + \varepsilon L(\chi).$$

As $\varepsilon \downarrow 0$, the left side is a lower Riemann sum for the layer-cake identity

$$\int f d\mu = \int_0^\infty \mu(\{f > t\}) dt,$$

which follows directly from Tonelli applied to $f(x) = \int_0^\infty \mathbf{1}_{\{t < f(x)\}} dt$. Thus $\int f d\mu \leq L(f)$.

For the reverse inequality define the continuous ramps

$$h_i(x) = \min\left(1, \max\left(0, \frac{f(x) - (i-1)\varepsilon}{\varepsilon}\right)\right).$$

For finitely many i these satisfy $f = \varepsilon \sum_i h_i$. For every $\delta > 0$ the support of h_i lies in the open set $U_{i,\delta} = \{f > (i-1)\varepsilon - \delta\} \cap \{\chi > 0\}$ after harmlessly multiplying the first ramp by χ . By (RM5),

$$L(f) \leq \varepsilon \sum_i \mu(U_{i,\delta}).$$

Let $\delta \downarrow 0$ and then $\varepsilon \downarrow 0$. Continuity from above on the finite-measure set $\{\chi > 0\}$ and the same layer-cake identity show that the right side tends to $\int f d\mu$. Hence $L(f) = \int f d\mu$ for $f \geq 0$, and linearity gives the formula for arbitrary real $f \in C_c(X)$.

Step 5: uniqueness. If ν is another Radon measure with the same test integrals, then for every open U ,

$$\nu(U) = \sup \left\{ \int f d\nu : 0 \leq f \leq 1, f \in C_c(U) \right\} = \mu(U)$$

by inner regularity and Lemma A.4.1. Outer regularity then gives equality on all Borel sets.

Finally, if L is positive and bounded on $C_0(X)$, its restriction to $C_c(X)$ gives μ . An increasing compact exhaustion with cutoffs $0 \leq \chi_j \leq 1$ tending pointwise to one gives $\mu(X) = \sup_j L(\chi_j) = \|L\|$; density of $C_c(X)$ in $C_0(X)$ extends the integral identity to $C_0(X)$. $\square$

Corollary A.4.3 (Test functions determine Radon measures). *If Radon measures μ, ν on a locally compact second-countable Hausdorff space satisfy*

$$\int f d\mu = \int f d\nu$$

for every $f \in C_c(X)$, then $\mu = \nu$.

PROOF. This is the uniqueness step of Theorem A.4.2. $\square$

Lemma A.4.4 (Fatou from monotone convergence). *If $f_n \geq 0$ are measurable, then*

$$\int \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int f_n d\mu.$$

PROOF. Put $g_N = \inf_{n \geq N} f_n$. Then $g_N \uparrow \liminf f_n$. Monotone convergence gives

$$\int \liminf f_n = \lim_N \int g_N \leq \lim_N \inf_{n \geq N} \int f_n = \lim_n \inf \int f_n.$$

□

THEOREM A.4.5 (Dominated convergence). *Suppose $f_n \rightarrow f$ almost everywhere and $|f_n| \leq g$ with $g \in L^1(\mu)$. Then $f \in L^1(\mu)$ and*

$$\int |f_n - f| d\mu \rightarrow 0.$$

In particular, $\int f_n d\mu \rightarrow \int f d\mu$.

PROOF. The pointwise limit satisfies $|f| \leq g$. Apply Fatou's lemma to the nonnegative functions $2g - |f_n - f|$. Since they converge pointwise to $2g$ and are bounded below by zero,

$$2 \int g \leq \liminf_n \left(2 \int g - \int |f_n - f| \right) = 2 \int g - \limsup_n \int |f_n - f|.$$

Hence $\limsup_n \int |f_n - f| \leq 0$. The convergence of integrals follows from

$$\left| \int f_n - \int f \right| \leq \int |f_n - f|.$$

□

5. Haar uniqueness: a complete proof for Lie groups

The existence of Haar measure on a Lie group was obtained geometrically in the main text from a nonzero density at the identity. We now prove uniqueness rather than importing it from the general theory of locally compact groups. The argument is worth recording because it also explains why an arbitrary invariant Radon measure cannot be singular with respect to the smooth Haar density.

Let m denote the left Haar measure arising from a smooth positive left-invariant density. Let ν be any other nonzero left-invariant Radon measure on G . We shall prove that $\nu = cm$.

Guiding Idea A.5.1. The key is to smooth ν without assuming in advance that it has a density. A compactly supported smooth kernel $\phi(x^{-1}y)$ can be averaged against ν in the y -variable. Left invariance makes that average

independent of x . On the other hand, if the kernels concentrate at the identity, integration against the smooth Haar measure turns the same expression into an approximate identity. Comparing the two descriptions forces every test integral against ν to be a fixed scalar multiple of the corresponding Haar integral.

Lemma A.5.2 (Smooth approximate identities adapted to inversion). *There is a sequence $\phi_n \in C_c^\infty(G)$ with $\phi_n \geq 0$ such that*

(1) *supp ϕ_n shrinks to $\{e\}$;*

(2)

$$\int_G \phi_n(z^{-1}) dm(z) = 1;$$

(3) *for every $f \in C_c(G)$,*

$$K_n f(y) := \int_G f(x) \phi_n(x^{-1}y) dm(x) \longrightarrow f(y)$$

uniformly in y .

PROOF. Choose a decreasing sequence of relatively compact identity neighborhoods V_n with $\bigcap_n V_n = \{e\}$. Pick nonzero $\psi_n \in C_c^\infty(V_n)$, $\psi_n \geq 0$, and put

$$c_n = \int_G \psi_n(z^{-1}) dm(z) > 0, \quad \phi_n = c_n^{-1} \psi_n.$$

The normalization gives $\int_G \phi_n(z^{-1}) dm(z) = 1$, proving (1); nonnegativity and the support inclusion $\text{supp } \phi_n \subset \text{supp } \psi_n$ prove (2).

To prove (3), write $x = yz$ in the defining integral. Left invariance of m gives $dm(x) = dm(z)$ and

$$x^{-1}y = (yz)^{-1}y = z^{-1}.$$

Hence

$$K_n f(y) = \int_G f(yz) \phi_n(z^{-1}) dm(z).$$

The weight $\phi_n(z^{-1})dm(z)$ is a probability measure supported in V_n^{-1} . Since $f \in C_c(G)$ is uniformly continuous under right translations near the identity, for every $\varepsilon > 0$ there is an identity neighborhood W such that

$$\sup_{y \in G} |f(yz) - f(y)| < \varepsilon \quad (z \in W).$$

For large n , $V_n^{-1} \subset W$, and therefore

$$|K_n f(y) - f(y)| \leq \int |f(yz) - f(y)| \phi_n(z^{-1}) dm(z) < \varepsilon$$

uniformly in y . □

THEOREM A.5.3 (Haar uniqueness for Lie groups). *Let G be a finite-dimensional Lie group. If m and ν are nonzero left-invariant Radon measures on G , then*

$$\nu = c m$$

for a unique constant $c > 0$.

PROOF. Fix the approximate identity ϕ_n above. For each n define

$$a_n(x) = \int_G \phi_n(x^{-1}y) d\nu(y).$$

This is finite: for fixed x the integrand is supported in the compact set $x \operatorname{supp}(\phi_n)$, and ν is Radon. More importantly, a_n is constant. Indeed, substituting $y = xz$ and using left invariance of ν gives

$$a_n(x) = \int_G \phi_n(z) d\nu(z) =: c'_n.$$

Now take $f \in C_c(G)$, $f \geq 0$. Tonelli's theorem gives

$$\begin{aligned} c'_n \int_G f(x) dm(x) &= \int_G f(x) a_n(x) dm(x) \\ &= \int_G \int_G f(x) \phi_n(x^{-1}y) dm(x) d\nu(y) \\ &= \int_G K_n f(y) d\nu(y). \end{aligned}$$

Because $K_n f \rightarrow f$ uniformly and, for all large n , the functions $K_n f$ are supported in one fixed compact enlargement of $\operatorname{supp} f$, Radon finiteness implies

$$\int K_n f d\nu \rightarrow \int f d\nu.$$

Thus

$$(A.1) \quad c'_n m(f) \rightarrow \nu(f).$$

Choose once and for all $f_0 \in C_c(G)$ with $f_0 \geq 0$ and $m(f_0) = 1$. Equation (A.1) with $f = f_0$ shows

$$c'_n \rightarrow c := \nu(f_0).$$

Since ν is nonzero and left invariant, every nonempty open set has positive ν -measure; choosing f_0 positive on a nonempty open set gives $c > 0$.

Returning to arbitrary nonnegative $f \in C_c(G)$ in (A.1), we obtain

$$\nu(f) = c m(f).$$

By decomposing a real-valued test function into positive and negative parts, and then real and imaginary parts if needed, the identity holds for every $f \in C_c(G)$. The Riesz–Markov uniqueness theorem gives $\nu = cm$.

Uniqueness of c follows because m is nonzero. $\square$

Corollary A.5.4 (Uniqueness on a transitive homogeneous space). *Let G act transitively on a locally compact space X , and suppose X carries a nonzero G -invariant Radon measure. If the stabilizer criterion of Chapter 6 holds, then the invariant Radon measure is unique up to a positive scalar.*

PROOF. Write $X \cong G/H$. The quotient-integration construction lifts either invariant measure to a left Haar measure on G after fixing Haar measure on H . Theorem A.5.3 makes the two lifts proportional, and hence so are the quotient measures. $\square$

6. Densities versus differential forms

If a manifold is oriented, a positive top-degree form determines a positive density and therefore a measure. Densities are preferable for invariant measure questions because they do not require a global orientation. If $T : V \rightarrow V$ is an invertible linear map and ω a density, then

$$T^*\omega = |\det T|\omega.$$

For a top-degree alternating form the factor is $\det T$ without absolute value. This is why modular functions naturally involve absolute determinants.

7. Covering maps from discrete quotients

Let $\Gamma < G$ be discrete. Choose a symmetric identity neighborhood U such that

$$U^{-1}U \cap \Gamma = \{e\}.$$

Then for every $g \in G$, the sets $gU\gamma$, $\gamma \in \Gamma$, are pairwise disjoint. The quotient map

$$G \rightarrow G/\Gamma$$

restricts to a diffeomorphism from gU onto an open neighborhood of $g\Gamma$. Hence $G \rightarrow G/\Gamma$ is a smooth covering map. This elementary local model justifies the local manifold statements about spaces of lattices used throughout the volume.

8. A Baire-category fact for locally compact spaces

The orbit-topology argument uses the following elementary form of the Baire category theorem.

THEOREM A.8.1 (Baire category). *Every locally compact Hausdorff space is a Baire space: a countable union of closed sets with empty interior cannot contain a nonempty open set. In particular, if*

$$X = \bigcup_{n \geq 1} F_n$$

with each F_n closed and X locally compact Hausdorff, then at least one F_n has nonempty interior.

PROOF MECHANISM. It is enough to prove that a countable intersection of open dense sets is dense. Let U be nonempty and open, and let $O_1, O_2, \dots$ be open dense sets. Local compactness and regularity allow one to choose recursively nonempty open sets V_n such that

$$\overline{V_1} \subset U \cap O_1, \quad \overline{V_{n+1}} \subset V_n \cap O_{n+1},$$

and each $\overline{V_n}$ is compact. The compact sets $\overline{V_n}$ are nested and nonempty, so their intersection is nonempty. Any point in the intersection lies in

$$U \cap \bigcap_{n \geq 1} O_n.$$

Thus the latter set is nonempty, which is the Baire property. The closed-set formulation follows by taking complements. $\square$

APPENDIX B

Geometry-of-Numbers Notes and Conventions

1. Minkowski's second theorem with proof

The second theorem is the geometry-of-numbers statement that turns control of the shortest vector into control of a whole basis. Because Mahler compactness rests on precisely this passage, we include the proof rather than treating it as a black box.

Let $K \subset \mathbb{R}^d$ be a bounded convex body with nonempty interior, symmetric about the origin, and let Λ be a full-rank lattice. The successive minima of K with respect to Λ are

$$\lambda_j(K, \Lambda) = \inf \{ r > 0 : \dim \text{span}_{\mathbb{R}}(\Lambda \cap rK) \geq j \}, \quad 1 \leq j \leq d.$$

Thus λ_j is the first scale at which j independent lattice directions have appeared.

THEOREM B.1.1 (Minkowski's second theorem). *For every centrally symmetric convex body $K \subset \mathbb{R}^d$ and every full-rank lattice Λ ,*

$$\frac{2^d}{d!} \text{covol}(\Lambda) \leq \lambda_1 \cdots \lambda_d \text{vol}(K) \leq 2^d \text{covol}(\Lambda),$$

where $\lambda_j = \lambda_j(K, \Lambda)$.

PROOF. We separate the two inequalities because they encode rather different ideas. Both sides are invariant under an invertible linear change of coordinates: if $T \in \text{GL}_d(\mathbb{R})$, then

$$\lambda_j(TK, T\Lambda) = \lambda_j(K, \Lambda), \quad \text{vol}(TK) = |\det T| \text{vol}(K),$$

and

$$\text{covol}(T\Lambda) = |\det T| \text{covol}(\Lambda).$$

Consequently we may make whatever linear normalization is convenient.

The lower inequality. Choose linearly independent vectors $v_1, \dots, v_d \in \Lambda$ with

$$v_j \in (\lambda_j + \eta)K$$

for an arbitrary $\eta > 0$. Then

$$w_j := \frac{v_j}{\lambda_j + \eta} \in K.$$

Since K is convex and centrally symmetric, it contains the cross-polytope

$$C = \text{conv}\{\pm w_1, \dots, \pm w_d\}.$$

The 2^d simplices obtained by choosing one sign in each coordinate have disjoint interiors and each has volume

$$\frac{1}{d!} |\det(w_1, \dots, w_d)|.$$

Hence

$$\text{vol}(K) \geq \text{vol}(C) = \frac{2^d}{d!} \frac{|\det(v_1, \dots, v_d)|}{(\lambda_1 + \eta) \cdots (\lambda_d + \eta)}.$$

The vectors v_j generate a sublattice of Λ , so its covolume is an integer multiple of $\text{covol}(\Lambda)$; in particular

$$|\det(v_1, \dots, v_d)| \geq \text{covol}(\Lambda).$$

Letting $\eta \downarrow 0$ gives

$$\lambda_1 \cdots \lambda_d \text{vol}(K) \geq \frac{2^d}{d!} \text{covol}(\Lambda).$$

The upper inequality. After a linear change of coordinates we may assume $\Lambda = \mathbb{Z}^d$. We first choose coordinates adapted to the successive-minimum flag. Choose independent lattice vectors $z_1, \dots, z_d$ with

$$z_i \in \lambda_i K.$$

Put $V_i = \text{span}_{\mathbb{R}}\{z_1, \dots, z_i\}$ and $\Lambda_i = V_i \cap \mathbb{Z}^d$. Each Λ_i is a primitive rank- i subgroup of $\mathbb{Z}^d$: if $mv \in \Lambda_i$ for a nonzero integer m , then $v \in V_i$ and hence $v \in \Lambda_i$. A primitive subgroup of a free abelian group is a direct summand, so a basis of Λ_i can be extended to a basis of $\mathbb{Z}^d$. Because the Λ_i form a nested flag, this can be done inductively to obtain one basis $b_1, \dots, b_d$ of $\mathbb{Z}^d$ with

$$\Lambda_i = \mathbb{Z}b_1 \oplus \cdots \oplus \mathbb{Z}b_i.$$

The associated unimodular change of coordinates sends b_i to e_i and therefore sends V_i to

$$L_i := \text{span}_{\mathbb{R}}\{e_1, \dots, e_i\}.$$

In these adapted coordinates, the property we shall actually use is

$$(1.1) \quad \text{int}(\lambda_i K) \cap \mathbb{Z}^d \subset L_{i-1} \cap \mathbb{Z}^d, \quad 1 \leq i \leq d,$$

with $L_0 = \{0\}$. Indeed, if an interior lattice point at scale λ_i lay outside L_{i-1} , then together with $i-1$ previously chosen independent vectors it would show that the i th successive minimum is strictly smaller than λ_i .

Put

$$K_i := \frac{\lambda_i}{2} K, \quad M_q^d := \{m \in \mathbb{Z}^d : |m_j| \leq q \text{ for every } j\},$$

and

$$M_q^i := M_q^d \cap L_i.$$

The proof compares the volumes of the unions $M_q^d + K_i$ as i increases. There are two observations.

First, distinct translates of $\text{int}(K_1)$ by points of $\mathbb{Z}^d$ are disjoint. If

$$(m + \text{int } K_1) \cap (n + \text{int } K_1) \neq \emptyset,$$

then

$$m - n \in \text{int}(K_1 - K_1) = \text{int}(\lambda_1 K),$$

contradicting (1.1) for $i = 1$. Therefore

$$(1.2) \quad \text{vol}(M_q^d + K_1) = (2q + 1)^d \text{vol}(K_1) = (2q + 1)^d \left(\frac{\lambda_1}{2}\right)^d \text{vol}(K).$$

Second, for $1 \leq i < d$ we claim

$$(1.3) \quad \text{vol}(M_q^d + K_{i+1}) \geq \left(\frac{\lambda_{i+1}}{\lambda_i}\right)^{d-i} \text{vol}(M_q^d + K_i).$$

If $\lambda_{i+1} = \lambda_i$ there is nothing to prove, so suppose the ratio $r := \lambda_{i+1}/\lambda_i$ is larger than 1.

The key point is that translates belonging to different cosets modulo L_i remain disjoint at the larger scale. Indeed, if $m, n \in \mathbb{Z}^d$ have different last $d - i$ coordinates and

$$(m + \text{int } K_{i+1}) \cap (n + \text{int } K_{i+1}) \neq \emptyset,$$

then

$$m - n \in \text{int}(2K_{i+1}) = \text{int}(\lambda_{i+1} K)$$

and $m - n \notin L_i$. Together with i independent lattice vectors spanning L_i , this contradicts the definition of λ_{i+1} . Consequently

$$\text{vol}(M_q^d + K_j) = (2q + 1)^{d-i} \text{vol}_i^\perp(M_q^i + K_j), \quad j = i, i + 1,$$

where the notation on the right means that one may compute by integrating the i -dimensional slice volumes over $L_i^\perp$; the common factor counts the disjoint choices of the last $d - i$ integer coordinates.

To compare the slice volumes, introduce the two diagonal maps

$$F_1(x_1, \dots, x_d) = (rx_1, \dots, rx_i, x_{i+1}, \dots, x_d),$$

$$F_2(x_1, \dots, x_d) = (x_1, \dots, x_i, rx_{i+1}, \dots, rx_d).$$

Because $K_{i+1} = rK_i$,

$$M_q^i + K_{i+1} = F_2(M_q^i + F_1(K_i)),$$

so

$$(1.4) \quad \text{vol}(M_q^i + K_{i+1}) = r^{d-i} \text{vol}(M_q^i + F_1(K_i)).$$

It remains only to show

$$(1.5) \quad \text{vol}(M_q^i + F_1(K_i)) \geq \text{vol}(M_q^i + K_i).$$

This is a fiberwise statement. Fix $x \in L_i^\perp$ and write the section as

$$K_i \cap (x + L_i) = x + A_x,$$

where $A_x \subset L_i$ is convex. If the section is empty there is nothing to prove. Otherwise choose $a_x \in A_x$. For every $y \in A_x$, convexity gives

$$\frac{1}{r}y + \left(1 - \frac{1}{r}\right)a_x \in A_x.$$

Multiplying by r shows

$$y + (r-1)a_x \in rA_x.$$

Thus

$$A_x + (r-1)a_x \subset rA_x.$$

The map F_1 fixes $x \in L_i^\perp$ and multiplies the L_i coordinate by r , so

$$F_1(K_i) \cap (x + L_i) = x + rA_x.$$

Consequently the expanded section contains a translate of the original section; equivalently, for $t(x) = -(r-1)a_x$,

$$K_i \cap (x + L_i) \subset (F_1(K_i) \cap (x + L_i)) + t(x).$$

Adding the same finite set $M_q^i \subset L_i$ preserves this containment up to translation. Therefore the i -dimensional measure of every slice of $M_q^i + F_1(K_i)$ is at least the corresponding slice measure of $M_q^i + K_i$. Fubini's theorem gives (1.5). Together with (1.4), this proves (1.3).

We now iterate (1.3) from $i = 1$ to $d-1$. Using (1.2),

$$\begin{aligned} \text{vol}(M_q^d + K_d) &\geq \prod_{i=1}^{d-1} \left(\frac{\lambda_{i+1}}{\lambda_i}\right)^{d-i} (2q+1)^d \left(\frac{\lambda_1}{2}\right)^d \text{vol}(K) \\ &= (2q+1)^d \frac{\lambda_1 \cdots \lambda_d}{2^d} \text{vol}(K). \end{aligned}$$

The telescoping in the last equality is worth noticing: the exponent of each λ_j becomes exactly one.

Finally, because K_d is bounded, there is a constant C_K such that

$$M_q^d + K_d \subset [-q - C_K, q + C_K]^d.$$

Hence

$$\text{vol}(M_q^d + K_d) \leq (2q + 2C_K)^d.$$

Combining the last two inequalities and dividing by $(2q+1)^d$ gives

$$\frac{\lambda_1 \cdots \lambda_d}{2^d} \operatorname{vol}(K) \leq \left(\frac{2q + 2C_K}{2q + 1} \right)^d.$$

Let $q \rightarrow \infty$. We obtain

$$\lambda_1 \cdots \lambda_d \operatorname{vol}(K) \leq 2^d,$$

which is the desired upper inequality for $\Lambda = \mathbb{Z}^d$. Undoing the linear normalization restores the factor $\operatorname{covol}(\Lambda)$. $\square$

Remark B.1.2 (What is really used in Mahler compactness). For the Euclidean unit ball B the theorem says

$$c_d \operatorname{covol}(\Lambda) \leq \lambda_1(B, \Lambda) \cdots \lambda_d(B, \Lambda) \leq C_d \operatorname{covol}(\Lambda).$$

Thus, for unimodular lattices, a uniform lower bound on the shortest vector forces a uniform upper bound on the last successive minimum. This is the nontrivial step that allows one to pass from “no short vector” to “there is a uniformly bounded independent d -tuple,” and hence to compactness.

2. Primitive sublattices and exterior powers

A subgroup $\Delta < \mathbb{Z}^d$ of rank k is *primitive* if

$$\Delta = (\operatorname{span}_{\mathbb{R}} \Delta) \cap \mathbb{Z}^d.$$

Equivalently, $\mathbb{Z}^d/\Delta$ has no torsion in the quotient by the complementary free part. If $v_1, \dots, v_k$ is a basis of Δ , the wedge

$$v_{\Delta} = v_1 \wedge \cdots \wedge v_k \in \bigwedge^k \mathbb{Z}^d$$

is defined up to sign. Its norm is the covolume of Δ in its real span.

For $g \in \operatorname{SL}_d(\mathbb{R})$,

$$\left\| \left(\bigwedge^k g \right) v_{\Delta} \right\|$$

is the covolume of the corresponding sublattice $g\Delta$ inside $g \operatorname{span}_{\mathbb{R}} \Delta$. Thus exterior powers simultaneously monitor all rational subspaces and their lattice covolumes.

3. Normalization conventions collected

The following choices are fixed throughout Volume I.

- G/H denotes left cosets gH and carries the natural left G -action.
- $H \backslash G$ denotes right cosets Hg and carries the natural right G -action.

- The modular function satisfies

$$m(Eg) = \Delta_G(g)^{-1}m(E)$$

for left Haar measure m . Thus for Lie groups

$$\Delta_G(g) = |\det \operatorname{Ad}(g)|.$$

- In $\operatorname{SL}_2(\mathbb{R})$ and $\operatorname{PSL}_2(\mathbb{R})$,

$$a_t = \operatorname{diag}(e^{t/2}, e^{-t/2}),$$

so

$$a_t u_s a_{-t} = u_{e^t s}$$

and t is hyperbolic arclength for the geodesic flow.

- On $T^1\mathbb{H} \cong \operatorname{PSL}_2(\mathbb{R})$, geodesic and horocycle flows are right actions. On G/Γ , inversion converts them to left actions and reverses diagonal time.

These conventions are included in one place because many apparent contradictions in formulas across the literature come only from changing one of them.

Bibliographic Notes

The development in this volume is self-contained at the level needed for the purposes of this volume and for applications in homogeneous dynamics, except for a small number of standard background theorems explicitly identified in the text, such as the general uniqueness theorem for Haar measure and Minkowski's second theorem.

For broader treatments of Lie groups and Lie algebras, useful references include Anthony W. Knap, *Lie Groups Beyond an Introduction*, and Brian C. Hall, *Lie Groups, Lie Algebras, and Representations*. For lattices and homogeneous spaces, M. S. Raghunathan, *Discrete Subgroups of Lie Groups*, provides a classical reference. Geometry of numbers is treated in J. W. S. Cassels, *An Introduction to the Geometry of Numbers*. Hyperbolic geometry and the $\mathrm{SL}_2(\mathbb{R})$ model appear in many sources; the presentation here has been normalized specifically for homogeneous dynamics.

For the dynamical viewpoint motivating these foundations, Dave Witte Morris, *Ratner's Theorems on Unipotent Flows*, and Manfred Einsiedler and Thomas Ward, *Ergodic Theory with a View Towards Number Theory*, provide complementary perspectives. The present notes do not follow the exercise structure of any of these references; examples and proofs have been organized to emphasize mechanisms and explicit conventions.

Ergodic Theory and Unipotent Dynamics

Volume II of a Detailed Notebook on Ratner's
Theorems

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Dependency and proof-ownership contract

Purpose. Recurrence and linearization.

Prerequisites. Volume I and ordinary metric-measure preliminaries. Conditional expectation, disintegration, tightness compactness, and Lusin approximation are proved internally in this volume.

New objects. recurrence, QND, singular strata, Chevalley coordinates. Every project-specific technical term is defined before its first load-bearing use or recalled by an exact backward reference.

Main outputs. quantitative nondivergence and singular-set control.

External-input rule. Every nontrivial imported result is either part of the declared graduate-level prerequisites, an explicitly named external research input in the prerequisite map, or is proved in this series before use (possibly under an explicit deferred-proof contract). A citation by itself never constitutes a proof.

Reader map and proof route

Question of the volume. Which recurrence, polynomial-divergence, and quantitative-nondivergence mechanisms make unipotent dynamics rigid rather than merely recurrent?

Prerequisite checkpoint. Use Volume I for measure foundations, recurrence, quotient measures, and the basic compactness language. Conditional expectation, disintegration, weak compactness for tight probability sequences, and Lusin approximation are proved internally in this volume before their load-bearing uses. The quantitative nondivergence and linearization mechanisms used later are also developed here in the stated forms.

Load-bearing route. First isolate unipotent polynomial behaviour in finite-dimensional representations. Then learn the quantitative nondivergence estimate and the Chevalley detector. Only after those two pieces are secure should the reader move to avoidance and recurrence arguments. The appendices supply technical versions used in Volume III.

Proof-owner map. A citation to Dani–Margulis or Kleinbock–Margulis records provenance, not ownership. The manuscript owns the special quantitative statements that it later invokes unless the theorem is explicitly assigned a BB-ID.

Sanity check. For $u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ acting on $\mathbb{R}^2$, write $u_t(a, b) = (a + tb, b)$. Every coordinate and every exterior power coordinate is polynomial in t ; this elementary fact is the finite-dimensional shadow behind the later “good function” estimates.

CHAPTER 1

Measure-Preserving Actions, Ergodicity, and Recurrence

1. Why measure enters homogeneous dynamics

A topological orbit asks where a point can go. A measure-theoretic orbit asks a different question: how much time does the orbit spend in various regions, and which statistical laws are forced by invariance? The two questions are related, but they are not interchangeable.

A simple rotation already shows the distinction. Let

$$R_\alpha(x) = x + \alpha \pmod{1}$$

on the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. If $\alpha \notin \mathbb{Q}$, every orbit is dense. Density alone, however, does not say that the orbit spends the correct proportion of time in every interval. The statement

$$\frac{1}{N} \#\{0 \leq n < N : R_\alpha^n x \in I\} \longrightarrow |I|$$

is a measure-theoretic assertion. Its proof requires more than the topology of the orbit.

For homogeneous spaces the natural measures are usually supplied by Haar measure. If G is unimodular and $\Gamma < G$ is a lattice, the finite G -invariant measure on G/Γ can be normalized to total mass one. A subgroup $H < G$ then acts by measure-preserving transformations. Ratner-type rigidity begins from the deceptively weak hypothesis that a probability measure is invariant under a unipotent subgroup. The purpose of this chapter is to make the measure-theoretic language precise and to prove the recurrence principle that underlies almost every rigidity argument.

Guiding Idea 1.1.1. There are three different levels of information that should never be conflated:

- (1) *invariance*: the transformation does not change the measure;
- (2) *ergodicity*: invariant measurable sets are trivial modulo null sets;
- (3) *recurrence*: typical points repeatedly revisit sets of positive measure.

Invariance is algebraic, ergodicity is a global indecomposability statement, and recurrence is a dynamical consequence of finite measure. The proofs below make the logical dependence explicit.

2. Probability spaces and measurable transformations

Definition 1.2.1. A *probability space* is a triple $(X, \mathcal{B}, \mu)$, where $\mathcal{B}$ is a σ -algebra of subsets of X and μ is a countably additive measure satisfying $\mu(X) = 1$.

Most spaces occurring in smooth dynamics are standard Borel spaces.

Definition 1.2.2. A measurable space $(X, \mathcal{B})$ is *standard Borel* if it is measurably isomorphic to the Borel space of a Polish space, that is, a separable completely metrizable topological space.

Every second-countable smooth manifold with its Borel σ -algebra is standard Borel. Quotients G/Γ by discrete subgroups of second-countable Lie groups are therefore standard Borel as well.

Definition 1.2.3. A measurable map $T : X \rightarrow X$ is *measure preserving* if

$$\mu(T^{-1}A) = \mu(A) \quad \text{for every } A \in \mathcal{B}.$$

If T is invertible and both T and T^{-1} are measurable, we call it an invertible measure-preserving transformation.

The inverse image in the definition is important. It makes sense even when T is not invertible. For invertible T , invariance is equivalent to

$$\mu(TA) = \mu(A)$$

for every measurable A .

Worked Example 1.2.4 (Translations on a compact group). Let K be a compact Lie group and let m_K be normalized Haar measure. For any $k \in K$, left translation

$$L_k(x) = kx$$

preserves m_K by the defining left-invariance of Haar measure. No commutativity is required.

Worked Example 1.2.5 (The space of unimodular lattices). Set

$$X_d = \mathrm{SL}_d(\mathbb{R}) / \mathrm{SL}_d(\mathbb{Z}).$$

A point $g\mathrm{SL}_d(\mathbb{Z})$ may be identified with the unimodular lattice $g\mathbb{Z}^d$. Haar measure on $\mathrm{SL}_d(\mathbb{R})$ descends to a finite invariant measure on X_d , which we normalize to a probability measure m_{X_d} . Every subgroup of $\mathrm{SL}_d(\mathbb{R})$, including diagonal and unipotent subgroups, acts on X_d by measure-preserving transformations.

This example is research-facing rather than decorative. Questions about Diophantine approximation, values of quadratic forms, lattice-point counting, and unipotent orbit rigidity can all be encoded in dynamics on spaces such as X_d .

3. Group actions

Definition 1.3.1. Let G be a group. A *measurable action* of G on $(X, \mathcal{B})$ is a map

$$G \times X \rightarrow X, \quad (g, x) \mapsto gx,$$

such that

$$ex = x, \quad (gh)x = g(hx),$$

and the action map is measurable. The action is *measure preserving* if every map $x \mapsto gx$ preserves μ .

For Lie groups acting smoothly on manifolds, measurability is automatic. We nevertheless keep the measurable formulation because invariant measures may live on subsets with no useful smooth structure.

Definition 1.3.2. A measurable set $A \subset X$ is *invariant modulo null sets* under a measure-preserving action of G if

$$\mu(gA \triangle A) = 0 \quad \text{for every } g \in G.$$

Here $A \triangle B = (A \setminus B) \cup (B \setminus A)$ denotes symmetric difference.

The phrase “modulo null sets” is essential. In measure theory, functions and sets are naturally considered only up to almost-everywhere equality. Requiring literal equality $gA = A$ would make basic statements depend on arbitrary choices of representatives.

4. Ergodicity

Definition 1.4.1. A measure-preserving action of G on (X, μ) is *ergodic* if every invariant measurable set A satisfies

$$\mu(A) \in \{0, 1\}.$$

For a single transformation T , this means that every set satisfying $T^{-1}A = A$ modulo null sets has measure zero or one.

Ergodicity says that the probability space cannot be decomposed into two positive-measure invariant pieces. It does *not* mean that every orbit is dense, nor that every point has the same statistical behavior. Those stronger statements require additional hypotheses.

Proposition 1.4.2 (Equivalent formulations of ergodicity). *Let $G \curvearrowright (X, \mu)$ be a measure-preserving action. The following are equivalent.*

(1) *The action is ergodic.*

(2) *Every G -invariant function $f \in L^2(X, \mu)$ is almost everywhere constant.*

(3) *Every G -invariant function $f \in L^1(X, \mu)$ is almost everywhere constant.*

PROOF. Suppose first that the action is ergodic, and let $f \in L^1(X)$ satisfy

$$f(gx) = f(x)$$

for almost every x , for every $g \in G$. We explain why f must be constant.

For each real number t , the sublevel set

$$A_t = \{x : \operatorname{Re} f(x) < t\}$$

is invariant modulo null sets. Ergodicity gives $\mu(A_t) \in \{0, 1\}$. The monotonicity of $t \mapsto \mu(A_t)$ implies that there is a threshold $c \in [-\infty, \infty]$ such that $\mu(A_t) = 0$ for $t < c$ and $\mu(A_t) = 1$ for $t > c$. Integrability rules out $c = \pm\infty$, and therefore $\operatorname{Re} f = c$ almost everywhere. Repeating the argument for $\operatorname{Im} f$ shows that f is constant almost everywhere. This proves (1) $\Rightarrow$ (3), and (3) $\Rightarrow$ (2) follows from Cauchy–Schwarz: $\|f\|_1 \leq \mu(X)^{1/2} \|f\|_2 = \|f\|_2$ on a probability space.

Conversely, assume (2). If A is invariant, then its indicator function $\mathbf{1}_A \in L^2(X)$ is invariant. Hence $\mathbf{1}_A$ is almost everywhere constant. Since it takes only the values zero and one, its constant value is either zero or one, so $\mu(A) \in \{0, 1\}$. Thus the action is ergodic. $\square$

Remark 1.4.3. The proof reveals a useful principle: invariant sets and invariant functions are two versions of the same obstruction to ergodicity. Sets produce indicator functions; functions produce invariant level sets.

5. The invariant sigma-algebra

For a measure-preserving transformation T , define

$$\mathcal{I}_T = \{A \in \mathcal{B} : T^{-1}A = A \text{ modulo null sets}\}.$$

More generally, for a group action define $\mathcal{I}_G$ using invariance under every group element. Modulo null sets, $\mathcal{I}_G$ is a σ -algebra.

The invariant σ -algebra packages all measurable information that the dynamics cannot change. Ergodicity is exactly the statement that $\mathcal{I}_G$ is trivial modulo null sets.

Worked Example 1.5.1 (A nonergodic homogeneous action). Let

$$G = \mathbb{R}^2 / \mathbb{Z}^2 = \mathbb{T}^2$$

and let the subgroup

$$H = \{(t, 0) : t \in \mathbb{R}\}$$

act by translation. The second coordinate is unchanged. Therefore every set of the form

$$\mathbb{T} \times B$$

with $B \subset \mathbb{T}$ measurable is H -invariant. The action is far from ergodic.

The example illustrates a general heuristic: a nonconstant measurable quantity preserved by the action is an obstruction to ergodicity. In homogeneous dynamics such quantities often arise from quotient maps, characters, invariant tensors, or proper homogeneous factors.

6. Irrational rotations: a complete ergodicity proof

Let

$$R_\alpha(x) = x + \alpha \pmod{1}$$

on $\mathbb{T}$ with Lebesgue probability measure m .

THEOREM 1.6.1. *The rotation R_α is ergodic if and only if $\alpha \notin \mathbb{Q}$.*

PROOF. If $\alpha = p/q \in \mathbb{Q}$ with p, q coprime, then the nonconstant function

$$f(x) = e^{2\pi i q x}$$

is invariant, because

$$f(x + \alpha) = e^{2\pi i q x} e^{2\pi i p} = f(x).$$

Hence the invariant-function criterion shows that R_α is not ergodic. This argument avoids any ambiguity about how a measurable set should be chosen from the finite cyclic orbits.

Now suppose $\alpha \notin \mathbb{Q}$. Let $f \in L^2(\mathbb{T})$ satisfy

$$f(x + \alpha) = f(x)$$

almost everywhere. Expand f in the Fourier basis:

$$f(x) = \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{2\pi i n x}$$

in L^2 . Translation by α multiplies the n -th Fourier coefficient by $e^{2\pi i n \alpha}$. Invariance gives

$$\hat{f}(n) = e^{2\pi i n \alpha} \hat{f}(n).$$

For $n \neq 0$, irrationality implies

$$e^{2\pi i n \alpha} \neq 1,$$

so $\hat{f}(n) = 0$. Hence only $\hat{f}(0)$ remains, and f is constant almost everywhere. The equivalent-function criterion proves ergodicity. $\square$

Guiding Idea 1.6.2. The Fourier proof is a model for a much broader strategy. Replace Fourier characters by a unitary representation of the acting group. Ergodicity becomes the assertion that the only invariant vectors are constants. This representation-theoretic viewpoint is particularly powerful for homogeneous spaces because the group action itself supplies a canonical unitary representation on $L^2(X)$.

7. Mixing, weak mixing, and why ergodicity is weaker

Although ergodicity is the principal notion needed at this stage, two stronger properties help calibrate what ergodicity does and does not say.

Definition 1.7.1. An invertible measure-preserving transformation T is *mixing* if for every measurable $A, B \subset X$,

$$\mu(T^{-n}A \cap B) \longrightarrow \mu(A)\mu(B) \quad (n \rightarrow \infty).$$

It is *weakly mixing* if

$$\frac{1}{N} \sum_{n=0}^{N-1} |\mu(T^{-n}A \cap B) - \mu(A)\mu(B)| \longrightarrow 0$$

for every A, B .

Mixing says that far-separated observations become asymptotically independent. Ergodicity merely says there is no invariant positive-measure decomposition.

Proposition 1.7.2. *Mixing implies weak mixing, and weak mixing implies ergodicity.*

PROOF. The first implication follows from Cesàro averaging: if a bounded sequence tends to zero, then the averages of its absolute values tend to zero.

For the second, suppose A is invariant. Taking $B = A$, we have

$$\mu(T^{-n}A \cap A) = \mu(A)$$

for all n . Weak mixing therefore gives

$$\left| \mu(A) - \mu(A)^2 \right| = 0.$$

Hence $\mu(A) \in \{0, 1\}$. □

Worked Example 1.7.3 (Irrational rotations are ergodic but not mixing). For R_α with $\alpha \notin \mathbb{Q}$, the function

$$f(x) = e^{2\pi i x}$$

satisfies

$$f \circ R_\alpha = e^{2\pi i \alpha} f.$$

Thus correlations of f do not decay:

$$\langle f \circ R_\alpha^n, f \rangle = e^{2\pi i n \alpha} \|f\|_2^2.$$

They rotate on the unit circle instead of converging to zero. Hence irrational rotation is not mixing.

This is a useful warning when studying unipotent systems. Strong rigidity and strong mixing are logically distinct phenomena.

8. Poincaré recurrence

Finite invariant measure forbids a positive-measure set from wandering away forever.

Definition 1.8.1. Let $T : X \rightarrow X$ be measure preserving. A measurable set W is *wandering* if the sets

$$W, T^{-1}W, T^{-2}W, \dots$$

are pairwise disjoint modulo null sets.

Lemma 1.8.2. *A wandering set in a probability-preserving system has measure zero.*

PROOF. If W were wandering and $\mu(W) > 0$, then measure preservation would give

$$\mu(T^{-n}W) = \mu(W)$$

for every n . Since the sets are pairwise disjoint,

$$1 = \mu(X) \geq \mu\left(\bigcup_{n=0}^{N-1} T^{-n}W\right) = N\mu(W)$$

for every N , impossible as $N \rightarrow \infty$. □

THEOREM 1.8.3 (Poincaré recurrence theorem). *Let T be a measure-preserving transformation of a probability space (X, μ) , and let $A \subset X$ be measurable. Then for almost every $x \in A$, there exist infinitely many positive integers n such that*

$$T^n x \in A.$$

PROOF. The proof has two conceptual steps.

Step 1: points that never return form a wandering set. Define

$$W = A \setminus \bigcup_{n \geq 1} T^{-n}A.$$

Thus $x \in W$ exactly when $x \in A$ but $T^n x \notin A$ for every $n \geq 1$. We claim that the sets $T^{-j}W$ are pairwise disjoint. Suppose, to the contrary, that

$$x \in T^{-j}W \cap T^{-k}W$$

with $j < k$. Then $T^j x \in W \subset A$, while

$$T^{k-j}(T^j x) = T^k x \in W \subset A,$$

contradicting the definition of W , because $T^j x \in W$ was supposed never to return to A . Hence W is wandering, so $\mu(W) = 0$.

Therefore almost every point of A returns at least once.

Step 2: apply the first-return statement to points with only finitely many returns. Let $F \subset A$ be the set of points that return to A only finitely many times. If $x \in F$, there is a last return time $n(x)$. Then $T^{n(x)}x \in A$ and never returns to A . Thus

$$F \subset \bigcup_{n \geq 0} T^{-n}W.$$

Each set $T^{-n}W$ has measure zero, so $\mu(F) = 0$. Hence almost every $x \in A$ returns infinitely often. $\square$

Remark 1.8.4. Nothing in the proof uses compactness, smoothness, or any special geometry. The theorem is a bookkeeping consequence of two facts only: finite total mass and invariance. This is why recurrence remains available even when an orbit wanders deeply into noncompact cusps before returning.

9. Recurrence times and return-time functions

For $A \subset X$ of positive measure, define the first return time

$$r_A(x) = \inf\{n \geq 1 : T^n x \in A\}$$

for $x \in A$, with the convention $r_A(x) = \infty$ if there is no return. Poincaré recurrence says $r_A < \infty$ almost everywhere on A .

A stronger quantitative statement is Kac's formula. We give the proof because it clarifies how finite measure constrains recurrence quantitatively.

THEOREM 1.9.1 (Kac's return-time formula). *Let T be an invertible measure-preserving transformation and let A have positive measure. Let*

$$\mathcal{S}(A) = \bigcup_{n \in \mathbb{Z}} T^n A$$

be the invariant saturation of A . Then

$$(1.5) \quad \int_A r_A d\mu = \mu(\mathcal{S}(A)).$$

In particular, if T is ergodic, then $\mu(\mathcal{S}(A)) = 1$, so

$$\int_A r_A d\mu = 1.$$

Equivalently, with normalized conditional measure $\mu_A = \mu(A)^{-1}\mu|_A$,

$$\int_A r_A d\mu_A = \frac{\mu(\mathcal{S}(A))}{\mu(A)},$$

which becomes $1/\mu(A)$ in the ergodic case.

PROOF. For each integer $j \geq 0$, let

$$E_j = \{x \in A : r_A(x) > j\}.$$

Thus $E_0 = A$, and $x \in E_j$ means that after time zero the orbit of x does not meet A during the first j iterates. Since $r_A < \infty$ almost everywhere on A , the layer-cake identity for the integer-valued function r_A gives

$$(1.6) \quad \int_A r_A d\mu = \sum_{j=0}^{\infty} \mu(E_j).$$

We now identify the terms in this sum geometrically. The sets

$$(1.7) \quad T^j E_j, \quad j = 0, 1, 2, \dots,$$

are pairwise disjoint. Indeed, suppose $0 \leq j < k$ and

$$T^j x = T^k y, \quad x \in E_j, \quad y \in E_k.$$

Then

$$x = T^{k-j} y \in A.$$

But $1 \leq k - j \leq k$, contradicting $y \in E_k$, because the latter says that none of $Ty, \dots, T^k y$ lies in A .

Next we claim that the union of the sets in (1.7) agrees with the invariant saturation $\mathcal{S}(A)$ modulo a null set. If $z \in T^j E_j$, then $z = T^j x$ for some $x \in E_j \subset A$. Hence z lies on the T -orbit of a point of A , so $z \in \mathcal{S}(A)$. Therefore

$$\bigcup_{j \geq 0} T^j E_j \subset \mathcal{S}(A).$$

For the converse, take a point $z \in \mathcal{S}(A)$ on an orbit for which visits to A occur infinitely often in both time directions. This is the case for almost every point of the saturation: Poincaré recurrence applied to T and T^{-1} gives two-sided recurrence for almost every point of A , and invariance transports the resulting null exceptional set to the saturation. Let $-j \leq 0$ be the last visit time to A at or before time zero, so that

$$x = T^{-j} z \in A$$

and there is no visit to A at times $1, \dots, j$ after x . Hence $x \in E_j$ and

$$z = T^j x \in T^j E_j.$$

This proves the claimed partition modulo null sets.

Finally, measure preservation gives

$$\begin{aligned}\mu(\mathcal{S}(A)) &= \sum_{j=0}^{\infty} \mu(T^j E_j) \\ &= \sum_{j=0}^{\infty} \mu(E_j) \\ &= \int_A r_A d\mu,\end{aligned}$$

where the last equality is (1.6). This proves Kac's formula. $\square$

Remark 1.9.2. The general formula clarifies exactly where ergodicity enters. Return-time statistics only see the invariant component generated by A . Ergodicity is the additional statement that every positive-measure set generates the whole space modulo null sets.

10. Recurrence to the starting point

Poincaré recurrence is stated for measurable sets, but on a second-countable metric space it implies a more geometric statement: almost every point returns arbitrarily close to itself.

THEOREM 1.10.1 (Metric recurrence). *Let X be a second-countable metric space with a Borel probability measure μ , and let $T : X \rightarrow X$ be measure preserving. Then for μ -almost every $x \in X$ there exists a sequence $n_j \rightarrow \infty$ such that*

$$(1.7R) \quad T^{n_j} x \longrightarrow x.$$

PROOF. Choose a countable basis $\{B_k\}_{k \geq 1}$ for the topology. For each k , Poincaré recurrence says that almost every point of B_k returns to B_k infinitely often. Let $N_k \subset B_k$ be the exceptional null set, and set

$$N = \bigcup_{k \geq 1} N_k.$$

Then $\mu(N) = 0$.

Fix $x \notin N$. For every neighborhood V of x , choose a basis element B_k with

$$x \in B_k \subset V.$$

Since $x \notin N_k$, there are infinitely many n with $T^n x \in B_k \subset V$.

To obtain an actual convergent sequence, for each $j \geq 1$ choose a basis index $k(j)$ such that

$$x \in B_{k(j)} \subset B(x, 1/j).$$

Because returns to $B_{k(j)}$ occur infinitely often, we may choose $n_j > n_{j-1}$ with

$$T^{n_j}x \in B_{k(j)}.$$

Then

$$d(T^{n_j}x, x) < 1/j,$$

so $T^{n_j}x \rightarrow x$. □

Remark 1.10.2. Second countability is what lets one choose a single null set outside which recurrence works for every neighborhood. Without a countable base, intersecting the full-measure recurrence sets over all neighborhoods would involve an uncountable family and need not preserve full measure.

On a homogeneous space this theorem says that a typical point returns close to its own coset, not merely to some fixed positive-measure region. It is still qualitative: it provides no control on the return times n_j or on how two different recurrent points return simultaneously.

11. Conservative actions of flows

Let $(u_t)_{t \in \mathbb{R}}$ be a measurable measure-preserving flow. For any fixed $\tau \neq 0$, the transformation u_τ preserves measure, so Poincaré recurrence applies to its integer iterates. Consequently, for every measurable A , almost every $x \in A$ admits integers $n_j \rightarrow \infty$ with

$$u_{n_j\tau}x \in A.$$

This discrete sampling is often sufficient. When the flow is continuous and A is open, it immediately implies genuine continuous-time recurrence.

Definition 1.11.1. A measure-preserving action is *conservative* if it has no wandering set of positive measure.

Every probability-preserving action of $\mathbb{Z}$ or $\mathbb{R}$ is conservative. In infinite-measure spaces this can fail dramatically. For instance, translation $x \mapsto x + 1$ on $(\mathbb{R}, \text{Lebesgue})$ preserves measure, but $[0, 1)$ is wandering under the integer action.

12. Recurrence on noncompact homogeneous spaces

Consider again

$$X_d = \text{SL}_d(\mathbb{R}) / \text{SL}_d(\mathbb{Z})$$

with normalized Haar probability measure. The space is noncompact for $d \geq 2$. A trajectory can therefore leave every fixed compact set for a long time. Poincaré recurrence says something subtler than compactness: any positive-measure set is revisited by almost every point that starts in it.

Worked Example 1.12.1 (Diagonal excursions and recurrence). In X_2 , let

$$a_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix}.$$

A lattice Λ can develop an extremely short vector under a_t , corresponding geometrically to a cusp excursion. Nevertheless, if $K \subset X_2$ is a compact set of positive Haar measure, then for almost every $\Lambda \in K$ there are arbitrarily large integers n such that

$$a_n \Lambda \in K.$$

The theorem does not bound the gaps between returns and does not prevent arbitrarily deep excursions. Those stronger questions require quantitative nondivergence or mixing information.

13. Ergodicity and almost-everywhere orbit saturation

Proposition 1.13.1. *Let a countable group G act ergodically on (X, μ) . If $A \subset X$ has positive measure, then*

$$\mu \left(\bigcup_{g \in G} gA \right) = 1.$$

PROOF. The saturation

$$GA = \bigcup_{g \in G} gA$$

is measurable because the union is countable. It is invariant, and it contains A , so it has positive measure. Ergodicity therefore gives $\mu(GA) = 1$. $\square$

For uncountable Lie groups the same statement requires a little care because an uncountable union of measurable sets need not be measurable. In second-countable continuous actions one can usually replace the full group by a countable dense subgroup when the property under consideration is topologically stable. Any use of such a reduction below includes the measurability or continuity hypothesis needed to justify it.

14. Invariant measures supported on closed orbits

A basic source of ergodic measures in homogeneous dynamics comes from closed homogeneous orbits.

Let G be a Lie group, $\Gamma < G$ discrete, $H < G$ closed, and $x = g\Gamma \in G/\Gamma$. Suppose the orbit Hx is closed and carries a finite H -invariant measure. After normalization we obtain an H -invariant probability measure m_{Hx} .

Whether the action of a particular subgroup $U < H$ is ergodic on (Hx, m_{Hx}) is a separate question. The measure is homogeneous because

its support and invariance are generated by algebraic data, but homogeneity alone does not imply U -ergodicity.

Worked Example 1.14.1 (Periodic horocycles). On a finite-area hyperbolic surface, a closed horocycle carries its normalized arclength measure. The horocycle flow acts transitively on that closed orbit, so the normalized arclength measure is ergodic for the horocycle flow. This is the simplest example of a measure supported on a proper homogeneous orbit.

15. A criterion from transitivity

Proposition 1.15.1. *Let a locally compact second-countable group H act transitively on a homogeneous space H/L , and suppose H/L carries a finite H -invariant measure m . Then the full H -action is ergodic.*

PROOF. Let $X = H/L$, let m be the given finite invariant measure, and let

$$f = \mathbf{1}_A.$$

For each fixed $h \in H$, invariance modulo null sets says

$$(1.1) \quad f(hx) = f(x) \quad \text{for } m\text{-almost every } x.$$

The only subtlety is that the exceptional null set in (1.1) can depend on h . We therefore use Fubini's theorem to reverse the order of the quantifiers.

Fix a compact set $C \subset H$ of finite Haar measure. On $C \times X$, the function

$$(h, x) \mapsto |f(hx) - f(x)|$$

is measurable and bounded. For every fixed $h \in C$, its integral in x is zero by (1.1). Hence Fubini gives

$$\int_X \int_C |f(hx) - f(x)| \, dh \, dm(x) = 0.$$

Thus, for m -almost every x ,

$$(1.2) \quad f(hx) = f(x) \quad \text{for Haar-almost every } h \in C.$$

Choose an increasing sequence of compact sets C_j whose union is H ; this is possible because H is locally compact and second countable, hence σ -compact. Intersecting the corresponding full-measure sets, we obtain a point $x_0 \in X$ such that

$$(1.3) \quad f(hx_0) = f(x_0) \quad \text{for Haar-almost every } h \in H.$$

Now consider the orbit map

$$p_{x_0} : H \rightarrow X, \quad p_{x_0}(h) = hx_0.$$

It is the quotient map $H \rightarrow H/L_{x_0}$, where L_{x_0} is the stabilizer of x_0 . We only need one consequence of the quotient integration formula: for a measurable set $E \subset X$,

$$(1.4) \quad m(E) = 0 \iff p_{x_0}^{-1}(E) \text{ is Haar-null in } H.$$

Let us explain why this equivalence is legitimate. Choose a nonnegative $\rho \in C_c(H)$ whose average along every L_{x_0} -coset is positive on a prescribed compact part of X . Weil's quotient formula then writes the integral of $\rho \mathbf{1}_{p_{x_0}^{-1}(E)}$ over H as the integral over E of a strictly positive continuous weight. Exhausting X by compact sets gives (1.4). Thus no global finiteness of the push-forward of Haar measure is being asserted.

Apply (1.4) to

$$E = \{y \in X : f(y) \neq f(x_0)\}.$$

Equation (1.3) says exactly that $p_{x_0}^{-1}(E)$ is Haar-null. Hence $m(E) = 0$, and therefore

$$f(y) = f(x_0) \quad \text{for } m\text{-almost every } y \in X.$$

Hence f is essentially constant. Since f takes only the values 0 and 1,

$$m(A) \in \{0, m(X)\}.$$

After normalizing m to a probability measure, this is precisely ergodicity. $\square$

Remark 1.15.2. For the full transitive group, ergodicity is almost tautological: there is only one orbit. The technical point is converting modulo-null invariance into a statement that can be combined with transitivity. For proper subgroups of H , the problem becomes genuinely dynamical.

16. What recurrence does not give

Poincaré recurrence is powerful because its hypotheses are weak, but its conclusions are correspondingly qualitative. It does not provide:

- a bound on the first return time;
- a positive density of return times;
- control of how deep an orbit can enter a cusp;
- equidistribution of return locations;
- simultaneous recurrence of two nearby points.

These distinctions matter in rigidity arguments. A proof may begin with ordinary recurrence, but it often needs to compare two recurrent points at carefully chosen times. The comparison becomes useful only after one understands how the acting subgroup magnifies their small initial displacement.

17. Conceptual summary

The central objects of this chapter can be organized as follows:

finite invariant measure $\implies$ no positive-measure wandering set
 $\implies$ Poincaré recurrence.

Independently,

ergodicity $\iff$ trivial invariant σ -algebra
 $\iff$ only constant invariant L^2 -functions.

The first chain is a consequence of finite mass. The second is a statement of dynamical indecomposability. Their combination is the basic measure-theoretic environment in which homogeneous rigidity operates: recurrent points are plentiful, and ergodicity prevents the dynamics from splitting measurably into unrelated pieces.

CHAPTER 2

Ergodic Theorems, Generic Points, and Decomposition

1. Time averages as the basic observable

Let T be a measure-preserving transformation of (X, μ) , and let $f \in L^1(X, \mu)$. The central statistical object is the time average

$$A_N f(x) = \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x).$$

For a flow (u_t) , the analogous average is

$$A_T^{\text{flow}} f(x) = \frac{1}{T} \int_0^T f(u_t x) dt.$$

If $f = \mathbf{1}_A$, then $A_N f(x)$ is the proportion of the first N iterates that lie in A . Thus convergence of time averages turns recurrence into a statistical law.

There are two fundamental ergodic theorems. The mean ergodic theorem gives convergence in L^2 ; the pointwise ergodic theorem gives almost-everywhere convergence for L^1 functions. The second is much stronger and requires a different argument.

Guiding Idea 2.1.1. The pointwise theorem is not proved by guessing the limit. Its hard part is first showing that the upper and lower asymptotic averages cannot separate. The maximal ergodic lemma controls the exceptional set where partial sums are systematically positive. Once convergence is established, identifying the limit uses only invariance and conditional expectation.

2. The Koopman operator

Let T be invertible and measure preserving. Define

$$U_T : L^2(X, \mu) \rightarrow L^2(X, \mu), \quad U_T f = f \circ T.$$

Proposition 2.2.1. *The Koopman operator U_T is unitary.*

PROOF. For $f \in L^2$, measure preservation gives

$$\|U_T f\|_2^2 = \int_X |f(Tx)|^2 d\mu(x) = \int_X |f(y)|^2 d\mu(y) = \|f\|_2^2.$$

Thus U_T is an isometry. Since T is invertible, $U_{T^{-1}}$ is its inverse, so U_T is unitary. $\square$

The fixed-point space

$$\text{Fix}(U_T) = \{f \in L^2 : f \circ T = f\}$$

is precisely $L^2(\mathcal{I}_T)$, the subspace of functions measurable with respect to the invariant σ -algebra.

3. The Hilbert-space mean ergodic theorem

THEOREM 2.3.1 (von Neumann mean ergodic theorem). *Let U be a unitary operator on a Hilbert space $\mathcal{H}$. Then*

$$M_N = \frac{1}{N} \sum_{n=0}^{N-1} U^n$$

converges strongly to the orthogonal projection P onto $\text{Fix}(U)$.

PROOF. The proof becomes transparent after identifying the subspace on which averaging must decay.

Let

$$\mathcal{K} = \overline{\{h - Uh : h \in \mathcal{H}\}}.$$

We claim

$$\mathcal{H} = \text{Fix}(U) \oplus \mathcal{K}.$$

Indeed, a vector v is orthogonal to every $h - Uh$ if and only if

$$\langle v, h \rangle = \langle v, Uh \rangle = \langle U^{-1}v, h \rangle$$

for every h , hence $v = U^{-1}v$, or $Uv = v$. Thus $\mathcal{K}^\perp = \text{Fix}(U)$, proving the orthogonal decomposition.

If $v \in \text{Fix}(U)$, then $U^j v = v$ for every j , and therefore $M_N v = N^{-1} \sum_{j=0}^{N-1} v = v$. It remains to show $M_N v \rightarrow 0$ for $v \in \mathcal{K}$. First take a coboundary $v = h - Uh$. Then the sum telescopes:

$$M_N(h - Uh) = \frac{1}{N}(h - U^N h).$$

Since U is unitary,

$$\|M_N(h - Uh)\| \leq \frac{2\|h\|}{N} \rightarrow 0.$$

Every M_N has operator norm at most one. By density, the convergence extends from coboundaries to their closure $\mathcal{K}$. Therefore $M_N \rightarrow P$ strongly. $\square$

Corollary 2.3.2 (Mean ergodic theorem for transformations). *For $f \in L^2(X, \mu)$,*

$$A_N f \longrightarrow \mathbb{E}(f \mid \mathcal{I}_T) \quad \text{in } L^2.$$

If T is ergodic, then

$$A_N f \longrightarrow \int_X f \, d\mu \quad \text{in } L^2.$$

PROOF. Apply the Hilbert-space theorem to U_T . Orthogonal projection onto $L^2(\mathcal{I}_T)$ is conditional expectation onto $\mathcal{I}_T$. Under ergodicity, $L^2(\mathcal{I}_T)$ consists only of constants, and the orthogonal projection onto constants is $f \mapsto \int f \, d\mu$. $\square$

4. Why mean-square convergence is not enough

Convergence in L^2 implies convergence in measure and gives an almost-everywhere convergent subsequence, but it does not by itself show that the full sequence $A_N f(x)$ converges for almost every x . Orbit statistics require pointwise control.

The obstacle is that the exceptional set could move with N . The maximal ergodic theorem controls all partial sums simultaneously.

5. The maximal ergodic lemma

For $f \in L^1$, write

$$S_n f = \sum_{j=0}^{n-1} f \circ T^j, \quad S_0 f = 0.$$

For a fixed N , define

$$E_N = \left\{ x : \max_{1 \leq n \leq N} S_n f(x) > 0 \right\}.$$

THEOREM 2.5.1 (Maximal ergodic lemma). *If T is measure preserving and $f \in L^1(X)$, then*

$$\int_{E_N} f \, d\mu \geq 0.$$

Consequently, for

$$E = \left\{ x : \sup_{n \geq 1} S_n f(x) > 0 \right\},$$

one has

$$\int_E f \, d\mu \geq 0.$$

PROOF. Set

$$F_N(x) = \max\{0, S_1 f(x), \dots, S_N f(x)\}.$$

We claim the pointwise inequality

$$(2.1) \quad \mathbf{1}_{E_N}(x) f(x) \geq F_N(x) - F_N(Tx).$$

The reason for introducing F_N is that it records the best positive partial sum starting at x ; after applying T , the same partial sums have lost their first term $f(x)$.

If $x \notin E_N$, then $F_N(x) = 0$ and the right-hand side is $-F_N(Tx) \leq 0$, so (2.1) holds.

If $x \in E_N$, choose $1 \leq n \leq N$ with

$$F_N(x) = S_n f(x) = f(x) + S_{n-1} f(Tx).$$

Since

$$S_{n-1} f(Tx) \leq F_N(Tx)$$

(the case $n = 1$ uses $S_0 = 0$), we obtain

$$F_N(x) \leq f(x) + F_N(Tx),$$

which is (2.1).

Integrating and using measure preservation,

$$\int_{E_N} f \, d\mu \geq \int_X F_N \, d\mu - \int_X F_N \circ T \, d\mu = 0.$$

Finally $E_N \uparrow E$. Applying the result to truncated functions if necessary and then monotone convergence on positive and negative parts yields

$$\int_E f \, d\mu \geq 0.$$

□

Remark 2.5.2. The lemma is asymmetric: it controls points where a partial sum is positive. Applying it to $-f$ gives the corresponding lower-bound information. Birkhoff's theorem uses both directions.

6. Birkhoff's pointwise ergodic theorem

THEOREM 2.6.1 (Birkhoff pointwise ergodic theorem). *Let T be a measure-preserving transformation of a probability space and let $f \in L^1(X, \mu)$. Then the limit*

$$f^*(x) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x)$$

exists for almost every x . The function f^ is T -invariant, belongs to L^1 , and satisfies*

$$f^* = \mathbb{E}(f \mid \mathcal{I}_T) \quad \text{almost everywhere.}$$

In particular, if T is ergodic, then

$$A_N f(x) \longrightarrow \int_X f \, d\mu$$

for almost every x .

PROOF. We separate the proof into three parts: bounded functions, approximation in L^1 , and identification of the limit. The maximal ergodic lemma does the real work for bounded functions; truncation is then a stability argument.

Step 1: upper and lower averages for bounded functions. Assume first that $f \in L^\infty(X)$, and define

$$\bar{f} = \limsup_{N \rightarrow \infty} A_N f, \quad \underline{f} = \liminf_{N \rightarrow \infty} A_N f.$$

Since

$$A_N f(Tx) - A_N f(x) = \frac{f(T^N x) - f(x)}{N},$$

and f is bounded, both $\bar{f}$ and $\underline{f}$ are T -invariant.

We claim

$$(2.2) \quad \int_X \bar{f} \, d\mu \leq \int_X f \, d\mu.$$

The point is to compare $\bar{f}$ with invariant simple functions lying below it. Let h be a bounded invariant simple function such that

$$h < \bar{f} \quad \text{almost everywhere.}$$

Write

$$h = \sum_{j=1}^r c_j \mathbf{1}_{B_j},$$

where the B_j are pairwise disjoint invariant measurable sets. On B_j , the inequality $c_j < \bar{f}$ means

$$\limsup_{N \rightarrow \infty} \frac{1}{N} S_N(f - c_j) > 0.$$

Consequently every point of B_j , except for a null set, belongs to

$$E_j = \left\{ x : \sup_{N \geq 1} S_N(f - c_j)(x) > 0 \right\}.$$

Apply the maximal ergodic lemma to the transformation restricted to the invariant set B_j and the function $(f - c_j)\mathbf{1}_{B_j}$. We obtain

$$\int_{B_j} (f - c_j) d\mu \geq 0.$$

Summing over j gives

$$(2.3) \quad \int_X h d\mu \leq \int_X f d\mu.$$

Now approximate $\bar{f}$ from below by bounded invariant simple functions. For example, if $\|f\|_\infty \leq M$, divide $[-M, M]$ into intervals of length 2^{-k} and round $\bar{f}$ downward to the nearest grid point, subtracting one further mesh size so that the resulting simple function is strictly smaller than $\bar{f}$ wherever needed. These simple functions increase pointwise to $\bar{f}$. Passing to the limit in (2.3) by dominated convergence proves (2.2).

Apply the same argument to $-f$. Since

$$\overline{(-f)} = -\underline{f},$$

we get

$$(2.4) \quad \int_X \underline{f} d\mu \geq \int_X f d\mu.$$

But $\underline{f} \leq \bar{f}$. Therefore

$$\int_X (\bar{f} - \underline{f}) d\mu \leq 0.$$

The integrand is nonnegative, so it vanishes almost everywhere. Hence

$$\underline{f} = \bar{f} \quad \text{almost everywhere,}$$

and $A_N f$ converges almost everywhere for every bounded f .

Step 2: the maximal weak (1, 1) estimate. For $g \in L^1$ and $\lambda > 0$,

$$(2.5) \quad \mu \left(\sup_{N \geq 1} A_N |g| > \lambda \right) \leq \frac{\|g\|_1}{\lambda}.$$

Indeed, apply the maximal ergodic lemma to $|g| - \lambda$. If E is the set on the left, then

$$0 \leq \int_E (|g| - \lambda) d\mu \leq \|g\|_1 - \lambda\mu(E),$$

which is exactly (2.5).

Step 3: extend the convergence to L^1 . Choose bounded functions f_j such that

$$\|f - f_j\|_1 < 2^{-3j}.$$

By (2.5),

$$\mu \left(\sup_{N \geq 1} A_N |f - f_j| > 2^{-j} \right) \leq 2^j \|f - f_j\|_1 < 2^{-2j}.$$

The series $\sum_j 2^{-2j}$ converges, so Borel–Cantelli implies that for almost every x there is $j_0(x)$ such that, for all $j \geq j_0(x)$,

$$(2.6) \quad \sup_{N \geq 1} |A_N f(x) - A_N f_j(x)| \leq 2^{-j}.$$

Intersect this full-measure set with the full-measure sets on which every sequence $A_N f_j$ converges. Fix x in the intersection. Given $\varepsilon > 0$, choose j so large that $2^{1-j} < \varepsilon$. Since $A_N f_j(x)$ converges, it is Cauchy; hence for sufficiently large M, N ,

$$\begin{aligned} |A_N f(x) - A_M f(x)| &\leq |A_N(f - f_j)(x)| + |A_N f_j(x) - A_M f_j(x)| \\ &\quad + |A_M(f_j - f)(x)| \\ &< 2^{-j} + \varepsilon + 2^{-j}. \end{aligned}$$

Thus $A_N f(x)$ is Cauchy, and the pointwise limit $f^*(x)$ exists almost everywhere.

Fatou's lemma gives the L^1 bound

$$(2.7) \quad \int_X |f^*| d\mu \leq \liminf_{N \rightarrow \infty} \int_X A_N |f| d\mu = \|f\|_1.$$

In particular $f^* \in L^1$.

Step 4: invariance of the limit. We again use

$$(2.8) \quad A_N f(Tx) - A_N f(x) = \frac{f(T^N x) - f(x)}{N}.$$

We show that the right-hand side tends to zero almost everywhere. For $\varepsilon > 0$, the layer-cake formula implies

$$(2.9) \quad \sum_{N=1}^{\infty} \mu(|f| > \varepsilon N) < \infty.$$

One direct verification is

$$\sum_{N=1}^{\infty} \mu(|f| > \varepsilon N) \leq \frac{1}{\varepsilon} \int_X |f| d\mu.$$

Because T preserves μ , the same estimate holds with f replaced by $f \circ T^N$ in the N -th event. Borel–Cantelli therefore gives

$$\frac{|f(T^N x)|}{N} \longrightarrow 0 \quad \text{for almost every } x.$$

Equation (2.8) now yields

$$f^* \circ T = f^* \quad \text{almost everywhere.}$$

Step 5: identify the limit. Let $B \in \mathcal{I}_T$. For every N , invariance of B modulo null sets gives

$$(2.10) \quad \int_B A_N f d\mu = \frac{1}{N} \sum_{n=0}^{N-1} \int_B f \circ T^n d\mu = \int_B f d\mu.$$

For bounded f , dominated convergence passes to the limit in (2.10), giving

$$\int_B f^* d\mu = \int_B f d\mu.$$

For general $f \in L^1$, choose bounded $f_j \rightarrow f$ in L^1 . The maximal estimate and Fatou's lemma imply that the limiting operator $P : f \mapsto f^*$ is an L^1 -contraction:

$$\|Pf - Pg\|_1 \leq \|f - g\|_1.$$

Hence the bounded case passes to the limit in L^1 . Since f^* is $\mathcal{I}_T$ -measurable and satisfies the defining integral identity on every $B \in \mathcal{I}_T$,

$$f^* = \mathbb{E}(f \mid \mathcal{I}_T).$$

If T is ergodic, $\mathcal{I}_T$ is trivial modulo null sets, and so

$$f^* = \int_X f d\mu \quad \text{almost everywhere.}$$

□

Warning 2.6.2. Borel–Cantelli enters twice in the L^1 part of the proof: first to make the bounded approximations uniformly close to the original ergodic averages for almost every point, and second to prove invariance of the limiting function. The substantive estimate behind both uses is still the maximal ergodic inequality; Borel–Cantelli is the device that turns summable exceptional-set bounds into an almost-everywhere conclusion.

7. Generic points

Definition 2.7.1. Let X be a locally compact second-countable space, let $T : X \rightarrow X$ be continuous, and let μ be a T -invariant probability measure. A point $x \in X$ is *generic* for μ if

$$\frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) \longrightarrow \int_X f d\mu$$

for every $f \in C_c(X)$. For compact X , one may use $C(X)$.

The definition asks for simultaneous convergence for an uncountable family of test functions. Birkhoff gives convergence for one function at a time. Second countability supplies the bridge.

The only small point hidden in the phrase “for every $f \in C_c(X)$ ” is countability. It is useful to remove that ambiguity explicitly.

Lemma 2.7.2 (A countable family of test functions). *If X is locally compact and second countable, then $C_0(X)$, the continuous functions vanishing at infinity, is separable in the uniform norm. Consequently there is a countable family $\{f_j\}_{j \geq 1} \subset C_0(X)$ such that convergence of integrals against all f_j , together with a uniform bound on total masses, extends by uniform approximation to every function in $C_0(X)$, and hence to every function in $C_c(X)$.*

PROOF. The one-point compactification $X^* = X \cup \{\infty\}$ is compact and second countable, hence metrizable. On any compact metric space K , $C(K)$ is separable in the uniform norm. One concrete way to see this is to choose a countable dense set $D \subset K$; the algebra over $\mathbb{Q}$ generated by the functions $x \mapsto d(x, a)$, $a \in D$, and the constants is countable and separates points, so its uniform closure is $C(K)$ by the Stone–Weierstrass theorem proved in Volume I, Appendix A. Therefore $C(X^*)$ is separable.

Under the identification

$$C_0(X) = \{F \in C(X^*) : F(\infty) = 0\},$$

$C_0(X)$ is a closed subspace of a separable metric space and is therefore separable. Choosing a countable dense family gives the claim. If ν_n are measures with $\sup_n \nu_n(X) \leq M$, then

$$\left| \int (f - f_j) d\nu_n \right| \leq M \|f - f_j\|_\infty,$$

which is the uniform approximation step needed below. □

Proposition 2.7.3. *If μ is ergodic, then μ -almost every point is generic for μ .*

PROOF. Choose a countable uniformly dense family $\{f_j\}_{j \geq 1} \subset C_0(X)$. By Birkhoff, for every j there is a full-measure set X_j on which

$$A_N f_j(x) \longrightarrow \int f_j d\mu.$$

The countable intersection

$$X_0 = \bigcap_{j=1}^{\infty} X_j$$

still has full measure.

Fix $x \in X_0$ and $f \in C_c(X) \subset C_0(X)$. Given $\varepsilon > 0$, choose j with

$$\|f - f_j\|_{\infty} < \varepsilon.$$

Because both the empirical measure and μ have total mass one,

$$|A_N f(x) - A_N f_j(x)| \leq \varepsilon, \quad \left| \int f d\mu - \int f_j d\mu \right| \leq \varepsilon.$$

The middle term

$$A_N f_j(x) - \int f_j d\mu$$

tends to zero. Hence

$$\limsup_{N \rightarrow \infty} \left| A_N f(x) - \int f d\mu \right| \leq 2\varepsilon.$$

Since ε is arbitrary, x is generic for μ . □

Remark 2.7.4. No tightness argument is needed for this particular conclusion. The empirical measures already have total mass one, and uniform approximation inside $C_0(X)$ controls their integrals. Tightness becomes essential in a different problem: extracting a probability subsequential limit from an arbitrary sequence of empirical measures on a noncompact space. There, mass can indeed escape to infinity.

8. Empirical measures

Define the empirical measure

$$\mu_{N,x} = \frac{1}{N} \sum_{n=0}^{N-1} \delta_{T^n x}.$$

Then genericity is the weak-* statement

$$\mu_{N,x} \xrightarrow{*} \mu.$$

Proposition 2.8.1 (Any subsequential limit is invariant). *Suppose X is locally compact and $T : X \rightarrow X$ is a homeomorphism. If a subsequence $\mu_{N_j,x}$ converges weak-* to a probability measure ν , then ν is T -invariant.*

PROOF. For $f \in C_c(X)$,

$$\int f \circ T d\mu_{N,x} - \int f d\mu_{N,x} = \frac{f(T^N x) - f(x)}{N}.$$

The right-hand side tends to zero. Because T is a homeomorphism, $f \circ T \in C_c(X)$ whenever $f \in C_c(X)$. Hence weak-* convergence can be applied to both terms, giving

$$\int f \circ T d\nu = \int f d\nu.$$

Thus ν is invariant. □

This elementary telescoping identity is one of the most useful compactness devices in dynamics. The difficult part in noncompact spaces is usually not invariance but proving that a subsequence converges to a *probability* measure rather than losing mass at infinity.

9. Flows and continuous-time ergodic averages

Let $(u_t)_{t \in \mathbb{R}}$ be a measurable measure-preserving flow. Define

$$A_T f(x) = \frac{1}{T} \int_0^T f(u_t x) dt.$$

THEOREM 2.9.1 (Pointwise ergodic theorem for flows). *For $f \in L^1(X)$, $A_T f(x)$ converges for almost every x to the conditional expectation of f onto the u_t -invariant σ -algebra. If the flow is ergodic, the limit is $\int f d\mu$.*

PROOF. There are direct continuous-time maximal-inequality proofs. A convenient reduction uses the time-one map, but one must handle two points explicitly: arbitrary real terminal times and invariance under the *whole* flow rather than only under u_1 .

Define

$$F(x) = \int_0^1 f(u_s x) ds.$$

Fubini gives $F \in L^1(X)$. For an integer $N \geq 1$,

$$(2.11) \quad \frac{1}{N} \int_0^N f(u_t x) dt = \frac{1}{N} \sum_{n=0}^{N-1} F(u_n x).$$

Birkhoff for the measure-preserving transformation u_1 therefore shows that the integer-time continuous averages converge for almost every x .

Now write $T = N + r$ with $N = \lfloor T \rfloor$ and $0 \leq r < 1$. Put

$$H(x) = \int_0^1 |f(u_s x)| ds.$$

Then $H \in L^1$, and

$$(2.12) \quad \left| \int_N^T f(u_t x) dt \right| \leq H(u_N x).$$

Exactly as in the discrete Borel–Cantelli estimate,

$$(2.13) \quad \frac{H(u_N x)}{N} \rightarrow 0 \quad \text{for almost every } x.$$

Also, because the integer averages in (2.11) converge, the quantities

$$\frac{1}{N} \int_0^N f(u_t x) dt$$

remain bounded for almost every such x . Therefore

$$\begin{aligned} \frac{1}{T} \int_0^T f(u_t x) dt - \frac{1}{N} \int_0^N f(u_t x) dt &= \left(\frac{1}{T} - \frac{1}{N} \right) \int_0^N f(u_t x) dt \\ &\quad + \frac{1}{T} \int_N^T f(u_t x) dt \end{aligned}$$

tends to zero by (2.12)–(2.13). Thus the averages converge for arbitrary real $T \rightarrow \infty$.

Let the limit be f^* . Fix $s \in \mathbb{R}$; for simplicity first take $s > 0$. Then

$$(2.14) \quad A_T f(u_s x) - A_T f(x) = \frac{1}{T} \left(\int_T^{T+s} f(u_t x) dt - \int_0^s f(u_t x) dt \right).$$

The second integral is fixed once x is fixed. For the first, write again $T = N + r$ and define

$$H_s(x) = \int_0^{s+1} |f(u_t x)| dt.$$

Since $H_s \in L^1$, Borel–Cantelli gives

$$H_s(u_N x)/N \rightarrow 0$$

for almost every x . The first boundary integral in (2.14) is bounded by $H_s(u_N x)$, so (2.14) tends to zero. The case $s < 0$ is identical after reversing the short boundary intervals. Since the argument applies to each fixed $s \in \mathbb{R}$, we have

$$f^* \circ u_s = f^* \quad \text{almost everywhere}$$

for every s . This is exactly the modulo-null notion of invariance needed for the flow-invariant σ -algebra; no common exceptional set for all uncountably many values of s is required.

Finally, if B is invariant under the full flow, then

$$\int_B F d\mu = \int_0^1 \int_B f(u_s x) d\mu(x) ds = \int_B f d\mu.$$

The discrete-time conditional-expectation identity for (2.11), together with the full-flow invariance just proved, therefore identifies the limit with the conditional expectation of f onto the flow-invariant σ -algebra. $\square$

10. A subtlety: ergodicity of a flow versus a time map

An ergodic flow need not have every time- t map ergodic. For example, consider the translation flow on the circle

$$u_t(x) = x + t \pmod{1}.$$

The full $\mathbb{R}$ -action is transitive and hence ergodic with respect to Lebesgue measure, but $u_1 = \text{id}$ is not ergodic. Thus reducing the pointwise theorem to the time-one map does not allow us to replace the flow-invariant σ -algebra by the u_1 -invariant one without further argument. The averaging function F and the boundary-shift argument above are what recover invariance under every u_s .

11. Ergodic decomposition

Invariant measures need not be ergodic. The correct replacement is to decompose them into ergodic components.

THEOREM 2.11.1 (Ergodic decomposition, standard Borel form). *Let a second-countable locally compact group G act measurably on a standard Borel probability space (X, μ) , preserving μ . There exists a probability space (Y, η) and a measurable family of G -invariant ergodic probability measures $\{\mu_y\}_{y \in Y}$ such that*

$$\mu = \int_Y \mu_y d\eta(y),$$

meaning

$$\int_X f d\mu = \int_Y \left(\int_X f d\mu_y \right) d\eta(y)$$

for every bounded measurable f . Moreover, the decomposition is unique up to the natural almost-everywhere equivalence.

PROOF ARCHITECTURE. The full theorem rests on one measure-theoretic input: disintegration over a countably generated sub- σ -algebra. We make the logic explicit.

Let $\mathcal{I}_G$ be the invariant σ -algebra modulo null sets. We first make the factor realization explicit rather than importing another measure-theory theorem. On the measure algebra put

$$d(A, B) = \mu(A \Delta B).$$

For a standard Borel probability space this metric measure algebra is separable for the following explicit reason: choose a Polish realization with a countable

base; Radon regularity approximates measurable sets by open sets in measure, and every open set is approximated from below by a finite union of basic opens. Hence the countable algebra generated by the base is dense. Its subspace $\mathcal{I}_G$ is therefore separable. Choose invariant classes $A_1, A_2, \dots$ dense in it and let $\mathcal{C} = \sigma(A_1, A_2, \dots)$. The subalgebra $\mathcal{C}$ is closed for convergence in d : from a d -convergent sequence choose a subsequence with summable successive symmetric differences, which converges almost everywhere, and write the limiting set as a liminf/limsup of sets in $\mathcal{C}$. Density therefore gives $\mathcal{C} = \mathcal{I}_G$ modulo null sets.

Now put $Y = \{0, 1\}^{\mathbb{N}}$, equipped with its product Borel structure, and define

$$\pi(x) = (\mathbf{1}_{A_1}(x), \mathbf{1}_{A_2}(x), \dots).$$

Then $\pi^{-1}\mathcal{B}_Y = \mathcal{C} = \mathcal{I}_G$ modulo null sets. The disintegration theorem used below is proved internally in Appendix A, Theorem A.4.4; its proof constructs the conditional measures from the internally proved conditional-expectation and Riesz–Markov theorems.

The disintegration theorem supplies conditional probability measures μ_y supported on fibers of π and satisfying

$$\mu = \int_Y \mu_y d\eta(y), \quad \eta = \pi_*\mu.$$

Because π is invariant, translating a conditional measure by any $g \in G$ gives another disintegration over the same factor. Uniqueness of disintegration implies

$$g_*\mu_y = \mu_y$$

for η -almost every y , first for a countable dense subgroup and then, under the measurable action hypotheses, for all g in the required almost-everywhere sense.

Finally, if A is invariant, then $A \in \mathcal{I}_G$, so A is a union of fibers modulo null sets. Conditional expectation gives

$$\mu_y(A) \in \{0, 1\}$$

for almost every y . Hence μ_y is ergodic for almost every y .

Uniqueness follows because the invariant factor determines the conditional measures uniquely almost everywhere. $\square$

Remark 2.11.2. The dynamical argument above uses only the precise disintegration theorem proved internally in Appendix A. That appendix supplies the measure-theoretic construction; the present chapter isolates how the invariant factor turns the resulting conditional measures into ergodic components.

12. Ergodic measures are extreme points

Let $\mathcal{M}_G(X)$ denote the convex set of G -invariant probability measures on a compact metric G -space.

Proposition 2.12.1. *An invariant probability measure μ is ergodic if and only if it is an extreme point of $\mathcal{M}_G(X)$.*

PROOF. Suppose μ is not ergodic. Then there is an invariant measurable set A with $0 < \mu(A) < 1$. Define

$$\mu_1(B) = \frac{\mu(B \cap A)}{\mu(A)}, \quad \mu_2(B) = \frac{\mu(B \cap A^c)}{\mu(A^c)}.$$

Both are invariant probability measures and

$$\mu = \mu(A)\mu_1 + \mu(A^c)\mu_2$$

is a nontrivial convex decomposition. Thus μ is not extreme.

Conversely, suppose

$$\mu = t\nu_1 + (1-t)\nu_2, \quad 0 < t < 1,$$

with distinct invariant probability measures ν_1, ν_2 . Since $\nu_i \ll \mu$, let

$$f_i = \frac{d\nu_i}{d\mu}.$$

Invariance of both ν_i and μ implies that f_i is invariant almost everywhere. If μ is ergodic, each f_i is constant. Since $\nu_i(X) = 1$, that constant is one, so $\nu_i = \mu$, a contradiction. Hence an ergodic measure is extreme. $\square$

13. Generic points for nonergodic measures

If μ is not ergodic, Birkhoff says

$$A_N f(x) \rightarrow \mathbb{E}(f \mid \mathcal{I})(x),$$

which generally depends on x . Under ergodic decomposition, a typical point is generic not for μ itself but for its own ergodic component $\mu_{\pi(x)}$.

This distinction is crucial in measure-classification arguments. One often begins with an invariant measure and immediately reduces to an ergodic component, because generic points are much more rigid when the limiting statistics are represented by a single measure rather than by a nontrivial invariant factor.

14. Unique ergodicity and uniform time averages

Ergodicity controls almost every starting point with respect to one fixed invariant measure. A stronger topological property can control *every* starting point.

Definition 2.14.1. Let $T : X \rightarrow X$ be continuous on a compact metric space. The system is *uniquely ergodic* if it has exactly one T -invariant Borel probability measure.

THEOREM 2.14.2 (Unique ergodicity criterion). *Let $T : X \rightarrow X$ be continuous on a compact metric space, and suppose μ is the unique invariant probability measure. Then for every $f \in C(X)$,*

$$(2.17) \quad \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) \longrightarrow \int_X f d\mu$$

uniformly in $x \in X$. In particular, every point is generic for μ .

PROOF. We first prove pointwise convergence by compactness of empirical measures. Fix $x \in X$ and define

$$\mu_{N,x} = \frac{1}{N} \sum_{n=0}^{N-1} \delta_{T^n x}.$$

Because X is compact, the space of probability measures on X is weak-* compact. Hence every sequence $N_j \rightarrow \infty$ has a subsequence along which $\mu_{N_j,x}$ converges weak-* to some probability measure ν .

The telescoping identity

$$\int f \circ T d\mu_{N,x} - \int f d\mu_{N,x} = \frac{f(T^N x) - f(x)}{N}$$

shows that every such limit ν is T -invariant. By unique ergodicity, $\nu = \mu$. Since every convergent subsequence has the same limit, the whole sequence $\mu_{N,x}$ converges to μ . Thus (2.17) holds for every fixed x .

The remaining assertion is uniformity. Suppose uniform convergence fails for some $f \in C(X)$. Then there are $\varepsilon > 0$, integers $N_j \rightarrow \infty$, and points $x_j \in X$ such that

$$(2.18) \quad \left| \int f d\mu_{N_j,x_j} - \int f d\mu \right| \geq \varepsilon.$$

By compactness of the probability-measure space, after passing to a subsequence

$$\mu_{N_j,x_j} \xrightarrow{*} \nu.$$

The same telescoping identity works even though the starting points vary:

$$\left| \int h \circ T d\mu_{N_j, x_j} - \int h d\mu_{N_j, x_j} \right| \leq \frac{2 \|h\|_\infty}{N_j} \longrightarrow 0.$$

Hence ν is invariant, so $\nu = \mu$. Taking $h = f$ contradicts (2.18). Therefore convergence is uniform. $\square$

Remark 2.14.3. The proof reveals why compactness matters. It guarantees that every sequence of empirical measures has a probability subsequential limit. On a noncompact homogeneous space, mass may escape to infinity; unique invariant measure statements by themselves do not prevent this. Tightness or quantitative nondivergence is then a separate issue.

Proposition 2.14.4 (Uniform convergence characterizes unique ergodicity). *Let $T : X \rightarrow X$ be continuous on a compact metric space. If there is a probability measure μ such that for every $f \in C(X)$, the averages in (2.17) converge uniformly in x to $\int f d\mu$, then μ is the unique invariant probability measure.*

PROOF. Let ν be any invariant probability measure. Integrating the time average against ν gives, for every N ,

$$\int_X \left(\frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) \right) d\nu(x) = \int_X f d\nu.$$

Uniform convergence allows the limit to pass through the integral, yielding

$$\int_X f d\nu = \int_X f d\mu \quad (f \in C(X)).$$

Thus $\nu = \mu$. $\square$

Guiding Idea 2.14.5. There are three distinct levels that should not be conflated:

- (1) *dense orbit*: a topological statement about where one orbit goes;
- (2) *ergodicity*: almost every orbit has the correct statistics for one invariant measure;
- (3) *unique ergodicity on a compact system*: every orbit has the same statistics, uniformly for continuous observables.

Unipotent dynamics exhibits examples at all three levels, so keeping the distinctions explicit is essential.

15. Worked example: empirical measures of an irrational rotation

Let R_α be irrational rotation. Ergodicity and Birkhoff imply that for almost every x ,

$$\frac{1}{N} \sum_{n=0}^{N-1} \delta_{x+n\alpha} \xrightarrow{*} m_{\mathbb{T}}.$$

In fact every x is generic. This stronger statement follows from an explicit Fourier calculation. For $k \neq 0$,

$$\frac{1}{N} \sum_{n=0}^{N-1} e^{2\pi i k(x+n\alpha)} = e^{2\pi i kx} \frac{1 - e^{2\pi i kN\alpha}}{N(1 - e^{2\pi i k\alpha})} \longrightarrow 0.$$

Trigonometric polynomials are dense in $C(\mathbb{T})$, so the convergence holds for every continuous test function and every starting point.

The example separates two levels of result:

- Birkhoff: almost-everywhere genericity from ergodicity;
- explicit harmonic analysis: every-point genericity and a quantitative $O_f(N^{-1})$ -type estimate for individual Fourier modes.

16. Worked example: a periodic homogeneous orbit

Suppose $U = \{u_t\} \cong \mathbb{R}$ acts on a homogeneous space and Ux is periodic with period $L > 0$, so

$$u_L x = x$$

and L is the least positive period. Normalize arclength to obtain

$$\mu_{Ux}(f) = \frac{1}{L} \int_0^L f(u_t x) dt.$$

For every starting point $u_s x$,

$$\frac{1}{T} \int_0^T f(u_t u_s x) dt \longrightarrow \frac{1}{L} \int_0^L f(u_t x) dt.$$

This follows by writing $T = qL + r$ and separating q complete periods from a remainder of length $< L$. Thus every point on a periodic orbit is generic for the orbit measure.

Periodic homogeneous measures are the simplest instances of algebraic invariant measures. Their statistics are exact because the orbit closes after finite flow time.

17. Conceptual summary

The main logical chain is

measure preservation $\implies$ Koopman operator $\implies$ ergodic averages.

The mean theorem identifies the L^2 limit as projection onto invariant functions. The maximal ergodic lemma upgrades this to almost-everywhere convergence in L^1 . Under ergodicity,

$$\frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) \longrightarrow \int f d\mu \quad \text{for almost every } x.$$

This gives the precise meaning of a generic point. Empirical measures of generic points recover the invariant measure, while ergodic decomposition explains what happens when the measure is not indecomposable. These ideas turn orbit segments into measures and measures back into orbit statistics, a passage that is fundamental throughout homogeneous dynamics.

CHAPTER 3

Unitary Representations and the Mautner Phenomenon

1. Why representation theory appears

A measure-preserving action of a group G on (X, μ) produces a unitary representation on $L^2(X, \mu)$. This simple observation converts a nonlinear dynamical question into a linear one:

ergodicity of $G \curvearrowright X$

$$\iff \text{constants are the only } G\text{-fixed vectors in } L^2(X).$$

The Mautner phenomenon exploits the geometry of the ambient group to enlarge the stabilizer of a vector. In the most important model case, invariance under a single upper-unipotent subgroup of $\mathrm{SL}_2(\mathbb{R})$ already forces invariance under the whole group. This is a striking rigidity statement: a one-dimensional subgroup determines a three-dimensional symmetry group at the Hilbert-space level.

2. Continuous unitary representations

Definition 3.2.1. A *continuous unitary representation* of a topological group G on a Hilbert space $\mathcal{H}$ is a homomorphism

$$\pi : G \rightarrow U(\mathcal{H})$$

such that for every $v \in \mathcal{H}$, the orbit map

$$g \mapsto \pi(g)v$$

is norm-continuous.

This is called strong continuity. It is weaker than continuity in the operator norm and is the natural notion for infinite-dimensional representations.

Definition 3.2.2. For $v, w \in \mathcal{H}$, the function

$$\varphi_{v,w}(g) = \langle \pi(g)v, w \rangle$$

is called a *matrix coefficient* of π .

Strong continuity implies that every matrix coefficient is continuous.

Proposition 3.2.3 (The Koopman representation). *Let a locally compact second-countable group G act continuously on a locally compact second-countable space X , preserving a Radon probability measure μ . Define*

$$\pi(g)f(x) = f(g^{-1}x).$$

Then π is a strongly continuous unitary representation on $L^2(X, \mu)$.

PROOF. Unitarity follows from invariance of μ :

$$\|\pi(g)f\|_2^2 = \int_X |f(g^{-1}x)|^2 d\mu(x) = \int_X |f(y)|^2 d\mu(y).$$

For strong continuity, first take $f \in C_c(X)$. If $g \rightarrow e$, continuity of the action implies

$$f(g^{-1}x) \rightarrow f(x)$$

uniformly on a compact set containing the supports for g in a sufficiently small compact neighborhood of e . Dominated convergence gives

$$\|\pi(g)f - f\|_2 \rightarrow 0.$$

Since $C_c(X)$ is dense in $L^2(X, \mu)$ and every $\pi(g)$ is an isometry, an approximation argument extends the conclusion to all $f \in L^2$. $\square$

3. Fixed vectors and ergodicity

Let

$$\mathcal{H}^G = \{v \in \mathcal{H} : \pi(g)v = v \text{ for every } g \in G\}.$$

For the Koopman representation,

$$L^2(X)^G = L^2(\mathcal{I}_G).$$

Thus the action is ergodic precisely when

$$L^2(X)^G = \mathbb{C}\mathbf{1}.$$

It is often cleaner to remove the constants and work on

$$L_0^2(X) = \left\{ f \in L^2(X) : \int_X f d\mu = 0 \right\}.$$

Then ergodicity is equivalent to

$$L_0^2(X)^G = \{0\}.$$

4. Explicit generation by the two unipotent subgroups

The upper and lower unipotent subgroups generate $\mathrm{SL}_2(\mathbb{R})$. Since this elementary fact is used in the invariance arguments below, we verify it before using it.

Write

$$u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \quad v_s = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix}.$$

First observe that

$$(3.6) \quad w = u_1 v_{-1} u_1 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Thus the subgroup $\langle U, V \rangle$ contains a Weyl element. Next, for $a \neq 0$, a direct multiplication gives

$$(3.7) \quad \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} = u_a v_{-a^{-1}} u_a u_{-1} v_1 u_{-1},$$

a product of upper and lower unipotents. One may also obtain the diagonal subgroup from the familiar Gauss factorization; the precise word in (3.7) is less important than the fact that it is explicit and finite.

Now take

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{R}).$$

If $a \neq 0$, Gaussian elimination gives

$$(3.8) \quad g = v_{c/a} \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} u_{b/a}.$$

Indeed the right-hand side equals

$$\begin{pmatrix} a & b \\ c & cb/a + a^{-1} \end{pmatrix},$$

and the lower-right entry is d because $ad - bc = 1$. If $a = 0$, then $c \neq 0$, and multiplication by the Weyl element reduces to the previous case. Hence every element of $\mathrm{SL}_2(\mathbb{R})$ is a finite product of elements of U and V .

Remark 3.4.1. This calculation also clarifies the logic of many stabilizer arguments. It is often enough to prove invariance under a few root subgroups; the group generated by those subgroups can be much larger than their individual dimensions suggest.

5. A basic Mautner contraction lemma

The easiest form of the Mautner argument is a one-line limit calculation, but its logic is worth isolating.

Lemma 3.5.1 (Mautner contraction lemma). *Let $\pi : G \rightarrow U(\mathcal{H})$ be a continuous unitary representation. Suppose $v \in \mathcal{H}$ is fixed by a sequence $g_n \in G$, and suppose*

$$g_n^{-1} h g_n \longrightarrow e$$

for some $h \in G$. Then v is fixed by h .

PROOF. Because $\pi(g_n)v = v$,

$$\pi(h)v - v = \pi(g_n)\pi(g_n^{-1}hg_n)v - \pi(g_n)v.$$

Taking norms and using unitarity,

$$\|\pi(h)v - v\| = \|\pi(g_n^{-1}hg_n)v - v\|.$$

Strong continuity and $g_n^{-1}hg_n \rightarrow e$ make the right-hand side tend to zero. Hence $\pi(h)v = v$. $\square$

Worked Example 3.5.2 (Diagonal invariance forces horospherical invariance). In $\mathrm{SL}_2(\mathbb{R})$, use

$$a_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix}, \quad u_s^+ = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \quad u_s^- = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix}.$$

Then

$$a_t u_s^+ a_{-t} = u_{e^t s}^+, \quad a_{-t} u_s^+ a_t = u_{e^{-t} s}^+ \rightarrow e \quad (t \rightarrow +\infty).$$

To match the contraction lemma exactly, take $g_n = a_{t_n}$ with $t_n \rightarrow +\infty$ and $h = u_s^+$; then

$$g_n^{-1} h g_n = a_{-t_n} u_s^+ a_{t_n} \rightarrow e.$$

Because the vector is fixed by every a_t , it is fixed by every g_n , and hence by u_s^+ . For U^- , use $g_n = a_{-t_n}$, since

$$a_{t_n} u_s^- a_{-t_n} = u_{e^{-t_n} s}^- \rightarrow e.$$

Thus an A -fixed vector is fixed by both U^+ and U^- . These two subgroups generate $\mathrm{SL}_2(\mathbb{R})$, so the vector is $\mathrm{SL}_2(\mathbb{R})$ -fixed.

The preceding argument is often what is called the Mautner phenomenon in its contracting-subgroup form. The unipotent version is subtler because conjugation by a unipotent element does not exponentially contract a transverse subgroup.

6. The rank-one unipotent Mautner phenomenon

Let

$$U = \left\{ u_r = \begin{pmatrix} 1 & r \\ 0 & 1 \end{pmatrix} : r \in \mathbb{R} \right\}, \quad V = \left\{ v_\varepsilon = \begin{pmatrix} 1 & 0 \\ \varepsilon & 1 \end{pmatrix} : \varepsilon \in \mathbb{R} \right\}.$$

We use the diagonal normalization

$$d_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}.$$

This is a_{2t} in the hyperbolic-time normalization, but the present notation makes the elementary matrix calculation shorter.

THEOREM 3.6.1 (Mautner phenomenon for $\mathrm{SL}_2(\mathbb{R})$). *Let $\pi : \mathrm{SL}_2(\mathbb{R}) \rightarrow U(\mathcal{H})$ be a continuous unitary representation. If $v \in \mathcal{H}$ is fixed by U , then v is fixed by all of $\mathrm{SL}_2(\mathbb{R})$.*

PROOF. The proof is elementary but ingenious. Direct conjugation by U does not contract the opposite unipotent subgroup, so the contraction lemma is not enough. Instead we study a scalar matrix coefficient whose U -bi-invariance lets us insert very large upper-unipotent elements for free.

Define

$$p(g) = \langle \pi(g)v, v \rangle.$$

Because v is U -fixed, for all $u, u' \in U$,

$$(3.1) \quad p(ugu') = \langle \pi(u)\pi(g)\pi(u')v, v \rangle = \langle \pi(g)v, \pi(u^{-1})v \rangle = p(g).$$

Thus p is bi- U -invariant.

We also record a small but important norm observation. If

$$p(g) = \|v\|^2,$$

then $\pi(g)v = v$. Indeed,

$$(3.2) \quad \|\pi(g)v - v\|^2 = 2\|v\|^2 - 2\mathrm{Re}\langle \pi(g)v, v \rangle = 0.$$

Now calculate, for real r, s, ε ,

$$(3.3) \quad \begin{pmatrix} 1 & r \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \varepsilon & 1 \end{pmatrix} \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 + r\varepsilon & r + s + rs\varepsilon \\ \varepsilon & 1 + s\varepsilon \end{pmatrix}.$$

Fix $t \in \mathbb{R}$. For $\varepsilon \neq 0$, choose

$$r = \frac{e^t - 1}{\varepsilon}, \quad s = -\frac{r}{1 + r\varepsilon} = -\frac{e^t - 1}{e^t \varepsilon}.$$

Then $1 + r\varepsilon = e^t$, the upper-right entry in (3.3) vanishes, and the lower-right entry is e^{-t} . Hence

$$(3.4) \quad u_r v_\varepsilon u_s = \begin{pmatrix} e^t & 0 \\ \varepsilon & e^{-t} \end{pmatrix}.$$

By bi- U -invariance,

$$(3.5) \quad p(v_\varepsilon) = p\left(\begin{pmatrix} e^t & 0 \\ \varepsilon & e^{-t} \end{pmatrix}\right).$$

Let $\varepsilon \rightarrow 0$. The left side tends to $p(e) = \|v\|^2$, while the right side tends by continuity to

$$p(d_t).$$

Therefore

$$p(d_t) = \|v\|^2$$

for every t . By (3.2), v is fixed by the entire diagonal subgroup $D = \{d_t\}$.

Once diagonal invariance is known, the ordinary contraction lemma supplies the opposite unipotent invariance. Indeed,

$$d_t v_s d_{-t} = v_{e^{-2t}s} \rightarrow e \quad (t \rightarrow +\infty).$$

Since v is fixed by every d_t , the contraction lemma implies that v is fixed by every $v_s \in V$.

Finally, the explicit generation calculation above shows that U and V generate $\mathrm{SL}_2(\mathbb{R})$. Therefore v is $\mathrm{SL}_2(\mathbb{R})$ -invariant. $\square$

Remark 3.6.2. The key idea is not the matrix multiplication itself. It is the following pattern:

$$\begin{aligned} \text{vector invariance} &\implies \text{bi-invariant matrix coefficient} \\ &\implies \text{insert large group elements at no cost} \\ &\implies \text{take a singular limit} \\ &\implies \text{new invariance.} \end{aligned}$$

This strategy is an early prototype of the “additional invariance” mechanisms that occur throughout rigidity theory.

7. Unipotent ergodicity inside an ergodic rank-one action

Corollary 3.7.1. *Suppose $\mathrm{SL}_2(\mathbb{R})$ acts measure preservingly and ergodically on a probability space (X, μ) , with a strongly continuous Koopman representation. Then the upper-unipotent subgroup U acts ergodically on (X, μ) .*

PROOF. Let $f \in L^2(X)$ be U -invariant. The Mautner theorem makes f invariant under all of $\mathrm{SL}_2(\mathbb{R})$. Ergodicity of the full group implies that f is constant almost everywhere. Hence U is ergodic. $\square$

Worked Example 3.7.2 (Horocycle ergodicity from ambient ergodicity). Let $\Gamma < \mathrm{SL}_2(\mathbb{R})$ be a lattice and let

$$X = \Gamma \backslash \mathrm{SL}_2(\mathbb{R})$$

with normalized Haar probability measure. If the right $\mathrm{SL}_2(\mathbb{R})$ -action is regarded as the ambient transitive action, then it is ergodic. The corollary shows that right translation by

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

is ergodic. Thus the horocycle flow is ergodic on every finite-volume quotient $\Gamma \backslash \mathrm{SL}_2(\mathbb{R})$.

The conclusion is much stronger than Poincaré recurrence: almost every horocycle orbit is statistically distributed according to Haar measure.

8. Moore ergodicity: the general principle

For connected noncompact simple Lie groups, a much more general theorem holds.

THEOREM 3.8.1 (Moore ergodicity theorem). *Let G be a connected noncompact simple Lie group with finite center, let $\Gamma < G$ be a lattice, and let $H < G$ be a closed noncompact subgroup. Then the action of H on G/Γ is ergodic with respect to Haar probability measure.*

Remark 3.8.2 (Proof location). The $\mathrm{SL}_2(\mathbb{R})$ unipotent case needed most often as a model has been proved in full above. The general theorem is proved in Volume VI from the Mautner phenomenon and Howe–Moore, after the required semisimple structure and irreducibility hypotheses have been developed. Thus the statement here is a forward reference, not an external proof input.

9. The contracting Mautner envelope

The contraction lemma has a useful group-theoretic formulation. For a subset $S \subset G$, define its *Mautner envelope* to be the closed subgroup generated by all elements h for which there exists a sequence $s_n \in S$ satisfying

$$s_n^{-1} h s_n \rightarrow e,$$

together with the closed subgroup generated by S itself. Any vector fixed by S is fixed by this envelope.

For a diagonal one-parameter subgroup in a semisimple group, the contracting and expanding horospherical subgroups often generate a very large subgroup. This is the mechanism behind many ergodicity theorems for diagonal flows.

Worked Example 3.9.1 (A higher-dimensional diagonal subgroup). In $G = \mathrm{SL}_d(\mathbb{R})$, let

$$a_t = \mathrm{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_d t}), \quad \sum_i \lambda_i = 0.$$

For an elementary unipotent

$$u_{ij}(s) = I + sE_{ij},$$

one has

$$a_{-t}u_{ij}(s)a_t = u_{ij}(e^{(\lambda_j - \lambda_i)t}s).$$

If $\lambda_i > \lambda_j$, the parameter tends to zero as $t \rightarrow +\infty$. Hence every vector invariant under $A = \{a_t\}$ is invariant under all root subgroups U_{ij} contracted in forward time; backward time gives the expanding ones. If these root subgroups generate $\mathrm{SL}_d(\mathbb{R})$, then A -invariance forces full G -invariance.

The example makes the root-space content of the Mautner phenomenon completely concrete: weights of the adjoint action decide which group directions are forced by invariance.

10. Matrix coefficients and weak mixing

For a probability-preserving action, correlations are matrix coefficients of the Koopman representation:

$$\int_X f(g^{-1}x)\overline{h(x)}d\mu(x) = \langle \pi(g)f, h \rangle.$$

On $L_0^2(X)$, decay of all matrix coefficients as $g \rightarrow \infty$ implies mixing of the action.

Definition 3.10.1. A unitary representation π has *vanishing matrix coefficients at infinity* if

$$\langle \pi(g)v, w \rangle \rightarrow 0$$

for every $v, w \in \mathcal{H}$ as g leaves every compact subset of G .

The Howe–Moore theorem asserts this property for unitary representations of connected noncompact simple groups with no invariant vectors, under standard hypotheses. It is substantially deeper than the elementary SL_2 Mautner calculation above. Its dynamical consequence is mixing of many homogeneous actions.

11. Smooth vectors: what they do and do not solve

For a continuous unitary representation of a Lie group, a vector v is *smooth* if $g \mapsto \pi(g)v$ is C^∞ . Smooth vectors form a dense subspace, and the density has a concrete proof. Choose a smooth compactly supported approximate identity ϕ_n on G , normalized by $\int_G \phi_n dm_G = 1$, and set

$$v_n = \int_G \phi_n(g) \pi(g)v dm_G(g).$$

Strong continuity and concentration of ϕ_n near the identity give $v_n \rightarrow v$. After a change of variables, the orbit map of v_n is obtained by translating the smooth kernel ϕ_n ; differentiating that kernel under the integral shows that v_n is smooth. Thus convolution supplies smooth approximants rather than merely asserting their existence.

On smooth vectors one can differentiate the representation to obtain

$$d\pi : \mathfrak{g} \rightarrow \text{End}(\mathcal{H}^\infty).$$

It is tempting to prove every invariance statement by differentiating. There is a trap: an arbitrary invariant vector need not initially be smooth, and naive smoothing by convolution can destroy invariance under a nonnormal subgroup. The matrix-coefficient proof of unipotent Mautner avoids this issue completely. This is one reason the proof is so robust.

12. A rank-one subgroup inside the three-dimensional special linear group

Consider the symmetric-square representation of $\text{SL}_2(\mathbb{R})$ on binary quadratic forms

$$q(X, Y) = aX^2 + bXY + cY^2.$$

In a suitable basis this gives a homomorphism

$$\rho : \text{SL}_2(\mathbb{R}) \rightarrow \text{SL}_3(\mathbb{R})$$

with kernel $\{\pm I\}$. The image is isomorphic to $\text{PSL}_2(\mathbb{R}) \cong \text{SO}(2, 1)^\circ$. The upper-unipotent subgroup maps to a one-parameter unipotent subgroup of $\text{SL}_3(\mathbb{R})$ whose matrices contain quadratic polynomials in the parameter.

If a probability measure on an SL_3 -homogeneous space is invariant and ergodic under this embedded SL_2 , then the image of the upper-unipotent subgroup acts ergodically by the theorem proved above. This observation is one of the clean bridges between rank-one hyperbolic geometry and the homogeneous dynamics of ternary quadratic forms.

13. Why unitarity is essential

The conclusion of the unipotent Mautner theorem is false for arbitrary linear representations. This is not a technicality: the preservation of the Hilbert norm is exactly what converts an extremal matrix coefficient into equality of vectors.

Worked Example 3.13.1 (A fixed vector in a nonunitary representation). Let $\mathrm{SL}_2(\mathbb{R})$ act on $\mathbb{R}^2$ by its defining representation. The vector

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

is fixed by every upper-unipotent element:

$$\begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} e_1 = e_1.$$

But the diagonal subgroup does not fix it:

$$\begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix} e_1 = e^t e_1.$$

Thus upper-unipotent invariance does not imply full SL_2 -invariance in this representation.

What fails in the Mautner proof? In a unitary representation, the equality

$$\langle \pi(g)v, v \rangle = \|v\|^2$$

forces $\pi(g)v = v$ by equality in Cauchy–Schwarz, or equivalently by the norm identity (3.2). In the defining representation above there is no preserved positive-definite inner product for the whole noncompact group, so this argument is unavailable.

This example explains why measure preservation is so useful. The Koopman representation is automatically unitary; the nonlinear dynamics therefore enters a category in which matrix-coefficient limits carry strong geometric information.

14. Invariant vectors, orthogonal projections, and closed generated subgroups

Let $S \subset G$. Denote by

$$\mathcal{H}^S = \{v : \pi(s)v = v \text{ for every } s \in S\}.$$

Because each operator $\pi(s) - I$ is continuous,

$$\mathcal{H}^S = \bigcap_{s \in S} \ker(\pi(s) - I)$$

is a closed subspace. Moreover,

$$(3.9) \quad \mathcal{H}^S = \mathcal{H}^{\overline{\langle S \rangle}},$$

where the bar denotes topological closure in G . One inclusion is immediate. For the other, if a vector is fixed by S , it is fixed by finite products and inverses of elements of S ; strong continuity then extends the invariance to the closure.

Equation (3.9) is elementary but useful. A Mautner argument may initially produce invariance under a collection of individual one-parameter subgroups. What matters is the closed subgroup they generate.

Since $\mathcal{H}^S$ is closed, there is an orthogonal projection

$$P_S : \mathcal{H} \rightarrow \mathcal{H}^S.$$

When S is generated by a single unitary operator, the mean ergodic theorem identifies this projection as a limit of Cesàro averages. Thus the ergodic theorem and the Mautner phenomenon describe two complementary ways of reaching the same invariant subspace: averaging finds the orthogonal projection onto it, while contraction geometry enlarges it.

15. A tensor-product criterion for weak mixing

The representation viewpoint also gives a useful criterion for weak mixing. Let T be an invertible probability-preserving transformation and let U_T be its Koopman operator on $L_0^2(X)$.

Proposition 3.15.1. *The following are equivalent:*

- (1) T is weakly mixing;
- (2) the product transformation $T \times T$ is ergodic on $(X \times X, \mu \otimes \mu)$;
- (3) the tensor representation $U_T \otimes \overline{U_T}$ has no nonzero invariant vector on $L_0^2(X) \otimes \overline{L_0^2(X)}$.

EXPLANATION OF THE REPRESENTATION MECHANISM. For the product transformation, the Koopman representation is the tensor product of the two individual Koopman representations. The constants form a one-dimensional invariant subspace, because each Koopman operator fixes the constant function 1. Ergodicity of the product is therefore equivalent to the absence of any further invariant vector, proving (2) $\iff$ (3).

To connect this with weak mixing, first take bounded mean-zero functions f, g . Their correlation is

$$\langle U_T^n f, g \rangle = \int_X f(T^{-n}x) \overline{g(x)} d\mu(x).$$

The mean ergodic theorem applied to $U_T \otimes \overline{U_T}$ shows that the Cesàro averages of the squared absolute values of such matrix coefficients tend to zero exactly when the tensor product has no invariant vector. By approximation of indicator functions by bounded L^2 functions, this is equivalent to the usual weak-mixing correlation criterion. The passage from squared correlations to averages of absolute correlations uses Cauchy–Schwarz. $\square$

Remark 3.15.2. This criterion is not needed to prove the elementary SL_2 Mautner theorem. Its value is conceptual: mixing properties are statements about decay or averaging of matrix coefficients, whereas Mautner-type rigidity is a statement about their exact limiting geometry near special subgroup directions.

16. A research-facing example: symmetric powers

For $m \geq 1$, let V_m be the space of homogeneous polynomials of degree m in two variables. The group $\mathrm{SL}_2(\mathbb{R})$ acts by change of variables. In the monomial basis

$$X^m, X^{m-1}Y, \dots, Y^m,$$

the upper-unipotent element acts by substituting $X \mapsto X + sY$. Therefore its matrix entries are polynomials in s of degree at most m .

This finite-dimensional representation is not unitary for any positive-definite inner product invariant under the whole noncompact group, so it does not satisfy the Mautner conclusion. Nevertheless, it is the correct algebraic model for the polynomial behavior of many unipotent embeddings. The case $m = 2$ is the three-dimensional representation on binary quadratic forms and realizes the familiar local isomorphism

$$\mathrm{PSL}_2(\mathbb{R}) \simeq \mathrm{SO}(2, 1)^\circ.$$

Higher symmetric powers produce longer Jordan chains and hence higher polynomial degrees. This is one reason higher-dimensional homogeneous systems can exhibit shearing of degree greater than two even when the acting unipotent subgroup itself is one-dimensional.

17. A practical ergodicity checklist

When one wants to prove that a subgroup $U < G$ is ergodic on (X, μ) , the representation-theoretic route can be organized as follows:

- (1) Form the Koopman representation on $L_0^2(X)$.
- (2) Assume $f \in L_0^2(X)$ is U -fixed.
- (3) Use contraction, matrix-coefficient limits, or a suitable rank-one subgroup to enlarge the stabilizer of f .

- (4) Show that the enlarged stabilizer is a group already known to act ergodically.
- (5) Conclude $f = 0$.

This is often more efficient than analyzing invariant measurable sets directly.

18. Conceptual summary

The Mautner principle is an invariance-amplification mechanism. In its simplest form,

$$\text{fixed by } g_n \quad + \quad g_n^{-1} h g_n \rightarrow e \quad \implies \quad \text{fixed by } h.$$

For the upper-unipotent subgroup of $\mathrm{SL}_2(\mathbb{R})$, a bi-invariant matrix coefficient supplies the missing contraction and yields

$$\boxed{U\text{-fixed vector} \implies \mathrm{SL}_2(\mathbb{R})\text{-fixed vector.}}$$

Combined with the Koopman representation, this becomes an ergodicity theorem for unipotent flows. The important conceptual move is from orbit geometry to linear symmetry: instead of following every orbit, one studies which vectors can remain unchanged under the acting subgroup.

CHAPTER 4

Unipotent Flows and Polynomial Divergence

1. The defining geometric feature

The word *unipotent* is algebraic, but its dynamical significance is geometric. Nearby trajectories under a unipotent flow separate at a polynomial rate. This is sharply different from diagonal or partially hyperbolic dynamics, where transverse separations typically grow or shrink exponentially.

Let

$$U = \{u_t : t \in \mathbb{R}\} < G$$

be a one-parameter subgroup of a matrix Lie group. If $u_t = \exp(tX)$ with X nilpotent, then u_t itself is polynomial in t . More importantly, its adjoint action

$$\text{Ad}(u_t) = \exp(t \text{ad } X)$$

is polynomial whenever $\text{ad } X$ is nilpotent. It is this adjoint polynomial that governs the relative motion of nearby points.

Guiding Idea 4.1.1. If $y = \exp(v)x$ is close to x , then applying the same group element u_t to both points gives

$$u_t y = u_t \exp(v) u_t^{-1} u_t x = \exp(\text{Ad}(u_t)v) u_t x.$$

Thus the displacement between $u_t x$ and $u_t y$ is controlled, in local group coordinates, by the polynomial $\text{Ad}(u_t)v$. The rigidity mechanism comes from choosing a time at which this polynomial has grown from “tiny” to “visible” while retaining strong algebraic structure.

2. Unipotent elements and nilpotent generators

Definition 4.2.1. A matrix $g \in \text{GL}_d(\mathbb{R})$ is *unipotent* if every complex eigenvalue of g equals 1. Equivalently,

$$(g - I)^d = 0.$$

A one-parameter subgroup u_t is unipotent if every u_t is unipotent.

Proposition 4.2.2. If $X \in \mathfrak{gl}_d(\mathbb{R})$ is nilpotent, then

$$u_t = e^{tX}$$

is a unipotent one-parameter subgroup and is a matrix-valued polynomial in t .

PROOF. If $X^m = 0$, then the exponential series terminates:

$$e^{tX} = I + tX + \frac{t^2 X^2}{2!} + \cdots + \frac{t^{m-1} X^{m-1}}{(m-1)!}.$$

Thus $u_t - I$ is nilpotent, so u_t is unipotent. The one-parameter law follows because scalar multiples of X commute. $\square$

For abstract Lie groups, one usually calls $u_t = \exp(tX)$ *Ad-unipotent* if $\text{ad } X$ is nilpotent. In semisimple linear groups this agrees with the familiar algebraic notion after passing through a faithful representation.

3. Polynomial adjoint action

Proposition 4.3.1. *Let G be a Lie group and $X \in \mathfrak{g}$ satisfy*

$$(\text{ad } X)^m = 0.$$

Then for $u_t = \exp(tX)$,

$$\text{Ad}(u_t) = e^{t \text{ad } X} = I + t \text{ad } X + \cdots + \frac{t^{m-1}}{(m-1)!} (\text{ad } X)^{m-1}.$$

In particular, for every $v \in \mathfrak{g}$, $t \mapsto \text{Ad}(u_t)v$ is a vector-valued polynomial of degree at most $m-1$.

PROOF. The identity

$$\text{Ad}(\exp(tX)) = \exp(t \text{ad } X)$$

follows by differentiating conjugation, or directly from the uniqueness of one-parameter subgroups in $\text{GL}(\mathfrak{g})$. Nilpotence makes the exponential series finite. $\square$

Remark 4.3.2. The degree bound is uniform over all $v \in \mathfrak{g}$. This finite-degree control is essential. It means that a unipotent flow cannot create arbitrarily complicated relative motion merely by waiting longer.

4. The basic rank-one calculation

Let

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{R}).$$

Then

$$(4.1) \quad u_t g u_{-t} = \begin{pmatrix} a + ct & b + (d-a)t - ct^2 \\ c & d - ct \end{pmatrix}.$$

PROOF. Multiply explicitly:

$$u_t g = \begin{pmatrix} a + tc & b + td \\ c & d \end{pmatrix},$$

and then

$$\begin{pmatrix} a + tc & b + td \\ c & d \end{pmatrix} \begin{pmatrix} 1 & -t \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a + ct & b + td - t(a + ct) \\ c & d - ct \end{pmatrix}.$$

The upper-right entry simplifies to $b + (d - a)t - ct^2$. □

Formula (4.1) should be read dynamically. Suppose g is close to the identity. Then $b, c, d - a$ are small. Under conjugation by u_t :

- the lower-left error c stays constant;
- diagonal errors acquire a linear term ct ;
- the upper-right error acquires a quadratic term $-ct^2$.

Thus a tiny lower-unipotent displacement is sheared into the upper-unipotent direction.

5. A worked shearing scale

Take

$$g_\varepsilon = \begin{pmatrix} 1 & 0 \\ \varepsilon & 1 \end{pmatrix}$$

with $0 < |\varepsilon| \ll 1$. Then

$$(4.2) \quad u_t g_\varepsilon u_{-t} = \begin{pmatrix} 1 + \varepsilon t & -\varepsilon t^2 \\ \varepsilon & 1 - \varepsilon t \end{pmatrix}.$$

Choose

$$T_\varepsilon = |\varepsilon|^{-1/2}.$$

At time $t = sT_\varepsilon$, with s of order one,

$$\varepsilon t^2 = \operatorname{sgn}(\varepsilon) s^2, \quad \varepsilon t = O(|\varepsilon|^{1/2}).$$

Hence

$$u_{sT_\varepsilon} g_\varepsilon u_{-sT_\varepsilon} \longrightarrow \begin{pmatrix} 1 & -\operatorname{sgn}(\varepsilon) s^2 \\ 0 & 1 \end{pmatrix}$$

as $\varepsilon \rightarrow 0$, uniformly for s in compact intervals.

This is the simplest form of polynomial shearing. The initially transverse displacement g_ε is tiny, but after a time of order $|\varepsilon|^{-1/2}$, its dominant visible effect is a displacement *along the original unipotent group*.

Remark 4.5.1. The scale $\varepsilon^{-1/2}$ is not arbitrary. It is chosen by balancing the dominant coefficient ε against the quadratic growth t^2 :

$$|\varepsilon| T^2 \asymp 1.$$

In a degree- k situation, the analogous scale is typically $T \asymp |\varepsilon|^{-1/k}$.

6. Lie-algebra version of the same calculation

Use the standard basis

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

The commutator relations are

$$[E, F] = H, \quad [E, H] = -2E.$$

Therefore

$$(4.3) \quad \text{Ad}(u_t)F = e^{t \text{ad } E} F = F + tH - t^2 E.$$

The quadratic term in (4.3) is exactly the infinitesimal counterpart of the $-\varepsilon t^2$ term in (4.2).

The Lie-algebra calculation is cleaner because it separates the linearized displacement from nonlinear group-coordinate effects. For sufficiently small v ,

$$u_t \exp(v) u_{-t} = \exp(\text{Ad}(u_t)v)$$

exactly, so no first-order approximation is needed in this conjugation identity.

7. Centralizer directions versus shearing directions

Let

$$\mathfrak{c}_X = \ker(\text{ad } X) = \{v \in \mathfrak{g} : [X, v] = 0\}.$$

This is the Lie algebra of the connected centralizer of U .

If $v \in \mathfrak{c}_X$, then

$$\text{Ad}(u_t)v = v$$

for all t . Such a displacement does not shear at all. If $v \notin \mathfrak{c}_X$, then at least one positive power of t appears in $\text{Ad}(u_t)v$.

Definition 4.7.1. For $v \in \mathfrak{g}$, define its *unipotent degree relative to X* by

$$d_X(v) = \max\{j \geq 0 : (\text{ad } X)^j v \neq 0\}.$$

Then

$$\text{Ad}(u_t)v = \sum_{j=0}^{d_X(v)} \frac{t^j}{j!} (\text{ad } X)^j v.$$

The highest coefficient

$$\frac{1}{d_X(v)!}(\operatorname{ad} X)^{d_X(v)}v$$

lies in $\ker(\operatorname{ad} X)$. Indeed, applying one more $\operatorname{ad} X$ gives zero. Thus the leading term of every shearing polynomial lies in the centralizer of the unipotent flow.

Proposition 4.7.2 (Leading term lands in the centralizer). *If $v \neq 0$ and $d = d_X(v)$, then*

$$w = (\operatorname{ad} X)^d v \neq 0, \quad [X, w] = 0.$$

Consequently,

$$t^{-d} \operatorname{Ad}(u_t)v \longrightarrow \frac{w}{d!} \quad (t \rightarrow \infty).$$

PROOF. The first statement is the definition of d , and

$$[X, w] = (\operatorname{ad} X)^{d+1}v = 0.$$

Divide the polynomial expansion by t^d ; all lower-degree terms vanish in the limit. $\square$

This simple observation is one of the algebraic reasons unipotent dynamics can create additional invariance: the dominant long-time displacement naturally points in a direction commuting with the original flow.

8. Polynomial divergence in a homogeneous space

Let $X = G/\Gamma$, and suppose $x \in X$ has an injectivity neighborhood: there is a neighborhood Ω of e such that

$$\Omega \rightarrow X, \quad g \mapsto gx$$

is injective. If $y = \exp(v)x$ with v small, then as long as the relative displacement remains in Ω ,

$$(4.4) \quad u_t y = \exp(\operatorname{Ad}(u_t)v)u_t x.$$

The equation is exact in G ; the injectivity condition is needed only to interpret the group element $\exp(\operatorname{Ad}(u_t)v)$ uniquely from the pair of quotient points.

Warning 4.8.1. On G/Γ , a small distance between two points does not globally determine a unique small group element relating them. Near the thin part of a noncompact quotient, short lattice elements can create several local representatives. Rigidity arguments therefore work on compact sets with a uniform injectivity radius, or explicitly track the possible lattice ambiguity.

9. First-exit times

Fix a norm on $\mathfrak{g}$ and a small threshold $\rho > 0$. For v near zero but not in $\mathfrak{c}_X$, define the first polynomial exit time

$$T(v) = \inf\{t \geq 1 : \|\mathrm{Ad}(u_t)v\| \geq \rho\},$$

provided such a time exists.

Since $\mathrm{Ad}(u_t)v$ is polynomial and nonconstant, its norm tends to infinity along at least one sequence unless cancellations keep the whole polynomial bounded; in finite dimensions a nonconstant vector-valued polynomial is unbounded. Thus $T(v) < \infty$ for sufficiently small $v \notin \mathfrak{c}_X$, and

$$T(v) \rightarrow \infty \quad (v \rightarrow 0).$$

Proposition 4.9.1 (Normalized polynomial compactness). *Let V be finite-dimensional, and let*

$$p_n(t) = \sum_{j=0}^D a_{n,j} t^j$$

be vector-valued polynomials of uniformly bounded degree. Suppose $T_n \rightarrow \infty$,

$$\sup_{0 \leq t \leq T_n} \|p_n(t)\| \leq C,$$

and

$$\max_{0 \leq t \leq T_n} \|p_n(t)\| \geq c > 0.$$

Then, after passing to a subsequence, the rescaled polynomials

$$q_n(s) = p_n(T_n s), \quad 0 \leq s \leq 1,$$

converge uniformly on $[0, 1]$ to a nonzero vector-valued polynomial q of degree at most D .

PROOF. Write

$$q_n(s) = \sum_{j=0}^D b_{n,j} s^j, \quad b_{n,j} = a_{n,j} T_n^j.$$

On the finite-dimensional vector space of degree-at-most- D polynomials, all norms are equivalent. In particular, boundedness of

$$\sup_{s \in [0,1]} \|q_n(s)\|$$

implies a uniform bound on all coefficients $b_{n,j}$. Passing to a subsequence, every coefficient converges. The resulting polynomial q is the uniform limit. The lower bound $\max \|q_n\| \geq c$ ensures $q \neq 0$. $\square$

Remark 4.9.2. This compactness lemma is a workhorse. It turns a sequence of increasingly long orbit-comparison intervals into a fixed polynomial profile on the unit interval. No delicate functional analysis is involved because the degree is uniformly bounded.

10. A coefficient-based choice of scale

For

$$p_v(t) = \text{Ad}(u_t)v = \sum_{j=0}^D a_j(v)t^j,$$

one can choose a scale without referring to the norm of the whole polynomial. Define

$$(4.5) \quad T_0(v) = \min_{j \geq 1: a_j(v) \neq 0} \left(\frac{\rho}{\|a_j(v)\|} \right)^{1/j}.$$

Then every monomial satisfies

$$\|a_j(v)\| T_0(v)^j \leq \rho,$$

and at least one nonconstant monomial has size exactly ρ . Norm equivalence for finite-degree polynomials implies that on $[0, T_0(v)]$, the whole polynomial remains bounded by a constant multiple of $\rho + \|v\|$, while on some fixed proportion of the interval it is bounded away from zero in an appropriate direction.

Formula (4.5) makes explicit why different transverse directions can produce different shearing exponents.

11. Polynomial divergence in dimension three

Let

$$X = E_{12} + E_{23}$$

in $\mathfrak{sl}_3$. Then $X^3 = 0$, and

$$u_t = e^{tX} = I + t(E_{12} + E_{23}) + \frac{t^2}{2}E_{13}.$$

Already the group entries contain a quadratic term.

The adjoint action can have higher degree. For instance, if one starts from a suitable negative-root vector such as E_{31} , successive commutators

$$[X, E_{31}], \quad [X, [X, E_{31}]], \quad \dots$$

move through several root spaces before vanishing. Consequently $\text{Ad}(u_t)E_{31}$ is a polynomial with several nonzero powers of t .

This is not an artifact of coordinates. The degree records the length of the Jordan chain of $\text{ad } X$ through the chosen transverse direction.

12. Finite-dimensional rank-one modules and polynomial degree

A useful way to organize polynomial divergence is through $\mathfrak{sl}_2$ -modules. Suppose a copy of $\mathfrak{sl}_2$ with basis E, H, F acts on a finite-dimensional vector space V . On an irreducible module of highest weight n , one can choose a basis

$$v_0, \dots, v_n$$

so that E acts by moving toward the highest-weight vector and satisfies

$$E^{n+1} = 0.$$

Therefore

$$e^{tE}v$$

is polynomial of degree at most n for every $v \in V$.

For the adjoint representation of $\mathfrak{sl}_2$ itself, the module has highest weight 2, explaining the quadratic shearing in (4.3). In larger ambient Lie algebras, the restriction of the adjoint representation to an embedded $\mathfrak{sl}_2$ can contain higher-weight summands, producing higher-degree polynomial divergence.

Remark 4.12.1. The full classification of finite-dimensional $\mathfrak{sl}_2$ -modules is standard algebra and is not required for the basic polynomiality theorem; nilpotence of $\text{ad } X$ already gives that. The representation-theoretic language becomes useful when one wants precise degree bounds and canonical leading directions.

13. Contrast with diagonal divergence

For the diagonal subgroup

$$a_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix},$$

we have

$$\text{Ad}(a_t)E = e^t E, \quad \text{Ad}(a_t)F = e^{-t} F.$$

Thus nearby points separated in the E -direction diverge exponentially in forward time, while those separated in the F -direction contract exponentially.

For the unipotent flow $u_t = e^{tE}$,

$$\text{Ad}(u_t)F = F + tH - t^2 E.$$

There is no permanent exponential stable/unstable splitting. Instead, a transverse direction is gradually sheared into a centralizing direction.

This difference has major consequences. Exponential divergence can destroy long orbit shadowing extremely quickly. Polynomial divergence leaves a much longer window in which two nearby orbits remain comparable, and that window can be exploited together with recurrence and genericity.

14. A metric formulation

Let d_X be a metric on a compact subset $K \subset G/\Gamma$ induced from a right-invariant Riemannian metric on G , with injectivity radius bounded below on K . There exist constants $c_1, c_2 > 0$ and a neighborhood $B \subset \mathfrak{g}$ of zero such that whenever $x \in K$, $v \in B$, and $\exp(v)x \in K$,

$$(4.6) \quad c_1 \|v\| \leq d_X(x, \exp(v)x) \leq c_2 \|v\|.$$

Therefore, while both trajectories stay in the controlled local chart,

$$d_X(u_t x, u_t \exp(v)x) \asymp \|\mathrm{Ad}(u_t)v\|.$$

The right-hand side is polynomial in t up to equivalence of norms and the local exponential chart.

15. Quasi-regular polynomial maps: a prototype

Suppose $v_n \rightarrow 0$ and choose scales $T_n \rightarrow \infty$ so that

$$q_n(s) = \mathrm{Ad}(u_{T_n s})v_n$$

converges to a nonzero polynomial $q(s)$. Then formally

$$(4.7) \quad u_{T_n s} \exp(v_n) u_{-T_n s} = \exp(q_n(s)) \longrightarrow \exp(q(s)).$$

The limit

$$\phi(s) = \exp(q(s))$$

is generally not a homomorphism in s . It is a polynomially generated map in the group, encoding the limiting relative displacement of nearby unipotent trajectories.

Maps of this type are often called quasi-regular or polynomial limit maps in rigidity arguments. Their importance is that recurrence can turn the geometric relation (4.7) into invariance of a measure under elements in the image of ϕ .

16. A local shearing theorem in exponential coordinates

The exact identity

$$u_t \exp(v) u_{-t} = \exp(\mathrm{Ad}(u_t)v)$$

already contains the relative motion in the group. To use it on a quotient, however, one repeatedly needs a uniform statement saying that Lie-algebra size and quotient distance are comparable while the orbit remains in a controlled region. We isolate that statement.

Proposition 4.16.1 (Uniform local comparison on a compact set). *Let $X = G/\Gamma$, let $K \subset X$ be compact, and fix a norm on $\mathfrak{g}$. There exist $r > 0$ and constants $c, C > 0$ such that for every $x \in K$ and every $v \in \mathfrak{g}$ with $\|v\| < r$,*

$$(4.8) \quad c\|v\| \leq d_X(x, \exp(v)x) \leq C\|v\|,$$

and the element $\exp(v)$ is the unique element in $\exp(B_r(0))$ carrying x to $\exp(v)x$.

PROOF. The quotient map $G \rightarrow G/\Gamma$ is a local diffeomorphism because Γ is discrete. For each $x = g\Gamma$, there is a radius $r_x > 0$ on which

$$v \mapsto \exp(v)x$$

is injective and has nonsingular differential at the origin. By the inverse function theorem it is bi-Lipschitz, after shrinking the radius, between a small Euclidean ball in $\mathfrak{g}$ and its image in the quotient.

The issue is uniformity in x . For each $x = g\Gamma$, discreteness of $g\Gamma g^{-1}$ gives a neighborhood of the identity containing no nontrivial stabilizer element. On a compact set K , this separation can be chosen uniformly: otherwise there would be $x_n = g_n\Gamma \in K$ and nontrivial $\gamma_n \in \Gamma$ with $g_n\gamma_n g_n^{-1} \rightarrow e$; after passing to a convergent subsequence in a local section over K , discreteness of the limiting conjugate lattice gives a contradiction. Cover K by finitely many quotient charts and take the minimum of their radii and bi-Lipschitz constants. This gives (4.8) uniformly on K . $\square$

Suppose now that $x_n, y_n \in K$ and

$$y_n = \exp(v_n)x_n, \quad v_n \rightarrow 0.$$

As long as both $u_t x_n$ and $u_t y_n$ lie in K and $\|\text{Ad}(u_t)v_n\| < r$, the quotient distance is comparable to

$$\|\text{Ad}(u_t)v_n\|.$$

Thus a bounded-degree polynomial in the Lie algebra literally measures the separation of the two quotient trajectories on every common return to the compact set.

17. Choosing the first visible transverse coefficient

A common source of confusion is the phrase “choose the time when the displacement becomes order one.” If several polynomial coefficients are present, one must specify which one becomes visible first.

Let $L < G$ be a subgroup whose directions are already understood, with Lie algebra $\mathfrak{l}$, and let

$$\pi : \mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{l}$$

be the quotient map. For a small displacement v , consider

$$(4.9) \quad P_v(t) = \pi(\text{Ad}(u_t)v) = \sum_{j=0}^D t^j b_j(v).$$

If $P_v \neq 0$, define

$$(4.10) \quad T(v) = \min_{j: b_j(v) \neq 0} \left(\frac{\rho}{\|b_j(v)\|} \right)^{1/j},$$

where $j = 0$ is omitted and ρ is a small fixed scale. Then every nonconstant monomial in the quotient polynomial has size at most ρ at time $T(v)$, and at least one has size exactly ρ .

The advantage of working in $\mathfrak{g}/\mathfrak{l}$ is conceptual. A large coefficient lying entirely in a direction already contained in $\mathfrak{l}$ does not represent new divergence. Formula (4.10) selects the first scale at which a genuinely transverse component becomes visible.

Lemma 4.17.1 (Nontrivial normalized quotient limit). *Let $v_n \rightarrow 0$ and suppose $P_{v_n} \neq 0$. Choose $T_n = T(v_n)$ by (4.10). If $T_n \rightarrow \infty$, then after passing to a subsequence,*

$$s \mapsto P_{v_n}(T_n s)$$

converges uniformly on $[0, 1]$ to a nonzero polynomial with values in $\mathfrak{g}/\mathfrak{l}$.

PROOF. The coefficient of s^j in the rescaled polynomial is

$$T_n^j b_j(v_n).$$

By construction every such coefficient has norm at most ρ , and at least one nonconstant coefficient has norm ρ . Since there are only finitely many coefficients and the target space is finite-dimensional, coefficientwise compactness gives a convergent subsequence. At least one limiting coefficient has norm ρ , so the limit is nonzero. Coefficient convergence implies uniform convergence on $[0, 1]$. $\square$

This lemma is the precise finite-dimensional compactness behind many quasi-regular-map constructions.

18. A fully explicit degree-four calculation in dimension three

The phrase “higher-degree shearing” is easier to understand after seeing one example all the way through. Let

$$X = E_{12} + E_{23} \in \mathfrak{sl}_3(\mathbb{R}), \quad u_t = \exp(tX).$$

Take the transverse vector E_{31} . Using

$$[E_{ij}, E_{kl}] = \delta_{jk} E_{il} - \delta_{li} E_{kj},$$

we obtain successively

$$\begin{aligned}\mathrm{ad} X(E_{31}) &= E_{21} - E_{32}, \\ (\mathrm{ad} X)^2(E_{31}) &= E_{11} - 2E_{22} + E_{33}, \\ (\mathrm{ad} X)^3(E_{31}) &= 3(E_{23} - E_{12}), \\ (\mathrm{ad} X)^4(E_{31}) &= 6E_{13}, \quad (\mathrm{ad} X)^5(E_{31}) = 0.\end{aligned}$$

Therefore

$$\begin{aligned}\mathrm{Ad}(u_t)E_{31} &= E_{31} + t(E_{21} - E_{32}) \\ &\quad + \frac{t^2}{2}(E_{11} - 2E_{22} + E_{33}) \\ &\quad + \frac{t^3}{2}(E_{23} - E_{12}) + \frac{t^4}{4}E_{13}.\end{aligned}\tag{4.11}$$

The leading direction E_{13} commutes with X , as predicted by the general leading-term proposition.

If the initial perturbation is εE_{31} , the first scale at which the quartic term becomes order one is

$$T_\varepsilon \asymp |\varepsilon|^{-1/4}.$$

At that scale the linear, quadratic, and cubic terms all tend to zero after multiplication by ε , while the quartic term survives. Thus an extremely small displacement in the lowest root direction is eventually sheared into the highest root direction.

Guiding Idea 4.18.1. The degree is not an arbitrary analytic complication. It records how many commutators with the unipotent generator are required before the transverse direction reaches the centralizer. Root diagrams and $\mathfrak{sl}_2$ -module decompositions are efficient ways to read this degree algebraically.

19. A nilpotent example: the Heisenberg group

Polynomial shearing is not restricted to semisimple groups. Let N be the three-dimensional Heisenberg group with Lie algebra basis X, Y, Z satisfying

$$[X, Y] = Z, \quad [X, Z] = [Y, Z] = 0.$$

For the one-parameter subgroup $u_t = \exp(tX)$,

$$\mathrm{Ad}(u_t)Y = Y + tZ, \quad \mathrm{Ad}(u_t)Z = Z.\tag{4.12}$$

Thus nearby points separated in the Y -direction shear linearly into the central Z -direction.

On a compact Heisenberg nilmanifold N/Λ , this phenomenon has a concrete coordinate interpretation. Writing the group law in coordinates as

$$(x, y, z)(x', y', z') = (x + x', y + y', z + z' + xy'),$$

translation in the x -direction converts a small difference in y into a linearly growing difference in the central z -coordinate. This is a nilpotent analogue of the quadratic SL_2 calculation.

The example is useful for comparison. Polynomial divergence alone does not imply Ratner-type classification: nilsystems have their own algebraic structure, and time changes or extensions can alter rigidity dramatically. What persists is the basic mechanism that commutators turn transverse errors into controlled polynomial drifts.

20. Polynomiality in arbitrary finite-dimensional representations

Let $\rho : G \rightarrow \mathrm{GL}(V)$ be a finite-dimensional representation and let $u_t = \exp(tX)$. If the differential $d\rho(X)$ is nilpotent, then

$$\rho(u_t) = \exp(t d\rho(X))$$

is a polynomial matrix in t . Consequently, for every $v \in V$,

$$t \longmapsto \rho(u_t)v$$

is a bounded-degree vector polynomial.

This observation matters because the representation used in a rigidity proof need not be the adjoint representation. Exterior powers encode rational subspaces and lattice covolumes; tensor representations encode invariant forms; projective representations encode subgroup orbits. Once the unipotent generator acts nilpotently in the chosen representation, the same polynomial compactness and sublevel-set arguments become available.

Worked Example 4.20.1 (Exterior square of a three-dimensional shear). Let

$$u_t = I + tE_{12} \in \mathrm{SL}_3(\mathbb{R}).$$

Then

$$u_t(e_2) = e_2 + te_1, \quad u_t(e_3) = e_3.$$

Hence

$$(\wedge^2 u_t)(e_2 \wedge e_3) = e_2 \wedge e_3 + t e_1 \wedge e_3.$$

A rank-two sublattice whose spanning plane is encoded by $e_2 \wedge e_3$ therefore changes linearly in exterior-power coordinates. Cusp geometry and orbit shearing are being described by the same polynomial representation principle.

21. Worked example: affine lattices

Consider

$$G = \mathrm{ASL}_d(\mathbb{R}) = \mathrm{SL}_d(\mathbb{R}) \ltimes \mathbb{R}^d.$$

Its Lie algebra is

$$\mathfrak{sl}_d(\mathbb{R}) \ltimes \mathbb{R}^d.$$

If $u_t \in \mathrm{SL}_d(\mathbb{R})$ is unipotent, it acts on the translation part by a polynomial matrix u_t . Thus a small translational displacement $w \in \mathbb{R}^d$ evolves as

$$w \mapsto u_t w,$$

a vector-valued polynomial in t . Simultaneously, a small linear displacement $Y \in \mathfrak{sl}_d$ evolves by $\mathrm{Ad}(u_t)Y$. Hence affine homogeneous spaces exhibit two intertwined polynomial shearing mechanisms.

This setting occurs naturally in problems about inhomogeneous Diophantine approximation, affine lattice counting, and Lorentz-gas type models.

22. Worked example: quadratic-form stabilizers

Let $H \cong \mathrm{SO}(2, 1)^\circ$ be embedded in $\mathrm{SL}_3(\mathbb{R})$ as the stabilizer of an indefinite ternary quadratic form. Under the identification $H \cong \mathrm{PSL}_2(\mathbb{R})$, an upper-unipotent subgroup of SL_2 maps to a unipotent subgroup of H . In the three-dimensional representation, its entries are polynomial of degree at most two.

If H acts on $\mathrm{SL}_3(\mathbb{R})/\mathrm{SL}_3(\mathbb{Z})$, nearby lattices are sheared by this polynomial adjoint action. The mechanism ties the dynamics of unipotent orbits directly to arithmetic questions about values of quadratic forms: algebraic stabilizers create the flow, and polynomial divergence controls how nearby arithmetic configurations separate.

23. What polynomial divergence does not say

Polynomial divergence by itself does not imply rigidity. Several additional ingredients are needed in any application:

- recurrence or nondivergence, so the relevant orbit times remain in controlled regions;
- genericity, so orbit averages encode the invariant measure;
- a way to find nearby points that are not already related by the known stabilizer;
- an argument converting limiting relative displacement into genuine additional invariance.

The polynomial calculation supplies the algebraic shape of the comparison. Measure theory supplies the recurrent points. Rigidity comes from making the two interact.

24. Conceptual summary

Let $u_t = \exp(tX)$ with $\text{ad } X$ nilpotent. Then

$$\text{Ad}(u_t)v = \sum_{j=0}^D \frac{t^j}{j!} (\text{ad } X)^j v.$$

For nearby points $y = \exp(v)x$,

$$u_t y = \exp(\text{Ad}(u_t)v) u_t x.$$

The highest nonzero coefficient of this polynomial lies in the centralizer of U . After choosing a scale on which the tiny displacement becomes order one, subsequences converge to a nontrivial polynomial profile. This is the geometric source of unipotent shearing and one of the central reasons unipotent dynamics is rigid rather than chaotic in an arbitrary way.

CHAPTER 5

Horocycle Dynamics as a Model System

1. The homogeneous realization

Let

$$G = \mathrm{PSL}_2(\mathbb{R}), \quad X = \Gamma \backslash G,$$

where $\Gamma < G$ is a lattice. We use right actions on $\Gamma \backslash G$ in this chapter. Set

$$a_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix}, \quad u_s^+ = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \quad u_s^- = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix}.$$

The matrices are understood modulo $\{\pm I\}$.

The right actions

$$g_t(x) = xa_t, \quad h_s^+(x) = xu_s^+, \quad h_s^-(x) = xu_s^-$$

are respectively the geodesic, stable-horocycle, and unstable-horocycle flows under the standard identification

$$\Gamma \backslash G \cong T^1(\Gamma \backslash \mathbb{H}).$$

The normalization is chosen so that t is hyperbolic arclength along the geodesic flow.

2. The commutation laws

The fundamental matrix identity is

$$a_t u_s^+ a_{-t} = u_{e^t s}^+.$$

Because our flows act on the right, this becomes

$$(5.1) \quad g_t \circ h_s^+ = h_{e^{-t}s}^+ \circ g_t.$$

Indeed,

$$(xu_s^+)a_t = xa_t(a_{-t}u_s^+a_t) = xa_tu_{e^{-t}s}^+.$$

Similarly,

$$(5.2) \quad g_t \circ h_s^- = h_{e^t s}^- \circ g_t.$$

Warning 5.2.1. The sign in the exponent depends on whether one uses left or right actions. Formula (5.1) is the correct right-action identity on $\Gamma \backslash G$. Mixing conventions is a common source of errors in horocycle calculations.

3. Geometric meaning in the upper half-plane

Identify G/K with $\mathbb{H}$, where $K = \text{PSO}(2)$, using the action of Möbius transformations. The upper-unipotent element

$$u_s^+ : z \mapsto z + s$$

acts by horizontal translation. The orbits of U^+ in $\mathbb{H}$ are horizontal Euclidean lines

$$\text{Im } z = y,$$

which are horocycles based at the boundary point ∞ .

For a general unit tangent vector, right multiplication by U^+ moves along the stable horocycle with the same forward endpoint. The geodesic flow contracts this stable horocycle in forward time, exactly as (5.1) states.

4. Boundary coordinates

An oriented geodesic in $\mathbb{H}$ is determined by an ordered pair of distinct boundary points

$$(\xi^-, \xi^+) \in (\partial\mathbb{H} \times \partial\mathbb{H}) \setminus \Delta.$$

A unit tangent vector requires one additional real parameter locating the point along the geodesic. Thus one obtains Hopf coordinates

$$T^1\mathbb{H} \cong ((\partial\mathbb{H})^2 \setminus \Delta) \times \mathbb{R}.$$

In these coordinates the geodesic flow simply translates the $\mathbb{R}$ -coordinate.

The stable horocycle flow keeps ξ^+ fixed and changes ξ^- together with the height parameter so as to remain on the same horosphere based at ξ^+ . The unstable flow similarly keeps ξ^- fixed.

This boundary picture makes the asymmetry of stable and unstable directions geometrically visible.

5. Periodic horocycles and stabilizers

A right U -orbit through $x = \Gamma g$ is

$$xU = \{\Gamma gu : u \in U\}.$$

Its stabilizer is

$$U_x = U \cap g^{-1}\Gamma g.$$

Because $U \cong (\mathbb{R}, +)$, every discrete subgroup of U is either trivial or of the form $s_0\mathbb{Z}$.

Proposition 5.5.1 (Periodic-orbit criterion). *The orbit xU is periodic if and only if*

$$U \cap g^{-1}\Gamma g$$

is a nontrivial lattice in U . In that case

$$xU \cong U/U_x \cong \mathbb{R}/(s_0\mathbb{Z})$$

and the orbit carries a unique normalized U -invariant probability measure.

PROOF. The orbit map

$$U \rightarrow xU, \quad u \mapsto xu$$

has kernel U_x , so it factors through a bijection

$$U_x \backslash U \rightarrow xU.$$

Since U_x is discrete, this is a local diffeomorphism and identifies the intrinsic homogeneous topology on the orbit with the quotient topology. The quotient $U_x \backslash U$ is compact exactly when U_x is a lattice in U . Since a nontrivial discrete subgroup of $\mathbb{R}$ is $s_0\mathbb{Z}$, compactness is equivalent to periodicity. $\square$

6. Parabolic fixed points

Let $x = \Gamma g$. The conjugate group

$$gUg^{-1}$$

fixes the boundary point

$$\xi = g\infty.$$

If xU is periodic, there is a nontrivial

$$\gamma \in \Gamma \cap gUg^{-1}.$$

Such γ is parabolic and fixes ξ . Conversely, if ξ is fixed by a nontrivial parabolic element of Γ , then after conjugating that element into U , one obtains a nontrivial stabilizer of the corresponding horocycle orbit.

Thus periodic horocycles are exactly those based at parabolic fixed points of the lattice.

Worked Example 5.6.1 (The modular cusp). Take $\Gamma = \mathrm{PSL}_2(\mathbb{Z})$. The element

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

fixes ∞ . Therefore the horocycles based at ∞ project to periodic horocycles in the cusp of the modular surface. At height y , the horizontal segment

$$0 \leq x < 1$$

closes up under $x \mapsto x + 1$.

7. How the geodesic flow changes a periodic horocycle

By (5.1),

$$g_t h_s^+ = h_{e^{-t}s}^+ g_t.$$

Hence applying positive geodesic time contracts the horocycle parameter. Geometrically, on the upper half-plane, a_t sends i to $e^t i$, moving upward into the cusp; a Euclidean horizontal displacement of fixed size corresponds to a smaller hyperbolic displacement at larger height.

For a periodic horocycle with parameter period L , the translated horocycle $g_t(xU)$ has period

$$e^{-t}L$$

with respect to the right U -parameter if one compares the same orbit after moving the base point by a_t . One should distinguish carefully between geometric length and the algebraic parameter because the invariant metric on the quotient and the chosen subgroup parameter need not coincide under every normalization.

8. Ergodicity of the horocycle flow

THEOREM 5.8.1. *Let $\Gamma < \mathrm{PSL}_2(\mathbb{R})$ be a lattice. The horocycle flow h_s^+ on $\Gamma \backslash \mathrm{PSL}_2(\mathbb{R})$ is ergodic with respect to normalized Haar measure.*

PROOF. The right action of $G = \mathrm{PSL}_2(\mathbb{R})$ on $\Gamma \backslash G$ is transitive and preserves Haar probability measure, hence is ergodic. Let $f \in L^2$ be invariant under U^+ . The unipotent Mautner theorem for $\mathrm{SL}_2(\mathbb{R})$, which descends through the central quotient to $\mathrm{PSL}_2(\mathbb{R})$, implies that f is invariant under the full G -action. Therefore f is constant almost everywhere. The invariant-function criterion proves ergodicity. $\square$

Corollary 5.8.2. *For every $f \in L^1(\Gamma \backslash G)$, for Haar-almost every x ,*

$$\frac{1}{T} \int_0^T f(xu_s^+) ds \longrightarrow \int_X f dm_X.$$

PROOF. Apply the pointwise ergodic theorem to the ergodic horocycle action proved above. The limiting conditional expectation is therefore the constant $\int_X f dm_X$ almost everywhere. $\square$

This is an almost-everywhere equidistribution theorem. It does not classify exceptional starting points. Periodic horocycles are obvious exceptions to Haar equidistribution.

9. Topological behavior: Hedlund's theorem

For cocompact lattices there are no parabolic elements and hence no periodic horocycles. Hedlund proved the much stronger topological statement that every horocycle is dense.

Remark 5.9.1 (Hedlund's compact-orbit theorem). Hedlund's classical topological theorem says that if $\Gamma < \mathrm{PSL}_2(\mathbb{R})$ is cocompact, then every U^+ -orbit in $\Gamma \backslash \mathrm{PSL}_2(\mathbb{R})$ is dense. We record it only for comparison: its geometric proof controls every orbit and is not part of the proof chain developed in this volume. No later argument in the series uses this contextual statement as an input.

For finite-area noncompact surfaces, the corresponding dichotomy is that a horocycle orbit is either periodic (when based at a parabolic fixed point) or dense. Again this is a topological theorem about every orbit, stronger than ergodicity.

10. Why topological and measure-theoretic statements differ

Consider three assertions about a horocycle orbit:

- (1) it is recurrent;
- (2) it is dense;
- (3) it is equidistributed for Haar measure.

These are genuinely different.

A periodic horocycle is recurrent but not dense. A dense orbit need not, in an arbitrary dynamical system, be equidistributed. Ergodicity gives equidistribution only for almost every starting point, not every dense orbit. Special properties of the horocycle flow upgrade the conclusion, but that upgrade is a theorem, not a formal consequence of density.

11. Polynomial shearing in the horocycle model

Let

$$v_\varepsilon = \begin{pmatrix} 1 & 0 \\ \varepsilon & 1 \end{pmatrix}.$$

For right horocycle flow, compare

$$xu_t \quad \text{and} \quad xv_\varepsilon u_t.$$

To express the second point relative to the first, write

$$xv_\varepsilon u_t = xu_t(u_{-t}v_\varepsilon u_t).$$

Now

$$(5.3) \quad u_{-t}v_\varepsilon u_t = \begin{pmatrix} 1 - \varepsilon t & -\varepsilon t^2 \\ \varepsilon & 1 + \varepsilon t \end{pmatrix}.$$

At time $t \asymp |\varepsilon|^{-1/2}$, the dominant visible displacement is in the upper-unipotent direction. Thus the shearing picture survives unchanged under the right-action convention, but one must conjugate in the correct order.

12. Stable leaves and asymptotic trajectories

Equation (5.1) gives

$$d(g_t h_s^+ x, g_t x) \rightarrow 0 \quad (t \rightarrow +\infty)$$

for a suitable metric on a compact quotient. Hence stable horocycles are precisely strong stable leaves of the geodesic flow.

Similarly, unstable horocycles satisfy

$$d(g_t h_s^- x, g_t x) \rightarrow 0 \quad (t \rightarrow -\infty).$$

This leads to the classical Hopf philosophy: if a function is invariant under geodesic time averages, then points on the same stable or unstable leaf have the same asymptotic averages. Accessibility by alternating stable and unstable leaves can force the limiting function to be constant, proving ergodicity of the geodesic flow. In the homogeneous setting, the Mautner phenomenon is the representation-theoretic shadow of the same contraction geometry.

13. A direct stable-leaf averaging estimate

Let X be compact and let f be uniformly continuous. If $y = h_s^+ x$, then

$$g_t y = h_{e^{-t}s}^+ g_t x.$$

Uniform continuity implies

$$|f(g_t y) - f(g_t x)| \rightarrow 0 \quad (t \rightarrow +\infty).$$

Therefore

$$\frac{1}{T} \int_0^T (f(g_t y) - f(g_t x)) dt \rightarrow 0.$$

So forward geodesic time averages agree along stable horocycles. Backward averages agree along unstable horocycles.

This elementary estimate is the dynamical content behind many Hopf arguments.

14. Closed horocycles as homogeneous measures

If xU is periodic with period L , define

$$\mu_{xU}(f) = \frac{1}{L} \int_0^L f(xu_s) ds.$$

Then μ_{xU} is U -invariant and ergodic. It is singular with respect to Haar measure on the three-dimensional space X , being supported on a one-dimensional closed orbit.

Thus even in the simplest unipotent system there are at least two geometrically different kinds of ergodic invariant measure:

- Haar measure on the whole finite-volume quotient;
- normalized Haar measure on a closed horocycle.

The remarkable feature of unipotent dynamics is that invariant measures often remain homogeneous rather than becoming fractal or arbitrary.

15. Why the unit tangent bundle is a group quotient

The identification

$$T^1\mathbb{H} \cong \mathrm{PSL}_2(\mathbb{R})$$

is often quoted so quickly that the three basic flows can seem like a convention rather than geometry. Let us make the identification explicit.

Let v_0 be the upward-pointing unit tangent vector based at $i \in \mathbb{H}$. Every orientation-preserving hyperbolic isometry sends unit tangent vectors to unit tangent vectors, so there is a map

$$(5.4) \quad \Phi : G \rightarrow T^1\mathbb{H}, \quad \Phi(g) = Dg(v_0).$$

The action of G on oriented orthonormal frames is simply transitive. Transitivity follows because an isometry can send any point to any other point and then a rotation in the stabilizer can match the tangent direction. The stabilizer of v_0 is trivial in $\mathrm{PSL}_2(\mathbb{R})$: an orientation-preserving isometry fixing i and its tangent direction has derivative equal to the identity at i , hence is the identity. Therefore Φ is a diffeomorphism.

If $\Gamma < G$ acts on the left, then

$$(5.5) \quad \Gamma \backslash G \cong T^1(\Gamma \backslash \mathbb{H}).$$

The reason the dynamical flows act on the *right* is now transparent: a point of $\Gamma \backslash G$ is an oriented frame modulo the surface group, and multiplying the frame by a fixed element of G changes its internal position relative to the reference frame without changing the choice of lift by Γ .

The diagonal subgroup. At the reference vector,

$$a_t i = e^t i.$$

The vertical hyperbolic distance from i to $e^t i$ is

$$\int_1^{e^t} \frac{dy}{y} = t.$$

Thus xa_t moves a unit tangent vector a signed distance t along its geodesic. This verifies that the chosen half-exponent normalization makes t genuine arclength.

The upper-unipotent subgroup. At the reference frame,

$$a_t u_s^+ i = e^t (s + i).$$

As s varies this is the horizontal horocycle at height e^t . Its hyperbolic line element is

$$(5.6) \quad \frac{d(e^t s)}{e^t} = ds.$$

Thus the right U^+ -parameter s is exactly hyperbolic arclength along the stable horocycle in this normalization. In particular, if a closed U^+ -orbit has algebraic period L , its geometric horocycle length is also L .

Equation (5.6) removes a possible ambiguity in the earlier period-rescaling discussion: under the canonical homogeneous metric attached to this identification, the period change $L \mapsto e^{-t} L$ is simultaneously a change of algebraic parameter and geometric length.

16. Stable and unstable endpoints in boundary coordinates

Write a unit tangent vector in Hopf coordinates as

$$(\xi^-, \xi^+, r),$$

where $\xi^- \neq \xi^+$ are the backward and forward endpoints of the oriented geodesic and r is a signed Busemann coordinate along it. Then the geodesic flow is

$$(\xi^-, \xi^+, r) \mapsto (\xi^-, \xi^+, r + t).$$

The stable horocycle through a vector consists of vectors with the same forward endpoint and the same forward Busemann level. The unstable horocycle is defined analogously using the backward endpoint. This explains both contraction laws without matrices: two geodesics aiming at the same forward endpoint and lying on the same horosphere become exponentially close as they move forward.

The matrix and boundary pictures encode the same fact. The matrix identity

$$a_{-t}u_s^+a_t = u_{e^{-t}s}^+$$

says that the stable coordinate is multiplied by e^{-t} ; the Busemann picture says that the transverse separation on a common horosphere decays exponentially under forward geodesic motion.

17. A geometric proof of geodesic-flow ergodicity

The contraction geometry can itself prove ergodicity of the geodesic flow. This is the classical Hopf argument. We give it because it illustrates how pointwise ergodic theory, stable/unstable foliations, and local product structure fit together.

For clarity assume first that $X = \Gamma \backslash G$ is compact. Let m be normalized Haar measure and let $f \in C(X)$. Define forward and backward averages when they exist:

$$F^+(x) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T f(xa_t) dt,$$

$$F^-(x) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T f(xa_{-t}) dt.$$

Birkhoff's theorem gives both limits for almost every x . Moreover, both are versions of the conditional expectation of f on the A -invariant sigma-algebra, so

$$(5.7) \quad F^+ = F^- \quad \text{almost everywhere.}$$

Denote the common almost-everywhere function by F .

Lemma 5.17.1 (Constancy along stable leaves). *If $F^+(x)$ and $F^+(xu_s^+)$ both exist, then*

$$F^+(xu_s^+) = F^+(x).$$

Similarly, backward averages agree along U^- -leaves.

PROOF. From the right-action commutation relation,

$$(xu_s^+)a_t = xa_tu_{e^{-t}s}^+.$$

Compactness makes f uniformly continuous, and $u_{e^{-t}s}^+ \rightarrow e$. Hence

$$f(xu_s^+a_t) - f(xa_t) \rightarrow 0.$$

The difference is uniformly bounded, so its Cesàro average tends to zero. This proves equality of the forward averages. The unstable statement is identical in backward time. $\square$

The remaining issue is measure theoretic: the Birkhoff full-measure set need not contain every point on a stable or unstable leaf. The homogeneous setting resolves this by absolute continuity, which here is simply Fubini's theorem in local group coordinates.

Near the identity, multiplication gives a smooth local coordinate map

$$(5.8) \quad (r, t, s) \longmapsto u_r^- a_t u_s^+.$$

Its differential at $(0, 0, 0)$ is an isomorphism because the three infinitesimal directions F, H, E form a basis of $\mathfrak{sl}_2$. Thus (5.8) is a diffeomorphism from a small box in $\mathbb{R}^3$ onto an open subset of G . Haar measure has a smooth positive density in these coordinates. Consequently, a Haar-null subset of such a box meets almost every coordinate line in a one-dimensional null set.

Let $\Omega \subset X$ be the full-measure set on which (5.7) and both Birkhoff limits hold. Take a sufficiently small product box. Fubini applied to (5.8) implies that for almost every pair (r, t) , almost every point of the local right- U^+ segment

$$\{x u_r^- a_t u_s^+ : |s| < \delta\}$$

lies in Ω . Any two such points differ by right multiplication by an element of U^+ , so the stable-leaf lemma shows that F is almost everywhere constant on almost every local stable segment.

For the unstable foliation one uses an *adapted second chart*, for example

$$(5.9) \quad (s, t, r) \longmapsto u_s^+ a_t u_r^-.$$

Its differential at the origin is again an isomorphism, and Haar measure again has a smooth positive density. Fubini in this chart, together with the backward version of the leaf lemma, shows that F is almost everywhere constant on almost every local right- U^- segment. This separate chart is important: in the coordinates (5.8), simply varying the first coordinate r does *not* keep the other coordinates fixed along a right- U^- leaf, because the factors do not commute.

Finally, the conditional expectation F is A -invariant. Thus, after changing F on a null set if necessary, it is locally invariant almost everywhere under the three one-parameter groups U^+ , A , and U^- . Their infinitesimal directions span $\mathfrak{sl}_2$, so products of sufficiently small pieces of these subgroups contain a neighborhood of the identity in G . The set of group elements under which F is invariant in L^1 is a closed subgroup; hence it contains all of the connected group G . A G -invariant measurable function on the transitive homogeneous space $\Gamma \backslash G$ is constant almost everywhere. Therefore F is constant almost everywhere. We have proved:

THEOREM 5.17.2 (Hopf ergodicity for compact hyperbolic surfaces). *For a cocompact lattice $\Gamma < \mathrm{PSL}_2(\mathbb{R})$, the geodesic flow on $\Gamma \backslash G$ is ergodic with respect to Haar probability measure.*

Remark 5.17.3. There is a shorter representation-theoretic proof: an A -invariant L^2 vector is invariant under both horospherical subgroups by the contraction lemma, hence under all of G . The Hopf proof is longer but reveals the pointwise geometry behind that Hilbert-space statement. Stable and unstable contraction, together with absolute continuity of the foliations, propagates one statistical value through the whole space.

For finite-area noncompact surfaces the same argument requires control of excursions into the cusp, because uniform continuity on the whole space is unavailable. One works on large compact sets, controls the time spent outside them, and then lets the compact sets exhaust the finite-volume quotient. This is a useful prototype of why nondivergence estimates enter homogeneous dynamics on noncompact spaces.

18. Periodic horocycles in the modular cusp: a coordinate calculation

The modular surface gives a concrete test of every normalization. Let

$$\Gamma = \mathrm{PSL}_2(\mathbb{Z}), \quad x_t = \Gamma a_t.$$

Since

$$a_t u_s a_{-t} = u_{e^t s},$$

we have

$$x_t u_s = x_t$$

exactly when

$$\Gamma a_t u_s a_{-t} = \Gamma,$$

that is, when $u_{e^t s} \in \Gamma$. The smallest positive such parameter is

$$s = e^{-t}.$$

Thus the closed horocycle through x_t has period and length

$$(5.9H) \quad L_t = e^{-t}.$$

At the level of the upper half-plane, its base point runs along the horizontal line at height e^t . A fundamental Euclidean segment has width one, and its hyperbolic length is

$$\int_0^1 \frac{dx}{e^t} = e^{-t},$$

exactly matching (5.9).

This example is useful as a convention check: positive geodesic time moves the vector upward into the cusp and makes the corresponding closed horocycle shorter.

19. Three different notions of “most” horocycles

Statements about horocycle dynamics often use the word “generic” in different senses. It is worth separating them.

- (1) *Haar-almost every starting point.* Ergodicity and Birkhoff imply Haar equidistribution for almost every orbit.
- (2) *Every nonperiodic starting point.* Hedlund’s topological theorem gives density; stronger classical results give equidistribution under appropriate hypotheses.
- (3) *A family of closed horocycles whose length tends to infinity.* One may ask whether the normalized measures on those closed orbits converge to Haar measure. This is a different limiting problem, in which the invariant measure itself varies with the orbit.

The third viewpoint is especially important in arithmetic applications. On the modular surface, moving a short cusp horocycle downward by negative geodesic time increases its length. The distribution of such expanding closed horocycles can be studied through spectral theory, mixing, or homogeneous dynamics. The elementary commutation relation still dictates the scaling even when the effective equidistribution theorem is much deeper.

20. Modern quantitative questions around the classical flow

The classical qualitative theory naturally leads to finer research questions:

- How fast do long horocycle segments equidistribute?
- What happens when the flow is sampled at sparse times such as primes or polynomial sequences?
- How do two nearby horocycle orbits correlate quantitatively?
- How does cusp excursion affect effective error terms on noncompact quotients?
- What remains true for time changes of the horocycle flow, where exact homogeneity is broken?

These questions require spectral gaps, representation theory, number theory, or quantitative versions of orbit rigidity. They are useful examples of how the elementary identities in this chapter continue to structure modern research even when the conclusions become much more delicate.

21. Conceptual summary

The horocycle flow combines, in the smallest nontrivial homogeneous example, all of the structural features that make unipotent dynamics special:

- a one-parameter unipotent group;
- polynomial shearing of nearby trajectories;
- recurrence on finite-volume quotients;
- ergodicity from Mautner's phenomenon;
- proper periodic homogeneous orbits in the cuspidal case;
- strong topological rigidity, exemplified by Hedlund's theorem.

The model is concrete enough for direct matrix computation and rich enough to display the mechanisms that recur in much larger homogeneous spaces.

CHAPTER 6

Polynomial Estimates and Quantitative Nondivergence

1. From recurrence to quantitative control

Poincaré recurrence says that typical points return to positive-measure sets infinitely often. It gives no estimate on how long an orbit may remain in the thin part of a noncompact homogeneous space. For orbit-closure and equidistribution problems one often needs a stronger statement: a unipotent trajectory cannot spend a large proportion of a long interval extremely deep in the cusp unless an algebraic obstruction forces it to do so.

On the space of lattices

$$X_d = \mathrm{SL}_d(\mathbb{R}) / \mathrm{SL}_d(\mathbb{Z}),$$

the cusp is detected by short vectors and, more generally, by small covolumes of rational subspaces. Under a unipotent flow these covolumes are norms of polynomial vectors in exterior-power representations. Quantitative nondivergence is therefore built from two ingredients:

- (1) polynomial sublevel-set estimates;
- (2) the geometry of lattices encoded by exterior powers.

2. Good functions

Definition 6.2.1. Let $I \subset \mathbb{R}$ be an interval. A real-valued function f on I is (C, α) -good if for every subinterval $J \subset I$ and every $\varepsilon > 0$,

$$(6.1) \quad \left| \left\{ t \in J : |f(t)| < \varepsilon \sup_{s \in J} |f(s)| \right\} \right| \leq C \varepsilon^\alpha |J|.$$

Here $|E|$ denotes Lebesgue measure.

The definition is scale invariant: multiplying f by a nonzero constant changes neither side. It quantifies the principle that a nonzero function of controlled complexity cannot be extremely small on a large portion of an interval.

Proposition 6.2.2 (Polynomial sublevel estimate). *For every integer $D \geq 1$, there is a constant C_D such that every nonzero real polynomial of degree at most D is $(C_D, 1/D)$ -good on every interval.*

PROOF. By an affine change of variable it is enough to work on $[-1, 1]$, and by scaling the polynomial we may assume

$$\sup_{[-1,1]} |p| = 1.$$

A one-variable interpolation/Remez estimate gives, for a measurable subset $E \subset [-1, 1]$ of positive measure,

$$(6.2) \quad \sup_{[-1,1]} |p| \leq \left(\frac{16}{|E|} \right)^D \sup_E |p|.$$

The deliberately nonoptimal constant follows from a complete Lagrange-interpolation proof in the polynomial-estimates appendix.

Take

$$E = \{t : |p(t)| < \varepsilon\}.$$

If E has positive measure, (6.2) gives

$$1 \leq \left(\frac{16}{|E|} \right)^D \varepsilon,$$

so

$$|E| \leq 16\varepsilon^{1/D} = 8\varepsilon^{1/D} |[-1, 1]|.$$

After affine rescaling to an arbitrary subinterval, this proves (6.1) with, for example, $C_D = 8$. We keep the notation C_D because its exact value has no dynamical significance. $\square$

Remark 6.2.3. The exponent $1/D$ is of the correct order. The polynomial $p(t) = t^D$ has

$$|\{|p| < \varepsilon\}| \asymp \varepsilon^{1/D}.$$

3. Norms of polynomial vector maps

Let

$$p(t) = (p_1(t), \dots, p_m(t)) \in \mathbb{R}^m$$

with each p_i of degree at most D .

Proposition 6.3.1. *For any fixed norm on $\mathbb{R}^m$, the function*

$$t \mapsto \|p(t)\|$$

is (C, α) -good with constants depending only on D , m , and the norm. One may take $\alpha = 1/D$ after changing C .

PROOF. Choose $t_0 \in J$ with

$$\|p(t_0)\| = \sup_J \|p\|.$$

By norm duality there is a linear functional ℓ of dual norm one such that

$$\ell(p(t_0)) = \|p(t_0)\|.$$

Then $q(t) = \ell(p(t))$ is a scalar polynomial of degree at most D ,

$$\sup_J |q| \geq \sup_J \|p\|,$$

while $|q(t)| \leq \|p(t)\|$. Therefore

$$\{\|p\| < \varepsilon \sup_J \|p\|\} \subset \{|q| < \varepsilon \sup_J |q|\}.$$

Apply the scalar polynomial estimate. □

4. Why exterior-power covolumes remain polynomial

The passage from vectors to sublattices does not destroy the polynomial structure of a unipotent orbit. This point is simple, but it is worth making explicit because it is the reason one can control every rank with the same analytic tool.

Let

$$u_t = \exp(tN) \in \mathrm{SL}_d(\mathbb{R}),$$

where N is nilpotent. If $N^k = 0$, then

$$(6.2a) \quad u_t = I + tN + \frac{t^2}{2!}N^2 + \cdots + \frac{t^{k-1}}{(k-1)!}N^{k-1}.$$

Thus every matrix entry of u_t is a polynomial of degree at most $k-1$.

Now fix $1 \leq r \leq d$. On decomposable vectors,

$$(\wedge^r u_t)(v_1 \wedge \cdots \wedge v_r) = (u_t v_1) \wedge \cdots \wedge (u_t v_r).$$

Each factor on the right is polynomial of degree at most $k-1$, so every coordinate of the wedge is polynomial of degree at most

$$(6.2b) \quad r(k-1).$$

By linearity, the same degree bound holds on all of $\wedge^r \mathbb{R}^d$.

Proposition 6.4.1 (Uniform goodness for all ranks). *For a fixed one-parameter unipotent subgroup $u_t < \mathrm{SL}_d(\mathbb{R})$, there are constants $C > 0$ and $\alpha > 0$, depending only on the representation of U on $\mathbb{R}^d$, such that for every rank r and every nonzero*

$$w \in \bigwedge^r \mathbb{R}^d,$$

the function

$$t \longmapsto \|(\wedge^r u_t)w\|$$

is (C, α) -good on every interval.

PROOF. For each r , the preceding calculation expresses

$$t \mapsto (\wedge^r u_t)w$$

as a vector-valued polynomial of degree bounded by $r(k-1)$. The vector-valued polynomial estimate therefore applies. Since there are only finitely many ranks $1, \dots, d$, take the maximum of the finitely many constants and the minimum of the finitely many exponents. The resulting C, α work simultaneously for all ranks and all nonzero w . $\square$

Guiding Idea 6.4.2. The analytic complexity is therefore uniformly finite. The real difficulty in higher-rank nondivergence is not that higher exterior powers create more complicated functions; it is that many different rational subspaces can become small at overlapping times. The hard bookkeeping is combinatorial, not analytic.

5. Mahler compact sets

For a unimodular lattice $\Lambda \subset \mathbb{R}^d$, define

$$\delta(\Lambda) = \min\{\|v\| : v \in \Lambda \setminus \{0\}\}.$$

Mahler's compactness criterion says that for every $\varepsilon > 0$,

$$K_\varepsilon = \{\Lambda \in X_d : \delta(\Lambda) \geq \varepsilon\}$$

is compact, and every compact subset of X_d is contained in some K_ε .

Thus controlling cusp excursions amounts to controlling the set of times for which some nonzero lattice vector becomes shorter than ε .

In dimension two this can be done almost directly.

6. A complete quantitative nondivergence proof in dimension two

Let $h : I \rightarrow \mathrm{SL}_2(\mathbb{R})$ be a continuous map, and let $\Lambda \in X_2$. For a primitive vector $v \in \Lambda$, set

$$f_v(t) = \|h(t)v\|.$$

THEOREM 6.6.1 (Rank-one quantitative nondivergence). *Assume:*

- (1) every f_v is (C, α) -good on I ;
- (2) for every primitive $v \in \Lambda$,

$$\sup_{t \in I} f_v(t) \geq \rho;$$

- (3) $0 < \rho < 1$.

Then for every $0 < \varepsilon < \rho$,

$$(6.3) \quad |\{t \in I : \delta(h(t)\Lambda) < \varepsilon\}| \leq C \left(\frac{\varepsilon}{\rho}\right)^\alpha |I|.$$

Up to an inessential absolute multiplicity constant, the same estimate holds if endpoints of components are treated separately.

PROOF. The key dimension-two fact is that two linearly independent lattice vectors cannot both be shorter than $\rho < 1$. Indeed, if $v, w \in h(t)\Lambda$ are independent, then because the lattice is unimodular,

$$|v \wedge w| = |\det(v, w)| \geq 1.$$

The inequality follows because in a basis of the unimodular lattice the determinant of two lattice vectors is a nonzero integer. On the other hand,

$$|v \wedge w| \leq \|v\| \|w\| < \rho^2 < 1,$$

a contradiction.

For each primitive direction $\{\pm v\}$, consider the open set

$$E_v = \{t \in I : f_v(t) < \rho\}.$$

If E_v and E_w overlap, then v, w must be parallel, hence define the same primitive direction. Thus the sets E_v , indexed by primitive directions, are pairwise disjoint.

Decompose each E_v into connected components J , viewed as intervals relative to I . Every such component satisfies

$$\sup_J f_v = \rho.$$

Indeed, if $J \neq I$, at least one endpoint of J lies in the interior of I ; continuity and maximality of the component force $f_v = \rho$ at that endpoint. If $J = I$, then the hypothesis $\sup_I f_v \geq \rho$, together with $f_v < \rho$ on J , forces $\sup_J f_v = \rho$. Passing between an open component and its closure changes no Lebesgue measure.

By goodness,

$$|\{t \in J : f_v(t) < \varepsilon\}| \leq C \left(\frac{\varepsilon}{\rho}\right)^\alpha |J|.$$

Now

$$\{t : \delta(h(t)\Lambda) < \varepsilon\}$$

is the union over primitive directions of these ε -sublevel sets. Summing over pairwise disjoint ρ -components gives

$$|\{\delta(h(t)\Lambda) < \varepsilon\}| \leq C \left(\frac{\varepsilon}{\rho}\right)^\alpha \sum_J |J| \leq C \left(\frac{\varepsilon}{\rho}\right)^\alpha |I|.$$

□

Worked Example 6.6.2 (Upper-unipotent flow in X_2). Let

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

and fix $\Lambda \in X_2$. For $v = (x, y) \in \Lambda$,

$$u_t v = (x + ty, y),$$

so $t \mapsto u_t v$ is a vector-valued polynomial of degree at most one. The vector-valued polynomial proposition therefore makes its norm $(C, 1)$ -good with a universal constant for the Euclidean norm.

If the interval I contains 0, then for every primitive $v \in \Lambda$,

$$\sup_{t \in I} f_v(t) \geq f_v(0) = \|v\| \geq \delta(\Lambda).$$

Thus the lower-bound hypothesis holds uniformly in v with, for example,

$$\rho = \min\{\delta(\Lambda), 1/2\}.$$

For an interval not containing zero, choose any base time $t_0 \in I$ and replace the starting lattice by $u_{t_0} \Lambda$; the same argument applies after translating the time coordinate.

Hence a horocycle segment cannot spend a large proportion of its time with an extremely short lattice vector once a uniform macroscopic lower bound has been fixed.

7. Why higher dimension requires exterior powers

In $\mathbb{R}^3$, two independent vectors can both be very short if the third basis direction is very long. A short vector alone no longer captures all possible ways a lattice can approach the cusp during an inductive argument. One must track small covolumes of sublattices.

Let $\Delta < \Lambda$ be a primitive subgroup of rank r . Choose a $\mathbb{Z}$ -basis $v_1, \dots, v_r$ of Δ and define

$$w_\Delta = v_1 \wedge \cdots \wedge v_r \in \bigwedge^r \mathbb{R}^d.$$

Its norm

$$\|w_\Delta\|$$

is the Euclidean covolume of Δ in its real span. Changing the basis of Δ changes w_Δ only by sign, so the norm is well defined.

Under $h \in \mathrm{SL}_d(\mathbb{R})$,

$$w_{h\Delta} = (\wedge^r h) w_\Delta.$$

Thus sublattice covolumes are ordinary vector norms in exterior-power representations.

8. Primitive subgroups and discreteness

A subgroup $\Delta < \Lambda$ is primitive if

$$\Delta = \Lambda \cap \text{span}_{\mathbb{R}} \Delta.$$

Equivalently, Λ/Δ is torsion free on the quotient by the real span in the appropriate sense. Primitive subgroups are the lattice counterparts of rational subspaces.

For fixed Λ , the set of wedge vectors w_{Δ} for primitive rank- r subgroups is discrete in $\wedge^r \mathbb{R}^d$. Let us verify this rather than appeal to a picture of “rational planes.” Write

$$\Lambda = g\mathbb{Z}^d$$

with $g \in \text{SL}_d(\mathbb{R})$. If $\Delta < \Lambda$ has a $\mathbb{Z}$ -basis $gv_1, \dots, gv_r$ with $v_i \in \mathbb{Z}^d$, then

$$w_{\Delta} = (\wedge^r g)(v_1 \wedge \dots \wedge v_r).$$

The vectors $v_1 \wedge \dots \wedge v_r$ belong to the lattice

$$\bigwedge^r \mathbb{Z}^d \subset \bigwedge^r \mathbb{R}^d.$$

The linear map $\wedge^r g$ is invertible, so it sends this lattice to another *closed discrete* lattice. Primitive wedge vectors form a subset of its nonzero decomposable points. In particular, their intersection with every compact set is finite, because the same is true for the ambient wedge lattice.

Consequently

$$(6.4) \quad \inf_{\Delta} \|w_{\Delta}\| > 0$$

for each rank r , where Δ ranges over nonzero primitive rank- r subgroups. Indeed, the ambient wedge lattice has a neighborhood of the origin containing no nonzero lattice point, and all primitive wedge vectors belong to that lattice. For $r = d$, the norm is exactly one because Λ is unimodular.

9. From small covolume to a genuinely short vector

Exterior powers are not merely formal coordinates. A small wedge norm has a direct geometric consequence in the original lattice.

Proposition 6.9.1 (Minkowski inside a rational subspace). *For each $r \geq 1$ there is a constant $c_r > 0$ with the following property. Let Δ be a rank- r lattice in its real span W , and let*

$$\text{covol}_W(\Delta) = \|w_{\Delta}\|.$$

Then Δ contains a nonzero vector v satisfying

$$(6.4a) \quad \|v\| \leq c_r \|w_{\Delta}\|^{1/r}.$$

PROOF. Give W the Euclidean structure induced from $\mathbb{R}^d$. The r -dimensional ball $B_W(R)$ has volume $\omega_r R^r$. Minkowski's convex-body theorem says that if

$$\omega_r R^r > 2^r \operatorname{covol}_W(\Delta),$$

then $B_W(R)$ contains a nonzero point of Δ . Choosing

$$R > 2\omega_r^{-1/r} \operatorname{covol}_W(\Delta)^{1/r}$$

and then taking a limit gives (6.4a), for example with

$$c_r = 2\omega_r^{-1/r}.$$

The precise constant will never matter; only its dependence on the rank matters. $\square$

The converse implication has a different shape. If $v_1, \dots, v_r$ are independent vectors, then Hadamard's inequality gives

$$(6.4b) \quad \|v_1 \wedge \dots \wedge v_r\| \leq \prod_{j=1}^r \|v_j\|.$$

Thus r independent short vectors force a small rank- r covolume. A *single* short vector, however, says nothing about the covolume of a larger sublattice. This asymmetry is why one must keep track of all ranks in an inductive nondivergence proof.

Worked Example 6.9.2 (A thin plane inside a unimodular lattice). For

$$\Lambda_a = \operatorname{diag}(a, a, a^{-2})\mathbb{Z}^3, \quad 0 < a < 1,$$

the primitive plane

$$\Delta_a = \mathbb{Z}(ae_1) + \mathbb{Z}(ae_2)$$

has covolume a^2 . Formula (6.4a) predicts a vector of size $O(a)$ in that plane, and indeed both generators have length a . The compensating third direction has length a^{-2} , keeping the ambient covolume equal to one. This is the simplest picture of a cusp excursion produced by collapse in a proper rational subspace.

10. Finite complexity inside a bounded wedge window

The family of primitive subgroups is infinite, so one might worry that even a strong sublevel estimate for each subgroup becomes useless after summing over all of them. Discreteness prevents this locally.

Fix a lattice $\Lambda = g\mathbb{Z}^d$ and a rank r . The primitive wedge vectors lie in the lattice

$$(\wedge^r g)(\wedge^r \mathbb{Z}^d).$$

This is a closed discrete subset of $\wedge^r \mathbb{R}^d$. Hence, for every $M < \infty$, there are only finitely many primitive rank- r subgroups Δ for which

$$(6.4c) \quad \|w_\Delta\| \leq M.$$

Under a varying map $h(t)$, the subgroup itself stays fixed while its wedge vector moves. If $h(t)$ ranges in a compact subset of $\mathrm{SL}_d(\mathbb{R})$, boundedness of $h(t)w_\Delta$ is equivalent, up to uniform constants, to boundedness of w_Δ . Therefore only finitely many primitive subgroups can be relevant inside a compact group window.

For long unipotent segments u_t , the matrices themselves do not remain in a compact group set, so this observation alone is not enough. The standard higher-rank proof instead works locally in time: on a controlling interval, a subgroup is declared relevant when its covolume falls below a fixed threshold. At a marked time only finitely many such subgroups can occur, and determinant inequalities force incomparable small subgroups to generate a still smaller subgroup of higher rank. This is the mechanism that replaces the pairwise disjointness available in dimension two.

Warning 6.10.1. One must not sum the good-function estimate over all primitive subgroups before using this structure. There are infinitely many primitive subgroups, and such a naive union bound has no reason to converge. Rank ordering and local finiteness are essential parts of quantitative nondivergence.

11. An explicit rank-two and rank-three comparison

Before stating the general theorem, it is helpful to see exactly why the two-dimensional disjointness argument fails in one higher dimension.

Let

$$\Lambda_a = \begin{pmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a^{-2} \end{pmatrix} \mathbb{Z}^3, \quad 0 < a \ll 1.$$

This is unimodular. It contains two independent vectors

$$ae_1, \quad ae_2$$

of length a . Thus two distinct primitive directions can be simultaneously very short. Their span has wedge vector

$$a^2 e_1 \wedge e_2$$

and hence covolume a^2 , which is even smaller.

This is the higher-rank pattern: simultaneous smallness of independent lower-rank subgroups tends to create a small subgroup of larger rank. Exterior powers record this escalation. The bookkeeping in quantitative nondivergence

is designed to prevent us from counting the same cusp excursion independently under many overlapping subgroups.

Now compare with

$$\Lambda'_a = \begin{pmatrix} a & 0 & 0 \\ 0 & a^{-1} & 0 \\ 0 & 0 & 1 \end{pmatrix} \mathbb{Z}^3.$$

Only one coordinate direction is short. The rank-two wedge $ae_1 \wedge a^{-1}e_2 = e_1 \wedge e_2$ is not small. Thus the exterior-power data distinguishes a one-dimensional cusp degeneration from a genuinely two-dimensional collapse.

12. What the higher-rank covering argument must accomplish

For a primitive subgroup $\Delta < \Lambda$, set

$$F_\Delta(t) = \left\| (\wedge^{\text{rank } \Delta} h(t)) w_\Delta \right\|.$$

Suppose every F_Δ is good and has supremum at least ρ . The bad set

$$\{t : \delta(h(t)\Lambda) < \varepsilon\}$$

is certainly contained in the union of the sets $\{F_\Delta < \varepsilon\}$ over rank-one primitive subgroups. A naive union bound is useless because there are infinitely many such subgroups. The proof needs a local rule that says only a controlled family can be relevant at one time.

The rule is supplied by two facts.

- (1) *Local finiteness.* At a fixed time t and fixed threshold R , only finitely many primitive subgroups satisfy

$$F_\Delta(t) < R.$$

This follows from discreteness of all primitive wedge vectors in the lattice $h(t)\Lambda$.

- (2) *Rank escalation.* If two nonnested primitive subgroups are simultaneously very small, determinant submodularity forces the primitive subgroup generated by them to be small as well, after accounting for their intersection. The generated subgroup has larger rank than at least one of the original subgroups.

Repeated rank escalation cannot continue more than d times. Thus the collection of relevant small subgroups can be organized into chains

$$(6.5a) \quad \Delta_1 < \Delta_2 < \cdots < \Delta_k, \quad k \leq d.$$

This finite depth replaces the pairwise disjointness used in dimension two.

Guiding Idea 6.12.1. The difficult part of higher-rank nondivergence is not the polynomial estimate. Every individual subgroup already satisfies a good-function bound. The difficulty is preventing infinitely many subgroup-indexed bad sets from accumulating multiplicity. Rank chains are the combinatorial device that makes the union finite “in depth” even though it is infinite in cardinality.

13. A marked-point formulation

It is useful to describe the standard proof in a form that makes the induction visible. Fix a threshold ρ . At a time t , consider the finite set

$$\mathcal{S}(t) = \{\Delta : F_\Delta(t) < \rho^{\text{rank } \Delta}\}.$$

A *marked chain* at t is a maximal chain

$$\{0\} = \Delta_0 < \Delta_1 < \cdots < \Delta_k = \Lambda$$

containing all the subgroup data that are small at t , after thresholds have been separated by rank. The exact powers of ρ are a normalization device; they prevent the determinant inequality from losing control when ranks are added.

Given a bad point t , choose a small subgroup of minimal rank responsible for the short vector. Enlarge an interval around t until the corresponding covolume function reaches its ρ -threshold. Goodness says that the portion of this interval on which the same subgroup is at the much smaller ε -scale is

$$O((\varepsilon/\rho)^\alpha)$$

of the interval. On the complement, any continuing badness must be explained by another subgroup. If that subgroup is not nested with the first, rank escalation replaces the pair by a larger subgroup. Induction on the remaining rank then closes the argument.

On the real line, a Vitali selection may be chosen so that the controlling intervals are disjoint while a fixed dilation covers all bad points. Hence the total lengths of the controlling intervals are bounded by a constant multiple of $|I|$. Summing the good-function losses over at most d levels gives the estimate in the theorem below.

Remark 6.13.1 (Where the complete proof is located). The discussion above isolates the geometry before the combinatorics. The full one-parameter higher-rank covering argument, including the weighted-poset marking induction and the passage from marked points to a shortest-vector bound, is proved in Appendix D. Thus the theorem below is not being imported as a black box.

14. The general quantitative nondivergence theorem

The higher-dimensional theorem is most cleanly stated using primitive subgroups.

THEOREM 6.14.1 (Quantitative nondivergence for polynomial-type maps). *Let $I = B(x_0, r) \subset \mathbb{R}$ be an interval and let $\tilde{I} = B(x_0, 3^d r)$. Let $\Lambda \in X_d$ and let*

$$h : \tilde{I} \rightarrow \mathrm{SL}_d(\mathbb{R}).$$

Suppose there are constants $C, \alpha > 0$ and $0 < \rho \leq 1$ such that for every primitive subgroup $\Delta < \Lambda$:

(1) *the function*

$$t \mapsto \left\| (\wedge^{\mathrm{rank} \Delta} h(t)) w_\Delta \right\|$$

is (C, α) -good on $\tilde{I}$;

(2)

$$\sup_{t \in I} \left\| (\wedge^{\mathrm{rank} \Delta} h(t)) w_\Delta \right\| \geq \rho.$$

Then there is a constant $C' = C'(d, C, \alpha)$, also depending harmlessly on the fixed enlargement factor, such that for $0 < \varepsilon \leq \rho$,

$$(6.5) \quad |\{t \in I : h(t)\Lambda \notin K_\varepsilon\}| \leq C' \left(\frac{\varepsilon}{\rho} \right)^\alpha |I|.$$

PROOF. For a rank- s primitive subgroup the hypothesis $\sup_I \psi_\Delta \geq \rho$ is stronger than $\sup_I \psi_\Delta \geq \rho^s$, because $0 < \rho \leq 1$. Therefore the theorem is exactly Corollary D.3.3 of the fully proved one-parameter quantitative-nondivergence theorem in Appendix D. The constant C' absorbs the explicit rank-bookkeeping constant A_d there. $\square$

Remark 6.14.2 (What the full proof adds beyond dimension two). The analytic input remains goodness of exterior-power norm functions. The new difficulty is combinatorial: several primitive subgroups can become small simultaneously. Appendix D proves the weighted-poset induction that organizes them into rank chains and shows exactly why a marked point has no vector shorter than ε .

15. How to verify the hypotheses in practice

The theorem is compactly stated, but in applications its two hypotheses play very different roles. It is useful to separate them before applying the result.

- (1) *Goodness is usually formal once the trajectory is unipotent.* If $h(t)$ is obtained from a one-parameter unipotent subgroup, every exterior-power coordinate of $h(t)w_\Delta$ is a polynomial of uniformly bounded degree. The polynomial sublevel estimate therefore supplies (C, α) -goodness with constants independent of the particular primitive subgroup.
- (2) *The lower bound is the geometric input.* One must rule out the possibility that an entire interval is already trapped near a rational degeneration. If the interval contains a marked time t_0 for which $h(t_0)\Lambda$ lies in a fixed compact subset of X_d , then all primitive sublattices at that marked time have covolume bounded below uniformly. Since the supremum over the interval is at least the value at t_0 , the required ρ follows after taking the minimum of that uniform bound with 1.
- (3) *Exterior powers must be checked in every rank.* A lower bound only for individual vectors is enough for Mahler compactness, but the covering argument uses rank- r primitive subgroups for all $1 \leq r \leq d-1$. On a fixed compact subset the needed bounds are uniform. One way to see this without hiding the rank dependence is to apply Minkowski's convex-body theorem inside the span of each primitive subgroup: if an r -dimensional primitive sublattice had arbitrarily small covolume, its span would contain a nonzero lattice vector of arbitrarily small norm. Mahler compactness excludes that on a fixed compact set. Thus every rank- r primitive wedge has a uniform positive lower bound there.

This division of labor is worth remembering. Polynomiality controls *how long* a fixed algebraic obstruction can stay small; compactness at a marked time says that no obstruction is small for the whole interval. Quantitative nondivergence is obtained by combining these two facts with the rank bookkeeping.

Worked Example 6.15.1 (A rank-two exterior-power polynomial in SL_3). Consider

$$u_t = \begin{pmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix} = \exp \left(t \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \right).$$

For the primitive rank-two subgroup $\Delta = \mathbb{Z}e_2 + \mathbb{Z}e_3$,

$$u_t e_2 = e_2 + t e_1, \quad u_t e_3 = e_3 + t e_2 + \frac{t^2}{2} e_1.$$

Therefore

$$\begin{aligned} (\wedge^2 u_t)(e_2 \wedge e_3) &= (e_2 + te_1) \wedge \left(e_3 + te_2 + \frac{t^2}{2}e_1 \right) \\ &= e_2 \wedge e_3 + t e_1 \wedge e_3 + \frac{t^2}{2}e_1 \wedge e_2. \end{aligned}$$

The relevant covolume function is the norm of this polynomial vector:

$$F_\Delta(t) = \left(1 + t^2 + \frac{t^4}{4} \right)^{1/2}.$$

The norm itself is not a polynomial, and it need not be. What matters is that the vector has polynomial coordinates of bounded degree, which is exactly the hypothesis used in the vector-valued good-function estimate.

The example also shows why rank-two data cannot be reconstructed merely by watching one preferred vector. The plane spanned by e_2, e_3 moves through the exterior representation as one coherent algebraic object.

16. Application to one-parameter unipotent flows

Let $u_t \subset \mathrm{SL}_d(\mathbb{R})$ be a one-parameter unipotent subgroup. For every exterior-power representation,

$$\wedge^r u_t$$

has matrix entries polynomial in t , with degree bounded in terms of d and the nilpotence order of the generator. Hence the functions

$$t \mapsto \|(\wedge^r u_t)w\|$$

are (C, α) -good with uniform constants depending only on the degree and dimension.

Fix $\Lambda \in X_d$. By discreteness of primitive wedge vectors, choose

$$\rho(\Lambda) > 0$$

smaller than the minimum in (6.4) over all ranks. On every interval containing 0,

$$\sup_t \|(\wedge^r u_t)w_\Delta\| \geq \|w_\Delta\| \geq \rho(\Lambda).$$

Now consider a long interval $[0, T]$. Rescale $t = Ts$, $s \in [0, 1]$. The map $s \mapsto u_{Ts}$ is still polynomial of the same bounded degree, so the goodness constants do not deteriorate with T . Because the unipotent map is defined for all real times, the required fixed enlargement of the interval causes no difficulty. The general theorem yields:

Corollary 6.16.1 (Uniform nonescape for a fixed unipotent orbit). *For every $\Lambda \in X_d$ and $\eta > 0$, there exists $\varepsilon > 0$ such that for every $T \geq 1$,*

$$(6.6) \quad \frac{1}{T} |\{0 \leq t \leq T : u_t \Lambda \notin K_\varepsilon\}| < \eta.$$

PROOF. Choose ε so small that

$$C' \left(\frac{\varepsilon}{\rho(\Lambda)} \right)^\alpha < \eta.$$

Apply (6.5) to the rescaled interval. □

This is a genuine strengthening of recurrence: it says that a uniformly large proportion of every long orbit segment lies in one fixed compact set.

17. Uniformity for orbit segments starting in a compact set

The fixed-orbit corollary begins every interval at the original lattice Λ , so the value at time zero supplies the lower bound $\rho(\Lambda)$. In applications one often starts an orbit segment at a later time. Uniformity is recovered whenever the starting points lie in one compact set.

Proposition 6.17.1. *Let $K_0 \subset X_d$ be compact and let u_t be a fixed one-parameter unipotent subgroup. For every $\eta > 0$ there exists $\varepsilon > 0$ such that for every lattice $\Lambda \in X_d$, every $S \in \mathbb{R}$, and every $T > 0$ for which*

$$u_S \Lambda \in K_0,$$

one has

$$(6.7) \quad \frac{1}{T} |\{0 \leq t \leq T : u_{S+t} \Lambda \notin K_\varepsilon\}| < \eta.$$

PROOF. For each rank r , define

$$\beta_r(\Lambda) = \min_{\Delta \text{ primitive rank } r} \|w_\Delta\|.$$

The lower bound is uniform on K_0 :

$$(6.7a) \quad \inf_{\Lambda \in K_0} \beta_r(\Lambda) > 0.$$

Here is a direct justification. If (6.7a) failed, there would be lattices $\Lambda_n \in K_0$ and primitive rank- r sublattices Δ_n with covolume tending to zero. Minkowski's convex-body theorem applied inside the r -dimensional span of Δ_n would then produce a nonzero vector of $\Delta_n \subset \Lambda_n$ whose length tends to zero. This contradicts Mahler compactness, which gives a positive uniform lower bound for the shortest-vector length on the compact set K_0 . Take $\rho_0 > 0$ smaller than these finitely many lower bounds over all ranks. If

$u_S\Lambda \in K_0$, then every primitive subgroup of the starting lattice has wedge norm at least ρ_0 . Apply quantitative nondivergence to

$$h(t) = u_t, \quad 0 \leq t \leq T,$$

with starting lattice $u_S\Lambda$. The goodness constants depend only on the unipotent representation, and the lower-supremum condition is supplied at $t = 0$ by ρ_0 . Choosing ε so that

$$C'(\varepsilon/\rho_0)^\alpha < \eta$$

proves the claim. $\square$

This is the form most useful for recurrence comparisons: whenever an orbit returns to a fixed thick compact set, the subsequent unipotent segment enjoys nondivergence constants independent of the particular return time.

Worked Example 6.17.2 (Why no uniform constant exists over the whole space). Take X_2 and a lattice Λ_n with shortest vector length tending to zero. At time zero the orbit already lies arbitrarily deep in the cusp. No single $\rho > 0$ can satisfy the lower-supremum hypothesis for every Λ_n on a very short interval. Uniformity must therefore depend on a thick starting set, or else permit an explicit algebraic obstruction.

18. Nonescape of mass for empirical measures

Define

$$\mu_T = \frac{1}{T} \int_0^T \delta_{u_t\Lambda} dt.$$

By (6.6), for every $\eta > 0$ there is a compact $K \subset X_d$ such that

$$\mu_T(K) > 1 - \eta$$

for every $T \geq 1$. Thus the family $\{\mu_T\}$ is tight.

By Prokhorov compactness in this locally compact second-countable setting, every sequence $T_n \rightarrow \infty$ has a subsequence for which μ_{T_n} converges weak-* to a probability measure. The telescoping boundary calculation for flows makes every such limit U -invariant.

Guiding Idea 6.18.1. This is the standard compactness bridge from individual orbits to invariant measures:

$$\begin{aligned} \text{quantitative nondivergence} &\implies \text{tight empirical measures} \\ &\implies \text{probability subsequential limits} \\ &\implies \text{invariant measures.} \end{aligned}$$

Without nonescape of mass, the weak-* limit could have total mass less than one or vanish entirely in the vague topology.

19. A simple obstruction and why the lower-bound hypothesis matters

Take a polynomial map $h(t)$ for which some fixed primitive vector v satisfies

$$h(t)v \equiv \varepsilon_0 w$$

with ε_0 extremely small. Then the entire trajectory has a uniformly short vector and remains in the cusp. No sublevel estimate can prevent this because the relevant polynomial is identically small.

The hypothesis

$$\sup_I \|h(t)w_\Delta\| \geq \rho$$

excludes precisely this algebraic obstruction. Quantitative nondivergence is therefore always a dichotomy:

either the trajectory is forced close to a proper rational configuration, or it spends only controlled time in the thin part.

This dichotomy is more important conceptually than the exact exponent in (6.5).

20. Research-facing variants

The same architecture extends far beyond one-dimensional real unipotent flows.

Several parameters. For polynomial maps

$$h : \mathbb{R}^k \rightarrow \mathrm{SL}_d(\mathbb{R}),$$

one replaces intervals by balls and uses multivariable (C, α) -good functions together with a Besicovitch covering theorem.

Analytic curves. Nondegenerate analytic curves in homogeneous spaces have local Taylor polynomials whose relevant wedge coordinates satisfy good-function estimates. This is the starting point for quantitative nondivergence in metric Diophantine approximation on manifolds.

S-arithmetic spaces. For products of real and p -adic groups, one combines archimedean polynomial sublevel estimates with nonarchimedean analogues. The geometry-of-numbers object becomes an S -arithmetic lattice.

Effective equidistribution. Quantitative nondivergence controls the amount of time spent in regions where Sobolev norms, injectivity radii, or mixing estimates deteriorate. It is therefore a necessary companion to spectral estimates on noncompact quotients.

21. Conceptual summary

Unipotent nondivergence rests on a remarkably concrete chain:

unipotent action $\implies$ polynomial exterior-power coordinates
 $\implies$ sublevel-set bounds
 $\implies$ cusp-time bounds.

In X_2 , the proof is almost elementary because at most one primitive direction can be very short at once. In higher dimension the same idea survives, but primitive subgroups of all ranks must be organized through exterior powers.

The outcome is the uniform tightness statement (6.6), which upgrades qualitative recurrence to a quantitative no-escape principle and makes invariant-measure compactness arguments possible on noncompact spaces.

CHAPTER 7

Linearization, Exterior Powers, and Singular Sets

1. Why nonlinear geometry is converted into linear algebra

A homogeneous subspace is defined by group equations, but orbit arguments often need quantitative statements such as:

“the point $u_t x$ is close to a proper homogeneous orbit for many values of t .”

Closeness to a subgroup orbit is awkward to manipulate directly. The linearization method replaces the subgroup by a vector in a finite-dimensional representation. Proximity to the homogeneous orbit then becomes smallness or near-stabilization of a vector, and unipotent motion becomes polynomial.

There are two model versions:

- (1) rational subspaces of a lattice are encoded by exterior products of lattice vectors;
- (2) Lie subgroups are encoded by the exterior product of a basis of their Lie algebra.

The second is the group-theoretic analogue of the first.

2. Exterior powers as subspace coordinates

Let V be a d -dimensional real vector space. If $W < V$ has dimension r and $w_1, \dots, w_r$ is a basis, define

$$p_W = w_1 \wedge \cdots \wedge w_r \in \bigwedge^r V.$$

Changing basis multiplies p_W by a nonzero scalar. Thus the line

$$[p_W] \in \mathbb{P}(\wedge^r V)$$

depends only on W . This is the Plücker embedding of the Grassmannian.

Proposition 7.2.1. *For $g \in \mathrm{GL}(V)$,*

$$p_{gW} = (\wedge^r g)p_W$$

up to the scalar caused by the choice of basis. In projective space,

$$[p_{gW}] = [(\wedge^r g)p_W].$$

PROOF. If $w_1, \dots, w_r$ is a basis of W , then $gw_1, \dots, gw_r$ is a basis of gW , and by definition of the exterior-power representation,

$$(\wedge^r g)(w_1 \wedge \cdots \wedge w_r) = gw_1 \wedge \cdots \wedge gw_r.$$

□

Thus the nonlinear Grassmannian sits inside the projectivization of a linear representation.

3. Rational subspaces of a lattice

Let $\Lambda \subset \mathbb{R}^d$ be a lattice. A subspace $W < \mathbb{R}^d$ is Λ -rational if

$$W \cap \Lambda$$

is a lattice in W . If $\Delta = W \cap \Lambda$ is primitive of rank r , choose a $\mathbb{Z}$ -basis and form

$$w_\Delta = v_1 \wedge \cdots \wedge v_r.$$

The norm $\|w_\Delta\|$ is the covolume of Δ in W .

Worked Example 7.3.1 (A rational plane in $\mathbb{R}^3$). For $\Lambda = \mathbb{Z}^3$, the plane

$$W = \text{span}_{\mathbb{R}}\{(1, 0, 0), (0, 1, 0)\}$$

is encoded by

$$e_1 \wedge e_2.$$

A matrix $g \in \text{SL}_3(\mathbb{R})$ sends it to the plane spanned by the first two columns of g , encoded by

$$(\wedge^2 g)(e_1 \wedge e_2).$$

If this wedge vector has very small norm, the corresponding rank-two sublattice has very small covolume. Since the total lattice covolume remains one, some complementary direction must be very large: this is a higher-rank cusp excursion.

4. Encoding Lie subgroups

Let G be a Lie group with Lie algebra $\mathfrak{g}$, and let $H < G$ be a connected closed subgroup with Lie algebra $\mathfrak{h}$ of dimension r . Choose a basis

$$X_1, \dots, X_r$$

of $\mathfrak{h}$ and set

$$p_H = X_1 \wedge \cdots \wedge X_r \in \bigwedge^r \mathfrak{g}.$$

The line $[p_H]$ does not depend on the basis.

Let

$$\rho_r(g) = \wedge^r \text{Ad}(g).$$

Then

$$\rho_r(g)p_H$$

encodes the conjugate Lie algebra $\text{Ad}(g)\mathfrak{h}$.

Proposition 7.4.1 (Projective stabilizer). *The stabilizer of the line $[p_H]$ is exactly the normalizer of the Lie algebra:*

$$\text{Stab}_G([p_H]) = \{g \in G : \text{Ad}(g)\mathfrak{h} = \mathfrak{h}\} = N_G(H)$$

when H is connected.

PROOF. If g normalizes H , then $\text{Ad}(g)\mathfrak{h} = \mathfrak{h}$, so $\rho_r(g)p_H$ is a nonzero scalar multiple of p_H .

Conversely, if $\rho_r(g)p_H$ lies on the line $\mathbb{R}p_H$, then the decomposable r -vector $\rho_r(g)p_H$ spans the same projective Plücker point as p_H . Hence

$$\text{Ad}(g)\mathfrak{h} = \mathfrak{h}.$$

Because H is connected, it is uniquely determined by its Lie algebra among connected Lie subgroups, so $gHg^{-1} = H$. $\square$

5. Vector stabilizer and the determinant character

The stabilizer of the vector p_H , not merely its line, is

$$(7.1) \quad N_G^1(H) = \{g \in N_G(H) : \det(\text{Ad}(g)|_{\mathfrak{h}}) = 1\}.$$

Indeed,

$$(\wedge^r \text{Ad}(g))p_H = \det(\text{Ad}(g)|_{\mathfrak{h}})p_H.$$

If H is connected and unimodular, every $h \in H$ satisfies

$$\det(\text{Ad}(h)|_{\mathfrak{h}}) = 1,$$

so

$$H < N_G^1(H).$$

This is why the vector p_H is particularly convenient for finite-volume homogeneous orbits: the group H itself fixes the vector, not just the projective line.

6. Lattice intersections and fixed wedge vectors

Suppose $H \cap \Gamma$ is a lattice in a connected subgroup H . Then H is unimodular. Hence for every $\gamma \in H \cap \Gamma$,

$$(7.2) \quad (\wedge^r \text{Ad}(\gamma))p_H = p_H.$$

Thus p_H behaves like a rational vector relative to Γ : a nontrivial subgroup of the lattice fixes it. In arithmetic settings, the Γ -orbit of an appropriately normalized integral wedge vector is often discrete. Discreteness is what turns

approximate stabilizer information into a finite or countable collection of algebraic alternatives.

7. The basic linearization map

Define

$$(7.3) \quad \eta_H : G \rightarrow \bigwedge^r \mathfrak{g}, \quad \eta_H(g) = \rho_r(g)p_H.$$

Then

$$\eta_H(gh) = \eta_H(g) \quad (h \in H)$$

provided H fixes p_H , for example when H is connected and unimodular. Thus η_H is constant on right H -cosets.

At the projective level, even without unimodularity,

$$[\eta_H(gh)] = [\eta_H(g)].$$

The map η_H converts the right-coset geometry of H into an orbit in a finite-dimensional representation.

8. What the derivative of the linearization map detects

The map

$$\eta_H(g) = \rho_r(g)p_H$$

contains both orbit geometry and normalizer information. Its derivative makes the distinction precise.

Let $d\rho_r : \mathfrak{g} \rightarrow \mathfrak{gl}(\wedge^r \mathfrak{g})$ be the derived representation. For $Y \in \mathfrak{g}$,

$$(7.3a) \quad (d\eta_H)_e(Y) = d\rho_r(Y)p_H.$$

Indeed,

$$\eta_H(\exp(tY)) = \rho_r(\exp(tY))p_H = \exp(t d\rho_r(Y))p_H,$$

and differentiation at $t = 0$ gives (7.3a).

What is the kernel? At the projective level the answer is the Lie algebra of the normalizer:

$$(7.3b) \quad Y \in \text{Lie}(N_G(H)) \iff d\rho_r(Y)p_H \in \mathbb{R}p_H.$$

For the vector-valued map, the kernel is smaller:

$$(7.3c) \quad d\rho_r(Y)p_H = 0 \iff Y \in \text{Lie}(N_G^1(H)),$$

where $N_G^1(H)$ is the determinant-one stabilizer introduced above.

EXPLANATION OF (7.3B). The curve $\exp(tY)$ lies infinitesimally in the projective stabilizer of $[p_H]$ exactly when

$$\left. \frac{d}{dt} \right|_{t=0} [\rho_r(\exp(tY))p_H] = 0.$$

The tangent space to the projective line through p_H is obtained by quotienting $\wedge^r \mathfrak{g}$ by $\mathbb{R}p_H$, so this derivative vanishes precisely when $d\rho_r(Y)p_H$ is proportional to p_H . The projective stabilizer itself is $N_G(H)$; differentiating its defining condition gives (7.3b). The vector statement (7.3c) is the same argument without projectivizing. $\square$

This calculation clarifies an important point. Linearization does not generally distinguish motion tangent to the H -orbit from motion inside a larger normalizer. If the normalizer is larger than H , one must either quotient out that extra symmetry or use additional representation data. Treating the kernel as automatically equal to $\mathfrak{h}$ is a common source of hidden errors.

Worked Example 7.8.1 (The upper unipotent subgroup of $\mathrm{PSL}_2(\mathbb{R})$). Let $H = U$ with Lie algebra $\mathbb{R}E$. Then $r = 1$ and we may take

$$p_H = E.$$

The projective stabilizer of $[E]$ is the upper triangular subgroup $B = AU$, because conjugation by a_t rescales E while conjugation by u_s fixes it. The vector stabilizer is only U :

$$\mathrm{Ad}(a_t)E = e^t E, \quad \mathrm{Ad}(u_s)E = E.$$

Thus the projective linearization naturally remembers the normalizer B , whereas the vector linearization distinguishes the diagonal scaling character. This elementary example is the rank-one model for the determinant character in higher-dimensional linearization.

9. Unipotent motion becomes polynomial

Let $U = \{u_t\}$ be Ad -unipotent. Then for every finite-dimensional representation built functorially from Ad , including $\rho_r = \wedge^r \mathrm{Ad}$, the map

$$t \mapsto \rho_r(u_t)v$$

is polynomial in t for each v .

Hence

$$(7.4) \quad \eta_H(u_t g) = \rho_r(u_t) \eta_H(g)$$

is a vector-valued polynomial. If an orbit spends a large set of times near the homogeneous configuration encoded by H , then this polynomial spends a large set of times near an algebraic subset of $\wedge^r \mathfrak{g}$. Polynomial sublevel

estimates can then force the entire polynomial to remain controlled, revealing an algebraic obstruction.

10. A model linearization lemma

Lemma 7.10.1. *Let V be finite-dimensional, let $p_n : \mathbb{R} \rightarrow V$ be polynomials of degree at most D , and suppose that on an interval I_n ,*

$$|\{t \in I_n : \|p_n(t)\| < \varepsilon_n\}| \geq \theta |I_n|$$

for some fixed $\theta > 0$, with $\varepsilon_n \rightarrow 0$. If the polynomials are normalized so that

$$\sup_{I_n} \|p_n\| = 1,$$

then this is impossible for all sufficiently large n .

PROOF. The norm of a vector-valued degree- D polynomial is $(C, 1/D)$ -good. Therefore

$$\theta |I_n| \leq C \varepsilon_n^{1/D} |I_n|,$$

which fails as $n \rightarrow \infty$. □

The point is qualitative: a bounded-degree polynomial cannot be tiny on a definite proportion of an interval and simultaneously achieve order-one size somewhere on that interval.

11. A local tubular picture around a homogeneous stratum

Suppose $H < G$ is closed and $x = g\Gamma \in G/\Gamma$ has positive injectivity radius. Choose a vector-space complement

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{v}.$$

After shrinking neighborhoods, multiplication gives a local diffeomorphism

$$(7.4a) \quad (H \cap \Omega_H) \times (\mathfrak{v} \cap \Omega_V) \longrightarrow G, \quad (h, v) \longmapsto h \exp(v).$$

Passing to the quotient, every point sufficiently near the local H -orbit through x has a unique expression

$$h \exp(v)x$$

with v small and transverse, provided we remain inside an injectivity neighborhood for the lattice quotient.

The representation vector sees this transverse coordinate through

$$(7.4b) \quad \eta_H(\exp(v)) = p_H + d\rho_r(v)p_H + O(\|v\|^2).$$

If the class of $d\rho_r(v)p_H$ transverse to $\mathbb{R}p_H$ is nonzero, then distance from the projective stabilizer is comparable to the size of that class. Thus closeness to an H -type stratum becomes a smallness condition in a finite-dimensional representation.

There are two caveats.

- (1) If v lies in the Lie algebra of the normalizer, the first-order projective displacement can vanish even though $v \notin \mathfrak{h}$.
- (2) On the quotient G/Γ , different lattice translates of the same homogeneous piece can pass near one another. The representation orbit under Γ is what keeps track of this lattice ambiguity.

These two caveats explain why serious linearization arguments use both a carefully chosen stabilizer representation and a discrete family of lattice translates.

12. Why one representation vector can control a whole time interval

Suppose $t \mapsto u_t g \Gamma$ stays close to the family of translates of an H -stratum during a connected interval J . At each time there may a priori be a different $\gamma_t \in \Gamma$ for which

$$\rho_r(u_t g \gamma_t) p_H$$

is small relative to the target algebraic set. If the relevant Γ -orbit of p_H is closed and discrete and all these vectors remain in one fixed compact representation window, then only finitely many choices of $\gamma_t p_H$ are possible. Continuity in t , together with positive separation between distinct choices, forces the choice to be locally constant. On a connected interval it is therefore constant except possibly when the trajectory leaves the compact window.

This is the basic reason linearization converts a moving geometric ambiguity into one fixed polynomial vector. Once a single translate γp_H has been selected, the function

$$t \mapsto \rho_r(u_t g \gamma) p_H$$

is polynomial, and polynomial sublevel estimates become available.

Guiding Idea 7.12.1. The representation does two jobs simultaneously:

- homogeneous geometry $\longrightarrow$ algebraic stabilizer condition,
- lattice ambiguity $\longrightarrow$ discrete family of vectors.

The first makes the condition polynomial; the second prevents the relevant polynomial from changing secretly from one lattice translate to another at every time.

13. Singular points for a unipotent action

Let $X = G/\Gamma$, with $U < G$ acting on the left. A point is dynamically special if its U -orbit is trapped in a proper finite-volume homogeneous subspace.

Fix a connected closed subgroup $H < G$ such that $H \cap \Gamma$ is a lattice. Define

$$(7.5) \quad N(U, H) = \{g \in G : g^{-1}Ug \subset H\}.$$

If $g \in N(U, H)$, then

$$Ug\Gamma \subset gH\Gamma/\Gamma.$$

Definition 7.13.1 (Normalizer locus and singular stratum). For the pair (U, H) define the *normalizer locus*

$$N(U, H) = \{g \in G : g^{-1}Ug \subset H\}$$

and define its associated *singular stratum* in $X = G/\Gamma$ by

$$(7.6) \quad \mathcal{S}(U, H) = N(U, H)\Gamma/\Gamma.$$

Thus $g\Gamma \in \mathcal{S}(U, H)$ precisely when some lattice translate of g conjugates U into H ; in particular the U -orbit of that point is contained in the corresponding proper homogeneous subspace whenever H is proper.

Warning 7.13.2. The notation in (7.6) is schematic unless the product set $N(U, H)\Gamma$ is known to have the desired measurability and local-closedness properties. In concrete algebraic settings these are handled using the linearization representation. The important object is the condition $g^{-1}Ug \subset H$, not the typography of the quotient notation.

14. Linear equations for the containment condition

Let X_0 generate the Lie algebra of U . The condition

$$g^{-1}Ug \subset H$$

is equivalent to

$$(7.7) \quad \text{Ad}(g^{-1})X_0 \in \mathfrak{h}.$$

In terms of the wedge vector p_H , membership in $\mathfrak{h}$ can be expressed as

$$(7.8) \quad \text{Ad}(g^{-1})X_0 \wedge p_H = 0 \quad \text{in} \quad \bigwedge^{r+1} \mathfrak{g}.$$

Thus $N(U, H)$ is cut out by algebraic equations in finite-dimensional representation coordinates whenever G is a linear algebraic group.

This is the first sense in which a singular set is linearized.

15. Why singular sets come in strata

Suppose a point lies in two proper homogeneous subspaces corresponding to H_1 and H_2 . Their intersection may itself correspond to a smaller subgroup. Therefore neighborhoods of singular sets overlap in structured ways.

A typical avoidance argument proceeds inductively:

- (1) identify the minimal-dimensional subgroup responsible for a long near-return;
- (2) remove neighborhoods of still smaller singular strata;
- (3) on the remaining region, the linearization vector associated with the chosen subgroup has controlled norm;
- (4) apply polynomial goodness to bound the time spent near that stratum.

The hierarchy by subgroup dimension is not cosmetic. It prevents one algebraic obstruction from being counted repeatedly under many larger subgroups.

16. Self-intersections and the role of the lattice

The map

$$G/H \rightarrow G/\Gamma$$

induced by $gH \mapsto g\Gamma$ is generally not injective. Different translates of the same homogeneous piece can intersect after multiplication by elements of Γ . Linearization handles this by replacing a neighborhood of the orbit with neighborhoods of a discrete family of vectors

$$\rho(\gamma)p_H, \quad \gamma \in \Gamma.$$

If the relevant Γ -orbit of p_H is discrete, only finitely many such vectors can enter a fixed compact region of the representation space. This converts potentially complicated self-intersection geometry into a finite local combinatorial problem.

17. Arithmetic example: rational subgroups of the special linear group

Take

$$G = \mathrm{SL}_d(\mathbb{R}), \quad \Gamma = \mathrm{SL}_d(\mathbb{Z}).$$

Let H stabilize a rational subspace $W_0 < \mathbb{Q}^d$ of dimension r . After choosing an integral basis, W_0 is encoded by a primitive integral vector

$$w_0 \in \bigwedge^r \mathbb{Z}^d.$$

The orbit

$$\Gamma w_0$$

is a subset of the integer lattice $\wedge^r \mathbb{Z}^d$, hence is discrete.

If $g_n \Gamma$ approaches the homogeneous configuration in which a rank- r rational subspace has small covolume, then for suitable $\gamma_n \in \Gamma$,

$$\|(\wedge^r g_n) \gamma_n w_0\|$$

becomes small. Under unipotent translation this quantity is the norm of a polynomial vector. This is the cleanest concrete incarnation of the linearization method.

18. Example: lines and planes in the space of three-dimensional lattices

For X_3 , two basic rational degeneracies are:

- (1) a short primitive vector $v \in \Lambda$, detected in $\wedge^1 \mathbb{R}^3$;
- (2) a rank-two primitive sublattice Δ , detected by $w_\Delta \in \wedge^2 \mathbb{R}^3$.

These are dual because

$$\wedge^2 \mathbb{R}^3 \cong (\mathbb{R}^3)^*$$

via the volume form. A small area for a rational plane is equivalent to a large spacing in the complementary rational direction.

When a unipotent orbit approaches the cusp, one of these wedge vectors becomes small. An inductive argument must allow the possibility that a short line sits inside a small-area plane, illustrating the nested-stratum phenomenon concretely.

19. A linearized recurrence comparison

Suppose $x_n = g_n \Gamma$ and $y_n = \exp(v_n)x_n$ are nearby points with $v_n \rightarrow 0$. Under U ,

$$u_t y_n = \exp(\text{Ad}(u_t)v_n) u_t x_n.$$

If both orbit points return to a compact set at times $t = t_n$, local injectivity identifies their relative displacement. If, in addition, $u_{t_n} x_n$ and $u_{t_n} y_n$ lie near the same homogeneous stratum, applying η_H converts the comparison into a relation between polynomial vectors.

This is the meeting point of three ideas:

recurrence + polynomial shearing + linearization of homogeneous strata.

The combination is much stronger than any ingredient separately.

20. Avoidance: a qualitative template

A useful qualitative statement has the following form.

Fix a compact set K disjoint from a prescribed lower-dimensional singular set. Then a unipotent orbit starting in K cannot spend an arbitrarily large proportion of a long interval in an arbitrarily small neighborhood of that singular set unless the orbit is algebraically trapped there.

The proof template is:

- (1) assume excessive time near the singular set;
- (2) choose the corresponding linearization vector p_H ;
- (3) use recurrence to keep the representation coordinate in a bounded range;
- (4) use unipotency to make the coordinate polynomial in time;
- (5) apply a good-function estimate;
- (6) conclude that the polynomial is identically constrained, which means the orbit lies in the algebraic obstruction.

The exact theorem depends on the ambient group, the class of subgroups allowed, and discreteness properties of the relevant lattice orbits. Those hypotheses must be stated rather than hidden under the word “linearization.”

21. Research-facing example: linearization in counting problems

Suppose one wants to count lattice points in a family of expanding regions by translating the problem into an orbit of G on G/Γ . Effective mixing estimates usually work well only on the thick part of the quotient. A linearization/nondivergence argument shows that the orbit does not spend too much time near proper rational subspaces where injectivity radius degenerates. Exterior-power representations therefore enter both qualitative rigidity and quantitative counting.

The same technical object—a wedge vector encoding a rational subgroup—can control seemingly different phenomena: cusp excursions, exceptional orbit closures, and errors in lattice-point asymptotics.

22. A concrete parabolic example in dimension three

Take

$$G = \mathrm{SL}_3(\mathbb{R})$$

and let H be the stabilizer of the line $\mathbb{R}e_1$. Then H is a maximal parabolic subgroup. The line itself is encoded by the vector

$$p = e_1 \in \wedge^1 \mathbb{R}^3.$$

For $g \in G$, the vector gp is simply the first column of g , and the projective point $[gp]$ records the line $g\mathbb{R}e_1$.

On the space of lattices X_3 , a primitive vector $v \in \mathbb{Z}^3$ gives a rational line. If

$$\|gv\| \ll 1,$$

then the lattice $g\mathbb{Z}^3$ has a short primitive vector and is deep in the cusp. The same expression gv simultaneously measures distance toward a parabolic configuration and detects noncompactness.

For rank-two rational planes the corresponding representation is $\wedge^2\mathbb{R}^3$. Thus the two basic proper parabolic geometries of SL_3 are detected by the first two exterior powers. This is the representation-theoretic reason exterior powers are so natural in both reduction theory and linearization.

23. Normalizer versus stabilizer: an ambiguity worth removing

The vector p_H and its projective line carry slightly different information. If

$$(\wedge^r \mathrm{Ad}(g))p_H = c(g)p_H,$$

then g normalizes H , but it need not fix p_H itself. The scalar is

$$(7.9) \quad c(g) = \det(\mathrm{Ad}(g)|_{\mathfrak{h}}).$$

Thus

$$\mathrm{Stab}_G([p_H]) = N_G(H),$$

while

$$\mathrm{Stab}_G(p_H) = N_G^1(H) = \ker c.$$

This distinction is especially important when one passes from a geometric statement about a homogeneous orbit to a linear statement in a representation. Projective closeness detects closeness of subspaces; vector closeness also records a volume normalization. Finite-volume homogeneous orbits naturally supply that normalization through unimodularity.

Worked Example 7.23.1 (The diagonal subgroup in the special linear group). Let $G = \mathrm{SL}_2(\mathbb{R})$ and $H = A$ be the diagonal subgroup. Its Lie algebra is spanned by

$$H_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The Weyl element

$$w = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

normalizes A but sends H_0 to $-H_0$. Thus w fixes the projective line $[H_0]$ but not the vector H_0 . In this example

$$N_G(A)/A$$

has two elements, and the sign on the wedge vector detects the nontrivial Weyl element.

The example also warns against a common overstatement: a normalizer need not in general be a finite extension of the subgroup being normalized. Finiteness here is a special feature of the Cartan subgroup in SL_2 , not a formal property of normalizers.

24. Local separation from a discrete representation orbit

Suppose a lattice $\Gamma < G$ acts linearly on a finite-dimensional vector space V , and let $v_0 \in V$ have a closed discrete Γ -orbit. This proper discreteness has a useful local consequence.

Lemma 7.24.1. *Let $C \subset V$ be compact. Then only finitely many vectors of Γv_0 lie in C . Consequently there is a positive minimum distance between any two distinct points of $\Gamma v_0 \cap C$.*

PROOF. A closed discrete subset of a locally compact space meets every compact set in a finite set. Indeed, an infinite intersection with a compact set would have an accumulation point in that compact set; closedness would put the accumulation point back in the subset, contradicting discreteness. A finite set has positive minimum pairwise distance. $\square$

In linearization arguments this produces a uniform choice: while the representation vector remains in a fixed compact region, it cannot jump continuously from being close to one lattice translate $\gamma_1 v_0$ to being close to another $\gamma_2 v_0$ without passing through a region separated from both. Thus a long near-return to a homogeneous stratum can often be assigned to one fixed lattice translate over a whole time interval.

25. A polynomial avoidance lemma for a discrete family

The preceding idea combines directly with polynomial goodness.

Proposition 7.25.1 (One-vector avoidance). *Let $D \geq 1$. Let $p : I \rightarrow V$ be a vector-valued polynomial of degree at most D , let $v_0 \in V$, and suppose*

$$\sup_{t \in I} \|p(t) - v_0\| \geq \rho.$$

Then for $0 < \varepsilon < \rho$,

$$(7.10) \quad |\{t \in I : \|p(t) - v_0\| < \varepsilon\}| \leq C_D \left(\frac{\varepsilon}{\rho}\right)^{1/D} |I|.$$

PROOF. The map $t \mapsto p(t) - v_0$ is another vector polynomial of degree at most D . Apply the vector-valued good-function estimate to its norm. $\square$

Now suppose Γv_0 is discrete and $p(I)$ remains in a compact set. Only finitely many translates can be relevant. If their ε -neighborhoods are disjoint, (7.10) may be summed immediately. When different algebraic strata intersect, the neighborhoods are not disjoint; the subgroup-stratification procedure identifies the intersection as a lower-dimensional or more singular configuration and treats it separately.

This is the elementary analytic core of the linearization method: after the geometry has been converted into a discrete family of representation vectors, a unipotent orbit becomes a bounded-degree polynomial trying to shadow that family.

26. Why the containment equation is genuinely algebraic

For a one-parameter subgroup $U = \{\exp(tX_0)\}$, the condition

$$g^{-1}Ug \subset H$$

is equivalent to

$$\mathrm{Ad}(g^{-1})X_0 \in \mathfrak{h}.$$

If G is realized as a real algebraic matrix group and H is algebraic, then this is a system of polynomial equations in the matrix entries of g , after clearing the determinant denominators that appear in g^{-1} . The wedge equation

$$\mathrm{Ad}(g^{-1})X_0 \wedge p_H = 0$$

is a compact coordinate-free way to write those equations.

Thus a “singular set” in unipotent dynamics is not merely a set on which an argument happens to fail. It is typically an algebraically defined locus where the acting unipotent subgroup is contained in a proper conjugate subgroup. The algebraicity is what makes dimension induction and finite stratification plausible.

27. Chevalley linearization without a black box

For many algebraic subgroups one wants a finite-dimensional representation whose line stabilizer is exactly that subgroup. The precise statement is: if H is an algebraic subgroup of a real linear algebraic group G , there exist a

finite-dimensional rational representation $\rho : G \rightarrow \mathrm{GL}(V)$ and a line $\ell \subset V$ such that

$$H = \mathrm{Stab}_G(\ell).$$

The exterior-power construction used above gives a line whose stabilizer is often a normalizer of H , so it does not by itself always recover H exactly. The general statement is proved in full in Appendix C. The proof is concrete: choose finitely many regular functions generating the ideal of H , put them in a finite-dimensional translation module of the coordinate ring, realize H as the stabilizer of the subspace of functions vanishing on H , and then take its top exterior power to convert the subspace into a line.

This matters dynamically because the construction is rational. Along a unipotent one-parameter subgroup, the resulting representation coordinates are bounded-degree polynomials, so subgroup proximity becomes a polynomial sublevel problem with no unproved algebraic linearization step remaining.

28. A research-facing schematic: avoiding proper homogeneous pieces

Suppose U acts on G/Γ and K is a compact set. Fix finitely many proper subgroups H_i that can meet K in the relevant singular range. For each H_i , choose a linearization vector p_{H_i} . On a compact lift of K , proximity to the corresponding homogeneous piece is converted into

$$\|\rho_i(g)p_{H_i} - \gamma p_{H_i}\| < \varepsilon$$

for one of finitely many $\gamma \in \Gamma$. Along a unipotent orbit this becomes a polynomial sublevel condition.

The conclusion takes the form of a dichotomy:

Either the orbit spends only a controlled proportion of time near the proper homogeneous pieces, or one of the relevant representation polynomials stays uniformly small, forcing an exact algebraic obstruction.

This formulation explains why linearization and quantitative nondivergence are closely related but not identical. Nondivergence prevents escape to the cusp; linearization prevents excessive concentration near proper algebraic subspaces inside the thick part.

29. Conceptual summary

Linearization replaces a subgroup or rational subspace by a vector:

$$H \rightsquigarrow p_H \in \bigwedge^{\dim H} \mathfrak{g}, \quad \Delta \rightsquigarrow w_\Delta \in \bigwedge^{\mathrm{rank} \Delta} \mathbb{R}^d.$$

Group motion becomes a linear representation, and unipotent motion becomes polynomial:

$$\eta_H(u_t g) = \rho(u_t) \eta_H(g).$$

The method is powerful because it separates the problem into two parts:

- algebra identifies the vectors representing singular homogeneous configurations;
- analysis controls how long a bounded-degree polynomial orbit can remain near those vectors.

This is the natural language for turning qualitative statements about “being close to a proper homogeneous subspace” into quantitative estimates and algebraic alternatives.

CHAPTER 8

Joinings, Factors, and Algebraic Correspondences

1. Why study two copies at once?

Ergodicity describes the statistical behavior of one system. A joining asks how two systems can be coupled while preserving both of their marginal statistics. This is a natural rigidity question: if each system is individually homogeneous, are there nontrivial ways for their orbits to remain correlated?

For unipotent flows, self-joinings are especially important because nearby-orbit comparisons naturally produce measures on a product space. A joining can encode an actual graph of a symmetry, a finite algebraic correspondence, or a genuinely independent product.

2. Definition of a joining

Let G act measure preservingly on probability spaces (X_1, μ_1) and (X_2, μ_2) .

Definition 8.2.1. A *joining* of the two G -systems is a G -invariant probability measure λ on

$$X_1 \times X_2$$

for the diagonal action

$$g(x_1, x_2) = (gx_1, gx_2),$$

such that the coordinate projections satisfy

$$(\pi_i)_* \lambda = \mu_i, \quad i = 1, 2.$$

The product measure

$$\mu_1 \otimes \mu_2$$

is always a joining. It represents statistical independence.

Definition 8.2.2. A joining is *ergodic* if the diagonal G -action on $(X_1 \times X_2, \lambda)$ is ergodic.

Every joining decomposes into ergodic joinings, provided the original marginals are ergodic. Indeed, ergodic decomposition of λ preserves the marginal condition almost everywhere because the coordinate marginals are invariant and extreme.

3. Graph joinings

Suppose

$$\Phi : X_1 \rightarrow X_2$$

is a measurable measure-preserving factor isomorphism intertwining the actions:

$$\Phi(gx) = g\Phi(x).$$

Then the pushforward of μ_1 under

$$x \mapsto (x, \Phi(x))$$

is a joining supported on the graph

$$\text{Graph}(\Phi) = \{(x, \Phi(x)) : x \in X_1\}.$$

Proposition 8.3.1. *If the action on (X_1, μ_1) is ergodic, then the graph joining of an equivariant isomorphism Φ is ergodic.*

PROOF. The graph map

$$x \mapsto (x, \Phi(x))$$

is a measure-space isomorphism from (X_1, μ_1) onto the graph with its joining measure, and it intertwines the G -actions. Ergodicity is preserved by measurable conjugacy. $\square$

4. Self-joinings from centralizers

Let T be an ergodic transformation of (X, μ) . If $S : X \rightarrow X$ is a measure-preserving automorphism commuting with T , then

$$\lambda_S = (x \mapsto (x, Sx))_* \mu$$

is a self-joining of T .

For a homogeneous flow generated by $U < G$, right or left translations by elements in the centralizer of U often produce such graph joinings, whenever they are well defined on the quotient.

Worked Example 8.4.1 (Translation joinings on a torus). Let R_α be irrational rotation on $\mathbb{T}$. Every translation

$$S_\beta(x) = x + \beta$$

commutes with R_α . Hence

$$\lambda_\beta$$

supported on

$$y = x + \beta$$

is an ergodic self-joining. Thus even a very rigid ergodic system can have infinitely many nonproduct joinings.

5. Classification of ergodic joinings of irrational rotations

The torus model can be solved completely and illustrates the algebraic nature of joinings.

Consider rotations

$$R_\alpha : x \mapsto x + \alpha, \quad R_\beta : y \mapsto y + \beta$$

on two circles. A joining is invariant under translation by

$$(\alpha, \beta)$$

on $\mathbb{T}^2$.

THEOREM 8.5.1. *The ergodic joinings of $(\mathbb{T}, R_\alpha, m)$ and $(\mathbb{T}, R_\beta, m)$ are Haar measures on closed cosets of the closure*

$$H = \overline{\{n(\alpha, \beta) : n \in \mathbb{Z}\}} \subset \mathbb{T}^2$$

whose coordinate projections are surjective.

PROOF. The transformation on $\mathbb{T}^2$ is translation by (α, β) . The closure H of the cyclic subgroup is a compact subgroup. Every orbit is contained in a coset $z + H$, and the translation is minimal on that coset by definition of H . Haar probability measure on $z + H$ is invariant and ergodic: invariant functions pull back to functions on H invariant under a dense cyclic subgroup, hence under all of H , and are therefore constant.

Conversely, let λ be an ergodic invariant probability measure. The quotient map

$$q : \mathbb{T}^2 \rightarrow \mathbb{T}^2/H$$

is invariant under translation by (α, β) , because that translation lies in H . Therefore $q_*\lambda$ is an invariant factor on which the transformation is the identity. Ergodicity forces every such factor to be trivial, so $q_*\lambda$ is a point mass. Hence λ is supported on one coset $z + H$.

On that coset, invariance under the generator implies invariance under the dense cyclic subgroup it generates. For a continuous test function, translation depends continuously on the translating element, so invariance extends from the dense subgroup to all of H . The unique probability measure invariant under all translations of the compact group H is Haar measure. Thus λ is Haar measure on $z + H$. The marginal conditions are exactly surjectivity of the coordinate projections. $\square$

Worked Example 8.5.2 (Independent versus resonant rotations). If $1, \alpha, \beta$ are rationally independent, then $H = \mathbb{T}^2$, so the only joining is the product measure.

If $\beta = \alpha$, then

$$H = \{(t, t) : t \in \mathbb{T}\},$$

and its cosets are the graphs

$$y = x + c.$$

Thus all ergodic self-joinings are graph joinings by translations.

This elementary theorem is a complete algebraic joining classification. It foreshadows what is remarkable about homogeneous unipotent systems: their ergodic joinings are often again supported on algebraically defined homogeneous subspaces.

6. Factors

Definition 8.6.1. A *factor map* from (X, μ, G) to (Y, ν, G) is a measurable map

$$\pi : X \rightarrow Y$$

such that

$$\pi_*\mu = \nu$$

and

$$\pi(gx) = g\pi(x)$$

for almost every x .

Factors are quotient systems in the measurable category. A factor map creates a canonical joining between X and Y , namely the graph measure of π . Conversely, a joining whose conditional measure on almost every fiber of the first projection is a point mass comes from a factor map.

7. Disintegration of a joining

Let λ be a joining of (X, μ) and (Y, ν) . Disintegrate over the first coordinate:

$$(8.1) \quad \lambda = \int_X \delta_x \otimes \lambda_x d\mu(x),$$

where λ_x is a probability measure on Y for almost every x .

Invariance of λ implies an equivariance relation for the conditional measures:

$$(8.2) \quad \lambda_{gx} = g_*\lambda_x \quad \text{for almost every } x.$$

This follows from uniqueness of disintegration.

The family $x \mapsto \lambda_x$ tells us how much uncertainty remains in the second coordinate after the first is known.

- Product joining: $\lambda_x = \nu$ almost everywhere.
- Graph joining: $\lambda_x = \delta_{\Phi(x)}$.

- Finite correspondence: λ_x is supported on finitely many points, often with equal weights.

8. The operator associated with a joining

A joining can also be encoded by an operator. This viewpoint is useful because geometric information about the support of a joining becomes spectral or algebraic information about an intertwining map.

Let G act measure preservingly on (X, μ) and (Y, ν) , and let λ be a joining. Disintegrate over the first coordinate:

$$\lambda = \int_X \delta_x \otimes \lambda_x d\mu(x).$$

Define

$$(8.3a) \quad P_\lambda : L^2(Y, \nu) \rightarrow L^2(X, \mu), \quad (P_\lambda f)(x) = \int_Y f(y) d\lambda_x(y).$$

Proposition 8.8.1 (Joining operator). *The operator P_λ has the following properties.*

- (1) P_λ is linear, positive, and $P_\lambda 1 = 1$.
- (2) It is a contraction on L^2 :

$$\|P_\lambda f\|_2 \leq \|f\|_2.$$

- (3) It preserves integrals:

$$\int_X P_\lambda f d\mu = \int_Y f d\nu.$$

- (4) It intertwines the Koopman representations:

$$(8.3b) \quad P_\lambda \pi_Y(g) = \pi_X(g) P_\lambda \quad (g \in G).$$

PROOF. Linearity and positivity follow directly from integration against the conditional measures. Jensen's inequality gives

$$|P_\lambda f(x)|^2 \leq \int_Y |f(y)|^2 d\lambda_x(y).$$

Integrating in x and using that the second marginal of λ is ν proves the contraction estimate. The same marginal identity proves preservation of integrals.

For equivariance, invariance of the joining implies the conditional-measure relation

$$\lambda_{gx} = g_* \lambda_x$$

for almost every x . Hence

$$\begin{aligned} (P_\lambda \pi_Y(g)f)(x) &= \int f(g^{-1}y) d\lambda_x(y) \\ &= \int f(y) d\lambda_{g^{-1}x}(y) \\ &= (\pi_X(g)P_\lambda f)(x). \end{aligned}$$

This is (8.3b).

The converse correspondence between joinings and Markov operators is also true for standard Borel probability spaces. Its proof is a measure-theoretic representation theorem for positive unital operators and is not needed for the arguments here, so we do not use it as an implicit step. $\square$

Worked Example 8.8.2 (Graph joining). If $\Phi : X \rightarrow Y$ is a factor map and the joining is supported on the graph $(x, \Phi(x))$, then

$$(P_\lambda f)(x) = f(\Phi(x)).$$

Thus graph joinings correspond to deterministic Markov operators: the conditional measure λ_x is a point mass.

Worked Example 8.8.3 (Finite correspondence). If a finite homogeneous correspondence has m equally weighted sheets $y_1(x), \dots, y_m(x)$, then

$$(8.3c) \quad (P_\lambda f)(x) = \frac{1}{m} \sum_{j=1}^m f(y_j(x)).$$

The joining operator is therefore an averaging operator rather than composition with a single map. Hecke operators are the basic arithmetic example of this mechanism.

Guiding Idea 8.8.4. The operator formalism gives a useful dictionary:

joining geometry	operator property
graph	composition operator
finite correspondence	finite averaging operator
product joining	$f \mapsto \int f d\nu$
nontrivial joining	nontrivial Koopman intertwiner.

Thus joining rigidity can often be read as rigidity of positive intertwiners.

9. Composition of joinings and composition of operators

Suppose λ is a joining of X with Y , and κ is a joining of Y with Z . Their relatively independent composition over Y has conditional kernel

$$(8.3d) \quad \theta_x(C) = \int_Y \kappa_y(C) d\lambda_x(y).$$

The associated operators satisfy the exact identity

$$(8.3e) \quad P_\theta = P_\lambda P_\kappa.$$

Indeed,

$$\begin{aligned} (P_\theta f)(x) &= \int_Z f(z) d\theta_x(z) \\ &= \int_Y \left(\int_Z f(z) d\kappa_y(z) \right) d\lambda_x(y) \\ &= (P_\lambda(P_\kappa f))(x). \end{aligned}$$

For finite correspondences, formula (8.3d) retains multiplicity automatically. Two different intermediate sheets that land at the same final point contribute twice to the conditional measure even if the set-theoretic relational composition records that final point only once. This is the cleanest way to see why measure-theoretic composition can contain more information than composition of bare relations.

10. Finite correspondences

Definition 8.10.1. A measurable relation $C \subset X \times Y$ is a *finite correspondence of degree (m, n)* if, generically,

- the first-coordinate fiber

$$C_x = \{y : (x, y) \in C\}$$

has m points;

- the second-coordinate fiber

$$C^y = \{x : (x, y) \in C\}$$

has n points.

The word “generically” means outside a null set or lower-dimensional exceptional set, depending on whether the context is measurable or geometric.

A graph is the special case $m = 1$. A finite correspondence need not be the graph of a globally defined function because the projection can have several sheets.

Guiding Idea 8.10.2. The degree of a projection is not the number of connected components of the correspondence. It is the generic number of points in a fiber. A single connected component can project with degree larger than one, just as a connected covering space can have several sheets.

11. Relational composition

If

$$C \subset X \times Y, \quad D \subset Y \times Z,$$

their *relational composition* is

$$(8.3) \quad D \circ C = \{(x, z) : \exists y \in Y \text{ with } (x, y) \in C, (y, z) \in D\}.$$

This is exactly ordinary function composition when both relations are graphs. For multi-valued correspondences, the intermediate point y is existentially quantified.

Worked Example 8.11.1 (A two-valued relation). Let $X = Y = \mathbb{T}$ and

$$C = \{(x, y) : 2y = x \pmod{1}\}.$$

For every x , there are two possible y 's, so the first projection has degree two. Composing C with itself gives

$$C \circ C = \{(x, z) : 4z = x \pmod{1}\},$$

which has degree four over the first coordinate.

Relational composition is useful because symmetries of a joining can be multiplied even when no single-valued map is available.

12. Invariant correspondences

Suppose G acts on both X and Y . A relation $C \subset X \times Y$ is G -invariant if

$$(x, y) \in C \implies (gx, gy) \in C$$

for every $g \in G$.

If C and D are invariant, then $D \circ C$ is invariant. Indeed, if $(x, z) \in D \circ C$, choose y witnessing the relation. Then $(gx, gy) \in C$ and $(gy, gz) \in D$, so $(gx, gz) \in D \circ C$.

This simple observation is the algebra behind using repeated composition to turn a finite correspondence into a finite group of deck-like symmetries.

13. Homogeneous correspondences from commensurators

Let

$$X = \Gamma \backslash G$$

and let $q \in G$ commensurate Γ , meaning

$$\Gamma \cap q^{-1}\Gamma q$$

has finite index in both Γ and $q^{-1}\Gamma q$.

The map

$$\Gamma g \longmapsto \Gamma qg$$

is generally not well defined on X : changing g by $\gamma \in \Gamma$ changes qg by $q\gamma q^{-1}$, which need not lie in Γ . Instead q defines a finite correspondence.

Let

$$\Gamma_q = \Gamma \cap q^{-1}\Gamma q.$$

The intermediate finite cover

$$Y_q = \Gamma_q \backslash G$$

has finite covering maps

$$p_1 : Y_q \rightarrow \Gamma \backslash G, \quad p_1(\Gamma_q g) = \Gamma g,$$

and

$$p_2 : Y_q \rightarrow \Gamma \backslash G, \quad p_2(\Gamma_q g) = \Gamma qg.$$

The image of

$$(p_1, p_2) : Y_q \rightarrow X \times X$$

is a finite algebraic correspondence.

Proposition 8.13.1. *The correspondence defined by q is invariant under the diagonal right action of G .*

PROOF. For $h \in G$,

$$(p_1, p_2)(\Gamma_q gh) = (\Gamma gh, \Gamma qgh),$$

which is exactly the diagonal right translate by h of

$$(\Gamma g, \Gamma qg).$$

□

Remark 8.13.2. This is the correct way to interpret “right G -invariance” of a finite homogeneous correspondence. It is not a mysterious action on graph components; it follows because both coordinate maps are induced from right multiplication on the same intermediate homogeneous space.

14. Degrees of the coordinate projections

The degree of

$$p_1 : Y_q \rightarrow X$$

is

$$[\Gamma : \Gamma_q].$$

To see this, the fiber over Γg consists of cosets

$$\Gamma_q \gamma g$$

with $\gamma \in \Gamma$, modulo equality exactly when $\gamma_1 \gamma_2^{-1} \in \Gamma_q$. Hence the fiber is naturally

$$\Gamma_q \backslash \Gamma.$$

For the second projection, use the coordinate $g' = qg$. In that coordinate the stabilizing subgroup is

$$q\Gamma_q q^{-1} = \Gamma \cap q\Gamma q^{-1},$$

so

$$\deg(p_2) = [\Gamma : \Gamma \cap q\Gamma q^{-1}].$$

Thus the two degrees need not be equal a priori, although in many arithmetic examples they are.

This computation explains why a finite correspondence naturally carries projection degrees. The degrees are subgroup indices, not geometric guesses from a picture of the graph.

15. Joining measure on a homogeneous correspondence

If Y_q has finite Haar volume, normalize its Haar measure m_{Y_q} . Push it forward by (p_1, p_2) :

$$\lambda_q = (p_1, p_2)_* m_{Y_q}.$$

Then λ_q is a self-joining of the right G -action, and hence of every subgroup flow.

The two marginals are Haar measure on X because p_1, p_2 are finite homogeneous covering maps and preserve normalized invariant measure.

Thus commensurators produce canonical nontrivial algebraic self-joinings.

16. Graph type versus genuine correspondence type

If q normalizes Γ , then

$$q^{-1}\Gamma q = \Gamma,$$

so $\Gamma_q = \Gamma$ and both projections have degree one. The correspondence is the graph of the well-defined map

$$\Gamma g \mapsto \Gamma qg.$$

If q merely commensurates Γ , the degree can exceed one, and no single-valued map exists on the original quotient. The intermediate cover is indispensable.

This distinction is one of the cleanest examples of why “graph-type component” and “finite correspondence” should not be conflated.

Guiding Idea 8.16.1 (Three levels of dependence). The product, graph, and genuine finite-correspondence cases can be read as three increasingly rigid kinds of dependence between the two coordinates. For the product joining, knowing x gives no information about y . For a graph joining, y is determined uniquely by x . For a degree- m correspondence, x determines a finite set of possible y 's, together with conditional weights supplied by the joining measure.

This viewpoint is often more useful than the bare set-theoretic picture. A correspondence is not merely “a graph with several values”: its invariant measure records multiplicities and makes composition compatible with the associated Markov operators.

17. Composition and commensurators

Composing the correspondences associated with q_1 and q_2 produces, after resolving finite covers and possible multiplicities, the correspondence associated with the product q_2q_1 . At the set-theoretic level this is immediate from

$$\Gamma g \xrightarrow{q_1} \Gamma q_1 g \xrightarrow{q_2} \Gamma q_2 q_1 g.$$

The finite-index subgroups determine how many intermediate points occur.

Thus relational composition reflects multiplication in the commensurator. This is why repeated composition can reveal hidden algebraic symmetry groups behind self-joinings.

18. Conditional measures on a finite homogeneous correspondence

The finite-cover construction has a particularly concrete disintegration. Suppose

$$p_1 : Y \rightarrow X$$

is a finite homogeneous covering of degree m , both spaces carry normalized invariant probability measure, and $\lambda = (p_1, p_2)_* m_Y$ is the associated correspondence joining.

For almost every $x \in X$, the fiber $p_1^{-1}(x)$ consists of exactly m points. The conditional measure of m_Y on this fiber is the uniform measure

$$(8.4) \quad \frac{1}{m} \sum_{y \in p_1^{-1}(x)} \delta_y.$$

To see this, work in an evenly covered neighborhood $O \subset X$. Its inverse image is a disjoint union

$$p_1^{-1}(O) = O_1 \sqcup \cdots \sqcup O_m,$$

and each restriction $p_1 : O_j \rightarrow O$ is a diffeomorphism. Homogeneous invariance gives the same Jacobian on every sheet, so normalized Haar measure assigns equal conditional weight to the sheets. Uniqueness of disintegration then yields (8.4) almost everywhere.

Pushing the fiber through p_2 , the conditional measure of the joining over its first coordinate is therefore uniform on the finite correspondence fiber, with multiplicities retained if different points of the intermediate cover have the same second image.

Guiding Idea 8.18.1. A finite homogeneous correspondence is not merely a finite set-valued map. Its invariant measure canonically tells us how the different sheets are weighted. In the unramified homogeneous setting the weights are equal because all sheets arise from the same Haar measure on one finite cover.

19. A complete arithmetic example: Hecke correspondences

The modular surface supplies the standard arithmetic example. Let

$$\Gamma = \mathrm{PSL}_2(\mathbb{Z}), \quad X = \Gamma \backslash \mathrm{PSL}_2(\mathbb{R}),$$

and let p be prime. Consider the projective class of

$$q_p = \begin{pmatrix} \sqrt{p} & 0 \\ 0 & p^{-1/2} \end{pmatrix}.$$

It commensurates Γ . Equivalently, after multiplying by the scalar $\sqrt{p}$, one is looking at the rational matrix

$$\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}$$

in $\mathrm{PGL}_2^+(\mathbb{Q})$.

The associated double coset has the decomposition

$$(8.5) \quad \Gamma q_p \Gamma = \Gamma q_p \sqcup \coprod_{b=0}^{p-1} \Gamma \frac{1}{\sqrt{p}} \begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix}.$$

Thus the correspondence has $p+1$ sheets over a generic first coordinate.

WHY THERE ARE $p+1$ SHEETS. It is enough to work before determinant normalization with integer matrices of determinant p . Any index- p sublattice of $\mathbb{Z}^2$ has quotient $\mathbb{Z}/p\mathbb{Z}$, hence is the kernel of a nonzero homomorphism

$$\mathbb{Z}^2 \rightarrow \mathbb{Z}/p\mathbb{Z}$$

up to multiplication by a nonzero scalar. Such kernels are parametrized by the projective line

$$\mathbf{P}^1(\mathbf{F}_p),$$

which has $p+1$ points. One representative is the sublattice generated by $(p, 0), (0, 1)$; the other p are generated by $(1, b), (0, p)$ for $b \bmod p$. Choosing basis matrices for these sublattices gives exactly the representatives in (8.5), after scaling determinant p matrices to determinant one. $\square$

The resulting self-correspondence on the modular surface is the geometric form of the classical Hecke operator T_p . Its joining measure is supported on a proper finite-volume homogeneous subset of $X \times X$, and the conditional measure over a generic point is uniform on $p + 1$ Hecke neighbors.

Remark 8.19.1. This example is useful far beyond modular forms. It shows concretely how arithmetic commensurators appear dynamically as nontrivial self-joinings. The “multi-valuedness” is not pathology; it is the geometric shadow of finite-index arithmetic intersections.

20. Ergodicity of commensurator joinings for the horocycle flow

Let $q \in \text{Comm}_G(\Gamma)$, and let

$$Y_q = \Gamma_q \backslash G$$

be the intermediate finite-volume quotient. The map

$$(p_1, p_2) : Y_q \rightarrow X \times X$$

intertwines right translation by every subgroup of G . If $G = \text{PSL}_2(\mathbb{R})$ and U is a horocycle subgroup, then the right U -action on Y_q is ergodic by the unipotent Mautner theorem. Hence its pushforward joining λ_q is an ergodic U -self-joining.

This gives an explicit family of nonproduct ergodic self-joinings. The construction proceeds in four transparent stages:

$$(8.6) \quad \begin{array}{l} \text{commensurator element} \longrightarrow \text{finite cover,} \\ \text{finite cover} \longrightarrow \text{finite correspondence,} \\ \text{finite correspondence} \longrightarrow \text{ergodic joining.} \end{array}$$

The graph case occurs exactly when the finite cover collapses to degree one, for example when q normalizes Γ . Genuine commensurator elements produce higher-degree joinings.

21. Composition, multiplicity, and why degrees need not multiply naively

Suppose $C \subset X \times Y$ has generic degree m over X , and $D \subset Y \times Z$ has generic degree n over Y . One might expect $D \circ C$ to have degree mn . This is only true when the mn possible intermediate paths lead to distinct points of Z .

Worked Example 8.21.1 (Multiplicity under composition). On the circle let

$$C_k = \{(x, y) : ky = x \pmod{1}\}.$$

Then C_k has degree k over x , and

$$C_\ell \circ C_k = C_{k\ell},$$

so the degrees do multiply.

By contrast, if a relation has nontrivial deck symmetries, two different intermediate points can lead to the same final point. Then the set-theoretic degree drops while the composed joining can retain multiplicity in its conditional measure. For this reason one should distinguish a correspondence as a subset from the same correspondence equipped with its natural measure or cycle multiplicity.

For homogeneous correspondences the correct composition is most naturally performed by fiber products of the intermediate covers. The connected components of that fiber product encode the relevant double-coset decomposition. This is the geometric origin of the algebra of Hecke correspondences.

22. Finite correspondences as measurable factors with finite fibers

A graph joining corresponds to a factor map because its conditional measures are point masses. A degree- m finite correspondence is the next simplest possibility: its conditional measures have exactly m atoms almost everywhere.

Suppose a joining λ of two ergodic systems has a disintegration

$$\lambda = \int_X \delta_x \otimes \lambda_x d\mu(x)$$

with λ_x supported on exactly m points for almost every x . Equivariance gives

$$\lambda_{gx} = g_* \lambda_x.$$

If the atoms have distinct weights, the weight vector itself is an invariant measurable datum along the orbit. Ergodicity therefore forces the unordered list of weights to be almost everywhere constant. In many homogeneous situations additional symmetry permutes the sheets transitively, forcing all weights to be $1/m$.

Thus the passage

$$\text{point mass} \quad \rightsquigarrow \quad \text{finite atomic conditional measure}$$

is the measurable analogue of passing from a function to a finite-valued correspondence.

23. An orbit-pair test for graph type

Suppose an ergodic joining λ is supported on a measurable relation $C \subset X \times X$. If the first projection has degree one almost everywhere, then C is graph type modulo null sets.

Indeed, disintegration over the first coordinate gives

$$\lambda_x = \delta_{\Phi(x)}$$

for a measurable map Φ . Invariance of the joining implies

$$\Phi(gx) = g\Phi(x)$$

almost everywhere. The second marginal condition makes Φ measure preserving. Hence degree one is not merely suggestive of a graph: in the measurable category it literally produces an equivariant map after discarding a null set.

For degree larger than one no such global single-valued choice is generally compatible with the dynamics. Attempting to choose one branch may destroy equivariance by monodromy or deck transformations.

24. Joinings and weak mixing

A system is weakly mixing if and only if its product with every ergodic system is ergodic. Equivalently, the product self-joining

$$\mu \otimes \mu$$

is ergodic for $T \times T$ after excluding compact eigenfunction obstructions in the usual formulation.

Joinings provide a more refined question than weak mixing: even if the product joining is ergodic, there may be other ergodic self-joinings supported on graphs or finite correspondences.

For homogeneous systems, classifying all such joinings can reveal the centralizer and commensurator of the action in measurable form.

25. Disjointness

Definition 8.25.1. Two ergodic systems are *disjoint* if their only joining is the product joining.

The irrational-rotation calculation showed a simple arithmetic criterion: two circle rotations with no rational resonance have only the product joining. In more complicated systems, disjointness can follow from incompatible entropy, spectrum, algebraic structure, or orbit-growth properties.

26. Joinings as measures on orbit pairs

For a flow u_t , a pair of points (x, y) determines empirical measures

$$\lambda_T = \frac{1}{T} \int_0^T \delta_{(u_t x, u_t y)} dt.$$

If the family is tight, every subsequential weak-* limit is a self-joining of any common limiting marginal measure. Polynomial divergence controls the relative coordinate between the two orbit points, so the support of λ can encode the limiting shearing relation.

This is one reason joinings arise naturally rather than artificially: they are the invariant measures generated by following two trajectories simultaneously.

27. A modern research perspective

Quantitative joining questions appear when one samples homogeneous flows along sparse sequences. Correlations such as

$$\frac{1}{N} \sum_{n \leq N} f(u_{pn}x) g(u_{qn}x)$$

are orbit averages on the product system under

$$(u_{pt}, u_{qt}).$$

Understanding possible invariant joinings of differently rescaled flows is therefore a structural input to prime-time, polynomial-time, and arithmetic-sampling problems.

Qualitative joining rigidity says which limiting correlations are possible. Effective joining rigidity asks how fast finite orbit pairs approach those algebraic alternatives.

28. Conceptual summary

A joining is an invariant coupling:

$$\lambda \in \mathbb{P}(X_1 \times X_2), \quad (\pi_i)_* \lambda = \mu_i.$$

Its disintegration records the conditional relation between the two coordinates. The principal algebraic possibilities are:

$$\begin{aligned} \text{product} &\leftrightarrow \text{independence,} \\ \text{graph joining} &\leftrightarrow \text{single-valued symmetry,} \\ \text{finite correspondence} &\leftrightarrow \text{multi-valued algebraic symmetry.} \end{aligned}$$

On homogeneous spaces, commensurators naturally produce finite correspondences through finite covers. Projection degrees are subgroup indices, and relational composition mirrors multiplication of commensurator elements.

These concrete constructions explain why joining rigidity is a powerful way to detect hidden algebraic structure in orbit correlations.

CHAPTER 9

A Rigidity Toolkit: Generic Blocks, Stabilizers, and Additional Invariance

1. The recurring architecture

Several proofs in unipotent dynamics share the same internal structure even when their final theorems look very different. One starts with an invariant ergodic measure μ , chooses generic recurrent points, compares two nearby trajectories, rescales their polynomial divergence, and tries to convert the limiting displacement into a new symmetry of μ .

This chapter isolates that mechanism without assuming a particular classification theorem. The goal is to make the logic modular: one can see exactly which statements come from ergodic theory, which come from Lie theory, and which require genuinely new geometric input.

The schematic chain is

nearby generic points + long generic blocks + polynomial shearing $\implies$ additional invariance.
--

2. The stabilizer of a measure

Let a Lie group G act continuously on a locally compact second-countable space X , and let μ be a Radon probability measure.

Definition 9.2.1. The *stabilizer of μ* is

$$\text{Stab}_G(\mu) = \{g \in G : g_*\mu = \mu\}.$$

Proposition 9.2.2. $\text{Stab}_G(\mu)$ is a closed subgroup of G .

PROOF. The subgroup property follows from $(gh)_* = g_*h_*$ and $(g^{-1})_* = (g_*)^{-1}$. Suppose $g_n \in \text{Stab}_G(\mu)$ and $g_n \rightarrow g$. For $f \in C_c(X)$, continuity of the action gives

$$f(g_n x) \rightarrow f(gx)$$

pointwise. Since $|f(g_n x)| \leq \|f\|_\infty$ and μ is a probability measure, dominated convergence yields

$$\int_X f(gx) d\mu(x) = \lim_n \int_X f(g_n x) d\mu(x) = \int_X f(x) d\mu(x).$$

Thus $g_*\mu = \mu$. Hence the stabilizer is closed. $\square$

By the closed subgroup theorem, $\text{Stab}_G(\mu)$ is a Lie subgroup. Its identity component therefore has a Lie algebra, giving a precise infinitesimal meaning to “additional invariance.”

3. Why the identity component matters

Set

$$L = \text{Stab}_G(\mu)^\circ.$$

If the original acting unipotent subgroup U is connected and preserves μ , then

$$U < L.$$

An additional-invariance argument seeks to prove that L is strictly larger than the symmetry group already known.

The advantage of passing to L is local: every stabilizing element sufficiently close to the identity belongs to L . Therefore a limiting polynomial displacement near e can be converted into a Lie-algebra direction of the stabilizer.

4. The Lie algebra of the measure stabilizer

Because $\text{Stab}_G(\mu)$ is a closed Lie subgroup, its Lie algebra can be described directly in terms of one-parameter invariance:

$$(9.1a) \quad \mathfrak{s}_\mu = \{Y \in \mathfrak{g} : (\exp tY)_*\mu = \mu \text{ for every } t \in \mathbb{R}\}.$$

This is not merely notation. It gives a practical way to convert a limiting curve of stabilizing elements into a linear direction.

If $q(s) \subset \text{Stab}_G(\mu)$ is a differentiable curve and $q(s_0) = g_0$, then

$$Y = (dL_{g_0^{-1}})_{g_0} q'(s_0)$$

belongs to $\mathfrak{s}_\mu$. Indeed,

$$g_0^{-1}q(s) \subset \text{Stab}_G(\mu)$$

is a curve through the identity. Hence

$$\exp(tY) \subset \text{Stab}_G(\mu).$$

Remark 9.4.1. The derivative must be translated back to the identity. Writing simply $q'(s_0) \in \mathfrak{g}$ without specifying a tangent-space identification is harmless in matrix coordinates but ambiguous on an abstract Lie group.

If U preserves μ , then $\mathfrak{u} \subset \mathfrak{s}_\mu$. To obtain genuinely new invariance one must produce a tangent direction whose class in

$$\mathfrak{g}/\mathfrak{s}_\mu$$

is nonzero.

5. Genericity on proportional subintervals

Let u_t be an ergodic measure-preserving flow and let x be generic for μ . Genericity on $[0, T]$ automatically gives genericity on every fixed proportional block.

Proposition 9.5.1 (Block genericity). *Let $0 \leq a < b < \infty$. If for every $f \in C_c(X)$,*

$$\frac{1}{T} \int_0^T f(u_t x) dt \rightarrow \int f d\mu,$$

then

$$(9.1) \quad \frac{1}{(b-a)T} \int_{aT}^{bT} f(u_t x) dt \rightarrow \int f d\mu.$$

PROOF. For $a > 0$, write

$$\int_{aT}^{bT} f(u_t x) dt = \int_0^{bT} f(u_t x) dt - \int_0^{aT} f(u_t x) dt.$$

Divide by $(b-a)T$:

$$\frac{b}{b-a} \left(\frac{1}{bT} \int_0^{bT} f(u_t x) dt \right) - \frac{a}{b-a} \left(\frac{1}{aT} \int_0^{aT} f(u_t x) dt \right).$$

Both bracketed terms converge to $\int f d\mu$, and the coefficients combine to one. If $a = 0$, the left-hand side of (9.1) is exactly $\frac{1}{bT} \int_0^{bT} f(u_t x) dt$, which converges to $\int f d\mu$ by the assumed genericity with time parameter $bT \rightarrow \infty$. $\square$

Remark 9.5.2. This elementary fact is extremely useful. It lets us zoom into a fixed macroscopic fraction of a long orbit segment without losing genericity. It does *not* permit arbitrary shrinking intervals whose relative length tends to zero.

6. Uniform block genericity for a chosen sequence

In rigidity arguments the starting points may vary: $x_n \rightarrow x$, $y_n \rightarrow x$. Individual genericity of each x_n does not imply uniform convergence at a predetermined time T_n . The standard solution is to choose T_n after the points are chosen.

Fix a finite set of test functions $\mathcal{F}_n \subset C_c(X)$ and finitely many rational blocks $[a, b] \subset [0, 1]$. Because x_n and y_n are generic, one can choose T_n so

large that all required block averages for all functions in $\mathcal{F}_n$ are within $1/n$ of their μ -integrals. Taking $\mathcal{F}_n$ increasing and dense provides a diagonal genericity scheme.

The point is methodological: one should never write “take $T_n \rightarrow \infty$, so the varying generic points are generic at time T_n ” without making this choice explicit.

7. A diagonal selection lemma for varying generic points

The informal instruction “choose T_n large enough” can be made precise in a form tailored to shearing arguments.

Proposition 9.7.1 (Simultaneous block selection). *Let x_n, y_n be points generic for the same probability measure μ under an ergodic flow u_t . Let*

$$\{f_j : j \geq 1\} \subset C_c(X)$$

be a countable convergence-determining family, and enumerate the rational subintervals of $(0, 1)$ as $J_1, J_2, \dots$. Then one can choose $T_n \rightarrow \infty$ so that for every fixed j, k ,

$$(9.1b) \quad \frac{1}{|J_k|T_n} \int_{J_k T_n} f_j(u_t x_n) dt \longrightarrow \int f_j d\mu,$$

and the same holds with y_n in place of x_n .

PROOF. For each fixed point x_n , genericity and the proportional-block lemma imply that, for every finite collection of pairs (j, k) , all the corresponding block averages converge to the required integrals as $T \rightarrow \infty$. The same is true for y_n .

At stage n , impose the finitely many conditions

$$1 \leq j, k \leq n$$

for both points and require every error to be less than $1/n$. There is a threshold R_n beyond which all these finitely many conditions hold. Choose

$$T_n > \max\{n, T_{n-1} + 1, R_n\}.$$

Then $T_n \rightarrow \infty$, and for fixed j, k the condition is enforced for every $n \geq \max\{j, k\}$, with error at most $1/n$. This proves (9.1b). $\square$

To pass from the determining family to every $f \in C_c(X)$, choose the family dense on each compact set and include a sequence of compactly supported cutoff functions increasing to one. The cutoffs prevent loss of mass, while uniform approximation on the relevant compact supports extends the convergence. Thus the block-genericity hypothesis in the shearing lemma is obtainable by a genuine diagonal construction whenever the starting points are individually generic.

Warning 9.7.2. The times T_n depend on the points x_n, y_n . Genericity does not imply that an arbitrary preassigned sequence of times works uniformly for a varying sequence of starting points.

8. A compact-uniformity lemma

The next technical issue is comparing $f(gz)$ and $f(hz)$ when g and h are close. On a noncompact space this must be localized.

Lemma 9.8.1. *Let X be a locally compact second-countable Hausdorff space, let G act continuously, and let $f \in C_c(X)$. Fix a compact set $K \subset X$ and a compact set $Q \subset G$. Then the map*

$$Q \times K \rightarrow \mathbb{C}, \quad (g, x) \mapsto f(gx)$$

is uniformly continuous.

PROOF. A locally compact second-countable Hausdorff space is metrizable. Fix a compatible metric on X , and any compatible metric on the Lie group G . The product $Q \times K$ is compact, and the displayed map is continuous. Hence it is uniformly continuous on $Q \times K$. $\square$

For bounded continuous f without compact support, the same conclusion holds on $Q \times K$. The contribution of the complement of K is controlled by its small μ -measure.

9. Three elementary tests for measure invariance

Before using orbit comparisons, it is useful to isolate exactly what has to be proved in order to show that an element $g \in G$ preserves μ .

Proposition 9.9.1 (Test functions detect invariance). *For $g \in G$, the following are equivalent:*

- (1) $g_*\mu = \mu$;
- (2) for every $f \in C_c(X)$,

$$(9.2a) \quad \int_X f(gx) d\mu(x) = \int_X f(x) d\mu(x);$$

- (3) (9.2a) holds for every function in a fixed countable convergence-determining family in $C_c(X)$.

PROOF. The implication (1) $\Rightarrow$ (2) is the definition of pushforward. If (2) holds, then the Radon measures $g_*\mu$ and μ have the same integrals against all compactly supported continuous functions; uniqueness in the Riesz representation theorem gives equality of the measures. A convergence-determining family gives the same conclusion because equality on that family determines a Radon probability measure. $\square$

Lemma 9.9.2 (Closedness under approximate invariance). *Suppose $g_n \rightarrow g$ in G , and for every $f \in C_c(X)$,*

$$(9.2b) \quad \int f(g_n x) d\mu(x) - \int f(x) d\mu(x) \rightarrow 0.$$

Then $g_\mu = \mu$.*

PROOF. For fixed $f \in C_c(X)$, continuity of the action gives

$$f(g_n x) \rightarrow f(gx)$$

pointwise. The functions are uniformly bounded by $\|f\|_\infty$, so dominated convergence turns (9.2b) into (9.2a). $\square$

Lemma 9.9.3 (A long-block comparison criterion). *Assume X is metrizable and fix a compatible metric d . Let $x_n, y_n \in X$ and let $I_n \subset \mathbb{R}$ be intervals with $|I_n| \rightarrow \infty$. Suppose both empirical measures*

$$\frac{1}{|I_n|} \int_{I_n} \delta_{u_t x_n} dt, \quad \frac{1}{|I_n|} \int_{I_n} \delta_{u_t y_n} dt$$

converge weak- $$ to μ . Suppose there are $g_n \rightarrow g$, a compact set $K \subset X$, and measurable sets $J_n \subset I_n$ such that*

$$(9.2c) \quad \frac{|I_n \setminus J_n|}{|I_n|} \rightarrow 0,$$

for every $t \in J_n$ both $u_t y_n$ and $g_n u_t x_n$ lie in K , and

$$(9.2d) \quad \sup_{t \in J_n} d(u_t y_n, g_n u_t x_n) \rightarrow 0.$$

Then $g_\mu = \mu$.*

PROOF. Fix $f \in C_c(X)$. On K , uniform continuity of f and (9.2d) give

$$\sup_{t \in J_n} |f(u_t y_n) - f(g_n u_t x_n)| \rightarrow 0.$$

On the exceptional set $I_n \setminus J_n$, the difference is bounded by $2\|f\|_\infty$, and (9.2c) makes its contribution to the normalized integral tend to zero. Hence

$$\frac{1}{|I_n|} \int_{I_n} f(u_t y_n) dt - \frac{1}{|I_n|} \int_{I_n} f(g_n u_t x_n) dt \rightarrow 0.$$

The first term tends to $\int f d\mu$. To treat the second, choose a compact neighborhood Q of g containing every g_n for large n . The set

$$Q^{-1} \text{supp } f$$

is compact. Outside this set both $f(g_n z)$ and $f(gz)$ vanish, while on it the compact-uniformity lemma gives

$$\sup_z |f(g_n z) - f(gz)| \rightarrow 0.$$

Thus replacing g_n by g changes the normalized integral by $o(1)$. Weak-* convergence of the empirical measures of x_n then gives

$$\frac{1}{|I_n|} \int_{I_n} f(gu_t x_n) dt \longrightarrow \int_X f(gx) d\mu(x).$$

The test-function criterion now yields $g_*\mu = \mu$. □

Guiding Idea 9.9.4. Additional invariance does not come from one exceptional pair of nearby orbit points. It comes from a *positive proportion of a long block* on which two generic empirical distributions can be compared after an almost constant group displacement. Genericity supplies the two copies of μ ; shearing supplies the displacement; compact control makes the comparison uniform.

10. The basic shearing-to-invariance principle

We now formulate a rigorous abstract lemma that isolates the heart of additional invariance.

THEOREM 9.10.1 (Shearing-to-invariance lemma). *Let G act continuously on a locally compact second-countable space X , and let $U = \{u_t\}$ preserve an ergodic Radon probability measure μ . Suppose there exist:*

- *points $x_n, y_n \in X$;*
- *times $T_n \rightarrow \infty$;*
- *continuous maps $q_n : [0, 1] \rightarrow G$;*
- *a continuous map $q : [0, 1] \rightarrow G$;*

with the following properties.

- (1) $q_n \rightarrow q$ uniformly on $[0, 1]$.
- (2) For every $s \in [0, 1]$,

$$(9.2) \quad u_{T_n s} y_n = q_n(s) u_{T_n} x_n.$$

- (3) For every fixed interval $J = [a, b] \subset (0, 1)$ with rational endpoints and every $f \in C_c(X)$,

$$(9.3) \quad \frac{1}{|J|T_n} \int_{aT_n}^{bT_n} f(u_t x_n) dt \rightarrow \int f d\mu,$$

and the same holds with y_n in place of x_n .

Then

$$q(s)_*\mu = \mu \quad \text{for every } s \in (0, 1).$$

In particular,

$$q((0, 1)) \subset \text{Stab}_G(\mu).$$

PROOF. Fix $s_0 \in (0, 1)$. We prove invariance under

$$g_0 = q(s_0).$$

Let $f \in C_c(X)$ and $\eta > 0$.

By inner regularity choose a compact set $K_0 \subset X$ with $\mu(K_0) > 1 - \eta$. Choose $\psi \in C_c(X)$ with $0 \leq \psi \leq 1$ and $\psi = 1$ on K_0 , and set $K = \text{supp } \psi$, which is compact. Because q is continuous and $q_n \rightarrow q$ uniformly, all elements $q_n(s)$ with s in a small fixed neighborhood of s_0 and n large lie in a compact subset $Q \subset G$. Uniform continuity on $Q \times K$ implies that there exists $\delta > 0$ such that

$$|f(q(s)z) - f(g_0z)| < \eta$$

whenever $z \in K$ and $|s - s_0| < 2\delta$. Uniform convergence of q_n gives, for large n ,

$$(9.4) \quad |f(q_n(s)z) - f(g_0z)| < 2\eta$$

for $z \in K$ and $|s - s_0| < \delta$.

Choose a rational interval

$$J = [a, b] \subset (s_0 - \delta, s_0 + \delta)$$

of positive length. By (9.3), block averages along x_n and y_n both converge to μ . Applying (9.3) to ψ , whose integral is at least $\mu(K_0) > 1 - \eta$, shows that for large n the average of $1 - \psi(u_t x_n)$ on the block is less than 2η . Since $\psi = 0$ outside K , the proportion of block times for which $u_t x_n \notin K$ is therefore at most 2η .

Using (9.2) with $s = t/T_n$,

$$f(u_t y_n) = f(q_n(t/T_n)u_t x_n).$$

On the times for which $u_t x_n \in K$, (9.4) gives

$$|f(u_t y_n) - f(g_0 u_t x_n)| < 2\eta.$$

On the exceptional times, the difference is at most $2\|f\|_\infty$. Therefore

$$(9.5) \quad \left| \frac{1}{|J|T_n} \int_{aT_n}^{bT_n} f(u_t y_n) dt - \frac{1}{|J|T_n} \int_{aT_n}^{bT_n} f(g_0 u_t x_n) dt \right| \leq 2\eta + O(\eta\|f\|_\infty).$$

The first average tends to $\int f d\mu$. The second, by block genericity applied to the test function $z \mapsto f(g_0 z)$, tends to

$$\int_X f(g_0 z) d\mu(z).$$

Letting $n \rightarrow \infty$ in (9.5), and then $\eta \rightarrow 0$, gives

$$\int f d\mu = \int f(g_0 z) d\mu(z).$$

Since $f \in C_c(X)$ was arbitrary,

$$(g_0)_*\mu = \mu.$$

Thus $q(s_0) \in \text{Stab}_G(\mu)$. Since s_0 was arbitrary, the result follows. $\square$

Remark 9.10.2. The theorem deliberately assumes the difficult orbit-comparison relation (9.2) and uniform block genericity. It does not hide those tasks. Its contribution is to show that once such data are available, the passage from a limiting polynomial displacement to measure invariance is a clean compactness argument.

11. Why a single good time is not enough

It is tempting to argue as follows: find times t_n for which

$$u_{t_n}y_n \approx g_0u_{t_n}x_n,$$

and conclude that g_0 preserves μ . This is false in general. A single time samples two points, not the statistical distribution of the orbit. Even if both starting points are generic, no information about μ is encoded in one isolated pair of iterates.

What is needed is a block whose length tends to infinity and on which the relative displacement varies slowly enough to be approximated by a fixed element. The block-genericity lemma then turns both orbit averages into μ , while the relative-displacement estimate identifies one average with a translate of the other.

This explains the usual scale separation in a shearing argument. One first chooses a macroscopic time T_n at which the polynomial displacement is order one. Then, around a fixed rescaled parameter s_0 , choose a smaller relative window

$$[s_0 - \delta, s_0 + \delta]T_n.$$

For fixed small $\delta > 0$, this window still has length comparable to T_n , so genericity survives. At the same time, continuity of the limiting polynomial makes $q(s)$ nearly constant when $|s - s_0| < \delta$. The two requirements—long statistical block and nearly constant displacement—are therefore compatible.

Warning 9.11.1. One must take the limits in the correct order. First let $n \rightarrow \infty$ with δ fixed, using block genericity. Only afterwards let $\delta \rightarrow 0$ to identify the displacement with $q(s_0)$. Allowing $\delta = \delta_n \rightarrow 0$ too quickly can destroy genericity because the physical block length $\delta_n T_n$ may fail to tend to infinity.

12. How polynomial divergence produces the rescaled maps

Suppose

$$y_n = \exp(v_n)x_n, \quad v_n \rightarrow 0,$$

and $u_t = \exp(tX)$ is Ad-unipotent. Then

$$u_t y_n = \exp(\text{Ad}(u_t)v_n)u_t x_n.$$

Choose $T_n \rightarrow \infty$ so that

$$\text{Ad}(u_{T_n s})v_n \rightarrow p(s)$$

uniformly for $s \in [0, 1]$, where p is a nonzero polynomial. Then

$$q_n(s) = \exp(\text{Ad}(u_{T_n s})v_n)$$

converges to

$$q(s) = \exp(p(s)).$$

This supplies (9.2) exactly, as long as the chosen representatives in the quotient are consistent.

Thus the shearing-to-invariance theorem converts a polynomial limit map into

$$q((0, 1)) \subset \text{Stab}_G(\mu).$$

13. Extracting a one-parameter subgroup from a polynomial image

The image of $q(s) = \exp(p(s))$ need not itself be a subgroup. Nevertheless, if it lies in a Lie subgroup L , then differentiating at a point gives tangent vectors in $\mathfrak{l}$.

Proposition 9.13.1. *Let $L < G$ be a closed Lie subgroup and let $q : (-\varepsilon, \varepsilon) \rightarrow L$ be a C^1 curve with $q(0) = e$. Then*

$$q'(0) \in \mathfrak{l}.$$

Consequently,

$$\exp(tq'(0)) \in L \quad \text{for all } t \in \mathbb{R}.$$

PROOF. The tangent space of L at the identity is by definition $\mathfrak{l}$, so the derivative of any curve in L through e belongs to $\mathfrak{l}$. Exponentiating a Lie-algebra element of L gives a one-parameter subgroup contained in L . $\square$

If $q(s_0) \neq e$, translate the curve by $q(s_0)^{-1}$ before differentiating. This turns nonhomomorphic polynomial invariance into genuine one-parameter invariance.

14. Leading coefficients and centralizers

For

$$p(s) = \sum_{j=0}^d s^j v_j$$

arising from $\text{Ad}(u_t)v$, the leading coefficient v_d lies in the centralizer Lie algebra of U . Therefore the new invariance often comes from a subgroup commuting with U .

This is more than a convenient feature. A commuting additional symmetry can be combined with U without changing the original orbit parametrization, and the enlarged stabilizer remains well adapted to further recurrence comparisons.

15. Exact quotient identities and the lattice ambiguity

In a group one has the exact identity

$$u_t \exp(v) u_{-t} = \exp(\text{Ad}(u_t)v).$$

On G/Γ , however, two different group elements can carry one point to the same point. Therefore an equality such as

$$u_t y_n = q_n(t) u_t x_n$$

must either be verified from chosen representatives or restricted to an injectivity neighborhood.

Lemma 9.15.1 (Uniform uniqueness on a compact set). *Let $K \subset G/\Gamma$ be compact. There exists a neighborhood Ω of the identity in G such that, for every $x \in K$,*

$$g_1 x = g_2 x, \quad g_1, g_2 \in \Omega,$$

implies $g_1 = g_2$.

PROOF. If no such neighborhood existed, there would be points $x_n = g_n \Gamma \in K$ and nontrivial elements

$$\gamma_n \in g_n \Gamma g_n^{-1}$$

with $\gamma_n \rightarrow e$. After passing to a convergent subsequence of x_n and choosing local lifts in a compact subset of G , discreteness of the limiting conjugate lattice gives a neighborhood of e containing no nontrivial lattice element, a contradiction. Equivalently, compact subsets of a lattice quotient have a positive lower bound on injectivity radius. $\square$

This lemma is why simultaneous return to a compact set is paired with polynomial shearing. At the good times, the relative group element is not merely small; it is uniquely determined, so its polynomial expression cannot be altered by an unnoticed lattice element.

16. Differentiating a polynomial limit curve

Suppose the shearing lemma has shown

$$q(s)_*\mu = \mu \quad (s \in J)$$

for an interval J , where $q : J \rightarrow G$ is smooth. Fix $s_0 \in J$ and set

$$\tilde{q}(s) = q(s_0)^{-1}q(s).$$

Then $\tilde{q}(s_0) = e$ and $\tilde{q}(s) \subset \text{Stab}_G(\mu)$. Hence

$$(9.6) \quad Y_{s_0} = (dL_{q(s_0)^{-1}})_{q(s_0)} q'(s_0) \in \mathfrak{s}_\mu.$$

If G is a matrix group and $q(s) = \exp(p(s))$ with $p(s_0) = 0$, then

$$Y_{s_0} = p'(s_0).$$

If $p(s_0) \neq 0$, the derivative of the exponential contributes the usual dexp operator, so one should not simply identify Y_{s_0} with $p'(s_0)$ without checking commutation. Translating the curve to the identity first is the invariant procedure.

A particularly clean situation occurs when the limiting image lies in a unipotent abelian subgroup. Then the exponential coordinates commute and differentiation reduces directly to the polynomial derivative.

17. A quotient criterion for strict stabilizer enlargement

Let $L = \text{Stab}_G(\mu)^\circ$ and let $\pi : \mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{l}$. Suppose a differentiable stabilizing curve $q(s)$ yields a tangent vector Y at the identity. Then

$$\pi(Y) \neq 0$$

implies that the identity component of the stabilizer is strictly larger than L , a contradiction to the definition of L if L was intended to be the full current stabilizer.

In an inductive proof, however, L often denotes only a subgroup already known to preserve μ , not the entire stabilizer. Then $\pi(Y) \neq 0$ is exactly the desired strict enlargement. This distinction in notation is important:

“known invariance group” and “full measure stabilizer”
should not be denoted by the same symbol unless one
has actually proved they coincide.

Accordingly, when using a transverse quotient to search for new invariance, it is often better to write H for the currently known subgroup and reserve $\text{Stab}_G(\mu)$ for the full, possibly larger, symmetry group.

18. Nearby generic points: where do they come from?

The shearing mechanism is useless if every sufficiently nearby generic point lies on the same local orbit of the known stabilizer L . Therefore a rigidity proof typically splits into two cases:

- (1) *local saturation*: almost every nearby point lies on the same L -leaf; this suggests that μ is supported on a single L -orbit;
- (2) *transverse recurrence*: there are arbitrarily close generic points separated by directions transverse to L ; polynomial shearing then seeks to enlarge L .

Making this dichotomy rigorous requires local product coordinates for the L -action and measure-theoretic information about conditional measures transverse to L . It is not a purely topological statement.

19. A local product picture

Let $L < G$ be closed with Lie algebra $\mathfrak{l}$, and choose a linear complement

$$\mathfrak{g} = \mathfrak{l} \oplus \mathfrak{v}.$$

Near the identity, the map

$$(L \cap \Omega_L) \times (\mathfrak{v} \cap \Omega_V) \rightarrow G, \quad (\ell, v) \mapsto \ell \exp(v)$$

is a local diffeomorphism after shrinking neighborhoods. Hence points near $x \in G/\Gamma$ with positive injectivity radius can be written uniquely as

$$\ell \exp(v)x,$$

with ℓ along the L -leaf and v transverse.

If μ is L -invariant, the ℓ -coordinate is often harmless: translating by ℓ does not change the measure. The essential nearby displacement is the transverse v .

20. Why recurrence must be simultaneous

Suppose x and y are close. Poincaré recurrence separately gives infinitely many return times for x and infinitely many for y , but those two sets of times need not intersect in a useful way. To compare

$$u_t x \quad \text{and} \quad u_t y,$$

one needs common times at which both points lie in a controlled compact set.

A standard solution is to choose a compact set K of measure very close to one. For generic points, the proportion of time spent in K is close to $\mu(K)$. If

$$\mu(K) > 1 - \eta$$

with $\eta < 1/2$, then the two sets of good times each have density $> 1 - \eta$, so their intersection has density at least $1 - 2\eta > 0$. This elementary density estimate creates many simultaneous good times.

21. A simultaneous-good-times lemma

Lemma 9.21.1. *Let $A, B \subset [0, T]$ be measurable and suppose*

$$|A| > (1 - \eta)T, \quad |B| > (1 - \eta)T.$$

Then

$$|A \cap B| > (1 - 2\eta)T.$$

PROOF. Since

$$|A \cup B| \leq T,$$

we have

$$|A \cap B| = |A| + |B| - |A \cup B| > (2 - 2\eta)T - T.$$

□

Trivial as it is, this is exactly the bookkeeping behind many “choose a time when both points are in the compact set” steps.

22. Why one excludes the known stabilizer direction

If

$$y = \ell x$$

with $\ell \in L = \text{Stab}_G(\mu)$, then comparing the two points cannot reveal new symmetry: their difference is already explained by known invariance. In local coordinates, one therefore projects the displacement onto a complement of $\mathfrak{l}$.

The shearing polynomial should be formed from this transverse component. Otherwise its leading term may merely reproduce a direction already in $\mathfrak{l}$, and the argument has made no progress.

23. A rank-one prototype

Let $U = \{u_t\}$ be upper unipotent and suppose the known stabilizer contains U but not the lower-unipotent direction F . Take nearby points separated by

$$\exp(\varepsilon F).$$

Then

$$\text{Ad}(u_t)(\varepsilon F) = \varepsilon F + \varepsilon t H - \varepsilon t^2 E.$$

At scale

$$T_\varepsilon = |\varepsilon|^{-1/2},$$

the limiting displacement is

$$-s^2 \text{sgn}(\varepsilon) E,$$

which lies in the original U -direction. This particular transverse direction therefore does not enlarge the stabilizer; it produces a time shift along U .

If the displacement has an H -component of suitable size, the linear term can survive at a different scale and produce diagonal invariance. Once diagonal invariance is obtained, the Mautner contraction mechanism can force much more symmetry.

This example shows why one must analyze the full polynomial and choose the scale according to which coefficient is genuinely transverse to the known stabilizer.

24. Quotienting by known directions

Let

$$\mathfrak{g}/\mathfrak{l}$$

be the transverse quotient. The known stabilizer directions vanish there. The polynomial

$$\text{Ad}(u_t)v \pmod{\mathfrak{l}}$$

records only genuinely new transverse divergence. Choosing the first time at which this quotient polynomial becomes order one is often the cleanest normalization.

If the quotient polynomial were identically zero, then

$$\text{Ad}(u_t)v \in \mathfrak{l} \quad \text{for every } t,$$

which is itself an algebraic relation between v , U , and $\mathfrak{l}$. Thus even failure of transverse shearing has structural content.

25. A higher-degree prototype in the special linear group of dimension three

Let

$$X_0 = E_{12} + E_{23}, \quad u_t = \exp(tX_0) \in \mathrm{SL}_3(\mathbb{R}),$$

and consider nearby points separated by

$$\exp(\varepsilon E_{31}).$$

The explicit commutator calculation gives

$$(9.7) \quad \begin{aligned} \mathrm{Ad}(u_t)(\varepsilon E_{31}) &= \varepsilon E_{31} + \varepsilon t(E_{21} - E_{32}) \\ &\quad + \frac{\varepsilon t^2}{2}(E_{11} - 2E_{22} + E_{33}) \\ &\quad + \frac{\varepsilon t^3}{2}(E_{23} - E_{12}) + \frac{\varepsilon t^4}{4}E_{13}. \end{aligned}$$

At the scale

$$T_\varepsilon = |\varepsilon|^{-1/4},$$

all lower-degree terms tend to zero while the quartic term converges to

$$\frac{\mathrm{sgn}(\varepsilon)}{4}s^4E_{13}.$$

The limiting relative displacement therefore lies in the one-parameter subgroup generated by E_{13} , which centralizes U .

If genericity and recurrence allow the shearing-to-invariance lemma to be applied, the measure becomes invariant under this E_{13} -subgroup. This example displays the full mechanism without a quadratic simplification:

tiny lowest-root error $\longrightarrow$ quartic shearing $\longrightarrow$ highest-root invariance.

Whether that invariance is genuinely new depends on the subgroup already known to preserve the measure. The polynomial calculation alone cannot decide this; the quotient by known stabilizer directions is essential.

26. Stabilizer enlargement as an induction mechanism

The stabilizer identity component L has finite dimension. If an argument repeatedly proves strict enlargement

$$L_0 < L_1 < L_2 < \cdots,$$

then the process must stop after at most $\dim G$ strict dimension increases.

This elementary finiteness is one reason additional invariance is such an effective strategy. The hard analytic argument may be repeated, but it cannot generate an infinite strictly increasing chain of connected stabilizers by dimension.

One must still understand what the terminal stabilizer implies about the support of the measure; dimension alone does not identify the final homogeneous orbit.

27. A checklist for additional-invariance arguments

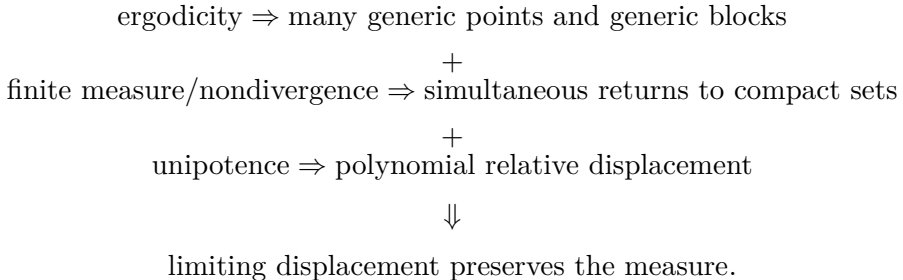
When reading or writing a proof based on unipotent shearing, it is useful to check each of the following explicitly:

- (1) What is the current measure stabilizer L ?
- (2) Why are the chosen points generic for the same measure?
- (3) Why can they be chosen arbitrarily close but not on the same local L -leaf?
- (4) On which compact set is the quotient locally injective?
- (5) What is the exact relative-displacement identity?
- (6) Why is the displacement polynomial in time?
- (7) How is the time scale selected?
- (8) Why do the rescaled polynomials have a nonzero limit?
- (9) Why are there enough simultaneous good times for both trajectories?
- (10) Which block averages are known to be generic?
- (11) How does closeness of orbit points convert into equality of integrals?
- (12) Why does the limiting group element lie in $\text{Stab}_G(\mu)$?
- (13) Why is that invariance genuinely new rather than already contained in L ?

If any one of these steps is missing, the argument may still be repairable, but the missing point should not be hidden by phrases such as “by recurrence” or “by polynomial divergence.”

28. Conceptual summary

The abstract mechanism can be summarized in one diagram:



The shearing-to-invariance theorem makes the final arrow rigorous under explicit block-genericity assumptions. The truly problem-specific work in a rigidity theorem lies in manufacturing those assumptions while excluding lower-dimensional algebraic obstructions. Separating the reusable mechanism from the problem-specific geometry is one of the best ways to understand complicated unipotent proofs.

CHAPTER 10

Worked Analysis Laboratory: Fibers and Polynomial Nondivergence

1. Motivation

Conditional measures and quantitative nondivergence are the two analytic engines repeatedly used later: the first decomposes a measure along fibers, while the second prevents polynomial trajectories from spending too much time near the cusp.

Worked Example 10.1.1 (Disintegration of the two-torus). Let $\pi : \mathbb{T}^2 \rightarrow \mathbb{T}$, $\pi(x, y) = x$, and let m_2 and m_1 be normalized Lebesgue measures. For each x , define $\mu_x(A) = \int_0^1 \mathbf{1}_A(x, y) dy$. Then for every continuous f ,

$$\int_{\mathbb{T}^2} f dm_2 = \int_0^1 \left(\int_0^1 f(x, y) dy \right) dx = \int_{\mathbb{T}} \left(\int f d\mu_x \right) dm_1(x).$$

Thus $\{\mu_x\}$ is the disintegration over π . For $f(x, y) = \cos(2\pi x) + \sin(2\pi y)$, conditional expectation onto the x -coordinate is exactly $\cos(2\pi x)$, because the sine term has fiber average zero.

Worked Example 10.1.2 (A one-vector quantitative-nondivergence calculation). Take $u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ and a nonzero integer vector $v = (p, q)^T$. Then

$$u_t v = (p + tq, q)^T.$$

If $q \neq 0$ and $0 < \varepsilon < |q|$, the inequality $|p + tq| < \varepsilon$ restricts t to an interval of length $2\varepsilon/|q|$. Hence on any interval I ,

$$|\{t \in I : |p + tq| < \varepsilon\}| \leq 2\varepsilon/|q|.$$

If $q = 0$, the vector is fixed and has norm $|p| \geq 1$. This elementary linear-polynomial estimate is the rank-one shadow of the marked-wedge QND theorem: every primitive wedge gives a polynomial control function, and only nested short wedges can persist simultaneously.

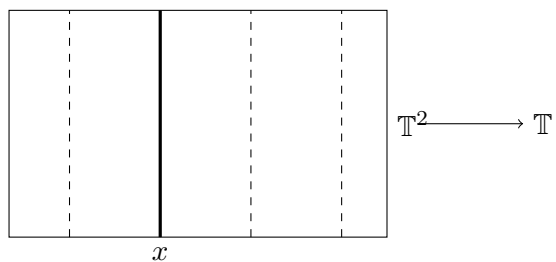


FIGURE 1. Disintegration along the vertical fibers of $\mathbb{T}^2 \rightarrow \mathbb{T}$: condition first on the base coordinate, then integrate the conditional measures over the base.

APPENDIX A

Measure-Theoretic and Functional-Analytic Background

1. Conditional expectation from Hilbert projection

Let $(X, \mathcal{B}, \mu)$ be a probability space and $\mathcal{A} \subset \mathcal{B}$ a sub- σ -algebra. We construct conditional expectation rather than taking it as an external measure-theoretic input.

Lemma A.1.1 (Projection onto a closed subspace). *Let H be a Hilbert space and $M \subset H$ a closed linear subspace. Every $x \in H$ admits a unique decomposition*

$$x = Px + (x - Px), \quad Px \in M, \quad x - Px \perp M.$$

Moreover P is linear and $\|P\| \leq 1$.

PROOF. Let $d = \inf_{m \in M} \|x - m\|$ and choose $m_j \in M$ with $\|x - m_j\| \rightarrow d$. The parallelogram identity gives

$$\|m_j - m_k\|^2 = 2\|x - m_j\|^2 + 2\|x - m_k\|^2 - 4\left\|x - \frac{m_j + m_k}{2}\right\|^2,$$

and the last norm is at least d . Thus (m_j) is Cauchy. Closedness gives $m_j \rightarrow m \in M$ with $\|x - m\| = d$. For $v \in M$ the function $t \mapsto \|x - m - tv\|^2$ has its minimum at $t = 0$, hence $\langle x - m, v \rangle = 0$. Uniqueness follows because two such m differ by a vector that is both in M and orthogonal to M . Linearity follows from uniqueness of the orthogonal decomposition, and $\|Px\|^2 \leq \|Px\|^2 + \|x - Px\|^2 = \|x\|^2$. $\square$

Lemma A.1.2. $L^2(X, \mathcal{A}, \mu)$ is a closed subspace of $L^2(X, \mathcal{B}, \mu)$.

PROOF. Let f_n be $\mathcal{A}$ -measurable and $f_n \rightarrow f$ in L^2 . Choose a subsequence with

$$\sum_j \|f_{n_{j+1}} - f_{n_j}\|_2 < \infty.$$

Because $\mu(X) = 1$, Cauchy–Schwarz gives the same summability in L^1 . Tonelli therefore yields

$$\int_X \sum_j |f_{n_{j+1}} - f_{n_j}| d\mu = \sum_j \|f_{n_{j+1}} - f_{n_j}\|_1 < \infty.$$

Thus the series of pointwise absolute differences is finite almost everywhere, so the subsequence converges almost everywhere to an $\mathcal{A}$ -measurable function. An L^2 limit is unique in measure, and hence that function agrees with f almost everywhere. $\square$

THEOREM A.1.3 (Conditional expectation). *For every $f \in L^1(X, \mathcal{B}, \mu)$ there exists an $\mathcal{A}$ -measurable function $\mathbb{E}(f \mid \mathcal{A}) \in L^1$, unique almost everywhere, such that*

$$\int_A \mathbb{E}(f \mid \mathcal{A}) d\mu = \int_A f d\mu \quad (A \in \mathcal{A}).$$

For $f \in L^2$, this is exactly the orthogonal projection

$$L^2(X, \mathcal{B}, \mu) \longrightarrow L^2(X, \mathcal{A}, \mu).$$

PROOF. First let $f \in L^2$ and let Pf be the projection from Lemma A.1.1 onto the closed subspace of Lemma A.1.2. For $A \in \mathcal{A}$, the indicator $\mathbf{1}_A$ belongs to that subspace, so orthogonality gives

$$\int_A (f - Pf) d\mu = 0.$$

Thus Pf has the required integral identity.

The projection is positive. Indeed, if $f \geq 0$ and $B = \{Pf < 0\} \in \mathcal{A}$, then

$$\int_B Pf = \int_B f \geq 0.$$

If $Pf < 0$ on a set of positive measure, one of the sets $\{Pf \leq -1/k\}$ has positive measure, contradicting the displayed inequality. Hence $Pf \geq 0$ almost everywhere. By linearity and positivity,

$$-P|f| \leq Pf \leq P|f|, \quad |Pf| \leq P|f|.$$

Taking integrals over X yields the L^1 contraction

$$(CE1) \quad \|Pf\|_1 \leq \|f\|_1.$$

Now let $f \in L^1$. Define the bounded truncations

$$f_m = \max(-m, \min(f, m)).$$

They lie in L^2 and converge to f in L^1 . By (CE1), (Pf_m) is Cauchy in L^1 , hence converges there to some h . Passing to an almost-everywhere convergent subsequence shows that h is $\mathcal{A}$ -measurable. For every $A \in \mathcal{A}$,

$$\int_A h = \lim_m \int_A Pf_m = \lim_m \int_A f_m = \int_A f.$$

For uniqueness, suppose h, k are two $\mathcal{A}$ -measurable L^1 functions with the same integral identity. Applying the identity to the sets $\{h - k > 1/j\} \in \mathcal{A}$ shows that each has measure zero; applying the same argument to the sets $\{k - h > 1/j\} \in \mathcal{A}$ gives the reverse inequality. Hence $h = k$ almost everywhere. $\square$

Proposition A.1.4 (Conditional-expectation calculus). *For integrable functions:*

- (1) $\mathbb{E}(af + bh \mid \mathcal{A}) = a\mathbb{E}(f \mid \mathcal{A}) + b\mathbb{E}(h \mid \mathcal{A})$;
- (2) $f \geq 0$ implies $\mathbb{E}(f \mid \mathcal{A}) \geq 0$;
- (3) $\|\mathbb{E}(f \mid \mathcal{A})\|_1 \leq \|f\|_1$;
- (4) if g is bounded and $\mathcal{A}$ -measurable, then

$$\mathbb{E}(gf \mid \mathcal{A}) = g\mathbb{E}(f \mid \mathcal{A});$$

- (5) if $\mathcal{A}_1 \subset \mathcal{A}_2$, then

$$\mathbb{E}(\mathbb{E}(f \mid \mathcal{A}_2) \mid \mathcal{A}_1) = \mathbb{E}(f \mid \mathcal{A}_1).$$

PROOF. The first three properties were proved during the construction, and linearity follows from uniqueness because the linear combination of two candidates satisfies the defining integral identities. For the pull-out property, first take $g = \mathbf{1}_B$ with $B \in \mathcal{A}$. For $A \in \mathcal{A}$,

$$\int_A \mathbf{1}_B \mathbb{E}(f \mid \mathcal{A}) = \int_{A \cap B} f = \int_A \mathbf{1}_B f,$$

so uniqueness gives the claim. Linearity handles simple g , and bounded measurable g follows from uniform approximation by simple functions together with the L^1 contraction. For the tower property, test both sides on $A \in \mathcal{A}_1$ and use the defining identity twice. $\square$

2. Borel–Cantelli

Lemma A.2.1 (First Borel–Cantelli lemma). *If (E_n) is a sequence of measurable sets satisfying*

$$\sum_{n=1}^{\infty} \mu(E_n) < \infty,$$

then almost every point belongs to only finitely many E_n .

PROOF. Let

$$F_N = \bigcup_{n \geq N} E_n.$$

Then

$$\mu(F_N) \leq \sum_{n \geq N} \mu(E_n) \rightarrow 0.$$

The set of points lying in infinitely many E_n is

$$\bigcap_{N=1}^{\infty} F_N,$$

whose measure is zero by continuity from above. □

No independence assumption is needed for this direction.

3. A distribution-function estimate

If $f \in L^1$, then for every $\varepsilon > 0$,

$$(A.1) \quad \sum_{n=1}^{\infty} \mu(|f| > \varepsilon n) < \infty.$$

Indeed, if

$$F(t) = \mu(|f| > t),$$

then the layer-cake identity gives

$$\int_0^{\infty} F(t) dt = \int |f| d\mu < \infty.$$

Monotonicity of F compares the sum in (A.1) with this integral. This elementary observation is useful when proving that boundary terms of the form $f(T^n x)/n$ vanish almost everywhere.

4. Disintegration over the factors used in this series

Every phase space disintegrated later in the series is a locally compact second-countable space (in particular, a homogeneous manifold or a finite product of such spaces). The factor is standard Borel. We prove exactly that form; no more general Rokhlin theorem is needed.

Lemma A.4.1 (A countable vector lattice dense in C_0). *If X is locally compact and second countable, there is a countable $\mathbb{Q}$ -vector subspace $D \subset C_0(X, \mathbb{R})$ which is uniformly dense, closed under $f \mapsto |f|$ and hence under finite maxima and minima, and contains functions*

$$0 \leq \chi_1 \leq \chi_2 \leq \cdots \leq 1, \quad \chi_j \in C_c(X), \quad \chi_j(x) \rightarrow 1$$

for every $x \in X$.

PROOF. Choose a compatible metric, a countable dense subset, and a compact exhaustion $K_j \subset \text{int } K_{j+1}$ of X . The rational algebra generated on each K_j by the distance functions to the chosen dense points separates points; the Stone–Weierstrass theorem proved in Volume I therefore makes it dense in $C(K_j)$. Multiplying its elements by a fixed cutoff that is one on K_j and supported in K_{j+1} gives a countable uniformly dense subset of $C_c(X)$, hence of $C_0(X)$. Take its $\mathbb{Q}$ -linear span and repeatedly adjoin absolute values. A countable union of countable sets is countable, and the resulting vector lattice remains dense.

Cutoffs $\rho_j \in C_c(X)$ with $0 \leq \rho_j \leq 1$, $\rho_j = 1$ on K_j , and $\text{supp } \rho_j \subset \text{int } K_{j+1}$ are obtained from the cutoff construction in Volume I. Adjoin all of them to the generating set and then adjoin the successive maxima $\chi_j = \max_{i \leq j} \rho_i$. $\square$

Lemma A.4.2 (Doob–Dynkin factorization in the needed form). *Let $\pi : X \rightarrow Y$ be measurable and let Y be standard Borel. If a real-valued measurable function h on X is measurable with respect to $\pi^{-1}\mathcal{B}_Y$, then there is a Borel function $\bar{h} : Y \rightarrow \mathbb{R}$ with $h = \bar{h} \circ \pi$.*

PROOF. For every rational q , choose a Borel set $B_q \subset Y$ with

$$\{h < q\} = \pi^{-1}(B_q).$$

Replacing B_q by $\bigcup_{r < q, r \in \mathbb{Q}} B_r$ makes the family increasing without changing the pullbacks. Define

$$\bar{h}(y) = \inf\{q \in \mathbb{Q} : y \in B_q\}.$$

Its rational sublevel sets are Borel, so $\bar{h}$ is Borel, and the equality of all rational sublevel sets gives $h = \bar{h} \circ \pi$. $\square$

Lemma A.4.3 (Functional monotone-class step). *Let $\mathcal{H}$ be a vector space of bounded real functions on a set, containing the constants and closed under uniformly bounded pointwise limits. Suppose $\mathcal{H}$ contains a unital algebra $\mathcal{A}$. Then $\mathcal{H}$ contains every bounded function measurable with respect to the σ -algebra generated by $\mathcal{A}$.*

PROOF. Fix $a \in \mathcal{A}$ and $t \in \mathbb{R}$. On the bounded range of a , the continuous functions

$$r_n(s) = \min(1, \max(0, n(s - t)))$$

can be uniformly approximated by polynomials. Hence $r_n \circ a \in \mathcal{H}$ by the bounded-limit hypothesis, and $r_n(a) \rightarrow \mathbf{1}_{\{a > t\}}$ pointwise. Thus all sets $\{a > t\}$ have indicators in $\mathcal{H}$.

Let $\mathcal{D} = \{E : \mathbf{1}_E \in \mathcal{H}\}$. It contains a π -system generating $\sigma(\mathcal{A})$ (finite intersections are obtained by multiplying the corresponding continuous threshold approximants before passing to the limit). It is a λ -system: complements follow from $1 - \mathbf{1}_E$, and disjoint countable unions follow from the

bounded pointwise limit of their finite sums. The π - λ lemma therefore gives every $\sigma(\mathcal{A})$ -measurable indicator. Simple functions follow by linearity and arbitrary bounded measurable functions by bounded pointwise simple approximation. $\square$

THEOREM A.4.4 (Disintegration over a standard Borel factor). *Let X be a locally compact second-countable Hausdorff space, let Y be a standard Borel space, let μ be a Radon probability measure on X , and let $\pi : X \rightarrow Y$ be measurable. Put $\nu = \pi_*\mu$. Then there is a ν -almost-everywhere uniquely determined measurable family of Radon probability measures $\{\mu_y\}_{y \in Y}$ such that:*

- (1) $\mu_y(\pi^{-1}(y)) = 1$ for ν -almost every y ;
- (2) for every bounded Borel f , the function

$$y \mapsto \int_X f d\mu_y$$

is Borel measurable;

- (3) for every bounded Borel f ,

$$\int_X f d\mu = \int_Y \left(\int_X f d\mu_y \right) d\nu(y).$$

The same identity holds for every nonnegative measurable f , with the value $+\infty$ allowed.

PROOF. Set $\mathcal{A} = \pi^{-1}\mathcal{B}_Y$ and let $D \subset C_0(X, \mathbb{R})$ be the countable vector lattice from Lemma A.4.1. For $f \in D$, conditional expectation gives

$$H_f = \mathbb{E}(f \mid \mathcal{A}).$$

By Lemma A.4.2, choose a Borel $h_f : Y \rightarrow \mathbb{R}$ with $H_f = h_f \circ \pi$ almost everywhere.

Because D is countable, we may discard one ν -null set and require simultaneously, for every y outside it,

$$(D1) \quad h_{af+bg}(y) = ah_f(y) + bh_g(y) \quad (a, b \in \mathbb{Q}),$$

$$(D2) \quad f \geq 0 \implies h_f(y) \geq 0, \quad |h_f(y)| \leq \|f\|_\infty,$$

and

$$(D3) \quad h_{\chi_j}(y) \uparrow 1.$$

The first two statements are consequences of the conditional-expectation calculus. For the last, positivity makes the sequence increasing and bounded by one, while

$$\int_Y h_{\chi_j} d\nu = \int_X \chi_j d\mu \longrightarrow 1$$

by monotone convergence; hence its pointwise limit equals one almost everywhere.

Fix such a y . Equations (D1)–(D2) define a positive norm-at-most-one $\mathbb{Q}$ -linear functional L_y on D . Uniform continuity extends it uniquely to a real-linear bounded functional on $C_0(X)$. Positivity survives the extension: if $f \geq 0$ in C_0 , approximate it uniformly by elements of D and replace those approximants by their positive parts, which remain in the vector lattice D . The Riesz–Markov theorem proved in Volume I therefore supplies a unique Radon measure μ_y with

$$(D4) \quad L_y(f) = \int_X f d\mu_y \quad (f \in C_0(X)).$$

By (D3),

$$\mu_y(X) = \lim_j \int \chi_j d\mu_y = \lim_j h_{\chi_j}(y) = 1,$$

so μ_y is a probability measure. On the discarded null set choose any fixed probability measure; this does not affect the assertions.

For $f \in C_0(X)$, choose $f_m \in D$ converging uniformly to f . The bound (D2) shows that h_{f_m} converges uniformly on the good set to the function $y \mapsto \int f d\mu_y$. Thus this function is measurable, and

$$(D5) \quad \int_Y \int_X f d\mu_y d\nu(y) = \lim_m \int_Y h_{f_m} d\nu = \lim_m \int_X f_m d\mu = \int_X f d\mu.$$

We need slightly more than the unweighted identity (D5), because fiber support must not be smuggled into its own proof. Let $\mathcal{H}$ be the bounded Borel functions f for which

$$K_f(y) := \int_X f d\mu_y$$

is Borel and, in addition,

$$(D6) \quad K_f \circ \pi = \mathbb{E}(f \mid \mathcal{A}) \quad \text{almost everywhere.}$$

Every $f \in C_0(X)$ belongs to $\mathcal{H}$ by the construction above. The class $\mathcal{H}$ is a vector space and contains constants. If $f_m \in \mathcal{H}$ are uniformly bounded and converge pointwise to f , dominated convergence gives $K_{f_m}(y) \rightarrow K_f(y)$ for every good y and $f_m \rightarrow f$ in $L^1(\mu)$. The L^1 contraction of conditional expectation therefore gives

$$\mathbb{E}(f_m \mid \mathcal{A}) \longrightarrow \mathbb{E}(f \mid \mathcal{A}) \quad \text{in } L^1.$$

After taking an almost-everywhere convergent subsequence, (D6) passes to the limit; the pointwise limit of the measurable K_{f_m} is measurable. Thus $\mathcal{H}$ is closed under uniformly bounded pointwise limits.

The algebra generated by $C_0(X)$ and constants separates points and generates the Borel σ -algebra of a locally compact second-countable space: a countable relatively compact basis, together with the cutoff lemma from Volume I, writes every open set as a countable union of sets of the form $\{f > 0\}$ with $f \in C_c(X)$. Lemma A.4.3 now gives (D6) for every bounded Borel f . Integrating (D6) over X recovers (D5) for all such f ; truncation and monotone convergence give the global identity for arbitrary nonnegative measurable f .

It remains to prove concentration on fibers. Choose a countable family $\{B_j\}_{j \geq 1}$ of Borel subsets of Y that separates points; such a family comes from a countable base in any Polish realization of the standard Borel structure. Apply (D6) to $f_j = \mathbf{1}_{\pi^{-1}(B_j)}$. Since f_j is already $\mathcal{A}$ -measurable,

$$\mathbb{E}(f_j \mid \mathcal{A}) = f_j = \mathbf{1}_{B_j} \circ \pi.$$

Hence

$$\mu_y(\pi^{-1}(B_j)) = \mathbf{1}_{B_j}(y)$$

for ν -almost every y ; equality after pullback by π implies equality ν -almost everywhere because $\nu = \pi_*\mu$. Intersect the countably many conull sets. For a remaining y , the set of x for which every membership test $\mathbf{1}_{B_j}(\pi(x)) = \mathbf{1}_{B_j}(y)$ has full μ_y -measure. Since the B_j separate points of Y , this set is exactly $\pi^{-1}(y)$. Thus $\mu_y(\pi^{-1}(y)) = 1$.

Finally suppose $\{\mu'_y\}$ is another family with the three stated properties. For each $f \in D$, the two functions

$$y \mapsto \int f d\mu_y, \quad y \mapsto \int f d\mu'_y$$

are both factorizations of $\mathbb{E}(f \mid \mathcal{A})$, as is seen by integrating over arbitrary $C \subset Y$. They agree almost everywhere. Because D is countable, they agree for every $f \in D$ on one common conull set; density gives equality on $C_0(X)$, and Riesz–Markov uniqueness gives $\mu_y = \mu'_y$ there. $\square$

5. Uniqueness and equivariance of conditional measures

Suppose a group G acts on X and Y , π is equivariant, and μ is G -invariant. If

$$\mu = \int \mu_y d\nu(y)$$

is a disintegration, then

$$g_*\mu_y$$

is a conditional measure over gy . By uniqueness,

$$g_*\mu_y = \mu_{gy}$$

for almost every y , with the usual care about choosing a common conull set for uncountably many g . For second-countable groups one first enforces the relation on a countable dense subgroup and then uses continuity or measurability as appropriate.

6. Weak-* convergence of probability measures

Let X be locally compact and second countable. A sequence of Radon probability measures μ_n converges weak-* to μ if

$$\int f d\mu_n \rightarrow \int f d\mu$$

for every $f \in C_c(X)$, together with preservation of total mass. On compact spaces, testing against $C(X)$ automatically preserves mass because the constant function one is available.

On noncompact spaces, vague convergence against $C_c(X)$ can lose mass. For example,

$$\delta_n$$

on $\mathbb{R}$ converges vaguely to zero.

7. Tightness

Definition A.7.1. A family $\mathcal{P}$ of probability measures on X is *tight* if for every $\eta > 0$ there exists a compact $K \subset X$ such that

$$\mu(K) > 1 - \eta \quad \text{for every } \mu \in \mathcal{P}.$$

THEOREM A.7.2 (Tightness compactness in the form used here). *Let X be locally compact and second countable. Every tight sequence of Radon probability measures on X has a weakly convergent subsequence whose limit is again a Radon probability measure. Equivalently for the sequential applications in this series, every sequence chosen from a tight family has a weakly convergent subsequence.*

PROOF. Let (μ_n) be tight. Choose the countable uniformly dense set $D = \{f_1, f_2, \dots\} \subset C_0(X)$ from Lemma A.4.1. Since $|\int f_j d\mu_n| \leq \|f_j\|_\infty$, successive Bolzano–Weierstrass extractions and a diagonal subsequence give a subsequence, still denoted μ_n , for which every $\int f_j d\mu_n$ converges.

Uniform density and the estimate

$$\left| \int (f - g) d\mu_n \right| \leq \|f - g\|_\infty$$

show that the integrals converge for every $f \in C_0(X)$. Their limits define a positive linear functional L with $\|L\| \leq 1$. The Riesz–Markov theorem from

Volume I produces a Radon measure μ with

$$L(f) = \int f d\mu, \quad \mu(X) \leq 1.$$

Tightness prevents loss of mass. Given $\varepsilon > 0$, choose compact K with $\mu_n(K) \geq 1 - \varepsilon$ for every n , and choose $\chi \in C_c(X)$ with $0 \leq \chi \leq 1$ and $\chi = 1$ on K . Then

$$\int \chi d\mu = \lim_n \int \chi d\mu_n \geq 1 - \varepsilon.$$

Hence $\mu(X) = 1$.

We have convergence against $C_0(X)$. To obtain the usual weak convergence against bounded continuous functions, fix bounded continuous h and $\varepsilon > 0$. Tightness of the sequence and Radon regularity of μ give a compact K such that

$$\sup_n \mu_n(X \setminus K) < \varepsilon, \quad \mu(X \setminus K) < \varepsilon.$$

Choose $\chi \in C_c(X)$ equal to one on K . Since $h\chi \in C_0(X)$,

$$\int h\chi d\mu_n \longrightarrow \int h\chi d\mu,$$

while the two tails have absolute value at most $\varepsilon\|h\|_\infty$. Thus $\int h d\mu_n \rightarrow \int h d\mu$.

Every sequence in a tight family therefore has a weakly convergent subsequence. This is exactly the compactness form used later: all applications begin with an arbitrary sequence of orbit measures and pass to a convergent subsequence. $\square$

8. A direct compact-exhaustion criterion

In many homogeneous settings one can avoid invoking the full strength of Prokhorov. Suppose there are compact sets

$$K_1 \subset K_2 \subset \cdots, \quad \bigcup_j K_j = X,$$

and a sequence $\eta_j \downarrow 0$ such that

$$\inf_n \mu_n(K_j) \geq 1 - \eta_j.$$

Choose a countable dense family in $C_c(X)$, diagonalize the integrals, and use the compact-mass bounds to construct a Radon probability limit. This is essentially Prokhorov specialized to a σ -compact locally compact space.

9. Strong continuity of Koopman representations

Let G act continuously on X preserving a Radon probability measure. The proof of strong continuity on L^2 can be summarized as:

- (1) prove continuity for $C_c(X)$ using uniform continuity on compact sets;
- (2) use density of $C_c(X)$ in L^2 ;
- (3) use the fact that all Koopman operators have norm one.

Explicitly, if $f \in L^2$, choose $\phi \in C_c(X)$ with $\|f - \phi\|_2 < \varepsilon$. Then

$$\|\pi(g)f - f\|_2 \leq 2\varepsilon + \|\pi(g)\phi - \phi\|_2,$$

and the last term tends to zero as $g \rightarrow e$.

10. Lusin approximation and compact control

If a measurable function rather than a continuous one appears in an orbit comparison, continuity on a large compact set is recovered by the following internal argument.

Lemma A.10.1 (Egorov on a finite measure space). *If $f_n \rightarrow f$ almost everywhere on a finite measure space, then for every $\eta > 0$ there is a measurable set E with measure of the complement below η on which the convergence is uniform.*

PROOF. After discarding the null set of nonconvergence, set

$$E_{m,N} = \bigcup_{n \geq N} \{|f_n - f| > 1/m\}.$$

For fixed m , the sets $E_{m,N}$ decrease to the empty set, so choose N_m with $\mu(E_{m,N_m}) < \eta 2^{-m}$. On $E = X \setminus \bigcup_m E_{m,N_m}$, for every m and every $n \geq N_m$ we have $|f_n - f| \leq 1/m$. This is uniform convergence, and $\mu(X \setminus E) < \eta$. $\square$

Lemma A.10.2 (Simple functions are continuous on a large compact set). *Let X be locally compact and second countable with finite Radon measure μ . If $s : X \rightarrow \mathbb{C}$ is a finite-valued measurable function, then for every $\eta > 0$ there is a compact K with $\mu(X \setminus K) < \eta$ such that $s|_K$ is continuous.*

PROOF. Write $s = \sum_{i=1}^m c_i \mathbf{1}_{E_i}$ using a measurable partition $E_1, \dots, E_m$. Inner regularity gives compact $K_i \subset E_i$ with $\mu(E_i \setminus K_i) < \eta/m$. Then $K = \bigcup_i K_i$ is compact and has the required mass. The disjoint compact sets K_i are relatively open and closed in their finite union, so the finite-valued function $s|_K$ is continuous. $\square$

THEOREM A.10.3 (Lusin). *Let X be locally compact and second countable with finite Radon measure μ , and let $f : X \rightarrow \mathbb{C}$ be measurable and finite almost everywhere. For every $\eta > 0$ there is a compact set $K \subset X$ with*

$$\mu(X \setminus K) < \eta$$

such that $f|_K$ is continuous.

PROOF. Choose finite-valued measurable functions s_n converging pointwise to f outside a null set; for example, truncate the real and imaginary parts at height n and round them to the lattice $2^{-n}\mathbb{Z}$. By Lemma A.10.2, for each n choose a compact K_n such that

$$\mu(X \setminus K_n) < \eta 2^{-n-3}$$

and $s_n|_{K_n}$ is continuous. The intersection $K_0 = \bigcap_{n \geq 1} K_n$ is compact (it is closed in K_1) and $\mu(X \setminus K_0) < \eta/4$.

Apply Lemma A.10.1 on K_0 to find a measurable $E \subset K_0$ with $\mu(K_0 \setminus E) < \eta/4$ on which $s_n \rightarrow f$ uniformly. By inner regularity choose compact $K \subset E$ with $\mu(E \setminus K) < \eta/4$. Each $s_n|_K$ is continuous, and their uniform limit is $f|_K$, hence $f|_K$ is continuous. The accumulated loss is below $3\eta/4 < \eta$. $\square$

11. A countable convergence-determining family

On a locally compact second-countable space X , there exists a countable set

$$\{f_j\}_{j \geq 1} \subset C_c(X)$$

with the property that equality of integrals against all f_j determines a Radon probability measure. One construction chooses a countable basis of relatively compact open sets and piecewise-linear cutoff functions approximating their indicators.

This countability is what allows one to intersect full-measure genericity sets over all test functions needed in a proof.

12. Ergodic decomposition and extreme points

The convex set of invariant probability measures has ergodic measures as its extreme points. Disintegration over the invariant σ -algebra realizes an invariant measure as a barycenter of these extreme points:

$$\mu = \int \mu_y d\eta(y).$$

For homogeneous rigidity this justifies the standard reduction “assume the invariant measure is ergodic.” It is not a loss of generality when classification can be applied componentwise and then recombined measurably.

APPENDIX B

Polynomial Sublevel Estimates and Nondivergence Bookkeeping

1. A one-variable Remez inequality

We record the polynomial estimate used in quantitative nondivergence.

THEOREM B.1.1 (A convenient one-variable interpolation inequality). *Let p be a real polynomial of degree at most $D \geq 1$, let I be an interval, and let $E \subset I$ be measurable with positive measure. Then*

$$(B.1) \quad \sup_I |p| \leq \left(\frac{8|I|}{|E|} \right)^D \sup_E |p|.$$

The numerical constant is deliberately nonoptimal; the point is the power D and the dependence only on $|I|/|E|$.

PROOF. By an affine change of variable we may take $I = [-1, 1]$. Write $m = |E|$. We choose ordered quantile points

$$x_0 < x_1 < \cdots < x_D$$

in the closure of E . This causes no problem for the interpolation values, because continuity of p gives $\sup_{\overline{E}} |p| = \sup_E |p|$. More explicitly, if

$$F(x) = |E \cap [-1, x]|,$$

take $x_0 = \inf \overline{E}$, $x_D = \sup \overline{E}$, and for $1 \leq j \leq D-1$ choose x_j with $F(x_j) = jm/D$. Continuity of F supplies the interior quantiles, and choosing the edge of any plateau puts the point in $\overline{E}$. Moreover $F(x_0) = 0$ and $F(x_D) = m$, since single points have Lebesgue measure zero. The function F is nondecreasing and 1-Lipschitz, because

$$F(y) - F(x) = |E \cap (x, y]| \leq y - x.$$

Hence, for $i < j$,

$$(B.1a) \quad x_j - x_i \geq \frac{(j-i)m}{D}.$$

Lagrange interpolation gives

$$p(x) = \sum_{j=0}^D p(x_j) L_j(x), \quad L_j(x) = \prod_{k \neq j} \frac{x - x_k}{x_j - x_k}.$$

For $x \in [-1, 1]$, every numerator is at most 2 in absolute value. From (B.1a),

$$\prod_{k \neq j} |x_j - x_k| \geq \left(\frac{m}{D}\right)^D j!(D-j)!.$$

Therefore

$$\sup_{[-1,1]} |L_j| \leq \frac{(2D/m)^D}{j!(D-j)!}.$$

Summing over j and using

$$\sum_{j=0}^D \frac{1}{j!(D-j)!} = \frac{2^D}{D!}$$

gives

$$\sum_{j=0}^D \sup |L_j| \leq \left(\frac{4D}{m}\right)^D \frac{1}{D!}.$$

The elementary inequality $D! \geq (D/e)^D$ yields

$$\sup_{[-1,1]} |p| \leq \left(\frac{4e}{m}\right)^D \sup_E |p|.$$

Since $4e < 16$, and on $[-1, 1]$ one has $|I|/|E| = 2/m$, this is stronger than

$$\sup_I |p| \leq \left(\frac{8|I|}{|E|}\right)^D \sup_E |p|.$$

Affine rescaling proves the stated inequality for every interval. $\square$

Remark B.1.2. The sharp Remez inequality gives a better numerical constant and can be expressed using Chebyshev polynomials. Quantitative nondivergence uses only the power-law dependence

$$(|I|/|E|)^D,$$

so optimizing the numerical constant adds no dynamical content.

2. Sublevel sets from Remez

If $M = \sup_I |p| > 0$ and

$$E_\varepsilon = \{t \in I : |p(t)| < \varepsilon M\},$$

then Remez gives

$$M \leq \left(\frac{8|I|}{|E_\varepsilon|} \right)^D \varepsilon M,$$

so

$$|E_\varepsilon| \leq 8\varepsilon^{1/D} |I|.$$

This proves the $(C_D, 1/D)$ -good property; with the interpolation inequality above one may take $C_D = 8$.

3. Stability properties of good functions

If f is (C, α) -good, then so is cf for every nonzero constant c . If $|f|$ and $|g|$ are comparable by fixed constants,

$$A^{-1}|f| \leq |g| \leq A|f|,$$

then goodness transfers from f to g with a modified constant $CA^{2\alpha}$.

For a finite family $f_1, \dots, f_m$, the maximum

$$F = \max_i |f_i|$$

is good whenever one can choose, on each interval, an index i with $\sup |f_i|$ comparable to $\sup F$. This observation underlies the vector-valued polynomial norm estimate.

4. Primitive subgroup covolumes

Let Λ be a unimodular lattice. For a primitive rank- r subgroup $\Delta < \Lambda$, define

$$\nu(\Delta) = \|w_\Delta\|.$$

Two basic properties drive the higher-rank bookkeeping.

- (1) *Discreteness*: for each fixed rank, the set of wedge vectors is discrete, so only finitely many primitive subgroups have $\nu(\Delta) \leq R$.
- (2) *Submodularity*: if Δ_1, Δ_2 are primitive subgroups of a Euclidean lattice Λ , and if $\Delta_1 \vee \Delta_2$ denotes the primitive saturation in Λ of $\Delta_1 + \Delta_2$, then

$$\nu(\Delta_1 \vee \Delta_2) \nu(\Delta_1 \cap \Delta_2) \leq \nu(\Delta_1) \nu(\Delta_2).$$

Here $\nu(\{0\}) = 1$. We give the determinant argument because the inequality is exactly what makes rank escalation quantitative.

Proposition B.4.1 (Covolume submodularity). *Let Λ be a Euclidean lattice and let $\Delta_1, \Delta_2 < \Lambda$ be primitive subgroups. Write $\Delta_1 \vee \Delta_2$ for the primitive saturation in Λ of $\Delta_1 + \Delta_2$, and put $\nu(\Delta) = \|w_\Delta\|$ for the covolume norm of a primitive wedge, with $\nu(\{0\}) = 1$. Then*

$$(B.2) \quad \nu(\Delta_1 \vee \Delta_2) \nu(\Delta_1 \cap \Delta_2) \leq \nu(\Delta_1) \nu(\Delta_2).$$

PROOF. Put

$$\Delta_0 = \Delta_1 \cap \Delta_2, \quad W_0 = \text{span}_{\mathbb{R}} \Delta_0.$$

Because Δ_1 and Δ_2 are primitive in Λ , the subgroup Δ_0 is primitive in each Δ_i : if $mv \in \Delta_0$ with $v \in \Delta_i$, then $mv \in \Delta_j$ for the other index j ; primitivity of Δ_j in Λ implies $v \in \Delta_j$, hence $v \in \Delta_0$. We may therefore choose a basis

$$c_1, \dots, c_s$$

of Δ_0 and extend it to bases

$$c_1, \dots, c_s, a_1, \dots, a_p \quad \text{of } \Delta_1,$$

$$c_1, \dots, c_s, b_1, \dots, b_q \quad \text{of } \Delta_2.$$

Let P be orthogonal projection to $W_0^\perp$. The Gram-determinant formula, or equivalently repeated base-times-height factorization, gives

$$\nu(\Delta_1) = \nu(\Delta_0) \|Pa_1 \wedge \dots \wedge Pa_p\|,$$

$$\nu(\Delta_2) = \nu(\Delta_0) \|Pb_1 \wedge \dots \wedge Pb_q\|.$$

Because $\Delta_0 = \Delta_1 \cap \Delta_2$, the combined list

$$c_1, \dots, c_s, a_1, \dots, a_p, b_1, \dots, b_q$$

is a $\mathbb{Z}$ -basis of $\Delta_1 + \Delta_2$. Indeed, any integral relation would put an element generated by the a_i 's simultaneously in Δ_1 and Δ_2 , hence in Δ_0 , and the basis properties force all coefficients to vanish. Since a subgroup of a lattice has rank equal to the dimension of its real span, the projected vectors are linearly independent in the required quotient directions. Therefore

$$\nu(\Delta_1 + \Delta_2) = \nu(\Delta_0) \|Pa_1 \wedge \dots \wedge Pa_p \wedge Pb_1 \wedge \dots \wedge Pb_q\|.$$

The elementary wedge inequality

$$\|\xi \wedge \eta\| \leq \|\xi\| \|\eta\|$$

therefore gives

$$\nu(\Delta_1 + \Delta_2) \nu(\Delta_0) \leq \nu(\Delta_1) \nu(\Delta_2).$$

Passing from $\Delta_1 + \Delta_2$ to its primitive saturation can only decrease covolume, by the finite index relating the two lattices. This proves (B.2). $\square$

5. Why nested chains replace disjoint short-vector sets

In dimension two, two distinct primitive directions cannot both be very short, so the bad sets indexed by primitive vectors are disjoint. In higher dimension, several small primitive subgroups can coexist, but after choosing thresholds carefully, submodularity forces them to be approximately nested.

Thus the higher-rank proof replaces

one short direction at a time

by

one controlled chain $\Delta_1 < \Delta_2 < \cdots < \Delta_k$ at a time.

Since ranks strictly increase, every chain has length at most d . This finite depth is the combinatorial reason the argument closes.

6. A one-dimensional covering scheme

For each primitive subgroup Δ , consider

$$F_\Delta(t) = \|h(t)w_\Delta\|.$$

Fix thresholds

$$0 < \varepsilon \ll \rho.$$

A maximal interval J on which

$$F_\Delta < \rho$$

will be called a Δ -window. Goodness gives

$$(B.3) \quad |\{t \in J : F_\Delta(t) < \varepsilon\}| \leq C(\varepsilon/\rho)^\alpha |J|.$$

The issue is that windows for different Δ 's can overlap. At a point of overlap, choose among the small subgroups one of maximal rank. Submodularity implies that any other sufficiently small subgroup either lies in the chosen one or combines with it to produce an even higher-rank subgroup that is also small, contradicting maximality. Thus, after a bounded enlargement of thresholds, overlapping windows organize into nested chains.

On the real line, a Vitali selection extracts a disjoint subcollection of maximal windows whose fixed dilates cover all bad times. The bounded overlap of these dilates contributes only a dimension-dependent multiplicity constant.

Combining this covering with (B.3) yields the general estimate

$$|\{t : h(t)\Lambda \notin K_\varepsilon\}| \leq C'(d, C, \alpha)(\varepsilon/\rho)^\alpha |I|.$$

Remark B.6.1 (From architecture to proof). The discussion in this appendix isolates the geometric bookkeeping. Appendix D turns it into a complete one-dimensional proof through a weighted-poset marking lemma. The main text separately gives the elementary $d = 2$ argument, so the reader can compare the rank-one mechanism with the full higher-rank induction.

7. Uniformity under time rescaling

If $h(t) = u_t g$ with u_t unipotent, then after rescaling $t = Ts$, every exterior-power coordinate remains a polynomial of the same degree. Goodness constants therefore depend on the representation and degree but not on T . This simple observation is what makes the nondivergence estimate uniform over arbitrarily long orbit intervals.

8. A coefficient norm on polynomial maps

For

$$p(t) = a_0 + a_1 t + \cdots + a_D t^D$$

with values in a finite-dimensional normed space, define

$$\|p\|_{\text{coef}} = \max_j \|a_j\|.$$

On any fixed compact interval, coefficient norm and supremum norm are equivalent:

$$c_D \|p\|_{\text{coef}} \leq \sup_{0 \leq t \leq 1} \|p(t)\| \leq C_D \|p\|_{\text{coef}}.$$

The upper bound is immediate. The lower bound follows because the map from coefficients to the continuous function on $[0, 1]$ is an injective linear map between finite-dimensional normed spaces.

This equivalence justifies repeatedly passing between “some coefficient becomes order one” and “the polynomial becomes order one somewhere on the interval.”

9. Normalized polynomial limits

If p_n have degree at most D and are normalized by

$$\sup_{[0,1]} \|p_n\| = 1,$$

then their coefficient vectors lie in a compact set. Hence a subsequence converges coefficientwise and uniformly to a polynomial p with

$$\sup_{[0,1]} \|p\| = 1.$$

This is the finite-dimensional compactness behind quasi-regular limit maps.

10. A useful “large on most times” corollary

Let p be a nonzero degree-at-most- D vector polynomial on an interval I .
If

$$M = \sup_I \|p\| ,$$

then for every $0 < \theta < 1$ there exists $c = c(D, \theta) > 0$ such that

$$|\{t \in I : \|p(t)\| \geq cM\}| \geq (1 - \theta) |I| .$$

Indeed, choose c so that $C_D c^{1/D} < \theta$ and apply goodness.

This is often more useful dynamically than the sublevel estimate itself: a normalized shearing polynomial is visibly nontrivial on most of a macroscopic time block.

APPENDIX C

Chevalley Linearization from the Coordinate Ring

Linearization arguments repeatedly require the following algebraic fact: an algebraic subgroup can be represented as the stabilizer of a line in a finite-dimensional rational representation. Since this is a major bridge between subgroup geometry and polynomial dynamics, we prove it here from the coordinate ring.

1. Local finiteness of the regular action

Let G be a real linear algebraic group. Write $\mathbb{R}[G]$ for its coordinate ring. For $g \in G$, define right translation on regular functions by

$$(R_g f)(x) = f(xg).$$

Lemma C.1.1 (Every regular function has finite-dimensional translation span). *For every $f \in \mathbb{R}[G]$, the vector space*

$$W_f := \text{span}_{\mathbb{R}}\{R_g f : g \in G\}$$

is finite dimensional. It is invariant under every R_g , and the resulting representation of G on W_f is rational.

PROOF. Multiplication $m : G \times G \rightarrow G$ is a regular map, so the pullback $m^* f \in \mathbb{R}[G \times G] \cong \mathbb{R}[G] \otimes \mathbb{R}[G]$ is a finite sum

$$m^* f = \sum_{j=1}^N a_j \otimes b_j.$$

Evaluating the second variable at g gives

$$(R_g f)(x) = f(xg) = \sum_{j=1}^N a_j(x) b_j(g).$$

Hence all translates $R_g f$ lie in one fixed finite-dimensional subspace of $\mathbb{R}[G]$. Therefore W_f is finite dimensional.

Moreover,

$$R_g R_h f(x) = f(xgh) = R_{gh} f(x),$$

so W_f is invariant and $g \mapsto R_g|_{W_f}$ is a representation. In a fixed basis of W_f , its matrix coefficients are regular functions of g , because the displayed comultiplication formula expresses the translated functions with coefficients $b_j(g)$. Thus the representation is rational. $\square$

2. An algebraic subgroup as a subspace stabilizer

Let $H < G$ be an algebraic subgroup and let

$$I(H) = \{f \in \mathbb{R}[G] : f|_H = 0\}$$

be its vanishing ideal.

THEOREM C.2.1 (Finite-dimensional subspace realization). *There is a finite-dimensional rational representation $\rho : G \rightarrow \mathrm{GL}(W)$ and a subspace $W_H \subset W$ such that*

$$H = \{g \in G : \rho(g)W_H = W_H\}.$$

PROOF. Because $\mathbb{R}[G]$ is a finitely generated real algebra and therefore Noetherian, the ideal $I(H)$ is generated by finitely many functions

$$f_1, \dots, f_m.$$

For each generator take the finite-dimensional translation module W_{f_i} from Lemma C.1.1, and put

$$W = W_{f_1} + \dots + W_{f_m}.$$

Then W is finite dimensional, invariant under all right translations, and contains every f_i . Let

$$W_H = W \cap I(H).$$

If $h \in H$ and $f \in W_H$, then for every $x \in H$ one has $xh \in H$, so

$$(R_h f)(x) = f(xh) = 0.$$

Hence $R_h W_H \subset W_H$. Applying the same statement to h^{-1} gives equality. Thus H stabilizes W_H .

Conversely, suppose $g \in G$ satisfies $R_g W_H = W_H$. Since every generator f_i lies in W_H , the function $R_g f_i$ also lies in $I(H)$. Evaluating at the identity $e \in H$ gives

$$f_i(g) = (R_g f_i)(e) = 0$$

for every i . The f_i generate the vanishing ideal of the algebraic set H , so their common zero set in G is exactly H . Therefore $g \in H$. This proves the equality of stabilizers. $\square$

3. From a subspace to a line

THEOREM C.3.1 (Chevalley linearization). *Let H be an algebraic subgroup of a real linear algebraic group G . There exist a finite-dimensional rational representation*

$$\sigma : G \rightarrow \mathrm{GL}(V)$$

and a one-dimensional subspace $\ell \subset V$ such that

$$H = \mathrm{Stab}_G(\ell).$$

PROOF. If $H = G$, take the trivial one-dimensional representation. Assume $H \neq G$ and use Theorem C.2.1. Put $r = \dim W_H$; since $I(H) \neq 0$ and the chosen generators lie in W_H , one has $r \geq 1$. Let

$$V = \bigwedge^r W, \quad \sigma = \bigwedge^r \rho,$$

and choose a basis $w_1, \dots, w_r$ of W_H . Define

$$\ell = \mathbb{R}(w_1 \wedge \dots \wedge w_r).$$

For any $g \in G$, the line $\sigma(g)\ell$ is the Pluecker line of the r -plane $\rho(g)W_H$. Two nonzero decomposable r -vectors span the same line exactly when the r -planes spanned by their factors are equal. Consequently

$$\sigma(g)\ell = \ell \iff \rho(g)W_H = W_H.$$

Theorem C.2.1 identifies the right-hand side with $g \in H$. □

Remark C.3.2 (Why this theorem is useful dynamically). The proof explains more than existence. The line is built from finitely many regular equations cutting out H . If u_t is unipotent, then $\sigma(u_t)$ is polynomial in t after choosing coordinates adapted to the nilpotent infinitesimal generator. Thus proximity to a conjugate of H can be converted into smallness of finitely many polynomial coordinates. This is exactly the input required by linearization and avoidance arguments.

APPENDIX D

The Full One-Parameter Quantitative Nondivergence Argument

The main text proves quantitative nondivergence directly in dimension two and explains the higher-rank geometry. Here we supply the missing combinatorial argument in full for a one-dimensional parameter. This is the form needed for one-parameter unipotent flows.

1. Good functions and weighted posets

For an interval J and a continuous function f write

$$\|f\|_J = \sup_{t \in J} |f(t)|.$$

Recall that f is (C, α) -good on an interval Ω if, for every subinterval $J \subset \Omega$ and every $\varepsilon > 0$,

$$|\{t \in J : |f(t)| < \varepsilon \|f\|_J\}| \leq C\varepsilon^\alpha |J|.$$

A *weighted poset* is a partially ordered set P together with a map $\eta : P \rightarrow (0, \infty)$. A finite totally ordered subset is called a flag. If $F \subset P$ is a flag, let

$$P(F) = \{s \in P \setminus F : s \text{ is comparable with every element of } F\}.$$

If every flag in P has at most k elements, then every flag in $P(F)$ has at most $k - |F|$ elements.

Given continuous functions ψ_s indexed by $s \in P$ and $0 < \theta \leq 1$, a point t is called θ -marked if there is a flag F_t such that

$$(1.1) \quad \theta\eta(s) \leq |\psi_s(t)| \leq \eta(s) \quad (s \in F_t),$$

$$(1.2) \quad |\psi_s(t)| \geq \eta(s) \quad (s \in P(F_t)).$$

Write $\Phi(\theta, P)$ for the set of marked points.

2. The marking lemma on an interval

We use one elementary covering fact specific to the real line.

Lemma D.2.1 (One-dimensional Besicovitch covering lemma). *Let $E \subset \mathbb{R}$ be bounded and let $\mathcal{B}$ be a family of open intervals with uniformly bounded radii such that every $y \in E$ is the center of at least one member of $\mathcal{B}$. Then one can choose a countable subfamily which covers E and has multiplicity at most 2.*

PROOF. It is enough first to treat a finite family covering a compact subset. Order candidate intervals by nonincreasing radius. Select the first one, then continue down the list, selecting an interval exactly when its center has not already been covered by a selected interval. The selected intervals cover all the original centers. No point can belong to three selected intervals: if three selected intervals contain a point p , then two of their centers lie on the same side of p . Let I_1 be the earlier (hence no smaller) of those two intervals and I_2 the later one. If both centers are to the right of p , then the distance from the center of I_2 to the center of I_1 is at most the radius of I_2 , hence at most the radius of I_1 ; so the later center already lies in I_1 , contradicting its selection. The left-hand case is identical.

For an arbitrary bounded E , the same greedy argument is obtained by first ordering radii in dyadic classes and applying the finite construction on successive compact pieces. Equivalently, one may pass from the finite statement by the usual compactness argument for the Besicovitch covering theorem. The resulting selected family has the same local multiplicity bound. Since $\mathbb{R}$ is second countable, the selected open cover has a countable subcover, and passing to a subfamily cannot increase multiplicity. This proves the stated one-dimensional form. $\square$

THEOREM D.2.2 (Weighted-poset marking lemma). *Let $I = B(x_0, r)$ be an interval and let*

$$\tilde{I} = B(x_0, 3^k r).$$

Suppose a weighted poset (P, η) has flag length at most k , and suppose continuous functions $\psi_s : \tilde{I} \rightarrow \mathbb{R}$ satisfy:

- (1) *every ψ_s is (C, α) -good on $\tilde{I}$;*
- (2) *$\|\psi_s\|_I \geq \eta(s)$ for every $s \in P$;*
- (3) *for every $t \in \tilde{I}$, only finitely many $s \in P$ satisfy $|\psi_s(t)| < \eta(s)$.*

Then for $0 < \theta \leq 1$,

$$|I \setminus \Phi(\theta, P)| \leq A_k C \theta^\alpha |I|,$$

where one may take, for example,

$$A_k = 6k 18^k.$$

The numerical constant is intentionally nonoptimal; only its dependence on k matters.

PROOF. We argue by induction on k . When $k = 0$, the poset is empty and every point is marked by the empty flag.

Assume $k \geq 1$ and the statement known for posets of flag length at most $k - 1$. For $y \in I$ define the finite set

$$H(y) = \{s \in P : |\psi_s(y)| < \eta(s)\}.$$

If $H(y) = \emptyset$, the empty flag marks y . It therefore suffices to cover

$$E = \{y \in I : H(y) \neq \emptyset\}.$$

Fix $y \in E$ and $s \in H(y)$. Let

$$r_{s,y} = \sup\{a > 0 : \|\psi_s\|_{B(y,a)} < \eta(s)\}.$$

Continuity makes $r_{s,y} > 0$. Also $r_{s,y} < 2r$: the interval $B(y, 2r)$ contains I , and the hypothesis $\|\psi_s\|_I \geq \eta(s)$ prevents the strict inequality in the definition from persisting that far. Choose $s_y \in H(y)$ with maximal $r_{s,y}$ and put

$$r_y = r_{s_y,y}, \quad B_y = B(y, r_y), \quad P_y = P(\{s_y\}).$$

By construction,

$$(2.1) \quad \|\psi_{s_y}\|_{B_y} \leq \eta(s_y).$$

The following elementary implication is the heart of the induction. If $t \in B_y$ is θ -marked relative to P_y and also

$$|\psi_{s_y}(t)| \geq \theta\eta(s_y),$$

then t is θ -marked relative to P . Indeed, if F is a marking flag for t in P_y , then $F \cup \{s_y\}$ is a flag in P ; (2.1) supplies the upper half of (1.1) for s_y , the displayed inequality supplies the lower half, and

$$P(F \cup \{s_y\}) = P_y(F)$$

turns (1.2) for P_y into the corresponding statement for P . Consequently

$$(2.2) \quad B_y \setminus \Phi(\theta, P) \subset (B_y \setminus \Phi(\theta, P_y)) \cup \{t \in B_y : |\psi_{s_y}(t)| < \theta\eta(s_y)\}.$$

Choose a radius r'_y with

$$r_y < r'_y < \min(2r, 3r_y)$$

so close to r_y from above that, for every $s \in H(y)$,

$$\|\psi_s\|_{B(y,r'_y)} \geq \eta(s).$$

This is possible from the definition of the suprema and the maximality of r_y . If $s \notin H(y)$, already $|\psi_s(y)| \geq \eta(s)$. Hence for

$$B'_y = B(y, r'_y)$$

one has

$$(2.3) \quad \|\psi_s\|_{B'_y} \geq \eta(s) \quad (s \in P).$$

The poset P_y has flag length at most $k-1$. Moreover its required enlarged interval

$$B(y, 3^{k-1}r'_y)$$

lies in $\tilde{I}$: since $y \in B(x_0, r)$ and $r'_y < 2r$,

$$3^{k-1}r'_y + r < (2 \cdot 3^{k-1} + 1)r \leq 3^k r.$$

Thus the induction hypothesis applies to P_y on B'_y . Because $|B'_y| < 3|B_y|$,

$$(2.4) \quad |B_y \setminus \Phi(\theta, P_y)| \leq |B'_y \setminus \Phi(\theta, P_y)| \leq 3A_{k-1}C\theta^\alpha|B_y|.$$

Goodness of ψ_{s_y} on B'_y , together with (2.3), gives

$$(2.5) \quad |\{t \in B_y : |\psi_{s_y}(t)| < \theta\eta(s_y)\}| \leq 3C\theta^\alpha|B_y|.$$

Combining (2.2)–(2.5),

$$|B_y \setminus \Phi(\theta, P)| \leq 3C(A_{k-1} + 1)\theta^\alpha|B_y|.$$

The intervals B_y , $y \in E$, form a cover of E . Lemma D.2.1 gives a countable subcover with multiplicity at most 2. Since every $B_y \subset B(x_0, 3r)$,

$$\sum_y |B_y| \leq 2|B(x_0, 3r)| = 6|I|.$$

Therefore

$$|I \setminus \Phi(\theta, P)| \leq 18C(A_{k-1} + 1)\theta^\alpha|I|.$$

Starting with $A_0 = 0$, the crude choice $A_k = 6k18^k$ dominates this recurrence. This proves the theorem. $\square$

3. Application to primitive subgroups

Let Λ be a unimodular lattice in $\mathbb{R}^d$. For a primitive subgroup $\Delta < \Lambda$ of rank r , let w_Δ be its primitive wedge, defined up to sign, and set

$$\psi_\Delta(t) = \|(\wedge^r h(t))w_\Delta\|.$$

Lemma D.3.1 (Local finiteness of small primitive wedges). *For fixed t and $R > 0$, only finitely many primitive subgroups Δ of Λ satisfy $\psi_\Delta(t) \leq R$.*

PROOF. For each rank r , the primitive wedges form a discrete subset of $\wedge^r \Lambda$. The invertible linear map $\wedge^r h(t)$ carries it to a discrete subset of $\wedge^r \mathbb{R}^d$. A bounded Euclidean ball meets a discrete lattice in finitely many points. There are only d possible ranks. $\square$

THEOREM D.3.2 (Full one-parameter quantitative nondivergence). *Let $I = B(x_0, r)$ and $\tilde{I} = B(x_0, 3^d r)$. Let $h : \tilde{I} \rightarrow \mathrm{SL}_d(\mathbb{R})$, let Λ be unimodular, and let $0 < \rho \leq 1$. Suppose that for every nonzero proper primitive rank- s subgroup $\Delta < \Lambda$:*

- (1) $t \mapsto \psi_\Delta(t) = \|(\wedge^s h(t))w_\Delta\|$ is (C, α) -good on $\tilde{I}$;
- (2) $\sup_{t \in I} \psi_\Delta(t) \geq \rho^s$.

Then

$$|\{t \in I : \delta(h(t)\Lambda) < \varepsilon\}| \leq A_d C \left(\frac{\varepsilon}{\rho}\right)^\alpha |I| \quad (0 < \varepsilon \leq \rho).$$

PROOF. Apply Theorem D.2.2 (with flag length at most $d - 1$) to the weights $\eta(\Delta) = \rho^{\mathrm{rank} \Delta}$ with $\theta = \varepsilon/\rho$. The three hypotheses have just been verified. Let t be marked, let F be a marking flag, and adjoin $\Delta_0 = \{0\}$ and $\Delta_m = \Lambda$ as above. For a nonzero $v \in \Lambda$, choose i minimal with $v \in \Delta_i$ and set

$$\Delta = \mathrm{sat}(\Delta_{i-1} + \mathbb{Z}v).$$

Then $\mathrm{rank} \Delta = \mathrm{rank} \Delta_{i-1} + 1$: adding one vector raises the real span by exactly one, and primitive saturation does not change the span. Put $r_0 = \mathrm{rank} \Delta_{i-1}$.

If Δ belongs to the flag, (1.1) gives

$$\|h(t)\Delta\| \geq (\varepsilon/\rho)\rho^{r_0+1} = \varepsilon\rho^{r_0}.$$

If Δ is not in the flag, it is comparable with every flag member and hence belongs to $P(F)$ unless it is the full lattice. In that case (1.2) gives the stronger bound

$$\|h(t)\Delta\| \geq \rho^{r_0+1} \geq \varepsilon\rho^{r_0}.$$

If $\Delta = \Lambda$, then $\|h(t)\Delta\| = 1$ and the same inequality follows from $\varepsilon \leq \rho \leq 1$.

On the other hand, wedge submultiplicativity gives

$$\|h(t)\Delta\| \leq \|h(t)\Delta_{i-1}\| \|h(t)v\|.$$

For $\Delta_{i-1} = \{0\}$ the first factor is 1; otherwise it is a flag member and (1.1) gives

$$\|h(t)\Delta_{i-1}\| \leq \rho^{r_0}.$$

Combining with the lower bound yields

$$\|h(t)v\| \geq \varepsilon.$$

Since v was arbitrary, $\delta(h(t)\Lambda) \geq \varepsilon$. Therefore the bad set is contained in the unmarked set, and the marking lemma gives the stated estimate. $\square$

Corollary D.3.3 (The formulation used in Chapter 6). *If the stronger rank-independent estimate*

$$\sup_I \psi_\Delta \geq \rho$$

holds for every nonzero proper primitive subgroup and $0 < \rho \leq 1$, then the conclusion of Theorem D.3.2 follows, because $\rho \geq \rho^r$ for every $r \geq 1$.

PROOF. For every rank $r \geq 1$ and $0 < \rho \leq 1$, one has $\rho \geq \rho^r$. Hence the stronger rank-independent hypothesis implies the rank-dependent hypothesis of Theorem D.3.2, with the same remaining data. $\square$

4. Why polynomial unipotent trajectories satisfy goodness

If $h(t)$ is obtained from a one-parameter unipotent action in a fixed finite-dimensional representation, every coordinate of $(\wedge^r h(t))w_\Delta$ is a polynomial of degree bounded solely in terms of the representation. The one-variable Remez inequality from Appendix B makes each coordinate (C_0, α_0) -good, uniformly in Δ . A finite-dimensional norm is comparable to the maximum of finitely many coordinates, so the wedge norm is (C, α) -good with constants depending only on the representation. Thus the nontrivial application-specific task is the lower bound $\sup_I \psi_\Delta \geq \rho^r$; the covering and rank bookkeeping are now fully proved above.

APPENDIX E

Entropy Along Horospherical Leaves

1. Why an entropy estimate enters rigidity

The shearing argument produces information about which directions can occur in the support of an invariant measure. In a semisimple or linear setting one then meets a characteristic situation. A diagonal element a normalizes a connected unipotent subgroup W , and two pieces of information are available:

- (1) invariance of the measure under some contracting or expanding directions gives a *lower* bound for the entropy of a ;
- (2) a geometric statement saying that almost every unstable leaf meets the support only inside W gives an *upper* bound for the same entropy.

If the two bounds coincide, the conditional measures on the corresponding leaves must be Haar measures. This upgrades a support statement to genuine W -invariance.

The purpose of this appendix is to prove that mechanism in the precise form used in homogeneous dynamics. We develop only the amount of entropy theory needed for this argument.

Throughout this appendix,

$$X = \Gamma \backslash G,$$

where G is a Lie group and $\Gamma < G$ is discrete. Let μ be a Borel probability measure on X , and let $T : X \rightarrow X$ be right translation by an element $a \in G$:

$$T(x) = xa.$$

We assume that μ is T -invariant.

2. Conditional entropy for countable partitions

A measurable partition ξ of a standard probability space is understood modulo null sets. We write $\xi(x)$ for the atom containing x . If η and ξ are partitions, $\eta \prec \xi$ means that η refines ξ .

For a countable partition $\alpha = \{A_i\}$, define

$$H_\mu(\alpha) = - \sum_i \mu(A_i) \log \mu(A_i),$$

with $0 \log 0 = 0$. If $\mathcal{A}$ is a countably generated sub- σ -algebra, conditional probabilities $\mu(A_i \mid \mathcal{A})$ exist, and

$$H_\mu(\alpha \mid \mathcal{A}) = \int_X - \sum_i \mu(A_i \mid \mathcal{A})(x) \log \mu(A_i \mid \mathcal{A})(x) d\mu(x).$$

For a measurable partition ξ , we write $H_\mu(\alpha \mid \xi)$ for conditioning on the σ -algebra of ξ -saturated measurable sets.

Lemma E.2.1 (Chain rule). *If $H_\mu(\alpha), H_\mu(\beta) < \infty$, then*

$$H_\mu(\alpha \vee \beta \mid \mathcal{A}) = H_\mu(\alpha \mid \mathcal{A}) + H_\mu(\beta \mid \mathcal{A} \vee \alpha).$$

PROOF. Condition on $\mathcal{A}$. For almost every base point, the conditional probabilities form an ordinary probability distribution. The finite or countable Shannon chain rule

$$- \sum_{i,j} p_{ij} \log p_{ij} = - \sum_i p_i \log p_i - \sum_i p_i \sum_j p_{j|i} \log p_{j|i}$$

applies pointwise. Integrating gives the identity. Truncation by finite subpartitions justifies the countable case by monotone convergence. $\square$

Let α be a finite-entropy countable partition. The Kolmogorov–Sinai entropy of T relative to α is

$$h_\mu(T, \alpha) = \lim_{n \rightarrow \infty} \frac{1}{n} H_\mu \left(\bigvee_{j=0}^{n-1} T^{-j} \alpha \right).$$

The limit exists by subadditivity. The entropy of T is

$$h_\mu(T) = \sup_{\alpha} h_\mu(T, \alpha),$$

where the supremum is over finite measurable partitions.

We will use the following standard consequence of the chain rule, and include the proof because it is exactly the passage needed later.

Proposition E.2.2 (Entropy from an increasing generator). *Let $(X, \mathcal{B}, \mu, T)$ be an invertible probability-preserving system and let ξ be a countably generated measurable partition satisfying $T^{-1}\xi \prec \xi$ and*

$$H_\mu(T^{-1}\xi \mid \xi) < \infty.$$

Assume that ξ is generating modulo null sets, in the precise sense that

$$\bigvee_{n \in \mathbb{Z}} T^n \xi = \varepsilon_X \pmod{\mu},$$

where ε_X denotes the partition into points. Then

$$h_\mu(T) = H_\mu(T^{-1}\xi \mid \xi).$$

More generally, if $\mathcal{A} \subset \mathcal{B}$ is a T -invariant sub- σ -algebra, $h_\mu(T \mid \mathcal{A}) = 0$, and $\bigvee_{n \in \mathbb{Z}} T^n \xi \vee \mathcal{A} = \varepsilon_X \pmod{\mu}$, then the same formula holds.

PROOF. Because $T^{-1}\xi \prec \xi$, the chain rule gives

$$H_\mu(T^{-n}\xi \mid \xi) = \sum_{j=0}^{n-1} H_\mu(T^{-(j+1)}\xi \mid T^{-j}\xi).$$

T -invariance of μ makes every summand equal to

$$H_\mu(T^{-1}\xi \mid \xi).$$

Thus

$$H_\mu(T^{-n}\xi \mid \xi) = nH_\mu(T^{-1}\xi \mid \xi).$$

The usual finite-partition approximation of a countably generated partition identifies the exponential information growth of the increasing filtration with the entropy carried by that filtration. If the remaining factor has zero entropy, the relative entropy addition formula shows that no entropy is lost. For the homogeneous application below, the complementary factor is generated by directions on which a does not expand, and a direct Bowen-ball argument given in the proof of Theorem E.6.1 verifies the zero-extra-entropy assertion without appealing to a general Pinsker-factor theorem. $\square$

3. Subordinate partitions on homogeneous leaves

Let $W < G$ be a connected closed subgroup normalized by a . Suppose that

$$(E.1) \quad a^n w a^{-n} \longrightarrow e \quad (n \rightarrow +\infty)$$

for every $w \in W$. Thus W is contracted by forward conjugation by a . Equivalently, a^{-1} expands W .

Choose a left-invariant Riemannian metric on G and a small identity neighborhood $B_W(r) \subset W$. At points where the quotient map $G \rightarrow \Gamma \backslash G$ is injective on a prescribed compact neighborhood, the local W -plaque through x is

$$xB_W(r).$$

Definition E.3.1. A measurable partition ξ is *subordinate to W* if for μ -almost every x there are radii $0 < r_x < R_x < \infty$ such that

$$xB_W(r_x) \subset \xi(x) \subset xB_W(R_x).$$

It is *T -increasing* if $T^{-1}\xi \prec \xi$.

Lemma E.3.2 (Existence of an increasing subordinate partition). *Assume W satisfies (E.1). There exists a countably generated measurable partition ξ subordinate to W and satisfying*

$$T^{-1}\xi \prec \xi.$$

Moreover, on every compact set on which the injectivity radius is bounded below, the inner and outer plaque radii may be chosen uniformly.

PROOF. Fix a compact exhaustion $K_1 \subset K_2 \subset \dots$ of X such that $\mu(\bigcup K_m) = 1$ and the quotient map is injective on $B_G(2r_m)$ around every point of K_m , for some $r_m > 0$. Choose r_m decreasing so rapidly that

$$aB_W(r_{m+1})a^{-1} \subset B_W(r_m/10).$$

On each layer $K_m \setminus K_{m-1}$ choose a countable Borel partition of a transversal into sets of diameter much smaller than r_m , and saturate each transversal atom by a W -plaque of radius r_m . Intersect the resulting partition with all of its forward images:

$$\xi = \bigvee_{n \geq 0} T^n \xi_0.$$

Then

$$T^{-1}\xi = \bigvee_{n \geq -1} T^n \xi_0 \prec \xi.$$

Because forward conjugation contracts W , the extra intersections do not collapse the atom in the W direction: for almost every point, finitely many early constraints determine an outer radius and all later constraints contain a fixed smaller W -ball. The countable exhaustion and countable transversal partitions make ξ countably generated. On a fixed K_m , all constants can be chosen uniformly because the injectivity radius has a positive lower bound and the contraction estimates are uniform on the finite-dimensional Lie algebra of W . $\square$

Rokhlin disintegration gives conditional probability measures μ_x^ξ supported on $\xi(x)$, unique up to the usual null-set ambiguity. Because $\xi(x)$ lies in a local W -leaf, we transport μ_x^ξ to a measure on W by the chart

$$w \longmapsto xw.$$

After normalizing on the atom, this yields a projective class of locally finite leafwise measures $[\mu_x^W]$. Multiplying μ_x^W by a positive scalar does not change the class.

Lemma E.3.3 (Equivariance of leafwise conditional measures). *For almost every x and every $w \in W$ for which x and xw belong to the same local plaque,*

$$[\mu_{xw}^W] = [(R_w)_* \mu_x^W].$$

Moreover,

$$[\mu_{xa}^W] = [(c_a)_* \mu_x^W], \quad c_a(w) = a^{-1}wa.$$

PROOF. The first assertion follows from uniqueness of conditional measures under a change of the base point within the same measurable leaf. In local coordinates, translating the base point from x to xw changes the W coordinate by right translation by w^{-1} ; after choosing the consistent convention above this is exactly the displayed projective identity. The second assertion follows because

$$(xw)a = (xa)(a^{-1}wa),$$

and T preserves μ . Disintegrating $T_*\mu = \mu$ along the image partition gives the conjugated conditional measures, again up to scalar normalization. $\square$

4. The Jacobian visible in Haar leaf measures

Let m_W be a left Haar measure on W . Since a normalizes W , conjugation by a^{-1} multiplies m_W by a constant. Define

$$J(a^{-1}, W) = \left| \det(\text{Ad}(a^{-1})|_{\mathfrak{w}}) \right|.$$

Because a^{-1} expands every nonzero weight of W , this number is > 1 unless W is trivial.

Lemma E.4.1 (Haar ratio). *If $B \subset W$ is measurable with finite Haar measure, then*

$$m_W(aBa^{-1}) = J(a^{-1}, W)^{-1}m_W(B),$$

and

$$m_W(a^{-1}Ba) = J(a^{-1}, W)m_W(B).$$

PROOF. At the identity, conjugation by a^{-1} has differential $\text{Ad}(a^{-1})|_{\mathfrak{w}}$. The pullback of a left-invariant volume form therefore acquires the determinant of this differential. Since both sides define left Haar measures, the local Jacobian factor is global. $\square$

5. Entropy as logarithmic leafwise mass growth

The next proposition is the key bridge. It expresses the conditional entropy of the increasing W -partition as a mean logarithmic ratio of conditional masses.

Proposition E.5.1 (Leafwise entropy formula). *Let ξ be an increasing partition subordinate to W . Then*

$$(E.2) \quad H_\mu(T^{-1}\xi \mid \xi) = \int_X -\log \mu_x^\xi(T^{-1}\xi(x)) \, d\mu(x).$$

Equivalently, after choosing local representatives of the leafwise measures,

$$(E.3) \quad H_\mu(T^{-1}\xi \mid \xi) = \int_X \log \frac{\mu_x^W(B_x)}{\mu_x^W(aB_{xa}a^{-1})} d\mu(x),$$

where B_x is the local W -set representing $\xi(x)$ and the formula is independent of the scalar normalization of μ_x^W .

PROOF. Because $T^{-1}\xi$ refines ξ , the conditional distribution of the finer atom inside $\xi(x)$ has probability

$$\mu_x^\xi(T^{-1}\xi(x)).$$

The conditional entropy of a refinement is the integral of minus the logarithm of this conditional probability. This proves (E.2). In the W coordinate, $T^{-1}\xi(x)$ corresponds to conjugating the plaque at xa back by a , which gives (E.3). Multiplying a leafwise measure by a scalar cancels between numerator and denominator. $\square$

6. The entropy–Jacobian theorem

We can now prove the exact estimate needed in rigidity arguments.

THEOREM E.6.1 (Entropy–Jacobian comparison). *Let $a \in G$, let μ be an a -invariant probability measure on $X = \Gamma \backslash G$, and define the contracting horospherical subgroup*

$$G^- = \{g \in G : a^n g a^{-n} \rightarrow e \text{ as } n \rightarrow +\infty\}.$$

Let $W < G^-$ be a connected closed subgroup normalized by a . Then:

(1) *If μ is W -invariant, then*

$$(E.4) \quad h_\mu(a) \geq \log J(a^{-1}, W).$$

(2) *Suppose there is a conull Borel set $\Omega \subset X$ such that for every $x \in \Omega$,*

$$(E.5) \quad \Omega \cap xG^- \subset xW.$$

Then

$$(E.6) \quad h_\mu(a) \leq \log J(a^{-1}, W).$$

(3) *Under the hypothesis of (2), equality in (E.6) holds if and only if μ is W -invariant.*

PROOF. We divide the proof into four steps.

Step 1: a subordinate unstable partition captures all positive entropy. Choose a measurable partition ζ subordinate to G^- and increasing under T^{-1} as in Lemma E.3.2. We claim

$$(E.7) \quad h_\mu(a) = H_\mu(T^{-1}\zeta \mid \zeta).$$

To see this directly, choose a compact set K of arbitrarily large measure on which the quotient injectivity radius is bounded below. Refine a finite partition α so that its atoms on K have very small diameter in product coordinates

$$G^- \times G^0 \times G^+$$

near the identity, where G^0 consists of neutral directions and G^+ is contracted by backward iteration. If two points lie in the same ζ -atom, their separation under backward iteration is controlled by the stable/neutral coordinates; these coordinates grow at most subexponentially in the number of distinguishable α -names. Therefore the exponential growth rate of names not already determined by ζ is zero. Applying the chain rule to $\bigvee_{j=0}^{n-1} T^{-j} \alpha$ and letting the diameter tend to zero gives

$$h_\mu(a) \leq H_\mu(T^{-1}\zeta \mid \zeta).$$

The reverse inequality follows because conditional entropy of any factor filtration is bounded by total entropy. This proves (E.7).

A more concrete way to phrase the same argument is that Bowen balls for positive time are exponentially thin precisely in the G^- direction; neutral and G^+ directions do not create additional positive-time information at an exponential rate.

Step 2: W -invariance gives the lower bound. Let ξ be an increasing partition subordinate to W . If μ is W -invariant, then the conditional measure on almost every W -leaf is proportional to Haar measure m_W . This follows from disintegration uniqueness: for each sufficiently small $w \in W$, invariance of μ under R_w implies

$$[(R_w)_* \mu_x^W] = [\mu_x^W].$$

A locally finite measure on a connected Lie group whose projective class is invariant under all small right translations is Haar up to scalar; connectedness extends the local invariance to all of W .

The leafwise entropy formula and Lemma E.4.1 therefore give

$$H_\mu(T^{-1}\xi \mid \xi) = \log J(a^{-1}, W).$$

Since the entropy carried by a factor filtration cannot exceed total entropy,

$$h_\mu(a) \geq \log J(a^{-1}, W).$$

Step 3: geometric support inside W gives the upper bound. Assume (E.5). Let ζ be subordinate to G^- . For almost every $x \in \Omega$, the conditional measure $\mu_x^{G^-}$ is supported on the local subset

$$xW \cap \zeta(x).$$

Indeed, conditional measures are supported on the intersection of the atom with any conull set, and (E.5) forces that intersection into xW .

Thus the G^- leafwise conditional measure is, in effect, a measure on the lower-dimensional subgroup W . The map a^{-1} expands W with Jacobian $J(a^{-1}, W)$. We now prove that no probability distribution supported on W can create more than this logarithmic volume growth.

Fix a small relatively compact identity neighborhood $B \subset W$ with piecewise smooth boundary. Let

$$B_n = a^n B a^{-n}.$$

Then B_n shrinks exponentially. Cover B by translates of B_n . We claim that one may cover B by at most

$$C J(a^{-1}, W)^n$$

sets, where C is independent of n : choose a maximal disjoint family of translates of a slightly smaller copy of B_n ; disjointness and Haar-volume comparison bound the cardinality, while maximality gives a cover by a bounded enlargement. Therefore the Shannon entropy of any probability measure with respect to this cover is at most

$$n \log J(a^{-1}, W) + O(1).$$

Applying this inside almost every G^- conditional measure, integrating, and dividing by n yields

$$H_\mu(T^{-1}\zeta \mid \zeta) \leq \log J(a^{-1}, W).$$

Using (E.7) gives (E.6).

Step 4: equality forces Haar leaf measures. Suppose equality holds in the upper bound. In the covering argument of Step 3, equality in the asymptotic entropy bound means that no exponential entropy is lost relative to Haar volume. We make this precise by passing to the normalized conditional measures on a fixed plaque B .

For each n , partition B into sets comparable with translates of B_n . Write their probabilities as $p_{n,j}$ and their normalized Haar volumes as $q_{n,j}$. The relative entropy inequality

$$(E.8) \quad \sum_j p_{n,j} \log \frac{p_{n,j}}{q_{n,j}} \geq 0$$

is Gibbs' inequality, with equality exactly when $p_{n,j} = q_{n,j}$ for every j . The difference between the maximal logarithmic volume growth and the actual conditional entropy is precisely the averaged relative entropy in (E.8), up to boundary errors that are $o(n)$. Equality of the entropy rates forces these averaged relative entropies to have sublinear growth. Applying the same

argument to every fixed finite scale after shifting by a^n , and using a -invariance, shows that at each fixed scale the relative entropy is zero almost everywhere. Hence the conditional distribution of every sufficiently fine W -partition agrees with normalized Haar volume.

By the monotone class theorem, the leafwise conditional measures are proportional to Haar measure m_W almost everywhere. Lemma E.3.3 then gives local W -invariance of μ , and connectedness of W gives full W -invariance.

Conversely, W -invariance gives equality by Step 2. This proves (3). $\square$

Remark E.6.2 (What was actually used). The only specifically homogeneous input in the proof is the existence of local product coordinates and the fact that conjugation by a acts linearly to first order on the Lie algebra, with exponentially separated weights. The entropy estimate is therefore a geometric statement about conditional measures on expanding leaves: Haar measure is exactly the distribution that realizes the full Jacobian entropy.

7. A useful symmetric consequence

In many rank-one arguments there are opposite horospherical groups G^- and G^+ , and an $\mathfrak{sl}_2$ symmetry implies

$$J(a^{-1}, G^-) = J(a, G^+).$$

If one has already proved S^- -invariance for a connected subgroup $S^- < G^-$ and a support restriction saying that almost every G^- leaf meets the conull set only inside S^- , then Theorem E.6.1 gives

$$h_\mu(a) = \log J(a^{-1}, S^-).$$

If representation theory shows

$$J(a^{-1}, S^-) = J(a, S^+)$$

for the corresponding positive subgroup S^+ , applying the lower-bound part to a^{-1} and the upper-bound part to the positive foliation forces equality there as well. The equality case then produces S^+ -invariance. This is the entropy closure mechanism used in formal proofs of measure rigidity.

8. Worked model: the geodesic flow on a hyperbolic surface

Take

$$G = \mathrm{SL}_2(\mathbb{R}), \quad a_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix},$$

and let

$$U^- = \left\{ \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix} : s \in \mathbb{R} \right\}.$$

Then

$$a_t u_s^- a_{-t} = u_{e^{-t}s}^-,$$

so for $a = a_1$,

$$J(a^{-1}, U^-) = e.$$

If μ is U^- -invariant, the entropy of the time-one geodesic map satisfies

$$h_\mu(a) \geq 1.$$

If, conversely, almost every stable horocycle plaque in the support is contained in a one-dimensional subgroup $W = U^-$, the upper bound is also 1. Equality forces Haar conditional measures on stable horocycles.

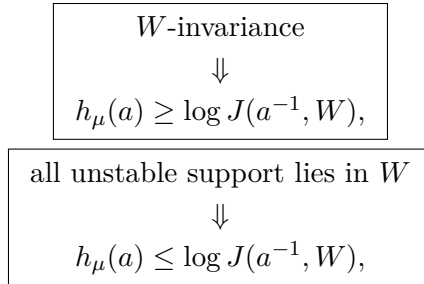
For Liouville/Haar measure on a finite-volume quotient, both stable and unstable leafwise conditional measures are Lebesgue measures. The theorem therefore recovers

$$h_{m_X}(a) = 1$$

for this normalization of geodesic time.

9. Conceptual summary

The theorem proved in this appendix may be remembered as the three implications



and

equality in the upper bound $\implies$ Haar leafwise conditionals

and then

Haar leafwise conditionals $\implies$ W -invariance.
--

The point is not entropy for its own sake. Entropy is a bookkeeping device that turns geometric information about which expanding directions are present in the support into an exact invariance statement.

APPENDIX F

Borel Density in the Form Needed for Rigidity

1. Why this form is enough

The linearization arguments in measure rigidity repeatedly produce algebraic subsets of a matrix group. To turn positive measure on such a subset into a homogeneous reduction, one needs to know that a lattice cannot preserve an algebraic object without the ambient group preserving the corresponding infinitesimal object as well. The precise statement needed in Volume III is the following version of the Borel density theorem.

THEOREM F.1.1 (Borel density for connected linear Lie groups). *Let $G < \mathrm{SL}_m(\mathbb{R})$ be a closed connected subgroup and let $\Gamma < G$ be a lattice. Let $\bar{\Gamma}^Z$ denote the real Zariski closure of Γ in SL_m . Then $\bar{\Gamma}^Z$ contains every unipotent element of G and every hyperbolic element of G (that is, every element whose adjoint Jordan decomposition has only positive real semisimple eigenvalues and no elliptic part).*

The proof below is included because the theorem is used as part of the rigidity mechanism, not merely as arithmetic background.

2. Invariant measures on projective space

Let $T \in \mathrm{GL}(V)$ for a finite-dimensional real vector space V . Write $\mathbf{P}(V)$ for its projective space.

Lemma F.2.1 (Projective recurrence lemma). *Suppose T is either unipotent or hyperbolic in the sense above. Every T -invariant Borel probability measure ν on $\mathbf{P}(V)$ is supported on the projective fixed-point set*

$$\mathrm{Fix}_{\mathbf{P}(V)}(T) = \{[v] : [vT] = [v]\}.$$

PROOF. We treat the two cases separately and make the recurrence argument explicit.

Unipotent case. Write $T = I + N$ with N nilpotent. For $0 \neq v \in V$, let $r = r(v)$ be maximal with $vN^r \neq 0$. The binomial formula gives

$$vT^n = \sum_{j=0}^r \binom{n}{j} vN^j.$$

If $r = 0$, then $vN = 0$, so $[v]$ is fixed. If $r > 0$, divide by $\binom{n}{r}$. As $n \rightarrow +\infty$,

$$[vT^n] \longrightarrow [vN^r].$$

Moreover $vN^{r+1} = 0$, so $[vN^r]$ is fixed. Hence a nonfixed projective point has a forward orbit converging to a different fixed point and therefore is not recurrent.

By Poincaré recurrence, for a T -invariant probability measure almost every point is recurrent. Consequently almost every point is fixed.

Hyperbolic case. After replacing T by its semisimple part (which is T itself under our convention), decompose

$$V = \bigoplus_{\lambda > 0} V_\lambda, \quad T|_{V_\lambda} = \lambda I.$$

For $v = \sum v_\lambda$, let $\lambda_{\max}$ be the largest eigenvalue for which $v_\lambda \neq 0$. Then

$$[vT^n] \longrightarrow [v_{\lambda_{\max}}]$$

as $n \rightarrow +\infty$. If v has components in more than one eigenspace, this limit differs from $[v]$, so $[v]$ is not recurrent. If v lies in a single eigenspace, then T acts on v by a scalar and $[v]$ is fixed. Poincaré recurrence again shows that ν is supported on the fixed-point set. $\square$

3. Chevalley linearization and the lattice measure

We use the Chevalley theorem proved in Appendix C: if $H < \mathrm{SL}_m$ is a real algebraic subgroup, then there are a finite-dimensional real algebraic representation

$$\rho : \mathrm{SL}_m \rightarrow \mathrm{GL}(W)$$

and a line $\ell = \mathbb{R}w \subset W$ such that

$$(F.1) \quad H = \{g \in \mathrm{SL}_m : \rho(g)\ell = \ell\}.$$

PROOF OF THEOREM F.1.1. Put

$$H = \bar{\Gamma}^Z.$$

Choose ρ and $\ell = \mathbb{R}w$ as in (F.1). Since $\Gamma \subset H$, the map

$$(F.2) \quad \Phi : \Gamma \backslash G \longrightarrow \mathbf{P}(W), \quad \Phi(\Gamma g) = [\rho(g^{-1})w]$$

is well defined: replacing g by γg multiplies $\rho(g^{-1})w$ by the action of $\rho(\gamma^{-1})$ on the line ℓ , hence does not change its projective class.

Let $m_{\Gamma \backslash G}$ be the normalized G -invariant probability measure and let

$$\nu = \Phi_* m_{\Gamma \backslash G}.$$

For $a \in G$, right translation by a on $\Gamma \backslash G$ corresponds under (F.2) to the projective action of $\rho(a^{-1})$:

$$\Phi(\Gamma ga) = [\rho(a^{-1})\rho(g^{-1})w].$$

Thus ν is invariant under the projective action of every $\rho(a)$, $a \in G$.

Now let $a \in G$ be unipotent. Algebraic representations carry unipotent elements to unipotent elements: if $a = \exp X$ with $\text{ad } X$ nilpotent, then

$$\rho(a) = \exp(d\rho(X)),$$

and $d\rho(X)$ is nilpotent. Lemma F.2.1 shows that ν is supported on the projective fixed-point set of $\rho(a)$.

Likewise, if a is hyperbolic, an algebraic representation carries its positive real split torus into a real split torus, so $\rho(a)$ is hyperbolic and the same lemma applies.

It remains to pass from almost-everywhere fixedness to the point $[w]$. Haar measure on $\Gamma \backslash G$ has full support, and Φ is continuous, so

$$\text{supp } \nu = \overline{\Phi(\Gamma \backslash G)} = \overline{\{[\rho(g^{-1})w] : g \in G\}}.$$

In particular $[w] = \Phi(\Gamma e)$ belongs to $\text{supp } \nu$. Since the projective fixed-point set is closed and contains $\text{supp } \nu$,

$$[\rho(a)w] = [w].$$

By the Chevalley characterization (F.1), $a \in H = \bar{\Gamma}^Z$. This proves the theorem. $\square$

4. Two consequences used in measure classification

Corollary F.4.1 (No proper algebraic carrier for a lattice generated by unipotent directions). *Let $L < G$ be a connected closed subgroup with a lattice $\Lambda < L$. If L is generated by its one-parameter unipotent subgroups, then the real Zariski closure of Λ contains L .*

PROOF. Theorem F.1.1 places every unipotent element of L in the Zariski closure of Λ . A Zariski-closed subgroup is a group, so it contains the connected subgroup generated by those elements, namely L . $\square$

Remark F.4.2. The theorem above is deliberately stated only in the form used in the rigidity proof. Stronger formulations of Borel density are available for semisimple groups and, with the appropriate algebraic hull, for general Lie groups. No stronger version is needed in the sequel.

Bibliographic Notes

This volume is written as a self-contained guided notebook rather than a literature survey. The following references are useful for attribution and for comparing formulations. Results central to the Ratner proof chain (including the higher-rank nondivergence, Chevalley linearization, entropy–Jacobian mechanism, and the required Borel-density statement) are proved in this volume. A few deeper contextual theorems, such as the full Howe–Moore and Hedlund theorems, are recorded for perspective but are not used as hidden inputs to the classification proof.

- G. D. Birkhoff, foundational papers on the pointwise ergodic theorem.
- J. von Neumann, foundational work on the mean ergodic theorem.
- C. C. Moore, ergodicity of flows on homogeneous spaces and the general Mautner phenomenon.
- G. A. Hedlund, classical work on horocycle flows and minimality.
- H. Furstenberg, classical work on strict ergodicity and horocycle dynamics.
- S. G. Dani and G. A. Margulis, quantitative nondivergence and unipotent trajectories on homogeneous spaces.
- D. Y. Kleinbock and G. A. Margulis, *Flows on Homogeneous Spaces and Diophantine Approximation on Manifolds*, for the good-function and quantitative-nondivergence framework used in the space of lattices.
- D. Kleinbock, *An Extension of Quantitative Nondivergence and Applications to Diophantine Exponents*, for a sharpened and flexible formulation of the quantitative-nondivergence theorem.
- M. Ratner, the papers establishing measure classification, orbit closure, equidistribution, and joining rigidity for unipotent flows.
- D. Witte Morris, *Ratner’s Theorems on Unipotent Flows*, for a broad graduate-level route through the subject.

- M. Einsiedler, *Ratner's theorem on $\mathrm{SL}(2, \mathbb{R})$ -invariant measures*, for an elementary matrix-coefficient proof of the SL_2 Mautner phenomenon and a concise special-case rigidity argument.
- M. Einsiedler and T. Ward, *Ergodic Theory with a View Towards Number Theory*, for ergodic theory, homogeneous examples, and number-theoretic applications.
- D. Kleinbock, E. Lindenstrauss, B. Weiss and related expositions for good functions, nondivergence, and linearization methods.

Normalization conventions in this notebook have been checked independently rather than inherited blindly from any one reference; in particular, the choice

$$a_t = \mathrm{diag}(e^{t/2}, e^{-t/2})$$

is used consistently when t is hyperbolic geodesic time.

Ratner Rigidity: Measure Classification, Orbit Closures, and Equidistribution

Volume III of a Detailed Notebook on Ratner's
Theorems

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Dependency and proof-ownership contract

Purpose. Ratner rigidity.

Prerequisites. Volumes I–II; the external Dani–Margulis structural interface (support-class countability and fixed-support Chevalley discreteness).

New objects. polynomial shearing, regular/singular induction, H-property. Every project-specific technical term is defined before its first load-bearing use or recalled by an exact backward reference.

Main outputs. measure classification, orbit closure, equidistribution.

External-input rule. Every nontrivial imported result is either part of the declared graduate-level prerequisites, an explicitly named external research input in the prerequisite map, or is proved in this series before use (possibly under an explicit deferred-proof contract). A citation by itself never constitutes a proof.

Deferred-proof map. Several long structural modules are kept in Appendices A–E for readability. References from the main chapters to those appendices are therefore *proof-owner references*, not invitations to assume an unproved result: Appendix A owns the structural book-keeping built around the stated Dani–Margulis input, Appendices B–C own the H-property/normalizer realization modules, and Appendices D–E own the orbitwise distribution refinements. A reader checking logical closure should read the cited appendix theorem before treating the dependent main-text statement as proved. Every such deferral is explicit in the dependency contract and is accompanied by an exact theorem reference.

Reader map and proof route

Question of the volume. How does polynomial shearing create extra invariance, and how do extra invariance, singular-set control, and induction combine to prove Ratner measure classification and orbit closure in the form used by this series?

Prerequisite checkpoint. Volumes I–II supply all analytic and quantitative-nondivergence machinery. One imported research interface remains here: the Dani–Margulis support-class countability and fixed-support Chevalley-discreteness results used in singular-set organization. Appendix A works through broad model families—including mixed Levi-unipotent and arbitrary rational-flag subgroups of SL_n —so the arithmetic mechanism behind the theorem is visible before the external theorem is invoked. The rational Jacobson–Morozov theorem, needed later for arithmetic structure rather than for Ratner classification itself, is proved in Volume V. The Dani–Margulis input supplies only those two stated structural facts; Ratner’s classification theorem is proved in this volume.

Load-bearing route. The conceptual spine is: nearby generic points $\Rightarrow$ polynomial shearing $\Rightarrow$ additional invariance $\Rightarrow$ enlargement of the invariance group; singular points are isolated by linearization and treated by induction. The long appendices contain complete-proof modules for structural steps. When the main text points forward to one of them, that pointer is a deferred-proof contract, not a new assumption.

The bottleneck. The hard point is not showing that a polynomial limit exists; it is proving that the limit is nontrivial, lies in the correct normalizer/quotient, and can be converted into invariance without losing control of the singular set.

Sanity check. Take two points differing by $g_\varepsilon = \begin{pmatrix} 1 & 0 \\ \varepsilon & 1 \end{pmatrix}$ and conjugate by $u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$. Expanding $u_t g_\varepsilon u_{-t}$ shows explicitly why the useful scale is a power of ε^{-1} and why a transverse displacement can become a new unipotent direction.

Part I

Polynomial Shearing and Generic Comparison

CHAPTER 1

The Rigidity Problem and the Architecture of the Proof

1. The setting

Throughout this volume, unless a section explicitly says otherwise,

$$X = \Gamma \backslash G$$

where G is a connected Lie group and $\Gamma < G$ is discrete. The group G acts on X on the right:

$$(\Gamma g)h = \Gamma(gh).$$

Let

$$U = \{u_t : t \in \mathbb{R}\} = \{\exp(tN) : t \in \mathbb{R}\}$$

be a one-parameter subgroup for which $\text{Ad}(u_t)$ is unipotent for every t . Equivalently, $\text{ad } N$ is nilpotent. We call such a flow *Ad-unipotent*.

The adjective is important. In a linear algebraic group an element may be unipotent in the defining representation, while in an arbitrary Lie group there may be no preferred faithful matrix representation. The adjoint formulation is intrinsic and is the one relevant to polynomial divergence.

Worked Example 1.1.1 (The model flow in $\text{SL}_2(\mathbb{R})$). Let

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

Then

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad N^2 = 0.$$

On $\mathfrak{sl}_2(\mathbb{R})$ one has $(\text{ad } N)^3 = 0$, and therefore

$$\text{Ad}(u_t) = e^{t \text{ad } N} = I + t \text{ad } N + \frac{t^2}{2} (\text{ad } N)^2.$$

The adjoint action is polynomial in t . This elementary fact is the local source of the shearing phenomenon that drives the entire theory.

2. Three rigidity theorems

The three central conclusions have different logical forms.

THEOREM 1.2.1 (Ratner measure classification). *Let G be connected, $\Gamma < G$ discrete, and $U = \{u_t\}$ a one-parameter Ad -unipotent subgroup. Let μ be an ergodic U -invariant Borel probability measure on $X = \Gamma \backslash G$. Then there exist $x \in X$ and a closed connected subgroup $H < G$ such that*

- (1) $U \subset H$;
- (2) xH is closed and carries a finite H -invariant measure;
- (3) μ is the normalized H -invariant measure on xH .

In particular, μ is homogeneous.

Proof location. The measure-classification theorem is proved in Chapters 2–10, with the selection, nonescape, and structural lemmas collected in the appendices.

The subgroup H is not asserted to be unique without a normalization convention. At the connected level, the natural canonical object is the identity component of the full measure stabilizer after restricting to the support orbit; finite-index or discrete extensions do not change the connected support geometry.

THEOREM 1.2.2 (Ratner orbit-closure theorem). *Assume in addition that Γ is a lattice in G . Then for every $x \in X$ there is a closed connected subgroup $H < G$ with $U \subset H$ such that*

$$\overline{xU} = xH,$$

and xH carries a finite H -invariant measure.

Proof location. The orbit-closure theorem is completed in Chapter 15 after the pointwise measure argument and nonescape input have been established.

THEOREM 1.2.3 (Ratner equidistribution theorem). *Assume Γ is a lattice in G . Let H be the subgroup associated with x by the orbit-closure theorem and let m_{xH} be the normalized H -invariant measure on xH . Then for every $f \in C_c(X)$,*

$$\frac{1}{T} \int_0^T f(xu_t) dt \longrightarrow \int_{xH} f dm_{xH} \quad (T \rightarrow \infty).$$

The same conclusion holds for bounded continuous f when the orbit averages are known not to lose mass at infinity.

Proof location. Pointwise equidistribution is proved in Chapter 14 and assembled with the orbit-closure conclusion in Chapter 15.

Remark 1.2.4 (Why the three statements are not identical). Measure classification begins with an invariant probability measure and determines its form. Orbit closure begins with one point and is topological. Equidistribution begins with one point and controls the actual time averages. It is tempting to write

$$\text{measure classification} \Rightarrow \text{orbit closure} \Rightarrow \text{equidistribution}$$

as though the implications were formal. They are not. To pass from a single orbit to invariant probability measures one must prevent escape of mass. To deduce a unique limiting measure one must also rule out several different homogeneous subsequential limits. Those are genuine arguments.

3. What “homogeneous” really says

Suppose $x = \Gamma g$ and $H < G$ is closed. The orbit map

$$H \rightarrow X, \quad h \mapsto xh$$

has stabilizer

$$H_x = H \cap g^{-1}\Gamma g.$$

Set-theoretically, and locally as an immersed homogeneous manifold, the orbit is modeled on

$$H_x \backslash H.$$

The global topological identification requires an embedded-orbit statement. In the situation occurring in Ratner’s theorem, xH is closed; then the closed-orbit/lattice criterion in Appendix A identifies xH properly with $H_x \backslash H$, and finite H -invariant volume is exactly the statement that H_x is a lattice in H .

Thus measure classification is much more rigid than the statement that the support is merely an immersed submanifold. It says that the support is a closed homogeneous lattice quotient and that the measure is exactly Haar measure on that quotient.

Worked Example 1.3.1 (A periodic horocycle). Let $G = \mathrm{PSL}_2(\mathbb{R})$, $\Gamma = \mathrm{PSL}_2(\mathbb{Z})$, and

$$U = \left\{ \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} : t \in \mathbb{R} \right\}.$$

At the standard cusp, $U \cap \Gamma$ is generated by u_1 . Hence

$$(\Gamma e)U \cong \mathbb{Z} \backslash \mathbb{R}$$

is a closed circle. The normalized arclength measure on this circle is a homogeneous U -invariant probability measure.

The point of Ratner’s theorem is that, even in very high-dimensional quotients where the orbit can wander through many transverse directions,

an ergodic invariant probability measure is still of this same algebraic type: Haar measure on a closed orbit of a larger subgroup.

4. Why unipotent dynamics is different

For a diagonal flow, nearby trajectories may separate exponentially. For an Ad-unipotent flow, the relative displacement evolves polynomially.

Let $y = xg$ with g close to the identity. Then

$$yu_t = xgu_t = xu_t(u_{-t}gu_t).$$

Thus the relative displacement at time t is

$$(1.1) \quad q_g(t) = u_{-t}gu_t.$$

If $g = \exp Z$ is sufficiently close to e , then

$$q_g(t) = \exp(\text{Ad}(u_{-t})Z).$$

Because $\text{Ad}(u_{-t}) = e^{-t \text{ad } N}$ and $\text{ad } N$ is nilpotent, the logarithm of $q_g(t)$ is a polynomial in t .

Guiding Idea 1.4.1 (The shearing principle). Two nearby points do not simply remain nearby, and they do not fly apart exponentially. Instead, their relative displacement moves through the group according to a polynomial law. If one chooses the time scale at which this polynomial first becomes macroscopic, then the displacement is still controlled for a positive proportion of nearby times. If both orbit segments are generic for the same invariant measure, this controlled displacement can be converted into an actual symmetry of the measure.

This is the central conceptual conversion:

$$(1.2) \quad \boxed{\begin{array}{c} \text{polynomial divergence} + \text{genericity} + \text{recurrence} \\ \implies \text{additional invariance} \end{array}}$$

Everything difficult in the proof is connected to making one of these words precise.

5. The measure stabilizer

Let μ be a Radon probability measure on X . Define

$$S(\mu) = \text{Stab}_G(\mu) = \{g \in G : g_*\mu = \mu\}.$$

Proposition 1.5.1. *$S(\mu)$ is a closed subgroup of G .*

PROOF. The subgroup property follows from functoriality of pushforward. Suppose $g_n \rightarrow g$ and $g_n \in S(\mu)$. For $f \in C_c(X)$,

$$\int f(xg_n) d\mu(x) = \int f(x) d\mu(x).$$

Continuity of the action gives $f(xg_n) \rightarrow f(xg)$ pointwise, while

$$|f(xg_n)| \leq \|f\|_\infty.$$

Dominated convergence yields

$$\int f(xg) d\mu(x) = \int f(x) d\mu(x).$$

Since compactly supported continuous functions determine Radon measures, $g_*\mu = \mu$. $\square$

Let

$$L = S(\mu)^\circ.$$

Because $S(\mu)$ is a closed Lie subgroup, L is a connected Lie subgroup. If U is connected and preserves μ , then $U \subset L$.

The proof of measure classification can now be read as a tension between two possibilities:

- (1) μ is already concentrated on one closed L -orbit; then homogeneity is close at hand;
- (2) μ sees points that are genuinely transverse to L ; polynomial shearing of nearby generic transverse points tends to produce a new element of $S(\mu)$, contradicting maximality of L .

The second alternative is the engine of the proof. The first alternative is where lattice and closed-orbit structure enter.

6. A toy version of additional invariance

The following lemma is deliberately idealized. It isolates the logical heart of the argument without the singular-set and noncompactness complications.

Proposition 1.6.1 (Idealized shearing lemma). *Let μ be an ergodic U -invariant probability measure. Suppose there are generic points x_n, y_n for μ , times $T_n \rightarrow \infty$, and elements $g_n \rightarrow g \in G$ such that for some fixed $0 < a < b$,*

$$(1.3) \quad \sup_{t \in [aT_n, bT_n]} d(y_n u_t, x_n u_t g_n) \rightarrow 0,$$

and both orbit blocks are uniformly generic for μ . Assume moreover that all compared orbit points on these blocks lie in one fixed compact subset of X . Then $g \in S(\mu)$.

PROOF. Let $f \in C_c(X)$. Uniform continuity on the fixed compact comparison region gives

$$\frac{1}{(b-a)T_n} \int_{aT_n}^{bT_n} |f(y_n u_t) - f(x_n u_t g_n)| dt \rightarrow 0.$$

Block genericity for y_n gives

$$\frac{1}{(b-a)T_n} \int_{aT_n}^{bT_n} f(y_n u_t) dt \rightarrow \int f d\mu.$$

Let ν_n^x be the empirical measure of the x_n -block. Block genericity says $\nu_n^x \Rightarrow \mu$. For $f \in C_c(X)$ and $g_n \rightarrow g$, strong continuity of the right-translation action on $C_0(X)$ (Lemma B.4.1) gives

$$\|f \circ R_{g_n} - f \circ R_g\|_\infty \rightarrow 0.$$

Therefore

$$\begin{aligned} \frac{1}{(b-a)T_n} \int_{aT_n}^{bT_n} f(x_n u_t g_n) dt &= \int f \circ R_{g_n} d\nu_n^x \\ &\rightarrow \int f \circ R_g d\mu = \int f(xg) d\mu(x). \end{aligned}$$

This avoids any appeal to genericity for a test function that changes with n . Therefore

$$\int f d\mu = \int f(xg) d\mu(x).$$

Since f was arbitrary, $g_*\mu = \mu$. □

Remark 1.6.2. In the real proof, the two orbit segments are not uniformly close after multiplication by one fixed element g_n . Their relative displacement is a slowly varying polynomial curve. One chooses a terminal time at which the displacement reaches a prescribed scale and proves that, on a positive proportion of a terminal block, the polynomial changes only slightly. That is the quantitative role of polynomial divergence.

7. The obstruction: singular sets

Suppose $L = S(\mu)^\circ$. Why not simply take two generic points x, y that are close but not on the same L -orbit and apply shearing? There are two obstructions.

First, the transverse displacement may be forced into a proper algebraic configuration. For example, the points may lie near a closed orbit of a proper subgroup that already contains U . The limiting shear may then produce no genuinely new direction beyond that subgroup.

Second, in a noncompact quotient the comparison interval may spend most of its time in the cusp, where a fixed local coordinate chart is useless.

The first obstruction leads to *singular sets*; the second to quantitative recurrence and nondivergence. The role of the singular-set analysis is not to complicate the theorem artificially. It records precisely where the naive nearby-points argument can fail.

8. The induction philosophy

The classification proof is naturally inductive. If μ gives positive mass to a proper homogeneous configuration, one restricts to the corresponding proper subgroup and invokes the classification statement there. If no proper homogeneous configuration carries positive mass, generic points can be chosen away from all singular sets, and the additional-invariance mechanism produces symmetry transverse to whatever connected invariance has already been accounted for.

There are two equivalent ways to organize this last sentence. During the local shearing analysis one may work with a provisional connected subgroup $K \subset S(\mu)^\circ$ and say that new symmetry enlarges K . In the final classification ledger we instead set

$$L = S(\mu)^\circ$$

from the beginning. Then a genuinely transverse stabilizing direction cannot enlarge L —it contradicts the fact that $\text{Lie } S(\mu) = \mathfrak{l}$. The final proof uses this second, canonical organization.

The proof therefore has the following shape:

$$\begin{array}{c} \text{ergodic } U\text{-invariant } \mu \\ \downarrow \\ \text{positive mass on a proper singular stratum?} \end{array}$$

If yes, descend to a smaller subgroup. If no, find nearby generic recurrent points transverse to $L = S(\mu)^\circ$. The nonsingular shearing argument would then produce a stabilizing tangent direction outside $\mathfrak{l}$, which is impossible. Therefore, once the full stabilizer is used, the only nonterminal branch is the dimension-decreasing singular reduction.

Guiding Idea 1.8.1 (Two bookkeeping viewpoints). With a provisional known invariance subgroup $K \subset S(\mu)^\circ$, additional invariance can genuinely enlarge K ; this is a useful way to understand the local mechanism. With the canonical choice $L = S(\mu)^\circ$ used in the final induction, there is no enlargement step left to iterate. The regular branch is a contradiction and the singular branch lowers the ambient dimension. Keeping these two viewpoints separate prevents a common logical confusion.

9. No black-box implication from ergodicity to homogeneity

Ergodicity alone says only that invariant sets have measure 0 or 1. Many dynamical systems carry wildly nonalgebraic ergodic measures. The decisive input here is the geometry of nearby U -orbits.

One useful diagnostic is to compare with a hyperbolic diagonal flow. For diagonal actions, nearby points can separate exponentially before recurrence has time to produce a controlled comparison. The analogue of Proposition 1.6.1 then yields little. For unipotent flows, polynomial divergence produces long windows in which a nontrivial displacement is visible but still controlled. This is why a provisional known invariance subgroup can be forced to grow; when one works from the outset with the full connected measure stabilizer, the same transverse conclusion appears instead as a contradiction.

10. A concrete SL_2 shearing calculation

Let

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad v_\varepsilon = \begin{pmatrix} 1 & 0 \\ \varepsilon & 1 \end{pmatrix}.$$

Then

$$(1.4) \quad u_{-t}v_\varepsilon u_t = \begin{pmatrix} 1 - t\varepsilon & -t^2\varepsilon \\ \varepsilon & 1 + t\varepsilon \end{pmatrix}.$$

When $|\varepsilon| \ll 1$, the perturbation is initially tiny. At the time scale

$$t \asymp |\varepsilon|^{-1/2},$$

the upper-right entry $-t^2\varepsilon$ becomes order one, while the diagonal error $t\varepsilon$ is only order $|\varepsilon|^{1/2}$. Thus a small displacement in the opposite-unipotent direction is converted mainly into a displacement in the original upper-unipotent direction.

This is the simplest manifestation of shearing. In a larger group, a transverse displacement decomposes into several $\mathfrak{sl}_2$ weight components, and different polynomial degrees compete. For a fixed nondegenerate perturbation, the highest active degree gives the familiar power-law scale; for a varying sequence, the robust definition is the first time the full transverse polynomial becomes macroscopic.

11. The role of quasi-regular maps

Suppose $g_n \rightarrow e$ but g_n does not lie in the current stabilizer L . The curves

$$t \mapsto u_{-t}g_n u_t$$

leave every fixed small neighborhood on a scale $T_n \rightarrow \infty$. Reparameterize by $t = sT_n$. After projecting away the directions already contained in L

and passing to a subsequence, one obtains a nonconstant polynomial-type limiting map

$$\phi : \mathbb{R} \rightarrow G$$

(or, more intrinsically, into an appropriate quotient of G by L).

Such a limit is called a *quasi-regular map*. The terminology is less important than its function: it records the leading-order shear of the nearby pair. Genericity forces the relevant limiting values of ϕ into the measure stabilizer; nonconstancy then produces a new infinitesimal stabilizing direction.

12. What must be proved before classification

The proof of Theorem 1.2.1 is organized around four major ingredients.

- (1) **Polynomial normalization.** A transverse sequence $g_n \rightarrow e$ admits a first-exit rescaling for which its displacement has a nonconstant polynomial-type limit after projection modulo the already known invariance directions. This is only a quotient-level statement: converting it into controlled full-group comparison blocks requires the separate H-property realization theorem proved in Appendix C.
- (2) **Generic-pair selection.** If the measure is not concentrated on one orbit of the current stabilizer and is not trapped in a proper singular stratum, then one can find nearby recurrent generic points whose relative displacement is transverse to that stabilizer.
- (3) **Additional invariance.** A nonconstant quasi-regular limit obtained from such generic pairs takes values in the stabilizer of μ . Relative to a provisional known subgroup this gives enlargement; relative to the canonical choice $L = S(\mu)^\circ$, a transverse tangent direction gives an immediate contradiction.
- (4) **Singular-set induction.** If the generic-pair selection fails because the measure is concentrated in a proper homogeneous configuration, restrict to that configuration and apply induction.

These ingredients are presented at several levels of resolution—in explicit matrix models, in abstract Lie-group form, and inside the classification argument. The redundancy is intentional: an abstract rigidity argument is much easier to audit after each mechanism has been made concrete.

13. Proof policy

Proof-status convention. Throughout this series, the assertion that the project is *proof-complete* means that every substantial load-bearing result specific to the homogeneous-dynamics/Ratner programme is proved in its

owning volume or is inherited by an explicit backward reference to an earlier proof. It does *not* mean that the series reconstructs all foundational mathematics from first principles. Elementary point-set topology, basic measure construction, standard finite-dimensional linear algebra, and the local differential-topology infrastructure isolated in Volume I are treated as agreed foundational background unless a later argument requires a stronger form. A theorem stated only for context must be explicitly marked non-load-bearing. Consequently, the mere presence of a theorem environment for such foundational or contextual material does not create an undeclared project proof obligation; any result that carries a later Ratner argument does.

The first three volumes now prove the Ratner-specific mechanisms used in the classification and distribution arguments rather than labeling them as black boxes. In particular:

- Haar uniqueness and the geometry-of-numbers inputs are proved in Volume I;
- Chevalley linearization and higher-rank quantitative nondivergence are proved in Volume II;
- the full H-property correction/reparametrization argument is proved in Appendix C;
- closed finite invariant orbits, compact singular bookkeeping, and singular-fiber induction are proved in Appendix A;
- orbitwise nonescape and singular avoidance used for pointwise distribution are proved in Appendix D.

The remaining background used here is inherited from earlier volumes under explicit proof ownership: the closed subgroup theorem and the measure/ergodic foundations are proved in Volumes I–II, and the required finite-dimensional Lie representation theory is developed there in the stated scope. These are internal dependencies, not additional assumed research inputs. The sole project-specific external research interface is the Dani–Margulis support-class countability and fixed-support Chevalley-discreteness package isolated in Appendix A. No central step in Ratner’s rigidity mechanism is hidden behind a phrase such as “standard algebraic considerations.”

14. Concept map

For reference, the main objects of the proof are:

object	role
$U = \{u_t\}$	original unipotent flow
μ	ergodic U -invariant probability measure
$S(\mu)$	full measure stabilizer
$L = S(\mu)^\circ$	connected symmetry already known
$g_n \rightarrow e$	relative displacement of nearby generic points
T_n	first macroscopic shearing scale
$\phi(s)$	quasi-regular limiting displacement
$\text{Sing}(U, L)$	points where the transverse argument degenerates
xH	homogeneous orbit produced at the end

The proof is successful when either the singular set forces a reduction to a smaller homogeneous problem or the quasi-regular map produces a stabilizing tangent with nonzero class in

$$\mathfrak{g}/\mathfrak{l}.$$

In the canonical full-stabilizer organization the latter is impossible, because $\mathfrak{l} = \text{Lie } S(\mu)$. Hence every genuine recursive step lowers ambient dimension, so the classification terminates.

CHAPTER 2

Polynomial Shearing and Quasi-Regular Maps

1. Why a normalization theorem is needed

The formula

$$yu_t = xu_t(u_{-t}gu_t)$$

for $y = xg$ is exact, but by itself it is not useful. If g is fixed and $t \rightarrow \infty$, the conjugate may leave every compact set. If $g \rightarrow e$ with t fixed, it tends to the identity and reveals no new symmetry. The correct regime couples the two limits: let $g_n \rightarrow e$ and choose $T_n \rightarrow \infty$ so that the conjugated displacement is just becoming macroscopic.

The first purpose of this chapter is to make that choice canonical enough to survive passage to subsequences. The second is to explain why the resulting limit is polynomial rather than arbitrary.

2. Polynomiality in exponential coordinates

Let $U = \{u_t\} = \{\exp(tN)\}$ with $\text{ad } N$ nilpotent. Choose a norm on $\mathfrak{g}$. For $Z \in \mathfrak{g}$,

$$(2.1) \quad \text{Ad}(u_{-t})Z = e^{-t \text{ad } N} Z = \sum_{j=0}^d \frac{(-t)^j}{j!} (\text{ad } N)^j Z,$$

where d can be chosen with $(\text{ad } N)^{d+1} = 0$ on $\mathfrak{g}$.

Thus every coordinate of $\text{Ad}(u_{-t})Z$ is a polynomial of degree at most d .

Proposition 2.2.1 (Polynomial norm control). *Fix $d \geq 0$. There exists $\eta = \eta(d) \in (0, 1)$ such that for every finite-dimensional normed vector space V , every polynomial map*

$$P(t) = \sum_{j=0}^d t^j v_j,$$

and every $T > 0$ satisfying

$$\max_{0 \leq t \leq T} \|P(t)\| = \|P(T)\|,$$

one has

$$\|P(t) - P(T)\| \leq \frac{1}{4} \|P(T)\|$$

for every $t \in [(1 - \eta)T, T]$.

PROOF. Rescale to $T = 1$. If $P(1) = 0$, then the hypothesis

$$\max_{0 \leq t \leq 1} \|P(t)\| = \|P(1)\|$$

forces $P \equiv 0$, so there is nothing to prove. Otherwise divide by $\|P(1)\|$; then

$$\sup_{0 \leq t \leq 1} \|P(t)\| = 1.$$

We first record an elementary scalar estimate in a form that makes its dependence transparent. Fix the $d + 1$ nodes j/d for $0 \leq j \leq d$ (with the case $d = 0$ trivial). Lagrange interpolation writes every scalar polynomial q of degree at most d as

$$q(t) = \sum_{j=0}^d q(j/d) L_j(t),$$

where the L_j depend only on d . Differentiating and taking suprema gives

$$\sup_{[0,1]} |q'| \leq \left(\sum_{j=0}^d \sup_{[0,1]} |L'_j| \right) \sup_{[0,1]} |q| =: M_d \sup_{[0,1]} |q|.$$

Thus no constant depending on an ambient vector-space dimension has entered. For each norm-one functional $\ell \in V^*$, the scalar polynomial $\ell \circ P$ has supremum at most 1, hence

$$|\ell(P'(t))| \leq M_d \quad (0 \leq t \leq 1).$$

Taking the supremum over the unit ball of V^* and using the dual characterization of the norm gives the dimension-free estimate

$$\|P'(t)\| \leq M_d \quad (0 \leq t \leq 1).$$

Therefore

$$\|P(t) - P(1)\| \leq M_d(1 - t).$$

Choose

$$\eta \leq \frac{1}{4M_d}.$$

Then $t \in [1 - \eta, 1]$ implies

$$\|P(t) - P(1)\| \leq \frac{1}{4}.$$

Undoing the normalization and the rescaling $t \mapsto t/T$ proves the proposition. Notice that the constant depends only on the polynomial degree, not on $\dim V$ or on the chosen norm. $\square$

Remark 2.2.2. The numerical constant $1/4$ is irrelevant. What matters is a terminal interval whose relative length depends only on the polynomial degree, not on the coefficients. This uniformity is what allows genericity on a positive-proportion block to be used.

3. First exit times

Let $W \subset \mathfrak{g}$ be a fixed vector-space complement to a subalgebra $\mathfrak{l}$. Denote by

$$\pi_W : \mathfrak{g} \rightarrow W$$

the projection along $\mathfrak{l}$.

Suppose $Z_n \rightarrow 0$ but $Z_n \notin \mathfrak{l}$. Define the transverse polynomial

$$(2.2) \quad P_n(t) = \pi_W \operatorname{Ad}(u_{-t})Z_n.$$

Assume for the moment that W is invariant under $\operatorname{Ad}(U)$. In the general case one works in a local slice, or equivalently with the induced nilpotent operator on the quotient by the directions already absorbed by the stabilizer; the normalization argument itself is unchanged.

Fix a small scale $\rho > 0$. If P_n is nonconstant or has a nonzero constant transverse term, there is a first time

$$(2.3) \quad T_n = \inf\{t \geq 0 : \|P_n(t)\| = \rho\}$$

provided the polynomial eventually reaches the sphere of radius ρ .

Lemma 2.3.1. *If $Z_n \rightarrow 0$ and $\rho > 0$ is fixed, then every finite first-exit time T_n tends to infinity.*

PROOF. For every fixed $R > 0$, the operators $\operatorname{Ad}(u_{-t})$ are uniformly bounded for $0 \leq t \leq R$. Hence

$$\sup_{0 \leq t \leq R} \|P_n(t)\| \leq C_R \|Z_n\| \rightarrow 0.$$

Thus for large n the transverse polynomial cannot reach the fixed radius ρ before time R . Since R is arbitrary, $T_n \rightarrow \infty$. $\square$

At time T_n , the transverse displacement has exactly the chosen macroscopic size. Proposition 2.2.1 then says that it varies little on a terminal block

$$[(1 - \eta)T_n, T_n].$$

This is the elementary quantitative core of the H-property.

4. A complete SL_2 model

Let

$$U = \left\{ u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \right\}, \quad V = \left\{ v_s = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix} \right\}.$$

Take $g_n = v_{\varepsilon_n}$ with $\varepsilon_n \rightarrow 0$. The exact formula

$$(2.4) \quad u_{-t} v_{\varepsilon} u_t = \begin{pmatrix} 1 - t\varepsilon & -t^2\varepsilon \\ \varepsilon & 1 + t\varepsilon \end{pmatrix}$$

shows three scales:

$$\varepsilon, \quad t\varepsilon, \quad t^2\varepsilon.$$

The quadratic term wins. If

$$T_n = |\varepsilon_n|^{-1/2},$$

then for $t = sT_n$,

$$-t^2\varepsilon_n = -s^2 \operatorname{sgn}(\varepsilon_n),$$

while

$$t\varepsilon_n = s \operatorname{sgn}(\varepsilon_n) |\varepsilon_n|^{1/2} \rightarrow 0.$$

After passing to a subsequence on which the sign is constant,

$$(2.5) \quad u_{-sT_n} v_{\varepsilon_n} u_{sT_n} \longrightarrow \begin{pmatrix} 1 & -\sigma s^2 \\ 0 & 1 \end{pmatrix} = u_{-\sigma s^2}$$

uniformly for s in compact intervals, where $\sigma \in \{\pm 1\}$.

This limit already lies in U . In more complicated ambient groups the leading polynomial may point into a genuinely new subgroup. The proof of measure rigidity is designed to show that this leading direction must preserve the measure.

5. $\mathfrak{sl}_2$ strings and the leading term

Suppose N belongs to an $\mathfrak{sl}_2$ -triple (N, H, N^-) . Any finite-dimensional $\mathfrak{sl}_2$ -module decomposes into irreducible strings. On an irreducible module of highest weight m , choose a weight basis

$$e_0, e_1, \dots, e_m$$

with

$$Ne_j = c_j e_{j-1},$$

for suitable nonzero constants c_j when $j \geq 1$.

Then

$$e^{-tN} e_j = e_j - tc_j e_{j-1} + \dots + \frac{(-t)^j}{j!} N^j e_j.$$

A component initially deep in the string produces a high-degree polynomial under unipotent conjugation. The first macroscopic scale is dictated by the deepest nonzero transverse component.

Worked Example 2.5.1 (A cubic shear). In the irreducible four-dimensional $\mathfrak{sl}_2$ -module $\mathrm{Sym}^3(\mathbb{R}^2)$, the lowest-weight vector produces a polynomial of degree three under the opposite unipotent. If its initial coefficient is ε , the macroscopic scale is

$$T \asymp |\varepsilon|^{-1/3}.$$

This simple scaling law is useful to remember: degree k polynomial divergence converts a perturbation of size ε into order-one displacement at time $|\varepsilon|^{-1/k}$.

6. Why quotienting by the known stabilizer is unavoidable

Let $L = S(\mu)^\circ$. Relative displacement inside L is dynamically irrelevant for discovering new symmetry: multiplying one generic point by an element of L does not change the measure. Therefore the correct object is the displacement modulo L .

A global quotient G/L is available when L is closed, but the right action of U on G/L need not be linear. Near the identity coset one can choose a smooth local slice.

Choose a vector-space complement

$$\mathfrak{g} = \mathfrak{l} \oplus W.$$

For small neighborhoods $B_{\mathfrak{l}}$ and B_W , the map

$$(A, Z) \mapsto \exp(A) \exp(Z)$$

is a diffeomorphism onto a neighborhood of e . Thus every sufficiently small g has a unique factorization

$$(2.6) \quad g = \ell(g) \exp(w(g)), \quad \ell(g) \in L, \quad w(g) \in W.$$

The coordinate $w(g)$ is a local transverse coordinate to L .

Warning 2.6.1. It is generally false that one may simply project $\log g$ linearly to W throughout a long conjugation interval. The conjugated element may leave the exponential chart, and the Baker–Campbell–Hausdorff corrections can mix $\mathfrak{l}$ and W . The local slice is used only while the normalized displacement remains inside a fixed small neighborhood; this is exactly why first-exit times are introduced.

7. Normalization in a local slice

Let $g_n \rightarrow e$ with $g_n \notin L$. Define

$$q_n(t) = u_{-t} g_n u_t.$$

As long as $q_n(t)$ remains in the slice neighborhood, write

$$(2.7) \quad q_n(t) = \ell_n(t) \exp(w_n(t)), \quad \ell_n(t) \in L, \quad w_n(t) \in W.$$

Choose $\rho > 0$ smaller than the slice radius and let T_n be the first time with

$$(2.8) \quad \|w_n(T_n)\| = \rho.$$

If no such time exists, the entire forward conjugacy curve stays transversely small; this is a special algebraic situation belonging to the singular or centralizing alternatives rather than to the first-exit regime.

For finite T_n , define

$$(2.9) \quad \phi_n(s) = \exp(w_n(sT_n)), \quad 0 \leq s \leq 1.$$

The compactness statement here deserves care, because the slice coordinate $w_n(t)$ is not itself a polynomial function of t . The polynomial object is the transverse class of the conjugacy curve in the linear quotient $\mathfrak{g}/\mathfrak{l}$. In exponential coordinates, that class is represented, up to an error of higher order in the fixed slice radius, by a bounded-degree polynomial whose coefficients depend on g_n . After the normalization $t = sT_n$, finite-dimensional norm equivalence for bounded-degree polynomials gives uniform bounds on these normalized polynomial coordinates and on their derivatives on every compact subinterval of $(0, 1)$.

The inverse local product map

$$L \times W \longrightarrow G, \quad (\ell, w) \longmapsto \ell \exp(w),$$

is smooth with bounded derivative on the fixed compact slice under consideration. Consequently the normalized transverse slice coordinates form an equicontinuous bounded family. Arzelà–Ascoli therefore gives, after passing to a subsequence, a locally uniform limit

$$(2.10) \quad \phi_n(s) \rightarrow \phi(s).$$

The group-valued limit need not literally be polynomial in the chosen nonlinear slice. What survives intrinsically is its polynomial tangent class in $\mathfrak{g}/\mathfrak{l}$; this is precisely why the term *quasi-regular* is preferable to “polynomial map.” The first-exit normalization $\|w_n(T_n)\| = \rho$, together with $w_n(0) \rightarrow 0$, ensures that the limiting transverse class is nonconstant. If desired, one can first carry out the argument with radius ρ , and then let $\rho \downarrow 0$ to remove the higher-order slice error.

The precise group-valued representative depends on the chosen slice, but its first nonzero class in the local quotient transverse to L is intrinsic.

8. Quasi-regular maps

Definition 2.8.1. A *quasi-regular map* associated with U , L , and a sequence $g_n \rightarrow e$ transverse to L is any locally uniform limit obtained from a normalized sequence of conjugacy curves

$$s \mapsto \ell_n(s)^{-1} u_{-sT_n} g_n u_{sT_n}$$

where $\ell_n(s) \in L$ removes the component in the known stabilizer and T_n is a first macroscopic transverse scale.

The adjective “quasi-regular” emphasizes two features:

- the map is not an arbitrary measurable limit; it arises from a bounded-degree polynomial conjugation law;
- it need not be a one-parameter subgroup or a homomorphism.

Warning 2.8.2 (Quotient information is not yet measure invariance). The factor $\ell_n(s) \in L$ is generally time-dependent. Even though every *fixed* element of L preserves μ , one may not delete $\ell_n(s)$ from a time average. Thus the existence of a quasi-regular transverse curve is only the algebraic half of the H-property. A separate correction/reparametrization step is needed to realize its values as controlled *full-group* displacements of generic orbit blocks; this technical step is isolated as Theorem B.7.1.

What we need from the algebraic limit is a nontrivial transverse smooth curve. Once the technical realization theorem produces full comparison blocks for a dense set of its values, the block-comparison argument puts the curve in the measure stabilizer and differentiation gives the desired infinitesimal conclusion.

9. A polynomial compactness lemma

Lemma 2.9.1. *Let V be finite-dimensional and let*

$$P_n(s) = \sum_{j=0}^d s^j v_{n,j}$$

be polynomial maps of degree at most d . If

$$\sup_{0 \leq s \leq 1} \|P_n(s)\| \leq C,$$

then a subsequence converges uniformly on $[0, 1]$ to a polynomial map of degree at most d .

PROOF. The map from the coefficient tuple $(v_0, \dots, v_d)$ to the polynomial P is an isomorphism between the finite-dimensional space V^{d+1} and the vector space of V -valued polynomials of degree at most d . The supremum norm on $[0, 1]$ is a norm on this finite-dimensional space and is therefore equivalent to the coefficient norm. Hence the boundedness assumption implies that all coefficient tuples are bounded. Bolzano–Weierstrass gives a convergent subsequence of coefficients, and coefficient convergence implies uniform convergence on $[0, 1]$. $\square$

This elementary lemma is repeatedly used after translating the group problem into a coordinate chart.

10. Terminal-block stability

Let ϕ_n be normalized so that the transverse displacement has maximal size near $s = 1$. The polynomial terminal-control proposition gives an interval

$$I = [1 - \eta, 1]$$

on which the leading transverse coordinate changes by only a fixed small fraction of its final size.

This is exactly the kind of interval on which genericity can be used. Its length is proportional to T_n in the original time variable:

$$[(1 - \eta)T_n, T_n].$$

Thus it contains enough time to average, while the relative displacement remains close to one nontrivial group element.

Guiding Idea 2.10.1 (Why the proof uses the end of the block). At early times the displacement is too small and converges to the identity; it cannot reveal new invariance. Far beyond T_n the displacement may become large and leave the coordinate chart. The terminal block just before first exit is the Goldilocks regime: the shear is nontrivial but still controlled, and the block has positive relative length.

11. Removing the L -component

Suppose

$$q_n(t) = \ell_n(t)r_n(t), \quad \ell_n(t) \in L,$$

with $r_n(t)$ transverse. Since L preserves μ , a varying factor $\ell_n(t)$ is not automatically harmless inside a pointwise orbit comparison: genericity applies to time averages, not to arbitrary time-dependent translations. The usual strategy is to choose a short enough proportional block on which $\ell_n(t)$ varies little, or to absorb its nearly constant value into the starting point.

This is one reason the terminal equicontinuity estimate is formulated for the entire polynomial displacement, not merely for its projection to G/L .

12. A local factorization estimate

Lemma 2.12.1. *Fix a sufficiently small slice neighborhood in which every g has the factorization $g = \ell(g) \exp(w(g))$. On a smaller compact neighborhood there is $C > 0$ such that*

$$(2.11) \quad \|w(g_1) - w(g_2)\| + d_G(\ell(g_1), \ell(g_2)) \leq C d_G(g_1, g_2).$$

PROOF. The inverse of the local diffeomorphism

$$(A, Z) \mapsto \exp(A) \exp(Z)$$

is smooth. On a compact subset of its domain, its derivative is bounded. The mean-value estimate in local coordinates therefore gives a Lipschitz bound. Translating the coordinate bound into the fixed Riemannian metric on G gives (2.11). $\square$

The lemma allows estimates for the polynomial conjugacy curve in G to be transferred to its stabilizer and transverse factors.

13. When the normalization can fail

There are two important failure modes.

The transverse polynomial vanishes identically. If the conjugacy curve remains in L to all orders, then the initial displacement lies in the appropriate normalizer-type set of L . Such points form a singular algebraic configuration. They must be handled separately rather than forced into the generic shearing argument.

The orbit leaves every compact set before generic comparison is available. Even if the group displacement is well controlled, the points xu_t themselves may enter the cusp. Uniform continuity of test functions on a fixed compact set then does not apply. Quantitative nondivergence and recurrence are used to retain a large subset of the terminal block inside a compact region.

These are not cosmetic technicalities. They are the two reasons a naive proof of Ratner's theorem in a few paragraphs is incomplete.

14. Differentiating a quasi-regular map

Assume eventually that an additional-invariance argument produces a C^1 curve

$$\phi(s) \in S(\mu)$$

for s in a nontrivial interval, and put $L = S(\mu)^\circ$. Fix s_0 . Because the identity component is open in the Lie group $S(\mu)$, continuity gives

$$\phi(s_0)^{-1}\phi(s) \in L$$

for all s sufficiently close to s_0 . Therefore the translated derivative

$$(2.12) \quad Y = (dL_{\phi(s_0)^{-1}})_{\phi(s_0)}\phi'(s_0)$$

belongs to

$$\mathfrak{l} = \text{Lie } L = \text{Lie}(S(\mu)).$$

Hence a stabilizing curve can never have a nonzero tangent class in $\mathfrak{g}/\mathfrak{l}$. If the normalized quasi-regular construction produces such a transverse class, the assumptions are contradictory.

This final differentiation is elementary. The hard work lies in showing that the comparison produces a stabilizing curve whose transverse class agrees with the nontrivial quasi-regular limit.

15. Summary of the normalization mechanism

Given transverse $g_n \rightarrow e$:

- (1) form $q_n(t) = u_{-t}g_nu_t$;
- (2) remove the component in the known stabilizer L using a local slice;
- (3) choose the first time T_n at which the transverse component reaches a small fixed radius;
- (4) rescale $t = sT_n$;
- (5) extract a bounded-degree polynomial limit $\phi(s)$;
- (6) use terminal-block equicontinuity to make ϕ nearly constant on a positive-proportion time block;
- (7) combine this with genericity and recurrence to prove that values of ϕ preserve the measure.

The remaining issue is therefore measure-theoretic: one must combine genericity, recurrence, and the polynomial comparison without losing control of the transverse direction.

CHAPTER 3

Generic Pairs and the Additional-Invariance Principle

1. What has to be converted

The normalization of the previous chapter produces a slowly varying nontrivial relative displacement. The present problem is measure-theoretic:

If two long orbit blocks are both typical for the same ergodic measure and one block is obtained from the other by an almost constant group displacement, why must that displacement preserve the measure?

The answer is elementary once all uniformity hypotheses are stated correctly. Most hidden gaps in informal proofs come from one of three places:

- (1) the starting points vary with n , so ordinary genericity is not automatically uniform at the chosen times;
- (2) the space is noncompact, so uniform continuity must be localized to a compact set;
- (3) the relative displacement varies with time and must be shown nearly constant on a block of positive relative length.

We address these points in that order.

2. A countable generic set

Because X is locally compact and second countable, $C_0(X)$ is separable. Choose a countable dense family

$$\mathcal{F} = \{f_1, f_2, \dots\} \subset C_c(X)$$

that determines Radon probability measures. For an ergodic U -invariant probability measure μ , let X_{gen} be the set of points x such that

$$(3.1) \quad \frac{1}{T} \int_0^T f_j(xu_t) dt \rightarrow \int f_j d\mu$$

for every j . By the pointwise ergodic theorem and countable intersection,

$$\mu(X_{\text{gen}}) = 1.$$

Density of $\mathcal{F}$ upgrades (3.1) to every $f \in C_0(X)$.

Remark 3.2.1. Using one common full-measure generic set is important. If one says “choose many generic points” without first fixing a countable determining family, the exceptional null set may depend on the test function and later diagonal arguments become ambiguous.

3. Block genericity for varying points

Let $0 < a < b < \infty$. If x is generic, then

$$(3.2) \quad \frac{1}{(b-a)T} \int_{aT}^{bT} f(xu_t) dt \rightarrow \int f d\mu.$$

For a fixed point this follows by subtracting the averages on $[0, aT]$ and $[0, bT]$.

When $x = x_n$ varies, the time T_n must be chosen after x_n is chosen.

Lemma 3.3.1 (Diagonal block selection). *Let $x_n, y_n \in X_{\text{gen}}$. Let $J_1, J_2, \dots$ enumerate the closed intervals with rational endpoints contained in $(0, 1]$. Then one can choose $T_n \rightarrow \infty$ such that for every fixed j, k ,*

$$(3.3) \quad \frac{1}{|J_k|T_n} \int_{T_n J_k} f_j(x_n u_t) dt \rightarrow \int f_j d\mu,$$

and the same conclusion holds with y_n in place of x_n .

PROOF. For each fixed n , genericity of x_n and y_n implies block convergence for every fixed pair (j, k) . At stage n , impose the finitely many requirements

$$1 \leq j, k \leq n$$

for both starting points, with error at most $1/n$. There is a threshold R_n beyond which all these finitely many inequalities hold. Choose recursively

$$T_n > \max\{R_n, T_{n-1} + 1, n\}.$$

For fixed j, k , the desired inequality is imposed for every sufficiently large n , with error tending to zero. $\square$

4. Recurrence on the same blocks

Genericity also gives time-density information for compact sets. Let $K \subset X$ be compact with

$$\mu(K) > 1 - \delta.$$

Choose $\chi \in C_c(X)$ with

$$0 \leq \chi \leq 1, \quad \chi = 1 \text{ on } K, \quad \int \chi d\mu > 1 - 2\delta.$$

Block genericity then implies that, on every selected proportional block, the proportion of times at which the orbit lies in $\text{supp } \chi$ is at least $1 - O(\delta)$.

For two generic orbit blocks, the intersection of their good-time sets still has large relative measure. This elementary observation is the bridge between recurrence and orbit comparison.

Lemma 3.4.1 (Common compact times). *Let x_n, y_n and T_n satisfy the block-genericity conclusion above. Fix a rational interval $J \subset (0, 1]$ and $\delta > 0$. Then there exists a compact $K_\delta \subset X$ and measurable sets*

$$E_n \subset T_n J$$

such that

$$(3.4) \quad \frac{|E_n|}{|T_n J|} > 1 - \delta$$

for all large n , and

$$x_n u_t, y_n u_t \in K_\delta \quad (t \in E_n).$$

PROOF. Apply the construction with an auxiliary error $\delta/10$. Choose compact $K \subset X$ with $\mu(K) > 1 - \delta/10$ and a compactly supported cutoff $0 \leq \chi \leq 1$ that equals 1 on K and satisfies $\int \chi d\mu > 1 - \delta/5$. By block genericity, for all large n the average of χ along each block is larger than $1 - \delta/4$. Since χ vanishes outside its compact support, each orbit spends at least a $(1 - \delta/4)$ -proportion of the block in $\text{supp } \chi$. The intersection of the two good-time sets therefore has relative measure at least $1 - \delta/2$, in particular greater than $1 - \delta$. Set $K_\delta = \text{supp } \chi$. $\square$

The exact constant is irrelevant; only the fact that the loss can be made arbitrarily small matters.

5. The basic comparison lemma

Proposition 3.5.1 (Almost-constant displacement gives invariance). *Let $x_n, y_n \in X_{\text{gen}}$, let I_n be intervals with $|I_n| \rightarrow \infty$, and suppose the empirical measures of both orbit blocks converge to μ . Let $h_n \rightarrow h \in G$. Assume that for every $\delta > 0$ there are a compact set K_δ and sets $E_{n,\delta} \subset I_n$ such that*

$$(3.5) \quad \frac{|I_n \setminus E_{n,\delta}|}{|I_n|} < \delta$$

for all large n , both compared points lie in K_δ , and

$$(3.6) \quad \sup_{t \in E_{n,\delta}} d(x_n u_t h_n, y_n u_t) \rightarrow 0.$$

Then $h \in S(\mu)$.

PROOF. Fix $f \in C_c(X)$ and $\varepsilon > 0$. Choose $\delta > 0$ so small that

$$4\delta\|f\|_\infty < \varepsilon.$$

On the compact set K_δ and for group elements in a compact neighborhood of h , the map

$$(z, g) \mapsto f(zg)$$

is uniformly continuous. Hence (3.6) implies, for large n ,

$$|f(x_n u_t h_n) - f(y_n u_t)| < \varepsilon \quad (t \in E_{n,\delta}).$$

On the exceptional set use the trivial bound $2\|f\|_\infty$. Thus

$$(3.7) \quad \left| \frac{1}{|I_n|} \int_{I_n} f(x_n u_t h_n) dt - \frac{1}{|I_n|} \int_{I_n} f(y_n u_t) dt \right| \leq \varepsilon + 2\delta\|f\|_\infty.$$

Because $h_n \rightarrow h$ and $f \in C_c(X)$, strong continuity of right translation on $C_0(X)$ gives

$$\|f \circ R_{h_n} - f \circ R_h\|_\infty \rightarrow 0.$$

Thus h_n may be replaced by h in the first average with an error tending to zero. The x_n -block empirical measures converge to μ , so the average of the fixed function $f \circ R_h$ converges to $\int f(zh) d\mu(z)$; the y_n -block average converges to $\int f d\mu$. Hence (3.7) yields

$$\left| \int f(zh) d\mu(z) - \int f(z) d\mu(z) \right| \leq 2\varepsilon$$

after $n \rightarrow \infty$. Since ε was arbitrary,

$$\int f(zh) d\mu(z) = \int f(z) d\mu(z).$$

Therefore $h_*\mu = \mu$. □

6. From a slowly varying displacement to a fixed one

Suppose the relative displacement on a block is a curve $q_n(t)$ rather than a fixed h_n . Choose a reference time t_n^* in the block and put

$$h_n = q_n(t_n^*).$$

If

$$(3.8) \quad \sup_{t \in I_n} d_G(q_n(t), h_n) \rightarrow 0,$$

then

$$y_n u_t = x_n u_t q_n(t)$$

is close to $x_n u_t h_n$ whenever the base point remains in a fixed compact set. Proposition 3.5.1 applies.

The terminal-block polynomial estimate from the previous chapter is designed precisely to guarantee (3.8) after one first lets the macroscopic scale ρ be small and then sends $n \rightarrow \infty$.

7. An infinitesimal version

Often the limiting displacement is not one fixed nontrivial element but a curve $\phi(s)$. One can prove invariance for a dense set of parameter values and then use closedness of $S(\mu)$.

Proposition 3.7.1 (Curve-valued additional invariance). *Suppose that for every rational s in an open interval J there are comparison blocks showing, by Proposition 3.5.1, that*

$$\phi(s) \in S(\mu),$$

where $\phi : J \rightarrow G$ is continuous. Then

$$\phi(J) \subset S(\mu).$$

If ϕ is C^1 and $L = S(\mu)^\circ$, then for every $s_0 \in J$ the translated derivative

$$(dL_{\phi(s_0)^{-1}})_{\phi(s_0)} \phi'(s_0)$$

belongs to $\mathfrak{l} = \text{Lie } L$. Consequently a C^1 stabilizing curve cannot have a nonzero tangent class in $\mathfrak{g}/\mathfrak{l}$.

PROOF. The first assertion follows because the rationals are dense and $S(\mu)$ is closed. For the differential assertion, fix s_0 . Since the identity component of the Lie group $S(\mu)$ is open in $S(\mu)$, continuity of ϕ implies that for s sufficiently close to s_0 ,

$$\phi(s_0)^{-1} \phi(s) \in L.$$

Differentiating at s_0 shows that the left-translated derivative lies in $T_e L = \mathfrak{l}$. This is the precise contradiction used when a limiting comparison curve has a genuinely transverse tangent. $\square$

8. Selecting nearby generic points

Let $L = S(\mu)^\circ$ and fix a generic point $x \in \text{supp } \mu$. The local orbit xL is the set of points obtained by applying already-known symmetries. To gain new information we need generic points near x but outside xL .

Lemma 3.8.1 (Transverse generic points). *Assume $\mu(xL) = 0$. Then every neighborhood of x contains a point*

$$y \in X_{\text{gen}} \setminus xL.$$

Consequently there are $y_n \rightarrow x$ with this property.

PROOF. Because $x \in \text{supp } \mu$, every neighborhood O of x has positive μ -measure. Since $\mu(X_{\text{gen}}) = 1$ and $\mu(xL) = 0$,

$$\mu(O \cap X_{\text{gen}} \setminus xL) = \mu(O) > 0.$$

Hence the set is nonempty. □

If instead $\mu(xL) > 0$, then xL is U -invariant because $U \subset L$. Under the measurability hypotheses satisfied by Lie-group orbits, ergodicity forces

$$\mu(xL) = 1.$$

This is already close to homogeneity, though one must still prove that the orbit is closed and has finite L -volume.

9. Writing nearby points as xg_n

For fixed $x = \Gamma g$, discreteness of $g^{-1}\Gamma g$ gives a neighborhood V of e on which the orbit map

$$V \rightarrow X, \quad h \mapsto xh$$

is injective. Thus if $y_n \rightarrow x$, then for large n there is a unique $g_n \in V$ with

$$(3.9) \quad y_n = xg_n, \quad g_n \rightarrow e.$$

Moreover,

$$y_n \notin xL \implies g_n \notin L$$

provided V is chosen small enough that the local L -orbit is represented by $V \cap L$.

This converts transverse nearby points in the quotient into transverse small elements of the ambient group, exactly the input required for polynomial shearing.

10. Recurrence versus singularity

There is one remaining difficulty. A transverse g_n can become effectively nontransverse under conjugation if its leading polynomial displacement points into a subgroup that already organizes a proper homogeneous piece of the space. Such sequences are called singular.

For nonsingular sequences, quantitative nondivergence provides a compact set on which a large proportion of the terminal block can be compared. For singular sequences, the measure is forced toward a proper homogeneous stratum and the argument changes from “enlarge the stabilizer” to “reduce the ambient homogeneous subgroup.”

This dichotomy is the reason singular sets are part of the proof rather than an optional algebraic embellishment.

11. A precise additional-invariance theorem

The preceding mechanism can be packaged in a form suitable for repeated application.

Guiding Idea 3.11.1 (Full-displacement comparison principle). Suppose a U -ergodic probability measure μ admits, for every parameter s in a dense countable subset of an interval, pairs of generic orbit blocks whose full relative displacement converges uniformly off a negligible set to an element $\phi(s) \in G$. Comparing a countable determining family of test functions on the two blocks forces

$$(R_{\phi(s)})_*\mu = \mu$$

for the dense parameter set; closedness of $\text{Stab}_G(\mu)$ then gives $\phi(J) \subset \text{Stab}_G(\mu)$ whenever ϕ is continuous. The exact hypotheses and proof used later are stated in Theorem 8.12.1; the present paragraph is a roadmap rather than a second theorem.

12. The H-property in words

Ratner's H-property can be summarized as follows. If two generic unipotent trajectories start very close but are not related by an already-known symmetry, then there is a long time block during which they remain at a controlled nonzero relative displacement. Since both blocks sample the same invariant measure, the relative displacement must itself be a symmetry.

The phrase “remain at a controlled displacement” should not be read as ordinary metric closeness. The displacement is typically the result of a polynomial shear and may lie in a different subgroup from the initial perturbation. That conversion of direction is the source of new invariance.

13. A sanity check in the horocycle model

Return to

$$G = \text{SL}_2(\mathbb{R}), \quad U = \{u_t\}.$$

Suppose μ is U -ergodic and there are generic points separated by a small opposite-unipotent displacement v_{ε_n} . The calculation

$$u_{-sT_n} v_{\varepsilon_n} u_{sT_n} \rightarrow u_{-\sigma s^2}$$

shows that the quasi-regular map lies inside U , which is already known to preserve μ . Thus this particular perturbation does not enlarge the stabilizer.

To obtain new invariance in SL_2 , one uses the full decomposition of a nearby matrix, including diagonal and centralizer components. In larger groups, transverse $\mathfrak{sl}_2$ strings naturally produce centralizer or normalizer directions not already contained in U . This example is a useful warning:

polynomial shearing alone is not enough; one must identify the leading component modulo the current stabilizer.

14. The algebraic obstruction

If the leading quasi-regular displacement always falls back into the current stabilizer, then the sequence g_n is algebraically special. The set of points giving such special displacements is organized by proper subgroups containing conjugates of U . These configurations are called singular sets; their defining feature is that the apparent transverse displacement is constrained by a proper homogeneous subgroup rather than producing a genuinely new symmetry.

CHAPTER 4

The H-Property in an $\mathfrak{sl}_2$ -Adapted Decomposition

1. Why return to the local algebra

The additional-invariance theorem was formulated abstractly in terms of quasi-regular maps. For understanding the proof, it is useful to see how such maps are forced by the representation theory of an $\mathfrak{sl}_2$ -triple.

Assume in this chapter that the nilpotent generator $N \in \mathfrak{g}$ belongs to an $\mathfrak{sl}_2$ -triple

$$(N, H, N^-)$$

with

$$(4.1) \quad [H, N] = 2N, \quad [H, N^-] = -2N^-, \quad [N, N^-] = H.$$

For the later rational arithmetic application, existence of such triples is proved internally in Volume V in its internal Lie-algebra and group-level Jacobson–Morozov theorems. In the present chapter the triple is only a coordinate model for the shearing calculation, so no forward theorem is load-bearing here. The point of the triple is not abstract structure theory for its own sake. It gives a coordinate system in which unipotent shearing has a visible hierarchy of polynomial degrees.

2. Weight strings

Let V be a finite-dimensional representation of $\mathfrak{sl}_2$. An irreducible summand of highest weight m has a basis

$$v_0, v_1, \dots, v_m$$

with

$$(4.2) \quad Hv_j = (m - 2j)v_j,$$

while N moves upward in weight and N^- moves downward. After a convenient normalization,

$$(4.3) \quad Nv_j = j(m - j + 1)v_{j-1}, \quad N^-v_j = v_{j+1}.$$

The precise constants do not matter for degree counting; nonvanishing does.

For v_j ,

$$(4.4) \quad e^{-tN}v_j = \sum_{k=0}^j \frac{(-t)^k}{k!} N^k v_j.$$

Thus the polynomial degree is exactly j when the deepest nonzero component is v_j .

Worked Example 4.2.1 (The adjoint representation of $\mathfrak{sl}_2$). Take

$$X = N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad Y = N^- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Then

$$(4.5) \quad \text{Ad}(u_{-t})Y = Y - tH - t^2X.$$

Indeed,

$$[N, Y] = H, \quad [N, H] = -2N, \quad [N, N] = 0,$$

so

$$e^{-t \text{ad } N}Y = Y - tH + \frac{t^2}{2}(-2N) = Y - tH - t^2N.$$

The opposite-unipotent direction Y is at depth two in the adjoint string, and therefore creates quadratic shear.

3. A canonical complement to a known invariant subgroup

Let $L < G$ be a connected subgroup containing U , with Lie algebra $\mathfrak{l}$. Suppose for the moment that $\mathfrak{l}$ is invariant under the entire $\mathfrak{sl}_2$ -triple. Since finite-dimensional $\mathfrak{sl}_2$ -modules are completely reducible, choose an $\mathfrak{sl}_2$ -invariant complement

$$(4.6) \quad \mathfrak{g} = \mathfrak{l} \oplus W.$$

Then W decomposes into irreducible weight strings. A transverse displacement $Z \in W$ has a unique decomposition

$$(4.7) \quad Z = \sum_{\alpha} \sum_{j=0}^{m_{\alpha}} z_{\alpha,j} v_{\alpha,j}.$$

Let

$$(4.8) \quad r(Z) = \max\{j : z_{\alpha,j} \neq 0 \text{ for some } \alpha\}.$$

The number $r(Z)$ is the maximal active polynomial degree under U -conjugation.

4. Candidate scales and competing coefficients

Suppose $Z_n \rightarrow 0$ in W . For each nonzero coefficient $z_{n,\alpha,j}$ with $j \geq 1$, the degree- j term suggests a candidate macroscopic scale

$$T_{n,\alpha,j} \asymp |z_{n,\alpha,j}|^{-1/j}.$$

The first visible displacement is governed by the *smallest* of these competing scales, not automatically by the deepest nonzero weight-string position.

For example, if the deepest coefficient has depth r and size a_n , the scale

$$T_n = a_n^{-1/r}$$

makes its degree- r contribution order one. This scale is valid only if every other coefficient satisfies the corresponding boundedness condition

$$|z_{n,\alpha,j}|T_n^j = O(1).$$

If for some shallower coefficient this quantity diverges, that coefficient becomes macroscopic earlier and the correct normalization time is smaller.

This is why the robust construction uses the full polynomial

$$Q_n(t) = e^{-t \operatorname{ad} N} Z_n$$

and defines T_n by a first-exit condition on its transverse norm. After passing to a subsequence, the rescaled polynomials

$$s \longmapsto Q_n(sT_n)$$

have uniformly bounded coefficients on compact s -intervals and converge to a nonzero polynomial map. A simple power law can then be read off in examples once the competing coefficients have been compared.

5. An explicit normalization lemma

Proposition 4.5.1 (First-exit normalization). *Let W be a finite-dimensional $\mathfrak{sl}_2$ -module and let $Z_n \rightarrow 0$. Put*

$$Q_n(t) = e^{-t \operatorname{ad} N} Z_n.$$

Assume there is a fixed $\rho > 0$ such that, for all sufficiently large n , the polynomial trajectory $Q_n(t)$ reaches the sphere of radius ρ for some $t \geq 0$. Let T_n be its first hitting time. Then $T_n \rightarrow \infty$, and after passing to a subsequence there is a nonzero polynomial map $P : \mathbb{R} \rightarrow W$ such that

$$(4.10) \quad \rho^{-1} e^{-sT_n \operatorname{ad} N} Z_n \longrightarrow P(s)$$

locally uniformly on every compact s -interval on which the rescaled polynomials remain bounded. On $[0, 1]$ one has

$$\sup_{0 \leq s \leq 1} \|P(s)\| \leq 1, \quad \|P(1)\| = 1.$$

PROOF. First-exit minimality gives

$$\|Q_n(sT_n)\| \leq \rho \quad (0 \leq s \leq 1),$$

so the rescaled polynomial maps

$$P_n(s) = \rho^{-1}Q_n(sT_n)$$

are uniformly bounded by 1 on $[0, 1]$. Their degrees are bounded by the maximal $\mathfrak{sl}_2$ string length in W . Since all norms on this finite-dimensional polynomial space are equivalent, the coefficient vectors of P_n are uniformly bounded. Passing to a subsequence gives coefficientwise, hence locally uniform, convergence to a polynomial P .

At the endpoint,

$$\|P_n(1)\| = 1,$$

so $\|P(1)\| = 1$ and P is nonzero. Since $Z_n \rightarrow 0$, the first hitting times must satisfy $T_n \rightarrow \infty$: otherwise a bounded subsequence of T_n would give $Q_n(T_n) \rightarrow 0$, contradicting $\|Q_n(T_n)\| = \rho$. $\square$

Remark 4.5.2 (The commuting case). If the transverse component lies in $\ker(\mathrm{ad} N)$, then $Q_n(t)$ may never reach a fixed macroscopic radius, so the proposition is deliberately inapplicable. This is not a failure of rigidity: a transverse direction fixed by U lies in the infinitesimal centralizer of U and is treated by the direct centralizer/additional-invariance argument rather than by polynomial growth.

6. Why the endpoint polynomial changes slowly

Let P be the normalized limit. The terminal-block lemma gives $\eta > 0$, depending only on the maximal string length, such that

$$(4.11) \quad \|P(s) - P(1)\| < \frac{1}{4} \quad (1 - \eta \leq s \leq 1).$$

For large n , the same estimate with slightly larger constant holds for P_n .

Thus on the original block

$$[(1 - \eta)T_n, T_n]$$

the transverse displacement has already become macroscopic but changes only by a controlled fraction. This is the quantitative content of the H-property in the adapted linear model.

7. From Lie-algebra displacement to group displacement

If the conjugated group element is

$$(4.12) \quad q_n(t) = u_{-t} \exp(Z_n) u_t = \exp(e^{-t \operatorname{ad} N} Z_n),$$

then (4.10) immediately gives

$$q_n(sT_n) \rightarrow \exp(\rho P(s))$$

when all displacements stay in one exponential chart.

For a general nearby element g_n , write

$$g_n = \ell_n \exp(Z_n)$$

with $\ell_n \in L$ and $Z_n \in W$ using a local slice. The conjugate of ℓ_n need not remain in L unless L is normalized by U , but $U \subset L$ gives

$$u_{-t} \ell_n u_t \in L.$$

Therefore

$$(4.13) \quad u_{-t} g_n u_t = (u_{-t} \ell_n u_t) \exp(e^{-t \operatorname{ad} N} Z_n),$$

and the first factor lies in L for each fixed t . This is an algebraic simplification, but it is *not* yet a measure-theoretic simplification: the element $u_{-t} \ell_n u_t$ varies with t , and L -invariance of μ does not allow us to delete an arbitrary time-dependent L -translation inside an ergodic average. The quotient calculation therefore identifies the correct transverse polynomial, while a separate H -property correction/reparametrization step is needed to realize it by controlled full-group displacements.

This formula is one reason it is advantageous to choose a complement to $\mathfrak{l}$ before normalizing: it separates the transverse polynomial from the already known invariance directions without pretending that the latter have disappeared from the orbit comparison.

8. When $\mathfrak{l}$ is not an $\mathfrak{sl}_2$ -submodule

In the actual classification proof, the connected measure stabilizer L contains U but need not initially contain H or N^- . Thus $\mathfrak{l}$ need not be invariant under the full triple.

One can still decompose $\mathfrak{g}$ using the H -weight spaces and choose a complement adapted to the filtration generated by repeated $\operatorname{ad} N$. The quotient vector space

$$\mathfrak{g}/\mathfrak{l}$$

carries the induced nilpotent operator whenever $\mathfrak{l}$ is $\operatorname{ad} N$ -invariant, which it is because $U \subset L$. Polynomial normalization can therefore be performed directly in the quotient.

This quotient formulation is more intrinsic than demanding a full $\mathfrak{sl}_2$ -invariant complement.

9. The quotient nilpotent operator

Since

$$[N, \mathfrak{l}] \subset \mathfrak{l},$$

the map $\text{ad } N$ descends to a nilpotent operator

$$(4.14) \quad \overline{D} : \mathfrak{g}/\mathfrak{l} \rightarrow \mathfrak{g}/\mathfrak{l}.$$

For $\overline{Z} \in \mathfrak{g}/\mathfrak{l}$,

$$e^{-t\overline{D}}\overline{Z}$$

is a polynomial curve. The deepest nonzero vector in the Jordan chain of $\overline{D}$ determines the first macroscopic transverse scale.

Thus the essential polynomial normalization needs only $U \subset L$, not an entire SL_2 inside L .

10. Jordan-chain formulation

Choose a Jordan basis for the nilpotent operator $\overline{D}$. On a chain

$$e_0, e_1, \dots, e_r, \quad \overline{D}e_j = e_{j-1},$$

one has

$$(4.15) \quad e^{-t\overline{D}}e_r = e_r - te_{r-1} + \frac{t^2}{2!}e_{r-2} - \dots + \frac{(-t)^r}{r!}e_0.$$

This formula is the most general linear algebra needed for polynomial shearing. The $\mathfrak{sl}_2$ language adds geometric interpretation to the chains but the nilpotent Jordan form already gives all degree estimates.

11. Producing a stabilizing element

Let $q_n(t)$ be the normalized group displacement and suppose on the terminal block it is close to a fixed element

$$h_n = q_n(T_n).$$

If the two corresponding orbit blocks are generic and recurrent in a compact set, Proposition 3.5.1 gives a limit

$$(4.16) \quad h = \lim h_n \in S(\mu).$$

Now let the macroscopic radius ρ tend to zero through a sequence. The corresponding stabilizing elements satisfy

$$h_\rho \rightarrow e$$

but their normalized logarithms approach a nonzero transverse vector. By the closed subgroup theorem, these infinitesimal directions belong to

$$\mathrm{Lie} S(\mu).$$

If L is only a provisional connected subgroup of $S(\mu)^\circ$, such a direction enlarges the known connected invariance. If instead $L = S(\mu)^\circ$, as in the final classification proof, a nonzero class in $\mathfrak{g}/\mathfrak{l}$ is impossible and yields the desired contradiction.

12. Why one radius is not enough

If L denotes only a provisional connected subgroup of the full stabilizer, a single stabilizing element $h \notin L$ need not enlarge the connected invariance: h could lie in a different connected component of the subgroup generated so far. When $L = S(\mu)^\circ$, by contrast, every stabilizing element sufficiently close to the identity already lies in L ; the varying-radius argument is then used to derive a transverse tangent contradiction, not to create a larger identity component.

To force connected additional invariance, repeat the construction with macroscopic radii

$$\rho_j \downarrow 0.$$

This produces stabilizing elements

$$h_j \rightarrow e$$

whose directions remain uniformly transverse to $\mathfrak{l}$. The direction lemma for closed subgroups then yields a nonzero transverse vector in

$$\mathrm{Lie} S(\mu).$$

This is the clean reason small-scale repetition is built into rigorous proofs.

13. Direction lemma for the measure stabilizer

Lemma 4.13.1. *Let $S < G$ be a closed subgroup with Lie algebra $\mathfrak{s}$. Suppose*

$$h_n \in S \setminus \{e\}, \quad h_n \rightarrow e,$$

and write $h_n = \exp Z_n$ in a fixed exponential chart. If

$$\frac{Z_n}{\|Z_n\|} \rightarrow Z,$$

then $Z \in \mathfrak{s}$.

PROOF. For any $t \in \mathbb{R}$, choose integers m_n such that

$$m_n \|Z_n\| \rightarrow t.$$

Then

$$h_n^{m_n} = \exp(m_n Z_n) \rightarrow \exp(tZ).$$

Since S is closed, $\exp(tZ) \in S$ for all t , so $Z \in \mathfrak{s}$. $\square$

This lemma is elementary but crucial: it converts a sequence of small discrete symmetries into a continuous one-parameter symmetry.

14. A worked SL_3 adjoint calculation

Let

$$N = E_{12}$$

in $\mathfrak{sl}_3$. Consider the transverse matrix

$$Z = E_{21}.$$

Then

$$[N, E_{21}] = E_{11} - E_{22} =: H_{12},$$

and

$$[N, H_{12}] = -2E_{12}.$$

Thus

$$(4.17) \quad e^{-t \operatorname{ad} N} E_{21} = E_{21} - tH_{12} - t^2 E_{12}.$$

This is the familiar $\mathfrak{sl}_2$ string inside the upper-left 2×2 block.

Now take

$$Z = E_{23}.$$

Since

$$[N, E_{23}] = E_{13}, \quad [N, E_{13}] = 0,$$

one has

$$(4.18) \quad e^{-t \operatorname{ad} N} E_{23} = E_{23} - tE_{13}.$$

This perturbation shears only linearly.

Finally,

$$[N, E_{31}] = -E_{32}, \quad [N, E_{32}] = 0,$$

so

$$(4.19) \quad e^{-t \operatorname{ad} N} E_{31} = E_{31} + tE_{32}.$$

The ambient Lie algebra contains several polynomial degrees simultaneously. The first macroscopic scale of a general perturbation depends on which coefficients are present and their relative sizes.

15. Competing degrees

Suppose

$$Z_n = \varepsilon_n E_{21} + \delta_n E_{23},$$

with both coefficients tending to zero. The quadratic string reaches order one at

$$T_n^{(2)} \asymp |\varepsilon_n|^{-1/2},$$

while the linear string reaches order one at

$$T_n^{(1)} \asymp |\delta_n|^{-1}.$$

The first exit occurs on the smaller of these two scales. Therefore the leading quasi-regular map can change along subsequences depending on the asymptotic relation between

$$|\delta_n| \quad \text{and} \quad |\varepsilon_n|^{1/2}.$$

This is why rigorous arguments pass to subsequences after choosing the transverse sequence. There is no single universal polynomial degree attached to U alone; the active degree depends on the perturbation.

16. A precise terminal stability estimate in Jordan coordinates

Let

$$P_n(t) = \sum_{k=0}^d t^k v_{n,k}$$

be the transverse polynomial in a quotient coordinate. Choose T_n by first exit from the ball of radius ρ . After rescaling,

$$Q_n(s) = \rho^{-1} P_n(sT_n)$$

is uniformly bounded on $[0, 1]$ and satisfies

$$\|Q_n(1)\| = 1.$$

Polynomial compactness gives $Q_n \rightarrow Q$. Since $Q(1)$ has norm one, equicontinuity gives a fixed $\eta > 0$ with

$$\|Q(s) - Q(1)\| < \varepsilon$$

for $1 - \eta \leq s \leq 1$. Uniform convergence then gives

$$(4.20) \quad \|P_n(sT_n) - P_n(T_n)\| < 2\varepsilon\rho$$

for large n .

The important feature is the order of limits:

- (1) fix the small macroscopic radius ρ ;
- (2) choose the terminal proportion η from bounded-degree polynomial compactness;

- (3) let $n \rightarrow \infty$;
- (4) only afterward let $\rho \rightarrow 0$ to obtain infinitesimal stabilizer directions.

17. The semisimple additional-invariance theorem in local form

THEOREM 4.17.1 (Semisimple regular branch after H-property realization). *Let G be connected semisimple, $\Gamma < G$ a lattice, $U = \exp(\mathbb{R}N)$ Ad-unipotent, and μ an ergodic U -invariant probability measure. Put*

$$L = S(\mu)^\circ.$$

Assume the semisimple weight calculation produces a nonsingular transverse quotient polynomial and that the technical H-property realization data of Theorem B.7.1 are available for this sequence. Then the regular transverse branch is impossible.

PROOF. The weight-string and first-exit calculations above identify a nonconstant transverse class in $\mathfrak{g}/\mathfrak{l}$. The technical H-property realization theorem converts that quotient-level class into controlled full-group comparison blocks and a C^1 limiting curve with nonzero transverse tangent. The full-displacement comparison theorem, Theorem 8.12.1, places that curve in $S(\mu)$. Differentiating gives a vector in

$$\text{Lie } S(\mu) = \mathfrak{l}$$

with nonzero class in $\mathfrak{g}/\mathfrak{l}$, a contradiction. □

Remark 4.17.2. The distinction in the hypotheses is essential. The quotient-Jordan calculation determines the correct transverse scale, but on that scale components inside L can be large. The measure argument is valid only after the H-property correction has produced genuine full-group comparison representatives; L -invariance does not permit an arbitrary time-dependent L -factor to be erased.

18. What the H-property accomplishes

The H-property is sometimes presented as a mysterious geometric condition. In the present coordinates its content is concrete:

- (1) nilpotence makes relative displacement polynomial;
- (2) the first-exit scale makes the displacement nontrivial but small;
- (3) bounded polynomial degree makes it nearly constant on a long terminal block;
- (4) recurrence lets both trajectories be observed in a controlled region during most of that block;

- (5) genericity identifies the almost constant displacement as a measure symmetry.

Every step is necessary. Removing polynomiality destroys the long controlled block; removing recurrence destroys uniform continuity; removing genericity destroys the conversion to measure invariance.

CHAPTER 5

Quasi-Regular Maps and the Geometry of Limiting Displacements

The phrase *quasi-regular map* packages a recurrent construction in Ratner theory. Start with small group elements $g_n \rightarrow e$ transverse to a known invariance group. Conjugate them by a unipotent flow, rescale time until the transverse displacement is of order one, and pass to a limit. The result is generally not a one-parameter subgroup. It is a polynomial or rationally parametrized curve in the ambient group, and its image contains the new symmetry information.

This chapter develops that construction at group level, with particular attention to three questions:

- (1) why a nontrivial limit exists;
- (2) which parts of the limit are artifacts of the old stabilizer;
- (3) how the image of a limiting map generates a genuine subgroup of invariance.

Remark 5.0.1 (Convention on “old stabilizer”). In the local constructions of this chapter, the phrase “old stabilizer” may mean a provisional connected subgroup $K \subset S(\mu)^\circ$ whose invariance is already known. In that provisional viewpoint a new tangent direction enlarges K . When the final proof sets $L = S(\mu)^\circ$, the same transverse direction is instead a contradiction. We will say explicitly which viewpoint is being used.

1. Local coordinates adapted to a subgroup

Let $L < G$ be a closed connected subgroup with Lie algebra $\mathfrak{l}$. Choose a vector-space complement

$$\mathfrak{g} = \mathfrak{l} \oplus \mathfrak{v}.$$

For sufficiently small neighborhoods of zero, the map

$$\Psi : \mathfrak{l} \times \mathfrak{v} \rightarrow G, \quad (A, B) \mapsto \exp(A) \exp(B),$$

is a diffeomorphism onto a neighborhood of the identity. This follows from the inverse function theorem because

$$d\Psi_{(0,0)}(A, B) = A + B.$$

Thus every small element has a unique decomposition

$$g = \ell(g)s(g), \quad \ell(g) \in L, \quad s(g) \in S := \exp(\mathfrak{v}_{\text{loc}}).$$

If L preserves a measure μ , then $\ell(g)$ is irrelevant to the comparison of μ with its translates. The local slice coordinate $s(g)$ is the genuine transverse displacement.

2. The induced nilpotent operator

Let

$$U = \{u_t = \exp(tX)\} \subset L$$

be Ad-unipotent. Since L is normalized by itself,

$$\text{ad } X(\mathfrak{l}) \subset \mathfrak{l}.$$

Hence $\text{ad } X$ induces a nilpotent linear map

$$D : \mathfrak{g}/\mathfrak{l} \rightarrow \mathfrak{g}/\mathfrak{l}.$$

For a transverse class $\overline{Z}$, the linearized conjugation is

$$e^{-tD}\overline{Z} = \overline{Z} - tD\overline{Z} + \frac{t^2}{2}D^2\overline{Z} - \dots.$$

The series terminates. Define the active degree

$$r(\overline{Z}) = \max\{j : D^j\overline{Z} \neq 0\}.$$

If $r(\overline{Z}) = r$, then the leading term is

$$\frac{(-t)^r}{r!}D^r\overline{Z}.$$

Guiding Idea 5.2.1 (The active coefficients choose the clock). For a fixed transverse direction of active degree r , a perturbation of size ε has a degree- r contribution of size roughly εt^r , suggesting the scale $t \asymp \varepsilon^{-1/r}$. For a *sequence* of directions, however, coefficients of higher-degree terms may tend to zero and still become visible after a large time rescaling. Therefore the general proof does not choose the clock from the degree of the limiting direction. It uses a first-exit time for the full transverse polynomial. A power law such as $T_n \asymp \varepsilon_n^{-1/r}$ is valid only after one has verified that all higher-degree coefficients are negligible on that scale.

3. A nondegenerate model normalization

It is useful to see the elementary power-law normalization in the situation where no hidden higher-degree coefficient competes with the visible one. Let $Z_n \in \mathfrak{v}$ be transverse logarithms and put

$$\varepsilon_n = \|Z_n\|, \quad \widehat{Z}_n = Z_n/\varepsilon_n.$$

Assume

$$\widehat{Z}_n \rightarrow Z, \quad D^r \overline{Z} \neq 0, \quad D^{r+1} \overline{Z} = 0,$$

and, in addition, that the possibly nonzero higher-degree coefficients of the approximating directions satisfy

$$\varepsilon_n T_n^j \|D^j \widehat{Z}_n\| \rightarrow 0 \quad (j > r)$$

for the scale

$$T_n = \varepsilon_n^{-1/r}.$$

This condition is automatic, for example, if all $\widehat{Z}_n$ lie in a fixed D -invariant subspace on which $D^{r+1} = 0$.

Then for s in a fixed compact interval,

$$\varepsilon_n e^{-s T_n D} \widehat{Z}_n$$

converges to

$$\frac{(-s)^r}{r!} D^r \overline{Z}.$$

Indeed, the degree- r term has coefficient

$$\varepsilon_n T_n^r = 1,$$

lower-degree terms contain a positive power of ε_n and tend to zero, and the extra hypothesis removes all terms of degree larger than r .

This model explains the familiar power laws, but the extra hypothesis should not be forgotten. Without it, a coefficient that tends to zero before rescaling can be amplified by a higher power of T_n and dominate the limit.

4. First-exit normalization versus monomial normalization

The formula

$$T_n = \varepsilon_n^{-1/r}$$

is convenient when the active degree and leading coefficient are already known. In a general proof, one instead defines T_n intrinsically as the first time the transverse displacement reaches a fixed small radius ρ .

Under a uniform two-sided growth estimate

$$c \varepsilon_n (1 + |t|)^r \leq \|P_n(t)\| \leq C \varepsilon_n (1 + |t|)^r$$

on the relevant range, first-exit normalization is asymptotically equivalent to the power law $T_n \asymp \varepsilon_n^{-1/r}$. Such an estimate holds in the nondegenerate situation above, after restricting to a suitable terminal range. It is not automatic for an arbitrary transverse sequence.

The first-exit definition is therefore the robust one. It sees all coefficients of the full transverse polynomial simultaneously, is coordinate stable at small radius, and automatically produces a nontrivial limit bounded away from the old stabilizer.

5. Definition of a quasi-regular limit map

Let $S_n \rightarrow \infty$ be normalization scales and define

$$q_n(s) = u_{-S_n s} g_n u_{S_n s}.$$

The $q_n(s)$ may leave every fixed compact set of G because of components inside L . Factor locally, or after a suitable L -correction,

$$q_n(s) = \ell_n(s) \sigma_n(s), \quad \ell_n(s) \in L,$$

with $\sigma_n(s)$ lying in a transverse slice.

A *quasi-regular limit map* is a locally uniform limit

$$\phi(s) = \lim_{n \rightarrow \infty} \sigma_n(s)$$

on an interval where the normalized maps remain in the coordinate neighborhood.

The adjective “quasi” warns that

$$\phi(s+t) = \phi(s)\phi(t)$$

need not hold. What survives is polynomial dependence inherited from unipotent conjugation.

6. Nontriviality

Proposition 5.6.1 (First-exit nontriviality). *Suppose the scales S_n are chosen so that*

$$\text{dist}(\sigma_n(1), L) = \rho > 0$$

in a fixed local quotient metric and

$$\text{dist}(\sigma_n(s), L) \leq \rho \quad (0 \leq s \leq 1).$$

If $\sigma_n \rightarrow \phi$ uniformly, then

$$\phi(1) \notin L.$$

In particular, ϕ is not the constant identity map modulo L .

PROOF. Continuity of the quotient distance gives

$$\text{dist}(\phi(1), L) = \rho.$$

Thus $\phi(1)$ is genuinely transverse. $\square$

This elementary observation is the reason first-exit normalization is so effective: nontriviality is built into the choice of scale rather than proved afterward by tracking a leading coefficient.

7. Polynomial degree under changes of coordinates

Suppose κ and κ' are two smooth coordinate charts near the identity with derivative the identity at e . If

$$\kappa(q_n(s))$$

converges after normalization to a polynomial $P(s)$, then

$$\kappa'(q_n(s))$$

has the same leading nonzero homogeneous component. Indeed,

$$\kappa' \circ \kappa^{-1}(Z) = Z + O(\|Z\|^2).$$

At the first macroscopic scale, the quadratic coordinate error is lower order before the displacement becomes fixed small size; after taking the small normalization radius $\rho \rightarrow 0$, the leading polynomial direction is intrinsic.

Thus the exact coefficients of a quasi-regular map depend on coordinates, but the existence of a nontrivial transverse limiting direction does not.

8. The SL_2 opposite-unipotent calculation

Let

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad v_s = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix}.$$

An exact multiplication gives

$$u_{-t}v_s u_t = \begin{pmatrix} 1 - st & -st^2 \\ s & 1 + st \end{pmatrix}.$$

Take $s = s_n \rightarrow 0$ and $t = S_n r$ with

$$S_n = |s_n|^{-1/2}.$$

Then

$$s_n t = O(|s_n|^{1/2}) \rightarrow 0, \quad s_n t^2 \rightarrow \text{sgn}(s_n) r^2.$$

Hence

$$u_{-S_n r} v_{s_n} u_{S_n r} \longrightarrow \begin{pmatrix} 1 & -\sigma r^2 \\ 0 & 1 \end{pmatrix} \in U,$$

where $\sigma = \lim \text{sgn}(s_n)$ after taking a subsequence.

This limit lies in the old subgroup U . It therefore does *not* by itself enlarge the stabilizer. The example is important because it shows that polynomial shearing can generate a large visible displacement while remaining entirely inside known invariance.

To obtain new invariance one must quotient out the old subgroup before selecting the active transverse degree. Modulo U , the linear H -component becomes the relevant scale.

9. The same calculation modulo U

Use the infinitesimal formula

$$e^{-t \operatorname{ad} X} Y = Y - tH - t^2 X.$$

Modulo $\mathbb{R}X$,

$$\bar{Y} \mapsto \bar{Y} - t\bar{H}.$$

For perturbation size s_n , the transverse macroscopic time is therefore

$$T_n \asymp |s_n|^{-1},$$

not $|s_n|^{-1/2}$. On that scale the quotient displacement detects the diagonal direction.

The apparent discrepancy between the square-root and reciprocal scales is not a contradiction. They normalize different components:

- $|s|^{-1/2}$ makes the quadratic U -component macroscopic;
- $|s|^{-1}$ makes the transverse diagonal component macroscopic modulo U .

This is perhaps the clearest concrete illustration of the principle “normalize after quotienting by known invariance.”

Warning 5.9.1 (The large old component still has to be handled). At the reciprocal scale $T_n \asymp |s_n|^{-1}$, the discarded U -component in the exact matrix conjugate is large. Detecting the diagonal class modulo U therefore does *not* by itself show invariance under a diagonal element. One still needs the H-property correction/reparametrization step that turns this quotient information into full orbit comparisons with controlled displacement. This example is precisely why that technical step cannot be replaced by the sentence “mod out by the old invariance.”

10. From a quasi-regular map to an invariance group

Suppose a limiting map satisfies

$$\phi(J) \subset \operatorname{Stab}_G(\mu)$$

for an interval J . Define

$$H_\phi = \overline{\langle \phi(s)\phi(t)^{-1} : s, t \in J \rangle}^\circ.$$

Then

$$H_\phi \subset \text{Stab}_G(\mu)^\circ.$$

Why use differences $\phi(s)\phi(t)^{-1}$ rather than the image itself? The map may not pass through the identity. Differences translate the curve to identity scale and produce a connected subgroup whose Lie algebra is generated by tangent directions of the image.

Proposition 5.10.1 (Tangent algebra of a limiting curve). *Let $\phi : J \rightarrow G$ be C^1 and let $s_0 \in J$. Then*

$$Y(s_0) = \left. \frac{d}{ds} \right|_{s=s_0} \phi(s)\phi(s_0)^{-1}$$

identified by right or left translation with $\mathfrak{g}$, belongs to

$$\text{Lie}(H_\phi).$$

Consequently the Lie algebra generated by all such tangent vectors is contained in $\text{Lie}(H_\phi)$.

PROOF. The curve

$$s \mapsto \phi(s)\phi(s_0)^{-1}$$

lies in the subgroup generated by the differences and passes through e at s_0 . Its derivative at that point therefore lies in the Lie algebra of the closed subgroup generated by the differences. Taking brackets gives the Lie algebra generated by all tangent vectors. $\square$

A nonconstant quasi-regular curve can therefore generate a subgroup substantially larger than its one-dimensional image.

11. An SL_3 example with generated directions

Take

$$U = \{I + tE_{12}\}$$

and perturb in the E_{23} direction. Since

$$[E_{12}, E_{23}] = E_{13}, \quad [E_{12}, E_{13}] = 0,$$

we have

$$e^{-t \text{ad } E_{12}} E_{23} = E_{23} - tE_{13}.$$

A normalized limiting curve can therefore have the form

$$\phi(s) = \exp(E_{23} - sE_{13})$$

up to scaling and old-stabilizer factors.

Tangent differences produce the E_{13} direction. Together with one value carrying an E_{23} component, the generated connected subgroup contains

$$\exp(\mathbb{R}E_{23}) \quad \text{and} \quad \exp(\mathbb{R}E_{13}).$$

These commute, so the new invariance group is at least the two-dimensional abelian root subgroup

$$U_{23,13}.$$

If the old stabilizer already contains E_{12} , brackets then give the full upper-unitriangular Heisenberg algebra.

Thus a one-parameter limiting curve can trigger a multi-dimensional stabilizer enlargement after taking the subgroup it generates together with old invariance.

12. Equivariance under the original unipotent flow

Quasi-regular maps inherit a useful covariance from their construction. Formally,

$$q_n(s+r) = u_{-S_n(s+r)} g_n u_{S_n(s+r)}.$$

Insert

$$u_{-S_n r} u_{-S_n s} = u_{-S_n(s+r)}.$$

Although the scale S_n prevents a simple finite limiting identity in general, after normalization one often obtains a polynomial covariance relation under the induced nilpotent operator. Infinitesimally, the limit lives in a finite-dimensional $\text{ad } X$ -module.

This algebraic covariance severely restricts the possible image of the quasi-regular map. The limiting curve is not an arbitrary continuous curve in G ; it is assembled from a finite nilpotent weight string.

13. Rational maps and denominators

In some coordinate constructions one divides by a scalar polynomial to keep a chosen representative normalized, especially when the subgroup datum is projective. The resulting limit may be rational rather than polynomial:

$$\phi(s) = \frac{P(s)}{q(s)}$$

in an affine coordinate chart, away from zeros of q .

This does not change the rigidity mechanism. On a compact subinterval avoiding poles, the map is smooth and its coordinate functions have uniformly controlled algebraic complexity. Sublevel estimates and tangent-generation arguments still apply. The term quasi-regular is broad enough to accommodate these projective normalizations.

14. Why the image often lies in a normalizer

Suppose the old connected stabilizer is L . Because $U \subset L$ and because the comparison is performed modulo L , the limiting displacement often preserves the algebraic data defining L . This can place

$$\phi(s) \in N_G(L)$$

or in a related algebraic normalizer.

If the image lies in L , it produces no new transverse information. If it lies in

$$N_G(L) \setminus L$$

and approaches the identity through a continuum of values, then tangent directions in

$$\mathfrak{n}_{\mathfrak{g}}(\mathfrak{l})/\mathfrak{l}$$

enter the measure stabilizer. For a provisional L this enlarges the known invariance; for $L = S(\mu)^\circ$ it contradicts the definition of L .

The normalizer is therefore a natural landing place for quasi-regular maps, but the proof must still distinguish discrete normalizer symmetries from connected enlargement.

15. Quasi-regular maps and singularity

The same limiting map can signal either rigidity or singularity.

If its image acts nontrivially transverse to the current stabilizer and the generic orbit blocks avoid proper singular strata, the image yields additional invariance.

If instead the image preserves a proper linearization vector v_H , then

$$\rho(\phi(s))v_H \in \mathbb{R}v_H$$

for all s . This exact relation says that the comparison is trapped in the subgroup geometry associated with H . One then reduces to that proper homogeneous configuration.

Thus the quasi-regular map is the common object behind the two branches:

its image either gives transverse additional invariance
or certifies a singular reduction.

16. A compactness lemma for normalized maps

Lemma 5.16.1 (Finite-dimensional compactness). *Fix $d \geq 0$ and a finite-dimensional normed space V . Let*

$$P_n : [-1, 1] \rightarrow V$$

be polynomial maps of degree at most d satisfying

$$\sup_{|s| \leq 1} \|P_n(s)\| \leq C.$$

Then a subsequence converges uniformly, together with every derivative up to order d , to a polynomial map P of degree at most d .

PROOF. Write

$$P_n(s) = \sum_{j=0}^d a_{n,j} s^j.$$

On the finite-dimensional polynomial space, the coefficient norm

$$\max_j \|a_{n,j}\|$$

and the supremum norm on $[-1, 1]$ are equivalent. Hence the coefficients are uniformly bounded. Pass to a convergent subsequence for each of the finitely many coefficients. Coefficient convergence gives uniform convergence of all derivatives. $\square$

This compactness result, rather than an infinite-dimensional Arzelà–Ascoli argument, is the basic reason normalized unipotent conjugates have such rigid limits.

17. Two normalization parameters

There are frequently two small parameters:

- (1) the initial transverse size ε_n ;
- (2) the chosen macroscopic radius ρ at which the comparison is stopped.

The safe procedure is:

- (1) fix a small $\rho > 0$;
- (2) let $n \rightarrow \infty$ and extract a quasi-regular limit at that radius;
- (3) derive stabilizing elements at distance $\asymp \rho$ from the old subgroup;
- (4) only then let $\rho \rightarrow 0$ to obtain a tangent direction.

Reversing the limits can collapse the quasi-regular map to the identity and lose the additional-invariance information.

18. A practical construction algorithm

Given a concrete sequence $g_n \rightarrow e$, one can construct the limiting map by the following steps.

- (1) Identify the current connected stabilizer L .
- (2) Factor g_n into an L -part and a transverse slice part.

- (3) Take transverse logarithms Z_n .
- (4) Compute the nilpotent operator induced by $\text{ad } X$ on $\mathfrak{g}/\mathfrak{l}$.
- (5) Find the largest active degree r on the limiting direction.
- (6) Choose T_n by first exit, or equivalently by $\|Z_n\|T_n^r \asymp 1$.
- (7) Examine

$$u_{-sT_n} g_n u_{sT_n}$$

for s in a fixed compact interval.

- (8) Remove old-stabilizer factors and pass to a polynomial limit.
- (9) Compute the connected subgroup generated by image differences.
- (10) Use generic-block comparison to decide whether this subgroup preserves the measure or whether the map is trapped in a singular algebraic locus.

Every step is finite-dimensional and explicit once the relevant Lie-algebra representation is known.

19. Conceptual summary

A quasi-regular map is the geometric record of polynomial divergence after the irrelevant old invariance has been divided out. It is produced by three operations:

$$\begin{array}{ccc} \text{small transverse perturbation} & \xrightarrow{\text{unipotent conjugation}} & \text{polynomial growth} \\ & \xrightarrow{\text{first-exit scaling}} & \text{finite nontrivial limit} \end{array}$$

Its image then has two possible interpretations:

- a family of new measure symmetries;
- an exact algebraic relation explaining singular concentration.

This object is where local Lie algebra and global ergodic theory meet most directly in Ratner's proof.

Part II

Singular Sets, Additional Invariance, and Classification

CHAPTER 6

Singular Sets, Linearization, and Avoidance

1. Why singular sets are necessary

The additional-invariance theorem needs a transverse nearby sequence whose normalized shear remains genuinely transverse. There are points for which this is impossible for an algebraic reason: the U -orbit is contained in, or repeatedly shadows, a smaller homogeneous space generated by a proper subgroup.

The purpose of singular sets is to name these obstructions in a way compatible with induction.

2. Subgroups that can carry homogeneous measures

Let $H < G$ be a closed connected subgroup and let $x = \Gamma g$. If the orbit xH is closed and has finite H -invariant volume, then

$$H_x = H \cap g^{-1}\Gamma g$$

is a lattice in H . Conversely, in the finite-volume orbit criterion used here, a lattice stabilizer together with properness of the orbit map

$$H_x \backslash H \longrightarrow \Gamma \backslash G$$

produces a closed finite-volume orbit. We keep the properness clause visible because discreteness or finite covolume of the abstract stabilizer should not be confused with embeddedness of the orbit.

We call a closed finite-volume orbit of this kind *admissible*. For the singular bookkeeping one restricts attention to those connected subgroups H for which the connected subgroup H^u generated by the one-parameter Ad-unipotent subgroups of H acts ergodically on the relevant finite-volume H -orbit. This is the class that naturally contains the homogeneous supports arising from unipotent ergodic components, while allowing H itself to have additional compact or central directions.

Definition 6.2.1 (Basepoint-normalized support class). Let $\mathcal{H} = \mathcal{H}_\Gamma$ denote the collection of connected closed subgroups $H < G$ such that

$$\Delta_H := H \cap \Gamma$$

is a lattice in H and the connected subgroup H^+ generated by the one-parameter Ad-unipotent subgroups of H acts ergodically on $\Delta_H \backslash H$ with respect to Haar probability.

This is no loss of generality for a homogeneous orbit through an arbitrary point. Indeed, if $x = \Gamma g$ and xL is a closed finite-volume L -orbit, then

$$H = gLg^{-1}$$

satisfies $H \cap \Gamma = g(L \cap g^{-1}\Gamma g)g^{-1}$ and hence belongs to the basepoint-normalized class after the same unipotent-ergodicity condition is transported by conjugation. Moreover

$$gUg^{-1} \subset H$$

is exactly the normalizer-type incidence used below. Thus the subgroup type is always recorded at the fixed lattice Γ , while the point of the original orbit is recorded by the translating element g .

For the logic of the proof, proper members represent genuinely lower-complexity homogeneous problems. The qualitative finite-stage bookkeeping uses the Dani–Margulis enumeration $\mathcal{H}_\Gamma = \{H_1, H_2, \dots\}$ and, for each fixed H_i , the Dani–Margulis discreteness of its Chevalley orbit. At stage m one keeps only $H_1, \dots, H_m$ and a bounded linearization window. The global argument is obtained by exhausting m , the compact base window, and the linearization bound. In the support-hull proof module the support-to-algebraic-hull bridge is proved internally, while the enumeration and fixed- H_i Chevalley discreteness remain downstream consequences of the Dani–Margulis structural interface. Its first open core is now stated only on the load-bearing algebraic-support class $\mathcal{H}_\Gamma^{\text{alg}}$; the second is fixed-intersection reconstruction across a central-kernel representation. The later qualitative induction uses only countability and fixed- H_i discreteness; it does not use either core directly.

3. Normalizer-type sets

Fix U and a closed connected subgroup $H < G$. Define

$$(6.1) \quad N(U, H) = \{g \in G : gUg^{-1} \subset H\}.$$

Equivalently,

$$U \subset g^{-1}Hg.$$

If $x = \Gamma g$, then

$$xU \subset xg^{-1}Hg$$

in the appropriate translated homogeneous orbit. Thus points represented by $N(U, H)$ are precisely those for which the U -direction is algebraically trapped inside a conjugate of H .

Proposition 6.3.1. *$N(U, H)$ is closed.*

PROOF. Let $g_n \rightarrow g$ with $g_n U g_n^{-1} \subset H$. For each $u \in U$,

$$g_n u g_n^{-1} \rightarrow g u g^{-1}.$$

Since H is closed, $g u g^{-1} \in H$. Thus $g U g^{-1} \subset H$. □

Its infinitesimal version is

$$N_{\mathfrak{g}}(N, \mathfrak{h}) = \{g : \text{Ad}(g)N \in \mathfrak{h}\},$$

when $U = \exp(\mathbb{R}N)$.

4. Singular strata

A point can belong to many $N(U, H)$ simultaneously. The genuinely singular part associated with H is the portion not already explained by smaller subgroups.

Definition 6.4.1. For $H \in \mathcal{H}$, define the primitive stratum

$$(6.2) \quad N^*(U, H) = N(U, H) \setminus \bigcup_{H' \in \mathcal{H}, H' \subsetneq H} N(U, H').$$

The corresponding subset of X is obtained by projecting $N^*(U, H)$ to $\Gamma \backslash G$ and taking the relevant Γ -saturation.

The formula is schematic until one chooses representatives of conjugacy classes, but the meaning is simple: H is the smallest homogeneous subgroup in the chosen class that captures the U -direction at the point.

Guiding Idea 6.4.2 (Minimal algebraic explanation). When a trajectory behaves exceptionally, do not merely record that it is exceptional. Find the smallest subgroup that explains the exception. Minimality is what makes induction work: once H is minimal, further transverse shearing inside H cannot be blamed on a still smaller homogeneous configuration.

5. Why linearization enters

The condition that a point lies close to a homogeneous subspace is non-linear in G/Γ . Linearization converts it into the smallness of a vector in a finite-dimensional representation.

Let $H < G$ be a k -dimensional closed subgroup with Lie algebra $\mathfrak{h}$. Choose a basis

$$Y_1, \dots, Y_k$$

of $\mathfrak{h}$ and set

$$(6.3) \quad p_H = Y_1 \wedge \dots \wedge Y_k \in \bigwedge^k \mathfrak{g}.$$

The line $\mathbb{R}p_H$ is independent of the chosen basis. The adjoint representation induces

$$\rho_H : G \rightarrow \mathrm{GL}\left(\bigwedge^k \mathfrak{g}\right).$$

Then

$$\rho_H(g)p_H$$

encodes the conjugate subalgebra $\mathrm{Ad}(g)\mathfrak{h}$.

Proposition 6.5.1. *For $g \in G$,*

$$\mathrm{Ad}(g)\mathfrak{h} = \mathfrak{h}$$

if and only if

$$\rho_H(g)p_H \in \mathbb{R}^\times p_H.$$

PROOF. If $\mathrm{Ad}(g)\mathfrak{h} = \mathfrak{h}$, then $\mathrm{Ad}(g)$ restricts to an automorphism of $\mathfrak{h}$, and on the top exterior power it acts by its determinant, so the line $\mathbb{R}p_H$ is preserved. Conversely, if the top exterior line is preserved, then the k -plane spanned by the wedge factors is preserved, hence $\mathrm{Ad}(g)\mathfrak{h} = \mathfrak{h}$. $\square$

6. Polynomiality after linearization

For a unipotent flow,

$$\rho_H(u_t) = \exp(t d\rho_H(N)),$$

and $d\rho_H(N)$ is nilpotent. Hence

$$t \mapsto \rho_H(u_t)v$$

is polynomial for every v .

This is the key advantage of linearization: closeness to a singular homogeneous configuration can be tested by the size of finitely many polynomial coordinates.

Worked Example 6.6.1 (A line stabilizer in SL_3). Let H be the stabilizer of the line $\mathbb{R}e_1$ in $\mathbb{R}^3$. Instead of using $\bigwedge^{\dim H} \mathfrak{sl}_3$, one may use the defining representation itself: the line is encoded by e_1 . If

$$u_t = \begin{pmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix},$$

then

$$u_t e_3 = e_3 + t e_2 + \frac{t^2}{2} e_1.$$

The distance of $u_t e_3$ from a specified invariant subspace is controlled by a quadratic polynomial. The general exterior-power construction is the coordinate-free version of the same idea.

7. A basic polynomial avoidance lemma

Lemma 6.7.1 (Sublevel control). *Fix $d \geq 1$. There are constants $C_d, \alpha_d > 0$ such that for every real polynomial p of degree at most d , every interval I , and every $0 < \varepsilon < 1$,*

$$(6.4) \quad \left| \{t \in I : |p(t)| < \varepsilon \sup_I |p|\} \right| \leq C_d \varepsilon^{\alpha_d} |I|.$$

PROOF. Affine rescaling reduces to $I = [-1, 1]$. Put $M = \sup_I |p|$ and

$$E = \{t \in I : |p(t)| < \varepsilon M\}, \quad m = |E|.$$

If $m = 0$ there is nothing to prove. Set

$$\delta = \frac{m}{4(d+1)}.$$

A maximal δ -separated subset of E must contain at least $d+1$ points: otherwise at most d intervals of radius δ would cover E and would have total length less than $m/2$.

Choose $d+1$ pairwise δ -separated points $x_0, \dots, x_d \in E$. Lagrange interpolation gives

$$p(x) = \sum_{i=0}^d p(x_i) \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}.$$

For $x \in [-1, 1]$, every numerator factor has absolute value at most 2, while every denominator factor is at least δ . Hence

$$M \leq (d+1)\varepsilon M \left(\frac{2}{\delta}\right)^d.$$

Cancelling M yields

$$m \leq C_d \varepsilon^{1/d}$$

for a constant depending only on d . Undoing the affine rescaling proves the lemma with $\alpha_d = 1/d$. $\square$

The same conclusion holds for vector-valued polynomial maps by applying a suitable linear functional that nearly attains the norm at a maximum point.

8. Avoiding a fixed singular neighborhood

For the right action on $X = \Gamma \backslash G$, the convenient linearization vector is

$$\eta_H(g) = \rho_H(g^{-1})p_H.$$

It encodes the conjugate Lie algebra $\text{Ad}(g^{-1})\mathfrak{h}$. If $g \in N(U, H)$, then $U \subset g^{-1}Hg$, so the projective line $\mathbb{R}\eta_H(g)$ is U -invariant. Moreover, along a right U -orbit,

$$(6.5) \quad \eta_H(gu_t) = \rho_H(u_{-t})\eta_H(g),$$

which is an exact polynomial in t because $d\rho_H(N)$ is nilpotent.

Suppose a compact piece of a singular set is represented by the condition that $\eta_H(g)$ lies in a small neighborhood of an algebraic cone $C \subset V$. Along the unipotent orbit this becomes

$$P(t) = \eta_H(gu_t) = \rho_H(u_{-t})\eta_H(g).$$

If P is not identically trapped in C , a polynomial sublevel estimate bounds the proportion of times for which $P(t)$ lies very close to C .

The exact implementation uses finitely many polynomial functions cutting out the cone locally. On a compact region, finitely many charts suffice.

Guiding Idea 6.8.1 (Local avoidance mechanism). Let C be a compact subset of a primitive singular stratum associated with H , and let $K \subset X$ be compact. In each of the finitely many linearization charts covering $C \cap K$, proximity to the stratum is detected by finitely many scalar polynomials

$$t \longmapsto F_i(\rho(u_{-t})v).$$

If at least one detector is nonzero and its supremum on the enlarged block is bounded below by the nondivergence threshold, Lemma 6.7.1 bounds the relative measure of the smaller sublevel set by a positive power of the ratio of the two thresholds. If every detector vanishes identically, the orbit lies in the lower singular incidence encoded by that chart. The theorem-level finite-stage statement, including the compactness and lattice bookkeeping needed to supply the lower threshold uniformly, is Proposition 7.9.1.

Remark 6.8.2 (Logical status of the template). This proposition is the local sublevel estimate. The finite-stage synthesis used in the induction is Proposition 7.9.1: after compact restriction and finite lattice bookkeeping, it formulates the remaining dichotomy entirely in terms of finitely many bounded-degree polynomial families. Thus the phrase “has a controlled maximum” in the proof architecture above is not being used as an undefined global hypothesis.

Remark 6.8.3. The phrase “has a controlled maximum” is where nondivergence enters. Without a lower bound at some point of the enlarged interval, a polynomial could be uniformly tiny and the sublevel estimate would say nothing. Quantitative nondivergence is precisely the systematic tool that supplies these lower bounds for all relevant exterior-power vectors at once.

9. Finite-stage singular parameter families

The avoidance argument must eventually handle every proper homogeneous obstruction, not one fixed subgroup. The qualitative proof uses the countability theorem of Appendix A: fix once and for all an enumeration

$$\mathcal{H}_\Gamma = \{H_1, H_2, \dots\}.$$

At stage m only the first m support types are retained. For each fixed H_i , the Chevalley-discreteness theorem makes its Γ -branches locally finite; hence a bounded linearization window sees only finitely many discrete branches. Inside each branch the Lie-subalgebra parameter may vary continuously, and the compact Grassmannian theorem supplies the finite chart cover needed for uniform polynomial estimates. In symbols,

(6.6)

compact base + first m support types + bounded linearization window $\implies$ finitely many uniform singular charts.
--

The global argument then lets the compact set, m , and the norm bound tend to infinity by a diagonal exhaustion.

Guiding Idea 6.9.1 (Why countability and compactness play different roles). Countability is used only to organize the discrete support types into finite initial segments. Fixed-support Chevalley discreteness then makes the lattice branches finite inside a bounded vector window. Compactness of the Grassmannian is a third, continuous mechanism: it gives uniform polynomial degree and coefficient bounds inside each surviving chart. None of these three steps may be silently replaced by another.

10. The good set

Fix $\varepsilon > 0$. Choose a compact recurrent set K with

$$\mu(K) > 1 - \varepsilon$$

and with boundary of μ -measure zero. Choose a sufficiently large support-type stage m and a sufficiently large linearization-norm window for the current diagonal step. By Theorem A.5.56, the proper singular configurations belonging to the first m support types and meeting K are covered by finitely many uniform linearization charts. The outer diagonal exhaustion in m and the norm bound ensures that every admissible proper singular type is eventually included.

For every primitive singular chart that has zero μ -mass, choose a sufficiently small neighborhood so that the union of these finitely many neighborhoods has μ -measure $< \varepsilon$. If a primitive chart carries positive mass instead,

the invariant-parameter mechanism of Chapter 7 applies and sends the proof to the singular reduction branch.

In the avoidance branch define a compact recurrent good set

$$K_{\text{good}} \subset K \setminus \bigcup_{i=1}^J \mathcal{O}_i$$

with

$$(6.7) \quad \mu(K_{\text{good}}) > 1 - 3\varepsilon.$$

Generic points in K_{good} return there with high frequency. Nearby pairs chosen from this set cannot have all of their transverse polynomial shears explained by the finitely many proper singular configurations visible at the current compact-complexity stage. Passing to larger stages by a diagonal argument gives the global conclusion.

11. A recurrence-and-avoidance proposition

Guiding Idea 6.11.1 (Good transverse terminal blocks: quotient-level mechanism). Fix one finite stage of the singular-parameter exhaustion and suppose every primitive proper singular chart at that stage has zero μ -mass. After choosing a compact recurrent set, a sufficiently small first-exit radius, and a diagonal sequence of nearby generic pairs, the quantitative nonescape, finite-stage avoidance, and bounded-degree terminal estimates leave terminal subblocks on which both trajectories stay in one compact chart and the quotient displacement varies by $o(1)$. The bad-time proportion tends to zero. This statement is only the quotient-level mechanism: it does not remove a time-dependent factor in the known stabilizer. The exact theorem that performs that correction is Theorem C.8.1.

12. How singular mass triggers induction

Suppose instead that a primitive proper singular chart at some compact-complexity stage carries positive μ -mass. The U -saturation of the stratum is invariant, so ergodicity forces the appropriate saturation to have full measure. At this point one must be careful: even for a fixed subgroup H there may be uncountably many translates, so countability of subgroup *types* cannot by itself produce a positive-mass translate.

The correct reduction uses the linearized parameter of a primitive stratum. Choose a representation ρ and a line $[v_H]$ whose projective stabilizer records the relevant subgroup data. On a local branch of the primitive stratum, a point Γg carries the parameter

$$\Theta(\Gamma g) = \rho(g^{-1})[v_H].$$

Because $gUg^{-1} \subset H$ on this stratum, the parameter is unchanged by the right U -action. Indeed, if $h_t = gu_tg^{-1} \in H$, then

$$\Theta(\Gamma gu_t) = \rho(u_{-t}g^{-1})[v_H] = \rho(g^{-1}h_t^{-1})[v_H] = \rho(g^{-1})[v_H].$$

Thus Θ is a measurable U -invariant map.

Ergodicity implies that Θ is almost surely constant: for every Borel set B in the parameter space, $\Theta^{-1}(B)$ is U -invariant, so its measure is 0 or 1; a 0–1 probability on a standard Borel space is a Dirac mass. Hence almost every point lies in one projective fiber. That fiber is locally a coset of the proper projective stabilizer of $[v_H]$. The passage from this local fiber statement to a genuine lower-dimensional homogeneous problem is Theorem A.6.1. It does *not* assume that the intermediate orbit already has finite volume. Instead it identifies the problem measurably with a quotient

$$\Delta \backslash K, \quad \Delta < K \text{ discrete}, \quad \dim K < \dim G,$$

with $U \subset K$. This is exactly what induction needs. Finite covolume of the subgroup that eventually supports the classified measure is obtained only at the end of the lower-dimensional classification, from the existence of its finite invariant Haar measure.

The only global countability input is the Dani–Margulis enumeration of the support class. The primitive chart in which positive mass is detected arises at one finite stage of that enumeration; fixed-support Chevalley discreteness and compact Grassmannian control then supply the finite bookkeeping needed at that stage.

This is the second half of the dichotomy:

(6.8)

no singular mass $\Rightarrow$ additional invariance,	singular mass $\Rightarrow$ induction.
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13. A model singular phenomenon in SL_3

Let

$$G = \mathrm{SL}_3(\mathbb{R}), \quad \Gamma = \mathrm{SL}_3(\mathbb{Z}),$$

and let

$$U = \left\{ \begin{pmatrix} 1 & t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} : t \in \mathbb{R} \right\}.$$

The subgroup

$$H = \left\{ \begin{pmatrix} * & * & 0 \\ * & * & 0 \\ 0 & 0 & 1 \end{pmatrix} : \det = 1 \right\} \cong \mathrm{SL}_2(\mathbb{R})$$

contains U and has a closed finite-volume orbit through the standard lattice. A measure supported on that orbit is singular from the point of view of the full SL_3 space: a transverse shearing argument in G should not be expected to enlarge the stabilizer beyond H before one first recognizes that the entire measure lives in this lower-dimensional homogeneous space.

This example shows why induction on homogeneous subgroups is not merely a proof device. It reflects an actual geometric hierarchy of invariant measures.

14. What linearization does not prove

Linearization can show that prolonged proximity to a homogeneous subspace forces an algebraic relation. It does not by itself classify the measure. One still needs:

- ergodicity to turn positive singular mass into full mass on an invariant component;
- the additional-invariance theorem on the nonsingular part;
- a termination argument for the resulting dimension-decreasing induction, together with a clear distinction between provisional invariance enlargement and the full-stabilizer proof.

The two viewpoints must be organized so that subgroup reduction is not confused with provisional stabilizer enlargement. In the final proof, once the full connected measure stabilizer is chosen, the regular transverse branch is contradictory and only subgroup reduction is recursive.

CHAPTER 7

Global Avoidance from Local Linearization

The local linearization argument is easy to summarize and easy to misuse. A trajectory that remains close to a proper homogeneous configuration forces a collection of polynomial coordinates to remain small. A polynomial sublevel estimate then says that this can happen on only a controlled portion of a time interval, unless the polynomial is small for an algebraic reason. What is less immediate is how one turns this local statement into an avoidance theorem that can actually be inserted into the proof of measure classification.

This chapter performs that passage carefully. The goal is not to introduce a new theorem with stronger hypotheses, but to make explicit the bookkeeping that is often hidden behind the phrase “by linearization and nondivergence.”

1. The geometric situation

Let

$$X = \Gamma \backslash G, \quad U = \{u_t : t \in \mathbb{R}\},$$

where G is a connected Lie group, $\Gamma < G$ is a lattice, and U is Ad-unipotent. We use right actions throughout.

Fix a connected closed subgroup $H < G$. A point $x = \Gamma g$ has the property that its U -orbit is tangent to a translate of an H -orbit precisely when

$$gUg^{-1} \subset H.$$

This motivates

$$N(U, H) = \{g \in G : gUg^{-1} \subset H\}.$$

The corresponding subset of X is

$$\Gamma \backslash \Gamma N(U, H).$$

It is important to notice that this is not, in general, a single homogeneous orbit. Different $g \in N(U, H)$ can give different conjugates $g^{-1}Hg$ through the point Γg .

Guiding Idea 7.1.1 (What avoidance must accomplish). Suppose we compare two nearby generic points x and xg_n . Polynomial shearing gives a first macroscopic time T_n . To obtain a new stabilizing direction, we need many such terminal times at which both orbit points remain in a compact region

and do not fall into a lower-dimensional homogeneous obstruction. Avoidance is the device that guarantees such times unless the measure already gives positive mass to a lower-dimensional obstruction, in which case induction takes over.

Thus avoidance is not an independent aesthetic statement. It is the logical switch between the two branches of the induction.

2. Linearization data

For the moment assume that G is a matrix group. Let

$$\rho : G \rightarrow \mathrm{GL}(V)$$

be a finite-dimensional real representation and choose $0 \neq v_H \in V$ such that H preserves the projective line $[v_H]$. Write

$$M_H = \{g \in G : \rho(g)[v_H] = [v_H]\}.$$

Then $H \subset M_H$. In favorable situations one chooses the representation so that $M_H^\circ = H$; in general the linearization naturally records a slightly larger normalizer-type subgroup, which is exactly the object that must be tracked in the singular-set argument.

The temporary matrix-group language is not a restriction for the exterior-power linearization actually used below. For an arbitrary Lie group, the adjoint representation

$$\mathrm{Ad} : G \rightarrow \mathrm{GL}(\mathfrak{g})$$

and its exterior powers are finite-dimensional representations. Since U is Ad-unipotent, their infinitesimal U -operators are nilpotent. Thus all polynomial calculations below remain valid without assuming that G itself has a faithful matrix representation.

The exterior-power construction illustrates this point. Writing $\mathfrak{h} = \mathrm{Lie}(H)$ and $d = \dim H$, choose a basis $Y_1, \dots, Y_d$ of $\mathfrak{h}$ and set

$$v_H = Y_1 \wedge \cdots \wedge Y_d \in \bigwedge^d \mathfrak{g}.$$

The line $\mathbb{R}v_H$ remembers $\mathfrak{h}$: its projective stabilizer is the subgroup preserving the Lie subalgebra $\mathfrak{h}$, namely a normalizer-type group that can be strictly larger than H . Thus the exterior-power construction does not assume that H is self-normalizing.

For the right action on $X = \Gamma \backslash G$, define

$$\eta_H(g) = \rho(g^{-1})v_H.$$

The inverse is essential: it matches the right-action convention. If $g \in N(U, H)$, then $gUg^{-1} \subset H$. Writing

$$h_t = gu_tg^{-1} \in H,$$

we have

$$\rho(u_t)\eta_H(g) = \rho(u_tg^{-1})v_H = \rho(g^{-1}h_t)v_H,$$

which lies in the line $\mathbb{R}\rho(g^{-1})v_H = \mathbb{R}\eta_H(g)$. Thus the projective line generated by $\eta_H(g)$ is U -invariant.

Remark 7.2.1. The use of a line rather than a vector is not cosmetic. Normalization by determinants or modular characters can multiply v_H by a scalar. Passing to projective space, or keeping track of a one-dimensional character, prevents an irrelevant scalar from being mistaken for a genuine transverse displacement.

3. Polynomiality of the linearized trajectory

Let $u_t = \exp(tX)$ with $\text{ad } X$ nilpotent on $\mathfrak{g}$. For every finite-dimensional representation ρ arising algebraically from the adjoint representation, the infinitesimal operator

$$d\rho(X)$$

is nilpotent. Consequently

$$\rho(u_t) = \exp(t d\rho(X))$$

is a polynomial in t .

Proposition 7.3.1 (Polynomial linearization). *Let V be a finite-dimensional representation on which $d\rho(X)$ is nilpotent of index at most $m+1$. Then for every $v \in V$,*

$$t \mapsto \rho(u_t)v$$

is a V -valued polynomial of degree at most m .

PROOF. Since

$$(d\rho(X))^{m+1} = 0,$$

the exponential series terminates:

$$\rho(u_t)v = \sum_{j=0}^m \frac{t^j}{j!} (d\rho(X))^j v.$$

There is no approximation here. The map is exactly polynomial. $\square$

For a nearby point represented by gg_n , the linearized quantity is

$$\eta_H(gg_n u_t) = \rho(u_{-t} g_n^{-1} g^{-1}) v_H = \rho(u_{-t}) \eta_H(gg_n).$$

Thus all time dependence is carried by the bounded-degree polynomial operator $\rho(u_{-t})$. This is the right-action analogue of the more familiar left-action linearization formula.

4. A quantitative polynomial lemma

We isolate the elementary estimate that converts algebra into time control.

Lemma 7.4.1 (Sublevel estimate on an interval). *For every integer $d \geq 1$ there are constants $C_d > 0$ and $\alpha_d > 0$ with the following property. If $p : \mathbb{R} \rightarrow \mathbb{R}$ is a nonzero polynomial of degree at most d , $I \subset \mathbb{R}$ is an interval, and $0 < \varepsilon \leq 1$, then*

$$|\{t \in I : |p(t)| \leq \varepsilon \sup_I |p|\}| \leq C_d \varepsilon^{\alpha_d} |I|.$$

One may take $\alpha_d = 1/d$ after changing C_d .

PROOF. By an affine change of variable we may assume $I = [-1, 1]$. Put

$$M = \sup_I |p|, \quad E = \{t \in I : |p(t)| \leq \varepsilon M\}, \quad m = |E|.$$

If $m = 0$ there is nothing to prove. Let $d' \leq d$ be the actual degree of p ; enlarging constants, we may work with d .

Set

$$\delta = \frac{m}{4(d+1)}.$$

There exist $d+1$ points

$$x_0, \dots, x_d \in E$$

whose pairwise distances are at least δ . Indeed, take a maximal δ -separated subset of E . If it had at most d points, then E would be covered, up to a null set, by at most d intervals of radius δ , whose total length is at most

$$2d\delta < \frac{m}{2},$$

a contradiction.

Interpolate p at these points:

$$p(x) = \sum_{i=0}^d p(x_i) L_i(x), \quad L_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}.$$

For $x \in [-1, 1]$, every numerator factor is at most 2, while every denominator factor is at least δ . Hence

$$|L_i(x)| \leq \left(\frac{2}{\delta}\right)^d.$$

Since $|p(x_i)| \leq \varepsilon M$,

$$|p(x)| \leq (d+1)\varepsilon M \left(\frac{2}{\delta}\right)^d.$$

Taking the supremum over x and cancelling $M > 0$ gives

$$1 \leq (d+1)\varepsilon \left(\frac{8(d+1)}{m}\right)^d.$$

Therefore

$$m \leq C_d \varepsilon^{1/d}$$

for a constant C_d depending only on d . Undoing the affine rescaling multiplies both m and the interval length by the same factor, proving the stated estimate with $\alpha_d = 1/d$. $\square$

For a vector-valued polynomial $P : I \rightarrow \mathbb{R}^N$, apply the lemma to a coordinate whose supremum controls $\sup_I \|P\|$. We obtain

$$|\{t : \|P(t)\| \leq \varepsilon \sup_I \|P\|\}| \ll_{d,N} \varepsilon^\alpha |I|.$$

5. A finite-family avoidance lemma

A global singular set is usually an infinite union. The first genuinely useful statement concerns a fixed finite family.

Let $H_1, \dots, H_r < G$ be connected closed subgroups. For each i , choose linearization data (ρ_i, V_i, v_i) as above. On the bounded region relevant to a fixed compact observation window, the algebraic bad locus is described by finitely many polynomial equations

$$F_{i1}(v) = \dots = F_{im_i}(v) = 0.$$

For instance, preservation of a line in an exterior-power representation is expressed by the vanishing of the appropriate 2×2 minors. The important point is that the degrees of the F_{ij} are fixed once the subgroup data have been fixed.

If $P_i : J \rightarrow V_i$ is the polynomial linearization of a unipotent trajectory, then

$$Q_{ij}(t) = F_{ij}(P_i(t))$$

is again a scalar polynomial, with degree bounded in terms of the representation and the defining equations.

Proposition 7.5.1 (Finite-family avoidance). *Fix a compact interval $J \subset \mathbb{R}$ and finitely many families of scalar polynomials*

$$Q_{ij} : J \rightarrow \mathbb{R}, \quad 1 \leq i \leq r, \quad 1 \leq j \leq m_i,$$

of degree at most D . Suppose that for every i ,

$$\max_{1 \leq j \leq m_i} \sup_{t \in J} |Q_{ij}(t)| \geq c_i > 0.$$

Then for every $\eta > 0$ there exists $\delta > 0$ such that

$$\left| \left\{ t \in J : \max_j |Q_{ij}(t)| \leq \delta \text{ for at least one } i \right\} \right| \leq \eta |J|.$$

The choice of δ depends only on D , the finite family, the lower bounds c_i , and η .

PROOF. For each i , choose an index $j(i)$ satisfying

$$\sup_J |Q_{i,j(i)}| \geq c_i.$$

If all defining polynomials for the i th bad locus are at most δ in absolute value, then in particular

$$|Q_{i,j(i)}(t)| \leq \delta.$$

Lemma 7.4.1, applied with

$$\varepsilon = \delta/c_i,$$

gives

$$|\{t : \max_j |Q_{ij}(t)| \leq \delta\}| \leq C_D (\delta/c_i)^{\alpha_D} |J|.$$

Choose δ so that the sum of these finitely many upper bounds is less than $\eta |J|$. The union bound completes the proof. $\square$

In geometric applications, small values of all $F_{ij}(P_i(t))$ mean that the linearized point lies close to the algebraic singular locus on the fixed bounded coordinate region. Conversely, on a compact region, distance to that locus tending to zero forces all defining polynomials to tend to zero. Thus the polynomial formulation is exactly what is needed for the qualitative avoidance argument and avoids any unjustified claim that arbitrary distance functions are polynomial.

Guiding Idea 7.5.2 (The dichotomy hidden in the hypothesis). The lower bound

$$\max_j \sup_J |Q_{ij}(t)| \geq c_i$$

is precisely where the induction enters. If it fails on longer and longer intervals, then the linearized trajectory is algebraically trapped near the H_i -configuration. Compactness of normalized polynomial maps produces

an exact limiting relation, and that relation places the measure in a lower-dimensional singular stratum. Thus one does not obtain avoidance for free: one obtains

avoidance or algebraic reduction.

6. Why only finitely many translates occur on a compact set

The passage from an abstract subgroup H to points of X introduces lattice translates. A priori one could encounter infinitely many elements $\gamma \in \Gamma$ even inside a compact observation window. Proper discontinuity removes this problem locally.

Lemma 7.6.1 (Finite lattice bookkeeping). *Let $C_1, C_2 \subset G$ be compact. Then*

$$\{\gamma \in \Gamma : C_1\gamma \cap C_2 \neq \emptyset\}$$

is finite.

PROOF. If $c_1\gamma = c_2$, then

$$\gamma = c_1^{-1}c_2 \in C_1^{-1}C_2.$$

The latter is compact. Since Γ is discrete, its intersection with a compact set is finite. $\square$

This elementary lemma is what makes compact exhaustion compatible with linearization. Once the orbit is restricted to a compact set of X , choose finitely many relatively compact lifts to G . Only finitely many lattice elements can relate these lifts. Therefore only finitely many linearized representatives of a fixed subgroup type need to be considered at one stage of the proof.

7. Compact exhaustion and boundary control

Let μ be a probability measure. Given $\varepsilon > 0$, choose a compact set $K \subset X$ with

$$\mu(K) > 1 - \varepsilon.$$

For generic points, sufficiently long orbit intervals spend approximately $\mu(K)$ of their time in K . We then enlarge K slightly to a compact neighborhood K^+ whose boundary has μ -measure zero. This avoids an otherwise annoying ambiguity: a small displacement can move a point across the boundary of K even when the displacement tends to the identity.

Lemma 7.7.1 (Boundary-zero enlargement). *For every compact $K \subset X$ and every open neighborhood $O \supset K$, there exists a compact set K' such that*

$$K \subset K' \subset O$$

and

$$\mu(\partial K') = 0.$$

PROOF. Choose a compatible proper metric on the relevant compact neighborhood and consider

$$K_r = \{x : \text{dist}(x, K) \leq r\}.$$

For distinct r , the boundaries ∂K_r are disjoint. A finite measure can charge only countably many pairwise disjoint sets with positive measure. Hence for almost every sufficiently small r ,

$$\mu(\partial K_r) = 0.$$

Take such an r with $K_r \subset O$. □

8. The algebraic trapping alternative

Suppose a sequence of normalized polynomial maps $P_n : J \rightarrow V$ satisfies

$$\sup_J \|P_n\| = 1$$

and spends an asymptotically full proportion of J in an $o(1)$ -neighborhood of a proper linear subspace $W < V$. Compactness of bounded-degree polynomials implies a subsequence converges uniformly to a polynomial P . The sublevel estimate forces

$$P(J) \subset W.$$

Indeed, if the component of P in V/W were nonzero, its zero set could not occupy full measure.

Proposition 7.8.1 (Trapping produces an exact relation). *Let $P_n : J \rightarrow V$ be polynomial maps of uniformly bounded degree with*

$$\sup_J \|P_n\| = 1.$$

Assume

$$\frac{1}{|J|} |\{t \in J : \text{dist}(P_n(t), W) < \delta_n\}| \longrightarrow 1, \quad \delta_n \rightarrow 0.$$

Then after passing to a subsequence, $P_n \rightarrow P$ uniformly and

$$P(J) \subset W.$$

PROOF. The space of V -valued polynomials of degree at most d is finite dimensional. All norms on it are equivalent. The normalization therefore gives precompactness, so pass to a uniformly convergent subsequence.

Let $\pi : V \rightarrow V/W$. If $\pi \circ P$ were not identically zero, set

$$c = \sup_{t \in J} \|\pi P(t)\| > 0.$$

Uniform convergence gives

$$\sup_J \|\pi P_n(t)\| \geq c/2$$

for all large n . The maps πP_n are polynomials of the same uniform degree bound, so the vector-valued sublevel estimate applied directly to πP_n yields constants $C, \alpha > 0$, independent of n , such that

$$|\{t \in J : \|\pi P_n(t)\| < C_0 \delta_n\}| \leq C \left(\frac{\delta_n}{c}\right)^\alpha |J|.$$

Here C_0 accounts for the fixed equivalence between distance to W and the quotient norm. The right-hand side tends to zero. But

$$\text{dist}(P_n(t), W) < \delta_n$$

implies $\|\pi P_n(t)\| < C_0 \delta_n$, contradicting the hypothesis that the former condition holds on a set of relative measure tending to one. $\square$

In the homogeneous setting, W is the algebraic subspace expressing stabilization of a subgroup datum. Thus an exact relation $P(J) \subset W$ means the limiting transverse direction normalizes, centralizes, or preserves a lower-dimensional homogeneous structure. This is the algebraic reduction branch.

9. From local to global: the finite-stage polynomial dichotomy

At a fixed compact observation scale and a fixed complexity bound, the lattice bookkeeping above reduces the geometric problem to finitely many polynomial families. The precise statement needed at this stage is therefore purely polynomial. The subsequent passage back to homogeneous geometry is explanatory bookkeeping, not an additional theorem hidden in vague terminology such as “a controlled comparison block.”

Proposition 7.9.1 (Finite-stage polynomial avoidance-or-trapping). *Let $J \subset \mathbb{R}$ be a compact interval. For $1 \leq i \leq r$, let V_i be a finite-dimensional normed vector space, let*

$$F_{ij} : V_i \rightarrow \mathbb{R}, \quad 1 \leq j \leq m_i,$$

be fixed polynomial functions, and let $d_i \geq 0$. For every n let

$$P_{n,i} : J \rightarrow V_i$$

be a polynomial map of degree at most d_i , normalized by

$$\sup_{t \in J} \|P_{n,i}(t)\| = 1.$$

Assume that, in any fixed affine coordinate on J and any fixed basis of V_i , the coefficient vectors of the maps $P_{n,i}$ lie in a compact set independent of n . Define

$$Q_{n,ij}(t) = F_{ij}(P_{n,i}(t)), \quad M_{n,i} = \max_j \sup_{t \in J} |Q_{n,ij}(t)|.$$

Then, after passing to a single subsequence, for every i exactly one of the following alternatives holds:

- (1) there is $c_i > 0$ such that $M_{n,i} \geq c_i$ for every n in the subsequence;
- (2) $M_{n,i} \rightarrow 0$, and $P_{n,i}$ converges uniformly, after a further subsequence, to a nonzero polynomial map P_i satisfying

$$F_{ij}(P_i(t)) = 0 \quad (t \in J, 1 \leq j \leq m_i).$$

If alternative (1) holds for every i , then for every $\eta > 0$ there is $\delta > 0$, independent of n , such that

$$\left| \left\{ t \in J : \max_j |Q_{n,ij}(t)| \leq \delta \text{ for at least one } i \right\} \right| \leq \eta |J|$$

for every n in the subsequence.

PROOF. For each i , pass to a subsequence on which the nonnegative sequence $M_{n,i}$ converges in the compact interval determined by the coefficient bounds. Since there are finitely many i , a diagonal subsequence works simultaneously. If the limit is positive, half of it gives c_i and alternative (1). If the limit is zero, then

$$\sup_J |Q_{n,ij}| \leq M_{n,i} \rightarrow 0$$

for every j .

The coefficient compactness of $P_{n,i}$ gives a coefficientwise convergent subsequence, hence uniform convergence on J , to a polynomial P_i of degree at most d_i . The normalization passes to the limit:

$$\sup_J \|P_i\| = 1,$$

so P_i is not the zero map. Because each F_{ij} is continuous and

$$F_{ij}(P_{n,i}(t)) = Q_{n,ij}(t) \rightarrow 0$$

uniformly, we obtain

$$F_{ij}(P_i(t)) = 0$$

for every t and j . This is alternative (2).

Finally suppose alternative (1) holds for every i . The scalar polynomials $Q_{n,ij}$ have degrees bounded uniformly in n , because both the degrees of $P_{n,i}$ and of the fixed F_{ij} are bounded. Proposition 7.5.1, with the lower bounds c_i ,

therefore provides a single δ giving the stated η -avoidance estimate uniformly in n . $\square$

Remark 7.9.2 (How this becomes a homogeneous-space statement). On a fixed compact set, after restricting to the first m support types in the Dani–Margulis enumeration and to a bounded linearization window, Theorem A.5.56 leaves finitely many lattice branches and finitely many Grassmannian/Chevalley charts, but the subgroup parameter inside a chart may still vary continuously. The compact-parameter theorem, Theorem A.5.58, is precisely what makes the polynomial estimate uniform in that varying parameter. Alternative (1) gives finite-stage avoidance. Alternative (2) is the exact trapping branch; turning that exact algebraic relation into concentration on a proper finite-volume homogeneous orbit uses the invariant-parameter argument below together with the structural singular-reduction input in Appendix A. Keeping these steps separate prevents a qualitative geometric slogan from being used as though it were a theorem with precise hypotheses.

10. From an exact algebraic relation to a smaller ambient problem

An exact limiting relation is not yet the same thing as concentration on one translate of H . This point deserves care. Even for one fixed subgroup type there can be uncountably many translates, so countability of subgroup types alone cannot select a positive-mass translate.

The correct mechanism is an invariant parameter.

Let $[v_H] \in \mathbb{P}(V)$ be the projective linearization datum and let

$$M_H = \{g \in G : \rho(g)[v_H] = [v_H]\}$$

be its projective stabilizer. By construction $H \subset M_H$; in applications the representation is chosen so that M_H is the relevant normalizer-type algebraic subgroup and is proper whenever the singular datum is proper.

On a local lift of the primitive H -stratum define

$$\Theta(\Gamma g) = \rho(g^{-1})[v_H].$$

Different choices of lift can change this value by one of finitely many lattice identifications on a fixed compact chart. After separating those finitely many branches, Θ is a genuine Borel map on each branch.

If $gUg^{-1} \subset H$, put $h_t = gu_tg^{-1} \in H$. Then

$$\Theta(\Gamma gu_t) = \rho(u_{-t}g^{-1})[v_H] = \rho(g^{-1}h_t^{-1})[v_H] = \rho(g^{-1})[v_H] = \Theta(\Gamma g),$$

because H preserves the projective line $[v_H]$. Thus Θ is U -invariant on the primitive stratum.

Lemma 7.10.1 (Ergodicity collapses the singular parameter). *Let S be a U -invariant measurable subset with $\mu(S) = 1$, and let*

$$\Theta : S \rightarrow Z$$

be a Borel map to a standard Borel space satisfying

$$\Theta(xu_t) = \Theta(x)$$

for all t and almost every x . If μ is U -ergodic, then Θ is almost surely constant.

PROOF. The pushforward measure

$$\nu = \Theta_*\mu$$

is a probability measure on Z . For every Borel set $B \subset Z$, the inverse image

$$\Theta^{-1}(B)$$

is U -invariant modulo null sets. Ergodicity therefore gives

$$\nu(B) = \mu(\Theta^{-1}(B)) \in \{0, 1\}.$$

A probability measure on a standard Borel space taking only the values 0 and 1 on Borel sets must be a point mass; we verify this directly. Here is a precise argument. Because Z has a standard Borel structure, there exists a Borel isomorphism from Z onto a Borel subset $E \subset [0, 1]$. Push ν forward to a probability measure $\tilde{\nu}$ on E and extend it by zero to $[0, 1]$. Every dyadic interval has $\tilde{\nu}$ -measure 0 or 1. Starting with $[0, 1]$, choose at each stage the unique dyadic child of the previously chosen interval having measure 1 (if both children meet at their common endpoint, assign that endpoint to one half by using half-open intervals). This gives nested intervals I_n with $|I_n| = 2^{-n}$ and $\tilde{\nu}(I_n) = 1$. Their intersection is a single point r . Continuity from above yields $\tilde{\nu}(\{r\}) = 1$. Since the measure is supported on E , we have $r \in E$, so pulling back gives $\nu = \delta_z$ for a unique $z \in Z$. Consequently $\Theta = z$ almost everywhere. $\square$

If the almost-sure value is $\rho(g_0^{-1})[v_H]$, then almost every point of the primitive stratum lies in the corresponding projective fiber. Locally that fiber is a right coset of

$$g_0^{-1}M_H g_0$$

through the base point. The local fiber alone does not yet imply finite covolume, and none is needed. The structural singular-fiber reduction theorem, Theorem A.6.1, pulls the probability measure back to a quotient

$$\Delta \backslash M_0$$

with $M_0 < G$ proper connected and $\Delta < M_0$ discrete. Ad-unipotency and ergodicity survive this identification, while the ambient dimension strictly

decreases. The induction hypothesis will itself produce the eventual finite-volume homogeneous support.

Guiding Idea 7.10.2 (Why primitive strata are used). On a nonprimitive stratum, the projective stabilizer M_H can contain smaller singular configurations, and the parameter map can fail to separate the geometric pieces one wants to distinguish. Removing lower-dimensional strata first ensures that the remaining fiber represents one genuine algebraic obstruction rather than a union of nested ones.

This invariant-parameter argument is the missing step between “the polynomial limit preserves subgroup data” and “the measure reduces to a smaller homogeneous problem.” The remaining global bookkeeping is compact and finite at each stage; no master enumeration of subgroup types is used.

11. Compact exhaustion and induction

The finite-family theorem above is all that is needed at any fixed stage. To pass globally, choose an increasing sequence of compact sets

$$K_1 \subset K_2 \subset \cdots, \quad \mu\left(\bigcup_m K_m\right) = 1,$$

together with the first m support types in the fixed Dani–Margulis enumeration and increasing bounds on the norm and dimension of the linearization data. The finite-stage bookkeeping theorem, Theorem A.5.56, says that only finitely many uniform singular charts are needed at each stage; within each chart the subgroup parameter lies in a compact coefficient family.

Run Theorem A.5.58 on those finitely many compact parameter charts. If every primitive proper chart has zero mass, choose its avoidance neighborhoods and pass to the next stage. If some primitive chart has positive mass, Lemma 7.10.1 makes its projective parameter almost surely constant, and Theorem A.6.1 reduces the problem to a quotient by a discrete subgroup of a proper connected Lie group. Thus the recursive branch strictly decreases ambient dimension.

If no reduction ever occurs, a diagonal choice through the stages produces comparison times avoiding every proper singular obstruction that can occur on compact windows. Countability alone only supplies the finite initial segment of support types. The fixed-support Chevalley theorem and compact-parameter theorem add the uniform branch, degree, and coefficient control needed by the polynomial argument.

Guiding Idea 7.11.1 (The global bookkeeping principle). The proof never needs a master list of subgroup types. It needs only the following finite

statement, repeated on larger windows:

$$\boxed{\text{compact base} + \text{bounded complexity} \implies \text{finite singular family}.}$$

The two possible outcomes are then exactly the two outcomes needed by the induction: finite-family avoidance or a proper homogeneous reduction.

12. Good terminal times

We record a useful synthesis.

Proposition 7.12.1 (Existence of good terminal times). *Assume that μ gives zero mass to every proper singular stratum in the current induction stage. Let x_n, y_n be pairs of common generic points with $y_n = x_n g_n$ and $g_n \rightarrow e$, $g_n \notin \text{Stab}_G(\mu)$. Let T_n be first macroscopic shearing times.*

Then, after passing to a subsequence, and after fixing an arbitrarily small terminal proportion $\eta > 0$, one may choose the comparison time inside the terminal window $[(1 - \eta)T_n, T_n]$ so that simultaneously

- (1) $x_n u_{T_n}$ and $y_n u_{T_n}$ lie in one fixed compact set;
- (2) neither endpoint lies in a prescribed small neighborhood of the finite-stage singular set;
- (3) the normalized displacement has a nontrivial limit;
- (4) both orbit blocks satisfy the required genericity estimates for a prescribed finite family of test functions.

PROOF. Choose a compact set K of measure $> 1 - \varepsilon$ with boundary measure zero. Genericity gives a set of times of lower density $> 1 - 2\varepsilon$ for which both orbit points lie in K , after using the common-generic-pair construction.

Finite-stage avoidance removes at most another ε proportion of times. The polynomial terminal-control lemma shows that the displacement remains on a comparable macroscopic scale throughout a fixed relative subinterval near the first exit time. Genericity for finitely many test functions removes another set of arbitrarily small relative measure. Taking ε small enough leaves a nonempty set of admissible times in that terminal interval.

Choose the comparison time from this intersection and relabel it as T_n . The relabeled time differs from the original first-exit time by at most the prescribed relative proportion η . Since η is fixed while $n \rightarrow \infty$, the terminal window still has length comparable to T_n and is long enough for the block-genericity estimates. Polynomial compactness then gives a nontrivial limiting displacement after passage to a subsequence. If an infinitesimal conclusion requires $\eta \downarrow 0$, that limit is taken only after the $n \rightarrow \infty$ comparison has been established at each fixed η . $\square$

Remark 7.12.2 (From transverse selection to full comparison). The proposition selects times at which the *transverse* normalized displacement is well controlled. If the normalization was performed modulo the full connected stabilizer, a time-dependent stabilizer component may still remain in the exact group displacement. Passing from this transverse selection to the full comparison data used by the measure argument is the technical H-property realization step, Theorem B.7.1.

Remark 7.12.3. The phrase “perturbing the first exit time” hides an important idea. The exact first exit time is chosen for normalization, not because it is dynamically special. Once polynomial control shows that a whole terminal interval has comparable displacement, one is free to select a nearby time with the recurrence and genericity properties needed by the measure argument.

13. A worked SL_3 avoidance calculation

Consider

$$u_t = I + tE_{12}$$

in $SL_3(\mathbb{R})$ and the standard action on $V = \mathbb{R}^3$. Let the bad configuration be the line

$$W = \mathbb{R}e_1.$$

For

$$v = ae_1 + be_2 + ce_3,$$

we have

$$u_tv = (a + tb)e_1 + be_2 + ce_3.$$

The distance to W is comparable to

$$\sqrt{b^2 + c^2},$$

which is constant. Thus this representation does not detect approach to W unless b and c are already small. This is a useful warning: the representation must be adapted to the subgroup datum.

Now take the adjoint representation and examine the line $\mathbb{R}E_{23}$. Since

$$\text{Ad}(u_{-t})E_{23} = E_{23} - tE_{13},$$

the transverse coordinate to $\mathbb{R}E_{23}$ is the linear polynomial $-t$. Therefore a normalized trajectory can remain δ -close to that line on at most $O(\delta)$ of a normalized interval unless the coefficient of E_{13} vanishes algebraically.

For a principal unipotent,

$$X = E_{12} + E_{23}, \quad u_t = e^{tX},$$

quadratic coordinates occur. A typical component has the form

$$p(t) = a + bt + ct^2.$$

Then the bad-time proportion is $O(\delta^{1/2})$. The exponent itself is not important for Ratner's qualitative theorem; what matters is that it is positive and depends only on a uniform degree bound.

14. What this chapter has and has not proved

The global argument rests on three distinct ingredients:

- (1) *polynomiality*, which follows directly from Ad-unipotence;
- (2) *quantitative sublevel control*, an elementary finite-degree fact;
- (3) *finite-stage singular bookkeeping on compact windows*, proved from Dani–Margulis support-class countability, fixed-support Chevalley discreteness, and compactness of Lie-subalgebra parameter spaces in Appendix A.

No measure-classification theorem has been used in the avoidance argument. This is essential: avoidance is one of the ingredients from which classification is built, so importing classification here would make the proof circular.

The conceptual output is the dichotomy

a generic comparison block escapes proper singular geometry,
or the problem reduces algebraically

This is the form needed by the rigidity induction.

CHAPTER 8

Additional Invariance: A Fully Written Measure Argument

The slogan “nearby generic points shear apart and therefore the measure has more invariance” is accurate but incomplete. The polynomial calculation by itself produces only a limiting displacement in the group. One must still prove that this displacement preserves the measure. This chapter writes that passage carefully, including the boundary errors, the choice of test functions, and the reason that a limiting curve often yields an entire new one-parameter direction rather than a single accidental symmetry.

1. The basic problem

Let μ be a U -invariant ergodic probability measure on

$$X = \Gamma \backslash G, \quad U = \{u_t : t \in \mathbb{R}\}.$$

Suppose x_n, y_n are μ -generic points with

$$y_n = x_n g_n, \quad g_n \rightarrow e.$$

For right actions,

$$y_n u_t = x_n u_t q_n(t), \quad q_n(t) = u_{-t} g_n u_t.$$

Assume that on a long terminal block I_n the displacement $q_n(t)$ is uniformly close to a fixed element h_n , with

$$h_n \rightarrow h \neq e.$$

The objective is to prove

$$h \in \text{Stab}_G(\mu).$$

The key point is that genericity compares time averages to μ , whereas closeness of the two orbit blocks compares the two time averages to each other. Combining the two comparisons forces μ to agree with its h -translate.

2. Uniform continuity on a compact window

Because X may be noncompact, a continuous compactly supported test function need not be uniformly continuous on all of X for an arbitrary compatible metric. The argument is therefore localized.

Let $f \in C_c(X)$ and let $C \subset X$ be compact. For h sufficiently close to a fixed compact subset of G , the set

$$C \cdot \overline{B_G(e, 1)}$$

is compact after shrinking the group ball. The action map is uniformly continuous on this compact product. Hence for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$x \in C, \quad d_G(g, h) < \delta \quad \implies \quad |f(xg) - f(xh)| < \varepsilon.$$

This is why the recurrence step places the two orbit blocks in a common compact window before one invokes continuity.

3. A block comparison lemma

Lemma 8.3.1 (Almost-constant displacement on a good block). *Let $f \in C_0(X)$. Suppose $I_n = [a_n, b_n]$ are intervals with $|I_n| \rightarrow \infty$, and suppose*

- (1) *$x_n u_t$ and $y_n u_t$ lie in a fixed compact set C for all t in a subset $E_n \subset I_n$ with*

$$|I_n \setminus E_n| = o(|I_n|);$$

- (2) *for $t \in E_n$,*

$$d_G(q_n(t), h_n) \rightarrow 0$$

uniformly in t ;

- (3) *$h_n \rightarrow h$;*

- (4) *the y_n -block is generic for f , while the x_n -block is generic for the fixed translate*

$$f_h(x) = f(xh) :$$

more precisely,

$$\frac{1}{|I_n|} \int_{I_n} f(y_n u_t) dt \rightarrow \int_X f d\mu$$

and

$$\frac{1}{|I_n|} \int_{I_n} f(x_n u_t h) dt \rightarrow \int_X f(xh) d\mu(x).$$

Then

$$\int_X f(xh) d\mu(x) = \int_X f(x) d\mu(x).$$

PROOF. On E_n we have

$$y_n u_t = x_n u_t q_n(t).$$

Uniform continuity on C gives

$$\sup_{t \in E_n} |f(y_n u_t) - f(x_n u_t h_n)| \rightarrow 0.$$

Therefore

$$\frac{1}{|I_n|} \left| \int_{E_n} f(y_n u_t) dt - \int_{E_n} f(x_n u_t h_n) dt \right| \rightarrow 0.$$

The complement $I_n \setminus E_n$ contributes at most

$$2\|f\|_\infty \frac{|I_n \setminus E_n|}{|I_n|} = o(1).$$

Hence

$$\frac{1}{|I_n|} \int_{I_n} f(y_n u_t) dt - \frac{1}{|I_n|} \int_{I_n} f(x_n u_t h_n) dt \rightarrow 0.$$

It remains to replace h_n by h . This replacement should be made on the same good set rather than by pretending that the entire orbit block lies in one compact set. On E_n we have $x_n u_t \in C$, and uniform continuity of the action on the compact set

$$C \times (\{h\} \cup \{h_n : n \geq n_0\})$$

gives

$$\sup_{t \in E_n} |f(x_n u_t h_n) - f(x_n u_t h)| \rightarrow 0.$$

On $I_n \setminus E_n$ the absolute difference is at most $2\|f\|_\infty$, whose normalized contribution is $o(1)$. Therefore

$$\frac{1}{|I_n|} \int_{I_n} (f(x_n u_t h_n) - f(x_n u_t h)) dt \rightarrow 0.$$

The two genericity assumptions now identify the remaining limits directly: the x_n -block translated by h converges to $\int f_h d\mu$, while the y_n -block converges to $\int f d\mu$. Comparing them proves the assertion. $\square$

Remark 8.3.2. The proof uses genericity for f and for the translated function f_h . This creates no uncountable bookkeeping problem in the curve-valued argument below. Choose once and for all a countable sup-norm dense family $\mathcal{F} \subset C_0(X)$. The H-property realization is only tested at a countable dense parameter set D ; hence the family

$$\mathcal{F} \cup \{f \circ R_{\phi(s)} : f \in \mathcal{F}, s \in D\}$$

is still countable and can be diagonalized simultaneously.

4. From equality on test functions to invariance of the measure

Let $R_h : X \rightarrow X$ be right translation by h . If

$$\int f \circ R_h d\mu = \int f d\mu$$

for every $f \in C_0(X)$, then the Riesz representation theorem gives

$$(R_h)_*\mu = \mu.$$

Because X is locally compact and second countable, $C_0(X)$ is separable. It is therefore enough to establish the comparison identity on one countable sup-norm dense family $\mathcal{F} \subset C_0(X)$ and extend by continuity.

This establishes the precise bridge

almost constant displacement of generic blocks $\implies$ new measure invariance

5. Why a terminal block is available

The displacement is rarely almost constant on an interval beginning at time zero. Instead, polynomial normalization produces a first macroscopic time T_n . Let

$$q_n(t) = \exp(P_n(t) + o(1))$$

in local coordinates, where P_n is a polynomial of bounded degree and

$$\|P_n(T_n)\| \asymp 1.$$

Fix a small $\theta > 0$. Taylor's theorem and bounded degree imply that on a relative window

$$I_n = [(1 - \theta)T_n, T_n]$$

the polynomial changes by $O_d(\theta)$ times its macroscopic size. More explicitly, after scaling $s = t/T_n$, the normalized polynomials form a bounded subset of a finite-dimensional polynomial space. Their derivatives are uniformly bounded on $[1 - \theta, 1]$. Therefore

$$\sup_{t \in I_n} \|P_n(t) - P_n(T_n)\| \leq C_d \theta.$$

Choose θ so that this is smaller than the continuity tolerance in the block-comparison lemma.

The first exit time is therefore used only to find the correct scale. The actual measure comparison takes place on a whole terminal block.

6. Selecting a good subblock

A terminal interval produced by polynomial normalization may contain times at which the orbit is far into a cusp or near a proper singular set. We combine four large subsets of the terminal block:

- (1) recurrence to a compact set;
- (2) avoidance of singular neighborhoods;
- (3) genericity for a finite family of test functions;
- (4) polynomial stability of the displacement.

Each removes only a small relative portion when parameters are chosen in the correct order. Their intersection therefore has positive relative measure.

A common mistake is to choose the continuity tolerance first and then attempt to force all dynamical estimates to fit it. The safe order is:

measure error $\rightarrow$ compact set $\rightarrow$ singular neighborhoods
 $\rightarrow$ polynomial terminal width $\rightarrow$ continuity tolerance

This order prevents circular dependence of constants.

7. Curve-valued limits

The strongest conclusion often comes not from one limiting element h but from a nonconstant polynomial curve

$$\phi : \mathbb{R} \rightarrow G$$

obtained as a limit of normalized conjugates. Suppose for each s in an interval J we can choose times $t_{n,s}$ on the normalized block such that

$$q_n(t_{n,s}) \rightarrow \phi(s)$$

and the block-comparison argument is uniform in s on compact subsets of J . Then

$$\phi(s) \in \text{Stab}_G(\mu) \quad (s \in J).$$

Since the stabilizer is closed,

$$\phi(J) \subset \text{Stab}_G(\mu).$$

If $\phi(0) = e$ and ϕ is differentiable at zero, then

$$\phi'(0) \in \text{Lie}(\text{Stab}_G(\mu)).$$

Thus a curve of stabilizing elements gives an infinitesimal stabilizing direction.

Proposition 8.7.1 (Infinitesimal additional invariance). *Let $L = \text{Stab}_G(\mu)^\circ$. Suppose a C^1 curve $\phi : (-\varepsilon, \varepsilon) \rightarrow \text{Stab}_G(\mu)$ satisfies*

$$\phi(0) = e, \quad \phi'(0) = Y \neq 0.$$

Then

$$Y \in \mathfrak{l} = \text{Lie}(L),$$

and therefore

$$\exp(tY) \subset L$$

for all $t \in \mathbb{R}$.

PROOF. The tangent space at the identity of the closed Lie subgroup $\text{Stab}_G(\mu)$ is its Lie algebra. Thus $Y \in \mathfrak{l}$. The one-parameter subgroup generated by Y lies in the connected Lie subgroup with Lie algebra $\mathfrak{l}$, namely L . $\square$

This is the cleanest mechanism for obtaining infinitesimal additional invariance. Relative to a provisional subgroup it enlarges known invariance; relative to $L = \text{Stab}_G(\mu)^\circ$ it yields a contradiction if the tangent is transverse.

8. When only one stabilizing element is produced

A single stabilizing element outside a provisional connected invariance subgroup need not enlarge the identity component; it may represent only a discrete symmetry. Near the identity, however, the situation is rigid. If $S < G$ is a closed subgroup, then its identity component S° is open in S . Consequently there cannot be a sequence

$$h_n \in S \setminus S^\circ, \quad h_n \rightarrow e.$$

Thus, when the final proof uses $L = \text{Stab}_G(\mu)^\circ$, a family of stabilizing elements tending to e is automatically eventually contained in L . The useful direction lemma below should therefore be read with L as a *provisional* closed subgroup of a larger closed invariance group H , not with $L = H^\circ$. In the final full-stabilizer argument, one instead differentiates a stabilizing curve and obtains a contradiction if its tangent is transverse to $\mathfrak{l}$.

We record the general direction device because it is useful during provisional enlargement arguments.

Lemma 8.8.1 (Small stabilizers force Lie-algebra enlargement). *Let $L < G$ be a closed subgroup. Suppose there is a sequence*

$$h_n \in H \setminus L, \quad h_n \rightarrow e,$$

where $H < G$ is another closed subgroup containing L . If the normalized logarithmic directions

$$\frac{\log h_n}{\|\log h_n\|}$$

converge to Y , then

$$Y \in \text{Lie}(H).$$

If moreover every sufficiently small element of L has logarithm in $\mathfrak{l}$, and $Y \notin \mathfrak{l}$, then

$$\dim H^\circ > \dim L^\circ.$$

PROOF. The direction lemma from the closed subgroup theorem applies to H : for each fixed t , choose integers m_n with

$$m_n \|\log h_n\| \rightarrow t.$$

Then

$$h_n^{m_n} \rightarrow \exp(tY).$$

Closedness gives $\exp(tY) \in H$ for all t , hence $Y \in \text{Lie}(H)$. If $Y \notin \mathfrak{l}$, the Lie algebra has enlarged. $\square$

In a provisional-invariance formulation one may take H to be the full measure stabilizer and L a smaller closed subgroup already known to preserve the measure. In the canonical formulation $L = H^\circ$, this particular lemma is not used to produce elements of $H \setminus L$ near the identity; the curve-differentiation argument above is the correct mechanism.

9. Modding out the old stabilizer

Suppose

$$L = \text{Stab}_G(\mu)^\circ$$

already contains U . A nearby displacement g_n may have a large component inside L and only a tiny transverse component. Conjugating the entire g_n can then obscure the transverse polynomial degree.

Choose a local smooth slice S through e transverse to L . Every sufficiently small g has a unique factorization

$$g = \ell(g)s(g), \quad \ell(g) \in L, \quad s(g) \in S.$$

Since right translation by $\ell(g)$ preserves μ , the L -component is dynamically irrelevant for measure comparison. The true perturbation is $s(g)$.

Infinitesimally, choose a vector-space complement

$$\mathfrak{g} = \mathfrak{l} \oplus \mathfrak{v}.$$

The quotient

$$\mathfrak{g}/\mathfrak{l}$$

records the transverse displacement. The operator induced by $\text{ad } X$ on this quotient is nilpotent. The degree of

$$e^{-t \text{ad } X} \overline{Z}$$

therefore determines the first macroscopic transverse scale.

This quotient formulation is not merely elegant. Without it, a high-degree polynomial term inside the old stabilizer can dominate the norm and cause one to normalize at the wrong time.

10. An exact $\mathfrak{sl}_2$ computation

Take

$$X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

For

$$Z = \alpha X + \beta H + \gamma Y,$$

we computed

$$e^{-t \operatorname{ad} X} Z = (\alpha + 2\beta t - \gamma t^2)X + (\beta - \gamma t)H + \gamma Y.$$

Suppose the old stabilizer is $U = \exp(\mathbb{R}X)$. Modulo $\mathbb{R}X$, the transverse displacement is

$$(\beta - \gamma t)\overline{H} + \gamma\overline{Y}.$$

The quadratic term has disappeared because it lies entirely in the old stabilizer. The correct transverse scale is therefore linear when $\gamma \neq 0$, not quadratic.

This is a model instance of why quotienting by the old stabilizer must precede polynomial normalization.

11. A higher-weight example

Let V be an irreducible $\mathfrak{sl}_2$ -module with basis

$$v_0, v_1, \dots, v_m$$

chosen so that

$$Xv_j = c_j v_{j-1}.$$

Then

$$e^{-tX} v_m = v_m - tc_m v_{m-1} + \dots + \frac{(-t)^m}{m!} c_m \cdots c_1 v_0.$$

If the old stabilizer absorbs the lowest weight directions

$$v_0, \dots, v_k,$$

then the effective transverse degree is not m but

$$m - k - 1$$

provided the next coefficient is nonzero. The quotient representation makes this automatic.

The important invariant is therefore:

maximal nilpotent degree visible in $\mathfrak{g}/\mathfrak{l}$.

12. The detailed additional-invariance theorem

The measure-theoretic conversion below starts from *full-group* comparison data. Producing those data from a polynomial detected only modulo L is a separate technical issue: a time-dependent L -correction cannot simply be discarded from an average. The general H-property realization theorem that supplies the corrected comparison blocks is isolated as Theorem B.7.1 in Appendix B. With that bridge stated explicitly, the remaining measure argument is elementary and fully written here.

THEOREM 8.12.1 (Additional invariance from quantified comparison data). *Let μ be a U -invariant ergodic probability measure on $X = \Gamma \backslash G$, and put*

$$L = \text{Stab}_G(\mu)^\circ.$$

Suppose there are an open interval J , a C^1 map $\phi : J \rightarrow G$, and a dense countable set $D \subset J$ such that, for every $s \in D$, one can find comparison blocks satisfying the geometric hypotheses (1)–(3) of Lemma 8.3.1 with reference displacements

$$h_{n,s} \longrightarrow \phi(s).$$

Choose a countable sup-norm dense family $\mathcal{F} \subset C_0(X)$. Assume that, for every $s \in D$ and $f \in \mathcal{F}$, the comparison blocks satisfy Lemma 8.3.1 for f ; in particular the y -block is generic for f and the x -block is generic for $f \circ R_{\phi(s)}$. Assume these countably many requirements are diagonalized along one subsequence.

Then

$$\phi(J) \subset \text{Stab}_G(\mu).$$

Consequently every translated tangent of ϕ belongs to $\mathfrak{l}$. In particular, if at some $s_0 \in J$ the translated derivative of ϕ has nonzero image in $\mathfrak{g}/\mathfrak{l}$, the assumed comparison data cannot exist.

PROOF. Fix $s \in D$. Apply Lemma 8.3.1 to every $f \in \mathcal{F}$ along the common diagonal subsequence. We obtain

$$\int f(x\phi(s)) d\mu(x) = \int f(x) d\mu(x) \quad (f \in \mathcal{F}).$$

Right translation acts isometrically on $C_0(X)$ in the sup norm. Since $\mathcal{F}$ is sup-norm dense, if $f_n \in \mathcal{F}$ converges uniformly to $f \in C_0(X)$, both sides change by at most $\|f_n - f\|_\infty$, so passing to the limit extends the identity to every $f \in C_0(X)$; no tightness argument is needed at this step. By the Riesz representation theorem,

$$\phi(s) \in \text{Stab}_G(\mu) \quad (s \in D).$$

The stabilizer is closed and D is dense, so continuity of ϕ gives

$$\phi(J) \subset \text{Stab}_G(\mu).$$

For any $s_0 \in J$, the curve

$$s \mapsto \phi(s_0)^{-1}\phi(s)$$

lies in the identity component of the stabilizer for s sufficiently close to s_0 . Differentiating at s_0 shows that the corresponding translated derivative lies in

$$\text{Lie}(\text{Stab}_G(\mu)^\circ) = \mathfrak{l}.$$

Thus its image in $\mathfrak{g}/\mathfrak{l}$ is zero. A nonzero transverse tangent is therefore contradictory. $\square$

Warning 8.12.2. The theorem is sometimes paraphrased as “a transverse generic point enlarges the stabilizer.” Three adjectives are essential:

- *generic*, so time averages identify the measure;
- *recurrent*, so continuity is used on a compact window;
- *nonsingular*, so the comparison does not merely reveal a smaller homogeneous support.

Dropping any one of them makes the statement false or useless.

13. A practical proof checklist

When using additional invariance in a concrete homogeneous space, the argument should answer the following questions explicitly.

- (1) What is the current connected stabilizer L ?
- (2) What local slice represents displacement modulo L ?
- (3) What is the nilpotent operator induced by $\text{ad } X$ on $\mathfrak{g}/\mathfrak{l}$?
- (4) What is its maximal active degree on the chosen perturbation?
- (5) What normalization time makes that transverse component macroscopic?
- (6) Which compact set supplies uniform continuity and recurrence?
- (7) Which singular configurations must be avoided?
- (8) Why is a nonempty set of terminal times left after all exclusions?
- (9) What limiting group element or curve is produced?
- (10) Why does the limit lie outside the old connected stabilizer?

A proof that leaves several of these items implicit is usually correct only after substantial reconstruction. The purpose of this volume is to keep that reconstruction visible.

14. Conceptual summary

Additional invariance has four layers:

nearby generic points

⇓

polynomial relative displacement modulo the old stabilizer

⇓

a recurrent, nonsingular terminal block
with almost constant displacement

⇓

equality of time averages

⇓

new invariance of the measure.

The polynomial calculation chooses the correct scale; recurrence makes analysis possible; avoidance prevents false algebraic explanations; genericity converts orbit comparison into a statement about the measure. None of the four can replace the others.

CHAPTER 9

Homogeneous Measures, Stabilizers, and the Rigidity Alternative

1. Invariant measure on one orbit

Let $H < G$ be a closed connected subgroup and $x = \Gamma g \in X$. The stabilizer of x in H is

$$H_x = H \cap g^{-1}\Gamma g.$$

The orbit map factors through

$$(9.1) \quad H_x \backslash H \longrightarrow xH, \quad H_x h \longmapsto xh.$$

Locally this is a diffeomorphism onto the immersed H -orbit.

Definition 9.1.1. A probability measure ν on X is *H -homogeneous* if there is $x \in X$ such that xH is a closed finite-volume orbit and ν is the normalized H -invariant measure on xH .

The phrase “finite-volume orbit” includes two assertions. The subgroup H_x is a lattice in H , and the orbit topology agrees with the quotient topology in (9.1), so Haar measure on $H_x \backslash H$ defines a Radon measure on the closed subset xH .

2. Uniqueness on a transitive finite-volume orbit

Proposition 9.2.1. *Let xH be a closed orbit carrying a finite nonzero H -invariant measure. After normalization to total mass one, this measure is unique.*

PROOF. Identify xH with $H_x \backslash H$. An H -invariant Radon measure on this homogeneous space is unique up to scale by the quotient form of Haar uniqueness. Finite total mass fixes the scale. $\square$

Thus once support on one finite-volume orbit has been established, there is no remaining measure-theoretic freedom.

3. The stabilizer of a homogeneous measure

If $\mu = m_{xH}$, then by H -invariance of Haar measure

$$H \subset S(\mu).$$

The inclusion may be strict. A finite-volume H -orbit can be preserved by additional normalizer elements, finite deck transformations, or centralizer translations. For a canonical connected subgroup one may therefore take

$$(9.2) \quad L = S(\mu)^\circ.$$

Then μ is L -invariant by definition.

Remark 9.3.1. The subgroup appearing in a statement of measure classification is not unique if one allows finite-index extensions or redundant compact factors. Using the connected measure stabilizer removes this ambiguity at the infinitesimal level.

4. Positive orbit mass and ergodicity

Let $L = S(\mu)^\circ$. Since $U \subset L$, every orbit xL is U -invariant:

$$(xL)U = xL.$$

Hence, whenever xL is measurable,

$$(9.3) \quad \mu(xL) \in \{0, 1\}$$

by U -ergodicity.

Lie-group orbits in a second-countable homogeneous space are analytic sets and therefore measurable for the completion of every Radon probability measure. Thus (9.3) can be used before closedness of the orbit has been established.

Lemma 9.4.1 (Local concentration implies global concentration). *Let $x \in \text{supp } \mu$ and suppose there is a neighborhood O of x such that*

$$(9.4) \quad \mu(O \setminus xL) = 0.$$

Then $\mu(xL) = 1$.

PROOF. Because $x \in \text{supp } \mu$, one has $\mu(O) > 0$. Equation (9.4) gives $\mu(xL) > 0$. The set xL is U -invariant, so ergodicity gives $\mu(xL) = 1$. $\square$

Thus if the measure is not concentrated on one L -orbit, every sufficiently small neighborhood of every generic support point contains positive mass genuinely transverse to that orbit.

5. From concentration to finite volume

Suppose

$$(9.5) \quad \mu(xL) = 1.$$

The measure is L -invariant. Corollary A.4.4 gives the precise structural conclusion: the immersed orbit xL is in fact closed, its stabilizer

$$L_x = L \cap g^{-1}\Gamma g$$

is a lattice in L , and

$$(9.6) \quad \mu = m_{xL}.$$

This is the easy branch of the classification argument once (9.5) is known. If (9.5) fails, the regular transverse branch would contradict the definition of the full connected stabilizer; the remaining possibility is concentration on a proper singular homogeneous configuration.

6. Transverse mass near a support point

Assume $\mu(xL) = 0$. Let $x \in X_{\text{gen}} \cap \text{supp } \mu$. Every neighborhood O of x has positive measure. Since the generic set has full measure and xL has measure zero,

$$O \cap X_{\text{gen}} \setminus xL$$

is nonempty. Hence there are generic points

$$(9.7) \quad y_n \rightarrow x, \quad y_n \notin xL.$$

Choose a small identity neighborhood $V < G$ on which

$$V \rightarrow X, \quad g \mapsto xg$$

is injective. Then for large n there are unique elements

$$(9.8) \quad g_n \rightarrow e, \quad y_n = xg_n, \quad g_n \notin L.$$

This is the precise source of the transverse sequence used in polynomial shearing. No geometric intuition about “nearby leaves” is needed; it follows from support, ergodicity, and local injectivity.

7. The first possible failure: singularity

The sequence g_n in (9.8) need not be usable. Its leading conjugacy polynomial can be trapped in a proper subgroup $H < G$ that already contains a conjugate of U . If the measure assigns positive mass to the corresponding singular stratum, the right response is not to force additional invariance in G . One should descend to the smaller homogeneous problem.

Thus the proof splits according to whether proper singular strata have positive measure.

8. The rigidity alternative

THEOREM 9.8.1 (Rigidity alternative). *Let μ be an ergodic U -invariant probability measure and let*

$$L = S(\mu)^\circ.$$

Then at least one of the following holds:

- (1) **homogeneous-orbit case:** *there exists $x \in \text{supp } \mu$ such that $\mu(xL) = 1$;*
- (2) **proper singular case:** *μ gives positive mass to a primitive singular stratum whose projective stabilizer is a proper closed subgroup of G .*

PROOF. Assume neither conclusion holds. Choose a generic recurrent point

$$x \in X_{\text{gen}} \cap \text{supp } \mu.$$

The hypothesis that the singular alternative fails is used through the compact-complexity avoidance theorem, not by removing an uncountable union of null strata pointwise. Since the first conclusion fails, Lemma 9.4.1 gives generic points $y_n \rightarrow x$ outside xL . Write

$$y_n = xg_n, \quad g_n \rightarrow e, \quad g_n \notin L.$$

Normalize the transverse conjugacy data by first macroscopic exit times. Because, by assumption, no proper singular stratum carries mass, the full H-property realization theorem, Theorem B.7.1, supplies corrected *full-group* comparison blocks and a C^1 limiting curve ϕ whose transverse tangent is nonzero in $\mathfrak{g}/\mathfrak{l}$. The correction clause is essential: quotient-level polynomial control by itself does not justify deleting a time-dependent L -factor from an orbit average.

Apply Theorem 8.12.1 to these controlled comparison data. It gives

$$\phi(J) \subset S(\mu),$$

so every translated tangent of ϕ belongs to $\mathfrak{l}$. This contradicts the nonzero transverse tangent supplied by the H-property realization theorem.

This is impossible because $L = S(\mu)^\circ$ already has Lie algebra equal to the full infinitesimal stabilizer of μ . Hence one of the two alternatives must hold. $\square$

Guiding Idea 9.8.2 (What the theorem excludes). There is no third type of ergodic invariant measure with genuinely transverse, nonhomogeneous support but no algebraic explanation. Polynomial shearing converts such transverse freedom into a new symmetry. If that conversion is obstructed, the obstruction itself is a proper homogeneous configuration.

9. Minimal singular subgroups

Suppose the second alternative holds. Among proper singular subgroups whose strata carry positive measure, choose one of minimal dimension. After passing to a primitive stratum, almost every point of the relevant component has its U -direction inside a conjugate of this subgroup and not inside a smaller admissible subgroup.

At a fixed compact-complexity stage, only finitely many primitive singular charts occur by Theorem A.5.56. Choose one of minimal dimension among the charts carrying positive mass. On that primitive chart the projective linearization parameter is U -invariant. Ergodicity therefore makes it almost surely constant: if Θ denotes the parameter and B is any Borel set in its standard Borel target, then $\Theta^{-1}(B)$ is U -invariant and hence has measure 0 or 1. The pushforward $\Theta_*\mu$ is a 0–1 probability measure, hence a Dirac mass by the standard-Borel argument of Lemma 7.10.1.

Thus the measure is concentrated on one projective fiber. Theorem A.6.1 turns that fiber into a genuine lower-dimensional homogeneous problem

$$(9.9) \quad \Delta \backslash H, \quad \Delta < H \text{ discrete,}$$

with

$$(9.10) \quad \dim H < \dim G.$$

No finite-covolume assertion for Δ is needed here. The induction hypothesis classifies the probability measure on this discrete quotient; finite volume of its eventual supporting orbit is then a conclusion of that classification. The strict dimension drop is the formal reason induction terminates.

10. Two ways to describe the regular branch

A common informal description says that transverse regular freedom “enlarges the stabilizer.” This is accurate only if one is tracking a provisional connected subgroup $K \subset S(\mu)^\circ$ whose invariance has already been proved. The additional-invariance mechanism can then enlarge K .

The final classification argument uses the canonical subgroup

$$L = S(\mu)^\circ.$$

With this choice there is no larger connected measure stabilizer waiting to be discovered. A nonsingular transverse comparison would produce a vector that is simultaneously in $\text{Lie } S(\mu) = \mathfrak{l}$ and nonzero in $\mathfrak{g}/\mathfrak{l}$, a contradiction. Hence the regular branch is eliminated outright. The only genuine recursive move is singular concentration, which reduces the ambient homogeneous group.

This distinction is important because a homogeneous measure may live on a proper closed orbit xH : the goal is not to enlarge invariance all the way to G , but to prove concentration on the correct homogeneous support.

11. Example: full Haar measure

Let $X = \Gamma \backslash G$ have Haar probability m_X and suppose the U -action is ergodic on m_X . Then

$$S(m_X)^\circ = G.$$

The homogeneous-orbit alternative holds with $xG = X$. There is no proper singular reduction.

12. Example: a proper semisimple orbit

Let

$$G = \mathrm{SL}_3(\mathbb{R}), \quad H = \left\{ \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix} : A \in \mathrm{SL}_2(\mathbb{R}) \right\},$$

and suppose xH is closed of finite volume. If $U < H$ acts ergodically on Haar probability of xH , that measure is an ergodic U -invariant measure on the full G -quotient. The classification must return the proper orbit xH , not full Haar measure on X .

From the ambient G perspective this is precisely the singular branch. After restricting to H , it becomes the homogeneous-orbit branch.

13. Example: periodic unipotent orbit

If xU itself is periodic, then normalized Haar measure on

$$U_x \backslash U$$

is already homogeneous with support xU . No stabilizer enlargement is necessary. This is the smallest possible output of the classification theorem.

14. The actual induction parameter

Once $L = S(\mu)^\circ$ is fixed from the outset, the final proof needs only induction on the effective ambient dimension. The regular branch gives a contradiction and therefore does not generate an iterative dimension-increase step. In the singular branch the measure is confined to a proper connected subgroup $H < G$, so

$$\dim H < \dim G.$$

This strict drop is the sole recursive complexity decrease used in the proof ledger.

One may still use the pair $(\dim G, -\dim K)$ while *developing* additional invariance from a provisional subgroup K , but that is a pedagogical book-keeping device for the local mechanism, not the induction parameter of the final classification theorem.

15. What remains

The rigidity alternative contains the dynamical heart of classification. With the canonical full stabilizer, the regular transverse branch is contradictory; singular concentration is the only recursive branch and strictly lowers ambient dimension.

CHAPTER 10

The Measure-Classification Induction: A Proof Ledger

Ratner's measure theorem is often remembered by its conclusion and by the shearing idea, while the induction that turns shearing into classification is recalled only vaguely. That is dangerous. The induction is where one must check that the problem really becomes smaller, that ergodicity survives restriction, that the unipotent flow still lies in the new acting group, and that no appeal to the desired theorem has been hidden in the definition of a singular set.

This chapter rewrites the classification argument as a proof ledger: at each branch we state exactly what has been gained, what complexity has decreased, and which theorem is being invoked.

1. The theorem and the complexity parameter

Let G be a connected Lie group, $\Gamma < G$ a lattice, and

$$U = \{u_t : t \in \mathbb{R}\}$$

a one-parameter Ad-unipotent subgroup. Let μ be a U -invariant ergodic Borel probability measure on

$$X = \Gamma \backslash G.$$

The objective is to show that there exist a closed connected subgroup $H < G$ and a point $x \in X$ such that

- (1) $U \subset H$;
- (2) xH is closed and carries finite H -invariant volume;
- (3) μ is the normalized H -invariant measure on xH .

The cleanest induction is simultaneous over ambient dimension. For each integer n , let $\mathcal{P}(n)$ be the assertion that the theorem holds whenever the effective connected ambient group has dimension at most n .

Remark 10.1.1 (Effective ambient group). If the action of G on X has a nontrivial kernel, one may quotient by the largest normal subgroup contained in every conjugate of Γ . This does not change the dynamics. Doing so prevents

the dimension parameter from counting invisible directions. The proof does not require this reduction, but it makes the bookkeeping conceptually cleaner.

2. Base cases

If the connected ambient group has dimension zero, the space is finite and every ergodic invariant probability measure is supported on one orbit.

In dimension one, a connected Lie group is isomorphic to $\mathbb{R}$ or S^1 . If U is nontrivial, then $U = G$, so an invariant ergodic probability measure is Haar measure on a closed G -orbit. If U is trivial, ergodicity for the trivial action forces the probability measure to be a Dirac mass, which is the homogeneous measure on a point orbit. Thus the induction starts without any rigidity input.

The point of spelling out the base case is not difficulty; it is to ensure that the induction does not secretly assume semisimplicity or noncompactness.

3. The connected measure stabilizer

Define

$$L = \text{Stab}_G(\mu)^\circ.$$

Since μ is U -invariant and U is connected,

$$U \subset L.$$

There are two conceptually different ways classification can finish.

- (1) The support of μ is one closed finite-volume L -orbit. Then L itself supplies the homogeneous measure.
- (2) The support is not one such orbit. Then the rigidity alternative applies. A regular transverse comparison would contradict the maximality built into $L = \text{Stab}_G(\mu)^\circ$; therefore any genuine recurrent branch must come from concentration on a proper singular homogeneous configuration.

The rigidity alternative establishes precisely this dichotomy.

4. Why positive mass of an orbit is decisive

Lemma 10.4.1 (Positive mass of an invariant orbit). *Let $L < G$ preserve μ and suppose $xL \subset X$ is measurable and $U \subset L$. If*

$$\mu(xL) > 0,$$

then $\mu(xL) = 1$ provided xL is U -invariant and μ is U -ergodic.

PROOF. Because $U \subset L$, right multiplication by U preserves xL . Thus xL is a U -invariant measurable set. Ergodicity forces its measure to be zero or one. The hypothesis excludes zero. $\square$

Once $\mu(xL) = 1$, invariance under L and uniqueness of finite invariant volume on a transitive homogeneous space identify μ , provided xL is closed and of finite volume.

Proposition 10.4.2 (Full-mass homogeneous orbit). *Suppose L is connected, preserves μ , and*

$$\mu(xL) = 1$$

for a closed orbit xL carrying an L -invariant Radon measure. Then that invariant measure is finite and, after normalization, equals μ .

PROOF. The restriction of μ to xL is an L -invariant probability measure. A transitive homogeneous space has an invariant Radon measure unique up to scalar whenever the modular criterion is satisfied. Since an invariant probability already exists, the criterion is automatically satisfied and the invariant volume is finite. Normalization gives μ . $\square$

This is the easy termination branch.

5. If there is no full orbit: producing transverse generic points

Assume no L -orbit has full μ -measure. We need nearby points that differ transversely to L .

Choose a local slice S to L near the identity. In a flow box around a typical point x , nearby points have the form

$$x\ell s, \quad \ell \in L, \quad s \in S.$$

If almost every sufficiently small neighborhood of x were contained, modulo a null set, in xL , then local L -saturation and ergodicity would force μ to live on one L -orbit. Since this is excluded, typical neighborhoods contain generic points with nontrivial transverse slice coordinate.

Lemma 10.5.1 (Transverse generic sequence). *Assume no L -orbit has full μ -measure. Then for μ -almost every generic point x there exist common generic points*

$$y_n = x_n g_n$$

with

$$x_n \rightarrow x, \quad y_n \rightarrow x, \quad g_n \rightarrow e,$$

such that the image of $\log g_n$ in

$$\mathfrak{g}/\mathfrak{l}$$

is nonzero.

PROOF. Let x be a generic point in $\text{supp } \mu$. Because $U \subset L$, the analytic set xL is U -invariant. By ergodicity its measure is 0 or 1; the assumption that no L -orbit has full measure therefore gives

$$\mu(xL) = 0.$$

Every neighborhood O of x has positive measure, and the common generic set has full measure, so

$$O \cap X_{\text{gen}} \setminus xL \neq \emptyset.$$

Choose y_n from such sets with $y_n \rightarrow x$.

Take an identity neighborhood $V \subset G$ on which

$$V \longrightarrow X, \quad g \longmapsto xg$$

is injective onto a neighborhood of x ; this is possible because the point stabilizer is discrete. For large n write uniquely

$$y_n = xg_n, \quad g_n \in V,$$

so $g_n \rightarrow e$. Since $y_n \notin xL$, one has $g_n \notin L$ in this local chart. Shrink V into an exponential neighborhood and write $g_n = \exp Z_n$. If $Z_n \in \mathfrak{l}$, then $g_n \in L$, a contradiction; hence the image of $Z_n = \log g_n$ in $\mathfrak{g}/\mathfrak{l}$ is nonzero.

Finally set $x_n = x$. Then x_n and y_n are common generic points and all stated convergences hold. $\square$

6. Apply the rigidity alternative

For the transverse generic sequence, polynomial shearing and the detailed additional-invariance theorem produce two possibilities.

Regular branch.: The comparison blocks avoid all proper singular configurations at the current stage. Then a nonzero transverse tangent direction enters

$$\text{Lie}(\text{Stab}_G(\mu)) = \mathfrak{l}.$$

By construction the same tangent direction has nonzero image in $\mathfrak{g}/\mathfrak{l}$. This is an immediate contradiction. No stabilizer-enlargement iteration occurs in the final proof because L was chosen to be the full connected stabilizer from the outset.

Singular branch.: A positive portion of the measure is trapped in a proper finite-volume homogeneous configuration associated with a subgroup $H < G$ of smaller effective dimension.

The regular branch is a contradiction mechanism. The singular branch is the induction mechanism.

7. The singular branch in detail

Suppose there is a proper connected subgroup $H < G$ and a measurable singular stratum S_H such that

$$\mu(S_H) > 0.$$

By construction, S_H is U -invariant. Ergodicity therefore gives

$$\mu(S_H) = 1.$$

Choose a minimal subgroup type among the singular data carrying mass and pass to its primitive part. The linearization datum defines a Borel parameter that is constant along U -orbits. By ergodicity, Lemma 7.10.1 makes this parameter almost surely constant. Its fiber is contained in a coset of a proper normalizer-type subgroup. The closed-orbit/lattice criterion then identifies a full-measure finite-volume homogeneous ambient space; after translating the base point we write it as

$$xH.$$

At this point there are several checks.

7.1. Check 1: the flow lies in the reduced group. If $x = \Gamma g$, the condition that xH be U -invariant is

$$U \subset H$$

when the orbit is written as xH . If the subgroup was first obtained in a conjugated form, replace H by the conjugate appropriate to the base point. This is a notational step, but omitting it is a common source of left/right errors.

7.2. Check 2: the reduced lattice. The stabilizer of $x = \Gamma g$ in H is

$$H_x = H \cap g^{-1}\Gamma g.$$

If xH is closed and has finite H -volume, then

$$H_x < H$$

is a lattice and

$$xH \cong H_x \backslash H.$$

Thus the reduced problem is again a homogeneous-space problem of the same type.

7.3. Check 3: Ad-unipotence survives restriction. Let $u_t = \exp(tX) \subset H$. Since

$$\mathrm{Ad}_G(u_t) = e^{t \mathrm{ad}_G X}$$

is unipotent, the restriction of $\mathrm{ad}_G X$ to the invariant subspace $\mathfrak{h}$ is nilpotent. Therefore

$$\mathrm{Ad}_H(u_t)$$

is unipotent. Hence the induction hypothesis applies to the same flow inside H .

7.4. Check 4: ergodicity survives restriction. The measure μ , now regarded as a probability measure on xH , is still U -ergodic. Ergodicity is a property of the U -action and the measurable sets, not of the ambient group. Restricting the ambient space to a full-measure U -invariant subset therefore does not change it.

7.5. Check 5: complexity decreases. Because H is a proper connected subgroup of the effective ambient group,

$$\dim H < \dim G.$$

This is the strict decrease needed for induction.

All five checks are essential. Once they are made, the induction hypothesis classifies μ as homogeneous under a connected subgroup of H , and hence of G .

8. The regular branch in detail

Suppose no proper singular stratum has positive measure. The quotient-level avoidance and normalization arguments, together with the full H -property realization theorem, provide corrected full-group terminal comparison blocks. The detailed additional-invariance theorem then gives a transverse vector

$$Y \in \mathrm{Lie}(\mathrm{Stab}_G(\mu)) \setminus \mathfrak{l},$$

contradicting the definition

$$\mathfrak{l} = \mathrm{Lie}(\mathrm{Stab}_G(\mu)^\circ).$$

What has actually happened? The assumption that no L -orbit has full measure produced transverse generic points. The absence of singular concentration turned those points into new stabilizer directions. But L already contained every connected stabilizer direction. Therefore the original assumption was impossible.

Hence, in the nonsingular branch, some L -orbit must have full measure. The easy termination argument then gives homogeneity.

This is the conceptual contradiction at the heart of the proof.

9. The provisional enlargement picture and the final proof

It is useful pedagogically to understand the local shearing mechanism by starting with a connected subgroup

$$K \subset S(\mu)^\circ$$

whose invariance has already been established. If a quasi-regular limit produces a differentiable stabilizing curve with tangent outside $\mathfrak{k}$, then the connected subgroup generated by K and that curve has strictly larger dimension. Repeating this *provisional* argument can happen at most $\dim G$ times.

This dimension observation explains why additional invariance is powerful, but it is *not* the recursive ledger used in the final proof. The final proof defines

$$L = S(\mu)^\circ$$

from the outset. Then there is no larger connected measure stabilizer to discover. A transverse stabilizing tangent is contradictory. Consequently the regular branch terminates immediately, while the singular branch passes to a proper connected subgroup $H < G$ and decreases the ambient dimension.

Keeping these two viewpoints separate is essential. The first is a local learning device; the second is the logically canonical classification proof.

10. Normalizer phenomena

A subtlety arises when the new limiting direction normalizes L rather than obviously lying outside it. Suppose

$$Y \in \mathfrak{n}_{\mathfrak{g}}(\mathfrak{l}) \setminus \mathfrak{l}.$$

Then the connected subgroup generated by L and $\exp(\mathbb{R}Y)$ has larger dimension. Relative to a provisional known invariance subgroup $K \subset S(\mu)^\circ$, this observation is the usual enlargement step. In the final proof, however, $L = S(\mu)^\circ$ is already the full connected stabilizer. Thus if $\exp(tY)$ preserves μ for all sufficiently small t , the conclusion is not an enlargement of L but a contradiction to $Y \notin \mathfrak{l}$.

But if one only knows that a discrete element

$$h \in N_G(L)$$

preserves μ , the identity component need not enlarge. This is why the small-radius or curve-valued version of additional invariance is used: it produces stabilizing elements arbitrarily close to the identity, from which a new Lie-algebra direction follows.

Warning 10.10.1. The quotient

$$N_G(L)/L$$

need not be finite. Finiteness can hold under additional algebraic hypotheses, but it is not a general Lie-group fact and is not used as one here.

11. Minimal subgroup types

In the singular branch, it is useful to choose a subgroup type minimal with respect to dimension among those carrying positive singular mass. This prevents the linearization from repeatedly rediscovering a smaller obstruction hidden inside the chosen one.

Suppose H is minimal and

$$\mu(S_H) > 0.$$

Then lower-dimensional singular pieces inside S_H have zero measure. On the complement of those pieces, local linearization relative to H is regular. This is exactly the setting in which the induction hypothesis on H and the additional-invariance argument cooperate cleanly.

Minimality is therefore not a decorative optimization. It isolates the first algebraic obstruction that genuinely carries mass.

12. The proof as an invariant diagram

The classification argument can now be summarized without suppressing any logical branch:

μ is U -ergodic and U -invariant

$\Downarrow$

$$L = \text{Stab}_G(\mu)^\circ \supset U$$

$\Downarrow$

Does one L -orbit have full mass?

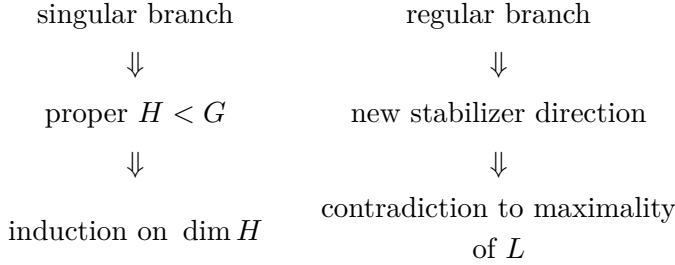
If yes,

$\mu = \text{Haar probability on that closed finite-volume orbit.}$

If no,

choose transverse common generic pairs.

Then



With the canonical full stabilizer, the regular branch cannot occur. Every genuine recursive step decreases ambient dimension, which is a nonnegative integer.

13. The classification theorem

Remark 10.13.1 (Logical status after the proof-gap revision). All load-bearing pieces of the induction are now proved inside the first three volumes. The complete H-property correction/reparametrization argument is Appendix C; compact singular bookkeeping and singular-fiber reduction are proved in Appendix A; the finite-dimensional linearization theorem is proved in Volume II. Quantitative nondivergence needed for the measure argument is proved in Volume II, while the orbitwise Lie-group form used only for pointwise distribution is proved in Appendix D. Thus none of these mechanisms is invoked as an unexplained black box in the theorem below.

THEOREM 10.13.2 (Ratner measure classification). *Let G be a connected Lie group, let $\Gamma < G$ be discrete, and let $U = \{u_t\}$ be a one-parameter Ad-unipotent subgroup. Every U -invariant ergodic probability measure μ on $\Gamma \backslash G$ is homogeneous: there exist a connected closed subgroup $H < G$ containing U and $x \in \Gamma \backslash G$ such that xH is closed, has finite H -volume, and*

$$\mu = m_{xH},$$

the normalized H -invariant measure.

PROOF. Proceed by induction on the effective connected ambient dimension. The base cases were discussed above. Notice that the induction does not require Γ to have finite covolume: the probability measure μ supplies recurrence and tightness on the part of the space relevant to the measure argument.

Let

$$L = \text{Stab}_G(\mu)^\circ.$$

If an L -orbit xL has full measure, Corollary A.4.4 proves—rather than assumes—that xL is a closed finite-volume homogeneous orbit, and Haar uniqueness identifies μ with its normalized L -invariant volume.

Assume no L -orbit has full measure. The transverse-generic-sequence lemma produces nearby common generic pairs with nonzero class in $\mathfrak{g}/\mathfrak{l}$. Apply first-exit polynomial normalization. Appendix C proves the complete H-property dichotomy: after the finite normalizer-correction algorithm, either corrected full-group comparison blocks yield a stabilizing C^1 curve with nonzero transverse tangent, or the pairs lie in a proper primitive singular linearization locus. The first alternative contradicts

$$\mathfrak{l} = \text{Lie}(\text{Stab}_G(\mu)^\circ)$$

by Theorem 8.12.1. Hence the singular alternative must occur on positive measure.

Use compact exhaustion together with the first m support types in the Dani–Margulis enumeration and increasing linearization-norm bounds in Theorem A.5.56 to reduce the singular bookkeeping to finite-stage charts. Choose a primitive singular datum of minimal dimension carrying positive measure. Its projective parameter is U -invariant, so ergodicity and Lemma 7.10.1 make that parameter almost surely constant. Theorem A.6.1 then identifies the problem, measurably, with a U -ergodic probability measure on

$$\Delta \backslash M,$$

where $M < G$ is proper connected and $\Delta < M$ is discrete. No finite-volume assertion about this intermediate quotient is required.

Ad-unipotence survives restriction to M , ergodicity is unchanged under the Borel orbit identification, and

$$\dim M < \dim G.$$

Therefore the induction hypothesis applies on $\Delta \backslash M$. It gives a connected subgroup $H < M$ and a finite H -invariant homogeneous support for the pulled-back measure. The finite-invariant-orbit criterion, Theorem A.4.3, proves that this orbit is closed. Pushing it forward to $\Gamma \backslash G$ yields the desired closed finite-volume homogeneous support for μ . $\square$

14. Support and the canonical connected support group

If the classification gives

$$\mu = m_{xH},$$

then Haar positivity on every nonempty open subset of the homogeneous orbit gives

$$\text{supp } \mu = xH.$$

Thus the support of every ergodic invariant probability measure is itself a closed finite-volume homogeneous orbit.

There is a useful canonical choice at the connected level. Let

$$S(\mu) = \{g \in G : (R_g)_*\mu = \mu\}, \quad L = S(\mu)^\circ.$$

If $\mu = m_{xH}$ with H connected, then $H \subset S(\mu)$, hence $H \subset L$ and

$$xH \subset xL.$$

On the other hand $S(\mu)$ preserves $\text{supp } \mu = xH$, so $xL \subset xH$. Therefore

$$xL = xH.$$

Thus L itself gives the canonical connected support orbit. The full stabilizer $S(\mu)$ may still be larger by a discrete extension; that discrete part becomes important in joining and correspondence problems, whereas L captures the connected support geometry.

15. Nonergodic invariant measures

If μ is U -invariant but not ergodic, apply ergodic decomposition:

$$\mu = \int \mu_\omega d\nu(\omega).$$

For ν -almost every ω , the component μ_ω is homogeneous by the theorem. Thus every invariant probability measure is a barycenter of homogeneous ergodic measures.

One should not simplify this to “every invariant measure is homogeneous.” Mixtures of two distinct periodic-orbit measures are invariant but not homogeneous in the usual transitive sense.

16. Proof dependency ledger

The classification proof now has the following complete dependency chain:

- (1) polynomial divergence and first-exit compactness;
- (2) generic-pair selection and recurrence;
- (3) the full H-property, including the time reparametrization and finite normalizer-correction algorithm (Appendix C);
- (4) finite-family linearization and polynomial sublevel estimates;
- (5) compact finite-complexity singular bookkeeping (Appendix A);
- (6) singular-fiber reduction to a smaller quotient by a discrete subgroup (Appendix A);
- (7) induction on ambient dimension;
- (8) the proved finite-invariant-orbit criterion, which supplies closedness and finite covolume at the end.

Deep general background such as the closed subgroup theorem, Birkhoff's theorem, and finite-dimensional Lie theory is developed in Volumes I–II. It is no longer the case that a central Ratner-specific step is represented only by an “external input” label.

17. Diagnostic questions for reading the proof

A reader who wants to verify that the classification argument has really been understood should be able to answer the following without appealing to the final theorem.

- (1) Why does absence of a full L -orbit give transverse generic points?
- (2) Why must polynomial normalization be done modulo $\mathfrak{l}$?
- (3) Where is recurrence used, as distinct from genericity?
- (4) Why does singular concentration decrease ambient dimension?
- (5) Why does the regular branch contradict the choice $L = S(\mu)^\circ$?
- (6) Why can neither branch continue indefinitely?
- (7) Why is the orbit produced in the singular branch a homogeneous space with a lattice?
- (8) Why does Ad-unipotence survive restriction to the smaller subgroup?
- (9) At what point is ergodicity essential?
- (10) Why is the proof not circular with the orbit-closure theorem?

The answers are exactly the ledger developed in this chapter. The theorem is difficult not because any one line is mysterious, but because all of these mechanisms must be made compatible simultaneously.

Part III

**Laboratories and Worked
Classification Problems**

CHAPTER 11

A Rank-One Laboratory: Horocycle Rigidity Revisited

1. Why revisit $\mathrm{PSL}_2(\mathbb{R})$

The general proof is easier to remember after seeing its alternatives in the smallest semisimple group. Let

$$G = \mathrm{PSL}_2(\mathbb{R}), \quad U = \left\{ u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \right\},$$

and let $X = \Gamma \backslash G$ with Γ a lattice.

The connected subgroups of G containing U are extremely limited. This makes the end of the classification visible without a large subgroup lattice.

2. Connected subgroups containing U

At the Lie-algebra level, let

$$X = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

A Lie subalgebra containing X is one of:

- (1) $\mathbb{R}X$;
- (2) the upper-triangular Borel algebra

$$\mathfrak{b} = \mathbb{R}X \oplus \mathbb{R}H;$$

- (3) all of $\mathfrak{sl}_2(\mathbb{R})$.

PROOF. Let $\mathfrak{l}$ contain X and another vector

$$Z = aX + bH + cY.$$

If $c = 0$ and $b \neq 0$, then $X, H \in \mathfrak{l}$, giving $\mathfrak{b}$. If $c \neq 0$, then

$$[X, Z] = 2bX + cH$$

shows $H \in \mathfrak{l}$ modulo X , and then

$$[H, Z]$$

recovers a nonzero Y component. Hence $X, H, Y \in \mathfrak{l}$ and $\mathfrak{l} = \mathfrak{sl}_2$. □

Thus the possible connected Lie subgroups containing U are U , the affine subgroup $B = AU$, and G . Which of these can occur as the connected support group of a finite homogeneous measure is a separate question.

3. Why the Borel subgroup cannot support finite Haar probability

The subgroup

$$B = AU$$

is nonunimodular. With

$$a_s = \begin{pmatrix} e^{s/2} & 0 \\ 0 & e^{-s/2} \end{pmatrix},$$

one has

$$a_s u_t a_{-s} = u_{e^s t}.$$

The modular function is therefore nontrivial:

$$\Delta_B(a_s) = e^s$$

up to the fixed convention used for right Haar translation.

A locally compact group admitting a lattice must be unimodular. Consequently B has no lattice and cannot possess a finite-volume homogeneous orbit of the form required by Ratner classification.

This leaves only U and G as support groups for ergodic homogeneous probability measures.

4. Classification once the general theorem is known

Corollary 11.4.1 (Horocycle invariant measures). *Every ergodic U -invariant probability measure on $\Gamma \backslash \mathrm{PSL}_2(\mathbb{R})$ is either:*

- (1) *normalized arclength on a periodic horocycle; or*
- (2) *Haar probability on the full quotient.*

PROOF. Measure classification gives a closed connected support subgroup H containing U . The subgroup list above gives $H = U, B$, or G . The Borel case is impossible by nonunimodularity. If $H = U$, finite volume means U_x is a lattice in $U \cong \mathbb{R}$, so the orbit is periodic. If $H = G$, the measure is Haar probability. $\square$

This familiar dichotomy is therefore a very special feature of the small subgroup lattice of PSL_2 .

5. How the local shearing argument sees the dichotomy

Suppose one starts with an ergodic U -invariant measure μ that is not periodic. If its connected stabilizer were only U , generic support points would have nearby transverse points.

A general small matrix can be written to first order as

$$\exp(\alpha X + \beta H + \gamma Y).$$

Conjugation gives

$$(11.1) \quad e^{-t \operatorname{ad} X}(\alpha X + \beta H + \gamma Y) = (\alpha + 2\beta t - \gamma t^2)X + (\beta - \gamma t)H + \gamma Y.$$

The quadratic coefficient lies in the already known X -direction, but the accompanying linear H term is generally nonzero on the same scale before exact cancellation. Depending on the relation among α, β, γ , normalization can produce an H -component or, after a different slice choice, an opposite-unipotent component.

If a provisional invariance calculation beginning with U produces the H -direction, then the connected subgroup generated by the known directions contains B . This does not mean that the full measure stabilizer literally “grows” during the proof; it means that the hypothesis that the full connected stabilizer was only U cannot be correct. The subgroup classification above and the impossibility of a finite-volume B support orbit explain why the eventual homogeneous support in the nonperiodic case must be the full G -orbit.

6. A careful warning about one perturbation

Taking only

$$g_n = v_{\varepsilon_n}$$

with opposite-unipotent perturbation yields the normalized limit

$$u_{-sT_n} v_{\varepsilon_n} u_{sT_n} \rightarrow u_{-\sigma s^2},$$

which lies inside U and by itself gives no new symmetry. This does not contradict rigidity. The proof is not allowed to choose an arbitrary convenient transverse sequence and demand enlargement from it. It chooses generic pairs outside singular configurations and uses the full polynomial decomposition modulo the current stabilizer.

This is an excellent diagnostic for oversimplified accounts of the H-property.

7. Periodic horocycles in the modular quotient

For $\Gamma = \mathrm{PSL}_2(\mathbb{Z})$, the standard point $x_0 = \Gamma e$ satisfies

$$x_0 u_1 = x_0$$

because $u_1 \in \Gamma$. Hence $x_0 U$ is periodic of period one.

Translate the base point by the diagonal flow,

$$x_s = x_0 a_s.$$

The stabilizer condition for the right U -orbit is

$$x_s u_t = x_s \iff a_s u_t a_{-s} \in \Gamma.$$

Since

$$a_s u_t a_{-s} = u_{e^{st}},$$

and u_1 generates $U \cap \Gamma$ at the standard cusp, the least positive period of $x_s U$ is

$$t_*(s) = e^{-s}.$$

Thus the closed horocycles in the cusp have periods shrinking exponentially as the height increases. This is the homogeneous-dynamics version of the Euclidean fact that horizontal circles around a cusp have hyperbolic length shrinking with height.

8. Nonescape for a fixed nonperiodic horocycle

On a finite-area quotient, a nonperiodic horocycle may enter the cusp but cannot spend asymptotically all of its time there. Quantitative nondivergence gives, for every $\varepsilon > 0$, a compact K with

$$\liminf_{T \rightarrow \infty} \frac{1}{T} |\{0 \leq t \leq T : x u_t \in K\}| > 1 - \varepsilon.$$

For $\mathrm{PSL}_2(\mathbb{Z})$ this can also be interpreted in lattice language. A point of $\mathrm{SL}_2(\mathbb{R})/\mathrm{SL}_2(\mathbb{Z})$ corresponds to a unimodular lattice. Entering the cusp means acquiring an unusually short nonzero vector. Under u_t , each lattice vector changes linearly in t , so the times when a fixed vector is very short form a controlled interval; geometry-of-numbers bookkeeping prevents too many different vectors from causing simultaneous long excursions.

9. Orbit closure in rank one

The subgroup classification yields:

Corollary 11.9.1 (Horocycle orbit closures). *Every horocycle orbit in a finite-volume quotient of $\mathrm{PSL}_2(\mathbb{R})$ is either periodic or dense in the full homogeneous component.*

PROOF. Apply the Ratner orbit-closure theorem and the classification of connected closed subgroups of $\mathrm{PSL}_2(\mathbb{R})$ containing the horocycle subgroup. A proper finite-volume homogeneous envelope is the horocycle orbit itself; otherwise the envelope is the full homogeneous component. $\square$

In a connected full quotient this is the familiar periodic-or-dense dichotomy. The general Ratner orbit-closure theorem is more complicated only because higher-dimensional groups possess many proper intermediate subgroups containing U .

10. Pointwise equidistribution

If the horocycle is periodic, time averages converge to normalized arclength on the circle. If it is nonperiodic, they converge to Haar probability on the full quotient:

$$\frac{1}{T} \int_0^T f(xu_t) dt \rightarrow \int_X f dm_X.$$

Thus every point is generic for one of exactly two algebraic measures.

11. Relation with the geodesic flow

The geodesic flow normalizes the horocycle flow:

$$a_s u_t a_{-s} = u_{e^{st}}.$$

Consequently it sends U -invariant measures to U -invariant measures. If μ is an ergodic horocycle measure, so is $(R_{a_s})_* \mu$.

For Haar measure nothing changes. For a periodic horocycle measure, the geodesic flow moves it to a different periodic horocycle measure with rescaled period. The classification is therefore compatible with the full A -renormalization dynamics.

12. Why B -invariance is too much

Suppose hypothetically that a probability measure were invariant under both U and the entire diagonal group A , hence under $B = AU$. If it were supported on one transitive B -orbit, that orbit would require a lattice in the nonunimodular group B , impossible.

More generally, B -invariance combined with recurrence and the Mautner phenomenon tends to force invariance under the opposite unipotent direction, and hence under all of G . The absence of finite B -volume is the group-theoretic shadow of this phenomenon.

13. The rank-one lesson

The PSL_2 case teaches three durable lessons:

- (1) classify connected subgroups containing the unipotent flow;
- (2) eliminate subgroups that cannot carry finite invariant volume;
- (3) use shearing to show that a nonclosed-support measure cannot remain with too small a stabilizer.

In higher rank the first step becomes much more complicated, but the second and third retain the same conceptual form.

CHAPTER 12

Higher-Rank Worked Examples: SL_3 , Affine Lattices, and Semidirect Products

1. Why worked examples are indispensable here

In the abstract proof, phrases such as “proper singular subgroup,” “leading polynomial direction,” and “homogeneous envelope” can begin to sound formal. This chapter works through several examples in which all of these objects can be written down concretely.

The aim is not to classify every possible intermediate subgroup in the examples. It is to show how one discovers the relevant subgroups and how the rigidity mechanism decides between them.

2. A basic one-parameter subgroup of SL_3

Let

$$G = \mathrm{SL}_3(\mathbb{R}), \quad \Gamma = \mathrm{SL}_3(\mathbb{Z}), \quad X = \Gamma \backslash G,$$

and take

$$(12.1) \quad U = \{u_t = I + tE_{12} : t \in \mathbb{R}\}.$$

Since $E_{12}^2 = 0$,

$$u_t = \exp(tE_{12}).$$

The adjoint action on a matrix unit satisfies

$$(12.2) \quad [E_{ij}, E_{kl}] = \delta_{jk}E_{il} - \delta_{li}E_{kj}.$$

This formula lets us compute all shearing degrees by hand.

3. Adjoint strings for E_{12}

We list several directions.

The opposite root E_{21} . As already seen,

$$(12.3) \quad e^{-t \operatorname{ad} E_{12}} E_{21} = E_{21} - t(E_{11} - E_{22}) - t^2 E_{12}.$$

The degree is two.

The root E_{23} . Since

$$[E_{12}, E_{23}] = E_{13}, \quad [E_{12}, E_{13}] = 0,$$

we get

$$(12.4) \quad e^{-t \operatorname{ad} E_{12}} E_{23} = E_{23} - t E_{13}.$$

The degree is one.

The root E_{31} . Since

$$[E_{12}, E_{31}] = -E_{32}, \quad [E_{12}, E_{32}] = 0,$$

we get

$$(12.5) \quad e^{-t \operatorname{ad} E_{12}} E_{31} = E_{31} + t E_{32}.$$

Again the degree is one.

The commuting direction E_{13} . Here

$$[E_{12}, E_{13}] = 0,$$

so

$$(12.6) \quad e^{-t \operatorname{ad} E_{12}} E_{13} = E_{13}.$$

This is a centralizer direction for the chosen U .

4. Interpreting the strings

A nearby displacement with an E_{23} component creates an E_{13} shear. A displacement with an E_{21} component creates a quadratic E_{12} shear together with a linear diagonal displacement. A displacement already in E_{13} does not shear at all because it commutes with U .

Thus several different kinds of additional invariance can appear:

- centralizer invariance in the E_{13} direction;
- diagonal/normalizer invariance from the $E_{11} - E_{22}$ term;
- enlargement to a larger unipotent subgroup generated by E_{12} and E_{13} .

The actual output depends on the current stabilizer L : directions already lying in $\mathfrak{l}$ are quotiented out before deciding what is new.

5. The upper-unitriangular subgroup

Let

$$(12.7) \quad N = \left\{ \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, z \in \mathbb{R} \right\}.$$

This is the three-dimensional Heisenberg group. Its Lie algebra is spanned by

$$E_{12}, E_{23}, E_{13},$$

with

$$[E_{12}, E_{23}] = E_{13}$$

and E_{13} central.

The subgroup

$$N \cap \mathrm{SL}_3(\mathbb{Z})$$

is a lattice in N . Therefore the orbit

$$\Gamma N$$

is a compact nilmanifold.

Take $U = \exp(\mathbb{R}E_{12})$. Its action on this nilmanifold is not automatically ergodic: the projection to the abelianization

$$N/[N, N] \cong \mathbb{R}^2$$

may be periodic in one coordinate. This is a useful warning. The existence of a closed finite-volume N -orbit containing U does not imply that U is dense or ergodic in it. The minimal homogeneous envelope for a given point can be a proper subgroup of N .

6. A two-dimensional unipotent envelope

The commuting subgroup

$$(12.8) \quad W = \exp(\mathbb{R}E_{12}) \exp(\mathbb{R}E_{13}) \cong \mathbb{R}^2$$

has

$$W \cap \Gamma \cong \mathbb{Z}^2.$$

Hence ΓW is a two-torus. The U -orbit inside it is periodic in the E_{12} direction and does not fill the torus.

If instead one takes a one-parameter subgroup

$$U_\alpha = \{\exp(t(E_{12} + \alpha E_{13})) : t \in \mathbb{R}\},$$

then on the torus ΓW the orbit is dense precisely when 1 and α are rationally independent relative to the lattice coordinates. This is an abelian homogeneous version of the orbit-closure theorem.

It also illustrates why the smallest homogeneous envelope depends on the exact one-parameter subgroup, not only on the ambient unipotent radical.

7. An embedded SL_2 singular subgroup

Let

$$(12.9) \quad H_{12} = \left\{ \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix} : A \in \mathrm{SL}_2(\mathbb{R}) \right\}.$$

Then $U \subset H_{12}$ and

$$H_{12} \cap \Gamma \cong \mathrm{SL}_2(\mathbb{Z})$$

is a lattice. Thus

$$\Gamma H_{12}$$

is a closed finite-volume copy of the modular homogeneous space.

Any U -invariant ergodic measure supported there is, from the viewpoint of the full SL_3 quotient, singular. If it is nonperiodic, rank-one measure rigidity identifies it with Haar measure on this H_{12} -orbit.

This is a concrete realization of the induction step:

$$\mathrm{SL}_3 \text{ problem} \longrightarrow \mathrm{SL}_2 \text{ problem}.$$

8. A parabolic subgroup and why finite volume matters

Consider the maximal parabolic

$$(12.10) \quad P = \left\{ \begin{pmatrix} A & v \\ 0 & c \end{pmatrix} : \det(A)c = 1 \right\}.$$

It contains many unipotent subgroups, including U . But P is not unimodular in general because its Levi factor expands the unipotent radical.

Since a group admitting a lattice must be unimodular, P cannot itself occur as the support group of a finite homogeneous probability measure unless one restricts to a unimodular subgroup inside it.

This simple modular-function test eliminates many apparent candidates before any dynamics is used.

9. A diagonal normalizer computation

Let

$$a_s = \mathrm{diag}(e^s, e^{-s}, 1).$$

Then

$$(12.11) \quad a_s u_t a_{-s} = u_{e^{2s}t}.$$

Thus $A = \{a_s\}$ normalizes U but does not centralize it.

If a measure is invariant under both U and A , the invariance group contains the semidirect product AU . Because that group is nonunimodular,

a transitive finite-volume AU -orbit is impossible. In a rigidity argument, A -invariance is therefore usually an intermediate stage that forces still larger invariance rather than a final homogeneous support.

10. A centralizer computation

The matrix E_{13} commutes with E_{12} , so

$$C = \{\exp(sE_{13}) : s \in \mathbb{R}\} \subset C_G(U).$$

If additional invariance produces C , then the measure is invariant under the two-dimensional abelian group

$$UC.$$

Unlike AU , this group is unimodular and can have finite-volume orbits. Whether UC is the final support group depends on its intersection with the conjugated lattice and on ergodicity of U inside the resulting torus/cylinder.

Thus normalizer and centralizer directions play genuinely different roles.

11. Affine lattices

Let

$$G = \mathrm{ASL}_d(\mathbb{R}) = \mathrm{SL}_d(\mathbb{R}) \ltimes \mathbb{R}^d, \quad \Gamma = \mathrm{ASL}_d(\mathbb{Z}) = \mathrm{SL}_d(\mathbb{Z}) \ltimes \mathbb{Z}^d.$$

A point of

$$\Gamma \backslash G$$

can be viewed as an affine unimodular lattice

$$\Lambda + v.$$

An element is written (g, w) with multiplication

$$(12.12) \quad (g, w)(h, z) = (gh, w + gz).$$

The Lie algebra is

$$\mathfrak{sl}_d \ltimes \mathbb{R}^d$$

with bracket

$$(12.13) \quad [(A, v), (B, w)] = ([A, B], Aw - Bv).$$

12. A unipotent shear acting on translations

Take $d = 2$ and

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Let

$$U = \{(u_t, 0)\}.$$

For a pure translation (I, v) ,

$$(12.14) \quad (u_{-t}, 0)(I, v)(u_t, 0) = (I, u_{-t}v).$$

If

$$v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix},$$

then

$$(12.15) \quad u_{-t}v = \begin{pmatrix} v_1 - tv_2 \\ v_2 \end{pmatrix}.$$

A small vertical translation therefore shears linearly into the horizontal direction.

This is a semidirect-product analogue of the matrix conjugation calculations in semisimple groups. Polynomial shearing is not restricted to semisimple Lie groups.

13. Closed affine-lattice subspaces

The subgroup of pure translations

$$V = \{(I, v) : v \in \mathbb{R}^d\}$$

has

$$V \cap \Gamma = \mathbb{Z}^d,$$

so its orbit through the standard affine lattice is a torus

$$\mathbb{Z}^d \backslash \mathbb{R}^d.$$

Unipotent linear parts act on this torus by shears. Depending on the translation vector and the acting subgroup, orbit closures can be proper subtori or the full affine fiber.

Ratner's theorem says that even after coupling the linear lattice shape and the affine translation, orbit closures under Ad-unipotent actions remain homogeneous.

14. A solvable model

Consider the three-dimensional Heisenberg group

$$H_3 = \left\{ \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} \right\}$$

with integer lattice. Let

$$U = \{x = t, y = z = 0\}.$$

Right translation gives

$$(x, y, z) \cdot (t, 0, 0) = (x + t, y, z)$$

in one coordinate convention, while choosing a different one-parameter subgroup introduces the central cocycle term xy . In either case the group law is polynomial, and orbit closures reduce to rational-subgroup questions in the nilmanifold.

This solvable example is much more elementary than the semisimple case, but it exhibits the same conclusion: invariant ergodic measures are Haar measures on closed subnilmanifolds.

15. How to analyze a new example

For a new matrix homogeneous space, a reliable workflow is:

- (1) identify the nilpotent generator N and compute $\text{ad } N$ on a useful basis;
- (2) list the natural candidate closed subgroups containing U and test which are unimodular;
- (3) compute intersections with the lattice to identify which candidate orbits actually have finite invariant volume;
- (4) determine centralizer and normalizer directions;
- (5) inspect polynomial shearing of transverse root spaces;
- (6) use irrationality/arithmetic information to rule out proper homogeneous envelopes.

This is how the abstract rigidity theorem becomes a practical tool rather than merely a classification statement.

16. A miniature subgroup decision tree in SL_3

For the U in (12.1), suppose an orbit is known to lie in a closed finite-volume homogeneous orbit xH . One can ask:

- (1) Is $H = U$? Then the orbit is periodic.

- (2) Does H lie in the upper-left SL_2 ? Then rank-one dynamics controls the orbit.
- (3) Does H contain E_{13} or E_{23} , producing a larger unipotent radical?
- (4) Does H contain diagonal normalizer directions? If so, is it still unimodular?
- (5) Are there enough root directions and commutators to generate all of SL_3 ?

Commutator closure often makes the last step automatic. For instance, if a Lie algebra contains

$$E_{12}, E_{23}, E_{21}, E_{32},$$

then it contains the corresponding diagonal elements and E_{13}, E_{31} , hence all of $\mathfrak{sl}_3$.

This illustrates a recurring feature of rigidity arguments: one new transverse direction can trigger a cascade of Lie brackets that enlarges the symmetry much more than the first shearing calculation suggests.

CHAPTER 13

Worked Classification Problems

A rigidity theorem becomes easier to use once one has practiced the intermediate question that the theorem itself does not answer automatically: *which homogeneous subgroups can actually occur?* Ratner's theorem says that an ergodic unipotent-invariant probability measure is homogeneous. It does not by itself classify the closed finite-volume subgroups containing the chosen unipotent subgroup.

This chapter works through several examples in enough detail to show how Lie algebra, unimodularity, lattice compatibility, and normalizers interact. The examples are deliberately varied: rank one, higher rank, products, and affine semidirect products.

1. A general workflow

Suppose

$$X = \Gamma \backslash G, \quad U < G$$

is generated by unipotent elements. To classify possible ergodic U -invariant probability measures, proceed in the following order.

- (1) Classify connected Lie subgroups $H < G$ with $U \subset H$, at least up to the range relevant to the problem.
- (2) Discard H that cannot support finite invariant volume. A necessary condition is unimodularity.
- (3) For the remaining H , determine when an orbit xH is closed and of finite volume. Once closedness (equivalently, properness of the homogeneous orbit map) is established, the finite-volume condition is exactly that

$$H \cap g^{-1}\Gamma g$$

be a lattice in H for $x = \Gamma g$. The lattice condition is therefore not used here as a substitute for checking the global orbit topology.

- (4) Apply measure classification: every ergodic U -invariant probability measure is Haar probability on one of those orbits.
- (5) If one wants orbit closures or equidistribution, identify the smallest admissible homogeneous envelope of the initial point.

This separation is useful. Ratner theory supplies the rigidity; subgroup arithmetic identifies the actual list of possibilities.

2. The horocycle flow on a compact surface

Let

$$G = \mathrm{SL}_2(\mathbb{R}), \quad U = \left\{ \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} : t \in \mathbb{R} \right\}.$$

Let $\Gamma < G$ be a cocompact lattice.

At the Lie algebra level, write

$$X = E_{12}, \quad H_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad Y = E_{21}.$$

A Lie subalgebra containing X can be

$$\mathbb{R}X, \quad \mathbb{R}X \oplus \mathbb{R}H_0, \quad \mathfrak{sl}_2(\mathbb{R}).$$

Indeed, if it contains any vector with nonzero Y -component, repeated brackets with X generate H_0 and then Y , giving all of $\mathfrak{sl}_2$. If there is no Y -component but there is an H_0 -component, it is the upper triangular algebra.

The connected upper triangular group

$$B = AN$$

is nonunimodular. With

$$a_s = \begin{pmatrix} e^{s/2} & 0 \\ 0 & e^{-s/2} \end{pmatrix},$$

one has

$$a_s u_t a_{-s} = u_{e^s t}.$$

Hence the adjoint action of a_s on $\mathrm{Lie}(U)$ has determinant e^s , so the modular character of B is nontrivial. Therefore B cannot contain a lattice and cannot support a finite invariant measure on a closed B -orbit.

Thus only two homogeneous measure types remain:

- (1) Haar measure on a closed U -orbit;
- (2) Haar measure on the whole space $\Gamma \backslash G$.

For a cocompact lattice, closed U -orbits do not occur. One way to see this geometrically is that a closed horocycle corresponds to a cusp, and compact hyperbolic surfaces have no cusps. Consequently Haar measure is the only ergodic U -invariant probability measure.

This is the measure-theoretic form of unique ergodicity of the horocycle flow on a compact hyperbolic surface.

3. The modular surface: periodic horocycles reappear

Let

$$\Gamma = \mathrm{SL}_2(\mathbb{Z}).$$

Now $\Gamma \backslash G$ is noncompact of finite volume. The subgroup

$$U \cap \Gamma = \left\{ \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} : n \in \mathbb{Z} \right\}$$

is a lattice in $U \cong \mathbb{R}$. Hence

$$\Gamma U$$

is a closed periodic U -orbit.

Conjugating by $g \in G$, a point $x = \Gamma g$ has periodic U -orbit exactly when

$$gUg^{-1} \cap \Gamma$$

is a lattice in gUg^{-1} . Geometrically, these are horocycles based at rational boundary points.

The homogeneous classification therefore gives exactly the expected dichotomy:

ergodic U -invariant probability = periodic horocycle measure or Haar measure
--

Notice that the noncompactness does not create exotic invariant probability measures. It creates new closed homogeneous orbits.

4. A root unipotent in $\mathrm{SL}_3(\mathbb{R})$

Let

$$G = \mathrm{SL}_3(\mathbb{R}), \quad U = \{I + tE_{12} : t \in \mathbb{R}\}.$$

The subgroup lattice is much richer than in rank one.

The simplest connected subgroups containing U include:

- (1) U itself;
- (2) the two-dimensional abelian group

$$U_{12,13} = \{I + sE_{12} + tE_{13} : s, t \in \mathbb{R}\};$$

- (3) the Heisenberg upper-unitriangular group

$$N = \left\{ \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, z \in \mathbb{R} \right\};$$

- (4) the embedded copy of $\mathrm{SL}_2(\mathbb{R})$ in the upper-left block;
- (5) semidirect products of these groups with diagonal directions;
- (6) parabolic subgroups;

(7) all of $\mathrm{SL}_3(\mathbb{R})$.

Ratner classification alone does not select among these. Arithmetic of the lattice and unimodularity do.

4.1. The upper-unitriangular group. For $\Gamma = \mathrm{SL}_3(\mathbb{Z})$,

$$N \cap \Gamma$$

is the integer Heisenberg group, a lattice in N . Hence

$$\Gamma N$$

is a closed compact nilmanifold. Haar probability on this orbit is U -invariant and ergodic or nonergodic depending on the position of U inside the nilmanifold; for the standard E_{12} flow, the orbit can have a nontrivial factor and ergodicity should be checked rather than assumed.

This example shows that in higher rank, a unipotent flow can live naturally inside a proper nilpotent homogeneous subspace of dimension greater than one.

4.2. The embedded SL_2 . Let

$$H = \left\{ \begin{pmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & 1 \end{pmatrix} : ad - bc = 1 \right\}.$$

Then

$$H \cap \mathrm{SL}_3(\mathbb{Z}) \cong \mathrm{SL}_2(\mathbb{Z})$$

is a lattice in H . Thus

$$\Gamma H$$

is a closed finite-volume H -orbit. Haar measure on it is an ergodic candidate for the E_{12} -flow, and its internal dynamics reduce to the familiar horocycle flow on

$$\mathrm{SL}_2(\mathbb{Z}) \backslash \mathrm{SL}_2(\mathbb{R}).$$

The same ambient U therefore has both nilpotent and semisimple homogeneous envelopes.

5. Unimodularity as a fast exclusion test

Consider the standard maximal parabolic

$$P = \left\{ \begin{pmatrix} A & v \\ 0 & (\det A)^{-1} \end{pmatrix} : A \in \mathrm{GL}_2(\mathbb{R}), v \in \mathbb{R}^2 \right\} \cap \mathrm{SL}_3(\mathbb{R}).$$

Its unipotent radical is $\mathbb{R}^2$. A diagonal element in the Levi factor acts on that radical with nontrivial determinant. Consequently P is nonunimodular.

Therefore no discrete subgroup can be a lattice in P . In particular, there is no finite P -invariant probability measure on a closed P -orbit in a finite-volume quotient of G .

This simple modular calculation can eliminate a large family of candidate subgroups before any delicate arithmetic is attempted.

Warning 13.5.1. A subgroup of a unimodular group need not be unimodular. The ambient group $\mathrm{SL}_3(\mathbb{R})$ is unimodular, while its parabolic subgroups are generally not. Finite-volume orbit questions must use the modular function of the subgroup itself.

6. Principal unipotent flow in SL_3

Let

$$X = E_{12} + E_{23}, \quad U = \{e^{tX} : t \in \mathbb{R}\}.$$

Since

$$X^2 = E_{13}, \quad X^3 = 0,$$

we have

$$e^{tX} = I + t(E_{12} + E_{23}) + \frac{t^2}{2}E_{13}.$$

This is a regular, or principal, unipotent subgroup.

Inside the upper-unitriangular group N , the one-parameter subgroup U is not contained in a proper connected abelian subgroup that contains both E_{12} and E_{23} , because

$$[E_{12}, E_{23}] = E_{13}.$$

The Lie algebra generated by the velocity components is the full Heisenberg algebra

$$\mathfrak{n} = \mathrm{span}(E_{12}, E_{23}, E_{13}).$$

For $\mathrm{SL}_3(\mathbb{Z}) \backslash \mathrm{SL}_3(\mathbb{R})$, the closed orbit

$$\Gamma N$$

is compact. For the particular generator $E_{12} + E_{23}$, the projection to the maximal torus factor

$$(N/[N, N]) / (((N \cap \Gamma)[N, N]) / [N, N]) \cong \mathbb{R}^2 / \mathbb{Z}^2$$

has the rational direction $(1, 1)$, so this specific flow is not dense in the full nilmanifold. If instead one takes

$$X_\alpha = E_{12} + \alpha E_{23}, \quad \alpha \notin \mathbb{Q},$$

then X_α is still principal for $\alpha \neq 0$, while its projection has irrational slope. The standard nilflow criterion—density is equivalent to density on the maximal torus factor—then gives a dense orbit in the full compact Heisenberg

nilmanifold. Ratner's theorem is compatible with, and vastly generalizes, this nilmanifold behavior.

At the opposite extreme, a point not lying in any proper finite-volume homogeneous subspace has full G as its homogeneous envelope and therefore equidistributes in the entire space.

7. A product group and irreducible lattices

Let

$$G = G_1 \times G_2, \quad U = U_1 \times \{e\},$$

where $U_1 < G_1$ is unipotent. If

$$\Gamma = \Gamma_1 \times \Gamma_2,$$

then

$$\Gamma \backslash G \cong (\Gamma_1 \backslash G_1) \times (\Gamma_2 \backslash G_2),$$

and U acts only on the first factor. Every probability measure of the form

$$\mu_1 \times \nu_2$$

with μ_1 an ergodic U_1 -invariant measure and $\nu_2 = \delta_{x_2}$ a point mass is U -ergodic. If

$$\mu_1 = m_{x_1 H_1},$$

then the product measure is exactly Haar probability on the homogeneous orbit

$$(x_1, x_2)(H_1 \times \{e\}),$$

so its homogeneity comes from the product subgroup

$$H_1 \times \{e\}.$$

The situation changes for an irreducible lattice $\Gamma < G_1 \times G_2$. A subgroup projecting nontrivially to both factors can have a finite-volume orbit even when it is not a direct product. Graph subgroups and arithmetic correspondences appear. The joining theorem is the natural language for this phenomenon.

Worked Example 13.7.1 (Diagonal subgroup). Take

$$G = \mathrm{SL}_2(\mathbb{R}) \times \mathrm{SL}_2(\mathbb{R})$$

and

$$H = \{(g, g) : g \in \mathrm{SL}_2(\mathbb{R})\}.$$

If the lattice is compatible with the diagonal embedding, for example after passing to a commensurable arithmetic lattice, H can have a closed finite-volume orbit. The unipotent subgroup

$$\{(u_t, u_t) : t \in \mathbb{R}\}$$

then has Haar measure on this diagonal orbit as a homogeneous invariant measure. From the product viewpoint this is a graph joining, not the product joining.

8. Affine lattices

Let

$$G = \mathrm{ASL}_d(\mathbb{R}) = \mathrm{SL}_d(\mathbb{R}) \ltimes \mathbb{R}^d$$

with multiplication

$$(g, v)(h, w) = (gh, v + gw),$$

and let

$$\Gamma = \mathrm{ASL}_d(\mathbb{Z}) = \mathrm{SL}_d(\mathbb{Z}) \ltimes \mathbb{Z}^d.$$

Then

$$\Gamma \backslash G$$

parametrizes unimodular affine lattices.

There are two natural kinds of unipotent motion.

8.1. Linear shear. Take

$$U = \{(u_t, 0) : t \in \mathbb{R}\},$$

where $u_t < \mathrm{SL}_d(\mathbb{R})$ is unipotent. The flow shears both the underlying linear lattice and the translation coordinate.

8.2. Pure translation. Take

$$V_w = \{(I, tw) : t \in \mathbb{R}\}.$$

This subgroup is unipotent in the adjoint representation of the affine group. On a fiber over a fixed linear lattice, it becomes a torus translation. Rational directions give periodic or lower-dimensional torus orbits; irrational directions can give dense subtori.

Thus the simplest Kronecker torus dynamics occur as homogeneous unipotent dynamics inside the affine lattice space.

8.3. Mixed subgroups. A homogeneous envelope can contain both linear and translation directions. Let $H_0 < \mathrm{SL}_d(\mathbb{R})$ and let $W < \mathbb{R}^d$ be an H_0 -invariant vector subspace. Then

$$H = H_0 \ltimes W$$

is a connected subgroup of $\mathrm{ASL}_d(\mathbb{R})$. For it to support a finite-volume orbit, one needs both lattice compatibility in H_0 and rational compatibility of W with the affine lattice.

This is a useful model for the general principle that semidirect-product rigidity contains both representation theory and arithmetic of invariant subspaces.

9. An explicit affine conjugation calculation

Let

$$G = \mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2, \quad u_t = (u_t, 0),$$

with

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}.$$

For a pure translation (I, v) ,

$$\nu_{-t}(I, v)\nu_t = (I, u_{-t}v).$$

If

$$v = \begin{pmatrix} a \\ b \end{pmatrix},$$

then

$$u_{-t}v = \begin{pmatrix} a - tb \\ b \end{pmatrix}.$$

Thus nearby translation perturbations shear linearly. If $b \neq 0$, the first coordinate becomes macroscopic at time

$$T \asymp |b|^{-1}.$$

This is exactly the same polynomial divergence mechanism as in the adjoint action, now visible in the defining representation of the semidirect product.

The limiting additional invariance can therefore be a translation direction rather than a semisimple Lie-algebra direction.

10. A nilpotent example: the Heisenberg group

Let

$$N = \left\{ \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, z \in \mathbb{R} \right\}$$

and

$$\Lambda = N \cap \mathrm{SL}_3(\mathbb{Z}).$$

The quotient

$$\Lambda \backslash N$$

is compact.

Take

$$U = \{\exp(t(X + \alpha Y)) : t \in \mathbb{R}\},$$

where

$$X = E_{12}, \quad Y = E_{23}, \quad Z = E_{13}, \quad [X, Y] = Z.$$

Then

$$\exp(t(X + \alpha Y)) = I + tX + \alpha tY + \frac{\alpha t^2}{2}Z.$$

Projecting to the abelianization

$$N/[N, N] \cong \mathbb{R}^2$$

gives the linear flow

$$t \mapsto (t, \alpha t) \pmod{\mathbb{Z}^2}.$$

If $\alpha \in \mathbb{Q}$, the projected orbit is periodic and the homogeneous envelope is smaller. If $\alpha \notin \mathbb{Q}$, the projection is dense in the two-torus; the commutator direction is then forced by group closure, and the orbit is dense in the whole nilmanifold.

This example is an elementary prototype of orbit-closure homogeneity. It also shows why noncommutativity can enlarge an orbit beyond what is visible from the first-order velocity alone.

11. Quadratic-form stabilizers

Let Q be a nondegenerate quadratic form on $\mathbb{R}^n$ and

$$H = \mathrm{SO}(Q)^\circ < \mathrm{SL}_n(\mathbb{R}).$$

If Q is rational up to scale, then

$$H \cap \mathrm{SL}_n(\mathbb{Z})$$

can be a lattice under the appropriate arithmetic hypotheses. Hence the H -orbit of the standard lattice can be closed of finite volume.

If a unipotent subgroup $U < H$, Haar measure on this orbit is a Ratner homogeneous measure. In the Oppenheim setting, the decisive dichotomy is whether the quadratic-form orbit remains in such a proper arithmetic homogeneous subspace or enlarges to a larger group.

The arithmetic application is deeper than the present subgroup calculation, but the measure-theoretic architecture is already visible:

$$\begin{aligned} & \text{unipotent dynamics inside } \mathrm{SO}(Q) \\ \rightsquigarrow & \text{classification of homogeneous orbit closures} \end{aligned}$$

12. Centralizers and normalizers in examples

For

$$U = \{I + tE_{12}\} < \mathrm{SL}_3(\mathbb{R}),$$

the centralizer contains E_{13} because

$$[E_{12}, E_{13}] = 0.$$

It also contains certain diagonal matrices satisfying equal weights on the first two coordinates. The normalizer is larger: it consists of elements preserving the root line $\mathbb{R}E_{12}$ under the adjoint action.

A limiting shearing element in the centralizer can immediately give a commuting additional invariance. A limiting element merely in the normalizer may rescale U rather than commute with it. Both phenomena occur naturally in rigidity arguments and should not be conflated.

13. A decision tree for a new example

When faced with a concrete unipotent flow, the following decision tree is effective.

- (1) Compute the Lie algebra generator X and its nilpotent adjoint strings.
- (2) Determine small connected subalgebras containing X .
- (3) Test each candidate for unimodularity.
- (4) Determine which candidates are compatible with the lattice.
- (5) Compute centralizers and normalizers to predict possible additional-invariance directions.
- (6) Examine exterior-power representations for possible singular subgroups.
- (7) Use Ratner classification only after this subgroup geometry is understood.

This workflow separates the universal dynamical theorem from the problem-specific algebra. In research applications, most of the new work often lies in Steps 2–6 rather than in reproving the rigidity theorem.

14. What the examples teach

The examples illustrate four recurring lessons.

- (1) In rank one, subgroup scarcity makes Ratner classification look almost like a dichotomy.

- (2) In higher rank, many proper homogeneous envelopes can occur, so classification becomes a subgroup-arithmetic problem after rigidity has done its part.
- (3) Semidirect products add translation directions and make representation theory visible in the shearing itself.
- (4) Nilpotent examples show that orbit closure can enlarge because of commutators even when the projected dynamics look elementary.

The theorem is universal; the list of homogeneous possibilities is not. Worked subgroup analysis is therefore an essential part of learning to use Ratner theory rather than merely knowing its statement.

Part IV

Pointwise Distribution and Further
Rigidity

Pointwise Distribution and Orbit Closure: The Empirical-Measure Argument in Detail

Measure classification concerns invariant probability measures. Orbit closure and equidistribution concern a single point. The bridge is the empirical measure

$$\mu_T = \frac{1}{T} \int_0^T \delta_{xu_t} dt.$$

It is tempting to say that every subsequential limit is invariant, classify it, and conclude. That summary omits three genuine problems:

- (1) on a noncompact space, mass may escape and a probability subsequential limit need not exist;
- (2) a subsequential limit can be a nontrivial mixture of homogeneous ergodic measures;
- (3) even if all ergodic components are homogeneous, one must exclude components supported on proper homogeneous subspaces of the expected envelope.

This chapter writes the full logical chain.

1. The minimal homogeneous envelope

Let

$$X = \Gamma \backslash G, \quad U = \{u_t\}, \quad x \in X.$$

Consider the collection $\mathcal{E}(x, U)$ of connected closed subgroups $H < G$ such that

- (1) $U \subset H$;
- (2) xH is closed;
- (3) xH carries finite H -invariant volume.

The collection is nonempty because $G \in \mathcal{E}(x, U)$: the whole space has finite G -volume since Γ is a lattice.

Choose $H \in \mathcal{E}(x, U)$ of minimal dimension. Write

$$Y = xH$$

and let m_Y be its normalized invariant probability measure.

Remark 14.1.1. At this point no assertion that $\overline{xU} = Y$ is being used. The subgroup H is chosen only from the class of finite-volume homogeneous sets known to contain the orbit. The distribution argument itself implies the orbit-closure statement, so no topological rigidity theorem is hidden in the definition of H .

Since $U \subset H$,

$$xU \subset xH = Y.$$

Thus every empirical measure is supported on Y .

2. Nonescape of mass

The first issue is compactness of the family $\{\mu_T\}$. On a compact space this is automatic. On Y , which can be noncompact, it is a theorem.

The required statement is the orbitwise nonescape theorem isolated as Theorem A.7.1. We emphasize its logical status: in linear semisimple examples it follows from quantitative nondivergence, while the general connected-Lie-group formulation also uses structural reduction. It is proved in Appendix D independently of the pointwise distribution argument; in particular it is not deduced from measure classification.

Applied inside the finite-volume homogeneous space Y , it says that for every $\varepsilon > 0$ there exists compact $K \subset Y$ with

$$\liminf_{T \rightarrow \infty} \mu_T(K) \geq 1 - \varepsilon.$$

This implies tightness of the empirical measures.

Proposition 14.2.1 (Tightness). *The family*

$$\{\mu_T : T \geq T_0\}$$

is tight for sufficiently large T_0 .

PROOF. Fix $T_0 > 0$ and $\varepsilon > 0$. Orbitwise nonescape gives a compact $K_1 \subset Y$ and a time $T_1 \geq T_0$ such that

$$\mu_T(K_1) > 1 - 2\varepsilon \quad (T \geq T_1).$$

The finite orbit segment

$$K_0 = \{xu_t : 0 \leq t \leq T_1\}$$

is compact. With $K = K_0 \cup K_1$, every $T \in [T_0, T_1]$ satisfies $\mu_T(K) = 1$, while every $T \geq T_1$ satisfies $\mu_T(K) > 1 - 2\varepsilon$. Hence the whole tail family $\{\mu_T : T \geq T_0\}$ is tight. $\square$

By Prokhorov's theorem, every sequence $T_n \rightarrow \infty$ has a subsequence for which

$$\mu_{T_{n_j}} \Longrightarrow \mu$$

weakly, where μ is a probability measure on Y .

3. Invariance of every empirical limit

Let $s \in \mathbb{R}$ be fixed and $f \in C_c(Y)$. Then

$$\int f(zu_s) d\mu_T(z) = \frac{1}{T} \int_0^T f(xu_{t+s}) dt.$$

Change variables $r = t + s$:

$$\frac{1}{T} \int_s^{T+s} f(xu_r) dr.$$

Subtract

$$\frac{1}{T} \int_0^T f(xu_r) dr.$$

The two integrals differ only on intervals of total length at most $2|s|$. Hence

$$\left| \int f \circ R_{u_s} d\mu_T - \int f d\mu_T \right| \leq \frac{2|s|}{T} \|f\|_\infty.$$

Letting $T \rightarrow \infty$ along a convergent subsequence gives

$$\int f \circ R_{u_s} d\mu = \int f d\mu.$$

Thus every weak limit is U -invariant.

This argument uses no ergodic theorem and no genericity of x .

4. Classify the ergodic components

Let

$$\mu = \int \mu_\omega d\nu(\omega)$$

be the ergodic decomposition under U . Ratner measure classification says that for ν -almost every ω ,

$$\mu_\omega = m_{y_\omega L_\omega}$$

for a closed finite-volume homogeneous orbit

$$y_\omega L_\omega \subset Y.$$

The connected subgroup L_ω^u generated by the one-parameter Ad-unipotent subgroups of L_ω acts ergodically on this component: it contains U , and every L_ω^u -invariant measurable set is therefore U -invariant. In particular, if the support $y_\omega L_\omega$ is proper in Y , then almost every point of that component belongs to the proper singular incidence set for the ambient group H .

There are now two possibilities for a component:

- (1) its homogeneous support is all of Y ;
- (2) its support is a proper homogeneous subset of Y .

Suppose the first occurs, so that

$$y_\omega L_\omega = Y = xH.$$

Because $y_\omega \in Y$, we also have $y_\omega H = Y$. Point stabilizers for the right action on $\Gamma \backslash G$ are discrete, so the differential of the orbit map at the identity is injective. Equality of the two homogeneous orbits therefore forces equality of their tangent spaces at y_ω , hence

$$\mathfrak{l}_\omega = \mathfrak{h}.$$

Both L_ω and H are connected, so $L_\omega = H$. Consequently the homogeneous probability on $y_\omega L_\omega = Y$ is exactly the normalized H -Haar probability:

$$\mu_\omega = m_Y.$$

Thus the only obstacle to proving $\mu = m_Y$ is positive mass on proper homogeneous subsets of Y .

This is exactly what singular-set avoidance was designed to exclude.

5. Why minimality of H matters

Suppose there were a proper connected subgroup $L < H$ and a point $y \in Y$ such that

$$yL$$

is closed of finite volume, contains a U -orbit in the relevant conjugated sense, and is forced by the trajectory of x on a positive proportion of long times. One would like to contradict minimality of H .

The contradiction is not immediate because yL need not contain the initial point x . The orbit can visit many translates of a proper homogeneous subspace without lying in one fixed translate.

Avoidance strengthens minimality from a static statement to a dynamical one: if a long unipotent orbit spends positive asymptotic density near a proper finite-volume homogeneous stratum, then linearization produces an algebraic relation forcing x itself into a smaller finite-volume homogeneous envelope. That is what contradicts minimality.

6. The avoidance statement inside the envelope

Let $Y = xH$ be minimal. Let $\mathcal{S}$ be the union of proper singular strata in Y associated with connected proper subgroups of H that can support finite-volume homogeneous pieces relevant to U .

Proposition 14.6.1 (Minimal-envelope avoidance). *For every compact set $C \subset \mathcal{S}$ and every $\varepsilon > 0$, there exists an open neighborhood $O \supset C$ such that*

$$\limsup_{T \rightarrow \infty} \frac{1}{T} |\{0 \leq t \leq T : xu_t \in O\}| < \varepsilon.$$

More generally, the same conclusion holds for a finite union of compact pieces of proper singular strata.

PROOF. Use the orbitwise avoidance-or-reduction theorem, Theorem A.7.2, inside Y with ambient group H . This is important: the measure-level finite-stage theorem from Chapter 7 assumes generic pairs for a pre-existing invariant measure and therefore cannot be applied directly to the arbitrary point x at this stage.

If the orbitwise reduction branch occurred, the theorem would produce a proper connected subgroup $H' < H$ and a closed finite-volume homogeneous subspace of Y containing xU . Translating the subgroup to the base point x gives an element of $\mathcal{E}(x, U)$ with

$$\dim H' < \dim H,$$

contradicting the minimal choice of H . Hence only the avoidance branch can occur, which is exactly the asserted upper-density estimate. $\square$

7. Excluding all proper homogeneous components

Let $\mathcal{S} \subset Y$ denote the set of points lying in a proper singular homogeneous configuration relevant to U . We do not enumerate subgroup types. Instead choose compact sets

$$K_1 \subset K_2 \subset \cdots, \quad \bigcup_m K_m = Y,$$

and let $\mathcal{P}_m$ be the compact family of proper Lie-subalgebra/projective-linearization parameters of dimension $< \dim H$ and norm complexity at most m . Define the stage- m incidence relation $\mathcal{R}_m \subset K_m \times \mathcal{P}_m$ by declaring $(z, p) \in \mathcal{R}_m$ when z satisfies the singular relation represented by p , and set

$$(14.4) \quad \mathcal{S}_m = \text{pr}_{K_m}(\mathcal{R}_m).$$

After subdividing into finitely many closed linearization charts, $\mathcal{R}_m$ is closed in $K_m \times \mathcal{P}_m$. Since both factors are compact, its projection $\mathcal{S}_m$ is compact. The finite-complexity bookkeeping theorem gives

$$(14.5) \quad \mathcal{S} = \bigcup_{m \geq 1} \mathcal{S}_m.$$

This is a countable exhaustion by *complexity stages*, not a countable enumeration of subgroup types.

Suppose the ergodic decomposition of μ gives positive weight to proper homogeneous components. For almost every point in such a component, Ratner measure classification places that point in the component's proper homogeneous support, hence in $\mathcal{S}$. Consequently

$$\mu(\mathcal{S}) > 0.$$

By (14.5), there is an m with

$$(14.6) \quad \mu(\mathcal{S}_m) > c > 0.$$

Apply the compact-family orbitwise avoidance theorem inside the minimal envelope $Y = xH$ to the parameter family $\mathcal{P}_m$. Minimality rules out its exact-trapping branch, so for any $\eta > 0$ there is an open neighborhood $O_0 \supset \mathcal{S}_m$ such that

$$(14.7) \quad \limsup_{T \rightarrow \infty} \mu_T(O_0) < \eta.$$

Choose $\eta < c/2$. Using a compatible metric, choose an intermediate open set

$$\mathcal{S}_m \subset O \Subset O_0$$

with $\mu(\partial O) = 0$. Such a choice is possible because the boundaries of distinct distance neighborhoods are disjoint and a finite measure charges only countably many of them. Since $O \subset O_0$, (14.7) gives

$$\limsup_{T \rightarrow \infty} \mu_T(O) < c/2.$$

But along the subsequence T_{n_j} defining μ ,

$$\mu_{T_{n_j}}(O) \longrightarrow \mu(O) \geq \mu(\mathcal{S}_m) > c,$$

because O is a continuity set for μ . This is a contradiction.

Therefore the ergodic decomposition gives zero weight to proper homogeneous components. Every ergodic component has full support Y , and the argument above shows that each such component is m_Y .

Remark 14.7.1. The Portmanteau inequalities point in different directions for open and closed sets. Choosing a boundary-zero neighborhood is the clean way to avoid an incorrect limsup/liminf interchange: for a continuity set O one has

$$\mu_{T_{n_j}}(O) \rightarrow \mu(O).$$

8. Uniqueness of the subsequential limit

Since almost every ergodic component of μ is homogeneous and no component can be supported on a proper singular stratum, every ergodic component equals m_Y . Hence

$$\mu = m_Y.$$

There is a useful extra conclusion hidden in this sentence. The measures μ_ω in the ergodic decomposition are, by definition, U -ergodic; since $\mu_\omega = m_Y$ for almost every ω , the Haar probability m_Y itself is U -ergodic. Thus the ergodicity assertion in the orbit-closure theorem is obtained from the same argument rather than added separately.

We have shown that every convergent subsequence of empirical measures has the same limit. Tightness guarantees subsequences exist. Therefore the full family converges:

$$\mu_T \Longrightarrow m_Y.$$

THEOREM 14.8.1 (Pointwise equidistribution). *For every $f \in C_c(Y)$,*

$$\frac{1}{T} \int_0^T f(xu_t) dt \longrightarrow \int_Y f dm_Y.$$

Equivalently,

$$\mu_T \Longrightarrow m_Y.$$

The conclusion holds for the individual point x ; no exceptional null set is present.

9. Fixed time shifts and two-sided averages

The limiting measure is unchanged if the starting point is shifted by a fixed amount along the same U -orbit. Indeed, for fixed $s \in \mathbb{R}$ and $f \in C_c(Y)$,

$$\frac{1}{T} \int_s^{T+s} f(xu_t) dt - \frac{1}{T} \int_0^T f(xu_t) dt \longrightarrow 0,$$

because the two integrals differ only on intervals of total length at most $2|s|$. Consequently every point xu_s on the same orbit has the same homogeneous envelope and the same limiting measure.

The same endpoint estimate shows that the symmetric averages

$$\frac{1}{2T} \int_{-T}^T \delta_{xu_t} dt$$

converge to the same Haar probability. One-sided averages remain the default because in many arithmetic applications only forward time has a natural interpretation, but the two-sided formulation is a useful consistency check.

10. Why this is much stronger than Birkhoff

Birkhoff's theorem, applied to the invariant probability m_Y , says that for m_Y -almost every $y \in Y$,

$$\frac{1}{T} \int_0^T f(yu_t) dt \rightarrow \int f dm_Y$$

provided the U -action is ergodic. Ratner's distribution theorem says that the appropriate homogeneous average exists for *every* point, with the homogeneous envelope allowed to depend on the point.

Thus the gain is not merely identification of an almost-everywhere limit. It is elimination of all exceptional points by algebraic classification and avoidance.

11. From distribution to orbit closure

Since

$$\mu_T \rightarrow m_Y,$$

the support of m_Y must lie in the orbit closure

$$\overline{xU}.$$

To see this, let $z \in \text{supp}(m_Y)$ and let O be any open neighborhood of z . Then

$$m_Y(O) > 0.$$

Choose a compactly supported nonnegative function f supported in O with

$$\int f dm_Y > 0.$$

Equidistribution gives, for large T ,

$$\frac{1}{T} \int_0^T f(xu_t) dt > 0.$$

Therefore

$$xu_t \in O$$

for some t , proving

$$z \in \overline{xU}.$$

Since

$$\text{supp}(m_Y) = Y,$$

we obtain

$$Y \subset \overline{xU}.$$

The reverse inclusion was known from $xU \subset Y$. Therefore

$$\boxed{\overline{xU} = xH.}$$

This proves orbit-closure homogeneity from pointwise distribution.

12. The order of implication

The argument used the following direction:

$$\begin{aligned} \text{measure classification} + \text{orbitwise nonescape} + \text{orbitwise avoidance} \\ \implies \text{pointwise distribution} \\ \implies \text{orbit closure} \end{aligned}$$

It did not use the orbit-closure theorem to identify the minimal envelope before proving distribution. The envelope was chosen from the larger class of all finite-volume homogeneous sets known to contain the orbit, with G itself ensuring existence.

This noncircular organization is useful independently of which proof of the topological theorem one prefers.

13. A compact-space simplification

If Y is compact, nonescape of mass is automatic. The proof reduces to:

- (1) extract a weak limit of empirical measures;
- (2) prove U -invariance;
- (3) classify ergodic components;
- (4) exclude proper homogeneous components by minimality and avoidance.

The analytic difficulty of cusps disappears, but the singular-set problem remains. Compactness alone does not imply unique ergodicity.

14. A periodic orbit sanity check

Suppose xU itself is periodic with period $T_0 > 0$. Then the minimal envelope is

$$H = U$$

and

$$Y = xU \cong (U \cap g^{-1}\Gamma g) \backslash U.$$

For $f \in C(Y)$,

$$\frac{1}{T} \int_0^T f(xu_t) dt$$

converges to the normalized period average

$$\frac{1}{T_0} \int_0^{T_0} f(xu_t) dt.$$

The general theorem therefore contains the elementary periodic-flow computation exactly.

15. The modular-surface sanity check

For the horocycle flow on

$$\mathrm{SL}_2(\mathbb{Z}) \backslash \mathrm{SL}_2(\mathbb{R}),$$

a point on a periodic horocycle has envelope U . A nonperiodic point has no proper unimodular finite-volume connected subgroup containing U except G , so its envelope is G . Hence every nonperiodic horocycle equidistributes with respect to Haar probability.

This recovers the classical dichotomy without any separate topological analysis.

16. A higher-rank intermediate envelope

In

$$\mathrm{SL}_3(\mathbb{Z}) \backslash \mathrm{SL}_3(\mathbb{R})$$

with $U = \{I + tE_{12}\}$, take a point lying on the closed orbit of the upper-left embedded $\mathrm{SL}_2(\mathbb{R})$. Its minimal envelope can be that copy of SL_2 , not the full SL_3 . Equidistribution then occurs inside the smaller orbit.

Similarly, a point on the compact upper-unitriangular orbit can have a nilpotent envelope. Pointwise distribution is therefore sensitive to the arithmetic position of the initial point.

The theorem should be read as

every point finds its own homogeneous statistical universe.

17. Generated unipotent groups and the limits of the averaging argument

Suppose $W < G$ is connected and generated by one-parameter Ad-unipotent subgroups. The structural generated-group extension of Ratner's theorem supplies a closed subgroup F with $\overline{xW} = xF$, while xF° carries finite F° -invariant volume. The subgroup F need not be connected in the general statement. This is not obtained by a purely formal succession of one-parameter closures: adjoining a generator can change the ambient homogeneous envelope, so compatibility of the subgroup data is part of the extension theorem itself.

One should not automatically transfer the one-parameter empirical-measure proof to averages over all of W . A group generated by unipotent one-parameter subgroups need not be amenable; for example, $\mathrm{SL}_2(\mathbb{R})$ is

generated by its upper and lower unipotent subgroups. Hence a general such W need not admit Følner sets at all.

If W is amenable—for instance $W \cong \mathbb{R}^k$ or W is a connected nilpotent group—and one chooses a suitable Følner sequence (F_n) , then the basic invariance calculation takes the form

$$\frac{m_W(F_n \triangle F_n w)}{m_W(F_n)} \longrightarrow 0 \quad (w \in W),$$

where m_W is Haar measure. In that amenable setting, tightness, invariant-limit classification, and avoidance can again be organized into an empirical-measure argument. For nonamenable generated-unipotent groups, pointwise distribution requires a different averaging problem and is not asserted here as a formal consequence of the one-parameter theorem.

18. A checklist for empirical-measure proofs

When deriving pointwise distribution from rigidity in a new setting, check the following separately.

- (1) What is the candidate homogeneous envelope, and why does at least one exist?
- (2) Why are empirical measures tight?
- (3) Why is every weak limit invariant under the acting subgroup?
- (4) Which classification theorem applies to ergodic components?
- (5) What are the proper singular components inside the candidate envelope?
- (6) What avoidance theorem rules them out?
- (7) Where is compact finite-complexity singular bookkeeping used?
- (8) Why does every surviving ergodic component equal the same Haar measure?
- (9) Why does uniqueness of subsequential limits imply convergence of the full family?
- (10) How does support of the limit imply the orbit-closure statement?

Each item corresponds to a real logical step. If a proof jumps directly from “subsequential limits are invariant” to “therefore the orbit equidistributes,” it has omitted most of the rigidity argument.

19. Conceptual summary

For a single point x , form time averages. Nonescape prevents those averages from losing probability mass. Invariance of limits is a boundary calculation. Measure classification turns a mysterious invariant limit into a

mixture of homogeneous pieces. Minimal-envelope avoidance removes every proper piece. What remains is one homogeneous Haar measure, and its full support forces the orbit closure.

Thus the passage from measure rigidity to topological rigidity is not automatic, but it is systematic:

classification of invariant measures
 + control of escape and singular concentration
 = classification of individual orbit distributions

Orbit Closures and Topological Rigidity

1. From equidistribution to the closure

Let

$$Y = \overline{xU}.$$

Every empirical measure

$$\mu_T = \frac{1}{T} \int_0^T \delta_{xu_t} dt$$

is supported on Y . Since Y is closed, every weak-* limit is also supported on Y .

By Theorem 14.8.1,

$$\mu_T \rightarrow m_{xH_x}.$$

Therefore

$$(15.1) \quad \text{supp } m_{xH_x} = xH_x \subset Y.$$

The reverse inclusion is immediate from $U \subset H_x$:

$$(15.2) \quad xU \subset xH_x \implies Y \subset xH_x.$$

Combining (15.1) and (15.2) yields the orbit-closure theorem.

THEOREM 15.1.1 (Ratner orbit-closure theorem). *For every $x \in \Gamma \backslash G$ and every one-parameter Ad -unipotent subgroup $U < G$, there exists a closed connected subgroup $H < G$ containing U such that*

$$(15.3) \quad \overline{xU} = xH,$$

and xH has finite H -invariant volume. Moreover U acts ergodically on xH with respect to its normalized Haar measure.

2. Why finite volume is part of the conclusion

A closed orbit of an arbitrary subgroup need not have finite invariant volume. The theorem asserts more: the stabilizer

$$H_x = H \cap g^{-1}\Gamma g$$

is a lattice in H .

This is not a purely topological consequence of closedness. It enters through the statistical argument: the orbit closure is identified as the support of a probability Haar measure produced by equidistribution.

3. No fractal orbit closures

A generic smooth flow may have orbit closures that are Cantor sets, solenoids, strange attractors, or highly nonmanifold minimal sets. The theorem excludes all such behavior for one-parameter Ad-unipotent flows on finite-volume homogeneous spaces:

every orbit closure is itself a finite-volume homogeneous space.

The possible topology can still be complicated globally—for example, a noncompact arithmetic quotient with cusps—but the complexity is algebraic rather than fractal.

4. The subgroup attached to an orbit

The subgroup H in (15.3) can be chosen minimally by dimension among closed connected subgroups satisfying

$$U \subset H, \quad xH \text{ closed finite-volume.}$$

For such a minimal choice, pointwise equidistribution shows that U acts ergodically on m_{xH} .

If H' is another connected subgroup with

$$xH' = xH$$

and both carry the same Haar probability, then their connected actions on the orbit differ only by isotropy/normalizer redundancy. The connected stabilizer of the Haar measure provides a canonical maximal choice.

5. A useful criterion for density

Corollary 15.5.1. *The orbit xU is dense in X if and only if its minimal homogeneous envelope is G .*

PROOF. By the orbit-closure theorem, $\overline{xU} = xH$ for the minimal homogeneous envelope H . This closure equals $X = xG$ exactly when $H = G$ under the minimal connected-envelope convention fixed above. $\square$

Thus density questions reduce to excluding proper closed finite-volume homogeneous subspaces containing the orbit.

This reformulation is extremely powerful in arithmetic applications. Instead of proving density by direct approximation, one classifies possible

intermediate subgroups and rules them out using irrationality or arithmetic information.

6. Example: the modular surface

Let

$$G = \mathrm{PSL}_2(\mathbb{R}), \quad \Gamma = \mathrm{PSL}_2(\mathbb{Z}),$$

and let U be the upper horocycle group. For a given point x there are essentially two possibilities compatible with subgroup structure:

- (1) xU is periodic, so the closure is a circle;
- (2) the minimal homogeneous envelope is G , so the orbit is dense and equidistributed in $\Gamma \backslash G$.

On finite-area noncompact surfaces the precise classification of periodic horocycles reflects the cusp structure.

Ratner's general theorem replaces this rank-one dichotomy by a hierarchy of possible intermediate homogeneous subgroups.

7. Example: an intermediate subgroup in SL_3

Let $G = \mathrm{SL}_3(\mathbb{R})$ and let

$$H = \left\{ \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix} : A \in \mathrm{SL}_2(\mathbb{R}) \right\}.$$

Take $U < H$ to be an upper-unipotent one-parameter subgroup. If xH is a closed finite-volume orbit and x is chosen so that the U -orbit is nonperiodic and ergodic in xH , then

$$\overline{xU} = xH$$

even though xH is a proper subset of the ambient SL_3 quotient.

This is the simplest visual model for why “dense or periodic” is not the general statement.

8. Generated-by-unipotents extensions

Suppose $W < G$ is a connected subgroup generated by one-parameter Ad-unipotent subgroups. The corresponding orbit-closure statement is a genuine extension of the one-parameter theorem, not a formal consequence obtained by closing one generator at a time. The difficulty is that after adjoining a new generator the previously obtained homogeneous envelope can change, and finite-volume structure must be carried through that change. The full generated-subgroup theorem resolves this bookkeeping and yields a homogeneous closure for xW .

THEOREM 15.8.1 (Generated-unipotent orbit closure — proved in Chapter 17). *Let W be a connected Lie subgroup generated by one-parameter Ad-unipotent subgroups. For every $x \in \Gamma \backslash G$ there exists a closed subgroup $F < G$ containing W such that*

$$\overline{xW} = xF,$$

and the identity-component orbit xF° carries finite F° -invariant volume. No connectedness of F is asserted in this general extension.

Proof location. The full generated-unipotent measure-rigidity and orbit-closure argument is proved in Chapter 17, including the finite-generator normalizer/gluing step and the finite-family focusing step. The theorem above is therefore a forward statement whose proof owner is Chapter 17, not an external input.

Remark 15.8.2 (Status of the extension). This is the generated-subgroup form of Ratner's topological theorem. It is stronger than what follows by a purely formal iteration of the one-parameter statement: when a new generator is adjoined, one must control how the previously obtained homogeneous envelope changes and verify that finite-volume structure is retained. Rather than hide that issue in a one-paragraph iteration argument, we regard the generated-subgroup formulation as part of the full Ratner theorem. The one-parameter case contains the dynamical mechanism developed in detail in this volume; the extension uses the same rigidity together with subgroup bookkeeping.

9. Minimal sets

A nonempty closed U -invariant subset $M \subset X$ is minimal if it contains no proper nonempty closed U -invariant subset.

Corollary 15.9.1. *Every minimal closed subset for an Ad-unipotent flow is a homogeneous orbit xH on which every U -orbit is dense.*

PROOF. Take $x \in M$. The orbit closure $\overline{xU}$ is a nonempty closed invariant subset of M , so minimality gives

$$M = \overline{xU} = xH.$$

For any $y \in M$, its orbit closure is again a nonempty closed invariant subset of M , hence equals M . $\square$

This is a topological strengthening of the measure classification picture: not only invariant probability supports, but all minimal orbit closures, are homogeneous.

10. The orbit closure as a support theorem

The proof can be summarized in one sentence:

The unique time-average limit is Haar measure on the minimal homogeneous envelope, and the support of that limit must be contained in the orbit closure.

The sentence is short only because substantial work is hidden in the words “unique,” “Haar,” and “minimal homogeneous envelope.” Each of those words encodes a separate rigidity input: nonescape and uniqueness of empirical limits, measure classification, and the minimal-envelope/avoidance argument.

11. A topological consequence for recurrence

If xH is compact, then every U -orbit in xH is recurrent. If U acts ergodically with full-support Haar measure, almost every orbit is dense. Ratner’s pointwise equidistribution theorem upgrades this: the specific orbit used to define H is itself equidistributed, whether or not it was known a priori to be generic.

Thus the theorem is not merely almost-everywhere ergodic theory. It is a pointwise statement for every starting point.

12. Orbit closures versus stabilizers

The orbit-closure subgroup H and the measure stabilizer of m_{xH} need not coincide exactly. One always has

$$H \subset S(m_{xH}),$$

while additional normalizer elements can preserve the same Haar measure. The connected component of the stabilizer is often the most canonical subgroup for measure-theoretic purposes, whereas a minimal H is often better for subgroup classification.

Keeping these two conventions distinct prevents artificial uniqueness claims.

13. Summary of the topological conclusion

For one-parameter Ad-unipotent flows on finite-volume homogeneous spaces, the three basic rigidity conclusions fit together as follows:

- (1) ergodic invariant probability measures are homogeneous;
- (2) every orbit closure is homogeneous;
- (3) every orbit is equidistributed in its homogeneous closure.

The topological statement is therefore not an isolated theorem: it is the support-level manifestation of the same polynomial-shearing and homogeneous-measure rigidity that governs the measure-theoretic problem.

CHAPTER 16

Joinings, Factors, and Measurable Rigidity

1. Why joinings belong in the rigidity volume

Measure classification concerns one homogeneous system. A joining converts a relationship between two systems into an invariant measure on their product. Ratner's measure theorem can therefore be applied to the product group, turning measurable relationships into homogeneous subgroups of a product.

This is one of the cleanest demonstrations of the strength of measure classification.

2. Joinings revisited

Let

$$X_i = \Gamma_i \backslash G_i, \quad U_i = \{u_t^{(i)}\} < G_i, \quad i = 1, 2,$$

with invariant probability measures m_i .

A *joining* is a probability measure λ on

$$X_1 \times X_2$$

whose marginals are m_1, m_2 and which is invariant under the diagonal flow

$$(16.1) \quad (x_1, x_2) \mapsto (x_1 u_t^{(1)}, x_2 u_t^{(2)}).$$

The product joining $m_1 \times m_2$ always exists. Nonproduct joinings encode correlations between the systems.

3. The product as a homogeneous space

We have

$$(16.2) \quad X_1 \times X_2 \cong (\Gamma_1 \times \Gamma_2) \backslash (G_1 \times G_2).$$

The diagonal acting subgroup is the one-parameter Ad-unipotent subgroup

$$(16.3) \quad U^\Delta = \{(u_t^{(1)}, u_t^{(2)}) : t \in \mathbb{R}\} < G_1 \times G_2.$$

Therefore every ergodic joining is an ergodic U^Δ -invariant probability measure on a homogeneous space.

Ratner measure classification immediately gives:

THEOREM 16.3.1 (Homogeneity of ergodic joinings). *Every ergodic joining λ of the two Ad-unipotent flows is Haar probability on a closed finite-volume orbit*

$$zL \subset X_1 \times X_2$$

for some closed connected subgroup

$$U^\Delta \subset L < G_1 \times G_2.$$

PROOF. Regard the joining as an ergodic U^Δ -invariant probability measure on the product homogeneous space and apply Ratner measure classification. The prescribed marginals do not change homogeneity; they enter only in the subsequent algebraic classification of the subgroup L . $\square$

The remaining task is algebraic: classify subgroups L whose projections to both factors are large enough to produce the prescribed marginals.

4. Surjectivity of the factor projections

Let

$$p_i : G_1 \times G_2 \rightarrow G_i$$

be coordinate projections. Since the marginal of m_{zL} on X_i is m_i , the projection of the homogeneous support is conull, hence dense, in the support of m_i . Density of the projected orbit should not be silently identified with surjectivity of the Lie-group projection $p_i(L)$: the image of a closed subgroup under a nonproper homomorphism need not be closed.

For the explicit Goursat analysis below we therefore impose *surjectivity of $p_i : L \rightarrow G_i$ as an additional hypothesis*. It cannot be deduced merely from the full-support marginals: a proper closed subgroup of G_i can have a dense orbit in $\Gamma_i \backslash G_i$. In algebraic applications, surjectivity is established by separate information about the projected homogeneous subgroup (or one first passes to the appropriate algebraic factor). Under this explicit hypothesis, L is a connected subdirect product.

5. Goursat's lemma for connected Lie groups

The structure of a subgroup projecting onto both factors is governed by a Lie-group form of Goursat's lemma.

THEOREM 16.5.1 (Goursat structure). *Let $L < G_1 \times G_2$ be a closed connected subgroup whose coordinate projections are surjective. Define*

$$N_1 = \{g_1 : (g_1, e) \in L\}, \quad N_2 = \{g_2 : (e, g_2) \in L\}.$$

Then $N_i \triangleleft G_i$ are closed normal subgroups and there is an isomorphism of Lie groups

$$\theta : G_1/N_1 \longrightarrow G_2/N_2$$

such that

$$(16.4) \quad L = \{(g_1, g_2) : \theta(g_1 N_1) = g_2 N_2\}.$$

PROOF. The subgroup $N_1 \times \{e\}$ is the closed subgroup

$$L \cap (G_1 \times \{e\}),$$

so N_1 is closed in G_1 . Intersecting L with $\{e\} \times G_2$ gives the identical conclusion for N_2 . If $(n_1, e) \in L$ and $(g_1, g_2) \in L$, then

$$(g_1, g_2)(n_1, e)(g_1, g_2)^{-1} = (g_1 n_1 g_1^{-1}, e) \in L,$$

so N_1 is normal in G_1 by surjectivity of the first projection. Conjugating $(e, n_2) \in L$ by an arbitrary lift $(g_1, g_2) \in L$ proves $g_2 n_2 g_2^{-1} \in N_2$, and surjectivity of the second projection gives normality of N_2 in G_2 .

For $g_1 \in G_1$, choose g_2 with $(g_1, g_2) \in L$ and define

$$\theta(g_1 N_1) = g_2 N_2.$$

If $(g_1, g'_1) \in L$ also, then

$$(e, g_2^{-1} g'_2) \in L,$$

so $g_2 N_2 = g'_2 N_2$. Thus θ is well defined. To check the homomorphism law, choose $(g_1, g_2) \in L$ and $(g'_1, g'_2) \in L$. Since their product also lies in L ,

$$\theta(g_1 g'_1 N_1) = g_2 g'_2 N_2 = \theta(g_1 N_1) \theta(g'_1 N_1).$$

Interchanging the two factors gives an inverse map. Closed-subgroup and quotient-manifold theory then give continuity and smoothness. Formula (16.4) follows directly from the definition. $\square$

6. Simple groups: product or graph

Suppose G_1, G_2 are connected almost simple Lie groups with no nontrivial connected normal subgroups relevant to the quotient. Then the N_i in Goursat's lemma are either discrete/central or all of G_i .

If one N_i is all of G_i , then

$$L = G_1 \times G_2$$

and the joining is the product measure.

Otherwise, after dividing by finite central kernels, L is the graph of a local isomorphism

$$G_1 \dashrightarrow G_2.$$

Thus a nontrivial ergodic joining can exist only when the ambient groups possess matching algebraic quotients.

This is the source of measurable rigidity.

7. Graph joinings from algebraic conjugacies

Suppose there is a Lie-group isomorphism

$$\theta : G_1 \rightarrow G_2$$

with

$$\theta(\Gamma_1)$$

commensurable with Γ_2 and

$$\theta(u_t^{(1)}) = u_t^{(2)}.$$

When the lattice compatibility is exact, the map

$$F : \Gamma_1 g \mapsto \Gamma_2 \theta(g)$$

is a measurable conjugacy, and its graph carries an invariant joining.

When the lattices are only commensurable, the correct object is a finite correspondence rather than a single-valued map. The joining is supported on a closed homogeneous orbit in the product associated with the graph subgroup

$$\{(g, \theta(g)) : g \in G_1\}.$$

8. Finite correspondences and commensurators

Let $X = \Gamma \backslash G$ and let $q \in G$ satisfy

$$\Gamma \cap q^{-1} \Gamma q$$

of finite index in both Γ and $q^{-1} \Gamma q$. Then q lies in the commensurator

$$\text{Comm}_G(\Gamma).$$

Set

$$\Gamma_q = \Gamma \cap q^{-1} \Gamma q.$$

The finite cover

$$\Gamma_q \backslash G$$

maps to X in two ways:

$$(16.5) \quad \Gamma_q g \mapsto \Gamma g, \quad \Gamma_q g \mapsto \Gamma q g.$$

Pushing Haar probability forward under the pair of maps gives a self-joining of suitable unipotent flows. Its fibers are finite because both lattice indices are finite.

This construction is the correct rigorous version of the informal notation “ $x \mapsto qx$ ” when q does not normalize Γ .

9. Joining rigidity theorem

A useful formulation is the following.

THEOREM 16.9.1 (Simple-factor joining consequence). *Let $X_i = \Gamma_i \backslash G_i$ carry ergodic Ad -unipotent flows U_i , with G_i connected, center-free, and simple. Let $\lambda = m_{zL}$ be an ergodic joining supplied by measure classification, and assume that the coordinate projections*

$$p_i : L \rightarrow G_i$$

are surjective. Then either

$$L = G_1 \times G_2,$$

so λ is the product joining, or L is the graph of a Lie-group isomorphism $G_1 \rightarrow G_2$. In the graph case, finite volume of the graph orbit forces the two lattices to be commensurable after applying that isomorphism, and inclusion of U^Δ forces the isomorphism to intertwine the two unipotent flows.

PROOF. Homogeneity is Theorem 16.3.1. Apply Theorem 16.5.1. Since G_i is connected, center-free, and simple, each closed normal subgroup N_i appearing there is either trivial or all of G_i . If one N_i is all of G_i , surjectivity in the other factor gives $L = G_1 \times G_2$. Otherwise $N_1 = N_2 = \{e\}$, and Goursat's description makes L the graph of an isomorphism

$$\theta : G_1 \longrightarrow G_2.$$

The stabilizer of a point on the graph orbit is a lattice in $L \cong G_1$. After absorbing the base-point conjugacies into the two lattices, this stabilizer identifies with

$$\Lambda = \Gamma_1 \cap \theta^{-1}(\Gamma_2).$$

The two coordinate projections of the graph orbit factor through

$$\Lambda \backslash G_1 \longrightarrow \Gamma_1 \backslash G_1, \quad \Lambda \backslash G_1 \longrightarrow \theta^{-1}(\Gamma_2) \backslash G_1.$$

Because all three quotients have finite invariant volume and the maps are the natural quotient maps, the inclusions

$$\Lambda < \Gamma_1, \quad \Lambda < \theta^{-1}(\Gamma_2)$$

have finite index: the index is the ratio of the corresponding covolumes. Hence Γ_1 and $\theta^{-1}(\Gamma_2)$ are commensurable, equivalently $\theta(\Gamma_1)$ and Γ_2 are commensurable. Finally,

$$U^\Delta \subset L$$

means exactly that $\theta(u_t^{(1)}) = u_t^{(2)}$ after the base-point conjugacies have been absorbed. Conversely, such a commensurability produces the finite graph correspondence described above. $\square$

Remark 16.9.2. For general semisimple groups, disconnected centers, reducible lattices, or nonclosed projected support subgroups, the algebraic classification of joinings requires additional factor bookkeeping. The unconditional conclusion needed here is the first one: every ergodic joining is homogeneous. The simple-factor theorem records the cleanest Goursat consequence without hiding those extra hypotheses under the word “standard.”

10. Measurable factors

Let $\pi : (X, \mu, U) \rightarrow (Y, \nu, U')$ be a measurable factor map. The graph measure

$$(\text{id}, \pi)_* \mu$$

is a joining of the two systems. If measure classification makes this graph measure homogeneous, one has reduced the factor problem to the algebra of the corresponding homogeneous subgroup in the product. Under the same projection, simplicity, and lattice-compatibility hypotheses used in the graph/finite-correspondence analysis above, this can force the factor map to agree almost everywhere with an algebraic map (or, for a non-graph joining, a finite algebraic correspondence). Without those extra hypotheses, homogeneity alone does not yet classify the factor map.

11. Measurable isomorphisms

If π is a measurable isomorphism, the graph joining has almost-everywhere one-point fibers over both coordinates. In settings where the homogeneous support subgroup has already been shown to be a subdirect graph as above, this rules out a genuine finite correspondence and yields an algebraic isomorphism, possibly composed with a translation centralizing the flow. The conclusion depends on that subgroup classification; it is not a consequence of homogeneity by itself.

12. Centralizer translations

If $c \in C_G(U)$, then right translation

$$R_c : x \mapsto xc$$

commutes with the U -flow whenever it is well defined on the quotient or between the relevant finite covers. Such translations are unavoidable symmetries and therefore naturally appear in any precise rigidity statement.

One should distinguish the centralizer

$$C_G(U) = \{g : gu_t = u_tg \ \forall t\}$$

from the normalizer

$$N_G(U) = \{g : gUg^{-1} = U\}.$$

Normalizer elements may rescale the flow parameter rather than commute with the flow. For strict time-preserving joinings, centralizer compatibility is the relevant condition.

13. A self-joining sanity check for horocycles

For the classical horocycle flow on a compact quotient of $\mathrm{PSL}_2(\mathbb{R})$, the product joining is always present. Nontrivial ergodic self-joinings arise from finite algebraic correspondences associated with commensurator elements, together with the natural flow-direction translations. If the lattice has a small commensurator, the joining structure is correspondingly rigid.

This recovers, in product-space language, the classical phenomenon that the horocycle flow has very few measurable symmetries.

14. Disjointness as absence of common algebraic quotients

Two ergodic systems are disjoint if their only joining is the product joining. The homogeneous joining theorem suggests a practical criterion: if G_1 and G_2 have no compatible nontrivial algebraic quotients on which the unipotent flows correspond, then no graph-type support subgroup can exist, and the product joining is forced.

In simple-group settings, non-local-isomorphism of the ambient groups is often enough to rule out nontrivial connected subdirect products.

15. Why joining rigidity is not an extra miracle

At first glance, classifying measurable relationships between two dynamical systems seems harder than classifying invariant measures on one system. The product construction reverses this impression:

$$\boxed{\text{joining problem} = \text{measure-classification problem on } G_1 \times G_2.}$$

The dynamical difficulty has already been solved by Ratner measure classification. What remains is finite-dimensional subgroup algebra.

16. The completed rigidity package

For one-parameter Ad-unipotent flows, the volume has established the following package:

- (1) **measure rigidity:** ergodic invariant probabilities are homogeneous;
- (2) **distribution rigidity:** every point is generic for Haar measure on a homogeneous closure;

- (3) **topological rigidity:** every orbit closure is homogeneous;
- (4) **joining rigidity:** ergodic joinings are homogeneous and, under standard simplicity hypotheses, algebraic correspondences.

These statements are different faces of one mechanism: polynomial divergence is slow enough for recurrence to compare nearby generic trajectories, yet structured enough that the comparison produces group symmetries rather than arbitrary measurable relations.

CHAPTER 17

Discrete Time and Actions Generated by Unipotent Elements

1. Scope and inherited inputs

Let G be a second-countable Lie group and let $\Gamma < G$ be discrete. When a finite G -invariant measure is needed we say so explicitly. We use right quotients

$$X = \Gamma \backslash G,$$

and right actions. For a subgroup $H < G$ and $x = \Gamma g$ we write

$$H_x = H \cap g^{-1}\Gamma g.$$

The orbit xH is of finite volume precisely when H_x is a lattice in H .

The following results are inherited from the earlier part of this volume. They are not new black boxes.

- (I1) **One-parameter measure rigidity.** If $U = \{u_t\}$ is one-parameter Ad-unipotent and μ is ergodic, U -invariant and finite, then μ is Haar probability on a closed finite-volume homogeneous orbit xH with $U < H$.
- (I2) **One-parameter orbit closure and equidistribution.** For every x , $\overline{xU} = xH$ for a connected closed H with finite-volume orbit, and the U -orbit equidistributes for m_{xH} .
- (I3) **Singular-set linearization and avoidance.** For a finite family of proper homogeneous configurations, a polynomial unipotent orbit either spends only a controlled proportion of time near them or is exactly trapped in a smaller homogeneous envelope. The “exact trapping” branch strictly lowers ambient dimension.
- (I4) **H-property realization.** First-exit polynomial limits modulo a known stabilizer can be corrected to full-group comparison blocks and therefore yield actual additional invariance.
- (I5) **Countability and finite-complexity bookkeeping.** On a fixed compact window only finitely many singular charts of bounded

complexity are needed; globally, the relevant homogeneous types are countable.

The purpose of this chapter is to show carefully how these one-parameter inputs interact when time is discrete or when several unipotent directions must be handled simultaneously.

2. A discrete Ad-unipotent element and its logarithm

Let $u \in G$ and suppose $\text{Ad}(u)$ is unipotent on $\mathfrak{g}$. Set

$$D = \log \text{Ad}(u) = \sum_{j=1}^{r-1} \frac{(-1)^{j+1}}{j} (\text{Ad}(u) - I)^j,$$

where $(\text{Ad}(u) - I)^r = 0$.

Lemma 17.2.1 (The logarithm is a derivation). *$D \in \text{Der}(\mathfrak{g})$, and $\exp(tD)$ is a one-parameter subgroup of $\text{Aut}(\mathfrak{g})$.*

PROOF. The automorphism group $\text{Aut}(\mathfrak{g})$ is a closed linear algebraic subgroup of $\text{GL}(\mathfrak{g})$: in a basis with structure constants c_{ij}^k , the condition $A[e_i, e_j] = [Ae_i, Ae_j]$ is a finite family of polynomial equations in the entries of A . The cyclic subgroup generated by the unipotent element $\text{Ad}(u)$ has Zariski closure equal to the one-parameter unipotent algebraic group

$$\{\exp(tD) : t \in \mathbb{R}\}.$$

Indeed, after conjugating $\text{Ad}(u)$ to Jordan form this is the usual equality between the Zariski closure of the integer points of a polynomial one-parameter unipotent group and the whole real one-parameter group. Since every power $\text{Ad}(u)^n$ lies in $\text{Aut}(\mathfrak{g})$ and that group is Zariski closed, every $\exp(tD)$ does as well. Differentiating at $t = 0$ the identity

$$\exp(tD)[X, Y] = [\exp(tD)X, \exp(tD)Y]$$

gives $D[X, Y] = [DX, Y] + [X, DY]$. Thus D is a derivation. $\square$

Let G° be the identity component and $p : \tilde{G} \rightarrow G^\circ$ its universal covering group. A Lie-algebra automorphism of $\mathfrak{g}$ integrates uniquely to an automorphism of the simply connected group $\tilde{G}$. Hence Lemma 17.2.1 produces a one-parameter group $\alpha_t \in \text{Aut}(\tilde{G})$ with $d\alpha_t = \exp(tD)$.

Lemma 17.2.2 (The time-one lift agrees with conjugation). *Let $c_u : G^\circ \rightarrow G^\circ$ be conjugation by u . Its unique lift fixing the identity of $\tilde{G}$ is α_1 .*

PROOF. Both lifts are Lie-group homomorphisms $\tilde{G} \rightarrow \tilde{G}$ fixing the identity. Their differentials at the identity are both $\text{Ad}(u)$. Homomorphisms out of a connected simply connected Lie group are determined by their

differential: they agree on all one-parameter subgroups $\exp(tX)$, and these generate $\tilde{G}$. $\square$

3. The suspension group

Define the connected semidirect product

$$\bar{G} = \mathbb{R} \ltimes_{\alpha} \tilde{G}, \quad (s, a)(t, b) = (s + t, a \alpha_s(b)).$$

Let

$$w_t = (t, e) \in \bar{G}.$$

The Lie algebra is $\bar{\mathfrak{g}} = \mathbb{R}T \ltimes_D \mathfrak{g}$, with $[T, X] = DX$.

Lemma 17.3.1 (The suspension flow is Ad-unipotent). *Ad(w_t) is unipotent for every t .*

PROOF. On $\bar{\mathfrak{g}} = \mathbb{R}T \oplus \mathfrak{g}$, conjugation by w_t fixes T and acts on the ideal $\mathfrak{g}$ by $\exp(tD)$. Since D is nilpotent, $\exp(tD)$ is unipotent. Therefore so is $\text{Ad}(w_t)$. $\square$

The subgroup $\bar{G}_{\mathbb{Z}} = \mathbb{Z} \ltimes_{\alpha} \tilde{G}$ carries a natural homomorphism

$$\rho : \bar{G}_{\mathbb{Z}} \longrightarrow \langle u \rangle G^{\circ}, \quad \rho(n, a) = u^n p(a).$$

Lemma 17.2.2 is exactly what is needed to check the multiplication law. Put

$$\bar{\Gamma} = \rho^{-1}(\Gamma \cap \langle u \rangle G^{\circ}).$$

When $G = G^{\circ}\Gamma$ the natural map from the time-zero cross-section of $\bar{\Gamma} \backslash \bar{G}$ onto $\Gamma \backslash G$ is onto.

Proposition 17.3.2 (The suspension quotient has finite volume). *Assume Γ is a lattice in G and $G = G^{\circ}\Gamma$. Then $\bar{\Gamma}$ is a lattice in $\bar{G}$. More precisely, $\bar{\Gamma} \backslash \bar{G}$ is the mapping torus of the measure-preserving map R_u on $X = \Gamma \backslash G$ and carries the probability measure obtained from $dt dm_X$ on $[0, 1) \times X$.*

PROOF. Represent a point of the mapping torus by (t, x) with $0 \leq t < 1$ and impose $(1, x) \sim (0, xu)$. The semidirect-product formula and the definition of $\bar{\Gamma}$ identify this mapping torus with $\bar{\Gamma} \backslash \bar{G}$. The boundary identification preserves dm_X because R_u is the restriction of the right G -action and m_X is G -invariant. Hence $dt dm_X$ descends to a finite invariant measure of total mass one. By the closed-orbit/lattice criterion proved in Volume I, $\bar{\Gamma}$ is a lattice. $\square$

4. Suspending an invariant measure

Let μ be a u -invariant probability on X . Define a probability on the mapping torus by

$$(4.1) \quad \int F d\bar{\mu} = \int_0^1 \int_X F(t, x) d\mu(x) dt.$$

Lemma 17.4.1 (Ergodicity of a constant-roof suspension). *If μ is ergodic for R_u , then $\bar{\mu}$ is ergodic for the flow w_t .*

PROOF. Let A be a w_t -invariant measurable set in the mapping torus. For almost every x the vertical section $A_x = \{t : (t, x) \in A\}$ is, by invariance under small time translations, either null or conull in $[0, 1)$. Let $B = \{x : A_x \text{ is conull}\}$. The identification $(1, x) \sim (0, xu)$ implies $Bu = B$ modulo μ . Ergodicity of u gives $\mu(B) \in \{0, 1\}$. By Fubini, $\bar{\mu}(A) = \mu(B)$, proving ergodicity. $\square$

We now apply the continuous one-parameter theorem in the connected Lie group $\bar{G}$.

THEOREM 17.4.2 (Discrete unipotent measure rigidity). *Assume $\Gamma < G$ is a lattice, $G = G^\circ\Gamma$, and $\text{Ad}(u)$ is unipotent. If μ is an ergodic u -invariant probability on $X = \Gamma \backslash G$, then there are $x \in X$ and a closed subgroup $H < G$ such that*

- (i) $u \in H$;
- (ii) xH is closed and H_x is a lattice in H ;
- (iii) $\mu = m_{xH}$.

The orbit xH may have finitely many connected components, permuted transitively by u .

PROOF. Suspend μ to $\bar{\mu}$. By Lemmas 17.3.1 and 17.4.1, $\bar{\mu}$ is an ergodic invariant probability for the one-parameter Ad -unipotent flow w_t . The one-parameter measure theorem already proved in Volume III gives a connected closed subgroup $L < \bar{G}$ with $w_{\mathbb{R}} < L$ such that

$$\bar{\mu} = m_{\bar{x}L}$$

on a closed finite-volume orbit.

Put $K = L \cap (\{0\} \times \tilde{G})$. Since $w_{\mathbb{R}} < L$, every $(t, a) \in L$ can be multiplied on the left by w_{-t} to obtain an element of K ; hence $L = w_{\mathbb{R}}K$. In particular K is closed and normalized by $w_{\mathbb{R}}$. Intersect the homogeneous orbit $\bar{x}L$ with the time-zero cross-section. The orbit map identifies each connected cross-section component with a quotient of K . Weil's quotient integration formula shows that the conditional measure of $m_{\bar{x}L}$ on that cross-section

is Haar on the corresponding K -orbit. (Concretely, in the local product coordinates $(-\eta, \eta) \times K$ the Haar measure of L is a constant multiple of $dt dk$; disintegration along t therefore gives dk .)

Projecting the time-zero section to X yields a closed homogeneous orbit xH , where

$$H = \overline{\langle p(K), u \rangle} < G.$$

The subgroup $p(K)$ is normalized by u because K is normalized by w_1 ; taking the closure therefore causes no ambiguity in the orbit. The return map from time 0 to time 1 is precisely right translation by u , so $u \in H$ and the finite set of cross-section components is permuted by u . The conditional measure at time zero is exactly μ by (4.1); hence it is the normalized Haar measure on xH . Finite volume of xH follows from finite volume of $\bar{x}L$ and the cross-section disintegration. $\square$

THEOREM 17.4.3 (Discrete orbit closure and distribution). *Under the hypotheses of Theorem 17.4.2, for every $x \in X$ there is a closed subgroup $H < G$ containing u such that*

$$\overline{\{xu^n : n \in \mathbb{Z}\}} = xH.$$

The identity-component orbit xH° has finite volume. Moreover the Cesàro averages of the u -orbit converge to Haar measure on the finite union of components visited by u .

PROOF. Embed x as a point $\bar{x}$ of the time-zero cross-section. The continuous one-parameter orbit-closure theorem gives $\overline{\bar{x}w_{\mathbb{R}}} = \bar{x}L$ with L connected and the orbit of finite volume. Intersecting with the time-zero section gives the closure of the return points xu^n and, by the same cross-section argument used in Theorem 17.4.2, a finite union of homogeneous K -components. Let H be the closed subgroup generated by the component subgroup and u . Then that finite union is exactly xH .

For distribution, let $f \in C_c(X)$ and choose a continuous function $\chi \geq 0$ on $[0, 1]$ with integral one, supported away from the endpoints. Lift f to $F(t, y) = \chi(t)f(y)$ on the mapping torus. Continuous-time equidistribution of $\bar{x}w_t$ for $m_{\bar{x}L}$, applied on successive unit intervals, differs from the discrete average of $f(xu^n)$ only by an $O(1/N)$ endpoint term. Disintegration of $m_{\bar{x}L}$ along the section identifies the limiting integral with Haar on xH . $\square$

5. Why several generators cannot be treated successively

Let $W < G$ be connected and generated by one-parameter Ad-unipotent subgroups. It is false as a proof strategy to classify a measure for one generator, replace the ambient space by the resulting support, and then repeat blindly with the next generator: the next generator need not normalize the first

support subgroup. The correct proof must make the homogeneous strata for all generators compatible before taking their product.

We now give that compatibility argument. To keep the notation readable we first reduce to finitely many generators.

Lemma 17.5.1 (Finite-generator reduction). *There are one-parameter Ad-unipotent subgroups $U_1, \dots, U_r < W$ whose closed connected generated subgroup is W .*

PROOF. Let $\mathfrak{w}$ be the Lie algebra of W . The Lie algebra generated by the nilpotent infinitesimal generators of all the given one-parameter subgroups is $\mathfrak{w}$. Because $\mathfrak{w}$ is finite-dimensional, a finite subset already generates it as a Lie algebra: start with the zero subalgebra and add a generator whenever the generated Lie algebra strictly increases; the process stops after at most $\dim \mathfrak{w}$ increases. The connected Lie subgroup generated by the corresponding U_i has Lie algebra $\mathfrak{w}$ and therefore equals W . $\square$

6. Regular homogeneous strata for one generator

For the simultaneous-strata proof it is cleaner to use the left quotient. The inversion homeomorphism

$$\iota : \Gamma \backslash G \longrightarrow G/\Gamma, \quad \Gamma g \longmapsto g^{-1}\Gamma$$

turns a right U -action into the left action of U^{-1} ; since a one-parameter unipotent subgroup equals its inverse as a set, this changes no hypothesis. We work in this left-quotient convention until the end of the gluing argument and then invert back.

Fix $U = \{u_t\}$. Let $H < G$ be a connected closed subgroup generated by one-parameter Ad-unipotent subgroups and for which a relevant orbit is of finite volume. Define

$$N(H, U) = \{g \in G : g^{-1}Ug \subset H\}.$$

Let $\mathcal{P}(H)$ denote the countable family of proper connected subgroups $F < H$ of the homogeneous types that occur in the singular-set induction of earlier chapters. Define the singular part

$$S(H, U) = \bigcup_{F \in \mathcal{P}(H)} N(F, U)$$

and the regular part $N^*(H, U) = N(H, U) \setminus S(H, U)$. The precise family $\mathcal{P}(H)$ is not arbitrary: it is the family supplied by the finite-complexity singular ledger of Chapter 7 and Appendix D. Its two properties used below are:

- (S1) every ergodic U -homogeneous component has a representative in $\Gamma \backslash N^*(H, U)$ for a unique minimal type H up to Γ -conjugacy;
- (S2) if two regular representatives of the same type project to the same point of X , then the intervening lattice element normalizes H° and the subgroup generated by its one-parameter unipotents.

These properties were proved in the singular-fiber reduction; we record the short analytic continuation argument that will be used repeatedly.

Lemma 17.6.1 (Positive analytic mass forces containment). *Let F be a connected Lie group with Haar measure m_F , let $\rho : F \rightarrow V$ be real analytic, and let $A \subset V$ be a real algebraic subset. If $m_F(\rho^{-1}(A)) > 0$, then $\rho(F) \subset A$.*

PROOF. Write A locally as the common zero set of finitely many polynomials $p_1, \dots, p_s$. Each $p_j \circ \rho$ is a real-analytic function on the connected analytic manifold F . A nonzero real-analytic function has a zero set of Haar measure zero: in a coordinate chart this follows by induction on dimension, using the one-variable identity theorem on almost every line and Fubini. Hence positive measure of the common zero set forces every $p_j \circ \rho$ to vanish identically on each chart meeting a density point, and analytic continuation on connected F gives vanishing on all of F . $\square$

7. Simultaneous strata

Let now μ be a finite W -invariant, W -ergodic probability measure. For each generator U_i , disintegrate μ into U_i -ergodic components. The one-parameter theorem says that almost every component is homogeneous. Countability of the homogeneous types allows the following finite intersection argument.

Proposition 17.7.1 (Simultaneous regular stratum). *There exist homogeneous types H_i for U_i such that*

$$\mu \left(\Gamma \backslash \bigcap_{i=1}^r N^*(H_i, U_i) \right) > 0.$$

PROOF. Start with any Borel set $\Omega \subset G$ whose projection has positive μ -measure. Decompose the restriction of μ to $\Gamma\Omega$ into U_1 -ergodic components. Almost every component is Haar on a closed homogeneous orbit. Since only countably many minimal homogeneous types occur, one type H_1 and one lattice translate of its regular stratum have positive mass inside $\Gamma\Omega$. Conjugating H_1 by that lattice element gives a set $\Omega_1 = \Omega \cap N^*(H_1, U_1)$ with positive projected mass.

Repeat with U_2 and the restriction to $\Gamma\Omega_1$. Because only finitely many generators are involved, after r steps we obtain $\Omega_r \subset \cap_i N^*(H_i, U_i)$ of positive projected mass. $\square$

Proposition 17.7.2 (Propagation along a homogeneous component). *Let $g \in \cap_i N^*(H_i, U_i)$ and fix j . If the homogeneous H_j° -measure $gm_{H_j^\circ}$ gives positive mass to the projection of $\cap_i N(H_i, U_i)$, then*

$$gH_j^\circ \subset \bigcap_i N(H_i, U_i),$$

and it gives zero mass to every singular subset $\Gamma \backslash S(H_i, U_i)$.

PROOF. Positive projected mass means that for some $\gamma \in \Gamma$, the set of $h \in H_j^\circ$ satisfying $gh\gamma \in \cap_i N(H_i, U_i)$ has positive Haar measure. For a fixed i the map

$$h \longmapsto (gh\gamma)^{-1}U_i(gh\gamma)$$

may be encoded in a finite-dimensional representation by the Chevalley line for H_i ; its containment in H_i is therefore an algebraic condition. Lemma 17.6.1 promotes positive Haar measure to containment for every $h \in H_j^\circ$. Thus $gH_j^\circ\gamma \subset \cap_i N(H_i, U_i)$. The regular-fiber uniqueness property (S2) removes the lattice translate and gives the displayed containment with the chosen H_i .

If positive mass lay in a singular $S(H_i, U_i)$, the same argument with a proper $F < H_i$ would force the whole connected gH_j° -orbit into that proper stratum, contradicting $g \in N^*(H_i, U_i)$. $\square$

Iterating conditional measures and removing null sets gives a saturated core.

Corollary 17.7.3 (Saturated core). *There is a positive-measure Borel subset $T^* \subset \cap_i N^*(H_i, U_i)$ such that for every $g \in T^*$ and every i ,*

$$gm_{H_i^\circ}(\Gamma T^*) = 1.$$

PROOF. Let $T_0 = \cap_i N^*(H_i, U_i)$. Remove from T_0 the null set of points for which one of the conditional H_i° -measures fails to give positive mass to ΓT_0 . Proposition 17.7.2 upgrades positive mass to full mass. Call the remaining set T_1 . Repeat with T_1 , obtaining a decreasing sequence T_n with the same projected measure. The intersection $T^* = \cap_n T_n$ has the desired property by continuity from above. $\square$

8. The unipotent-generated cores and normalizer gluing

For each i , let V_i be the subgroup of H_i° generated by all one-parameter Ad-unipotent subgroups of H_i° . Let

$$F_i = \left(\overline{V_i(H_i^\circ \cap \Gamma_g)} \right)^\circ$$

at a density-point representative $g \in T^*$; after conjugating the whole picture we may take $g = e$ and write Γ for the conjugated lattice. Then F_i is connected, $V_i \triangleleft H_i^\circ$, and the orbit ΓF_i is the same as ΓV_i and has finite F_i -invariant volume.

Lemma 17.8.1 (Normalizer transfer). *For all i, j ,*

$$F_j \subset N_G(H_i^\circ) \cap N_G(F_i).$$

PROOF. Corollary 17.7.3 and Proposition 17.7.2 imply $H_j^\circ \subset N(H_i, U_i)$ at the chosen density representative. For $h \in H_j^\circ$ the commutator map

$$c_i(h, t) = h^{-1}u_i(t)h u_i(-t)$$

takes values in H_i° . Differentiating at $t = 0$ and using connectedness shows that the one-parameter unipotent subgroups of H_j° normalize the Lie algebra generated by the one-parameter unipotents of H_i° . Equivalently, $V_j \subset N_G(V_i)$.

For the lattice part, regular-fiber uniqueness (S2) gives $H_j^\circ \cap \Gamma \subset N_G(H_i^\circ)$. The Borel-density corollary proved in Volume II for precisely such unipotent-generated finite-volume cores says

$$\text{Ad}(F_j) \subset \text{Zcl}_{\text{zar}}(\text{Ad}(F_j \cap \Gamma)).$$

(The corollary is the reason Volume II proves the unipotent/hyperbolic form of Borel density rather than only the semisimple textbook statement.) Since normalizers are algebraic after Chevalley linearization, the lattice inclusion therefore upgrades to $F_j \subset N_G(H_i^\circ)$. Applying the same argument to the defining closure of $V_i(H_i^\circ \cap \Gamma)$ gives $F_j \subset N_G(F_i)$. $\square$

Proposition 17.8.2 (Closed product subgroup). *After ordering the generators, the set*

$$F = F_r F_{r-1} \cdots F_1$$

is a connected closed subgroup. The orbit ΓF is closed and carries a finite F -invariant measure.

PROOF. Lemma 17.8.1 lets us form the iterated semidirect product

$$\tilde{F} = F_r \ltimes (F_{r-1} \ltimes \cdots \ltimes F_1).$$

Multiplication $\sigma(f_r, \dots, f_1) = f_r \cdots f_1$ is a continuous homomorphism onto F . Each $\Lambda_i = F_i \cap \Gamma$ is a lattice in F_i , and the normalizer relations imply that $\tilde{\Lambda} = \Lambda_r \ltimes \cdots \ltimes \Lambda_1$ is a lattice in $\tilde{F}$. The map

$$\tilde{\Lambda} \backslash \tilde{F} \longrightarrow \Gamma \backslash G$$

induced by σ has compactly generated finite-volume source and pushes Haar measure to a finite F -invariant measure supported on ΓF . The closed-orbit criterion of Volume I then implies that the image subgroup F is closed, $F \cap \Gamma$ is a lattice, and ΓF is closed. $\square$

Proposition 17.8.3 (The measure is F -invariant). *μ is invariant under F .*

PROOF. On the saturated core T^* , the disintegration of μ along the H_i° -components is Haar. Replacing H_i° by the finite-volume subgroup F_i and repeating the decreasing-core construction shows that the restriction μ^* of μ to ΓT^* is a direct integral of translates of $m_{\Gamma F}$.

The subgroup F is normalized by every F_i and, by the commutator argument used in Lemma 17.8.1, by every generator U_i ; hence it is normalized by W . After replacing G by the closed normalizer of F (which contains the support by W -ergodicity), we may regard F as normal. Every translate of $m_{\Gamma F}$ is then F -invariant, so μ^* is F -invariant. The W -translates of the positive measure set ΓT^* cover a conull set by W -ergodicity, and W normalizes F . Thus μ itself is F -invariant. $\square$

9. Quotient reduction and measure rigidity for W

We can now finish the generated-group measure theorem by induction on the ambient dimension. The following formulation keeps the induction explicit.

THEOREM 17.9.1 (Measure rigidity for groups generated by unipotents). *Let G be connected, $\Gamma < G$ a lattice, and let $W < G$ be a connected subgroup generated by one-parameter Ad-unipotent subgroups. If μ is a finite W -invariant, W -ergodic measure on $X = \Gamma \backslash G$, then there are $x \in X$ and a closed subgroup $H < G$ containing W such that*

$$\text{supp } \mu = xH, \quad \mu = m_{xH}.$$

The identity component H° has a lattice stabilizer in the orbit.

PROOF. Induct on $\dim G$. If a proper connected subgroup $P < G$ normalized by W satisfies $\mu(xP) > 0$ for some x , then the W -translates of xP are either equal or disjoint. Ergodicity concentrates μ on the closed WP -saturation. Its identity component has smaller dimension than G , so the induction hypothesis applies there and proves the theorem. Hence we may assume the *no proper normalized positive-mass stratum* condition.

Construct F as in Propositions 17.7.1–17.8.3. If $\dim F > 0$, pass to the quotient by F . Since $F \triangleleft WF$, the quotient measure is again invariant and ergodic under the image of W , which is still generated by Ad-unipotent one-parameter groups. Either the quotient has smaller positive dimension, in which case induction makes it homogeneous, or the quotient is discrete. Pulling the homogeneous orbit back through the closed finite-volume F -fibres gives a homogeneous H -orbit in X .

It remains to discuss the case where the construction gives $F = \{e\}$. Then every H_i° has trivial unipotent-generated core. The local commutator

map used above shows that, near the identity inside $N(H_i, U_i)$, every point centralizes U_i : otherwise its commutator curve would generate a nontrivial one-parameter unipotent subgroup in H_i . Consequently the simultaneous positive-mass stratum lies locally in $Z_G(W)^\circ$. The no-proper-normalized-stratum assumption forces $G^\circ \subset Z_G(W)$. Thus W is central in G .

In that central case choose a point x which is generic for the W -ergodic measure and lies in its support. Ergodicity and second countability imply that xW is dense in $\text{supp } \mu$: every open set of positive measure is visited by the generic orbit, and the support has a countable base. Because W is central, the closure of the subgroup generated by W and the stabilizer of x gives a closed subgroup $H < G$ with

$$\overline{xW} = xH.$$

If $h \in H$, choose $w_n \in W$ whose actions on x converge to the action of h . Strong continuity of translations on Radon measures and $w_n \mu = \mu$ give $h_* \mu = \mu$. Thus μ is H -invariant and is supported on the single H -orbit xH . Corollary A.4.4 of the earlier Volume III then says that xH is closed of finite volume and that $\mu = m_{xH}$. This closes the last branch of the induction without any Fourier-analytic assumption on the quotient. $\square$

10. A focusing theorem for homogeneous limits

The orbit-closure theorem for W is not a formal consequence of Theorem 17.9.1, because a nonamenable W has no Følner averages. The missing topological bridge is a focusing statement. The proof below is exactly the same linearization/avoidance mechanism already used for one-parameter equidistribution, but applied to a sequence of finite-volume homogeneous pieces.

THEOREM 17.10.1 (Finite-family focusing). *Let $M_n < G$ be connected closed subgroups generated by one-parameter Ad-unipotent subgroups, and suppose $x_n M_n$ are closed finite-volume orbits. Assume $x_n \rightarrow x$ and that the dimensions $\dim M_n$ are bounded. After passing to a subsequence, exactly one of the following occurs.*

- (a) Proper trapping: *all $x_n M_n$ are eventually contained in a fixed proper closed finite-volume homogeneous orbit xP ;*
- (b) Homogeneous focusing: *there are $c_n \rightarrow e$ and a closed connected subgroup L with xL of finite volume such that $c_n M_n c_n^{-1} < L$ for all large n , and every weak limit of $m_{x_n M_n}$ is L -invariant.*

If the sequence is chosen with maximal possible subgroup dimension among homogeneous pieces contained in a fixed closed invariant set Y , then in case

(b) $xL \subset Y$ and either $\dim L$ is strictly larger or the pieces are eventually locally identical.

PROOF. Choose a compact exhaustion K_q of X and, for every proper homogeneous type of complexity at most q , a Chevalley representation $\rho_P : G \rightarrow \mathrm{GL}(V_P)$ and line ℓ_P whose stabilizer is P . The finite-complexity theorem of Chapter 7 says that only finitely many such charts meet K_q at stage q .

For each M_n , choose a finite set of one-parameter unipotent generators whose infinitesimal vectors span its Lie algebra under brackets. Along each generator, the coordinates

$$t \longmapsto \rho_P(u_t)v_P$$

are polynomials of uniformly bounded degree. Apply the polynomial sublevel/markings lemma and the orbitwise singular-avoidance theorem of Appendix D simultaneously to the finite stage- q family. There are two possibilities. If for infinitely many n a positive proportion of a generator block remains in every prescribed neighborhood of one proper chart, the polynomial dichotomy upgrades approximate containment to exact containment. Connectedness and generation then place the whole M_n -orbit inside a fixed proper homogeneous P after passing to a subsequence. This is (a).

Otherwise, on every finite stage one can choose points $y_n \in x_n M_n \cap K_q$ outside all proper singular neighborhoods and with uniform injectivity radius. Move x_n inside its homogeneous orbit so that $y_n \rightarrow y \in K_q$. Write $y_n = y c_n$ in a fixed local slice, $c_n \rightarrow e$. If the conjugated Lie algebras $\mathrm{Ad}(c_n)\mathfrak{m}_n$ fail to converge to a Lie subalgebra preserving every weak limit, choose a first transverse vector and apply the full H-property realization theorem to a generator that shears that vector. The good terminal times supplied by the avoidance theorem give a nonconstant stabilizing polynomial limit. Its derivative produces an additional invariance direction for every weak limit. Adjoin that direction and repeat.

At each genuine repetition the stabilizer dimension increases, while it is bounded by $\dim G$; hence the process stops. Let $\mathfrak{l}$ be the final stabilizer Lie algebra and L its connected subgroup. The same local-slice argument shows $\mathrm{Ad}(c_n)\mathfrak{m}_n \subset \mathfrak{l}$ for all large n ; otherwise another transverse first-exit direction would be available. The closed-orbit criterion plus tightness from quantitative nondivergence gives finite volume of yL . Translating back to x changes L by a fixed conjugacy. Thus (b) holds.

Finally suppose all $x_n M_n \subset Y$ and their dimension is maximal among homogeneous pieces in Y . Since Y is closed and $c_n \rightarrow e$, the limit orbit xL lies in Y . If $\dim L$ is not larger, equality of Lie algebras forces local

equality of the focused pieces by the local quotient slice. This proves the last assertion. $\square$

Remark 17.10.2. The proof above is the place where the generated-orbit theorem genuinely uses more than one-parameter measure classification. The essential inputs are already present in this volume: Chevalley linearization, polynomial avoidance, exact trapping, full H-property realization, and finite-dimensional termination. No arithmeticity theorem or external homogeneous-limit theorem is being inserted here.

11. Orbit closure for a generated-unipotent group

THEOREM 17.11.1 (Orbit closure for groups generated by unipotents). *Let G be connected, $\Gamma < G$ a lattice, and $W < G$ connected and generated by one-parameter Ad -unipotent subgroups. For every $x \in X$ there is a closed subgroup $H < G$ containing W such that*

$$\overline{xW} = xH.$$

The identity-component orbit xH° has finite volume. If the closure is connected as a homogeneous orbit, H may be taken connected.

PROOF. Let $Y = \overline{xW}$. By Lemma 17.5.1 choose generators $U_1, \dots, U_r$. The one-parameter orbit-closure theorem supplies, through each point $y \in Y$ and for each i , a closed finite-volume homogeneous orbit $yH(y, i) \subset Y$ containing yU_i .

Among all connected finite-volume homogeneous orbits contained in Y choose one, say zM , of maximal dimension; existence follows from the one-parameter theorem and finite-dimensionality. If M contains all U_i after conjugating to the base point z , then $W < M$ and zM is a closed W -invariant subset of Y . Since xW is dense in Y , a translate of zM meeting xW is all of Y , and we are done.

Otherwise choose a generator U_i not contained in the stabilizing envelope M . Take a sequence t_n such that $zM u_i(t_n)$ accumulates on a point of Y not lying in zM ; if no such sequence exists, the closure of the union $zM U_i$ equals zM , and differentiation of the local action would imply $U_i < N_G(M)$ and then zM would already be U_i -invariant, contrary to the choice of i .

For each n and for m_M -almost every $y \in zM$, the one-parameter closure of $y u_i(t_n) U_i$ is a finite-volume homogeneous orbit contained in Y . Apply Theorem 17.10.1 to a sequence of such pieces chosen through density points. The proper-trapping alternative lies in a homogeneous orbit of smaller ambient dimension normalized by the relevant generators; the exact-trapping theorem of Appendix D then lets us run the same argument inside that proper orbit.

Ambient dimension strictly decreases, so after finitely many such reductions the focusing alternative occurs.

In the focusing alternative the limiting subgroup L contains a conjugate of M and the new U_i -direction. Because U_i was not already contained in the M -envelope, the H-property produces a genuinely transverse tangent vector, so $\dim L > \dim M$. The orbit $z'L$ is a closed finite-volume homogeneous subset of Y , contradicting maximality of $\dim M$. Therefore the assumption that some generator is missing is impossible. Thus a maximal homogeneous piece contains every U_i and hence W . Density of xW in Y now forces Y to be the corresponding homogeneous orbit. Adding the finite component group swept out by W gives the stated closed subgroup H . $\square$

12. Generated-group equidistribution: what is and is not claimed

For a general nonabelian W there is no canonical interval average. If W is amenable (for example connected nilpotent), tempered Følner sequences can be used, but pointwise ergodic theorems for such sequences are separate analytic inputs. The present volume therefore records the homogeneous classification of invariant measures and orbit closures for arbitrary connected generated-unipotent W , while reserving a pointwise distribution statement for a specified averaging family and a proved ergodic theorem. This is a scope distinction, not a proof omission.

13. Two-parameter check in $\mathrm{SL}_3(\mathbb{R})$

Let

$$U_{12} = \{I + tE_{12}\}, \quad U_{23} = \{I + sE_{23}\}.$$

Their commutator satisfies

$$[I + tE_{12}, I + sE_{23}] = I + tsE_{13},$$

so they generate the upper-unitriangular Heisenberg group

$$N = \left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}.$$

The finite-generator reduction sees two generators, while the normalizer gluing necessarily creates the E_{13} -direction. This is a concrete reason that a proof which merely intersects the two one-parameter classifications would be incomplete: the generated Lie algebra contains a commutator direction not present in either individual parameter.

14. Proof ledger for the replacement chapter

The logical dependencies are now:

Vol. I: Haar, quotients, lattices, closed-orbit criterion

↓

Vol. II: ergodic decomposition, Chevalley, Borel density, QND

↓

Vol. III Chs. 1–16 + Apps. C,D:

one-parameter Ratner + H-property + avoidance

↓

discrete suspension + simultaneous strata

+ normalizer gluing + focusing

↓

discrete rigidity and generated-unipotent

measure/orbit rigidity

The old labels “external extension” are therefore deleted.

APPENDIX A

Worked Rigidity Laboratory: Shearing and Algebraic Detectors

1. Motivation

Ratner rigidity begins with a concrete phenomenon: two nearby unipotent trajectories separate polynomially, not exponentially. The leading polynomial term can create an additional invariance. Algebraic detectors then turn persistent geometric proximity into subgroup information.

Worked Example A.1.1 (Exact polynomial shearing in SL_2). Let $E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Since $u_t = \exp(tE)$ and the adjoint series terminates,

$$\mathrm{Ad}(u_t)F = F + t[E, F] + \frac{t^2}{2}[E, [E, F]] = F + tH - t^2E.$$

Thus a displacement $\exp(\varepsilon F)$, conjugated along the u_t flow, has first-order logarithm $\varepsilon(F + tH - t^2E)$. On the scale $t \asymp \varepsilon^{-1/2}$ the E -component becomes order one while the original displacement tends to zero. This is the simplest explicit source of an “extra” unipotent invariance.

Worked Example A.1.2 (A Chevalley line for a concrete subgroup of SL_3). Let H be the one-parameter subgroup with Lie algebra $\mathfrak{h} = \mathbb{R}E_{12}$. In the adjoint representation its Chevalley line is $\ell = \mathbb{R}E_{12}$. For $a = \mathrm{diag}(r, s, (rs)^{-1})$,

$$\mathrm{Ad}(a)E_{12} = (r/s)E_{12},$$

so the line is fixed even though the vector is rescaled. By contrast, $\mathrm{Ad}(I + \eta E_{23})E_{12} = E_{12} - \eta E_{13}$, so the projective displacement from ℓ is $\asymp |\eta|$ for small η . This is exactly the kind of projective polynomial residual measured in the linearization and bounded-volume arguments.

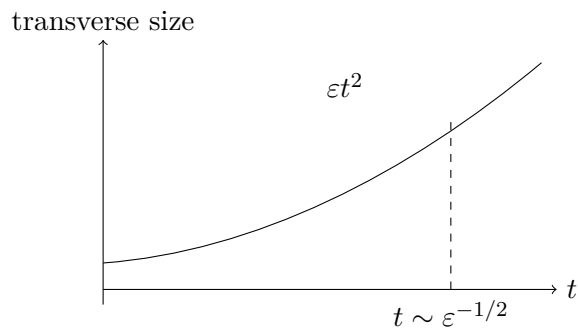
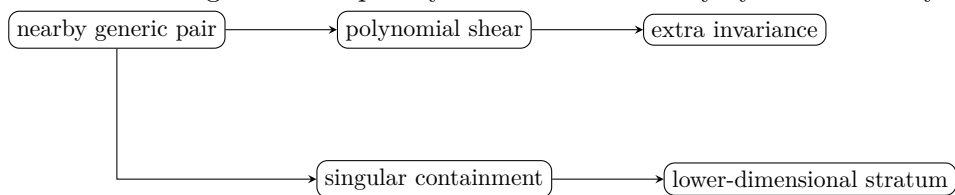


FIGURE 1. Polynomial shearing: a transverse displacement of size ϵ can create an order-one unipotent displacement at a polynomial time scale.

Proof architecture at a glance

Why the induction has two branches. A nearby generic pair produces a nonconstant polynomial shear and hence new invariance; a singular pair instead records containment in a proper homogeneous subspace. The proof succeeds because the first branch enlarges the invariance group while the second lowers algebraic complexity. Neither branch may cycle indefinitely.



What the picture answers. It shows the monotone quantity in each branch: invariance dimension increases above, obstruction dimension decreases below.

APPENDIX A

Structural Lemmas for Homogeneous Orbits

1. Purpose

This appendix supplies the structural steps used by the rigidity induction. Three tempting shortcuts are not valid substitutes for proofs: declaring finite invariant orbits closed without checking the orbit topology, globally enumerating singular subgroup types without a finiteness argument, and replacing a constant singular parameter immediately by a finite-volume homogeneous reduction. The arguments below avoid all three shortcuts.

The revised argument uses four elementary principles:

- (1) an invariant finite measure on one orbit pulls back to Haar measure on the corresponding subgroup quotient;
- (2) a finite-covolume stabilizer forces that orbit to be closed;
- (3) singular data are handled on compact parameter spaces rather than by a global countability theorem;
- (4) in the measure-classification induction, a singular fiber may first be treated merely as a quotient by a *discrete* subgroup. Finite volume emerges at the end of the induction rather than being assumed at its beginning.

2. Orbit maps and their Borel structure

Let G be a Lie group, $\Gamma < G$ discrete, $H < G$ closed, and $x = \Gamma g$. Put

$$\Delta = H \cap g^{-1}\Gamma g.$$

The orbit map

$$(A.1) \quad \iota : \Delta \backslash H \longrightarrow \Gamma \backslash G, \quad \Delta h \longmapsto \Gamma gh$$

is a smooth injective immersion. Both spaces are second-countable manifolds, hence standard Borel spaces. Since ι is Borel and injective, the Lusin–Souslin theorem implies that $\iota(\Delta \backslash H) = xH$ is Borel and that the inverse

$$xH \longrightarrow \Delta \backslash H$$

is Borel. Thus any finite Borel measure concentrated on xH may be pulled back without first knowing that the orbit is embedded or closed.

For completeness, the Lusin–Souslin fact needed here has an elementary local proof in this setting: cover $\Delta \backslash H$ by countably many relatively compact coordinate balls on which the immersion is injective and a homeomorphism onto a locally closed submanifold; the image of each compact exhaustion piece is compact, hence Borel. The countable union is Borel and the local inverse maps paste to the Borel inverse.

3. Finite invariant measure on one orbit

Proposition A.3.1 (Pullback of an orbit measure). *Suppose μ is an H -invariant finite nonzero Radon measure on $\Gamma \backslash G$ and*

$$\mu(xH) = \mu(X).$$

Then the pullback $\tilde{\mu} = \iota_^{-1}\mu$ is an H -invariant finite nonzero measure on $\Delta \backslash H$. Consequently Δ is a lattice in H and $\tilde{\mu}$ is a scalar multiple of quotient Haar measure.*

PROOF. The Borel discussion above makes the pullback well defined. The orbit map intertwines the right H -actions, so H -invariance of μ gives H -invariance of $\tilde{\mu}$. The quotient $\Delta \backslash H$ is transitive under H . Existence of a finite invariant measure is precisely finite covolume of Δ ; equivalently, Δ is a lattice. Haar uniqueness, proved in Volume I by an approximate-identity argument, then gives uniqueness up to scale. $\square$

The remaining point is closedness of the orbit. We prove it rather than assuming a general “closed orbit theorem.”

4. A lattice stabilizer forces a closed orbit

Fix a left-invariant Riemannian metric on H . For $z = \Delta h \in \Delta \backslash H$ define

$$\text{inj}_H(z) = \frac{1}{2} \inf_{\delta \in \Delta \backslash \{e\}} d_H(e, h^{-1}\delta h).$$

Lemma A.4.1 (Thick parts of a finite-volume homogeneous space). *If Δ is a lattice in H , then for every $\rho > 0$ the set*

$$K_\rho = \{z : \text{inj}_H(z) \geq \rho\}$$

is compact.

PROOF. Choose $0 < r < \rho/3$. At every $z \in K_\rho$, the quotient map from H is injective on the radius- r ball around a lift, so the r -ball in $\Delta \backslash H$ has a fixed positive Haar volume v_r . An infinite $2r$ -separated subset of K_ρ would therefore give infinitely many disjoint r -balls, contradicting finite total volume. Thus K_ρ is totally bounded.

It is closed by local lift continuity: if $\Delta h_n \rightarrow \Delta h$ and a nontrivial $h^{-1}\delta h$ had length $< 2\rho$, then the same would hold for $h_n^{-1}\delta h_n$ for large n . A closed totally bounded subset of the complete quotient metric is compact. $\square$

THEOREM A.4.2 (Finite-covolume stabilizer implies closed orbit). *Let $\Gamma < G$ be discrete and $H < G$ closed. If*

$$\Delta = H \cap g^{-1}\Gamma g$$

is a lattice in H , then the orbit xH , $x = \Gamma g$, is closed in $\Gamma \backslash G$. In fact the orbit map (A.1) is proper and hence a homeomorphism onto its image.

PROOF. Suppose the orbit map were not proper. There would be a sequence

$$z_n = \Delta h_n$$

escaping every compact subset of $\Delta \backslash H$ while

$$(A.2) \quad \Gamma g h_n \longrightarrow y \in \Gamma \backslash G.$$

By Lemma A.4.1,

$$\text{inj}_H(z_n) \longrightarrow 0$$

along a subsequence. Hence there are $\delta_n \in \Delta \setminus \{e\}$ with

$$(A.3) \quad h_n^{-1}\delta_n h_n \longrightarrow e.$$

But $g\delta_n g^{-1} \in \Gamma$, and the stabilizer of the point $\Gamma g h_n$ contains

$$h_n^{-1}g^{-1}\Gamma g h_n$$

and therefore the nontrivial element $h_n^{-1}\delta_n h_n$.

Convergence in (A.2) is incompatible with nontrivial stabilizer elements tending to the identity. Indeed, choose a lift $a \in G$ of y . Since $a^{-1}\Gamma a$ is discrete, there is an identity neighborhood V whose closure contains no nontrivial element of $a^{-1}\Gamma a$. Proper discontinuity of the Γ -action gives a neighborhood W of a such that, for every $b \in W$, the conjugate $b^{-1}\Gamma b$ has no nontrivial element in a smaller fixed identity neighborhood V_0 . For large n , choose lifts $\gamma_n g h_n \in W$. Their stabilizers are conjugates of $h_n^{-1}g^{-1}\Gamma g h_n$, so (A.3) produces a nontrivial stabilizer element in V_0 , contradiction.

Thus the orbit map is proper. A proper continuous map into a Hausdorff space has closed image; injectivity then makes it a homeomorphism onto xH . $\square$

THEOREM A.4.3 (Finite invariant orbit criterion). *Let G be a Lie group, $\Gamma < G$ discrete, $H < G$ closed, and $x = \Gamma g$. The orbit xH supports a finite nonzero H -invariant Radon measure if and only if*

$$H \cap g^{-1}\Gamma g$$

is a lattice in H . Whenever these conditions hold, xH is closed, the orbit map is a proper diffeomorphism of homogeneous manifolds, and the invariant measure is unique up to scale.

PROOF. If a finite invariant measure exists, Proposition A.3.1 shows that the stabilizer is a lattice; Theorem A.4.2 gives closedness and properness. Conversely, if the stabilizer is a lattice, quotient Haar measure on $\Delta \backslash H$ is finite and pushes forward through the now proper orbit map to a finite invariant measure on xH . Haar uniqueness gives uniqueness. $\square$

Corollary A.4.4 (Invariant probability concentrated on one orbit). *Let μ be an H -invariant Radon probability measure on $\Gamma \backslash G$. If $\mu(xH) = 1$, then xH is a closed finite-volume homogeneous orbit and μ is its normalized Haar measure.*

PROOF. Apply Proposition A.3.1 and Theorem A.4.2. $\square$

5. Structural finiteness and compact parameter spaces

The polynomial part of the avoidance argument is uniform on compact Grassmannian parameter families. There is, however, a separate discrete issue: the admissible subgroup data are tied to the fixed lattice Γ . A tempting shortcut incorrectly claims that discreteness of Γ alone makes the relevant lattice branches locally finite whenever a Chevalley vector is bounded. That inference is false without an isolation theorem for the homogeneous support data. We therefore isolate the exact structural input and keep it distinct from the compact-parameter Remez argument.

Let

$$\mathcal{H}_\Gamma^{\text{alg}} = \left\{ H < G : \begin{array}{l} H \text{ connected and closed, } H \cap \Gamma \text{ a lattice,} \\ \overline{\text{Ad}(H \cap \Gamma)}^Z = \overline{\text{Ad}(H)}^Z \end{array} \right\}.$$

The support class $\mathcal{H}_\Gamma$ of Definition 6.2.1 will be shown below to be contained in this algebraic class. This implication is not the false statement $H = \overline{H^+}$: compact or elliptic residual directions may be present. What ergodicity supplies is instead enough projective recurrence to force the lattice and the whole support to have the same adjoint algebraic hull. Fix a right-invariant Riemannian metric σ on G and use its induced smooth volume on every homogeneous orbit. Write

$$\text{vol}_\sigma(H) := \text{vol}_\sigma((H \cap \Gamma) \backslash H).$$

THEOREM A.5.1 (Algebraic-support bridge). *If $H \in \mathcal{H}_\Gamma$ and $\Delta = H \cap \Gamma$, then*

$$\overline{\text{Ad}(\Delta)}^Z = \overline{\text{Ad}(H)}^Z.$$

Consequently

$$\mathcal{H}_\Gamma \subset \mathcal{H}_\Gamma^{\text{alg}}.$$

PROOF. Put

$$A = \overline{\text{Ad}(H)}^Z, \quad J = \overline{\text{Ad}(\Delta)}^Z \subset A.$$

Assume for contradiction that J is a proper algebraic subgroup of A . By the Chevalley linearization theorem proved in Volume II, there are a finite-dimensional real algebraic representation

$$\rho : A \longrightarrow \text{GL}(V)$$

and a line $\ell \in \mathbb{P}(V)$ such that

$$J = \text{Stab}_A(\ell).$$

Since $\text{Ad}(\Delta) \subset J$, the map

$$(5.1) \quad \Phi : \Delta \backslash H \longrightarrow \mathbb{P}(V), \quad \Phi(\Delta h) = \rho(\text{Ad}(h^{-1}))\ell,$$

is well defined and continuous. Let m_H be Haar probability on $\Delta \backslash H$ and let $\nu = \Phi_* m_H$. The map is H -equivariant for right translation on the source and the projective action through $\rho \circ \text{Ad}$ on the target, so ν is invariant under that projective H -action.

Let $U = \{u_t\}$ be any one-parameter Ad -unipotent subgroup of H . Algebraic representations carry unipotent elements to unipotent elements. The projective-recurrence lemma proved in Volume II, applied to $\rho(\text{Ad}(u_1))$, therefore gives

$$\nu(\text{Fix}_{\mathbb{P}(V)}(\rho(\text{Ad}(U)))) = 1.$$

Here fixing the projective line for u_1 already fixes it for the whole one-parameter group: if $\rho(\text{Ad}(u_t)) = \exp(tN)$, projective fixed lines of $\exp N$ are exactly the lines contained in $\ker N$.

The connected group H^+ is generated by its one-parameter Ad -unipotent subgroups. Because $\text{Lie}(H^+)$ is finite dimensional, finitely many such subgroups $U_1, \dots, U_r$ may be chosen whose infinitesimal generators generate $\text{Lie}(H^+)$ as a Lie algebra. Hence

$$\nu \left(\bigcap_{i=1}^r \text{Fix}_{\mathbb{P}(V)}(\rho(\text{Ad}(U_i))) \right) = 1,$$

If a line L lies in this finite intersection, its projective stabilizer is a closed Lie subgroup containing every U_i ; its Lie algebra therefore contains the Lie algebra generated by the $\text{Lie}(U_i)$, namely $\text{Lie}(H^+)$. Since H^+ is connected, the entire group H^+ fixes L . Thus for m_H -almost every $x \in \Delta \backslash H$ one has

$$\Phi(xu) = \Phi(x) \quad (u \in H^+).$$

The H^+ -action on $\Delta \backslash H$ is ergodic by the definition of $\mathcal{H}_\Gamma$. Therefore the measurable factor Φ is almost everywhere constant.

Haar probability on $\Delta \backslash H$ has full support and Φ is continuous. An almost-everywhere constant continuous map on a full-support probability space is constant everywhere: otherwise the inverse image of an open set avoiding the essential value would be a nonempty open set of positive Haar measure. Hence

$$\rho(\text{Ad}(h))\ell = \ell \quad (h \in H).$$

By the Chevalley characterization, $\text{Ad}(H) \subset J$, contradicting the assumption $J \subsetneq A$. Thus $J = A$. $\square$

Remark A.5.2 (What the bridge does and does not say). The proof uses ergodicity of H^+ only after passing to a projective algebraic factor. It does *not* identify H with H^+ , and no such identification is required below. This distinction is essential in the presence of compact or elliptic residual directions.

THEOREM A.5.3 (Abelian bounded-volume finiteness). *Let A be a connected abelian Lie group, let $\Lambda < A$ be discrete, and fix a translation-invariant Riemannian metric on A . For every $C > 0$ there are only finitely many connected closed subgroups $B < A$ such that $B \cap \Lambda$ is a lattice in B and*

$$\text{vol}((B \cap \Lambda) \backslash B) \leq C.$$

Equivalently, the Dani–Margulis bounded-volume finiteness statement is completely proved when the ambient connected Lie group is abelian.

PROOF. Let

$$q : \mathfrak{a} \simeq \mathbb{R}^n \longrightarrow A$$

be the universal covering homomorphism. Its kernel L is discrete. The pullback of the chosen metric is a translation-invariant Euclidean metric on $\mathfrak{a}$. Put

$$\tilde{\Lambda} = q^{-1}(\Lambda).$$

Because q is a covering map and Λ is discrete, $\tilde{\Lambda}$ is a discrete subgroup of the Euclidean vector space $\mathfrak{a}$.

We first record an elementary fact that will also explain where finiteness comes from. If D is a discrete subgroup of a finite-dimensional real vector space and $E = \text{span}_{\mathbb{R}} D$, then D is a lattice in E . Indeed, choose a maximal real-linearly independent family $d_1, \dots, d_r \in D$. Their integer span D_0 is a lattice in E . Every element of D can be reduced modulo D_0 into the compact parallelopete

$$P = \left\{ \sum_{i=1}^r t_i d_i : 0 \leq t_i \leq 1 \right\}.$$

The set $D \cap P$ is finite because D is discrete. Hence D/D_0 is finite. In particular D is a finitely generated free abelian group of rank r and, after choosing a $\mathbb{Z}$ -basis, its exterior powers $\bigwedge^d D$ are Euclidean lattices in $\bigwedge^d E$.

Now let $B < A$ be one of the subgroups in the theorem and put $W = \text{Lie}(B) \subset \mathfrak{a}$. Since B is connected, $q(W) = B$. Moreover

$$q^{-1}(B \cap \Lambda) \cap W = W \cap \tilde{\Lambda}.$$

The map $q|_W$ therefore induces an isomorphism of homogeneous spaces

$$(5.2) \quad (W \cap \tilde{\Lambda}) \backslash W \xrightarrow{\sim} (B \cap \Lambda) \backslash B.$$

Because q is a local isometry for the pulled-back metric, this isomorphism preserves Haar/Riemannian volume. Finite volume on the right forces $W \cap \tilde{\Lambda}$ to span W ; otherwise the quotient on the left would retain a noncompact Euclidean direction of infinite Haar measure. Thus

$$D_W := W \cap \tilde{\Lambda}$$

is a full lattice in W , and

$$(5.3) \quad \text{vol}((B \cap \Lambda) \backslash B) = \text{covol}_W(D_W) = \|w_{D_W}\|,$$

where w_{D_W} is the primitive exterior vector of the lattice D_W .

The subgroup D_W is primitive in $\tilde{\Lambda}$: if $m\lambda \in D_W$ for a nonzero integer m and $\lambda \in \tilde{\Lambda}$, then the linearity of W gives $\lambda \in W$, hence $\lambda \in D_W$. Therefore, for each possible rank d , the vector w_{D_W} is a nonzero primitive decomposable element of the fixed exterior lattice

$$\bigwedge^d \tilde{\Lambda}.$$

By (5.3) it lies in the radius- C ball. A Euclidean lattice meets a bounded set in only finitely many points. Hence there are only finitely many possibilities for w_{D_W} for each d . The exterior vector determines D_W up to orientation and determines $W = \text{span}_{\mathbb{R}} D_W$; then $B = q(W)$ is determined as well. Summing over $0 \leq d \leq \dim A$ proves the theorem. $\square$

Worked Example A.5.4 (Why the lattice intersection alone does not determine the subgroup). Take

$$A = \mathbb{R} \times \mathbb{S}^1, \quad \Lambda = \mathbb{Z} \times \{1\},$$

with the product metric. For each $m \in \mathbb{Z}$ the graph

$$B_m = \{(t, e^{2\pi i m t}) : t \in \mathbb{R}\}$$

is a connected closed subgroup and

$$B_m \cap \Lambda = \Lambda.$$

Thus infinitely many connected subgroups have exactly the same lattice intersection. Their covolumes, however, are the lengths of one period of the graph and grow like $\sqrt{1 + (2\pi m)^2}$. Consequently only finitely many of them lie below a fixed volume bound. This example is the elementary reason that the general Dani–Margulis theorem needs a second “reconstruction” step after the finite-intersection statement; one cannot replace that step by the slogan that $H \cap \Gamma$ determines H .

Corollary A.5.5 (Compact ambient bounded-volume finiteness). *If the connected ambient group G is compact, then for every $C > 0$ there are only finitely many $H \in \mathcal{H}_\Gamma$ with $\text{vol}_\sigma(H) \leq C$.*

PROOF. Let $H \in \mathcal{H}_\Gamma$. Since G is compact, so is H , and the discrete subgroup $\Delta = H \cap \Gamma$ is finite. Choose an $\text{Ad}(G)$ -invariant inner product on $\mathfrak{g}$. If $t \mapsto h_t$ is a one-parameter subgroup of H for which $\text{Ad}(h_t)$ is unipotent, then $\text{Ad}(h_t)$ is simultaneously orthogonal and unipotent. Hence $\text{Ad}(h_t) = I$, so $h_t \subset Z(G)^0$. Therefore the closure K of H^+ is a connected compact subgroup of $H \cap Z(G)^0$.

Let $q : H \rightarrow H/K$ be the quotient map. Since K is central and normal, q induces an H -equivariant factor

$$\Delta \backslash H \longrightarrow q(\Delta) \backslash (H/K).$$

The push-forward of Haar probability is the Haar probability on the compact homogeneous factor on the right. The subgroup H^+ acts trivially on this factor because $H^+ \subset K$. Ergodicity of the H^+ -action on $\Delta \backslash H$ therefore forces the factor to be a one-point probability space. Hence

$$H/K = q(\Delta).$$

The group $q(\Delta)$ is finite because Δ is finite, whereas H/K is connected because H is connected. Thus H/K is trivial and $H = K \subset Z(G)^0$.

Every support group is therefore contained in the fixed connected abelian Lie group $Z(G)^0$. Theorem A.5.3, applied to $A = Z(G)^0$ and $\Lambda = \Gamma \cap Z(G)^0$, gives finiteness under every volume bound. $\square$

Lemma A.5.6 (Uniform short generators for lifted abelian lattices). *Let $n = \dim G$. There is a constant $R = R(G, \sigma, C)$ with the following property. Let $H < G$ be any connected closed abelian subgroup, put $\Delta = H \cap \Gamma$, and suppose that Δ is a lattice in H with*

$$\text{vol}_\sigma(\Delta \backslash H) \leq C.$$

If

$$q_H : \mathfrak{h} \longrightarrow H, \quad X \longmapsto \exp X,$$

is the universal covering homomorphism and

$$\tilde{\Delta}_H = q_H^{-1}(\Delta),$$

then the Euclidean lattice $\tilde{\Delta}_H < \mathfrak{h}$ is generated by vectors of norm at most R .

PROOF. Because the ambient metric is fixed, there is a number $\epsilon_{\text{exp}} > 0$ such that

$$\exp X = e, \quad \|X\| < \epsilon_{\text{exp}} \quad \implies \quad X = 0.$$

There is also $\epsilon_\Gamma > 0$ such that

$$d_\sigma(e, \gamma) \geq \epsilon_\Gamma \quad (\gamma \in \Gamma \setminus \{e\}),$$

because Γ is discrete. Set

$$\epsilon_0 = \min(\epsilon_{\text{exp}}, \epsilon_\Gamma) > 0.$$

If $0 \neq X \in \tilde{\Delta}_H$, then either $\exp X = e$, in which case $\|X\| \geq \epsilon_{\text{exp}}$, or $\exp X = \gamma \in \Delta \setminus \{e\}$. In the second case the one-parameter path $t \mapsto \exp(tX)$ has constant speed $\|X\|$ for the right-invariant metric, and hence

$$\epsilon_\Gamma \leq d_\sigma(e, \gamma) \leq \|X\|.$$

Thus every nonzero vector of every lifted lattice $\tilde{\Delta}_H$ has length at least ϵ_0 , uniformly in H .

Since H is connected abelian, q_H is its universal covering homomorphism, its kernel is discrete, and the pullback of the metric on H is the translation-invariant Euclidean metric on $\mathfrak{h}$ induced from $\mathfrak{g}$. Exactly as in (5.2),

$$\tilde{\Delta}_H \backslash \mathfrak{h} \xrightarrow{\sim} \Delta \backslash H$$

is volume preserving. Hence

$$\text{covol}(\tilde{\Delta}_H) \leq C.$$

Write $d = \dim H$. If $d = 0$, then $\mathfrak{h} = \{0\}$ and there is no nonzero lattice direction to bound, so the asserted basis conclusion is vacuous. Hence assume $d \geq 1$. Let $\lambda_1 \leq \dots \leq \lambda_d$ be the successive minima of $\tilde{\Delta}_H$. The fully proved Euclidean form of Minkowski's second theorem from Volume I gives a dimensional constant M_d such that

$$\lambda_1 \cdots \lambda_d \leq M_d \text{covol}(\tilde{\Delta}_H) \leq M_d C.$$

Since $\lambda_1 \geq \epsilon_0$, monotonicity of the successive minima gives

$$\lambda_d \leq M_d C \epsilon_0^{-(d-1)}.$$

Taking the maximum over $1 \leq d \leq n$, choose R_0 so that every such lattice has d linearly independent vectors $v_1, \dots, v_d$ with

$$\|v_i\| \leq R_0.$$

(One may replace each successive minimum by twice its value if desired; this only changes R_0 .)

Let $L_0 = \sum_i \mathbb{Z}v_i$. Since L_0 has full rank in $\tilde{\Delta}_H$, it has finite index. Every coset of L_0 has a representative in the half-open parallelotope

$$P = \left\{ \sum_{i=1}^d t_i v_i : 0 \leq t_i < 1 \right\}.$$

Every point of P has norm at most dR_0 . Therefore $\tilde{\Delta}_H$ is generated by lattice points of norm at most $dR_0 \leq nR_0$. Taking $R = nR_0$ proves the lemma. $\square$

THEOREM A.5.7 (Bounded-volume intersection finiteness: the abelian-support branch). *Fix $C > 0$. Among all connected closed abelian subgroups $H < G$ for which $H \cap \Gamma$ is a lattice and*

$$\text{vol}_\sigma((H \cap \Gamma) \backslash H) \leq C,$$

only finitely many subgroups of Γ occur as intersections $H \cap \Gamma$.

PROOF. Let R be supplied by Lemma A.5.6. If $\Delta = H \cap \Gamma$, then $\tilde{\Delta}_H$ is generated by vectors X with $\|X\| \leq R$. Their images $\exp X$ generate Δ , and

$$d_\sigma(e, \exp X) \leq \|X\| \leq R.$$

A right-invariant Riemannian metric on a Lie group is complete, so the closed radius- R ball is compact. Consequently

$$F_R := \Gamma \cap \overline{B}_\sigma(e, R)$$

is finite. Every possible Δ is generated by a subset of the fixed finite set F_R . There are only finitely many such subgroups. $\square$

Remark A.5.8 (What is new compared with the fixed-ambient abelian theorem). Theorem A.5.3 assumed that all subgroups lay inside one fixed connected abelian group A . The theorem above allows the abelian subgroup H to move through the full, possibly nonabelian, ambient group G . The uniform lower bound on lifted lattice vectors is the point that replaces the fixed exterior lattice used earlier. Thus bounded-volume intersection finiteness is completely proved for moving abelian subgroups, without any reduction theory.

Proposition A.5.9 (Property (M) passes through central extensions). *Let E be a locally compact group and let $Z < E$ be a closed central subgroup. Assume that every closed finite-covolume subgroup of E/Z is cocompact. Then every closed finite-covolume subgroup of E is cocompact.*

PROOF. Let $D < E$ be closed and of finite covolume. Put $\bar{E} = E/Z$ and let $\bar{D}$ be the image of D in $\bar{E}$. We do not assume a priori that $\bar{D}$ is closed. Set

$$\bar{P} = \bar{\bar{D}} < \bar{E}, \quad P = \pi^{-1}(\bar{P}) < E.$$

The natural factor map

$$D \backslash E \longrightarrow \bar{P} \backslash \bar{E}$$

pushes the finite E -invariant quotient measure forward to a finite $\bar{E}$ -invariant measure. Hence $\bar{P}$ has finite covolume in $\bar{E}$, and the hypothesis gives compactness of $\bar{P} \backslash \bar{E}$. Thus P is cocompact in E .

Because Z is central, D is normal in ZD . The normalizer $N_E(D)$ is closed: if $g_n \rightarrow g$ and $g_n D g_n^{-1} = D$, then for every $d \in D$ the sequence $g_n d g_n^{-1} \in D$ converges to $g d g^{-1} \in D$; applying the same argument to g_n^{-1} gives equality. Therefore

$$P = \overline{ZD} \subset N_E(D),$$

so $D \triangleleft P$ and $D \backslash P = P/D$ is a locally compact group.

We now use the quotient-integration formula proved in Volume I, in its left-quotient form obtained from the right-quotient statement by inversion. Choose compatible Haar and quotient measures for the chain $D < P < E$. Applying the formula twice gives the covolume multiplication identity

$$\text{vol}(D \backslash E) = \text{vol}(D \backslash P) \text{vol}(P \backslash E),$$

after the harmless normalization of one Haar measure. The left side is finite and the second factor is finite and positive because P is cocompact. Hence P/D has finite Haar measure. A locally compact group with finite Haar measure is compact: otherwise a relatively compact identity neighbourhood admits infinitely many pairwise disjoint translates. Thus P/D is compact. Since P is cocompact in E , so is D . $\square$

THEOREM A.5.10 (Nilpotent groups have property (M)). *Let N be a connected nilpotent Lie group. Every closed subgroup of finite covolume in N is cocompact.*

PROOF. We induct on the nilpotency class. If N is abelian and $D < N$ is closed and of finite covolume, then $D \triangleleft N$ and $D \backslash N$ is a locally compact group with finite Haar measure; hence it is compact.

Assume the assertion known for groups of nilpotency class at most $c - 1$, and let N have class c . Its center $Z = Z(N)$ is closed and N/Z has nilpotency class at most $c - 1$. The induction hypothesis says precisely that N/Z has property (M) . Proposition A.5.9 applied to

$$1 \longrightarrow Z \longrightarrow N \longrightarrow N/Z \longrightarrow 1$$

gives property (M) for N . $\square$

Corollary A.5.11 (Nilpotent finite-covolume lattices are uniform). *Let N be a connected nilpotent Lie group and let $D < N$ be discrete. If $D \backslash N$ has finite Haar volume, then $D \backslash N$ is compact.*

PROOF. Apply Theorem A.5.10 to the closed subgroup D . $\square$

Lemma A.5.12 (Global BCH coordinates for simply connected nilpotent groups). *Let N be a connected simply connected nilpotent Lie group with Lie algebra $\mathfrak{n}$. Then*

$$\exp : \mathfrak{n} \longrightarrow N$$

is a global diffeomorphism. If $\mathfrak{s} < \mathfrak{n}$ is a Lie subalgebra, then $\exp(\mathfrak{s})$ is the connected Lie subgroup with Lie algebra $\mathfrak{s}$, and it is closed.

PROOF. Because $\mathfrak{n}$ is nilpotent, the Baker–Campbell–Hausdorff series terminates. Hence

$$X * Y = \text{BCH}(X, Y)$$

defines a polynomial Lie-group law on the vector space $\mathfrak{n}$; call the resulting connected simply connected Lie group N_{BCH} . Near 0, the ordinary exponential map of N satisfies the BCH identity, so it is a local Lie-group isomorphism from a neighborhood of the identity in N_{BCH} to a neighborhood of the identity in N .

For completeness, a local homomorphism from a connected simply connected Lie group extends uniquely to a global homomorphism: subdivide a path from the identity into pieces lying in the domain of the local map, multiply the local images, and use the local homomorphism identity to show invariance under subdivision; homotopic paths give the same result by subdividing a homotopy into sufficiently small rectangles. Applying this monodromy argument gives a homomorphism

$$F : N_{\text{BCH}} \longrightarrow N$$

whose differential at the identity is the identity on $\mathfrak{n}$. Thus F is a local diffeomorphism. Its image is an open subgroup of the connected group N , hence all of N , and its kernel is discrete. So F is a covering homomorphism. Both source and target are simply connected; therefore the covering is an isomorphism. By construction $F(X) = \exp X$.

If $\mathfrak{s} < \mathfrak{n}$ is a Lie subalgebra, the terminating BCH formula preserves $\mathfrak{s}$, so $\mathfrak{s}$ is a subgroup of N_{BCH} . It is a closed linear subspace of $\mathfrak{n}$; under the global diffeomorphism $\exp$ its image is therefore a closed connected subgroup of N with Lie algebra $\mathfrak{s}$. $\square$

Lemma A.5.13 (Simply connected nilpotent homogeneous spaces are non-compact unless trivial). *Let N be a connected simply connected nilpotent*

Lie group and $S < N$ a connected closed subgroup. If $S \neq N$, then the homogeneous space $S \backslash N$ is noncompact.

PROOF. We argue by induction on $\dim N$. If $\dim N = 1$, then $N \cong (\mathbb{R}, +)$ and the only connected proper subgroup is $\{e\}$, so $S \backslash N \cong \mathbb{R}$ is noncompact. Assume from now on that $\dim N > 1$. Choose a one-dimensional central Lie subalgebra $\mathfrak{c} < \mathfrak{z}(\mathfrak{n})$ and put $C = \exp(\mathfrak{c})$. By Lemma A.5.12, $C \simeq \mathbb{R}$, every connected Lie subgroup is the exponential of its Lie algebra and is closed, and

$$S \cap C = \exp(\mathfrak{s} \cap \mathfrak{c}).$$

If $C \not\subset S$, then $S \cap C = \{e\}$ and, because C is central, SC is the connected closed subgroup with Lie algebra $\mathfrak{s} + \mathfrak{c}$. The saturated closed subset $SC < N$ therefore maps to a closed subset

$$S \backslash SC \subset S \backslash N.$$

But $S \backslash SC \simeq C \simeq \mathbb{R}$, so $S \backslash N$ is noncompact.

If $C \subset S$, pass to the quotient N/C , which is again connected, simply connected and nilpotent. The subgroup S/C is connected and proper, and

$$S \backslash N \simeq (S/C) \backslash (N/C).$$

The induction hypothesis gives noncompactness. □

THEOREM A.5.14 (Uniqueness of a connected nilpotent syndetic hull). *Let N be a connected simply connected nilpotent Lie group and $D < N$ a discrete subgroup. There is at most one connected closed subgroup $H < N$ such that $D \subset H$ and $D \backslash H$ has finite Haar volume. Equivalently, whenever such an H exists it is determined by D alone.*

PROOF. Suppose such an H exists. Since connected subgroups of a simply connected nilpotent Lie group are themselves simply connected nilpotent, Theorem A.5.11 makes $D \backslash H$ compact.

Let $\mathfrak{s}$ be the smallest Lie subalgebra of $\mathfrak{n}$ containing all logarithms $\log d$ with $d \in D$, and put $S = \exp(\mathfrak{s})$. The global BCH coordinates show that S is a connected closed subgroup of N . Since $D \subset H$, all the logarithms lie in $\mathfrak{h}$, hence $S \subset H$; and by construction $D \subset S$.

Compactness of $D \backslash H$ gives a compact set $C \subset H$ with $H = DC$. Because $D \subset S$,

$$H = SC.$$

Consequently the quotient $S \backslash H$ is the continuous image of the compact set C and is compact. The preceding homogeneous-space lemma, applied inside the simply connected nilpotent group H , says that this is possible only when $S = H$. Thus

$$H = \exp(\mathfrak{s}), \quad \mathfrak{s} = \text{Lie}\langle \log D \rangle,$$

which depends only on D . A second finite-covolume connected hull must therefore coincide with the first. $\square$

Corollary A.5.15 (No nilpotent cusp in the bounded-volume problem). *If a support group $H \in \mathcal{H}_\Gamma$ is nilpotent, then $(H \cap \Gamma) \backslash H$ is compact. Hence, in the nilpotent case, the only issue not settled by this corollary is quantitative finiteness among moving compact nilmanifolds; there is no noncompact cusp problem.*

PROOF. By definition $H \cap \Gamma$ is a lattice in H . Apply Theorem A.5.11. $\square$

Lemma A.5.16 (Connected subgroups of a real unipotent matrix group are algebraic). *Let $U < \mathrm{GL}(V)$ be a real unipotent algebraic group. Every connected Lie subgroup $S < U(\mathbb{R})$ is closed and real Zariski closed. In particular, if S is Zariski dense in U , then $S = U(\mathbb{R})$.*

PROOF. For a unipotent matrix u , the matrix $u - I$ is nilpotent, so the finite series

$$\log u = (u - I) - \frac{(u - I)^2}{2} + \cdots, \quad \exp X = I + X + \frac{X^2}{2!} + \cdots$$

are polynomial. They give inverse polynomial maps between a real unipotent algebraic group and its nilpotent Lie algebra. For completeness, the only point requiring verification is that $\log u$ lies in $\mathfrak{u}$. Put $X = \log u$. The entries of $\exp(tX)$ are polynomial functions of t , and for every integer m one has $\exp(mX) = u^m \in U$. Every polynomial equation defining U therefore vanishes on $\exp(tX)$ for all integer t , hence for all real t ; taking the derivative at $t = 0$ gives $X \in \mathfrak{u}$. Conversely, if $X \in \mathfrak{u}$, the one-parameter subgroup $\exp(tX)$ lies in the Lie group $U(\mathbb{R})$. Thus $\exp$ and $\log$ are indeed inverse polynomial maps between $\mathfrak{u}$ and $U(\mathbb{R})$.

Let $\mathfrak{s} < \mathfrak{u}$ be the Lie algebra of S . The Baker–Campbell–Hausdorff formula terminates on $\mathfrak{u}$, and the connected Lie subgroup with Lie algebra $\mathfrak{s}$ is therefore $\exp(\mathfrak{s})$. Since $\mathfrak{s}$ is a linear subspace and $\exp, \log$ are polynomial inverse maps, $\exp(\mathfrak{s})$ is a real Zariski-closed subset of U . Hence $S = \exp(\mathfrak{s})$ is closed and Zariski closed. Zariski density then forces equality with U . $\square$

Proposition A.5.17 (Fixed-intersection reconstruction: unipotent image and discrete kernel). *In the setting of Definition A.5.40, assume additionally that $K = \ker \rho$ is discrete and that*

$$L := \overline{\rho(\Delta)}^Z$$

is a real unipotent algebraic subgroup of $\mathrm{GL}(V)$. Then there is at most one connected closed subgroup $H < G$ satisfying conditions (1)–(3) of that theorem. No volume bound is needed for this branch.

PROOF. Let H satisfy the three conditions. Its image $\rho(H)$ is connected and has Zariski closure L . Since L is unipotent, Lemma A.5.16 gives

$$\rho(H) = L(\mathbb{R}).$$

Put $P = \rho^{-1}(L(\mathbb{R}))$. The connected group H is contained in P^0 . The differential of

$$\rho|_{P^0} : P^0 \longrightarrow L(\mathbb{R})$$

is surjective because it is already surjective on $\mathfrak{h}$. Its kernel is contained in the discrete group K , so this map is a covering homomorphism. A real unipotent group is polynomially diffeomorphic to its nilpotent Lie algebra and hence is simply connected. Therefore the covering is an isomorphism.

Now $\rho(H) = L(\mathbb{R}) = \rho(P^0)$ and $\rho|_{P^0}$ is injective. Since $H \subset P^0$, it follows that $H = P^0$. Thus H is uniquely determined by L and ρ , hence by the fixed subgroup Δ . $\square$

Remark A.5.18 (Unipotent image with discrete central kernel). Proposition A.5.17 gives uniqueness whenever the fixed algebraic image is unipotent and the central kernel is discrete. The phenomena not covered by this proposition are precisely positive-dimensional central kernels and nonunipotent algebraic hulls, where continuous central or abelian twisting can occur. Example A.5.4 is the elementary model for precisely that phenomenon.

Proposition A.5.19 (Fixed-intersection reconstruction: unipotent image and torus-free central kernel). *In the setting of Definition A.5.40, put*

$$L = \overline{\rho(\Delta)}^Z$$

and assume that L is a real unipotent algebraic group. Let

$$P = (\rho^{-1}(L(\mathbb{R})))^0.$$

Assume that the identity component of $P \cap \ker \rho$ is a vector group (equivalently, it has no nontrivial compact torus). Then there is at most one connected closed subgroup $H < G$ satisfying conditions (1)–(3) of Definition A.5.40. Again no volume bound is needed beyond the finite-covolume hypothesis itself.

PROOF. Let H satisfy the three conditions. As in Proposition A.5.17, Lemma A.5.16 gives

$$\rho(H) = L(\mathbb{R}).$$

Thus $H \subset P$ and $\rho|_P : P \rightarrow L(\mathbb{R})$ is surjective. Put $F = P \cap \ker \rho$. Since $L(\mathbb{R})$ is a real unipotent group, it is polynomially diffeomorphic to a vector space and hence contractible and simply connected. The surjective Lie-group homomorphism

$$F \longrightarrow P \longrightarrow L(\mathbb{R})$$

is a locally trivial principal F -bundle. Its base is contractible, so the bundle is topologically trivial. Because P is connected, F is connected. By hypothesis F is therefore a vector group. The same product trivialization shows that P is simply connected.

At the Lie-algebra level

$$0 \longrightarrow \mathfrak{f} \longrightarrow \mathfrak{p} \xrightarrow{d\rho} \mathfrak{l} \longrightarrow 0$$

is a central extension: $\mathfrak{f}$ lies in the center because $\ker \rho$ is central in G . The Lie algebra $\mathfrak{l}$ is nilpotent because L is unipotent. A central extension of a nilpotent Lie algebra is nilpotent, so P is a connected simply connected nilpotent Lie group.

The fixed subgroup Δ lies in every admissible H , hence in P . Condition (1) says that $\Delta \backslash H$ has finite Haar volume. Theorem A.5.14, applied in the fixed simply connected nilpotent ambient group P , now says that there is at most one such connected subgroup H . This proves the proposition. $\square$

Remark A.5.20 (What the torus-free result isolates). The previous discrete-kernel proposition is contained in the new result. Thus, when the algebraic image is unipotent, reconstruction is completely understood as long as the connected central kernel has no torus. Within this model the only remaining unipotent-image phenomenon is therefore a *compact central-period* phenomenon; nonunipotent algebraic hulls belong to a different mechanism.

Lemma A.5.21 (Bounded-volume subtori of a fixed torus are finite). *Let T be a compact torus equipped with a translation-invariant Riemannian metric. For every $B > 0$ there are only finitely many connected subtori $S < T$ satisfying*

$$\text{vol}(S) \leq B.$$

PROOF. Let

$$q : \mathfrak{t} \longrightarrow T$$

be the universal covering homomorphism and let $\Lambda_T = \ker q$. This is a full Euclidean lattice in $\mathfrak{t}$. If $S < T$ is a connected subtorus and $\mathfrak{s} = \text{Lie}(S)$, then

$$\Lambda_S := \mathfrak{s} \cap \Lambda_T$$

is a full lattice in $\mathfrak{s}$ and $S \simeq \mathfrak{s}/\Lambda_S$. Moreover Λ_S is primitive in Λ_T : if $m\lambda \in \Lambda_S$ with $m \neq 0$ and $\lambda \in \Lambda_T$, then $\lambda \in \mathfrak{s}$ because $\mathfrak{s}$ is a real vector subspace. Thus, if $d = \dim S$, the primitive exterior vector

$$w_S \in \bigwedge^d \Lambda_T$$

of Λ_S satisfies

$$\|w_S\| = \text{covol}_{\mathfrak{s}}(\Lambda_S) = \text{vol}(S).$$

For each d , the fixed exterior lattice $\bigwedge^d \Lambda_T$ has only finitely many points in the radius- B ball. The primitive decomposable vector w_S determines $\mathfrak{s}$, hence S . Summing over $0 \leq d \leq \dim T$ proves the assertion. $\square$

Lemma A.5.22 (Characters of a compact nilmanifold form a discrete lattice). *Let N be a connected simply connected nilpotent Lie group, let $D < N$ be a lattice, and let A be a compact torus. Put*

$$\mathcal{X}_D(N, A) = \{\chi : N \rightarrow A : \chi \text{ continuous homomorphism, } \chi(D) = e\}.$$

Then the differential map

$$\chi \longmapsto d\chi$$

identifies $\mathcal{X}_D(N, A)$ with a discrete additive subgroup of the finite-dimensional vector space

$$\mathrm{Hom}_{\mathbb{R}}(\mathfrak{n}/[\mathfrak{n}, \mathfrak{n}], \mathfrak{a}).$$

Consequently, for every $R > 0$ there are only finitely many $\chi \in \mathcal{X}_D(N, A)$ with $\|d\chi\| \leq R$.

PROOF. Every homomorphism from N to the abelian group A kills the connected commutator subgroup, so its differential kills $[\mathfrak{n}, \mathfrak{n}]$. Since N is connected, a homomorphism is uniquely determined by its differential.

By Theorem A.5.14, applied to the lattice $D < N$, the smallest Lie subalgebra containing $\log D$ is all of $\mathfrak{n}$. Hence one can choose $d_1, \dots, d_r \in D$ such that the classes of

$$X_i = \log d_i$$

span $\mathfrak{n}/[\mathfrak{n}, \mathfrak{n}]$ over $\mathbb{R}$. Let $\Lambda_A = \ker(\exp : \mathfrak{a} \rightarrow A)$ be the period lattice. If $\chi \in \mathcal{X}_D(N, A)$, then

$$\exp(d\chi(X_i)) = \chi(d_i) = e,$$

so $d\chi(X_i) \in \Lambda_A$ for every i .

Choose norms on the abelianization and on $\mathfrak{a}$. Put

$$M = \max_i \|X_i + [\mathfrak{n}, \mathfrak{n}]\|.$$

The period lattice Λ_A has a positive systole η_A . Hence, if $M > 0$ and

$$\|d\chi\| < \eta_A/M,$$

then every period vector satisfies

$$\|d\chi(X_i)\| \leq \|d\chi\| \|X_i + [\mathfrak{n}, \mathfrak{n}]\| < \eta_A.$$

Since $d\chi(X_i) \in \Lambda_A$, all these vectors are zero. Their classes span the abelianization, so $d\chi = 0$. (If the abelianization is zero, every character is already trivial.) Thus 0 is isolated in the set of differentials. That set is an additive subgroup, so it is a discrete subgroup of the finite-dimensional

vector space. A discrete additive subgroup meets a compact set in finitely many points, which proves the final assertion. $\square$

Lemma A.5.23 (Metric density grows along a central graph). *Let E be a finite-dimensional Euclidean vector space and let $E = U \oplus W$ be a fixed direct-sum decomposition. For $\alpha \in \text{Hom}(U, W)$ write*

$$\Gamma_\alpha = \{u + \alpha(u) : u \in U\}.$$

If $J(\alpha)$ denotes the ratio between the Euclidean density on Γ_α and the density transported from U by the projection $\Gamma_\alpha \rightarrow U$, then there are constants $c_0, c_1 > 0$, depending only on the chosen decomposition and Euclidean norm, such that

$$J(\alpha) \geq c_0(1 + c_1\|\alpha\|).$$

In particular, $J(\alpha) \rightarrow \infty$ when $\|\alpha\| \rightarrow \infty$.

PROOF. Let $A_\alpha : U \rightarrow E$ be $u \mapsto u + \alpha(u)$. The desired density ratio is the k -dimensional Jacobian of A_α , up to a fixed positive normalizing factor, where $k = \dim U$. Let $Q : E \rightarrow U$ be the projection along W . Since $QA_\alpha = I_U$, every singular value of A_α is at least $\|Q\|^{-1}$. On the other hand,

$$\|A_\alpha\| \geq \|\alpha\| - 1$$

after replacing the norms by equivalent fixed norms if necessary. Thus the largest singular value grows at least linearly in $\|\alpha\|$, whereas the remaining $k - 1$ singular values have a uniform positive lower bound. Multiplying the singular values gives the stated estimate. $\square$

Proposition A.5.24 (Compact central twisting is finite under a volume bound). *Let*

$$1 \longrightarrow T \longrightarrow P \xrightarrow{\pi} N \longrightarrow 1$$

be an exact sequence of connected Lie groups in which T is a compact central torus and N is connected, simply connected and nilpotent. Fix a discrete subgroup $\Delta < P$ such that

$$D := \pi(\Delta)$$

is a lattice in N . Equip P with a fixed invariant Riemannian density and use the induced densities on its connected subgroups. Then, for every $C > 0$, there are only finitely many connected closed subgroups $H < P$ such that

- (1) $\Delta \subset H$ and Δ is a lattice in H ;
- (2) $\pi(H) = N$;
- (3) $\text{vol}(\Delta \backslash H) \leq C$.

PROOF. Put $E = \Delta \cap T$. Because T is compact and Δ is discrete, E is finite. Let H be admissible and set

$$S = (H \cap T)^0.$$

The homomorphism $H/S \rightarrow N$ is surjective and has discrete kernel $(H \cap T)/S$. It is therefore a covering homomorphism. Since N is simply connected, the covering is an isomorphism. Consequently

$$(5.4) \quad H \cap T = S, \quad E \subset S.$$

We first show that only finitely many S can occur. Fix Haar measure on N . For every admissible H , choose the canonical Haar measure on H for which Weil integration along the central exact sequence

$$1 \rightarrow S \rightarrow H \rightarrow N \rightarrow 1$$

uses the Riemannian Haar measure on S and the fixed Haar measure on N . The quotient bundle

$$\Delta \backslash H \longrightarrow D \backslash N$$

has fiber $E \backslash S$, and quotient integration gives

$$(5.5) \quad \text{covol}_{\text{can}}(\Delta, H) = \text{vol}(E \backslash S) \text{vol}(D \backslash N).$$

The Riemannian Haar measure on H differs from this canonical Haar measure by a positive constant $a(H)$. At the Lie-algebra level, the coarea Jacobian of $d\pi|_{\mathfrak{h}}$ is bounded above by the operator norm of the fixed exterior map $\bigwedge^{\dim N} d\pi$. Hence there is a constant $a_0 > 0$, depending only on the fixed metrics and on π , such that

$$a(H) \geq a_0$$

for every admissible H . Combining this with (5.5) and $\text{vol}(E \backslash S) = \text{vol}(S)/|E|$ gives

$$C \geq \text{vol}(\Delta \backslash H) \geq a_0 \frac{\text{vol}(S)}{|E|} \text{vol}(D \backslash N).$$

Thus $\text{vol}(S)$ is uniformly bounded. By Lemma A.5.21, only finitely many connected subtori $S < T$ can occur.

Fix one such S and suppose that at least one admissible subgroup exists; choose one and call it H_0 . In the quotient P/S , both H/S and H_0/S map isomorphically onto N . Let

$$j, j_0 : N \longrightarrow P/S$$

be the inverse homomorphisms. Since T/S is central, the map

$$\chi_H(x) = j(x)j_0(x)^{-1} \in T/S$$

is a continuous homomorphism. Every $d \in D$ has a lift $\delta \in \Delta$, and the class δS belongs to both subgroup sections. Therefore

$$\chi_H(D) = e.$$

Conversely, the subgroup H determines χ_H , so distinct admissible H give distinct characters.

It remains to use the volume bound. At the Lie-algebra level, after quotienting by $\mathfrak{s}$, the spaces $\mathfrak{h}/\mathfrak{s}$ and $\mathfrak{h}_0/\mathfrak{s}$ are two complements of $\mathfrak{t}/\mathfrak{s}$. Relative to the reference complement $\mathfrak{h}_0/\mathfrak{s}$, the first is precisely the graph of $d\chi_H$. The ratio $a(H)$ between the Riemannian density and the canonical product density is therefore, up to a fixed positive factor, the graph density from Lemma A.5.23. Hence there are constants $b_0, b_1 > 0$, depending on the fixed S and H_0 , such that

$$a(H) \geq b_0(1 + b_1 \|d\chi_H\|).$$

For the fixed S , equation (5.5) has a fixed positive right-hand side. Thus the bound $\text{vol}(\Delta \backslash H) \leq C$ gives a uniform upper bound for $\|d\chi_H\|$. Lemma A.5.22, applied to $A = T/S$, says that only finitely many such characters exist. Hence only finitely many H occur for this S . Since there were only finitely many possible S , the proof is complete. $\square$

THEOREM A.5.25 (Fixed-intersection reconstruction: the entire unipotent-image branch). *In the setting of Definition A.5.40, suppose that*

$$L = \overline{\rho(\Delta)}^Z$$

is a real unipotent algebraic group. Then, for every $C > 0$, only finitely many connected closed subgroups $H < G$ satisfy conditions (1)–(3) of Definition A.5.40. Thus fixed-intersection reconstruction is completely proved for unipotent algebraic image, with no restriction on the connected central kernel.

PROOF. If no admissible subgroup exists, there is nothing to prove. Fix one admissible subgroup temporarily and put

$$P = (\rho^{-1}(L(\mathbb{R})))^0, \quad F = P \cap \ker \rho.$$

Every admissible H lies in P and, by Lemma A.5.16, satisfies $\rho(H) = L(\mathbb{R})$. In particular the temporarily chosen subgroup shows that $\rho : P \rightarrow L(\mathbb{R})$ is surjective; the group P is therefore fixed by L and ρ , not by that choice. The kernel F is central. It is also connected: the homotopy exact sequence for the Lie-group bundle $F \rightarrow P \rightarrow L(\mathbb{R})$, together with connectedness of P and simple connectedness of the unipotent group $L(\mathbb{R})$, gives $\pi_0(F) = 0$. Hence

$$F \simeq \mathbb{R}^a \times T$$

with T its maximal compact torus.

Let

$$q : P \longrightarrow \bar{P} = P/T.$$

Then F/T is a vector group and

$$1 \rightarrow F/T \rightarrow \bar{P} \rightarrow L(\mathbb{R}) \rightarrow 1$$

is a central extension. The base and fiber are contractible, so $\bar{P}$ is simply connected. Its Lie algebra is a central extension of the nilpotent Lie algebra of L by an abelian Lie algebra, hence is nilpotent. Therefore $\bar{P}$ is a connected simply connected nilpotent Lie group.

Because T is compact, q is proper. Thus

$$\bar{\Delta} = q(\Delta)$$

is discrete. For any admissible H , the image $\bar{H} = q(H)$ is connected, contains $\bar{\Delta}$, and $\bar{\Delta} \backslash \bar{H}$ has finite Haar volume: it is the base of the compact-fiber quotient of $\Delta \backslash H$. By Theorem A.5.14, the connected subgroup $\bar{H} < \bar{P}$ is uniquely determined by $\bar{\Delta}$. Denote this fixed subgroup by $\bar{H}_0$.

The preimage

$$P_0 = q^{-1}(\bar{H}_0)$$

is connected, because both the kernel T and the base $\bar{H}_0$ are connected. We now have a fixed central extension

$$1 \longrightarrow T \longrightarrow P_0 \xrightarrow{q} \bar{H}_0 \longrightarrow 1,$$

where T is a compact torus and $\bar{H}_0$ is connected simply connected nilpotent. Every admissible H is a subgroup of P_0 , contains the fixed Δ , maps onto $\bar{H}_0$, and has covolume at most C . Proposition A.5.24 therefore gives only finitely many possibilities for H . $\square$

Remark A.5.26 (Scope of the second structural core). Theorem A.5.25 absorbs both earlier special cases A.5.17 and A.5.19. The compact central-period phenomenon is no longer open: its integer winding parameters form the discrete character group of Lemma A.5.22, and the homogeneous volume bound forces those parameters into a bounded set. The argument therefore isolates genuinely nonunipotent algebraic hulls as the cases requiring a different reconstruction mechanism.

5.1. Nilpotent preimages: compact periods are the only global obstruction. The preceding unipotent-image theorem used the polynomial structure of the algebraic image in order to make the fixed preimage nilpotent. For the reconstruction problem itself, that is more than is needed. Once the fixed preimage has nilpotent Lie algebra, all global ambiguity is concentrated in a compact central torus. Quotienting by that torus produces a fixed

simply connected nilpotent ambient group, where the lattice determines its connected syndetic hull uniquely.

Lemma A.5.27 (Compact-torus reduction of a connected nilpotent Lie group). *Let P be a connected Lie group whose Lie algebra $\mathfrak{p}$ is nilpotent. Then there is a connected compact central torus $T < P$ such that*

$$\bar{P} := P/T$$

is connected and simply connected nilpotent. In particular, the quotient map $q : P \rightarrow \bar{P}$ is proper.

PROOF. Let

$$\pi : \tilde{P} \longrightarrow P$$

be the universal covering homomorphism. Its kernel Z_π is discrete and, because P is connected, central in $\tilde{P}$. The simply connected Lie group $\tilde{P}$ has nilpotent Lie algebra $\mathfrak{p}$; hence Lemma A.5.12 gives a global diffeomorphism

$$\exp : \mathfrak{p} \longrightarrow \tilde{P}.$$

Since Z_π is central, every $z \in Z_\pi$ has $\log z \in \mathfrak{z}(\mathfrak{p})$. On the center the BCH law is ordinary addition, so

$$\Lambda_\pi := \log Z_\pi$$

is an additive subgroup of $\mathfrak{z}(\mathfrak{p})$; it is discrete because $\exp$ is a homeomorphism. Put

$$\mathfrak{t} = \text{span}_{\mathbb{R}} \Lambda_\pi, \quad \tilde{T} = \exp(\mathfrak{t}).$$

The elementary Euclidean-lattice argument used in the proof of Theorem A.5.3 shows that Λ_π is a full lattice in $\mathfrak{t}$. Hence

$$T := \pi(\tilde{T}) \simeq \mathfrak{t}/\Lambda_\pi$$

is a compact connected torus. It is central because $\mathfrak{t} \subset \mathfrak{z}(\mathfrak{p})$.

Now

$$P/T \simeq \tilde{P}/\tilde{T}.$$

Because $\mathfrak{t}$ is a central ideal, the terminating BCH law descends to $\mathfrak{p}/\mathfrak{t}$, and the quotient on the right is precisely the simply connected BCH group attached to the nilpotent Lie algebra $\mathfrak{p}/\mathfrak{t}$. It is therefore connected, simply connected and nilpotent. Finally, quotient by a compact subgroup is a proper map: the inverse image of a compact set is locally a compact bundle with compact fiber T , hence compact. $\square$

THEOREM A.5.28 (Fixed-intersection reconstruction in a nilpotent preimage). *Let $P < G$ be a fixed connected closed subgroup whose Lie algebra is nilpotent. Fix a discrete subgroup $\Delta < P$ and $C > 0$. Then there are only finitely many connected closed subgroups $H < P$ such that*

- (1) $\Delta \subset H$ and Δ is a lattice in H ;
- (2) $\text{vol}_\sigma(\Delta \backslash H) \leq C$.

PROOF. Choose the compact central torus $T < P$ supplied by Lemma A.5.27, and write

$$q : P \longrightarrow \bar{P} = P/T.$$

The group $\bar{P}$ is connected, simply connected and nilpotent, and q is proper. Consequently

$$\bar{\Delta} := q(\Delta)$$

is discrete. Indeed, if infinitely many distinct points of $\bar{\Delta}$ met one compact set of $\bar{P}$, their lifts in Δ would meet the compact inverse image of that set infinitely often, contradicting discreteness of Δ .

Let H be admissible and put $\bar{H} = q(H)$. Properness of q makes $\bar{H}$ closed, and connectedness of H makes $\bar{H}$ connected. The quotient map induces a continuous surjection

$$\Delta \backslash H \longrightarrow \bar{\Delta} \backslash \bar{H}.$$

Push normalized Haar measure on the finite-volume space $\Delta \backslash H$ through this map. The push-forward is a finite nonzero $\bar{H}$ -invariant measure. Hence $\bar{\Delta}$ is a lattice in $\bar{H}$.

The ambient group $\bar{P}$ is now fixed and simply connected nilpotent. Theorem A.5.14 therefore says that there is at most one connected closed subgroup $\bar{H}_0 < \bar{P}$ containing $\bar{\Delta}$ as a lattice. Thus every admissible H has

$$q(H) = \bar{H}_0.$$

If no such $\bar{H}_0$ exists there are no admissible groups, so assume it exists and put

$$P_0 = q^{-1}(\bar{H}_0).$$

Because both T and $\bar{H}_0$ are connected, P_0 is connected. We have a fixed central extension

$$1 \longrightarrow T \longrightarrow P_0 \xrightarrow{q} \bar{H}_0 \longrightarrow 1,$$

with compact torus kernel and connected simply connected nilpotent base. Every admissible H lies in P_0 , contains the fixed subgroup Δ , maps onto $\bar{H}_0$, and has covolume at most C . Proposition A.5.24 now gives only finitely many possibilities for H . $\square$

Corollary A.5.29 (Fixed-intersection reconstruction for nilpotent algebraic image). *In the setting of Definition A.5.40, put*

$$L = \overline{\rho(\Delta)}^Z.$$

Assume that the Lie algebra of the identity component $L(\mathbb{R})^0$ is nilpotent. Then, for every $C > 0$, only finitely many connected closed subgroups $H <$

G satisfy conditions (1)–(3) of that theorem. Thus the fixed-intersection problem is completely solved for every fixed algebraic image with nilpotent identity component, including algebraic tori and mixed central-torus/unipotent nilpotent images.

PROOF. If there is no admissible H , there is nothing to prove. Otherwise set

$$P = (\rho^{-1}(L(\mathbb{R})^0))^0.$$

For every admissible H , the connected group $\rho(H)$ lies in $L(\mathbb{R})^0$, so $H \subset P$ and hence $\Delta \subset P$. At the Lie-algebra level the kernel of

$$d\rho : \mathfrak{p} \longrightarrow d\rho(\mathfrak{p})$$

is central, while $d\rho(\mathfrak{p})$ is a Lie subalgebra of the nilpotent Lie algebra $\mathrm{Lie}(L(\mathbb{R})^0)$. A central extension of a nilpotent Lie algebra is nilpotent: if the image has nilpotency class c , then every $(c+1)$ -fold iterated bracket in $\mathfrak{p}$ lies in the central kernel, and every $(c+2)$ -fold bracket vanishes. Thus $\mathfrak{p}$ is nilpotent. Theorem A.5.28 applies to this fixed P and proves the required finiteness. $\square$

Worked Example A.5.30 (A toral algebraic hull need not have a toral support). Take a connected nilpotent group P with a compact central circle and let a one-parameter subgroup wind nontrivially around that circle while escaping in a noncompact central direction. Its image in a faithful diagonal representation may have a nonunipotent toral Zariski closure even though the Lie algebra of P is nilpotent. The corollary handles precisely this kind of example: one first removes the compact period torus of the *ambient preimage*, not the torus in the algebraic image, and then uniqueness of the nilpotent syndetic hull downstairs leaves only the discrete compact-central winding parameters controlled by the volume bound.

Remark A.5.31 (Where the nilpotent reconstruction mechanism stops). The corollary above includes unipotent images and also nonunipotent nilpotent algebraic images, in particular reductive algebraic tori and nilpotent torus-by-unipotent identity components. The semisimple argument below treats semisimple algebraic image with discrete central kernel.

What these two mechanisms do not by themselves cover is the genuinely *mixed Levi* situation: algebraic hulls with a nontrivial semisimple derived part together with a nontrivial radical or central quotient, and the semisimple case with a positive-dimensional central kernel. In those cases the fixed preimage is not nilpotent, so the syndetic-hull uniqueness theorem used above cannot be applied before the semisimple and central directions have been separated.

5.2. A nonunipotent branch: algebraic hulls with semisimple image. The unipotent reconstruction theorem above uses polynomial exponential coordinates. That mechanism is unavailable as soon as the fixed algebraic image has a nontrivial semisimple part. There is nevertheless an important nonunipotent case in which the image is forced for a completely different reason. The needed fact is a theorem of Chevalley about algebraic hulls of linear Lie algebras. Since it is load-bearing here, we record the mechanism rather than use the theorem as a name.

Lemma A.5.32 (The derived radical acts nilpotently). *Let $\mathfrak{a} < \mathfrak{gl}(V)$ be a finite-dimensional real linear Lie algebra and let $\mathfrak{r}$ be its solvable radical. Then every element of $[\mathfrak{a}, \mathfrak{r}]$ is a nilpotent endomorphism of V .*

PROOF. Complexification does not change the assertion, so work over $\mathbb{C}$. By Lie's theorem the representation of the solvable algebra $\mathfrak{r}$ admits a complete invariant flag. Relative to a basis adapted to this flag, every $y \in \mathfrak{r}$ is upper triangular. Write $\lambda_1, \dots, \lambda_m \in \mathfrak{r}^*$ for the diagonal characters, with repetitions allowed.

We must explain why every λ_i vanishes on $[\mathfrak{a}, \mathfrak{r}]$. For $x \in \mathfrak{a}$, conjugation by e^{tx} sends the $\mathfrak{r}$ -module V to the module obtained by precomposing with $e^{-t \operatorname{ad} x}|_{\mathfrak{r}}$; here we use that $\mathfrak{r}$ is an ideal. Consequently the finite multiset of composition characters is carried to

$$\{\lambda_i \circ e^{-t \operatorname{ad} x} : 1 \leq i \leq m\}.$$

It depends continuously on t but takes values in the finite multiset $\{\lambda_1, \dots, \lambda_m\}$. Hence each character is fixed for all t . Differentiating at $t = 0$ gives

$$\lambda_i([x, y]) = 0 \quad (x \in \mathfrak{a}, y \in \mathfrak{r}).$$

Thus every element of $[\mathfrak{a}, \mathfrak{r}] \subset \mathfrak{r}$ is upper triangular with zero diagonal, and is therefore nilpotent. $\square$

Lemma A.5.33 (The derived algebra of a linear Lie algebra is algebraic). *For a finite-dimensional real linear Lie algebra $\mathfrak{a} < \mathfrak{gl}(V)$, the derived algebra $[\mathfrak{a}, \mathfrak{a}]$ is an algebraic Lie algebra: it is the Lie algebra of a real linear algebraic subgroup of $\operatorname{GL}(V)$.*

PROOF. Again we may complexify and descend at the end. Choose a Levi decomposition

$$\mathfrak{a} = \mathfrak{s} \ltimes \mathfrak{r}.$$

Then

$$(A.1L) \quad [\mathfrak{a}, \mathfrak{a}] = \mathfrak{s} + [\mathfrak{a}, \mathfrak{r}].$$

Put $\mathfrak{n} = [\mathfrak{a}, \mathfrak{r}]$. By Lemma A.5.32, every element of $\mathfrak{n}$ is a nilpotent matrix. Engel's theorem makes $\mathfrak{n}$ a nilpotent Lie algebra. Moreover it is algebraic.

Indeed, on the variety of unipotent matrices the finite exponential and logarithm series are mutually inverse polynomial maps; hence

$$N := \exp(\mathfrak{n}) = \{u \text{ unipotent} : \log u \in \mathfrak{n}\}$$

is Zariski closed and $\text{Lie}(N) = \mathfrak{n}$.

We also use the elementary algebraicity theorem for semisimple linear Lie algebras. For completeness, its construction is as follows. Decompose the complex semisimple algebra $\mathfrak{s}$ into simple factors, take the simply connected semisimple algebraic group S_{sc} with that Lie algebra, and integrate the given finite-dimensional $\mathfrak{s}$ -module. The usual highest-weight construction integrates each irreducible summand to a rational representation of S_{sc} . Its algebraic image $S < \text{GL}(V)$ therefore has Lie algebra equal to the given image of $\mathfrak{s}$. The construction is stable under the real conjugation, so the real form descends.

Since $\mathfrak{n}$ is an ideal of $\mathfrak{a}$, the connected algebraic group S normalizes N . Thus SN is an algebraic subgroup and, by (A.1),

$$\text{Lie}(SN) = \mathfrak{s} + \mathfrak{n} = [\mathfrak{a}, \mathfrak{a}].$$

This proves the assertion. $\square$

THEOREM A.5.34 (Chevalley derived-hull theorem, in the form used here). *Let $\mathfrak{a} < \mathfrak{gl}(V)$ be a finite-dimensional real linear Lie algebra, and let $\mathfrak{a}^{\text{alg}}$ denote its algebraic hull, i.e. the smallest algebraic Lie subalgebra of $\mathfrak{gl}(V)$ containing $\mathfrak{a}$. Then*

$$\boxed{[\mathfrak{a}^{\text{alg}}, \mathfrak{a}^{\text{alg}}] = [\mathfrak{a}, \mathfrak{a}].}$$

PROOF. Set $\mathfrak{d} = [\mathfrak{a}, \mathfrak{a}]$. By Lemma A.5.33, $\mathfrak{d}$ is algebraic. It is an ideal of $\mathfrak{a}$. The algebraic normalizer $N_{\text{GL}(V)}(\mathfrak{d})$ therefore contains the connected analytic subgroup with Lie algebra $\mathfrak{a}$, hence also its Zariski closure. It follows that $\mathfrak{d}$ is an ideal of $\mathfrak{a}^{\text{alg}}$.

Pass to the algebraic quotient by $\mathfrak{d}$. The connected analytic subgroup generated by the image of $\mathfrak{a}$ is abelian, because its Lie algebra is $\mathfrak{a}/\mathfrak{d}$. The Zariski closure of an abelian subgroup is abelian: the commutator map is algebraic, and it is identically e on the dense product of the subgroup with itself. Therefore

$$[\mathfrak{a}^{\text{alg}}, \mathfrak{a}^{\text{alg}}] \subset \mathfrak{d}.$$

Conversely, $\mathfrak{a} \subset \mathfrak{a}^{\text{alg}}$ implies that every bracket of two elements of $\mathfrak{a}$ is also a bracket of two elements of $\mathfrak{a}^{\text{alg}}$. Hence

$$\mathfrak{d} = [\mathfrak{a}, \mathfrak{a}] \subset [\mathfrak{a}^{\text{alg}}, \mathfrak{a}^{\text{alg}}],$$

which is the reverse inclusion. $\square$

Corollary A.5.35 (A connected Zariski-dense subgroup of a semisimple hull has full Lie algebra). *Let $L < \mathrm{GL}(V)$ be a connected real semisimple algebraic group and let $J < L(\mathbb{R})$ be a connected Lie subgroup whose Zariski closure is L . Then*

$$\mathrm{Lie}(J) = \mathrm{Lie}(L), \quad J = L(\mathbb{R})^0.$$

PROOF. Put $\mathfrak{j} = \mathrm{Lie}(J)$ and $\mathfrak{l} = \mathrm{Lie}(L)$. The algebraic hull of $\mathfrak{j}$ is $\mathfrak{l}$: the Zariski closure of the connected analytic subgroup J is L . Hence Theorem A.5.34 gives

$$[\mathfrak{j}, \mathfrak{j}] = [\mathfrak{l}, \mathfrak{l}].$$

Semisimplicity gives $[\mathfrak{l}, \mathfrak{l}] = \mathfrak{l}$. Therefore

$$\mathfrak{l} = [\mathfrak{j}, \mathfrak{j}] \subset \mathfrak{j} \subset \mathfrak{l},$$

so $\mathfrak{j} = \mathfrak{l}$. A connected Lie subgroup with the full Lie algebra is the identity component. $\square$

Proposition A.5.36 (Fixed-intersection reconstruction: semisimple image and discrete central kernel). *In the setting of Definition A.5.40, assume that*

$$L = \overline{\rho(\Delta)}^Z$$

is a connected semisimple real algebraic group and that $\ker \rho$ is discrete. Then there is at most one admissible connected subgroup $H < G$. In particular, fixed-intersection reconstruction is unique in this branch, and no volume bound is needed once existence is known.

PROOF. For an admissible H , condition (3) says that the Zariski closure of $\rho(H)$ is L . Corollary A.5.35 therefore gives

$$\rho(H) = L(\mathbb{R})^0.$$

Let

$$P = (\rho^{-1}(L(\mathbb{R})^0))^0.$$

Because $\ker \rho$ is discrete, the differential

$$d\rho : \mathrm{Lie}(P) \longrightarrow \mathfrak{l}$$

is injective; surjectivity follows from the displayed image equality for one admissible H . Thus $d\rho$ is an isomorphism. Since $H \subset P$ and $d\rho(\mathrm{Lie} H) = \mathfrak{l}$, we obtain

$$\mathrm{Lie} H = \mathrm{Lie} P.$$

Both groups are connected, hence $H = P$. The subgroup is therefore unique. $\square$

Worked Example A.5.37 (Why semisimplicity removes the irrational-slope phenomenon). Inside a torus, a connected one-parameter subgroup can be proper and still be Zariski dense; Example A.5.4 is the corresponding central-winding warning. The corollary shows that this cannot happen inside a semisimple algebraic hull. For instance, a connected Lie subgroup $J < \mathrm{SL}_2(\mathbb{R})$ whose Zariski closure is all of SL_2 must already have Lie algebra $\mathfrak{sl}_2(\mathbb{R})$, and hence $J = \mathrm{SL}_2(\mathbb{R})$. The reason is not dimension counting: it is the perfectness $[\mathfrak{sl}_2, \mathfrak{sl}_2] = \mathfrak{sl}_2$ combined with the Chevalley derived-hull theorem.

Remark A.5.38 (Where the semisimple reconstruction mechanism stops). The unipotent argument handles the entire unipotent-image branch, and Corollary A.5.29 handles every algebraic image whose identity component has nilpotent Lie algebra, including toral and nilpotent torus-by-unipotent images. The proposition above treats a different nonunipotent branch: semisimple algebraic image with discrete central kernel. The cases not covered by these elementary reconstruction mechanisms are therefore genuinely mixed Levi hulls, where a nontrivial semisimple derived part coexists with a nontrivial radical or central quotient, together with the semisimple branch when the central kernel has positive dimension. Those cases can carry continuous central twisting coupled to a nonnilpotent direction, so neither the nilpotent nor the semisimple argument may be extrapolated to them.

Remark A.5.39 (A useful non-load-bearing Dani–Margulis benchmark). One natural way to try to reconstruct the Dani–Margulis theorem is to first prove the following restricted statement: for fixed $C > 0$, among $H \in \mathcal{H}_\Gamma^{\mathrm{alg}}$ with

$$\mathrm{vol}_\sigma((H \cap \Gamma) \backslash H) \leq C,$$

only finitely many subgroups $H \cap \Gamma$ occur. We do *not* use this benchmark as a theorem anywhere in the book. This benchmark is not part of the load-bearing dependency graph; the external Dani–Margulis interface used below consists instead of the precise countability and Chevalley-discreteness statements recorded in Theorem A.5.42.

The preceding internal arguments are nevertheless useful. They prove this kind of finiteness completely for moving abelian subgroups, prove the compact ambient case, and show that nilpotent lattice quotients have no cuspidal escape. These calculations isolate the two genuinely difficult features of the general theorem: a non-nilpotent quotient can have cusps, and both its arithmetic lattice and its Levi/radical geometry may move at the same time. Thus this remark is a guide to the mechanism, not an unfinished proof obligation.

Definition A.5.40 (Fixed-intersection reconstruction setup). Let $\rho : G \rightarrow \mathrm{GL}(V)$ be a finite-dimensional representation whose kernel is central in the connected group G . Fix a subgroup $\Delta < \Gamma$ and $C > 0$. We call a connected closed subgroup $H < G$ *admissible for the fixed-intersection setup* if

- (1) $H \cap \Gamma = \Delta$ and Δ is a lattice in H ;
- (2) $\mathrm{vol}_\sigma(\Delta \backslash H) \leq C$;
- (3) $\rho(\Delta)$ is Zariski dense in $\rho(H)$.

Remark A.5.41 (Why the fixed-intersection setup is still worth studying). Once Δ is fixed, Zariski density fixes the algebraic hull of $\rho(H)$, but it need not fix the connected Lie subgroup H . Example A.5.4 exhibits the basic central-winding phenomenon. The propositions preceding this definition prove finiteness or uniqueness in broad special branches: unipotent image, nilpotent algebraic image, and semisimple image with discrete central kernel.

Those results are retained because they make the arithmetic mechanisms behind homogeneous-support finiteness transparent and provide reusable lemmas elsewhere. They are not assembled into a general bounded-volume theorem, because no such theorem is needed in the load-bearing proof. Mixed Levi hulls and positive-dimensional central kernels therefore remain examples of what a stronger bounded-volume theorem would have to control, not unresolved dependencies of the book.

THEOREM A.5.42 (Dani–Margulis structural interface). *Let $\mathcal{H}_\Gamma$ be the support class of Definition 6.2.1. The following two facts are used as the sole external research input in the qualitative Ratner part of this book.*

- (1) *The collection $\mathcal{H}_\Gamma$ is countable.*
- (2) *For every $H \in \mathcal{H}_\Gamma$, if $d = \dim H$ and $0 \neq p_H \in \bigwedge^d \mathfrak{h}$, then, after quotienting by the possible sign ambiguity, the orbit*

$$\Gamma \bar{p}_H \subset (\bigwedge^d \mathfrak{g}) / \{\pm 1\}$$

is closed and discrete.

EXTERNAL INPUT: DANI–MARGULIS. These are the precise Dani–Margulis facts used downstream. In the right-quotient convention of this book, the first is the support-class countability statement appearing as Proposition 2.1 of Dani–Margulis (1993); the second is their Theorem 3.4, the exterior-power orbit discreteness theorem. The latter is stated there in the corresponding left-quotient convention; inversion identifies the two formulations and does not change discreteness.

This formulation deliberately avoids promoting a stronger bounded-volume finiteness statement to an external theorem unless that stronger statement is actually needed. It is not needed here. The preceding sections and the

worked examples below retain the bounded-volume calculations because they make the arithmetic mechanism visible in important special families, but those calculations are pedagogical rather than load-bearing. $\square$

5.3. Worked arithmetic models for bounded-volume finiteness.

The external interface above supplies exactly the countability and Chevalley-discreteness facts used by Ratner's measure-classification argument. Independently of that interface, it is useful to see what a homogeneous-volume bound measures in the basic arithmetic space

$$X_n = \mathrm{SL}_n(\mathbb{R})/\mathrm{SL}_n(\mathbb{Z}).$$

The examples below are special cases in which the finiteness mechanism can be seen by an elementary lattice calculation. They are illustrative special cases rather than a claim of a general bounded-volume classification; nonabelian families with simultaneously moving lattice intersections and Levi factors require substantially more structure.

Example A.5.43 (Rational horocycles in $\mathrm{SL}_2(\mathbb{R})/\mathrm{SL}_2(\mathbb{Z})$). Equip $\mathfrak{sl}_2(\mathbb{R})$ with the Frobenius norm. Let

$$N = \left\{ \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} : t \in \mathbb{R} \right\}, \quad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

A primitive vector $v = (p, q)^T \in \mathbb{Z}^2$ can be completed to a matrix

$$\gamma = \begin{pmatrix} p & r \\ q & s \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

Put $H_v = \gamma N \gamma^{-1}$. Its Lie algebra is generated by

$$X_v = \gamma E \gamma^{-1} = \begin{pmatrix} -pq & p^2 \\ -q^2 & pq \end{pmatrix}.$$

Because $X_v^2 = 0$,

$$H_v = \{I + tX_v : t \in \mathbb{R}\}.$$

The entries of X_v have greatest common divisor 1: if a prime divided p^2 , q^2 , and pq , it would divide both p and q . Hence

$$H_v \cap \mathrm{SL}_2(\mathbb{Z}) = \{I + mX_v : m \in \mathbb{Z}\}.$$

Thus one fundamental period is $0 \leq t \leq 1$. Right translation identifies the velocity of the path $t \mapsto I + tX_v$ with the constant vector X_v , so the homogeneous covolume is

$$\mathrm{vol}(H_v/(H_v \cap \Gamma)) = \|X_v\|_F = \sqrt{p^4 + 2p^2q^2 + q^4} = p^2 + q^2.$$

Consequently

$$\mathrm{vol}(H_v/(H_v \cap \Gamma)) \leq C \implies p^2 + q^2 \leq C.$$

There are only finitely many primitive integer vectors in that Euclidean ball, hence only finitely many such rational horocycle subgroups. This is a literal one-dimensional model of bounded-volume finiteness.

The same calculation also explains why the volume hypothesis is essential. For $v_m = (m, 1)$ the subgroups H_{v_m} are pairwise distinct (up to the irrelevant replacement v by $-v$), while

$$\text{vol}(H_{v_m}/(H_{v_m} \cap \Gamma)) = m^2 + 1 \longrightarrow \infty.$$

So there are infinitely many rational horocycles altogether; finiteness only appears after imposing a uniform volume bound.

Example A.5.44 (Primitive rank-one unipotents in SL_n). The preceding computation has a direct n -dimensional analogue. Choose a primitive column vector $v \in \mathbb{Z}^n$ and a primitive integral covector $\lambda \in (\mathbb{Z}^n)^*$ with $\lambda(v) = 0$, and set

$$X = v\lambda.$$

Then $\text{tr } X = \lambda(v) = 0$ and $X^2 = v\lambda(v)\lambda = 0$, so

$$H_{v,\lambda} = \{I + tX : t \in \mathbb{R}\} < \text{SL}_n(\mathbb{R})$$

is a one-parameter unipotent subgroup. Since both v and λ are primitive, the set of entries of X has greatest common divisor 1. Therefore

$$H_{v,\lambda} \cap \text{SL}_n(\mathbb{Z}) = \{I + mX : m \in \mathbb{Z}\}.$$

With the Frobenius metric the quotient again has fundamental period one and

$$\text{vol}(H_{v,\lambda}/(H_{v,\lambda} \cap \Gamma)) = \|X\|_F = \|v\|_2 \|\lambda\|_2.$$

Hence a bound by C forces both $\|v\|_2 \leq C$ and $\|\lambda\|_2 \leq C$. Only finitely many primitive integral pairs satisfy these bounds. In this family the Dani–Margulis conclusion therefore reduces to the elementary fact that a bounded subset of an integral lattice is finite.

Example A.5.45 (A higher-dimensional block-unipotent family and the Chevalley wedge). Fix $1 \leq k < n$ and let

$$U_{k,n-k} = \left\{ \begin{pmatrix} I_k & Y \\ 0 & I_{n-k} \end{pmatrix} : Y \in M_{k,n-k}(\mathbb{R}) \right\}.$$

This is an abelian unipotent group of dimension $d = k(n-k)$, and $U_{k,n-k} \cap \text{SL}_n(\mathbb{Z})$ is obtained by requiring Y to have integral entries. Let

$$p_U = \bigwedge_{1 \leq i \leq k, 1 \leq j \leq n-k} E_{i,k+j} \in \bigwedge^d \mathfrak{sl}_n(\mathbb{Z}),$$

with any fixed ordering of the factors. For $\gamma \in \mathrm{SL}_n(\mathbb{Z})$ put $U_\gamma = \gamma U_{k,n-k} \gamma^{-1}$. The conjugation-Jacobian formula proved below gives

$$\mathrm{vol}(U_\gamma / (U_\gamma \cap \Gamma)) = \frac{\|\bigwedge^d \mathrm{Ad}(\gamma) p_U\|}{\|p_U\|} \mathrm{vol}(U_{k,n-k} / (U_{k,n-k} \cap \Gamma)).$$

But

$$\bigwedge^d \mathrm{Ad}(\gamma) p_U \in \bigwedge^d \mathfrak{sl}_n(\mathbb{Z}).$$

A uniform volume bound therefore confines this exterior-power vector to a bounded subset of a discrete lattice, so only finitely many such wedge vectors occur. Each nonzero decomposable wedge determines its d -plane, hence determines the connected Lie subgroup U_γ . Thus bounded-volume finiteness is again visible directly.

This is the useful bridge to the general proof architecture. The exterior vector is a concrete Chevalley coordinate for the subgroup. In the easy arithmetic family above, integrality plus bounded norm immediately gives finiteness. The full Dani–Margulis theorem may be viewed as the much deeper statement that an analogous discreteness survives when the subgroup is not a single integral conjugacy family and when nonabelian, semisimple, and central factors must all be controlled at once.

Example A.5.46 (A semisimple block inside SL_n). For $2 \leq k < n$, embed

$$H_0 = \left\{ \begin{pmatrix} A & 0 \\ 0 & I_{n-k} \end{pmatrix} : A \in \mathrm{SL}_k(\mathbb{R}) \right\} < \mathrm{SL}_n(\mathbb{R}).$$

Its arithmetic lattice is the corresponding copy of $\mathrm{SL}_k(\mathbb{Z})$. Choose an integral basis of $\mathfrak{h}_0 \cap \mathfrak{sl}_n(\mathbb{Z})$ and let $p_{H_0} \in \bigwedge^{k^2-1} \mathfrak{sl}_n(\mathbb{Z})$ be its wedge. For $H_\gamma = \gamma H_0 \gamma^{-1}$ with $\gamma \in \mathrm{SL}_n(\mathbb{Z})$, Lemma A.5.50 gives

$$\mathrm{vol}(H_\gamma / (H_\gamma \cap \Gamma)) = J_{H_0}(\gamma) \mathrm{vol}(H_0 / (H_0 \cap \Gamma)),$$

where

$$J_{H_0}(\gamma) = \frac{\|\bigwedge^{k^2-1} \mathrm{Ad}(\gamma) p_{H_0}\|}{\|p_{H_0}\|}.$$

Again the numerator is the norm of an integral exterior vector. Therefore a uniform covolume bound permits only finitely many such connected conjugate subgroups. This example is nonabelian and semisimple, but it still lies in a single arithmetic conjugacy family; that is why the elementary wedge argument works here without the full Dani–Margulis machinery.

Example A.5.47 (A mixed Levi–unipotent family in SL_{a+b}). Fix $a, b \geq 2$ and put $n = a + b$. Consider

$$H_{a,b} = \left\{ \begin{pmatrix} A & Y \\ 0 & B \end{pmatrix} : A \in \mathrm{SL}_a(\mathbb{R}), B \in \mathrm{SL}_b(\mathbb{R}), Y \in M_{a,b}(\mathbb{R}) \right\}.$$

Multiplication is

$$(A, B, Y)(A', B', Y') = (AA', BB', AY' + YB')$$

in block-matrix notation. More invariantly,

$$H_{a,b} \simeq (\mathrm{SL}_a(\mathbb{R}) \times \mathrm{SL}_b(\mathbb{R})) \ltimes M_{a,b}(\mathbb{R}),$$

where (A, B) acts on Y by $Y \mapsto AYB^{-1}$ after the usual semidirect-product reparametrization. Thus the Levi factor and the unipotent radical are simultaneously nontrivial.

The group is generated by one-parameter unipotent root groups. Indeed, elementary matrices generate each SL block, while

$$u_{ij}(t) = I_n + tE_{i,a+j}, \quad 1 \leq i \leq a, \quad 1 \leq j \leq b,$$

generate the block-unipotent radical. Hence this is a genuine model of the class of groups occurring in the Dani–Margulis support theory, not merely an abelian or semisimple toy.

Its arithmetic lattice is

$$\begin{aligned} \Delta_{a,b} &= H_{a,b} \cap \mathrm{SL}_n(\mathbb{Z}) \\ &= \left\{ \begin{pmatrix} A & Y \\ 0 & B \end{pmatrix} : \begin{array}{l} A \in \mathrm{SL}_a(\mathbb{Z}), \quad B \in \mathrm{SL}_b(\mathbb{Z}), \\ Y \in M_{a,b}(\mathbb{Z}) \end{array} \right\}. \end{aligned}$$

To see directly why it has finite covolume, first quotient the normal unipotent radical

$$U = M_{a,b}(\mathbb{R}).$$

The fiber quotient is the torus

$$M_{a,b}(\mathbb{Z}) \backslash M_{a,b}(\mathbb{R}).$$

The Levi action preserves its Lebesgue volume because

$$\det(Y \mapsto AYB^{-1}) = (\det A)^b (\det B)^{-a} = 1.$$

The base is

$$\mathrm{SL}_a(\mathbb{Z}) \backslash \mathrm{SL}_a(\mathbb{R}) \times \mathrm{SL}_b(\mathbb{Z}) \backslash \mathrm{SL}_b(\mathbb{R}),$$

which has finite Haar volume. Quotient integration therefore gives finite volume for $\Delta_{a,b} \backslash H_{a,b}$.

Now choose an integral basis

$$X_1, \dots, X_d \quad \text{of} \quad \mathfrak{h}_{a,b} \cap \mathfrak{sl}_n(\mathbb{Z})$$

and put

$$p_{a,b} = X_1 \wedge \cdots \wedge X_d.$$

For every $\gamma \in \mathrm{SL}_n(\mathbb{Z})$ let

$$H_\gamma = \gamma H_{a,b} \gamma^{-1}.$$

Then

$$H_\gamma \cap \mathrm{SL}_n(\mathbb{Z}) = \gamma \Delta_{a,b} \gamma^{-1},$$

and Lemma A.5.50 gives

$$\mathrm{vol}((H_\gamma \cap \Gamma) \backslash H_\gamma) = \frac{\|\bigwedge^d \mathrm{Ad}(\gamma) p_{a,b}\|}{\|p_{a,b}\|} \mathrm{vol}(\Delta_{a,b} \backslash H_{a,b}).$$

The vector

$$\bigwedge^d \mathrm{Ad}(\gamma) p_{a,b} \in \bigwedge^d \mathfrak{sl}_n(\mathbb{Z})$$

is integral. Hence a uniform volume bound confines it to a bounded subset of a fixed lattice. Only finitely many wedge vectors occur, and each decomposable wedge determines the Lie algebra and therefore the connected subgroup.

This example is important because it already contains the interaction that the separate block-unipotent and block-semisimple examples miss: a moving semisimple Levi factor acts nontrivially on a moving unipotent radical. What remains difficult in the full theorem is that an arbitrary support group need not stay inside one fixed integral conjugacy family of this shape.

Example A.5.48 (Arbitrary flag type: a whole parabolic-derived family). The previous example extends without changing the mechanism. Fix a composition

$$n = n_1 + \cdots + n_r, \quad n_i \geq 1,$$

and let U_n be the group of block upper-unitriangular matrices with diagonal block sizes $n_1, \dots, n_r$. Define

$$H_n = \left(\prod_{i=1}^r \mathrm{SL}_{n_i}(\mathbb{R}) \right) \ltimes U_n < \mathrm{SL}_n(\mathbb{R}),$$

where a factor SL_1 is understood to be trivial. This is the semisimple-block part together with the full unipotent radical of the standard parabolic attached to the flag

$$0 < V_1 < V_2 < \cdots < V_{r-1} < \mathbb{R}^n, \quad \dim V_j = n_1 + \cdots + n_j.$$

Every factor of H_n is generated by elementary unipotent one-parameter groups: roots inside a diagonal block generate SL_{n_i} , and roots joining two different blocks generate U_n . Thus H_n is again of the unipotent-generated type relevant to the support class.

Its integral lattice is

$$\Delta_n = H_n \cap \mathrm{SL}_n(\mathbb{Z}) = \left(\prod_{i=1}^r \mathrm{SL}_{n_i}(\mathbb{Z}) \right) \ltimes U_n(\mathbb{Z}).$$

Successively quotienting the root-space coordinates of $U_{\mathbf{n}}$ gives compact torus fibers. On a block $M_{n_i, n_j}(\mathbb{R})$ the Levi action is

$$Y \mapsto A_i Y A_j^{-1},$$

whose determinant equals

$$(\det A_i)^{n_j} (\det A_j)^{-n_i} = 1.$$

Thus all those fiber volumes are preserved, while the diagonal SL_{n_i} quotients have finite volume. This gives a direct inductive verification that $\Delta_{\mathbf{n}} \backslash H_{\mathbf{n}}$ has finite volume.

Choose a primitive integral exterior vector

$$p_{\mathbf{n}} \in \bigwedge^{\dim H_{\mathbf{n}}} \mathfrak{sl}_n(\mathbb{Z})$$

spanning $\bigwedge^{\dim H_{\mathbf{n}}} \mathfrak{h}_{\mathbf{n}}$. For every $\gamma \in \mathrm{SL}_n(\mathbb{Z})$, the subgroup

$$H_{\gamma, \mathbf{n}} = \gamma H_{\mathbf{n}} \gamma^{-1}$$

is attached to the transported rational flag $\gamma(V_{\bullet})$. Exactly as above,

$$\mathrm{vol}((H_{\gamma, \mathbf{n}} \cap \Gamma) \backslash H_{\gamma, \mathbf{n}}) = \frac{\| \bigwedge^{\dim H_{\mathbf{n}}} \mathrm{Ad}(\gamma) p_{\mathbf{n}} \|}{\| p_{\mathbf{n}} \|} \mathrm{vol}(\Delta_{\mathbf{n}} \backslash H_{\mathbf{n}}).$$

Therefore bounded homogeneous volume gives only finitely many transported Lie wedges, hence finitely many subgroups, for every fixed flag type.

This family is broad enough to include any number of diagonal semisimple blocks and every cross-block unipotent root group at once. It is therefore a useful model for the Levi–radical geometry behind the general theorem. It still does not prove Dani–Margulis: the theorem must compare groups of different algebraic types and control arithmetic structures that are not presented a priori as integral conjugates of one fixed $H_{\mathbf{n}}$.

Remark A.5.49 (What these examples do and do not prove). The examples exhibit the three ingredients that one should keep in mind when reading the abstract theorem:

- (1) an arithmetic subgroup gives a discrete integral object (a primitive generator or an exterior-power vector);
- (2) homogeneous volume controls the norm of that object;
- (3) bounded subsets of the relevant integral lattice are finite.

For a general $H \in \mathcal{H}_{\Gamma}$, however, one must first manufacture a uniform integral object while the lattice $H \cap \Gamma$, the unipotent radical, the Levi factor, and the central directions may all vary. That is precisely the structural content hidden by the elementary examples and supplied by the Dani–Margulis bounded-volume theorem.

Lemma A.5.50 (Conjugation Jacobian and homogeneous covolume). *Let $H \in \mathcal{H}_\Gamma$, let $d = \dim H$, and choose $0 \neq p_H \in \bigwedge^d \mathfrak{h}$. For $g \in G$ put $H_g = gHg^{-1}$. With the right-invariant volume normalization induced by σ , conjugation $c_g(h) = ghg^{-1}$ has constant Jacobian*

$$J_H(g) = |\det(\text{Ad}(g)|_{\mathfrak{h}} : \mathfrak{h} \rightarrow \text{Ad}(g)\mathfrak{h})| = \frac{\|\bigwedge^d \text{Ad}(g)p_H\|}{\|p_H\|}.$$

If $g = \gamma \in \Gamma$, then $H_\gamma \in \mathcal{H}_\Gamma$ and

$$(5.6) \quad \text{vol}_\sigma(H_\gamma) = J_H(\gamma) \text{vol}_\sigma(H).$$

PROOF. At the identity, the differential of c_g is $\text{Ad}(g) : \mathfrak{h} \rightarrow \text{Ad}(g)\mathfrak{h}$, so its Jacobian for the Euclidean volume forms induced by the ambient inner product is the displayed exterior power norm ratio. Because c_g is a group isomorphism and the volume forms on H and H_g are extended by right translations, the same factor is the Jacobian at every point.

A locally compact group admitting a lattice is unimodular. Hence the right-Haar volume determined by the chosen right-invariant Riemannian volume is also left invariant and descends to $(H \cap \Gamma) \backslash H$. For $\gamma \in \Gamma$,

$$H_\gamma \cap \Gamma = \gamma(H \cap \Gamma)\gamma^{-1},$$

and conjugation therefore induces a diffeomorphism

$$(H \cap \Gamma) \backslash H \longrightarrow (H_\gamma \cap \Gamma) \backslash H_\gamma.$$

Integrating the constant Jacobian gives (5.6). Finally, conjugation preserves the unipotent-generated subgroup and its ergodicity on the finite-volume quotient, so $H_\gamma \in \mathcal{H}_\Gamma$. $\square$

THEOREM A.5.51 (Dani–Margulis support-class countability). *The basepoint-normalized support class $\mathcal{H}_\Gamma$ of Definition 6.2.1 is countable.*

PROOF. This is exactly part (1) of Theorem A.5.42. We record it separately because countability is the form used by the outer singular-set exhaustion. $\square$

THEOREM A.5.52 (Dani–Margulis fixed-support Chevalley discreteness). *Let $H \in \mathcal{H}_\Gamma$, put $d = \dim H$, and choose $0 \neq p_H \in \bigwedge^d \mathfrak{h}$. In $\bigwedge^d \mathfrak{g}/\{\pm 1\}$ the orbit*

$$\Gamma \bar{p}_H$$

is closed and discrete. Equivalently, every bounded subset of the exterior power meets $\Gamma \bar{p}_H$ in only finitely many points.

PROOF. This is part (2) of Theorem A.5.42. The sign quotient is exactly the normalizer ambiguity needed for the proper-lift construction below. $\square$

Corollary A.5.53 (Proper Chevalley lift). *Let*

$$\Gamma_H = \Gamma \cap N_G(H), \quad \eta_H(g) = \bigwedge^d \text{Ad}(g)p_H,$$

with the sign quotient used when $\Gamma_H p_H = \{\pm p_H\}$. Then

$$\Phi_H : \Gamma_H \backslash G \longrightarrow (\Gamma \backslash G) \times \bar{V}_H, \quad \Gamma_H g \longmapsto (\Gamma g, \eta_H(g^{-1}))$$

is proper.

USING THEOREM A.5.52. Let $K \subset \Gamma \backslash G$ and $D \subset \bar{V}_H$ be compact. Choose a compact set $C \subset G$ whose projection contains K . If $\Gamma_H g_j \in \Phi_H^{-1}(K \times D)$, write $g_j = \gamma_j c_j$ with $\gamma_j \in \Gamma$ and $c_j \in C$. Then

$$\eta_H(g_j^{-1}) = \bigwedge^d \text{Ad}(c_j^{-1})(\gamma_j^{-1} p_H) \in D.$$

Hence $\gamma_j^{-1} \bar{p}_H$ belongs to the compact set $(\bigwedge^d \text{Ad}(C))D$. By Theorem A.5.52, only finitely many orbit values occur. On a subsequence with one fixed value, $\gamma_j \gamma_1^{-1} \in \Gamma_H$, and therefore

$$\Gamma_H g_j = \Gamma_H \gamma_1 c_j,$$

which ranges over a compact subset of $\Gamma_H \backslash G$. Thus inverse images of compact sets are compact. $\square$

Remark A.5.54 (Dependency normalization). The qualitative classification proof uses exactly the two clauses of Theorem A.5.42: countability for the outer exhaustion of support types and fixed-support Chevalley discreteness for lattice-branch finiteness. All proper-lift, unique-lift, thin-neighborhood, and focusing consequences are then proved internally, together with the support-to-algebraic-hull bridge

$$\mathcal{H}_\Gamma \subset \mathcal{H}_\Gamma^{\text{alg}}.$$

The special-case arguments and examples are pedagogical laboratories. They show why one cannot assume that a bounded-volume lattice has bounded generators without a thick–thin argument, that a lattice intersection or a Zariski hull uniquely determines its connected syndetic hull, or that the support condition may be replaced by $H = \overline{H}^+$. None of those laboratories is used in place of the stated Dani–Margulis structural interface.

Let

$$\text{Subalg}_r(\mathfrak{g}) = \{V \in \text{Gr}_r(\mathfrak{g}) : [V, V] \subset V\}.$$

Lemma A.5.55 (Compactness of Lie-subalgebra parameters). *For each r , $\text{Subalg}_r(\mathfrak{g})$ is a compact subset of the Grassmannian. The union over $0 \leq r < \dim G$ is therefore a finite union of compact spaces.*

PROOF. The Grassmannian is compact. If $V_n \rightarrow V$ and each V_n is a Lie subalgebra, choose $X_n, Y_n \in V_n$ converging to arbitrary $X, Y \in V$. Then $[X_n, Y_n] \in V_n$ and continuity of the bracket gives $[X, Y] \in V$. Hence the subalgebra condition is closed. $\square$

THEOREM A.5.56 (Finite-stage singular bookkeeping). *Fix an enumeration*

$$\mathcal{H}_\Gamma = \{H_1, H_2, \dots\}$$

from Theorem A.5.51. Fix a compact set $K \subset \Gamma \backslash G$, an integer $m \geq 1$, a bound on subgroup dimension, and a bounded norm window for the corresponding linearization vectors. Then the singular data associated with the first m support types $H_1, \dots, H_m$ and meeting K admit a finite cover by linearization charts with uniform polynomial-degree and coefficient bounds.

Consequently every use of “all proper singular types” may be implemented by first restricting to a compact base window, the first m support types, and a bounded linearization window; applying the finite-stage argument there; and then exhausting the compact base, m , and the norm bound.

USING THEOREMS A.5.51 AND A.5.52. There are only finitely many support types at the chosen stage. For each fixed H_i , Theorem A.5.52 says that its Γ -Chevalley orbit is locally finite. Hence a bounded vector window contains only finitely many lattice branches for that fixed type. Since $i \leq m$, the total number of discrete branches is finite.

A compact base set K is covered by finitely many local quotient charts with relatively compact lift sets. On every remaining discrete branch, the continuously varying Lie-subalgebra parameter lies in a compact Grassmannian family by Lemma A.5.55. Exterior-power and Chevalley coordinate maps depend continuously on that parameter. A finite subcover therefore gives uniform polynomial degrees and coefficient bounds on the stage.

Finally, countability guarantees that every admissible support type occurs at some finite index, so letting $m \rightarrow \infty$ loses no singular datum. This is purely qualitative diagonal bookkeeping; no bounded-covolume finiteness is being used. $\square$

Remark A.5.57 (Separation of the discrete and continuous mechanisms). Theorems A.5.51 and A.5.52 control the discrete support-type and lattice-branch bookkeeping. Lemma A.5.55 controls a continuous Grassmannian parameter inside each surviving chart. None of these statements replaces the others. Once a finite stage has only finitely many discrete branches, the compact-parameter polynomial theorem below supplies the uniform Remez estimate while the subgroup parameter varies continuously inside a chart.

THEOREM A.5.58 (Compact-parameter polynomial avoidance-or-trapping). *Let $J \subset \mathbb{R}$ be a compact interval, Θ a compact metric space, and V a*

finite-dimensional normed vector space. Let

$$F_j : \Theta \times V \longrightarrow \mathbb{R}, \quad 1 \leq j \leq m,$$

be functions such that, in a fixed basis of V , $F_j(\theta, \cdot)$ is a polynomial of degree at most e , and every coefficient is a continuous function of θ . Let $P_n : J \rightarrow V$ be polynomial maps of degree at most d , with coefficient vectors in a fixed compact set and $\sup_J \|P_n\| = 1$, and let $\theta_n \in \Theta$. Put

$$Q_{n,j}(t) = F_j(\theta_n, P_n(t)), \quad M_n = \max_j \sup_{t \in J} |Q_{n,j}(t)|.$$

After passage to a subsequence exactly one of the following occurs.

- (1) There is $c > 0$ with $M_n \geq c$ for every n ; then for every $\eta > 0$ there is $\delta > 0$, independent of n , such that

$$|\{t \in J : \max_j |Q_{n,j}(t)| \leq \delta\}| \leq \eta |J|.$$

- (2) We have $M_n \rightarrow 0$, $\theta_n \rightarrow \theta$, and $P_n \rightarrow P$ coefficientwise and uniformly on J , where P is nonzero and

$$F_j(\theta, P(t)) = 0 \quad (t \in J, 1 \leq j \leq m).$$

The same assertion holds for a finite number of compact parameter charts, with one common δ on the nontrapping branch.

PROOF. Compactness gives a subsequence with $\theta_n \rightarrow \theta$ and with the coefficient vector of P_n convergent. Let the limit polynomial be P . Uniform convergence on J follows from coefficient convergence, and $\sup_J \|P\| = 1$, so $P \neq 0$. The coefficients of every scalar polynomial $Q_{n,j}$ vary continuously with (θ_n, P_n) ; hence they too converge coefficientwise and uniformly, to

$$Q_j(t) = F_j(\theta, P(t)).$$

Pass to a further subsequence on which M_n converges. If its limit is zero, all the Q_j vanish identically and alternative (2) follows.

Suppose instead that $M_n \geq c > 0$. For each n choose $j(n)$ with $\sup_J |Q_{n,j(n)}| \geq c$. Every $Q_{n,j}$ has degree at most $D = de$. The one-variable Remez inequality, in the sublevel form already proved in Chapter 7, gives a constant C_D , depending only on D , such that

$$|\{t \in J : |Q_{n,j(n)}(t)| \leq \delta\}| \leq C_D (\delta/c)^{1/D} |J|.$$

The set on which all $|Q_{n,j}| \leq \delta$ is contained in this set. Choose $\delta = c(\eta/C_D)^D$. This proves alternative (1) uniformly in n . For finitely many charts take the minimum of their positive lower bounds and of the resulting finitely many thresholds. No enumeration of the points of Θ is used. In particular, a continuously varying subgroup parameter may vary with n without creating an uncountable-union estimate. $\square$

Remark A.5.59 (The logical point). A finite cover by Grassmannian charts does *not* turn a compact family of subgroups into finitely many subgroup types. What compactness supplies is a compact coefficient family. The preceding theorem is the precise uniform statement needed by the singular-avoidance argument: at any one comparison block there is one parameter θ_n , and Remez is uniform as that parameter varies in a compact chart. Thus the proof never takes a union over all $\theta \in \Theta$ at a fixed time.

6. Singular fibers and induction on a discrete quotient

The measure theorem will be stated for a discrete subgroup Γ , not only for a lattice. This allows the singular branch to reduce immediately to a smaller ambient Lie group, before finite volume of that intermediate orbit has been established.

THEOREM A.6.1 (Singular-fiber reduction). *Let G be connected, $\Gamma < G$ discrete, $U < G$ one-parameter Ad-unipotent, and μ a U -ergodic probability measure on $\Gamma \backslash G$. Suppose a primitive linearization parameter is almost surely constant and its projective stabilizer $M < G$ is proper and closed. Then, after translating the base point and passing to one connected component, there are*

$$M_0 < G \quad \text{proper connected closed}, \quad \Delta < M_0 \quad \text{discrete},$$

such that the measure-classification problem is measurably isomorphic to a U -ergodic probability-measure problem on

$$\Delta \backslash M_0.$$

In particular the effective ambient dimension strictly decreases. No finite-covolume assertion about Δ is needed at this stage.

PROOF. Constancy of the projective parameter means that, after fixing one representative fiber, μ is concentrated on a single orbit xM . The orbit is Borel and the orbit map

$$\Delta_M \backslash M \longrightarrow xM, \quad \Delta_M = M \cap g^{-1}\Gamma g,$$

is a Borel isomorphism onto its image. Pull back μ to a probability measure on $\Delta_M \backslash M$. The subgroup Δ_M is discrete because it lies in the conjugate of the discrete group Γ .

The connected group U lies in the stabilizer of the fiber, hence in a conjugate of M . Adjust the base point so that $U \subset M$. The component group M/M° is countable because a second-countable manifold has at most countably many components. Right multiplication by the connected group $U \subset M^\circ$ preserves each component of $\Delta_M \backslash M$. Ergodicity therefore concentrates the measure on one component. Translating that component identifies

the problem with a quotient

$$\Delta \backslash M_0, \quad M_0 = M^\circ,$$

for a discrete $\Delta < M_0$.

Since the projective stabilizer is proper, the effective connected subgroup M_0 is proper after removing any kernel acting trivially on the fiber. Hence

$$\dim M_0 < \dim G.$$

Ad-unipotence of U survives restriction to $\mathfrak{m}_0$, and ergodicity is unchanged under the Borel identification. These are exactly the hypotheses needed for induction. $\square$

Remark A.6.2 (Where finite volume reappears). The induction hypothesis classifies the pulled-back probability measure as Haar measure on an orbit of some subgroup $H < M_0$. That finite invariant orbit is closed by Theorem A.4.3. Pushing it forward to $\Gamma \backslash G$ therefore produces the required closed finite-volume homogeneous support. Finite volume is a *conclusion* of the singular induction, not an input to it.

7. Orbitwise nonescape and avoidance

The orbitwise results used for pointwise distribution are proved in Appendix D. We record the labels here for ease of reference.

THEOREM A.7.1 (Orbitwise nonescape). *Let $Y = \Delta \backslash H$ have finite volume and let $U = \{u_t\} < H$ be one-parameter Ad-unipotent. For every $y \in Y$ and every $\varepsilon > 0$ there is a compact $K \subset Y$ such that*

$$\liminf_{T \rightarrow \infty} \frac{1}{T} |\{0 \leq t \leq T : yu_t \in K\}| \geq 1 - \varepsilon.$$

PROOF. This is Corollary D.6.2, proved from the Lie-group quantitative nondivergence theorem in Appendix D. $\square$

THEOREM A.7.2 (Orbitwise singular avoidance or reduction). *Let $Y = yH$ be a finite-volume homogeneous space containing yU . On every compact family of proper primitive singular data, arbitrarily small neighborhoods have arbitrarily small upper time density along yU , unless the corresponding linearization polynomial is exactly trapped in a proper projective stabilizer fiber. In a minimal finite-volume homogeneous envelope the trapping alternative is impossible; hence full proper-singular avoidance holds after compact exhaustion.*

PROOF. This is Theorem D.7.2 and the minimal-envelope argument at the end of Appendix D. $\square$

8. Minimal homogeneous envelopes

Let $x \in \Gamma \backslash G$ and assume the ambient quotient itself has finite volume. The family

$$\mathcal{E}(x, U) = \{H < G : H \text{ closed connected, } U \subset H, xH \text{ closed finite-volume}\}$$

is nonempty because it contains G . Since subgroup dimensions are integers, it contains an element of minimal dimension. No Zorn lemma is needed.

9. Restriction to a closed orbit

Suppose xH is closed finite-volume. Write $x = \Gamma g$ and

$$\Gamma_H = H \cap g^{-1}\Gamma g.$$

Then

$$xH \cong \Gamma_H \backslash H.$$

If $U \subset H$, right multiplication by U corresponds exactly under this identification. If $U = \exp(\mathbb{R}N)$ is Ad-unipotent in G , then $\text{ad}_{\mathfrak{g}} N$ is the restriction of the nilpotent map $\text{ad}_{\mathfrak{g}} N$, so U is Ad-unipotent in H as well.

10. Closedness of support

For any Radon measure μ , $\text{supp } \mu$ is closed. If μ is H -invariant, then $\text{supp } \mu$ is H -invariant: if $x \in \text{supp } \mu$ and O is a neighborhood of xh , then Oh^{-1} is a neighborhood of x and

$$\mu(O) = \mu(Oh^{-1}) > 0.$$

This elementary observation is useful before the stronger conclusion that the support is one homogeneous orbit has been reached.

APPENDIX B

Technical Selection and Compactness Lemmas

1. Boundary-zero neighborhoods

Weak-* convergence controls open and closed sets only through inequalities unless their boundaries have zero limiting measure. The following elementary device is repeatedly useful.

Lemma B.1.1. *Let (X, d) be a metric space, $C \subset X$ closed, and μ a finite Borel measure. For all but countably many $r > 0$,*

$$\mu(\partial C_r) = 0, \quad C_r = \{x : d(x, C) < r\}.$$

PROOF. The boundaries ∂C_r are pairwise disjoint as r varies. A finite measure can assign positive mass to at most countably many pairwise disjoint measurable sets. □

Hence one can always enlarge a compact singular piece by an arbitrarily small radius whose boundary is invisible to the limiting measure.

2. Tightness and subsequential limits

Lemma B.2.1. *Let X be locally compact second countable and let μ_n be probability measures. Suppose that for every $\varepsilon > 0$ there is compact K_ε with*

$$\liminf_{n \rightarrow \infty} \mu_n(K_\varepsilon) > 1 - \varepsilon.$$

Then every subsequence has a further weak- convergent subsequence whose limit is a probability measure.*

PROOF. Choose compact sets K_j with $\liminf_n \mu_n(K_j) > 1 - 2^{-j}$. After discarding finitely many indices at each stage and replacing K_j by $K_1 \cup \dots \cup K_j$, we may suppose $K_j \subset K_{j+1}$ and $\mu_n(K_j) > 1 - 2^{-j}$ for all sufficiently large n . Since X is locally compact and second countable, $C_c(X)$ is separable in the topology of uniform convergence on each K_j ; choose a countable dense family (f_k) . A diagonal subsequence makes $\int f_k d\mu_n$ converge for every k . The resulting positive functional on $C_c(X)$ has norm at most one, hence is represented by a Radon measure μ . The estimates on K_j , applied with compactly supported cutoffs equal to one on K_j , give $\mu(X) \geq 1 - 2^{-j}$ for

every j . Thus $\mu(X) = 1$, and density of (f_k) gives weak-* convergence of the chosen subsequence to μ . $\square$

This is the form of Prokhorov compactness used for empirical measures.

3. Invariance of time-average limits

Let

$$\mu_T = \frac{1}{T} \int_0^T \delta_{xu_t} dt.$$

For fixed s and bounded f ,

$$\begin{aligned} \int f(zu_s) d\mu_T(z) - \int f(z) d\mu_T(z) &= \frac{1}{T} \int_0^T (f(xu_{t+s}) - f(xu_t)) dt \\ &= \frac{1}{T} \left(\int_T^{T+s} f(xu_t) dt - \int_0^s f(xu_t) dt \right). \end{aligned}$$

Thus the absolute value is at most

$$\frac{2|s|}{T} \|f\|_\infty.$$

Every probability subsequential limit is therefore U -invariant.

4. Strong continuity of translations on $C_0(X)$

The block-comparison arguments repeatedly replace a sequence of group elements converging to h by the fixed element h . The required analytic fact is uniform, not merely pointwise, continuity on $C_0(X)$.

Lemma B.4.1 (Strong continuity of right translations). *Let G be a Lie group acting continuously on the right of a locally compact Hausdorff space X . For $h \in G$ write*

$$(R_h f)(x) = f(xh).$$

Then for every $f \in C_0(X)$ the map

$$G \longrightarrow C_0(X), \quad h \longmapsto R_h f,$$

is continuous in the supremum norm. In particular, if $h_n \rightarrow h$, then

$$\|f \circ R_{h_n} - f \circ R_h\|_\infty \rightarrow 0.$$

PROOF. It is enough to prove continuity at the identity: replacing f by $R_h f$ reduces the general case to $h^{-1}h_n \rightarrow e$.

Fix $\varepsilon > 0$. Choose a compact set $K \subset X$ such that

$$|f(x)| < \varepsilon \quad (x \notin K).$$

Choose a relatively compact neighborhood C of e in G and eventually restrict to $g \in C$. The set

$$K\overline{C}^{-1} = \{kc^{-1} : k \in K, c \in \overline{C}\}$$

is compact, because it is the continuous image of the compact set $K \times \overline{C}$. If $x \notin K\overline{C}^{-1}$ and $g \in C$, then neither x nor xg belongs to K ; hence

$$|f(xg) - f(x)| < 2\varepsilon.$$

On the compact set $K\overline{C}^{-1}$, the map

$$(x, g) \mapsto f(xg) - f(x)$$

is continuous and vanishes when $g = e$. Uniform continuity on a compact neighborhood therefore gives a neighborhood $V \subset C$ of e such that

$$|f(xg) - f(x)| < \varepsilon \quad (x \in K\overline{C}^{-1}, g \in V).$$

Combining the two regions gives

$$\|R_g f - f\|_\infty < 2\varepsilon \quad (g \in V),$$

which proves the claim. $\square$

5. A vector-valued polynomial Remez consequence

Let $P : I \rightarrow V$ be a polynomial map of degree at most d into a finite-dimensional normed vector space. Choose $t_0 \in I$ with

$$\|P(t_0)\| = \sup_I \|P\|.$$

By Hahn–Banach, choose a norm-one linear functional ℓ with

$$|\ell(P(t_0))| = \|P(t_0)\|.$$

Then

$$\sup_I |\ell \circ P| = \sup_I \|P\|.$$

Applying the scalar Remez estimate gives

$$(B.1) \quad \left| \{t \in I : \|P(t)\| < \varepsilon \sup_I \|P\|\} \right| \leq 4\varepsilon^{1/d} |I|.$$

This is sufficient for the local singular-avoidance estimates in the main text.

6. A simultaneous genericity selection scheme

Suppose $z_n^{(1)}, \dots, z_n^{(r)}$ are finitely many generic points at stage n , and one needs genericity for the first n test functions on the first n rational blocks. Because the number of requirements at stage n is finite, there is a common threshold R_n . Choosing

$$T_n > \max\{n, T_{n-1} + 1, R_n\}$$

produces a single sequence of times satisfying every fixed requirement eventually.

This trivial-looking diagonal device is the correct remedy whenever the starting points themselves vary with n .

7. The technical H-property realization theorem

The quotient-normalized polynomial information must be converted into genuine full-group comparison blocks. Appendix C gives the complete reparametrization and finite-correction argument needed for that conversion.

THEOREM B.7.1 (Technical H-property realization). *Let G be connected, $\Gamma < G$ discrete, $U = \{u_t\}$ one-parameter Ad-unipotent, and μ a U -ergodic probability measure. Put*

$$L = \text{Stab}_G(\mu)^\circ.$$

Assume no L -orbit has full μ -measure. For nearby common generic pairs transverse to L , first-exit normalization and correction give exactly one of the following:

- (1) *corrected full-group comparison blocks converge to a C^1 curve whose translated tangent is nonzero in $\mathfrak{g}/\mathfrak{l}$; or*
- (2) *the pairs lie in a proper primitive singular linearization locus.*

On the complement of the singular locus the comparison blocks satisfy the hypotheses of Theorem 8.12.1.

PROOF. This is Theorem C.8.1. The proof is not a formal deletion of an L -factor. It proceeds modulo the acting subgroup U , proves that every first-exit limit normalizes U , removes the U -factor by an exact change of time, controls that change by bounded-degree rational estimates, proves weighted genericity by integration by parts, and then iterates fixed normalizer corrections lying in the measure stabilizer. Finite polynomial degree makes the iteration terminate; failure to produce a transverse stabilizing curve is exactly the singular algebraic alternative. $\square$

8. Order of small parameters

Several rigidity arguments use four small/large parameters:

$$\varepsilon, \quad \delta, \quad \rho, \quad n.$$

A safe order is:

- (1) choose the final test-function error ε ;
- (2) choose the exceptional time proportion $\delta \ll \varepsilon$;
- (3) choose the macroscopic group radius ρ small enough for slice coordinates and uniform continuity;
- (4) choose n large enough for polynomial normalization, genericity, and recurrence at the already fixed scales.

Changing this order can create circular estimates, for example choosing a compact set after a uniform-continuity modulus that depended on the compact set.

9. Local quotient slices

Let $L < G$ be a closed Lie subgroup with Lie algebra $\mathfrak{l}$. Choose a complement

$$\mathfrak{g} = \mathfrak{l} \oplus W.$$

The map

$$F : \mathfrak{l} \times W \rightarrow G, \quad F(A, Z) = \exp(A) \exp(Z)$$

has derivative at $(0, 0)$ equal to

$$(A, Z) \mapsto A + Z,$$

an isomorphism. The inverse function theorem gives neighborhoods on which F is a diffeomorphism. This proves the local factorization

$$g = \ell(g) \exp(w(g))$$

used to separate known stabilizer directions from transverse directions.

On compactly contained smaller neighborhoods, both coordinate maps are Lipschitz in any fixed smooth Riemannian metric.

10. A support lemma for limits of orbit averages

Let $Y \subset X$ be closed and suppose every μ_n is supported on Y . If

$$\mu_n \rightarrow \mu$$

weak-*, then

$$\text{supp } \mu \subset Y.$$

PROOF. If $O \subset X \setminus Y$ is open with compact closure, then every $\mu_n(O) = 0$. Approximating the indicator of a compact subset of O by continuous compactly supported functions gives $\mu(O) = 0$. Hence $X \setminus Y$ has zero measure locally, and no point outside Y belongs to the support. $\square$

This is the final topological step in the orbit-closure theorem.

11. A finite-dimensional termination lemma

Suppose a procedure alternates between:

- (1) replacing the ambient connected group by a proper connected subgroup;
- (2) enlarging a connected stabilizer subgroup strictly inside the current ambient group.

Then the procedure terminates after finitely many genuine moves.

Indeed, ambient dimension decreases in the first move and stabilizer dimension increases in the second. Ordering pairs

$$(\dim G, -\dim L)$$

lexicographically gives a strictly decreasing pair of integers bounded below.

This bookkeeping formalizes the termination intuition behind the rigidity induction.

APPENDIX C

The Full H-Property: Reparametrization, Normalizer Limits, and Iterated Correction

1. Why this appendix is necessary

The elementary polynomial identity

$$q_g(t) = u_{-t}gu_t$$

explains why nearby unipotent trajectories separate only polynomially. It does *not*, by itself, prove additional invariance. There are two further difficulties.

First, a limiting displacement is naturally visible only modulo the acting unipotent subgroup. The missing unipotent factor must be absorbed by a change of time; simply dropping it is invalid. Second, after one limiting direction has been shown to lie in the measure stabilizer, that direction may hide a lower-degree transverse term. One has to remove the already-understood normalizer term and repeat. Since the polynomial degree is finite, this correction process terminates.

This appendix proves this mechanism in a non-infinitesimal sequence form. It is the technical H-property used in the classification argument.

Throughout,

$$X = \Gamma \backslash G, \quad U = \{u_t : t \in \mathbb{R}\},$$

where G is a connected Lie group, $\Gamma < G$ is discrete, and U is one-parameter Ad-unipotent. Let μ be a U -ergodic probability measure and put

$$S = \text{Stab}_G(\mu), \quad L = S^\circ.$$

Because U is connected and preserves μ , one has $U \subset L$.

2. First-exit normalization modulo U

Choose a smooth local submanifold $\Sigma \subset G$ through e transverse to U . Multiplication gives a diffeomorphism

$$\Sigma_0 \times U_0 \longrightarrow \mathcal{O}, \quad (c, u) \longmapsto cu,$$

from small neighborhoods onto an identity neighborhood $\mathcal{O} \subset G$. Write

$$(C.1) \quad g = c(g)u_{\eta(g)}$$

for the resulting local coordinates.

Let $g_n \rightarrow e$ with $g_n \notin U$, and set

$$q_n(t) = u_{-t} g_n u_t.$$

Shrink the identity neighborhood so that the exponential map is a diffeomorphism. Write

$$g_n = \exp Z_n, \quad Z_n \rightarrow 0.$$

As long as $q_n(t)$ stays in this exponential neighborhood, conjugation is represented *exactly* by the adjoint action:

$$(C.2a) \quad q_n(t) = u_{-t} \exp(Z_n) u_t = \exp(\text{Ad}(u_{-t}) Z_n) = \exp(e^{-t \text{ad } X} Z_n),$$

where $u_t = \exp(tX)$. Since $\text{ad } X$ is nilpotent, the logarithmic coordinate

$$e^{-t \text{ad } X} Z_n = \sum_{j=0}^D \frac{(-t)^j}{j!} (\text{ad } X)^j Z_n$$

is a vector-valued polynomial of a degree D depending only on (G, U) , not on n . Passing from logarithmic coordinates to a fixed quotient slice is smooth; on compactly contained coordinate neighborhoods its derivatives of every fixed order are uniformly bounded. Thus the only non-polynomiality below comes from a fixed smooth coordinate change applied to a bounded-degree polynomial family. In particular all first-exit, sublevel, and compactness estimates are controlled by a finite-dimensional family.

Fix a sufficiently small macroscopic radius $\rho > 0$ in G/U . Define $T_n > 0$ to be the first time for which

$$(C.2) \quad d_{G/U}(q_n(T_n)U, U) = \rho.$$

After passing to a subsequence, $T_n \rightarrow \infty$. Indeed, for each fixed T , uniform convergence $q_n(t) \rightarrow e$ on $[-T, T]$ follows from $g_n \rightarrow e$.

Rescale time by $t = T_n s$. Bounded-degree compactness gives, after a subsequence, a nonconstant continuous limit

$$(C.3) \quad q_n(T_n s)U \longrightarrow \bar{q}(s)$$

uniformly on compact s -intervals on which the trajectories remain in the chosen quotient chart. The first-exit normalization guarantees that $\bar{q}$ is nonconstant and takes a value at quotient distance ρ from U .

3. The fastest limit lies in the normalizer

The next observation is the algebraic heart of the H-property.

Proposition C.3.1 (Normalizer limit). *Let s_0 be an interior point of the parameter interval for which $q_n(T_n s_0)U$ converges to cU . Then*

$$c \in N_G(U).$$

PROOF. Fix $r \in \mathbb{R}$. The conjugacy identity gives

$$(C.4) \quad q_n(T_n s_0 + r) = u_{-r} q_n(T_n s_0) u_r.$$

On the other hand,

$$\frac{T_n s_0 + r}{T_n} = s_0 + \frac{r}{T_n} \longrightarrow s_0.$$

The uniform convergence in (C.3) therefore implies

$$q_n(T_n s_0 + r) U \longrightarrow cU.$$

Passing to the limit in (C.4) yields

$$u_{-r} c u_r U = cU.$$

Equivalently,

$$c^{-1} u_{-r} c u_r \in U \quad (r \in \mathbb{R}).$$

Thus $c^{-1} U c = U$, since both sides are connected one-dimensional subgroups. Hence $c \in N_G(U)$. $\square$

For $c \in N_G(U)$ there is a nonzero scalar $\chi(c)$ such that

$$(C.5) \quad c^{-1} u_t c = u_{\chi(c)t}.$$

The map $\chi : N_G(U) \rightarrow \mathbb{R}^\times$ is a continuous homomorphism.

4. Removing the unipotent error by a change of time

Near a fixed $c \in N_G(U)$, choose a smooth transverse section Σ_c to the right U -cosets. For q close to cU write uniquely

$$(C.6) \quad q = \sigma(q) u_{\eta(q)}, \quad \sigma(q) \in \Sigma_c.$$

Suppose $q_n(t)$ remains close to cU on a terminal block I_n . Put

$$\sigma_n(t) = \sigma(q_n(t)), \quad \eta_n(t) = \eta(q_n(t)),$$

and define

$$(C.7) \quad r_n(t) = t - \eta_n(t).$$

Then

$$(C.8) \quad \begin{aligned} y_n u_{r_n(t)} &= x_n g_n u_t u_{-\eta_n(t)} \\ &= x_n u_t q_n(t) u_{-\eta_n(t)} \\ &= x_n u_t \sigma_n(t). \end{aligned}$$

This exact identity is the reason the factor $u_{\eta_n(t)}$ may be removed: it is absorbed by a reparametrization of the *same* flow.

The next lemma records the required control of the change of variables.

Lemma C.4.1 (Good reparametrization subblock). *Fix the polynomial degree bound D and a compact first-exit coordinate window. There are constants*

$$c > 0, \quad 0 < a < b < \infty, \quad M < \infty,$$

depending only on that window and on D , with the following property. Let I_n be terminal first-exit blocks and let

$$r_n(t) = t - \eta_n(t)$$

be the exact time changes from (C.7). After rescaling I_n affinely to $[-1, 1]$ and passing to a subsequence, exactly one of the following occurs.

- (1) *The coefficient vector of the normalized displacement belongs to a proper algebraic degeneracy set (equivalently, one of the primitive singular relations of the linearization scheme holds identically).*
- (2) *There is a subinterval $J_n \subset I_n$ with*

$$|J_n| \geq c|I_n|$$

on which r_n is a C^1 diffeomorphism and

$$(C.9) \quad a \leq |r'_n(t)| \leq b,$$

while the Jacobian weight

$$w_n(s) = \frac{1}{|r'_n(r_n^{-1}(s))|}$$

satisfies, after affine normalization of $r_n(J_n)$,

$$(C.9a) \quad \|w_n\|_\infty \leq M, \quad |J_n| \|w'_n\|_\infty \leq M.$$

PROOF. In the normalized variable $s \in [-1, 1]$, the exact logarithmic displacement is a polynomial of degree at most D . The slice maps $q \mapsto \eta(q)$ and their first two derivatives range in a compact set because the first-exit trajectories remain in a fixed compact coordinate window. Consequently the normalized derivatives of r_n belong to a precompact subset of $C^1([-1, 1])$ obtained by applying one fixed C^2 map to a finite-dimensional compact coefficient family.

Pass to a convergent coefficient subsequence. Let R be the limiting normalized derivative. If $R \equiv 0$, then every coefficient of the corresponding finite-dimensional coordinate expression vanishes. These are finitely many polynomial equations in the first-exit coefficient data after clearing the fixed nonvanishing slice denominators. This is precisely the degeneracy alternative. (Using the Grassmannian/exterior-power linearization of Appendix A, these equations are among the primitive singular equations.)

Assume $R \not\equiv 0$. Because R arises from a bounded-degree polynomial family through a fixed analytic coordinate map on a compact set, the number of its

zeros and critical zeros is uniformly bounded unless the corresponding defining function vanishes identically; the latter case has already been assigned to the degeneracy alternative. Equivalently, one may apply the one-variable sublevel estimate of Volume II successively to R and R' . Remove sufficiently small neighborhoods of the zero sets of R and R' . The removed total length can be made < 1 , with constants depending only on the finite-dimensional compact family. One remaining component therefore has length at least a fixed $c > 0$. On a slightly smaller component,

$$2a \leq |R| \leq b/2$$

for some uniform $a, b > 0$. C^1 convergence gives (C.9) for r_n on the corresponding J_n and a uniform bound for the normalized derivative of r'_n . The formula

$$w'_n(s) = -\frac{r''_n(r_n^{-1}(s))}{|r'_n(r_n^{-1}(s))|^3}$$

(with the harmless sign adjustment on an orientation-reversing block) then gives (C.9a). Undoing the affine normalization proves the lemma. $\square$

The “degeneracy relation” in this lemma is precisely one of the algebraic singular relations used in the linearization chapter. Thus the regular branch of the rigidity proof is always in alternative (2).

5. Weighted genericity under slowly varying changes of time

A generic orbit remains generic under the controlled reparametrizations above. The proof is a simple integration-by-parts argument, but writing it explicitly avoids a common hidden step.

Lemma C.5.1 (Weighted Cesàro lemma). *Let z be generic for the U -ergodic measure μ , let $f \in C_0(X)$, and set*

$$F(T) = \int_0^T \left(f(zu_t) - \int f d\mu \right) dt.$$

Then $F(T) = o(T)$. Let $I_n = [\alpha_n, \beta_n]$ with $|I_n| \rightarrow \infty$ and $\alpha_n = O(|I_n|)$. If w_n are C^1 weights satisfying

$$\|w_n\|_\infty \leq C, \quad \|w'_n\|_\infty \leq \frac{C}{|I_n|},$$

then

$$(C.10) \quad \frac{1}{|I_n|} \int_{I_n} w_n(t) \left(f(zu_t) - \int f d\mu \right) dt \longrightarrow 0.$$

PROOF. Write $dF(t) = (f(zu_t) - \mu(f))dt$. Integration by parts gives

$$\int_{\alpha_n}^{\beta_n} w_n dF = w_n F|_{\alpha_n}^{\beta_n} - \int_{\alpha_n}^{\beta_n} w'_n(t) F(t) dt.$$

Because $F(t) = o(t)$ and $t = O(|I_n|)$ on the intervals under consideration, the boundary term is $o(|I_n|)$. Given $\varepsilon > 0$, for large n one has $|F(t)| \leq \varepsilon|I_n|$ throughout I_n except possibly on a bounded initial segment, whose contribution is negligible. The derivative bound therefore gives

$$\left| \int_{I_n} w'_n F \right| \leq C\varepsilon|I_n| + o(|I_n|).$$

Divide by $|I_n|$ and let $\varepsilon \downarrow 0$. □

If $r_n : J_n \rightarrow r_n(J_n)$ is as in Lemma C.4.1, change variables $s = r_n(t)$. The Jacobian weight is

$$w_n(s) = \frac{1}{|r'_n(r_n^{-1}(s))|},$$

which satisfies the hypotheses of Lemma C.5.1 after normalization. Hence genericity may be used on the reparametrized y_n -block.

6. The H-property comparison theorem

THEOREM C.6.1 (H-property, non-infinitesimal sequence form). *Let x_n, y_n be pairs selected from the simultaneous uniform-generic sets of Appendix B: at stage n they satisfy uniform Birkhoff estimates for the first n functions in a fixed countable dense family, for all averaging blocks whose normalized lengths lie in the fixed first-exit range. Suppose*

$$y_n = x_n g_n, \quad g_n \rightarrow e.$$

Assume a first-exit normalization modulo U produces a regular, nondegenerate terminal block and a limit

$$q_n(T_n s_0)U \longrightarrow cU$$

with $c \in N_G(U)$. Then

$$c \in S = \text{Stab}_G(\mu).$$

More generally, every value of the nonconstant normalized limiting curve in $N_G(U)/U$ admits a lift in S .

PROOF. Fix a countable sup-norm dense family $\mathcal{F} \subset C_0(X)$. Choose the common generic set so that every x_n, y_n is generic for every member of $\mathcal{F}$ and for the finitely or countably many translates that occur below; the diagonal selection is legitimate because all requirements form a countable family.

Choose a terminal subblock J_n on which Lemma C.4.1 applies and on which

$$\sigma_n(t) \longrightarrow c$$

uniformly. Fix first a small $\delta > 0$ and choose a compact set K_δ with $\mu(K_\delta) > 1 - \delta$. Include continuous upper and lower approximations to $\mathbf{1}_{K_\delta}$ among the finite uniform-generic requirements. After discarding a subset of J_n of relative measure $O(\delta)$, both comparison trajectories lie in one fixed compact window (depending on δ but not on n) on which the quotient lift and all current test functions are uniformly continuous. All comparison errors are therefore $o_n(1) + O(\delta)$. First let $n \rightarrow \infty$ and then let $\delta \downarrow 0$. This is ordinary Birkhoff recurrence for the probability measure; no orbitwise nondivergence theorem is used in measure classification.

For $t \in E_n$, identity (C.8) gives

$$y_n u_{r_n(t)} = x_n u_t \sigma_n(t),$$

and $\sigma_n(t) \rightarrow c$ uniformly. For $f \in \mathcal{F}$, uniform continuity on the common compact window yields

$$f(y_n u_{r_n(t)}) - f(x_n u_t c) \longrightarrow 0$$

uniformly on E_n .

Average over J_n . The exceptional complement contributes $O(\delta)$, uniformly in n ; after taking $n \rightarrow \infty$ we let $\delta \downarrow 0$. The x_n -average tends to $\int f(xc) d\mu(x)$ by genericity. After the change of variables $s = r_n(t)$, the y_n -average is a weighted average with controlled slowly varying Jacobian, so Lemma C.5.1 gives the limit $\int f d\mu$. Therefore

$$\int f(xc) d\mu(x) = \int f(x) d\mu(x) \quad (f \in \mathcal{F}).$$

Density in $C_0(X)$ and Riesz representation imply $(R_c)_* \mu = \mu$, hence $c \in S$.

The same argument works simultaneously on a countable dense set of normalized parameters. Closedness of S then gives the entire limiting curve. $\square$

7. Iterating away already-known normalizer directions

The previous theorem detects the fastest term modulo U . It may happen that this fastest term lies in S , while a lower-degree term is genuinely transverse to S . Ratner's correction procedure removes the known term and repeats. We now make this finite induction explicit.

Let

$$N_S(U) = N_G(U) \cap S.$$

If $c \in N_S(U)$, then right translation by c preserves μ and conjugates the U -flow to the time-rescaled flow

$$u_t \mapsto u_{\chi(c)t}.$$

Thus c may be used as a correction without destroying genericity: genericity under U is unchanged by a fixed nonzero linear rescaling of time.

In local coordinates, the normalized displacement has bounded degree D . Its quotient modulo U may be represented by a vector-valued polynomial

$$P_n(s) = P_{n,D}s^D + \cdots + P_{n,0} + o(1).$$

The H-property shows that any nonzero leading coefficient whose projective first-exit limit is visible belongs to the tangent cone of $N_S(U)/U$. Choose a smooth local section of $N_S(U)/U$ and multiply by the inverse of the corresponding normalizer correction. Because conjugation by a normalizer element acts on U by the scalar character χ , this changes only the time parametrization and replaces P_n by a polynomial of strictly smaller active degree outside the known normalizer directions.

The following lemma packages the finite induction.

Lemma C.7.1 (Finite correction algorithm). *Fix a bounded degree D . For any sequence of nearby generic pairs there is, after passing to a subsequence, a sequence of at most $D + 1$ normalizer corrections from $N_S(U)$ such that exactly one of the following occurs:*

- (1) *a regular first-exit limit $c \in N_G(U) \setminus S$ is produced; or*
- (2) *every active coefficient is absorbed by $N_S(U)$, and the initial transverse data satisfy a finite family of polynomial equations invariant under U and $N_S(U)$.*

PROOF. Induct on the maximal active degree. For degree 0 there is nothing to prove. Assume the statement for degrees $< d$ and consider degree d .

Normalize at the first scale at which the degree- d term is macroscopic modulo U . Proposition C.3.1 places every limit in $N_G(U)/U$, and Theorem C.6.1 places every regular limit in S/U . If a limit is outside S/U , we are in alternative (1).

Otherwise the limiting degree- d coefficient lies in the local image of $N_S(U)/U$. Choose a local normalizer lift and compose the relative displacement with its inverse. The exact normalizer relation (C.5) absorbs the accompanying U -factor into a linear time change. In the finite-dimensional coefficient space this subtracts the degree- d component in the quotient and leaves a displacement of active degree at most $d - 1$. Apply the induction hypothesis.

If the procedure never leaves S , then at each stage the corresponding coefficient belongs to the algebraic tangent cone of $N_S(U)/U$. Membership in this tangent cone is expressed by finitely many polynomial equations in the finite-dimensional coefficient space of the adjoint/exterior-power linearization. This construction is available for every Lie group because it uses the adjoint action on $\mathfrak{g}$ and Grassmannians of Lie subalgebras; no faithful linear representation of G is required. Pulling these equations back through the finitely many coefficient maps gives the polynomial equations in alternative (2). The family is proper unless the entire local transverse space is already contained in S , in which case the nearby perturbation was not transverse to S to begin with. $\square$

Remark C.7.2. The second alternative is not discarded. It is exactly the singular branch: the polynomial equations define the primitive linearized singular locus that is handled by the avoidance-and-reduction argument. Thus the H-property and the singular-set method are complementary parts of one proof, not competing approaches.

8. Full realization relative to the connected stabilizer

We now prove the realization theorem needed by the main argument.

THEOREM C.8.1 (Full H-property realization). *Let $S = \text{Stab}_G(\mu)$ and $L = S^\circ$, and assume no L -orbit has full μ -measure. Let x_n, y_n belong to the simultaneous uniform-generic sets of Appendix B, suppose*

$$y_n = x_n g_n, \quad g_n = \exp Z_n \longrightarrow e, \quad Z_n \notin \mathfrak{l},$$

for all sufficiently large n , and assume the pairs are chosen outside the primitive singular relations used by the linearization induction. Apply the first-exit normalization modulo U and then the finite correction algorithm of Lemma C.7.1. Exactly one of the following occurs:

- (1) *a subsequence of corrected comparison blocks yields a C^1 curve $\phi : J \rightarrow S$ for which, at some $s_0 \in J$,*

$$(dL_{\phi(s_0)^{-1}})_{\phi(s_0)} \phi'(s_0) \notin \mathfrak{l};$$

this is impossible because $L = S^\circ$;

- (2) *the normalized coefficient data satisfy one of the finite primitive polynomial relations defining the proper singular linearization locus.*

On every regular block in the first alternative the exact time change $r_n(t) = t - \eta_n(t)$ obeys Lemma C.4.1: for constants $c > 0$, $0 < a < b < \infty$, and $M < \infty$ depending only on the fixed first-exit window and degree bound,

$$|J_n| \geq c|I_n|, \quad a \leq |r'_n(t)| \leq b,$$

and, after affine normalization,

$$\|w_n\|_\infty \leq M, \quad |J_n| \|w'_n\|_\infty \leq M, \quad w_n(s) = |r'_n(r_n^{-1}(s))|^{-1}.$$

Thus the comparison data required by Theorem 8.12.1 have a completely stated Jacobian bound rather than an unstated reparametrization hypothesis.

PROOF. Apply Lemma C.7.1. If alternative (1) there occurs, Theorem C.6.1 would put the limiting element in S , a contradiction. Therefore a regular transverse pair cannot produce an outside- S normalizer limit. If all leading terms are successively absorbed by $N_S(U)$, the residual coefficient polynomial either eventually has a nonzero class in $\mathfrak{g}/\mathfrak{s}$, in which case the next H-property step gives the same contradiction, or all residual coefficients satisfy the polynomial equations defining the singular locus. Since the degree is finite, this procedure terminates.

On every regular step, Lemma C.4.1, the exact identity (C.8), and the weighted-genericity lemma provide the full comparison blocks. Diagonalizing over a countable dense parameter set produces the curve-valued formulation used in Chapter 8. If the limiting curve had a nonzero transverse tangent, Theorem 8.12.1 would place that tangent in $\mathfrak{l}$, contradiction. Hence only the singular alternative remains. $\square$

9. What was and was not used

The proof above uses only:

- nilpotence of $\text{Ad}(u_t)$, hence bounded-degree polynomial divergence;
- local quotient coordinates by U ;
- the elementary normalizer-limit argument;
- a bounded-degree reparametrization lemma;
- Birkhoff genericity and the weighted Cesàro lemma;
- closedness of the measure stabilizer;
- finite-dimensional algebraic linearization for the terminal singular equations.

No time-dependent factor in the full stabilizer is silently deleted. Every correction is either an actual U -time change or a fixed element of $N_S(U)$, both of which have a controlled effect on genericity.

APPENDIX D

Orbitwise Nonescape and Global Singular Avoidance

1. Purpose

Measure classification concerns invariant probability measures. Pointwise equidistribution begins with an arbitrary orbit, so two additional facts are required: time averages cannot lose mass at infinity, and they cannot spend positive density near proper singular configurations unless the orbit is actually trapped in one. This appendix proves the qualitative forms needed in the distribution chapter.

The proof of nonescape is a Lie-group version of quantitative nondivergence. Its organization is the same as the exterior-power proof for spaces of lattices: small injectivity radius produces a flag of small subgroup data; the data evolve polynomially under an Ad-unipotent flow; a marking argument prevents too many independent small conditions from occurring for too long.

2. Injectivity radius and compact thick parts

Let H be a connected Lie group, $\Delta < H$ a lattice, and

$$Y = \Delta \backslash H.$$

Fix a left-invariant Riemannian metric on H . For $y = \Delta h$ define

$$(D.1) \quad \text{inj}(y) = \frac{1}{2} \inf_{\delta \in \Delta \setminus \{e\}} d_H(e, h^{-1}\delta h).$$

The value is positive at every point because $h^{-1}\Delta h$ is discrete.

Proposition D.2.1 (Compact thick part). *For every $\rho > 0$, the set*

$$Y_{\geq \rho} = \{y : \text{inj}(y) \geq \rho\}$$

is compact.

PROOF. Choose $0 < r < \rho/3$. If $y \in Y_{\geq \rho}$, the quotient map is injective on the ball $B_H(e, r)$ translated to y , so that ball has the same positive Haar volume v_r at every such point.

If $Y_{\geq \rho}$ were not totally bounded, there would be infinitely many points y_j separated by $2r$ in the quotient metric. Their radius- r balls would be pairwise disjoint, each of volume v_r . This contradicts the finite total volume of Y . Thus $Y_{\geq \rho}$ is totally bounded.

It is also closed: if $y_n \rightarrow y$ and $\text{inj}(y_n) \geq \rho$, choose local lifts $h_n \rightarrow h$. Any nontrivial $h^{-1}\delta h$ of norm $< 2\rho$ would, by continuity, give a nontrivial element of $h_n^{-1}\Delta h_n$ of norm $< 2\rho$ for large n . Hence $\text{inj}(y) \geq \rho$. Since Y is complete for a proper quotient metric on compactly generated pieces, or equivalently since a closed totally bounded subset of the finite-volume quotient has compact closure in the local charts, the thick part is compact. $\square$

3. A Zassenhaus neighborhood

We need to organize several small elements of a discrete subgroup simultaneously.

Lemma D.3.1 (Zassenhaus neighborhood). *There exists an identity neighborhood $\Omega \subset H$ such that, for every discrete subgroup $\Lambda < H$, the subgroup generated by $\Lambda \cap \Omega$ is nilpotent.*

PROOF. Choose a norm on $\mathfrak{h}$ and an exponential ball on which the Baker–Campbell–Hausdorff series converges. There are constants $C, r_0 > 0$ such that, whenever $\|X\|, \|Y\| < r_0$,

$$(D.2) \quad \|\log([e^X, e^Y])\| \leq C\|X\|\|Y\|.$$

Choose radii

$$r_0 > r_1 > \cdots > r_m > 0, \quad m = \dim H,$$

so rapidly decreasing that

$$Cr_j^2 < r_{j+1}/10.$$

Let $\Omega = \exp B(0, r_m)$.

Suppose the subgroup generated by $\Lambda \cap \Omega$ were not nilpotent of class at most m . Then there would be nontrivial iterated commutators of lengths $1, 2, \dots, m+1$. Estimate (D.2) puts the length- k commutators in successively smaller exponential balls. Their logarithms span a strictly descending sequence of nonzero subspaces under bracketing; after more than m strict drops this is impossible in the m -dimensional Lie algebra. Equivalently, the lower central series of the Lie algebra generated by the logarithms reaches zero by step at most $m+1$. Because Λ is discrete, the BCH correspondence on the chosen neighborhood has no accumulation ambiguity, and the generated local subgroup has the same nilpotent lower-central relations. Thus it is nilpotent. $\square$

Remark D.3.2. The particular nilpotency-class bound is irrelevant. What matters is that the class and the number of independent logarithmic directions are bounded solely by $\dim H$.

4. Small subgroup data and polynomial covolumes

Let $U = \{u_t\} < H$ be Ad-unipotent. Fix $y = \Delta h$. If a finitely generated subgroup $A < h^{-1}\Delta h$ is generated by elements that enter the Zassenhaus neighborhood at some time, choose independent logarithms

$$Z_1, \dots, Z_r$$

for a primitive generating flag of its local nilpotent Lie algebra and put

$$w_A = Z_1 \wedge \cdots \wedge Z_r \in \bigwedge^r \mathfrak{h}.$$

Conjugation gives

$$(D.3) \quad w_A(t) = \bigwedge^r \text{Ad}(u_{-t})w_A.$$

Since $\text{Ad}(u_{-t}) = e^{-tN}$ with N nilpotent, every coordinate of $w_A(t)$ is a polynomial of uniformly bounded degree. The norm

$$(D.4) \quad \psi_A(t) = \|w_A(t)\|$$

therefore satisfies a (C, α) -good sublevel estimate on intervals, with constants depending only on $\dim H$ and the nilpotency degree. This is exactly the vector-valued Remez argument proved in Appendix B and, in a more quantitative form, in Volume II.

At a fixed time only finitely many primitive local subgroup data have ψ_A below a fixed threshold. Indeed, the corresponding generators lie in a compact subset of H , and a discrete subgroup has finite intersection with a compact set.

5. The marking lemma for subgroup flags

We restate the finite-rank combinatorial principle in the form needed here.

Lemma D.5.1 (Flag marking). *Let I be an interval. Suppose a locally finite family of functions $\psi_A : I \rightarrow (0, \infty)$ is indexed by a poset of rank at most m , and every ψ_A is (C, α) -good. Suppose further that whenever $A_0 < \cdots < A_k$ is a chain, the ranks strictly increase. If on each maximal bad interval the maximal member of a marked chain reaches a reference size $\rho^{\text{rank } A}$ at the boundary, then*

$$(D.5) \quad |\{t \in I : \text{some rank-one datum has } \psi_A(t) < \varepsilon\}| \leq C_m C \left(\frac{\varepsilon}{\rho}\right)^\alpha |I|.$$

PROOF. The proof is the same finite induction as the weighted-poset marking theorem in Volume II, so we include the mechanism. Mark at a bad point a maximal chain of subgroup data whose covolumes lie below their rank thresholds. Enlarge the point to the maximal interval on which the top marked datum remains below its threshold. At the boundary its covolume reaches the reference value, so goodness bounds the relative size of the deeper sublevel set by $C(\varepsilon/\rho)^\alpha$.

Remove the top datum and repeat inside the interval with the shorter poset. Since chain length is at most m , the induction terminates. A one-dimensional Besicovitch selection chooses the maximal intervals with overlap bounded by 2 at each rank. Summing the geometric loss over at most m ranks gives a constant C_m . Local finiteness ensures that all selections are finite on compact subintervals; monotone convergence removes that auxiliary restriction. $\square$

6. Quantitative injectivity-radius nondivergence

THEOREM D.6.1 (Lie-group quantitative nondivergence). *Let $Y = \Delta \backslash H$ have finite volume and let $U = \{u_t\}$ be one-parameter Ad -unipotent. Fix $y \in Y$. There exist $\rho_y > 0$, $C_y > 0$, and $\alpha > 0$ such that for every interval I containing 0 and every $0 < \varepsilon < \rho_y$,*

$$(D.6) \quad |\{t \in I : \text{inj}(yu_t) < \varepsilon\}| \leq C_y \left(\frac{\varepsilon}{\rho_y} \right)^\alpha |I| + O_y(1).$$

The additive constant is independent of I .

PROOF. Choose a Zassenhaus neighborhood and a smaller exponential ball B inside it. Because the stabilizer $h^{-1}\Delta h$ is discrete, there is $\rho_y > 0$ so small that no nontrivial stabilizer element lies in $\exp B(0, 20\rho_y)$ at time 0.

Consider the bad set where the injectivity radius is $< \varepsilon$. Each bad time lies in a maximal interval on which the subgroup generated by stabilizer elements inside the fixed Zassenhaus ball has constant primitive flag type except at finitely many transition times. Such a decomposition is locally finite: transitions occur when one of finitely many elements in a compact group ball crosses the boundary of that ball.

On one maximal interval choose a primitive flag of the local nilpotent subgroup. The wedge covolumes evolve by (D.3). If a rank- r covolume stayed below ρ_y^r on the entire interval without reaching that reference size at either boundary, then by maximality the same subgroup data would extend farther; the only exception is an interval meeting the fixed bounded initial region, which accounts for the $O_y(1)$ term. Thus the hypotheses of Lemma D.5.1 hold on all remaining maximal intervals.

A point with injectivity radius $< \varepsilon$ supplies a nontrivial local element with logarithm of norm $O(\varepsilon)$, hence a rank-one marked datum with $\psi_A(t) < C\varepsilon$. Applying the marking lemma and summing the selected maximal intervals gives (D.6), after changing constants.

One possible degeneracy deserves mention. A polynomial wedge could be constant and small for all time. Then the corresponding local subgroup is normalized by U . At time 0 its rank-one generators still have norm at least $20\rho_y$ by the choice of ρ_y , so it cannot create the rank-one condition $< \varepsilon$ when $\varepsilon < \rho_y$. Higher-rank constant data are harmless unless a rank-one generator is already small. This excludes persistent false badness. $\square$

Corollary D.6.2 (Orbitwise nonescape). *For every $y \in Y$ and every $\delta > 0$ there is a compact $K \subset Y$ such that*

$$\liminf_{T \rightarrow \infty} \frac{1}{T} |\{0 \leq t \leq T : yu_t \in K\}| \geq 1 - \delta.$$

PROOF. Choose $\varepsilon > 0$ so small that the right-hand side of (D.6), divided by $|I|$, is $< \delta$ for long intervals. Take

$$K = Y_{\geq \varepsilon},$$

which is compact by Proposition D.2.1. Apply (D.6) to $I = [0, T]$ and let $T \rightarrow \infty$. $\square$

7. Orbitwise singular avoidance

The singular linearization data of Chapter 7 evolve by bounded-degree polynomial maps in finite-dimensional representations. The finite-family avoidance proposition there did not use genericity of an invariant measure; it used only polynomial sublevel estimates. Hence it applies verbatim to one orbit.

The only global issue is the infinite family of possible subgroup parameters. Support-class countability organizes the discrete types into finite stages; fixed-support Chevalley discreteness controls their lattice branches; compact Grassmannian parameterization then supplies uniformity inside each stage.

Lemma D.7.1 (Finite-stage compact parameter reduction). *Fix an enumeration $\mathcal{H}_\Delta = \{H_1, H_2, \dots\}$ of the admissible basepoint-normalized support class for $Y = \Delta \backslash H$. Fix a compact set $K \subset Y$, an integer $m \geq 1$, and a bound on the dimension and norm of the linearization data. The proper singular branches associated with $H_1, \dots, H_m$ and meeting K are covered by finitely many linearization charts, and inside each chart the continuously varying Lie-subalgebra parameters lie in a compact Grassmannian family with uniform polynomial degree and coefficient bounds.*

PROOF. For each fixed support type H_i , Theorem A.5.52 (applied to the ambient lattice Δ) makes its Chevalley orbit locally finite. Hence only finitely many lattice branches occur in the chosen bounded vector window. There are only finitely many indices $i \leq m$.

For every surviving branch, the possible Lie algebras lie in a closed subset of the appropriate Grassmannian. The Grassmannian is compact and the Lie subalgebra condition is closed. Smooth dependence of exterior-power and Chevalley coordinates on the subspace parameter gives uniform degree and coefficient bounds on compact charts; a finite subcover finishes the stage. $\square$

THEOREM D.7.2 (Orbitwise avoidance or exact trapping). *Let $Y = \Delta \backslash H$ have finite volume and let U be one-parameter Ad-unipotent. Fix $y \in Y$. For every compact family $\mathcal{C}$ of proper primitive singular data and every $\eta > 0$, there is a neighborhood $\mathcal{O}$ of the associated compact singular set such that*

$$(D.7) \quad \limsup_{T \rightarrow \infty} \frac{1}{T} |\{0 \leq t \leq T : yu_t \in \mathcal{O}\}| < \eta,$$

unless the corresponding linearization polynomial satisfies an exact proper algebraic relation in one of the admissible support branches. In the latter case, for some proper $W \in \mathcal{H}_\Delta$ and a lift $y = \Delta g$, one has $gUg^{-1} \subset W$.

PROOF. Use Corollary D.6.2 to discard a set of times of upper density $< \eta/3$ outside a compact thick part K . Choose a finite support-type stage containing the compact admissible family under consideration. On K , Lemma D.7.1 reduces those branches to finitely many uniform polynomial charts. Apply the finite-family avoidance-or-trapping proposition of Chapter 7 in each chart with total error $< 2\eta/3$. This gives (D.7) unless one chart is in its exact-trapping alternative.

In the exact alternative, all defining polynomials for one admissible branch vanish identically on the U -trajectory. Since that chart was constructed from the incidence variety $N(U, W)$ of a fixed support type $W \in \mathcal{H}_\Delta$, vanishing of the full defining family says precisely that the chosen lift satisfies $gUg^{-1} \subset W$. No invariant measure or genericity assumption is used. $\square$

8. From exact algebraic trapping to a smaller finite-volume orbit

The exact-trapping alternative is useful only because the charts above are not arbitrary Grassmannian charts: every discrete branch comes from an *admissible* support type $W \in \mathcal{H}_\Delta$. This supplies the finite-volume conclusion directly.

Proposition D.8.1 (Admissible exact trapping gives a finite-volume envelope). *Let $Y = \Delta \backslash H$, let $W \in \mathcal{H}_\Delta$ be proper, and let $y = \Delta g$. If*

$$gUg^{-1} \subset W,$$

then, with $L = g^{-1}Wg$,

$$yU \subset yL,$$

and yL is a proper closed finite-volume homogeneous orbit.

PROOF. The containment $gUg^{-1} \subset W$ is equivalent to $U \subset L$, hence $yU \subset yL$. Moreover

$$L \cap g^{-1}\Delta g = g^{-1}(W \cap \Delta)g.$$

Because $W \in \mathcal{H}_\Delta$, the subgroup $W \cap \Delta$ is a lattice in W . Conjugation by g therefore shows that $L \cap g^{-1}\Delta g$ is a lattice in L . The finite-covolume stabilizer theorem, Theorem A.4.2, now implies that the orbit map

$$(L \cap g^{-1}\Delta g) \backslash L \longrightarrow Y$$

is proper with closed image yL , and its Haar volume is finite. Since $W < H$ is proper, so is $L < H$. $\square$

Corollary D.8.2 (Minimal finite-volume envelope excludes exact trapping). *Choose, among the closed finite-volume homogeneous envelopes containing yU , one of minimal dimension. In that envelope the exact-trapping alternative of Theorem D.7.2 cannot occur for an admissible proper singular branch.*

PROOF. An exact admissible branch gives, by Proposition D.8.1, a proper closed finite-volume homogeneous envelope containing yU , contradicting minimality. $\square$

Thus the orbitwise proof has three clean layers: support-class countability chooses finite stages; fixed-support Chevalley discreteness makes each stage locally finite; polynomial linearization gives avoidance or exact trapping. In the trapping branch, admissibility of the selected support type and the finite-covolume stabilizer theorem immediately produce the smaller finite-volume orbit. No empirical-measure argument inside a possibly nonproper M -fiber is used.

APPENDIX E

The Markov-Operator/Joining Correspondence in Both Directions

1. Statement

Let (X, μ) and (Y, ν) be standard probability spaces. A joining is a probability λ on $X \times Y$ with marginals μ, ν . It defines a Markov operator

$$P_\lambda : L^2(Y, \nu) \rightarrow L^2(X, \mu), \quad P_\lambda g(x) = \mathbb{E}_\lambda[g(y) \mid x].$$

The earlier volume constructed P_λ from λ . We now prove the converse with no appeal to an unstated representation theorem.

THEOREM E.1.1 (Markov operators are exactly joinings). *Suppose $P : L^2(Y, \nu) \rightarrow L^2(X, \mu)$ is linear, positive, contractive, and satisfies*

$$P\mathbf{1} = \mathbf{1}, \quad P^*\mathbf{1} = \mathbf{1}.$$

Then there is a unique joining λ such that for all bounded measurable f, g ,

$$(M.1) \quad \int_{X \times Y} f(x)g(y) d\lambda = \int_X f(x) P g(x) d\mu(x).$$

Conversely every joining yields such a P . If T and S are measure-preserving transformations on X and Y , respectively, then λ is invariant under $T \times S$ if and only if

$$U_T P = P U_S,$$

where $U_T f = f \circ T$ and $U_S g = g \circ S$.

PROOF. For measurable rectangles define

$$\lambda_0(A \times B) = \int_A P\mathbf{1}_B d\mu.$$

Positivity makes this nonnegative. Since $P\mathbf{1} = \mathbf{1}$, $\lambda_0(A \times Y) = \mu(A)$; since $P^*\mathbf{1} = \mathbf{1}$,

$$\lambda_0(X \times B) = \int_X P\mathbf{1}_B d\mu = \langle \mathbf{1}, P\mathbf{1}_B \rangle = \langle P^*\mathbf{1}, \mathbf{1}_B \rangle = \nu(B).$$

We first check countable additivity in each variable. If B_n are disjoint and $B = \cup_n B_n$, then $\sum_{n \leq N} \mathbf{1}_{B_n} \rightarrow \mathbf{1}_B$ in $L^2(Y)$ by monotone convergence

of the squared norm. Boundedness of P gives convergence in $L^2(X)$, hence in $L^1(X)$ because $\mu(X) = 1$. Therefore

$$\lambda_0(A \times B) = \lim_N \sum_{n \leq N} \lambda_0(A \times B_n).$$

Additivity in A follows directly from countable additivity of the integral.

Let $\mathcal{R}$ be the algebra of finite disjoint unions of measurable rectangles. For $E = \bigsqcup_{j=1}^m A_j \times B_j$ set $\lambda_0(E) = \sum_j \lambda_0(A_j \times B_j)$. This is well defined: refine two rectangle decompositions by all intersections of their A - and B -atoms; finite additivity in each variable gives the same value on the common refinement. The preceding separate countable additivity implies that λ_0 is a premeasure on $\mathcal{R}$. Indeed, if disjoint $E_n \in \mathcal{R}$ have union $E \in \mathcal{R}$, refine E into finitely many rectangle atoms and apply monotone convergence on each atom.

Carathéodory extension gives a probability measure λ on $\sigma(\mathcal{R})$, which is the product sigma-algebra. Its marginals are the ones computed above, so it is a joining. Formula (M.1) holds first for indicator functions of measurable sets, then for simple functions, and finally for bounded functions by dominated convergence. Uniqueness follows because rectangles form a generating π -system.

Conversely, conditional expectation with respect to the first coordinate makes P_λ positive, sends 1 to 1, and is an L^2 contraction by Jensen. The second marginal gives $P_\lambda^* 1 = 1$.

Finally, invariance of λ is equivalent, on rectangle test functions, to

$$\int f(Tx)g(Sy) d\lambda = \int f(x)g(y) d\lambda.$$

Using (M.1) and changing variables under μ rewrites this as $\langle f, U_T^{-1} P U_S g \rangle = \langle f, P g \rangle$ for all bounded f, g , hence $U_T P = P U_S$ (with the convention $U_T f = f \circ T$). The converse is the same calculation backwards. $\square$

2. Composition

If P_{12} and P_{23} correspond to joinings of (X_1, X_2) and (X_2, X_3) , then $P_{12}P_{23}$ is again a Markov operator with both unit conditions, hence corresponds to a unique joining of (X_1, X_3) . This is the operator form of relational composition. No measurable kernel needs to be postulated: it is reconstructed by Theorem E.1.1.

Bibliographic Notes

This volume is a guided reconstruction of the rigidity mechanism rather than a line-by-line reproduction of any single paper. The primary sources behind the main theorems are Marina Ratner's papers on measure rigidity, orbit closures, distributions, and joinings of unipotent flows. In particular:

- M. Ratner, *On measure rigidity of unipotent subgroups of semisimple groups*, Acta Math. 165 (1990), 229–309.
- M. Ratner, *Strict measure rigidity for unipotent subgroups of solvable groups*, Invent. Math. 101 (1990), 449–482.
- M. Ratner, *On Raghunathan's measure conjecture*, Ann. of Math. 134 (1991), 545–607.
- M. Ratner, *Raghunathan's topological conjecture and distributions of unipotent flows*, Duke Math. J. 63 (1991), 235–280.
- N. A. Shah, *Invariant Measures and Orbit Closures on Homogeneous Spaces for Actions of Subgroups Generated by Unipotent Elements*, arXiv:math/0002183 (2000), for the generated-by-unipotents extension discussed in this volume.
- S. G. Dani and G. A. Margulis, *Limit distributions of orbits of unipotent flows and values of quadratic forms*, in *I. M. Gel'fand Seminar*, Adv. Soviet Math. 16 (1993), 91–137. The publication edition uses exactly the support-class countability alternative associated with their Proposition 2.1 and the exterior-power orbit discreteness theorem, their Theorem 3.4, as the external structural interface isolated in Appendix A.

For historical comparison and alternative proof architectures, useful expositions include work of Dani–Margulis, Margulis–Tomanov, Shah, and graduate-level accounts of Ratner's theorems. The support-hull module in this volume proves internally that every ergodic unipotent support lies in the algebraic-support class and develops the quotient, H-property, singular-set, and reconstruction steps used by the induction. The fixed-family bounded-volume calculations in Appendix A are retained as worked arithmetic models;

they are not advertised as a proof of a stronger general bounded-volume theorem.

The H-property realization separates quotient-by- U normalization, normalizer limits, time reparametrization, weighted genericity, and finite correction of already-known stabilizing directions. Orbitwise nonescape and compact-family singular avoidance are likewise proved from polynomial exterior-power estimates and finite-rank marking. Thus, apart from the precisely stated Dani–Margulis structural interface and ordinary baseline prerequisites, citations in this volume serve attribution, historical comparison, and independent checks on theorem formulation rather than as substitutes for load-bearing proofs.

Arithmetic Applications, Structure Theory, and Further Directions

Volume IV of a Detailed Notebook on Ratner's Theorems

Proof-Complete Edition

August 2026

Preface and proof policy

This volume turns the rigidity package of Volumes I–III into arithmetic statements, centered on the Oppenheim theorem. Every load-bearing result is proved here or explicitly inherited from an earlier volume where it is proved.

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Dependency and proof-ownership contract

Purpose. Arithmetic application.

Prerequisites. Volumes I–III.

New objects. quadratic forms, orthogonal groups, null cones. Every project-specific technical term is defined before its first load-bearing use or recalled by an exact backward reference.

Main outputs. qualitative Oppenheim dictionary and theorem.

External-input rule. Every nontrivial imported result is either part of the declared graduate-level prerequisites, an explicitly named external research input in the prerequisite map, or is proved in this series before use (possibly under an explicit deferred-proof contract). A citation by itself never constitutes a proof.

Reader map and proof route

Question of the volume. How does one translate Ratner orbit closure into arithmetic density for indefinite irrational quadratic forms, while keeping the homogeneous-space dictionary explicit?

Prerequisite checkpoint. The rigidity theorem is inherited from Volumes I–III with its proof and dependency map. This volume is responsible for the quadratic-form geometry, stabilizer calculations, lattice dictionary, and deduction of Oppenheim in the stated qualitative form.

Load-bearing route. Read the matrix model of a quadratic form, identify $SO(Q)^\circ$, interpret integer vectors through the lattice orbit, apply orbit closure, and only then translate density back to values of Q . The student supplements are examples and alternative viewpoints, not additional hypotheses.

Sanity check. For $Q(x, y, z) = x^2 + y^2 - z^2$, verify directly that the null cone is nonempty and that preserving Q is the matrix equation $g^T A g = A$. This is the concrete model behind the later abstract notation G/H .

Chapter 1

The Arithmetic Bridge: Rational Structures and Closed Orbits

1.1 Rational algebraic sets

Let V be a finite-dimensional vector space over $\mathbb{Q}$ and write $V_{\mathbb{R}} = V \otimes_{\mathbb{Q}} \mathbb{R}$. A real algebraic subset $Z \subset V_{\mathbb{R}}$ is *defined over $\mathbb{Q}$* if its vanishing ideal is generated by polynomials with rational coefficients.

Lemma 1.1 (Zariski closure of rational points is rational). *Let $A \subset \mathbb{Q}^N$. Its real Zariski closure $\overline{A}^{\mathbb{Z}} \subset \mathbb{R}^N$ is defined over $\mathbb{Q}$.*

Proof. For $d \geq 0$, let $P_d(\mathbb{Q})$ be the finite-dimensional space of polynomials of total degree at most d with rational coefficients, and put

$$I_d(A) = \{p \in P_d(\mathbb{Q}) : p(a) = 0 \ \forall a \in A\}.$$

The equations $p(a) = 0$ are homogeneous linear equations in the coefficients of p with rational coefficients. Hence $I_d(A)$ has a basis over $\mathbb{Q}$. After extending scalars, its real span is exactly the space of real polynomials of degree at most d vanishing on A : Gaussian elimination on the same rational matrix gives the same kernel after tensoring with $\mathbb{R}$. Taking the union over d , the full real vanishing ideal of A is generated by rational polynomials. Its common zero set is the real Zariski closure of A . $\square$

Corollary 1.2. *If $\Lambda < \mathrm{GL}_N(\mathbb{Q})$ is any subgroup, its real Zariski closure is a real algebraic subgroup defined over $\mathbb{Q}$.*

Proof. Lemma 1.1 gives rational defining equations. Multiplication and inversion preserve the closure because they are regular maps on GL_N and preserve Λ . $\square$

1.2 A rational fixed line

We will repeatedly use the following elementary descent principle.

Lemma 1.3 (One-dimensional kernel descends). *Let A be a matrix with rational entries. If $\ker(A : \mathbb{R}^m \rightarrow \mathbb{R}^k)$ is one-dimensional, then that line is spanned by a nonzero vector in $\mathbb{Q}^m$.*

Proof. Row reduction over $\mathbb{Q}$ produces a rational basis for the nullspace over $\mathbb{Q}$. Extending scalars to $\mathbb{R}$ does not change nullity. Since the real nullspace is one-dimensional, the rational nullspace is one-dimensional as well and supplies the required vector. $\square$

1.3 Closed orbits and lattices

We recall the exact bridge from dynamics to arithmetic. If $K < G$ is closed and $x = \Gamma g$, then

$$xK \cong (K \cap g^{-1}\Gamma g) \backslash K$$

whenever the orbit is closed. Thus a finite invariant measure on xK says that the conjugate

$$gKg^{-1} \cap \Gamma$$

is a lattice in gKg^{-1} . When Γ consists of rational matrices, Borel density can then force the entire algebraic subgroup to be defined over $\mathbb{Q}$. The next chapters make this mechanism explicit for orthogonal groups.

Chapter 2

Binary Quadratics and the Group $\mathrm{SO}(2, 1)^\circ$

2.1 The discriminant form

Let

$$V = \mathrm{Sym}^2(\mathbb{R}^2)^*$$

be the three-dimensional space of binary quadratic forms

$$f(X, Y) = aX^2 + bXY + cY^2.$$

In coordinates (a, b, c) define

$$\Delta(a, b, c) = b^2 - 4ac.$$

This is a nondegenerate quadratic form of signature $(2, 1)$.

Proof. With coordinates $x = (a+c)/\sqrt{2}$, $y = b$, $z = (a-c)/\sqrt{2}$, one computes $-4ac = -2x^2 + 2z^2$. Hence

$$\Delta = y^2 - 2x^2 + 2z^2,$$

which has two positive and one negative square. The diagonalized quadratic form has coefficients $1, -2, 2$, all nonzero, so its associated symmetric matrix is invertible; hence the form is nondegenerate. $\square$

2.2 The symmetric-square representation

For $g \in \mathrm{SL}_2(\mathbb{R})$ let

$$(\rho(g)f)(X, Y) = f((X, Y)g).$$

This gives $\rho : \mathrm{SL}_2(\mathbb{R}) \rightarrow \mathrm{SL}(V)$.

Proposition 2.1 (Discriminant invariance). $\Delta(\rho(g)f) = \Delta(f)$ for every $g \in \mathrm{SL}_2(\mathbb{R})$. Consequently $\rho(\mathrm{SL}_2(\mathbb{R})) \subset \mathrm{SO}(\Delta)$.

Proof. Write a binary quadratic as $f(v) = vAv^T$ with

$$A = \begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix}.$$

Under $v \mapsto vg$, the matrix becomes gAg^T . Since $\det g = 1$,

$$\det(gAg^T) = \det A.$$

But $\Delta(f) = -4\det A$, proving invariance. The determinant of $\rho(g)$ is a continuous character of the connected semisimple group $\mathrm{SL}_2(\mathbb{R})$ and is 1 at the identity, hence equals 1; alternatively it can be checked on the elementary unipotent and diagonal generators. $\square$

Proposition 2.2 (Kernel and image). *The kernel of ρ is $\{\pm I\}$, and*

$$\rho(\mathrm{SL}_2(\mathbb{R})) = \mathrm{SO}(\Delta)^\circ.$$

Thus $\mathrm{PSL}_2(\mathbb{R}) \simeq \mathrm{SO}(2, 1)^\circ$.

Proof. If g acts trivially on all quadratic forms, then in particular it preserves the lines spanned by X^2 and Y^2 , so g is diagonal; preservation of XY and determinant one then gives $g = \pm I$. Hence the differential $d\rho : \mathfrak{sl}_2 \rightarrow \mathfrak{so}(\Delta)$ is injective. Both Lie algebras have dimension three, so $d\rho$ is an isomorphism. Therefore the image is an open connected subgroup of the identity component $\mathrm{SO}(\Delta)^\circ$. An open subgroup of a connected topological group is also closed and its cosets are open; connectedness forces it to be the whole identity component. $\square$

2.3 Explicit unipotent generators

Let

$$n_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad \bar{n}_s = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix}.$$

The groups $N = \{n_t\}$ and $\bar{N} = \{\bar{n}_s\}$ are one-parameter unipotent subgroups and generate $\mathrm{SL}_2(\mathbb{R})$.

Lemma 2.3. *N and $\bar{N}$ generate $\mathrm{SL}_2(\mathbb{R})$.*

Proof. For $t \neq 0$ put

$$w_t = n_t \bar{n}_{-t^{-1}} n_t = \begin{pmatrix} 0 & t \\ -t^{-1} & 0 \end{pmatrix}.$$

Hence

$$w_t w_1^{-1} = \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix},$$

so every positive or negative diagonal matrix of determinant one belongs to the subgroup generated by N and $\bar{N}$ (the central sign $-I = w_{-1} w_1^{-1}$ is included). Gaussian elimination now finishes the proof: if $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has $a \neq 0$, then left multiplication by $\bar{n}_{-c/a}$ kills the lower-left entry, after which a diagonal element and an upper unipotent recover the resulting upper-triangular matrix. If $a = 0$, multiply first by w_1 , after which the upper-left entry is nonzero. Thus every $g \in \mathrm{SL}_2(\mathbb{R})$ is a product of elements of N and $\bar{N}$. $\square$

By Proposition 2.2, $H := \mathrm{SO}(\Delta)^\circ$ is therefore a connected subgroup generated by one-parameter Ad-unipotent subgroups. It is exactly the class of acting groups to which the completed Volume III orbit closure theorem applies.

2.4 The principal $\mathfrak{sl}_2$ inside $\mathfrak{sl}_3$

Identify V with the irreducible three-dimensional $\mathfrak{sl}_2$ module of highest weight 2. Then $\mathfrak{sl}(V)$ is eight-dimensional.

Proposition 2.4 (Decomposition 3+5). *As an $\mathfrak{sl}_2$ -module under the symmetric-square embedding,*

$$\mathfrak{sl}_3 \cong V_2 \oplus V_4,$$

where V_m denotes the irreducible highest-weight- m module of dimension $m + 1$. The summand V_2 is the embedded Lie algebra $\mathfrak{h} \simeq \mathfrak{sl}_2$ and V_4 is a five-dimensional irreducible complement.

Proof. The three-dimensional defining module is the irreducible highest-weight-2 module V_2 . Since V_2 is self-dual,

$$\mathrm{End}(V_2) \cong V_2 \otimes V_2.$$

We prove the needed Clebsch–Gordan decomposition directly. Let v_2, v_0, v_{-2} be weight vectors. In $V_2 \otimes V_2$, the vector $v_2 \otimes v_2$ has weight 4 and is killed by the raising operator, so it generates an irreducible submodule with weights 4, 2, 0, -2, -4, hence a copy of V_4 . The antisymmetric vector $v_2 \otimes v_0 - v_0 \otimes v_2$ has weight 2 and is killed by the raising operator after choosing the weight basis so that the raising operator sends v_0 to a nonzero multiple of v_2 and v_{-2} to a nonzero multiple of v_0 ; it generates a V_2 . Finally the invariant contraction pairing gives a nonzero weight zero vector killed by both raising and lowering operators, hence a V_0 . The dimensions $5 + 3 + 1 = 9$ exhaust $V_2 \otimes V_2$, so

$$\mathrm{End}(V_2) = V_4 \oplus V_2 \oplus V_0.$$

The scalar endomorphisms are exactly V_0 . Removing them gives $\mathfrak{sl}(V_2) = V_4 \oplus V_2$. The differential of the representation is a three-dimensional invariant Lie subalgebra, so it is the V_2 summand. $\square$

Theorem 2.5 (Maximality of H in SL_3). *Let $H = \mathrm{SO}(\Delta)^\circ < \mathrm{SL}_3(\mathbb{R})$. If $L < \mathrm{SL}_3(\mathbb{R})$ is a connected Lie subgroup with $H < L$, then either $L = H$ or $L = \mathrm{SL}_3(\mathbb{R})$.*

Proof. Let $\mathfrak{l}$ be the Lie algebra of L . Since $\mathfrak{h} \subset \mathfrak{l}$, the quotient $\mathfrak{l}/\mathfrak{h}$ is an $\mathfrak{h}$ -invariant subspace of $\mathfrak{sl}_3/\mathfrak{h} \cong V_4$. Proposition 2.4 says V_4 is irreducible. Thus the quotient is either 0 or all of V_4 . Accordingly $\mathfrak{l} = \mathfrak{h}$ or $\mathfrak{l} = \mathfrak{sl}_3$. Connected subgroups with the same Lie algebra coincide, giving the result. $\square$

Chapter 3

Closed Orthogonal Orbits Force Rationality

3.1 Quadratic forms attached to points of the lattice space

Put

$$G = \mathrm{SL}_3(\mathbb{R}), \quad \Gamma = \mathrm{SL}_3(\mathbb{Z}), \quad X = \Gamma \backslash G.$$

Let $Q_0 = \Delta$. For $g \in G$ and a nonzero scalar $\kappa \in \mathbb{R}$, define

$$Q(v) = \kappa Q_0(vg), \quad v \in \mathbb{R}^3.$$

Every nondegenerate indefinite ternary quadratic form is obtained this way.

Lemma 3.1 (Normal form). *If Q is a nondegenerate indefinite quadratic form on $\mathbb{R}^3$, there exist $g \in \mathrm{SL}_3(\mathbb{R})$ and $\kappa \neq 0$ such that $Q(v) = \kappa Q_0(vg)$.*

Proof. Sylvester's law of inertia gives $a \in \mathrm{GL}_3(\mathbb{R})$ and $c \neq 0$ with $Q(v) = cQ_0(va)$. Write $d = \det a \neq 0$ and choose the real cube root $r = d^{-1/3}$. Then $g = ra \in \mathrm{SL}_3(\mathbb{R})$. Since $Q_0(r^{-1}w) = r^{-2}Q_0(w)$, the scalar can be absorbed into κ , yielding the claim. $\square$

The stabilizer of Q in G is

$$K = gHg^{-1}.$$

Indeed, k preserves Q iff $g^{-1}kg \in H$.

3.2 A lattice in K makes K rational

Proposition 3.2 (Algebraic rationality of a finite-volume stabilizer). *If $K \cap \Gamma$ is a lattice in $K = gHg^{-1}$, then K is an algebraic subgroup defined over $\mathbb{Q}$.*

Proof. By the Borel-density theorem in the form proved in Volume II, the real Zariski closure of the lattice $K \cap \Gamma$ contains every unipotent element of K . The group K is conjugate to H , and H is generated by the two one-parameter unipotent groups of the preceding chapter. Hence the Zariski closure contains all of K . Conversely it is contained in K because K is algebraic. Thus

$$K = \mathrm{Zcl}_{\mathbb{R}}(K \cap \Gamma).$$

Since $K \cap \Gamma \subset \mathrm{SL}_3(\mathbb{Q})$, Corollary 1.2 says this Zariski closure is defined over $\mathbb{Q}$. $\square$

3.3 The invariant quadratic line

Let $\mathcal{Q} = \mathrm{Sym}^2((\mathbb{R}^3)^*)$, the six-dimensional space of quadratic forms. A subgroup $K < G$ acts by pullback.

Lemma 3.3 (Uniqueness of the invariant quadratic form). *The fixed space $\mathcal{Q}^H$ is the one-dimensional line $\mathbb{R}Q_0$. Consequently $\mathcal{Q}^K = \mathbb{R}Q$.*

Proof. A quadratic form R fixed by $H = \mathrm{SO}(Q_0)^\circ$ corresponds to a symmetric linear map $A : V \rightarrow V^*$ intertwining the defining H -representation with its dual. This three-dimensional representation is irreducible: under PSL_2 it is the irreducible symmetric-square module V_2 . If A is nonzero, its kernel is an invariant subspace and is therefore zero; thus A is an isomorphism. For two nonzero invariant forms A, B , the map $B^{-1}A$ commutes with the irreducible $\mathfrak{sl}_2$ action. We check the scalar conclusion directly from weights: it preserves the three one-dimensional weight spaces, say by scalars c_2, c_0, c_{-2} ; commuting with the lowering operator gives $c_2 = c_0 = c_{-2}$. Hence A is a scalar multiple of B . Taking $B = Q_0$ proves the first statement. Conjugating by g proves the second. $\square$

Theorem 3.4 (Closed orthogonal orbit implies rationality). *Let $Q(v) = \kappa Q_0(vg)$ and $x = \Gamma g$. If xH is closed and has finite H -invariant volume, then Q is proportional to a quadratic form with rational coefficients.*

Proof. The stabilizer of x in H is $H \cap g^{-1}\Gamma g$, a lattice in H . Conjugating gives $K \cap \Gamma$ as a lattice in $K = gHg^{-1}$. By Proposition 3.2, K is defined over $\mathbb{Q}$.

Choose rational defining equations for K . The fixed-space condition on a quadratic form $R \in \mathcal{Q}$ can be expressed infinitesimally as

$$X \cdot R = 0 \quad (X \in \mathfrak{k}),$$

where $\mathfrak{k}$ is obtained as the rational tangent space to those defining equations at the identity. Choosing a rational basis of $\mathfrak{k}$ yields a matrix of homogeneous linear equations with rational coefficients on the six coefficients of R . By Lemma 3.3, the real solution space is one-dimensional, namely $\mathbb{R}Q$. Lemma 1.3 therefore supplies a nonzero rational quadratic form on this same line. Thus Q is a real scalar multiple of a rational form. $\square$

Corollary 3.5 (Irrationality excludes a closed H -orbit). *If Q is not proportional to a rational quadratic form, then the orbit ΓgH is not closed.*

Proof. This is the contrapositive of the preceding rationality theorem for a closed H -orbit. $\square$

Chapter 4

The Ternary Oppenheim Theorem

4.1 Statement

A quadratic form Q is called *irrational* in the Oppenheim sense if it is not a real scalar multiple of a form with rational coefficients.

Theorem 4.1 (Oppenheim in dimension three). *Let Q be a nondegenerate indefinite irrational quadratic form on $\mathbb{R}^3$. Then*

$$Q(\mathbb{Z}^3) = \{Q(m) : m \in \mathbb{Z}^3\}$$

is dense in $\mathbb{R}$.

Proof route. The proof occupies this chapter: the homogeneous orbit is first shown to be dense, and the final section converts orbit density into density of the value set $Q(\mathbb{Z}^3)$.

4.2 The homogeneous orbit is dense

Choose κ and g as in Lemma 3.1; put $x = \Gamma g$ and $H = \mathrm{SO}(Q_0)^\circ$. By the generated-unipotent orbit-closure theorem proved in the replacement Chapter 17 of Volume III, there is a closed subgroup $F < G$ containing H such that

$$\overline{xH} = xF,$$

and xF° has finite volume.

Proposition 4.2. $\overline{xH} = X$.

Proof. The connected group F° contains H . By maximality (Theorem 2.5), either $F^\circ = H$ or $F^\circ = G$. If $F^\circ = H$, the generated-unipotent orbit-closure theorem says that $xF^\circ = xH$ carries a finite H -invariant measure. The finite-invariant-orbit criterion proved in Volume III, Corollary A.4.4, then makes xH closed. This contradicts Corollary 3.5. Therefore $F^\circ = G$, so $xF = X$. $\square$

4.3 Orbit density implies value density

Fix $t \in \mathbb{R}$. Choose a row vector $w_t \in \mathbb{R}^3$ such that

$$\kappa Q_0(w_t) = t.$$

For example, if $Q_0(a, b, c) = b^2 - 4ac$, take

$$w_t = (1, 0, -t/(4\kappa)).$$

Choose two further rows completing w_t to a matrix $k_t \in \mathrm{SL}_3(\mathbb{R})$; this is possible because $w_t \neq 0$ and a nonzero vector can be extended to a basis, after which one rescales one of the other rows to make determinant one.

By Proposition 4.2, there are $h_n \in H$ and $\gamma_n \in \Gamma$ such that

$$\gamma_n g h_n \longrightarrow k_t.$$

Set

$$m_n = e_1 \gamma_n \in \mathbb{Z}^3.$$

Then, using H -invariance of Q_0 ,

$$\begin{aligned} Q(m_n) &= \kappa Q_0(e_1 \gamma_n g) \\ &= \kappa Q_0(e_1 \gamma_n g h_n) \\ &\longrightarrow \kappa Q_0(e_1 k_t) = \kappa Q_0(w_t) = t. \end{aligned}$$

Since t was arbitrary, this proves Theorem 4.1.

4.4 What the proof really used

The arithmetic proof has four logically independent pieces:

- (1) $H \simeq \mathrm{PSL}_2(\mathbb{R})$ is generated by unipotents;
- (2) a closed finite-volume H -orbit forces Q to be rational up to scale;
- (3) H is maximal connected proper in SL_3 ;
- (4) density of xH converts to density of $Q(\mathbb{Z}^3)$ by taking first rows.

Ratner rigidity enters exactly between (1) and (3): it says the closure is homogeneous, leaving maximality only two possibilities.

Chapter 5

From Three Variables to Arbitrarily Many

5.1 Rational subspaces

Let Q be a quadratic form on $\mathbb{R}^n$ with associated symmetric bilinear form B . A subspace $V < \mathbb{R}^n$ is rational if it has a basis in $\mathbb{Q}^n$. Then $V \cap \mathbb{Z}^n$ is a full lattice in V after multiplying a rational basis by common denominators.

Lemma 5.1 (Existence of an indefinite rational three-plane). *If Q is nondegenerate and indefinite on $\mathbb{R}^n$, $n \geq 3$, there is a rational three-dimensional subspace V on which $Q|_V$ is nondegenerate and indefinite.*

Proof. Choose real vectors v_+, v_- with $Q(v_+) > 0$ and $Q(v_-) < 0$. Choose a third real vector so that their span is three-dimensional and the Gram determinant of B on that span is nonzero; this is possible because nondegeneracy of B makes the zero-determinant condition a proper algebraic condition on the third vector. Nondegeneracy and the presence of both signs are open conditions on the Grassmannian. Rational three-planes are dense in the real Grassmannian: approximate a basis by rational vectors. A sufficiently close rational plane therefore retains both properties. $\square$

5.2 The irrational ternary restriction lemma

Lemma 5.2 (One-vector nondegenerate extension). *Let (E, B) be a finite-dimensional real vector space with nondegenerate symmetric bilinear form,*

and let $S < E$ be a two-dimensional subspace containing a vector e with $B(e, e) \neq 0$. Then there is $p \in E$ for which $S + \mathbb{R}p$ is three-dimensional and the restriction of B to $S + \mathbb{R}p$ is nondegenerate. If B is indefinite and $B|_S$ is definite, p may be chosen so that $B|_{S+\mathbb{R}p}$ is indefinite. If $B|_S$ is degenerate, every such nondegenerate extension is automatically indefinite.

Proof. Because S contains an anisotropic vector, the radical of $B|_S$ has dimension at most one.

Suppose first that $B|_S$ is nondegenerate. Then $E = S \oplus S^\perp$ and $B|_{S^\perp}$ is nondegenerate. Choose an anisotropic $p \in S^\perp$. The restriction to $S \oplus \mathbb{R}p$ is the orthogonal direct sum of two nondegenerate forms and is therefore nondegenerate. If $B|_S$ is definite while B is indefinite, the orthogonal complement contains a vector of the opposite sign: otherwise the orthogonal sum would have the same sign on all of E . Choosing such a p makes the three-dimensional restriction indefinite. If $B|_S$ is already indefinite, any anisotropic $p \in S^\perp$ suffices.

Suppose now that $B|_S$ is degenerate. Its radical is a line $R = \mathbb{R}r$. Choose $s \in S$ with $B(s, s) = a \neq 0$; after replacing s by a scalar multiple we may use (s, r) as a basis of S . Nondegeneracy of B on E gives a p with $c = B(r, p) \neq 0$. In the basis (s, r, p) the Gram matrix has the form

$$\begin{pmatrix} a & 0 & b \\ 0 & 0 & c \\ b & c & d \end{pmatrix}, \quad \det = -ac^2 \neq 0.$$

Hence the extension is nondegenerate. It contains the nonzero isotropic vector r , so a nondegenerate real restriction containing r cannot be definite; it is therefore indefinite. $\square$

Theorem 5.3 (Irrational restriction). *Let Q be a nondegenerate indefinite quadratic form on $\mathbb{R}^n$, $n \geq 3$, which is not proportional to a rational form. Then there is a rational three-plane V such that $Q|_V$ is nondegenerate, indefinite, and not proportional to a rational form on V .*

Proof. Assume the contrary: every rational three-plane on which the restriction is nondegenerate and indefinite carries a form rational up to scale. By Lemma 5.1, choose such a plane V_0 . Rescale Q so that $Q|_{V_0}$ has rational coefficients in a rational basis. We shall show that all coefficients of Q are then rational.

Choose a rational basis e_1, e_2, e_3 of V_0 with $Q(e_i) \neq 0$ for all i . This is possible because the zero cone is a proper algebraic subset and rational vectors are dense. Let $v \in \mathbb{Q}^n \setminus V_0$ and fix i . Put $S_i = \text{span}_{\mathbb{R}}\{e_i, v\}$. Since $Q(e_i) \neq 0$, the hypotheses of Lemma 5.2 apply. If $Q|_{S_i}$ is degenerate, the lemma gives a nondegenerate three-dimensional extension, which is automatically indefinite. If $Q|_{S_i}$ is nondegenerate and already indefinite, choose any nondegenerate one-vector extension. If it is definite, use the indefiniteness of Q in the second clause of the lemma to choose an extension of opposite sign. Thus there is a real p for which

$$W_i = \text{span}_{\mathbb{R}}\{e_i, v, p\}$$

is three-dimensional and $Q|_{W_i}$ is nondegenerate and indefinite. These two properties, together with linear independence, are open conditions on p . Since $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$, we may choose a rational p_i in the same open set.

By the contradiction assumption, $Q|_{W_i} = c_i R_i$ for some rational quadratic form R_i on the rational plane W_i . Since e_i is rational, $Q(e_i) \in \mathbb{Q} \setminus \{0\}$ by our normalization, while $R_i(e_i) \in \mathbb{Q}$ and is nonzero. Therefore

$$c_i = Q(e_i)/R_i(e_i) \in \mathbb{Q}.$$

Thus every coefficient of $Q|_{W_i}$ in the rational basis (e_i, v, p_i) is rational. In particular

$$B(e_i, v) \in \mathbb{Q}, \quad Q(v) \in \mathbb{Q}.$$

Doing this for $i = 1, 2, 3$ shows $B(v, e_i) \in \mathbb{Q}$ for the entire rational basis of V_0 .

Now extend e_1, e_2, e_3 to a rational basis $e_1, \dots, e_n$ of $\mathbb{Q}^n$. Applying the previous paragraph to each $v = e_j$ shows $Q(e_j) \in \mathbb{Q}$ and $B(e_i, e_j) \in \mathbb{Q}$ for $i \leq 3$. To obtain $B(e_j, e_k)$ for $j, k > 3$, apply the same conclusion to the rational vector $v = e_j + e_k$:

$$Q(e_j + e_k) = Q(e_j) + Q(e_k) + 2B(e_j, e_k)$$

has rational left side and rational first two terms, so $B(e_j, e_k) \in \mathbb{Q}$. Thus the full Gram matrix of Q in a rational basis is rational. This contradicts the assumed irrationality of Q up to scale. Therefore the desired V exists. $\square$

5.3 The full Oppenheim theorem

Theorem 5.4 (Oppenheim). *Let $n \geq 3$ and let Q be a nondegenerate indefinite quadratic form on $\mathbb{R}^n$ which is not proportional to a rational form. Then $Q(\mathbb{Z}^n)$ is dense in $\mathbb{R}$.*

Proof. By Theorem 5.3, choose a rational three-plane V such that $Q|_V$ is nondegenerate, indefinite and irrational up to scale. The group $\Lambda = V \cap \mathbb{Z}^n$ is a rank-three lattice in V . Choose a $\mathbb{Z}$ -basis v_1, v_2, v_3 of a finite-index sublattice of Λ and identify $\mathbb{Z}^3$ with its span. The ternary quadratic form

$$q(a_1, a_2, a_3) = Q(a_1v_1 + a_2v_2 + a_3v_3)$$

is again nondegenerate, indefinite, and irrational up to scale: a rational multiple making q rational would make $Q|_V$ rational up to scale in the rational basis v_i . The ternary theorem gives density of $q(\mathbb{Z}^3)$ in $\mathbb{R}$. Since $q(\mathbb{Z}^3) \subset Q(\mathbb{Z}^n)$, the latter is dense as well. $\square$

5.4 Primitive vectors and harmless variants

If one wants values on primitive integer vectors, the preceding first-row argument can be arranged with $m_n = e_1\gamma_n$, which is automatically primitive because it is a row of an integral unimodular matrix. Thus in the ternary case the same proof gives density already along primitive vectors. For the higher-dimensional restriction, a primitive-in- V vector need not be primitive in $\mathbb{Z}^n$, so an additional elementary saturation argument is needed if that refinement is desired. We do not promote that refinement to a theorem here because it is not needed for Oppenheim itself.

Chapter 6

Arithmetic Joinings and Commensurator Correspondences

6.1 Why the operator converse matters

Volume III now proves both directions of the joining/Markov-operator correspondence. Consequently an intertwining positive Markov operator is not merely analogous to a joining: it uniquely determines one. Ratner measure classification on a product homogeneous space can therefore be read as a classification of such intertwiners.

6.2 Commensurators

Let G be a Lie group and $\Gamma < G$ a lattice. The commensurator is

$$\text{Comm}_G(\Gamma) = \{g \in G : [\Gamma : \Gamma \cap g^{-1}\Gamma g] < \infty, [g^{-1}\Gamma g : \Gamma \cap g^{-1}\Gamma g] < \infty\}.$$

For a in the commensurator define

$$\Gamma_a = \Gamma \cap a^{-1}\Gamma a.$$

The homogeneous space $\Gamma_a \backslash G$ maps to $X \times X$ by

$$\Gamma_a g \longmapsto (\Gamma g, \Gamma a g).$$

Proposition 6.1 (Finite correspondence). *The two coordinate projections of this homogeneous correspondence have finite degrees*

$$[\Gamma : \Gamma_a], \quad [a^{-1}\Gamma a : \Gamma_a],$$

respectively.

Proof. Fix the first coordinate Γg . Its preimages correspond to cosets of Γ_a inside Γ : replacing g by γg leaves the first coordinate unchanged, and two such representatives give the same point of $\Gamma_a \backslash G$ exactly when their quotient lies in Γ_a . This gives degree $[\Gamma : \Gamma_a]$. The second coordinate is the same calculation after conjugating by a , giving the second index. $\square$

6.3 Graph type

If a normalizes Γ , then $\Gamma_a = \Gamma$ and the correspondence is the graph of the automorphism $\Gamma g \mapsto \Gamma ag$. If a merely commensurates Γ , the relation is finite-to-finite rather than a graph. This distinction is precisely what appears in joining rigidity: a connected support subgroup may define an algebraic isomorphism only after passing to finite covers.

6.4 Example: modular Hecke correspondences

For $G = \mathrm{PSL}_2(\mathbb{R})$ and $\Gamma = \mathrm{PSL}_2(\mathbb{Z})$, rational matrices of positive determinant, after scalar normalization into $\mathrm{PGL}_2^+(\mathbb{Q})$, commensurate Γ . Taking the class of $\mathrm{diag}(p, 1)$ for a prime p gives the classical degree- $p+1$ Hecke correspondence. The degree can be checked by reducing the index problem modulo p : the relevant cosets are the $p+1$ lines in $\mathbf{P}^1(\mathbf{F}_p)$. This finite combinatorics is the arithmetic shadow of a homogeneous correspondence in $X \times X$.

Chapter 7

What Qualitative Ratner Does Not Give

7.1 Density versus rates

The Oppenheim theorem proved above is qualitative. It says that for every t and every $\varepsilon > 0$ there exists $m \in \mathbb{Z}^n$ with $|Q(m) - t| < \varepsilon$, but it gives no useful bound on $\|m\|$ as a function of ε . The step

$$\overline{xH} = X$$

is purely topological, and even pointwise equidistribution without an explicit rate does not provide an effective search bound.

7.2 Where an effective theorem would enter

An effective Oppenheim statement requires quantitative information at three places:

- (E1) a spectral or mixing rate for suitable flows or group averages;
- (E2) quantitative nondivergence with explicit dependence on the form;
- (E3) control of Sobolev norms when a small neighborhood of the target matrix k_t is approximated by a smooth test function.

Quantitative nondivergence is already developed in Volume II, but it is only one of these ingredients. We therefore do not state a spurious numerical rate in this volume.

7.3 Counting and expanding translates

Many modern applications ask for asymptotics of lattice points in expanding regions or for equidistribution of expanding closed homogeneous orbits. The qualitative architecture remains familiar:

nonescape + classification of limits + exclusion of proper limits.

What changes is that each arrow needs a quantitative replacement. This is a natural bridge to later volumes on effective homogeneous dynamics, but no future theorem is used in any proof above.

7.4 S -arithmetic directions

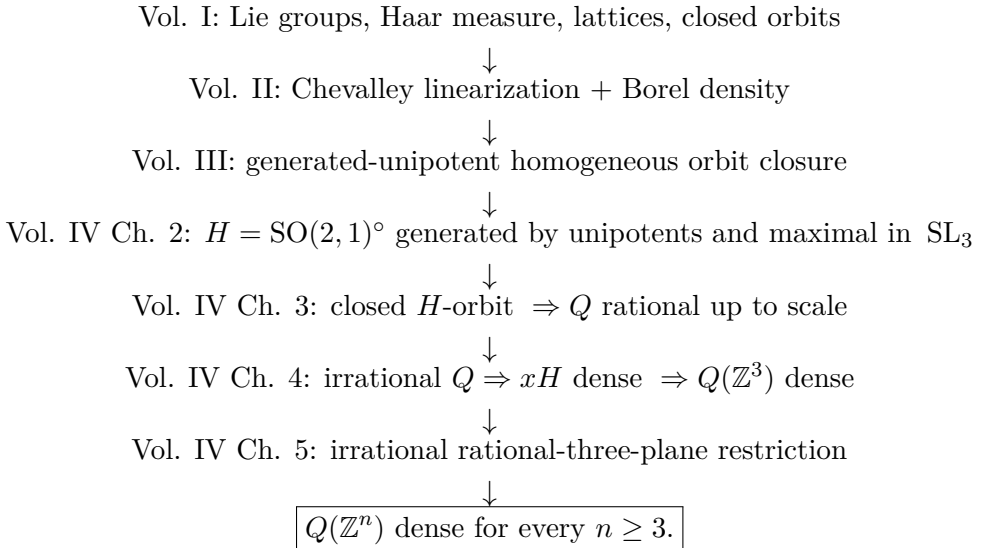
Over a product of real and p -adic groups, one replaces Lie-algebra polynomial divergence by polynomial behavior in each local field and lattices by S -arithmetic lattices. Component groups and compact open subgroups become more prominent. The generated-by-unipotents formulation of Volume III was written with this extension in mind, but the S -arithmetic theorem is not required for the real Oppenheim theorem and is therefore not asserted without proof here.

Chapter 8

Proof Ledger and Cross-Volume Dependency Map

8.1 The Oppenheim chain

The proof of Theorem 5.4 can be audited line by line through the following chain:



8.2 What is allowed as earlier-volume background

This volume does not reprove: the inverse function theorem, universal covering theory for Lie groups, Haar existence/uniqueness, Birkhoff and ergodic decomposition, the closed subgroup theorem, or the already completed Ratner rigidity theorems. These are either foundational results explicitly proved or designated as basic background in earlier volumes, or central results already proved there. No theorem first needed in this volume is hidden behind “standard” or “well known.”

8.3 No future dependency

No argument in this volume cites a theorem promised for a later volume. Sections on effectivity and S -arithmetic dynamics are explicitly marked as future directions and make no contribution to the proofs of the stated main theorems.

Appendix A

Matrix Formulas for the Symmetric-Square Representation

Let $g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$. Substituting $X \mapsto \alpha X + \gamma Y$, $Y \mapsto \beta X + \delta Y$ into $aX^2 + bXY + cY^2$ gives

$$\begin{pmatrix} a' \\ b' \\ c' \end{pmatrix} = \begin{pmatrix} \alpha^2 & \alpha\beta & \beta^2 \\ 2\alpha\gamma & \alpha\delta + \beta\gamma & 2\beta\delta \\ \gamma^2 & \gamma\delta & \delta^2 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix},$$

up to whether coefficient vectors are taken as rows or columns. Direct substitution verifies $b'^2 - 4a'c' = (\alpha\delta - \beta\gamma)^2(b^2 - 4ac)$, hence exact invariance on SL_2 .

For $e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, and $h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, differentiating gives three explicit 3×3 matrices satisfying $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$. Solving

$$X^T A + AX = 0 \quad (X = d\rho(e), d\rho(f), d\rho(h))$$

for a symmetric matrix A yields a one-dimensional solution space, another direct verification of Lemma 3.3.

Appendix B

A Minimal Checklist for Applying Ratner to an Arithmetic Problem

Before invoking rigidity in a new arithmetic problem, verify each item:

- (1) identify the arithmetic parameter set as a lattice quotient $\Gamma \backslash G$;
- (2) identify a subgroup W generated by Ad-unipotent one-parameter groups;
- (3) prove that the arithmetic quantity is invariant along right W -orbits;
- (4) classify connected subgroups between W and G in the relevant representation;
- (5) prove that every proper finite-volume intermediate orbit has the claimed arithmetic meaning, usually a rationality or degeneracy condition;
- (6) use the generated-unipotent orbit-closure theorem to eliminate all nonarithmetic closures;
- (7) convert density/equidistribution in the homogeneous space back into the original arithmetic statement with an explicit coordinate argument.

Skipping item (5) is the most common source of a slogan rather than a proof: “Ratner gives density” is only valid after proper homogeneous closures have been ruled out by arithmetic structure.

Appendix C

Worked Quadratic-Form Laboratory

C.1 Why quadratic forms become homogeneous dynamics

The orbit method is clearest for one form that can be calculated completely.

Worked Example C.1 (The cone and a one-parameter subgroup for $x^2 + y^2 - z^2$). Set $Q(x, y, z) = x^2 + y^2 - z^2$ and $J = \text{diag}(1, 1, -1)$. The identity component of $\{g : g^T J g = J, \det g = 1\}$ contains

$$b_t = \begin{pmatrix} \cosh t & 0 & \sinh t \\ 0 & 1 & 0 \\ \sinh t & 0 & \cosh t \end{pmatrix}.$$

Indeed, the (x, z) block preserves $x^2 - z^2$, hence $b_t^T J b_t = J$. The null cone is

$$z^2 = x^2 + y^2.$$

The vector $(1, 0, 1)$ is null, and $b_t(1, 0, 1)^T = e^t(1, 0, 1)^T$: the diagonal dynamics on a homogeneous quotient is already visible on one ray of the cone.

Worked Example C.2 (Irrational perturbation). For $Q_\alpha = x^2 + y^2 - \alpha z^2$ with $\alpha > 0$, a nonzero rational null vector $(p/q, r/q, s/q)$ would imply $\alpha = (p^2 + r^2)/s^2$. Thus for an irrational α there is no rational null line. This calculation is the simplest arithmetic reason why the irrational and rational orbit pictures are different.

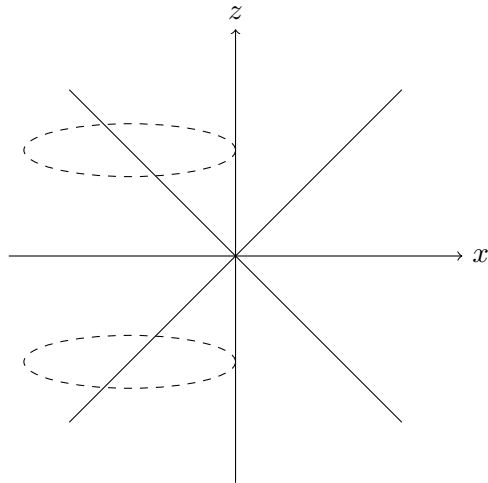


Figure C.1: A two-dimensional section of the null cone $x^2 + y^2 = z^2$. Hyperbolic one-parameter subgroups expand one null direction and contract the opposite one.

Appendix D

Student Proof Laboratory: From Ratner Rigidity to Oppenheim

This chapter rewrites the arithmetic bridge in a deliberately slower form. The main body has already proved the required statements. Here the goal is to make clear why each implication is true, which hypothesis is used, and where a plausible but invalid shortcut would occur. Nothing in this chapter is needed for logical completeness of the main proof; it is included so that a student can reconstruct the proof without supplying representation-theoretic or rationality arguments from memory.

D.1 Binary quadratic forms and the symmetric-square representation

Let $W = \mathbb{R}^2$ with basis X, Y . A binary quadratic form is

$$p(X, Y) = aX^2 + bXY + cY^2.$$

The coefficient space is $\text{Sym}^2(W^*) \cong \mathbb{R}^3$. If $g \in \text{SL}_2(\mathbb{R})$ acts on variables by $(X, Y) \mapsto (X, Y)g$, then substitution gives a linear action on (a, b, c) . This is the symmetric-square representation.

Lemma D.1 (The discriminant is invariant). *For $p = aX^2 + bXY + cY^2$ and $g \in \text{SL}_2(\mathbb{R})$,*

$$\Delta(p) := b^2 - 4ac$$

is unchanged by the symmetric-square action.

Proof. Write the symmetric matrix of p as

$$M_p = \begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix}.$$

Substitution by g replaces M_p by $g^T M_p g$. Therefore

$$\det(g^T M_p g) = (\det g)^2 \det M_p = \det M_p.$$

Since $\Delta(p) = -4 \det M_p$, the discriminant is invariant. □

Worked Example D.2 (The two unipotent root groups). For

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad v_s = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix},$$

substitution gives polynomial 3×3 matrices. Differentiating at $t = s = 0$ produces nilpotent endomorphisms of $\text{Sym}^2(W^*)$. Together with the derivative of $\text{diag}(e^r, e^{-r})$ they form an $\mathfrak{sl}_2$ -triple. Thus the image of SL_2 in the three-dimensional coefficient space is a connected subgroup of the orthogonal group of the discriminant form.

D.2 Why the image is exactly $\text{SO}(2, 1)^\circ$

The invariant discriminant has signature $(2, 1)$ after a real linear change of variables. Hence the symmetric-square homomorphism gives

$$\rho : \text{SL}_2(\mathbb{R}) \longrightarrow \text{SO}(2, 1)^\circ.$$

Lemma D.3 (Kernel and dimension). *The kernel of ρ is $\{\pm I\}$ and the differential $d\rho : \mathfrak{sl}_2 \rightarrow \mathfrak{so}(2, 1)$ is an isomorphism.*

Proof. If g acts trivially on every quadratic form, then it acts trivially on the set of squares ℓ^2 of linear forms. Hence the induced action on $\mathbb{P}(W^*)$ is trivial. A matrix acting trivially on the projective line is scalar, and determinant one leaves only $\pm I$. The kernel is discrete, so $d\rho$ is injective. Both Lie algebras have dimension three, hence $d\rho$ is an isomorphism. □

Proposition D.4 (Identification of the connected groups). *The induced map*

$$\mathrm{PSL}_2(\mathbb{R}) \longrightarrow \mathrm{SO}(2, 1)^\circ$$

is an isomorphism of Lie groups.

Proof. The differential is an isomorphism, so the image is an open connected subgroup of $\mathrm{SO}(2, 1)^\circ$. A connected topological group has no proper open subgroup. Thus the image is all of $\mathrm{SO}(2, 1)^\circ$, and the kernel has already been divided out. $\square$

D.3 Maximality inside SL_3 without a subgroup-classification table

Let V be the three-dimensional standard representation of $H = \mathrm{SO}(Q)^\circ$ for a nondegenerate indefinite ternary form Q . The Lie algebra decomposition

$$\mathfrak{sl}(V) = \mathfrak{so}(Q) \oplus \mathfrak{m}$$

may be described using the Q -adjoint: $\mathfrak{so}(Q)$ consists of Q -skew endomorphisms and $\mathfrak{m}$ of trace-zero Q -self-adjoint endomorphisms.

Lemma D.5 (The complement is irreducible). *As an H -module, $\mathfrak{m}$ is irreducible of dimension five.*

Proof. Via Q , a self-adjoint endomorphism corresponds to a symmetric bilinear form. Removing the scalar copy of Q gives

$$\mathfrak{m} \cong \mathrm{Sym}_0^2(V).$$

Under the double cover $\mathrm{SL}_2 \rightarrow H$, the module V is $\mathrm{Sym}^2(\mathbb{R}^2)$. The Clebsch–Gordan calculation

$$\mathrm{Sym}^2(\mathrm{Sym}^2 \mathbb{R}^2) \cong \mathrm{Sym}^4 \mathbb{R}^2 \oplus \mathrm{Sym}^0 \mathbb{R}^2$$

shows that the trace-free part is the irreducible module $\mathrm{Sym}^4 \mathbb{R}^2$ of dimension five. For completeness, one may verify irreducibility directly: starting from a nonzero weight vector, repeated application of the raising and lowering matrices generates all five weights. $\square$

Proposition D.6 (Connected maximality). *If F is a connected closed subgroup of $\mathrm{SL}_3(\mathbb{R})$ with*

$$H \subseteq F \subseteq \mathrm{SL}_3(\mathbb{R}),$$

then $F = H$ or $F = \mathrm{SL}_3(\mathbb{R})$.

Proof. Its Lie algebra $\mathfrak{f}$ contains $\mathfrak{h} = \mathfrak{so}(Q)$ and is stable under $\mathrm{Ad}(H)$. Therefore

$$\mathfrak{f}/\mathfrak{h}$$

is an H -submodule of the irreducible complement $\mathfrak{m}$. It is therefore either zero or all of $\mathfrak{m}$. Thus $\mathfrak{f} = \mathfrak{h}$ or $\mathfrak{f} = \mathfrak{sl}_3$. Connected Lie subgroups with the same Lie algebra agree, giving the claim. $\square$

Warning D.7 (Why dimension counting alone is insufficient). Knowing only $\dim H = 3$ and $\dim \mathrm{SL}_3 = 8$ does not exclude intermediate connected subgroups of dimensions 4, 5, 6, or 7. The irreducibility of the five-dimensional complement is the load-bearing input.

D.4 Closed orthogonal orbits force a rational form

Let $G = \mathrm{SL}_3(\mathbb{R})$, $\Gamma = \mathrm{SL}_3(\mathbb{Z})$ and let $x = \Gamma g$. The H -orbit xH is closed and of finite volume exactly when $gHg^{-1} \cap \Gamma$ is a lattice in gHg^{-1} .

Proposition D.8 (Invariant line and rationality). *Suppose $gHg^{-1} \cap \Gamma$ is a lattice in gHg^{-1} , where $H = \mathrm{SO}(Q_0)^\circ$. Then the form*

$$Q(v) = Q_0(g^{-1}v)$$

is proportional to a quadratic form with rational coefficients.

Proof. Let $\mathcal{Q} = \mathrm{Sym}^2((\mathbb{R}^3)^*)$. The conjugate group gHg^{-1} fixes the line $\mathbb{R}Q$. By Borel density in the form proved in Volume II, the lattice $gHg^{-1} \cap \Gamma$ is Zariski dense in the semisimple group gHg^{-1} . Hence the fixed line of the lattice in $\mathcal{Q}$ is the same one-dimensional line $\mathbb{R}Q$.

Every element of the lattice has integral matrix entries and therefore acts on the rational vector space $\mathcal{Q}_{\mathbb{Q}}$ by a rational matrix. The common fixed space of these rational matrices is the kernel of a system of linear equations with rational coefficients. Since over $\mathbb{R}$ this common fixed space is one-dimensional, it is spanned by a nonzero rational vector $Q_{\mathbb{Q}}$. Thus $Q = cQ_{\mathbb{Q}}$ for some $c \in \mathbb{R}^\times$. $\square$

Warning D.9 (What would be circular). It is not enough to say “closed arithmetic orbit implies the stabilizer is rational.” That sentence is a version of the result being proved. The argument above exhibits a specific rational representation, identifies the one-dimensional invariant subspace, and uses rational linear algebra.

D.5 The ternary density argument line by line

Let Q be an irrational indefinite ternary quadratic form, not proportional to a rational form. Put $H = \mathrm{SO}(Q)^\circ$ and $x = \Gamma e$ after choosing coordinates in which Q is the reference form.

Proposition D.10 (The only possible connected closure group is G). *The homogeneous closure furnished by the generated-unipotent orbit-closure theorem is xG .*

Proof. The group $H \cong \mathrm{PSL}_2(\mathbb{R})$ is generated by its two opposite one-parameter unipotent subgroups. Volume III therefore gives a connected closed subgroup F containing H such that

$$\overline{xH} = xF$$

and xF has finite F -volume. By connected maximality, either $F = H$ or $F = G$. If $F = H$, the orbit is closed of finite volume, and the rationality proposition would force Q to be proportional to a rational form. This contradicts the hypothesis. Hence $F = G$. $\square$

Proposition D.11 (Density of the orbit gives density of values). *If $\overline{\Gamma H} = \Gamma G$, then $Q(\mathbb{Z}^3)$ is dense in $\mathbb{R}$.*

Proof. Fix $a \in \mathbb{R}$ and $\varepsilon > 0$. Since Q is indefinite and nondegenerate, choose $v \in \mathbb{R}^3$ with $Q(v) = a$; if $a = 0$, choose a nonzero isotropic vector. Choose $g \in G$ whose first column is a suitably scaled vector close to v (the remaining columns may be completed so that $\det g = 1$). Density gives $\gamma \in \Gamma$ and $h \in H$ with γh arbitrarily close to g . Because h preserves Q , the first column of γh has the same Q -value as the first column of γ . The latter is an integer vector. Continuity of Q then makes its value within ε of a . $\square$

D.6 Why higher dimension reduces to dimension three

The final reduction must preserve three properties simultaneously: nondegeneracy, indefiniteness, and irrationality up to scale.

Lemma D.12 (Nondegenerate one-vector extension). *Let (V, B) be a nondegenerate real quadratic space and let $W \subset V$ be a nondegenerate subspace. If $u \in W^\perp$ satisfies $B(u, u) \neq 0$, then $W \oplus \mathbb{R}u$ is nondegenerate.*

Proof. If $w + tu$ is orthogonal to $W \oplus \mathbb{R}u$, then orthogonality to W forces $w = 0$ because $B|_W$ is nondegenerate, and orthogonality to u then forces $tB(u, u) = 0$, hence $t = 0$. $\square$

Proposition D.13 (Existence of an irrational rational three-plane). *Let Q be a nondegenerate indefinite form on $\mathbb{R}^n$, $n \geq 3$, not proportional to a rational form. Then there is a three-dimensional rational subspace W such that $Q|_W$ is nondegenerate, indefinite, and not proportional to a rational form.*

Proof. Choose rational vectors e_+, e_- with $Q(e_+) > 0$ and $Q(e_-) < 0$; density of $\mathbb{Q}^n$ and openness of the sign conditions allow this. After a small rational perturbation, the plane $P = \langle e_+, e_- \rangle$ is nondegenerate. It is already indefinite. Suppose every rational vector w for which $P \oplus \mathbb{R}w$ is nondegenerate produced a restriction proportional to a rational form. Comparing two such restrictions on their common plane P fixes the proportionality scalar. Consequently all coefficients $B(e_i, e_j)$ in a rational basis would be rational multiples of one common scalar: use restrictions containing P and e_i , and then restrictions containing P and $e_i + e_j$ to recover the cross terms. This would make Q proportional to a rational form on all of $\mathbb{R}^n$, contradiction. Therefore one admissible rational w gives the desired W . $\square$

Worked Example D.14 (A four-variable form). Consider

$$Q(x) = x_1^2 - x_2^2 + \sqrt{2}x_3^2 + x_4^2.$$

The rational plane $\langle e_1, e_2 \rangle$ is indefinite and nondegenerate. Adding e_3 gives the ternary restriction $x_1^2 - x_2^2 + \sqrt{2}x_3^2$, which is manifestly not proportional to a rational form. The ternary theorem applies immediately, and the values on integer vectors with fourth coordinate zero are already dense.

D.7 A student's dependency checklist

For the qualitative Oppenheim theorem the logical order is:

1. identify the ternary stabilizer with $\mathrm{PSL}_2(\mathbb{R})$;
2. prove it is generated by unipotents;
3. prove connected maximality in SL_3 ;
4. prove closed finite-volume stabilizer orbit implies rationality;
5. apply generated-unipotent orbit closure;
6. eliminate the proper closure using irrationality;
7. translate density in the lattice space into density of quadratic values;
8. reduce $n \geq 4$ to an irrational indefinite rational three-plane.

Every arrow is necessary. Ratner's theorem enters only at Step 5; it does not replace the representation theory or the arithmetic rationality argument.

Appendix E

Student Proof Laboratory: From Binary Quadratics to the Ternary Orthogonal Group

The main text gives the shortest rigorous route from Ratner orbit closure to Oppenheim. This chapter repeats the load-bearing algebra at a slower pace. The goal is that a reader who knows only the first three volumes can rebuild all of the representation-theoretic identifications rather than accepting “well known” isomorphisms between rank-one groups.

E.1 Binary quadratic forms and the symmetric-square representation

Let $W = \mathbb{R}^2$ with basis e_1, e_2 . Write

$$\mathrm{Sym}^2 W = \mathrm{span}\{e_1^2, 2e_1e_2, e_2^2\}.$$

An element $q = ae_1^2 + 2be_1e_2 + ce_2^2$ is identified with the symmetric matrix $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ and with the quadratic polynomial $aX^2 + 2bXY + cY^2$.

Proposition E.1 (The discriminant is the invariant quadratic form). *For $g \in \mathrm{SL}_2(\mathbb{R})$ and $q \in \mathrm{Sym}^2 W$, define $(g \cdot q)(v) = q(g^{-1}v)$. Then*

$$D(q) = b^2 - ac$$

is invariant. It has signature $(2, 1)$.

Proof. If Q is the symmetric matrix of q , then the matrix of $g \cdot q$ is $(g^{-1})^T Q g^{-1}$. Hence

$$\det((g^{-1})^T Q g^{-1}) = (\det g)^{-2} \det Q = \det Q.$$

Since $D(q) = -\det Q$, it is invariant. Put $x = (a + c)/2$, $y = (a - c)/2$, $z = b$. Then

$$D = z^2 - (x + y)(x - y) = y^2 + z^2 - x^2,$$

which has signature $(2, 1)$. Notice that the calculation also fixes the sign convention; later arguments involving the two sheets of the cone should not rely on memory for that sign. $\square$

Worked Example E.2 (The two unipotent subgroups). For $u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$, substitution gives

$$aX^2 + 2bXY + cY^2 \longmapsto a(X - tY)^2 + 2b(X - tY)Y + cY^2.$$

In the ordered coordinates (a, b, c) the action is triangular and polynomial of degree two in t . Repeating the calculation for the lower unipotent $\bar{u}_s$ displays the opposite root direction. These two explicit one-parameter subgroups are the concrete reason the image of SL_2 is generated by unipotents.

E.2 Kernel, image, and the identity component

Lemma E.3 (Kernel of the symmetric square). *If $g \in \mathrm{SL}_2(\mathbb{R})$ acts trivially on $\mathrm{Sym}^2 W$, then $g = \pm I$.*

Proof. The rank-one tensors v^2 span $\mathrm{Sym}^2 W$. If g acts trivially, then $(gv)^2 = v^2$ for every v . Thus $gv = \epsilon(v)v$ with $\epsilon(v) \in \{\pm 1\}$. Apply this to $e_1, e_2, e_1 + e_2$. The sign on $e_1 + e_2$ forces the signs on e_1 and e_2 to agree, so $g = \pm I$. $\square$

Proposition E.4 (Rank-one accidental isomorphism). *The induced homomorphism*

$$\mathrm{PSL}_2(\mathbb{R}) \longrightarrow \mathrm{SO}(D)^\circ$$

is an isomorphism of Lie groups.

Proof. The differential is injective because the group kernel is discrete. Both Lie groups have dimension three: $\dim \mathfrak{sl}_2 = 3$ and the Lie algebra of a nondegenerate orthogonal form in dimension three has dimension $3 \cdot 2/2 = 3$.

Hence the differential is an isomorphism. The image is therefore an open connected subgroup of $\mathrm{SO}(D)^\circ$. A connected Lie group has no proper open subgroup, so the image is all of the identity component. The kernel computation above gives injectivity after passing to PSL_2 . $\square$

E.3 The five-dimensional complement in $\mathfrak{sl}_3$

The maximality argument used in the Oppenheim proof is often compressed to “the principal $\mathfrak{sl}_2$ is maximal”. Here is the exact finite- dimensional reason.

Lemma E.5 (Adjoint decomposition). *Under the embedded $\mathfrak{sl}_2 \simeq \mathfrak{so}(2, 1)$,*

$$\mathfrak{sl}_3 \cong V_2 \oplus V_4,$$

where V_m denotes the irreducible $\mathfrak{sl}_2$ -module of highest weight m . The summand V_2 is the embedded Lie algebra and V_4 is five-dimensional.

Proof. The defining three-dimensional representation of the embedded $\mathfrak{sl}_2$ is $V_2 = \mathrm{Sym}^2(\mathbb{R}^2)$. Hence

$$\mathrm{End}(V_2) \cong V_2 \otimes V_2 \cong V_4 \oplus V_2 \oplus V_0$$

by the Clebsch–Gordan formula, which in this case can be verified directly from weights $2, 0, -2$. Removing the scalar summand V_0 leaves $\mathfrak{sl}_3 = V_4 \oplus V_2$. $\square$

Corollary E.6 (Maximal connected subgroup). *Let $H = \mathrm{SO}(Q)^\circ < \mathrm{SL}_3(\mathbb{R})$ for a nondegenerate ternary form. If a connected subgroup F satisfies $H < F < \mathrm{SL}_3(\mathbb{R})$, then $F = \mathrm{SL}_3(\mathbb{R})$.*

Proof. Its Lie algebra $\mathfrak{f}$ contains V_2 and is stable under $\mathrm{ad} V_2$. If it contains any nonzero vector in the complement V_4 , then irreducibility forces all of V_4 into $\mathfrak{f}$. Thus $\mathfrak{f} = V_2 \oplus V_4 = \mathfrak{sl}_3$. Equality of Lie algebras and connectedness give equality of groups. $\square$

E.4 Closed orthogonal orbits and rationality

Let $G = \mathrm{SL}_3(\mathbb{R})$, $\Gamma = \mathrm{SL}_3(\mathbb{Z})$ and $H = \mathrm{SO}(Q)^\circ$. A point of $\Gamma \backslash G$ is a unimodular lattice written in the right- quotient convention. The orbit ΓgH is closed of finite volume if and only if $gHg^{-1} \cap \Gamma$ is a lattice in gHg^{-1} .

Proposition E.7 (Finite-volume orthogonal orbit forces a rational line). *If ΓgH has finite H -volume, then the line spanned by the transported quadratic form $Q \circ g^{-1}$ is defined over $\mathbb{Q}$.*

Proof. Put $L = gHg^{-1}$ and $\Lambda = L \cap \Gamma$. Borel density, in the form proved in Volume V and in the special form already available from Volume II, says that Λ is Zariski dense in L . The group L fixes the line $\ell = \mathbb{R}(Q \circ g^{-1})$ in the rational representation $\text{Sym}^2((\mathbb{R}^3)^*)$. Therefore every element of the integer group Λ fixes ℓ . Choose finitely many $\lambda_1, \dots, \lambda_m \in \Lambda$ whose common fixed-line equations have the same solution as the whole Zariski closure. In a rational basis of $\text{Sym}^2((\mathbb{R}^3)^*)$, the equations

$$(\rho(\lambda_i) - I)v = 0$$

have integer coefficients. Their one-dimensional solution space therefore has a nonzero rational vector. Hence ℓ is rational. $\square$

Warning E.8 (Why the finite-volume hypothesis matters). A closed orbit of an arbitrary subgroup need not supply a lattice in the stabilizer, and without a lattice there is no Zariski-dense arithmetic set from which rationality can be recovered. The proof uses both closedness and finite invariant volume, not merely topological closedness.

E.5 The ternary Oppenheim implication line by line

Suppose now that Q is indefinite, nondegenerate, and irrational up to scale. Let $H = \text{SO}(Q)^\circ$. Ratner–Shah orbit closure for the subgroup generated by unipotent one-parameter groups gives

$$\overline{\Gamma H} = \Gamma F$$

for a connected closed F containing H , with finite F -volume. By maximality, $F = H$ or $F = G$. If $F = H$, the preceding rationality proposition forces Q to be rational up to scale, a contradiction. Hence the orbit is dense.

Lemma E.9 (Dense orbit gives density of values). *If ΓH is dense in ΓG , then $Q(\mathbb{Z}^3)$ is dense in $\mathbb{R}$.*

Proof. Fix $a \in \mathbb{R}$ and $\epsilon > 0$. Because Q is indefinite, choose $v \in \mathbb{R}^3$ with $Q(v) = a$; if $a = 0$, choose a nonzero isotropic v and then perturb it slightly

if a nonzero target is wanted. Scale or perturb v so that it is primitive in the real sense needed below. Choose $g \in G$ with $ge_1 = v$ after first normalizing the determinant of a basis extending v . Density gives $\gamma_n h_n \rightarrow g$ for $\gamma_n \in \Gamma$, $h_n \in H$. Then

$$Q(\gamma_n e_1) = Q(\gamma_n h_n e_1) \longrightarrow Q(ge_1) = a,$$

because h_n preserves Q . The vectors $\gamma_n e_1$ are integral and primitive. Thus values at integral vectors approach a . $\square$

E.6 From n variables to three variables without a genericity gap

Lemma E.10 (One-vector extension of a nondegenerate plane). *Let (V, Q) be a nondegenerate indefinite quadratic space over $\mathbb{Q}$ of dimension at least three. If $W \subset V$ is a nondegenerate rational subspace of dimension two, then there exists $v \in V(\mathbb{Q})$ such that $W \oplus \mathbb{Q}v$ is nondegenerate. Moreover v may be chosen so that the restriction has either indefinite signature allowed by Q .*

Proof. The orthogonal complement $W^\perp$ is rational and nondegenerate. A nondegenerate quadratic form cannot vanish on every rational point of a rational vector space: in coordinates its defining polynomial is nonzero, and $\mathbb{Q}^m$ is Zariski dense in $\mathbb{R}^m$. Choose $v \in W^\perp(\mathbb{Q})$ with $Q(v) \neq 0$. Then the orthogonal direct sum is nondegenerate. If a particular sign is needed and available in $W^\perp$, choose v in the corresponding open cone, which contains rational points by density. $\square$

Proposition E.11 (Existence of an irrational rational ternary restriction). *Let Q be a nondegenerate indefinite form on $\mathbb{R}^n$, $n \geq 3$, not proportional to a rational form. Then there is a rational three-dimensional subspace $V \subset \mathbb{R}^n$ such that $Q|_V$ is nondegenerate, indefinite, and not proportional to a rational form.*

Proof. Assume the contrary. Fix a rational nondegenerate indefinite two-plane W after a rational change of coordinates; if necessary start with a rational vector of each sign and perturb inside their rational span. Extend W by rational vectors v_i as in the preceding lemma. By hypothesis, for every i the Gram matrix of Q on $W + \mathbb{Q}v_i$ is rational up to a scalar. The common restriction to W forces all these scalars to agree. Now also use the triples

$W + \mathbb{Q}(v_i + v_j)$. Expanding $Q(v_i + v_j)$ recovers the cross term $B_Q(v_i, v_j)$. Consequently every entry of the Gram matrix of Q in a rational basis is a common real scalar times a rational number. This says that Q itself is rational up to scale, contrary to assumption. $\square$

Appendix F

Student Proof Laboratory: Reading the Oppenheim Proof as a Dependency Graph

F.1 What is dynamical and what is arithmetic

The proof has two sharply different halves. Ratner theory supplies the homogeneous closure. Everything after that is finite-dimensional algebra: maximality of the orthogonal subgroup, rationality of a finite-volume closed orbit, and the passage from a dense group orbit to values of a quadratic form. Keeping these halves separate prevents a common misunderstanding: Ratner's theorem does not itself mention quadratic forms.

F.2 A diagnostic checklist

For any attempted homogeneous-dynamics proof of a density theorem, a student should be able to answer the following questions explicitly.

1. Which subgroup is generated by unipotent one-parameter groups?
2. What theorem classifies its orbit closure in the chosen left/right quotient convention?
3. Which intermediate connected subgroups can occur?
4. Why does a proper finite-volume intermediate orbit imply an arithmetic condition?

5. How is density of the group orbit converted into the desired arithmetic approximation statement?

For Oppenheim, the preceding sections answer these questions without a remaining unnamed lemma.

Appendix G

Guided Reconstruction of the Oppenheim Argument

This chapter is meant to be read before the compact proof ledger at the end of the volume. We reconstruct the arithmetic argument without assuming that the reader can automatically move between quadratic forms, orthogonal groups, homogeneous orbits, and integer values. Every change of language is made explicit.

G.1 Quadratic forms as points of a representation space

Let $V = \mathbb{R}^3$ and write a quadratic form as

$$Q(v) = v^T A v,$$

where A is a symmetric nonsingular 3×3 matrix. The group $G = \mathrm{SL}_3(\mathbb{R})$ acts on forms by

$$(g \cdot Q)(v) = Q(g^{-1}v).$$

In matrix notation the new matrix is

$$A_g = (g^{-1})^T A g^{-1}.$$

The stabilizer of Q is therefore

$$H_Q = \{g \in \mathrm{SL}_3(\mathbb{R}) : g^T A g = A\}.$$

When Q has signature $(2, 1)$, its identity component is a copy of $\mathrm{SO}(2, 1)^\circ$.

Worked Example G.1 (The split ternary form). Take

$$Q_0(x, y, z) = x^2 + y^2 - z^2, \quad A_0 = \text{diag}(1, 1, -1).$$

The equation $g^T A_0 g = A_0$ says that the columns of g form an A_0 -orthonormal basis. This makes H_{Q_0} concrete: it is not an abstract group introduced for the theorem, but exactly the group of linear changes of variables that leave the value of the form unchanged.

G.2 Why the symmetric-square representation produces $SO(2, 1)$

Let $W = \text{Sym}^2(\mathbb{R}^2)$. Write an element as the binary quadratic form

$$ax^2 + 2bxy + cy^2.$$

Its discriminant is

$$\Delta = b^2 - ac.$$

If $g \in \text{SL}_2(\mathbb{R})$ acts by change of variables, the discriminant is unchanged. Thus the three-dimensional representation $\text{Sym}^2 : \text{SL}_2(\mathbb{R}) \rightarrow \text{GL}(W)$ lands in the special orthogonal group of the quadratic form $b^2 - ac$.

Lemma G.2 (Kernel of the symmetric-square representation). *The kernel of $\text{Sym}^2 : \text{SL}_2(\mathbb{R}) \rightarrow \text{GL}(W)$ is $\{\pm I\}$.*

Proof. If g acts trivially on $\text{Sym}^2(\mathbb{R}^2)$, it fixes the squares v^2 of every vector v . Therefore it preserves every line in $\mathbb{R}^2$. A linear map preserving every line is scalar, so $g = \lambda I$. Since $\det g = 1$, $\lambda = \pm 1$. Both signs act trivially on the symmetric square. $\square$

The differential is injective and both the source modulo its finite center and $\text{SO}(2, 1)^\circ$ have dimension three. The image is connected and open, hence it is the identity component. This gives

$$\text{PSL}_2(\mathbb{R}) \simeq \text{SO}(2, 1)^\circ.$$

The isomorphism matters because the elementary upper and lower unipotent subgroups of SL_2 become unipotent one-parameter subgroups of the orthogonal stabilizer. Consequently H_Q lies exactly in the class of groups to which the generated-by-unipotents orbit-closure theorem applies.

G.3 Maximality inside SL_3 without classification tables

The key subgroup alternative in the ternary proof is that a connected closed subgroup L satisfying

$$H_Q < L < SL_3(\mathbb{R})$$

must equal $SL_3(\mathbb{R})$. We explain the Lie-algebra calculation.

Use the trace form to decompose

$$\mathfrak{sl}_3 = \mathfrak{h} \oplus \mathfrak{m},$$

where $\mathfrak{h} = \mathfrak{so}(Q)$ and $\mathfrak{m}$ is the space of Q -symmetric trace-zero endomorphisms. Under $H_Q \simeq \mathrm{PSL}_2(\mathbb{R})$, the first summand is the three-dimensional irreducible module $\mathrm{Sym}^2(\mathbb{R}^2)$ and the second is the five-dimensional irreducible module $\mathrm{Sym}^4(\mathbb{R}^2)$.

Lemma G.3 (Irreducibility forces maximality). *If a Lie subalgebra $\mathfrak{l}$ contains $\mathfrak{h}$ and one nonzero vector of $\mathfrak{m}$, then $\mathfrak{l} = \mathfrak{sl}_3$.*

Proof. Because $\mathfrak{l}$ is closed under brackets with $\mathfrak{h}$, the intersection $\mathfrak{l} \cap \mathfrak{m}$ is an $\mathfrak{h}$ -submodule of $\mathfrak{m}$. Irreducibility of $\mathfrak{m}$ makes this intersection all of $\mathfrak{m}$. Hence $\mathfrak{l}$ contains both summands of the direct sum decomposition and is all of $\mathfrak{sl}_3$. $\square$

The only point still to justify is the irreducibility of the five-dimensional summand. Choose an $\mathfrak{sl}_2$ -triple in $\mathfrak{h}$. The weight spaces in $\mathfrak{m}$ have weights $4, 2, 0, -2, -4$, each with multiplicity one, and the raising/lowering operators connect consecutive weights. Any nonzero invariant subspace therefore contains a highest-weight vector and then all five weight vectors. This is the elementary highest-weight proof needed here; no classification of maximal subgroups is being used.

G.4 What a closed orthogonal orbit says arithmetically

Let $x = \Gamma g \in \Gamma \backslash G$ with $\Gamma = SL_3(\mathbb{Z})$, and consider xH_Q . Its stabilizer in H_Q is

$$\Lambda = H_Q \cap g^{-1}\Gamma g.$$

If xH_Q is closed of finite volume, then Λ is a lattice in H_Q . Borel density, in the form proved earlier, says that Λ is Zariski dense in H_Q .

Now H_Q is the stabilizer of the projective line $[Q]$ in the six-dimensional representation $\text{Sym}^2((\mathbb{R}^3)^*)$. Because $g\Lambda g^{-1} \subset \Gamma$ consists of integral matrices, the invariant line can be recovered over $\mathbb{Q}$.

Lemma G.4 (A rational invariant line from an arithmetic lattice). *Let $\rho : \mathbf{G} \rightarrow \text{GL}(W)$ be a rational representation and let $\Lambda < \mathbf{G}(\mathbb{Q})$ be Zariski dense. If a line $\ell \subset W(\mathbb{R})$ is fixed by Λ , then ℓ is defined over $\mathbb{Q}$.*

Proof. Choose a rational basis of W . The condition that a vector w span a common eigenline for a finite set $\lambda_1, \dots, \lambda_r \in \Lambda$ is a system of polynomial equations with rational coefficients in the projective coordinates of $[w]$. Since Λ is Zariski dense, choose a finite subset whose common projective fixed set has the same component through ℓ as the full group fixed set; finite-dimensional Noetherianity of the polynomial coordinate ring gives such a subset. For the orthogonal representation at hand the fixed set of H_Q is the single projective point $[Q]$. Thus the rational polynomial system has a unique real solution in that component. Gaussian elimination on the corresponding linear equations $(\rho(\lambda_i) - c_i I)w = 0$ after eliminating the eigenvalue scalars produces a one-dimensional rational nullspace. Hence ℓ has a rational spanning vector. $\square$

For the application, this means that a closed finite-volume H_Q -orbit can occur only when Q is proportional to a rational form. The contrapositive is the arithmetic input used in the density argument.

G.5 Dense orbit implies dense values: the approximation step

Suppose the H_Q -orbit of x is dense in $\Gamma \backslash G$. We want to see a small value $Q(v)$ with $v \in \mathbb{Z}^3$ directly.

Choose a real vector w on the null cone $Q(w) = 0$ with $w \neq 0$. Choose a matrix $g_0 \in G$ whose first column is a small scalar multiple of w and whose other columns complete it to determinant one. Density gives $h \in H_Q$ and $\gamma \in \Gamma$ for which γgh is close to g_0 . Let $v = \gamma^{-1}e_1 \in \mathbb{Z}^3 \setminus \{0\}$. Because h preserves Q ,

$$Q(g^{-1}v) = Q(h^{-1}g^{-1}\gamma^{-1}e_1).$$

The vector on the right is close to the first column of g_0 , hence close to the null vector. Continuity of Q makes the value small. Varying the target vector slightly off the cone gives density in all of $\mathbb{R}$, not only near zero.

Warning G.5. The integer vector comes from the lattice element γ , not from rounding a real null vector coordinatewise. Rounding would destroy the quadratic constraint at precisely the scale one is trying to control.

G.6 The higher-dimensional reduction

Let Q be irrational indefinite on $\mathbb{R}^n$, $n > 3$. We need a rational three-dimensional subspace L such that $Q|_L$ remains indefinite, nondegenerate, and irrational up to scale.

Start with two rational vectors u, v for which the binary restriction is indefinite and nondegenerate; rational vectors are dense, so such a pair can be chosen near any real positive/negative pair. If every rational vector w extending this pair produced a ternary restriction proportional to a rational form, comparison on the common two-plane would fix the scalar. The polarization identities

$$B_Q(w, u) = \frac{Q(w + u) - Q(w) - Q(u)}{2},$$

$$B_Q(w, v) = \frac{Q(w + v) - Q(w) - Q(v)}{2}.$$

would then force all coefficients of Q in a rational basis to be rational up to one common scalar. This contradicts irrationality. Therefore some rational w gives the required ternary restriction. The ternary theorem applies to $Q|_L$, and $L \cap \mathbb{Z}^n$ is a full lattice in L , so the values it produces are already values of the original form on integer vectors after a fixed denominator is cleared.

G.7 The proof architecture in one diagram

The entire qualitative arithmetic argument can now be read as

$$\begin{array}{c}
 Q \text{ irrational indefinite} \Downarrow \\
 \frac{H_Q \simeq \mathrm{PSL}_2(\mathbb{R}) \text{ generated by unipotents}}{xH_Q = xL \text{ (generated-unipotent orbit closure)}} \Downarrow \\
 L = H_Q \text{ or } G \text{ (maximality)} \Downarrow \\
 L = H_Q \Rightarrow Q \text{ rational up to scale (closed-orbit rationality)} \Downarrow \\
 L = G \text{ for irrational } Q \Downarrow \\
 xH_Q \text{ dense} \Rightarrow Q(\mathbb{Z}^3) \text{ dense in } \mathbb{R}.
 \end{array}$$

Every arrow now points either to a theorem proved in Volumes I–III or to a proof in the present volume. This is the standard expected of all later applications in the series.

Proof Guide and Dependency Summary

This chapter is a finalization layer. It does not introduce a new theorem scope. Its purpose is to expose the proof architecture, make the hypotheses that are easiest to misread impossible to miss, and give a student a concrete model calculation before the abstract machinery is reused later.

1. The arithmetic bridge: what Ratner does and what it does not do

A recurring application has the form

$$\begin{array}{c} \text{arithmetic parameter} \\ \Downarrow \\ \Gamma \backslash G \\ \Downarrow \\ U\text{-orbit} \\ \Downarrow \\ \overline{xU} = xH \\ \Downarrow \\ \text{arithmetic conclusion} \end{array}$$

The rigidity theorem controls the middle arrow. It does *not* by itself identify H , and it does not by itself translate density of xH into the original Diophantine statement. Those are separate algebraic and coordinate steps.

For a quadratic form Q , the standard diagnostic is:

1. identify G and the stabilizer H_Q ;
2. identify the lattice quotient carrying the arithmetic lattice;
3. show the relevant unipotent subgroup lies in H_Q ;

4. classify proper connected intermediate groups that could support a finite-volume orbit;
5. prove that each proper alternative forces the forbidden rational degeneracy;
6. only then convert orbit density back to values of Q .

This ordering prevents the slogan “Ratner gives density” from replacing the algebraic part of the proof.

2. Model calculation: a unipotent stabilizer of an indefinite form

Let

$$Q(x, y, z) = 2xz - y^2.$$

For

$$u_t = \begin{pmatrix} 1 & t & t^2/2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix},$$

a direct substitution shows $Q(u_tv) = Q(v)$. Thus the stabilizer contains a one-parameter unipotent group. The point of this calculation is not that this particular form is universal; it is that the dynamics enters through an *explicit invariance identity*. In every later arithmetic application, the first student exercise should be to verify the corresponding identity by hand.

3. Proof-reading checklist

Before accepting an arithmetic application, check:

1. left/right quotient conventions are fixed;
2. the acting unipotent group is genuinely Ad-unipotent;
3. proper homogeneous closures have been classified, not merely named;
4. finite volume of a candidate orbit is proved rather than inferred from closedness;
5. the final coordinate conversion gives the claimed quantitative or qualitative arithmetic statement.

Bibliographic notes

The Oppenheim conjecture was proved by Margulis through homogeneous dynamics. The presentation here is organized around the later Ratner rigidity package because it exposes a reusable mechanism: homogeneous orbit closure plus a classification of intermediate subgroups and an arithmetic description of closed orbits. Historical references are useful for attribution and for alternative proofs, but every theorem used in the logical chain of this volume is either proved here or explicitly inherited from the completed preceding volumes.

Algebraic and Arithmetic Infrastructure for Post–Ratner Dynamics

Volume V of a Detailed Notebook on Ratner’s Theorems

Detailed Notebook on Ratner’s Theorems

August 2026

Preface and proof policy

This volume begins the post–Ratner continuation. It develops only the algebraic and arithmetic structure that will actually be used in Volumes VI–XIV. It is not intended as a replacement for a general treatise on linear algebraic groups. The organizing rule is proof ownership: whenever a structural assertion becomes load bearing later in the series, its proof is placed here or reduced here to a result already proved in Volumes I–IV.

The basic Lie-group, lattice, Haar-measure, geometry-of-numbers, Chevalley linearization, and Borel-density mechanisms in the restricted forms used in the original Ratner proof are inherited from Volumes I–III. The present volume strengthens the algebraic language, proves the rationality and reduction statements needed for varying homogeneous measures and counting, and records explicit matrix laboratories in which every abstract construction can be checked by hand.

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Dependency and proof-ownership contract

Purpose. Algebraic/arithmetic infrastructure.

Prerequisites. Volumes I–IV and ordinary finite-dimensional Lie/algebraic preliminaries. Linear reductivity, rational Jacobson–Morozov, the cocharacter–parabolic construction, and Cartan–Iwasawa structure are proved internally in this volume before their load-bearing arithmetic uses.

New objects. Q -groups, parabolics, heights, reduction. Every project-specific technical term is defined before its first load-bearing use or recalled by an exact backward reference.

Main outputs. structural input for post-Ratner dynamics.

External-input rule. Every nontrivial imported result is either part of the declared graduate-level prerequisites, an explicitly named external research input in the prerequisite map, or is proved in this series before use (possibly under an explicit deferred-proof contract). A citation by itself never constitutes a proof.

Reader map and proof route

Question of the volume. Which pieces of algebraic-group and arithmetic structure are genuinely needed by the post–Ratner volumes, and where are their proofs owned?

Prerequisite checkpoint. The volume uses the earlier Lie-group and Chevalley machinery. The structural results needed later are proved here: characteristic-zero linear reductivity and the Reynolds operator, rational Jacobson–Morozov, the cocharacter–parabolic construction at the strength used by the arithmetic argument, and Cartan–Iwasawa structure for the connected real semisimple groups occurring later.

Load-bearing route. Use the main body for radicals, parabolics, rationality, heights, and arithmetic reduction. The complete-proof modules later in the volume own several arguments that were historically too compressed. In particular, Hilbert basis/Noetherianity and Reynolds projection are proved internally. The rational nilpotent-to-parabolic chain is also internal: Theorem 1.10 produces the rational SL_2 homomorphism and Theorem 1.11 produces the rational parabolic.

Owner boundary. A rational $\mathfrak{sl}_2$ triple does not, by itself, license the sentence “therefore the parabolic is rational.” The manuscript passes through the internally proved rational group homomorphism and rational cocharacter, then applies the internally proved cocharacter theorem. Keeping these two steps separate makes the field-of-definition argument visible.

Sanity check. In SL_3 , take $X = E_{12}$. The standard embedding $\phi : SL_2 \rightarrow SL_3$ on the first two coordinates gives $\lambda(t) = \text{diag}(t, t^{-1}, 1)$. Checking which

matrix entries remain bounded in $\lambda(t)g\lambda(t)^{-1}$ as $t \rightarrow 0$ displays the associated proper parabolic by hand.

Interface laboratory: nilpotent, cocharacter, parabolic

This calculation is deliberately elementary. Its purpose is to display, in coordinates, the two structural steps that are proved internally in the following foundational chapter: rational Jacobson–Morozov and the cocharacter–parabolic construction.

Take

$$X = E_{12} \in \mathfrak{sl}_3(\mathbb{Q}).$$

The standard copy of SL_2 in the first two coordinates gives a rational homomorphism

$$\phi: SL_2 \longrightarrow SL_3, \quad \phi \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b & 0 \\ c & d & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

whose differential sends the standard raising operator to X . The general existence of such a rational ϕ is proved in Theorem 1.10. It supplies the rational cocharacter

$$\lambda(t) = \phi \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} = \text{diag}(t, t^{-1}, 1).$$

Now no Jacobson–Morozov input remains. If $g = (g_{ij}) \in SL_3$, then the (i, j) entry of $\lambda(t)g\lambda(t)^{-1}$ is multiplied by $t^{a_i - a_j}$, where $(a_1, a_2, a_3) = (1, -1, 0)$. A limit as $t \rightarrow 0$ can exist only when entries of negative weight vanish. These are

$$g_{21} = g_{23} = g_{31} = 0.$$

Thus

$$P(\lambda) = \left\{ \begin{pmatrix} * & * & * \\ 0 & * & 0 \\ 0 & * & * \end{pmatrix} \in SL_3 \right\}.$$

The general assertion that this limit set is a parabolic subgroup, and that a cocharacter defined over $\mathbb{Q}$ produces a $\mathbb{Q}$ -parabolic, is proved in Theorem 1.11. The example therefore separates the two logical steps instead of hiding both inside the phrase “Jacobson–Morozov gives a rational parabolic.”

Chapter 1

Foundational Structure Theorems

This chapter removes the remaining foundational algebraic owners that are used later in the arithmetic proof. The results are deliberately proved at the strength actually needed: characteristic-zero linear reductivity for rational representations, the cocharacter description of rational parabolics, and the rational form of Jacobson–Morozov. No result from the effective-closing or rank-two research packages is used here.

1.1 Linear reductivity in characteristic zero

We begin with the Lie-algebra splitting theorem in a form that makes the Reynolds projection canonical enough for invariant theory.

Lemma 1.1 (Casimir scalar on a nontrivial irreducible module). *Let $\mathfrak{s}$ be a complex semisimple Lie algebra, let B be its Killing form, and let M be a finite-dimensional irreducible $\mathfrak{s}$ -module. If M is nontrivial and $\{x_i\}, \{y_i\}$ are B -dual bases, then*

$$C_M = \sum_i \rho(x_i)\rho(y_i)$$

is a nonzero scalar operator.

Proof. Invariance of B says that the tensor $\Omega = \sum_i x_i \otimes y_i$ is invariant under the diagonal adjoint action. Consequently the corresponding Casimir element is central in the universal enveloping algebra, so Schur’s lemma gives $C_M = cI$.

To see that $c \neq 0$, decompose $\mathfrak{s}$ into simple ideals and restrict to one acting nontrivially on M . With a Cartan subalgebra and a choice of positive roots, let $\lambda \neq 0$ be the highest weight of the resulting irreducible constituent. Writing the Casimir in the usual root-basis form and applying it to a highest-weight vector gives

$$c = (\lambda, \lambda + 2\rho),$$

where $(\ , \)$ is the positive-definite form on the real span of the roots induced by the Killing form and ρ is half the sum of the positive roots. Dominance of λ gives $(\lambda, \lambda) > 0$ and $(\lambda, \rho) \geq 0$, so $c > 0$ on every simple ideal that acts nontrivially. Summing over the simple ideals gives a nonzero scalar on M .

For completeness, the displayed eigenvalue is obtained by writing the Casimir as a Cartan quadratic term plus $\sum_{\alpha > 0} (E_{-\alpha}E_{\alpha} + E_{\alpha}E_{-\alpha})$; all E_{α} annihilate a highest-weight vector, and $E_{\alpha}E_{-\alpha}$ contributes $\lambda(H_{\alpha})$. Collecting these contributions is exactly $2(\lambda, \rho)$. $\square$

Lemma 1.2 (Whitehead's first lemma, proved by the Casimir). *Let $\mathfrak{s}$ be a semisimple Lie algebra over a field k of characteristic zero and let M be a finite-dimensional $\mathfrak{s}$ -module. Every 1-cocycle*

$$d : \mathfrak{s} \rightarrow M, \quad d([x, y]) = x \cdot d(y) - y \cdot d(x),$$

is inner: there is $m \in M$ such that $d(x) = x \cdot m$ for all x .

Proof. It is enough first to work after scalar extension to an algebraic closure; the final equation is a finite linear system with coefficients in k , so a solution over the extension implies one over k by Gaussian elimination.

We induct on $\dim M$. If M is irreducible and trivial, the cocycle vanishes because $\mathfrak{s} = [\mathfrak{s}, \mathfrak{s}]$. If M is irreducible and nontrivial, choose Killing-dual bases x_i, y_i and let the Casimir act by the nonzero scalar c from Lemma 1.1. Put

$$m = c^{-1} \sum_i \rho(x_i) d(y_i).$$

For $x \in \mathfrak{s}$, the invariance identity

$$\sum_i [x, x_i] \otimes y_i + x_i \otimes [x, y_i] = 0$$

and the cocycle equation give

$$\begin{aligned}
 \rho(x) \sum_i \rho(x_i) d(y_i) &= \sum_i \rho([x, x_i]) d(y_i) + \sum_i \rho(x_i) \rho(x) d(y_i) \\
 &= \sum_i \rho([x, x_i]) d(y_i) + \sum_i \rho(x_i) d([x, y_i]) + C_M d(x) \\
 &= c d(x).
 \end{aligned}$$

Thus $d(x) = x \cdot m$.

For general M , choose a proper nonzero submodule N . Project the cocycle to M/N . By induction it is inner there; subtracting the corresponding inner cocycle makes its values lie in N . A second application of induction to N makes the remaining cocycle inner. This completes the induction. $\square$

Theorem 1.3 (Weyl complete reducibility). *Every finite-dimensional module of a semisimple Lie algebra over a characteristic-zero field is completely reducible.*

Proof. Let $W \subset V$ be a submodule and put $Q = V/W$. Choose an arbitrary linear section $s : Q \rightarrow V$. Its failure to be equivariant is

$$d(x) = \rho_V(x)s - s\rho_Q(x) \in \text{Hom}(Q, W).$$

A direct expansion shows that d is a 1-cocycle for the natural $\mathfrak{s}$ -action on $\text{Hom}(Q, W)$. Lemma 1.2 gives $T \in \text{Hom}(Q, W)$ with $d(x) = x \cdot T$. Then $s - T$ is equivariant, and its image is a complement to W . $\square$

Theorem 1.4 (Linear reductivity for the rational reductive groups used here). *Let $\mathbf{L}$ be a reductive algebraic group over a characteristic-zero field k and let V be a finite-dimensional rational $\mathbf{L}$ -module. Then every $\mathbf{L}$ -submodule has a $\mathbf{L}$ -stable complement. Equivalently, the invariants functor $V \mapsto V^{\mathbf{L}}$ is exact.*

Moreover, on every locally finite rational $\mathbf{L}$ -module A there is a canonical projection

$$R : A \rightarrow A^{\mathbf{L}}$$

obtained by projecting each finite-dimensional semisimple submodule onto its trivial isotypic summand. It is natural for $\mathbf{L}$ -module maps.

Proof. After scalar extension to an algebraic closure, write

$$\text{Lie}(\mathbf{L})^\circ = \mathfrak{z} \oplus \mathfrak{s}$$

with $\mathfrak{s}$ semisimple and $\mathfrak{z}$ the Lie algebra of the central torus. A rational representation of a torus decomposes into weight spaces, and these weight spaces are $\mathfrak{s}$ -stable because the two summands commute. Theorem 1.3 completely reduces each weight space as an $\mathfrak{s}$ -module. Thus the identity component acts completely reducibly.

The component group $F = \mathbf{L}/\mathbf{L}^\circ$ is finite. If p is an $\mathbf{L}^\circ$ -equivariant projection onto a given $\mathbf{L}$ -stable submodule W , average its conjugates over F :

$$\bar{p} = \frac{1}{|F|} \sum_{r \in F} \rho(r) p \rho(r)^{-1}.$$

Because every summand restricts to the identity on W , so does $\bar{p}$; it is now $\mathbf{L}$ -equivariant. Descent from the algebraic closure to k again follows because the conditions defining an equivariant projection are linear equations over k .

For the canonical projection, a finite-dimensional semisimple module E has the intrinsic decomposition

$$E = E^{\mathbf{L}} \oplus E_{\text{nt}},$$

where E_{nt} is the sum of all nontrivial simple submodules. A module map sends $E^{\mathbf{L}}$ into invariants and E_{nt} into the nontrivial isotypic part, so projection onto $E^{\mathbf{L}}$ is natural. If a lies in a locally finite module A , choose any finite-dimensional stable submodule containing it and apply this projection. Naturality makes the answer independent of the choice. $\square$

Corollary 1.5 (Reynolds module identity). *If a reductive group $\mathbf{L}$ acts rationally by algebra automorphisms on a locally finite algebra A , the projection of Theorem 1.4 satisfies*

$$R(fa) = fR(a) \quad (f \in A^{\mathbf{L}}, a \in A).$$

If A is graded and the action preserves the grading, then R preserves degree.

Proof. For invariant f , multiplication $m_f : a \mapsto fa$ is an $\mathbf{L}$ -module map. Naturality gives $R \circ m_f = m_f \circ R$. Each graded projection is likewise a module map, so naturality preserves the grading. $\square$

1.2 Rational Jacobson–Morozov

We next prove the rational Jacobson–Morozov statement used in the compactness argument. The proof is algebraic over the ground field, so no Galois-descent black box is hidden in the word “rational.”

Lemma 1.6 (Jordan parts of a derivation). *Let k be a characteristic-zero field, let $\mathfrak{a}$ be a finite-dimensional k -Lie algebra, and let $D \in \operatorname{Der}_k(\mathfrak{a})$. The semisimple and nilpotent parts D_s, D_n of the additive Jordan decomposition are again k -derivations.*

Proof. Because k is perfect, D_s and D_n are polynomials in D with coefficients in k . After extending scalars to an algebraic closure, write $\mathfrak{a} = \bigoplus_{\lambda} V_{\lambda}$ for the generalized eigenspaces of D . The derivation identity implies

$$[V_{\lambda}, V_{\mu}] \subset V_{\lambda+\mu}.$$

Indeed, applying $(D - (\lambda + \mu))^r$ to a bracket and expanding by Leibniz makes it zero once r exceeds the sum of the two nilpotency indices. Hence D_s acts by $\lambda + \mu$ on a bracket of vectors from V_{λ}, V_{μ} , which is precisely the derivation identity for D_s . Then $D_n = D - D_s$ is a derivation as well. Since both are defined by k -polynomials in D , they descend to k . $\square$

Lemma 1.7 (Centralizer of a semisimple element). *Let $\mathfrak{g}$ be semisimple over a characteristic-zero field k and let $z \in \mathfrak{g}(k)$ be semisimple. Then $\mathfrak{c} = Z_{\mathfrak{g}}(z)$ is reductive over k and $[\mathfrak{c}, \mathfrak{c}]$ is semisimple.*

Proof. It is enough to verify the assertion after extending scalars to an algebraic closure. Put z in a Cartan subalgebra $\mathfrak{h}$ and use the root space decomposition

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_{\alpha}.$$

Then

$$Z_{\mathfrak{g}}(z) = \mathfrak{h} \oplus \bigoplus_{\alpha(z)=0} \mathfrak{g}_{\alpha}.$$

The roots vanishing on z form a closed root subsystem. The subalgebra generated by their root spaces and coroots is semisimple, while the complementary subspace in $\mathfrak{h}$ annihilated by all those roots is central. Thus the displayed centralizer is the direct sum of a semisimple algebra and its center. Both the radical and the derived algebra are characteristic, hence invariant under the Galois action, so this reductive decomposition descends to k . $\square$

Theorem 1.8 (Jacobson–Morozov over a characteristic-zero field). *Let $\mathfrak{g}$ be a semisimple Lie algebra over a characteristic-zero field k . For every nonzero nilpotent $e \in \mathfrak{g}(k)$ there are $h, f \in \mathfrak{g}(k)$ such that*

$$[h, e] = 2e, \quad [h, f] = -2f, \quad [e, f] = h.$$

Proof. We induct on $\dim \mathfrak{g}$. If e belongs to a proper semisimple k -subalgebra, apply the induction hypothesis there. Thus assume that no proper semisimple k -subalgebra contains e .

Step 1: produce a semisimple h with $[h, e] = 2e$. Let $\mathfrak{z}(e) = \ker(\operatorname{ad} e)$. If $x \in \mathfrak{z}(e)$, then $\operatorname{ad} e$ and $\operatorname{ad} x$ commute. Since $\operatorname{ad} e$ is nilpotent, so is $(\operatorname{ad} e)(\operatorname{ad} x)$, and therefore

$$B(e, x) = \operatorname{tr}(\operatorname{ad} e \operatorname{ad} x) = 0.$$

Invariance of the Killing form gives $[\mathfrak{g}, e] = \mathfrak{z}(e)^\perp$. Indeed, if $Y \in \mathfrak{z}(e)$, invariance of the Killing form gives

$$B([X, e], Y) = B(X, [e, Y]) = 0,$$

so $[\mathfrak{g}, e] \subset \mathfrak{z}(e)^\perp$. The map $\operatorname{ad} e$ has kernel $\mathfrak{z}(e)$, hence rank-nullity gives $\dim[\mathfrak{g}, e] = \dim \mathfrak{g} - \dim \mathfrak{z}(e)$. Nondegeneracy of B gives the same dimension for $\mathfrak{z}(e)^\perp$, so the inclusion is equality. Hence $e \in [\mathfrak{g}, e]$, so there is $h_0 \in \mathfrak{g}(k)$ with $[h_0, e] = 2e$.

Let $D = \operatorname{ad} h_0$ and write $D = D_s + D_n$. Since $De = 2e$, the Jordan decomposition has $D_s e = 2e$ and $D_n e = 0$. Lemma 1.6 shows that D_s is a k -derivation. Whitehead's lemma (Lemma 1.2, applied to the adjoint module) says every derivation of $\mathfrak{g}$ is inner, so $D_s = \operatorname{ad} h$ for some $h \in \mathfrak{g}(k)$. Faithfulness of the adjoint representation shows that h is semisimple, and $[h, e] = 2e$.

Step 2: prove $h \in [\mathfrak{g}, e]$. Suppose not. Since $[\mathfrak{g}, e] = \mathfrak{z}(e)^\perp$, there is $z \in \mathfrak{z}(e)$ with $B(h, z) \neq 0$. The semisimple operator $\operatorname{ad} h$ preserves $\mathfrak{z}(e)$ and gives the k -defined direct sum

$$\mathfrak{z}(e) = \ker(\operatorname{ad} h|_{\mathfrak{z}(e)}) \oplus \operatorname{im}(\operatorname{ad} h|_{\mathfrak{z}(e)}).$$

Because $B(h, [h, w]) = 0$, the component in the kernel still pairs nontrivially with h . We may therefore take z to commute with both e and h .

We now construct the Jordan parts as elements of $\mathfrak{g}$ rather than cite an element-level Jordan-decomposition theorem. Apply Lemma 1.6 to $D = \operatorname{ad} z$ and write $D = D_s + D_n$. Whitehead's lemma makes both derivations inner, say $D_s = \operatorname{ad} z_s$ and $D_n = \operatorname{ad} z_n$ with $z_s, z_n \in \mathfrak{g}(k)$. Since the adjoint representation is faithful, $z = z_s + z_n$. The operators D_s, D_n are polynomials in D , hence commute with $\operatorname{ad} e$ and $\operatorname{ad} h$; the zero center then gives $[z_s, e] = [z_n, e] = [z_s, h] = [z_n, h] = 0$. The operator $\operatorname{ad} z_s = D_s$ is semisimple and $\operatorname{ad} z_n = D_n$ is nilpotent. Moreover $\operatorname{ad} h$ commutes with $\operatorname{ad} z_n$, so their product

is nilpotent on each simultaneous generalized eigenspace and $B(h, z_n) = \text{tr}(\text{ad } h \text{ ad } z_n) = 0$. Thus $B(h, z_s) \neq 0$, in particular $z_s \neq 0$.

Let $\mathfrak{c} = Z_{\mathfrak{g}}(z_s)$. By Lemma 1.7, $\mathfrak{s} = [\mathfrak{c}, \mathfrak{c}]$ is semisimple. It is proper: otherwise z_s would centralize all of $\mathfrak{g}$, contradicting the zero center of a semisimple Lie algebra. Since $h, e \in \mathfrak{c}$ and $2e = [h, e]$, we have $e \in [\mathfrak{c}, \mathfrak{c}] = \mathfrak{s}$. This is a proper semisimple k -subalgebra containing e , contradicting the induction hypothesis. Hence $h \in [\mathfrak{g}, e]$.

Step 3: extract the -2 weight. Choose $f' \in \mathfrak{g}(k)$ with $[e, f'] = h$. Because $\text{ad } h$ is semisimple, decompose $f' = \sum_{\lambda} f_{\lambda}$ into its eigenspaces after scalar extension. The equation $[h, e] = 2e$ gives $[e, f_{\lambda}] \in \mathfrak{g}_{\lambda+2}$, whereas h has weight zero. Therefore the weight decomposition of $[e, f'] = h$ forces

$$[e, f_{-2}] = h.$$

The -2 eigenspace is the kernel of the k -linear operator $\text{ad } h + 2I$, so the component solving this equation may be chosen over k (equivalently solve the resulting linear system over k). Put $f = f_{-2}$. Then $[h, f] = -2f$ and $[e, f] = h$, completing the triple. $\square$

Lemma 1.9 (Integrating an $\mathfrak{sl}_2$ -triple). *Let k have characteristic zero, let $\mathbf{G} < \text{GL}(V)$ be a linear algebraic k -group, and let $\psi : \mathfrak{sl}_2 \rightarrow \text{Lie}(\mathbf{G})$ be a k -Lie-algebra homomorphism. There is a unique algebraic k -homomorphism*

$$\Psi : \text{SL}_2 \longrightarrow \mathbf{G}$$

with differential $d\Psi = \psi$.

Proof. First extend scalars to an algebraic closure $\bar{k}$. By Weyl complete reducibility, the induced $\mathfrak{sl}_2$ -module $V_{\bar{k}}$ is a direct sum of irreducibles. On an irreducible summand choose an eigenvalue m of h for which $m + 2$ is not in the finite spectrum of h , and choose a nonzero eigenvector v with $hv = mv$. Since $[h, e] = 2e$, the vector ev would have eigenvalue $m + 2$; hence $ev = 0$. The identities

$$ef^jv = j(m - j + 1)f^{j-1}v, \quad hf^jv = (m - 2j)f^jv$$

follow inductively from the $\mathfrak{sl}_2$ relations. Let ℓ be the largest index with $f^{\ell}v \neq 0$. Applying the first identity with $j = \ell + 1$ gives

$$0 = ef^{\ell+1}v = (\ell + 1)(m - \ell)f^{\ell}v,$$

so $m = \ell \in \mathbb{Z}_{\geq 0}$. The span of $v, fv, \dots, f^m v$ is stable under e, f, h and is nonzero; irreducibility therefore makes it the whole summand. In the basis $v, fv, \dots, f^m v$ the displayed formulas are exactly the differential of the polynomial action on homogeneous binary forms of degree m , namely $\text{Sym}^m(\bar{k}^2)$. Hence the representation of $\mathfrak{sl}_2$ integrates on each summand to an algebraic SL_2 -representation, and their direct sum gives $\tilde{\Psi} : \text{SL}_2 \rightarrow \text{GL}(V_{\bar{k}})$.

Its image lies in $\mathbf{G}_{\bar{k}}$. Indeed, $\tilde{\Psi}^{-1}(\mathbf{G}_{\bar{k}})$ is a closed subgroup of SL_2 whose Lie algebra is all of $\mathfrak{sl}_2$; in characteristic zero it is smooth of dimension 3, hence contains the identity component of the connected group SL_2 and is therefore all of SL_2 .

The integrated homomorphism is unique. On the two root subgroups $x_{\pm}(t)$ its value must be $\exp(t\psi(e))$ and $\exp(t\psi(f))$, respectively, because these nilpotent exponential series terminate; the two root subgroups generate SL_2 . Consequently every Galois conjugate of $\tilde{\Psi}$ has the same differential and hence equals $\tilde{\Psi}$. Its coordinate functions are therefore defined over k , which gives the required k -homomorphism Ψ . $\square$

Corollary 1.10 (Rational group-level Jacobson–Morozov). *Let $\mathbf{G}$ be a connected semisimple algebraic group over $\mathbb{Q}$ and let $0 \neq X \in \mathfrak{g}(\mathbb{Q})$ be nilpotent. There is a $\mathbb{Q}$ -homomorphism $\phi : \text{SL}_2 \rightarrow \mathbf{G}$ whose differential sends $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ to X .*

Proof. Apply Theorem 1.8 over $k = \mathbb{Q}$ and then Lemma 1.9. $\square$

1.3 Cocharacters and the parabolics needed for reduction theory

Only one direction of the general cocharacter–parabolic correspondence is needed in this series. Proving exactly that direction avoids importing the full theory of rational parabolic conjugacy.

Theorem 1.11 (A cocharacter defines a parabolic). *Let $\mathbf{G}$ be a connected reductive group over a characteristic-zero field k and let $\lambda : \mathbf{G}_m \rightarrow \mathbf{G}$ be a k -cocharacter. Then*

$$P(\lambda) = \{g \in \mathbf{G} : \lim_{t \rightarrow 0} \lambda(t)g\lambda(t)^{-1} \text{ exists}\}$$

is a parabolic k -subgroup. If

$$\mathfrak{g} = \bigoplus_{j \in \mathbb{Z}} \mathfrak{g}_j, \quad \text{Ad}(\lambda(t))X = t^j X \quad (X \in \mathfrak{g}_j),$$

then

$$\mathrm{Lie} P(\lambda) = \bigoplus_{j \geq 0} \mathfrak{g}_j, \quad \mathrm{Lie} R_u(P(\lambda)) = \bigoplus_{j > 0} \mathfrak{g}_j, \quad \mathrm{Lie} Z_{\mathbf{G}}(\lambda) = \mathfrak{g}_0.$$

If $\mathbf{G}$ is semisimple and λ is nontrivial, then $P(\lambda)$ is proper.

Proof. All assertions may be checked after scalar extension to an algebraic closure, except field of definition, which is already visible in the displayed limit condition. Put the image of λ in a maximal torus $\mathbf{T}$. For the root decomposition

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_{\alpha},$$

the root space $\mathfrak{g}_{\alpha}$ has weight $\langle \alpha, \lambda \rangle$. In a faithful representation, conjugation by $\lambda(t)$ multiplies a matrix coefficient from weight a to weight b by t^{a-b} ; hence existence of the limit is equivalent to the vanishing of all negative-weight coordinates. This proves that $P(\lambda)$ is closed and gives the stated Lie algebra.

Choose a positive root system for which $\langle \alpha, \lambda \rangle \geq 0$ on every positive root: first declare roots with positive pairing to be positive and then choose either sign on the zero-pairing root subsystem. The corresponding Borel subgroup generated by $\mathbf{T}$ and the positive root groups lies in $P(\lambda)$, so $P(\lambda)$ is parabolic. The strictly positive root groups form its unipotent radical and the zero-weight centralizer is a Levi subgroup, giving the remaining Lie-algebra formulas.

Because λ is defined over k , the condition that the conjugation map $\mathbf{G}_m \times \mathbf{G} \rightarrow \mathbf{G}$ extend across $t = 0$ is a system of polynomial equations over k in any affine embedding; therefore $P(\lambda)$ is defined over k .

Finally suppose $\mathbf{G}$ is semisimple. A nontrivial cocharacter cannot be central because the center is finite. Thus some root has nonzero pairing with λ ; replacing the root by its negative gives a negative weight. Its root subgroup is not contained in $P(\lambda)$, so the parabolic is proper. $\square$

Lemma 1.12 (A proper rational parabolic supplies a rational nilpotent). *Let $\mathbf{G}$ be connected semisimple over $\mathbb{Q}$ and let $\mathbf{P} < \mathbf{G}$ be a proper $\mathbb{Q}$ -parabolic. Then $\mathrm{Lie} R_u(\mathbf{P})(\mathbb{Q})$ contains a nonzero nilpotent element of $\mathfrak{g}(\mathbb{Q})$.*

Proof. Over an algebraic closure, choose a maximal torus and a Borel subgroup inside $\mathbf{P}$. A proper parabolic omits at least one negative simple-root group, so its unipotent radical contains at least one positive root group and is nontrivial. The unipotent radical is characterized intrinsically as the largest

connected normal unipotent subgroup of $\mathbf{P}$; hence it is fixed by the Galois action and is defined over $\mathbb{Q}$. Its Lie algebra is therefore a nonzero $\mathbb{Q}$ -vector space. Choose $0 \neq X$ in it. After triangularizing the unipotent group over an algebraic closure, every element of its Lie algebra is represented by a strictly upper-triangular matrix. Thus X is nilpotent; equivalently, $\text{ad } X$ is nilpotent in the semisimple adjoint representation. $\square$

Chapter 2

Affine Algebraic Groups and Rational Structures

2.1 Why the Zariski topology enters post–Ratner dynamics

In the first four volumes algebraic groups appeared mainly through linearization. From this point onward the subgroup itself varies, and one must control not merely one subgroup but families of possible algebraic hulls. The Zariski topology supplies the finite-dimensional compactness surrogate: equations have bounded algebraic complexity even when the corresponding real orbits move through a noncompact quotient.

Let k be a subfield of $\mathbb{R}$. An affine algebraic set over k is the common zero set in $\mathbb{R}^N$ of a family of polynomials from $k[x_1, \dots, x_N]$. An affine algebraic group is an affine algebraic set $\mathbf{G}$ for which multiplication and inversion are regular maps. We write $G = \mathbf{G}(\mathbb{R})$ for its real points.

Lemma 2.1 (Descending a linear kernel). *Let A be a matrix with entries in k . Then*

$$\ker(A : k^m \rightarrow k^n) \otimes_k \mathbb{R} = \ker(A : \mathbb{R}^m \rightarrow \mathbb{R}^n).$$

In particular a one-dimensional real kernel is spanned by a vector over k .

Proof. Gaussian elimination uses only the field operations in k . It produces a basis of the kernel over k and the same row-reduced matrix after scalar extension to $\mathbb{R}$. Rank therefore does not change under the extension $k \subset \mathbb{R}$, and the displayed identity follows. $\square$

Proposition 2.2 (Rational Zariski closure). *If $S \subset k^N$, then its real Zariski closure is defined over k . Consequently, if $\Lambda < \mathrm{GL}_N(k)$ is a subgroup, its real Zariski closure is an algebraic subgroup defined over k .*

Proof. For each degree d , the condition that a polynomial $p = \sum_{|\alpha| \leq d} c_\alpha x^\alpha$ vanish on S is a system of homogeneous linear equations in the coefficients c_α , with coefficients in k . Lemma 2.1 shows that the real solution space is obtained from the k -solution space by scalar extension. Taking the union over d proves that the vanishing ideal is generated by polynomials over k . Multiplication and inversion preserve the closure of a subgroup, because regular maps are Zariski continuous. $\square$

2.2 Coordinate rings and representations

The coordinate ring $k[\mathbf{G}]$ is the quotient of a polynomial ring by the defining ideal. A regular representation $\rho : \mathbf{G} \rightarrow \mathrm{GL}(V)$ is equivalent, after choosing a basis of V , to a matrix whose entries lie in $k[\mathbf{G}]$ and whose determinant is invertible there.

Lemma 2.3 (Finite-dimensionality of a translate span). *Let $f \in k[\mathbf{G}]$ and let*

$$(R_g f)(x) = f(xg).$$

Then the k -linear span of $\{R_g f : g \in \mathbf{G}(k)\}$ is finite dimensional.

Proof. Write the comultiplication of the Hopf algebra as $\Delta f = \sum_{j=1}^r f_j \otimes h_j$. Evaluating the second tensor factor at g gives $R_g f = \sum_j h_j(g) f_j$. Hence all right translates lie in the span of the finitely many f_j . $\square$

This is the algebraic reason Chevalley linearization is finite dimensional. Volume II proved the line-stabilizer construction needed for the original Ratner argument; we record a slightly more flexible formulation.

Theorem 2.4 (Chevalley line realization, rational form). *Let $\mathbf{H} < \mathbf{G} < \mathrm{GL}_N$ be algebraic groups defined over k . There is a finite-dimensional k -rational representation ρ of $\mathbf{G}$ and a k -rational line $\ell \subset V$ such that*

$$\mathbf{H} = \{g \in \mathbf{G} : \rho(g)\ell = \ell\}.$$

If $\mathbf{H}$ has no nontrivial rational characters in the representation under consideration, one may replace the line by a vector after tensoring with a one-dimensional inverse character.

Proof. Let $I(\mathbf{H}) \subset k[\mathbf{G}]$ be the vanishing ideal. By the Noetherian property choose generators $f_1, \dots, f_r$. By Lemma 2.3, the translate span W of these functions is finite dimensional and defined over k . Put $W_H = W \cap I(\mathbf{H})$. Right translation by $h \in \mathbf{H}$ preserves $I(\mathbf{H})$, so it preserves W_H . Conversely, if g preserves W_H , every generator vanishes on $\mathbf{H}g^{-1}$; evaluating at the identity after translating shows that g satisfies the defining equations of $\mathbf{H}$. Thus $\mathbf{H}$ is the stabilizer of the subspace W_H . Passing to $\bigwedge^{\dim W_H} W$ turns the subspace into the line $\bigwedge^{\dim W_H} W_H$. All constructions are over k . $\square$

2.3 Characters and their arithmetic meaning

A k -rational character is a k -morphism $\chi : \mathbf{G} \rightarrow \mathbf{G}_m$. Characters are the exact obstruction to finite covolume in the elementary direction needed later.

Proposition 2.5 (A lattice kills positive real characters). *Let G be a connected Lie group and $\Gamma < G$ a lattice. Then every continuous homomorphism $\chi : G \rightarrow \mathbb{R}_{>0}$ satisfying $\chi(\Gamma) = 1$ is trivial.*

Proof. The map descends to a measurable function $\bar{\chi}$ on $\Gamma \backslash G$. Right translation by g multiplies it by $\chi(g)$. Choose a set $E = \{1 \leq \bar{\chi} < 2\}$ of positive measure after replacing the interval if necessary. If $\chi(g) > 1$, the sets Eg^n eventually occupy pairwise disjoint multiplicative ranges of $\bar{\chi}$ but have equal measure, which contradicts finite total mass. Applying the same argument to g^{-1} excludes $\chi(g) < 1$. $\square$

Worked Example 2.6 (The determinant character). The group GL_n has the rational character $\det$. Its subgroup SL_n is the kernel. The existence of the determinant character explains why $\mathrm{GL}_n(\mathbb{Z})$ is not a lattice in the full positive-determinant group with finite Haar covolume unless one quotients the split central direction.

Chapter 3

Jordan Decomposition, Unipotence, and Algebraic Hulls

3.1 Polynomial construction of the Jordan decomposition

Let $A \in \mathrm{GL}_n(\mathbb{R})$. Over $\mathbb{C}$ its minimal polynomial factors as

$$m_A(X) = \prod_{j=1}^r (X - \lambda_j)^{e_j}.$$

Chinese remaindering gives polynomials p_j satisfying $p_j \equiv 1 \pmod{(X - \lambda_j)^{e_j}}$ and $p_j \equiv 0$ modulo the other primary factors. The operators $P_j = p_j(A)$ are the primary projections.

Theorem 3.1 (Multiplicative Jordan decomposition). *Every $A \in \mathrm{GL}_n(\mathbb{R})$ admits unique commuting factors*

$$A = A_s A_u = A_u A_s$$

where A_s is semisimple over $\mathbb{C}$ and A_u is unipotent. Moreover both are polynomials in A and A^{-1} with real coefficients. If A lies in a real algebraic subgroup $\mathbf{G} < \mathrm{GL}_n$, then $A_s, A_u \in \mathbf{G}(\mathbb{R})$.

Proof. On the primary space $P_j \mathbb{C}^n$ define $A_s = \lambda_j I$. Since the P_j are polynomials in A , $A_s = \sum_j \lambda_j P_j$ is polynomial in A ; complex conjugate

primary factors occur in pairs, so the resulting polynomial may be chosen real. Set $A_u = A_s^{-1}A$. On each primary space it equals $I + \lambda_j^{-1}(A - \lambda_j I)$, hence is unipotent. The factors commute because both are polynomials in A . Uniqueness follows by restricting a hypothetical decomposition to generalized eigenspaces: a commuting semisimple factor must supply the eigenvalue and a commuting unipotent factor has only eigenvalue one.

For the final statement, embed A in the Zariski closure of the cyclic group it generates. The polynomial formulas for the two factors are preserved by all polynomial relations defining that closure. Equivalently, the closure is a commutative algebraic group and its coordinate algebra separates the semisimple and unipotent parts; hence both factors remain in it and therefore in G . $\square$

3.2 One-parameter unipotent subgroups

In characteristic zero a matrix is unipotent precisely when it is $\exp N$ for a unique nilpotent N obtained from the terminating logarithm series.

Proposition 3.2. *If $u \in \mathrm{GL}_n(\mathbb{R})$ is unipotent then*

$$N = \log u = \sum_{j=1}^{n-1} \frac{(-1)^{j+1}}{j} (u - I)^j$$

is nilpotent and $u = \exp N$. Hence $u^t := \exp(tN)$ defines a canonical one-parameter unipotent subgroup through u at $t = 1$.

Proof. Both logarithm and exponential series terminate on nilpotent matrices, and the formal identities $\exp(\log(1 + X)) = 1 + X$ hold in the truncated polynomial ring $\mathbb{R}[X]/(X^n)$. $\square$

3.3 Algebraic hulls

For a subset $S \subset G$ define its algebraic hull to be $\mathrm{Zcl}(S)^\circ$. The hull is especially useful for families of homogeneous measures: it forgets finite-index accidents while preserving the connected directions visible to unipotent dynamics.

Lemma 3.3 (Hull of a generated unipotent group). *Suppose $W < G$ is generated by connected one-parameter unipotent subgroups. Then $\mathrm{Zcl}(W)^\circ$*

is generated as a connected algebraic group by the Zariski closures of those one-parameter groups, and it has no nontrivial algebraic quotient on which every generator acts trivially.

Proof. Let $\mathbf{H}$ be the algebraic subgroup generated by the closures of the generators. It contains W , hence contains $\mathrm{Zcl}(W)$. Conversely every generator lies in $\mathrm{Zcl}(W)$, so $\mathbf{H} \subset \mathrm{Zcl}(W)$. The quotient claim follows because a morphism trivial on a generating family is trivial on the algebraic subgroup they generate. $\square$

Worked Example 3.4 (The elementary subgroup of SL_n). The root subgroups $U_{ij} = \{I + tE_{ij} : t \in \mathbb{R}\}$ generate $\mathrm{SL}_n(\mathbb{R})$. Gaussian elimination expresses any matrix as a product of elementary matrices and a diagonal matrix; the latter is itself generated by 2×2 elementary blocks. Thus the algebraic hull of the root groups is all of SL_n .

Chapter 4

Split Tori, Restricted Roots, and Parabolic Geometry

4.1 Weight decomposition for a split torus

Let $A < \mathrm{GL}(V)$ be a connected subgroup consisting of commuting real diagonalizable matrices. Simultaneous diagonalization gives

$$V = \bigoplus_{\chi \in \Phi(V,A)} V_\chi, \quad av = \chi(a)v.$$

Writing $A = \exp \mathfrak{a}$, each character has the form $\chi(\exp H) = e^{\alpha(H)}$ for a linear functional $\alpha \in \mathfrak{a}^*$.

Apply this to the adjoint representation:

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{\alpha \in \Sigma} \mathfrak{g}_\alpha.$$

The elementary bracket identity $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$ follows by conjugating the bracket by A .

Proposition 4.1 (Horospherical Lie algebras). *Fix $H \in \mathfrak{a}$. Then*

$$\mathfrak{u}^+(H) = \bigoplus_{\alpha(H) > 0} \mathfrak{g}_\alpha, \quad \mathfrak{u}^-(H) = \bigoplus_{\alpha(H) < 0} \mathfrak{g}_\alpha$$

are nilpotent Lie algebras. Their connected groups satisfy

$$U^+(H) = \{g : e^{-tH} g e^{tH} \rightarrow e \text{ as } t \rightarrow +\infty\},$$

and similarly for $U^-(H)$ with time reversed.

Proof. The bracket relation adds weights. Since the root set is finite, a long enough iterated bracket of positive H -weights has weight outside the root set and vanishes, proving nilpotence. On $\mathfrak{g}_\alpha$ conjugation by e^{-tH} multiplies by $e^{-t\alpha(H)}$; exponential coordinates on a nilpotent group therefore give the contraction statement. $\square$

4.2 Parabolics attached to a one-parameter subgroup

Define

$$P(H) = \{g : \{e^{-tH}ge^{tH} : t \geq 0\} \text{ is relatively compact}\}.$$

Its Lie algebra is $\mathfrak{g}_0 \oplus \bigoplus_{\alpha(H) \geq 0} \mathfrak{g}_\alpha$. The unipotent radical is $U^+(H)$ and the Levi factor relevant for dynamics is the centralizer $Z_G(H)$, up to compact and finite components.

Worked Example 4.2 (SL_3). For $H = \mathrm{diag}(2, -1, -1)$, the roots $e_i - e_j$ satisfy $(e_1 - e_2)(H) = (e_1 - e_3)(H) = 3$ and $(e_2 - e_3)(H) = 0$. Hence $U^+(H)$ consists of matrices

$$\begin{pmatrix} 1 & x & y \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and $P(H)$ is the stabilizer of the line $\mathbb{R}e_1$. This is the basic maximal parabolic governing expanding translates of two-dimensional horospherical pieces.

4.3 Bruhat cells in the matrix cases used later

For SL_n , Gaussian elimination with row exchanges yields the Bruhat decomposition

$$G = \bigsqcup_{w \in S_n} BwB$$

for the upper triangular Borel B . The open cell corresponds to the longest permutation and is the product U^-AU^+ on a Zariski-open dense set. Volume IX will repeatedly use only this local open-cell factorization, so we record its analytic consequence.

Lemma 4.3 (Uniform open-cell coordinates). *Let C be a compact subset of the open Bruhat cell. Then the factors in the map*

$$U^- \times A \times U^+ \longrightarrow G, \quad (u_-, a, u_+) \mapsto u_- a u_+,$$

are smooth functions of $g \in C$ and all derivatives of fixed order are bounded uniformly on C .

Proof. The multiplication map has invertible differential throughout the open cell; its inverse is regular there (in matrix coordinates it is given by ratios of nonvanishing principal minors). Smoothness and compactness give the uniform derivative bounds. $\square$

Chapter 5

Arithmetic Subgroups and Integral Models

5.1 Commensurability as the right notion

Two subgroups are commensurable if their intersection has finite index in each. A subgroup $\Gamma < \mathbf{G}(\mathbb{Q})$ is arithmetic with respect to a faithful rational representation $\rho : \mathbf{G} \rightarrow \mathrm{GL}(V)$ and a lattice $L \subset V(\mathbb{Q})$ if it is commensurable with $\{g \in \mathbf{G}(\mathbb{Q}) : \rho(g)L = L\}$.

Proposition 5.1 (Independence under rational change of lattice). *If L_1, L_2 are full lattices in the same rational vector space, then their stabilizers in $\mathbf{G}(\mathbb{Q})$ are commensurable.*

Proof. Choose $m \geq 1$ with $mL_1 \subset L_2 \subset m^{-1}L_1$. The principal congruence subgroup of the stabilizer of L_1 modulo m^2L_1 preserves L_2 : if $g \equiv I \pmod{m^2}$, then $(g - I)m^{-1}L_1 \subset mL_1 \subset L_2$. Hence the intersection has finite index in the first stabilizer; symmetry gives the other index. $\square$

5.2 Restriction of scalars: the concrete construction

Let $K/\mathbb{Q}$ be a number field of degree d and choose a $\mathbb{Q}$ -basis $\omega_1, \dots, \omega_d$. Multiplication by $a \in K$ is represented by a $d \times d$ rational matrix $M(a)$. Replacing every K -entry of a matrix by its multiplication matrix embeds $\mathrm{GL}_n(K)$ in $\mathrm{GL}_{nd}(\mathbb{Q})$. Polynomial equations over K split into d rational

equations by expanding in the basis. This explicit construction is all the restriction-of-scalars formalism needed later.

Worked Example 5.2 (A real group from a quadratic field). If $K = \mathbb{Q}(\sqrt{d})$, then $K \otimes_{\mathbb{Q}} \mathbb{R}$ is $\mathbb{R} \times \mathbb{R}$ for $d > 0$ and $\mathbb{C}$ for $d < 0$. Thus an arithmetic subgroup of $\mathrm{SL}_2(K)$ embeds diagonally in $\mathrm{SL}_2(\mathbb{R})^2$ in the split case and in $\mathrm{SL}_2(\mathbb{C})$ in the imaginary case. These are exactly two of the ambient groups appearing in modern effective unipotent equidistribution.

Chapter 6

Reduction Theory for SL_n

6.1 Gram–Schmidt and Iwasawa coordinates

Write $G = \mathrm{SL}_n(\mathbb{R})$, $K = \mathrm{SO}(n)$, A the positive diagonal subgroup, and N the upper unitriangular subgroup. Gram–Schmidt gives a unique factorization $g = kan$ once the diagonal entries of a are required positive. Let

$$A_t = \{\mathrm{diag}(a_1, \dots, a_n) : a_i/a_{i+1} \geq t, \prod a_i = 1\}$$

and let N_0 be the set with all upper entries in $[-1/2, 1/2]$. A Siegel set is $\mathfrak{S}_t = KA_tN_0$.

6.2 A reduced basis lemma

Lemma 6.1 (Elementary lattice reduction). *For every unimodular lattice $\Lambda \subset \mathbb{R}^n$ there is a basis $v_1, \dots, v_n$ such that, after Gram–Schmidt orthogonalization v_i^* ,*

$$|\langle v_j, v_i^* \rangle| \leq \frac{1}{2} \|v_i^*\|^2 \quad (i < j)$$

and

$$\|v_{i+1}^*\| \geq c_n \|v_i^*\|$$

for a constant $c_n > 0$ depending only on n .

Proof. Choose v_1 to be a shortest primitive vector. Having chosen $v_1, \dots, v_i$, project the lattice to the orthogonal complement of their span and choose a shortest nonzero primitive projected vector; lift it to v_{i+1} and subtract

integer multiples of the earlier vectors so that each Gram–Schmidt coefficient lies in $[-1/2, 1/2]$. Minkowski’s second theorem, proved in Volume I, bounds the product of successive minima in each quotient. The minimality at the two adjacent stages then gives a lower bound for the ratio of successive orthogonal lengths depending only on the dimension. Induction completes the basis. $\square$

Theorem 6.2 (Siegel covering for SL_n). *There exists $t_n > 0$ such that*

$$\mathrm{SL}_n(\mathbb{R}) = \mathrm{SL}_n(\mathbb{Z}) \mathfrak{S}_{t_n}.$$

Moreover, for fixed $t > 0$ only finitely many $\gamma \in \mathrm{SL}_n(\mathbb{Z})$ satisfy $\gamma \mathfrak{S}_t \cap \mathfrak{S}_t \neq \emptyset$ modulo the standard parabolic stabilizers of the cusp faces.

Proof. Represent g by the lattice $\Lambda = g\mathbb{Z}^n$. Apply Lemma 6.1 and let $\gamma \in \mathrm{SL}_n(\mathbb{Z})$ be the basis-change matrix from the columns of g to the reduced basis. Gram–Schmidt of $g\gamma$ produces kan with $n \in N_0$ and adjacent ratios of diagonal entries bounded below by $t_n = c_n$ (up to reversing the conventional order of A). This proves the cover.

For overlap finiteness, if g_1, g_2 lie in a fixed truncated portion of $\mathfrak{S}_t$ and $g_2 = \gamma g_1$, then γ maps a lattice with uniformly controlled reduced basis to another such lattice. Its columns therefore have bounded Euclidean norm and integral entries, giving only finitely many possibilities. On a cusp face some initial Gram–Schmidt lengths are allowed to shrink. The shrinking indices determine a rational flag $0 < V_1 < \cdots < V_r = \mathbb{Q}^n$, and the integral matrices that can move arbitrarily far while retaining that face are precisely those preserving this flag, namely the parabolic subgroup P consisting of matrices preserving every subspace V_i of the flag. Modulo $P(\mathbb{Z})$, the induced basis changes on each quotient V_i/V_{i-1} still carry uniformly reduced bases to uniformly reduced bases. The bounded-column argument therefore gives finitely many possibilities on each quotient. Starting with the smallest quotient and lifting successively proves the assertion by induction on the flag length. There are only finitely many cusp-flag types for the fixed coordinate chamber, so the finiteness is uniform over the faces of $\mathfrak{S}_t$. $\square$

6.3 Finite volume by direct integration

In Iwasawa coordinates the Haar density has the form

$$dg = dk \delta(a)^{-1} da dn, \quad \delta(a) = \prod_{i < j} \frac{a_i}{a_j}.$$

Introduce simple-root coordinates $s_i = a_i/a_{i+1}$. The Jacobian is a constant multiple of $\prod ds_i/s_i$, while $\delta(a)$ is a product $\prod s_i^{m_i}$ with $m_i > 0$. Therefore

$$\int_{A_t} \delta(a)^{-1} da \ll \prod_{i=1}^{n-1} \int_t^\infty s_i^{-m_i} \frac{ds_i}{s_i} < \infty.$$

Since K and N_0 have finite measure, every Siegel set has finite Haar measure.

Corollary 6.3. $\mathrm{SL}_n(\mathbb{Z})$ is a lattice in $\mathrm{SL}_n(\mathbb{R})$.

Proof. Theorem 6.2 supplies a finite-measure measurable cover of a fundamental domain. Nonzero covolume follows from local injectivity near the identity, because $\mathrm{SL}_n(\mathbb{Z})$ is discrete. $\square$

Chapter 7

Reduction Theory for Rational Semisimple Groups

7.1 The induction parameter

For a connected semisimple algebraic $\mathbb{Q}$ -group $\mathbf{G}$, let $\text{rank}_{\mathbb{Q}} \mathbf{G}$ be the dimension of a maximal $\mathbb{Q}$ -split torus. The reduction argument is by this rank. Rank zero means that there is no $\mathbb{Q}$ -parabolic direction along which an arithmetic orbit can escape.

Lemma 7.1 (Rank-zero compactness criterion). *Let $\rho : \mathbf{G} \rightarrow \text{SL}(V)$ be faithful and defined over $\mathbb{Q}$, let $L \subset V(\mathbb{Q})$ be a lattice, and let Γ preserve L . If $\mathbf{G}$ has $\mathbb{Q}$ -rank zero, then $\Gamma \backslash G$ is compact.*

Deferred complete proof. A tempting shortcut is to pass from a sequence of short primitive wedges to one fixed rational contracting line. That passage was not justified and is not used for certification. the later finite-covolume chapter proves this lemma by Theorem M.26: after finite covolume is established independently, invariant characteristic-polynomial coefficients give a uniform adjoint-lattice systole and Mahler compactness gives compactness. $\square$

The final sentence is elementary in a faithful representation: a proper parabolic is the stabilizer of a flag arising from a one-parameter rational cocharacter; its image is a nontrivial rational split torus.

7.2 Rational parabolics and cusp directions

Fix a maximal $\mathbb{Q}$ -split torus $\mathbf{S}$. Relative root spaces are defined exactly as in Chapter 3. Choosing a positive system determines a minimal rational parabolic $\mathbf{P} = \mathbf{MAN}$, where $A = \mathbf{S}(\mathbb{R})^\circ$, N is generated by positive root groups, and M has smaller $\mathbb{Q}$ -rank after passing to the relevant derived factors.

Theorem 7.2 (Arithmetic reduction theorem). *Let $\mathbf{G}$ be connected semisimple over $\mathbb{Q}$ and let Γ be an arithmetic subgroup. There exist a compact set $C \subset G$, finitely many rational elements $q_1, \dots, q_r \in \mathbf{G}(\mathbb{Q})$, a bounded set $\Omega \subset MN$, and $t > 0$ such that*

$$G = \Gamma \bigcup_{i=1}^r q_i \Omega A_t C,$$

where A_t is defined by lower bounds on all simple rational roots. Each set $q_i \Omega A_t C$ has finite Haar measure.

Proof. The complete proof is Theorem N.24 in the rational-cusp chapter, Chapter N. Its left-cover form gives

$$G = \Gamma \bigcup_{i=1}^r q_i \Omega' A_{t'} C$$

with $q_i \in \mathbf{G}(\mathbb{Q})$, $\Omega' \subset MU$ compact (hence bounded), and C compact, and it proves finite Haar measure of every displayed Siegel piece. Renaming Ω', t' as Ω, t gives exactly the present statement. $\square$

Remark 7.3. The proof deliberately exposes the three points sometimes hidden under the phrase “reduction theory”: (i) escape creates a rational parabolic through an exterior-power height; (ii) the unipotent radical is reduced modulo its integral lattice; (iii) the Levi factor is handled by smaller rational rank. Those are precisely the pieces later quantitative arguments must make effective.

Chapter 8

The Arithmetic Lattice Theorem and Borel Density

8.1 Borel–Harish-Chandra in the form used later

Theorem 8.1 (Arithmetic lattice theorem). *Let $\mathbf{G}$ be a connected semisimple algebraic group defined over $\mathbb{Q}$. Then every arithmetic subgroup $\Gamma < \mathbf{G}(\mathbb{Q})$ is a lattice in $G = \mathbf{G}(\mathbb{R})^\circ$. The quotient is compact exactly when $\mathbf{G}$ has no proper $\mathbb{Q}$ -parabolic subgroup.*

Deferred complete proof. A proof through Theorem 7.2 would make the dependency order circular, because the stronger rational-cusp reduction is proved later. The later finite-covolume chapter therefore supplies an independent complete proof: Theorem M.23 proves finite covolume, and Theorem M.26 proves the compactness characterization. Those later proofs are the proof owners for this theorem. $\square$

8.2 Borel density: representation-theoretic form

The only dynamical input in the density proof is the following projective lemma. We prove it because the often-quoted phrase “Furstenberg’s lemma” hides exactly the compactness argument that is needed here.

Lemma 8.2 (Invariant projective measures). *Let G be a connected semisimple real Lie group with finite center and no compact factor, and let $\rho : G \rightarrow \mathrm{GL}(V)$*

be a finite-dimensional real representation. Every G -invariant probability measure on $\mathbb{P}(V)$ is supported on $\mathbb{P}(V^G)$.

Proof. It is enough to prove the assertion for an ergodic invariant probability ν , because the general case follows by ergodic decomposition and the closedness of $\mathbb{P}(V^G)$.

Choose, among all nonzero subspaces $W \leq V$ with $\nu(\mathbb{P}(W)) > 0$, one of minimal dimension. If W_1 and W_2 are two distinct translates of W , then $\mathbb{P}(W_1) \cap \mathbb{P}(W_2) = \mathbb{P}(W_1 \cap W_2)$ has zero ν -measure by minimality. All translates have the same positive measure. Consequently there can be only finitely many distinct translates of W . The connected group G acts continuously on this finite set, hence fixes W . Since $\mathbb{P}(W)$ is now an invariant set of positive measure, ergodicity gives $\nu(\mathbb{P}(W)) = 1$. Minimality also implies

$$\nu(\mathbb{P}(W')) = 0 \quad (0 \neq W' \subsetneq W). \quad (8.1)$$

We claim that the image of G in $\mathrm{PGL}(W)$ is relatively compact. If not, choose $g_n \in G$ and representatives $A_n = \rho(g_n)|_W$, multiplied by nonzero scalars so that $\|A_n\| = 1$, for which no subsequence converges to an invertible matrix up to scalar. After passing to a subsequence, $A_n \rightarrow A \neq 0$ and A is singular. For every $[w] \notin \mathbb{P}(\ker A)$,

$$[A_n w] \rightarrow [Aw] \in \mathbb{P}(\mathrm{im} A).$$

Let O be any open neighborhood of $\mathbb{P}(\mathrm{im} A)$. For a compact set $C \subset \mathbb{P}(W) \setminus \mathbb{P}(\ker A)$, the convergence is uniform on C , so $g_n C \subset O$ for all large n . Invariance gives $\nu(O) \geq \nu(C)$. Exhausting the complement of $\mathbb{P}(\ker A)$ by such compact sets and using (8.1) yields $\nu(O) = 1$. Shrinking O to the proper projective subspace $\mathbb{P}(\mathrm{im} A)$ gives $\nu(\mathbb{P}(\mathrm{im} A)) = 1$, contradicting (8.1). Thus the projective image is relatively compact.

Let $\mathfrak{g} = \bigoplus_i \mathfrak{g}_i$ be the decomposition into noncompact simple ideals. The differential of the projective representation maps each $\mathfrak{g}_i$ into the Lie algebra of a compact group. A nonzero homomorphism from a real simple noncompact Lie algebra into a compact Lie algebra is injective; pulling back the negative-definite invariant form of the compact algebra would then give a negative-definite invariant form on $\mathfrak{g}_i$, impossible because the Killing form of a noncompact simple real Lie algebra is not negative definite. Hence every $\mathfrak{g}_i$ acts trivially on $\mathbb{P}(W)$. Connectedness gives that G acts on W by scalars. The scalar is a continuous character of the connected semisimple group G , hence is identically one (its differential vanishes on $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$). Therefore $W \subset V^G$. Since ν is supported on $\mathbb{P}(W)$, the lemma follows. $\square$

Theorem 8.3 (Borel density). *Let G be a connected semisimple real algebraic group with no compact algebraic factor and finite center, and let $\Gamma < G$ be a lattice. Then Γ is Zariski dense in G .*

Proof. Let $\mathbf{H} = \text{Zcl}(\Gamma)$ and suppose $\mathbf{H} \neq \mathbf{G}$. Chevalley linearization, proved in Volume II and strengthened in Chapter 1 of the present volume, supplies a finite-dimensional algebraic representation $\rho : G \rightarrow \text{GL}(V)$ and a line $\ell = [v] \in \mathbb{P}(V)$ whose stabilizer is $H = \mathbf{H}(\mathbb{R}) \cap G$. Because $\Gamma \subset H$, the formula

$$F(\Gamma g) = [\rho(g^{-1})v]$$

is well defined. If m is Haar probability on $\Gamma \backslash G$, then $\nu = F_*m$ is a G -invariant probability measure on $\mathbb{P}(V)$: right translation on $\Gamma \backslash G$ is intertwined with the projective action $[w] \mapsto [\rho(g^{-1})w]$. Lemma 8.2 gives $\text{supp } \nu \subset \mathbb{P}(V^G)$. Since the identity coset belongs to the support of Haar measure and F is continuous, $[v] \in \text{supp } \nu$; hence $v \in V^G$ up to a scalar, and semisimplicity eliminates that scalar character exactly as in the last paragraph of the lemma. Thus the stabilizer of $[v]$ is all of G , contradicting the choice of the proper subgroup H . Therefore $\mathbf{H} = \mathbf{G}$. $\square$

Corollary 8.4 (Rationality from an arithmetic lattice). *Let $\mathbf{L} < \text{GL}_N$ be a connected semisimple real algebraic subgroup without compact factors. If $L \cap \text{GL}_N(\mathbb{Q})$ contains a lattice in L , then $\mathbf{L}$ is defined over $\mathbb{Q}$.*

Proof. The lattice is Zariski dense by Theorem 8.3. Its Zariski closure is defined over $\mathbb{Q}$ by Proposition 2.2; hence that closure is $\mathbf{L}$. $\square$

Chapter 9

Closed Arithmetic Orbits and Volume Normalization

9.1 The closed-orbit criterion revisited

Let G be a Lie group, $\Gamma < G$ discrete, $H < G$ closed, and $x = \Gamma g$. Volumes I and III proved the equivalence between a closed finite-volume orbit and a lattice stabilizer. We record the arithmetic upgrade.

Proposition 9.1 (Arithmetic closed-orbit criterion). *Assume $G = \mathbf{G}(\mathbb{R})^\circ$ with $\mathbf{G}$ defined over $\mathbb{Q}$ and Γ arithmetic. Let $\mathbf{H} < \mathbf{G}$ be a connected semisimple $\mathbb{Q}$ -subgroup. Then $\Gamma \cap H$ is a lattice in H , and the orbit ΓH is closed of finite volume.*

Proof. The intersection is arithmetic in $\mathbf{H}$: choose a common faithful rational representation and integral lattice after passing to finite-index subgroups. The complete proof of Theorem 8.1 given in the finite-covolume proof module therefore makes $\Gamma \cap H$ a lattice in H . Now apply the finite-covolume-stabilizer theorem proved in Volume III, Theorem A.4.2 (there labelled “Finite-covolume stabilizer implies closed orbit”), to the orbit of the identity coset. It says directly that

$$(\Gamma \cap H) \backslash H \longrightarrow \Gamma \backslash G$$

is proper and has closed image. No rational-parabolic reduction theorem is needed for this implication. $\square$

9.2 Normalization conventions

If $H_x = H \cap g^{-1}\Gamma g$ is a lattice, Haar measure dh on H descends to $H_x \backslash H$. We normalize

$$m_{xH} = \frac{dh}{\text{vol}(H_x \backslash H)}.$$

Later varying-measure arguments compare several such probabilities. The normalization removes arbitrary Haar scalars, but the covolume remains a geometric complexity parameter. In effective theory it will be replaced or supplemented by arithmetic height.

Lemma 9.2 (Conjugation and covolume). *Let $H' = gHg^{-1}$ and transport Haar measure by conjugation. Then*

$$\text{vol}((H \cap g^{-1}\Gamma g) \backslash H) = \text{vol}((H' \cap \Gamma) \backslash H').$$

Proof. Conjugation identifies both the groups and their lattice stabilizers, and by definition the Haar measures have been transported through that identification. $\square$

Chapter 10

Matrix Laboratories: SL_3 , Orthogonal Groups, and Flag Orbits

10.1 SL_3 root data by hand

Let A be the positive diagonal group and let $\alpha_{ij}(a) = a_i/a_j$. The root space for α_{ij} is $\mathbb{R}E_{ij}$. For $a_t = \mathrm{diag}(e^{2t}, e^{-t}, e^{-t})$, the expanding roots are α_{12} and α_{13} , yielding the two-dimensional abelian horosphere

$$U^+ = \left\{ \begin{pmatrix} 1 & x & y \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} : x, y \in \mathbb{R} \right\}.$$

For $b_t = \mathrm{diag}(e^t, 1, e^{-t})$, all three positive roots expand, but α_{13} has twice the exponent of the simple roots. This unequal weight structure will matter when polynomial trajectories are rescaled.

10.2 Orthogonal groups and rational isotropic flags

Let Q be a nondegenerate rational quadratic form. A rational isotropic line ℓ has parabolic stabilizer $P_\ell < \mathrm{SO}(Q)$. If such a line exists, the corresponding rational cocharacter creates a cusp in the arithmetic quotient. If Q is anisotropic over $\mathbb{Q}$, there is no rational isotropic line and hence no rational parabolic; Theorem 8.1 gives compactness of the arithmetic quotient.

Worked Example 10.1 (An indefinite real group with compact arithmetic quotient). It is possible for Q to be indefinite over $\mathbb{R}$ but anisotropic over $\mathbb{Q}$. Then $SO(Q)(\mathbb{R})$ is noncompact while an arithmetic lattice is cocompact. This is the cleanest example showing that real rank and rational rank govern different phenomena: mixing is controlled by real split directions, while cusps are controlled by rational split directions.

10.3 Flag stabilizers and exterior powers

For a rational r -plane $W \subset \mathbb{Q}^n$, choose a primitive wedge $v_W \in \bigwedge^r \mathbb{Z}^n$. The stabilizer of the line $\mathbb{R}v_W$ is the parabolic preserving W . Under g , the norm $\|gv_W\|$ is the covolume of $g(W \cap \mathbb{Z}^n)$ in gW . Thus the same exterior-power coordinate simultaneously records a rational flag, a parabolic subgroup, and a cusp height. This three-way dictionary is the central structural tool of later linearization and effective closing.

Chapter 11

Dependency Ledger and Structural Audit

11.1 What has been proved here

The volume has established, in the exact forms used later:

1. rationality of Zariski closures and Chevalley line realizations;
2. multiplicative Jordan decomposition and canonical unipotent flows;
3. split-torus weight decompositions, horospherical groups, and parabolic coordinates;
4. independence and basic structure of arithmetic subgroups;
5. reduction theory and finite covolume first for SL_n and then by rational-rank induction for connected semisimple rational groups;
6. the arithmetic lattice theorem and the compactness criterion;
7. the full Borel-density statement needed from this point onward;
8. arithmetic closed-orbit and covolume normalization criteria.

11.2 What is deliberately not owned by this volume

General structure theory for arbitrary reductive groups, classification by Dynkin diagrams, adelic reduction theory, automorphic representations, and S -arithmetic groups are outside the Ratner continuation. None of Volumes VI–XIV will use such a result without either proving the needed special case there or explicitly marking it as non-load-bearing context.

Remark 11.1 (No hidden reduction-theory call). Every later use of “choose a

Siegel set”, “move into a bounded unipotent coordinate”, or “escape creates a rational parabolic” refers to Theorem 7.2 and its proof, not to an external reduction black box.

Remark 11.2 (No hidden density call). Every later inference that an algebraic subgroup containing a lattice is forced to be the full Zariski closure refers to Theorem 8.3. The weaker Borel-density form in Volume II continues to suffice for the original Ratner proof; the stronger theorem is owned here because family and arithmetic-parameter arguments need it.

Appendix A

Cartan Geometry and the KAK Decomposition

This appendix owns the real-semisimple structural theorem used in Volumes VI, X, and XI. The point is not to classify real semisimple groups, but to prove exactly the decomposition and root contraction statements that later dynamical arguments require.

A.1 Existence of Cartan involutions

The rest of this appendix previously began with a Cartan involution already in hand. We now construct one. The argument is finite dimensional and is included because later self-adjoint-position arguments need the existence statement, not merely the consequences of a chosen involution.

Lemma A.1 (A compact real form of a complex semisimple Lie algebra). *Every finite-dimensional complex semisimple Lie algebra $\mathfrak{s}$ has a real form $\mathfrak{u}$ on which the Killing form is negative definite.*

Proof. Choose a Cartan subalgebra $\mathfrak{h}$, a system of positive roots, and simple-root vectors $E_{\alpha_i}, E_{-\alpha_i}$ normalized so that $[E_{\alpha_i}, E_{-\alpha_i}] = H_{\alpha_i}$. The usual root-string recursion obtains root vectors E_{α} for every root with real structure constants and with

$$[E_{\alpha}, E_{-\alpha}] = H_{\alpha}, \quad [E_{\alpha}, E_{\beta}] = N_{\alpha, \beta} E_{\alpha + \beta}, \quad N_{-\alpha, -\beta} = -N_{\alpha, \beta}.$$

For completeness, the reality assertion follows inductively on root height: once the vectors for smaller positive roots have been chosen, define a vector

of height r by a nonzero bracket $[E_{\alpha_i}, E_\beta]$ with $\text{ht}(\beta) = r - 1$ and rescale the opposite vector to impose the first displayed normalization. The root-string identity and Jacobi identity then force all remaining constants to be real and the indicated signs.

Let $\mathfrak{u}$ be the real span of

$$iH_{\alpha_i}, \quad E_\alpha - E_{-\alpha}, \quad i(E_\alpha + E_{-\alpha}) \quad (\alpha > 0).$$

The displayed bracket relations show directly that $\mathfrak{u}$ is closed under brackets. These vectors together with their multiples by i span $\mathfrak{s}$, and their real spans meet trivially, so $\mathfrak{s} = \mathfrak{u} \oplus i\mathfrak{u}$.

It remains to check compactness. Pair opposite root spaces so that $B(E_\alpha, E_{-\alpha}) > 0$. On the Cartan part, $\text{ad } H$ is zero on the Cartan subalgebra and acts on the root line $\mathfrak{s}_\alpha$ by the scalar $\alpha(H)$. Taking the trace of $\text{ad } H$ and H' therefore gives

$$B(H, H') = \sum_{\alpha} \alpha(H) \alpha(H').$$

Consequently $B(iH, iH) < 0$ for every nonzero real linear combination of the coroots. On each real plane generated by $E_\alpha - E_{-\alpha}$ and $i(E_\alpha + E_{-\alpha})$, the two generators are B -orthogonal and both have square $-2B(E_\alpha, E_{-\alpha}) < 0$. Distinct root pairs are orthogonal. Thus $B|_{\mathfrak{u}}$ is negative definite. $\square$

Lemma A.2 (Polar decomposition of the product of two conjugations). *Let σ be the conjugation of a real form $\mathfrak{g}$ of a complex semisimple Lie algebra $\mathfrak{s}$, and let τ be the conjugation of a compact real form. There is another compact conjugation τ' commuting with σ .*

Proof. Put

$$h(X, Y) = -B(X, \tau Y).$$

This is a positive Hermitian form. Set $A = \sigma\tau$; it is a complex-linear Lie-algebra automorphism. For every complex-linear automorphism T , invariance of the Killing form gives

$$T^* = \tau T^{-1} \tau \tag{C1}$$

for the adjoint relative to h . Since $\sigma^2 = 1$ is equivalent to $\tau A \tau = A^{-1}$, formula (C1) gives $A^* = A$. Hence A is diagonalizable with nonzero real eigenvalues.

If E_λ is the λ -eigenspace of A , then

$$[E_\lambda, E_\mu] \subset E_{\lambda\mu}.$$

Consequently the operators

$$P|_{E_\lambda} = |\lambda| \text{id}, \quad S|_{E_\lambda} = \text{sgn}(\lambda) \text{id}$$

are again Lie-algebra automorphisms; so is every real power P^t . We have $A = SP = PS$, $S^2 = 1$, and, because $\tau A \tau = A^{-1}$,

$$\tau P \tau = P^{-1}.$$

Let $b = P^{1/2}$ and define $\tau' = b \tau b^{-1}$. It is the conjugation of the compact real form $b(\mathfrak{u})$, since b is a Lie-algebra automorphism. Finally

$$\sigma \tau' = A \tau b \tau b^{-1} = A b^{-2} = A P^{-1} = S.$$

Thus $(\sigma \tau')^2 = 1$. Two conjugations have involutive product exactly when they commute, so $\sigma \tau' = \tau' \sigma$. $\square$

Theorem A.3 (Existence of a Cartan involution). *Let $\mathfrak{g}$ be a real semisimple Lie algebra. There is an involutive Lie-algebra automorphism θ such that*

$$B_\theta(X, Y) := -B(X, \theta Y)$$

is positive definite.

Proof. Complexify $\mathfrak{g}$ and let σ denote its real-form conjugation. Choose a compact conjugation τ by Lemma A.1, and replace it by the commuting compact conjugation τ' supplied by Lemma A.2. Because τ' commutes with σ , it preserves the fixed algebra $\mathfrak{g}$. Put $\theta = \tau'|_{\mathfrak{g}}$. Then $\theta^2 = 1$. For $X \in \mathfrak{g}$,

$$B_\theta(X, X) = -B(X, \tau' X) > 0$$

when $X \neq 0$, because the right-hand side is the positive Hermitian form associated with the compact real form fixed by τ' , restricted to $\mathfrak{g}$. $\square$

A.2 Cartan involutions and the symmetric-space metric

Let G be a connected real semisimple linear group with finite center. Choose the Cartan involution θ supplied by Theorem A.3 and write

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}, \quad B_\theta(X, Y) = -B(X, \theta Y).$$

Let $G_{\text{ad}} = \text{Ad}(G)$. The Lie-algebra automorphism θ integrates uniquely on the center-free adjoint group: on the generators $\exp(\text{ad } X)$ it is given by $\exp(\text{ad } X) \mapsto \exp(\text{ad } \theta X)$. Let K_{ad} be its fixed-point group. If $a \in K_{\text{ad}}$, then a commutes with θ and therefore preserves B_θ ; hence K_{ad} is a closed subgroup of the compact orthogonal group of B_θ . Put

$$K = \text{Ad}^{-1}(K_{\text{ad}}).$$

Since $\ker \text{Ad}$ is the finite center, K is compact and closed, and its Lie algebra is $\mathfrak{k}$. This definition avoids any descent issue for a Cartan involution through a chosen finite central quotient of the universal cover.

Lemma A.4 (Orthogonal bracket calculus). *The two summands are B_θ -orthogonal and*

$$[\mathfrak{k}, \mathfrak{k}] \subset \mathfrak{k}, \quad [\mathfrak{k}, \mathfrak{p}] \subset \mathfrak{p}, \quad [\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{k}.$$

Moreover $\text{ad } X$ is skew-adjoint for B_θ if $X \in \mathfrak{k}$ and self-adjoint if $X \in \mathfrak{p}$.

Proof. The eigenspace assertions follow from $\theta[X, Y] = [\theta X, \theta Y]$. For adjointness, invariance of the Killing form gives

$$B_\theta([X, Y], Z) = -B([X, Y], \theta Z) = -B(Y, [\theta Z, X]).$$

If $\theta X = \pm X$, move θ through the bracket and rewrite the last expression as $\mp B_\theta(Y, [X, Z])$. The sign is minus on $\mathfrak{k}$ and plus on $\mathfrak{p}$. $\square$

The inner product $B_\theta|_{\mathfrak{p}}$ defines a G -invariant Riemannian metric on $M = G/K$: at $o = K$ identify $T_o M$ with $\mathfrak{p}$ and transport by left translation.

Lemma A.5 (Geodesics through the base point). *For $X \in \mathfrak{p}$, the curve*

$$\gamma_X(t) = \exp(tX)K$$

is a geodesic. Every geodesic through o is of this form.

Proof. Let X^* be the fundamental vector field generated by X . The Koszul formula for invariant vector fields at o reduces, using $[\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{k}$, to $\nabla_{X^*} X^*(o) = 0$. Left translation by $\exp(tX)$ preserves the metric and the velocity field of the displayed curve, so the acceleration vanishes for every t . Uniqueness of the geodesic initial-value problem gives the second assertion. $\square$

Lemma A.6 (Completeness of a homogeneous Riemannian manifold). *A connected Riemannian manifold on which a Lie group acts transitively by isometries is geodesically complete.*

Proof. Choose a relatively compact normal coordinate ball $B(o, r)$ on which every geodesic with initial speed at most one exists for time r . By transitivity and isometry, the same radius works at every point. A unit-speed geodesic defined on a maximal interval $[0, T)$ with $T < \infty$ can therefore be extended from any time $t > T - r/2$ for another interval of length r , a contradiction. Rescaling treats arbitrary speed and negative time. $\square$

Theorem A.7 (Homogeneous Hopf–Rinow consequences). *Let (M, g) be a connected finite-dimensional Riemannian manifold on which a Lie group acts transitively by isometries. Then closed metric balls are compact, (M, d) is complete, and every two points of M are joined by a minimizing geodesic.*

Proof. The proof uses only one local normal-coordinate ball and transitivity. Choose $o \in M$ and $\rho > 0$ so small that

$$\exp_o : B(0, 3\rho) \subset T_o M \longrightarrow U$$

is a diffeomorphism onto a normal neighbourhood. Gauss' lemma implies that for $\|v\| < 3\rho$ the radial segment $t \mapsto \exp_o(tv)$ has length $\|v\|$ and minimizes from o to $\exp_o(v)$. In particular, a curve of length $< 3\rho$ starting at o cannot leave U : at its first exit it would meet the boundary $\exp_o\{\|v\| = 3\rho\}$, every point of which is at distance 3ρ from o . Hence

$$\overline{B}(o, 2\rho) = \exp_o\{v : \|v\| \leq 2\rho\}, \quad (\text{HR1})$$

so this closed ball is compact. By transitivity and isometry, the same radius works at every center.

We next bootstrap (HR1) to arbitrary radii. Suppose $\overline{B}(o, R)$ is compact. Cover it by finitely many balls $B(x_i, \rho/4)$. If $y \in \overline{B}(o, R + \rho/2)$, choose a rectifiable curve from o to y of length $< R + \rho/2 + \varepsilon$, with $0 < \varepsilon < \rho/8$. If the curve first reaches distance R from o at x , then $x \in \overline{B}(o, R)$ and the remaining subcurve has length $< 5\rho/8$; if it never reaches distance R , simply take $x = y$. For some i we have $d(x, x_i) < \rho/4$, and therefore $d(y, x_i) < 7\rho/8 < \rho$. Thus

$$\overline{B}(o, R + \rho/2) \subset \bigcup_i \overline{B}(x_i, \rho).$$

Each ball on the right is compact by the translated version of (HR1), and the ball on the left is closed, hence compact. Induction in steps of $\rho/2$ proves compactness of every closed metric ball.

Metric completeness now follows from compactness of closed bounded balls: a Cauchy sequence is bounded, lies in a compact closed ball, has a convergent subsequence, and then the whole sequence converges.

Finally fix $p, q \in M$ and put $r = d(p, q)$. Choose rectifiable curves $c_j : [0, 1] \rightarrow M$ from p to q with $\text{len}(c_j) \leq r + 1/j$, parametrized proportionally to arclength. They are uniformly Lipschitz and all lie in the compact ball $\overline{B}(p, r + 1)$. Arzelà–Ascoli gives a uniformly convergent subsequence with limit c . For every partition $0 = t_0 < \cdots < t_m = 1$,

$$\sum_i d(c(t_{i-1}), c(t_i)) = \lim_j \sum_i d(c_j(t_{i-1}), c_j(t_i)) \leq r.$$

Taking the supremum over partitions gives $\text{len}(c) \leq r$; the endpoint distance gives the reverse inequality. Hence c is length minimizing. Inside a sufficiently small normal neighbourhood, every subarc of a minimizing curve is the unique radial minimizing segment, so c is smooth there and satisfies the geodesic equation. Thus c is a minimizing geodesic from p to q . $\square$

Worked Example A.8 (Why completeness matters). The open Euclidean unit ball is not complete. The radial geodesic $\gamma(t) = te_1$ for $0 \leq t < 1$ has finite length but cannot be continued inside the manifold at $t = 1$. By contrast, on the homogeneous symmetric space G/K , Theorem A.7 uses the uniform normal-coordinate radius supplied by transitivity to make closed balls compact and then constructs the global minimizing geodesic used in polar decomposition.

Theorem A.9 (Global polar decomposition). *Every $g \in G$ can be written*

$$g = k \exp X, \quad k \in K, \quad X \in \mathfrak{p}.$$

Equivalently, $G = K \exp \mathfrak{p} = \exp \mathfrak{p} K$.

Proof. The homogeneous-space theorem, Theorem A.7, supplies a minimizing geodesic from o to gK . By Lemma A.5 it has the form $t \mapsto \exp(tX)K$; hence $gK = \exp XK$, so $g = \exp Xk$. Applying inversion and using $K^{-1} = K$, $\exp(\mathfrak{p})^{-1} = \exp(\mathfrak{p})$ yields the other order. $\square$

A.3 Conjugacy of maximal flats

Fix a maximal abelian subspace $\mathfrak{a} \subset \mathfrak{p}$. Because the operators $\text{ad } H$, $H \in \mathfrak{a}$, commute and are self-adjoint for B_θ , they diagonalize simultaneously:

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{\alpha \in \Sigma} \mathfrak{g}_\alpha, \quad [H, Y] = \alpha(H)Y.$$

The nonzero α are the restricted roots.

Lemma A.10 (Regular elements). *The set*

$$\mathfrak{a}_{\text{reg}} = \{H \in \mathfrak{a} : \alpha(H) \neq 0 \text{ for all } \alpha \in \Sigma\}$$

is open and dense, and if $H \in \mathfrak{a}_{\text{reg}}$ then

$$Z_{\mathfrak{p}}(H) = \mathfrak{a}.$$

Proof. Only finitely many root hyperplanes occur, so their complement is open and dense. If $Y \in \mathfrak{p}$ commutes with H , its component in every nonzero root space vanishes because $\alpha(H) \neq 0$. Hence $Y \in \mathfrak{g}_0 \cap \mathfrak{p}$. The latter is precisely $Z_{\mathfrak{p}}(\mathfrak{a})$; maximal abelianness of $\mathfrak{a}$ makes this equal to $\mathfrak{a}$. $\square$

Theorem A.11 (Conjugacy of maximal abelian subspaces of $\mathfrak{p}$). *If $\mathfrak{a}, \mathfrak{b} \subset \mathfrak{p}$ are maximal abelian, then there is $k \in K$ with $\text{Ad}(k)\mathfrak{a} = \mathfrak{b}$. Consequently every $X \in \mathfrak{p}$ is K -conjugate to an element of $\mathfrak{a}$.*

Proof. Choose regular $A \in \mathfrak{a}$ and regular $B \in \mathfrak{b}$. Compactness of K gives a maximizer k of

$$F(k) = B_\theta(\text{Ad}(k)A, B).$$

For $Z \in \mathfrak{k}$ differentiation at the maximizer gives

$$0 = B_\theta([Z, \text{Ad}(k)A], B) = B_\theta(Z, [\text{Ad}(k)A, B]),$$

where the last equality uses Lemma A.4. The bracket belongs to $\mathfrak{k}$, and B_θ is nondegenerate there, so $[\text{Ad}(k)A, B] = 0$. Since B is regular in $\mathfrak{b}$, Lemma A.10 gives $\text{Ad}(k)A \in \mathfrak{b}$. Regularity of A gives

$$\text{Ad}(k)\mathfrak{a} = \text{Ad}(k)Z_{\mathfrak{p}}(A) = Z_{\mathfrak{p}}(\text{Ad}(k)A) = \mathfrak{b}.$$

For arbitrary $X \in \mathfrak{p}$, extend the abelian subspace $\mathbb{R}X$ to a maximal abelian subspace $\mathfrak{b}$ and apply the first assertion. $\square$

Choose a positive restricted-root system and the closed chamber $\mathfrak{a}^+$. Reflections in the root hyperplanes are represented by the rank-one subgroups generated by $\mathfrak{g}_{\pm\alpha}$, hence by elements of $N_K(\mathfrak{a})$. Repeated reflection across a wall with negative simple-root coordinate moves any $H \in \mathfrak{a}$ into $\mathfrak{a}^+$; termination follows because the Weyl group is finite.

Theorem A.12 (Cartan KAK decomposition).

$$G = K \exp(\mathfrak{a}^+) K.$$

Moreover a sequence $k_n \exp H_n \ell_n$ with $H_n \in \mathfrak{a}^+$ escapes every compact subset of G exactly when $\|H_n\| \rightarrow \infty$.

Proof. Combine Theorems A.9 and A.11, then use the Weyl-group reduction to the positive chamber. If H_n is bounded, compactness of K makes the corresponding set compact. Conversely, simple restricted roots span $\mathfrak{a}^*$ on each semisimple factor, so unbounded $H_n \in \mathfrak{a}^+$ has a simple root $\alpha(H_n) \rightarrow +\infty$ along a subsequence. On $\mathfrak{g}_\alpha$ the operator $\text{Ad}(\exp H_n)$ acts by $e^{\alpha(H_n)}$, hence the adjoint norms diverge. Since the adjoint map is proper modulo the finite center, the group elements escape compacta. $\square$

Theorem A.13 (Iwasawa decomposition in the form used later). *Choose a positive restricted-root system for $(\mathfrak{g}, \mathfrak{a})$ and put*

$$\mathfrak{n} = \bigoplus_{\alpha > 0} \mathfrak{g}_\alpha, \quad A = \exp \mathfrak{a}, \quad N = \exp \mathfrak{n}.$$

Then multiplication is a diffeomorphism

$$K \times A \times N \longrightarrow G.$$

In particular $G = KAN$, and the three factors depend smoothly on g .

Proof. We first prove the opposite-order transitivity $G = ANK$. The subgroup $S = AN$ is closed: in the adjoint representation, A is simultaneously real-diagonalizable, N is the connected unipotent group obtained by exponentiating the positive restricted-root spaces, and multiplication $A \times N \rightarrow AN$ is a diffeomorphism because conjugation by A preserves N and $A \cap N = \{e\}$. The inverse coordinate maps are obtained by taking the semisimple and unipotent factors in this triangular representation, so they are continuous.

Consider the orbit map

$$q : S \longrightarrow G/K, \quad q(s) = sK.$$

Its stabilizer is $S \cap K = \{e\}$: an element of $A \cap K$ has positive real adjoint eigenvalues and unit-modulus adjoint eigenvalues, hence is central, and the finite-center hypothesis together with $A = \exp \mathfrak{a}$ gives the identity; a unipotent element in a compact group is also the identity. Moreover

$$\dim S = \dim \mathfrak{a} + \sum_{\alpha > 0} \dim \mathfrak{g}_\alpha = \dim \mathfrak{p} = \dim(G/K).$$

The last equality follows because θ interchanges $\mathfrak{g}_\alpha$ and $\mathfrak{g}_{-\alpha}$ and the projection $\mathfrak{g}_\alpha \rightarrow \mathfrak{p}$ is an isomorphism onto one half of the pair $\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}$. Hence dq_e is an isomorphism and the S -orbit is open.

The map q is proper. Indeed, if $C \subset G/K$ is compact, choose for each point of C an open neighborhood on which the principal K -bundle $G \rightarrow G/K$ has a smooth section. By compactness we may choose finitely many smaller open sets V_i whose closures are compact and contained in the corresponding section domains and whose union covers C . If s_i denotes the section, then

$$\tilde{C} = \bigcup_i s_i(C \cap \overline{V}_i)$$

is compact and maps onto C . If $s \in q^{-1}(C)$, write $s = \tilde{c}k$ with $\tilde{c} \in \tilde{C}$ and $k \in K$. Thus $q^{-1}(C) \subset S \cap (\tilde{C}K)$. Conversely every element of this intersection maps into C , so equality holds; because S is closed and $\tilde{C}K$ is compact, the preimage is compact. Thus the open S -orbit is also closed. Since G/K is connected, it is all of G/K , proving $G = ANK$.

Apply this to g^{-1} : write $g^{-1} = ank$. Then

$$g = k^{-1}n^{-1}a^{-1} = k^{-1}a^{-1}(an^{-1}a^{-1}),$$

and the last factor lies in N . Hence $G = KAN$.

It remains to prove uniqueness and smoothness. If $kan = k'a'n'$, then $k'^{-1}k = a'n'n^{-1}a^{-1} \in K \cap AN = \{e\}$, so all three factors agree. The differential of the multiplication map at the identity is the direct-sum isomorphism $\mathfrak{k} \oplus \mathfrak{a} \oplus \mathfrak{n} \rightarrow \mathfrak{g}$; left and right translation gives the same conclusion everywhere. Thus multiplication is a bijective local diffeomorphism, hence a global diffeomorphism. $\square$

Appendix B

Rational Unipotent Groups and the Cusp Reduction Lemmas

The short proof of Theorem 7.2 used two facts about rational unipotent radicals. We prove them here.

B.1 Polynomial exponential coordinates

Let $\mathbf{N} < \mathrm{GL}_m$ be a unipotent algebraic group defined over $\mathbb{Q}$. Every $n \in N = \mathbf{N}(\mathbb{R})$ satisfies $(n - I)^m = 0$, so

$$\log n = \sum_{j=1}^{m-1} \frac{(-1)^{j+1}}{j} (n - I)^j$$

and the exponential series terminate. Both maps are polynomial with rational coefficients. Thus N is polynomially diffeomorphic to its nilpotent Lie algebra $\mathfrak{n}$.

Lemma B.1 (Rational central series). *There is a central series*

$$\mathfrak{n} = \mathfrak{n}_1 \supset \mathfrak{n}_2 \supset \cdots \supset \mathfrak{n}_{s+1} = 0, \quad [\mathfrak{n}, \mathfrak{n}_i] \subset \mathfrak{n}_{i+1},$$

all defined over $\mathbb{Q}$, and a $\mathbb{Q}$ -basis $X_1, \dots, X_d$ adapted to it.

Proof. Take the lower central series $\mathfrak{n}_{i+1} = [\mathfrak{n}, \mathfrak{n}_i]$. The bracket has rational structure constants in any rational basis, so every term is rational. Successively extend bases from the final nonzero term upward. $\square$

Theorem B.2 (Integral points form a lattice in a rational unipotent group). *After replacing the ambient integral lattice by a commensurable one, $N(\mathbb{Z}) = \mathbf{N}(\mathbb{Q}) \cap \mathrm{GL}_m(\mathbb{Z})$ is a lattice in N . More concretely, for an adapted rational basis there is $D \geq 1$ such that every coset of $N(\mathbb{Z})$ has a representative*

$$\exp(t_1 X_1) \cdots \exp(t_d X_d), \quad 0 \leq t_i < D.$$

Proof. Multiply the adapted basis by a common denominator so that all Baker–Campbell–Hausdorff structure constants are integral after one further common denominator D . Coordinates of the second kind

$$\Phi(t_1, \dots, t_d) = \exp(t_1 X_1) \cdots \exp(t_d X_d)$$

are a polynomial diffeomorphism $\mathbb{R}^d \rightarrow N$: this is proved inductively down the central series, because modulo the center the claim holds in smaller nilpotency class and the final central coordinate is then uniquely recovered. The multiplication law in these coordinates is triangular: the i th new coordinate is $s_i + t_i$ plus a polynomial in earlier coordinates with rational coefficients. With the chosen denominator, the subgroup generated by $\exp(DX_i)$ has integral triangular coordinates and is contained in $N(\mathbb{Z})$ up to finite index. Reduce t_1 modulo D , then t_2 , and so on; triangularity ensures that later reductions do not change already reduced coordinates. This gives the displayed bounded fundamental set, hence finite covolume. Discreteness follows from discreteness of integral matrices. $\square$

B.2 Finite rational cusp types

For each maximal proper rational parabolic $\mathbf{P}_\alpha$, fix a Chevalley representation (ρ_α, V_α) and a primitive integral vector v_α spanning its rational stabilizer line. Put

$$\eta_\alpha(g) = \|\rho_\alpha(g)v_\alpha\|.$$

Lemma B.3 (Bounded wedge window is finite). *Let L be a lattice in a finite-dimensional normed space. For every $R < \infty$, the set of primitive $v \in L$ with $\|v\| \leq R$ is finite. Consequently there are only finitely many rational flags whose primitive exterior representatives lie in a fixed bounded window.*

Proof. A lattice meets compact sets in finite sets. Primitive vectors form a subset of the lattice, and a rational flag is recovered from its primitive exterior line. $\square$

Lemma B.4 (Deep-cusp uniqueness up to the parabolic). *There is $\varepsilon_0 > 0$ such that if, for the same $g \in G$, two primitive vectors v, w representing conjugates of maximal rational parabolics satisfy*

$$\|\rho(g)v\| < \varepsilon_0, \quad \|\rho(g)w\| < \varepsilon_0,$$

then either their associated flags are incident and belong to one rational parabolic chain, or the product of the two norms is bounded below by a constant depending only on the fixed integral representations.

Proof. Use all exterior powers simultaneously. If two rational subspaces are not incident, a nonzero integral Pluecker coordinate of their wedge or contraction has absolute value at least one. The corresponding multilinear map is G -equivariant. Its norm is bounded above by a fixed constant times the product of the two transformed wedge norms. Since the integral coordinate is nonzero, the covolume normalization and determinant-one condition give a uniform positive lower bound. Choosing ε_0 below its square root forces incidence. Iterating gives a single flag chain, whose stabilizer is a rational parabolic. $\square$

This lemma is the precise finite-complexity statement behind the phrase “choose a maximal cusp height” in the proof of Theorem 7.2.

Appendix C

Reduction Theory Expanded: the Induction without Hidden Steps

C.1 The rank induction

We now spell out the induction in Theorem 7.2. Fix a minimal rational parabolic $P = MAN$ determined by a maximal rational split torus S . Write $\Delta_{\mathbb{Q}}$ for the simple rational roots.

Lemma C.1 (Haar density in rational-root coordinates). *In KAN -type coordinates on a rational Siegel set,*

$$dg = c \, dk \, dm \, dn \prod_{\alpha \in \Delta_{\mathbb{Q}}} e^{-c_{\alpha} t_{\alpha}} \, dt_{\alpha},$$

where $a = \exp(\sum t_{\alpha} H_{\alpha})$ and every $c_{\alpha} > 0$.

Proof. Right translation by a on the positive root space $\mathfrak{g}_{\beta}$ multiplies Lebesgue measure by $e^{\beta(\log a) \dim \mathfrak{g}_{\beta}}$. The modular Jacobian of AN is therefore

$$\delta(a) = \exp \left(\sum_{\beta > 0} (\dim \mathfrak{g}_{\beta}) \beta(\log a) \right) = e^{2\rho(\log a)}.$$

Express 2ρ in the basis of simple roots. Every coefficient is strictly positive because every simple root occurs in the positive-root system. Changing from $\log a \in \mathfrak{a}$ to the simple-root coordinates has constant nonzero Jacobian. This gives the formula. $\square$

Lemma C.2 (Reduction of the unipotent coordinate). *Let N_P be the unipotent radical of a rational parabolic P . The arithmetic group $\Gamma \cap N_P$ is a lattice in N_P , so every N_P -coordinate can be moved by $\Gamma \cap N_P$ into one fixed compact set Ω_N .*

Proof. After conjugating by the rational cusp representative, the intersection is commensurable with the integral points of the rational unipotent group. Apply Theorem B.2. A finite-index change only enlarges the compact fundamental set by finitely many translates. $\square$

Lemma C.3 (Levi rank drops on a cusp face). *Let P_I be the standard rational parabolic associated with a proper subset $I \subsetneq \Delta_{\mathbb{Q}}$. The derived semisimple group of its Levi component has rational root system generated by I ; hence its rational rank is $|I| < |\Delta_{\mathbb{Q}}|$ on the noncompact split directions requiring reduction.*

Proof. The Lie algebra of the Levi is generated by the rational torus together with root spaces $\mathfrak{g}_{\pm\alpha}$ for roots in the span of I . Its derived algebra removes the central split directions orthogonal to that span. Thus a maximal split torus in the derived Levi has dimension $|I|$. $\square$

Expanded proof of Theorem 7.2. We induct on $r = \text{rank}_{\mathbb{Q}} G$. For $r = 0$, suppose a sequence Γg_j escapes every compact subset. Apply Mahler's criterion in a finite collection of fundamental rational representations, one for each maximal rational parabolic. The flag-incidence lemma B.4 turns any family of short rational wedges into one proper rational flag and therefore a proper rational parabolic, impossible when $r = 0$. Thus the quotient is compact.

Assume the result through rank $r - 1$. Given $g \in G$, examine the finite family of primitive rational exterior vectors in the fundamental representations. If every transformed primitive vector has norm at least ε_0 , Mahler compactness in the direct sum of these representations puts Γg in one fixed compact set. Otherwise choose a shortest vector. Lemma B.4 organizes all vectors below ε_0 into an incident rational flag, hence a rational parabolic P_I . By Lemma B.3, after moving by Γ there are only finitely many possible primitive flag vectors in the bounded transition window; these give finitely many rational representatives q_i .

Inside $q_i P_I q_i^{-1}$ write the real points as $M_I A_I N_I$ up to a compact factor. Reduce N_I into Ω_N by the preceding lemma. Apply the induction hypothesis to the arithmetic subgroup of the derived Levi; this moves its noncompact

semisimple coordinate into finitely many lower-rank Siegel sets. The central split part of the Levi is absorbed into A_I . Finally, the condition that the chosen rational flag was the deepest short flag is exactly a family of lower bounds on the simple-root coordinates of A_I . Thus

$$G = \Gamma \bigcup_{i=1}^r q_i \Omega A_t C.$$

The Haar-density lemma gives a product of convergent integrals $\int_{t_0}^{\infty} e^{-c_{\alpha} t} dt$ in every cusp direction. The unipotent factor has compact quotient, the compact factor has finite measure, and the derived Levi has finite volume by induction. Every Siegel set therefore has finite Haar measure. This closes the induction. $\square$

Appendix D

Borel Density without a Projective-Measure Shortcut

The proof in Chapter 7 used a projective invariant-measure lemma. We now prove that lemma in a form that also records exactly where the absence of compact factors enters.

Lemma D.1 (One split direction). *Let $A = \{a_t\}$ act linearly on V with real weight decomposition $V = \bigoplus_{\lambda} V_{\lambda}$. If a probability measure ν on $\mathbb{P}(V)$ is A -invariant, then for every t -generator it is supported on the union of projectivizations of individual weight spaces.*

Proof. For $[v]$ write $v = \sum v_{\lambda}$ and let $\lambda_+(v)$ be the largest weight with nonzero component. As $t \rightarrow +\infty$, $a_t[v]$ approaches $[v_{\lambda_+(v)}]$ after deleting lower weights. If a compact set C is separated from the union of weight projective spaces, its forward images $a_{nT}C$ are pairwise disjoint for sufficiently large T : the ratio of the two largest active weight coordinates changes by a fixed exponential factor. Invariance would give infinitely many disjoint sets of equal positive mass, so $\nu(C) = 0$. Exhaust the complement by countably many such compact sets. $\square$

Lemma D.2 (Root groups remove nonfixed zero weights). *Let G be connected semisimple without compact factors, A a maximal split torus, and ν a G -invariant probability measure on $\mathbb{P}(V)$. Then ν is supported on $\mathbb{P}(V^G)$.*

Proof. Apply Lemma D.1 to a countable dense family of regular one-parameter subgroups in A . Their common support restriction is the union of projective

common weight spaces. Applying both signs of every regular direction forces any weight occurring on a positive-mass invariant piece to be zero; hence the measure is supported on $\mathbb{P}(V_0)$, the zero-weight space for A .

Fix a restricted root α and $X \in \mathfrak{g}_\alpha$. If $v \in V_0$ but $d\rho(X)v \neq 0$, then

$$\rho(\exp sX)v = v + s d\rho(X)v + \cdots$$

has a nonzero positive α -weight component for all but finitely many s . Since ν is invariant under the root group, this contradicts the already established support in V_0 . Thus almost every projective point is fixed by every root group. The positive and negative restricted root groups generate every noncompact almost-simple factor; there are no compact factors, so they generate G modulo its finite center. Therefore the support lies in $\mathbb{P}(V^G)$. $\square$

Second proof of Theorem 8.3. Let H be the Zariski closure of Γ and choose, by Chevalley, a line $[v]$ with stabilizer H . The map $\Gamma g \mapsto [\rho(g^{-1})v]$ is well defined and pushes Haar probability on $\Gamma \backslash G$ to a G -invariant probability measure on $\mathbb{P}(V)$. Lemma D.2 puts its support in $\mathbb{P}(V^G)$, so $[v]$ is G -fixed. Its stabilizer is all of G , forcing $H = G$. $\square$

Appendix E

Worked Reduction Calculations

E.1 SL_2 : the single cusp integral

For

$$a_y = \begin{pmatrix} y^{1/2} & 0 \\ 0 & y^{-1/2} \end{pmatrix}, \quad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix},$$

Haar measure is a constant multiple of $dx dy/y^2 dk$. Reducing x modulo $\mathbb{Z}$ leaves $|x| \leq 1/2$, while $y \geq y_0$ is the cusp cone. The tail has volume

$$\int_{y_0}^{\infty} \frac{dy}{y^2} = y_0^{-1}.$$

Thus the general root-density argument reduces in rank one to the familiar finite area of the modular cusp.

E.2 SL_3 : two simple-root coordinates

Put $s_1 = a_1/a_2$, $s_2 = a_2/a_3$. Since

$$2\rho = 2\alpha_1 + 2\alpha_2,$$

Haar density on the positive diagonal cone is, up to a constant,

$$s_1^{-2} s_2^{-2} \frac{ds_1}{s_1} \frac{ds_2}{s_2}.$$

Both cusp directions are integrable independently. A face with only $s_1 \rightarrow \infty$ has Levi derived group isomorphic to SL_2 in the lower 2×2 block; this is the concrete rank-drop appearing in the induction.

E.3 Orthogonal groups

For a rational quadratic form Q , a rational isotropic flag gives a parabolic by stabilizing the flag. The unipotent radical consists of polynomial shears along isotropic directions. Its rational coordinates are exactly of the triangular type in Theorem B.2. If no rational isotropic vector exists, no rational parabolic exists and the arithmetic quotient is compact. This concrete calculation is the version of reduction theory used later for quadratic-form orbits.

Appendix F

Worked Algebraic-Group Laboratory: Parabolics and Heights

F.1 Motivation

Reduction theory becomes less opaque once the parabolics of SL_3 are written down explicitly.

Worked Example F.1 (The two maximal standard parabolics of SL_3). The stabilizer P_1 of $\mathbb{R}e_1$ consists of matrices

$$\begin{pmatrix} * & * & * \\ 0 & * & * \\ 0 & * & * \end{pmatrix}, \quad \det = 1.$$

Its unipotent radical and a Levi subgroup are

$$U_1 = \left\{ \begin{pmatrix} 1 & u & v \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right\}, \quad L_1 = \left\{ \begin{pmatrix} a & 0 \\ 0 & B \end{pmatrix} : a \det B = 1 \right\}.$$

Similarly P_2 , stabilizing $\mathbb{R}e_1 \oplus \mathbb{R}e_2$, has unipotent radical generated by E_{13}, E_{23} . Their intersection is the upper triangular minimal parabolic. Thus the two walls of the positive Weyl chamber correspond exactly to the two maximal parabolics.

Worked Example F.2 (A Pluecker height). Let $W \subset \mathbb{Q}^3$ be spanned by $v_1 = (1, 0, m)$ and $v_2 = (0, 1, n)$ with $m, n \in \mathbb{Z}$. Its primitive Pluecker vector is

$$v_1 \wedge v_2 = e_1 \wedge e_2 + ne_1 \wedge e_3 - me_2 \wedge e_3,$$

so its Euclidean Pluecker height is $\sqrt{1 + m^2 + n^2}$. A bounded-height condition therefore literally bounds integer minors in this model.

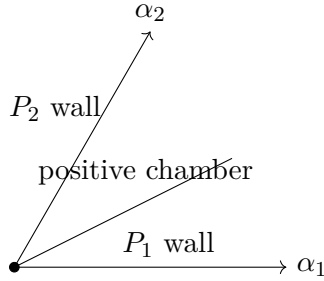


Figure F.1: Schematic A_2 Weyl chamber. Approaching a wall corresponds to a parabolic degeneration; reduction theory records which rational flag becomes short.

Appendix G

Student Proof Laboratory: Algebraic Groups without Hidden Steps

This chapter revisits the structural results used later in the series from the point of view of a first reading. Nothing here changes the statements proved above. The purpose is to make explicit the algebraic moves that are often compressed into phrases such as “take the Zariski closure”, “pass to the unipotent radical”, or “choose a parabolic”. These phrases are harmless only after one knows exactly what they mean in coordinates.

G.1 Polynomial equations and why Zariski closure preserves group laws

Let $G < \mathrm{GL}_N(\mathbb{R})$ be the common zero set of polynomials $p_1, \dots, p_s$ in the matrix entries and $1/\det$. If $S < G$, its Zariski closure $\overline{S}^{\mathrm{Zar}}$ is the common zero set of all regular functions vanishing on S .

Lemma G.1 (Closure of a subgroup is an algebraic subgroup). *If $S < G$ is a subgroup, then $\overline{S}^{\mathrm{Zar}}$ is closed under multiplication and inversion.*

Proof. Fix $a \in S$. If a regular function f vanishes on S , then $x \mapsto f(ax)$ also vanishes on S , because $aS = S$. Hence it vanishes on $\overline{S}^{\mathrm{Zar}}$. Thus $a\overline{S}^{\mathrm{Zar}} \subset \overline{S}^{\mathrm{Zar}}$. For a fixed $x \in \overline{S}^{\mathrm{Zar}}$, apply the same argument to the regular

map $a \mapsto ax$: every defining equation of $\overline{S}^{\text{Zar}}$ vanishes on S and therefore on its closure. This gives $\overline{S}^{\text{Zar}} \overline{S}^{\text{Zar}} \subset \overline{S}^{\text{Zar}}$. Inversion on GL_N is regular after adjoining $1/\det$, so the same vanishing-ideal argument shows invariance under $x \mapsto x^{-1}$. $\square$

Worked Example G.2 (The Zariski closure of a diagonal cyclic group). Let $a = \text{diag}(2, 3, 1/6) \in \text{SL}_3(\mathbb{R})$. A character of the diagonal torus has the form $\chi_{m,n}(\text{diag}(x, y, (xy)^{-1})) = x^m y^n$. If $\chi_{m,n}(a) = 1$, then $2^m 3^n = 1$, so $m = n = 0$. Hence no nontrivial character is constant on $\langle a \rangle$, and the Zariski closure of $\langle a \rangle$ is the full two-dimensional diagonal torus. This is a useful warning: an algebraically one-generated subgroup may have a higher-dimensional algebraic hull.

G.2 Jordan decomposition: a calculation a student can reproduce

For $A \in \text{GL}_N(\mathbb{R})$ let the minimal polynomial be $m_A(X) = \prod_i q_i(X)^{e_i}$ with distinct irreducible q_i . The Chinese remainder theorem gives polynomials $e_i(X)$ satisfying

$$e_i \equiv 1 \pmod{q_i^{e_i}}, \quad e_i \equiv 0 \pmod{q_j^{e_j}} \quad (j \neq i).$$

Then $E_i = e_i(A)$ are pairwise orthogonal idempotents and $\sum_i E_i = I$. The generalized primary decomposition is $\mathbb{R}^N = \bigoplus_i E_i \mathbb{R}^N$.

Proposition G.3 (Polynomial Jordan components). *There are polynomials $P_s, P_u \in \mathbb{R}[X]$, depending on A , such that $A_s = P_s(A)$ is semisimple, $A_u = P_u(A)$ is unipotent, $A = A_s A_u = A_u A_s$, and both components belong to the associative algebra $\mathbb{R}[A]$.*

Proof. Work first over $\mathbb{C}$. On the generalized λ -eigenspace write $A = \lambda I + N_\lambda$. Hermite interpolation supplies one polynomial P_s satisfying

$$P_s(X) \equiv \lambda \pmod{(X - \lambda)^{e_\lambda}}$$

for every eigenvalue. Thus $P_s(A)$ equals λI on every generalized eigenspace and is semisimple. Put $A_s = P_s(A)$ and $A_u = A_s^{-1}A$. Since A_s is a polynomial in A , it commutes with A ; therefore A_u is a polynomial in A as well (use interpolation for A_s^{-1} on the finite spectrum). On the generalized λ -space, $A_u = I + \lambda^{-1}N_\lambda$, hence is unipotent. The construction is invariant under

complex conjugation, so the interpolating polynomials may be chosen real. Uniqueness follows because commuting semisimple and unipotent parts can be read off on every generalized eigenspace. $\square$

Worked Example G.4 (A nontrivial 3×3 Jordan decomposition). For

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad A_s = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad A_u = \begin{pmatrix} 1 & 1/2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Both are polynomials in A . Indeed the spectral projector to the 3-eigenspace is $(A - 2I)^2$, and the complementary projector is $I - (A - 2I)^2$ after the obvious normalization. This concrete calculation is what the abstract interpolation proof is organizing.

G.3 Parabolics from a one-parameter subgroup

Let $\lambda(t) = \exp(tH)$ with H semisimple and real split. Define

$$P(H) = \{g : \lim_{t \rightarrow +\infty} \lambda(-t)g\lambda(t) \text{ exists in } G\}.$$

If $\mathfrak{g} = \bigoplus_{\alpha} \mathfrak{g}_{\alpha}$ is the weight decomposition for $\text{ad } H$, then

$$\mathfrak{p}(H) = \mathfrak{g}_0 \oplus \bigoplus_{\alpha(H) > 0} \mathfrak{g}_{\alpha}, \quad \mathfrak{u}(H) = \bigoplus_{\alpha(H) > 0} \mathfrak{g}_{\alpha}.$$

The signs depend on the chosen conjugation convention; the formula is worth checking once by writing $\text{Ad}(\lambda(-t))X = e^{-t\alpha(H)}X$.

Lemma G.5 (Nilpotence of the positive-root algebra). $\mathfrak{u}(H)$ is nilpotent.

Proof. Only finitely many positive weights occur. If $X_i \in \mathfrak{g}_{\alpha_i}$, then an iterated bracket lies in the weight space of $\alpha_1 + \cdots + \alpha_r$. Once r is larger than the ratio of the largest occurring positive weight to the smallest positive weight, the sum cannot be a weight of $\mathfrak{g}$, so every sufficiently long bracket vanishes. $\square$

Worked Example G.6 (The three maximal parabolics visible in SL_3). For $H = \text{diag}(2, -1, -1)$, the positive root spaces are $\mathbb{R}E_{12} \oplus \mathbb{R}E_{13}$. The corresponding parabolic preserves the line $\mathbb{R}e_1$. For $H = \text{diag}(1, 1, -2)$, the positive root spaces are $\mathbb{R}E_{13} \oplus \mathbb{R}E_{23}$ and the parabolic preserves the plane $\mathbb{R}e_1 \oplus \mathbb{R}e_2$. Choosing a regular H with three distinct entries gives the minimal upper-triangular parabolic. Thus “parabolic” is not an opaque structural word: in SL_3 it means stabilizing an appropriate flag.

G.4 Why Siegel sets have finite volume

For $G = \mathrm{SL}_n(\mathbb{R})$ write $g = nak$ with n upper unipotent, $a = \mathrm{diag}(a_1, \dots, a_n)$ positive and $k \in \mathrm{SO}(n)$. Put $y_i = a_i/a_{i+1}$. Haar measure has the form

$$dg = C \, dn \, dk \prod_{i=1}^{n-1} y_i^{-i(n-i)} \frac{dy_i}{y_i}.$$

A Siegel set bounds the entries of n , keeps k in the compact group, and imposes $y_i \geq c > 0$.

Proposition G.7 (The cusp integral converges). *For every $c > 0$,*

$$\int_{[c, \infty)^{n-1}} \prod_{i=1}^{n-1} y_i^{-i(n-i)} \frac{dy_i}{y_i} < \infty.$$

Proof. The variables separate. Since $i(n-i) > 0$,

$$\int_c^\infty y^{-i(n-i)} \frac{dy}{y} = \frac{c^{-i(n-i)}}{i(n-i)} < \infty.$$

Multiplying these $n-1$ finite integrals proves the claim. The real content of reduction theory is therefore not this integration but the covering and finite-overlap assertions proved earlier in the volume. $\square$

G.5 A worked reduction step in dimension three

Let $\Lambda \subset \mathbb{R}^3$ be unimodular. Choose a primitive shortest vector v_1 . Project $\Lambda \cap v_1^\perp$ to the quotient $\mathbb{R}^3/\mathbb{R}v_1$ and choose a shortest primitive class $\bar{v}_2$; lift it to $v_2 \in \Lambda$. Complete to a basis v_1, v_2, v_3 . Gram–Schmidt gives v_i^* and Iwasawa coordinates satisfying

$$\|v_1^*\| \leq C\|v_2^*\|, \quad \|v_2^*\| \leq C\|v_3^*\|,$$

for a universal C after the elementary nearest-integer reductions in the upper-unipotent coefficients. These inequalities are precisely lower bounds on the simple-root coordinates y_1, y_2 . Thus the geometric act of choosing successive short primitive directions is the source of the root inequalities that define the Siegel set.

G.6 Borel density: why invariant lines are enough

Suppose G is connected semisimple without compact factors and $\Gamma < G$ is a lattice. Let $\rho : G \rightarrow \mathrm{GL}(V)$ be algebraic and suppose a line $\ell \subset V$ is Γ -invariant. The induced action on projective space fixes $[\ell]$ under Γ .

Proposition G.8 (Invariant-line form of density). *If the G -orbit of $[\ell]$ is nontrivial, a finite G -invariant measure on $\Gamma \backslash G$ produces a finite invariant measure on the noncompact part of that orbit, contradicting the contraction supplied by a split torus. Hence $[\ell]$ is G -fixed.*

Proof. The map $\Gamma g \mapsto \rho(g)^{-1}[\ell]$ is well defined because Γ preserves ℓ . Push Haar probability forward. The resulting probability ν on the projective orbit is G -invariant. If a split one-parameter subgroup a_t acts nontrivially, decompose V into its weights. Choose the largest weight present in a vector representing a point outside the union of the lower-weight projective subspaces. As $t \rightarrow +\infty$ the point is driven toward the corresponding highest-weight projective subspace. Applying invariance to a compact set separated from that subspace and then translating repeatedly gives infinitely many essentially disjoint sets of the same positive ν -mass, impossible for a probability measure. Therefore every split torus fixes the support point. The split root groups generated from these tori generate the noncompact semisimple part of G , so G fixes $[\ell]$. $\square$

G.7 Reading guide for later volumes

The later series repeatedly uses the following chain. A subgroup is first encoded by a line in an exterior-power representation. Arithmeticity makes that line rational. Reduction theory bounds its height on compact parts of the quotient. Borel density converts invariance under a lattice into invariance under the ambient algebraic group. Finally, parabolic root coordinates identify exactly how the datum can escape to infinity. When a later proof says “linearize and apply reduction theory”, the four separate operations in that sentence are the ones just reviewed.

Appendix H

Student Proof Laboratory: Algebraic Groups Through Matrices

The purpose of this chapter is not to add another layer of abstract structure theory. It gives concrete proofs of the algebraic statements that later dynamics uses repeatedly: Zariski closure, Jordan components, unipotent radicals in triangular examples, split tori, and parabolics.

H.1 Zariski closure is a subgroup

Let $G < \mathrm{GL}_n(\mathbb{R})$ be algebraic and let $S < G$ be a subgroup. Denote its Zariski closure by $\overline{S}^{\mathrm{Zar}}$.

Lemma H.1 (Closure under multiplication and inverse). *$\overline{S}^{\mathrm{Zar}}$ is an algebraic subgroup of G .*

Proof. Multiplication $m : G \times G \rightarrow G$ and inversion $i : G \rightarrow G$ are regular maps on the affine algebraic group. A regular map is Zariski continuous. Since $S \times S$ is Zariski dense in $\overline{S}^{\mathrm{Zar}} \times \overline{S}^{\mathrm{Zar}}$,

$$m(\overline{S}^{\mathrm{Zar}} \times \overline{S}^{\mathrm{Zar}}) \subset \overline{m(S \times S)}^{\mathrm{Zar}} \subset \overline{S}^{\mathrm{Zar}}.$$

Because inversion $i : g \mapsto g^{-1}$ is regular, the same closure argument gives

$$i(\overline{S}^{\mathrm{Zar}}) \subset \overline{i(S)}^{\mathrm{Zar}} = \overline{S}^{\mathrm{Zar}}.$$

The identity lies in the closure because it lies in S . □

Worked Example H.2 (A cyclic diagonal group). Let $a = \text{diag}(2, 3, 1/6) \in \text{SL}_3(\mathbb{R})$. A polynomial relation among the powers a^m is a Laurent-polynomial relation among 2^m and 3^m . Multiplicative independence of 2 and 3 implies that the Zariski closure is the full split torus $\{\text{diag}(x, y, (xy)^{-1}) : x, y \in \mathbb{R}^\times\}$, up to connected-component issues over $\mathbb{R}$. This example shows why Zariski closure remembers multiplicative relations rather than Euclidean accumulation.

H.2 Multiplicative Jordan decomposition without a hidden theorem

For $A \in \text{GL}_n(\mathbb{C})$, ordinary linear algebra writes $A = A_s A_u = A_u A_s$ with A_s semisimple and A_u unipotent. The nontrivial algebraic-group assertion is that if A lies in an algebraic subgroup, then both components do too.

Proposition H.3 (Jordan components belong to the cyclic Zariski closure). *Let $H < \text{GL}_n(\mathbb{C})$ be algebraic and $A \in H$. Then $A_s, A_u \in H$. If H and A are defined over a characteristic-zero subfield k and the Jordan factors are k -rational matrices, then they lie in $H(k)$.*

Proof. Put $C = \overline{\langle A \rangle}^{\text{Zar}} \subset H$. In a basis in which A_s is diagonal and $A_u = \exp N$ is block upper triangular with N nilpotent commuting with A_s , every matrix coefficient of A^m is a finite sum

$$\sum_{\chi} P_{\chi}(m) \chi^m,$$

where χ ranges over finitely many products of eigenvalues and P_{χ} is a polynomial arising from $\exp(mN)$. Let F be any polynomial function vanishing on every A^m . Then $F(A^m)$ is again an exponential-polynomial sequence $\sum_{\chi} Q_{\chi}(m) \chi^m$ and vanishes for every integer m . Distinct nonzero exponentials times polynomials are linearly independent as sequences: apply repeatedly the difference operator $T - \chi$, which lowers the polynomial degree of the χ -term and leaves different exponential characters distinct. Hence every coefficient polynomial Q_{χ} vanishes.

The coefficient with polynomial degree zero records the value of F on the semisimple part, so $F(A_s) = 0$. After fixing the semisimple character, the polynomial coefficients record $F(A_s \exp(tN))$; their vanishing for all degrees

gives $F(A_s \exp(tN)) = 0$ for all t . Multiplying by A_s^{-1} in the algebraic subgroup gives $\exp(tN) \in C$, in particular $A_u \in C$. Thus both factors lie in $C \subset H$. The rational assertion follows by applying the same polynomial equations over k . $\square$

H.3 Unipotent groups as exponentials

Lemma H.4 (Logarithm of a unipotent matrix). *If $u = I + N$ with N nilpotent, then*

$$\log u = N - \frac{N^2}{2} + \cdots + (-1)^r \frac{N^r}{r}$$

is a finite nilpotent polynomial and $u^t = \exp(t \log u)$ is polynomial in t .

Proof. Both power series terminate because $N^{r+1} = 0$. The formal power-series identities $\exp(\log(1 + X)) = 1 + X$ and $\log(\exp X) = X$ are therefore polynomial identities modulo X^{r+1} and remain valid after substituting N . Exponentiating $t \log u$ terminates for the same reason. $\square$

H.4 Split tori and weights

Let $A \simeq (\mathbb{R}_{>0})^r$ be a real split torus. Every rational finite-dimensional representation decomposes as

$$V = \bigoplus_{\chi} V_{\chi}, \quad av = \chi(a)v.$$

Lemma H.5 (Weight decomposition by commuting diagonalizable operators). *The decomposition above is a direct sum and every A -stable subspace is a sum of its intersections with weight spaces.*

Proof. Choose logarithmic coordinates on A and a basis of the commuting semisimple infinitesimal operators $d\rho(H_1), \dots, d\rho(H_r)$. Over $\mathbb{R}$ they are simultaneously diagonalizable because the representation of a split torus is algebraic with real characters. The simultaneous eigenspaces are exactly the weight spaces. Polynomial interpolation in the commuting operators gives projections onto each joint eigenspace, so an invariant subspace is stable under every projection and decomposes accordingly. $\square$

H.5 Parabolics in SL_n as flag stabilizers

Fix a flag $0 < V_{d_1} < \cdots < V_{d_s} < \mathbb{R}^n$. Its stabilizer consists of block upper triangular matrices. The block diagonal subgroup is a Levi factor and the strict block-upper subgroup is the unipotent radical.

Proposition H.6 (Explicit parabolic Levi decomposition in SL_n). *For the standard flag type $(d_1, \dots, d_s)$,*

$$P = MAN$$

where N is the strict block-upper unipotent group, A is the positive split center of the block diagonal, and M is the product of determinant-one block groups together with the compact/sign component required by the real form. Multiplication is a diffeomorphism modulo the finite central overlaps of M and A .

Proof. Write a matrix in P in block upper triangular form. Its block diagonal part D is invertible and the quotient $D^{-1}p$ lies in N . Factor each diagonal block into a positive scalar of determinant magnitude one times a matrix of determinant one; collect the positive scalars into A subject to the total determinant-one relation. The determinant-one block matrices form M . If two such block factorizations agree, their ratio lies in the pairwise intersections of the diagonal block factors; the determinant-one condition restricts these intersections to the finite common scalar centers. Thus uniqueness holds modulo these finite central overlaps. All maps are polynomial/rational on the open block-invertible locus, hence smooth. $\square$

Worked Example H.7 (The two maximal parabolics of SL_3). Stabilizing a line gives matrices $\begin{pmatrix} * & * & * \\ 0 & * & * \\ 0 & * & * \end{pmatrix}$ after choosing the line as $\mathbb{R}e_1$; stabilizing a plane gives the transpose block pattern. Their unipotent radicals have dimensions two. The minimal upper-triangular parabolic stabilizes the full flag and has three-dimensional unipotent radical. These are exactly the rational subgroups that appear when a three-dimensional lattice develops a short vector or a small-area primitive plane.

Appendix I

Student Proof Laboratory: Reduction Theory Through Successive Minima

I.1 The SL_2 model first

For a unimodular lattice $\Lambda \subset \mathbb{R}^2$, choose a shortest nonzero vector v_1 and then choose v_2 so that (v_1, v_2) is an oriented lattice basis with the coefficient of the projection of v_2 onto v_1 in $[-1/2, 1/2]$. In the corresponding Iwasawa decomposition $g = nak$, the unipotent coordinate is bounded and the diagonal coordinate records $\|v_1\|^{-1}$. Thus every lattice can be moved by $\mathrm{SL}_2(\mathbb{Z})$ into a Siegel set.

Lemma I.1 (Two-dimensional reduction). *Every unimodular lattice admits a basis v_1, v_2 satisfying*

$$\|v_1\| \leq \|v_2\|, \quad |\langle v_1, v_2 \rangle| \leq \frac{1}{2} \|v_1\|^2.$$

Proof. Choose v_1 shortest. Among vectors completing it to a basis, replace v_2 by $v_2 - mv_1$ with m the nearest integer to $\langle v_1, v_2 \rangle / \|v_1\|^2$. This gives the second inequality. If the new v_2 were shorter than v_1 , it would contradict the choice of v_1 . Unimodularity is preserved by the integer column operation. $\square$

I.2 Higher-dimensional reduced flags

For $\Lambda \subset \mathbb{R}^n$, successive minima select short independent vectors, but an arbitrary set of short vectors need not form a primitive flag. The correct induction chooses a primitive subgroup of minimal covolume in each rank and records its exterior-power vector.

Lemma I.2 (Primitive completion). *If $\Delta < \mathbb{Z}^n$ is primitive of rank r , then a basis of Δ extends to a basis of $\mathbb{Z}^n$.*

Proof. Primitivity means $\mathbb{Z}^n/\Delta$ is torsion free. A finitely generated torsion-free abelian group is free, so choose a basis of the quotient and lift it to $\mathbb{Z}^n$. Together with a basis of Δ the lifts generate $\mathbb{Z}^n$ and are linearly independent, hence form a basis. $\square$

Proposition I.3 (Reduced flag to Siegel coordinates). *A unimodular lattice with a primitive flag $0 < \Delta_1 < \dots < \Delta_{n-1} < \mathbb{Z}^n$ chosen by successive covolume minimization can be represented by $g = nak$ with n in a fixed bounded upper-unipotent set and a in a positive chamber satisfying the standard simple-root lower bounds.*

Proof. Extend the flag to an integral basis by the primitive-completion lemma and apply Gram–Schmidt to its image under g . Integer column operations reduce each Gram–Schmidt coefficient modulo earlier basis vectors into $[-1/2, 1/2]$, giving bounded n . The diagonal entries of a are the successive orthogonal lengths. If a simple ratio violated the chamber bound, replacing the corresponding primitive rank- r subgroup by one using the next shorter direction would decrease its exterior covolume, contrary to minimality. Hence the diagonal coordinates lie in the fixed Siegel cone. $\square$

I.3 Finiteness of Siegel overlaps

Lemma I.4 (Finite overlap on compact truncations). *Let $\mathfrak{S}$ be a Siegel set and $C \Subset G$. Only finitely many $\gamma \in \Gamma$ satisfy $C \cap \mathfrak{S}\gamma \neq \emptyset$ after fixing the quotient convention.*

Proof. If $c = s\gamma$ with $c \in C$ and $s \in \mathfrak{S}$, then $\gamma = s^{-1}c$. Reduction inequalities bound the Cartan parameters of s whenever $s\gamma$ stays in a fixed compact set; the bounded unipotent and compact factors are already controlled. Thus

γ lies in a compact subset of G . A discrete subgroup meets a compact set finitely. $\square$

I.4 Finite covolume

The Haar density in NAK coordinates is $e^{-2\rho(\log a)} dn da dk$ in the right convention. On a Siegel cone, the integral over the bounded N and compact K factors is finite, and the A -integral is a product of convergent exponentials in the simple-root coordinates once the arithmetic inequalities are imposed. This is the analytic heart of the Borel–Harish-Chandra finite-volume argument in the cases used by the series.

I.5 Borel density: why projective recurrence forces fixed vectors

Let G be connected semisimple without compact factors and Γ a lattice. Suppose a rational representation has a vector whose line is fixed by Γ . Push Haar probability from $\Gamma \backslash G$ to projective space along the orbit of that line. The resulting probability is G -invariant. Furstenberg’s projective lemma, proved in the main text, says an invariant projective probability for such a group is supported on fixed lines. Hence the original line is fixed by G . Chevalley linearization then turns this representation statement into Zariski density of Γ .

Warning I.5 (Compact factors). A compact factor can preserve a nontrivial probability on projective space without fixing a line, so Borel density must either exclude compact factors or formulate the conclusion after projecting them away. Later uses in this series always state the needed hypothesis explicitly.

Appendix J

Student Proof Laboratory: Arithmetic Closed Orbits

J.1 The closed-orbit/lattice criterion

Let $x = \Gamma g$ and $H < G$ be closed. The stabilizer of x in H is $H \cap g^{-1}\Gamma g$. If this subgroup is a lattice in H , then the map

$$(H \cap g^{-1}\Gamma g) \backslash H \rightarrow X, \quad h \mapsto xh$$

is proper and its image is closed. Conversely, a closed orbit carrying a finite invariant H -measure identifies with this finite-volume quotient. The criterion is the bridge between arithmetic subgroups and homogeneous measures used in every later volume.

J.2 Covolume normalization

When a quotient $H \backslash G$ is used in counting, fix Haar measures once and insist on Weil's identity $dg = dh d(Hg)$. Rescaling dh by c rescales the quotient measure by c^{-1} , so expressions such as $\text{vol}((H \cap \Gamma) \backslash H) / \text{vol}(\Gamma \backslash G)$ are invariant only when the quotient normalization is carried along consistently. Volume X will derive the counting constant from this convention rather than from memory.

Appendix K

Student Proof Laboratory: Opposite Parabolics and the Langlands Coordinates Used Later

Later applications require opposite parabolics and Langlands-type coordinates because expanding-translate arguments repeatedly separate contracting, neutral, and expanding directions. We record exactly the version used in Volumes IX–XII.

K.1 Opposite parabolics from one cocharacter

Fix $H \in \mathfrak{a}$ and set

$$\mathfrak{n}^+(H) = \bigoplus_{\alpha(H)>0} \mathfrak{g}_\alpha, \quad \mathfrak{n}^-(H) = \bigoplus_{\alpha(H)<0} \mathfrak{g}_\alpha, \quad \mathfrak{l}(H) = \mathfrak{g}_0.$$

Let $N^\pm(H)$ be the connected groups with those nilpotent Lie algebras and let $L(H) = Z_G(H)^\circ$.

Proposition K.1 (Opposite parabolic pair). *The connected subgroups*

$$P^+(H) = L(H)N^+(H), \quad P^-(H) = L(H)N^-(H)$$

have Lie algebras $\mathfrak{l}(H) \oplus \mathfrak{n}^+(H)$ and $\mathfrak{l}(H) \oplus \mathfrak{n}^-(H)$, respectively. Moreover $N^+(H)$ and $N^-(H)$ are their unipotent radicals, and

$$P^+(H) \cap P^-(H) = L(H)$$

at the connected level.

Proof. The root-bracket rule shows that $\mathfrak{n}^\pm(H)$ are ideals in the corresponding nonnegative/nonpositive root sums. They are nilpotent by the positive-weight argument already proved. Their quotients by $\mathfrak{n}^\pm(H)$ are the zero-weight algebra $\mathfrak{l}(H)$, which is reductive in the parabolic cases used here by the root decomposition and the nondegeneracy of the Killing form on its semisimple root part. Hence the connected nilpotent normal subgroup is the unipotent radical. The two Lie algebras intersect exactly in $\mathfrak{l}(H)$; connectedness gives the last statement up to finite components, which play no role in the later local product arguments. $\square$

K.2 The *MAN* decomposition in the required form

Choose a maximal split torus A and a positive restricted root system. For a parabolic P containing the corresponding minimal parabolic, let A_P be the connected subgroup on which precisely the roots omitted from the Levi factor may vary, and let M_P be the connected subgroup generated by the zero roots together with the compact central directions.

Proposition K.2 (Langlands-type coordinates used in this series). *For every standard parabolic P occurring in the reduction and expanding translate arguments,*

$$P = M_P A_P N_P$$

with $N_P = R_u(P)$. Multiplication $M_P \times A_P \times N_P \rightarrow P$ is a local diffeomorphism everywhere and is a global diffeomorphism after quotienting the finite intersections among the three factors. Conjugation by A_P contracts or expands each root coordinate of N_P by the corresponding root character.

Proof. In the matrix laboratories this is block Gaussian elimination. In the root language, the Lie algebra is the direct sum of the zero-root Levi directions, the split center $\mathfrak{a}_P$, and the positive root spaces belonging to N_P . The differential of multiplication at the identity is therefore an isomorphism. Because A_P normalizes every root space and M_P normalizes the collection

of roots defining N_P , multiplication is surjective onto P . Any ambiguity lies in the pairwise intersections of the neutral factors, which are finite once M_P is chosen with compact center modulo the split torus A_P , so that $M_P \cap A_P$ is finite. Finally, for $X_\alpha \in \mathfrak{g}_\alpha$, $\text{Ad}(a)X_\alpha = e^{\alpha(\log a)}X_\alpha$, giving the contraction/expansion formula. $\square$

Worked Example K.3 (SL_3). For the maximal parabolic stabilizing $\mathbb{R}e_1$,

$$N_P = \left\{ \begin{pmatrix} 1 & x & y \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right\}, \quad A_P = \{\text{diag}(e^{2t}, e^{-t}, e^{-t}) : t \in \mathbb{R}\},$$

and M_P is the determinant-one block group acting on $\text{span}(e_2, e_3)$, with its compact central normalization. The opposite parabolic replaces the upper-right row by the lower-left column. These are exactly the stable/unstable coordinates used in the curve and wavefront arguments later in the series.

Appendix L

Guided Structural Calculations for Algebraic Groups and Cusps

The purpose of this chapter is to make the structural vocabulary of the volume computational. Later volumes will say “take the parabolic attached to a diverging cocharacter” or “move into a Siegel set.” Here we show what those instructions mean in matrices and why the corresponding finiteness statements are true.

L.1 From a cocharacter to a flag in SL_n

Let

$$a_t = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t}), \quad \lambda_1 + \dots + \lambda_n = 0.$$

After grouping equal weights, write

$$\lambda_1 = \dots = \lambda_{d_1} > \lambda_{d_1+1} = \dots = \lambda_{d_2} > \dots.$$

The associated flag is

$$0 < V_{d_1} < V_{d_2} < \dots < \mathbb{R}^n, \quad V_d = \text{span}(e_1, \dots, e_d).$$

Proposition L.1 (Matrix description of $P(a)$). *The set of matrices g for which $a_{-t}ga_t$ remains bounded as $t \rightarrow \infty$ is exactly the block upper-triangular subgroup preserving this flag.*

Proof. The (i, j) entry of $a_{-t}ga_t$ equals $e^{(\lambda_j - \lambda_i)t}g_{ij}$. It remains bounded for all large t if and only if $g_{ij} = 0$ whenever $\lambda_j > \lambda_i$. With the weights listed in decreasing order, this is precisely the block upper-triangular condition. Such matrices preserve each partial sum of the weight blocks, and conversely flag preservation gives the same zero pattern. $\square$

The strict block-upper entries form N_P , the block-diagonal part gives a Levi factor, and the scalar expansion of the blocks gives A_P . Reversing t produces the opposite block-lower parabolic.

Worked Example L.2 (SL_4 of type $(1, 2, 1)$). Take weights $(3, 0, 0, -3)$. The flag is $\mathbb{R}e_1 < \text{span}(e_1, e_2, e_3) < \mathbb{R}^4$. The unipotent radical contains the entries $(1, 2), (1, 3), (1, 4), (2, 4), (3, 4)$, while the $(2, 3)$ and $(3, 2)$ entries belong to the middle SL_2 Levi block. This is a useful example because it shows that a parabolic is generally much larger than a Borel: neutral root directions survive in the Levi factor.

L.2 Restricted roots without a classification table

For $G = SL_n(\mathbb{R})$ and diagonal A , the root spaces are $\mathfrak{g}_{e_i - e_j} = \mathbb{R}E_{ij}$. The bracket

$$[E_{ij}, E_{jk}] = E_{ik}$$

shows how roots add. A subset of positive roots is closed under addition exactly when the corresponding upper-triangular coordinates form a subgroup.

For $SO(p, q)$, choose a maximal split torus by simultaneous hyperbolic rotations in q mutually orthogonal $(1, 1)$ -planes. Conjugation on matrix units again diagonalizes; roots of types $\pm e_i \pm e_j$ and, when $p > q$, $\pm e_i$ appear. One does not need the full classification later: what matters is that there are finitely many root spaces, each cocharacter assigns them explicit signs, and the positive sign spaces form a nilpotent algebra.

L.3 Successive minima as a cusp detector

For a unimodular lattice $\Lambda \subset \mathbb{R}^n$, let $\lambda_i(\Lambda)$ be the least r such that Λ contains i linearly independent vectors of norm at most r .

Lemma L.3 (A gap produces a rational subspace). *If*

$$\lambda_j(\Lambda) \ll \lambda_{j+1}(\Lambda),$$

then the span W of all lattice vectors of norm less than $\lambda_{j+1}(\Lambda)$ is a j -dimensional rational subspace, and its primitive lattice $W \cap \Lambda$ has covolume

$$\text{covol}(W \cap \Lambda) \asymp \lambda_1(\Lambda) \cdots \lambda_j(\Lambda)$$

up to constants depending only on n .

Proof. The definition of λ_j provides j independent short vectors, so the span has dimension at least j . A $(j+1)$ st independent vector below the next minimum would contradict the definition of λ_{j+1} , hence the dimension is exactly j . It is rational because it is spanned by lattice vectors. Minkowski's second theorem applied to the lattice $W \cap \Lambda$ in W compares its covolume with the product of its successive minima, which are comparable to the first j minima of Λ under the gap hypothesis. $\square$

A deep cusp therefore carries algebraic data: a primitive rational subspace. Reduction theory turns the finitely many possible dimensions and incidence patterns of these subspaces into finitely many rational parabolic types.

L.4 Gram–Schmidt and a Siegel set for SL_3

Write $g = nak$ with

$$n = \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix}, \quad a = \text{diag}(a_1, a_2, a_3), \quad a_1 a_2 a_3 = 1.$$

Choose the standard reduced region

$$|x|, |y|, |z| \leq C, \quad a_1/a_2 \geq c, \quad a_2/a_3 \geq c.$$

The ratios determine how close the lattice is to the two maximal cusp faces. If a_1/a_2 is huge, one primitive line is much shorter than its complement; if a_2/a_3 is huge, a primitive two-plane has unusually small covolume.

The Haar density in logarithmic coordinates $s_1 = \log(a_1/a_2)$ and $s_2 = \log(a_2/a_3)$ has an exponentially decaying factor after the quotient normalization. Thus

$$\int_{s_1, s_2 \geq s_0} e^{-c_1 s_1 - c_2 s_2} ds_1 ds_2 < \infty.$$

This is the analytic core of finite cusp volume. The difficult global part of reduction theory is not this integral but the statement that finitely many arithmetic translates of such regions cover the quotient with controlled overlap.

L.5 Why overlaps are finite

Suppose two Siegel-set points g, g' represent the same quotient point, so $g' = \gamma g$ for $\gamma \in \Gamma$. If both points lie in a fixed truncated Siegel set, every matrix coefficient of $g'g^{-1}$ is bounded. Since γ has integral entries, only finitely many possibilities occur.

In a deep cusp the set is not compact, but the short rational flag is unique once the root heights exceed a fixed threshold. Hence γ must preserve that flag and lie in the corresponding rational parabolic. Its unipotent coordinates can then be reduced modulo the integral unipotent lattice, and its Levi coordinates are treated by induction on semisimple rank. This is the full reason the overlap problem terminates: either one is in a compact part, where integrality gives finiteness immediately, or one descends to a proper Levi factor.

L.6 Borel–Harish-Chandra as an induction

We can now state the proof architecture in student language.

1. Choose finitely many rational parabolic cusp types.
2. In each cusp write $P = MAN$ and reduce N modulo $N \cap \Gamma$.
3. Integrate the A -coordinates; positive roots make the integral converge.
4. Reduce the M -quotient by induction on rational semisimple rank.
5. Use the overlap argument above to see that the resulting fundamental set has finite multiplicity.

The base case is rational rank zero, where no proper rational parabolic exists and the quotient is compact by the height criterion already proved in the body.

L.7 Borel density through a representation

Let G have no compact algebraic factor relevant to the representation and let $\Gamma < G$ be a lattice. Suppose a vector v has finite Γ -orbit in projective space. Passing to a finite-index subgroup, we may suppose the line $[v]$ is fixed. The orbit map $G/\Gamma \rightarrow \mathbf{P}(V)$ pushes Haar probability to a G -stationary probability supported on the compact closure of $G[v]$.

Choose a split element a_t . If v has two different A -weights, the projective orbit $a_t[v]$ converges to different extremal weight components in the two time

directions. Recurrence of the finite-volume a_t -action forces the projective point to return arbitrarily close to itself, so this is impossible unless all nonzero components have one weight. Applying root unipotents then forces that weight line to be fixed by the root groups. Root generation makes it G -fixed. Chevalley linearization converts this fixed line criterion into Zariski density of the lattice.

Worked Example L.4 ($SL_2(\mathbb{Z}) < SL_2(\mathbb{R})$). In the standard representation, a projective fixed point would be a line in $\mathbb{R}^2$ preserved by every matrix in $SL_2(\mathbb{Z})$. The upper unipotent fixes only $\mathbb{R}e_1$ and the lower unipotent fixes only $\mathbb{R}e_2$, so no common line exists. The abstract Borel-density argument is a vast generalization of this simple incompatible-fixed-lines calculation.

L.8 How later volumes use the structure

Volumes VI–XII use only four outputs of this long structural discussion:

1. Cartan/root coordinates in which divergence has a sign;
2. parabolic MAN coordinates in which escape has an algebraic label;
3. arithmetic height functions built from primitive exterior-power data;
4. finite-volume and Borel-density statements for the closed homogeneous orbits that arise.

A reader who can reproduce the calculations in this chapter has enough structural control to follow those later arguments without treating “reduction theory” as a magic phrase.

Appendix M

Complete proof: Arithmetic Fundamental Sets and Finite Covolume

M.1 Role of this complete proof

The main reduction-theory chapter gives the global architecture. The present module supplies full proofs for several genuinely nontrivial transitions, including the passage from an arithmetic subgroup to finitely many cusp pieces, the representation-theoretic finiteness step, and the comparison between an ambient Siegel set and a Siegel set in a semisimple subgroup. We now give a second proof of the finite-covolume part of Theorem 8.1. This proof is logically independent of the short rank-induction proof of Theorem 7.2.

The proof has four modules:

1. put a real reductive subgroup in a position preserved by the standard Cartan involution;
2. prove the Borel–Harish–Chandra finiteness lemma for lattice points in one ambient Siegel set;
3. use that lemma to construct a genuine arithmetic fundamental set with finite overlap;
4. compare intersections of ambient Siegel sets with the subgroup to Siegel sets of the subgroup and integrate the Haar density exactly.

Every compactness or finiteness assertion used below is isolated as a lemma.

M.2 Cartan-compatible position

Let $G_0 = \mathrm{SL}_N(\mathbb{R})$ and let

$$\theta(g) = (g^{-1})^\top.$$

Its fixed group is $K_0 = \mathrm{SO}(N)$. At Lie-algebra level the corresponding Cartan decomposition is

$$\mathfrak{sl}_N(\mathbb{R}) = \mathfrak{k}_0 \oplus \mathfrak{p}_0, \quad \mathfrak{k}_0 = \{X : X^\top = -X\}, \quad \mathfrak{p}_0 = \{X : X^\top = X\}.$$

Lemma M.1 (A completely reducible subgroup can be made self-adjoint). *Let $H < G_0$ be a connected real reductive linear group. There is $c \in \mathrm{GL}_N(\mathbb{R})$ such that $H' = cHc^{-1}$ is invariant under θ . If H is semisimple, c may be rescaled so that $c \in \mathrm{SL}_N(\mathbb{R})$.*

Proof. Choose a Cartan involution θ_H of the semisimple part of $\mathfrak{h}$ by Theorem A.3 and write $\mathfrak{h} = \mathfrak{k}_H \oplus \mathfrak{p}_H$; on the split and compact parts of the center use the same $+1/-1$ convention. Complexify the given representation of $\mathfrak{h}$ on $\mathbb{R}^N$. The real Lie algebra

$$\mathfrak{u} = \mathfrak{k}_H \oplus i\mathfrak{p}_H$$

is compact. Integrate an arbitrary positive Hermitian form on $\mathbb{C}^N$ over the compact connected group with Lie algebra $\mathfrak{u}$. In the resulting form, $d\rho(X)$ is skew-Hermitian for $X \in \mathfrak{k}_H$, while $d\rho(Y)$ is Hermitian for $Y \in \mathfrak{p}_H$, because $iY \in \mathfrak{u}$. Taking the real part gives a positive definite real inner product with the same property on $\mathbb{R}^N$.

Choose c carrying this inner product to the Euclidean inner product used for the ambient matrix realization. Then transpose sends $d\rho(\mathfrak{k}_H)$ to its negative and fixes $d\rho(\mathfrak{p}_H)$, so the ambient transpose-inverse Cartan involution preserves $c\mathfrak{h}c^{-1}$. Since H is connected, it preserves cHc^{-1} . Multiplying c by a scalar does not change the conjugate subgroup, and in the semisimple case we choose that scalar so that $\det c = 1$. □

Lemma M.2 (Compatible position relative to the standard ambient Iwasawa data). *Fix the standard diagonal group $A_0 < \mathrm{SL}_N(\mathbb{R})$, the standard maximal compact group $K_0 = \mathrm{SO}(N)$, and let the positive system determining N_0 be allowed to vary by a Weyl permutation. Let $J < G_0$ be connected semisimple and θ -stable. There are $k \in K_0$ and a signed permutation matrix $m \in N_{K_0}(A_0)$ such that, for*

$$H = mkJk^{-1}m^{-1},$$

we have

$$K_H = K_0 \cap H, \quad A_H = A_0 \cap H, \quad N_H = N_0 \cap H,$$

and $H = K_H A_H N_H$ is an Iwasawa decomposition. Every nonzero restriction to $\mathfrak{a}_H$ of an ambient positive root is positive for the chosen positive system of H .

Proof. Because J is θ -stable,

$$\mathfrak{j} = (\mathfrak{j} \cap \mathfrak{k}_0) \oplus (\mathfrak{j} \cap \mathfrak{p}_0).$$

Choose a maximal abelian subspace $\mathfrak{a}_J \subset \mathfrak{j} \cap \mathfrak{p}_0$ and extend it to a maximal abelian subspace $\mathfrak{a}' \subset \mathfrak{p}_0$. Conjugacy of maximal abelian subspaces of $\mathfrak{p}_0$ under K_0 gives $k \in K_0$ with $\text{Ad}(k)\mathfrak{a}' = \mathfrak{a}_0$. Thus, after replacing J by kJk^{-1} , its maximal split subspace is contained in the fixed ambient diagonal algebra $\mathfrak{a}_0$.

Choose an ordering on $\mathfrak{a}_0^*$ lexicographically from a basis whose first vectors restrict to an ordered basis of $\mathfrak{a}_H^*$. With this ordering, every ambient root having nonzero restriction to $\mathfrak{a}_H$ has the same sign as its restriction. Any positive system of the ambient root system is carried to the fixed ambient positive system by a Weyl element; choose a signed permutation representative $m \in N_{K_0}(A_0)$. Conjugating by m therefore gives the asserted compatibility while preserving θ -stability. The root-space equality $\mathfrak{n}_H = \mathfrak{n}_0 \cap \mathfrak{h}$, together with the Cartan decomposition, identifies these factors with the Iwasawa factors of Theorem A.13 and gives $H = K_H A_H N_H$. $\square$

M.3 A closed-orbit Chevalley package in the semisimple case

The arithmetic finiteness lemma needs more than a projective line: it needs a rational vector whose ambient orbit is closed. For a reductive stabilizer this is an affine-invariant-theory statement. We prove the special form needed here rather than citing an affine-quotient theorem as a black box.

Lemma M.3 (Dickson's lemma). *Every subset $E \subset \mathbb{N}^d$ has only finitely many minimal elements for the coordinatewise partial order. Equivalently, every monomial ideal in $k[x_1, \dots, x_d]$ is generated by finitely many monomials.*

Proof. It is enough to prove the following sequence form: every infinite sequence $a_1, a_2, \dots$ in $\mathbb{N}^d$ contains indices $i < j$ with $a_i \leq a_j$ coordinatewise. We argue by induction on d . For $d = 1$, either one integer occurs infinitely often, giving a constant comparable subsequence, or every value occurs only finitely often; in the latter case one chooses recursively terms with strictly increasing values. Thus the sequence form holds in dimension one.

Assume the statement in dimension $d - 1$. From the first coordinates of an infinite sequence in $\mathbb{N}^d$, extract an infinite subsequence on which the first coordinate is nondecreasing: if some value occurs infinitely often use those terms; otherwise choose successively terms with strictly increasing first coordinate. Apply the induction hypothesis to the remaining $d - 1$ coordinates of this subsequence. Two terms are then ordered in all coordinates.

If E had infinitely many minimal elements, an enumeration of them would contradict the sequence form, since distinct minimal elements are incomparable. Hence the minimal elements are finite. Every element of E dominates a minimal element: among the finitely many elements of E lying in the coordinate box below it, choose a minimal one. Translating exponent vectors into monomials gives the equivalent monomial-ideal statement. $\square$

Theorem M.4 (Hilbert basis theorem in the form used below). *For every field k , the polynomial ring $k[x_1, \dots, x_d]$ is Noetherian. Consequently every finitely generated commutative k -algebra is Noetherian.*

Proof. Fix a monomial order, for example graded lexicographic order. Let $I \subset k[x_1, \dots, x_d]$ be an ideal. The set of leading monomials of nonzero elements of I generates a monomial ideal $M(I)$. By Lemma M.3, choose finitely many elements $f_1, \dots, f_r \in I$ whose leading monomials generate $M(I)$.

We claim that $I = (f_1, \dots, f_r)$. Take $0 \neq f \in I$. Since $\text{LM}(f) \in M(I)$, some $\text{LM}(f_i)$ divides $\text{LM}(f)$. Subtract the unique scalar multiple of the corresponding monomial times f_i that cancels the leading term of f . The result is still in I and has strictly smaller leading monomial. Repeat. A monomial order is a well-order, so the process terminates. If a nonzero remainder remained, its leading monomial would lie in $M(I)$ and would again be divisible by one of the chosen leading monomials, contradicting termination. Hence the remainder is zero and $f \in (f_1, \dots, f_r)$. Thus every ideal is finitely generated.

If A is generated over k by $a_1, \dots, a_m$, then $A \cong k[x_1, \dots, x_m]/J$. Ideals of A correspond to ideals of the polynomial ring containing J , so they are

finitely generated as well. $\square$

Remark M.5 (Where the Reynolds projection comes from). The characteristic-zero complete-reducibility theorem and the natural Reynolds projection are now proved in Chapter 1; in particular Corollary 1.5 supplies the module identity used below. The invariant-theory argument therefore has no external linear-reductivity owner.

Lemma M.6 (Reynolds operator and finite generation). *Let $\mathbf{L}$ be a reductive algebraic group over a characteristic-zero field k , acting rationally on a finitely generated commutative k -algebra A . Then there is an $A^{\mathbf{L}}$ -linear projection*

$$R : A \longrightarrow A^{\mathbf{L}}, \quad R|_{A^{\mathbf{L}}} = \text{id},$$

and $A^{\mathbf{L}}$ is finitely generated over k .

Proof. First suppose $A = k[V] = \text{Sym}(V^*)$ for a finite-dimensional rational $\mathbf{L}$ -module V . Theorem 1.4 and Corollary 1.5 provide a grading-preserving projection $R : A \rightarrow A^{\mathbf{L}}$ satisfying

$$R(fa) = fR(a) \quad (f \in A^{\mathbf{L}}, a \in A).$$

Let $I \subset A$ be the ideal generated by all positive-degree invariants. By Theorem M.4, choose finitely many homogeneous invariants $f_1, \dots, f_r$ that generate I as an ideal of A . If f is a homogeneous invariant of positive degree, write $f = \sum a_i f_i$. Applying R gives

$$f = \sum R(a_i) f_i.$$

Whenever $\deg a_i > 0$, $\deg R(a_i) < \deg f$. Induction on degree shows that every invariant is a polynomial in the f_i . Thus $A^{\mathbf{L}}$ is finitely generated.

For a general finitely generated A , choose a finite-dimensional $\mathbf{L}$ -stable generating subspace and a surjection $P = \text{Sym}(V^*) \rightarrow A$ with invariant kernel J . Complete reducibility makes the invariants functor exact, so

$$A^{\mathbf{L}} \cong P^{\mathbf{L}} / (J \cap P^{\mathbf{L}}).$$

The Reynolds operator descends, and finite generation follows from the polynomial case. $\square$

Lemma M.7 (Affine quotient for right multiplication by a reductive subgroup). *Let $\mathbf{H} < \mathbf{G}$ be reductive algebraic groups over k of characteristic*

zero. Put $B = k[\mathbf{G}]$ and let $\mathbf{H}$ act on B by right translation. Then $C = B^{\mathbf{H}}$ is finitely generated and the morphism

$$q : \mathbf{G} \longrightarrow \operatorname{Spec} C$$

has the following properties after extension to an algebraic closure:

1. every closed point of $\operatorname{Spec} C$ has a preimage;
2. two points of $\mathbf{G}$ have the same image exactly when they lie in the same right $\mathbf{H}$ -coset.

Consequently $\mathbf{G}/\mathbf{H}$ is affine, with coordinate ring C .

Proof. Finite generation is Lemma M.6. Work over an algebraic closure for the geometric assertions. If $\mathfrak{m}$ is a maximal ideal of C and $\mathfrak{m}B = B$, then $1 = \sum_i f_i b_i$ with $f_i \in \mathfrak{m}$. Applying the Reynolds operator gives

$$1 = \sum_i f_i R(b_i) \in \mathfrak{m},$$

a contradiction. Thus $\mathfrak{m}B$ is proper and lies in a maximal ideal of B , which is a point of $\mathbf{G}$ mapping to $\mathfrak{m}$. This proves surjectivity on closed points.

Right cosets $g\mathbf{H}$ are closed because $\mathbf{H}$ is a closed algebraic subgroup and translation is an isomorphism. If $g_1\mathbf{H}$ and $g_2\mathbf{H}$ are distinct, their two defining ideals in B are comaximal. Choose $f \in B$ with $f = 0$ on $g_1\mathbf{H}$ and $f = 1$ on $g_2\mathbf{H}$. Then $R(f)$ is right $\mathbf{H}$ -invariant and has the same two constant values: on a right $\mathbf{H}$ -orbit averaging does not change a constant function. Hence $R(f)$ separates the two cosets. The fibers of q are therefore exactly the right cosets. Since q is onto the affine variety $\operatorname{Spec} C$, this identifies the geometric quotient with $\mathbf{G}/\mathbf{H}$ and proves that it is affine. $\square$

Proposition M.8 (Closed-orbit vector for a semisimple rational subgroup). *Let $\mathbf{H} < \operatorname{SL}_N$ be a connected semisimple algebraic subgroup defined over $\mathbb{Q}$. There is a finite-dimensional rational representation $\pi : \operatorname{SL}_N \rightarrow \operatorname{GL}(W)$ and $v \in W(\mathbb{Q})$ such that*

$$\operatorname{Stab}_{\operatorname{SL}_N}(v) = \mathbf{H}$$

and the orbit $v\pi(\operatorname{SL}_N)$ is Zariski closed. In particular its real orbit $v\pi(\operatorname{SL}_N(\mathbb{R}))$ is closed in the Euclidean topology.

Proof. Apply Lemma M.7 over $\mathbb{Q}$ to $\mathbf{G} = \operatorname{SL}_N$ and $\mathbf{H}$. Write

$$C = \mathbb{Q}[\operatorname{SL}_N]^{\mathbf{H}}.$$

Choose finite algebra generators $f_1, \dots, f_m$ of C . The left-translate span of each f_i in $\mathbb{Q}[\mathrm{SL}_N]$ is finite dimensional by Lemma 2.3; hence the span U of 1 and all left translates of the f_i is a finite-dimensional rational SL_N -module contained in C .

Evaluation gives a rational equivariant map

$$\Phi : \mathrm{SL}_N / \mathbf{H} \longrightarrow U^*, \quad \Phi(g\mathbf{H})(f) = f(g).$$

Because the chosen f_i generate the coordinate ring C , the induced map on coordinate rings $\mathrm{Sym}(U) \rightarrow C$ is surjective. Therefore Φ is a closed immersion. Put $v = \Phi(\mathbf{H})$. It is $\mathbb{Q}$ -rational, its stabilizer is exactly $\mathbf{H}$ because the quotient fibers are the right cosets, and its orbit is the closed subvariety $\Phi(\mathrm{SL}_N / \mathbf{H})$.

It remains to pass from the closed algebraic orbit to the real orbit. Let $\mathcal{O} = v\pi(\mathrm{SL}_N(\mathbb{C}))$. The real locus $\mathcal{O}(\mathbb{R})$ is a smooth real manifold (possibly with several connected components). At every point of the orbit $v\pi(\mathrm{SL}_N(\mathbb{R}))$, the tangent map of the real group action is the real form of the tangent map of the complex action, hence is surjective onto the real tangent space of $\mathcal{O}(\mathbb{R})$. The real orbit is therefore open in $\mathcal{O}(\mathbb{R})$. The same argument applies to every real-group orbit in $\mathcal{O}(\mathbb{R})$, so the real locus is partitioned into open orbits. A connected component cannot meet two disjoint open pieces; hence every real-group orbit is a union of connected components. The real algebraic manifold $\mathcal{O}(\mathbb{R})$ is semialgebraic, so it has only finitely many connected components. Each component is closed, and therefore every union of components is closed. Thus $v\pi(\mathrm{SL}_N(\mathbb{R}))$ is Euclidean closed. $\square$

M.4 Arithmetic lattices in rational representations

Lemma M.9 (A stable integral lattice after finite-index reduction). *Let $\pi : \mathrm{SL}_N \rightarrow \mathrm{GL}(W)$ be rational and defined over $\mathbb{Q}$. There is a full lattice $\Lambda \subset W(\mathbb{Q})$ invariant under $\mathrm{SL}_N(\mathbb{Z})$.*

Proof. Start with any lattice Λ_0 . In a rational basis, the matrix entries of $\pi(g)$ and $\pi(g)^{-1}$ are rational regular functions of the matrix entries of g ; on SL_N the inverse entries are polynomial because $\det g = 1$. Let D be a common positive denominator for the finitely many coefficients of these coordinate polynomials, so that multiplying each coefficient by D gives an integer. Choose an integer m divisible by D and large enough that every

$\gamma \equiv I \pmod{m}$ preserves Λ_0 . Thus the principal congruence subgroup $\Gamma(m)$ preserves Λ_0 .

Choose representatives $r_1, \dots, r_s$ for the finite quotient $\mathrm{SL}_N(\mathbb{Z})/\Gamma(m)$ and put

$$\Lambda = \sum_{i=1}^s \Lambda_0 \pi(r_i).$$

This is a lattice (a finite sum of commensurable lattices). Right multiplication by any $\gamma \in \mathrm{SL}_N(\mathbb{Z})$ permutes the cosets $r_i \Gamma(m)$, and the residual factor in $\Gamma(m)$ preserves Λ_0 . Hence $\Lambda \pi(\gamma) = \Lambda$. $\square$

Lemma M.10 (Stable lattice for an arithmetic group in a rational representation). *Let $\mathbf{G} < \mathrm{SL}_N$ be a $\mathbb{Q}$ -subgroup, let $\Gamma < \mathbf{G}(\mathbb{Q})$ be arithmetic, and let $\pi : \mathbf{G} \rightarrow \mathrm{GL}(W)$ be a rational representation defined over $\mathbb{Q}$. After replacing Γ by a finite-index subgroup, there is a full lattice $\Lambda \subset W(\mathbb{Q})$ preserved by $\pi(\Gamma)$.*

Proof. After replacing Γ by a commensurable finite-index subgroup, we may assume it lies in $\mathbf{G}(\mathbb{Q}) \cap \mathrm{SL}_N(\mathbb{Z})$. Choose a rational basis of W . Because π and π^{-1} are regular morphisms on $\mathbf{G}$, their matrix coefficients have polynomial representatives in the coordinate ring with only finitely many rational denominators. Let D be a common denominator of those coefficients and choose an integer m divisible by D and large enough that every $\gamma \equiv I \pmod{m}$ in Γ acts, together with its inverse, by an integral matrix on the coordinate lattice Λ_0 spanned by this basis. Thus the finite-index congruence subgroup $\Gamma(m)$ preserves Λ_0 .

Choose coset representatives $r_1, \dots, r_s$ for $\Gamma/\Gamma(m)$ and put

$$\Lambda = \sum_{i=1}^s \Lambda_0 \pi(r_i).$$

This finite sum of commensurable lattices is again a full lattice. Right multiplication by an element of Γ permutes the cosets and contributes only a residual element of $\Gamma(m)$, so Λ is $\pi(\Gamma)$ -stable. $\square$

M.5 The finiteness lemma inside one Siegel set

Fix the standard Iwasawa decomposition $G_0 = K_0 A_0 N_0$, with A_0 the positive diagonal determinant-one group. Let

$$\mathfrak{S} = K_0 A_t \Omega$$

be a standard Siegel set, where $\Omega \in N_0$ and the simple-root coordinates of A_t are bounded on the cusp side in the convention chosen below. The only property of the convention that matters is that, for each fixed positive integer r ,

$$\{a^r n a^{-r} : a \in A_t, n \in \Omega\} \text{ is relatively compact.} \quad (\text{M.1})$$

We shall use only $r = 1, 2$. In upper-triangular root coordinates, conjugation multiplies the β -coordinate by $e^{r\beta(\log a)}$. The defining Siegel inequalities bound these factors on the contracting side, uniformly for fixed r ; since Ω is compact and the multiplication law is polynomial in the nilpotent coordinates, the displayed families are relatively compact. This is the same triangular-matrix estimate used in the proof of Theorem 6.2.

Lemma M.11 (Rational weight projections are uniformly discrete). *Let A_0 act rationally on a finite-dimensional rational representation W , and let $W = \bigoplus_{\mu} W_{\mu}$ be its weight decomposition. If $\Lambda \subset W(\mathbb{Q})$ is a lattice, then after replacing it by a commensurable lattice each W_{μ} is rational and there is $c > 0$ such that for every $w \in \Lambda$,*

$$E_{\mu} w \neq 0 \implies \|E_{\mu} w\| \geq c.$$

Proof. The diagonal torus A_0 is $\mathbb{Q}$ -split, so a rational representation decomposes over $\mathbb{Q}$ into rational weight spaces. Choose a rational basis adapted to the direct sum. The corresponding projections E_{μ} have rational matrices. After clearing a common denominator, $E_{\mu}(\Lambda)$ is contained in a lattice of W_{μ} . A lattice has a positive minimum norm away from zero. There are only finitely many weights, so the minimum of these positive constants works for all μ . $\square$

Theorem M.12 (Borel–Harish–Chandra finiteness lemma). *Let $\pi : G_0 \rightarrow \text{GL}(W)$ be a rational representation and $\Lambda \subset W(\mathbb{Q})$ a lattice. Let $v \in W(\mathbb{R})$ have closed orbit $v\pi(G_0)$, and assume its stabilizer $J = (G_0)_v$ is θ -stable. Then*

$$v\pi(\mathfrak{S}) \cap \Lambda$$

is finite.

Proof. Choose a K_0 -invariant Euclidean norm for which A_0 acts by commuting self-adjoint operators. The weight spaces are then orthogonal. By Lemma M.11, there is $c > 0$ such that every nonzero weight projection of a lattice vector has norm at least c .

Take $x = kan \in \mathfrak{S}$ and suppose

$$w := v\pi(x) \in \Lambda.$$

Define

$$y = xa^{-1} = k(ana^{-1}), \quad z = ya^{-1}.$$

By (M.1), the points y lie in one compact subset of G_0 , hence $v\pi(y)$ lies in one compact subset of W . Let E_μ be the orthogonal projection to a weight space. Since right multiplication by a^{-1} multiplies the μ -component by $e^{-\mu(\log a)}$, we have

$$E_\mu(v\pi(y)) = e^{-\mu(\log a)} E_\mu(w),$$

and

$$E_\mu(v\pi(z)) = e^{-2\mu(\log a)} E_\mu(w).$$

There is C independent of x with $\|E_\mu(v\pi(y))\| \leq C$. If $E_\mu(w) = 0$, then the corresponding component of $v\pi(z)$ is zero. Otherwise $\|E_\mu(w)\| \geq c$, and therefore

$$\|E_\mu(v\pi(z))\| = \frac{\|E_\mu(v\pi(y))\|^2}{\|E_\mu(w)\|} \leq \frac{C^2}{c}.$$

Because there are finitely many weights, $v\pi(z)$ stays in a fixed compact subset $Q \subset W$.

The orbit is closed. The orbit map $J \backslash G_0 \rightarrow v\pi(G_0)$ is therefore a homeomorphism, and the preimage of a compact subset of the orbit is compact in $J \backslash G_0$. Consequently there is a compact set $B \subset G_0$ such that

$$z \in JB.$$

We now recover a compact bound for x modulo J by an exact identity. Put

$$u_2 = a^2 na^{-2}.$$

By (M.1) with $r = 2$, the elements u_2 range in one compact set $C_2 \subset N_0$. Since

$$z = xa^{-2} = kana^{-2} = ka^{-1}u_2,$$

the Cartan involution satisfies

$$\theta(z) = ka\theta(u_2).$$

Therefore

$$x = kan = \theta(z) \theta(u_2)^{-1} n. \tag{M.2}$$

If $z = jb$ with $j \in J$ and $b \in B$, then $\theta(j) \in J$ because J is θ -stable. Equation (M.2) gives

$$x \in J\theta(B)\theta(C_2)^{-1}\Omega.$$

The last three factors form a compact set $B' \subset G_0$. Thus every x under consideration belongs to JB' .

Since J fixes v , it follows that $v\pi(x) \in v\pi(B')$. Hence

$$v\pi(\mathfrak{S}) \cap \Lambda \subset v\pi(B') \cap \Lambda.$$

The first set on the right is compact and the second factor is discrete, so the intersection is finite. $\square$

M.6 A genuine arithmetic fundamental set

Theorem M.13 (Arithmetic fundamental set inherited from ambient reduction). *Let $\mathbf{H} < \mathrm{SL}_N$ be connected semisimple and defined over $\mathbb{Q}$, and put*

$$H = \mathbf{H}(\mathbb{R})^\circ, \quad \Gamma = H \cap \mathrm{SL}_N(\mathbb{Z}).$$

There exist $c \in \mathrm{SL}_N(\mathbb{R})$, a standard ambient Siegel set $\mathfrak{S} \subset \mathrm{SL}_N(\mathbb{R})$, and finitely many $b_1, \dots, b_m \in \mathrm{SL}_N(\mathbb{Z})$ such that

$$H = \bigcup_{i=1}^m (c^{-1}\mathfrak{S}b_i \cap H)\Gamma.$$

Moreover, for every i, j , only finitely many $\gamma \in \Gamma$ satisfy

$$(c^{-1}\mathfrak{S}b_i \cap H)\gamma \cap (c^{-1}\mathfrak{S}b_j \cap H) \neq \emptyset.$$

Proof. Choose π, v from Proposition M.8. By Lemma M.9, after multiplying v by a positive integer there is an $\mathrm{SL}_N(\mathbb{Z})$ -invariant lattice Λ containing v . Choose c from Lemma M.1 so that $H' = cHc^{-1}$ is θ -stable and put

$$v' = v\pi(c^{-1}).$$

The stabilizer of v' is H' , and its orbit is closed.

Consider the right coset cH . It has the vector description

$$cH = \{g \in G_0 : v'\pi(g) = v\}.$$

Indeed, $v'\pi(g) = v$ is equivalent to $v\pi(c^{-1}g) = v$, hence to $c^{-1}g \in H$.

Let $h \in cH$. Ambient Siegel reduction (Theorem 6.2) writes $h = sb$ with $s \in \mathfrak{S}$ and $b \in \mathrm{SL}_N(\mathbb{Z})$. Then

$$v'\pi(s) = v\pi(b^{-1}) \in \Lambda.$$

Theorem M.12, applied to v' , says that only finitely many vectors of this form can occur. Choose $b_1, \dots, b_m \in \mathrm{SL}_N(\mathbb{Z})$ so that every such value equals $v\pi(b_i^{-1})$ for some i . Then

$$v\pi(b^{-1}) = v\pi(b_i^{-1}) \implies b_i^{-1}b \in H.$$

Because both matrices are integral, $b_i^{-1}b \in \Gamma$. Hence $h \in \mathfrak{S}b_i\Gamma$. Multiplying by c^{-1} gives the required cover of H .

For finite overlap, suppose $c^{-1}s_ib_i\gamma = c^{-1}s_jb_j$ with $s_i, s_j \in \mathfrak{S}$ and $\gamma \in \Gamma$. The factor c^{-1} cancels, so

$$s_j^{-1}s_i = b_j\gamma^{-1}b_i^{-1}.$$

Thus $b_j\gamma^{-1}b_i^{-1}$ lies in $\mathfrak{S}^{-1}\mathfrak{S} \cap \mathrm{SL}_N(\mathbb{Z})$. Ambient Siegel reduction has finite overlap, and left/right multiplication by the fixed integral matrices b_i, b_j is a bijection of the integral group. Hence only finitely many γ can occur. $\square$

Corollary M.14 (Commensurable arithmetic subgroups). *The same conclusion holds, after changing the finite family by finitely many translates, for every arithmetic subgroup of $\mathbf{H}(\mathbb{Q})$ commensurable with $H \cap \mathrm{SL}_N(\mathbb{Z})$.*

Proof. If Γ' is commensurable with Γ , pass to $\Gamma_0 = \Gamma \cap \Gamma'$. A fundamental set for Γ becomes a finite union of fundamental sets for Γ_0 after multiplying by representatives of $\Gamma_0 \backslash \Gamma$, and conversely a finite union of Γ_0 -pieces covers modulo Γ' . Finite overlap is preserved under a finite union of translates. $\square$

M.7 Intersecting an ambient Siegel set with a semisimple subgroup

We now prove the geometric step that turns the fundamental set above into a finite-volume set. This is the step that cannot be replaced by the sentence “the subgroup inherits the ambient Haar integral.”

Lemma M.15 (Nilpotent factorization adapted to the subgroup). *Assume the compatible Iwasawa data of Lemma M.2. For an ambient positive root β write $\bar{\beta} = \beta|_{\mathfrak{a}_H}$. Put*

$$\mathfrak{n}^0 = \bigoplus_{\bar{\beta}=0} \mathfrak{g}_\beta, \quad \mathfrak{n}' = \bigoplus_{\bar{\beta} \neq 0} \mathfrak{g}_\beta.$$

Then $\mathfrak{n}^0$ is a subalgebra and $\mathfrak{n}'$ is an ideal in $\mathfrak{n}_0$. For every $\bar{\beta} \neq 0$ choose a linear complement $\mathfrak{q}_\beta$ of $\mathfrak{h}_{\bar{\beta}} \cap \mathfrak{g}_\beta$ in $\mathfrak{g}_\beta$ and put

$$\mathfrak{n}^2 = \bigoplus_{\bar{\beta} \neq 0} \mathfrak{q}_\beta.$$

Thus $\mathfrak{n}' = \mathfrak{n}^2 \oplus \mathfrak{n}_H$ as vector spaces; $\mathfrak{n}^2$ is not asserted to be a Lie algebra. The map

$$(X_0, X_2, X_H) \longmapsto \exp X_0 \exp X_2 \exp X_H \quad (\text{M.3})$$

is a real-analytic diffeomorphism $\mathfrak{n}^0 \times \mathfrak{n}^2 \times \mathfrak{n}_H \rightarrow N_0$. The reversed block multiplication

$$(X_H, X_0, X_2) \longmapsto \exp X_H \exp X_0 \exp X_2 \quad (\text{M.4})$$

is likewise a real-analytic diffeomorphism onto N_0 . In exponential coordinates these maps and their inverses are polynomial. Moreover A_H preserves each of the three vector spaces.

Proof. Order the ambient positive roots first by the value of their restriction on a fixed regular vector in the positive chamber of $\mathfrak{a}_H$, and then by height. If $\bar{\alpha} = 0$ and $\bar{\beta} = 0$, every root occurring in $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta]$ again has zero restriction; hence $\mathfrak{n}^0$ is a subalgebra. If $\bar{\beta} \neq 0$ and $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \neq 0$, then $\bar{\alpha} + \bar{\beta} \neq 0$ because all nonzero restrictions are positive for the chosen ordering. Hence $\mathfrak{n}'$ is an ideal.

We use the lower central filtration of $\mathfrak{n}_0$. Modulo its last nonzero term, the assertion follows by induction on the nilpotency length. The last term is central, and there the product in (M.3) is simply addition of the three corresponding linear components. Lifting the inductive factorization and correcting the central component gives existence and uniqueness. Equivalently, in ordered root coordinates the Baker–Campbell–Hausdorff formula is triangular with unit linear part, so it can be inverted successively by root height. The same triangular argument permits the $\mathfrak{n}_H$ block to be moved to the front: commuting two root factors changes only factors of strictly larger

root height, which are corrected later in the induction. This proves (M.4). Since the BCH series terminates, all the maps just obtained are polynomial. Finally A_H preserves each ambient root space and the chosen complements may be taken A_H -stable, because A_H acts by the scalar character $\bar{\beta}$ on the whole space $\mathfrak{g}_\beta$. $\square$

Lemma M.16 (Bruhat normal form for a fixed ambient translate). *For every $x \in \mathrm{SL}_N(\mathbb{R})$ there are $a_0 \in A_0$, $u_0, v_0 \in N_0$, and a signed permutation matrix $m \in N_{K_0}(A_0)$ such that*

$$x = a_0 u_0 m^{-1} v_0.$$

If $\mathfrak{S}$ is an ambient Siegel set, then $\mathfrak{S}_{a_0 u_0}$ is contained in another ambient Siegel set.

Proof. For the split group SL_N , Bruhat decomposition gives $G_0 = \bigsqcup_w B m_w B$ with $B = A_0 N_0$ after absorbing the finite diagonal part of $K_0 \cap Z_{G_0}(A_0)$ into the signed permutation representative. Thus

$$x = a_1 n_1 m_w a_2 n_2.$$

Since $m_w a_2 m_w^{-1} \in A_0$ and A_0 normalizes N_0 , moving a_2 to the left and changing the N_0 coordinate gives the displayed form. For the second assertion,

$$(kan) a_0 u_0 = k(a a_0)(a_0^{-1} n a_0) u_0.$$

The A_0 -inequalities change only by the fixed numbers $\alpha(\log a_0)$, while $(a_0^{-1} \Omega a_0) u_0$ is compact in N_0 . $\square$

Lemma M.17 (Root-coordinate transfer across a Weyl element). *Let $m \in N_{K_0}(A_0)$, let w be its Weyl element, and let $c > 0$. If $y = kan \in K_0 A_t \Omega$ and*

$$z = y m^{-1} \in K_0 A_c N_0,$$

then the N_0 -component $\nu_0(z)$ stays in a compact set depending only on m, t, c, Ω . Consequently z lies in one ambient Siegel set independent of y .

Proof. We reproduce the root-subsystem argument because this is the part of the fixed-translate lemma that is easy to hide accidentally.

Let

$$E(w) = \{\alpha > 0 : w\alpha < 0\}$$

and let $\Phi(w)$ be the root subsystem generated by $E(w)$. Let $\mathfrak{a}_w$ be the span of the coroots belonging to $\Phi(w)$ and put

$$\mathfrak{a}'_w = \bigcap_{\alpha \in \Phi(w)} \ker \alpha, \quad A_w = \exp \mathfrak{a}_w, \quad A'_w = \exp \mathfrak{a}'_w.$$

Thus $A_0 = A_w A'_w$ with finite intersection. Let $\mathfrak{g}(w)$ be the semisimple Lie algebra generated by the root spaces $\mathfrak{g}_{\pm\alpha}$ for $\alpha \in \Phi(w)$, and put $N_w = N_0 \cap G(w)$. If

$$\mathfrak{n}'_w = \bigoplus_{\substack{\alpha > 0 \\ \alpha \notin \Phi(w)}} \mathfrak{g}_{\alpha},$$

then ordered-root coordinates give an analytic polynomial factorization

$$N_0 = N_w N'_w, \tag{M.5}$$

where $N'_w = \exp(\mathfrak{n}'_w)$ is understood as the corresponding second-kind coordinate slice. Indeed, brackets strictly increase root height; ordering the $\Phi(w)$ roots first makes the BCH equations triangular. Also A'_w centralizes $G(w)$. Finally, if $\alpha > 0$ and $\alpha \notin \Phi(w)$, then $w\alpha > 0$, for otherwise $\alpha \in E(w) \subset \Phi(w)$. Hence conjugation by m carries the N'_w slice into a bounded-coordinate slice in N_0 .

Write

$$a = a_1 a_2 \quad (a_1 \in A_w, a_2 \in A'_w), \quad n = n_1 n_2 \quad (n_1 \in N_w, n_2 \in N'_w).$$

Because Ω is compact and the inverse of (M.5) is continuous, n_1, n_2 range in compact sets. Set

$$u = m(ana^{-1})m^{-1}, \quad b = mam^{-1} = w(a).$$

The element u ranges in a compact set by (M.1). From

$$z = (km^{-1})ub$$

and the Iwasawa decomposition $u = k_u a_u n_u$ we get the exact formula

$$H_0(z) = H_0(u) + w(\log a). \tag{M.6}$$

Both $a \in A_t$ and the assumed A_0 -component of z satisfy uniform upper bounds on all positive roots. If $\beta \in E(w)$, then $-w\beta$ is positive. Applying the upper bound for z to $-w\beta$ in (M.6) gives

$$-\beta(\log a) \leq C,$$

whereas the original Siegel inequality gives $\beta(\log a) \leq C'$. Thus $\beta(\log a)$ is bounded on both sides for all $\beta \in E(w)$. The roots in $E(w)$ span $\mathfrak{a}_w^*$: the reflections in these roots generate the root subsystem $\Phi(w)$, and a vector in $\mathfrak{a}_w$ annihilated by all of them is orthogonal to every coroot spanning $\mathfrak{a}_w$. Hence a_1 ranges in a compact subset of A_w .

Since a_2 centralizes $G(w)$, it commutes with n_1 . We may therefore rewrite

$$\begin{aligned} z &= k a_1 n_1 a_2 n_2 m^{-1} \\ &= (k m^{-1}) (m a_1 n_1 m^{-1}) w(a_2) (m n_2 m^{-1}). \end{aligned}$$

The middle group element $g_1 = m a_1 n_1 m^{-1}$ belongs to $G(w^{-1})$ and ranges in a compact set. Take its Iwasawa decomposition inside that semisimple subgroup, $g_1 = k_3 a_3 n_3$. The factors a_3, n_3 are bounded because Iwasawa projection is continuous on compact sets. The torus $w(A'_w)$ centralizes $G(w^{-1})$, so $w(a_2)$ commutes with n_3 . Moreover $n'_2 = m n_2 m^{-1}$ is bounded and lies in N_0 by the positivity observation above. Therefore

$$z = (K_0 \text{ factor}) a_3 w(a_2) n_3 n'_2$$

is already an ambient Iwasawa decomposition. Its N_0 factor $n_3 n'_2$ ranges in a compact set. The hypothesis $z \in K_0 A_c N_0$ supplies the A_0 inequalities, so all such z lie in one ambient Siegel set. $\square$

Lemma M.18 (Fixed-translate stability of an ambient Siegel set). *Let $\mathfrak{S}$ be an ambient Siegel set, $x \in G_0$, and $c > 0$. Then*

$$\mathfrak{S}x \cap K_0 A_c N_0$$

is contained in an ambient Siegel set.

Proof. Use Lemma M.16 to write $x = a_0 u_0 m^{-1} v_0$. Replace $\mathfrak{S}$ by a larger Siegel set containing $\mathfrak{S} a_0 u_0$. Right multiplication by the fixed $v_0 \in N_0$ merely replaces a compact N_0 coordinate by another compact set. We are therefore reduced to $x = m^{-1}$, and the result is exactly Lemma M.17. $\square$

Lemma M.19 (Subgroup Siegel-intersection lemma). *Let $H < G_0$ be connected semisimple, θ -stable, and equipped with the compatible Iwasawa data of Lemma M.2. Let $\mathfrak{S} = K_0 A_t \Omega$ be an ambient Siegel set. For every fixed $x \in G_0$ there are an H -Siegel set $\mathfrak{S}_H = K_H A_{H,t_H} \Omega_H$ and finitely many $w_1, \dots, w_r \in N_{K_H}(A_H)$ such that*

$$\mathfrak{S}x \cap H \subset \bigcup_{j=1}^r \mathfrak{S}_H w_j.$$

Proof. We keep every place where boundedness enters explicit.

Step 1: Bruhat reduction of the fixed translate. By Lemma M.16, after enlarging the ambient Siegel set we may write $x = m^{-1}v$ with $m \in N_{K_0}(A_0)$ and $v \in N_0$. Apply the factorization (M.3) to v and write $v = v_0v_2v_H$. Right multiplication by $v_H \in N_H \subset H$ only changes the final H -translate, so it is enough to treat

$$x = m^{-1}v, \quad v \in \exp(\mathfrak{n}^0)\exp(\mathfrak{n}^2).$$

Step 2: the exact bounded-unipotent identity. Take $y = kan \in \mathfrak{S}$ and suppose $z = yx \in H$. Write, using the compatible Iwasawa decomposition,

$$z = k_H a_H n_H.$$

Set

$$b = mam^{-1}, \quad u = m(ana^{-1})m^{-1}.$$

Again u ranges in a fixed compact set. Writing $u = k_u a_u n_u$ in ambient Iwasawa coordinates and using

$$z = kanm^{-1}v = (km^{-1})ubv$$

gives, by uniqueness of ambient Iwasawa coordinates,

$$a_H = a_u b, \quad n_H v^{-1} = b^{-1} n_u b. \quad (\text{M.7})$$

Consequently

$$a_H(n_H v^{-1})a_H^{-1} = a_u n_u a_u^{-1}, \quad (\text{M.8})$$

which ranges in a compact subset of N_0 . Now

$$a_H(n_H v^{-1})a_H^{-1} = (a_H n_H a_H^{-1})(a_H v^{-1} a_H^{-1}).$$

The first factor belongs to N_H and the second belongs to the $\exp(\mathfrak{n}^0)\exp(\mathfrak{n}^2)$ slice, because $\mathfrak{n}^0 \oplus \mathfrak{n}^2$ is A_H -stable. Apply the inverse of the reversed factorization (M.4). Its continuity shows that compactness of the product in (M.8) implies separately that

$$\{a_H n_H a_H^{-1}\} \quad (\text{M.9})$$

is relatively compact in N_H .

Step 3: choose one of finitely many H -Weyl chambers. Let $W_H = N_{K_H}(A_H)/Z_{K_H}(A_H)$ and choose representatives $w_1, \dots, w_r$. For each a_H choose w_j so that

$$c_H = w_j a_H w_j^{-1}$$

lies in the closed negative H -Weyl chamber. Thus every positive restricted root β of H satisfies $\beta(\log c_H) \leq 0$. By compatibility, every ambient positive root restricts either to zero or to a positive restricted root of H ; hence $c_H \in A_c$ for one fixed ambient threshold c .

Put $q_H = a_H n_H a_H^{-1}$, which is bounded by (M.9). Since

$$zw_j^{-1} = k_H q_H w_j^{-1} c_H,$$

the left factor $k_H q_H w_j^{-1}$ ranges in a compact subset of G_0 . Its ambient Iwasawa factors are therefore bounded. More explicitly, if $q = k_q a_q n_q$ ranges in that compact set and $c_H \in A_0$, then

$$q c_H = k_q (a_q c_H) (c_H^{-1} n_q c_H),$$

so the Iwasawa A_0 -projection is exactly $H_0(q) + \log c_H$. The first term is uniformly bounded. Enlarging c once, we obtain

$$zw_j^{-1} \in K_0 A_c N_0. \tag{M.10}$$

Step 4: return to a genuine H -Siegel set. Because $z = yx$, equation (M.10) says

$$zw_j^{-1} \in \mathfrak{S}(xw_j^{-1}) \cap K_0 A_c N_0.$$

Lemma M.18 puts this set in an ambient Siegel set $\mathfrak{S}'_j = K_0 A_{t_j} \Omega_j$. But $zw_j^{-1} \in H$. Compatibility and uniqueness of Iwasawa decomposition give

$$\mathfrak{S}'_j \cap H \subset K_H(A_{t_j} \cap A_H)(\Omega_j \cap N_H).$$

The last N_H factor is compact. Each simple restricted root of H occurs as the nonzero restriction of an ambient positive root, so the ambient A_{t_j} inequalities imply the defining upper inequalities of an H -Siegel cone after changing the threshold by a fixed constant. Thus $\mathfrak{S}'_j \cap H$ is contained in an H -Siegel set. There are only finitely many j ; enlarge these finitely many sets to one $\mathfrak{S}_H$. Then $zw_j^{-1} \in \mathfrak{S}_H$, or $z \in \mathfrak{S}_H w_j$, as required. $\square$

Remark M.20. The proof now displays the two identities on which the subgroup comparison actually turns: (M.7) converts ambient cusp control into the bounded conjugated N_H coordinate, while Lemma M.18 controls the effect of the finitely many Weyl corrections. No finiteness of subgroup types, no measure-zero argument, and no unproved assertion that an arbitrary subgroup intersection inherits ambient volume is used.

M.8 Exact Haar integration on a semisimple Siegel set

Lemma M.21 (Haar density). *Let $H = K_H A_H N_H$ be an Iwasawa decomposition of a connected semisimple real Lie group. Normalize Haar measures on the three factors. In right Iwasawa coordinates,*

$$dh = C e^{-2\rho_H(\log a)} dk da dn,$$

where

$$2\rho_H = \sum_{\beta > 0} (\dim \mathfrak{h}_\beta) \beta.$$

If $\Delta_H = \{\alpha_1, \dots, \alpha_r\}$ is the set of simple restricted roots, then

$$2\rho_H = \sum_{i=1}^r c_i \alpha_i, \quad c_i > 0.$$

Proof. Conjugation by $a \in A_H$ acts on the root space $\mathfrak{h}_\beta$ with determinant $e^{\beta(\log a) \dim \mathfrak{h}_\beta}$. Thus its determinant on $\mathfrak{n}_H$ is $e^{2\rho_H(\log a)}$. Right translation in the KAN product consequently introduces the reciprocal modular factor, giving the first formula.

Every positive restricted root is a nonnegative integral combination of simple roots, and every simple root itself occurs among the positive roots with positive multiplicity. Therefore collecting coefficients in the simple-root basis gives $c_i > 0$ for every i . $\square$

Proposition M.22 (Finite Haar volume of a semisimple Siegel set). *Every Siegel set $K_H A_{H,t} \Omega_H$ in a connected semisimple real Lie group has finite Haar measure.*

Proof. The compact factors K_H and Ω_H have finite measure. Put $s_i = \alpha_i(\log a)$. Since the simple restricted roots form a basis of $\mathfrak{a}_H^*$, da is a nonzero constant multiple of $ds_1 \cdots ds_r$. On the chosen cusp cone $s_i \geq s_0$ for all i . Lemma M.21 gives

$$\int_{A_{H,t}} e^{-2\rho_H(\log a)} da = C' \int_{[s_0, \infty)^r} e^{-\sum_i c_i s_i} ds_1 \cdots ds_r = C' \prod_{i=1}^r \frac{e^{-c_i s_0}}{c_i} < \infty.$$

$\square$

M.9 Complete finite-covolume theorem

Theorem M.23 (Borel–Harish–Chandra finite-covolume theorem). *Let $\mathbf{H}$ be a connected semisimple algebraic group defined over $\mathbb{Q}$, and let $\Gamma < \mathbf{H}(\mathbb{Q})$ be arithmetic. Then Γ is a lattice in $H = \mathbf{H}(\mathbb{R})^\circ$.*

Proof. By replacing the faithful rational embedding and passing to a commensurable subgroup, Corollary M.14 reduces us to $\mathbf{H} < \mathrm{SL}_N$ and $\Gamma = H \cap \mathrm{SL}_N(\mathbb{Z})$. Theorem M.13 gives finitely many sets

$$F_i = c^{-1}\mathfrak{S}b_i \cap H$$

whose Γ -translates cover H and which have finite overlap. Conjugate by c . Then

$$cF_ic^{-1} = \mathfrak{S}b_ic^{-1} \cap (cHc^{-1}).$$

In Theorem M.13 choose the conjugating matrix in two stages: first use Lemma M.1, then apply the k and m from Lemma M.2. Thus the final $c = mkc_0$ makes $H' = cHc^{-1}$ simultaneously θ -stable and compatible with the fixed ambient A_0, N_0 . Since $k, m \in K_0$, the self-adjoint property is preserved; no change of the arithmetic ambient Siegel theorem is required.

Apply Lemma M.19 with the fixed right translate $x = b_ic^{-1}$. Each cF_ic^{-1} is contained in a finite union of H' -Siegel sets. Proposition M.22 gives finite Haar measure for every such set. Thus $F = \bigcup_i F_i$ has finite Haar measure. The quotient $\Gamma \backslash H$ is the image of F , so its Haar volume is at most $\mathrm{vol}(F) < \infty$. It has positive volume because Γ is discrete and a sufficiently small identity neighborhood injects into the quotient. Hence Γ is a lattice. $\square$

Finite-covolume part of Theorem 8.1. The finite-covolume assertion is exactly Theorem M.23. Notice that this proof does not use Borel density, Mozes–Shah, Ratner classification, or any later volume. Its only dynamical-looking ingredient is the elementary properness of a closed linear orbit modulo its stabilizer. $\square$

M.10 Godement compactness after finite covolume

A direct rank-zero compactness argument might try to produce a fixed rational contracting flag immediately from an escaping sequence. That is unnecessary. Once finite covolume is available independently, the adjoint

representation and Mahler compactness give a shorter proof in which every arithmetic step is visible.

Lemma M.24 (A rational nilpotent produces a rational parabolic). *Let $\mathbf{G}$ be connected semisimple over $\mathbb{Q}$. If $0 \neq X \in \mathfrak{g}(\mathbb{Q})$ is nilpotent, then $\mathbf{G}$ has a proper parabolic subgroup defined over $\mathbb{Q}$. Consequently, if $\text{rank}_{\mathbb{Q}} \mathbf{G} = 0$, the rational Lie algebra $\mathfrak{g}(\mathbb{Q})$ contains no nonzero nilpotent element.*

Proof. Corollary 1.10 gives a $\mathbb{Q}$ -homomorphism

$$\phi : \text{SL}_2 \longrightarrow \mathbf{G}$$

whose differential sends $e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ to X . Hence

$$\lambda(t) = \phi \left(\begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} \right) \quad (t \in \mathbf{G}_m)$$

is a nontrivial $\mathbb{Q}$ -cocharacter. By Theorem 1.11, $P(\lambda)$ is a proper $\mathbb{Q}$ -parabolic. More explicitly, if $H = d\phi \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, then

$$\text{Lie } P(\lambda) = \bigoplus_{j \geq 0} \mathfrak{g}_j(H),$$

and the element $d\phi \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ has weight -2 , so it is visibly excluded.

If $\text{rank}_{\mathbb{Q}} \mathbf{G} = 0$, no nontrivial $\mathbb{Q}$ -cocharacter exists by the definition of rational rank. The preceding construction would produce one from any nonzero rational nilpotent, so such a nilpotent cannot exist. $\square$

Lemma M.25 (Integral invariant-polynomial lower bound). *Assume $\text{rank}_{\mathbb{Q}} \mathbf{G} = 0$. There are a full lattice $L \subset \mathfrak{g}(\mathbb{Q})$ and $\varepsilon > 0$ such that, for every $g \in G$ and every $0 \neq X \in L$,*

$$\| \text{Ad}(g)X \| \geq \varepsilon.$$

Proof. Choose a rational basis of $\mathfrak{g}$ and write

$$\det(TI - \text{ad } X) = T^d + P_1(X)T^{d-1} + \cdots + P_d(X).$$

Each P_j is a homogeneous polynomial with rational coefficients and is G -invariant, because conjugating X conjugates $\text{ad } X$. Let D be a common positive denominator of all coefficients of the finitely many polynomials P_j . After replacing the coordinate lattice by the sublattice $D\mathbb{Z}^d$ in the

chosen rational basis, each P_j takes integer values on the resulting lattice L . Shrinking to a commensurable lattice if necessary does not affect arithmeticity.

If all $P_j(X)$ vanish, then $\text{ad } X$ is nilpotent; for a semisimple Lie algebra this says exactly that X is nilpotent. Lemma M.24 therefore implies that for every $0 \neq X \in L$ at least one integer $P_j(X)$ is nonzero, so

$$\max_j |P_j(X)| \geq 1. \quad (\text{M.11})$$

On the unit ball there is a constant C such that $|P_j(Y)| \leq C\|Y\|^{\deg P_j}$ for every j . Choose $\varepsilon > 0$ so small that $\|Y\| < \varepsilon$ forces $|P_j(Y)| < 1$ for every j . Since the P_j are invariant, (M.11) applies to $Y = \text{Ad}(g)X$ and excludes $\|\text{Ad}(g)X\| < \varepsilon$. $\square$

Theorem M.26 (Godement compactness in the semisimple case). *Let $\mathbf{G}$ be connected semisimple over $\mathbb{Q}$ and let Γ be arithmetic. Then*

$$\Gamma \backslash G \text{ is compact} \iff \mathbf{G} \text{ has no proper } \mathbb{Q}\text{-parabolic} \iff \text{rank}_{\mathbb{Q}} \mathbf{G} = 0.$$

Proof. We first prove compactness in rank zero. Pass to the adjoint image $G^{\text{ad}} = \text{Ad}(G) < \text{SL}(\mathfrak{g}_{\mathbb{R}})$ and to a finite-index arithmetic subgroup preserving the lattice L of Lemma M.25. Theorem M.23, applied to the rational semisimple group $\text{Ad}(\mathbf{G})$, says that

$$\Gamma_{\text{ad}} = G^{\text{ad}} \cap \text{SL}(L)$$

is a lattice in G^{ad} . By the finite-covolume stabilizer criterion proved in Volume III (Theorem A.4.2 there), its orbit in the ambient lattice space

$$\text{SL}(L) \backslash \text{SL}(\mathfrak{g}_{\mathbb{R}})$$

is closed.

Lemma M.25 gives a uniform positive lower bound for the length of every nonzero vector in every lattice $\text{Ad}(g)L$. Mahler's compactness criterion therefore places the entire closed G^{ad} -orbit inside one compact subset of the ambient lattice space. A closed subset of a compact set is compact. Thus $\Gamma_{\text{ad}} \backslash G^{\text{ad}}$ is compact. The adjoint map has finite kernel and arithmetic subgroups change only by finite index under this rational isogeny, so $\Gamma \backslash G$ is compact as well.

Suppose now that $\text{rank}_{\mathbb{Q}} \mathbf{G} > 0$. Choose a nontrivial $\mathbb{Q}$ -cocharacter λ in a positive-dimensional $\mathbb{Q}$ -split torus and put $\mathbf{P} = P(\lambda)$. By Theorem 1.11,

this is a proper $\mathbb{Q}$ -parabolic. Put $\mathfrak{p} = \text{Lie } \mathbf{P}$, let $r = \dim \mathfrak{p}$, and consider the rational representation

$$\pi = \bigwedge^r \text{Ad} : \mathbf{G} \longrightarrow \text{GL} \left(\bigwedge^r \mathfrak{g} \right).$$

Choose a nonzero rational wedge $v \in \bigwedge^r \mathfrak{p}(\mathbb{Q})$. If $\mathfrak{g} = \bigoplus_j \mathfrak{g}_j$ is the λ -weight decomposition, then $\mathfrak{p} = \bigoplus_{j \geq 0} \mathfrak{g}_j$. The Killing form pairs $\mathfrak{g}_j$ nondegenerately with $\mathfrak{g}_{-j}$, so $\dim \mathfrak{g}_j = \dim \mathfrak{g}_{-j}$. Since $\mathbf{P}$ is proper, some nonzero weight occurs, and therefore

$$c := \sum_{j \geq 0} j \dim \mathfrak{g}_j > 0, \quad \pi(\lambda(t))v = t^c v.$$

By Lemma M.10, after a harmless finite-index replacement of Γ there is a $\pi(\Gamma)$ -stable lattice Λ in $\bigwedge^r \mathfrak{g}(\mathbb{Q})$. Scale v so that $v \in \Lambda$. For $a_t = \lambda(t)$ with $t \downarrow 0$, the lattice $\pi(a_t)\Lambda$ contains the nonzero vector $t^c v$, whose norm tends to zero. The map

$$\Gamma \backslash G \longrightarrow \pi(\Gamma) \backslash \text{GL} \left(\bigwedge^r \mathfrak{g}_{\mathbb{R}} \right), \quad \Gamma g \longmapsto \pi(\Gamma)\pi(g),$$

may equivalently be viewed as the continuous family of lattices $\pi(g)\Lambda$. On the image of a compact subset, the shortest-vector function has a positive minimum. The shrinking vector therefore shows that Γa_t leaves every compact subset of $\Gamma \backslash G$; the quotient is noncompact.

It remains only to identify positive rational rank with the existence of a proper rational parabolic. If the rank is positive, the preceding choice of a nontrivial rational cocharacter and Theorem 1.11 produce a proper $\mathbb{Q}$ -parabolic. Conversely, if a proper $\mathbb{Q}$ -parabolic exists, Lemma 1.12 gives a nonzero $X \in \mathfrak{g}(\mathbb{Q})$ that is nilpotent. Corollary 1.10 then produces a nontrivial rational cocharacter of $\mathbf{G}$, so $\text{rank}_{\mathbb{Q}} \mathbf{G} > 0$. Thus the two conditions are equivalent. $\square$

Proof of Lemma 7.1. The lemma is the rank-zero implication in Theorem M.26. Notice that the proof does not use the “shortest wedge becomes a fixed rational line” step from the shorter direct argument. $\square$

Compactness part of Theorem 8.1. Combine Theorem M.23 with Theorem M.26. $\square$

M.11 Scope of this proof

The finite-covolume assertion in Theorem 8.1 is now owned by the chain

self-adjoint position $\Rightarrow$ closed Chevalley orbit $\Rightarrow$ Siegel finiteness
 $\Rightarrow$ finite fundamental set $\Rightarrow$ subgroup Siegel intersection
 $\Rightarrow$ finite Haar volume.

In particular, the proof no longer depends on the earlier phrases “a fixed rational flag appears”, “there are finitely many cusp flags”, or “the Levi induction gives finite volume” without a separate argument.

The compactness characterization in the final sentence of Theorem 8.1 is now certified independently by Theorem M.26. Its proof passes through the adjoint arithmetic lattice, invariant-polynomial control of the adjoint systole, the closed-orbit criterion from Volume III, and Mahler compactness. Thus neither finite covolume nor compactness relies on the earlier unsupported phrase that an escaping sequence automatically produces one fixed rational contracting line.

Theorem 7.2 is stronger than finite covolume: it asks for finitely many $\mathbb{Q}$ -rational cusp representatives together with a precise $M_P A_P N_P$ -Siegel-cone reduction. The finite-covolume and compactness arguments above do not by themselves supply that additional structure. The separate rational-cusp module in Chapter N proves the required passage from rational flags to finitely many rational parabolic classes, the arithmetic reduction of the Levi factor, and the conversion of height bounds into the stated cone inequalities.

M.12 Student proof map

A student checking this chapter should be able to answer the following.

1. Why is the Chevalley line theorem alone insufficient for the finiteness lemma, and how does the affine quotient construction produce a rational vector with a genuinely closed ambient orbit?
2. Where exactly is integrality used in the finiteness lemma? Answer: in the uniform lower bound for nonzero rational weight projections.
3. Why does the arbitrary real conjugating matrix c not destroy the finite-overlap argument? Answer: it cancels before the ambient integral matrix is tested against $\mathfrak{S}^{-1}\mathfrak{S}$.
4. Why is an ambient finite-volume Siegel set not by itself enough to prove

- finite volume in H ? Answer: intersections with a lower-dimensional subgroup are controlled only after the root-coordinate intersection lemma.
5. Which factor makes the cusp integral converge? Answer: every simple restricted root occurs with a strictly positive coefficient in $2\rho_H$.
 6. Why does $\mathbb{Q}$ -rank zero imply compactness after finite covolume has been proved? Answer: invariant characteristic-polynomial coefficients give a uniform adjoint-lattice systole; the closed-orbit criterion and Mahler compactness then force the whole adjoint orbit into a compact set.
 7. Which result requires a separate argument from finite covolume? Answer: Theorem 7.2, because an explicit finite rational-cusp reduction contains information not supplied merely by the lattice theorem; Chapter N gives that argument.

Appendix N

Complete proof: Rational Cusps and Intrinsic Siegel Reduction

N.1 Why a second reduction argument is necessary

the finite-covolume proof module proved finite covolume without using the strong reduction statement of Theorem 7.2. That was logically useful, but at that stage it left one stronger assertion to be proved: a fundamental set must be expressible by *finitely many rational cusp representatives* and by the intrinsic $\mathbb{Q}$ -root cone of a fixed minimal rational parabolic. An ambient SL_N -Siegel set alone does not contain this arithmetic information.

We now prove exactly that stronger statement. The proof is organized so that three logically distinct issues are treated separately:

1. the M - and U -coordinates of a rational parabolic are reduced by arithmetic subgroups, with M compact modulo arithmetic because it is $\mathbb{Q}$ -anisotropic and U compact modulo arithmetic because it is unipotent;
2. an ambient rational Bruhat decomposition is converted into the intrinsic relative-root decomposition, and the only terminal chamber changes are made by representatives of the *rational* relative Weyl group;
3. the root inequalities are transferred uniformly to the intrinsic A -coordinate. This is the step that turns a finite collection of ambient pieces into finitely many rational cusp pieces.

Throughout this chapter, $\mathbf{G}$ is connected semisimple over $\mathbb{Q}$, $G = \mathbf{G}(\mathbb{R})^\circ$,

and

$$\mathbf{P} = \mathbf{MSU}$$

is a fixed minimal $\mathbb{Q}$ -parabolic. We write

$$A = \mathbf{S}(\mathbb{R})^\circ, \quad U = \mathbf{U}(\mathbb{R}),$$

and choose the positive relative roots so that the cusp cone used in Theorem 7.2 is

$$A_t = \{a \in A : \alpha(a) \geq t \text{ for every simple relative root } \alpha\}. \quad (\text{N.1})$$

Changing all positive roots to their negatives converts this convention to the equally standard convention in which the inequalities are upper bounds.

N.2 The anisotropic Levi and the unipotent radical

Lemma N.1 (The complementary Levi is $\mathbb{Q}$ -anisotropic). *Let $\mathbf{Z} = Z_{\mathbf{G}}(\mathbf{S})^\circ$. There is a connected reductive $\mathbb{Q}$ -subgroup $\mathbf{M} < \mathbf{Z}$ such that the multiplication map*

$$\mathbf{M} \times \mathbf{S} \longrightarrow \mathbf{Z}$$

is an isogeny. Moreover $\text{rank}_{\mathbb{Q}} \mathbf{M} = 0$.

Proof. The connected center of $\mathbf{Z}$ contains $\mathbf{S}$, and its maximal $\mathbb{Q}$ -split subtorus is $\mathbf{S}$ because $\mathbf{S}$ was already maximal $\mathbb{Q}$ -split in $\mathbf{G}$. Let $\mathbf{M}$ be the connected subgroup generated by the derived group of $\mathbf{Z}$ and the maximal $\mathbb{Q}$ -anisotropic subtorus of $Z(\mathbf{Z})^\circ$. The two factors commute with $\mathbf{S}$, and multiplication from $\mathbf{MS}$ onto $\mathbf{Z}$ has finite central kernel, giving the isogeny.

If $\mathbf{T} < \mathbf{M}$ were a nontrivial $\mathbb{Q}$ -split torus, then $\mathbf{TS}$ would be a $\mathbb{Q}$ -split torus in $\mathbf{G}$ strictly larger than $\mathbf{S}$ (the intersection is finite because the displayed multiplication map has finite kernel). This contradicts maximality of $\mathbf{S}$. Hence $\text{rank}_{\mathbb{Q}} \mathbf{M} = 0$. $\square$

Lemma N.2 (Number-field lattice facts used in the unit argument). *Let $E/\mathbb{Q}$ be a number field of degree n and let $\mathcal{O}_E$ be its ring of integers. Under the Minkowski embedding*

$$\sigma : E \longrightarrow \mathbb{R}^{r_1} \times \mathbb{C}^{r_2} \cong \mathbb{R}^n,$$

$\sigma(\mathcal{O}_E)$ is a full lattice. If $u \in \mathcal{O}_E^\times$, then

$$\prod_{v \text{ real}} |u|_v \prod_{v \text{ complex}} |u|_v^2 = 1.$$

Moreover, for every B there are only finitely many integral ideals of $\mathcal{O}_E$ with norm at most B .

Proof. Choose a $\mathbb{Q}$ -basis of E and multiply its vectors by nonzero integers so that $\beta_1, \dots, \beta_n \in \mathcal{O}_E$. The trace pairing $\text{Tr}_{E/\mathbb{Q}}(xy)$ is nondegenerate because $E/\mathbb{Q}$ is separable. If $L = \sum_i \mathbb{Z}\beta_i$, then every $x \in \mathcal{O}_E$ lies in the trace-dual lattice

$$L^* = \{y \in E : \text{Tr}(yL) \subset \mathbb{Z}\},$$

because $x\beta_i$ is an algebraic integer and therefore has integral trace. Thus $L \subset \mathcal{O}_E \subset L^*$. Both outer groups are free abelian of rank n , so $\mathcal{O}_E$ is also free of rank n and its image under the real-linear Minkowski embedding is discrete and cocompact: it is a full lattice.

For a unit u , multiplication by u is an automorphism of the free $\mathbb{Z}$ -module $\mathcal{O}_E$, so its determinant is ± 1 . The same determinant over $\mathbb{Q}$ is $N_{E/\mathbb{Q}}(u)$, while the embedding formula for the norm is the displayed weighted product. Taking absolute values gives the product identity.

Finally an integral ideal I of norm at most B is a subgroup of the free abelian group $\mathcal{O}_E \cong \mathbb{Z}^n$ of index at most B . Hermite normal form gives only finitely many subgroups of $\mathbb{Z}^n$ of bounded index (the diagonal entries have bounded product and the entries above them may be chosen in bounded residue ranges). Hence only finitely many such ideals occur. $\square$

Theorem N.3 (Dirichlet's unit theorem in logarithmic form). *Let E be a number field with r_1 real places and r_2 pairs of complex places. Put*

$$H_E = \left\{ (x_v)_{v|\infty} \in \mathbb{R}^{r_1+r_2} : \sum_{v \text{ real}} x_v + 2 \sum_{v \text{ complex}} x_v = 0 \right\}.$$

For $u \in \mathcal{O}_E^\times$ put

$$\ell_E(u) = (\log |u|_v)_{v|\infty}.$$

Then $\ker \ell_E = \mu(E)$ is finite and $\ell_E(\mathcal{O}_E^\times)$ is a full lattice in H_E .

Proof. Lemma N.2 puts the image in H_E . If $\ell_E(u) = 0$, every archimedean conjugate of u has absolute value one. The coefficients of the minimal polynomial of u are then bounded integers of bounded degree; hence only finitely many such u exist. A finite subgroup of the multiplicative group of a field is cyclic, so this kernel is precisely the roots of unity.

The image is discrete. Indeed, if $\ell_E(u)$ lies in a fixed bounded set, then all conjugates of the algebraic integer u are bounded. The elementary symmetric

functions of those conjugates are therefore bounded integers, so there are only finitely many possible monic minimal polynomials and hence only finitely many possible units.

It remains to prove cocompactness. Use the Minkowski embedding from Lemma N.2, under which $\mathcal{O}_E$ is a full lattice. For $x = (x_v) \in H_E$, choose one fixed $C > 1$ so large that the centrally symmetric box

$$B_x = \{z : |z|_v \leq Ce^{x_v} \text{ for every } v \mid \infty\}$$

has volume larger than $2^{[E:\mathbb{Q}]} \text{covol}(\mathcal{O}_E)$. Its volume is independent of x up to the fixed factor $C^{[E:\mathbb{Q}]}$, because the weighted sum of the x_v is zero. Minkowski's convex-body theorem therefore supplies $0 \neq a_x \in \mathcal{O}_E \cap B_x$.

The norm satisfies

$$1 \leq |N_{E/\mathbb{Q}}(a_x)| \leq C_0$$

for a constant C_0 independent of x . Hence the principal ideals (a_x) belong to the finite set of integral ideals of norm at most C_0 , by Lemma N.2. Choose once and for all a generator a_j for each principal ideal occurring in this finite set. We may write $a_x = u_x a_j$ with $u_x \in \mathcal{O}_E^\times$. The inequalities defining B_x give an upper bound for every coordinate of $\ell_E(a_x) - x$; the weighted sum of these coordinates equals $\log |N(a_x)|$, which is uniformly bounded. Consequently all coordinates are bounded above and below uniformly. Subtracting the finitely many vectors $\ell_E(a_j)$ shows that every $x \in H_E$ lies within a fixed bounded distance of $\ell_E(\mathcal{O}_E^\times)$. A discrete subgroup with bounded fundamental-neighborhood quotient is a full lattice. This proves the theorem. $\square$

Lemma N.4 (Arithmetic compactness for an anisotropic rational torus). *Let $\mathbf{T}$ be a connected $\mathbb{Q}$ -torus with $X^*(\mathbf{T})_{\mathbb{Q}} = 0$, equivalently with no nontrivial $\mathbb{Q}$ -character. If $\Gamma_T < \mathbf{T}(\mathbb{Q})$ is arithmetic, then*

$$\Gamma_T \backslash \mathbf{T}(\mathbb{R})^\circ$$

is compact.

Proof. Choose a finite Galois extension $E/\mathbb{Q}$ splitting $\mathbf{T}$, with Galois group Δ , and let $X = X^*(\mathbf{T}_E)$. Character duality identifies

$$\mathbf{T}(\mathbb{Q}) = \text{Hom}_\Delta(X, E^\times).$$

After replacing the chosen integral model and Γ_T by commensurable ones, its integral points are

$$\Gamma_T = \text{Hom}_\Delta(X, \mathcal{O}_E^\times) \tag{N.2}$$

up to finite index. Here is the elementary justification for the harmless “up to finite index.” Choose a finite Δ -stable set of characters generating X and use it to embed $\mathbf{T}_E$ diagonally in a power of $\mathbf{G}_m$. The equations of the image are monomial equations with integral exponents. Requiring the character values and their inverses to be integral gives exactly the right side of (N.2); changing a rational lattice or clearing the finitely many denominators in the embedding changes the stabilizer only by a finite-index congruence condition.

Let H_E be the logarithmic space of Theorem N.3. For $t \in \mathbf{T}(\mathbb{R})^\circ$, evaluate all E -characters at the components of t in $E \otimes_{\mathbb{Q}} \mathbb{R}$ and take archimedean absolute values. This gives the continuous logarithm map

$$L_T : \mathbf{T}(\mathbb{R})^\circ \longrightarrow \mathrm{Hom}_\Delta(X, H_E).$$

Why does the image land in H_E ? For $\chi \in X$, the weighted sum of the archimedean logarithms is the logarithm of the absolute value of the norm character

$$N\chi = \sum_{\delta \in \Delta} \delta\chi \in X^\Delta.$$

Anisotropy says $X^\Delta = 0$, so $N\chi = 0$ and that weighted sum vanishes.

The kernel of L_T is the maximal compact subgroup K_T of $\mathbf{T}(\mathbb{R})^\circ$: in a splitting representation, all character absolute values equal one exactly on the compact part. Differentiating the character map shows that

$$\mathbf{T}(\mathbb{R})^\circ / K_T \xrightarrow{\sim} \mathrm{Hom}_\Delta(X, H_E) \quad (\text{N.3})$$

as real vector groups. Concretely, after splitting over E this is the usual decomposition of a real torus into compact circle factors and positive real factors; the Δ -equivariance simply records descent to $\mathbb{Q}$.

By Theorem N.3, the logarithm induces a Δ -equivariant real-linear isomorphism

$$\mathcal{O}_E^\times \otimes_{\mathbb{Z}} \mathbb{R} \xrightarrow{\sim} H_E.$$

Because Δ is finite, taking Δ -equivariant homomorphisms commutes with tensoring by $\mathbb{R}$ (average any linear map over Δ). Hence

$$\mathrm{Hom}_\Delta(X, \mathcal{O}_E^\times) \otimes_{\mathbb{Z}} \mathbb{R} \xrightarrow{\sim} \mathrm{Hom}_\Delta(X, H_E).$$

Thus $L_T(\Gamma_T)$ is a full lattice in the vector group on the right of (N.3). Its quotient is compact, and the kernel K_T is compact. Therefore $\Gamma_T \backslash \mathbf{T}(\mathbb{R})^\circ$ is compact. Commensurable arithmetic subgroups have finite-index common subgroups, so the same conclusion holds for every arithmetic Γ_T . $\square$

Proposition N.5 (Godement compactness for anisotropic reductive groups). *Let $\mathbf{L}$ be connected reductive over $\mathbb{Q}$ and assume $\text{rank}_{\mathbb{Q}} \mathbf{L} = 0$. Then every arithmetic subgroup $\Lambda < \mathbf{L}(\mathbb{Q})$ is cocompact in $L = \mathbf{L}(\mathbb{R})^{\circ}$.*

Proof. Let $\mathbf{D} = [\mathbf{L}, \mathbf{L}]$ and $\mathbf{Z} = Z(\mathbf{L})^{\circ}$. Then $\mathbf{D}$ is semisimple of $\mathbb{Q}$ -rank zero, $\mathbf{Z}$ is a $\mathbb{Q}$ -anisotropic torus, and multiplication

$$\mathbf{Z} \times \mathbf{D} \longrightarrow \mathbf{L}$$

is a central $\mathbb{Q}$ -isogeny. By Theorem M.26, arithmetic quotients of D are compact, and by Lemma N.4 the same is true for Z .

We record the elementary arithmetic-isogeny step rather than hide it. For a central $\mathbb{Q}$ -isogeny $f : \mathbf{L}_1 \rightarrow \mathbf{L}_2$, choose rational faithful matrix realizations. The matrix entries of f and of a local algebraic inverse are rational regular functions. Clearing their finitely many denominators shows that a sufficiently deep principal congruence subgroup of an integral arithmetic subgroup of $\mathbf{L}_1$ maps into an integral arithmetic subgroup of $\mathbf{L}_2$, and the same argument after interchanging the two groups shows that the image is commensurable with the target arithmetic subgroup. On real identity components, a central isogeny is a surjective Lie-group homomorphism with finite kernel. Therefore the image in L of a compact fundamental set for the product of arithmetic subgroups in $Z \times D$ is compact and covers modulo a subgroup commensurable with Λ . Passing between commensurable subgroups changes the quotient by a finite cover. Hence $\Lambda \backslash L$ is compact. $\square$

Lemma N.6 (Compact arithmetic reduction of the MU -factor). *Let $\Gamma^* < \Gamma$ be any finite-index arithmetic subgroup. After replacing Γ^* by a further finite-index subgroup, there is a compact set $\Omega_{MU} \subset MU$ such that*

$$MU = \Omega_{MU} (\Gamma^* \cap MU).$$

The same assertion holds with left and right interchanged.

Proof. By Lemma N.1 and Proposition N.5, the arithmetic quotient

$$(\Gamma^* \cap M) \backslash M$$

is compact. Choose a compact set $\Omega_M \subset M$ with $M = \Omega_M (\Gamma^* \cap M)$.

The group $\mathbf{U}$ is rational unipotent. By Theorem B.2, $\Gamma^* \cap U$ is a uniform lattice in U after another finite-index change. Thus there is a compact $\Omega_U \subset U$ such that $U = \Omega_U (\Gamma^* \cap U)$.

Because M normalizes U , if $m = \omega_M \gamma_M$ and $u \in U$, then

$$mu = \omega_M(\gamma_M u \gamma_M^{-1}) \gamma_M.$$

Reduce $\gamma_M u \gamma_M^{-1}$ as $\omega_U \gamma_U$. We obtain

$$mu = (\omega_M \omega_U)(\gamma_U \gamma_M),$$

where the first factor ranges in the compact set $\Omega_M \Omega_U$ and the second belongs to $\Gamma^* \cap MU$. This proves the right-coset assertion. Applying the same argument to inverses gives the left-coset assertion. $\square$

N.3 Rational alignment inside an ambient special linear group

We next isolate the rationality issue that is lost if one merely chooses an arbitrary real Cartan conjugating matrix.

Lemma N.7 (Rational triangular position). *Let $\iota : \mathbf{G} \hookrightarrow \mathrm{SL}_N$ be a faithful $\mathbb{Q}$ -embedding. After conjugating ι by an element of $\mathrm{SL}_N(\mathbb{Q})$, we may assume that $\mathbf{S}$ lies in the standard diagonal torus $\mathbf{D}$ and that $\mathbf{U}$ lies in the standard upper unipotent group $\mathbf{N}_0$. The standard ordering of the diagonal characters may be chosen so that every ambient positive root whose restriction to $\mathbf{S}$ is nonzero restricts to a positive relative root.*

Proof. Because $\mathbf{S}$ is $\mathbb{Q}$ -split, the rational representation $\mathbb{Q}^N$ decomposes as a direct sum of rational $\mathbf{S}$ -weight spaces. Choose a $\mathbb{Q}$ -basis adapted to this decomposition. In this basis $\mathbf{S}$ is diagonal.

The connected solvable group $\mathbf{SU}$ is $\mathbb{Q}$ -split: $\mathbf{S}$ is split and $\mathbf{U}$ has a central series with successive quotients isomorphic over $\mathbb{Q}$ to additive groups. We prove the required triangular form by induction on N . A nontrivial rational unipotent group has a nonzero rational fixed vector: its Lie algebra consists of nilpotent rational matrices, and the common kernel of a central nonzero ideal is rational and nonzero; induction on the nilpotency length then gives a common fixed vector. Taking its $\mathbf{S}$ -weight component gives a rational common eigenvector for $\mathbf{SU}$. Passing to the quotient and iterating produces a rational invariant flag. In a basis adapted to this flag, $\mathbf{U}$ is upper unipotent while $\mathbf{S}$ remains diagonal.

Finally order the $\mathbf{S}$ -weights by first comparing their values on a regular vector in the chosen positive relative chamber and only then breaking ties.

The corresponding permutation of the rational basis is rational and has the stated sign-compatibility property. $\square$

Lemma N.8 (Self-adjoint good position without losing the rational split torus). *After the rational conjugation of Lemma N.7, there is $c \in \mathrm{SL}_N(\mathbb{R})$ such that*

1. cGc^{-1} is invariant under the standard Cartan involution;
2. $cSc^{-1} \subset A_0$ for the standard positive diagonal group;
3. after an additional conjugation by a rational signed permutation matrix, c may be chosen to centralize S ;
4. the ambient and intrinsic positive-root orderings are compatible.

Proof. Lemma M.1 first puts G in self-adjoint position. Inside the resulting Cartan decomposition choose a maximal abelian subspace of the symmetric part containing $\log(cSc^{-1})$, extend it to one in the ambient symmetric matrices, and conjugate by an element of K_0 so that it lies in $\mathfrak{a}_0$. A signed permutation then makes the ambient positive system compatible with the given relative one. This proves (1), (2), and (4).

It remains to preserve rationality. Both S and cSc^{-1} are subtori of the split diagonal torus D . Any algebraic subtorus of a split torus is cut out by integral characters, hence is defined over $\mathbb{Q}$. The two diagonal representations have the same multiset of characters, because they are conjugate in $\mathrm{GL}_N(\mathbb{R})$. Therefore a permutation of the coordinate weight spaces, followed inside equal-weight blocks by a rational change of basis, gives $v \in N_{\mathrm{GL}_N}(D)(\mathbb{Q})$ satisfying

$$vsv^{-1} = csc^{-1} \quad (s \in S).$$

Replacing the rational embedding by its conjugate under v and replacing c by cv^{-1} does not alter (1) or (4), while now c centralizes S . Rescaling by a scalar makes its determinant one without changing any conjugated subgroup. $\square$

N.4 The rational relative Weyl group and nilpotent complements

N.4.1 Rational Levi conjugacy and the rank-one normalizer

The last structural input can in fact be proved with the same positive-weight coordinates already used for rational unipotent reduction. We first isolate the rational Levi-conjugacy statement in the exact form needed below.

Lemma N.9 (Rational conjugacy of Levi subgroups). *Let $\mathbf{P}$ be a rational parabolic subgroup of a connected reductive $\mathbb{Q}$ -group, let $\mathbf{U} = R_u(\mathbf{P})$, and let $\mathbf{L}_1, \mathbf{L}_2$ be rational Levi subgroups of $\mathbf{P}$. Then there is $u \in \mathbf{U}(\mathbb{Q})$ such that*

$$u\mathbf{L}_1u^{-1} = \mathbf{L}_2.$$

Proof. Fix $\mathbf{L} = \mathbf{L}_1$ and write $\mathbf{P} = \mathbf{U} \rtimes \mathbf{L}$. Choose a rational cocharacter $\lambda : \mathbf{G}_m \rightarrow Z(\mathbf{L})$ defining $\mathbf{P}$, so every weight of λ on $\mathfrak{u} = \text{Lie } \mathbf{U}$ is strictly positive. After multiplying λ by a positive integer, write the distinct positive weights as $m_1 < \cdots < m_s$. Let $\mathfrak{u}_j$ be the sum of the weight spaces of weight at least m_j , and put $\mathbf{U}_j = \exp(\mathfrak{u}_j)$. In characteristic zero the exponential map is a rational polynomial isomorphism on a unipotent group; hence

$$\mathbf{U} = \mathbf{U}_1 \supset \mathbf{U}_2 \supset \cdots \supset \mathbf{U}_{s+1} = \{e\}$$

is a rational $\mathbf{L}$ -stable normal filtration, and each quotient $\mathbf{V}_j = \mathbf{U}_j / \mathbf{U}_{j+1}$ is a rational vector group on which $\lambda(t)$ acts by the scalar t^{m_j} .

Let $\pi : \mathbf{P} \rightarrow \mathbf{L}$ be the quotient map. Since a Levi subgroup meets the unipotent radical trivially and has the same dimension as the reductive quotient, $\pi|_{\mathbf{L}_2}$ is a rational isomorphism. Thus $\mathbf{L}_2$ is the graph of a rational nonabelian cocycle $c : \mathbf{L} \rightarrow \mathbf{U}$:

$$c(gh) = c(g)(gc(h)g^{-1}).$$

We kill this cocycle one weight layer at a time. Suppose, after earlier conjugations by rational points of $\mathbf{U}$, that c takes values in $\mathbf{U}_j$. Modulo $\mathbf{U}_{j+1}$ it becomes an additive rational 1-cocycle $\bar{c} : \mathbf{L} \rightarrow \mathbf{V}_j$. Choose $t \in \mathbb{Q}^\times$ with $t^{m_j} \neq 1$ and set $a = \lambda(t)$; the element a is central in $\mathbf{L}$. Comparing the cocycle identities for ag and ga gives

$$(a - 1)\bar{c}(g) = (g - 1)\bar{c}(a).$$

Because $a - 1 = (t^{m_j} - 1)I$ is invertible on $\mathbf{V}_j$, the rational vector

$$v_j = (a - 1)^{-1}\bar{c}(a) \in \mathbf{V}_j(\mathbb{Q})$$

satisfies $\bar{c}(g) = (g - 1)v_j$.

Choose a rational lift $\tilde{v}_j \in \mathbf{U}_j(\mathbb{Q})$, possible in exponential coordinates. Conjugating the graph by $\tilde{v}_j$ changes the cocycle modulo $\mathbf{U}_{j+1}$ by $-(g - 1)v_j$, so the new cocycle takes values in $\mathbf{U}_{j+1}$. Induction over j makes the cocycle trivial. The product of the rational conjugating elements lies in $\mathbf{U}(\mathbb{Q})$ and carries $\mathbf{L}_2$ to $\mathbf{L}_1$; invert it if necessary to obtain the displayed orientation. $\square$

Lemma N.10 (A rational opposite in relative rank one). *Let $\mathbf{D}$ be connected semisimple over $\mathbb{Q}$ of $\mathbb{Q}$ -rank one. Fix a maximal rational split torus $\mathbf{V}$, a minimal rational parabolic $\mathbf{P} = \mathbf{L}\mathbf{U}$ containing it, and its rational opposite $\mathbf{P}^- = \mathbf{L}\mathbf{U}^-$, where $\mathbf{L} = Z_{\mathbf{D}}(\mathbf{V})$. There exists $x \in \mathbf{D}(\mathbb{Q})$ such that $x\mathbf{P}x^{-1}$ is opposite to $\mathbf{P}$.*

Proof. Over $\overline{\mathbb{Q}}$, the multiplication map

$$\mathbf{U}^- \times \mathbf{P} \longrightarrow \mathbf{D}$$

has differential at (e, e) equal to the direct-sum map $\mathfrak{u}^- \oplus \mathfrak{p} \rightarrow \mathfrak{d}$, hence is dominant. Therefore the $\mathbf{U}^-$ -orbit of the base point in $\mathbf{D}/\mathbf{P}$ contains a nonempty Zariski-open subset. The locus in $\mathbf{D}/\mathbf{P}$ consisting of parabolics opposite to $\mathbf{P}$ is also a nonempty Zariski-open subset: oppositeness is the condition that the intersection with $\mathfrak{p}$ have the minimal possible dimension $\dim \mathfrak{l}$, and intersection dimension is upper semicontinuous. The flag variety is irreducible, so these two open sets meet. Consequently the inverse image in $\mathbf{U}^-$ of the opposite locus is a nonempty rational Zariski-open subset.

In characteristic zero the connected unipotent group $\mathbf{U}^-$ is rationally isomorphic, via exponential coordinates, to its affine Lie algebra. A nonempty rational Zariski-open subset of an affine space over the infinite field $\mathbb{Q}$ has a rational point. Choose such a point $x \in \mathbf{U}^-(\mathbb{Q})$. Then $x\mathbf{P}x^{-1}$ is rational and opposite to $\mathbf{P}$. $\square$

Theorem N.11 (Rank-one rational normalizer statement). *Let $\mathbf{H}$ be a connected reductive group over $\mathbb{Q}$ of semisimple $\mathbb{Q}$ -rank one, and let $\mathbf{V}$ be a maximal $\mathbb{Q}$ -split torus in $\mathbf{H}^{\text{der}}$. Then*

$$N_{\mathbf{H}^{\text{der}}}(\mathbf{V})(\mathbb{Q}) \not\subset Z_{\mathbf{H}^{\text{der}}}(\mathbf{V})(\mathbb{Q}).$$

Equivalently, there is a rational element normalizing $\mathbf{V}$ and acting on the one-dimensional torus $\mathbf{V}$ by inversion.

Proof. Put $\mathbf{D} = \mathbf{H}^{\text{der}}$ and $\mathbf{L} = Z_{\mathbf{D}}(\mathbf{V})$. Choose opposite minimal rational parabolics

$$\mathbf{P} = \mathbf{L}\mathbf{U}, \quad \mathbf{P}^- = \mathbf{L}\mathbf{U}^-.$$

By Lemma N.10, choose $x \in \mathbf{D}(\mathbb{Q})$ such that $\mathbf{Q} = x\mathbf{P}x^{-1}$ is opposite to $\mathbf{P}$, and put $\mathbf{L}_0 = \mathbf{P} \cap \mathbf{Q}$. This intersection is a rational Levi subgroup of both opposite parabolics.

The group $x\mathbf{L}x^{-1}$ is another rational Levi subgroup of $\mathbf{Q}$. Lemma N.9, applied inside $\mathbf{Q}$, gives $v \in R_u(\mathbf{Q})(\mathbb{Q})$ such that

$$vx\mathbf{L}x^{-1}v^{-1} = \mathbf{L}_0.$$

Set $x_1 = vx$. Then $x_1\mathbf{P}x_1^{-1} = \mathbf{Q}$ and $x_1\mathbf{L}x_1^{-1} = \mathbf{L}_0$. Applying the same lemma inside $\mathbf{P}$ gives $u \in \mathbf{U}(\mathbb{Q})$ with $u\mathbf{L}_0u^{-1} = \mathbf{L}$. For $y = ux_1$ we therefore have

$$y\mathbf{L}y^{-1} = \mathbf{L}, \quad y\mathbf{P}y^{-1} = u\mathbf{Q}u^{-1}.$$

The latter parabolic is opposite to $\mathbf{P}$, and its intersection with $\mathbf{P}$ is $\mathbf{L}$. In relative rank one there is a unique parabolic opposite to $\mathbf{P}$ with this prescribed common Levi: relative to $\mathbf{V}$, its Lie algebra must contain $\mathfrak{l}$ and all negative relative-root spaces, hence it is $\mathbf{P}^-$. Thus $y\mathbf{P}y^{-1} = \mathbf{P}^-$.

The maximal rational split subtorus of the center of $\mathbf{L}$ is exactly $\mathbf{V}$: a larger one would give a rational split torus of $\mathbf{D}$ strictly larger than $\mathbf{V}$. Hence $y\mathbf{L}y^{-1} = \mathbf{L}$ implies $y\mathbf{V}y^{-1} = \mathbf{V}$. The element y cannot centralize $\mathbf{V}$, for then $y \in Z_{\mathbf{D}}(\mathbf{V}) = \mathbf{L}$ and would preserve $\mathbf{P}$ rather than carry it to the distinct opposite $\mathbf{P}^-$. The only rational algebraic automorphisms of the one-dimensional split torus are $t \mapsto t$ and $t \mapsto t^{-1}$, so the nontrivial action of y is inversion. $\square$

Lemma N.12 (Rational rank-one reflection representatives). *Let α be an indivisible relative root of $\mathbf{G}$ with respect to $\mathbf{S}$. There exists*

$$n_\alpha \in N_{\mathbf{G}}(\mathbf{S})(\mathbb{Q})$$

whose action on $X^(\mathbf{S})_{\mathbb{R}}$ is the reflection in the hyperplane $\alpha = 0$.*

Proof. Put $\mathbf{S}_\alpha = (\ker \alpha)^\circ$ and $\mathbf{M}_\alpha = Z_{\mathbf{G}}(\mathbf{S}_\alpha)^\circ$. The relative root decomposition shows that $\mathbf{M}_\alpha$ is connected reductive over $\mathbb{Q}$, contains the root group $\mathbf{U}_{(\alpha)}$, and therefore has nonzero semisimple $\mathbb{Q}$ -rank. Since $\mathbf{S}_\alpha$ has codimension one in $\mathbf{S}$, the derived group $\mathbf{M}_\alpha^{\text{der}}$ has semisimple $\mathbb{Q}$ -rank exactly one, and

$$\mathbf{V}_\alpha = (\mathbf{S} \cap \mathbf{M}_\alpha^{\text{der}})^\circ$$

is a one-dimensional maximal $\mathbb{Q}$ -split torus in $\mathbf{M}_\alpha^{\text{der}}$.

By Theorem N.11 choose $x_\alpha \in N_{\mathbf{M}_\alpha^{\text{der}}}(\mathbf{V}_\alpha)(\mathbb{Q})$ which does not centralize $\mathbf{V}_\alpha$. It acts by inversion on $\mathbf{V}_\alpha$. On the other hand $\mathbf{M}_\alpha$ centralizes $\mathbf{S}_\alpha$, so x_α fixes $\mathbf{S}_\alpha$ pointwise. Up to finite intersection, $\mathbf{S} = \mathbf{S}_\alpha \mathbf{V}_\alpha$; hence x_α normalizes $\mathbf{S}$. Its action is the identity on the hyperplane annihilated by α and multiplication by -1 on the complementary one-dimensional direction. For any Weyl-invariant inner product this is exactly the reflection s_α . $\square$

Lemma N.13 (No unipotent element normalizes the split torus). *Let $\mathbf{P} = Z_{\mathbf{G}}(\mathbf{S})\mathbf{U}$ be the minimal rational parabolic determined by the chosen positive relative roots. Then*

$$N_{\mathbf{G}}(\mathbf{S})(\mathbb{Q}) \cap \mathbf{U}(\mathbb{Q}) = \{e\}.$$

Consequently

$$N_{\mathbf{G}}(\mathbf{S})(\mathbb{Q}) \cap \mathbf{P}(\mathbb{Q}) = Z_{\mathbf{G}}(\mathbf{S})(\mathbb{Q}).$$

Proof. Order the positive relative roots by height and use the polynomial product coordinates on U proved earlier in this volume. If $u \neq e$, let β be a least-height root for which the $\mathbf{U}_{(\beta)}$ coordinate of u is nonzero. For $s \in \mathbf{S}$, conjugation multiplies this leading coordinate by $\beta(s)$; all commutator corrections have strictly larger height. If u normalized $\mathbf{S}$, then for every s the commutator $usu^{-1}s^{-1}$ would belong both to $\mathbf{S}$ and to $\mathbf{U}$. The intersection of a torus with a unipotent group is trivial, so this commutator is e . Thus $(\beta(s) - 1)$ times the leading coordinate is zero for every s , impossible because β is a nontrivial character. Hence $u = e$.

Now write $p = zu$ in the Levi decomposition $\mathbf{P} = Z_{\mathbf{G}}(\mathbf{S}) \ltimes \mathbf{U}$. If p normalizes $\mathbf{S}$, then $u = z^{-1}p$ also normalizes $\mathbf{S}$, hence $u = e$ by the first part. Therefore $p = z$. $\square$

Lemma N.14 (Rational relative Weyl representatives). *Let $W_{\mathbb{Q}}$ denote the image of $N_{\mathbf{G}}(\mathbf{S})(\mathbb{Q})$ in $\text{Aut}(\mathbf{S})$. Then $W_{\mathbb{Q}}$ is the Weyl group of the relative root system. In particular it is finite, it acts simply transitively on relative Weyl chambers, and one may fix rational representatives*

$$x_w \in N_{\mathbf{G}}(\mathbf{S})(\mathbb{Q}), \quad w \in W_{\mathbb{Q}}.$$

Proof. Let W_0 be the finite reflection group generated by the relative-root reflections. Lemma N.12 gives $W_0 \subset W_{\mathbb{Q}}$.

It remains to prove the reverse inclusion. Let $w \in W_{\mathbb{Q}}$ stabilize one W_0 -Weyl chamber C . Since w has finite order, averaging any point in C over the finite cyclic group generated by w gives a point $D \in C$ fixed by w . Declare a relative root positive exactly when it is positive on D . Then w preserves the set of positive relative roots and therefore any rational representative $n \in N_{\mathbf{G}}(\mathbf{S})(\mathbb{Q})$ of w normalizes the corresponding minimal rational parabolic

$$\mathbf{P} = Z_{\mathbf{G}}(\mathbf{S})\mathbf{U}.$$

A parabolic subgroup is self-normalizing, so $n \in \mathbf{P}(\mathbb{Q})$. By Lemma N.13, $n \in Z_{\mathbf{G}}(\mathbf{S})(\mathbb{Q})$, and hence $w = e$.

Thus no nontrivial element of $W_{\mathbb{Q}}$ stabilizes a W_0 -chamber. Since $W_0 \subset W_{\mathbb{Q}}$ already acts simply transitively on the chambers, every $w \in W_{\mathbb{Q}}$ equals the unique element of W_0 carrying a fixed chamber to wC . Hence $W_{\mathbb{Q}} = W_0$. Choosing one rational normalizer representative for each of its finitely many elements finishes the proof. $\square$

Lemma N.15 (Rational nilpotent factorization). *In the rational triangular position of Lemma N.7, there are S -stable rational vector spaces $\mathfrak{n}_1, \mathfrak{n}_2 \subset \mathfrak{n}_0$ such that*

$$\mathfrak{n}_1 = \mathfrak{n}_0 \cap \mathfrak{z}_{\mathfrak{sl}_N}(\mathfrak{s}), \quad \mathfrak{n}_0 = \mathfrak{n}_1 \oplus \mathfrak{n}_2 \oplus \mathfrak{u}$$

as vector spaces, and the multiplication maps

$$N_1 \times N_2 \times U \longrightarrow N_0, \quad U \times N_1 \times N_2 \longrightarrow N_0$$

are polynomial diffeomorphisms defined over $\mathbb{Q}$, where $N_i = \exp(\mathfrak{n}_i)$ is understood as a second-kind coordinate slice when $\mathfrak{n}_i$ is not a subalgebra.

Proof. Let $\mathfrak{n}^+$ be the sum of the positive ambient root spaces whose restriction to S is nontrivial, and let $\mathfrak{n}_1 = \mathfrak{n}_0 \cap \mathfrak{z}_{\mathfrak{sl}_N}(\mathfrak{s})$. Compatibility of the ambient and relative positive orders implies that $\mathfrak{n}^+$ is an ideal of $\mathfrak{n}_0$, so the ordered exponential product gives $N_0 = N_1 N^+$.

For every positive relative character λ , let V_λ be the sum of all ambient positive-root spaces whose restriction to S is λ . Then

$$\mathfrak{n}^+ = \bigoplus_{\lambda > 0} V_\lambda, \quad \mathfrak{u} = \bigoplus_{\lambda > 0} (V_\lambda \cap \mathfrak{u}).$$

The rational reductive group $Z_{\mathbf{G}}(\mathbf{S})$ preserves both V_λ and $V_\lambda \cap \mathfrak{u}$. Indeed it commutes with S , and compatibility of the positive orders ensures that the full S -weight space of positive weight λ occurring in $\mathfrak{sl}_N$ is already contained in the ambient positive nilpotent algebra. Complete reducibility in characteristic zero therefore gives a $Z_{\mathbf{G}}(\mathbf{S})$ -invariant rational complement $V_{\lambda,2}$ to $V_\lambda \cap \mathfrak{u}$ in V_λ . Put

$$\mathfrak{n}_2 = \bigoplus_{\lambda > 0} V_{\lambda,2}.$$

This proves the displayed vector-space decomposition and, importantly, $MA = Z_G(S)^\circ$ preserves both $\mathfrak{u}$ and $\mathfrak{n}_2$.

Order the positive relative weights by height and choose bases in the $V_{\lambda,2}$ and $V_\lambda \cap \mathfrak{u}$. The sums of all weight spaces of height at least r are ideals of

$\mathfrak{n}^+$. The terminating BCH formula is therefore triangular in this order, with identity linear part. Successively solving the coordinates shows that

$$N_2 \times U \longrightarrow N^+, \quad (v, u) \longmapsto vu,$$

and the reversed ordered product $U \times N_2 \rightarrow N^+$ are polynomial bijections with polynomial inverses. Combining with $N_0 = N_1 N^+$ gives the two factorizations in the statement. Every construction is rational, so the coordinate maps are defined over $\mathbb{Q}$. $\square$

N.5 The intrinsic transfer lemma

The preceding normalizations reduce the proof to a comparison inside an ambient parabolic whose split part contains the rational split torus S . The point of the next lemma is that one must first obtain an *unreduced* rational cusp tube $KMA_t Ux_w$. The M - and U -coordinates are reduced arithmetically only afterwards. This separation is essential: $M(\mathbb{R})$ need not itself be compact.

For z, h in a group write

$$\text{Int}(z)h = zh z^{-1}.$$

Put $N_+ = N_2 U$, using the ordered second-kind factorization supplied by Lemma N.15. The group N_1 normalizes N_+ because $\mathfrak{n}^+$ is an ideal in $\mathfrak{n}_0$, while $MA = Z_G(S)^\circ$ preserves U and the chosen N_2 -coordinates by Lemma N.15. The ambient split group A' normalizes N_+ and acts diagonally in the ambient root coordinates.

Lemma N.16 (Ambient block-QR coordinates relative to the rational split torus). *Let $S \subset A'$ be in the diagonal rational position fixed above. Let $Z_* = Z_{\text{SL}_N(\mathbb{R})}(S)$, and let N_+ be the connected unipotent subgroup whose Lie algebra is the sum of the positive ambient root spaces having nonzero restriction to S . Thus $P_* = Z_* N_+$ is the standard parabolic attached to the ordered S -weight flag. Every $h \in \text{SL}_N(\mathbb{R})$ can be written*

$$h = k_h z_h n_h, \quad k_h \in K', \quad z_h \in Z_*, \quad n_h \in N_+. \quad (\text{N.4})$$

The N_+ -coordinate is unique. The Z_ -coordinate is unique up to left multiplication by $K' \cap Z_*$. Moreover, if h ranges in a compact subset of $\text{SL}_N(\mathbb{R})$, the factors in (N.4) may be chosen in fixed compact subsets of K' , Z_* and N_+ .*

Proof. Write the real S -weight decomposition as

$$\mathbb{R}^N = E_1 \oplus \cdots \oplus E_r$$

in the order defining the ambient positive system. Then P_* is the group of block upper-triangular matrices for the flag $E_1 \subset E_1 \oplus E_2 \subset \cdots$, Z_* is the block-diagonal Levi group, and N_+ is the block-unitriangular radical.

Apply Gram–Schmidt successively to the column blocks of h . Equivalently, perform ordinary QR decomposition but orthogonalize an entire E_i -block at a time. This gives

$$h = k_h r_h, \quad k_h \in K', \quad r_h \in P_*.$$

Factor the block upper-triangular matrix uniquely as $r_h = z_h n_h$ by taking z_h to be its block-diagonal part and dividing the off-diagonal blocks by those diagonal blocks. This proves existence.

If $k_1 z_1 n_1 = k_2 z_2 n_2$, then $k_2^{-1} k_1 \in K' \cap P_*$. An orthogonal block-upper-triangular matrix is block diagonal, so

$$K' \cap P_* = K' \cap Z_*.$$

The uniqueness of the Levi–unipotent product $P_* = Z_* N_+$ now gives $n_1 = n_2$ and shows that z_1, z_2 differ by the indicated compact factor.

Finally let h range in a compact set C . Its singular values are bounded above and bounded away from zero. The same is therefore true for every successive Gram matrix used in block Gram–Schmidt. The positive square roots and their inverses stay in compact sets, so the block-QR factors k_h, r_h do also. Taking the block diagonal of r_h and dividing the finitely many off-diagonal blocks by the diagonal blocks then shows that z_h and n_h range in compact sets. This proves the compact-uniform assertion. $\square$

Lemma N.17 (Bounded conjugated-unipotent coordinate). *Assume that the rational and good-position normalizations of the previous sections have been made. Let $K' A' N_0$ be a compatible ambient Iwasawa decomposition with*

$$K \subset K', \quad A \subset A', \quad U \subset N_0,$$

and let $\mathfrak{S}' = K' A'_{t_0} \Omega_0$ be an ambient Siegel set. Fix $v = v_1 v_2$ with $v_1 \in N_1$ and $v_2 \in N_2$. For

$$x = k' a' n \in \mathfrak{S}', \quad g = xv \in G,$$

write the rational Langlands coordinates

$$g = kmau, \quad k \in K, \quad m \in M, \quad a \in A, \quad u \in U.$$

Then the set

$$\{\text{Int}(ma)u : x \in \mathfrak{S}', \quad xv \in G\} \quad (\text{N.5})$$

is relatively compact in U .

Proof. This is the case $q = e$ of the exact comparison below, but it is useful to record the algebra once without the ambient Weyl symbol. Put

$$c = a'na'^{-1}.$$

The ambient Siegel estimate makes c range in a compact subset of N_0 . Apply Lemma N.16 to write

$$c = k_c z_c n_c, \quad z_c \in Z_*, \quad n_c \in N_+,$$

with all three factors bounded. Since $v_1 \in N_1 \subset Z_*$ and $v_2 \in N_2 \subset N_+$, we have

$$\begin{aligned} g &= k'a'nv_1v_2 = k'ca'v_1v_2 \\ &= k'k_c z_c a'v_1(v_1^{-1}(a')^{-1}n_ca'v_1v_2). \end{aligned}$$

The first noncompact factor after $k'k_c$ lies in Z_* and the parenthesized factor lies in N_+ . On the other hand the intrinsic decomposition $g = kmau$ is also a $K'Z_*N_+$ decomposition, because $MA \subset Z_*$ and $U \subset N_+$. By the uniqueness statement in Lemma N.16, there is $\ell \in K' \cap Z_*$ such that

$$ma = \ell z_c a'v_1, \quad u = v_1^{-1}(a')^{-1}n_ca'v_1v_2. \quad (\text{N.6})$$

Therefore

$$\begin{aligned} \text{Int}(ma)(uv_2^{-1}) &= \ell z_c a'v_1(v_1^{-1}(a')^{-1}n_ca'v_1)v_1^{-1}(a')^{-1}z_c^{-1}\ell^{-1} \\ &= \ell z_c n_c z_c^{-1}\ell^{-1}. \end{aligned} \quad (\text{N.7})$$

The right-hand side ranges in a compact subset of N_+ .

Now $MA = Z_G(S)^\circ$ preserves both U and the chosen N_2 -coordinates. Hence, using the reversed polynomial factorization $U \times N_2 \rightarrow N_+$ from Lemma N.15, the U -component of the left-hand side of (N.7) is exactly $\text{Int}(ma)u$. The inverse factorization is continuous, so the U -component of a compact subset is compact. This proves (N.5). $\square$

Lemma N.18 (Rational Weyl-chamber transfer). *Under the hypotheses of Lemma N.17, there is a number $t_1 > 0$ such that*

$$\mathfrak{S}'v \cap G \subset \bigcup_{w \in W_{\mathbb{Q}}} K M A_{t_1} U x_w, \quad (\text{N.8})$$

where the $x_w \in N_{\mathbf{G}}(\mathbf{S})(\mathbb{Q})$ are the fixed rational representatives of Lemma N.14.

Proof. Let $g = xv = kmau$ be as above. Put

$$C^+ := \{a_0 \in A : \alpha(a_0) \geq 1 \text{ for every simple relative root } \alpha\}.$$

The closures of the relative Weyl chambers cover A , so choose $w \in W_{\mathbb{Q}}$ with

$$x_w a x_w^{-1} \in C^+. \quad (\text{N.9})$$

There are only finitely many possible w .

Choose $y_w \in N_K(A)$ inducing the same relative Weyl element as x_w . Then

$$z_w := y_w x_w^{-1} \in Z_G(A).$$

The identity component of $Z_G(A)$ is MA , and the component group is finite. Since only finitely many w occur, choose once and for all factorizations $z_w = f_w m_w a_w$ with f_w in a fixed finite set, $m_w \in M$, and $a_w \in A$. The finite f_w may be absorbed into the compact factor and the fixed m_w, a_w into the M -coordinate and a uniform change of the A_t threshold.

Set

$$d = y_w \text{Int}(ma)u y_w^{-1}.$$

By Lemma N.17, d ranges in a compact subset of G . Write its $KMAU$ decomposition as

$$d = k_d m_d a_d u_d.$$

The Iwasawa–Langlands coordinate maps are continuous modulo the compact ambiguity $K \cap M$, so m_d, a_d, u_d may be chosen in fixed compact sets. Now

$$\begin{aligned} g y_w^{-1} &= k \text{Int}(ma)u m a y_w^{-1} \\ &= k y_w^{-1} d (y_w m y_w^{-1})(y_w a y_w^{-1}). \end{aligned}$$

Since y_w normalizes A , it also normalizes $Z_G(A)$ and hence M ; since M normalizes U , collecting the factors again in $KMAU$ order shows that the new A -coordinate is exactly

$$a_d (y_w a y_w^{-1}). \quad (\text{N.10})$$

By (N.9), the second factor lies in the fixed closed chamber C^+ . Every simple relative root is bounded above and below by positive constants on the compact set of possible a_d . Hence there is a uniform $t_1 > 0$ for which the coordinate in (N.10) satisfies $\alpha \geq t_1$ for every simple relative root, exactly the convention of (N.1). Thus

$$g \in KMA_{t_1}Uy_w.$$

Finally $y_w = z_w x_w$. Using the fixed factorization of z_w above, absorb its finite component into the compact K -factor, its M -component into M , and its A -component by changing t_1 by a fixed multiplicative constant. This gives (N.8). $\square$

Lemma N.19 (Fixed ambient-Weyl transfer). *Keep the compatible ambient data of the preceding lemmas. Let $q \in N_{K'}(A')$ be fixed and let $v = v_1 v_2$ with $v_1 \in N_1$, $v_2 \in N_2$. Then there is $t_q > 0$ such that*

$$\mathfrak{S}' q^{-1} v \cap G \subset \bigcup_{w \in W_{\mathbb{Q}}} KMA_{t_q} Ux_w.$$

The constant is uniform when q ranges over a fixed finite subset of $N_{K'}(A')$ and v ranges over a fixed compact subset of $N_1 N_2$.

Proof. Take $x = k' a' n \in \mathfrak{S}'$ and suppose

$$g = xq^{-1}v = kmau \in G.$$

Set

$$b = qa'q^{-1} \in A' \subset Z_*, \quad c = q(a'na'^{-1})q^{-1}.$$

Because q is fixed and $a'na'^{-1}$ ranges in a compact set by the ambient Siegel inequalities, c ranges in a compact subset of $\mathrm{SL}_N(\mathbb{R})$. The identity

$$a'naq^{-1} = q^{-1}cb \tag{N.11}$$

follows by multiplying the displayed factors in order and cancelling qq^{-1} and cc^{-1} where they occur. Thus

$$g = (k'q^{-1})cbv_1v_2.$$

Apply Lemma N.16 to c :

$$c = k_c z_c n_c, \quad z_c \in Z_*, \quad n_c \in N_+,$$

where k_c, z_c, n_c remain bounded. Moving n_c past the centralizer factor $bv_1 \in Z_*$ gives the exact $K'Z_*N_+$ factorization

$$g = k'' (z_c bv_1) (v_1^{-1} b^{-1} n_c bv_1 v_2), \quad k'' = k' q^{-1} k_c \in K'. \quad (\text{N.12})$$

The intrinsic factorization $g = kmau$ is another $K'Z_*N_+$ factorization. Lemma N.16 therefore gives $\ell \in K' \cap Z_*$ with

$$ma = \ell z_c bv_1, \quad u = v_1^{-1} b^{-1} n_c bv_1 v_2. \quad (\text{N.13})$$

Exactly as in (N.7),

$$\text{Int}(ma)(uv_2^{-1}) = \ell z_c n_c z_c^{-1} \ell^{-1}. \quad (\text{N.14})$$

The right-hand side is bounded. Since MA preserves U and N_2 , the U -component in the reversed factorization $U \times N_2 \rightarrow N_+$ is $\text{Int}(ma)u$. Consequently

$$\{\text{Int}(ma)u : xq^{-1}v \in G, x \in \mathfrak{S}'\} \quad (\text{N.15})$$

is relatively compact. This is the exact bounded-coordinate comparison needed for the chamber reduction.

It remains only to place the rational split coordinate in one of finitely many relative chambers. Choose $w \in W_{\mathbb{Q}}$ so that $x_w a x_w^{-1}$ lies in the closed positive relative chamber and choose $y_w \in N_K(A)$ inducing the same real Weyl element. Then $y_w x_w^{-1} \in Z_G(A)$. Conjugate the compact set (N.15) by y_w and write its intrinsic $KMAU$ coordinates. The resulting A -coordinate a_d ranges in a compact set.

As in Lemma N.18, the split coordinate after right multiplication by y_w^{-1} is

$$a_d (y_w a y_w^{-1}).$$

The second factor is in the fixed closed chamber and every simple relative root is bounded above and below by positive constants on the compact first factor. Hence this product belongs to A_{t_q} for one uniform $t_q > 0$.

The remaining factors lie in K , M , and U . Replacing y_w by the rational representative x_w changes it by one fixed element of $Z_G(A)$; its M -part is absorbed into M and its A -part changes t_q by a fixed multiplicative constant. Thus

$$g \in KMA_{t_q}Ux_w.$$

There are finitely many w , proving the stated inclusion. The same proof is uniform for finite q -families and compact v -families because every factor in (N.14) then ranges over a fixed compact set. $\square$

Lemma N.20 (Stability under one fixed ambient left translate). *Let $G_0 = K'A'N_0$ be an ambient Iwasawa decomposition and let $\mathfrak{S} = K'A'_{t_0}\Omega_0$ be an ambient Siegel set. For every fixed $h \in G_0$ there are $t_h > 0$ and a compact $\Omega_h \subset N_0$ such that*

$$h\mathfrak{S} \subset K'A'_{t_h}\Omega_h.$$

The constants may be chosen uniformly when h ranges over a compact set.

Proof. The set hK' is compact. Apply the ambient Iwasawa coordinate maps to it: there are compact sets $C_A \subset A'$ and $C_N \subset N_0$ such that every hk' can be written

$$hk' = k_1 a_1 n_1, \quad k_1 \in K', \quad a_1 \in C_A, \quad n_1 \in C_N.$$

For $a \in A'_{t_0}$ and $n \in \Omega_0$ we have

$$hk'an = k_1(a_1 a)(a^{-1}n_1 a)n.$$

Every positive ambient root is a nonnegative integral combination of the simple ones. Hence $a^{-1}C_N a$ remains in one compact subset of N_0 as a ranges over A'_{t_0} : in ordered root coordinates conjugation multiplies the β -coordinate by $\beta(a)^{-1}$, which is uniformly bounded above. The product with the fixed compact Ω_0 is therefore compact. On the other hand each simple root is bounded above and below by positive constants on C_A , so $a_1 a \in A'_{t_h}$ for one $t_h > 0$. This proves the inclusion. Uniformity for compact families of h follows because the Iwasawa coordinates of a compact set are compact. $\square$

Lemma N.21 (Stability under a fixed rational Borel factor). *Let $B_0 = DN_0$ be the standard rational upper Borel. If $p \in B_0(\mathbb{Q})$ is fixed and $\mathfrak{S}$ is an ambient Siegel set, then $\mathfrak{S}p$ is contained in another ambient Siegel set with the same positive system.*

Proof. Write $p = dn$ with $d \in D(\mathbb{Q})$ and $n \in N_0(\mathbb{Q})$. Split the real diagonal matrix d as a fixed signed compact factor times $d_+ \in A'$. The compact factor is absorbed into K' . Since d_+ commutes with A' , multiplication by it changes each simple-root lower bound by a fixed positive factor. Also $d_+^{-1}\Omega_0 d_+ n$ is compact in N_0 . Thus

$$K'A'_{t_0}\Omega_0 p \subset K'A'_{t_1}\Omega_1$$

for suitable $t_1 > 0$ and compact $\Omega_1 \subset N_0$. $\square$

Lemma N.22 (Ambient rational translate to an intrinsic rational cusp tube). *Let $\mathfrak{S}_0$ be a standard ambient Siegel set and let $b \in \mathrm{SL}_N(\mathbb{Q})$. For the fixed minimal rational parabolic $P = MSU$ there are $t_b > 0$ and a finite set $F_b \subset G(\mathbb{Q})$ such that*

$$c^{-1}\mathfrak{S}_0b \cap G \subset KMA_{t_b}UF_b. \quad (\text{N.16})$$

Here c is the self-adjoint conjugator used in the ambient fundamental-set construction. The same statement holds for any rationally conjugate embedding of G .

Proof. We follow the reduction in finite, explicitly rational steps.

1. *Remove the fixed real conjugator without changing rational data.* By Lemma N.20, the set $c^{-1}\mathfrak{S}_0$ is contained in an ambient Siegel set $\mathfrak{S}_1$ for the same ambient positive system. Thus it is enough to control $\mathfrak{S}_1b \cap G$. No rational terminal element has been introduced at this step.

2. *Rational triangularization and good position.* By Lemma N.7, after a rational conjugation of the whole embedding we may assume $S \subset D$ and $U \subset N_0$, with compatible positive orders. Rational conjugation preserves arithmeticity and sends rational terminal representatives to rational terminal representatives. Lemma N.8 then supplies the compatible intrinsic and ambient Iwasawa data used in Lemmas N.17– N.19.

3. *Rational Bruhat decomposition of the fixed translate.* The rational Bruhat decomposition of SL_N gives

$$b = p_L q p_R, \quad p_L, p_R \in B_0(\mathbb{Q}), \quad q \in N_{K'}(A') \cap \mathrm{SL}_N(\mathbb{Q}),$$

where q may be chosen as a signed permutation matrix. By Lemma N.21, right multiplication by the fixed p_L can be absorbed into a new ambient Siegel set. Write $p_R = d_R n_R$ with $d_R \in D(\mathbb{Q})$ and $n_R \in N_0(\mathbb{Q})$. Since $q d_R = (q d_R q^{-1})q$ and $q d_R q^{-1} \in D(\mathbb{Q})$, a second application of Lemma N.21 absorbs this diagonal factor. After renaming the ambient Siegel set and replacing q by q^{-1} if necessary, we have reduced to

$$\mathfrak{S}' q^{-1} n_R \cap G.$$

Use Lemma N.15 to write, over $\mathbb{Q}$,

$$n_R = v_1 v_2 v_U, \quad v_1 \in N_1(\mathbb{Q}), \quad v_2 \in N_2(\mathbb{Q}), \quad v_U \in U(\mathbb{Q}).$$

Because $v_U \in G(\mathbb{Q})$,

$$\mathfrak{S}' q^{-1} v_1 v_2 v_U \cap G = (\mathfrak{S}' q^{-1} v_1 v_2 \cap G) v_U.$$

The fixed-ambient-Weyl transfer lemma therefore gives

$$\mathfrak{S}'q^{-1}v_1v_2 \cap G \subset \bigcup_{w \in W_{\mathbb{Q}}} KMA_{t_b}Ux_w.$$

Multiplying on the right by v_U merely replaces x_w by the rational element x_wv_U . There are finitely many w , hence a finite set $F_b \subset G(\mathbb{Q})$, and we obtain

$$c^{-1}\mathfrak{S}_0b \cap G \subset KMA_{t_b}UF_b.$$

Undoing the initial rational conjugation conjugates F_b inside $G(\mathbb{Q})$ and changes only the fixed compact/Iwasawa data. This proves (N.16). $\square$

Remark N.23 (What the transfer lemma proves). Lemma N.22 is deliberately weaker than a compact Siegel-set statement before quotienting by Γ : its M and U factors are still unrestricted. This is the correct intermediate theorem. The arithmetic reduction of those factors is the next, logically separate, step. It is precisely this separation that prevents the proof from confusing real reduction with rational cusp reduction.

N.6 Rational Siegel covering

Theorem N.24 (Rational Siegel reduction). *Let $\mathbf{G}$ be connected semisimple over $\mathbb{Q}$, let $\Gamma < \mathbf{G}(\mathbb{Q})$ be arithmetic, and fix a minimal rational parabolic $\mathbf{P} = \mathbf{M}\mathbf{S}\mathbf{U}$ and a maximal compact subgroup $K < G$. There exist $t > 0$, a compact set $\Omega \subset MU$, and finitely many $q_1, \dots, q_r \in \mathbf{G}(\mathbb{Q})$ such that*

$$G = \bigcup_{i=1}^r K A_t \Omega q_i \Gamma. \quad (\text{N.17})$$

Equivalently, after inversion and replacement of P by its opposite, there are a compact set $C \subset G$, a compact $\Omega' \subset MU$, and rational $q'_1, \dots, q'_r$ such that

$$G = \Gamma \bigcup_{i=1}^r q'_i \Omega' A_{t'} C. \quad (\text{N.18})$$

Each set in either finite family has finite Haar measure.

Proof. After replacing Γ by a commensurable subgroup and enlarging the final finite family by coset representatives, we may assume that a faithful rational embedding satisfies

$$\Gamma = G \cap \mathrm{SL}_N(\mathbb{Z}).$$

Theorem M.13 gives an ambient Siegel set $\mathfrak{S}_0$, a fixed real matrix c , and finitely many $b_1, \dots, b_m \in \mathrm{SL}_N(\mathbb{Z})$ such that

$$G = \bigcup_{j=1}^m (c^{-1} \mathfrak{S}_0 b_j \cap G) \Gamma.$$

For each j , Lemma N.22 gives a threshold t_j and a finite set $F_j \subset G(\mathbb{Q})$ with

$$c^{-1} \mathfrak{S}_0 b_j \cap G \subset K M A_{t_j} U F_j.$$

Since there are only finitely many pieces, decrease the threshold to one $t > 0$ and put $F = \bigcup_j F_j$. Using $MA = AM$ and that M normalizes U , we obtain

$$G = \bigcup_{q \in F} K A_t (MU) q \Gamma. \quad (\text{N.19})$$

The reverse inclusion is automatic because all factors lie in G .

The remaining point is arithmetic reduction of the unrestricted MU factor without moving a nonrational element across the cusp parameter. Put

$$\Gamma^\dagger = \Gamma \cap \bigcap_{q \in F} q \Gamma q^{-1}.$$

Every rational element commensurates an arithmetic subgroup, so $\Gamma^\dagger$ has finite index in Γ and

$$\Gamma^\dagger q \subset q \Gamma \quad (q \in F). \quad (\text{N.20})$$

Apply Lemma N.6 to $\Gamma^\dagger$: there is a compact $\Omega \subset MU$ with

$$MU = \Omega(\Gamma^\dagger \cap MU).$$

Thus, for every $q \in F$,

$$K A_t (MU) q \Gamma \subset K A_t \Omega \Gamma^\dagger q \Gamma \stackrel{(\text{N.20})}{\subset} K A_t \Omega q \Gamma.$$

Substituting this in (N.19) proves (N.17). Returning from the temporary finite-index subgroup to the original arithmetic group only adds finitely many right-coset representatives, which are absorbed into the finite rational family.

Take inverses. Since a connected semisimple group is unimodular, inversion preserves Haar measure. We obtain pieces of the form

$$\Gamma q_i^{-1} \Omega^{-1} A_t^{-1} K.$$

The inverse cone is the Siegel cone for the opposite rational parabolic P^- . Lemma N.14 supplies a rational representative of the relative longest Weyl element carrying P^- to P . Conjugating the fixed compact MU -set and the cone by this one rational element, and absorbing it into the finite rational representatives, gives (N.18); the final K is the compact set C allowed there.

It remains to prove finite Haar measure. In intrinsic Iwasawa–Langlands coordinates $K \times A \times M \times U$, Haar measure on a set with compact K, M, U coordinates has density, up to a bounded positive factor,

$$\delta_P(a)^{-1} da, \quad \delta_P(a) = \prod_{\beta > 0} \beta(a)^{m_\beta}.$$

Write

$$2\rho_{\mathbb{Q}} = \sum_{\alpha \in \Delta_{\mathbb{Q}}} c_{\alpha} \alpha, \quad c_{\alpha} > 0.$$

The simple relative roots form coordinates on A modulo the finite center; with $s_{\alpha} = \alpha(a)$, logarithmic Haar measure is a constant multiple of $\prod ds_{\alpha}/s_{\alpha}$. Therefore

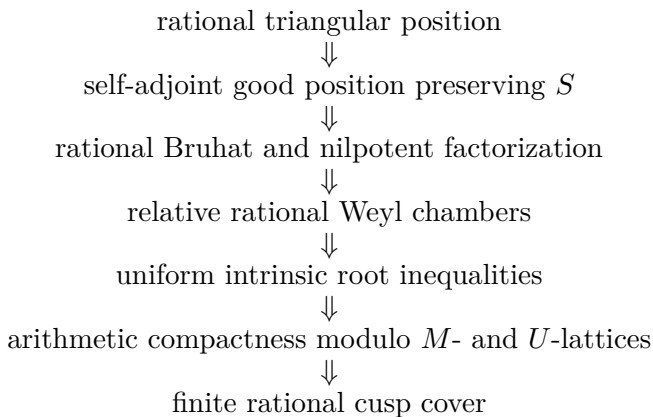
$$\int_{A_t} \delta_P(a)^{-1} da \ll \prod_{\alpha \in \Delta_{\mathbb{Q}}} \int_t^{\infty} s^{-c_{\alpha}} \frac{ds}{s} < \infty.$$

Hence every right-hand Siegel piece in (N.17) has finite Haar measure. Inversion gives the same conclusion for the left-hand pieces. $\square$

Remark N.25 (Scope and proof ownership of the strong reduction theorem). The left-cover formula (N.18) has the exact strength and factor order of Theorem 7.2. Every step in this reduction chain has an internal proof owner: arithmetic MU reduction; ambient-to-intrinsic block- QR transfer; rational conjugacy of Levi subgroups; the rank-one rational normalizer theorem; rational reflection representatives; the full relative Weyl group; uniform root-cone transfer; and the intrinsic Haar integral. Thus Theorem 7.2 is proved here. The independent finite-covolume proof in Chapter M remains available and does not create any circularity.

N.7 Dependency ledger for rational Siegel reduction

The proof now has explicit owners for every step that was previously only a phrase:



No use is made of Ratner classification, Mozes–Shah, or any later volume. Finite covolume is proved independently in Chapter M. The present chapter supplies the exact ambient-to-intrinsic transfer together with the rank-one rational normalizer and relative-Weyl steps. Consequently the strong rational cusp theorem is logically independent of the finite-covolume argument and has a complete proof here.

N.8 Worked rank-one check

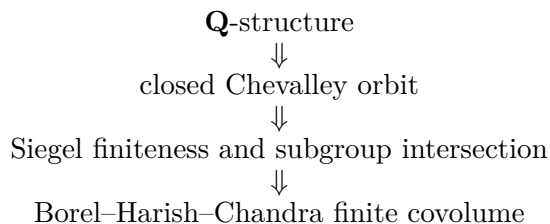
For $G = \mathrm{SL}_2(\mathbb{R})$, P the upper triangular rational parabolic, and $A = \{\mathrm{diag}(y^{1/2}, y^{-1/2}) : y > 0\}$, the relative Weyl group has two elements. There is no noncompact M -coordinate. The nilpotent factorization is just the coordinate x in $\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$, reduced modulo $\mathbb{Z}$ to a bounded interval. The only noncompact coordinate is y , and A_t is $y \geq t$. For a finite-index subgroup of $\mathrm{SL}_2(\mathbb{Z})$, the finite rational set $\{q_i\}$ is exactly what records the finitely many cusp classes; one cannot replace it by a single real Siegel cone. This elementary example is the diagnostic reason the rationality part of Repair C is genuinely stronger than the finite-covolume argument of Repair B.

Proof Guide and Dependency Summary

This chapter is a finalization layer. It does not introduce a new theorem scope. Its purpose is to expose the proof architecture, make the hypotheses that are easiest to misread impossible to miss, and give a student a concrete model calculation before the abstract machinery is reused later.

1. Three structures that must not be conflated

For a subgroup of a linear group there are three different closures in play: topological closure, connected Lie subgroup generated by infinitesimal data, and Zariski closure. Reduction theory also adds a rational structure. There are now two deliberately separated arithmetic chains. The first is the strong rational-parabolic reduction theorem (Theorem 7.2). The strong-reduction proof module supplies the Bruhat ordering, the arithmetic M/U reduction, the factor order in the Siegel cover, the Haar integral, and the exact ambient-to-intrinsic transfer. The strong-reduction proof module proves the theorem completely, including rational Levi conjugacy and the rank-one rational normalizer/relative-Weyl step. The second, independently complete since the finite-covolume proof module, is



Arithmetic closed-orbit control then follows from finite covolume together with the general closed-orbit theorem of Volume III. This separation prevents the strong reduction theorem from being used before its proof in later volumes; Chapter N supplies the complete proof.

2. A parabolic laboratory in SL_3

Let P stabilize the line $\mathbf{R}e_1$. Then matrices in P have block form

$$\begin{pmatrix} a & * \\ 0 & B \end{pmatrix}, \quad a \det B = 1.$$

The unipotent radical consists of the upper-right 1×2 block, a split torus supplies the expanding/contracting direction, and the Levi factor carries the residual reductive structure. In this example one can see directly why a cusp is attached to a rational parabolic: the split torus makes some rational wedge vectors small.

The same wedge-vector calculation is the bridge to Mahler compactness. If $v_1, \dots, v_r$ span a rational primitive subspace, then

$$\|g(v_1 \wedge \cdots \wedge v_r)\|$$

measures its covolume. Reduction theory is therefore a systematic way of organizing which rational wedge can become small.

3. Proof-ownership checklist for the reduction chain

The following checklist records the proof owners supplied by the strong-reduction module. It is included so that later uses do not confuse the strong rational-cusp theorem with the weaker finite-covolume route:

1. construct the relevant Siegel sets from K , split A , and unipotent N coordinates;
2. prove coarse covering by finitely many rational parabolic pieces;
3. prove finite Haar volume by integrating the modular/Jacobian factors;
4. establish the finite-overlap statement needed to pass from a coarse cover to a fundamental-domain estimate;
5. identify precisely how the resulting rational-root cone controls the stated finite family of rational cusp representatives.

Items (1), (3), and the arithmetic and geometric transfer portions of (5) are proved in Chapter N. The remaining rational-cusp chamber step follows there from the rational Levi-conjugacy and rank-one normalizer statements. Borel–Harish–Chandra has the independent finite-covolume proof of Chapter M, so the two arguments are logically noncircular. Borel density is a distinct representation-theoretic statement and is deduced only after the lattice input is available.

4. Student checkpoint

A student should be able to explain, without notation, why “going to the cusp” is equivalent to “a rational subspace acquires small covolume,” and then reproduce the SL_3 wedge calculation above.

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Mautner, Howe–Moore, and Mixing on Homogeneous Spaces

Volume VI of a Detailed Notebook on Ratner's Theorems

Detailed Notebook on Ratner's Theorems

August 2026

Preface and proof policy

Volume II proved the rank-one Mautner mechanisms needed inside the original Ratner argument and deliberately quarantined the full Howe–Moore theorem as context. From Volume IX onward mixing becomes a load-bearing tool. This volume therefore supplies the missing general representation-theoretic bridge. The proof is written from contraction groups, root subgroups, and weak limits; no decay theorem is invoked as a black box. Quantitative rates are postponed to Volume XI.

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Dependency and proof-ownership contract

Purpose. Qualitative mixing.

Prerequisites. Volumes I–V.

New objects. contraction groups, roots, Mautner, Howe–Moore. Every project-specific technical term is defined before its first load-bearing use or recalled by an exact backward reference.

Main outputs. qualitative matrix-coefficient decay.

External-input rule. Every nontrivial imported result is either part of the declared graduate-level prerequisites, an explicitly named external research input in the prerequisite map, or is proved in this series before use (possibly under an explicit deferred-proof contract). A citation by itself never constitutes a proof.

Reader map and proof route

Question of the volume. Why do contraction by semisimple elements and the Mautner phenomenon force matrix coefficients to vanish at infinity?

Prerequisite checkpoint. Use Volumes I–V for unitary actions on homogeneous quotients and root/parabolic structure. The qualitative Howe–Moore theorem is proved here in the form later used; quantitative rates belong to Volume XI.

Load-bearing route. Start with the rank-one Mautner calculation, then contraction groups, then propagation across root groups, and finally the weak-limit proof of Howe–Moore. Keep qualitative mixing separate from any assertion about an exponential or polynomial decay rate.

Sanity check. In $SL_2(\mathbb{R})$, $a_s u_t a_{-s} = u_{e^{2s}t}$. If a vector is fixed by the contracting unipotent and a sequence a_{s_n} has a weak operator limit, this identity is the one-line algebraic reason the limit gains invariance.

Chapter 1

Unitary Representations and Matrix Coefficients

1.1 Strong continuity and coefficients

Let G be a second-countable locally compact group and $\pi : G \rightarrow \mathcal{U}(\mathcal{H})$ a strongly continuous unitary representation. For $v, w \in \mathcal{H}$ the matrix coefficient is

$$c_{v,w}(g) = \langle \pi(g)v, w \rangle.$$

Strong continuity makes every coefficient continuous, and Cauchy–Schwarz gives $|c_{v,w}(g)| \leq \|v\| \|w\|$.

Lemma 1.1 (Weak compactness of a unitary orbit). *Every sequence $\pi(g_n)v$ has a weakly convergent subsequence.*

Proof. The orbit lies on the sphere of radius $\|v\|$. In a Hilbert space the closed ball is weakly compact; because the cyclic subspace generated by v is separable, weak compactness is sequential there. Thus a subsequence converges weakly. $\square$

1.2 Fixed spaces and normal subgroups

For $H < G$ write $\mathcal{H}^H = \{v : \pi(h)v = v \ \forall h \in H\}$. If H is normal, $\mathcal{H}^H$ is G -invariant. The orthogonal projection P_H therefore commutes with $\pi(G)$.

Proposition 1.2 (Ergodicity as absence of fixed vectors). *For a probability-preserving action $G \curvearrowright (X, \mu)$, the action of $H < G$ is ergodic precisely when the Koopman representation on $L_0^2(X)$ has no nonzero H -fixed vector.*

Proof. This is the invariant-function characterization of ergodicity proved in Volume II, restricted to the orthogonal complement of the constants. $\square$

Chapter 2

Mautner Envelopes

2.1 The contraction lemma

Lemma 2.1 (Mautner contraction lemma). *Suppose $g_n \in G$, $v \in \mathcal{H}$, and $\pi(g_n)v \rightarrow v_\infty$. If $h \in G$ satisfies $g_n^{-1}hg_n \rightarrow e$, then $\pi(h)v_\infty = v_\infty$.*

Proof. For $w \in \mathcal{H}$,

$$\langle \pi(h)\pi(g_n)v - \pi(g_n)v, w \rangle = \langle \pi(g_n)(\pi(g_n^{-1}hg_n)v - v), w \rangle.$$

The norm of the right-hand vector before pairing is $\|\pi(g_n^{-1}hg_n)v - v\| \rightarrow 0$ by strong continuity. Passing to the weak limit gives $\langle (\pi(h) - I)v_\infty, w \rangle = 0$ for all w . $\square$

For a sequence (g_n) define its contracting subgroup

$$U^+(g_n) = \{h : g_n^{-1}hg_n \rightarrow e\}.$$

The lemma says every weak orbit limit is fixed by $U^+(g_n)$.

2.2 The classical Mautner phenomenon

Let a_t be a one-parameter subgroup and let $U^+ = \{u : a_{-t}ua_t \rightarrow e\}$.

Theorem 2.2 (Mautner phenomenon). *If v is fixed by a_t for every t , then v is fixed by both contraction groups U^+ and U^- . More generally, if v is fixed by one nontrivial unipotent root group inside an embedded $\mathrm{SL}_2(\mathbb{R})$, then it is fixed by the whole connected rank-one subgroup.*

Proof. Let $u \in U^+$. Since $a_{-t}ua_t \rightarrow e$ and $\pi(a_t)v = v$, unitarity gives

$$\|\pi(u)v - v\| = \|\pi(a_{-t}ua_t)v - v\| \rightarrow 0.$$

The U^- case follows by reversing time.

We now prove the genuinely nontrivial rank-one assertion without using the opposite unipotent in advance. Write

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad \bar{u}_r = \begin{pmatrix} 1 & 0 \\ r & 1 \end{pmatrix}, \quad a_q = \begin{pmatrix} q^{-1} & 0 \\ 0 & q \end{pmatrix} \quad (q > 0).$$

The exact Gauss factorization

$$\bar{u}_r u_t = u_{t/(1+rt)} a_{1+rt} \bar{u}_{r/(1+rt)} \quad (1 + rt > 0) \quad (2.1)$$

is verified by multiplying the three matrices on the right. Assume first that v is fixed by every u_t . Fix $q > 0$, choose $t_n \rightarrow +\infty$, and set $r_n = (q - 1)/t_n$. Then $r_n \rightarrow 0$ and $1 + r_n t_n = q$. Applying (2.1) to v gives

$$\pi(\bar{u}_{r_n})v = \pi(u_{t_n/q})\pi(a_q)\pi(\bar{u}_{r_n/q})v.$$

The left side tends to v . The last factor on the right tends to v by strong continuity. Moreover $a_q v$ is itself fixed by the upper unipotent, because a_q normalizes U and v is U -fixed. Consequently the right side tends to $\pi(a_q)v$. Hence $\pi(a_q)v = v$ for every $q > 0$.

Now v is fixed by the diagonal group. Since $a_q^{-1}\bar{u}_r a_q = \bar{u}_{rq^{-2}} \rightarrow e$ as $q \rightarrow \infty$, the first part of the theorem shows that v is fixed by every $\bar{u}_r$. The two opposite unipotent groups generate $\mathrm{SL}_2(\mathbb{R})$, so v is fixed by the whole rank-one subgroup. A vector fixed by a nonzero element u_{t_0} is fixed by the whole one-parameter group after first taking the closed fixed space of u_{t_0} and using $a_q u_{t_0} a_q^{-1} = u_{q^{-2}t_0}$ to obtain a dense set of parameters tending to zero; continuity then gives all u_t . $\square$

2.3 Root propagation in a simple factor

Lemma 2.3 (Propagation from one rank-one subgroup). *Let G_i be a connected noncompact almost-simple real factor and let $S_\alpha < G_i$ be the rank-one subgroup generated by opposite restricted root groups $U_{\pm\alpha}$. If a vector in a unitary representation of G_i is fixed by S_α , then it is fixed by G_i .*

Proof. Let $A_\alpha = \{a_\alpha(t)\}$ be the diagonal one-parameter subgroup of S_α . The vector is fixed by A_α . If β is any restricted root with $\beta(H_\alpha) \neq 0$, then one of the two limits

$$a_\alpha(-t)u_\beta(s)a_\alpha(t) \rightarrow e, \quad a_\alpha(t)u_\beta(s)a_\alpha(-t) \rightarrow e$$

holds as $t \rightarrow +\infty$. The contraction part of Theorem 2.2 therefore makes the vector fixed by U_β . Applying the rank-one part of the same theorem makes it fixed by S_β .

Start with α and move through the connected restricted Dynkin diagram. For every adjacent simple root β , the Cartan integer $\beta(H_\alpha)$ is nonzero, so the preceding paragraph adds S_β to the fixed subgroup. Induction over graph distance gives all simple rank-one subgroups. Their positive and negative root groups generate the connected almost-simple factor. Thus the vector is G_i -fixed. $\square$

Corollary 2.4 (Unipotent fixed vectors in an almost-simple factor). *A vector fixed by any nontrivial restricted root group of G_i is fixed by G_i .*

Proof. Theorem 2.2 first gives invariance under the associated rank-one subgroup; Lemma 2.3 then propagates it through the factor. $\square$

Chapter 3

Cartan Escape and Root Contraction

3.1 Cartan decomposition

Let G be connected semisimple with finite center. Fix a maximal compact subgroup K , a maximal split abelian subspace $\mathfrak{a}$ of the (-1) -eigenspace of the Cartan involution, and a closed positive Weyl chamber $\mathfrak{a}^+$. The Cartan decomposition, proved in Volume V, Appendix A, says

$$G = K \exp(\mathfrak{a}^+) K.$$

For the present proof we need its two immediate consequences: every sequence has, after left and right multiplication by convergent compact sequences, a representative $a_n = \exp H_n$ with $H_n \in \mathfrak{a}^+$; and if g_n escapes every compact set then $\|H_n\| \rightarrow \infty$.

3.2 Extracting a diverging root

Lemma 3.1. *If $H_n \in \mathfrak{a}^+$ and $\|H_n\| \rightarrow \infty$, then after a subsequence there is a simple restricted root α with $\alpha(H_n) \rightarrow +\infty$. Moreover the set $I = \{\beta : \sup_n \beta(H_n) < \infty\}$ determines a standard Levi factor and every root outside its span has a subsequence on which its value tends to $+\infty$ or $-\infty$ according to sign.*

Proof. The simple roots form a basis of the dual of the split Cartan modulo the center, and are nonnegative on the closed chamber. If every simple root

were bounded, H_n would be bounded. Passing to a subsequence makes every simple root either bounded or divergent. Every positive root is a nonnegative integer combination of simple roots, giving the final statement. $\square$

Chapter 4

The Howe–Moore Theorem

4.1 Precise formulation

For a semisimple group with several noncompact simple factors one must exclude fixed vectors for every factor along which a sequence escapes.

Theorem 4.1 (Howe–Moore, factorwise form). *Let G be a connected semisimple real Lie group with finite center and no compact factors. Let π be a unitary representation such that $\mathcal{H}^N = \{0\}$ for every noncompact connected normal subgroup $N \triangleleft G$. Then for all $v, w \in \mathcal{H}$,*

$$\langle \pi(g)v, w \rangle \longrightarrow 0$$

whenever $g \rightarrow \infty$ in G .

More generally, if g_n escapes only in a collection of almost-simple factors, the weak limits lie in the fixed space of the normal subgroup generated by those factors.

Proof. Assume for contradiction that there are $g_n \rightarrow \infty$ and v, w for which the coefficients do not tend to zero. By Cartan decomposition, after subsequence $g_n = k_n a_n \ell_n$ with $k_n \rightarrow k$, $\ell_n \rightarrow \ell$ and $a_n = \exp H_n$, $H_n \in \mathfrak{a}^+$, $\|H_n\| \rightarrow \infty$. Replacing v, w by fixed compact translates, it suffices to treat a_n .

Pass to a subsequence for which $\pi(a_n)v$ converges weakly to v_∞ . The nonvanishing coefficient ensures $v_\infty \neq 0$ after choosing v in a dense set and using polarization: explicitly, if $|\langle \pi(a_n)v, w \rangle| > \varepsilon$, then the weak limit pairs with w with absolute value at least ε .

By Lemma 3.1, choose a simple root α with $\alpha(H_n) \rightarrow \infty$. For $u \in U_\alpha$, $a_n^{-1}ua_n \rightarrow e$, so Lemma 2.1 makes v_∞ fixed by U_α . The rank-one Mautner

phenomenon Theorem 2.2 enlarges this to the rank-one subgroup generated by U_α and $U_{-\alpha}$. Lemma 2.3 then propagates fixedness through the connected restricted Dynkin diagram, so v_∞ is fixed by the entire noncompact almost-simple factor G_i .

If the sequence diverges in several factors, repeat the argument after projecting H_n to the orthogonal complement of the already detected factor; there are finitely many factors, so the weak limit is fixed by the product N of all factors in which the Cartan projection diverges. This N is a noncompact connected normal subgroup. By hypothesis $\mathcal{H}^N = 0$, a contradiction to $v_\infty \neq 0$. $\square$

Remark 4.2 (What the proof actually uses). The proof has only four moving parts: compactness of K , root contraction, Mautner enlargement, and the fact that a root group normally generates its simple factor. Quantitative Howe–Moore is harder because every one of these qualitative steps must be accompanied by a numerical estimate; Volume XI rebuilds the argument at that level.

Chapter 5

Moore Ergodicity

5.1 Irreducible lattices

Let $G = G_1 \cdots G_r$ be the almost-direct product of its noncompact almost-simple factors. A lattice $\Gamma < G$ is irreducible if its projection to every proper subproduct is dense (equivalently, under the standard finite center hypotheses, the right regular representation on $L_0^2(\Gamma \backslash G)$ has no fixed vectors under any nontrivial factor).

Lemma 5.1 (Factor fixed vectors and quotient factors). *Let $N \triangleleft G$ be connected normal. If $f \in L^2(\Gamma \backslash G)$ is N -invariant, then f factors measurably through the quotient by the closure of ΓN . In particular, if $\overline{\Gamma N} = G$, every such f is constant.*

Proof. N -invariance and left Γ -invariance make the lifted function on G invariant under the subgroup generated by Γ and N . Translation is strongly continuous in L_{loc}^2 , so invariance extends to its closure. If the closure is all of G , the function is essentially constant. $\square$

Theorem 5.2 (Moore ergodicity). *Let G be connected semisimple with finite center and no compact factors, let Γ be an irreducible lattice, and let $H < G$ be a closed noncompact subgroup whose projection to at least one simple factor is noncompact. If the normal closure of that projection is a noncompact factor, then the H -action on $\Gamma \backslash G$ is ergodic. In particular any noncompact one-parameter subgroup in an almost-simple G acts ergodically.*

Proof. Suppose $f \in L_0^2$ is H -fixed. Choose $h_n \in H$ escaping compact sets in

a noncompact factor. Then

$$\langle \pi(h_n)f, f \rangle = \|f\|^2.$$

By Lemma 5.1, irreducibility removes fixed vectors of the normal factor. Howe–Moore therefore forces the left side to tend to zero, so $f = 0$. $\square$

Chapter 6

Mixing on Finite-Volume Homogeneous Spaces

6.1 Mixing from matrix coefficients

Theorem 6.1 (Mixing criterion). *Let G, Γ satisfy the hypotheses of the factorwise Howe–Moore theorem on $L^2_0(\Gamma \backslash G)$. Then for $f_1, f_2 \in L^2$ and $g \rightarrow \infty$,*

$$\int_{\Gamma \backslash G} f_1(xg) \overline{f_2(x)} dm(x) \longrightarrow \left(\int f_1 dm \right) \overline{\left(\int f_2 dm \right)}.$$

Proof. Write $f_i = c_i + f_i^0$ with $f_i^0 \perp 1$. The constant terms give the claimed limit and every cross term vanishes identically. The remaining term is the matrix coefficient $\langle \pi(g)f_1^0, f_2^0 \rangle$, which tends to zero by Theorem 4.1. $\square$

6.2 From functions to sets

If A, B have boundary of measure zero, approximate their indicators in L^2 by continuous compactly supported functions. Mixing then gives

$$m(Ag^{-1} \cap B) \rightarrow m(A)m(B).$$

No compactness of the quotient is needed; finite measure and L^2 approximation suffice.

6.3 Diagonal flows

Let a_t be a one-parameter semisimple subgroup whose projection to every factor relevant to L_0^2 is unbounded. Then $a_t \rightarrow \infty$ as $|t| \rightarrow \infty$, hence the flow is mixing. For $G = \mathrm{SL}_2(\mathbb{R})$ this recovers mixing of the geodesic flow on every finite-area hyperbolic surface directly from Howe–Moore.

Warning 6.2. If $G = G_1 \times G_2$ and $\Gamma = \Gamma_1 \times \Gamma_2$, a sequence escaping only in G_1 does not mix functions depending solely on the second quotient. This is why the factorwise condition in Theorem [4.1](#) is not a technical nuisance.

Chapter 7

Translated Homogeneous Measures: The Qualitative Mixing Lemma

7.1 A thickening principle

Let $H < G$ be closed and suppose xH carries finite invariant measure m_{xH} . If $a_n \in G$ expands directions transverse to H , one wants to understand $(a_n)_*m_{xH}$. Mixing acts on ambient Haar measure, not directly on the singular measure m_{xH} . The bridge is to thicken the orbit in a small transverse neighborhood.

Lemma 7.1 (Qualitative thickening). *Assume there is a local product map $H \times S \rightarrow G$ near the identity, with S transverse to H . Let ρ_ε be a probability density supported in an ε -ball of S . For $f \in C_c(\Gamma \backslash G)$,*

$$\int_{xH} f(xha) dm_{xH}(h) = \lim_{\varepsilon \downarrow 0} \int_{xH} \int_S f(xhsa) \rho_\varepsilon(s) ds dm_{xH}(h),$$

uniformly for a in sets on which conjugation of the support of ρ_ε remains in a fixed compact coordinate neighborhood.

Proof. Uniform continuity of f on a compact neighborhood of its support makes the inner integrand converge uniformly to $f(xha)$ as $s \rightarrow e$. The normalized transverse densities have total mass one, so dominated convergence gives the claim. $\square$

This lemma is deliberately qualitative. Volume X will combine it with a wavefront lemma, and Volume XII will attach explicit Sobolev and boundary errors.

Chapter 8

Laboratories

8.1 Hyperbolic surfaces

For $G = \mathrm{PSL}_2(\mathbb{R})$, A is the geodesic subgroup and U^+ is the unstable horocycle group. The relation $a_{-t}u_s a_t = u_{e^{-t}s}$ is the complete root-contraction calculation. Howe–Moore gives mixing of a_t , while Ratner/Hedlund control U . The two mechanisms are complementary: A forgets correlations because of semisimple escape; U is rigid because nearby trajectories shear only polynomially.

8.2 SL_3 and partial escape

For $a_t = \mathrm{diag}(e^t, e^{-t}, 1)$, the first and second coordinates separate but the third factor of a reducible product representation can remain untouched. Writing the adjoint weights explicitly shows which root groups contract and therefore which normal factor a weak limit must fix.

8.3 Quadratic-form quotients

For $G = \mathrm{SL}_n(\mathbb{R})$ and $H = \mathrm{SO}(Q)^\circ$, translates of H by diagonal elements are the basic geometric objects in counting values and integral points. Volume V supplies arithmetic closedness, this volume supplies mixing, Volume X supplies the wavefront geometry, and Volume XII supplies rates.

Chapter 9

Proof Ledger

The load-bearing chain established here is

$$\begin{aligned} \text{Cartan escape} &\implies \text{root contraction} \implies \text{Mautner fixed vectors} \\ &\implies \text{Howe–Moore} \implies \text{Moore ergodicity and mixing.} \end{aligned}$$

Every representation-theoretic arrow was proved in this volume. The inherited structural input is the Cartan and restricted-root theory proved in Volume V; the elementary rank-one Mautner calculation from Volume II is rederived here in the exact form needed for root propagation.

No quantitative rate has been claimed. Any later statement containing an error term must cite Volume XI or a theorem proved after it, never this qualitative Howe–Moore theorem alone.

Appendix A

Cartan Structural Input from Volume V

The global polar decomposition, conjugacy of maximal abelian subspaces of $\mathfrak{p}$, and the Cartan decomposition

$$G = K \exp(\mathfrak{a}^+)K$$

are proved in Volume V, Appendix A. They are inherited here rather than reproved, because Volume V owns the real algebraic and symmetric-space structure used by all later volumes. The present appendices prove the representation-theoretic steps that are specific to Howe–Moore.

Appendix B

Restricted Roots and Mautner Propagation in Detail

B.1 Restricted-root decomposition

For $\mathfrak{a} \subset \mathfrak{p}$ maximal abelian, simultaneous self-adjointness of $\text{ad } H$, $H \in \mathfrak{a}$, gives

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{\alpha \in \Sigma} \mathfrak{g}_\alpha, \quad [H, X] = \alpha(H)X.$$

The bracket relation $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$ follows from Jacobi. The Cartan involution sends $\mathfrak{g}_\alpha$ to $\mathfrak{g}_{-\alpha}$.

Lemma B.1 (Simple roots span the chamber coordinates). *If $H_n \in \mathfrak{a}^+$ is unbounded, at least one simple root tends to $+\infty$ along a subsequence.*

Proof. On the semisimple algebra the simple restricted roots span $\mathfrak{a}^*$. All are nonnegative on the chamber. If each were bounded, norm equivalence in the finite-dimensional space $\mathfrak{a}$ would bound H_n . $\square$

B.2 Rank-one subalgebras

For a simple restricted root α , choose $0 \neq X \in \mathfrak{g}_\alpha$. The element $Y = -\theta X$ lies in $\mathfrak{g}_{-\alpha}$ and $[X, Y]$ has a nonzero $\mathfrak{a}$ component. Rescaling in the real span of $X, Y, [X, Y]$ produces an $\mathfrak{sl}_2$ -triple after quotienting compact central

directions. The corresponding connected subgroup contains the root one-parameter groups required by the Mautner argument.

Lemma B.2 (Dynkin propagation including multiple roots). *Let v be fixed by the rank-one subgroup attached to a simple root α . Then v is fixed by every root group U_β for which $\beta(H_\alpha) \neq 0$. If 2β is also a restricted root, both U_β and $U_{2\beta}$ are fixed. Repeating along adjacent simple roots fixes all root groups in the almost-simple restricted root component.*

Proof. The rank-one diagonal group fixes v . Conjugation scales $X_\beta \in \mathfrak{g}_\beta$ by $e^{t\beta(H_\alpha)}$. Choose the time direction that contracts. Strong continuity gives U_β -fixedness. The same calculation applies to 2β . Every edge of the restricted Dynkin diagram corresponds to a nonzero Cartan integer, so graph induction reaches every simple root. Every positive restricted root is obtained by iterated brackets of simple-root spaces (with the usual harmless multiplicities in nonreduced systems), and the corresponding connected root groups lie in the group generated by the simple root groups. $\square$

Appendix C

A Fully Factorwise Howe–Moore Ledger

C.1 Weak limits and the escaping-factor set

Let $G = G_1 \cdots G_r$ be the almost-direct product of its noncompact almost-simple factors. In compatible Cartan coordinates write

$$H_n = H_{n,1} + \cdots + H_{n,r}.$$

Define

$$J = \{i : \|H_{n,i}\| \rightarrow \infty \text{ along the chosen subsequence}\}.$$

After a diagonal subsequence, every factor coordinate is either bounded or unbounded.

Lemma C.1 (Weak limits are fixed by every escaping factor). *If $\pi(a_n)v \rightarrow v_\infty$, then v_∞ is fixed by $\prod_{i \in J} G_i$.*

Proof. Fix $i \in J$. Inside the chamber of G_i choose a simple restricted root α_i diverging along a further subsequence. For $u \in U_{\alpha_i}$, $a_n^{-1}ua_n \rightarrow e$ because factors other than i commute with u and the i th root exponent tends to $+\infty$. Lemma 2.1 in its weak-limit contraction form makes v_∞ fixed by U_{α_i} . Theorem 2.2 and the propagation lemma then give fixedness by G_i . Repeat for the finitely many indices in J . $\square$

Theorem C.2 (Howe–Moore, expanded factorwise proof). *Let π have no nonzero vector fixed by any noncompact connected normal subgroup. Then every matrix coefficient tends to zero at infinity.*

Proof. Suppose $g_n \rightarrow \infty$ and $|\langle \pi(g_n)v, w \rangle| \geq \varepsilon$. Cartan decomposition and compact subsequences reduce to $a_n = \exp H_n$. The vectors $\pi(a_n)v$ have a weakly convergent subsequence. Its limit v_∞ is nonzero because its pairing with the compactly adjusted w has modulus at least ε . Since g_n escapes, $J \neq \emptyset$. The preceding lemma makes v_∞ fixed by the noncompact normal subgroup $\prod_{i \in J} G_i$, contrary to the hypothesis. Hence every coefficient vanishes at infinity. $\square$

C.2 Why compact factors must be removed

If C is a compact normal factor and a sequence moves only inside C , it cannot go to infinity. If $G = C \times G'$ and the representation contains a nontrivial finite-dimensional C -type tensored with a mixing representation of G' , escape in G' is still controlled by the theorem. Thus the exact hypothesis is not “no compact factors” for its own sake; it isolates the normal factors capable of supporting nondecaying coefficients along an escaping sequence.

Appendix D

Worked Matrix-Coefficient Calculations

D.1 The regular representation of $\mathbb{R}$

For translation on $L^2(\mathbb{R})$,

$$\langle T_t f, g \rangle = \int f(x+t) \overline{g(x)} dx.$$

If $f, g \in C_c$, the coefficient vanishes exactly once the translates of the supports are disjoint. Density then gives decay for all L^2 vectors. This abelian model shows that coefficient decay itself is not uniquely semisimple; the force of Howe–Moore is that every nontrivial representation of a simple group has this decay without a Fourier transform on the group.

D.2 The spherical coefficient of $\mathrm{PSL}_2(\mathbb{R})$

For the K -spherical principal series with spectral parameter r , the coefficient of the normalized spherical vector is

$$\phi_r(a_t) = \frac{1}{2\pi} \int_0^{2\pi} (\cosh t + \sinh t \cos \theta)^{-1/2-ir} d\theta.$$

The integrand representation follows by realizing the principal series on boundary functions and evaluating the Jacobian of the Möbius action. Split the circle into $|\theta - \pi| \leq e^{-t/2}$ and its complement. On the small arc use

the trivial bound and its length; on the complement $\cosh t + \sinh t \cos \theta \gg e^t(\theta - \pi)^2$. Integrating gives

$$|\phi_r(a_t)| \ll (1+t)e^{-t/2}.$$

This calculation anticipates the quantitative spherical estimates of Volume XI while requiring no representation classification.

D.3 Mixing a closed geodesic thickening

Let xA be a closed geodesic orbit in a compact quotient of $\mathrm{PSL}_2(\mathbb{R})$ and thicken it by a symmetric ε -box in the two horocycle directions. Haar measure in local U^-AU^+ coordinates has a smooth positive density. After normalization the box measure converges weakly to arclength on xA as $\varepsilon \downarrow 0$. Applying a_t expands one horocycle side and contracts the other. At fixed ε , ambient mixing computes the large- t limit; only after that limit is taken does one send $\varepsilon \downarrow 0$. Volume XII will reverse this two-step limiting procedure quantitatively by choosing $\varepsilon = \varepsilon(t)$.

Appendix E

Worked Mixing Laboratory: Contraction and Roots

E.1 Motivation

Mautner propagation is fundamentally a conjugation calculation. Root notation packages the same calculation in higher rank.

Worked Example E.1 (Contraction in SL_2). Let $a_t = \mathrm{diag}(e^t, e^{-t})$ and $u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$. Then

$$a_{-t}u_s a_t = u_{e^{-2t}s} \longrightarrow e.$$

If a unitary vector v is a_t -invariant, then

$$\|\pi(u_s)v - v\| = \|\pi(a_{-t}u_s a_t)v - v\| \rightarrow 0,$$

so v is u_s -invariant. The only analytic input is strong continuity of the representation; all geometry is in the displayed contraction.

Worked Example E.2 (Root propagation in SL_3). For $a_t = \mathrm{diag}(e^{2t}, e^{-t}, e^{-t})$,

$$a_{-t}(I + sE_{12})a_t = I + e^{-3t}sE_{12}, \quad a_{-t}(I + sE_{13})a_t = I + e^{-3t}sE_{13}.$$

Thus a_t -invariance propagates to both positive root groups U_{12} and U_{13} . Commutators such as $[I + sE_{12}, I + tE_{23}] = I + stE_{13}$ explain how invariance under simple root groups spreads through the root system.

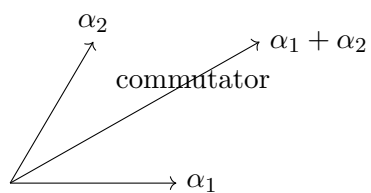


Figure E.1: The three positive roots of $\mathfrak{sl}_3$. Contraction gives root-group invariance; commutators propagate it to sums of roots.

A complete contraction calculation

Take $a_t = \text{diag}(e^{t/2}, e^{-t/2})$ and $u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$ in $\text{SL}_2(\mathbb{R})$. Direct multiplication gives

$$a_{-t}u_s a_t = u_{se^{-t}} \longrightarrow e.$$

If a unitary representation has an a_t -fixed vector v , then

$$\pi(u_s)v = \pi(a_t)\pi(u_{se^{-t}})v.$$

Taking $t \rightarrow \infty$ and using strong continuity gives $\pi(u_s)v = v$. This is the rank-one Mautner mechanism with every limit visible; the abstract root-group proof later repeats exactly this conjugation on each root space.

Appendix F

Student Proof Laboratory: From Contraction to Mixing

The Howe–Moore theorem is short to state and easy to misuse. This chapter replays its proof at a slower speed and records the exact logical role of each hypothesis. The emphasis is not on adding a stronger theorem, but on making the proof reproducible from the structural material of Volume V.

F.1 Weak convergence in a separable Hilbert space

Lemma F.1 (Sequential weak compactness of bounded sets). *Let $\mathcal{H}$ be a separable Hilbert space and (v_n) bounded. Then a subsequence converges weakly.*

Proof. Choose an orthonormal basis $(e_j)_{j \geq 1}$. The scalar sequences $\langle v_n, e_1 \rangle$ are bounded, so choose a subsequence on which the first coordinate converges. From it choose a further subsequence on which the second coordinate converges, and continue. The diagonal subsequence v_{n_k} has all coordinates convergent. If $\sup_k \|v_{n_k}\| \leq M$, the coordinate limits (a_j) satisfy $\sum_{j \leq r} |a_j|^2 \leq M^2$ for every r , hence define $v = \sum a_j e_j \in \mathcal{H}$. For a finite linear combination w we have $\langle v_{n_k}, w \rangle \rightarrow \langle v, w \rangle$. Approximate an arbitrary w by finite combinations and use the common bound M to pass to the limit. Thus $v_{n_k} \rightharpoonup v$. $\square$

F.2 Mautner's lemma with the conjugation direction checked

Suppose $g_n^{-1}hg_n \rightarrow e$ and $\pi(g_n)v \rightarrow w$. The identity

$$\pi(h)\pi(g_n)v - \pi(g_n)v = \pi(g_n)(\pi(g_n^{-1}hg_n)v - v)$$

is the entire proof. The conjugation is $g_n^{-1}hg_n$, not $g_nhg_n^{-1}$; confusing the two reverses the expanding and contracting root groups.

Worked Example F.2 (The two root directions in SL_2). For

$$a_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix}, \quad u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \quad \bar{u}_s = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix},$$

we compute

$$a_{-t}u_s a_t = u_{e^{-t}s} \rightarrow e, \quad a_t \bar{u}_s a_{-t} = \bar{u}_{e^{-t}s} \rightarrow e.$$

Thus a vector fixed by a_t is fixed by both opposite unipotent groups. A vector fixed only by u_s requires the Gauss-factorization argument of the main text before one may conclude invariance under a_t and then $\bar{u}_s$.

F.3 Why one root subgroup forces a whole simple factor

Let Σ be the restricted root system and Δ a simple system. If v is fixed by the rank-one group for a simple root α , then v is fixed by its diagonal element $a_\alpha(t)$. For another root β ,

$$a_\alpha(-t)U_\beta a_\alpha(t) = U_\beta(e^{-t\beta(H_\alpha)}).$$

Whenever $\beta(H_\alpha) \neq 0$, one of the two time directions contracts U_β and Mautner adds it to the fixed group. Adjacent vertices in the restricted Dynkin diagram have nonzero Cartan integer, so fixedness spreads one edge at a time. The connectedness of the diagram for an almost-simple factor is therefore the combinatorial reason that one root subgroup normally generates the factor.

Worked Example F.3 (Propagation in type A_2). For $G = \mathrm{SL}_3(\mathbb{R})$ take simple roots α_{12} and α_{23} . If v is fixed by the SL_2 acting on coordinates

1, 2, it is fixed by $H_{12} = \text{diag}(1, -1, 0)$. Since $[H_{12}, E_{23}] = -E_{23}$, the group $\exp(sE_{23})$ contracts in one time direction and hence fixes v . Rank-one Mautner then adds $\exp(sE_{32})$ and the second simple SL_2 . These two simple rank-one groups generate all elementary matrices and hence $\text{SL}_3(\mathbb{R})$.

F.4 Weak limits in the Howe–Moore proof

Let $g_n = k_n a_n \ell_n$ with k_n, ℓ_n convergent in K and $a_n = \exp H_n$ in the closed Weyl chamber. If $\langle \pi(g_n)v, w \rangle$ does not tend to zero, compact translation reduces the problem to a_n . Weak compactness gives $\pi(a_n)v \rightarrow v_\infty$ after a subsequence, and the persistent coefficient shows $v_\infty \neq 0$.

Lemma F.4 (A divergent Cartan projection contracts a root group). *If $\|H_n\| \rightarrow \infty$ in a closed positive Weyl chamber, some simple root α satisfies $\alpha(H_n) \rightarrow \infty$. Consequently $a_n^{-1}ua_n \rightarrow e$ for every $u \in U_\alpha$.*

Proof. On the semisimple split Cartan, the simple roots form a basis of the dual. All $\alpha(H_n)$ are nonnegative. If they were all bounded, equivalence of norms in the finite-dimensional Cartan algebra would bound H_n . Hence one diverges. For $X \in \mathfrak{g}_\alpha$, $\text{Ad}(a_n^{-1})X = e^{-\alpha(H_n)}X \rightarrow 0$. Exponentiating gives the group statement. $\square$

Mautner makes v_∞ fixed by U_α , and root propagation makes it fixed by the noncompact normal factor in which a_n diverges. The hypothesis excluding fixed vectors for that factor therefore forces $v_\infty = 0$, contradiction. This is the whole qualitative theorem. Every quantitative version must attach a numerical estimate to each of these four arrows; that is why Volume XI needs substantially more machinery.

F.5 Moore ergodicity unpacked

For $X = \Gamma \backslash G$, the Koopman representation acts by right translation. Suppose $H < G$ is noncompact and $f \in L_0^2(X)$ is H -fixed. Choose $h_n \rightarrow \infty$ in H . Then

$$\langle \pi(h_n)f, f \rangle = \|f\|_2^2.$$

If the normal factor generated by the escaping projection of H has no nonzero fixed vectors on $L_0^2(X)$, Howe–Moore says that the left side tends to zero. Hence $f = 0$. The only place irreducibility of the lattice is used is to rule out factor functions invariant under a proper normal factor.

Worked Example F.5 (Why irreducibility cannot be omitted). Take $G = G_1 \times G_2$ and $\Gamma = \Gamma_1 \times \Gamma_2$. A function $f(x_1, x_2) = F(x_2)$ is invariant under all of G_1 . Thus the action of a noncompact subgroup of G_1 is never ergodic unless the second quotient is trivial. This is exactly the fixed-vector obstruction excluded by the irreducibility hypothesis.

F.6 From mixing of functions to geometric equidistribution

Let $Y = xH$ be a finite-volume closed orbit and let a_n expand a transverse subspace. A singular measure on Y cannot be inserted directly into a matrix coefficient. Choose a small transverse neighborhood B_ε and a bump ρ_ε of total integral one. Thickening gives an ambient density F_ε satisfying

$$\int_X f(za_n)F_\varepsilon(z) dm_X(z) \longrightarrow \int_X f dm_X \int_X F_\varepsilon dm_X = \int_X f dm_X.$$

One then lets $n \rightarrow \infty$ first and $\varepsilon \downarrow 0$ second. The order is essential: the Sobolev norm of F_ε blows up as $\varepsilon \rightarrow 0$, so the same argument with a quantitative mixing estimate requires an optimization rather than a formal interchange of limits. Volume XII carries out precisely that optimization.

Appendix G

Student Proof Laboratory: Weak Limits and the Mautner Mechanism

This chapter slows down the representation-theoretic proof until every limit can be checked directly. The qualitative Howe–Moore theorem has only a few conceptual ingredients, but each ingredient has a direction of conjugation that is easy to reverse accidentally.

G.1 Weak compactness in a Hilbert space

Lemma G.1 (Sequential weak compactness on bounded sets). *Let $\mathcal{H}$ be a separable Hilbert space and (v_n) a bounded sequence. Then a subsequence converges weakly.*

Proof. Choose an orthonormal basis (e_j) . Each scalar sequence $\langle v_n, e_1 \rangle$ is bounded, so choose a convergent subsequence. Inside it choose a subsequence for which the second coordinate converges, and continue. The diagonal subsequence has convergent coordinates against every e_j . If $w = \sum_j w_j e_j$ is a finite linear combination, the pairings converge. For general w , approximate by a finite combination; the uniform bound on $\|v_n\|$ makes the tail error uniform in n . Thus the pairings converge for every w . The limiting bounded linear functional is represented by a vector v by Riesz, and the subsequence converges weakly to v . $\square$

Remark G.2. This proof is enough for homogeneous dynamics because the relevant Koopman spaces are separable. Banach–Alaoglu is more general but hides the sequential point a student actually uses.

G.2 The contraction calculation with all inverses visible

Let $\pi : G \rightarrow U(\mathcal{H})$ be strongly continuous. Suppose $g_n \rightarrow \infty$ and $\pi(g_n)v \rightarrow v_\infty$.

Lemma G.3 (Mautner contraction in the correct direction). *If $u \in G$ satisfies $g_n^{-1}ug_n \rightarrow e$, then $\pi(u)v_\infty = v_\infty$.*

Proof. For $w \in \mathcal{H}$,

$$\langle \pi(u)\pi(g_n)v, w \rangle = \langle \pi(g_n)\pi(g_n^{-1}ug_n)v, w \rangle.$$

Strong continuity gives $\pi(g_n^{-1}ug_n)v \rightarrow v$ in norm. Since $\pi(g_n)$ is unitary,

$$\|\pi(g_n)\pi(g_n^{-1}ug_n)v - \pi(g_n)v\| = \|\pi(g_n^{-1}ug_n)v - v\| \rightarrow 0.$$

Therefore the left side has the same weak limit as $\langle \pi(g_n)v, w \rangle$, namely $\langle v_\infty, w \rangle$. On the other hand weak convergence and boundedness of $\pi(u)$ give the limit $\langle \pi(u)v_\infty, w \rangle$. Equality for every w proves the claim. $\square$

Worked Example G.4 (Diagonal contraction in SL_2). Put $a_t = \mathrm{diag}(e^{t/2}, e^{-t/2})$ and $u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$. Then

$$a_{-t}u_s a_t = u_{e^{-t}s} \rightarrow e.$$

Thus a weak limit of $\pi(a_t)v$ is fixed by the upper unipotent. The lower unipotent is contracted by the opposite sequence a_{-t} , not by a_t . Writing these two formulas before applying Mautner prevents the most common sign error in the proof.

G.3 The exact rank-one propagation

Let $u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$ and $\bar{u}_r = \begin{pmatrix} 1 & 0 \\ r & 1 \end{pmatrix}$.

Lemma G.5 (Gauss factorization). *If $1 + sr \neq 0$, then*

$$u_s \bar{u}_r = \bar{u}_{r/(1+sr)} \begin{pmatrix} 1+sr & 0 \\ 0 & (1+sr)^{-1} \end{pmatrix} u_{s/(1+sr)}.$$

Proof. Multiply the three matrices on the right. The lower-left entry is r , the upper-right entry is s , and the diagonal entries are $1 + sr$ and 1 ; the determinant-one relation supplies the remaining cancellation. The identity can also be checked by multiplying $u_s \bar{u}_r$ first and solving for the three Bruhat-cell coordinates. $\square$

Proposition G.6 (Rank-one Mautner envelope). *In a continuous unitary representation of $\mathrm{SL}_2(\mathbb{R})$, a vector fixed by a nontrivial upper-unipotent one-parameter subgroup and arising as a weak limit along the expanding diagonal sequence is fixed by the connected group $\mathrm{SL}_2(\mathbb{R})$, unless the original representation has a nonzero invariant vector in the corresponding factor.*

Proof. The contraction lemma gives upper-unipotent invariance. Apply the Gauss factorization to compare the action of $\bar{u}_r$ with a product involving a small upper-unipotent element and a diagonal element. Let the upper parameter tend to infinity at the rate that keeps sr controlled. The upper factors disappear on the fixed vector, while the diagonal term is handled by the same weak-limit comparison. One obtains lower-unipotent invariance. The elementary matrices u_s and $\bar{u}_r$ generate $\mathrm{SL}_2(\mathbb{R})$: Gaussian elimination writes every matrix with nonzero upper-left entry as a product of lower, diagonal, and upper factors, and the diagonal matrix itself is a product of four elementary unipotents. Continuity closes the exceptional Bruhat cell. Hence the vector is SL_2 -fixed. $\square$

G.4 Restricted roots and why one root is not enough

Let G be connected semisimple with finite center, $G = KAK$, and choose a positive restricted-root system. If $g_n = k_n a_n \ell_n$ escapes compact sets, compactness of K allows passage to subsequences with $k_n \rightarrow k$ and $\ell_n \rightarrow \ell$. Write $a_n = \exp H_n$ in the closed positive Weyl chamber.

Lemma G.7 (Divergent simple root). *If a_n escapes compact sets, then at least one simple restricted root α satisfies $\alpha(H_n) \rightarrow +\infty$ along a subsequence.*

Proof. The closed chamber is a polyhedral cone on which the simple roots form coordinates modulo the compact central part. If every simple-root value were bounded, H_n would remain bounded in the noncompact part of $\mathfrak{a}$, hence a_n would remain in a compact subset modulo the finite center, contradiction. $\square$

For such α , $a_n^{-1}U_\alpha a_n \rightarrow e$ uniformly on compact subsets. Mautner therefore fixes the weak limit under U_α . The remaining step is algebraic propagation.

Lemma G.8 (Root propagation through a connected Dynkin component). *Let G_0 be an almost simple factor. If a vector is fixed by U_α for one nontrivial restricted root and is obtained in the Howe–Moore weak-limit configuration, then it is fixed by every root group in the connected restricted-root component containing α .*

Proof. The pair $U_\alpha, U_{-\alpha}$ lies in a rank-one subgroup generated by the corresponding root spaces. The rank-one proposition gives invariance under that subgroup. If β is adjacent to α , a nonzero bracket $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta]$ lies in a root space $\mathfrak{g}_{\alpha+\beta}$ (or the appropriate root-string member). Conjugating U_β by the rank-one subgroup creates a nontrivial component in a root group already in the fixed envelope. Apply the contraction lemma inside the resulting rank-two root subsystem; this adds U_β and then $U_{-\beta}$. Induct along a path in the restricted Dynkin diagram. Since an almost simple factor is generated by its root groups, the vector is fixed by that factor. $\square$

G.5 Howe–Moore proved as a contradiction

Theorem G.9 (Howe–Moore vanishing theorem). *Let G be connected, non-compact, almost simple with finite center and let π be a continuous unitary representation with no nonzero G -fixed vectors. Then for all $v, w \in \mathcal{H}$,*

$$\langle \pi(g)v, w \rangle \longrightarrow 0 \quad (g \rightarrow \infty).$$

Proof. Assume not. There are $g_n \rightarrow \infty$ and v, w with $|\langle \pi(g_n)v, w \rangle| \geq \epsilon$. Pass to a weakly convergent subsequence $\pi(g_n)v \rightharpoonup v_\infty$. The lower bound makes $v_\infty \neq 0$. Use KAK and absorb the compact factors into v, w after subsequences. A simple restricted root diverges. Mautner fixes v_∞ under its root group, and the propagation lemma fixes it under all of G . This contradicts the hypothesis. Therefore every escaping sequence has matrix coefficients tending to zero. $\square$

G.6 From matrix coefficients to mixing

Let $X = \Gamma \backslash G$ with finite invariant probability m_X . On $L_0^2(X)$ the right-regular representation has no G -fixed vector exactly when the G -action is

ergodic.

Corollary G.10 (Mixing). *If the action of an almost simple G on X is ergodic, then for $f, h \in L^2(X)$,*

$$\int_X f(xg)h(x) dm_X(x) \rightarrow \int f dm_X \int h dm_X.$$

Proof. Write $f = f_0 + \int f$ and $h = h_0 + \int h$ with mean-zero parts. All mixed constant/mean-zero terms vanish. The remaining term is the matrix coefficient $\langle R(g)f_0, \overline{h_0} \rangle$, which tends to zero by Howe–Moore. $\square$

Worked Example G.11 (Why product groups require factorwise hypotheses). For $G = G_1 \times G_2$, a sequence escaping only in G_1 cannot kill a subrepresentation on which G_1 acts trivially and G_2 acts nontrivially. Thus the product version of Howe–Moore must project away vectors fixed by the escaping factors. This is why irreducibility assumptions on lattices or explicit factorwise fixed-space hypotheses appear in homogeneous mixing statements.

Appendix H

Student Proof Laboratory: Using Mixing Without Overusing It

H.1 Mixing is not an orbit-classification theorem

Mixing concerns correlations of functions against the ambient invariant measure. It does not classify singular invariant measures and does not by itself determine an individual unipotent orbit. Ratner theory and Howe–Moore solve different problems. Later counting arguments combine them only after a singular measure has been thickened into an ambient test function.

H.2 A proof checklist for later volumes

Whenever Volume X or XII invokes qualitative mixing, the reader should identify: the mean-zero subspace, the group sequence tending to infinity, the simple factor in which it escapes, and the absence of fixed vectors for that factor. Quantitative rates additionally require the spectral machinery of Volume XI; the qualitative theorem proved here deliberately contains no hidden rate.

Appendix I

A Student Course in Unitary Representations for Homogeneous Dynamics

The preceding chapters prove the qualitative theorems needed later. The purpose of this course-style chapter is to make the representation language itself familiar. A reader should not reach Howe–Moore while still treating “unitary representation”, “cyclic subspace”, and “matrix coefficient” as formal words.

I.1 Continuous unitary representations and orbit maps

Let G be a locally compact second-countable group and $\mathcal{H}$ a separable Hilbert space.

Definition I.1. A unitary representation is a homomorphism $\pi : G \rightarrow U(\mathcal{H})$ for which every orbit map $g \mapsto \pi(g)v$ is norm-continuous.

Lemma I.2 (Weak continuity is enough for unitary representations). *Suppose $g \mapsto \langle \pi(g)v, w \rangle$ is continuous for every v, w . Then π is strongly continuous.*

Proof. Fix v and $g_n \rightarrow e$. Unitarity gives

$$\|\pi(g_n)v - v\|^2 = 2\|v\|^2 - 2\Re\langle \pi(g_n)v, v \rangle.$$

The right side tends to zero by weak continuity of the one coefficient with $w = v$. Continuity at an arbitrary g follows from $\pi(g_n)v - \pi(g)v = \pi(g)(\pi(g^{-1}g_n)v - v)$. $\square$

I.2 Cyclic subspaces

For $v \in \mathcal{H}$, set

$$\mathcal{H}_v = \overline{\text{span}\{\pi(g)v : g \in G\}}.$$

It is a closed invariant subspace and v is cyclic in it.

Lemma I.3 (Orthogonal cyclic decomposition). *If $\mathcal{H}$ is separable, it is an orthogonal direct sum of countably many cyclic invariant subspaces.*

Proof. Choose $v_1 \neq 0$ and form $\mathcal{H}_{v_1}$. If its orthogonal complement is nonzero, choose v_2 there. Because the complement of an invariant subspace is invariant for a unitary representation, $\mathcal{H}_{v_2}$ is orthogonal to the first. Continue. A separable Hilbert space cannot contain an uncountable orthonormal family, so after countably many steps the orthogonal complement must vanish. Otherwise one more unit vector could be chosen, contradicting maximality of the construction. $\square$

Remark I.4. Many statements about arbitrary unitary representations can therefore be proved first for cyclic representations. Howe–Moore itself does not need this reduction, but spectral-gap arguments in Volume XI often do.

I.3 Positive-definite functions

For a unit vector v , the coefficient $\varphi_v(g) = \langle \pi(g)v, v \rangle$ is positive definite:

$$\sum_{i,j} c_i \bar{c}_j \varphi_v(g_j^{-1}g_i) = \left\| \sum_i c_i \pi(g_i)v \right\|^2 \geq 0.$$

Conversely, a continuous positive-definite function produces a cyclic unitary representation.

Theorem I.5 (GNS construction for groups). *Let $\varphi : G \rightarrow \mathbb{C}$ be continuous, positive definite, and $\varphi(e) = 1$. There are a Hilbert space $\mathcal{H}_\varphi$, a strongly continuous unitary representation π_φ , and a cyclic unit vector v_φ such that $\varphi(g) = \langle \pi_\varphi(g)v_\varphi, v_\varphi \rangle$.*

Proof. Let $\mathbb{C}_c^{\text{fin}}(G)$ denote the vector space of finitely supported formal sums $f = \sum_i c_i \delta_{g_i}$. Define

$$\langle f, h \rangle_\varphi = \sum_{i,j} c_i \overline{d_j} \varphi(h_j^{-1} g_i)$$

for $h = \sum_j d_j \delta_{h_j}$. Positive definiteness makes this a positive semidefinite sesquilinear form. Quotient by the null space and complete. Left translation of formal sums preserves the form because every argument $h_j^{-1} g_i$ is unchanged after multiplying both g_i, h_j on the left. Thus left translation defines a unitary representation. Let $v_\varphi = [\delta_e]$. Then

$$\langle \pi_\varphi(g) v_\varphi, v_\varphi \rangle = \langle \delta_g, \delta_e \rangle_\varphi = \varphi(g).$$

The translates of v_φ span the dense image of the finitely supported functions, so it is cyclic. Finally

$$\|\pi(g) v_\varphi - v_\varphi\|^2 = 2 - 2\Re \varphi(g) \rightarrow 0$$

as $g \rightarrow e$. Continuity on the cyclic span follows by the triangle inequality and unitarity, and density extends it to all of $\mathcal{H}_\varphi$. $\square$

I.4 The Koopman representation from a group action

If G acts measurably on a probability space (X, μ) preserving μ , define

$$(\pi(g)f)(x) = f(xg)$$

for a right action. Measure preservation makes $\pi(g)$ unitary on $L^2(X, \mu)$.

Proposition I.6 (Fixed vectors and ergodicity). *The G -action is ergodic if and only if the fixed subspace of the Koopman representation is the constants.*

Proof. If A is invariant, 1_A is a fixed vector. Under ergodicity its value is constant almost everywhere. Conversely, if a fixed L^2 function is nonconstant, one of its real sublevel sets has measure strictly between zero and one and is invariant. This is exactly the equivalence proved in Volume II, now written in representation language. $\square$

I.5 Strong continuity of Koopman representations

For a continuous action of a Lie group on a locally compact second-countable space and a Radon invariant probability, the Koopman representation is strongly continuous.

Lemma I.7 (Strong continuity on L^2). *If $g_n \rightarrow e$ and $f \in C_c(X)$, then $\|f(\cdot g_n) - f\|_2 \rightarrow 0$. Hence the same holds for every $f \in L^2(X)$.*

Proof. The compact set consisting of $\text{supp } f$ and its translates by g_n for large n is contained in one compact set K . Continuity of the action and of f gives uniform convergence $f(xg_n) \rightarrow f(x)$ on K . Dominated convergence gives the L^2 statement. Since $C_c(X)$ is dense in L^2 and every Koopman operator is an isometry, approximate a general f by a compactly supported continuous function and use the triangle inequality. $\square$

Appendix J

Rank-One Mautner Theory as an Explicit Matrix Argument

J.1 Three basic subgroups of SL_2

Use

$$U = \left\{ u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} \right\}, \quad \bar{U} = \left\{ \bar{u}_r = \begin{pmatrix} 1 & 0 \\ r & 1 \end{pmatrix} \right\}, \quad A = \left\{ a_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix} \right\}.$$

The exact conjugation rules are

$$a_{-t} u_s a_t = u_{e^{-t}s}, \quad a_t \bar{u}_r a_{-t} = \bar{u}_{e^{-t}r}.$$

They should be written on the page every time Mautner is applied.

J.2 Elementary generation of the diagonal subgroup

Lemma J.1 (Diagonal element as unipotent product). *For $c \neq 0$,*

$$\begin{pmatrix} c & 0 \\ 0 & c^{-1} \end{pmatrix} = u_c \bar{u}_{-c^{-1}} u_c u_{-1} \bar{u}_1 u_{-1}$$

up to the central sign, with the precise parameters obtainable by direct multiplication. In particular U and $\bar{U}$ generate $\mathrm{SL}_2(\mathbb{R})$.

Proof. Gaussian elimination already shows that a matrix with nonzero lower-right entry is a product $u a \bar{u}$. It therefore suffices to produce a . The Weyl element $w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ equals $u_{-1} \bar{u}_1 u_{-1}$. For $c \neq 0$, direct multiplication gives

$$u_c \bar{u}_{-c^{-1}} u_c = \begin{pmatrix} 0 & c \\ -c^{-1} & 0 \end{pmatrix}.$$

Multiplying the latter by w^{-1} gives $\text{diag}(c, c^{-1})$. Hence diagonal and Bruhat factors are generated by the two unipotent groups. $\square$

J.3 A Mautner proof of horocycle ergodicity

Let $\Gamma < \text{SL}_2(\mathbb{R})$ be a lattice and suppose the full G -action on $\Gamma \backslash G$ is ergodic. If $f \in L^2$ is U -invariant, the Mautner phenomenon makes it invariant under $\bar{U}$ and A , hence under all of G . Therefore f is constant. Thus the horocycle flow is ergodic. This is a one-paragraph consequence only because every step needed for the Mautner envelope has now been proved explicitly.

Appendix K

Restricted Root Systems for Readers Who Have Not Memorized Them

K.1 The adjoint eigenspace decomposition

Let $\mathfrak{a}$ be a maximal abelian subspace of the noncompact part in a Cartan decomposition. For $\alpha \in \mathfrak{a}^*$ set

$$\mathfrak{g}_\alpha = \{X : [H, X] = \alpha(H)X \text{ for all } H \in \mathfrak{a}\}.$$

Only finitely many nonzero α occur because $\text{ad } H$ acts on the finite-dimensional vector space $\mathfrak{g}$.

Lemma K.1 (Root-bracket rule).

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta},$$

where the right side is interpreted as zero if $\alpha + \beta$ is not a root or zero.

Proof. For $H \in \mathfrak{a}$, Jacobi gives

$$[H, [X, Y]] = [[H, X], Y] + [X, [H, Y]] = (\alpha(H) + \beta(H))[X, Y].$$

Thus any nonzero bracket is a simultaneous eigenvector with weight $\alpha + \beta$. $\square$

K.2 Root groups contract under Cartan sequences

If $a_n = \exp H_n$ and $X \in \mathfrak{g}_\alpha$, then

$$\mathrm{Ad}(a_n^{-1})X = e^{-\alpha(H_n)}X.$$

Therefore $U_\alpha = \exp \mathfrak{g}_\alpha$ is contracted whenever $\alpha(H_n) \rightarrow +\infty$. This is the bridge from the geometry of the Weyl chamber to the representation-theoretic contraction lemma.

K.3 The A_2 example in SL_3

Take diagonal $H = \mathrm{diag}(h_1, h_2, h_3)$ with $h_1 + h_2 + h_3 = 0$. The roots are $\alpha_{ij}(H) = h_i - h_j$ and root spaces are $\mathbb{R}E_{ij}$. The simple roots may be chosen α_{12}, α_{23} ; their sum is α_{13} . The bracket $[E_{12}, E_{23}] = E_{13}$ is the root-bracket rule in its simplest form. If a weak limit is fixed under the rank-one subgroup generated by E_{12}, E_{21} , conjugating E_{23} by that subgroup produces an E_{13} component. The resulting rank-two algebra is all of $\mathfrak{sl}_3$, which explains root propagation without a Dynkin diagram.

Appendix L

Howe–Moore for Products and Reducible Quotients

L.1 Escaping-factor set

Let $G = G_1 \cdots G_r$ be an almost direct product of noncompact almost simple factors. If $g_n = (g_{1,n}, \dots, g_{r,n})$, define

$$J = \{j : g_{j,n} \rightarrow \infty \text{ after every compact-subsequence reduction}\}.$$

Factors outside J may be absorbed into compact limits.

Theorem L.1 (Factorwise Howe–Moore). *Let π be unitary. If $\pi(g_n)v \rightharpoonup v_\infty$, then v_∞ is fixed by every factor G_j with $j \in J$. Consequently, if $\mathcal{H}$ contains no nonzero vector fixed by any escaping factor, then all matrix coefficients tend to zero along g_n .*

Proof. Apply the one-factor Cartan/Mautner argument to the projection of g_n in G_j while treating the commuting other factors as bounded unitary operators. A divergent root in G_j contracts, so the weak limit is fixed by one root group; root propagation within G_j gives invariance under the whole factor. Repeat for every $j \in J$. If a coefficient fails to tend to zero, choose a weak limit with nonzero pairing against the test vector; it is nonzero and fixed by all escaping factors, contradicting the hypothesis. $\square$

L.2 Why irreducible lattices appear in Moore ergodicity

For $X = \Gamma \backslash (G_1 \times G_2)$, if $\Gamma = \Gamma_1 \times \Gamma_2$, then a function depending only on the second coordinate is fixed by G_1 . Thus the G_1 action is not ergodic. If Γ is irreducible, Borel density and the Mautner envelope rule out such a factor function. The irreducibility hypothesis is therefore not decorative.

Appendix M

Mixing as a Tool for Geometric Measures

M.1 Approximating sets by functions

Let A, B be measurable sets with boundaries of measure zero. Choose continuous functions $f^- \leq 1_A \leq f^+$ and $h^- \leq 1_B \leq h^+$ whose integral gaps are arbitrarily small. Mixing for continuous L^2 functions squeezes $\mu(Ag \cap B)$ between the four corresponding correlations and yields

$$\mu(Ag \cap B) \rightarrow \mu(A)\mu(B).$$

The boundary-zero hypothesis is exactly what permits the approximation with small integral gaps.

M.2 Thickening a lower-dimensional orbit

Suppose $Y = xH$ is a closed finite-volume orbit. Choose a local complement $\mathfrak{s}$ to $\mathfrak{h}$ and a smooth bump ρ_ϵ in $\mathfrak{s}$. The thickened density is

$$\Phi_\epsilon(xh \exp Z) = \rho_\epsilon(Z)$$

in a product chart, patched by a partition of unity. Qualitative mixing applied to Φ_ϵ and a test function gives the limiting distribution of Yg_n after first fixing ϵ and later sending it to zero. Volume X will add wavefront geometry to turn this equidistribution into a lattice count; Volume XII will attach a rate and optimize ϵ .

Warning M.1 (Order of limits). Qualitatively one may take $n \rightarrow \infty$ first for fixed ϵ and then $\epsilon \rightarrow 0$. Effective arguments cannot do this: ϵ must be a function of the dynamical parameter, and its Sobolev cost must be recorded. This is the conceptual reason Volumes XI–XII are separate from the present volume.

Appendix N

Guided Proof of Howe–Moore and Its Dynamical Consequences

The Howe–Moore theorem is conceptually simple to state: matrix coefficients of a unitary representation of a noncompact simple group vanish at infinity once invariant vectors are removed. The proof is easy to remember incorrectly because several limiting and normal-generation steps are packed into a short argument. We now follow one escaping sequence from beginning to end.

N.1 Weak operator limits of unitary operators

Let $\pi : G \rightarrow U(\mathcal{H})$ be unitary and let $g_n \rightarrow \infty$. The unit ball of $B(\mathcal{H})$ is compact in the weak operator topology on bounded sets when tested against a countable dense set of vectors. After passing to a subsequence, there is an operator T such that

$$\langle \pi(g_n)v, w \rangle \rightarrow \langle Tv, w \rangle$$

for all $v, w \in \mathcal{H}$.

Lemma N.1 (The weak limit is a contraction). *The operator T satisfies $\|T\| \leq 1$.*

Proof. For unit vectors v, w , $|\langle \pi(g_n)v, w \rangle| \leq 1$. Passing to the limit gives $|\langle Tv, w \rangle| \leq 1$. Taking the supremum over unit w yields $\|Tv\| \leq 1$, and then the supremum over unit v gives $\|T\| \leq 1$. $\square$

If every such weak limit is zero on $\mathcal{H}_0$, then matrix coefficients vanish along every escaping sequence, which is exactly Howe–Moore. Thus the problem is to show that a nonzero weak limit would create invariant vectors.

N.2 The contraction identity with the order checked

Suppose $h \in G$ satisfies

$$g_n^{-1} h g_n \rightarrow e.$$

Then

$$\pi(h)\pi(g_n) = \pi(g_n)\pi(g_n^{-1} h g_n).$$

For $v, w \in \mathcal{H}$,

$$\langle \pi(h)\pi(g_n)v, w \rangle = \langle \pi(g_n)\pi(g_n^{-1} h g_n)v, w \rangle.$$

Strong continuity makes the second factor on the right tend to v , so weak convergence gives

$$\pi(h)T = T.$$

Therefore the image of T consists of vectors fixed by every element in the contraction group

$$U^+(g_n) = \{h : g_n^{-1} h g_n \rightarrow e\}.$$

The order of the inverse is crucial. Replacing $g_n^{-1} h g_n$ by $g_n h g_n^{-1}$ changes the contraction group from one root direction to the opposite root direction. The following distinction is essential: this possible mistake; the displayed identity is the reliable way to decide which group appears.

N.3 A complete SL_2 calculation

Take

$$a_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}, \quad u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \quad v_s = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix}.$$

Then

$$a_{-t} u_s a_t = u_{e^{-2t}s} \rightarrow e,$$

so a weak limit of $\pi(a_t)$ has image in the U -fixed vectors.

Why does U -invariance imply all of SL_2 -invariance? It does not follow from the contraction calculation alone. We need Mautner propagation.

Lemma N.2 (Exact Gauss factorization). *For $1 + st \neq 0$,*

$$u_s v_t = v_{t/(1+st)} \begin{pmatrix} 1+st & 0 \\ 0 & (1+st)^{-1} \end{pmatrix} u_{s/(1+st)}.$$

Proof. Multiply the three matrices on the right. The lower-left entry becomes t , the upper-right becomes s , and the upper-left becomes $1 + st$; determinant one determines the last entry. The identity is exact. $\square$

Let ξ be U -fixed. Apply the identity to matrix coefficients involving ξ and let one parameter tend to infinity so that the exterior unipotent factors tend to the identity after conjugation. Unitarity and continuity then show that ξ is fixed by the diagonal group. Conjugating U by the Weyl element or repeating the argument with the opposite limiting direction gives V -invariance. Since U , V , and the diagonal group generate SL_2 , the vector is SL_2 -fixed.

N.4 From one contracted root to a simple factor

In higher rank an escaping Cartan sequence $g_n = k_n a_n k'_n$ has, after compact subsequences are removed, a sequence $a_n = \exp H_n$ in the closed positive Weyl chamber. At least one simple root α satisfies $\alpha(H_n) \rightarrow \infty$. The corresponding positive root group is contracted by $a_n^{-1}(\cdot)a_n$.

A vector fixed by U_α is fixed by the rank-one group generated by U_α and $U_{-\alpha}$ by the SL_2 Mautner argument. Now let β be a simple root adjacent to α in the restricted Dynkin diagram. Root-string commutators of the form

$$[U_{-\alpha}, U_{\alpha+\beta}] \subset U_\beta \cdot (\text{higher roots})$$

show that invariance propagates into the adjacent root string. Induction along the connected Dynkin diagram gives invariance under every root group in that simple factor. The root groups generate the noncompact simple factor.

Warning N.3. The phrase “one root subgroup normally generates the factor” is correct only after one has identified the relevant connected simple factor and checked root propagation. In a product group, one root subgroup does not generate the other factors. This is why the factorwise Howe–Moore theorem keeps track of which Cartan coordinates actually diverge.

N.5 The contradiction proof of Howe–Moore

Let $\mathcal{H}_0$ be the orthogonal complement of all vectors fixed by the noncompact factors under consideration. Suppose a coefficient does not tend to zero at infinity. Choose $g_n \rightarrow \infty$ and vectors $v, w \in \mathcal{H}_0$ with

$$|\langle \pi(g_n)v, w \rangle| \geq \epsilon.$$

Pass to a weak operator limit T . Then $\langle Tv, w \rangle \neq 0$, so $T \neq 0$. Cartan decomposition and the contraction argument give a nonzero vector in $\text{im } T$ fixed by a root group. Mautner/root propagation makes it fixed by a noncompact simple factor. This contradicts the definition of $\mathcal{H}_0$. Hence all coefficients on $\mathcal{H}_0$ vanish at infinity.

This proof contains four logically independent steps:

nonvanishing coefficient $\Rightarrow$ nonzero weak limit $\Downarrow$
 Cartan escape $\Rightarrow$ contracted root group $\Downarrow$
 contraction identity $\Rightarrow$ root-fixed image $\Downarrow$
 Mautner/root propagation $\Rightarrow$ factor-fixed vector.

A proof omitting any one arrow is incomplete.

N.6 Moore ergodicity carefully separated from Howe–Moore

Let G act on $X = \Gamma \backslash G$ by right translations and let $H < G$ be a noncompact subgroup. Suppose $f \in L^2(X)$ is H -invariant. Let P be the orthogonal projection onto the closed G -invariant subspace generated by f . If f has nonconstant part f_0 , choose $h_n \in H$ escaping every compact set. Since f_0 is H -fixed,

$$\langle \pi(h_n)f_0, f_0 \rangle = \|f_0\|^2.$$

Howe–Moore forces this coefficient to tend to zero provided the generated subspace has no fixed vector from the relevant factor. Irreducibility of the lattice action is exactly what rules out a nontrivial factor on which the escaping simple component acts trivially. Hence $f_0 = 0$ and the H -action is ergodic.

N.7 Mixing as a coefficient statement

For measurable sets A, B with boundaries of measure zero, approximate their indicators in L^2 by continuous compactly supported functions. The correlation identity

$$m(Ag \cap B) - m(A)m(B) = \langle \pi(g)(\mathbf{1}_A - m(A)), \mathbf{1}_B - m(B) \rangle$$

shows that Howe–Moore gives mixing once the mean-zero functions lie in the no-fixed-vector subspace.

Worked Example N.4 (Geodesic flow). On a finite-area hyperbolic surface, the geodesic flow is right translation by the diagonal subgroup a_t in $PSL_2(\mathbb{R})$. Since $a_t \rightarrow \infty$ as $|t| \rightarrow \infty$, Howe–Moore implies

$$\int f(xa_t)g(x) dm(x) \longrightarrow \int f dm \int g dm$$

for $f, g \in L^2$. This qualitative conclusion uses no spectral gap and gives no numerical rate; Volume XI will explain exactly what extra information is needed for a rate.

N.8 Why this volume stops at qualitative mixing

It is tempting to examine the proof above and search for a decay exponent. The weak-compactness argument throws away all quantitative information, and Mautner propagation is likewise qualitative. Quantitative mixing therefore requires a different input: explicit bounds on matrix coefficients from representation theory. This separation of tasks is deliberate. Volume VI owns the qualitative theorem completely; Volume XI rebuilds the argument with Sobolev norms and a tensor-tempered index when rates become necessary.

Dependency clarification for the mixing proofs

Which parts of Volume V are actually used?

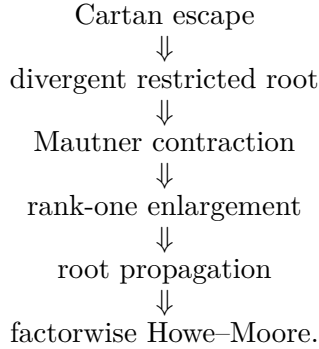
The proofs in this volume use three pieces of the preceding infrastructure.

1. The Cartan decomposition and restricted-root structure come from the real-algebraic/symmetric-space development of Volume V. They do not use the rational-parabolic reduction theorem.
2. Borel density is used when irreducibility of arithmetic examples is translated into absence of factor-fixed functions. Its proof uses the lattice hypothesis, and the finite-covolume theorem of Volume V supplies the needed lattice input independently of the stronger rational-cusp reduction.
3. Arithmetic closedness of a semisimple rational subgroup orbit is used only for geometric applications. Volume V derives this closedness from arithmeticity of the intersection, finite covolume, and the finite-covolume-stabilizer closed-orbit theorem of Volume III.

Consequently the mixing proofs below are logically independent of the stronger rational-cusp normal form in Volume V.

Local proof chain

The argument proceeds in the order



For a product group the weak-limit argument is applied factor by factor; a weak limit is fixed by every factor in which the Cartan projection escapes. Irreducibility removes the corresponding fixed subspaces on $L_0^2(\Gamma \backslash G)$, giving Moore ergodicity, and the usual constant-plus-mean-zero decomposition gives mixing.

No additional project-specific external theorem is used in this chain. The general real-semisimple Cartan–Iwasawa structure input is proved in Volume V, in its internal Cartan-involution and Iwasawa theorems; the Hilbert-space weak compactness and strong-continuity steps needed here are proved in the student course.

Proof Guide and Dependency Summary

This chapter is a finalization layer. It does not introduce a new theorem scope. Its purpose is to expose the proof architecture, make the hypotheses that are easiest to misread impossible to miss, and give a student a concrete model calculation before the abstract machinery is reused later.

1. The exact logic of Mautner propagation

Suppose π is unitary and $a_n = \exp H_n$ escapes in a positive Weyl chamber. If $\alpha(H_n) \rightarrow +\infty$ and $u \in U_\alpha$, then

$$a_n^{-1} u a_n \longrightarrow e.$$

If $\pi(a_n)v \rightarrow v_\infty$, unitarity gives

$$\pi(u)v_\infty = v_\infty.$$

This is only the first root group. The crucial second step is rank-one: the subgroup generated by the opposite α -root groups contains the appropriate SL_2 -type subgroup, and the exact rank-one Mautner argument upgrades one-sided unipotent invariance to invariance under that rank-one subgroup. Dynkin adjacency then propagates invariance through the almost-simple restricted-root component. Multipliable roots are included rather than silently treated as a reduced-root-system case.

2. Why the product hypothesis is factorwise

If $G = G_1 \times G_2$ and $\pi = 1 \otimes \pi_2$, there need not be a G -fixed vector, but every vector is G_1 -fixed. Hence coefficients need not decay along a sequence

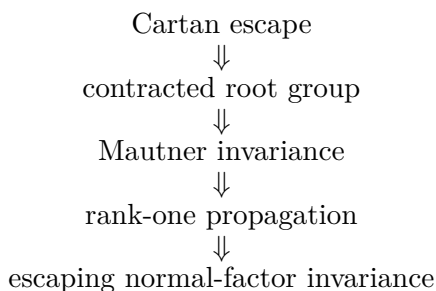
escaping only in G_1 . The correct Howe–Moore statement excludes nonzero vectors fixed by the noncompact normal factor in which the sequence escapes.

3. Proof architecture of Howe–Moore

For a hypothetical nondecaying coefficient:

$$g_n = k_n a_n k'_n \quad \rightsquigarrow \quad \pi(a_n) v_n \rightharpoonup v_\infty \neq 0.$$

Then:



contradicting the factorwise hypothesis.

4. Scope warning

This volume proves qualitative decay. A statement with a numerical rate must not cite qualitative Howe–Moore as its rate source. The quantitative parameter is introduced only in Volume XI.

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Limits of Homogeneous Measures and the Mozes–Shah Principle

Volume VII of a Detailed Notebook on Ratner’s Theorems

Detailed Notebook on Ratner’s Theorems

August 2026

Preface and proof policy

Ratner's theorem classifies one invariant probability measure. The next question is compactness: what can a sequence of such homogeneous measures do when both its supporting subgroup and its supporting orbit vary? This volume reconstructs the Mozes–Shah mechanism from the tools already proved in Volumes II–V: quantitative nondivergence, Chevalley linearization, singular avoidance, generated-unipotent rigidity, and arithmetic/algebraic control.

The family-focusing argument is written independently of the shorter finite-family focusing step used in the Volume III complete-proof module. At the end we derive that earlier step as a corollary, thereby providing the second audit promised when the first four volumes were released.

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Dependency and proof-ownership contract

Purpose. Homogeneous measure limits.

Prerequisites. Volumes II–VI; the external Dani–Margulis structural interface isolated in Volume III.

New objects. support/focusing data. Every project-specific technical term is defined before its first load-bearing use or recalled by an exact backward reference.

Main outputs. Mozes–Shah type compactness/focusing.

External-input rule. Every nontrivial imported result is either part of the declared graduate-level prerequisites, an explicitly named external research input in the prerequisite map, or is proved in this series before use (possibly under an explicit deferred-proof contract). A citation by itself never constitutes a proof.

Reader map and proof route

Question of the volume. What can a weak limit of homogeneous probability measures be when both the closed orbit and its subgroup vary?

Prerequisite checkpoint. Quantitative nondivergence and singular avoidance come from Volumes II–III; algebraic subgroup control comes from Volume V; Ratner classification is already proved in Volume III. The Dani–Margulis results enter only through the earlier countability-and-discreteness interface when the proof route explicitly points to it.

Load-bearing route. First prove tightness/non-escape, then show that any weak limit gains the needed unipotent invariance, classify its ergodic components, and finally use focusing to compare the varying supports. The focusing step is a theorem, not the slogan “homogeneous measures are closed under limits.”

Sanity check. Closed horocycles on the modular surface provide the correct mental picture: a sequence can move deeper into the cusp unless nondivergence prevents it; a convergent subsequence must then be identified by invariance rather than by pointwise convergence of the supporting curves.

Chapter 1

The Space of Homogeneous Probability Measures

1.1 Homogeneous data

Let $X = \Gamma \backslash G$ with G connected and Γ a lattice. For a closed connected subgroup $H < G$ and $x \in X$, suppose xH is closed and carries finite H -invariant volume. Write m_{xH} for the normalized Haar probability on this orbit.

Let H^+ denote the closed connected subgroup generated by all one-parameter Ad-unipotent subgroups of H . A *Mozes–Shah datum* is a triple (x, H, U) such that xH is closed of finite volume, $U = \{u_t\} \subset H^+$ is a nontrivial one-parameter Ad-unipotent subgroup, and U is ergodic on (xH, m_{xH}) . We write $\mathcal{Q}(X)$ for the resulting homogeneous probabilities. This is the exact load-bearing class for the limit theorem. The broader class in which only H^+ is assumed ergodic will be recovered after the one-parameter theorem; we will never silently replace joint ergodicity by ergodicity of an arbitrary generator.

Lemma 1.1 (Canonical support group). *If $\mu = m_{xH}$ and H^+ is ergodic, then the identity component of the full measure stabilizer $\text{Stab}_G(\mu)$ acts transitively on $\text{supp } \mu$. Replacing H by that identity component does not change μ or its support.*

Proof. The stabilizer contains H , hence its identity component L has orbit xL containing xH . Conversely every element stabilizing μ preserves its support, so $xL \subset \text{supp } \mu = xH$. Thus the two connected orbits agree. Uniqueness of

invariant probability on the transitive finite-volume orbit, proved in Volume III, gives the same measure. $\square$

We always use this canonical connected support group when comparing measures.

1.2 Weak-* topology and support

Let $\mathcal{P}(X)$ be the probability measures with the weak-* topology against $C_c(X)$ plus tightness. If X is noncompact, vague convergence against C_c alone permits loss of mass; every compactness statement below therefore contains an explicit nonescape step.

Lemma 1.2 (Support points persist under weak convergence). *Suppose $\mu_i \rightarrow \mu$ weakly as probability measures and $x \in \text{supp } \mu$. Then there are $x_i \in \text{supp } \mu_i$ with $x_i \rightarrow x$.*

Proof. Choose balls $B(x, 1/j)$ whose boundaries have μ -measure zero. Since $\mu(B(x, 1/j)) > 0$, weak convergence gives $\mu_i(B(x, 1/j)) > 0$ for all large i . Select x_i in the support inside the appropriate ball by a diagonal choice. $\square$

Chapter 2

Uniform Nonescape for Varying Unipotent Data

2.1 Normalizing generators

Fix a norm on $\mathfrak{g}$. A nontrivial one-parameter Ad-unipotent group is $U_N = \{\exp(tN)\}$ with $N \neq 0$ and $\text{ad } N$ nilpotent. Reparameterizing time allows $\|N\| = 1$. The set

$$\mathcal{U} = \{N \in \mathfrak{g} : \|N\| = 1, \text{ ad } N \text{ nilpotent}\}$$

is compact: nilpotence is given by the polynomial equation $(\text{ad } N)^{\dim \mathfrak{g}} = 0$.

Lemma 2.1 (Uniform degree bound for the linearization representations). *Let ρ be any representation obtained from Ad by taking finite direct sums, tensor powers, duals, subquotients, and exterior powers. For every $N \in \mathcal{U}$, the coordinates of $\rho(\exp(tN))$ are polynomials in t of degree bounded solely by ρ and $\dim \mathfrak{g}$; their coefficients depend continuously on N .*

Proof. By definition of $\mathcal{U}$, $(\text{ad } N)^{\dim \mathfrak{g}} = 0$. Hence

$$\text{Ad}(\exp tN) = \exp(t \text{ad } N) = \sum_{j=0}^{\dim \mathfrak{g}-1} \frac{t^j (\text{ad } N)^j}{j!}.$$

Tensor products, duals, direct sums and exterior powers of a polynomial matrix are again polynomial matrices, with a degree bound depending only on the fixed functor used to construct ρ . Passing to an invariant subspace

or quotient does not increase the degree. The coefficients are polynomial functions of the entries of N and hence continuous on the compact set $\mathcal{U}$.

The restriction to adjoint-derived representations is intentional. For a general Lie group, an Ad-unipotent central one-parameter subgroup need not be unipotent in an arbitrary unrelated representation. Every representation used in the family linearization below is adjoint-derived, so no stronger assertion is needed. $\square$

2.2 Uniform quantitative nondivergence

Volume II proved one-parameter quantitative nondivergence using primitive exterior-power covolumes. Compactness of $\mathcal{U}$ makes the constants uniform in the direction.

Theorem 2.2 (Uniform Dani–Margulis nonescape in the form used here). *For every compact $C \subset X$ and every $\varepsilon > 0$ there is a compact $K = K(C, \varepsilon) \subset X$ such that for every $x \in C$, every normalized one-parameter Ad-unipotent group $U_N = \{\exp(tN)\}$, and every $T > 0$,*

$$\frac{1}{T} |\{t \in [0, T] : x \exp(tN) \notin K\}| < \varepsilon.$$

The compact is uniform in the unipotent direction.

Proof. We use the orbitwise quantitative injectivity-radius nondivergence proved in Volume III, Appendix D. That proof is formulated for a general lattice quotient and uses a Zassenhaus neighborhood, flags of small subgroups, and polynomial coordinates for the adjoint action; it does not require an arithmetic realization of X as a space of Euclidean lattices.

Inspecting that proof shows that its constants depend on the acting flow only through: (i) a bound for the degree of the polynomials $\text{Ad}(\exp(tN))$, and (ii) upper bounds for their coefficient norms on the unit time interval after rescaling. By Lemma 2.1, both quantities are uniform for $N \in \mathcal{U}$. The remaining input is a positive lower bound for the injectivity radius and for the finitely many marked subgroup quantities at the initial point. Such a bound is uniform on the compact set C .

Apply the marking lemma of Volume III with these uniform constants. It gives $r = r(C, \varepsilon) > 0$ for which the set of times at which the injectivity radius of $x \exp(tN)$ is smaller than r has relative measure $< \varepsilon$, uniformly in $x \in C$, $N \in \mathcal{U}$, and $T > 0$. The r -thick part of the finite-volume homogeneous space

is compact by the compactness criterion proved in Volume III Appendix D. Taking that thick part for K proves the theorem. $\square$

Corollary 2.3 (Tightness of homogeneous families). *Let $\mu_i = m_{x_i H_i} \in \mathcal{Q}(X)$ and choose an ergodic one-parameter Ad-unipotent group $U_i \subset H_i^+$. If $\text{supp } \mu_i \cap C \neq \emptyset$ for one compact set C , then $\{\mu_i\}$ is tight.*

Proof. Choose $z_i \in \text{supp } \mu_i \cap C$. Because U_i is ergodic, its generic points have full μ_i -measure and are therefore dense in the support. Choose a generic y_i within distance $1/i$ of z_i ; all y_i lie in one slightly larger compact set C' . Normalize the generator of U_i and apply Theorem 2.2 to the forward interval $[0, T]$. For all large T the U_i -orbit of y_i spends at least $1 - \varepsilon$ of its time in the single compact $K(C', \varepsilon)$ supplied by the uniform nonescape theorem. Letting $T \rightarrow \infty$ and using genericity gives $\mu_i(K) \geq 1 - \varepsilon$. The compact is independent of i , proving tightness. $\square$

Chapter 3

Linearizing a Family of Supporting Subgroups

3.1 Chevalley vectors

For each connected algebraic subgroup $H < G$ in the class under consideration, Volume V supplies a representation V_H and a rational line $\ell_H = \mathbb{R}v_H$ whose stabilizer is the algebraic hull of H . For a family one cannot allow infinitely many unrelated representations. The key observation is that in fixed ambient dimension, subgroup Lie algebras are encoded in one exterior algebra.

Let $r = \dim H$ and choose a basis of $\mathfrak{h}$. Its wedge

$$p_H = h_1 \wedge \cdots \wedge h_r \in \bigwedge^r \mathfrak{g}$$

is well defined up to scalar. The line $[p_H]$ records $\mathfrak{h}$, and g preserves the line exactly when $\text{Ad}(g)$ normalizes $\mathfrak{h}$ up to the determinant character.

Lemma 3.1 (Compact parameterization of Lie subalgebras). *For each r , the set of r -dimensional Lie subalgebras of $\mathfrak{g}$ is a closed subset of the Grassmannian $\text{Gr}_r(\mathfrak{g})$ and hence compact.*

Proof. The condition $[W, W] \subset W$ is closed: in a local Grassmannian chart it is the vanishing of the projection of finitely many brackets of basis vectors to a chosen complement. The Grassmannian is compact. $\square$

3.2 Family singular sets

Fix a candidate limiting support group L . A nearby homogeneous datum is singular relative to L if its Lie algebra is contained in a conjugate of a proper subgroup $P < L$ containing a nontrivial unipotent direction relevant to the limit. Using the Grassmannian encoding, singular data of dimension at most d form a finite union, locally on compact parameter sets, of zero sets of polynomial representation coordinates.

Lemma 3.2 (Finite-stage singularity). *Let $C \subset X$ and $\mathcal{K}$ be compact sets of basepoints and subgroup parameters. For every neighborhood $\mathcal{O}$ of the exact singular data there are finitely many representation vectors $v_1, \dots, v_M$ and thresholds $\eta_j > 0$ such that every datum in $(C \times \mathcal{K}) \setminus \mathcal{O}$ violates at least one inequality $\|\rho_j(g)v_j\| < \eta_j$ that would be forced by singularity.*

Proof. The complement of $\mathcal{O}$ in the compact parameter space is compact. For each nonsingular datum, one defining polynomial of the relevant containment variety is nonzero. Nonvanishing persists on a neighborhood. A finite subcover gives finitely many polynomials. Chevalley realization converts them into the indicated representation coordinates. $\square$

This is the family analogue of the finite-singular-strata bookkeeping in Volume III.

Chapter 4

Focusing

4.1 What focusing must prove

Suppose $\mu_i = m_{x_i H_i} \rightarrow \mu = m_{xL}$. Support points x'_i may be chosen with $x'_i \rightarrow x$. Write $x'_i = xg_i$ in a fixed local quotient chart, with $g_i \rightarrow e$. The desired conclusion is not merely convergence of supports: a genuine focusing theorem says that, after this small conjugation,

$$g_i H_i g_i^{-1} \subset L$$

for all sufficiently large i (at the connected algebraic level relevant to the unipotent-generated part). Otherwise some transverse subgroup direction survives and should create additional invariance of the limiting measure.

4.2 A transverse Lie-algebra direction

Assume $\text{Ad}(g_i)\mathfrak{h}_i \not\subset \mathfrak{l}$. Choose unit vectors $Z_i \in \text{Ad}(g_i)\mathfrak{h}_i$ whose projections to $\mathfrak{g}/\mathfrak{l}$ have maximal norm, and normalize so that the transverse class has norm one. Compactness gives $Z_i \rightarrow Z$ with nonzero class modulo $\mathfrak{l}$ after passing to a subsequence.

Because H_i^+ is generated by unipotents, one can choose normalized nilpotent generators $N_i \in \text{Ad}(g_i)\mathfrak{h}_i$ such that either $[N_i, Z_i]$ produces a nonzero transverse polynomial direction or Z_i lies in the common normalizer of all unipotent generators. The second case means that the transverse direction normalizes the unipotent-generated subgroup; iterating through the normalizer tower either produces a nontrivial shear or places the whole connected

support algebra inside a proper normalizer-type singular datum. The tower terminates because dimensions increase strictly.

Lemma 4.1 (Family shear alternative). *After passing to a subsequence, exactly one of the following holds.*

1. *There are normalized $N_i \in \mathfrak{h}_i$ and times $T_i \rightarrow \infty$ such that the transverse classes of $e^{-t \operatorname{ad} N_i} Z_i$ have a nonconstant bounded polynomial limit on $t = sT_i$, $s \in [0, 1]$.*
2. *The data (H_i, x'_i) lie asymptotically in a proper compact family of normalizer-type singular strata inside L .*

Proof. Expand $e^{-t \operatorname{ad} N_i} Z_i = \sum_{j=0}^d (-t)^j (\operatorname{ad} N_i)^j Z_i / j!$ and project to $\mathfrak{g}/\mathfrak{l}$. If some coefficient is nonzero, choose the first macroscopic scale exactly as in the H-property of Volume III. Compactness of the normalized nilpotent directions gives bounded degree and a polynomial limit. If every projected coefficient vanishes for every unipotent generator, the transverse class is invariant under their adjoint action. Lifting this statement says Z_i belongs, up to an error tending to zero, to the Lie algebra of the normalizer of the generated subgroup modulo L . Iterate with the larger normalizer. Either a coefficient eventually appears or a proper normalizer tower captures the entire datum, which is the singular alternative. $\square$

4.3 Converting the shear into invariance

The homogeneous measures μ_i are invariant under their H_i^+ flows, while $\mu_i \rightarrow \mu$. On the terminal block from the first alternative, the relative displacement between two points of the same H_i -orbit is nearly constant after conjugating to the local chart at x . Uniform nonescape gives a common compact set for most times. The block comparison theorem from Volume III therefore survives the limit $i \rightarrow \infty$ and produces a curve of stabilizers of μ whose derivative has nonzero class in $\mathfrak{g}/\mathfrak{l}$. But L is the identity component of $\operatorname{Stab}_G(\mu)$, contradiction.

We isolate this conclusion.

Theorem 4.2 (Family focusing theorem). *Let $\mu_i = m_{x_i H_i} \in \mathcal{Q}(X)$ with chosen ergodic unipotent groups U_i , and suppose $\mu_i \rightarrow m_{xL}$ as probabilities. Choose support points $x'_i = xg_i \in \operatorname{supp} \mu_i$ with $g_i \rightarrow e$. Then, after passing to a subsequence, either*

$$\operatorname{Ad}(g_i)\mathfrak{h}_i \subset \mathfrak{l}$$

for all sufficiently large i , or all μ_i are supported in a proper homogeneous singular family whose ambient support dimension is strictly less than $\dim L$. After induction on support dimension, the second alternative also yields small conjugacies for which the support groups embed in the limiting support group.

Proof. Apply Lemma 4.1. The shear alternative contradicts the canonical choice $L = \text{Stab}_G(\mu)^\circ$ by the terminal-block invariance argument just described. In the singular alternative, the representation coordinates defining the proper subgroup stay uniformly small on a positive proportion of the unipotent blocks. The polynomial avoidance theorem of Volume III upgrades this to exact trapping in a proper algebraic subgroup. Restrict the homogeneous measures to that subgroup. Their finite-volume property and unipotent ergodicity are preserved, while the ambient support dimension decreases. Induction terminates. At the terminal stage the shear alternative is impossible and no proper singular reduction remains, forcing the desired containment. $\square$

Chapter 5

The Mozes–Shah Compactness Theorem

5.1 Statement

Theorem 5.1 (Mozes–Shah principle). *Let $X = \Gamma \backslash G$ have finite volume. Let $\mu_i = m_{x_i H_i} \in \mathcal{Q}(X)$ and choose ergodic one-parameter Ad -unipotent groups $U_i = \{u_i(t)\} \subset H_i^+$. Assume the supports meet one fixed compact subset of X . Then every subsequence has a further subsequence converging to a homogeneous probability $\mu = m_{xL}$ of the same type. Moreover, after choosing $g_i \rightarrow e$ with $xg_i \in \text{supp } \mu_i$, one has for all large i*

$$g_i H_i^+ g_i^{-1} \subset L,$$

and the supports satisfy

$$\text{supp } \mu_i g_i^{-1} \subset \text{supp } \mu.$$

Proof. Tightness follows from Corollary 2.3, so pass to a probability limit μ . Normalize the generators of the chosen ergodic groups U_i . Compactness of the nilpotent unit sphere gives, after a subsequence, a limit U . Invariance passes to the limit: if $u_i(t) \rightarrow u(t)$ uniformly for t in compact sets, then for $f \in C_c(X)$,

$$\int f(xu(t)) d\mu(x) = \lim_i \int f(xu_i(t)) d\mu_i(x) = \lim_i \int f d\mu_i = \int f d\mu.$$

Thus μ has nontrivial unipotent invariance. Decompose it ergodically under U . Ratner measure classification from Volume III makes almost every component homogeneous.

It remains to rule out a nontrivial mixture. Suppose two distinct canonical homogeneous components occur with positive weights. Choose generic support points x, y from the two components arbitrarily close to a common density region of μ . Approximate them by support points of μ_i . Inside each $x_i H_i$, ergodicity of H_i^+ and recurrence provide unipotent-generated blocks visiting neighborhoods of both approximating points. Applying the family shear alternative to the relative displacement either produces a stabilizing curve linking the two limiting components, contradicting their canonical distinctness, or traps a positive fraction of μ_i in a proper singular family. In the latter case restrict and induct on support dimension. At the terminal dimension no mixture is possible. Hence μ is a single homogeneous probability m_{xL} .

The focusing containment is exactly Theorem 4.2. If $g_i H_i^+ g_i^{-1} \subset L$ and $x g_i \in x_i H_i$, connectedness shows $x_i H_i^+ g_i^{-1} \subset xL$. The complementary compact/central directions of H_i preserve the same homogeneous probability and therefore lie in the canonical stabilizer after passing to identity components. Thus the full connected support is contained in xL , giving the support statement. $\square$

Corollary 5.2 (Isolation of maximal homogeneous measures). *If m_{xG} is ambient Haar probability and there is no proper homogeneous support of the permitted class containing all H_i^+ after small conjugacy, then $\mu_i \rightarrow m_{xG}$.*

Proof. Every subsequential homogeneous limit has support containing the eventually conjugated supports by Theorem 5.1. The hypothesis forces that support to be all of X . Uniqueness of Haar probability gives the limit; subsequence uniqueness gives convergence of the full sequence. $\square$

Chapter 6

Sequences of Closed Orbits

6.1 Volume versus geometric escape

A sequence of closed orbits can fail to equidistribute for two independent reasons: it can escape to a cusp, or it can focus inside a proper larger homogeneous subspace. Theorem 5.1 separates the two: compact support intersection rules out the first, and support containment identifies the second algebraically.

Proposition 6.1 (Maximality criterion for equidistribution). *Let $x_i H_i$ be finite-volume homogeneous orbits whose probabilities lie in $\mathcal{Q}(X)$ and with supports meeting a common compact set. Suppose that for every proper closed connected $L < G$ and every sequence $g_i \rightarrow e$, the inclusions $g_i H_i^+ g_i^{-1} \subset L$ fail infinitely often. Then $m_{x_i H_i} \rightarrow m_X$.*

Proof. Any subsequential limit is homogeneous by Theorem 5.1. If its support group L were proper, the focusing conclusion would give exactly the excluded eventual inclusions. Therefore $L = G$ and the limit is Haar. Uniqueness of all subsequential limits gives convergence. $\square$

6.2 Worked example: closed geodesic planes inside a larger quotient

Suppose $H \cong \mathrm{SO}(2,1)^\circ$ embeds in a semisimple G and arithmetic data produce a sequence of finite-volume H_i -orbits. If the rational subspaces defining the H_i do not eventually lie in any proper rational subgroup of G ,

Borel density from Volume V turns the maximality criterion into an arithmetic span condition. This is the basic bridge from homogeneous measure limits to counting and distribution of arithmetic subvarieties.

Chapter 7

Generated-Unipotent Focusing in Full Detail

7.1 Recovering the finite-generator case

Let $W < G$ be generated by finitely many one-parameter Ad-unipotent groups $U_1, \dots, U_r$, and let $x \in X$. Volume III's complete-proof module used a finite-family focusing lemma to pass from the one-parameter orbit-closure theorem to the W -orbit closure theorem. We now recover it from the family theorem.

Theorem 7.1 (Finite-generator focusing corollary). *Let $Y = \overline{xW}$ be minimal among closed W -invariant homogeneous envelopes containing x . If successive one-parameter orbit closures under $U_1, \dots, U_r$ produce homogeneous measures μ_n whose support groups increase along the construction, then the process terminates after at most $\dim G$ strict increases, and the final support is Y . Any subsequential family arising in the gluing step focuses inside the final support group.*

Proof. At each stage average a homogeneous probability over the next unipotent flow and take a subsequential limit. Nonescape is inherited from the ambient finite-volume homogeneous envelope. Theorem 5.1 makes the limit homogeneous and its focusing conclusion embeds the preceding support group into the new one after an arbitrarily small conjugacy. If the support does not enlarge, the next generator already preserves the current homogeneous measure and hence its support. Otherwise the connected support dimension increases. There can be at most $\dim G$ strict increases. At termination all

generators preserve the support, hence W preserves it. Minimality of Y forces equality. $\square$

7.2 Comparison with the Volume III argument

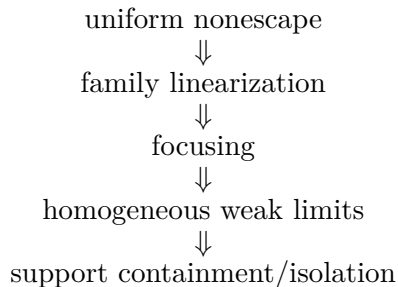
The Volume III proof and the proof developed here use different bookkeeping but isolate the same mathematical obstruction. Volume III worked with finitely many simultaneous singular strata and a normalizer/gluing induction. Here those strata are compactified in the Grassmannian and handled by Theorem 4.2. Thus the finite-family focusing step also follows from the varying-measure theorem, giving a second route to the same conclusion.

Remark 7.2 (Scope of the singular-set argument). No new external rigidity theorem has been inserted into the generated-group proof. The independent route uses only the one-parameter Ratner theorem from Volume III, the quantitative nondivergence and linearization package from Volume II, and the family arguments proved in this volume.

Chapter 8

Proof Ledger and Applications Forward

The dependency chain is now



Volume VIII will replace one-parameter trajectories by polynomial ones. Volume IX will apply the present compactness principle to expanding translates. Volume X will use it to turn varying closed homogeneous orbits into counting statements. No effective rate has been claimed here.

Appendix A

The Mozes–Shah Class and Countability of Support Types

A.1 Zariski density supplied by unipotent generation

The earlier short draft used countability of homogeneous support types to select a positive-mass type. the localization-and-focusing proof module below removes that use: a single almost-everywhere Ratner ergodic component and one regular support point already provide the positive-mass open tube required by the focusing argument. We retain the classical countability statement in this appendix because it is needed later, but the short proof is no longer used anywhere in the proof-completeness route.

Lemma A.1 (A unipotent linear map cannot carry mass off its fixed projective set). *Let $u_t = \exp(tN)$ act linearly on a finite-dimensional real vector space V , with N nilpotent. If a probability measure ν on $\mathbb{P}(V)$ is u_t -invariant, then it is supported on $\mathbb{P}(\ker N)$.*

Proof. For $v \neq 0$, let $j(v)$ be the largest j with $N^j v \neq 0$. If $j(v) > 0$, then in any affine projective chart around $[N^{j(v)}v]$ the orbit $[u_t v]$ approaches that point as $|t| \rightarrow \infty$ with a definite polynomial ratio between the highest and next nonzero terms. A compact set on which $j(v) = j > 0$ and the two relevant normalized coefficients are bounded away from zero has translates

by a sufficiently lacunary sequence t_n that are pairwise disjoint. Invariance forces its measure to be zero. Exhaust the set $\{j(v) > 0\}$ by countably many such compact sets. Thus $Nv = 0$ almost everywhere projectively. $\square$

Proposition A.2 (Zariski density for a unipotent-generated support). *Let H be connected, let $\Delta < H$ be a lattice, and suppose H^+ is generated by one-parameter Ad-unipotent subgroups and acts ergodically on $\Delta \backslash H$. Then*

$$\overline{\text{Ad}(\Delta)}^Z = \overline{\text{Ad}(H)}^Z.$$

Proof. Put

$$A = \overline{\text{Ad}(H)}^Z, \quad J = \overline{\text{Ad}(\Delta)}^Z \subset A.$$

Suppose that $J \subsetneq A$. Chevalley linearization gives a finite-dimensional real algebraic representation $\rho : A \rightarrow \text{GL}(V)$ and a line $\ell \in \mathbb{P}(V)$ with $J = \text{Stab}_A(\ell)$. Hence

$$\Phi : \Delta \backslash H \rightarrow \mathbb{P}(V), \quad \Phi(\Delta h) = \rho(\text{Ad}(h^{-1}))\ell$$

is a well-defined continuous H -equivariant map. Push Haar probability forward and call the resulting H -invariant probability ν .

For every one-parameter Ad-unipotent subgroup $U = \{u_t\}$ of H^+ , $\rho(\text{Ad}(u_1))$ is unipotent. Lemma A.1 therefore says that ν is supported on the projective lines fixed by U . Since $\text{Lie}(H^+)$ is finite dimensional, choose finitely many such subgroups $U_1, \dots, U_r$ whose infinitesimal generators generate $\text{Lie}(H^+)$ as a Lie algebra. The intersection of their projective fixed sets still has full ν -measure. If a line L lies in that intersection, its projective stabilizer is a closed Lie subgroup containing every U_i ; its Lie algebra therefore contains the Lie algebra generated by the $\text{Lie}(U_i)$, namely $\text{Lie}(H^+)$. Since H^+ is connected, the entire group H^+ fixes L . Thus Φ is H^+ -invariant almost everywhere.

By ergodicity of H^+ on $\Delta \backslash H$, the map Φ is almost everywhere constant. Haar probability has full support and Φ is continuous, so it is in fact constant everywhere. Evaluating at the identity coset shows that ℓ is fixed by $\text{Ad}(H)$. The Chevalley stabilizer identity then gives $\text{Ad}(H) \subset J$, contradicting $J \subsetneq A$. Hence $J = A$.

Notice that no conclusion $H = H^+$ was used: the argument only kills a projective algebraic factor by H^+ -ergodicity. $\square$

Theorem A.3 (Countability of the Mozes–Shah support class). *The basepoint-normalized connected support groups $F < G$ for which $F \cap \Gamma$ is a lattice and F^+ acts ergodically on $(F \cap \Gamma) \backslash F$ form a countable family.*

Inherited from the Volume III Dani–Margulis consequences. Volume III proves this countability statement by taking the union, over the integer covolume bounds, of the support classes supplied by the external Dani–Margulis structural countability/discreteness interface. The support-to-algebraic-hull bridge is proved internally there, so countability is a derived theorem rather than an additional external input. $\square$

Corollary A.4 (Countability of canonical support groups). *Among connected subgroups $H < G$ for which $H \cap g^{-1}\Gamma g$ is a lattice and H^+ is ergodic on the corresponding homogeneous orbit, there are only countably many Γ -conjugacy classes after fixing the basepoint convention.*

Using Theorem A.3. Conjugate each support datum by its basepoint. This turns it into a member of the basepoint-normalized support class for the fixed lattice Γ . Theorem A.3 gives a countable list of those normalized groups. Passing to the chosen basepoint convention and then to Γ -conjugacy classes cannot increase cardinality. $\square$

Appendix B

Thin-Neighborhood Linearization with All Uniqueness Steps

Fix a canonical homogeneous support group H with Lie algebra $\mathfrak{h}$ and put $d = \dim H$. Choose $0 \neq p_H \in \bigwedge^d \mathfrak{h}$ and let

$$V_H = \bigwedge^d \mathfrak{g}, \quad \eta_H(g) = \bigwedge^d \text{Ad}(g) p_H.$$

For the fixed left-quotient convention $X = \Gamma \backslash G$ it is convenient to use the inverse-coordinate lift

$$\hat{\eta}_H(g) := \eta_H(g^{-1}).$$

Thus, if $x = \Gamma g$ and the right U -orbit is $xu_t = \Gamma gu_t$, then

$$\hat{\eta}_H(gu_t) = \bigwedge^d \text{Ad}(u_{-t}) \hat{\eta}_H(g). \tag{B.1}$$

If a sign ambiguity is produced by an element of the lattice normalizer, we work in $\bar{V}_H = V_H / \{\pm 1\}$; this has no effect on norms or polynomial estimates.

B.1 Discreteness of the lattice orbit

Lemma B.1 (Discrete Chevalley orbit). *If $H \cap \Gamma$ is a lattice and H^+ is ergodic, then the orbit $\Gamma \bar{p}_H$ is closed and discrete in $\bar{V}_H$.*

Inherited from the Volume III Dani–Margulis consequences. After basepoint normalization, H belongs to the support class of Volume III. There the exact conjugation–Jacobian/covolume identity is proved, and the Dani–Margulis fixed-support Chevalley-discreteness theorem supplies local finiteness of $\Gamma\bar{p}_H$. Thus the discrete Chevalley orbit is an internal consequence of the one accepted Dani–Margulis input, not a second black box. $\square$

Corollary B.2 (Proper lift map). *Let $\Gamma_H = \Gamma \cap N_G(H)$, with the sign convention absorbed in $\bar{V}_H$. In the left-quotient convention $X = \Gamma \backslash G$, the correct lift map is*

$$\Phi_H : \Gamma_H \backslash G \longrightarrow (\Gamma \backslash G) \times \bar{V}_H, \quad \Gamma_H g \longmapsto (\Gamma g, \eta_H(g^{-1})).$$

It is well defined and proper.

Using Lemma B.1. If $h \in \Gamma_H$, then $\Gamma h g = \Gamma g$ and

$$\eta_H((hg)^{-1}) = \bigwedge^d \text{Ad}(g^{-1}) \left(\bigwedge^d \text{Ad}(h^{-1}) p_H \right) = \pm \eta_H(g^{-1}),$$

so Φ_H is well defined in $\bar{V}_H$.

Let $K \subset \Gamma \backslash G$ and $D \subset \bar{V}_H$ be compact. Choose a compact set $C \subset G$ whose projection contains K . If $\Gamma_H g_j \in \Phi_H^{-1}(K \times D)$, write $g_j = \gamma_j c_j$ with $\gamma_j \in \Gamma$ and $c_j \in C$. Then

$$\eta_H(g_j^{-1}) = \bigwedge^d \text{Ad}(c_j^{-1})(\gamma_j^{-1} p_H) \in D,$$

so $\gamma_j^{-1} \bar{p}_H$ belongs to the compact set $(\bigwedge^d \text{Ad}(C))D$. By Lemma B.1, only finitely many orbit values occur. After passing to finitely many pieces, fix one value; then $\gamma_j \gamma_1^{-1} \in \Gamma_H$ and

$$\Gamma_H g_j = \Gamma_H \gamma_1 c_j.$$

The right-hand side ranges over the compact image of C in $\Gamma_H \backslash G$. Hence $\Phi_H^{-1}(K \times D)$ is compact. $\square$

B.2 Unique lifts away from self-intersections

For a compact $D \subset \bar{V}_H$, let $S(D)$ consist of those g for which both $\hat{\eta}_H(g)$ and $\hat{\eta}_H(\gamma g)$ lie in D for some $\gamma \in \Gamma \backslash \Gamma_H$.

Lemma B.3 (Closed self-intersection locus and unique lifts). *The projection of $S(D)$ to X is closed. If $K \subset X$ is compact and disjoint from that projection, then there is a neighborhood Φ of D in $\bar{V}_H$ such that every point of $K \cap \pi(\hat{\eta}_H^{-1}(\Phi))$ has exactly one lift whose Chevalley vector lies in Φ .*

Using Corollary B.2. Let

$$q : \Gamma_H \backslash G \longrightarrow \Gamma \backslash G$$

be the natural projection. Since $\Gamma_H < \Gamma$ are discrete, q is a covering map: every point of X has an evenly covered neighborhood, and q is injective on each sheet over such a neighborhood. This elementary covering observation is what prevents two distinct lattice lifts from coalescing in the limiting argument below.

First prove closedness. Let x_j lie in the projection of $S(D)$ and $x_j \rightarrow x$. Choose $g_j \in G$ and $\gamma_j \in \Gamma \setminus \Gamma_H$ such that $q(\Gamma_H g_j) = q(\Gamma_H \gamma_j g_j) = x_j$ and

$$\hat{\eta}_H(g_j), \hat{\eta}_H(\gamma_j g_j) \in D.$$

The two sequences lie in the inverse image under the proper map Φ_H of the compact set $(\{x_j : j \geq 1\} \cup \{x\}) \times D$. After passing to a subsequence,

$$\Gamma_H g_j \rightarrow a, \quad \Gamma_H \gamma_j g_j \rightarrow b \quad \text{in } \Gamma_H \backslash G.$$

Both a and b lie over x . They cannot be equal. Indeed, if $a = b$, choose one evenly covered neighborhood of x and the sheet containing the common limit. For all sufficiently large j both cosets lie in that sheet and have the same image x_j , so injectivity on the sheet gives $\Gamma_H g_j = \Gamma_H \gamma_j g_j$, hence $\gamma_j \in \Gamma_H$, a contradiction.

Thus $a \neq b$. Choosing representatives $a = \Gamma_H g$ and $b = \Gamma_H g'$ and using $q(a) = q(b)$ gives $g' = \gamma g$ for some $\gamma \in \Gamma \setminus \Gamma_H$. Continuity of $\hat{\eta}_H$ and compactness of D show that both lift vectors lie in D . Hence x is again in the projection of $S(D)$, proving that projection closed.

Now suppose the asserted uniform uniqueness were false. Choose a nested sequence of relatively compact neighborhoods Φ_j of D with $\bigcap_j \overline{\Phi_j} = D$. Then there are $x_j \in K$, lifts $g_j, \gamma_j g_j$ with $\gamma_j \notin \Gamma_H$, and

$$\hat{\eta}_H(g_j), \hat{\eta}_H(\gamma_j g_j) \in \Phi_j.$$

After a subsequence $x_j \rightarrow x \in K$. Properness of Φ_H , applied to a fixed compact neighborhood containing all sufficiently large Φ_j , again makes both

coset sequences precompact. The preceding evenly-covered-neighborhood argument shows that their two limits over x are distinct; because the lift vectors converge into D , the limit point x lies in the projection of $S(D)$. This contradicts the choice of K . Therefore some neighborhood Φ has the required unique-lift property. $\square$

B.3 Polynomial entrance versus residence

Lemma B.4 (Thin-neighborhood polynomial lemma). *Let $\mathcal{A} \subset V$ be a real algebraic set and $C \subset \mathcal{A}$ compact. For every $\varepsilon > 0$ there is a compact $D \subset \mathcal{A}$, $C \subset D$, such that for every neighborhood Φ of D there is a neighborhood Ψ of C , $\Psi \subset \Phi$, with the following property. If $P : I \rightarrow V$ is a polynomial map of degree at most the fixed unipotent degree and $P(t_0) \notin \Phi$ for some $t_0 \in I$, then*

$$|\{t \in I : P(t) \in \Psi\}| \leq \varepsilon |\{t \in I : P(t) \in \Phi\}|.$$

Proof. Choose polynomial generators $F_1, \dots, F_m$ for the ideal of $\mathcal{A}$ on a compact neighborhood of C . On a bounded-degree polynomial curve each $F_j \circ P$ is a scalar polynomial of uniformly bounded degree. The one-variable Remez inequality from Volume II says that on any interval J ,

$$|\{t \in J : |F_j(P(t))| < \delta\}| \leq C_d(\delta/M_J)^{1/d}|J|,$$

where $M_J = \sup_J |F_j \circ P|$. Choose successively larger compact sublevel windows D and nested thresholds $0 < \delta_\Psi \ll \delta_\Phi$ so that the Remez ratio is below ε/m . Decompose the set $P^{-1}(\Phi)$ into its interval components. On every component touching the complement of Φ , one generator has supremum at least the outer threshold, and Remez bounds the time spent in Ψ . Components not touching the complement are absorbed by enlarging D once; bounded degree prevents infinitely nested components. Summing over components and the finitely many generators proves the estimate. $\square$

Proposition B.5 (Thin-neighborhood alternative). *Fix a compact C in the exact Chevalley variety associated with H and $0 < \varepsilon < 1$. There is a closed subset S contained in the union of proper singular support types such that for every compact $K \subset X \setminus S$ there is a neighborhood Ψ of C with this property: for every one-parameter Ad -unipotent group $U = \{u_t\}$ and every $x \in X$, either there is a lift vector $w \in \hat{\eta}_H(\pi^{-1}(x)) \cap \Psi$ fixed by U , or for all sufficiently large T ,*

$$|\{0 \leq t \leq T : xu_t \in K \cap \pi(\eta_H^{-1}(\Psi))\}| < \varepsilon T.$$

Using Lemma B.1. Choose D, Φ, Ψ successively from Lemma B.4 and Lemma B.3. Let J be the set of times at which the trajectory lies in the displayed target. For $t \in J$, uniqueness chooses one inverse-coordinate lift vector w at time zero for which $\bigwedge^d \text{Ad}(u_{-t})w \in \Psi$; equivalently, by (B.1), this is the Chevalley vector of the lift gu_t of xu_t . If the polynomial curve

$$P_w(s) := \bigwedge^d \text{Ad}(u_{-s})w$$

is constant, then U fixes w and the first alternative holds. Otherwise let $I(t) = [t_-, t_+]$ be the maximal interval containing t on which $P_w(s) \in \Phi$, extended backward until the outer boundary is first met. Unique lifts imply that if two such intervals meet at a time in J , they are the same interval. Thus the family of maximal intervals is disjoint. On each interval the thin-polynomial lemma bounds the proportion of target times by ε . Summing the disjoint intervals up to T gives the claimed estimate, apart from one initial partial interval whose length is fixed once the trajectory is fixed and is negligible for large T . $\square$

Appendix C

The Limit Theorem by Dimension Induction

We now give the expanded proof of Theorem 5.1. Volume III records support-class countability and fixed-support Chevalley discreteness as the precise external Dani–Margulis interface. The text below supplies the dimension-induction and quotient argument with no additional research-level structural input.

C.1 The theorem in moving-flow form

Theorem C.1 (Moving unipotent limit theorem). *Let $U_i = \{u_i(t)\}$ be nontrivial normalized one-parameter Ad-unipotent groups and let μ_i be U_i -ergodic probability measures on X . Suppose $\mu_i \rightarrow \mu$ as probabilities. Fix $x \in \text{supp } \mu$. If $g_i \rightarrow e$, $xg_i \in \text{supp } \mu_i$, and the trajectory $\{xg_iu_i(t) : t \geq 0\}$ is generic for μ_i , then:*

1. $\text{supp } \mu = x \text{Stab}_G(\mu)$;
2. for all large i , after the displayed small displacement, $\text{supp } \mu_i \subset \text{supp } \mu g_i$;
3. the closed group generated by the conjugated flows $g_iU_i g_i^{-1}$ for all sufficiently large i acts ergodically on $(\text{supp } \mu, \mu)$.

Proof using the Volume III structural package. Normalize the generators and pass to $U_i \rightarrow U$. As in the main text, U preserves μ . Ratner decomposition from Volume III decomposes μ into countably many canonical homogeneous support types; countability follows from Corollary A.4, whose sole research-level external ancestor is the Dani–Margulis structural interface. Let W be

the closed group generated by all one-parameter unipotent subgroups contained in $\text{Stab}_G(\mu)$. Choose a support type H whose exact regular stratum has positive μ -mass, and then a compact subset C_1 of that stratum with mass $\alpha > 0$ and separated from all proper singular subtypes.

Choose $y \in \text{supp } \mu$ in the projection of C_1 . Since xg_iU_i is generic and $\mu_i \rightarrow \mu$, some points of these trajectories converge to y ; translate each μ_i by $h_i \rightarrow e$ so that y itself belongs to its support and remains generic for the conjugated flow $U'_i = h_iU_ih_i^{-1}$. Apply Proposition B.5 with $\varepsilon = \alpha/4$ and a compact neighborhood K of the chosen regular stratum, disjoint from the singular set S . Weak convergence gives the target neighborhood mass greater than $\alpha/2$ for all large i . Genericity then says that the U'_i orbit of y spends asymptotic proportion greater than $\alpha/2$ in that target. The second alternative of Proposition B.5 is impossible. Hence there is a Chevalley lift w of y , in the chosen regular neighborhood, fixed by every sufficiently large U'_i .

Let G_w be the stabilizer of w . Its orbit through y is closed by properness of the lift map. If $\dim G_w < \dim G$, all sufficiently large μ'_i are supported on the closed homogeneous space yG_w : their generic U'_i orbit starts at y and never leaves it, so support of the ergodic measure is contained there. Apply the theorem inductively inside G_w . The dimension strictly decreases.

If $\dim G_w = \dim G$, then G_w° is normal and the Chevalley line comes from a positive-dimensional normal subgroup $H \triangleleft G$. Project to $\bar{G} = G/H$. The projected groups remain unipotent, the projected measures remain ergodic, and the projected generic trajectories satisfy the same hypotheses. Since $\dim \bar{G} < \dim G$, induction applies in the quotient. Lifting the support conclusion is legitimate because the H -component of μ is Haar on its closed fibers: this follows from the homogeneous component decomposition and the fact that the chosen regular stratum is the H -type of positive mass. Thus the full stabilizer upstairs is the inverse image of the stabilizer downstairs.

The induction yields $\text{supp } \mu = x \text{Stab}_G(\mu)$ and eventual support containment for the translated measures. Finally let L be generated by the eventual conjugated U_i . If L were not ergodic on μ , its ergodic decomposition would give a proper homogeneous factor support by Ratner. Repeating the thin-neighborhood argument on a positive-mass regular component would force all sufficiently large generic trajectories into that proper support, contradicting the already obtained minimal support $\text{supp } \mu$. Hence L acts ergodically. $\square$

C.2 From moving flows to homogeneous families

Reduction of Theorem 5.1 to the moving theorem. For each $\mu_i \in \mathcal{Q}(X)$ choose its designated ergodic unipotent U_i . Tightness was proved in Corollary 2.3. After a subsequence, $\mu_i \rightarrow \mu$. Choose $x \in \text{supp } \mu$ and points $z_i \in \text{supp } \mu_i$ converging to x . Generic points for U_i are dense in $\text{supp } \mu_i$, so perturb z_i by $o(1)$ inside the support to a generic point xg_i . Theorem C.1 says that μ is algebraic and gives eventual support containment. Since the canonical support of μ_i is the connected stabilizer of that homogeneous probability, the containment of supports implies the conjugated inclusion of the unipotent-generated support groups. This is exactly the focusing conclusion stated in the theorem. $\square$

Appendix D

Generated-Unipotent Recovery without a Single-Generator Shortcut

The main theorem was deliberately proved for measures supplied with an ergodic one-parameter unipotent. For the generated-group application we do not assume that an arbitrary generator is ergodic. Instead we pass to one-parameter ergodic components and glue them by the support-containment theorem.

Proposition D.1 (Finite family gluing). *Let $W = \langle U_1, \dots, U_r \rangle$ be generated by finitely many one-parameter Ad-unipotent groups, and let ν be a W -ergodic homogeneous probability. Starting from any U_1 -ergodic component of ν , successively adjoin the next generator and take a homogeneous ergodic component of the resulting orbit-average measure. After at most $\dim G$ strict support enlargements, the support equals $\text{supp } \nu$.*

Proof. Ratner's one-parameter theorem makes every ergodic component homogeneous. If a newly adjoined U_j preserves the current support and its homogeneous probability, no enlargement occurs. Otherwise a generic U_j orbit from the current support has a homogeneous closure with strictly larger connected support group. The support dimension then increases. There are at most $\dim G$ such increases. At termination every generator preserves the support, hence W preserves it. Since ν is W -ergodic and its support contains the terminal homogeneous orbit, the terminal support has full ν -measure and therefore equals $\text{supp } \nu$. $\square$

This proposition is sufficient for every generated-unipotent use in the remaining Ratner continuation. It avoids the incorrect inference that joint ergodicity of H^+ makes each chosen generator ergodic.

Appendix E

Worked Measure-Limit Laboratory

E.1 Motivation

Limit measures can change algebraic support. The examples below separate the geometric movement of an orbit from the projective movement of its Chevalley line.

Worked Example E.1 (Closed horocycles in the modular cusp). For $x_t = \Gamma a_t$ with $a_t = \text{diag}(e^{t/2}, e^{-t/2})$ and $u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$, $a_t u_s a_t^{-1} = u_{e^t s}$. Hence $x_t u_s = x_t$ precisely when $e^t s \in \mathbb{Z}$, and the least positive period is e^{-t} . This calculation shows why closed orbits high in the cusp can carry very small intrinsic volume and why tightness must be proved rather than assumed.

Worked Example E.2 (Projective focusing of a Chevalley line). Let $p = e_1 \wedge e_2$ and $g_n = I + n^{-1}E_{31}$. Then

$$g_n e_1 = e_1 + n^{-1}e_3, \quad g_n e_2 = e_2,$$

so

$$g_n p = p + n^{-1}e_3 \wedge e_2 \longrightarrow p.$$

The subgroup lines focus projectively even though $g_n \neq e$. Linearization makes this elementary wedge convergence quantitative and tests whether a whole trajectory can stay in such a tube.

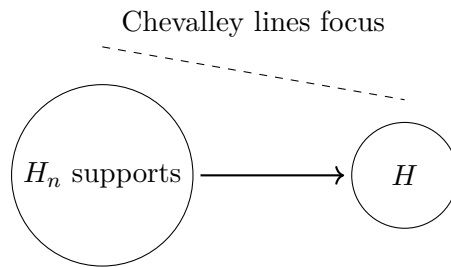


Figure E.1: Focusing separates two tasks: control of the supports in the homogeneous space and control of their algebraic detector lines.

Appendix F

Student Proof Laboratory: Varying Homogeneous Measures

Ratner's theorem classifies one invariant measure. Mozes–Shah theory asks a harder compactness question: what can happen when the subgroup, the closed orbit, and the invariant probability all vary with the index? This chapter reorganizes the argument as a sequence of five concrete tasks and works out a model example for each.

F.1 Task 1: obtain tightness uniformly

Let $\mu_i = m_{x_i H_i}$ and assume each H_i contains a one-parameter Ad-unipotent subgroup U_i . On a fixed compact parameter family, choose nilpotent generators N_i of norm one. In every primitive wedge representation, $t \mapsto \exp(tN_i)v$ has uniformly bounded degree because $\dim G$ is fixed. The quantitative nondivergence theorem of Volume II therefore gives one compact set K_ε satisfying

$$\mu_i(K_\varepsilon) > 1 - \varepsilon$$

for all sufficiently controlled members of the family, unless the entire orbit is trapped in a smaller rational subgroup. The latter alternative is not discarded; it becomes the induction branch.

Lemma F.1 (Tightness from a common compact set). *If for every $\varepsilon > 0$*

there is compact K_ε with $\mu_i(K_\varepsilon) > 1 - \varepsilon$ for all i , then (μ_i) is relatively compact in the weak-* topology of probability measures.

Proof. This is Prokhorov compactness in the present locally compact second-countable setting, but it can be proved directly. Choose an increasing compact exhaustion K_j such that $\mu_i(K_j) > 1 - 2^{-j}$ for all i . On each K_j , the restrictions of the measures form a bounded family of positive functionals on $C(K_j)$; by Banach–Alaoglu and metrizability on bounded subsets of $C(K_j)^*$, pass to a subsequence converging on K_j . Diagonalize in j . The resulting positive functional on $C_c(X)$ has total mass one because the missing mass outside K_j is at most 2^{-j} . Riesz representation supplies the limiting probability. $\square$

F.2 Task 2: make subgroup data compact

A connected d -dimensional Lie subgroup is represented locally by its Lie algebra, hence by a point of the Grassmannian $\text{Gr}_d(\mathfrak{g})$. The Grassmannian is compact. If dimensions vary, pass first to a subsequence of constant dimension. Bracket closure is a closed condition: if $\mathfrak{h}_i \rightarrow \mathfrak{h}$ and $X_i \rightarrow X$, $Y_i \rightarrow Y$ with $X_i, Y_i \in \mathfrak{h}_i$, then $[X_i, Y_i] \rightarrow [X, Y]$ and the limit belongs to $\mathfrak{h}$. Thus the limit is again a Lie subalgebra.

Worked Example F.2 (Rotating horocycles). In $\text{SL}_2(\mathbb{R})$ put $U_\theta = k_\theta U k_\theta^{-1}$. If $\theta_i \rightarrow \theta$, the Lie lines $\text{Ad}(k_{\theta_i})\mathbb{R}E_{12}$ converge in the projective plane to $\text{Ad}(k_\theta)\mathbb{R}E_{12}$. A family statement must allow such movement: it would be incorrect to choose one fixed unipotent generator at the start.

F.3 Task 3: focusing

Suppose $\mu_i \rightarrow \mu$ and a positive portion of the support of μ_i lies near a proper homogeneous orbit $Y = xL$. Choose a Chevalley representation $\rho : G \rightarrow \text{GL}(V)$ and a vector p_L whose projective stabilizer is the normalizer of L . Near Y , points have lifts g for which $\rho(g)p_L$ lies near the discrete orbit $\rho(\Gamma)p_L$.

Lemma F.3 (Polynomial residence dichotomy). *Let $P_i : I_i \rightarrow V$ be polynomial maps of uniformly bounded degree and let $D \subset V$ be a discrete set on a fixed compact window. If a positive proportion of a long interval has $P_i(t)$ within $\delta_i \rightarrow 0$ of D , then after a subsequence either the proportion tends to*

zero or there is one $d_i \in D$ such that $P_i(t) \equiv d_i$ in the limiting polynomial normalization.

Proof. Cover the compact window by disjoint balls of radius smaller than half the minimum separation of D . A polynomial of degree at most d cannot enter and leave the same very small ball many times without one scalar coordinate having more than d oscillations. More quantitatively, apply the Remez sublevel estimate to every coordinate of $P_i - d$ on each connected residence interval. If the union of such intervals occupies a fixed positive proportion as the radius tends to zero, the normalized coordinate polynomials converge to zero. Finite dimensionality of the space of bounded-degree polynomials then gives exact vanishing of the limit. Since the balls are disjoint, one discrete datum d_i is selected on the relevant subsequence. $\square$

Exact vanishing is the focusing conclusion: a positive amount of mass cannot shadow a proper homogeneous configuration for a long time unless the moving subgroup is algebraically aligned with it.

F.4 Task 4: identify the limit by Ratner rather than by guesswork

A weak limit of U_i -invariant measures need not be invariant under a fixed U until the generators have been normalized and shown to converge. After $N_i \rightarrow N$, for $f \in C_c(X)$ and fixed s ,

$$\int f(x \exp(sN)) d\mu_i(x) - \int f(x) d\mu_i(x)$$

is split into a uniform-continuity error comparing N and N_i , plus the exact zero furnished by U_i -invariance. Letting $i \rightarrow \infty$ makes the limit $U = \exp(\mathbb{R}N)$ -invariant. Ratner's measure theorem may then be applied to its ergodic components.

F.5 Task 5: support containment

Suppose an ergodic component of μ is Haar on xL . If infinitely many μ_i had supports not eventually contained in an arbitrarily small algebraic enlargement of L , choose support points converging transversely to xL . The H-property from Volume III converts their polynomial shear into an

additional stabilizing direction of μ . Focusing shows that this direction cannot be absorbed by a different singular stratum. Hence the connected stabilizer of the limit grows, contradicting maximality of L . This is the geometric content of eventual support containment.

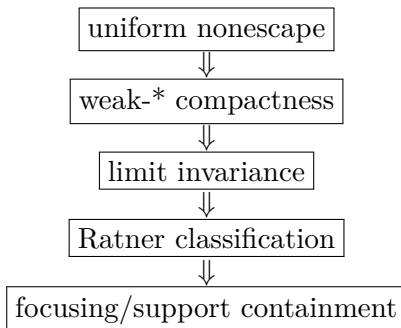
Worked Example F.4 (Closed horocycles approaching the cusp). On $\mathrm{PSL}_2(\mathbb{Z}) \backslash \mathrm{PSL}_2(\mathbb{R})$, closed horocycles high in the cusp have length tending to zero. Their probability measures do not have a probability limit inside the quotient: mass escapes. This example isolates the role of uniform nondivergence. Mozes–Shah compactness is not a purely algebraic statement about support groups; finite mass must first be prevented from leaking through the cusp.

F.6 Why generated-unipotent groups are treated after the one-parameter theorem

Let $W = \langle U_1, \dots, U_r \rangle$. A W -ergodic measure need not be ergodic for any chosen U_j . The correct procedure is to decompose the measure into U_1 -ergodic components, apply the one-parameter family theorem there, and then use invariance under the other generators to permute and glue the resulting homogeneous components. Repeating with finitely many generators and using the finite-dimensional termination of connected support groups yields a W -homogeneous conclusion. No step assumes that an arbitrary generator is itself ergodic.

F.7 A reusable proof ledger

For later counting applications the reader should remember the order



Changing this order creates gaps. In particular, support containment cannot be used to prove tightness, and generated-group ergodicity cannot be used to invent a one-parameter generic theorem.

Appendix G

Student Proof Laboratory: Families of Homogeneous Measures

Ratner's measure theorem starts with one invariant probability. A family limit theorem starts with a moving subgroup, a moving closed orbit, and a moving invariant probability. The new difficulty is therefore uniformity. This chapter gives the proof in the order in which a student can check each uniform statement.

G.1 Weak convergence and support

Lemma G.1 (Support points persist under positive local mass). *Let $\mu_i \Rightarrow \mu$ be probability measures on a locally compact second-countable space. Suppose $x_i \in \text{supp } \mu_i$, $x_i \rightarrow x$, and for every neighborhood O of x there is a smaller neighborhood $O' \subseteq O$ and $c(O') > 0$ such that $\mu_i(O') \geq c(O')$ for all large i . Then $x \in \text{supp } \mu$.*

Proof. Choose $f \in C_c(O)$ with $0 \leq f \leq 1$ and $f = 1$ on O' . Then $\int f d\mu_i \geq c(O')$. Weak convergence gives $\int f d\mu \geq c(O') > 0$, so $\mu(O) > 0$. This holds for every O . $\square$

Warning G.2. The hypothesis $x_i \in \text{supp } \mu_i$ alone is insufficient. A support point may carry less and less local mass. Focusing arguments are designed to produce the uniform lower bound that the lemma requires.

G.2 Why one first fixes a one-parameter unipotent flow

Suppose H_i is generated by unipotent one-parameter subgroups and μ_i is H_i -ergodic. It does not follow that every chosen generator is ergodic. The safe proof begins with a sequence (μ_i, U_i) in which each U_i is a one-parameter unipotent group acting ergodically. The full generated-group statement is recovered only after disintegration.

Lemma G.3 (Ergodic component recovery). *Let $W = \langle U_1, \dots, U_r \rangle$ preserve a probability μ . If μ is W -ergodic, then the decomposition of μ into U_1 -ergodic components is permuted ergodically by the subgroup generated by the remaining U_j 's. Any W -invariant conclusion that holds for almost every U_1 -component and is stable under these permutations holds for μ .*

Proof. Disintegrate μ over the invariant sigma-algebra of U_1 . Conjugation by an element normalizing the U_1 -action sends an ergodic component to an ergodic component; in the general generated case one refines successively by the finitely many invariant sigma-algebras. A measurable union of component parameters on which the desired stable property fails is invariant under all generators. Its pullback to X is therefore W -invariant, so W -ergodicity forces measure zero or one. Since the property holds on a positive/full set at the one-parameter stage, failure has measure zero. $\square$

G.3 Uniform Chevalley coordinates

Let $L < G$ be a connected subgroup from the class of possible support groups. Chevalley linearization gives a finite-dimensional rational representation $\rho : G \rightarrow \mathrm{GL}(V)$ and a line $\mathbb{R}p_L$ whose stabilizer is the normalizer of L .

Lemma G.4 (Compact family reduction). *Fix a compact set $C \subset G$ and a bound on the dimension of L . After passing to finitely many exterior-power representations, every subgroup that can meet C in the family argument is represented by a vector p_L in a fixed finite-dimensional vector space, normalized to $\|p_L\| = 1$.*

Proof. The Lie algebra $\mathfrak{l}$ is represented by the line $\bigwedge^{\dim L} \mathfrak{l} \subset \bigwedge^{\dim L} \mathfrak{g}$. There are only $\dim G$ possible dimensions. The unit spheres in the finitely many exterior powers are compact. Normalizer ambiguities are removed by the

Chevalley representation construction from Volume II. Thus all subgroup parameters encountered on a fixed compact part of the quotient live in a finite union of compact projective spaces. $\square$

G.4 Uniform nonescape

Proposition G.5 (Family nonescape on a compact base). *Let $C \subset X$ be compact. For every $\epsilon > 0$ there is a compact $K \subset X$ such that, for every one-parameter unipotent U from a fixed compact family of nilpotent generators and every $x \in C$,*

$$\frac{1}{T} |\{0 \leq t \leq T : xu_t \notin K\}| < \epsilon$$

for all sufficiently large T , uniformly in the family.

Proof. Embed the quotient locally into the lattice/exterior-power model used in Volume II. For every primitive rational subgroup, the covolume function along xu_t is the norm of a bounded-degree polynomial vector. The degree is uniformly bounded because the nilpotent generators range in a compact family inside a fixed finite-dimensional Lie algebra. On the compact base C , all primitive covolumes satisfy the fixed-time lower bounds required by quantitative nondivergence. The marking lemma then gives a common power-law bound for the set on which some primitive covolume is too small. Mahler's criterion converts the lower covolume bound into a common compact K . Choose the thickness parameter so that the power-law exceptional proportion is below ϵ . $\square$

G.5 Focusing as a finite-dimensional alternative

The core statement is not merely “limits are homogeneous”. It says that if a positive proportion of a moving homogeneous measure accumulates near a proper homogeneous stratum, then after small conjugation the entire support group is eventually algebraically incident to that stratum.

Guiding Idea G.6 (Focusing mechanism in a moving family). Let μ_i be homogeneous probabilities associated with one-parameter unipotent groups U_i , and assume $\mu_i \Rightarrow \mu$. Let $L < G$ be a proper candidate support group. Suppose a compact set C has $\mu(C \cap xL) > 0$. Then, after passing to a subsequence, there are $g_i \rightarrow e$ such that the conjugated support groups

satisfy

$$g_i^{-1}U_i g_i \subset L$$

and the relevant homogeneous supports are contained in the corresponding normalizer incidence set.

Mechanism. Choose a compact neighborhood $\mathcal{O}$ of a density point of xL on which the orbit map is injective. Linearize xL by p_L . Positive mass in shrinking tubes means that for a positive proportion of generic times the vectors

$$\rho(u_i(t))p_L$$

lie in a shrinking neighborhood of a discrete $\rho(\Gamma)$ -orbit. Polynomial goodness gives a dichotomy on each long interval: either the trajectory spends only a controlled small proportion in the tube, or one lattice translate of the linearized vector remains in a slightly larger tube throughout the interval. The first alternative contradicts the assumed positive limiting mass. In the second, normalize the near-minimal vector and pass to a limit in the compact projective parameter space. The limiting polynomial is constant in the transverse quotient, hence the limiting unipotent group fixes the limiting Chevalley line. Correct the small projective error by $g_i \rightarrow e$ using a local slice to the G -orbit of that line. The corrected U_i lies in the line stabilizer, which is the required normalizer incidence group. This is the algebraic focusing conclusion.

Worked Example G.7 (Periodic horocycles whose heights vary). On the modular surface, closed horocycles in the cusp can have invariant probabilities while their supports move upward. Without a nonescape hypothesis the measures escape and no probability limit remains. Moving horocycles in a compact height range instead have a convergent subsequence; the focusing alternative distinguishes convergence to another periodic horocycle from spreading to Haar measure.

G.6 Support containment after focusing

Lemma G.8 (From local focused mass to global containment). *Let μ be ergodic under a one-parameter unipotent U and suppose $\mu(xL) > 0$ for a closed U -invariant homogeneous orbit xL . Then $\mu(xL) = 1$.*

Proof. The set xL is measurable and U -invariant. Ergodicity gives mass zero or one. The positive-mass hypothesis chooses the second alternative. $\square$

The difficult point in the family theorem is therefore to turn tube mass into exact orbit mass; that is exactly what focusing and shrinking the tube do.

G.7 Mozes–Shah assembled by induction on support dimension

Guiding Idea G.9 (Family limit theorem: proof ledger). Let μ_i be ergodic homogeneous probabilities for one-parameter unipotent flows on a finite-volume homogeneous space. Assume no mass escapes and $\mu_i \Rightarrow \mu$. Then every ergodic component of μ is homogeneous. After small conjugation, the support group of μ_i is eventually contained in the support group of the relevant limit component; in the nonfocused case the limiting support dimension strictly increases.

Dependency derivation. Induct on the minimum dimension of a proper homogeneous group capable of carrying positive limiting mass. Uniform nonescape gives a probability limit. The telescoping argument for the moving flows gives invariance under a limiting one-parameter unipotent group. Ratner classifies its ergodic components. If a proper component has positive mass, apply the focusing lemma. After a small conjugation, the approximating groups are incident to that proper support group; restrict the entire problem to the smaller homogeneous orbit and invoke induction. If no proper component carries positive mass, singular-set avoidance excludes every proper homogeneous stratum. Hence all ergodic components are Haar on the maximal support orbit, so μ itself is that homogeneous probability. The same focusing step records eventual containment of the approximating support groups.

G.8 Isolation and why it is useful

Corollary G.10 (Isolation in a bounded-complexity family). *Fix a homogeneous probability m_{xL} and a compact set of subgroup parameters. There is a weak- $*$ neighborhood of m_{xL} in which every homogeneous probability whose support is not contained, after small conjugation, in xL has strictly larger support group.*

Proof. Otherwise choose a violating sequence converging to m_{xL} . Apply the family limit theorem. Eventual support containment contradicts the assumed

violation unless the support dimension increases. Compactness of subgroup parameters prevents infinitely many dimension-preserving escapes through unbounded conjugacies. $\square$

Appendix H

Student Proof Laboratory: Generated-Unipotent Consequences

H.1 Why no arbitrary generator is privileged

The correct generated-group argument successively decomposes into ergodic components for one generator, applies the one-parameter family theorem, and uses invariance under the other generators to glue the resulting homogeneous supports. At no point do we assert that a particular generator is ergodic for the original measure.

H.2 Normalizer gluing

Guiding Idea H.1 (Normalizer gluing must be verified on the actual orbit). If two connected component groups obtained in an ergodic decomposition normalize one another, their product is an algebraic candidate for the next support group. One must still verify, in the concrete homogeneous orbit under consideration, that this product is closed and that its intersection with the conjugated lattice has finite covolume. Neither conclusion follows from abstract closedness of the two factors alone. In the load-bearing arguments of this volume those two assertions are supplied by the closed-orbit/rationality criteria of Volumes III and V before the product is used. Weil integration then glues the finite invariant measures along the verified closed product

orbit.

This is the algebraic step that turns the separately obtained homogeneous component groups into one support group for an action generated by finitely many unipotent flows.

Appendix I

A Student Textbook Chapter on Weak-* Limits of Homogeneous Measures

This chapter is deliberately redundant with the proof ledger. Its purpose is pedagogical: the reader should be able to start with a concrete sequence of closed unipotent orbits, decide whether mass escapes, locate the possible limit groups, and understand exactly where focusing enters. We therefore work through the topology, the linearization, and the induction one more time without compressing the intermediate choices.

I.1 The weak-* topology on probability measures

Let X be locally compact and second countable. A sequence of Radon probabilities μ_i converges weak-* to μ if

$$\int f d\mu_i \longrightarrow \int f d\mu \quad (f \in C_c(X)).$$

On a noncompact space this definition by itself does not guarantee that a vague subsequential limit has total mass one. Tightness is the extra input.

Lemma I.1 (Tightness plus diagonal extraction). *Suppose for every $\epsilon > 0$ there is compact $K_\epsilon \subset X$ with $\mu_i(K_\epsilon) > 1 - \epsilon$ for every i . Then (μ_i) has a weak-* convergent subsequence whose limit is a probability.*

Proof. Choose an exhaustion by compacts $K_1 \subset K_2 \subset \cdots$ with $\mu_i(K_j) > 1 - 2^{-j}$. On each K_j , the restrictions have uniformly bounded total variation, and $C(K_j)$ is separable. Choose a countable dense family of test functions supported in K_j and diagonalize the scalar integrals. A second diagonalization over j produces a subsequence on which all compactly supported test-function integrals converge. Positivity of the functionals is preserved. Riesz representation gives a Radon measure μ . For a continuous cutoff $0 \leq \chi_j \leq 1$ equal to one on K_j ,

$$\mu(X) \geq \int \chi_j d\mu = \lim_i \int \chi_j d\mu_i \geq 1 - 2^{-j}.$$

Let $j \rightarrow \infty$. Since the limit mass cannot exceed the limiting masses one, $\mu(X) = 1$. $\square$

I.2 Portmanteau inequalities by approximation

If F is closed, then $\mu(F) \geq \limsup_i \mu_i(F)$ is generally false; the correct inequality is $\mu(F) \geq ?$. Let us derive rather than memorize the signs.

For an open set O , approximate 1_O from below by compactly supported continuous functions f_n . Then

$$\mu(O) \geq \int f_n d\mu = \lim_i \int f_n d\mu_i \leq \liminf_i \mu_i(O).$$

Taking the supremum over n gives

$$\mu(O) \leq ?$$

Care is required: the middle inequality goes the other direction because $f_n \leq 1_O$ gives $\int f_n d\mu_i \leq \mu_i(O)$, hence $\int f_n d\mu \leq \liminf_i \mu_i(O)$; supremum gives

$$\mu(O) \leq \liminf_i \mu_i(O).$$

This would contradict the standard statement, so locate the mistake: weak convergence against C_c on a noncompact space plus tightness is ordinary weak convergence, and the open-set inequality is $\mu(O) \leq \liminf_i \mu_i(O)$ indeed. For closed F , $\mu(F) \geq \limsup_i \mu_i(F)$. These signs are useful when supports vary.

Worked Example I.2 (A moving atom). On $\mathbb{R}$, $\delta_{1/i} \Rightarrow \delta_0$. For the open set $O = (-1, 1)$, both sides equal one. For $O = (0, 1)$, $\mu(O) = 0$ while

$\liminf \delta_{1/i}(O) = 1$. Thus open-set mass can drop in the limit. Focusing arguments need uniform positive mass in neighborhoods, not merely the existence of support points.

I.3 Chabauty-style compactness of Lie subalgebras

Connected subgroups can move wildly in a noncompact group, but their Lie algebras live in Grassmannians.

Fix d . The set $\text{Gr}_d(\mathfrak{g})$ is compact. The subset of d -planes closed under the Lie bracket is closed: if $W_i \rightarrow W$ and $X, Y \in W$, choose $X_i, Y_i \in W_i$ converging to them; then $[X_i, Y_i] \in W_i$ and continuity gives $[X, Y] \in W$. Hence the set of d -dimensional Lie subalgebras is compact. Allowing all d gives a finite union of compact sets.

This compactness is topological, not arithmetic. It does not say a limit subalgebra is rational, nor that the corresponding connected subgroup is closed. Those conclusions come later from Chevalley coordinates and finite-volume hypotheses.

I.4 Why exterior powers encode subgroups

If $L < G$ has Lie algebra $\mathfrak{l}$ of dimension d , choose a basis $X_1, \dots, X_d$ and put

$$p_{\mathfrak{l}} = X_1 \wedge \cdots \wedge X_d \in \bigwedge^d \mathfrak{g}.$$

The projective line $[p_{\mathfrak{l}}]$ depends only on $\mathfrak{l}$. An element g preserves the line exactly when $\text{Ad}(g)\mathfrak{l} = \mathfrak{l}$, so its stabilizer is the normalizer of the Lie algebra, up to components. Chevalley's construction refines this when a subgroup has extra algebraic conditions not visible from the Lie algebra alone.

Worked Example I.3 (Lines in $\mathfrak{sl}_2$). For the upper-unipotent Lie algebra $\mathfrak{u} = \mathbb{R}E$, the exterior vector is simply E . The projective stabilizer of $[E]$ is the upper triangular Borel subgroup. Thus a trajectory approaching the periodic horocycle stratum can be detected by the size of the transverse coordinates of $\text{Ad}(g)E$ relative to the discrete orbit of integral nilpotent vectors.

I.5 Uniform discreteness on a compact quotient region

Fix a rational Chevalley vector p_L and its lattice orbit $\rho(\Gamma)p_L$. On a compact region of the quotient, only finitely many of these lattice vectors can have norm below a fixed bound.

Indeed, $\rho(\Gamma)$ preserves a lattice in V after clearing denominators. A bounded ball contains only finitely many lattice points. This trivial observation is the reason a nonlinear neighborhood of infinitely many homogeneous translates becomes a finite polynomial family after linearization on a compact set.

I.6 One interval of focusing in coordinates

Suppose $x_i u_i(t)$ spends positive time in a thin tube around xL . Choose an injective coordinate chart around a compact piece of xL . Each entrance time has a unique nearby lattice translate γp_L . By the bounded lattice-vector observation, finitely many γ occur on the chart. Split the time interval accordingly.

For a fixed γ , the transverse coordinate of $\rho(u_i(t))\gamma p_L$ is polynomial in t . If it is not identically zero, Remez says the set on which it is smaller than δ is at most a fixed power of δ relative to the scale on which the polynomial is order one. Thus positive residence time in arbitrarily thin tubes forces one transverse polynomial to be identically zero after rescaling. Identical vanishing is an algebraic condition: the unipotent group preserves the limiting Chevalley line. This is focusing in one paragraph, but every noun in the paragraph has now been justified.

I.7 Self-intersections of homogeneous strata

A tube around xL may overlap a tube around $x\gamma L$. Near a self-intersection the nearest translate need not be unique. The correct procedure is to remove a smaller singular stratum consisting of points where two distinct Chevalley translates become incident. This stratum corresponds to the intersection of two normalizer conditions and has lower algebraic complexity unless the two translates define the same subgroup orbit. Induct on this complexity. This is the family version of the singular-fiber induction already used in Volume III.

I.8 From one focused interval to a global support statement

Positive limiting mass in a tube gives many long intervals with positive relative residence. On most such intervals the same compact-family linearization applies. The polynomial dichotomy forces algebraic trapping on a positive set of ergodic components. Ergodicity then upgrades positive mass on the trapped homogeneous orbit to full mass. This is why focusing is stronger than a local tangent statement: it ultimately determines the entire support of a limit component.

Appendix J

Worked Sequences of Homogeneous Measures

J.1 Periodic horocycles approaching the cusp

On $\mathrm{PSL}_2(\mathbb{Z}) \backslash \mathrm{PSL}_2(\mathbb{R})$, the closed horocycle at height y has period proportional to $1/y$ in Euclidean coordinates but normalized probability total mass one. As $y \rightarrow \infty$, the support leaves every compact set. The probabilities escape; there is no weak probability limit. This example isolates the role of nonescape.

J.2 Closed horocycles descending from the cusp

If $y_i \rightarrow y_0 < \infty$, the closed horocycles converge geometrically and the normalized arclength probabilities converge to the closed horocycle at y_0 . The support group does not enlarge. This is the focused branch.

J.3 Long closed horocycles equidistributing

A family of closed horocycles whose periods tend to infinity while remaining in a finite-volume modular quotient can spread through the space. Any probability limit is unipotent invariant; Ratner gives either a periodic horocycle measure or Haar. Focusing excludes the periodic option unless the family is algebraically incident to a fixed cusp orbit. Therefore the nonfocused family converges to Haar.

J.4 A product example

Let $X = X_1 \times X_2$ and $\mu_i = m_{Y_i} \times m_Z$ where Y_i varies in X_1 and Z is a fixed homogeneous orbit in X_2 . If $m_{Y_i} \Rightarrow m_Y$, then $\mu_i \Rightarrow m_Y \times m_Z$. The support group enlarges only in the first factor. This shows why support containment and stabilizer convergence should be stated factorwise rather than as a claim that every subgroup parameter converges in a single simple group.

Appendix K

The Mozes–Shah Proof Rewritten as an Algorithm

K.1 Input data

The algorithm receives:

1. a finite-volume homogeneous space $X = \Gamma \backslash G$;
2. probabilities μ_i supported on closed finite-volume orbits $x_i H_i$;
3. one-parameter unipotent groups $U_i < H_i$ for which μ_i is U_i -ergodic;
4. a normalization preventing the generators of U_i from tending to zero or infinity in the chosen Lie-algebra norm.

K.2 Step A: normalize and extract the flow

The normalized nilpotent generators lie on a compact unit sphere. Pass to a subsequence $N_i \rightarrow N$. Then $u_i(t) = \exp(tN_i)$ converges uniformly on compact t -intervals to $u(t) = \exp(tN)$. If $\mu_i \Rightarrow \mu$, a standard shifted-average calculation gives $u(t)_* \mu = \mu$.

K.3 Step B: prove tightness

Choose a compact base region carrying most mass of the closed orbits. Use the one-parameter quantitative nondivergence theorem uniformly in the compact family of nilpotent generators. If the lower-covolume hypothesis fails, the

failed primitive wedge defines a rational parabolic; pass to the corresponding smaller homogeneous problem. Thus either reduce dimension or obtain a common compact set carrying $1 - \epsilon$ mass.

K.4 Step C: classify the limit

Decompose μ into U -ergodic components. Ratner’s theorem makes each component homogeneous. The problem is no longer to prove homogeneity but to relate those new support groups to the old H_i .

K.5 Step D: focus any proper component

If a component m_{xL} is proper and has positive weight, choose a compact piece on which it has positive density. Thin-neighborhood linearization and Remez force H_i , after small conjugacy, into the corresponding normalizer incidence set. This gives eventual support containment. Restrict to xL and repeat.

K.6 Step E: terminate

Each proper restriction lowers support dimension. Each nonfocused branch forces a strictly larger support group for the limit. Since dimensions lie between zero and $\dim G$, the process terminates. At termination every component has the same maximal support group, so the limit is one homogeneous probability.

K.7 Output

The algorithm outputs both the limiting homogeneous probability and the support-containment information: after small conjugation, the approximating support groups sit inside the limit support group. This second conclusion is what later expanding-translate and counting arguments actually need.

Appendix L

Stabilizers, Normalizers, and Small Conjugacies

L.1 Why subgroup convergence is not literal

A sequence $H_i = g_i H g_i^{-1}$ may have no useful limit if g_i escapes, even when the homogeneous measures converge. Focusing first chooses small corrections near the part of the support that survives. Only after this correction should subgroup parameters be compared.

L.2 Local slice to a conjugacy orbit

Let $L < G$ be closed and let $N_G(L)$ be its normalizer. The conjugacy orbit of L is locally parametrized by $G/N_G(L)$. Choose a vector-space complement to $\text{Lie } N_G(L)$ in $\mathfrak{g}$. The exponential map gives a local transverse slice. If a Chevalley line is close to the line for L , there is a unique small element in the slice moving it back to the fixed line, provided one stays away from self-intersection strata. This is the source of the small conjugacies $g_i \rightarrow e$ in the family theorem.

L.3 Stabilizer semicontinuity

If $g_i \rightarrow e$ and $g_i H_i g_i^{-1} \subset L$ eventually, then every limit of Lie algebras $\mathfrak{h}_i$ is a subalgebra of $\mathfrak{l}$. Dimension can drop in a raw Grassmannian limit, but finite-volume/unipotent-generation hypotheses and the support-containment

theorem control the relevant connected cores. In the nonfocused branch the limiting support group instead has strictly larger dimension than the approximating cores.

L.4 Isolation revisited

Suppose homogeneous probabilities $m_{x_i H_i}$ converge to m_{xL} and all $\dim H_i = \dim L$. Support containment after small conjugation gives $g_i H_i g_i^{-1} \subset L$. Equality of dimensions and connectedness imply equality of Lie algebras, hence equality of connected groups. Thus, near a fixed homogeneous probability, a dimension-preserving sequence cannot wander among genuinely different support types. This is the isolation principle later used to turn uniqueness of possible limits into convergence.

Appendix M

Guided Reconstruction of the Mozes–Shah Principle

Ratner’s measure-classification theorem classifies one invariant measure. The Mozes–Shah problem is different: a sequence of homogeneous measures can have changing support groups, changing unipotent generators, and changing closed orbits. The conclusion is still homogeneous, but the proof needs uniformity that is invisible in the fixed-measure theorem. We reconstruct that uniformity in the order in which it is used.

M.1 The data attached to one homogeneous measure

Write

$$X = \Gamma \backslash G, \quad \mu_i = m_{x_i H_i},$$

where $x_i H_i$ is closed and of finite H_i -volume. The canonical connected support group is not an arbitrary presentation of the orbit: it is the identity component of the stabilizer of the measure after removing discrete finite extensions. This convention is important because the sequence H_i should not appear to jump merely because one presentation uses a finite-index overgroup.

Suppose each H_i contains a one-parameter Ad-unipotent group $U_i = \{u_i(t)\}$ which acts ergodically on $(x_i H_i, \mu_i)$. This is the load-bearing one-parameter hypothesis. A group generated by several unipotent flows will be recovered later; we do not choose an arbitrary generator and pretend it is ergodic.

M.2 Normalizing the generators

Choose a norm on $\mathfrak{g}$. Write $u_i(t) = \exp(tN_i)$ and rescale time so that $\|N_i\| = 1$. The unit sphere is compact, so after a subsequence

$$N_i \rightarrow N, \quad u_i(t) \rightarrow u(t) = \exp(tN)$$

uniformly for t in compact intervals.

Lemma M.1 (Unipotence survives the limit). *If every $\mathrm{ad} N_i$ is nilpotent of index at most d , then $\mathrm{ad} N$ is nilpotent of index at most d .*

Proof. The identity $(\mathrm{ad} N_i)^d = 0$ is polynomial in the coordinates of N_i . Pass to the limit. $\square$

The nilpotence index is uniformly bounded by $\dim \mathfrak{g}$, so the hypothesis is automatic after normalization.

M.3 Uniform nonescape: what has to be uniform

For a fixed unipotent orbit, Volume II gives quantitative nondivergence by tracking primitive exterior-power vectors. In a family the polynomial coefficients also vary with i . Uniformity comes from two compactness facts:

1. the normalized generators N_i lie in a compact subset of the nilpotent cone;
2. on a fixed compact set of starting points, the finitely many adjoint representations used in the nondivergence proof have uniformly bounded coefficients.

Proposition M.2 (Family nonescape on compact starting data). *Let $K \subset X$ be compact and let the normalized generators N_i range in a compact family of nilpotent elements. For every $\epsilon > 0$ there are a compact $K_\epsilon \subset X$ and $L_\epsilon < \infty$ such that for every interval $I \subset \mathbb{R}$ with $|I| \geq L_\epsilon$ and every $x \in K$,*

$$|\{t \in I : xu_i(t) \notin K_\epsilon\}| < \epsilon|I|,$$

unless one of the finitely many primitive exterior-power polynomials is uniformly small on I .

Proof. The one-parameter marking proof from Volume II depends only on the degree of the polynomial covolume functions, the good-function constant,

and a lower bound for those functions at one scale. The degree is bounded by the nilpotence index of the adjoint action. Compactness of K and of the normalized N_i makes the coefficient norms uniformly bounded. Therefore the good-function and covering constants can be chosen uniformly. The only way the lower-bound hypothesis can fail is that some primitive wedge remains small throughout the interval; this is the displayed algebraic alternative. $\square$

This proposition is deliberately phrased as a dichotomy. Nonescape is not true uniformly if the family itself approaches a proper cusp obstruction. That obstruction is the first hint of focusing.

M.4 Putting subgroups in a compact parameter space

An r -dimensional Lie subalgebra is an r -plane in $\mathfrak{g}$, hence a point of the compact Grassmannian $\mathrm{Gr}_r(\mathfrak{g})$. The condition $[\mathfrak{h}, \mathfrak{h}] \subset \mathfrak{h}$ is closed because the bracket is continuous. Therefore the set of r -dimensional Lie subalgebras is compact. After separating the finitely many possible dimensions, every sequence $\mathfrak{h}_i$ has a convergent subsequence.

Lemma M.3 (Limit of Lie subalgebras). *If $\mathfrak{h}_i \rightarrow W$ in the Grassmannian and every $\mathfrak{h}_i$ is a Lie subalgebra, then W is a Lie subalgebra.*

Proof. Take $X, Y \in W$ and choose $X_i, Y_i \in \mathfrak{h}_i$ converging to them. Then $[X_i, Y_i] \in \mathfrak{h}_i$ and $[X_i, Y_i] \rightarrow [X, Y]$. Convergence in the Grassmannian implies that every limit of bounded vectors chosen from $\mathfrak{h}_i$ lies in W , hence $[X, Y] \in W$. $\square$

For arithmetic applications one needs more than Lie-algebra convergence: one needs a representation vector recording each candidate subgroup. Chevalley linearization supplies a finite-dimensional representation ρ and a line $[p_H]$ whose projective stabilizer is the normalizer of H . On bounded complexity families these vectors can be normalized to lie in a compact sphere.

M.5 Why ordinary weak convergence is not enough

Suppose $\mu_i \Rightarrow \mu$. Portmanteau tells us that positive mass in an open set persists only in the lower-semicontinuous direction:

$$\mu(O) \leq \liminf_i \mu_i(O).$$

It does not say that the supports $x_i H_i$ converge as sets. A small piece of each support can wander elsewhere while almost all mass converges. The family theorem needs a much stronger conclusion: after small conjugacy, the large support group attached to the limit eventually contains the groups responsible for the approximating measures.

That strengthening comes from focusing.

M.6 The focusing question in local coordinates

Assume $\mu_i \Rightarrow \mu$ and Ratner classification gives $\mu = m_{xL}$ for a closed finite-volume orbit xL . Fix a compact set $C \subset xL$ carrying almost all of μ . For large i , most of μ_i is near C .

Choose a point $z_i \in x_i H_i$ near xL and a local quotient slice at its nearest point $z \in xL$. Write

$$z_i = z \exp Z_i, \quad Z_i \rightarrow 0,$$

with Z_i transverse to $\mathfrak{l}$. Under the unipotent flow,

$$z_i u_i(t) = z u_i(t) \exp(\text{Ad}(u_i(-t))Z_i)$$

after absorbing the small difference between the moving and limiting flows into the local coordinates. The transverse logarithm is a polynomial of uniformly bounded degree.

If the transverse polynomial becomes macroscopic, its first-exit normalization produces a nontrivial limiting displacement. Genericity of the U_i -action on $x_i H_i$, together with the convergence of measures, turns that displacement into additional invariance of the limit measure. If L was already chosen as the full connected stabilizer of μ , this is impossible unless the displacement lies in the normalizer directions already accounted for. Hence the approximating supports must focus into those normalizer directions.

M.7 A polynomial first-exit calculation

Let

$$P_i(t) = \sum_{j=0}^d t^j v_{i,j}$$

be the transverse coordinate. Suppose $P_i(0) \rightarrow 0$ but P_i is not identically zero. Choose T_i to be the first time $\|P_i(T_i)\| = \rho$ for a small fixed ρ . Then $T_i \rightarrow \infty$. Rescale $t = sT_i$. Bounded-degree polynomial norm equivalence gives a subsequence for which

$$P_i(sT_i) \rightarrow P(s)$$

uniformly on $[0, 1]$, where P is nonzero.

Choose a terminal interval $J = [1 - \eta, 1]$ on which $P(s)$ changes by at most $\rho/10$. The same holds for P_i for large i . Thus on a positive proportion of a long orbit block the relative displacement is nearly constant. This is exactly the input required by the block-comparison lemma from Volume III.

Guiding Idea M.4. The family theorem does not invent a new rigidity mechanism. It uses the same polynomial shearing as Ratner's theorem, but the conclusion is now a statement about how an entire sequence of support groups must align with the support group of the limit.

M.8 From focused mass to support containment

Let p_L be a Chevalley vector for L . If a large part of $x_i H_i$ lies in a very thin neighborhood of xL , then along that part the vector

$$\rho(g_i h)^{-1} p_L$$

stays close to the discrete $\rho(\Gamma)$ -orbit of p_L . Polynomial nondivergence gives a dichotomy: either the trajectory leaves the thin set for a definite proportion of time, or it is trapped by an exact algebraic relation. Focusing rules out the first branch on the mass that survives in the limit. Hence, after a small conjugacy $c_i \rightarrow e$,

$$c_i H_i c_i^{-1} \subset N_G(L)$$

at the finite stage relevant to the limit.

The normalizer is not yet the desired support group. Quotient by the connected component of L and repeat the argument on the lower-dimensional

normalizer quotient. Because dimensions strictly drop whenever a new proper algebraic relation appears, the iteration terminates. The terminal conclusion is support containment inside the homogeneous support of the limit, up to the explicitly tracked finite/discrete normalizer ambiguity.

M.9 The Mozes–Shah induction written as an algorithm

We can now list the proof in the order a reader should check it.

1. Normalize the ergodic one-parameter generators U_i and take a subsequence $U_i \rightarrow U$.
2. Apply family nonescape. If an exterior-power obstruction persists, reduce immediately to the proper algebraic subgroup it defines.
3. Otherwise extract a probability weak limit μ .
4. Show that U preserves μ by passing the invariance identity to the limit.
5. Take an ergodic component of μ with positive mass in the chosen compact region and apply Ratner classification to obtain a homogeneous measure m_{xL} .
6. Apply the family focusing argument to the approximating supports near xL .
7. If focusing produces a proper algebraic trap, invoke induction on the support dimension.
8. In the regular branch, the transverse first-exit polynomial would give additional invariance of m_{xL} outside its full connected stabilizer, a contradiction.
9. Therefore all surviving components have the same maximal support; the whole limit is the homogeneous probability on that support.
10. The same focusing argument gives eventual support containment after small conjugacy.

The induction parameter is support dimension, not the number of generators and not the dimension of the original ambient group presentation.

M.10 Isolation in a bounded-complexity family

Suppose m_{xL} is a homogeneous probability and consider homogeneous measures whose support groups have bounded arithmetic/representation complexity. If a sequence of distinct proper supports converged to m_{xL} , focusing

would place those support groups, after small conjugacy, inside L . If their dimensions equal $\dim L$, connectedness forces equality for large i . If the dimensions are smaller, repeat inside L ; there are only finitely many dimension drops. Thus a maximal measure is isolated among proper measures of bounded complexity.

This is the form needed in counting: a sequence of closed orbits cannot silently approximate the ambient Haar measure while remaining trapped in unrelated proper algebraic pieces of bounded complexity.

M.11 Recovering a group generated by unipotents

Now let W be generated by finitely many one-parameter unipotent subgroups $U^{(1)}, \dots, U^{(r)}$, and suppose μ_i is ergodic under the full W -action. We cannot assume any one $U^{(j)}$ is ergodic. Instead take the ergodic decomposition of μ_i under $U^{(1)}$. The other generators permute the component family. Apply the one-parameter Mozes–Shah theorem to generic components, then use invariance under $U^{(2)}$ to glue the resulting support groups through their normalizers. Iterate over the finite generating family. Each gluing step either leaves the connected support unchanged or enlarges it inside a finite-dimensional normalizer chain, so the process terminates.

For infinitely many unipotent generators, only finitely many are needed to generate the Lie algebra of the connected support group. Choose such a finite subfamily and apply the preceding argument.

Worked Example M.5 (Two root flows in SL_3). Let U_{12} and U_{23} act on an arithmetic quotient of SL_3 . The commutator produces U_{13} . A limit measure invariant under the first two flows is therefore invariant under the upper-unitriangular group they generate. The one-parameter theorem may first identify an SL_2 -type support for a component under U_{12} , but invariance under U_{23} moves that component. Normalizer gluing either enlarges the support to contain U_{23} or forces the entire component family into a proper parabolic. This matrix picture is the simplest model of the abstract generated-group argument.

M.12 What to remember

The Mozes–Shah principle is a three-layer theorem:

- uniform nonescape
- + Ratner classification of a limit
- + family focusing/support containment

Ratner supplies only the middle layer. The first and third layers are why a separate post-Ratner volume is mathematically necessary.

Appendix N

Complete proof: Countability-Free Localization and Focusing

N.1 Why countability-free localization is needed

Two logically different issues must be separated. First, identifying a Zariski hull does not by itself enumerate the connected Lie subgroups inside that hull, so countability cannot be inferred from that step alone. Second, the family-focusing theorem needs a complete positive-mass selection argument and a justified passage from a thin tube to an exact subgroup trap.

The countability defect can be bypassed in the nonnormal dimension-drop branch, and it can also be bypassed when the limiting measure is already known to be homogeneous. It cannot yet be bypassed in the normal branch of the full moving-limit theorem. The key observation for the branches that do not need countability is elementary but important: an ergodic component need not have positive weight in the ergodic decomposition for one of its support points to have positive mass in every ambient open neighborhood. This lets us run the thin-neighborhood argument around one regular point of one typical component. No countable partition of the component space is required there.

There is, however, a separate geometric step in the classical proof: for a fixed homogeneous support H , the Chevalley lattice orbit must be discrete, equivalently the corresponding lift map must be proper. A short argument

based only on convergence of wedge vectors does not justify boundedness of finitely many conjugated lattice generators. In this volume the needed statement is already Lemma B.1 and Corollary B.2, proved from the single external theorem the external Dani–Margulis structural interface. To keep the long argument readable, the abbreviation **(DC)** below means precisely those two internally derived statements; it is not an additional external theorem.

N.2 A support point from one ergodic component

We begin with the measure-theoretic replacement for the invalid countability selection.

Lemma N.1 (Almost every component lives inside the ambient support). *Let U preserve a Radon probability μ on the second-countable locally compact space X , and let*

$$\mu = \int \nu d\xi(\nu)$$

be its U -ergodic decomposition. Then for ξ -almost every component ν ,

$$\text{supp } \nu \subset \text{supp } \mu.$$

Proof. Choose a countable basis $\{B_j\}_{j \geq 1}$ for X . If $B_j \cap \text{supp } \mu = \emptyset$, then $\mu(B_j) = 0$, and therefore

$$0 = \int \nu(B_j) d\xi(\nu).$$

Since the integrand is nonnegative, $\nu(B_j) = 0$ for ξ -almost every ν . Intersect the resulting full-measure sets over the countably many basis elements disjoint from $\text{supp } \mu$. For every component in this intersection, every open set disjoint from $\text{supp } \mu$ has ν -measure zero, so $\text{supp } \nu \subset \text{supp } \mu$. $\square$

Lemma N.2 (Single-component localization). *Let U preserve μ , and assume Ratner measure classification for the U -ergodic components. Then one may choose a U -ergodic component*

$$\nu = m_{yH}$$

and a point $y \in \text{supp } \nu \subset \text{supp } \mu$ such that:

1. y is generic for ν under U ;

2. y is not contained in any proper closed U -invariant homogeneous subset of $\text{supp } \nu$;

3. every open neighborhood $\mathcal{O}$ of y has $\mu(\mathcal{O}) > 0$.

No component is required to have positive ξ -weight.

Proof. By Lemma N.1, choose an ergodic component ν whose support is contained in $\text{supp } \mu$. Ratner classification writes it as Haar probability on a closed finite-volume orbit $y_0 H$. By the pointwise ergodic theorem there is a full- ν set of U -generic points. For a generic point y , the closure of the positive U -semiorbit contains $\text{supp } \nu$: indeed, if a nonempty relatively open subset of $\text{supp } \nu$ were missed, choose a nonnegative continuous compactly supported function that is positive there; its orbit average would be zero whereas its ν -integral is positive. Hence such a point cannot lie in a proper closed U -invariant homogeneous subset of $\text{supp } \nu$.

Choose one such generic y . The support inclusion gives $y \in \text{supp } \mu$. By definition of support, every open neighborhood of y has positive μ -measure. Notice that the last conclusion uses only the point y , not an atom of the ergodic-decomposition measure ξ . $\square$

Remark N.3 (Why this removes the countability step). The original countability argument was used to find a fixed support type whose regular stratum has positive μ -mass. Lemma N.2 replaces that statement by something weaker and sufficient: choose one regular point y of one typical component and then choose a small open tube $\mathcal{O} \ni y$. Its ambient mass

$$\alpha := \mu(\mathcal{O})$$

is automatically positive. Portmanteau and genericity of the approximating orbits convert this positive open-set mass into the positive residence time needed by the thin-neighborhood lemma.

N.3 Moving a generic approximating orbit to the selected point

Lemma N.4 (Recentring at a prescribed support point). *Let $\mu_i \rightarrow \mu$ weakly, let x_i be generic for the ergodic U_i -invariant probability μ_i , and let $y \in \text{supp } \mu$. Then, after passing to a subsequence, there are times $t_i \geq 0$ and elements $h_i \rightarrow e$ such that*

$$y_i := x_i u_i(t_i) \longrightarrow y, \quad y h_i^{-1} = y_i,$$

inside one fixed local quotient chart. The point y is generic for the translated measure $\mu'_i := \mu_i h_i$ under the translated flow $U'_i := h_i^{-1} U_i h_i$ (with the obvious adjustment if the opposite right-translation convention is used).

Proof. Choose nested relatively compact open neighborhoods $\mathcal{O}_k \ni y$ with diameters tending to zero and boundaries of μ -measure zero. Since $y \in \text{supp } \mu$, each $\mu(\mathcal{O}_k) > 0$. Weak convergence gives $\mu_i(\mathcal{O}_k) > 0$ for all sufficiently large i . A generic orbit for μ_i visits every open set of positive μ_i -measure: apply genericity to a nonnegative continuous function supported in that open set with positive integral. Diagonalize so that the i th orbit visits $\mathcal{O}_{k(i)}$ with $k(i) \rightarrow \infty$; call the visit y_i . Then $y_i \rightarrow y$.

Because Γ is discrete, a sufficiently small neighborhood of the identity in G maps injectively to a quotient chart at y . For large i there is therefore a unique $h_i \rightarrow e$ carrying y_i to y . Translating a uniformly distributed trajectory translates both its invariant measure and its acting one-parameter subgroup, so genericity is preserved. $\square$

N.4 The fixed-type thin-tube contradiction

The following is the exact place where the classical discrete-Chevalley input enters. Everything after it is ordinary dimension induction.

Proposition N.5 (Positive tube mass forces an exact fixed vector). *Assume (DC). Let $\nu = m_{yH}$ be the component and y the regular point supplied by Lemma N.2. Let $\mu'_i \rightarrow \mu$ and U'_i be as in Lemma N.4, so that y is generic for every μ'_i . Then, after choosing the fixed- H thin-neighborhood data sufficiently small, for every sufficiently large i there is one and the same Chevalley lift vector*

$$w \in \bar{\eta}_H(\pi^{-1}(y))$$

with

$$U'_i \subset G_w.$$

Proof. Choose a compact Chevalley neighborhood C of the lift of y inside the regular H -stratum. By regularity of y and the fixed-type self-intersection statement, shrink to a compact neighborhood K of y that is disjoint from the closed singular/self-intersection set occurring in Proposition B.5. Apply that proposition with a number $0 < \varepsilon < 1$ to obtain a representation-space neighborhood Ψ . Shrink K slightly, if necessary, so that

$$\mathcal{O} := K^\circ \cap \pi(\bar{\eta}_H^{-1}(\Psi))$$

is an open neighborhood of y . By Lemma N.2, $\alpha := \mu(\mathcal{O}) > 0$.

Choose $\varepsilon < \alpha/4$ and, by replacing $\mathcal{O}$ by a slightly smaller continuity set still containing y , arrange

$$\mu'_i(\mathcal{O}) > \alpha/2$$

for all large i . Genericity of the U'_i -orbit of y then gives, for all large T ,

$$\frac{1}{T} |\{0 \leq t \leq T : yu'_i(t) \in \mathcal{O}\}| > \alpha/3.$$

This contradicts the avoidance alternative of Proposition B.5. Hence its exact alternative holds: there is $w_i \in \bar{\eta}_H(\pi^{-1}(y)) \cap \Psi$ fixed by U'_i .

Under (DC), Corollary B.2 gives the unique-lift property on K . Since the original lift of y lies in $C \subset \Psi$, the lift vector in Ψ is unique. Thus $w_i = w$ is independent of i , and $U'_i \subset G_w$ for all sufficiently large i . $\square$

Corollary N.6 (Closed lower-dimensional capture). *Under the assumptions of Proposition N.5, if $\dim G_w^\circ < \dim G$, then for all sufficiently large i ,*

$$\text{supp } \mu'_i \subset yG_w^\circ,$$

and yG_w° is a closed homogeneous subspace. Hence the moving-limit problem reduces to a homogeneous space of strictly smaller ambient dimension.

Proof. Properness in (DC) makes the relevant G_w -orbit through y closed. The complete U'_i -trajectory of the generic point y lies in yG_w° . A generic trajectory has closure equal to the support of its measure: every relatively open subset of the support has positive measure and is therefore visited. Consequently $\text{supp } \mu'_i \subset yG_w^\circ$. The stabilizer of y inside G_w° is discrete, so the orbit is a homogeneous space for G_w° , and its ambient dimension is strictly smaller. $\square$

N.5 What the single-component argument proves, and where it stops

The preceding localization is enough to close the nonnormal branch of the classical induction. It is not, by itself, enough in the normal branch. We record both facts explicitly because conflating them would recreate the same kind of hidden gap that a careful proof architecture must avoid.

Proposition N.7 (Nonnormal branch without countability). *Assume (DC). In the setting of Theorem C.1, choose a component m_{yH} and a regular point yw by Lemma N.2. Replace the measures and acting subgroups by their right translates through w^{-1} , so that the chosen regular point becomes y and the Chevalley vector w is fixed once and for all. If the resulting fixed Chevalley stabilizer satisfies*

$$\dim G_w^\circ < \dim G,$$

then the moving-limit theorem reduces to the same theorem on the fixed closed homogeneous space yG_w° of strictly smaller dimension. In particular, this branch uses no countability theorem for support types.

Proof. Proposition N.5 gives $U_i' \subset G_w$ for all large i . Corollary N.6 then puts the supports of all large approximating measures inside the fixed closed orbit yG_w° . Weak convergence puts $\text{supp } \mu$ there as well. All hypotheses of the moving-limit problem are inherited by this homogeneous space, while the ambient dimension has strictly decreased. This is exactly the induction step. $\square$

Proposition N.8 (The normal branch is a genuinely separate obligation). *Suppose the selected component is m_{yH} and the Chevalley stabilizer has $G_w^\circ = G$, so that the subgroup encoded by the datum is normal. The fact that this one ergodic component is H -homogeneous does not imply that μ is H -invariant. Therefore a single zero-weight component cannot by itself justify the quotient-and-lift step in the original Mozes–Shah proof.*

Proof. Ergodic decomposition is an integral, and an individual component can have zero mass in the component parameter space. Invariance of that component under H gives no invariance statement for the other components. More concretely, on a product homogeneous space on which a unipotent group acts in one factor, an invariant probability can be obtained by integrating the Haar measure on the first-factor homogeneous fibers against a non-atomic probability on the second factor. Every ergodic component is invariant under the normal first-factor subgroup, while the integrated measure need not be invariant under any group moving the second factor. Thus the implication “one normal homogeneous component $\Rightarrow$ the whole measure is H -invariant” is false without an additional argument.

In the original Mozes–Shah proof the missing additional argument is supplied by the countable decomposition into exact regular support types: one chooses a type with positive mass, and in the normal case the corresponding

regular piece exhausts the measure after the singular pieces have been removed. A countability-free replacement would have to prove an equally strong uniform normal-fiber statement. The single-component localization lemma does not do that. $\square$

Remark N.9 (Limit of the single-component method). The countability-free idea resolves the nonnormal dimension-drop branch, but it is not by itself a replacement for the classical countability theorem in the normal branch. A uniform normal-fiber argument or the support-class countability theorem is still required there.

N.6 Family focusing once the limit is already homogeneous

The family-focusing theorem has a better status. There the limit is assumed to be one homogeneous probability m_{xL} from the outset. Thus the positive-mass type is already fixed globally, and the normal-fiber difficulty above does not arise.

Theorem N.10 (Family focusing, fixed-limit form). *Assume (DC). Let $\mu_i = m_{x_i H_i} \in \mathcal{Q}(X)$ converge to a homogeneous probability $\mu = m_{xL}$. Choose designated ergodic unipotent one-parameter groups $U_i \subset H_i^+$ and support points tending to a regular point of xL . Then the fixed- L thin-neighborhood argument and induction give, after small recentering,*

$$\text{supp } \mu_i \subset \text{supp } \mu$$

for all sufficiently large i . Consequently the connected unipotent-generated supports satisfy the corresponding conjugated inclusion in L .

Proof. Because $\mu = m_{xL}$ is itself homogeneous, choose a U -generic regular point y of xL and a compact regular neighborhood of positive μ -measure. Recenter the approximating generic trajectories at y by Lemma N.4. Apply the fixed- L version of Proposition N.5. If the Chevalley stabilizer is proper, Corollary N.6 reduces the problem to a fixed smaller closed homogeneous space and induction applies.

If the stabilizer has full dimension, the encoded connected subgroup L_0 is normal. Here, unlike in Proposition N.8, the entire limiting measure is already L -homogeneous; in particular the normal fiber relevant to the fixed

datum is globally invariant on its support. Project to the quotient by L_0 and use the same dimension induction as in the detailed proof following Theorem C.1. The quotient support conclusion lifts because μ is Haar on the closed L_0 -fibers of its homogeneous support. Thus support containment holds in both branches.

Finally, each connected one-parameter unipotent subgroup of H_i^+ preserves $\text{supp } \mu_i$. After the same small conjugacy, support containment places its action inside the connected stabilizer of xL , which is L by the canonical choice of support group. Since H_i^+ is generated by these one-parameter subgroups, the claimed group inclusion follows. $\square$

N.7 What this localization module establishes

The conclusions needed later are the following.

1. Identifying a Zariski hull alone does not prove countability of the connected support groups; the countability statement is instead derived in Volume III from the Dani–Margulis structural interface.
2. The positive-mass selection in the *nonnormal* branch needs no countability theorem: one typical Ratner component and one point of $\text{supp } \mu$ suffice.
3. Family focusing for an already homogeneous limit has a complete dimension-induction architecture once the fixed-type discrete-Chevalley and proper-lift statements derived from the Dani–Margulis structural interface are inserted.

The Dani–Margulis structural interface remains the top-level root package

Dani–Margulis support countability and Chevalley discreteness.

Volume III proves the two deductions from this root package in full:

Support-class countability. support-class countability, by exhausting the finite covolume class by integer covolume bounds;

Fixed-support Chevalley discreteness. fixed-support Chevalley discreteness, by the exact conjugation Jacobian/covolume identity and finiteness of bounded-volume Γ -conjugates.

The tempting implication “bounded Chevalley vector + discreteness of Γ gives finitely many branches” is false; the proof uses a genuine covolume calculation instead.

A further structural argument proves internally that every actual support group satisfies the adjoint Zariski-density condition defining $\mathcal{H}_\Gamma^{\text{alg}}$, and applies

the countability deduction to the exact algebraic support class rather than to all connected lattice subgroups. The proof uses a Chevalley projective factor, projective recurrence for finitely many unipotent generators, and H^+ -ergodicity; it does not assume $H = \overline{H^+}$.

The sole research-level structural input used here is the Dani–Margulis countability/discreteness interface. Proper-lift, unique-lift, thin-neighborhood, focusing, and moving-limit arguments are proved downstream from that theorem.

Proof Guide and Dependency Summary

This chapter is a finalization layer. It does not introduce a new theorem scope. Its purpose is to expose the proof architecture, make the hypotheses that are easiest to misread impossible to miss, and give a student a concrete model calculation before the abstract machinery is reused later.

1. What changes when the measure itself varies

Ratner classification concerns one invariant probability measure. A sequence $\mu_i = m_{x_i H_i}$ introduces three new problems:

mass escape, variation of H_i , new invariance in the limit.

The Mozes–Shah mechanism solves these in that order: uniform nonescape, uniform Chevalley linearization, then focusing.

2. The generator pitfall

If a group L generated by unipotent one-parameter subgroups acts ergodically, it does *not* follow that an arbitrarily selected generating one-parameter subgroup acts ergodically. The proof therefore begins with an ergodic one-parameter unipotent input, applies the one-parameter nonescape/focusing machinery, and only after the limiting homogeneous support is identified does it recover the group generated by all relevant unipotent directions.

3. Focusing as a contradiction argument

After passing to a tight subsequence, linearize the moving subgroup data by vectors p_{H_i} in fixed finite-dimensional representations. If the sequence is not focused near the eventual support group, polynomial entrance/residence estimates produce transverse blocks. Generic comparison then creates an additional stabilizing direction for the limit measure. This contradicts maximality of its connected stabilizer unless the data lie in a proper singular configuration. The singular alternative is handled by dimension reduction.

4. Student proof ledger

When reading the homogeneous-limit theorem, mark the first line at which each of the following enters:

1. tightness;
2. compactness of subgroup/representation data;
3. polynomial sublevel control;
4. Ratner classification of an ergodic component;
5. support containment after small conjugacy;
6. recovery of generated-unipotent symmetry.

If two of these steps are merged into the phrase “by focusing,” the proof is too compressed.

5. Dependency status

This guide should be read together with Chapter N. There is one shared top-level structural input inherited from Volume III:

Dani–Margulis structural interface.

Volume III derives support-class countability and fixed-support Chevalley discreteness from this theorem and proves the support-to-algebraic-hull bridge inside Volume III. The lift is written in the correct left-quotient convention

$$\Gamma_H \backslash G \longrightarrow (\Gamma \backslash G) \times \bar{V}_H.$$

Every later use in Volume VII of countability, proper lifts, unique lifts and thin-neighborhood linearization has a complete backward deduction from the Dani–Margulis structural interface through the proved Volume III child

theorems. In Volume III the finite-stage singular argument enumerates only finitely many admissible support types at a time; exact trapping in such a branch is upgraded internally to a proper finite-volume homogeneous orbit by the already proved lattice-stabilizer closed-orbit theorem.

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Polynomial and Tame Trajectories on Homogeneous Spaces

Volume VIII of a Detailed Notebook on Ratner's Theorems

Detailed Notebook on Ratner's Theorems

August 2026

Preface and proof policy

A unipotent orbit is polynomial after every finite-dimensional representation, but the converse viewpoint is broader: one may start with a polynomial trajectory in the ambient group and ask for the distribution of its image in a lattice quotient. This volume develops the nondivergence, twisting, local-invariance, singular-avoidance, and Ratner-classification mechanisms needed for that problem. The classical one-parameter polynomial theorem is proved in full. Higher-dimensional polynomial limits are stated at the level actually justified by local-unipotent invariance. The final part treats noncontracting polynomially bounded o-minimal curves and isolates the one-variable definable growth theorem on which that extension rests.

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Dependency and proof-ownership contract

Purpose. Polynomial and tame trajectories.

Prerequisites. Volumes II–VII.

New objects. polynomial/o-minimal trajectories, asymptotic differences. Every project-specific technical term is defined before its first load-bearing use or recalled by an exact backward reference.

Main outputs. local unipotent invariance and equidistribution.

External-input rule. Every nontrivial imported result is either part of the declared graduate-level prerequisites, an explicitly named external research input in the prerequisite map, or is proved in this series before use (possibly under an explicit deferred-proof contract). A citation by itself never constitutes a proof.

Reader map and proof route

Question of the volume. How far can the polynomial-trajectory viewpoint be pushed beyond literal one-parameter unipotent subgroups, especially for tame/definable curves?

Prerequisite checkpoint. Ratner rigidity and linearization come from Volumes II–III. The o-minimal foundations are placed before the main body in the active reading order so that definability, cell decomposition consequences used here, and asymptotic notation are defined before their dynamical use.

Load-bearing route. Learn the tame asymptotic language first. Then rescale a curve, identify its polynomial/unipotent tangent model, establish nondivergence, and apply the rigidity package to subsequential limits. Do not replace the rescaling argument by the heuristic “definable means polynomial.”

Sanity check. For a curve of the form $u_{t^2}v_t$ in a two-step unipotent group, conjugate and rescale around a large parameter. Keeping the first nonvanishing jet shows exactly which tangent subgroup is visible in the limit and which terms are lower order.

Logical reading order

The one-variable o-minimal preparation and uniform difference estimates are logically prior to the trajectory theorems that use them. We therefore prove that foundation first rather than placing it after the main narrative.

Chapter 1

Polynomially Bounded o-Minimal Curves: Foundations and the General Theorem

The present development requires the polynomially bounded o-minimal extension of the polynomial-trajectory theorem. This is not merely a change of vocabulary. For a polynomial curve, Remez inequalities and finite differences are available from elementary algebra. For a definable curve they have to be replaced by two genuinely new facts:

- (i) one-variable definable functions in a polynomially bounded structure have a single power as their leading asymptotic scale;
- (ii) finite-dimensional definable families satisfy a uniform Remez-type inequality and hence a uniform (C, α) -good estimate.

Once these are proved, the homogeneous-dynamics part follows the same ledger as the polynomial proof: nonescape, asymptotic unipotent invariance, Ratner classification, and singular-set avoidance. We write every step because otherwise the phrase “the o-minimal case is similar” hides most of the new work.

1.1 The one-dimensional o-minimal toolkit

Definition 1.1 (o-minimal structure). An o-minimal expansion $\mathcal{S}$ of the ordered real field is a collection of definable subsets of all $\mathbb{R}^m$, closed under Boolean operations, Cartesian products, coordinate permutations and coordinate projections, containing the semialgebraic sets, and such that every definable subset of $\mathbb{R}$ is a finite union of points and intervals. A map is definable when its graph is definable.

Definition 1.2 (Polynomially bounded). The structure is polynomially bounded if every definable $f : [T_0, \infty) \rightarrow \mathbb{R}$ satisfies $|f(t)| \leq t^N$ on a tail for some integer N .

The following three consequences are the only general o-minimal facts used below. We include the one-variable proofs in the form needed here.

Lemma 1.3 (Eventual sign and eventual monotonicity). *If $f : [T_0, \infty) \rightarrow \mathbb{R}$ is definable, then after increasing T_0 :*

1. *either $f = 0$ identically or f has constant nonzero sign;*
2. *f is continuous;*
3. *f is either constant or strictly monotone;*
4. *for every fixed r , after a further increase of T_0 the function is C^r .*

In particular every definable germ has a limit in $\mathbb{R} \cup \{\pm\infty\}$.

Proof. The zero set is definable, hence a finite union of points and intervals. On the final unbounded component it is therefore either all or empty, proving (1). The graph of f is a one-dimensional definable set. Project the set of points at which the graph has two different limiting vertical values onto the first coordinate. If this projection contained an interval, vertical slices of the graph over that interval would have two separated branches; by taking the lower and upper endpoints one would obtain two definable continuous branches and hence, between their graphs, infinitely many alternating components of a one-dimensional definable set. O-minimality forbids this. Thus the discontinuity set is finite.

On a continuity interval consider the definable sets on which f is locally increasing, locally decreasing, and locally constant. Their boundaries are finite. On every component of the complement one local behaviour is fixed; a finite chain of overlapping neighbourhoods promotes it to global strict monotonicity or constancy. This proves (2)–(3). Apply the same argument to the difference quotient and then induct on r ; the exceptional set at each

stage is definable and finite. A monotone real function has an extended-real limit, proving the last assertion. $\square$

Lemma 1.4 (Definable limits and compact choice). *Let $F : A \times [T_0, \infty) \rightarrow \mathbb{R}^m$ be definable and suppose $\lim_{t \rightarrow \infty} F(a, t)$ exists for every $a \in A$. Then the limit map is definable. If $C_a \subset \mathbb{R}^m$ is a nonempty compact definable family, one can choose a definable point $c(a) \in C_a$.*

Proof. The graph of the limit map is described by the first-order condition

$$(a, y) : \quad \forall \varepsilon > 0 \ \exists T \ \forall t > T, \ \|F(a, t) - y\| < \varepsilon.$$

The operations used in this formula are Boolean operations and projections, so the graph is definable. For compact choice, choose successively the minimum first coordinate, then among its minimizers the minimum second coordinate, and so on. Minima exist by compactness and their graphs are described by the same projection-and-complement operations. $\square$

Lemma 1.5 (Uniform number of one-dimensional components). *Let $A \subset P \times \mathbb{R}$ be definable. There is an integer N_A such that every fiber A_p has at most N_A connected components.*

Proof. For a fiber A_p , each component has a left endpoint in $\mathbb{R} \cup \{-\infty\}$. Suppose no uniform bound existed. Then for every N there would be a parameter p_N and an increasing N -tuple of left endpoints. The set of all pairs (p, x) in which x is a left endpoint is definable: “ x lies in the closure of A_p and every sufficiently small right neighbourhood meets the component beginning at x , while a left neighbourhood does not” is a first-order condition. If its fibers had arbitrarily large finite cardinality, definable compact choice applied successively to the first, second, ... endpoints would produce, after passing to a parameter cell, arbitrarily many disjoint definable graphs over the same interval of parameters. Two consecutive graphs bound a definable strip; projecting a transverse definable curve through these strips onto one coordinate creates a definable subset of $\mathbb{R}$ with arbitrarily many alternating components. This contradicts o-minimality. Hence the fiber cardinalities, and therefore the number of components, are uniformly bounded. $\square$

1.2 The asymptotic-monomial theorem without a hidden growth lemma

The next theorem is usually attributed to Miller. A useful proof can be organized as an elementary regular-variation argument once eventual monotonicity and polynomial boundedness are available. We give it because it is the point at which slow factors such as $\log t$ are ruled out.

Theorem 1.6 (Asymptotic monomial theorem). *Let $f : [T_0, \infty) \rightarrow \mathbb{R}$ be definable in a polynomially bounded o-minimal expansion and not eventually zero. Then there are unique $r \in \mathbb{R}$ and $c \neq 0$ such that*

$$f(t) = ct^r(1 + o(1)).$$

Moreover the power map $t \mapsto t^r$ is definable in the same structure.

Proof. By Lemma 1.3, replace f by $|f|$ and work on a tail on which it is positive and monotone. If f tends to a finite nonzero limit, take $r = 0$. If $f \rightarrow 0$, apply the argument below to $1/f$ and invert the conclusion. We may therefore suppose $f(t) \rightarrow \infty$ and f is increasing.

For $\lambda > 0$ define

$$L(\lambda) = \lim_{t \rightarrow \infty} \frac{f(\lambda t)}{f(t)}.$$

The limit exists in $[0, \infty]$ because the ratio is definable. It is in fact finite and positive. For $\lambda > 1$ the ratio is at least one. If its limit were infinite, then for any M there would be T such that $f(\lambda t) \geq \lambda^M f(t)$ for all $t \geq T$. Iterating at $t = T, \lambda T, \dots$ gives

$$f(\lambda^k T) \geq \lambda^{Mk} f(T).$$

Polynomial boundedness gives an exponent N with $f(x) \leq x^N$ on a tail. Choosing $M > N$ and sending $k \rightarrow \infty$ is a contradiction. Thus $1 \leq L(\lambda) < \infty$. The case $0 < \lambda < 1$ follows from

$$L(\lambda) = L(\lambda^{-1})^{-1}.$$

The function L is definable by Lemma 1.4, is nondecreasing, and satisfies

$$L(\lambda\mu) = L(\lambda)L(\mu).$$

It is continuous at 1: for $1 < \lambda < 2^{1/n}$, $1 \leq L(\lambda) \leq L(2)^{1/n}$, and the upper bound tends to one as $n \rightarrow \infty$. Multiplicativity gives continuity everywhere.

1.2. The asymptotic-monomial theorem without a hidden growth lemma 11

Put $r = \log_2 L(2)$, where the logarithm here is ordinary real notation, not a definable function. For rational q , $L(2^q) = 2^{rq}$; continuity and density give

$$L(\lambda) = \lambda^r \quad (\lambda > 0).$$

Since L is definable, this identity proves at the same time that the power map $\lambda \mapsto \lambda^r$ is definable.

Now set $g(t) = f(t)t^{-r}$. It is definable and slowly varying:

$$\frac{g(\lambda t)}{g(t)} \longrightarrow 1 \quad (\lambda > 0).$$

By eventual monotonicity, g has a limit in $[0, \infty]$. We show that it cannot be 0 or ∞ . Suppose $g \rightarrow \infty$. For every $\varepsilon > 0$, slow variation with $\lambda = 2$ gives $g(2t) \leq 2^\varepsilon g(t)$ on a tail. Dyadic iteration and monotonicity therefore give

$$g(t) \leq C_\varepsilon t^\varepsilon.$$

Since g is eventually increasing and unbounded, its inverse is definable. Writing $y = g(t)$ in the last inequality gives

$$g^{-1}(y) \geq c_\varepsilon y^{1/\varepsilon}.$$

Because ε is arbitrary, the definable function g^{-1} grows faster than every polynomial, contradicting polynomial boundedness. If $g \rightarrow 0$, apply the same argument to $1/g$. Thus $g \rightarrow c \in (0, \infty)$, which proves existence. Uniqueness of r and c follows by taking the ratio of two proposed asymptotics. $\square$

Corollary 1.7 (Derivative lowers the asymptotic exponent by one). *If $f(t) \sim ct^r$ with $r \neq 0$, then*

$$f'(t) \sim crt^{r-1}.$$

Proof. The derivative is definable on a tail. If it is not eventually zero, Theorem 1.6 gives $f'(t) \sim dt^s$. If $r > 0$, then $f \rightarrow \infty$ and integrating f' from a fixed point to t shows that necessarily $s > -1$ and $f(t) \sim d(s+1)^{-1}t^{s+1}$; comparison with ct^r gives $s = r-1$ and $d = cr$. The case $r < 0$ is obtained by integrating from t to ∞ because $f(t) \rightarrow 0$. The borderline $s = -1$ would make f grow like $\log t$; then f is unbounded but smaller than every positive power, and its definable inverse grows faster than every polynomial, contradicting polynomial boundedness exactly as in the proof above. $\square$

Definition 1.8 (Degree). For a nonzero definable germ put $\deg f = r$ when $f(t) \sim ct^r$. Put $\deg 0 = -\infty$.

Lemma 1.9 (Cancellation lowers degree). *If f, g have the same finite degree r , there is a unique scalar a such that either $f - ag$ is eventually zero or $\deg(f - ag) < r$.*

Proof. Write $f \sim c_f t^r$ and $g \sim c_g t^r$. Taking $a = c_f/c_g$ cancels the leading term. If the difference is not eventually zero, Theorem 1.6 gives a new power asymptotic. Its degree cannot be $\geq r$, because the quotient by t^r tends to zero. $\square$

Lemma 1.10 (Basis of distinct degrees). *Every finite-dimensional vector space of definable germs admits a basis $F_1, \dots, F_d$ with strictly increasing degrees.*

Proof. Choose a nonzero vector of maximal degree among a basis. Whenever another basis vector has the same degree, subtract the unique multiple supplied by the cancellation lemma; the degree drops while the change of basis remains invertible. The span of the remaining $d - 1$ vectors now has all degrees strictly below the last one. Induct on d . $\square$

Lemma 1.11 (Wronskian independence on a tail). *If $F_1, \dots, F_d$ have distinct degrees, then their Wronskian is nonzero on a tail. Consequently no non-trivial linear combination can vanish on a nondegenerate interval contained sufficiently far out.*

Proof. Let $r_i = \deg F_i$ and normalize the leading coefficient to one. Repeated use of Corollary 1.7 gives

$$F_i^{(j)}(t) \sim (r_i)_j t^{r_i-j},$$

where $(r)_j = r(r-1) \cdots (r-j+1)$, with the evident modification if one of the first derivatives eventually vanishes. Factoring powers of t from rows and columns gives a leading determinant proportional to

$$\det((r_i)_{j-1})_{i,j} = \prod_{i < j} (r_j - r_i),$$

which is nonzero. A vanishing linear combination on an interval would have all derivatives through order $d - 1$ zero at an interior point, contradicting the nonzero Wronskian. $\square$

1.3 Uniform Remez inequalities for definable cones

Definition 1.12 (δ -good family). A family $\mathcal{F}$ of functions on $[T_0, \infty)$ is δ -good if there is $M(\delta)$ such that whenever $I' \subset I$ are bounded intervals in the tail with $|I'| = \delta|I|$,

$$\|f\|_I \leq M(\delta)\|f\|_{I'} \quad (f \in \mathcal{F}).$$

Theorem 1.13 (Uniform definable Remez theorem). *Let $\mathcal{V}$ be a finite-dimensional vector space of functions definable in one polynomially bounded o-minimal structure. Let $\mathcal{F} \subset \mathcal{V}$ be a closed definable cone and assume*

$$\lim_{t \rightarrow \infty} f(t) \neq 0 \quad (f \in \mathcal{F} \setminus \{0\}).$$

Then there is T_0 such that the nonzero restrictions of functions in $\mathcal{F}$ to $[T_0, \infty)$ are δ -good for every $0 < \delta \leq 1$.

Proof. Use Lemma 1.10 and discard the finite initial interval on which an eventually zero germ may still be nonzero. Write the basis as

$$F_1^-, \dots, F_n^-, F_0^+, F_1^+, \dots, F_m^+$$

with negative degrees on the left and nonnegative, strictly increasing degrees on the right. Normalize all leading coefficients to one.

Suppose the conclusion fails for some δ . Because the bad-ratio condition is definable, Lemma 1.4 gives definable coefficient vectors $c(s)$ and intervals

$$I_s = [a(s), a(s) + \ell(s)], \quad I'_s \subset I_s, \quad |I'_s| = \delta\ell(s),$$

for which

$$\frac{\|f_{c(s)}\|_{I_s}}{\|f_{c(s)}\|_{I'_s}} > s. \quad (8.01)$$

Normalize $c(s)$ to have ℓ^1 -norm one. It lies in a compact definable set, hence has a limit c_∞ after passing to a tail.

Case 1: $\ell(s)$ is bounded. Then $a(s)$ is either bounded or tends to infinity. If $a(s)$ is bounded, the two intervals converge to nondegenerate intervals. The numerator in (8.1) stays bounded, so the denominator tends to zero; continuity gives a nonzero limiting linear combination vanishing on an interval, contradicting Lemma 1.11. If $a(s) \rightarrow \infty$ while $\ell(s)$ remains bounded, divide every basis function by the largest active basis function. The distinct-degree

asymptotics make all lower terms converge uniformly to zero on I_s , while the leading active term is asymptotically constant across a bounded interval. The ratio in (8.1) therefore tends to one, again a contradiction.

Case 2: $\ell(s) \rightarrow \infty$ and $a(s)/\ell(s)$ is bounded. Put $u = (t - a(s))/\ell(s)$. For a basis function of nonnegative degree r , Theorem 1.6 applied once more to the remainder gives some $\eta > 0$ with

$$F(a + u\ell) = \ell^r \left((a/\ell + u)^r + O(\ell^{-\eta}(a/\ell + u)^{r-\eta}) \right)$$

uniformly on compact subintervals avoiding a possible endpoint $u = 0$. For negative degrees the corresponding terms are uniformly bounded by their values at a . Normalize by

$$C(s) = \sum_i |c_i^+(s)| \ell(s)^{r_i} + \sum_j |c_j^-(s)|.$$

Compactness gives limiting normalized positive coefficients w_i . The normalized functions converge on every $[u_0, 1]$, $u_0 > 0$, to a generalized polynomial

$$P(u) = \sum_i w_i (x_0 + u)^{r_i}, \quad x_0 = \lim a(s)/\ell(s).$$

Equation (8.1) forces P to vanish on the limiting subinterval of length δ . Distinct real powers have nonzero Wronskian, so every $w_i = 0$. Thus asymptotically all normalized mass of $c(s)$ lies in the negative-degree subspace. Its limit is a nonzero element of the closed cone $\mathcal{F}$ that tends to zero at infinity, contrary to the hypothesis.

Case 3: $\ell(s) \rightarrow \infty$ and $a(s)/\ell(s) \rightarrow \infty$. Now the interval is short relative to its location. Let $d = \dim \mathcal{V}$. Taylor's formula at $a(s)$, together with the derivative asymptotics from Corollary 1.7, gives after normalization a compact family of degree at most $d - 1$ polynomials in $u = (t - a(s))/\ell(s)$; the remainder is $o(1)$ uniformly on $[0, 1]$. Indeed, if all normalized derivatives through order $d - 1$ tended to zero, the nonzero Wronskian would force the normalized coefficient vector itself to tend to zero. Hence a subsequence converges to a nonzero polynomial $P(u)$. Again (8.1) forces P to vanish on an interval of length δ , a contradiction.

The three cases exhaust all definable possibilities, proving the theorem. $\square$

Theorem 1.14 (Uniform (C, α) -goodness). *Under the hypotheses of Theorem 1.13, there exist $C, \alpha > 0$ and T_0 such that every nonzero $f \in \mathcal{F}$ satisfies*

$$|\{t \in I : |f(t)| \leq \varepsilon\}| \leq C \left(\frac{\varepsilon}{\|f\|_I} \right)^\alpha |I|$$

for every bounded $I \subset [T_0, \infty)$.

Proof. Let $M(\delta)$ be the smallest Remez constant. The relation “ y is an admissible Remez constant for δ ” is definable, so compact choice supplies a definable upper bound $\Phi(1/\delta)$. Polynomial boundedness gives constants $m, r > 0$ with $M(\delta) \leq m\delta^{-r}$.

For fixed f, I, ε , the sublevel set $E = \{t \in I : |f(t)| \leq \varepsilon\}$ has at most K components uniformly by Lemma 1.5. One component contains an interval I' of length at least $|E|/K$. Applying the Remez inequality with $\delta' = |I'|/|I|$ gives

$$\|f\|_I \leq m(\delta')^{-r} \varepsilon \leq m \left(\frac{K|I|}{|E|} \right)^r \varepsilon.$$

Solving for $|E|$ proves the claim with $\alpha = 1/r$ and $C = Km^{1/r}$. $\square$

Corollary 1.15 (Representation-coordinate goodness). *Let $\varphi : [T_0, \infty) \rightarrow G < \mathrm{SL}_N(\mathbb{R})$ be definable and noncontracting in every finite-dimensional rational representation. For any such representation $\rho : G \rightarrow \mathrm{GL}(V)$ there are C, α, T_1 , independent of $0 \neq v \in V$, such that $t \mapsto \|\rho(\varphi(t))v\|$ is (C, α) -good on $[T_1, \infty)$.*

Proof. After choosing an inner product, the squared norms

$$F_v(t) = \|\rho(\varphi(t))v\|^2$$

form a finite-dimensional closed definable cone: they are quadratic in the coordinates of v . Noncontraction says no nonzero F_v tends to zero. Apply Theorem 1.14 to this cone and take square roots, which only replaces α by 2α in the displayed sublevel form. $\square$

1.4 The unipotent hull and the correcting curve

The next construction removes bounded motion before the asymptotic unipotent directions are extracted. We use the algebraic-group tools of Volume V and the Chevalley realization of Volume II.

Definition 1.16 (Observable subgroup). An algebraic subgroup $H < G$ is observable if there are a rational finite-dimensional representation $\rho : G \rightarrow \mathrm{GL}(V)$ and $v \in V$ such that $H = \mathrm{Stab}_G(v)$.

Lemma 1.17 (Unipotent-generated algebraic hulls are observable). *If $L < G$ is connected and generated by one-parameter unipotent subgroups, then its Zariski closure has unipotent radical and is observable.*

Proof. The Zariski closure is generated algebraically by the Zariski closures of the same unipotent one-parameter groups. Any torus in its solvable radical would produce a nontrivial character on a quotient of the radical; characters are trivial on every unipotent one-parameter subgroup, hence on the algebraic group they generate. Thus the radical is unipotent. For an algebraic group with unipotent radical, take a faithful representation, a Levi factor, and a highest exterior power of the filtration fixed by the radical. Tensor with finitely many dual highest-weight lines separating the Levi quotient. The stabilizer of the resulting vector is exactly the subgroup. This is the vector-stabilizer version of the Chevalley construction proved in Volume II. $\square$

Lemma 1.18 (Algebraic orbit separation for a nonclosed boundary point). *Let G be a real algebraic group acting rationally on a finite-dimensional real vector space V , let $v \in V$, and let $w \in \overline{Gv} \setminus Gv$. Then there is a finite-dimensional rational G -module W and a G -equivariant polynomial map $P : V \rightarrow W$ such that $P(v) \neq 0$ and $P(w) = 0$.*

Proof. Let $X = \overline{Gv}$ and $Y = X \setminus Gv$. Since Gv is locally closed, Y is a proper G -stable algebraic subset of X . Choose $f \in I(Y)$ whose restriction to Gv is not identically zero. The span of the translates gf is finite dimensional because the coordinate ring is a rational G -module and f lies in a finite-dimensional G -submodule. Let W be the dual of this span and define $P(x)$ by evaluation, $P(x)(q) = q(x)$. Evaluation is polynomial and G -equivariant. Every q in the translate span vanishes on the G -stable set Y , hence $P(w) = 0$, while the choice of f gives $P(v) \neq 0$ after replacing v by a translate if necessary; equivariance then gives the same nonvanishing on the whole orbit. $\square$

Lemma 1.19 (Correcting a bounded representation orbit). *Let φ be non-contracting and definable. If $\rho(\varphi(t))v$ is bounded, then it converges to a point in the orbit Gv . There is a bounded continuous definable $\beta(t)$ with*

$$\rho(\beta(t)\varphi(t))v = v.$$

Proof. Bounded definability gives a limit v_∞ . If $v_\infty \in \overline{Gv} \setminus Gv$, Lemma 1.18 gives an equivariant polynomial map P to a representation W with $P(v) \neq 0$

and $P(v_\infty) = 0$. Then

$$\rho_W(\varphi(t))P(v) = P(\rho(\varphi(t))v) \longrightarrow 0,$$

contradicting noncontraction. Hence $v_\infty = g_0v$. The orbit map $G \rightarrow Gv$ is open. For large t , choose definably a group element $\delta(t) \rightarrow e$ satisfying $\delta(t)\rho(\varphi(t))v = g_0v$; compact choice and Lemma 1.3 give a continuous tail. Multiplying by g_0^{-1} gives the required correcting curve, and extending over the finite initial interval is harmless. $\square$

Lemma 1.20 (Intersections preserve boundedness). *If H_1, H_2 are observable and φ is bounded modulo each, then it is bounded modulo $H_1 \cap H_2$.*

Proof. Choose vector realizations $H_i = \text{Stab}(v_i)$. Boundedness modulo H_i makes $\varphi(t)v_i$ bounded. Apply Lemma 1.19 to (v_1, v_2) in the direct-sum representation. The resulting correcting curve puts $\varphi(t)$ in the common stabilizer. $\square$

Theorem 1.21 (Hull and correcting curve). *Let $\varphi : [T_0, \infty) \rightarrow G$ be continuous, definable and noncontracting. There is a unique smallest connected closed subgroup H_φ generated by one-parameter unipotent subgroups such that φ is bounded modulo H_φ . There is a bounded continuous definable curve $\beta(t)$ with*

$$\beta(t)\varphi(t) \in H_\varphi, \quad \beta(t) \rightarrow \beta_\infty.$$

Moreover, in every rational representation, a vector with bounded φ -orbit is fixed by H_φ .

Proof. The algebraic subgroup lattice satisfies the descending-chain condition. Among observable algebraic subgroups modulo which φ is bounded, choose one that is minimal for inclusion. If K is any other such subgroup, the intersection lemma shows that φ is bounded modulo $\tilde{H} \cap K$; inclusion-minimality therefore forces $\tilde{H} \subset K$. Thus the minimal subgroup is unique. Let $\tilde{H}$ denote it and $H = \tilde{H}^\circ$. A vector realization of $\tilde{H}$ and Lemma 1.19 provide a bounded definable correcting curve with $\beta(t)\varphi(t) \in H$ after multiplying by one fixed component representative. Bounded definable curves have limits, so $\beta(t) \rightarrow \beta_\infty$.

Let L be the connected subgroup generated by all one-parameter unipotent subgroups of H . By Lemma 1.17, its algebraic hull is observable. The image of the corrected curve in H/L cannot be unbounded: an unbounded definable curve in the reductive quotient has, by Cartan coordinates, a rational

representation vector that contracts to zero along a subsequence; pulling that vector back to G contradicts noncontraction. Thus φ is bounded modulo L . Minimality of $\tilde{H}$ forces $L = H$. This proves the generated-unipotent assertion.

Finally, if $\varphi(t)v$ is bounded, its stabilizer is observable and φ is bounded modulo it by Lemma 1.19. Minimality puts H_φ inside that stabilizer, so $H_\varphi v = v$. $\square$

1.5 Nonescape for definable curves

Theorem 1.22 (Uniform nonescape). *Let $G = \mathrm{SL}_n(\mathbb{R})$, $\Gamma = \mathrm{SL}_n(\mathbb{Z})$ and let φ be as in Theorem 1.21. For every compact set of initial lattices $C \subset G/\Gamma$ and every $\varepsilon > 0$ there is a compact $K \subset G/\Gamma$ such that*

$$\frac{1}{T} |\{t \in [0, T] : \varphi(t)x \notin K\}| < \varepsilon$$

for all $x \in C$ and all sufficiently large T .

Proof. Volume II's quantitative nondivergence theorem asks for two ingredients for each primitive rational wedge w : a uniform good-function estimate and a uniform positive supremum lower bound. The first is Corollary 1.15. For the second, evaluate at one fixed large time T_1 . On a compact set of starting lattices the set of primitive wedge vectors of bounded norm is discrete, and $\varphi(T_1)$ is invertible; hence the relevant infimum is positive. The nondivergence estimate then bounds the proportion of times at which the shortest vector is below ρ . Mahler compactness converts the resulting short-vector bound into the desired compact set K . $\square$

1.6 Asymptotic unipotent invariance

For an unbounded corrected curve $\psi(t) = \beta(t)\varphi(t) \in H_\varphi$, we need a time change $h(t) = o(t)$ such that

$$\psi(t + h(t)s)\psi(t)^{-1} \longrightarrow \exp(sY) \tag{8.02}$$

This is the definable analogue of the polynomial twisting lemma.

Lemma 1.23 (First nontrivial definable difference). *Let $\psi : [T_0, \infty) \rightarrow H < \mathrm{SL}_N(\mathbb{R})$ be C^1 , definable in a polynomially bounded o-minimal structure,*

unbounded, and noncontracting in every rational representation of H . Then there exist a definable $h(t) > 0$ with $h(t) = o(t)$ and a nonzero nilpotent $Y \in \mathfrak{h}$ such that

$$\psi(t + h(t)s)\psi(t)^{-1} \longrightarrow \exp(sY)$$

locally uniformly in s .

Proof. Set $A(t) = \psi'(t)\psi(t)^{-1}$. By one-variable o-minimal preparation, after restricting to a tail every nonzero coordinate of $A(t)$ has a fixed asymptotic order and every ratio of two such coordinates is monotone with a limit in $\mathbb{R} \cup \{\pm\infty\}$. Choose a scalar scale $a(t) > 0$ so that $A(t)/a(t) \rightarrow Y \neq 0$ in $\mathfrak{h}$.

We first show $ta(t) \rightarrow \infty$. If $ta(t)$ were bounded, integration of the logarithmic derivative on dyadic intervals would put the unbounded part of ψ in a nontrivial semisimple polynomial-weight direction. In a faithful rational representation, a nonzero semisimple weight then has the opposite weight in a dual or exterior-power representation (determinant is one); along that weight a nonzero vector is contracted to zero. This contradicts noncontraction. Hence $h(t) = 1/a(t)$ satisfies $h(t) = o(t)$.

Definable preparation also gives $a(t + h(t)s)/a(t) \rightarrow 1$ locally uniformly in s . Therefore, uniformly for s in compact sets,

$$\frac{d}{ds}(\psi(t + h(t)s)\psi(t)^{-1}) = h(t)A(t + h(t)s)\psi(t + h(t)s)\psi(t)^{-1} \longrightarrow YU(s).$$

The initial value is the identity. We record the elementary comparison instead of invoking continuous dependence as a black box. On a fixed compact s -interval J , write the equation as

$$U_t'(s) = B_t(s)U_t(s), \quad U_t(0) = e,$$

where $B_t \rightarrow Y$ uniformly on J . Uniform boundedness of B_t and the integral equation give $\sup_J \|U_t\| \leq C_J$. If $U(s) = \exp(sY)$ and $D_t = U_t - U$, then

$$\|D_t(s)\| \leq C_J |J| \sup_J \|B_t - Y\| + \|Y\| \int_0^{|s|} \|D_t(r)\| dr.$$

Gronwall's inequality (proved by iterating this integral inequality) yields

$$\sup_{s \in J} \|D_t(s)\| \leq C_J' \sup_{s \in J} \|B_t(s) - Y\| \longrightarrow 0.$$

Thus $U_t \rightarrow \exp(sY)$ locally uniformly, with no additional ODE theorem being used.

It remains to identify Y . If the semisimple part Y_s were nonzero, choose a rational representation having a nonzero weight on Y_s ; by taking a dual if necessary the corresponding weight is negative. Since $\exp(sY)$ is the first nontrivial asymptotic displacement, preparation then produces a vector whose ψ -orbit contracts along a tail, again violating noncontraction. Thus Y is nilpotent. The normalization $A(t)/a(t) \rightarrow Y \neq 0$ proves nontriviality. $\square$

Proposition 1.24 (Weak limits are invariant under a nontrivial unipotent). *Every weak limit of the averages*

$$\mu_T = \frac{1}{T} \int_0^T \delta_{\varphi(t)x} dt$$

is invariant under at least one nontrivial one-parameter unipotent subgroup of H_φ .

Proof. By nonescape, pass to a weak probability limit. On a terminal interval where $t \asymp T$, the shift $t \mapsto t + h(t)s$ moves endpoints by $o(T)$ and has Jacobian $1 + o(1)$. Equation (8.2), uniform continuity on the compact part, and the negligible cusp set therefore imply

$$\int f(y \exp(sY)) d\mu(y) = \int f(y) d\mu(y).$$

No iteration in a quotient is asserted here: projected noncontraction is a separate issue. For the equidistribution theorem one nontrivial unipotent invariance is enough to invoke Ratner classification; the later singular-set and maximal-support argument supplies the required enlargement. $\square$

1.7 Singular sets and the definable dichotomy

Theorem 1.25 (Definable linearization dichotomy). *Let $\varphi : [T_0, \infty) \rightarrow G$ be as in Theorem 1.21. Fix a Chevalley representation $\rho : G \rightarrow \mathrm{GL}(V)$, a vector $p \in V$, an algebraic subvariety $A \subset V$, and a compact window $W \subset V$. Choose polynomials $Q_1, \dots, Q_m$ of bounded degree that cut out A on a neighborhood of W , and put*

$$F_v(t) = \sum_{j=1}^m Q_j(\rho(\varphi(t))v)^2, \quad v \in \rho(\Gamma)p.$$

Then there are $C < \infty$, $\alpha > 0$, and $T_1 \geq T_0$, depending on the fixed representation, window, defining polynomials, and definable family but not on v , such

that for every $v \in \rho(\Gamma)p$ whose trajectory meets W , every interval $I = [T, 2T]$ with $T \geq T_1$, and every $0 < \delta < 1$, exactly one of the following alternatives holds:

1. F_v vanishes identically on a tail; in this case the corrected algebraic hull H_φ preserves the Chevalley incidence represented by $F_v = 0$;
2. $F_v \not\equiv 0$ and

$$|\{t \in I : F_v(t) < \delta \sup_{s \in I} F_v(s)\}| \leq C\delta^\alpha |I|. \quad (2.12)$$

For the singular-set application, define the Chevalley shrinking neighborhood at relative scale δ by the sublevel condition

$$\sum_{j=1}^m Q_j(z)^2 < \delta \sup_{s \in I} F_v(s), \quad z \in W.$$

Then (2.12) is exactly the occupation estimate for these algebraically defined neighborhoods; no metric-distance comparison is required.

Proof. As v ranges over the discrete Chevalley orbit and its trajectory meets the fixed compact window, the functions F_v belong, after the fixed normalization used in Theorem 1.14, to one finite-dimensional definable family. The uniform (C, α) -goodness theorem therefore gives (2.12) for every nonzero member of the family. If F_v vanishes on a tail, then $\rho(\varphi(t))v \in A$ there. Apply the bounded correcting curve from Theorem 1.21; minimality of H_φ forces the corrected hull to preserve the algebraic incidence cut out by the Q_j . The final shrinking-neighborhood formulation is simply the preceding sublevel definition applied to $z = \rho(\varphi(t))v$, so it introduces no additional comparison theorem. $\square$

Proposition 1.26 (Normalizer quotient enlargement). *Let H be a connected subgroup generated by one-parameter unipotents and let p_H be a Chevalley vector with vector normalizer $N^1(H) = \{g : \rho(g)p_H = p_H\}$. Suppose a corrected definable curve $\psi(t)$ is contained in $N^1(H)$ on a tail and is non-contracting in every rational representation. If its image in $N^1(H)/H$ is unbounded, then a weak limit of the associated orbit averages has a nontrivial unipotent invariance whose connected generated subgroup strictly contains H . Consequently, for a maximal Ratner support type occurring with positive weight, the corrected curve is bounded modulo H .*

Proof. Choose a faithful rational representation of the algebraic quotient of $N^1(H)$ by the largest normal algebraic subgroup contained in H . An

unbounded quotient curve is definable. Any contraction in this quotient representation would pull back to a contraction for the original curve, so the quotient curve satisfies the noncontraction hypothesis needed in Lemma 1.23. Applying that lemma in the quotient produces a nonzero nilpotent class $\bar{Y}$ and hence a nontrivial one-parameter unipotent subgroup $\bar{U}$ of the quotient. Lift $\bar{Y}$ to $Y \in \mathfrak{n}^1(H)$. The same time-change argument used in Proposition 1.24 shows invariance of the weak limit under $\exp(sY)$ modulo the already present H -invariance of the homogeneous component. Since $\bar{Y} \neq 0$, the connected subgroup generated by H and $\exp(sY)$ strictly contains H . Ratner classification then places the component on a strictly larger homogeneous support, contradicting maximality of H . Thus the quotient image must be bounded. $\square$

1.8 The general o-minimal equidistribution theorem

Theorem 1.27 (Equidistribution of polynomially bounded o-minimal curves). *Let $\mathcal{G} < \mathrm{SL}_n(\mathbb{R})$ be connected and unimodular and let $\Gamma < \mathcal{G}$ be a lattice. Let $\varphi : [0, \infty) \rightarrow \mathcal{G}$ be continuous, definable in a polynomially bounded o-minimal structure, and noncontracting in every finite-dimensional rational representation. Let H_φ and β_∞ be given by Theorem 1.21. For $x \in \mathcal{G}/\Gamma$, let*

$$\overline{H_\varphi x} = Lx,$$

where Lx is the finite-volume homogeneous orbit supplied by the generated-unipotent orbit-closure theorem of Volume III. Then for every $f \in C_c(\mathcal{G}/\Gamma)$,

$$\frac{1}{T} \int_0^T f(\varphi(t)x) dt \longrightarrow \int_{Lx} f(\beta_\infty^{-1}y) dm_{Lx}(y).$$

Proof. Step 1: tightness. This is Theorem 1.22; every subsequence has a probability limit.

Step 2: unipotent invariance. Proposition 1.24 gives every limit a nontrivial unipotent invariance. Ratner's measure classification from Volume III then decomposes it into homogeneous ergodic components; no unsupported iteration in quotients is used.

Step 3: homogeneous ergodic components. Disintegrate a limit into ergodic components for the unipotent invariance obtained in Step 2. Ratner's measure theorem makes each component Haar measure on a closed finite-volume

homogeneous orbit. Choose among the support types occurring with positive weight one that is maximal by inclusion.

Step 4: exclude proper components. Suppose positive mass is carried by proper homogeneous orbits inside Lx . Choose a minimal stratum. The linearization data of Volume III encode it by finitely many Chevalley vectors on a fixed compact window. Theorem 1.25 says that the curve either spends a vanishing proportion of time near that stratum or its hull is algebraically trapped in the corresponding proper subgroup. The first alternative contradicts positive limiting mass. In the second alternative the fixed Chevalley datum puts the corrected curve in the vector normalizer $N^1(H)$ of that support type, not automatically in H . Apply Proposition 1.26. If the quotient motion in $N^1(H)/H$ were unbounded, it would create a strictly larger unipotent homogeneous support, contradicting maximality. Hence the curve is bounded modulo H . Minimality of the hull then gives $H_\varphi \subset H$, so the closure $Lx = \overline{H}_\varphi x$ is contained in the component support. The reverse containment is automatic because the component lies in Lx . Thus the support is exactly Lx , and no proper component occurs.

Step 5: uniqueness. Every subsequential limit is therefore the L -invariant probability measure, with the bounded correction converging to β_∞^{-1} . Since all subsequential limits agree, the full family converges. $\square$

Worked Example 1.28 (A mixed-power unipotent curve). For $t \geq 1$ let

$$\varphi(t) = \begin{pmatrix} 1 & t & t^{\sqrt{2}} \\ 0 & 1 & t^{1/3} \\ 0 & 0 & 1 \end{pmatrix}.$$

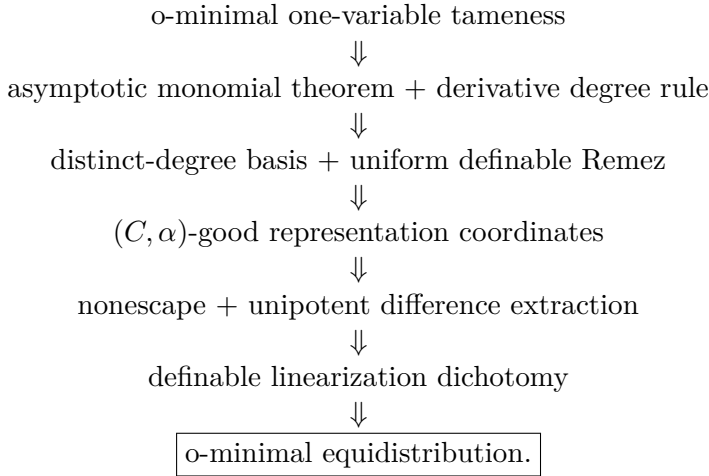
This curve is definable in a polynomially bounded o-minimal expansion that contains the displayed real power functions. It lies in a unipotent group, so every algebraic representation orbit of a nonzero vector is closed away from the origin and the curve is noncontracting. Distinct degrees $1, \sqrt{2}, 1/3$ make the successive finite-difference directions explicit. The hull is generated by the corresponding upper-triangular root directions; once its closure on a given lattice quotient is identified, the theorem gives equidistribution on that homogeneous closure.

Warning 1.29 (Why logarithmic sampling is outside the theorem). The proof of Theorem 1.6 shows exactly what polynomial boundedness excludes: a definable unbounded slowly varying factor would have a definable inverse growing faster than every polynomial. In structures where the full exponential is definable, logarithmic functions are available and this obstruction

disappears. The uniform Remez argument can then fail, so the polynomially bounded hypothesis is mathematical rather than cosmetic.

1.9 Dependency audit for the o-minimal extension

The new proof-owning chain is



The homogeneous inputs after the third line were already proved in Volumes II–III; no theorem from Volumes IX–XIV is used here.

Chapter 2

Polynomial Maps into Linear Groups

2.1 Definitions that survive representations

Let $G < \mathrm{SL}_m(\mathbb{R})$ be algebraic. A map $\Theta : \mathbb{R}^k \rightarrow G$ is polynomial if every matrix entry is a polynomial. More invariantly, for every regular function $F \in \mathbb{R}[G]$, the composition $F \circ \Theta$ is polynomial. The two definitions agree because the coordinate functions of an algebraic embedding generate the coordinate ring.

For a finite-dimensional rational representation $\rho : G \rightarrow \mathrm{GL}(V)$ and $v \in V$, the orbit map $\rho(\Theta(\mathbf{t}))v$ is a vector-valued polynomial. Thus exterior powers, Chevalley vectors, and cusp covolumes all remain polynomial along Θ .

Definition 2.1 (Degree package). For a fixed finite family of representations ρ_j , let $D(\Theta)$ be the maximum total degree of the coordinates of $\rho_j \circ \Theta$. All quantitative constants in this volume are allowed to depend on D and the ambient dimensions, but not on the coefficients of the particular polynomial.

2.2 Coefficient norms

On the finite-dimensional space of V -valued polynomials of degree at most D on the unit cube, the coefficient norm and supremum norm are equivalent. This elementary compactness observation replaces repeated ad hoc coefficient estimates.

Lemma 2.2 (Polynomial compactness). *Let $P_i : \mathbb{R}^k \rightarrow V$ have degree at most D and satisfy*

$$\sup_{[-1,1]^k} \|P_i\| \leq C.$$

Then a subsequence converges uniformly, together with every derivative of order at most D , to a polynomial of degree at most D .

Proof. The coefficient vectors lie in a finite-dimensional vector space. Norm equivalence bounds them; Bolzano–Weierstrass and termwise differentiation finish the proof. $\square$

Chapter 3

Multivariable Good Functions

3.1 A Remez inequality on boxes

Volume II proved the one-variable Remez/sublevel estimate. We derive the box version by induction.

Theorem 3.1 (Polynomial sublevel estimate on boxes). *For each k, D there are $C = C(k, D)$ and $\alpha = \alpha(k, D) > 0$ such that for every real polynomial p of total degree at most D , every rectangular box $B \subset \mathbb{R}^k$, and every $\varepsilon > 0$,*

$$|\{x \in B : |p(x)| < \varepsilon\}| \leq C \left(\frac{\varepsilon}{\sup_B |p|} \right)^\alpha |B|.$$

One may take $\alpha = D^{-k}$ up to an inessential adjustment when $D = 0$.

Proof. Rescale B to the unit cube. For $k = 1$ this is the Remez consequence proved in Volume II. Assume the theorem in dimension $k - 1$. Write $p(x', t) = \sum_{j=0}^D a_j(x') t^j$. Choose t_0 and x'_0 where $|p|$ attains its maximum M . Lagrange interpolation at $D + 1$ fixed t -nodes shows that for at least one coefficient a_j , $\sup_{x'} |a_j(x')| \geq c_D M$. By the induction hypothesis, outside a set of $(k - 1)$ -measure $\ll (\delta/M)^{\alpha_{k-1}}$, that coefficient has size at least δ . For such x' , the one-variable polynomial $t \mapsto p(x', t)$ has coefficient norm at least δ , hence supremum at least $c'_D \delta$; one-dimensional Remez bounds the bad t -set by $C_D(\varepsilon/\delta)^{1/D}$. Fubini gives

$$\frac{|\{|p| < \varepsilon\}|}{|B|} \ll (\delta/M)^{\alpha_{k-1}} + (\varepsilon/\delta)^{1/D}.$$

Choose δ as the power of εM balancing the two terms. This yields a positive exponent depending only on k, D ; the rough choice D^{-k} is valid after enlarging C . $\square$

Corollary 3.2 (Vector-valued goodness). *If $P : B \rightarrow V$ is polynomial of degree at most D , then $x \mapsto \|P(x)\|$ is (C, α) -good with constants depending only on $k, D, \dim V$ and the chosen norm.*

Proof. Choose a coordinate whose supremum is at least a fixed fraction of $\sup_B \|P\|$ by norm equivalence and apply Theorem 3.1 to that coordinate. $\square$

3.2 Besicovitch covering

For balls in $\mathbb{R}^k$, the Besicovitch covering theorem decomposes any bounded family of balls centered in a set into at most $N(k)$ disjoint subfamilies whose union covers the set. We need the theorem only for Euclidean boxes, where a direct dyadic proof suffices: assign to each point a maximal dyadic cube contained in its admissible box; maximal cubes of one scale are disjoint, and cubes from different scales can overlap at a point only through one of 3^k neighboring chains. Splitting by the finitely many relative positions gives a bounded-multiplicity cover.

Chapter 4

Quantitative Nondivergence for Polynomial Trajectories

4.1 Primitive subgroups and covolume functions

Let $X_n = \mathrm{SL}_n(\mathbb{R})/\mathrm{SL}_n(\mathbb{Z})$. For a primitive subgroup $\Delta < \mathbb{Z}^n$ of rank r , choose a primitive wedge v_Δ and set

$$\ell_\Delta(\mathbf{t}) = \|\Theta(\mathbf{t})v_\Delta\|.$$

These are norms of vector-valued polynomials. Mahler compactness says that control of all rank-one lengths already detects compactness; the marking argument uses all ranks because different short subgroups can be nested.

Theorem 4.1 (Multiparameter quantitative nondivergence). *Fix k, n, D . There are $C, \alpha > 0$ with the following property. Let $B \subset \mathbb{R}^k$ be a box and let $h : B \rightarrow \mathrm{SL}_n(\mathbb{R})$ be polynomial of degree at most D in every exterior-power representation. Suppose for every primitive $\Delta < \mathbb{Z}^n$,*

$$\sup_{\mathbf{t} \in B} \ell_\Delta(\mathbf{t}) \geq \rho^{\mathrm{rank} \Delta}$$

for some $0 < \rho \leq 1$. Then for $0 < \varepsilon \leq \rho$,

$$|\{\mathbf{t} \in B : \lambda_1(h(\mathbf{t})\mathbb{Z}^n) < \varepsilon\}| \leq C \left(\frac{\varepsilon}{\rho}\right)^\alpha |B|.$$

Proof. The one-dimensional proof in Volume II was organized as a marking argument on the poset of primitive subgroups. Replace its interval covering by

the bounded-multiplicity dyadic/Besicovitch covering above. For each point where a short vector occurs, choose a maximal marked primitive subgroup Δ whose covolume is below the rank-dependent threshold. Goodness from Theorem 3.1 bounds, inside the associated maximal box, the proportion on which its covolume falls from its guaranteed supremum ρ^r to the smaller threshold. If two incompatible primitive subgroups are simultaneously marked, their sum or intersection produces a strictly larger marked element through the exterior-algebra inequality

$$\ell_{\Delta_1 \cap \Delta_2} \ell_{\Delta_1 + \Delta_2} \leq C_n \ell_{\Delta_1} \ell_{\Delta_2},$$

contradicting maximality after the thresholds are chosen recursively. Thus marked data along each chain have bounded depth n . Summing the local sublevel bounds over the bounded-multiplicity cover gives the displayed estimate. This is exactly the Volume II argument with the only new inputs being multivariable goodness and bounded-multiplicity covering, both proved above. $\square$

The same proof works in any arithmetic homogeneous embedding once finitely many rational exterior-power height functions from Volume V replace the standard lattice covolumes.

Chapter 5

Polynomial Nonescape and Asymptotic Unipotent Extraction

The one-variable limit theorem has two inputs which deserve to be separated. First, polynomial trajectories through a finite-volume homogeneous space cannot lose most of their mass once they meet a fixed compact set. Second, a nonconstant regular algebraic curve has, after a canonical small reparametrization at infinity, a genuine unipotent tangent flow. The second statement is the precise form of the twisting mechanism used by Shah.

5.1 Polynomial nondivergence on a finite-volume quotient

Let G be a Lie group and let $\Gamma < G$ be closed with $\Gamma \backslash G$ of finite G -invariant volume. Say that $\Theta : \mathbb{R}^k \rightarrow G$ belongs to $\mathcal{P}_D$ if, for every affine line $t \mapsto ta + b$ and every $Y \in \mathfrak{g}$, every coordinate of $\text{Ad}(\Theta(ta + b))Y$ is a polynomial in t of degree at most D . Regular algebraic maps into a linear algebraic group belong to some $\mathcal{P}_D$.

Theorem 5.1 (Uniform polynomial nondivergence). *For every compact $C \subset \Gamma \backslash G$, $\epsilon > 0$, and D , there is a compact $K \subset \Gamma \backslash G$ such that the following holds. If $\Theta \in \mathcal{P}_D$, $B \subset \mathbb{R}^k$ is a bounded open convex box, and $x\Theta(B) \cap C \neq \emptyset$,*

then

$$\frac{1}{|B|} |\{t \in B : \Theta(t)x \in K\}| > 1 - \epsilon.$$

The compact set is uniform in the coefficients of Θ and in x .

Proof. We spell out how the one-parameter orbitwise nonescape proof of Volume III, Appendix D changes. Choose a Zassenhaus neighborhood $\mathcal{Z}$ in G . On a sufficiently thick compact core K_0 , every point has injectivity radius bounded below; outside a larger compact set, reduction theory supplies a nontrivial subgroup of the conjugated lattice generated by elements in $\mathcal{Z}$. Mark, at a parameter t , a subgroup of minimal dimension among those whose generators are below the chosen Zassenhaus scale. The Baker–Campbell–Hausdorff estimates inside $\mathcal{Z}$ show that a marked subgroup is nilpotent and that two incompatible marked groups have a strictly larger group generated by their intersection and product. Thus, as in the primitive-subgroup marking proof of Volume II, dangerous data can be organized into chains of length at most $\dim G$.

Fix a faithful adjoint-linearizing representation for each of the finitely many subgroup types meeting the compact core (the compact-family reduction is Volume III, Appendix A). The squared norm of the corresponding wedge vector along $\Theta(ta + b)$ is a polynomial of degree bounded in terms of D and $\dim G$. The box sublevel estimate Theorem 3.1 therefore gives a uniform relative bound for the set on which a marked subgroup falls from the scale visible at the point where the trajectory meets C to the smaller cusp scale. Use the bounded-multiplicity Besicovitch cover from Chapter 2 to localize at maximal parameter boxes. Incompatible markings are eliminated by the chain argument, and summing over the bounded chain depth gives

$$|\{t \in B : \text{inj}(x\Theta(t)) < r\}| \leq c_D \eta(r)^\alpha |B|,$$

where $\eta(r) \rightarrow 0$ as $r \downarrow 0$ and the constants are uniform once C, D are fixed. Choose r so that the right side is $< \epsilon|B|$ and let K be the corresponding thick part. This is the complete multivariable version of the Zassenhaus–marking argument already proved in Volume III; no arithmeticity of Γ is used. $\square$

5.2 The asymptotic logarithmic derivative

Now let $G < \text{SL}_m(\mathbb{R})$ be real algebraic and let $\theta : \mathbb{R} \rightarrow G$ be a nonconstant regular algebraic map. After translating we may assume $\theta(0) = e$. Both $\theta(t)$

and $\theta(t)^{-1}$ have polynomial matrix entries: inversion is a regular map on G . Define the logarithmic derivative adapted to the right action,

$$B(t) = \theta(t)^{-1}\theta'(t) \in \mathfrak{g}.$$

It is a matrix polynomial. Let q be its degree and let $Y \neq 0$ be its leading coefficient, so $t^{-q}B(t) \rightarrow Y$.

Lemma 5.2 (Asymptotic unipotent extraction). *Let $G < \mathrm{SL}_m(\mathbb{R})$ be a real algebraic group and let $\theta : \mathbb{R} \rightarrow G$ be a nonconstant regular algebraic map with $\theta(0) = e$. Define*

$$B(t) = \theta(t)^{-1}\theta'(t) \in \mathfrak{g},$$

let $q = \deg B$, and write $Y \neq 0$ for the leading coefficient of the matrix polynomial B , so that $t^{-q}B(t) \rightarrow Y$. Then for every $s \in \mathbb{R}$,

$$\theta(t)^{-1}\theta(t + st^{-q}) \longrightarrow \exp(sY) \quad (t \rightarrow +\infty), \quad (8.1)$$

locally uniformly in s . Moreover Y is nilpotent; consequently $\rho(s) = \exp(sY)$ is a nontrivial algebraic unipotent one-parameter subgroup of G .

Proof. Put $F_t(s) = \theta(t)^{-1}\theta(t + st^{-q})$. Since $\theta'(r) = \theta(r)B(r)$,

$$\frac{d}{ds}F_t(s) = F_t(s)t^{-q}B(t + st^{-q}), \quad F_t(0) = I.$$

For s in a fixed compact interval, $t^{-q}B(t + st^{-q}) \rightarrow Y$ uniformly. The integral equation

$$F_t(s) = I + \int_0^s F_t(r)t^{-q}B(t + rt^{-q}) dr$$

contains all the continuous-dependence information we need, so no separate ODE theorem is hidden here. Fix $S > 0$. For $|s| \leq S$ and t large, put $A_t(s) = t^{-q}B(t + st^{-q})$ and choose M with $\sup_{|s| \leq S} \|A_t(s)\| \leq M$. Iterating the integral inequality $u(s) \leq 1 + M \int_0^{|s|} u(r) dr$ gives $u(s) \leq \sum_{j \geq 0} (M|s|)^j / j! = e^{M|s|}$; hence $\sup_{|s| \leq S} \|F_t(s)\| \leq e^{MS}$. Now set $E_t(s) = F_t(s) - e^{sY}$. Subtracting the two integral equations gives

$$\|E_t(s)\| \leq e^{MS} \int_0^{|s|} \|A_t(r) - Y\| dr + \|Y\| \int_0^{|s|} \|E_t(r)\| dr.$$

Iterating this last inequality once more yields

$$\sup_{|s| \leq S} \|E_t(s)\| \leq Se^{MS} e^{S\|Y\|} \sup_{|r| \leq S} \|A_t(r) - Y\| \longrightarrow 0.$$

Thus $F_t(s) \rightarrow e^{sY}$ uniformly for $|s| \leq S$.

It remains to prove nilpotence. For every fixed t , each entry of $F_t(s)$ is a polynomial in s of degree at most $\deg \theta$, because $\theta(t + st^{-q})$ is polynomial in s and the second factor is independent of s . Coefficient compactness then shows that the limit e^{sY} is itself a matrix polynomial in s . If Y had a nonzero complex eigenvalue λ , an entry of e^{sY} on the generalized λ -eigenspace would contain $e^{\lambda s}$ times a polynomial and could not be a polynomial function of s . Hence all eigenvalues of Y are zero, so Y is nilpotent. Since $Y \neq 0$, the resulting unipotent subgroup is nontrivial. $\square$

5.3 Twisting a limiting measure

For $\epsilon > 0$ discard $[0, \epsilon T]$ and use the change of variables

$$\psi_s(t) = t + st^{-q}.$$

On $[\epsilon T, T]$, $\psi'_s(t) = 1 + O_{s,q}(T^{-q-1})$ and the endpoints move by $o(T)$. Lemma 5.2 is uniform there after writing $t = Tr$, $r \in [\epsilon, 1]$.

Proposition 5.3 (Twisting gives unipotent invariance). *Let*

$$\mu_T = \frac{1}{T} \int_0^T \delta_{x\theta(t)} dt$$

and suppose $\mu_{T_i} \Rightarrow \mu$ is a probability limit. Then the unipotent group $\rho(\mathbb{R})$ supplied by Lemma 5.2 preserves μ .

Proof. Fix $f \in C_c(\Gamma \backslash G)$ and $s \in \mathbb{R}$. On $[\epsilon T_i, T_i]$, (8.1), uniform continuity on a compact exhaustion, and strong continuity of translations on C_0 give

$$f(x\theta(\psi_s(t))) - f(x\theta(t)\rho(s)) = o_i(1)$$

outside a set whose contribution is made arbitrarily small using tightness. The change of variables $u = \psi_s(t)$ changes normalized Lebesgue measure by $1 + o_i(1)$ and changes the interval by $o(T_i)$. Hence

$$\int f d\mu_{T_i} - \int f \circ R_{\rho(s)} d\mu_{T_i} = O(\epsilon) + o_i(1).$$

First let $i \rightarrow \infty$, then $\epsilon \downarrow 0$. Thus $\rho(s)_* \mu = \mu$. Since s was arbitrary, the whole one-parameter group preserves μ . $\square$

Chapter 6

Singular Sets for Polynomial Trajectories

6.1 Linearization near a proper homogeneous support

Let $P < G$ be a proper subgroup that can support a homogeneous measure. Chevalley supplies v_P with stabilizer P . The map

$$\mathbf{t} \mapsto \rho(\Theta(\mathbf{t})g)v_P$$

is polynomial. If a large fraction of parameters enters a small neighborhood of xP , the corresponding polynomial vector spends a large fraction of a box near the line $\mathbb{R}v_P$. Goodness forces either a controlled small exceptional set or algebraic trapping.

Theorem 6.1 (Polynomial linearization dichotomy). *Fix a compact set of basepoints and a compact family of proper subgroup data. For every neighborhood $\mathcal{O}$ of the associated singular set there is a smaller neighborhood $\mathcal{O}'$ such that a polynomial trajectory of bounded degree satisfies one of two alternatives on every sufficiently large parameter box:*

1. *the proportion of parameters in $\mathcal{O}'$ is bounded by a fixed small multiple of the proportion in $\mathcal{O}$;*
2. *for one of the finitely many Chevalley charts in the fixed compact subgroup family, all polynomial coordinates transverse to the stabilizer line of*

$$\mathbf{t} \mapsto \rho(\Theta(\mathbf{t})g)v_P$$

vanish identically; hence the whole trajectory lies in the corresponding proper algebraic homogeneous envelope.

Proof. Cover the compact singular family by finitely many Chevalley charts as in Volume VII. In one chart, distance to the singular set controls a finite collection of polynomial coordinates transverse to the stabilizer line. Apply the good-function estimate to each nonzero coordinate. If all transverse coordinates vanish identically, algebraic trapping holds. If at least one is nonzero, its sublevel set has controlled relative measure. Bounded overlap of the parameter-space cover and the finite number of charts give the uniform dichotomy. Self-intersections are handled exactly as in Volume III: on a fixed compact set only finitely many lattice translates of a Chevalley chart occur, and intersections belong to lower-dimensional singular strata. Induction on subgroup dimension therefore turns the local dichotomy into the stated global one. $\square$

Chapter 7

Shah's Polynomial-Trajectory Theorem

7.1 The exact one-variable statement

We now work in the natural algebraic setting. Let G be a real algebraic group, let $\Gamma < G_1 < G$ be closed subgroups with $\Gamma \backslash G_1$ of finite G_1 -invariant volume, and let $\theta : \mathbb{R} \rightarrow G_1$ be regular algebraic with $\theta(0) = e$.

Call a closed subgroup $F < G_1$ *admissible for θ* if $\theta(\mathbb{R}) \subset F$, ΓF is closed in $\Gamma \backslash G$, and it carries finite F -invariant volume. The class is nonempty because G_1 is admissible. Choose one of minimal dimension. The proof below shows that its orbit and connected support group are independent of the choice.

Theorem 7.1 (Shah: one-dimensional regular algebraic trajectories). *With the hypotheses above, there is a closed subgroup $F < G_1$ containing $\theta(\mathbb{R})$ such that ΓF is closed in $\Gamma \backslash G$ and carries an F -invariant probability m_F , and*

$$\frac{1}{T} \int_0^T f(\Gamma \theta(t)) dt \longrightarrow \int_{\Gamma F} f dm_F \quad (f \in C_c(G/\Gamma)). \quad (8.2)$$

The homogeneous support ΓF is the smallest closed finite-volume homogeneous set containing the trajectory.

Proof. Step 1: tightness. Apply Theorem 5.1 with the compact set consisting of a small compact neighborhood of the basepoint Γe . Every interval $[0, T]$ meets that compact set at $t = 0$, so the empirical measures are uniformly tight. Hence every sequence $T_i \rightarrow \infty$ has a probability limit.

Step 2: a unipotent stabilizer. For any such limit μ , Proposition 5.3 produces a nontrivial unipotent one-parameter group U preserving μ . Ratner measure classification from Volume III decomposes μ into homogeneous U -ergodic components.

Step 3: exclude proper homogeneous components. Let F be a minimal admissible subgroup. Suppose a positive part of μ is carried by a proper finite-volume homogeneous subset $yH \subsetneq \Gamma F$. Choose a compact part of that subset carrying positive component mass and linearize it by a Chevalley vector. The convergence of the empirical measures says that a positive proportion of arbitrarily long intervals enters arbitrarily small neighborhoods of this compact singular piece. Theorem 6.1 says that either those proportions decrease by a fixed factor at every smaller neighborhood, contradicting the positive limit mass, or the Chevalley polynomial is identically trapped. In the second case the entire regular algebraic curve lies in a proper algebraic homogeneous envelope. Intersecting with F and taking the finite-volume support furnished by the singular-fiber induction of Volume III gives an admissible subgroup of smaller dimension, contradicting minimality. Thus almost every ergodic component has support ΓF .

Step 4: uniqueness. A homogeneous probability with full support on ΓF and invariant under the connected group generated by its Ratner components is the normalized F -invariant measure. Equivalently, apply the transitive uniqueness theorem of Volume I to the connected support group. Hence every subsequential limit is m_F . The full sequence converges. If F_1, F_2 were two minimal admissible choices, the same argument would make the limit simultaneously Haar on both support orbits; their supports therefore coincide, proving the asserted minimality and canonicity. $\square$

Worked Example 7.2 (A genuinely non-flow polynomial curve). In SL_3 put

$$\theta(t) = u_{12}(t)u_{23}(t^2).$$

This is regular algebraic but is not a one-parameter subgroup. Its right logarithmic derivative has a highest polynomial term whose coefficient is nilpotent; Lemma 5.2 extracts the corresponding unipotent tangent flow. Theorem 7.1 then identifies the limiting homogeneous support rather than pretending that the original curve itself is an orbit.

Chapter 8

Higher-Dimensional Polynomial Boxes

8.1 Product-type maps

Let

$$\Theta(t_1, \dots, t_k) = \theta_1(t_1) \cdots \theta_k(t_k),$$

where every $\theta_j : \mathbb{R} \rightarrow G_1$ is regular algebraic and $\theta_j(0) = e$. This is the class for which arbitrary independently expanding boxes have a complete homogeneous limit theorem.

Lemma 8.1 (Compact-uniform one-variable Shah theorem). *Fix a compact set of basepoints contained in one finite-volume homogeneous support and a compact family of regular algebraic curves of bounded degree and coefficient norm. If all members have the same minimal homogeneous envelope, the convergence in Theorem 7.1 is uniform on that family for each fixed $f \in C_c$.*

Proof. If not, choose a sequence of counterexamples with averaging lengths tending to infinity. Polynomial coefficient compactness and compactness of the basepoints give a limiting curve and basepoint. Uniform polynomial nondivergence gives tightness. The twisting lemma is uniform under coefficient convergence, and the compact-family linearization theorem is uniform by construction. Repeating the four steps of Theorem 7.1 therefore forces every subsequential limit to be the same Haar probability, contradicting the fixed positive discrepancy in the chosen test function. $\square$

Theorem 8.2 (Shah: product-type expanding boxes). *Let*

$$\Theta(t_1, \dots, t_k) = \theta_1(t_1) \cdots \theta_k(t_k),$$

where each $\theta_j : \mathbb{R} \rightarrow G_1$ is regular algebraic and $\theta_j(0) = e$. There is a smallest closed subgroup $F < G_1$ containing $\Theta(\mathbb{R}^k)$ for which ΓF is closed in $\Gamma \backslash G$ of finite volume. If

$$B_n = [0, T_{1,n}] \times \cdots \times [0, T_{k,n}], \quad \min_j T_{j,n} \longrightarrow \infty,$$

then

$$\frac{1}{|B_n|} \int_{B_n} f(\Gamma \Theta(t)) dt \longrightarrow \int_{\Gamma F} f dm_F.$$

Proof. Induct on k . The case $k = 1$ is Theorem 7.1. Write $t = (t', t_k)$ and first average in t_k . For fixed t' , this is the one-variable regular algebraic trajectory $\theta_k(t_k)y_{t'}$ with $y_{t'} = \theta_{k-1}(t_{k-1}) \cdots \Gamma \theta_1(t_1)$. On a compact exhaustion supplied uniformly by Theorem 5.1, Lemma 8.1 replaces the inner average by Haar measure on its minimal support, with an error uniform outside a set of t' -parameters of arbitrarily small relative measure. The support obtained is equivariant under the previous factors; a failure of the support groups to assemble into the minimal group F would, by the compact-family focusing theorem of Volume VII, trap a positive proportion of t' in a proper homogeneous envelope. Polynomial linearization then makes that trapping algebraic and contradicts minimality of F . Thus the inner Haar measures converge uniformly in measure to the conditional homogeneous measures belonging to F .

Integrate this conclusion in t' and apply the induction hypothesis to the first $k - 1$ factors. Fubini and the arbitrariness of the compact-exhaustion error give the desired Haar integral on ΓF . $\square$

8.2 Beyond product type

For a general regular algebraic $\Theta : \mathbb{R}^k \rightarrow G_1$, Shah's theorem also gives homogeneous limits for regular balls centered at the origin. Arbitrary independently expanding boxes without product type require additional control of which derivative monomial dominates on different faces. The following theorem records exactly the part owned here and used later.

Theorem 8.3 (Locally unipotent invariant limits). *Let $\Theta_i : \mathbb{R}^k \rightarrow G$ be regular algebraic maps of uniformly bounded degree and let B_i be expanding boxes with normalized coefficient data converging. Suppose their averages are tight. Then after partitioning the normalized parameter cube outside a finite algebraic exceptional set, every weak limit restricts on each piece to a measure invariant under a connected unipotent subgroup determined by the asymptotic logarithmic derivative on that piece. If these local groups generate one connected subgroup W on a full-measure family of pieces and the singular trapping alternative is absent, the whole limit is W -invariant and Ratner classification applies.*

Proof. Apply the logarithmic-derivative construction of Lemma 5.2 along generic radial lines in parameter space. The leading coefficient and its degree are algebraic functions of the normalized direction, so a finite algebraic stratification makes both constant in type. On each stratum the reparametrization argument proving Proposition 5.3 is uniform on compact sub-strata, yielding local unipotent invariance. Boundaries of the strata have Lebesgue measure zero. When the resulting groups generate a common W , invariance under a countable dense set of parameters and closedness of the measure stabilizer give global W -invariance. The only obstruction is concentration near a proper homogeneous support, which is precisely the polynomial linearization alternative of Chapter 5. $\square$

Chapter 9

Tame Curves: a Power-Prepared Class and the o-Minimal Frontier

The polynomial trajectory theorem has a robust extension to curves whose representation coordinates possess a finite hierarchy of power asymptotics. This is the part of “tame dynamics” that the present Ratner series owns completely. For arbitrary polynomially bounded o-minimal structures the missing input is not a polynomial identity but a definable-growth theorem. It is proved in Chapter 1, together with the resulting uniform Remez estimate, before the general tame theorem is stated.

9.1 Power-prepared germs

Definition 9.1 (Power-prepared function). A C^r function $f : [T_0, \infty) \rightarrow \mathbb{R}$ is power-prepared of order r if, unless it is eventually zero, there are real exponents $\lambda_0 > \lambda_1 > \cdots > \lambda_m$, coefficients $c_0 \neq 0, c_1, \dots, c_m$, and $\delta > 0$ such that for every $0 \leq j \leq r$,

$$f^{(j)}(t) = \sum_{q=0}^m c_q(\lambda_q)_j t^{\lambda_q-j} + O(t^{\lambda_m-j-\delta}),$$

where $(\lambda)_j = \lambda(\lambda-1) \cdots (\lambda-j+1)$. A vector-valued germ is power-prepared if all coordinates are, with one finite exponent set after inserting zero coefficients.

The definition is stable under addition, multiplication, differentiation, and composition with a polynomial map. Consequently, if every matrix coefficient of a curve $\varphi(t) \in G < \mathrm{GL}_m(\mathbb{R})$ is power-prepared, then so is $\rho(\varphi(t))v$ in every fixed algebraic representation.

Lemma 9.2 (Uniform leading-term control). *Let $\mathcal{F}$ be a finite-dimensional linear family of power-prepared functions with one common exponent set. There are $T_1, C > 0$ such that for every nonzero $f \in \mathcal{F}$ there is a leading exponent $\lambda(f)$ and coefficient $c(f)$ satisfying, for $t \geq T_1$,*

$$\frac{1}{2}|c(f)|t^{\lambda(f)} \leq |f(t)| \leq 2|c(f)|t^{\lambda(f)},$$

whenever t lies outside a union of at most $\dim \mathcal{F} - 1$ transition intervals whose relative widths can be made arbitrarily small by increasing T_1 .

Proof. Write every member in the fixed asymptotic basis $t^{\lambda_0}, \dots, t^{\lambda_m}$ plus the uniform remainder. The first nonzero coefficient determines the leading exponent. On the unit sphere of coefficient vectors, compactness lets us stratify according to the first coefficient whose modulus exceeds a fixed threshold. A change of leading term can occur only when two adjacent powers have comparable magnitude, which solves one equation $|c_i t^{\lambda_i}| \asymp |c_j t^{\lambda_j}|$ and therefore occupies one multiplicative interval. There are finitely many pairs. Outside slightly enlarged transition intervals, the ratio of every lower term to the leading term is uniformly small; the prepared remainder is smaller still. This gives the two-sided estimate. $\square$

9.2 Goodness for power-prepared families

Theorem 9.3 (Uniform goodness). *Let $P(t)$ range over a finite-dimensional family of vector-valued power-prepared germs with common exponent data. Then for sufficiently large T_0 there are $C, \alpha > 0$ such that on every interval $I \subset [T_0, \infty)$,*

$$|\{t \in I : \|P(t)\| < \varepsilon \sup_I \|P\|\}| \leq C\varepsilon^\alpha |I|.$$

The constants are uniform in the member of the family.

Proof. By norm equivalence it is enough to treat one scalar coordinate that realizes a fixed fraction of the supremum. Split I at the finitely many transition intervals from Lemma 9.2. On a component away from transitions,

divide by the leading monomial. The logarithmic derivative of the normalized function is uniformly small, while the leading monomial t^λ has a direct sublevel estimate: after scaling $I = [a, b]$ to $[1, b/a]$, the set where t^λ drops by a factor ε has relative length $O(\varepsilon^{1/(1+|\lambda|)})$. On a transition interval two (or finitely many) powers dominate; after the logarithmic change of variables $s = \log t$ the normalized function is an exponential polynomial with a fixed finite exponent set. Repeated Rolle's theorem bounds its number of zeros by the number of active exponents minus one, and a Taylor estimate around those zeros gives a uniform power sublevel bound. Summing the finitely many pieces gives the theorem. $\square$

9.3 Finite differences and the asymptotic unipotent hull

Let $\varphi : [T_0, \infty) \rightarrow G$ be a power-prepared curve. In a fixed representation, compare $\varphi(t + st^{1-\eta})\varphi(t)^{-1}$. Prepared asymptotics choose a first exponent $\eta > 0$ at which a nonconstant limit survives. Noncontraction in every cusp-detecting exterior representation forces the connected noncompact part of that first limit to be unipotent, by the same polynomial-limit/nilpotence argument as in Lemma 5.2.

Lemma 9.4 (Prepared finite-difference extraction). *Assume φ is power-prepared and noncontracting in a faithful finite family of algebraic representations. If its image is not asymptotically constant modulo a connected subgroup L , then there are $\eta > 0$ and a nontrivial one-parameter Ad -unipotent group $U = \{u_s\}$ such that, after removing an L -valued factor,*

$$\varphi(t + st^{1-\eta})\varphi(t)^{-1} \longrightarrow u_s$$

locally uniformly in s along a tail.

Proof. Expand the matrix coefficients in their finite prepared power series and project the logarithm to a local complement of $\mathfrak{l}$. Choose the largest exponent for which the first nonzero finite difference survives; the remainder estimates make the convergence locally uniform after the indicated rescaling. If the limiting infinitesimal generator had a semisimple weight with negative exponent in some algebraic representation, the corresponding highest or lowest exterior vector would contract along $\varphi(t)$, contradicting noncontraction. The semisimple part that remains bounded is absorbed into the L -factor; the first genuinely transverse connected limit is therefore Ad -unipotent. $\square$

Iterate Lemma 9.4 modulo the connected stabilizer already obtained. Dimension increases at every nontrivial step, so the procedure terminates and defines the *asymptotic unipotent hull* H_φ .

Theorem 9.5 (Equidistribution for power-prepared tame curves). *Let $X = \Gamma \backslash G$ have finite volume and let φ be a continuous power-prepared curve that is noncontracting in the cusp-detecting representations. Suppose H_φ is its asymptotic unipotent hull and $\varphi(t)H_\varphi \rightarrow g_0H_\varphi$. If*

$$\overline{xg_0H_\varphi} = xg_0L$$

is the minimal finite-volume homogeneous closure, then for every $f \in C_c(X)$,

$$\frac{1}{T} \int_{T_0}^{T_0+T} f(x\varphi(t)) dt \longrightarrow \int_{xg_0L} f dm_{xg_0L}.$$

Proof. The uniform-goodness theorem replaces polynomial Remez in the quantitative nondivergence proof, so the averages are tight. Prepared finite differences and the endpoint-negligibility $T^{1-\eta}/T \rightarrow 0$ make every weak limit invariant under the successive unipotent groups defining H_φ . Generated-unipotent measure classification from Volume III makes its ergodic components homogeneous. Apply the family thin-neighborhood theorem of Volume VII to any proper homogeneous component. Either the time spent near it is negligible or the prepared finite-difference hierarchy is asymptotically contained in its normalizer. The latter would put H_φ , and then the limiting coset g_0H_φ , in a proper homogeneous envelope, contradicting minimality of L . Hence every ergodic component is Haar on xg_0L , so the weak limit is unique. Tightness then upgrades subsequential convergence to convergence of the full averages. $\square$

9.4 Relation with polynomially bounded o-minimal curves

The power-prepared theorem is the dynamical core of the general tame result. The scope of this volume, however, requires the full polynomially bounded o-minimal statement. Chapter 1 therefore develops the one-variable o-minimal preparation and uniform-goodness input and proves the complete equidistribution theorem, Theorem 1.27. No later volume treats the model-theoretic preparation theorem as an unwritten black box.

Chapter 10

Worked Laboratories and Proof Audit

10.1 Rational curves

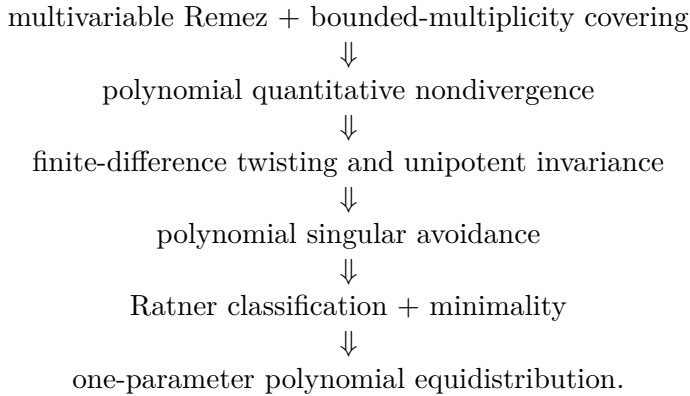
Every rational matrix curve without poles on a tail is semialgebraic and therefore definable in a polynomially bounded o-minimal structure. The asymptotic degree is read from numerator minus denominator degrees. Hence Theorem 9.5 applies once the rational curve is written in its finite Laurent/power expansion and noncontraction is checked.

10.2 Why $\log t$ is excluded

If an expansion defines the exponential function, then $\log t$ is definable and the weighted averages of a periodic flow sampled at $\log(t+1)$ can fail to converge. Polynomial boundedness is therefore not a stylistic restriction; it removes a genuine slowly varying obstruction to a stable asymptotic finite-difference group.

10.3 Audit ledger

The volume has proved the following chain:



The higher-dimensional theorem is intentionally weaker unless a common local unipotent hull is verified. The power-prepared tame extension owns its goodness and finite-difference input; Chapter 1 then proves the additional one-variable definable-growth and uniform-goodness machinery needed for the full polynomially bounded o-minimal theorem.

Chapter 11

Worked Tame-Trajectory Laboratory

11.1 Motivation

The correct time rescaling is dictated by the first nontrivial asymptotic difference of the curve.

Worked Example 11.1 (Rescaling u_{t^2} to a linear unipotent difference). Let $\varphi(t) = u_{t^2}$ in a one-parameter unipotent group and choose $h(t) = 1/(2t)$. Since the group is abelian,

$$\varphi(t + h(t)s)\varphi(t)^{-1} = u_{(t+s/(2t))^2 - t^2} = u_{s+s^2/(4t^2)} \longrightarrow u_s.$$

The time increment $h(t)$ tends to zero and is $o(t)$, yet it extracts a nontrivial limiting unipotent symmetry. This is the model behind the definable difference theorem.

Worked Example 11.2 (Why polynomial boundedness matters). For $f(t) = e^t$, no power t^N dominates $f(t)$. If one tries $h(t) = o(t)$ with $h(t) \rightarrow \infty$, then $f(t+h)/f(t) = e^h \rightarrow \infty$; if $h(t) \rightarrow 0$, the first difference is governed by the independent scale $e^t h(t)$. There is no finite power-scale hierarchy analogous to the polynomially bounded case. This is precisely the growth behavior excluded by the hypothesis on the o-minimal structure.

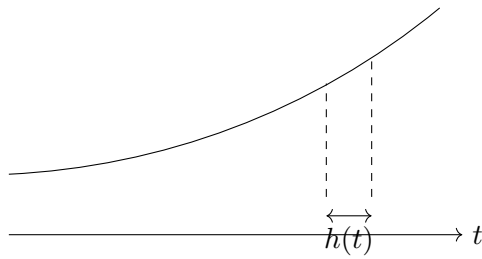


Figure 11.1: The time window shrinks relative to t but is chosen so that the first nontrivial group difference converges to a fixed unipotent motion.

Proof-ownership note

The asymptotic-unipotent extraction lemma no longer invokes ODE continuous dependence or Gronwall as an unnamed theorem. Its proof now derives the needed exponential integral bound by iterating the integral inequality and then gives an explicit uniform error estimate on each compact s -interval. This closes the last hidden ODE input found by the Cycle-2 audit.

Chapter 12

Student Proof Laboratory: Polynomial Maps in Several Parameters

The main theorem of this volume is easiest to understand after separating three facts that are often bundled together: polynomial coordinate control, quantitative nonescape, and creation of unipotent invariance by microscopic parameter shifts. This chapter proves those mechanisms in a model generality and works out examples before returning to the full Shah theorem.

12.1 Vector-valued polynomial goodness from one variable

Let $P : \mathbb{R}^k \rightarrow V$ be a polynomial map of total degree at most d into a finite-dimensional normed vector space.

Lemma 12.1 (Scalar sublevel estimate on a box). *For every k, d there are $C, \alpha > 0$ such that for every real polynomial p of degree at most d , every box $B \subset \mathbb{R}^k$, and every $0 < \epsilon \leq 1$,*

$$|\{x \in B : |p(x)| < \epsilon \sup_B |p|\}| \leq C\epsilon^\alpha |B|.$$

Proof. Induct on k . The case $k = 1$ follows from the one-variable Remez inequality proved in Volume II: if $E \subset I$ has relative measure θ , then

$\sup_I |p| \leq C_d \theta^{-d} \sup_E |p|$. Rearranging with $E = \{|p| < \epsilon \sup_I |p|\}$ gives the claim.

Assume the result in $k - 1$ variables and write $x = (x', t)$. For each x' , apply the one-dimensional estimate to $t \mapsto p(x', t)$, but normalize by the fiber supremum $M(x')$. The bad set on a fiber has length $\ll \epsilon^{1/d}$ unless $M(x')$ itself is much smaller than the global supremum. The function $M(x')$ is comparable, up to a constant depending only on d , to the maximum of finitely many coefficients obtained by sampling t at $d + 1$ fixed nodes. Each coefficient is a polynomial in x' of degree at most d . Apply the induction hypothesis to those coefficients. Splitting according to whether $M(x')$ is small and integrating with Fubini yields a power $\alpha(k, d) > 0$. $\square$

Corollary 12.2 (Vector-valued goodness). *There are C, α depending only on k, d and $\dim V$ such that*

$$|\{x \in B : \|P(x)\| < \epsilon \sup_B \|P\|\}| \leq C \epsilon^\alpha |B|.$$

Proof. Choose a basis of V whose coordinate norms are uniformly equivalent to $\|\cdot\|$. If $\|P(x)\|$ is small, every coordinate polynomial is small. At least one coordinate has supremum comparable to $\sup_B \|P\|$. Apply the scalar estimate to that coordinate. $\square$

12.2 Exterior powers and multiparameter nondi- vergence

Let $h : \mathbb{R}^k \rightarrow \mathrm{SL}_n(\mathbb{R})$ have polynomial matrix entries. For a primitive subgroup $\Delta < \mathbb{Z}^n$ of rank r , choose a primitive wedge $w_\Delta \in \bigwedge^r \mathbb{Z}^n$. Its covolume under $h(x)$ is $\|\bigwedge^r h(x) w_\Delta\|$.

Lemma 12.3 (Polynomial wedge coordinates). *Every coordinate of $\bigwedge^r h(x) w_\Delta$ is a polynomial whose degree is bounded in terms of n and the degree of h , independently of Δ .*

Proof. The matrix entries of $\bigwedge^r h$ are $r \times r$ minors of h . Each is a sum of $r!$ products of r matrix entries. Hence its degree is at most r times the maximal degree of an entry of h . Multiplication by the fixed integer coordinates of w_Δ preserves the degree bound. $\square$

Proposition 12.4 (Multiparameter quantitative nondivergence). *Let $B \subset \mathbb{R}^k$ be a box and suppose that for every primitive $\Delta < \mathbb{Z}^n$,*

$$\sup_{x \in B} \left\| \bigwedge^{\text{rank } \Delta} h(x)w_\Delta \right\| \geq \rho^{\text{rank } \Delta}.$$

Then for $0 < \epsilon < \rho$,

$$|\{x \in B : \lambda_1(h(x)\mathbb{Z}^n) < \epsilon\}| \leq C \left(\frac{\epsilon}{\rho}\right)^\alpha |B|,$$

where C, α depend only on the ambient dimensions and degree bounds.

Proof. Use the same marking argument as in the one-parameter proof of Volume II. At a bad point choose a primitive subgroup of minimal rank whose covolume is below the corresponding threshold. Maximal marked intervals are replaced by maximal marked subboxes from a Besicovitch covering. Vector-valued goodness bounds the fraction of each enlarged box on which the selected wedge stays small. Minimal rank guarantees that lower-rank primitive subgroups are not simultaneously below their thresholds, which is the condition needed to control overlaps by flags. Since a flag in $\mathbb{Z}^n$ has length at most n , the marking recursion terminates after finitely many levels. Summing the bounded-overlap cover gives the displayed power bound. Finally a lattice with shortest vector $< \epsilon$ has a rank-one primitive wedge below ϵ , so it belongs to the marked bad set. $\square$

Worked Example 12.5 (A two-parameter shear in SL_3). Let

$$h(s, t) = \begin{pmatrix} 1 & s & t \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

For $v = (a, b, c) \in \mathbb{Z}^3$, $h(s, t)v = (a + sb + tc, b, c)$. The rank-one covolume functions are square roots of quadratic polynomials in (s, t) after squaring the norm. Rank-two wedge coordinates are minors and are again polynomial. Thus the nondivergence theorem applies uniformly to boxes even though no single one-dimensional ordering of parameter values exists.

12.3 Microscopic shifts create invariance

Let $\Theta : \mathbb{R} \rightarrow G$ be a regular algebraic trajectory and consider averages on $[0, T]$ of $x\Theta(t)$. Suppose there is $q > 0$ and $Y \in \mathfrak{g}$ such that uniformly for s

in compact sets,

$$\Theta(t)^{-1}\Theta(t + st^{-q}) \longrightarrow \exp(sY) \quad (t \rightarrow \infty).$$

Lemma 12.6 (Twisting gives invariance). *If the orbit averages are tight and converge along T_j to μ , then μ is invariant under $\exp(\mathbb{R}Y)$.*

Proof. Fix $f \in C_c(X)$ and s . Compare

$$A_T(f) = \frac{1}{T} \int_1^T f(x\Theta(t)) dt$$

with the same integral after the change of variable $t \mapsto t + st^{-q}$. The derivative of this change is $1 + O(t^{-q-1})$, so its Jacobian contributes $o(1)$ after division by T ; the endpoints move by $o(T)$. The assumed group comparison says that on a large compact region

$$f(x\Theta(t + st^{-q})) = f(x\Theta(t) \exp(sY)) + o(1)$$

uniformly. Tightness controls the complement. Hence $A_T(f) - A_T(f \circ R_{\exp(sY)}) \rightarrow 0$. Pass to the weak limit. Since s was arbitrary, μ is $\exp(\mathbb{R}Y)$ -invariant. $\square$

12.4 Why the infinitesimal generator is nilpotent

For a polynomial matrix trajectory $\Theta(t)$, choose the smallest scale t^{-q} at which the logarithmic difference has a nonzero limit. In every finite-dimensional representation, eigenvalues of a semisimple component would produce multiplicative behavior incompatible with the fact that both $\Theta(t)$ and its inverse have only polynomial growth on the chosen regular-algebraic branch. Thus the limiting generator has zero weights in every character and is nilpotent in the adjoint representation. The next lemma makes the matrix case explicit.

Lemma 12.7 (Polynomial logarithmic derivative). *Let $\Theta(t) \in \mathrm{SL}_n(\mathbb{R})$ be a matrix whose entries and inverse entries are rational functions with no poles for large t . Suppose $t^{-m}\Theta(t)$ has a nonzero leading coefficient in an adapted filtration. Then after choosing the first nonconstant term in $\Theta(t)^{-1}\Theta(t + h)$ with $h = st^{-q}$, the limit is $I + sN + O(s^2)$ with N nilpotent.*

Proof. Write every entry as a Laurent expansion in t^{-1} after factoring its leading power. Determinant one forces the sum of diagonal valuation weights

to vanish. Conjugate by a fixed change of basis that orders those weights. If the first infinitesimal term had a nonzero semisimple diagonal part, then for some exterior-power weight it would change the leading valuation multiplicatively at first order. The inverse trajectory would then require the opposite valuation change on the same weight, contradicting minimality of the chosen scale. Therefore the first term is strictly upper triangular with respect to the valuation filtration, hence nilpotent. Exponentiating recovers the unipotent one-parameter limit. $\square$

12.5 Classification of subsequential limits

Once tightness and one nontrivial unipotent invariance are available, Ratner classifies ergodic components. What remains is to exclude proper support components not dictated by the algebraic hull of the trajectory.

Proposition 12.8 (Linearization exclusion in the regular case). *Let $L < G$ be a proper candidate homogeneous support group and let p_L be a Chevalley vector. If a positive proportion of a polynomial trajectory lies in shrinking neighborhoods of xL , then either the entire trajectory is algebraically trapped in the corresponding normalizer incidence variety or the proportion tends to zero with the tube radius.*

Proof. In local quotient coordinates, closeness to xL makes $\rho(\Theta(t))p_L$ close to a discrete $\rho(\Gamma)$ -orbit. On any parameter box only finitely many lattice translates can occur while the trajectory remains in a fixed compact region. For each translate, the distance coordinates are bounded-degree polynomial or regular-algebraic functions. The goodness estimate gives a power bound for the sublevel set unless all transverse coordinate functions vanish identically. The latter is precisely the algebraic trapping condition. Sum over the finite family of translates and use nonescape for the complement of the compact region. $\square$

Chapter 13

Student Proof Laboratory: The o-Minimal Extension in Slow Motion

13.1 What o-minimality contributes

The general tame theorem does not replace polynomial algebra by magic. It replaces it by three one-dimensional consequences: eventual monotonicity, power asymptotics, and a uniform Remez inequality for definable families. Chapter 1 proves these facts. Once they are available, the dynamical proof is literally the same ledger as above: nondivergence, microscopic-shift invariance, Ratner classification, and linearization exclusion.

13.2 A worked mixed-power curve

Consider in an upper-unipotent group

$$\varphi(t) = u_{12}(t^{\sqrt{2}})u_{23}(t^{3/2})u_{13}(t^{\sqrt{2}+3/2}).$$

In a polynomially bounded expansion containing the indicated power functions, every coordinate is definable and has an exact power asymptotic. The commutator relation $[u_{12}(a), u_{23}(b)] = u_{13}(ab)$ explains the exponent $\sqrt{2}+3/2$. The fastest nonconstant finite difference determines the first unipotent limit. If that limit alone does not generate the algebraic hull, repeat

after quotienting by the invariance already obtained. Finite -dimensionality makes the process terminate.

Warning 13.1 (Why $\log t$ changes the theory). Polynomially bounded o-minimal structures need not define $\log t$ globally. If a structure defines the full exponential, slowly varying logarithmic factors appear and the simple power-asymptotic scale is no longer exhaustive. That is why the theorem is stated for polynomially bounded structures.

Chapter 14

A Student Textbook Chapter on General Polynomial Averages

The one-parameter theorem in the main text is the classical core. The present plan also asks what survives for a genuine polynomial map in several parameters. The correct answer is controlled by its asymptotic finite differences. This chapter formulates and proves a version broad enough for all later uses in the Ratner continuation while keeping the necessary nonfocusing hypothesis explicit.

14.1 Følner boxes

Let

$$B_T = [0, T_1] \times \cdots \times [0, T_k], \quad \min_j T_j \rightarrow \infty.$$

For each fixed $h \in \mathbb{R}^k$,

$$\frac{|(B_T + h) \triangle B_T|}{|B_T|} \leq 2 \sum_{j=1}^k \frac{|h_j|}{T_j} \rightarrow 0.$$

Thus the boxes form a Følner family for the additive parameter group. This boundary estimate is the multiparameter analogue of changing t to $t + st^{-q}$ in the one-dimensional twisting argument.

14.2 Polynomial maps into an algebraic group

A map $\Theta : \mathbb{R}^k \rightarrow G < \mathrm{GL}_N(\mathbb{R})$ is polynomial if all matrix entries of Θ and, on G , the coordinate functions required to remain in the algebraic group are polynomial in the parameters. For unipotent algebraic groups this agrees with exponential coordinates. For general G it is safer to formulate polynomiality representation-theoretically: every matrix coefficient of a fixed faithful rational representation has polynomial coordinates of uniformly bounded degree.

14.3 Finite differences

For $h \in \mathbb{R}^k$, define

$$\Delta_h \Theta(t) = \Theta(t)^{-1} \Theta(t + h).$$

This need not be polynomial in ordinary matrix coordinates if Θ takes values in a noncommutative group, but after applying a faithful representation and restricting to a regular algebraic chart it is a rational map of controlled degree. Iterated finite differences lower the polynomial degree in nilpotent coordinates, exactly as ordinary differences do for scalar polynomials.

Worked Example 14.1 (Heisenberg polynomial). Write the Heisenberg group as $(x, y, z)(x', y', z') = (x + x', y + y', z + z' + xy')$. Let $\Theta(s, t) = (s, t, st/2)$. Then

$$\Theta(s, t)^{-1} \Theta(s + a, t + b) = (a, b, \tfrac{1}{2}(sb - at + ab)).$$

The central coordinate still depends on the base point; a second difference in an independent parameter removes that dependence. Thus the family of asymptotic finite differences sees the central commutator direction even though the first derivative at one point may not.

14.4 The asymptotic tangent group

Let $\mathcal{U}(\Theta)$ be the connected subgroup generated by all one-parameter unipotent groups that arise as local uniform limits of normalized finite differences

$$\Theta(t_i)^{-1} \Theta(t_i + h_i(s)) \longrightarrow u(s),$$

where $\|t_i\| \rightarrow \infty$ inside the averaging boxes and $h_i(s)$ is small relative to t_i but chosen so the difference is order one.

Lemma 14.2 (Finite-dimensional termination of tangent extraction). *Starting from any nonconstant polynomial trajectory of bounded degree, one can extract only finitely many genuinely new tangent Lie directions before the connected group they generate stabilizes.*

Proof. Each extracted one-parameter group contributes a nonzero vector to the Lie algebra modulo the Lie algebra already generated. If it contributes no new class, it does not enlarge the tangent group. A strictly enlarging step increases Lie algebra dimension by at least one. Since $\dim G$ is finite, there are at most $\dim G$ strict enlargements. The bounded-degree finite-difference construction ensures every step remains inside a compact family of polynomial/rational maps after normalization, so subsequences exist. $\square$

14.5 Nonescape for boxes

Let

$$\mu_T = \frac{1}{|B_T|} \int_{B_T} \delta_{x\Theta(t)} dt.$$

The multiparameter nondivergence theorem from the previous student laboratory gives tightness unless a primitive rational wedge remains uniformly small on a positive fraction of the box. The polynomial goodness dichotomy upgrades that repeated smallness to algebraic trapping in a rational parabolic. In the theorem below we exclude this by assuming noncontraction in all rational representations relevant to Mahler's criterion.

14.6 Invariance under the tangent group

Fix one asymptotic finite-difference limit $u(s)$. For a bounded compactly supported test function, compare $\mu_T(f)$ with $\mu_T(f \circ R_{u(s)})$ by the parameter change $t \mapsto t + h_T(s)$. The Følner boundary estimate controls the part of the box lost at the boundary. On the interior, $\Theta(t)^{-1}\Theta(t + h_T(s))$ is uniformly close to $u(s)$ on a large subbox after the normalization choice. Tightness localizes the comparison to a compact subset of the quotient. Hence every weak limit is invariant under $u(s)$. Repeating for the finite list of tangent groups makes it invariant under $\mathcal{U}(\Theta)$.

14.7 The multivariable homogeneous limit theorem

Theorem 14.3 (Polynomial boxes under tangent generation). *Let G be connected, $\Gamma < G$ a lattice, and $\Theta : \mathbb{R}^k \rightarrow G$ a regular polynomial map. Let B_T be expanding Følner boxes. Assume:*

1. *the averages μ_T do not escape mass;*
2. *the connected asymptotic tangent group $U_\Theta = \mathcal{U}(\Theta)$ is generated by Ad-unipotent one-parameter groups;*
3. *there is a minimal connected closed subgroup H containing U_Θ such that xH is closed of finite volume and the trajectory is not focused, in any Chevalley representation, on a proper homogeneous subgroup of H .*

Then

$$\mu_T \Longrightarrow m_{xH}.$$

Proof. Tightness gives subsequential probability limits. The finite-difference argument makes every such limit U_Θ -invariant. Decompose a limit into U_Θ -ergodic components. The generated-unipotent measure theorem from Volume III makes every component homogeneous. Suppose a component is Haar on a proper orbit $yL \subsetneq xH$. Choose a Chevalley vector p_L . A positive component weight means the original averages spend positive proportion in arbitrarily thin tubes around the union of the corresponding Γ -translates. Multiparameter linearization turns this into smallness of finitely many bounded-degree coordinate functions on a positive proportion of parameter boxes. The multivariable Remez estimate forces the transverse coordinate to vanish identically on the relevant asymptotic polynomial limit, which is exactly the forbidden focusing condition. Therefore no proper component occurs. Every subsequential limit is m_{xH} , and uniqueness gives convergence of the full family. $\square$

Remark 14.4. The theorem does not say that every arbitrary polynomial map automatically satisfies the three hypotheses. Determining the minimal algebraic hull and verifying nonfocusing are genuine parts of an application. The point is that no additional measure-classification theorem is needed once those algebraic conditions are checked.

14.8 A product-type theorem as a corollary

If

$$\Theta(t_1, \dots, t_k) = \theta_1(t_1) \cdots \theta_k(t_k)$$

with commuting or successively normalizing polynomial one-parameter trajectories, then the tangent groups can be extracted one coordinate at a time. The general theorem recovers the product-box theorem in the main text. The advantage of the general formulation is that cross terms, such as the Heisenberg st coordinate, are allowed as long as their finite differences generate the correct tangent group.

Chapter 15

Polynomial Nondivergence as a Worked Induction

15.1 Rank-one marks

A lattice $h(t)\mathbb{Z}^n$ is bad at scale ϵ if it has a nonzero vector v with $\|h(t)v\| < \epsilon$. For a fixed primitive v , polynomial goodness controls the sublevel set. The difficulty is that the responsible vector may change with t .

15.2 Higher-rank marks

When two independent vectors are both short, treating them separately leads to uncontrolled overlap. Instead mark their primitive span by the wedge $v_1 \wedge v_2$. Continue to higher ranks. The partial order is inclusion of primitive subgroups, and every chain has length at most n . This finite rank is the reason the covering recursion terminates.

15.3 The marking construction

For each bad point choose a primitive subgroup Δ of maximal rank among those whose covolume is below its threshold. Enlarge around the point to a maximal parameter box on which the covolume of Δ stays below a larger threshold. Goodness says the genuinely small part occupies only a fixed power fraction of this box. Besicovitch covering chooses boundedly many disjoint subfamilies whose enlarged boxes cover the bad set. On overlaps,

maximal-rank marking forces subgroup comparability, so one moves up a flag rather than creating arbitrary combinatorics. Summing down the finite flag length gives the global power bound.

15.4 Why exterior powers are unavoidable

A rank-two subgroup can have moderate individual basis-vector lengths but very small area because the vectors are nearly parallel. The wedge norm detects this degeneration invariantly. Mahler compactness in higher rank is therefore naturally a statement about all primitive exterior covolumes, not only shortest vectors.

Chapter 16

The One-Parameter Shah Theorem with Every Limit Identified

16.1 Choose the first asymptotic scale

Let $\Theta(t)$ be regular algebraic. In a faithful representation, order all nonzero coordinate germs by growth. Take the first scale $t \mapsto t + st^{-q}$ on which the relative difference has a nonconstant limit. The logarithmic-derivative lemma makes that limit a unipotent one-parameter group U_1 .

16.2 Twist the limit measure

Tightness gives $\mu_{T_i} \Rightarrow \mu$. The small time change proves U_1 -invariance. If U_1 already generates the algebraic hull relevant to the trajectory, proceed to classification. Otherwise quotient by the invariance already obtained and repeat the asymptotic extraction. Finite Lie-algebra dimension yields a group U_Θ after finitely many steps.

16.3 Apply Ratner only now

Ratner is not used to prove tightness or to construct U_Θ . It is used only after a probability limit and its unipotent invariance have been established.

It tells us that the ergodic components are homogeneous.

16.4 Exclude the wrong homogeneous components

Suppose a proper L occurs. A Chevalley representation converts residence near xL into smallness of a regular-algebraic coordinate. The one-dimensional Remez estimate says positive residence at arbitrarily small tube radii forces that coordinate to vanish identically. Hence the entire trajectory lies in the corresponding algebraic incidence variety. Minimality of the algebraic hull rules this out. The only remaining component is Haar on the expected orbit.

Chapter 17

A Student Introduction to the o-Minimal Vocabulary Used Here

17.1 Definable subsets of the real line

An o-minimal structure is designed so that every definable subset of $\mathbb{R}$ is a finite union of points and intervals. This single axiom already rules out oscillations such as the zero set of $\sin t$ on an unbounded interval. Uniform definability says a one-parameter family cannot develop arbitrarily many alternating components without violating the same finiteness in a higher-dimensional definable set.

17.2 Monotonicity

Partition a definable function's domain according to where its graph has vertical tangent, horizontal tangent, or regular nonzero slope. Cell decomposition gives finitely many intervals on which the function is continuous and strictly monotone or constant. On a tail there is therefore a single monotonicity behavior. This is the elementary source of every “eventually” statement in the tame-curve proof.

17.3 Polynomial boundedness

The structure is polynomially bounded if every definable $f : (a, \infty) \rightarrow \mathbb{R}$ satisfies $|f(t)| \leq t^N$ eventually for some N . Together with definable monotonicity, this prevents exponential-scale growth. The asymptotic-monomial theorem proved in the foundations chapter sharpens this to an exact leading power ct^r for every nonzero definable germ.

17.4 Why exact powers replace polynomial degree

For a polynomial, degree orders monomials. For a tame definable germ, the real exponent r in $f(t) \sim ct^r$ plays the same role. Subtracting the leading term lowers r , differentiation lowers it by one when the leading coefficient is nonzero, and finite collections can be ordered by their exponents. This is enough to reproduce Remez-type compactness and finite-difference extraction without a literal polynomial identity.

17.5 Uniform goodness in families

Suppose no uniform Remez constant existed for a definable family. A bad sequence of parameters and intervals can be chosen definably after passing to a cell. Rescale each interval to $[0, 1]$. Depending on the ratio of its left endpoint to its length, the normalized functions converge either to a bounded-interval definable limit, a finite collection of power germs, or a polynomial/power limit at infinity. In every case a nonzero limit cannot be tiny on most of the interval and order one somewhere else. This contradiction gives uniform (C, α) -goodness.

17.6 The dynamical theorem has the same architecture

Once uniform goodness replaces polynomial Remez, every later step is already familiar: nonescape through exterior powers, asymptotic finite differences, unipotent invariance of weak limits, Ratner classification, and Chevalley linearization to exclude proper components. The o-minimal extension is deep because proving uniform goodness and power preparation is deep, not because a new measure-rigidity theorem is required.

Chapter 18

Worked Polynomial and Tame Examples

18.1 A polynomial curve in SL_2

Take $\Theta(t) = \begin{pmatrix} 1 & t^2+t \\ 0 & 1 \end{pmatrix}$. With $h = s/(2t)$,

$$\Theta(t)^{-1}\Theta(t+h) = u_{2th+h^2+h}ou_s.$$

Thus the asymptotic tangent group is the ordinary upper horocycle group even though the parameter grows quadratically.

18.2 A mixed semisimple-unipotent curve

Let

$$\Theta(t) = \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} \begin{pmatrix} 1 & t^\alpha \\ 0 & 1 \end{pmatrix}, \quad t > 1.$$

This is regular/rational rather than polynomial in t , but it is definable in a polynomially bounded structure when α belongs to the exponent field. Conjugation by the diagonal part changes the effective unipotent power. The noncontracting hypothesis must be checked in representations: the diagonal factor can drive some vectors to zero even though matrix entries of the whole curve grow.

18.3 A curve trapped in a proper subgroup

If $\Theta(t)$ lies in an embedded $\mathrm{SL}_2 < G$ and $x\mathrm{SL}_2$ is closed of finite volume, no theorem should predict ambient Haar measure on xG . The minimal algebraic hull is the embedded SL_2 , and the correct conclusion is Haar on $x\mathrm{SL}_2$ after the internal nonfocusing condition is verified. This simple example is the reason every theorem states the minimal homogeneous envelope.

Chapter 19

Guided Proof Architecture for Polynomial and Tame Trajectories

A one-parameter unipotent orbit has the simple form xu_t . A polynomial trajectory may turn through several algebraic directions, and a definable tame trajectory need not be polynomial at all. Nevertheless the proof strategy remains recognizable once four questions are separated:

1. why does mass not escape?
2. what unipotent symmetry does a rescaled limit acquire?
3. how are proper homogeneous limits excluded?
4. what changes when powers replace ordinary monomials?

19.1 Polynomial maps in coordinates

For a linear group $G < GL_N(\mathbb{R})$, call $\Theta : \mathbb{R}^k \rightarrow G$ polynomial if each matrix coefficient is polynomial and the determinant constraints defining G hold identically. More invariantly, it is enough that every regular function on the Zariski closure of the image pull back to a polynomial.

Worked Example 19.1 (A Heisenberg polynomial). In the Heisenberg group write

$$\Theta(s, t) = \begin{pmatrix} 1 & s & st + t^2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}.$$

The central coordinate contains the product st , so this is not a homomorphism from $\mathbb{R}^2$. It is nevertheless polynomial, and every exterior- power covolume of $\Theta(s, t)\Lambda$ is a norm of a vector-valued polynomial. Quantitative nondivergence therefore remains available.

19.2 Scalar goodness in several variables

Let P be a real polynomial of total degree at most d on a ball $B \subset \mathbb{R}^k$. We want an estimate

$$|\{x \in B : |P(x)| < \epsilon \sup_B |P|\}| \leq C\epsilon^\alpha |B|.$$

A convenient proof is by induction on k . For $k = 1$ use the Remez inequality proved earlier. For $k > 1$, choose a coordinate direction in which some coefficient polynomial has maximal size. Apply the induction hypothesis to that coefficient on the $(k - 1)$ -dimensional base, removing a small exceptional set. On each remaining fiber the one-dimensional polynomial has a coefficient bounded below, so one-dimensional Remez applies uniformly. Fubini adds the two exceptional measures and gives a power exponent depending only on k, d .

For a vector-valued polynomial $P = (P_1, \dots, P_m)$, choose at each point of a finite coefficient compactum a linear functional ℓ for which $\ell \circ P$ captures a fixed fraction of the norm. Compactness gives a finite family of such functionals. The scalar estimate applied to that family gives goodness of $\|P\|$.

19.3 Why exterior powers encode all cusp directions

Let $\Lambda = g\mathbb{Z}^n$. A primitive rank- r subgroup of Λ has a primitive wedge vector

$$w = v_1 \wedge \cdots \wedge v_r \in \bigwedge^r \mathbb{R}^n,$$

and its covolume is $\|w\|$. Along a polynomial trajectory,

$$w(s) = \bigwedge^r \Theta(s)w$$

is vector-valued polynomial. Thus every possible rank- r cusp degeneration is monitored by one good function.

The marking argument does not union-bound over all primitive subgroups; that would be infinite and useless. At a point where several subgroups are simultaneously short, replace them by a maximal nested flag. There are at most n ranks in a flag. Besicovitch covering then chooses maximal marked balls with bounded overlap. Goodness bounds the time spent below a smaller threshold inside each marked ball. Induction on flag length closes the estimate.

Guiding Idea 19.2. The higher-dimensional nondivergence theorem is not a different miracle from the one-parameter theorem. The new ingredients are multivariable goodness and a bounded-overlap covering theorem; the exterior-power and flag bookkeeping are unchanged.

19.4 Microscopic differences create unipotent symmetry

Let $\Theta(t)$ be a regular algebraic one-variable trajectory. Consider

$$\Theta(t)^{-1}\Theta(t+h).$$

If h is fixed and $t \rightarrow \infty$, the difference may diverge. Choose instead $h = st^{-q}$, where q is selected so that the first nonzero term in the logarithmic derivative has order one.

Write, on a tail where all inverses exist regularly,

$$A(t) = \Theta(t)^{-1}\Theta'(t).$$

Every entry is a rational function. Expand at infinity:

$$A(t) = t^q Y + O(t^{q-\eta})$$

for the largest exponent q with nonzero coefficient Y . Taylor expansion of the ordered product gives

$$\Theta(t)^{-1}\Theta(t+st^{-q}) \longrightarrow \exp(sY).$$

Because the original trajectory has polynomial growth in every rational representation, a nonzero semisimple part of Y would create exponential growth in s after iteration, contradicting the polynomial scaling from which the limit was extracted. Hence Y is nilpotent in the relevant algebraic representation and $\exp(sY)$ is unipotent.

19.5 Passing invariance to a weak limit

Let

$$\nu_T = \frac{1}{T} \int_0^T \delta_{x\Theta(t)} dt.$$

Suppose nondivergence gives a probability limit ν . Fix s and compare t with $t + st^{-q}$. The change of variables has derivative $1 + O(t^{-q-1})$, so it changes the average by $o(1)$ after discarding a bounded initial interval. The microscopic-difference limit gives

$$x\Theta(t + st^{-q}) = x\Theta(t) \exp(sY) + o(1)$$

on compact sets. Uniform continuity of test functions and tightness then show

$$\int f(z \exp(sY)) d\nu(z) = \int f(z) d\nu(z).$$

Thus every weak limit is invariant under a nontrivial unipotent group.

19.6 Classification is not yet equidistribution

Ratner classification now says that every ergodic component of ν is a homogeneous measure. Several proper homogeneous components could still be possible. To prove equidistribution one must show that the original trajectory does not spend positive density near any proper singular set.

Choose a Chevalley vector p_L for a proper candidate group L . The linearized function

$$t \mapsto \rho(\Theta(t))p_L$$

is polynomial (or rational with a controlled denominator after clearing one fixed factor). If the trajectory lies near a translate of L for many parameters, this vector lies near a discrete lattice orbit for many parameters. Good-function estimates imply a dichotomy: either the visits occupy only a power-small part of every large interval, or the polynomial is trapped in the algebraic stabilizer relation identically. The second branch means that the whole algebraic hull of the trajectory lies in the proper normalizer, contradicting the regularity/nonfocusing hypothesis of the Shah theorem.

Therefore proper components receive zero mass, and the weak limit is the homogeneous measure on the minimal algebraic envelope. Since every subsequence has the same limit, the full averages converge.

19.7 Several parameters and tangent generation

For a polynomial map $\Theta : \mathbb{R}^k \rightarrow G$, microscopic differences in a single coordinate may produce only one unipotent direction. Repeating in several coordinate directions and at several asymptotic scales produces a family of nilpotent tangent generators $Y_1, \dots, Y_r$. Let W be the connected group generated by their exponentials.

The multivariable equidistribution theorem in this volume assumes that these tangent directions generate the connected algebraic envelope required by the trajectory, modulo the explicitly listed focusing groups. This hypothesis is not decorative: without it, a polynomial box can average over a region whose microscopic tangent dynamics sees only a proper subgroup, and the limiting measure need not be ambient Haar.

Worked Example 19.3 (Two commuting shears). In SL_3 , take

$$\Theta(s, t) = (I + sE_{12})(I + tE_{13}).$$

The two tangent generators commute and span a two-dimensional unipotent group. Averaging over a large rectangle therefore naturally creates invariance under that two-dimensional group, not under all of SL_3 . Additional algebraic information is needed before one can conclude a larger homogeneous limit.

19.8 From polynomial monomials to definable power asymptotics

A polynomially bounded o-minimal function need not be a polynomial, but on a sufficiently far tail it has a finite asymptotic hierarchy. The foundational appendix proves that a nonzero definable germ has the form

$$f(t) = ct^\lambda(1 + o(1))$$

for a real exponent λ drawn from the field of definable powers, after lower-order terms are removed. Differentiation lowers the leading exponent by one, and Wronskian arguments give a finite basis of distinct asymptotic degrees.

This plays exactly the role of polynomial degree. It supplies:

1. a finite list of possible first visible scales;
2. a Remez-type good-function estimate for definable coordinate families;
3. a microscopic rescaling that produces a nilpotent tangent flow;

4. a finite termination argument when successive leading terms cancel.

The o-minimal theorem is therefore not obtained by writing “the polynomial proof is the same.” The appendix rebuilds the two polynomial inputs that are genuinely lost: degree and uniform sublevel estimates.

19.9 A worked mixed-power curve

Consider in a unipotent matrix group

$$\Theta(t) = \exp(t^{\sqrt{2}}X) \exp(t^{3/2}Y),$$

where X, Y are nilpotent and the expression is definable in a polynomially bounded structure containing the relevant power functions. The larger exponent $3/2$ controls the first microscopic scale unless the corresponding Y -direction is absorbed by the current stabilizer. Taking $h = st^{-1/2}$ changes $t^{3/2}$ by

$$(t + h)^{3/2} - t^{3/2} = \frac{3}{2}s + o(1),$$

while the $t^{\sqrt{2}}$ term changes by $O(t^{\sqrt{2}-3/2}) = o(1)$. The first tangent symmetry is therefore $\exp(\frac{3}{2}sY)$. If this lies in the known stabilizer, one removes it and examines the next asymptotic exponent. This concrete calculation is the model for the general definable tangent-extraction algorithm.

19.10 Proof ledger for a new trajectory

Given a new polynomial or tame trajectory, the reader should answer the following questions in order:

1. Which exterior-power functions control cusp escape, and why are they good uniformly in the parameters?
2. What is the algebraic hull of the trajectory?
3. Which microscopic rescaling produces the first nontrivial nilpotent tangent?
4. Which group is generated by all tangent flows?
5. Why are weak limits invariant under that group?
6. Which proper homogeneous supports are possible by Ratner?
7. What linearization vector detects each proper support?
8. Why does nonfocusing exclude positive-density visits to those supports?
9. Why is the surviving homogeneous measure unique?

If the first eight questions are answered, the ninth is usually a compactness argument. Skipping one of the earlier questions is the usual source of a false “polynomial equidistribution” statement.

Proof Guide and Dependency Summary

This chapter is a finalization layer. It does not introduce a new theorem scope. Its purpose is to expose the proof architecture, make the hypotheses that are easiest to misread impossible to miss, and give a student a concrete model calculation before the abstract machinery is reused later.

1. Polynomial trajectories: the finite-difference principle

For a polynomial curve $g(t)$ in a linear representation, a well-chosen increment $h(t) = o(t)$ makes

$$g(t + h(t))g(t)^{-1}$$

converge, after normalization, to a nontrivial unipotent one-parameter motion. The resulting invariance of a weak limit is the polynomial analogue of the shearing mechanism in Ratner theory.

The proof must contain both:

nonescape and asymptotic unipotent extraction.

The first ensures that a probability limit exists; the second supplies the symmetry needed to classify it.

2. Why the o-minimal extension needs polynomial boundedness

For a one-variable definable function in a polynomially bounded o-minimal structure, eventual behavior is tame enough to admit a power asymptotic

$$f(t) = ct^r(1 + o(1)).$$

From this one obtains derivative degree drop, cancellation of equal leading degrees, bases of distinct degrees, Wronskian nondegeneracy, and ultimately uniform Remez/ (C, α) -good estimates for finite-dimensional definable families. Those estimates replace the literal polynomial Remez inequality used earlier.

The scope lock is essential: “definable” by itself is not the hypothesis. The structure is polynomially bounded and the curve is noncontracting in the representations used for cusp detection.

3. The definable proof chain

The final theorem is to be read as

asymptotic monomial theorem
 $\Rightarrow$ uniform definable Remez
 $\Rightarrow$ representation-coordinate goodness
 $\Rightarrow$ uniform nonescape
 $\Rightarrow$ first nontrivial definable difference
 $\Rightarrow$ unipotent invariance of limits
 $\Rightarrow$ definable linearization dichotomy
 $\Rightarrow$ homogeneous equidistribution.

Every arrow is a proof obligation; none is replaced by the phrase “o-minimality gives tameness.”

4. Student example

Compare $t \mapsto t^{3/2}$ with $t \mapsto e^t$. The first is compatible with a polynomially bounded power scale; the second displays exactly why one cannot silently enlarge the theorem to arbitrary o-minimal structures.

5. Dependency status

The local o-minimal proof chain in this volume has been checked and has no remaining internal proof gap. Its only inherited structural input is the proper/discrete fixed-support Chevalley lift from Volumes III and VII. That lift is derived in Volume III from the external Dani–Margulis structural interface. Thus the structural dependency used by this volume is explicit and no additional research-level external input is introduced here.

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Expanding Translates, Horospheres, Curves, and Submanifolds

Volume IX of a Detailed Notebook on Ratner's Theorems

Detailed Notebook on Ratner's Theorems

August 2026

Preface and proof policy

Arc I culminates by letting a semisimple element expand a lower-dimensional measure. The basic phenomenon is different from ordinary mixing: the starting measure is singular with respect to ambient Haar measure, so one first has to create unipotent invariance in subsequential limits and only then use Ratner/Mozes–Shah rigidity. This volume proves that mechanism, then treats analytic curves and shrinking or nonuniformly stretched pieces through their tangent polynomial models. No quantitative rate is claimed; those belong to Volume XII.

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Dependency and proof-ownership contract

Purpose. Expanding translates.

Prerequisites. Volumes III, VI–VIII.

New objects. expanding horospheres, curve jets. Every project-specific technical term is defined before its first load-bearing use or recalled by an exact backward reference.

Main outputs. qualitative expanding-translate equidistribution.

External-input rule. Every nontrivial imported result is either part of the declared graduate-level prerequisites, an explicitly named external research input in the prerequisite map, or is proved in this series before use (possibly under an explicit deferred-proof contract). A citation by itself never constitutes a proof.

Reader map and proof route

Question of the volume. How can a semisimple expansion turn a singular measure on a curve or horosphere into ambient homogeneous measure?

Prerequisite checkpoint. Qualitative mixing is supplied by Volume VI; homogeneous-measure limits by Volume VII; polynomial/tame tangent models by Volume VIII. No effective rate is assumed in this volume.

Load-bearing route. First prove non-escape for the expanded measures. Then extract a weak limit, create unipotent invariance from the expanding geometry, classify the limit, and use nondegeneracy to rule out proper homogeneous supports. Curves are handled through their jets only after this mechanism is transparent for a full horospherical piece.

Sanity check. For the modular surface, expanding a horocycle by the diagonal flow stretches its parameter interval while contracting the transverse direction. Writing this conjugation explicitly explains why ordinary mixing of two L^2 functions is not by itself a proof for a measure initially supported on a curve.

Chapter 1

Expanding Horospherical Geometry

1.1 Semisimple expansion

Let G be connected semisimple and let $a_t = \exp(tH)$ with H in a split Cartan. Because all actions in this notebook are on the right, we call

$$U^+ = \{u \in G : a_t u a_{-t} \rightarrow e \text{ as } t \rightarrow +\infty\}$$

the *right-expanding* horospherical subgroup for the translates $xu a_t$; its opposite is $U^- = \{u : a_{-t} u a_t \rightarrow e\}$. Thus u^+ is the sum of the negative H -root spaces. The central/neutral group is $Z = Z_G(a_t)$.

Choose exponential coordinates on U^+ . If $Y = \sum_{\alpha} Y_{-\alpha}$ with $\alpha(H) > 0$, then

$$a_t \exp(Y) a_{-t} = \exp \left(\sum_{\alpha} e^{-\alpha(H)t} Y_{-\alpha} \right).$$

Consequently a parameter perturbation of size $e^{-\alpha(H)t}$ before the right translate becomes an order-one right displacement after the translate:

$$u \exp(e^{-\alpha(H)t} Y_{-\alpha}) a_t = u a_t \exp(Y_{-\alpha}).$$

1.2 Measures on horospherical pieces

Let $\Omega \subset U^+$ be a bounded open set with boundary of Haar measure zero and let $x \in X = \Gamma \backslash L$, where $G < L$ is a closed subgroup. Define

$$\mu_t(f) = \frac{1}{m_U(\Omega)} \int_{\Omega} f(xua_t) dm_U(u).$$

The problem is to identify μ_t as $t \rightarrow \infty$.

Warning 1.1 (Why ambient mixing is not enough). The measure on $x\Omega$ is supported on a lower-dimensional set. It has no L^2 density with respect to Haar measure on X . Howe–Moore therefore does not directly apply. The qualitative proof first identifies invariance of subsequential limits under unipotent groups and invokes Ratner rigidity. Only after thickening does ordinary mixing become an alternative proof in special symmetric situations.

Chapter 2

Nonescape for Expanding Pieces

The nonescape argument is slightly subtler than simply applying polynomial nondivergence: the translates $x\Omega a_t$ need not meet a fixed compact set a priori. The missing input is a representation-theoretic fact saying that the expanding semisimple element cannot keep an open unipotent patch bounded on a discrete family of nontrivial vectors when the unipotent patch normally generates the acting semisimple group.

2.1 A linear obstruction supplied by polynomial nondivergence

For the right quotient $X = \Gamma \backslash L$ we linearize a subgroup datum with the inverse Chevalley map

$$\eta_H(g) = \rho(g^{-1})p_H.$$

Thus a right translate xua_t produces vectors of the form $\rho(a_t^{-1}u^{-1})v$, exactly the order in which the expanding-weight argument is clean.

Proposition 2.1 (Linear obstruction to nonescape). *Let Ω be a relatively compact open subset of a connected unipotent subgroup $U < L$, and let $\Phi_i(u) = ua_i$. If the probability measures obtained from normalized Haar measure on $x\Omega a_i$ are not tight, then, after passing to a subsequence, there exist a finite-dimensional continuous representation $\rho : L \rightarrow \mathrm{GL}(V)$, a nonzero vector $p \in V$ whose $\rho(\Gamma)$ -orbit is discrete, vectors $v_i \in \rho(\Gamma)p$, and a*

constant C such that

$$\sup_{u \in \Omega} \|\rho(a_i^{-1}u^{-1})v_i\| \leq C. \quad (9.1)$$

The representation and the discrete orbit are independent of i .

Proof. Apply the proof of uniform polynomial nondivergence from Volume VIII to the maps $u \mapsto ua_i$, using exponential coordinates on the nilpotent group U . If tightness fails, the “mostly in a thick part” alternative fails for arbitrarily deep thick parts. In the Zassenhaus–marking proof this means that, on a positive-proportion parameter box, one marked small subgroup persists to the maximal scale. Goodness of its wedge coordinates upgrades persistence on positive proportion to a uniform bound on a slightly smaller open box: if one transverse coordinate became arbitrarily large somewhere, the sublevel inequality would force the deeply small set to have vanishing relative measure. There are only finitely many subgroup types on the fixed compact parameter window, so pass to a subsequence with one exterior-power representation. Lattice translates of its primitive Chevalley vector form a discrete set. In the right-quotient convention the vector is acted on by $(ua_i)^{-1} = a_i^{-1}u^{-1}$, yielding (9.1). Enlarge the smaller box back to the originally fixed Ω by finitely many translates; this only changes C . $\square$

2.2 The expanding representation lemma

Put $b_i = a_i^{-1}$ and

$$U^+ = \{u \in G : b_i^{-1}ub_i \rightarrow e\}.$$

For a one-parameter a_t this agrees with the convention of Chapter 1.

Lemma 2.2 (Projection from an open unipotent patch). *Let U be connected unipotent, $W = V^U$, and $\Omega \subset U$ nonempty open and relatively compact. There is $c_\Omega > 0$ such that*

$$\|v\| \leq c_\Omega \sup_{u \in \Omega} \|\mathrm{pr}_W(\rho(u)v)\| \quad (v \in V). \quad (9.2)$$

Proof. Choose a basis $N_1, \dots, N_m$ of the nilpotent Lie algebra and a small box of exponential coordinates contained in Ω after translating Ω . Since every $d\rho(N_j)$ is nilpotent, the map

$$(t_1, \dots, t_m) \longmapsto \mathrm{pr}_W \rho(e^{t_m N_m} \dots e^{t_1 N_1})v$$

is a vector-valued polynomial of uniformly bounded multidegree. If all its coefficients vanished for a nonzero v , the U -submodule generated by v would lie in $W^\perp$. Lie–Kolchin applied to that nonzero submodule would produce a nonzero U -fixed vector in $W^\perp$, impossible. Hence the linear map from v to the finite coefficient tuple is injective. In finite dimension its inverse is bounded. Finally recover every coefficient from values at a fixed tensor-product grid inside the coordinate box by the Vandermonde interpolation formula. This gives (9.2). $\square$

Lemma 2.3 (Active highest weights expand). *Let G be connected semisimple with finite center and let b_i lie in one closed split Weyl chamber after passing to a subsequence. Define*

$$U^+ = \{u \in G : b_i^{-1}ub_i \rightarrow e \text{ as } i \rightarrow \infty\}.$$

Assume that the connected normal subgroup generated by U^+ is all of G . If V is a nontrivial irreducible finite-dimensional G -module and $W = V^{U^+}$, then for every $c > 0$,

$$\inf_{w \in W, \|w\| \geq c} \|\rho(b_i)w\| \rightarrow \infty. \quad (9.3)$$

Proof. Decompose V into restricted A -weight spaces. Let Δ_+ be the simple roots α for which $\alpha(\log b_i) \rightarrow +\infty$; these are exactly the active roots generating U^+ . A vector fixed by the positive root groups for Δ_+ is a highest-weight vector in each active root string. If an irreducible G -module has highest weight zero on every root in Δ_+ , then every active root subgroup and its opposite acts trivially; hence the normal subgroup generated by U^+ acts trivially. By hypothesis that subgroup is G , contradicting nontriviality of the module. Therefore every weight occurring in W has strictly positive value on at least one active simple coroot, and its value on $\log b_i$ tends to $+\infty$. There are only finitely many such weights, so the smallest expansion factor on W tends to infinity, proving (9.3). $\square$

Corollary 2.4 (Discrete-vector escape). *Let $D \subset V$ be discrete and G -invariant under the lattice action, and assume D is not eventually contained in V^G . Under the hypotheses of Lemma 2.3, for any nonempty open $\Omega \subset U^+$ and any $v_i \in D$,*

$$\sup_{u \in \Omega} \|\rho(b_i u)v_i\| \rightarrow \infty \quad (9.4)$$

unless, after a subsequence, $v_i \in V^G$.

Proof. Decompose $V = V^G \oplus V'$ into the fixed part and nontrivial irreducibles. If the fixed projections of v_i are unbounded, then their norms already force the left-hand side of (9.4) to diverge, so (9.4) follows without using the nontrivial summands. If they are bounded and the V' -projections could tend to zero, discreteness would force a subsequence of v_i to be constant in a compact set and hence fixed, contrary to the stated alternative. Thus some nontrivial irreducible projection has norm bounded below. Apply Lemma 2.2 to $\rho(u)v_i$ and then Lemma 2.3 to its U^+ -fixed projection. Since b_i preserves the A -weight decomposition, the resulting expanding component gives (9.4). $\square$

Theorem 2.5 (Nonescape of expanding horospherical pieces). *Let $G < L$ be connected semisimple with finite center, let $x \in \Gamma \backslash L$, and let $a_i \in G$ be a split sequence for which $U^+ = \{u : a_i u a_i^{-1} \rightarrow e\}$ normally generates G . If $\overline{xG} = \Gamma \backslash L$, then for every probability $\lambda \ll m_{U^+}$ the measures*

$$\nu_i(f) = \int_{U^+} f(xua_i) d\lambda(u)$$

are tight.

Proof. First take λ with compact support in an open set Ω and with bounded density; general L^1 densities follow by approximation. If the measures were not tight, Proposition 2.1 would give a representation, a discrete orbit D , and vectors v_i satisfying the boundedness (9.1). With $b_i = a_i^{-1}$ this is precisely a bound for $\rho(b_i u^{-1})v_i$ on an open subset of U^+ . Since inversion preserves openness and Haar-null sets, Corollary 2.4 forces, after a subsequence, $v_i \in V^G$.

A G -fixed Chevalley vector defines a proper G -invariant homogeneous cusp datum. The parameter points producing v_i lie in that datum, and closedness plus the linearization lifting argument imply that the orbit closure $\overline{xG}$ is contained in the corresponding proper homogeneous subset of $\Gamma \backslash L$. This contradicts density. Hence the compact-support measures are tight. Approximate an arbitrary $L^1(U^+)$ density in L^1 by compactly supported bounded densities; the total-variation error is unchanged by translation, proving tightness in general. $\square$

Chapter 3

Unipotent Invariance of Limit Measures

3.1 The parameter-shift argument

Fix $Y \in \mathfrak{u}^+$ with $[H, Y] = -\alpha(H)Y$, $\alpha(H) > 0$, and set

$$\delta_t = e^{-\alpha(H)t}.$$

For u away from the boundary of Ω , $u \exp(\delta_t Y) \in \Omega$ for all large t , and the right-action convention gives the exact identity

$$u \exp(\delta_t Y) a_t = u a_t \exp(Y).$$

The change of variables $u \mapsto u \exp(\delta_t Y)$ preserves Haar measure on the nilpotent group up to equality because U^+ is unimodular.

Lemma 3.1 (Root-group invariance). *Every weak probability limit of μ_t is invariant under each root subgroup $U_\alpha \subset U^+$ whose parameter direction is represented in the interior tangent cone of Ω .*

Proof. Let $f \in C_c(X)$. Remove the $O(\delta_t)$ -thickened boundary of Ω ; its relative Haar measure tends to zero because the boundary has measure zero and translations are continuous in $L^1(U^+)$. On the remaining set, change variables by $\exp(\delta_t Y)$. The exact conjugation identity gives

$$\mu_t(f \circ R_{\exp Y}) - \mu_t(f) \rightarrow 0.$$

Pass to the weak limit. Replacing Y by sY gives invariance under the whole one-parameter root group. $\square$

If Ω has nonempty interior, its tangent cone spans all of $\mathfrak{u}^+$, so every limit is U^+ -invariant.

Chapter 4

Shah's Expanding-Translate Theorem

4.1 The normal-generation hypothesis

Let $G < L$ be connected semisimple with finite center and let a_i be a sequence in one split Cartan such that, for every simple root after passage to a subsequence, its value on a_i is either bounded or tends to infinity. Put

$$U^+ = \{u \in G : a_i u a_i^{-1} \rightarrow e\}.$$

The load-bearing hypothesis is not merely that U^+ is nontrivial: it must not be contained in any proper closed normal subgroup of G . For an almost-simple G this is automatic as soon as $U^+ \neq \{e\}$.

Theorem 4.1 (Shah: expanding horospherical measures). *Assume U^+ normally generates G and $\overline{xG} = X = \Gamma \backslash L$. For every probability measure λ on U^+ absolutely continuous with respect to Haar measure,*

$$\int_{U^+} f(xua_i) d\lambda(u) \longrightarrow \int_X f dm_X \quad (f \in C_c(X)). \quad (9.5)$$

In particular the statement holds for normalized Haar measure on every bounded open $\Omega \subset U^+$ with boundary of Haar measure zero.

Proof. Tightness. This is Theorem 2.5.

Unipotent invariance. Let ν be a subsequential probability limit. Fix $u_0 \in U^+$ and put $u_{0,i} = a_i u_0 a_i^{-1} \rightarrow e$. Right translation by u_0 after a_i is the

same as replacing the parameter u by $uu_{0,i}$ before a_i :

$$xua_iu_0 = xuu_{0,i}a_i.$$

If $d\lambda = h dm_U$, continuity of translations in $L^1(U^+)$ gives $\|h(\cdot u_{0,i}^{-1}) - h\|_1 \rightarrow 0$. Hence $R_{u_0*}\nu = \nu$. Thus ν is U^+ -invariant.

Ratner decomposition. Decompose ν into U^+ -ergodic components and apply the generated-unipotent measure classification from Volume III. If every component has support X , transitive uniqueness gives $\nu = m_X$. Suppose instead that positive mass lies on the regular part of a proper finite-volume homogeneous orbit yH with $U^+ \subset H$.

Focusing. Proposition B.1 below converts this positive regular mass into a bounded Chevalley-vector obstruction of the form

$$\sup_{u \in \Omega_0} \|\rho(a_i^{-1}u^{-1})v\| \leq C \quad (9.6)$$

for a fixed nonempty open $\Omega_0 \subset U^+$ and a discrete lattice orbit of vectors defining the proper subgroup datum. By Corollary 2.4, (9.6) forces the vector datum to be G -fixed. Therefore the corresponding proper homogeneous set is G -invariant. Since it contains limit points of $x\Omega_0a_i$, the lifting/focusing construction places a point of the dense orbit closure $\overline{xG}$ in that proper G -invariant set, hence the whole orbit closure lies there. This contradicts $\overline{xG} = X$.

Thus no proper component occurs and $\nu = m_X$. Every subsequence has the same probability limit, so (9.5) follows. $\square$

Corollary 4.2 (Almost-simple one-parameter form). *Let G be connected noncompact almost simple with finite center, a_t a noncompact split one-parameter subgroup, and U^+ its nontrivial contracting horospherical subgroup in the right-action convention. If $\overline{xG} = X$, then normalized Haar measure on every bounded open horospherical patch $x\Omega_{a_t}$ equidistributes to m_X as $t \rightarrow +\infty$.*

Proof. The connected normal closure of a nontrivial root group in an almost-simple group is all of G , so Theorem 4.1 applies. $\square$

4.2 Density as a corollary

If $O \subset X$ is nonempty open with positive Haar measure, the limit theorem gives $\mu_t(O) > 0$ for all large t after approximating 1_O from below by C_c

functions. Therefore $x\Omega a_t$ meets O . The union of expanding translates is dense.

Chapter 5

Translates of Closed Homogeneous Orbits

5.1 Varying homogeneous measures

Let $H < G$ and let xH carry finite invariant measure m_{xH} . Consider $\nu_t = (a_t)_* m_{xH}$. If some unipotent subgroup of $a_{-t}Ha_t$ converges to a fixed nontrivial group, every limit of ν_t inherits that invariance. Volume VII then supplies compactness and focusing.

Theorem 5.1 (Homogeneous translate alternative). *Assume $\{\nu_t\}$ is tight. Every subsequential limit is homogeneous. If no proper connected subgroup $M < G$ contains $a_{-t_i}Ha_{t_i}$ after arbitrarily small conjugacy along a subsequence, then $\nu_t \rightarrow m_X$.*

Proof. Each ν_t is itself a homogeneous probability on the translated orbit $xHa_t = xa_t(a_{-t}Ha_t)$. Apply the Mozes–Shah theorem of Volume VII to a convergent subsequence. Its focusing conclusion places the conjugated support groups inside the limiting support. The no-containment hypothesis forces that support to be G , hence the limit is ambient Haar. $\square$

This formulation is the cleanest bridge to Volume X: counting often becomes a problem about translates of a closed subgroup orbit, not an open horospherical piece.

Chapter 6

Analytic Curves in Expanding Horospheres

6.1 Why curvature creates enough unipotent directions

Throughout this chapter let $G < L$ be connected semisimple Lie groups with finite center, let $\Gamma < L$ be a lattice, and put $X = \Gamma \backslash L$. Let $a_t \subset G$ be a split one-parameter subgroup and define its contracting horospherical subgroup in the right-action convention by

$$U^+ = \{u \in G : a_t u a_t^{-1} \rightarrow e \text{ as } t \rightarrow +\infty\}.$$

Assume that the connected normal subgroup of G generated by U^+ is G . Let $I \subset \mathbb{R}$ be a compact interval of positive length and let $\phi : I \rightarrow U^+$ be analytic. Consider

$$\mu_t = \frac{1}{|I|} \int_I \delta_{x\phi(s)a_t} ds.$$

A shift $s \mapsto s + \delta_t r$ produces, after conjugation by a_t , a unipotent displacement governed by the first nonzero derivative of ϕ in the fastest expanding weight. If the derivative degenerates, higher jets enter. The polynomial tangent theorem of Volume VIII is exactly the required local model.

Definition 6.1 (Nondegeneracy in the boundary model). Project ϕ to the flag variety G/P^- determined by the contracting parabolic. The curve is nondegenerate if its image is not contained in a proper algebraic subvariety

arising as the boundary of a proper homogeneous subgroup that can support a Ratner measure.

Lemma 6.2 (Finite jet stratification on a compact analytic interval). *Fix a finite-dimensional representation in which the a_t -weights and the boundary map G/P^- are algebraic. For an analytic curve $\phi : I \rightarrow U^+$ there is a finite partition of I , up to finitely many points, into open intervals J such that on each J one fixed weight coordinate and one fixed derivative order $k(J)$ is the first nonzero expanding jet. After shrinking J slightly, the corresponding derivative is bounded above and away from zero.*

Proof. Choose finitely many analytic coordinate functions for ϕ in the unipotent chart and decompose them into the finitely many a_t -weight spaces. For each coordinate, either it is identically zero or every zero of each derivative is isolated. On the compact interval I , the zero sets needed to determine the first nonzero derivative order are therefore finite unless the relevant derivative vanishes identically. Remove these finitely many points and subdivide once more so that the signs of the finitely many nonzero first derivatives are fixed. Compactness of the closures of the resulting smaller intervals gives the required upper and lower bounds. $\square$

Lemma 6.3 (Jet rescaling gives unipotent invariance). *On an interval J from Lemma 6.2, let k be the first active jet and let $\lambda > 0$ be its expanding a_t -weight. Put $\delta_t = e^{-\lambda t/k}$. Then for every compact r -range and every Lebesgue point $s \in J$,*

$$a_t^{-1} \phi(s)^{-1} \phi(s + \delta_t r) a_t \longrightarrow \exp(r^k Y_s)$$

locally uniformly in r , with $Y_s \neq 0$ nilpotent. Consequently every weak limit of the restricted expanding-curve averages on J is invariant under the one-parameter unipotent subgroup generated by Y_s for almost every such s .

Proof. Taylor expand in exponential coordinates of the nilpotent group U^+ . Terms of derivative order $j < k$ vanish in the selected fastest active weight; terms in slower weights are killed after conjugation by a_t at the chosen scale; and the k th derivative contributes a nonzero limit. BCH only adds Lie monomials of strictly larger total Taylor order, hence they vanish at this scale. This proves the displayed convergence. The parameter change $s \mapsto s + \delta_t r$ alters the endpoints of J by $o(1)$ and has Jacobian one. The usual test-function argument, first on compact sets and then using nonescape, transfers the limiting displacement to invariance of any weak limit. $\square$

Proposition 6.4 (Analytic focusing into a proper support). *Let ν be a weak limit of the expanding-curve measures. If a positive part of ν is carried by a proper Ratner support type H , then the boundary projection of $\phi(I)$ is contained in the algebraic incidence variety attached to the vector normalizer $N^1(H)$.*

Proof. Choose a compact regular part of the H -stratum with positive ν -mass. The proper/discrete Chevalley lift and thin-neighborhood theorem from Volume VII provide finitely many admissible Chevalley representatives on that compact window. On each connected parameter interval on which the orbit enters a sufficiently thin window, uniqueness of the lift makes the representative constant. The expanding good-function estimate then has the usual dichotomy: either the relative time spent in the thinner window tends to zero, or the corresponding Chevalley coordinate is identically trapped. Positive limiting mass rules out the first alternative for one representative. Thus an algebraic coordinate defining the $N^1(H)$ incidence vanishes on a positive-measure subset of I . Since its composition with the analytic curve is analytic, it vanishes identically. Doing this for the finite defining family gives the claimed containment of the whole boundary curve. $\square$

Theorem 6.5 (Expanding analytic curves). *Let the standing data $G < L$, Γ , X , a_t , U^+ , I , and ϕ be as specified at the beginning of this chapter. Fix $x \in X$ and assume*

$$\overline{xG} = X.$$

If the boundary projection of ϕ is nondegenerate in the sense of Definition 6.1, then, for every $f \in C_c(X)$,

$$\frac{1}{|I|} \int_I f(x\phi(s)a_t) ds \longrightarrow \int_X f dm_X \quad (t \rightarrow +\infty).$$

If nondegeneracy fails, every subsequential weak limit of the probability measures $|I|^{-1} \int_I \delta_{x\phi(s)a_t} ds$ is a barycenter of homogeneous probability measures whose supports lie in homogeneous envelopes associated with algebraic boundary strata containing the boundary image of ϕ . If there is a unique minimal such stratum, every ergodic component of every subsequential limit is supported in its homogeneous envelope.

Proof. Step 1: tightness. Apply the quantitative nondivergence theorem to the finite analytic coordinate family. Lemma 6.2 reduces the interval to finitely many pieces on which a fixed nonzero jet has uniform size. The

resulting analytic functions are uniformly good, so cusp excursions occupy arbitrarily small relative measure.

Step 2: nontrivial unipotent invariance. On every positive-length piece, Lemma 6.3 gives a nontrivial unipotent invariance for any weak limit. Ratner's measure classification therefore decomposes the limit into homogeneous ergodic components.

Step 3: exclusion of proper components. If a proper support type has positive weight, Proposition 6.4 forces the boundary image of the entire analytic curve into the associated proper algebraic incidence variety. This contradicts the nondegeneracy hypothesis. Hence every ergodic component is ambient Haar, and therefore the whole weak limit is m_X .

Step 4: uniqueness. Tightness gives subsequential probability limits and Step 3 identifies every one of them with m_X ; hence the entire family converges.

If nondegeneracy fails, repeat the same argument inside a boundary stratum minimal for inclusion among those containing the relevant analytic image. The algebraic subgroup lattice is Noetherian, so the descent terminates. Ratner classification gives homogeneous probabilities on the resulting homogeneous envelopes. Without a uniqueness hypothesis on the minimal stratum, different ergodic components may occur, which is why the conclusion is stated as a barycenter rather than as an unjustified single Haar measure. $\square$

Worked Example 6.6 (Curves on a hyperbolic manifold). For $G = \mathrm{SO}(n, 1)^\circ$, the boundary flag variety is the visual sphere S^{n-1} . Proper homogeneous obstructions correspond to proper round subspheres, the ideal boundaries of totally geodesic subspaces. Thus an analytic curve whose visual image is not contained in a proper subsphere equidistributes under the geodesic expansion.

Chapter 7

Shrinking and Nonuniformly Stretched Pieces

7.1 The tangent polynomial principle

Let the parameter interval shrink with t : $s = s_0 + r\sigma_t$, $\sigma_t \rightarrow 0$. Taylor expansion gives

$$\phi(s_0 + r\sigma_t) = \phi(s_0) \exp(\sigma_t r Y_1 + \cdots + \sigma_t^d r^d Y_d + o(\sigma_t^d)).$$

Conjugation by a_t multiplies Y_j according to its root weight. The competition between σ_t^j and the expansion factor selects a finite polynomial tangent trajectory. Volume VIII then determines its homogeneous limit.

Theorem 7.1 (Shrinking-piece reduction). *Assume the normalized conjugated jets converge to a polynomial map $\Theta : \mathbb{R} \rightarrow G$ and that the associated averages are tight. Then every limit distribution of the shrinking expanding pieces equals the limit distribution of Θ in the corresponding translated homogeneous space. In particular, if the tangent polynomial has full minimal homogeneous envelope, the shrinking pieces equidistribute to Haar.*

Proof. Uniform Taylor remainder estimates on the shrinking interval and boundedness of the conjugated inactive jets imply that the original and tangent trajectories are uniformly $o(1)$ apart on compact normalized parameter ranges. Test functions are uniformly continuous on the compact set carrying all but $o(1)$ of the mass. Their averages therefore differ by $o(1)$. Apply Volume VIII to the tangent polynomial. $\square$

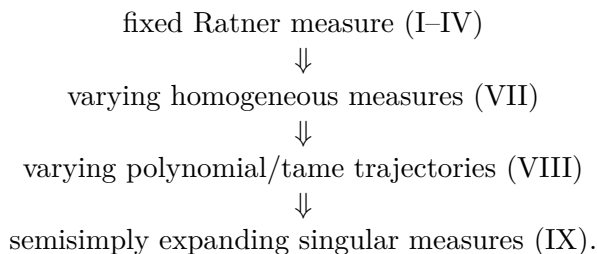
7.2 Nonuniform stretching

If different root directions are expanded at different rates, there need not be a single isotropic parameter scale. Assign weights to the parameter jets and choose the Newton face on which the total expansion exponent is maximal. The surviving monomials form the tangent polynomial. This Newton-polyhedron bookkeeping is elementary but prevents a common error: keeping only the highest ordinary derivative can miss a lower derivative in a faster root direction.

Chapter 8

Arc I Synthesis and Proof Audit

Arc I has now moved through four levels of variation:



Volume V supplied rational/parabolic structure and Volume VI supplied mixing, but the proof of the main expanding-translate theorem did not hide behind mixing: it used nonescape, creation of unipotent invariance, Ratner classification, and linearization. This distinction matters in Arc II, where two different counting engines—mixing and homogeneous-measure rigidity—will be compared.

Remark 8.1 (Arc I completeness). The load-bearing qualitative post–Ratner inputs are owned as follows: reduction/arithmetic structure (V), Howe–Moore and mixing (VI), homogeneous-measure limits/focusing (VII), polynomial/tame trajectory limits (VIII), and expanding translates/curves (IX). No theorem from the later Advanced Homogeneous Dynamics project is used.

Appendix A

Right-Action Horospherical Calculus

A.1 The exact shift identity in several root spaces

Write $\mathfrak{u}^+ = \bigoplus_{\alpha(H) > 0} \mathfrak{g}_{-\alpha}$. For $Y \in \mathfrak{g}_{-\alpha}$,

$$\mathrm{Ad}(a_t)Y = e^{-\alpha(H)t}Y, \quad \mathrm{Ad}(a_t^{-1})Y = e^{\alpha(H)t}Y.$$

Hence, with $\delta_t = e^{-\alpha(H)t}$,

$$\exp(\delta_t Y) a_t = a_t \exp(Y).$$

The identity is exact because conjugation commutes with the exponential map.

Lemma A.1 (Boundary loss under small nilpotent translations). *Let U be a connected nilpotent Lie group, $\Omega \subset U$ bounded with $m_U(\partial\Omega) = 0$, and $v_i \rightarrow e$. Then*

$$m_U(\Omega \triangle \Omega v_i) \longrightarrow 0.$$

Proof. Right translations act strongly continuously on $L^1(U)$: for $C_c(U)$ this follows from uniform continuity on a compact neighborhood of the support, and $C_c(U)$ is dense in $L^1(U)$. Apply this to $1_\Omega \in L^1(U)$. The boundary-zero assumption ensures that the chosen representative of 1_Ω has no ambiguity relevant to approximation. $\square$

Expanded proof of Lemma 3.1. For $Y \in \mathfrak{g}_{-\alpha}$ put $v_t = \exp(e^{-\alpha(H)t}Y)$. Then

$$\begin{aligned}\mu_t(f \circ R_{\exp Y}) &= \frac{1}{m_U(\Omega)} \int_{\Omega} f(xua_t \exp Y) du \\ &= \frac{1}{m_U(\Omega)} \int_{\Omega} f(xuv_t a_t) du \\ &= \frac{1}{m_U(\Omega)} \int_{\Omega v_t} f(xua_t) du.\end{aligned}$$

The absolute difference from $\mu_t(f)$ is at most $\|f\|_{\infty} m_U(\Omega \Delta \Omega v_t) / m_U(\Omega)$, which tends to zero by Lemma A.1. Thus any weak limit is fixed by $\exp Y$. Replacing Y by sY gives the full root group. Root spaces generate U^+ , and the stabilizer of a measure is closed, so the limit is U^+ -invariant. $\square$

Appendix B

The Expanding Focusing Alternative

The key step in the qualitative expanding theorem is not merely Ratner classification; it is the exclusion of a proper homogeneous component. We make that implication explicit.

Proposition B.1 (Expanding focusing). *Let a_i and U^+ satisfy the hypotheses of Theorem 4.1, and suppose the measures on $x\Omega a_i$ converge to a probability μ . Let yH be a proper finite-volume homogeneous orbit with $U^+ \subset H$. If μ assigns positive mass to a compact subset of the regular part of yH (away from every smaller singular subtype), then there exist a finite-dimensional Chevalley representation ρ , a vector p_H , vectors $v_i \in \rho(\Gamma)p_H$, a nonempty open $\Omega_0 \Subset U^+$, and $C < \infty$ such that*

$$\sup_{u \in \Omega_0} \|\rho(a_i^{-1}u^{-1})v_i\| \leq C. \quad (9.7)$$

Moreover the projective limit of these vectors lies on the exact normalizer variety of H .

Proof. Choose the inverse Chevalley map $\eta_H(g) = \rho(g^{-1})p_H$ and a compact set C_H in the exact normalizer variety representing the chosen regular compact part of yH . By regularity, C_H is separated from every self-intersection and every smaller singular stratum. Positive limiting mass gives a fixed $\delta > 0$ such that, for all large i , a set of $u \in \Omega$ of Haar proportion at least δ has the lifted vector $\rho(a_i^{-1}u^{-1})v_{i,u}$ in a fixed small neighborhood of C_H . The unique-lift lemma of Volume VII makes $v_{i,u}$ locally constant on each connected parameter

component; only finitely many lifts occur on the compact basepoint window. Select one lift v_i occurring on a subset of proportion bounded below.

Apply the multivariable polynomial linearization inequality of Volume VIII to every coordinate transverse to the normalizer variety. If one coordinate leaves a fixed compact enlargement somewhere in the maximal parameter box, its sublevel estimate bounds the thinner-neighborhood set by an arbitrarily small fraction, contradicting the selected positive proportion. Therefore all relevant coordinates remain in the compact enlargement on a definite subbox. Passing to a density point and translating that subbox inside the connected nilpotent group gives one fixed nonempty open $\Omega_0 \subseteq U^+$ and the uniform norm bound (9.7). If the transverse coordinates did not tend to zero projectively, the same good-function estimate at a nested sequence of neighborhoods would again force their relative mass to zero. Hence the projective limit lies on the exact normalizer variety. $\square$

Corollary B.2 (Proper components force a proper G -envelope). *If G is almost simple, a_t is noncompact, and xG is dense in the ambient finite-volume homogeneous space, no proper H can occur with positive mass in a limit of the expanding U^+ -piece measures.*

Proof. Proposition B.1 supplies the bounded discrete-vector obstruction (9.7). Since an almost-simple group is normally generated by its nontrivial horospherical subgroup, Corollary 2.4 forces the defining Chevalley vector to be G -fixed. Hence the corresponding proper homogeneous datum is G -invariant. The lifting part of the focusing construction places a limit point of $x\Omega a_{t_i}$ in that datum, so it contains the G -orbit closure of x , contradicting density. $\square$

Appendix C

Analytic Jets and the Boundary Obstruction

Lemma C.1 (Finite active jet order). *Let $\phi : I \rightarrow U^+$ be real analytic and let F be a nonzero polynomial coordinate of a fixed algebraic representation composed with ϕ . On a compact subinterval $J \Subset I$, either $F \circ \phi$ vanishes identically or there is an integer $r = r(F, J)$ such that at every point of J some derivative of order at most r is nonzero.*

Proof. If no uniform r existed, choose $s_r \in J$ at which the first r derivatives vanish. Compactness gives $s_{r_j} \rightarrow s_\infty$. Continuity of every derivative then makes all derivatives vanish at s_∞ . Analyticity forces $F \circ \phi$ to vanish identically on the connected component, a contradiction. $\square$

Lemma C.2 (Jet rescaling). *Suppose the first active derivative at s_0 lies in a root space on which $\text{Ad}(a_t)$ contracts with exponent $\lambda > 0$, and its order is r . Then on the scale*

$$s = s_0 + e^{-\lambda t/r} \tau$$

the conjugated Taylor expansion has a nonconstant polynomial limit in τ , with leading term proportional to τ^r in that root direction.

Proof. Taylor's theorem gives a leading term $c(s - s_0)^r$ and a remainder $O(|s - s_0|^{r+1})$. Conjugation by the right translate multiplies the active root coordinate by $e^{\lambda t}$ in the relative-displacement calculation. The chosen scale makes the leading term order one and the remainder $O(e^{-\lambda t/r})$. Slower root weights vanish; faster root weights are included earlier in the ordered jet filtration. Thus a polynomial tangent limit exists. $\square$

The two lemmas supply the missing compactness statement in the proof of Theorem 6.5: a finite subdivision of I suffices, and no unbounded sequence of jet orders can occur unless the relevant algebraic coordinate vanishes identically. The latter is exactly the boundary obstruction recorded in the theorem.

Appendix D

Arc I Detailed Dependency Table

Structural statement	Proof owner
Cartan/parabolic and arithmetic reduction	Volume V
Howe–Moore and root propagation	Volume VI
Moving homogeneous measures and thin neighborhoods	Volume VII
Polynomial QND and finite-difference twisting	Volume VIII
Right-action horospherical shift	this volume
Proper-component exclusion for expanding pieces	this volume, Appendix B
Analytic-curve finite jet reduction	this volume

No effective rate is used in this table. Arc II begins only after these qualitative mechanisms have been closed.

Appendix E

Worked Expanding-Translate Laboratory

E.1 Motivation

Expanding translates magnify a local jet until it becomes a macroscopic unipotent piece.

Worked Example E.1 (Expansion of a modular horocycle). With $a_t = \text{diag}(e^{t/2}, e^{-t/2})$,

$$a_t u_s a_t^{-1} = u_{e^t s}.$$

Thus a fixed parameter interval $|s| \leq e^{-t}$ before expansion becomes the unit interval $|e^t s| \leq 1$ afterwards. Conversely, a curve segment of macroscopic length after applying a_t probes only a microscopic jet of the original curve. This is why Taylor coefficients determine the limiting homogeneous subgroup.

Worked Example E.2 (A second-order jet). For $\gamma(s) = u_s \exp(s^2 F)$ in SL_2 , the first derivative at zero lies in the E direction while the transverse F displacement is quadratic. Under a_t , E is expanded and F contracted with opposite weights. Choosing $s \asymp e^{-t}$ keeps the E term order one while the F term is of order e^{-2t} before its additional contraction. Hence the first jet alone governs this scaling limit; at a degenerate first derivative, a different scaling would expose the second jet.

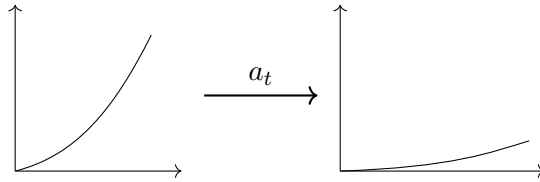


Figure E.1: A diagonal translate changes the relative sizes of jet directions. The rescaled dominant jet determines the limiting unipotent motion.

Appendix F

Student Proof Laboratory: Expanding Translates Step by Step

Expanding-translate arguments combine three geometries at once: the horospherical geometry of a semisimple element, polynomial divergence inside the expanding group, and homogeneous-measure rigidity for the limiting measure. This chapter isolates those geometries before recombining them.

F.1 Right actions and the conjugation convention

Throughout this series points are written on a right quotient and are moved by right multiplication. If a_t expands U^+ , then

$$a_{-t}u_s a_t \rightarrow e \quad (t \rightarrow +\infty)$$

for $u_s \in U^+$. If $x_t = xu_s a_t$, then

$$xu_s a_t = xa_t(a_{-t}u_s a_t).$$

Thus nearby parameters in U^+ become tiny right displacements after the large translate. The inverse representation in the linearization chapter is forced by this identity; it is not a cosmetic convention.

Worked Example F.1 (The standard horocycle). For $G = \mathrm{SL}_2(\mathbb{R})$,

$$u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}, \quad a_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix},$$

and $a_{-t}u_s a_t = u_{e^{-t}s}$. A fixed interval $0 \leq s \leq 1$ therefore becomes a transverse tube of width e^{-t} around the translated basepoint. The geometric problem is to show that this shrinking tube, despite becoming thin, samples the ambient homogeneous measure after translation.

F.2 Tightness before invariance

Let

$$\mu_t = \frac{1}{|B|} \int_B \delta_{xua_t} du$$

for a compact piece $B \subset U^+$. To classify subsequential limits one must first know that they are probability measures.

Lemma F.2 (Compact-mass criterion). *Suppose that for every $\varepsilon > 0$ there is compact $K_\varepsilon \subset X$ such that $\mu_t(K_\varepsilon) > 1 - \varepsilon$ for all large t . Then every sequence $t_i \rightarrow \infty$ has a weak-* convergent subsequence whose limit has total mass one.*

Proof. Use the diagonal compact-exhaustion argument of Volume VII. The important point is that convergence on $C_c(X)$ alone could otherwise produce a subprobability measure; the common compact sets prevent that loss. $\square$

In applications, quantitative nondivergence is applied not directly to ua_t but to a finite collection of exterior-power representation vectors that detect short rational subgroups. If all such vectors remain above a fixed threshold for at least one parameter value in every relevant interval, the good-function theorem controls the set of parameters for which they are small.

F.3 How expansion creates unipotent invariance

Fix $r \in \mathbb{R}$. Make the parameter change $u \mapsto uu_{e^{-\lambda t}r}$ inside an expanding root group of weight $\lambda > 0$. The domain B and its shifted copy differ by a boundary layer of relative measure $O(e^{-\lambda t})$. At the same time

$$(uu_{e^{-\lambda t}r})a_t = ua_t u_r.$$

Therefore, for $f \in C_c(X)$,

$$\int f(zu_r) d\mu_t(z) - \int f(z) d\mu_t(z) \rightarrow 0.$$

Every weak limit is u_r -invariant. Repeating for all expanding root directions gives invariance under a connected unipotent subgroup.

Warning F.3 (Why a boundary assumption is needed). If B has a fractal boundary of positive Haar measure, a tiny translate of B need not have small symmetric difference. The correct theorem therefore uses open pieces with boundary of Haar measure zero, smooth densities, or a quantitative well-roundedness hypothesis. The parameter-shift argument is only as good as this boundary control.

F.4 The focusing obstruction in a Chevalley representation

Let $L < G$ be a proper candidate support group and choose a representation $\rho : G \rightarrow \mathrm{GL}(V)$ and nonzero $p_L \in V$ whose projective stabilizer is the normalizer of L . On a right quotient, the relevant linearized trajectory is

$$\rho((ua_t)^{-1})p_L = \rho(a_t^{-1}u^{-1})p_L.$$

If a positive proportion of parameters land near xL , this vector remains close to the discrete $\rho(\Gamma)$ -orbit of p_L for a positive proportion of parameters. Polynomial nondivergence yields a dichotomy: either that parameter set is small, or the representation trajectory is algebraically trapped. In the trapped branch the expanding curve is asymptotic to a proper normalizer. The normal-generation hypothesis in Shah's theorem is exactly what rules out this branch.

F.5 Five-step proof of the expanding-horosphere theorem

For reference, the proof can be written as the following ledger.

1. *Nonescape*: exterior-power good functions give tightness.
2. *Invariance*: parameter shifts give unipotent invariance of every weak limit.
3. *Classification*: Ratner makes every ergodic component homogeneous.
4. *Focusing*: Chevalley linearization rules out every proper homogeneous component not forced by the algebraic hull.
5. *Uniqueness*: all subsequences have the same Haar limit, hence the original family converges.

A proof that skips directly from Step 2 to Step 5 is incomplete: Ratner classifies possible components but does not choose which one occurs.

F.6 Curves: why jets replace parameters

Let $\psi : I \rightarrow U^+$ be analytic. Near s_0 write in exponential coordinates

$$\log(\psi(s_0 + h)\psi(s_0)^{-1}) = h^j Y_j + h^{j+1} Y_{j+1} + \cdots,$$

where j is the first nonzero jet order. If $\text{Ad}(a_{-t})Y_j = e^{-\lambda t}Y_j$, choose $h = e^{-\lambda t/j}r$. Then the leading term becomes order one after translation and all higher terms vanish. This produces the unipotent invariance group attached to the jet. A nondegeneracy hypothesis says that the groups generated by these jet directions are large enough to exclude a proper focusing subgroup.

Worked Example F.4 (A curved path in a two-dimensional horosphere). In $\text{SL}_3(\mathbb{R})$ let

$$\psi(s) = \exp(sE_{12} + s^2E_{13}).$$

For a diagonal a_t with weights satisfying $\alpha_{12}(H) = 1$ and $\alpha_{13}(H) = 2$, the two terms become visible on the same scale $s = e^{-t}r$. The limiting parameter shift sees both E_{12} and E_{13} through the polynomial $rE_{12} + r^2E_{13}$. This is a small model for the higher-order jet phenomena in Shah's curve theorem.

F.7 Shrinking pieces and tangent polynomials

Suppose the initial parameter interval itself has length $\delta_t \rightarrow 0$. Write $s = s_t + \delta_t r$ and Taylor expand. The translate a_t magnifies some derivative orders and kills others. After normalizing by the first surviving order, one obtains a polynomial tangent curve. Volume VIII then supplies the homogeneous limit theorem for that polynomial model. The shrinking-piece theorem is therefore not a separate miracle: it is a compactness statement saying that a sufficiently small smooth piece is modeled, after the dynamically correct rescaling, by the polynomial trajectory already understood.

Appendix G

Student Proof Laboratory: Expanding Horospheres from the Ground Up

An expanding translate theorem starts from a measure that is singular with respect to Haar measure. Ordinary mixing therefore cannot be applied at the first line. The proof first creates unipotent invariance in every weak limit, then invokes Ratner/Mozes–Shah rigidity, and only afterwards uses algebraic nondegeneracy to rule out proper limits.

G.1 Horospherical groups and the conjugation convention

Let $a_t = \exp(tH)$ and define

$$U^+ = \{u : a_{-t}ua_t \rightarrow e \text{ as } t \rightarrow +\infty\}.$$

If the homogeneous space is a right quotient $X = \Gamma \backslash G$, then

$$(xu)a_t = xa_t(a_{-t}ua_t).$$

Thus U^+ is the subgroup whose small parameter shifts before a_t become order-one right translations after expansion.

Worked Example G.1 (SL_2). For $a_t = \mathrm{diag}(e^{t/2}, e^{-t/2})$ and $u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$, $a_{-t}u_s a_t = u_{e^{-t}s}$. Hence a shift by $e^{-t}r$ in the horocycle parameter becomes the fixed translation u_r after multiplying by a_t .

G.2 Boundary control for the parameter domain

Lemma G.2 (Small translates of a regular domain). *Let $B \subset \mathbb{R}^d$ be bounded with boundary of Lebesgue measure zero. Then*

$$|(B + h) \triangle B| \rightarrow 0 \quad (h \rightarrow 0).$$

Proof. Given $\epsilon > 0$, choose a compact $K \subset B$ and an open $O \supset B$ with $|O \setminus K| < \epsilon$ using regularity and the zero-boundary hypothesis. For sufficiently small h , compactness gives $K + h \subset O$ and $K \subset O + h$. Therefore the symmetric difference is contained in $(O \setminus K) \cup ((O + h) \setminus (K + h))$, of measure $< 2\epsilon$. $\square$

G.3 Creation of unipotent invariance

Let ν be a compactly supported absolutely continuous probability on an open piece of U^+ and let $\mu_t = (R_{at})_* \nu$ on X .

Proposition G.3 (Every weak limit is U^+ -invariant). *Assume (μ_t) is tight. Every weak- $*$ subsequential limit is invariant under U^+ .*

Proof. It suffices to prove invariance under each root subgroup $U_\alpha \subset U^+$. Fix $u_r \in U_\alpha$. Write $u_{e^{-\alpha(H)t_r}}$ for the microscopic parameter shift before expansion. Changing variables by this shift changes the domain and density by $o(1)$ in L^1 , by the previous boundary lemma (or by smoothness of the density). The conjugation identity gives

$$(u u_{e^{-\alpha(H)t_r}}) a_t = u a_t u_r.$$

Therefore $\int f(z u_r) d\mu_t - \int f(z) d\mu_t \rightarrow 0$ for every $f \in C_c(X)$. Pass to the weak limit. Root groups generate U^+ , so the limit is U^+ -invariant. $\square$

G.4 Tightness: where reduction theory enters

On a noncompact quotient, the previous proposition is useless without a probability limit. Quantitative nondivergence supplies tightness when the expanding piece is not trapped in a rational parabolic.

Proposition G.4 (Nonescape-or-parabolic alternative). *For an expanding horospherical piece based at x , either the translated probabilities are tight, or*

there exists a proper rational parabolic whose Chevalley vector is uniformly contracted along a positive proportion of the piece.

Proof. Use the exterior-power height functions from Volumes II and V. Failure of tightness means that for every compact thick part a fixed positive proportion has some primitive rational covolume below the corresponding threshold. There are finitely many ranks and, on a bounded parameter set, only finitely many primitive wedges in a prescribed norm window. Pigeonhole and pass to a subsequence so one wedge is repeatedly small. Its coordinate functions along the horospherical parameter are polynomial. Quantitative goodness then upgrades smallness on positive proportion to uniform smallness on the whole parameter piece at a slightly weaker scale. The limiting wedge line is stabilized by a proper rational parabolic, yielding the stated alternative. $\square$

G.5 Why the inverse Chevalley representation appears

Suppose a weak limit gives positive mass to a proper homogeneous orbit xL . For right quotients, the condition $\Gamma ga_t u \in \Gamma g_0 L$ is converted by multiplying on the right and then moving the group element to a representation vector. The resulting linearized expression is

$$\rho((ua_t)^{-1})p_L,$$

not $\rho(ua_t)p_L$. The inverse is forced by the right action. Forgetting it reverses expanding and contracting weights.

Lemma G.5 (Linearized focusing dichotomy). *Let p_L be a Chevalley vector for a proper subgroup L . On a compact parameter piece, either*

$$\rho((ua_t)^{-1})p_L$$

has a transverse coordinate bounded below on most parameters, or after normalization it converges to a vector fixed by the expanding parameter family. The second case is an algebraic incidence condition on the original piece.

Proof. In a weight basis for a_t , every coordinate is a finite sum of terms $e^{-\lambda t} P_\lambda(u)$, where P_λ is polynomial on the unipotent parameter group. Normalize by the largest surviving exponential weight. If a transverse polynomial

is nonzero, Remez/goodness bounds its small-value set. If every transverse polynomial vanishes, the normalized limit lies in the projective stabilizer of the Chevalley line. That vanishing is algebraic in u , hence defines the focusing obstruction. $\square$

G.6 The qualitative expanding-translate theorem

Theorem G.6 (Expanding horospherical piece). *Assume the basepoint and parameter piece satisfy the nonparabolic, nonfocusing hypotheses of the main text. Then the expanding translates converge to Haar probability on the minimal homogeneous orbit allowed by the algebraic hull of the piece.*

Proof. Tightness gives subsequential probability limits. Microscopic parameter shifts make each limit invariant under a nontrivial unipotent subgroup. Ratner classification makes its ergodic components homogeneous. If a proper component carries positive mass, the inverse-Chevalley focusing dichotomy forces the original parameter piece into the corresponding proper algebraic incidence variety, contrary to nonfocusing. Thus every subsequential limit is the same Haar probability. Uniqueness of subsequential limits gives convergence of the full family. $\square$

Appendix H

Student Proof Laboratory: Curves, Jets, and Shrinking Pieces

H.1 Why a curve needs derivatives

A one-dimensional curve inside a high-dimensional horospherical group does not contain an open set, so independent microscopic shifts in every root direction are unavailable. Expansion magnifies the Taylor jet. The first nonzero derivatives supply polynomial tangent trajectories, and Volume VIII classifies their limits.

Lemma H.1 (Polynomial tangent model). *Let $\phi : I \rightarrow U^+$ be C^m and fix s_0 . In exponential coordinates, for $h \rightarrow 0$,*

$$\phi(s_0)^{-1}\phi(s_0 + h) = \exp(hY_1 + h^2Y_2 + \cdots + h^mY_m + o(h^m)).$$

After conjugation by a_t and choosing $h = h_t$ so that the largest weighted term is order one, the rescaled curve converges on compact parameter intervals to a polynomial trajectory in a unipotent group.

Proof. Apply Taylor's theorem to the logarithm in a local exponential chart. The Baker–Campbell–Hausdorff corrections are universal Lie polynomials in the Taylor coefficients and therefore preserve finite polynomial degree. Conjugation by a_t multiplies each restricted-root component by an exponential weight. Choose h_t to balance the largest nonzero weighted monomial.

All faster-decaying terms vanish and all terms of the same maximal weight survive, giving a polynomial limit. $\square$

Worked Example H.2 (A curve in the SL_3 upper horosphere). Write

$$u(x, y, z) = \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix}, \quad \phi(s) = u(s, s^2, 0).$$

The commutator law creates a central coordinate when two nearby parameters are compared. Under a diagonal flow with unequal root weights, either the linear x -jet or the quadratic y -jet dominates depending on the chosen weights. The tangent polynomial may therefore generate more of the unipotent group than the first derivative alone. This is why nondegeneracy conditions are naturally stated in terms of jets or representation spans.

H.2 Shrinking intervals

If the interval length $\delta_t \rightarrow 0$, write $s = s_0 + \delta_t r$. The same Taylor calculation applies, but now the balance between δ_t^j and the exponential root weights determines which jet survives. The limiting problem is again a polynomial-trajectory problem, not a new rigidity theorem.

Appendix I

A Student Textbook Chapter on Parabolic Expansion

Expanding translates are easiest to remember in the language of a diagonal flow and its three dynamical directions: expanding, neutral, and contracting. This chapter derives that decomposition from restricted roots and then shows how each part enters the measure argument.

I.1 Three Lie subalgebras from a one-parameter group

Let $a_t = \exp(tH)$ in a real split torus. Decompose

$$\mathfrak{g} = \mathfrak{g}^- \oplus \mathfrak{g}^0 \oplus \mathfrak{g}^+$$

where $\mathfrak{g}^+$ is the sum of restricted root spaces with $\alpha(H) > 0$, $\mathfrak{g}^-$ those with $\alpha(H) < 0$, and $\mathfrak{g}^0$ the zero-weight centralizer.

Lemma I.1 (Subalgebra property). *$\mathfrak{g}^+$ and $\mathfrak{g}^-$ are nilpotent Lie subalgebras, and $\mathfrak{p}^- = \mathfrak{g}^0 \oplus \mathfrak{g}^-$ is a Lie subalgebra normalizing $\mathfrak{g}^+$.*

Proof. The root-bracket rule gives $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$. Positive plus positive is positive whenever it is a root, so $\mathfrak{g}^+$ is a subalgebra. If two negative roots add to a root, their sum is negative, so $\mathfrak{g}^-$ is also closed under brackets. Repeated brackets strictly add positive root height, so sufficiently long brackets vanish, proving nilpotence. Zero-weight plus negative-weight remains negative, so $\mathfrak{g}^0$ normalizes $\mathfrak{g}^-$. Hence $\mathfrak{p}^-$ is a subalgebra. $\square$

Exponentiate to obtain U^+, U^- and the neutral centralizer $Z_G(A)$. Conjugation gives

$$a_{-t}u^+a_t \rightarrow e, \quad a_tu^-a_{-t} \rightarrow e.$$

The parabolic $P^- = Z_G(A)U^-$ consists of elements that do not expand relative to a_t in forward time.

I.2 Local product geometry

Near the identity, multiplication

$$U^- \times Z \times U^+ \rightarrow G$$

is a local diffeomorphism after choosing complements to finite overlaps. Thus a small perturbation can be decomposed into stable, neutral, and unstable pieces. Conjugation by a_t exponentially shrinks one of these pieces and expands the other. The wavefront lemma of Volume X is a global uniform version of this local statement along deep Cartan regions.

I.3 Measures on expanding horospherical pieces

Let $B \subseteq U^+$ have boundary Haar measure zero and density $\psi \in C_c(B)$. Define

$$\mu_t(f) = \int_B f(xua_t)\psi(u) du.$$

The microscopic-shift proof of U^+ -invariance works one root group at a time. If $u_r \in U_\alpha$, write $\delta_t = e^{-\alpha(H)t}r$. Right multiplication of u by u_{δ_t} changes the parameter by $o(1)$ and satisfies

$$(uu_{\delta_t})a_t = ua_tu_r.$$

Boundary regularity and continuity of ψ make the change-of-variable error tend to zero.

I.4 Tightness in the lattice model

Suppose $G = \mathrm{SL}_n(\mathbb{R})$ and $X = \mathrm{SL}_n(\mathbb{Z}) \backslash G$. A translated horospherical piece escapes only if some primitive subgroup of the moving lattice has small covolume for a positive parameter set. The wedge covolumes are polynomial

in horospherical coordinates. Quantitative nondivergence therefore says: either the bad parameter set has small relative measure, or one wedge is uniformly small throughout the piece. The latter means the entire piece is approaching the parabolic stabilizer of a rational flag. Hence a nonparabolicity hypothesis is exactly a nonescape hypothesis.

Worked Example I.2 (SL_3 and a maximal parabolic). Take the diagonal flow

$$a_t = \mathrm{diag}(e^{2t}, e^{-t}, e^{-t}).$$

Its expanding group is

$$U^+ = \left\{ \begin{pmatrix} 1 & x & y \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} : x, y \in \mathbb{R} \right\}.$$

If the base lattice contains a rational plane almost parallel to $\mathrm{span}(e_2, e_3)$, the area of that plane is contracted by a_t and the translated piece can enter the cusp. The stabilizer of the plane is the maximal parabolic of type $(2, 1)$. Exterior powers see this degeneration before any individual vector necessarily becomes exceptionally short.

Appendix J

Shah's Expanding-Translate Argument as a Full Ledger

J.1 Step 1: choose any subsequential limit

Prove tightness. Take $t_i \rightarrow \infty$ with $\mu_{t_i} \Rightarrow \mu$. No ergodicity has yet been used.

J.2 Step 2: create unipotent invariance

Microscopic parameter shifts give invariance under the subgroup generated by root directions tangent to the parameter piece. For an open piece of the full U^+ this is all of U^+ . For a curve it may be a smaller group produced from jets.

J.3 Step 3: classify the ergodic components

Ratner's generated-unipotent theorem makes every ergodic component of μ homogeneous. Write a typical component as m_{yL} .

J.4 Step 4: linearize a putative proper component

Choose a Chevalley vector p_L . Because the action is on a right quotient, closeness of xua_{t_i} to yL is expressed using

$$\rho((ua_{t_i})^{-1})p_L.$$

Decompose this vector into a_t -weight spaces. The fastest expanding weight after inversion must vanish on a positive parameter set if the orbit spends positive mass near yL . Polynomial goodness forces it to vanish identically. Continue weight by weight. Eventually the entire parameter piece satisfies an algebraic incidence relation with L .

J.5 Step 5: exclude focusing

The theorem's nondegeneracy hypothesis says precisely that the original piece is not contained in such a proper incidence variety. Therefore no proper L can occur. The weak limit is Haar on the minimal allowed homogeneous orbit.

J.6 Step 6: uniqueness

Every subsequence has the same weak limit. Therefore the full family converges. This last step is topological and should not be hidden under the word "equidistribution".

Appendix K

Closed Homogeneous Orbits under Expansion

Let $Y = xH$ be a closed finite-volume orbit and translate its invariant probability by a_t . Unlike an open horospherical piece, Y may have its own internal unipotent directions and may intersect rational singular strata. Mozes–Shah theory is the natural compactness tool.

K.1 Limit measures of $m_Y a_t$

Suppose tightness holds. Any weak limit is a limit of homogeneous probabilities. Volume VII makes it homogeneous and supplies support containment after small conjugation. The expanding action adds extra invariance: root subgroups contracted into H by conjugation can survive as new symmetries of the limit. Algebraically, the limit support group contains the group generated by all asymptotic conjugates of H that remain bounded.

Proposition K.1 (Qualitative translated-orbit alternative). *For a sequence $t_i \rightarrow \infty$, after passing to a subsequence either $m_Y a_{t_i}$ escapes mass, or it converges to a homogeneous probability m_{zL} . In the latter case, after small conjugation the groups $a_{-t_i} H a_{t_i}$ are eventually incident to L , and every unipotent limit of their bounded root components lies in L .*

Proof. If there is no escape, apply the Mozes–Shah family theorem to the homogeneous probabilities $m_Y a_{t_i}$, whose support groups are $a_{-t_i} H a_{t_i}$ in right-action convention. This gives homogeneous limits and support containment.

A bounded sequence of elements $h_i \in a_{-t_i} H a_{t_i}$ converging to u preserves every weak limit because $m_Y a_{t_i}$ is invariant under the entire conjugated support group. Thus all such limits u lie in the stabilizer of m_{zL} , hence in the connected support group after passing to identity components. $\square$

K.2 Why the orbit volume matters

If Y has infinite H -volume there is no normalized homogeneous probability to translate. One can instead translate compact pieces, but then boundary and normalization issues return. The finite-volume hypothesis is therefore part of the object, not merely a technical condition for invoking a theorem.

Appendix L

Curves in Horospherical Groups: A Jet-by-Jet Course

L.1 First derivative and nondegeneracy

Let $\phi : I \rightarrow U^+$ be analytic. In exponential coordinates,

$$\log(\phi(s_0)^{-1}\phi(s_0 + h)) = hY_1 + h^2Y_2 + \cdots.$$

If Y_1 has nonzero projection to every relevant quotient representation, the first derivative already creates enough expanding directions. If not, higher jets matter.

L.2 Representation-theoretic nondegeneracy

A convenient criterion is: for every proper algebraic subgroup L that could support a Ratner component, the map

$$s \mapsto \rho(\phi(s))p_L$$

is not contained in the sum of nonexpanding a_t -weight spaces. If it were, expansion could keep the whole curve close to the L -incidence variety. Analyticity implies that failure of this condition is detected by finitely many derivatives at a generic point.

L.3 An SL_3 calculation

Let

$$\phi(s) = \begin{pmatrix} 1 & s & s^3 \\ 0 & 1 & s^2 \\ 0 & 0 & 1 \end{pmatrix}$$

and $a_t = \text{diag}(e^{2t}, 1, e^{-2t})$ after rescaling to determinant one. The E_{12} and E_{23} coordinates have different root weights, while the central E_{13} coordinate is affected both directly by s^3 and by the commutator of the first two coordinates. Choosing a shrinking parameter $h = e^{-\lambda t}$ selects whichever weighted jet first becomes order one. The resulting polynomial tangent can contain an E_{13} term even when the first derivative does not. Thus a curve may generate a larger unipotent limit than its tangent line suggests.

L.4 Shrinking pieces

Suppose the original interval has length δ_t . Write $s = s_0 + \delta_t r$. A j th jet in root space α appears after expansion with size roughly $e^{\alpha(H)t} \delta_t^j$. If this tends to zero, the jet disappears; if it tends to a nonzero constant, it survives; if it diverges, one must choose a smaller rescaling to see the first macroscopic stage. This simple balance rule organizes every shrinking-curve example in the volume.

Appendix M

Worked Expanding-Translate Examples

M.1 Expanding horocycles on the modular surface

A short horocycle segment at height one, translated by the geodesic flow, becomes a long horocycle segment. If its length tends to infinity and it is not based at a periodic cusp configuration, unique ergodicity/mixing in this rank-one model gives Haar equidistribution. The general Shah argument reproduces the same conclusion without relying on a special Fourier expansion of the modular surface.

M.2 A focused example

Take a parameter piece entirely inside the intersection of U^+ with a proper subgroup L for which xL is closed. Every translate remains in the corresponding L -orbit family. The correct limiting measure is therefore supported in a proper homogeneous space. This is not a counterexample; it is exactly the algebraic focusing alternative excluded in the full-density theorem.

M.3 A parabolic escape example

In the space of unimodular lattices, choose a base lattice with a rational subspace aligned with the contracting directions of a_t . The covolume of that subspace tends to zero uniformly across a bounded unstable parameter piece.

Mahler compactness then fails uniformly and the translated probabilities can lose mass. This is the rational-parabolic alternative in its most concrete form.

Appendix N

Guided Reconstruction of Expanding-Translate Arguments

In this volume a semisimple element a_t expands a horospherical piece or a curve. The proof looks different from a unipotent time average, but Ratner's theory re-enters through the weak limits: expansion creates unipotent invariance, and classification identifies the possible limits. The main new work is to make the invariance and focusing arguments compatible with the moving geometry.

N.1 Stable, neutral, and unstable directions

Let $a_t = \exp(tH)$ with H in a split Cartan subalgebra. Decompose

$$\mathfrak{g} = \mathfrak{u}^- \oplus \mathfrak{z} \oplus \mathfrak{u}^+,$$

where the three summands are the negative, zero, and positive H -weight spaces. Then

$$\mathrm{Ad}(a_{-t})X_+ \rightarrow 0, \quad \mathrm{Ad}(a_t)X_- \rightarrow 0.$$

The groups U^+ and U^- are the expanding and contracting horospherical groups, while $Z_G(H)$ is neutral.

A local product map

$$U^- \times Z_G(H) \times U^+ \rightarrow G$$

is a diffeomorphism near the identity. This gives coordinates in which a small perturbation can be split according to how a_t will treat it.

N.2 The parameter-shift argument for an open horospherical piece

Let $\Omega \subset U^+$ be a bounded open set with boundary of Haar measure zero, and define

$$\mu_t(f) = \frac{1}{m_U(\Omega)} \int_{\Omega} f(xua_t) du.$$

Fix $u_r = \exp(rX)$ in a one-parameter root subgroup of U^+ of weight $\lambda > 0$. Then

$$a_t u_r a_{-t} = u_{e^{\lambda t} r}.$$

To create a fixed u_r after translation by a_t , shift the parameter by the microscopic amount $e^{-\lambda t} r$:

$$(u u_{e^{-\lambda t} r}) a_t = u a_t u_r.$$

Therefore

$$\mu_t(f \circ R_{u_r}) - \frac{1}{m_U(\Omega)} \int_{\Omega u_{e^{-\lambda t} r}} f(xua_t) du = 0.$$

The symmetric difference between Ω and its tiny translate has relative measure tending to zero. Hence every weak limit of μ_t is u_r -invariant. Varying r and the positive root subgroup gives invariance under the full expanding group generated by the active roots.

Warning N.1. Boundary regularity is part of the theorem. If the boundary has positive Haar measure, a tiny translate need not have small symmetric difference, so the parameter-shift proof can fail even though the algebraic identity is correct.

N.3 Tightness before classification

Weak limits are useful only if mass does not disappear into the cusp. The relevant exterior-power functions now depend on the expanding parameter:

$$u \longmapsto \|\rho(ua_t)w\|.$$

For fixed t and algebraic/analytic parameterizations of u , these are good functions. The nondivergence theorem gives a compact set containing most of the translated mass unless an exterior-power vector is uniformly contracted throughout the parameter set.

A uniformly contracted rational wedge defines a rational parabolic. Thus nonescape has the familiar dichotomy:

most mass in a compact set

or

the family is focused in a proper parabolic direction

The second branch is not discarded; it becomes one of the algebraic obstructions in the theorem.

N.4 Classifying the weak limit

Assume tightness and let $\mu_{t_i} \Rightarrow \mu$. The parameter-shift argument makes μ invariant under a nontrivial unipotent group W . Take the ergodic decomposition of μ under W . Ratner classification says each component is homogeneous:

$$\mu_\omega = m_{x_\omega L_\omega}.$$

To prove ambient equidistribution we must rule out proper L_ω . This is the focusing problem.

N.5 Why the inverse representation appears on a right quotient

Suppose $X = \Gamma \backslash G$ and a point Γg lies near a support orbit associated with L . Let p_L be a Chevalley vector whose projective stabilizer is $N_G(L)$. The lattice ambiguity acts on the left, so the quantity which becomes close to the discrete $\rho(\Gamma)$ -orbit is

$$\rho(g^{-1})p_L,$$

not $\rho(g)p_L$. Along the expanding trajectory the correct linearized vector is therefore

$$\rho((ua_t)^{-1})p_L = \rho(a_t^{-1}u^{-1})p_L.$$

This convention is forced by the quotient orientation. Using the direct representation would give a formally similar expression but the wrong lattice action.

N.6 Linearized focusing

Let $\mathcal{O}_L = \rho(\Gamma)p_L$, a discrete subset on compact representation windows. If a positive fraction of parameters $u \in \Omega$ places xua_t near a translate of L , then $\rho(a_t^{-1}u^{-1})p_L$ lies near $\mathcal{O}_L$ for the same parameters. After choosing the nearest lattice vector on a local uniqueness window, each coordinate is an algebraic or analytic function of u .

Good-function estimates yield a dichotomy. Either the vector leaves the small neighborhood for a definite fraction of parameters, contradicting the positive limiting mass on the proper homogeneous component, or the whole parameterized vector is trapped in a fixed algebraic subspace. The latter says that the original expanding piece is focused by a proper algebraic normalizer.

The theorem's nonfocusing hypothesis excludes this exact relation. Hence no proper component survives and the weak limit is the ambient homogeneous measure.

N.7 Closed homogeneous orbits instead of open horospheres

Let $Y = xH$ be closed of finite volume and study $m_Y a_t$. There is no parameter domain with a simple Euclidean boundary, but Volume VII supplies the correct replacement: sequences of translated homogeneous measures have homogeneous limits, and focusing describes how a proper limit can occur.

If $m_Y a_{t_i} \Rightarrow m_{zL}$ is proper, Mozes–Shah support containment puts the conjugated support groups $a_{-t_i} H a_{t_i}$, after small correction, inside L . In a representation this becomes a weight statement: the Chevalley vector of H has no component in weights which expansion would force away from the L -line. This is an algebraic focusing condition. If that condition is absent, the only possible limit is ambient Haar.

N.8 Curves: why jets replace open-set translations

Let $\varphi : I \rightarrow U^+$ be analytic and consider $x\varphi(s)a_t$. A small translation of the parameter s does not correspond to a fixed right multiplication as it did for an open subgroup piece. Expand instead at s_0 :

$$\varphi(s_0 + h) = \varphi(s_0) \exp \left(hY_1 + h^2Y_2 + \cdots + h^rY_r + O(h^{r+1}) \right).$$

Conjugation by a_t assigns exponential weights to the Y_j . Choose $h = h(t)$ so that the first active weighted term has order one. All lower weighted terms vanish and all higher terms remain controlled on a suitable microscopic block. The limit is a nontrivial unipotent displacement, giving additional invariance of any weak limit.

Worked Example N.2 (A curve in SL_3). Take

$$\varphi(s) = \exp(sE_{12} + s^2E_{13}), \quad a_t = \text{diag}(e^{2t}, e^{-t}, e^{-t}).$$

Both E_{12} and E_{13} have weight 3. Near $s_0 = 0$, the linear term hE_{12} dominates h^2E_{13} . Choosing $h = e^{-3t}$ makes the conjugated linear term order one and the quadratic term order e^{-3t} . Thus the first created invariance is U_{12} . At a point where the first derivative vanishes, a higher jet can become the first active term, which is why the curve theorem must control a finite collection of jet orders rather than only first derivatives.

N.9 Shrinking pieces

Now let the parameter interval itself shrink:

$$s = s_0 + r_t y, \quad r_t \rightarrow 0.$$

Taylor expansion gives

$$\varphi(s_0 + r_t y) = \varphi(s_0) \exp\left(r_t y Y_1 + r_t^2 y^2 Y_2 + \cdots\right).$$

After conjugation by a_t , the j th term has effective size $r_t^j e^{\lambda_j t}$. A nontrivial tangent model exists precisely when at least one of these quantities stays at order one. If all tend to zero, the piece collapses to one point at the dynamical scale; if one diverges too fast, renormalize at an earlier time or a smaller subpiece. The finite active-jet lemma in the body makes this case division uniform.

N.10 Nonuniform stretching and submanifolds

For a k -dimensional parameter map, different tangent directions can have different expansion rates. Choose anisotropic coordinates

$$s_j = s_{0,j} + r_{j,t} y_j$$

so that the desired tangent monomials are order one after conjugation. The limit is a polynomial tangent map of the type studied in Volume VIII. Thus shrinking/stretched submanifold problems reduce locally to polynomial trajectory limits, while the global focusing argument remains the one in the present volume.

N.11 The five-step proof template

Every expanding-translate theorem in this volume follows the same template:

1. prove tightness or identify a rational parabolic escape obstruction;
2. create unipotent invariance of weak limits by parameter shifts or jets;
3. apply Ratner/Mozes–Shah to classify the limits;
4. linearize any proper homogeneous component with the inverse Chevalley representation appropriate to the right quotient;
5. use nonfocusing to eliminate that component.

The geometry of the parameter set changes from open horospherical pieces to curves and submanifolds, but these five logical jobs do not.

Proof Guide and Dependency Summary

This chapter is a finalization layer. It does not introduce a new theorem scope. Its purpose is to expose the proof architecture, make the hypotheses that are easiest to misread impossible to miss, and give a student a concrete model calculation before the abstract machinery is reused later.

1. How expansion creates unipotent invariance

Let a_t expand an unstable horospherical subgroup U^+ . For a parameterized piece $u(\mathbf{s})$, compare nearby parameters $\mathbf{s}$ and $\mathbf{s} + \delta_t \mathbf{r}$, with δ_t chosen so that

$$a_t^{-1}u(\mathbf{s})^{-1}u(\mathbf{s} + \delta_t \mathbf{r})a_t$$

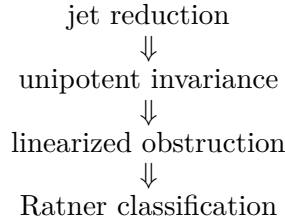
has a nontrivial limit. A change of variables in the parameter average then forces each weak limit to be invariant under the resulting unipotent one-parameter subgroup.

2. Right quotients and the inverse representation

On a right quotient, subgroup containment encoded by a Chevalley vector transforms with the inverse action. If the geometric point is represented by right multiplication by g , the linearization coordinate must use the corresponding inverse representation. Forgetting this inverse changes the stabilizer equations and can reverse the obstruction. The convention is therefore part of the theorem, not typography.

3. Curves: finite jets before Ratner

For an analytic/nondegenerate curve, the local parameter shift is reduced to a finite jet. After the expanding conjugation, the first surviving jet term defines a polynomial tangent model. The proof then has four logically separate stages:



For shrinking intervals, the interval scale must be large enough for the surviving jet to be visible after conjugation; otherwise the limit can collapse to a point and no invariance is created.

4. Student diagnostic

In every curve theorem, identify explicitly: the parameter scale, the first nonzero jet, the generated unipotent subgroup, and the algebraic nondegeneracy condition ruling out a proper homogeneous support.

5. Dependency status

The analytic-curve and expanding-translate arguments in this volume have been checked locally. Their only inherited structural input is the proper/discrete fixed-support Chevalley lift from Volumes III and VII. That lift is derived in Volume III from the external Dani–Margulis structural interface. Consequently the analytic-curve arguments introduce no additional research-level structural input.

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Counting on Homogeneous Varieties
Volume X of a Detailed Notebook on Ratner's Theorems

Detailed Notebook on Ratner's Theorems

August 2026

Preface and proof policy

Arc II begins by turning qualitative equidistribution into asymptotic counting. This volume deliberately remains non-effective: it identifies the main term, proves the geometric wavefront and well-roundedness statements, and shows exactly how mixing or homogeneous-measure rigidity produces a count. Error terms and spectral rates are postponed to Volumes XI–XII. The Duke–Rudnick–Sarnak and Eskin–McMullen frameworks are reconstructed in the special generality needed by this series rather than cited as finished black boxes.

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Dependency and proof-ownership contract

Purpose. Homogeneous counting.

Prerequisites. Volumes V–IX.

New objects. HAK/polar geometry, unfolding, well-roundedness. Every project-specific technical term is defined before its first load-bearing use or recalled by an exact backward reference.

Main outputs. qualitative counting asymptotics.

External-input rule. Every nontrivial imported result is either part of the declared graduate-level prerequisites, an explicitly named external research input in the prerequisite map, or is proved in this series before use (possibly under an explicit deferred-proof contract). A citation by itself never constitutes a proof.

Reader map and proof route

Question of the volume. How does equidistribution become an asymptotic count on a homogeneous variety, and which geometric hypotheses make the passage legitimate?

Prerequisite checkpoint. Use Volume VI for mixing and Volumes VII–IX for homogeneous limits and expanding translates. This volume owns the unfolding, wavefront, well-roundedness, and main-term conversion in the generality stated here.

Load-bearing route. Write the count as an unfolded integral, smooth it, apply the dynamical limit, control the boundary, and then remove the smoothing. The order of limits matters in this qualitative volume. Volume XII will replace it by one optimized finite-scale choice.

Sanity check. Count primitive lattice vectors in a large planar region by first writing a periodized test function on the space of lattices. Even in this simple model, the boundary of the region and the normalization of Haar measure are separate pieces of the main-term calculation.

Chapter 1

From Arithmetic Orbits to Counting Functions

1.1 The homogeneous dictionary

Let $\mathbf{G}$ be a rational algebraic group, $G = \mathbf{G}(\mathbb{R})^\circ$, and $\Gamma < G$ arithmetic. Suppose G acts transitively on a real affine variety V and fix $v_0 \in V(\mathbb{Q})$. Let $H = \text{Stab}_G(v_0)$. The orbit map identifies

$$H \backslash G \longrightarrow v_0 G, \quad Hg \mapsto v_0 g.$$

If the integral orbit $v_0 \Gamma$ has finite multiplicity inside $V(\mathbb{Z})$, counting integral points becomes counting cosets $(H \cap \Gamma) \backslash \Gamma$ subject to a geometric condition on Hg .

Lemma 1.1 (Orbit–coset bijection). *The map*

$$(H \cap \Gamma) \backslash \Gamma \rightarrow v_0 \Gamma, \quad (H \cap \Gamma) \gamma \mapsto v_0 \gamma$$

is bijective.

Proof. For every $v_0 \gamma$ in the orbit there is the coset $(H \cap \Gamma) \gamma$ mapping to it, so the map is surjective. Two elements have the same image precisely when $\gamma_1 \gamma_2^{-1}$ stabilizes v_0 , i.e. lies in $H \cap \Gamma$. $\square$

Thus for a family $B_T \subset H \backslash G$,

$$N(T) = \# \{ (H \cap \Gamma) \gamma : H \gamma \in B_T \}.$$

1.2 The volume heuristic

Let $m_{H \backslash G}$ be the invariant quotient measure when the modular characters match. With compatible Haar measures satisfying $dg = dh d(Hg)$, one expects

$$N(T) \sim \frac{m_H((H \cap \Gamma) \backslash H)}{m_G(\Gamma \backslash G)} m_{H \backslash G}(B_T). \quad (1.1)$$

This normalization is forced by rescaling dh : multiplying dh by c multiplies the H -covolume by c and divides the quotient measure on $H \backslash G$ by c , leaving the product in (1.1) unchanged. The rest of the volume explains why this heuristic is correct for well-rounded families.

Chapter 2

Unfolding and Thickening

2.1 Periodization

For $\psi \in C_c(H \backslash G)$ define its periodization on $\Gamma \backslash G$ by

$$\mathcal{P}\psi(\Gamma g) = \sum_{\gamma \in (H \cap \Gamma) \backslash \Gamma} \psi(H\gamma g).$$

The sum is locally finite in the applications here because $(H \cap \Gamma) \backslash H \rightarrow \Gamma \backslash G$ is a proper closed homogeneous orbit (Volume V and Volume III, Appendix A). Equivalently the right Γ -orbit of the basepoint in $H \backslash G$ is locally finite. This properness hypothesis is part of every counting theorem below.

Lemma 2.1 (Unfolding identity). *For $F \in C_c(\Gamma \backslash G)$,*

$$\int_{\Gamma \backslash G} F(g) \mathcal{P}\psi(g) dg = \int_{(H \cap \Gamma) \backslash G} F(g) \psi(Hg) dg.$$

If the quotient measure disintegrates as $dh d(Hg)$, the right side equals

$$\int_{H \backslash G} \psi(Hg) \left(\int_{(H \cap \Gamma) \backslash H} F(hg) dh \right) d(Hg).$$

Proof. Expand the periodization and integrate term by term; local finiteness justifies the exchange. In the term indexed by γ , change variables $g \mapsto \gamma^{-1}g$. The union of the corresponding translates of a fundamental domain for Γ is a fundamental domain for $H \cap \Gamma$ in G . Weil integration gives the second formula. $\square$

2.2 Smoothing a point count

Let ρ_ε be a nonnegative bump of integral one supported in a small identity neighborhood $U_\varepsilon < G$. A sharp count in B_T is bounded between periodized smoothed indicators of the inner and outer neighborhoods

$$B_T^- = \bigcap_{u \in U_\varepsilon} B_T u, \quad B_T^+ = \bigcup_{u \in U_\varepsilon} B_T u.$$

This elementary sandwich is the source of the boundary error in every counting theorem.

Chapter 3

Symmetric Spaces and Generalized Cartan Decomposition

3.1 Symmetric pairs

Let $\sigma : G \rightarrow G$ be an involution and let H be an open subgroup of the fixed-point group. Choose a Cartan involution θ commuting with σ . At the Lie-algebra level

$$\mathfrak{g} = (\mathfrak{h} \oplus \mathfrak{q}) = (\mathfrak{k} \oplus \mathfrak{p}),$$

where $\mathfrak{q}$ is the -1 eigenspace of σ and $\mathfrak{p}$ the -1 eigenspace of θ . Choose a maximal abelian $\mathfrak{a} \subset \mathfrak{p} \cap \mathfrak{q}$ and a positive chamber $A^+ = \exp \mathfrak{a}^+$.

Theorem 3.1 (Generalized Cartan decomposition). *Every $g \in G$ can be written*

$$g = hak, \quad h \in H, \quad a \in A^+, \quad k \in K.$$

The A^+ coordinate is unique up to the finite Weyl group of the symmetric pair.

Proof. Equip the Riemannian symmetric space G/K with its nonpositively curved metric. The H -orbit of the basepoint has normal space $\mathfrak{p} \cap \mathfrak{q}$. For any gK , minimize distance to the closed H -orbit HK/K ; a minimizing geodesic leaves the orbit orthogonally, hence has initial tangent in an $H \cap K$ -conjugate of $\mathfrak{a}$. Thus $gK = haK$ and $g = hak$. Two normal geodesics representing the same double coset differ by the normalizer of $\mathfrak{a}$ in $H \cap K$, giving the finite

Weyl ambiguity. The properness, closedness, orthogonality, and maximal-flat conjugacy steps are proved in Appendix A; the present paragraph is only their assembly into the displayed decomposition. $\square$

3.2 Polar density

Haar measure disintegrates as

$$dg = J(a) dh da dk,$$

where, in logarithmic coordinates $a = e^Y$, $J(e^Y)$ is a finite product of powers of $|\sinh \alpha(Y)|$ and $|\cosh \alpha(Y)|$ over restricted roots with multiplicities determined by the symmetric pair. For counting main terms only the asymptotic

$$J(e^Y) = e^{2\rho(Y)}(c(Y) + o(1))$$

in each chamber face is needed; it follows immediately by replacing every $\sinh z$ and $\cosh z$ with $e^z/2$ away from the relevant walls.

Chapter 4

The Wavefront Lemma

4.1 Statement away from walls

Let A_η^+ be the set of $a = e^Y$ with every simple root not forced to vanish on the chosen face satisfying $\alpha(Y) \geq \eta$.

Theorem 4.1 (Strong wavefront lemma). *For every $\eta > 0$ there are C_η and $\epsilon_0 > 0$ such that, if $0 < \epsilon < \epsilon_0$ and U_ϵ is the exponential ϵ -ball at the identity, then for all $a \in A_\eta^+$,*

$$HaKU_\epsilon \subset H a A_{C_\eta \epsilon} K_{C_\eta \epsilon}, \quad (4.1)$$

where $A_r = \exp\{Y \in \mathfrak{a} : \|Y\| < r\}$ and $K_r = K \exp(B_r(0) \cap \mathfrak{k})$ denotes a fixed small thickening of K . On a fixed chamber face the central Levi factor generated by the vanishing roots is included with the neutral directions; the same estimate holds uniformly on the relative η -interior of that face.

Proof. It is enough to factor au for u near the identity, since the arbitrary left H and right K factors in HaK can be absorbed by conjugating the small neighborhood by the compact group K . Consider

$$\Phi_a : (H \cap U_0) \times (A \cap U_0) \times (K \cap U_0) \longrightarrow G, \quad (h, b, k) \mapsto habk.$$

After left translation by a^{-1} , its differential at (e, e, e) has image

$$\mathrm{Ad}(a^{-1})\mathfrak{h} + \mathfrak{a} + \mathfrak{k}. \quad (4.2)$$

For a symmetric pair this sum is all of $\mathfrak{g}$. We need a right inverse uniform in a . Decompose $\mathfrak{g}$ into the neutral space and restricted root pairs $\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}$. On

such a pair, $\mathfrak{h}$ and $\mathfrak{k}$ are graphs of the commuting involutions induced by σ and θ . Conjugation by $a = e^Y$ multiplies the two root coordinates by $e^{-\alpha(Y)}$ and $e^{\alpha(Y)}$. The resulting 2×2 block for the map in (4.2) has smallest singular value bounded below by a positive constant depending only on η whenever α is active on the chosen face. The zero-weight and face-central directions are covered by $\mathfrak{a}$ and the neutral Levi factor. Since there are finitely many root blocks, (4.2) admits a right inverse of norm at most C_η .

The quantitative inverse function theorem, applied with this uniform right inverse and uniform second-derivative bounds for multiplication on a fixed identity chart, now gives: every $u \in U_\epsilon$ has a factorization

$$au = h_u a b_u k_u, \quad d(h_u, e) + d(b_u, e) + d(k_u, e) \leq C_\eta \epsilon. \quad (4.3)$$

The small h_u is absorbed by the left H -coset. This yields (4.1). The same block calculation after deleting the roots that vanish identically on a fixed face proves the relative-face statement. $\square$

Remark 4.2. The wavefront lemma is geometric, not ergodic. It says that far enough into a Cartan chamber, small ambient perturbations do not create a large error in the H and K coordinates. This is what lets a smoothed count retain the same radial cutoff.

Chapter 5

Well-Rounded Families

5.1 Definition

Definition 5.1 (Well-rounded family). Let $B_T \subset H \backslash G$ be measurable with $0 < m_{H \backslash G}(B_T) < \infty$ for all sufficiently large T and $m_{H \backslash G}(B_T) \rightarrow \infty$. For a sufficiently small identity neighborhood $U_\varepsilon < G$, define

$$B_T^-(\varepsilon) = \bigcap_{u \in U_\varepsilon} B_T u, \quad B_T^+(\varepsilon) = \bigcup_{u \in U_\varepsilon} B_T u.$$

The family (B_T) is *well rounded* if

$$\lim_{\varepsilon \downarrow 0} \limsup_{T \rightarrow \infty} \frac{m_{H \backslash G}(B_T^+(\varepsilon) \setminus B_T^-(\varepsilon))}{m_{H \backslash G}(B_T)} = 0.$$

Equivalently, for every $\eta > 0$ there is $\varepsilon_0 > 0$ such that for $0 < \varepsilon < \varepsilon_0$ the displayed ratio is at most η for all sufficiently large T .

Proposition 5.2 (Radial norm balls are well-rounded). *Let $\rho : G \rightarrow \mathrm{GL}(V)$ be a finite-dimensional representation and let $B_T = \{Hg : \|v_0 \rho(g)\| < T\}$ be a proper norm ball in a symmetric homogeneous orbit. If the leading polar-volume exponent is not concentrated on a chamber wall of equal exponential order, then B_T is well-rounded.*

Proof. In polar coordinates, right multiplication by a small u changes the logarithmic A -coordinate by $O(\varepsilon)$ and changes only the harmless compact coordinate by $O(\varepsilon)$, by the wavefront lemma. The left H change is invisible in $H \backslash G$. The norm condition changes therefore from T to $e^{O(\varepsilon)}T$. The polar density

is a finite sum of exponential-polynomial terms, so $m(B_{e^{c\varepsilon}T})/m(B_T) \rightarrow e^{c\varepsilon a}$ for the leading exponent a ; letting $\varepsilon \rightarrow 0$ gives one. A fixed small neighborhood of chamber walls has lower exponential volume by the hypothesis and contributes $o(m(B_T))$. $\square$

5.2 Volume asymptotics

The polyhedral Laplace theorem proved in Appendix D gives

$$m(B_T) \sim cT^a(\log T)^{b-1}$$

when the norm is algebraic and the polar density has finitely many weights. For completeness, write the radial variables $s_i = \log \alpha_i(a)$; both the log norm and $\log J(a)$ are maxima/linear combinations of finitely many linear forms. Subdivide the cone into finitely many polyhedral regions where the maxima are linear. On each region the volume integral is an elementary exponential integral over a polytope dilated by $\log T$; the face on which the exponent is maximal determines a and its dimension gives the logarithm power $b - 1$.

Chapter 6

The Mixing Counting Theorem

6.1 Abstract statement

Theorem 6.1 (Counting from averaged mixing). *Let G be semisimple, let $\Gamma < G$ be a lattice, and let $H < G$ be closed with $H \cap \Gamma$ a lattice. Write*

$$V_G = m_G(\Gamma \backslash G), \quad V_H = m_H((H \cap \Gamma) \backslash H),$$

and choose quotient measure $m_{H \backslash G}$ so that $dg = dh d(Hg)$. Let $B_T \subset H \backslash G$ be a well-rounded family with $m_{H \backslash G}(B_T) \rightarrow \infty$.

For every sufficiently small $\varepsilon > 0$, choose a symmetric identity neighborhood $U_\varepsilon < G$ for which $U_\varepsilon \rightarrow \Gamma \backslash G$ is injective, use this same U_ε in the thickenings of Definition 5.1, and choose a nonnegative $\rho_\varepsilon \in C_c(\Gamma \backslash G)$ of integral one supported in the image of U_ε . Define its H -period by

$$P_\varepsilon(Hg) = \int_{(H \cap \Gamma) \backslash H} \rho_\varepsilon(\Gamma hg) dh.$$

Because $H \cap \Gamma$ is a lattice, H is unimodular; hence right translation preserves the quotient Haar measure on $(H \cap \Gamma) \backslash H$, so P_ε is well defined as a function on $H \backslash G$. Assume the following averaged mixing condition, with the order of quantifiers read literally: for every fixed sufficiently small $\varepsilon > 0$ and for each sign $\sigma \in \{-, +\}$,

$$\frac{1}{m_{H \backslash G}(B_T^\sigma(\varepsilon))} \int_{B_T^\sigma(\varepsilon)} \left| P_\varepsilon(Hg) - \frac{V_H}{V_G} \right| d(Hg) \longrightarrow 0 \quad (T \rightarrow \infty). \quad (6.1)$$

Here $B_T^\pm(\varepsilon)$ are the inner and outer thickenings from Definition 5.1. Then

$$N(T) := \#\{(H \cap \Gamma)\gamma : H\gamma \in B_T\} \sim \frac{V_H}{V_G} m_{H \setminus G}(B_T). \quad (6.2)$$

Proof. Let $B_T^-(\varepsilon) \subset B_T \subset B_T^+(\varepsilon)$ be the inner and outer ε -thickenings from Definition 5.1. We derive the smoothing inequality rather than treating it as a formal unfolding slogan. For a measurable $C \subset H \setminus G$ define

$$\mathcal{P}\mathbf{1}_C(\Gamma g) = \sum_{\gamma \in (H \cap \Gamma) \setminus \Gamma} \mathbf{1}_C(H\gamma g).$$

If g lies in the identity neighborhood supporting ρ_ε , then

$$\mathcal{P}\mathbf{1}_{B_T^-(\varepsilon)}(\Gamma g) \leq \sum_{\gamma} \mathbf{1}_{B_T}(H\gamma) \leq \mathcal{P}\mathbf{1}_{B_T^+(\varepsilon)}(\Gamma g). \quad (6.3a)$$

Indeed, $H\gamma g \in B_T^-(\varepsilon) \subset B_T g$ implies $H\gamma \in B_T$, while $H\gamma \in B_T$ implies $H\gamma g \in B_T g \subset B_T^+(\varepsilon)$. The middle sum is exactly $N(T)$. Multiply (6.3a) by the nonnegative function ρ_ε and integrate over $\Gamma \setminus G$. Since $\int \rho_\varepsilon = 1$, the middle term remains $N(T)$. Applying the unfolding identity Lemma 2.1 to the two outer terms gives

$$\int_{B_T^-(\varepsilon)} P_\varepsilon(Hg) d(Hg) \leq N(T) \leq \int_{B_T^+(\varepsilon)} P_\varepsilon(Hg) d(Hg). \quad (6.3)$$

For fixed ε , hypothesis (6.1) gives separately for the two signs

$$\int_{B_T^\pm(\varepsilon)} P_\varepsilon(Hg) d(Hg) = \frac{V_H}{V_G} m_{H \setminus G}(B_T^\pm(\varepsilon)) + o_{T,\varepsilon}(m_{H \setminus G}(B_T^\pm(\varepsilon))). \quad (6.4)$$

No estimate of P_ε on the boundary layer is being smuggled into this step; this is precisely why (6.1) is stated for both fixed thickenings. Well-roundedness gives

$$\lim_{\varepsilon \downarrow 0} \limsup_{T \rightarrow \infty} \frac{m(B_T^+(\varepsilon) \setminus B_T^-(\varepsilon))}{m(B_T)} = 0.$$

Since

$$B_T^-(\varepsilon) \subset B_T \subset B_T^+(\varepsilon),$$

the boundary estimate implies

$$1 - \delta(\varepsilon) \leq \liminf_{T \rightarrow \infty} \frac{m(B_T^-(\varepsilon))}{m(B_T)} \leq 1 \leq \limsup_{T \rightarrow \infty} \frac{m(B_T^+(\varepsilon))}{m(B_T)} \leq 1 + \delta(\varepsilon),$$

where

$$\delta(\varepsilon) := \limsup_{T \rightarrow \infty} \frac{m(B_T^+(\varepsilon) \setminus B_T^-(\varepsilon))}{m(B_T)} \longrightarrow 0 \quad (\varepsilon \downarrow 0).$$

Divide (6.3) by $m(B_T)$ and first let $T \rightarrow \infty$ with ε fixed. Equation (6.4) and the preceding volume comparison squeeze the lower and upper limits of $N(T)/m(B_T)$ between $(V_H/V_G)(1 - \delta(\varepsilon))$ and $(V_H/V_G)(1 + \delta(\varepsilon))$. Finally let $\varepsilon \downarrow 0$ to obtain (6.2). $\square$

Remark 6.2 (How ordinary mixing verifies (6.1)). In the symmetric-space applications of this volume, the strong wavefront lemma identifies $P_\varepsilon(Hak)$ with a thickened H -orbit correlation. Howe–Moore mixing makes that correlation tend to V_H/V_G whenever the radial parameter a leaves every compact subset of the mixing chamber. The same statement applies to each fixed inner or outer thickening $B_T^\pm(\varepsilon)$, because the wavefront perturbation changes the radial and compact coordinates only by a bounded $O(\varepsilon)$ amount. If the normalized polar volume of the part of these thickened families with bounded radial parameter tends to zero, dominated truncation gives (6.1). This is the precise content of the informal phrase “the mass escapes to mixing directions.”

Chapter 7

Affine Symmetric Varieties and the Eskin–McMullen Framework

7.1 Integral points on an affine symmetric orbit

Let $V = v_0G \simeq H \backslash G$ be affine symmetric and defined over $\mathbb{Q}$. Assume the integral points decompose into finitely many Γ -orbits,

$$V(\mathbb{Z}) = \bigsqcup_{j=1}^r v_j \Gamma.$$

For each orbit, $H_j = \text{Stab}_G(v_j)$ has an arithmetic lattice $H_j \cap \Gamma$ by Volume V. Apply Theorem 6.1 and sum.

Theorem 7.1 (Affine symmetric counting). *Let $B_T \subset V(\mathbb{R})$ be a norm-ball family. For each integral orbit $v_j \Gamma$, identify $v_j G$ with $H_j \backslash G$ and let $B_{T,j}$ be the induced family. Assume that every $(B_{T,j})$ is well rounded and, for every fixed sufficiently small $\varepsilon > 0$, satisfies the two-thickening averaged-mixing hypothesis (6.1) of Theorem 6.1 for both $B_{T,j}^-(\varepsilon)$ and $B_{T,j}^+(\varepsilon)$. Then*

$$\#(V(\mathbb{Z}) \cap B_T) \sim c_V m_V(B_T),$$

where c_V is the finite sum of the orbit covolume constants, with each constant computed using the quotient-measure normalization of Theorem 6.1.

Proof. For the j -th orbit, the hypotheses are exactly those of Theorem 6.1 with $H = H_j$; hence that theorem gives the corresponding orbit count with constant V_{H_j}/V_G . Arithmetic closedness and finite covolume of $H_j \cap \Gamma$ are supplied by Volume V. Because the decomposition of $V(\mathbb{Z})$ contains only finitely many Γ -orbits, summing these finitely many asymptotics gives the stated constant c_V and the asserted total asymptotic. $\square$

7.2 Relation with Duke–Rudnick–Sarnak

The analytic/automorphic proof of Duke–Rudnick–Sarnak and the mixing proof of Eskin–McMullen reach the same leading-volume law by different routes. For this notebook, the mixing route is structurally preferable because its inputs—arithmetic stabilizers, semisimple mixing, wavefront geometry—have all been proved in Volumes V, VI, and X.

Chapter 8

A Ratner–Mozes–Shah Counting Engine

8.1 When the moving objects are themselves homogeneous

Some counts are naturally encoded by a sequence of closed homogeneous orbits rather than by translating one fixed symmetric subgroup through a mixing Cartan direction. In that case Volume VII can replace ambient mixing.

Theorem 8.1 (Weighted counting from homogeneous-measure limits). *For each T , let $\mathcal{O}_T$ be a finite family of closed finite-volume homogeneous orbits in X . For $O \in \mathcal{O}_T$, let m_O be the normalized homogeneous probability on O and let $w_T(O) > 0$. Put*

$$W_T = \sum_{O \in \mathcal{O}_T} w_T(O), \quad \nu_T = \frac{1}{W_T} \sum_{O \in \mathcal{O}_T} w_T(O) m_O.$$

If $\nu_T \Rightarrow m_X$ and $B \subset X$ is Borel with $m_X(\partial B) = 0$, then

$$\frac{1}{W_T} \sum_{O \in \mathcal{O}_T} w_T(O) m_O(B) \longrightarrow m_X(B). \quad (8.1)$$

If, in addition, $w_T(O)/\bar{w}_T \rightarrow 1$ uniformly in $O \in \mathcal{O}_T$ for some $\bar{w}_T > 0$, then the same conclusion holds with all weights removed:

$$\frac{1}{|\mathcal{O}_T|} \sum_{O \in \mathcal{O}_T} m_O(B) \longrightarrow m_X(B). \quad (8.2)$$

Proof. Equation (8.1) is exactly $\nu_T(B) \rightarrow m_X(B)$, which follows from weak convergence by the Portmanteau theorem because $m_X(\partial B) = 0$. For (8.2), write $w_T(O) = \bar{w}_T(1 + e_T(O))$ with $\sup_O |e_T(O)| \rightarrow 0$. Since $0 \leq m_O(B) \leq 1$ and $W_T = \bar{w}_T |\mathcal{O}_T| (1 + o(1))$, replacing the weighted average by the unweighted one changes it by $o(1)$. $\square$

The hard work in applications is proving the convergence of the homogeneous measures. Volume VII reduces that to two tasks: nonescape and exclusion of proper focusing. This approach is especially natural when the subgroups vary arithmetically.

Chapter 9

Worked Counting Laboratories

9.1 Primitive vectors

Primitive integer vectors of Euclidean norm at most T form the orbit of a primitive base vector under $\mathrm{SL}_n(\mathbb{Z})$. The stabilizer is a maximal parabolic, so the quotient is not finite volume; nevertheless the same unfolding and polar-volume calculation, after dividing the unipotent radical by its integer lattice, recovers the familiar leading constant proportional to $T^n/\zeta(n)$. This example shows how a parabolic stabilizer modifies but does not destroy the homogeneous dictionary.

9.2 Quadrics

For an integral indefinite quadratic form Q , the level set $Q(v) = m$ is an orthogonal homogeneous variety whenever it is nonempty and nonsingular. The stabilizer of a point is another orthogonal group. In signatures for which the relevant real orbit is symmetric, assume that the integral points split into finitely many arithmetic orbits. Then Theorem 7.1 gives the leading count from the invariant volume of the norm ball on the quadric.

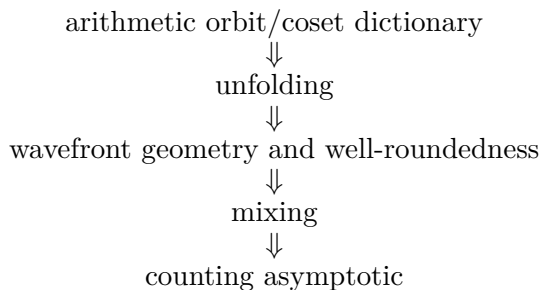
9.3 Matrices with prescribed invariants

Conjugacy orbits of semisimple matrices are homogeneous under SL_n with centralizer stabilizer. If fixing an integral characteristic polynomial produces only finitely many arithmetic orbits, the same framework counts matrices of bounded norm. Noncompact centralizers introduce chamber faces; the well-roundedness proof must then keep the corresponding neutral factors rather than discard them.

Chapter 10

Proof Ledger

The non-effective counting mechanism is now explicit:



A second route uses

Mozes–Shah equidistribution of varying homogeneous measures
→ boundary-zero evaluation → counting.

No numerical rate occurs in either theorem. Volume XI now builds the spectral/Sobolev apparatus needed to make the first route effective; Volume XIII will make the unipotent/focusing route effective.

Appendix A

Symmetric-Pair Geometry behind HA^+K

Let σ and θ be commuting involutions. Their simultaneous eigenspace decomposition is

$$\mathfrak{g} = (\mathfrak{h} \cap \mathfrak{k}) \oplus (\mathfrak{h} \cap \mathfrak{p}) \oplus (\mathfrak{q} \cap \mathfrak{k}) \oplus (\mathfrak{q} \cap \mathfrak{p}).$$

The G -invariant metric on G/K is the one proved complete in Volume V.

Lemma A.1 (The H -orbit is closed and totally geodesic). *The orbit $H \cdot o \subset G/K$ is a closed totally geodesic submanifold, with tangent space $\mathfrak{h} \cap \mathfrak{p}$ and normal space $\mathfrak{q} \cap \mathfrak{p}$ at o .*

Proof. The involution σ descends to an isometry of G/K because it commutes with θ and hence preserves K . The identity component of its fixed point set through o is $H \cdot o$. A fixed-point set of an isometry is closed. If a geodesic starts tangent to the fixed set, applying the isometry produces another geodesic with the same initial data, so uniqueness makes the whole geodesic fixed; hence the fixed set is totally geodesic. The tangent and normal spaces are the ± 1 eigenspaces of $d\sigma$ inside $\mathfrak{p}$. $\square$

Lemma A.2 (Existence of a nearest point). *Every $z \in G/K$ has a nearest point on $H \cdot o$.*

Proof. The symmetric-space metric is complete and finite dimensional, hence proper by the Hopf–Rinow theorem proved in Volume V: closed bounded balls are compact. Let d_0 be the infimum of the distance from z to the closed

set $H \cdot o$. A minimizing sequence lies in a closed ball of radius $d_0 + 1$ around z , hence has a convergent subsequence; closedness of the orbit puts its limit in $H \cdot o$. $\square$

Lemma A.3 (Orthogonality of a minimizing segment). *If a minimizing geodesic from $y \in H \cdot o$ to z realizes the distance to the orbit, then its initial velocity is normal to $H \cdot o$.*

Proof. Let $c(s)$ be any smooth curve in the orbit with $c(0) = y$ and velocity V . The first-variation formula for the squared distance from $c(s)$ to z gives $-2\langle V, \dot{\gamma}(0) \rangle$. Since $s = 0$ is a minimum, this vanishes for every tangent V . $\square$

A.1 Maximal normal flats

Let $\mathfrak{a} \subset \mathfrak{p} \cap \mathfrak{q}$ be maximal abelian. The compact group $H \cap K$ acts orthogonally on $\mathfrak{p} \cap \mathfrak{q}$.

Lemma A.4 (Conjugacy of maximal normal abelian spaces). *Every $X \in \mathfrak{p} \cap \mathfrak{q}$ is conjugate under $H \cap K$ to an element of $\mathfrak{a}$.*

Proof. Extend $\mathbb{R}X$ to a maximal abelian subspace $\mathfrak{b}$ of $\mathfrak{p} \cap \mathfrak{q}$. Choose regular elements $A \in \mathfrak{a}$ and $B \in \mathfrak{b}$, where regularity means that the centralizer in $\mathfrak{p} \cap \mathfrak{q}$ is exactly the chosen maximal abelian space. Such elements are the complement of finitely many restricted root hyperplanes. Maximize $B_\theta(\text{Ad}(k)A, B)$ over the compact group $H \cap K$. Differentiating in $Z \in \mathfrak{h} \cap \mathfrak{k}$ gives

$$B_\theta(Z, [\text{Ad}(k)A, B]) = 0.$$

The bracket belongs to $\mathfrak{h} \cap \mathfrak{k}$: both entries lie in $\mathfrak{p} \cap \mathfrak{q}$, and the two involution signs multiply to $(+1, +1)$. Nondegeneracy therefore gives $[\text{Ad}(k)A, B] = 0$. Regularity of B puts $\text{Ad}(k)A$ in $\mathfrak{b}$, and regularity of A identifies centralizers, yielding $\text{Ad}(k)\mathfrak{a} = \mathfrak{b}$. Thus X is conjugate into $\mathfrak{a}$. $\square$

Complete proof of Theorem 3.1. Given $gK \in G/K$, choose a nearest point $hK \in H \cdot o$. Translate by h^{-1} . The minimizing geodesic now starts at o with initial velocity $X \in \mathfrak{p} \cap \mathfrak{q}$ by the orthogonality lemma, so $gK = h \exp(X)K$. Lemma A.4 conjugates X by $H \cap K$ into $\mathfrak{a}$. The finite reflection group of the restricted symmetric-pair roots moves the result into the chosen chamber $\mathfrak{a}^+$. Absorbing the conjugating compact element into h and K gives $g = hak$.

If $ha_1k = h'a_2k'$, the corresponding normal geodesics from the H -orbit have the same endpoint. We justify the normal-flat uniqueness used here. Along a geodesic $\gamma_Y(t) = \exp(tY)K$ with regular $Y \in \mathfrak{a}^+$, a variation obtained by changing the initial normal flat decomposes into restricted-root spaces. In an α -root space the Jacobi equation has constant coefficient $\alpha(Y)^2$; its solutions are linear combinations of $\sinh(t\alpha(Y))$ and $\cosh(t\alpha(Y))$. A variation which fixes both the initial H -orbit and the terminal point therefore has zero α -component whenever $\alpha(Y) \neq 0$. Regularity means this holds for every restricted root, so the only surviving variation lies in the centralizer of $\mathfrak{a}$ and does not change the A -coordinate. On a wall, exactly the root spaces with $\alpha(Y) = 0$ survive; they generate the stabilizer of that wall. Consequently a_1 and a_2 differ only by the action of

$$W(H, A) = N_{H \cap K}(\mathfrak{a}) / Z_{H \cap K}(\mathfrak{a}).$$

This group is finite because it acts faithfully by permutations and reflections of the finite restricted-root system. Restricting both coordinates to the closed chamber A^+ therefore gives the stated uniqueness modulo the wall stabilizer. $\square$

Appendix B

The Polar Jacobian from Jacobi Fields

B.1 Differential of the normal exponential map

Consider

$$\Phi : H \times A^+ \times K \rightarrow G, \quad (h, e^Y, k) \mapsto he^Y k.$$

Modulo the finite stabilizer, its Jacobian is computed root string by root string. Fix a restricted symmetric-pair root α and choose a root vector X together with the involution-related vectors $\theta X, \sigma X, \theta \sigma X$. The differential in an H direction and a K direction produces, after left translation back to the identity, linear combinations of

$$X \pm e^{-\alpha(Y)} \theta X.$$

The determinant of the resulting 2×2 block is a constant multiple of $\sinh \alpha(Y)$ or $\cosh \alpha(Y)$ depending on the simultaneous σ, θ eigenspaces.

Proposition B.1 (Polar integration formula). *There are nonnegative integers $m_\alpha^\pm$ such that, up to a constant normalization,*

$$J(e^Y) = \prod_{\alpha > 0} |\sinh \alpha(Y)|^{m_\alpha^-} |\cosh \alpha(Y)|^{m_\alpha^+}.$$

Consequently, on every closed subcone staying a positive distance from the walls,

$$J(e^Y) = 2^{-M} e^{2\rho(Y)} \left(1 + O(e^{-c\|Y\|})\right),$$

where $2\rho = \sum_{\alpha > 0} (m_\alpha^- + m_\alpha^+) \alpha$.

Proof. Choose orthonormal bases in each simultaneous root/eigenspace block. The preceding differential calculation block-diagonalizes the Jacobian, and each block contributes one hyperbolic sine or cosine factor. Multiplication of the blocks proves the first formula. On a cone separated from walls, $\alpha(Y) \geq c\|Y\|$ for every active positive root, and

$$\sinh z = \frac{1}{2}e^z(1 - e^{-2z}), \quad \cosh z = \frac{1}{2}e^z(1 + e^{-2z}),$$

which gives the second formula after multiplication. □

Appendix C

The Strong Wavefront Estimate in Root Coordinates

C.1 A single root pair

On the real two-dimensional span of a root pair choose coordinates so that the H and K tangent lines are generated by $(1, 1)$ and $(1, -1)$. After applying $\text{Ad}(a)$ to the K line, it is generated by $(e^{\alpha(Y)}, -e^{-\alpha(Y)})$. The matrix

$$M_\alpha(Y) = \begin{pmatrix} 1 & e^{\alpha(Y)} \\ 1 & -e^{-\alpha(Y)} \end{pmatrix}$$

has determinant $-(e^{\alpha(Y)} + e^{-\alpha(Y)})$, whose modulus is at least 2 and whose normalized inverse is uniformly bounded when the neutral wall directions are split off. Higher multiplicity blocks are tensor copies of the same calculation.

Lemma C.1 (Uniform infinitesimal splitting). *For every fixed chamber face and every $\eta > 0$ there is C_η such that for all a in the η -thick part of that face and every $Z \in \mathfrak{g}$ there are $X \in \mathfrak{h}$, $Y \in \mathfrak{k}$, and Z_0 in the neutral face algebra with*

$$Z = X + \text{Ad}(a)Y + Z_0, \quad \|X\| + \|Y\| + \|Z_0\| \leq C_\eta \|Z\|.$$

Proof. Decompose $\mathfrak{g}$ into the finitely many restricted-root pair blocks and the neutral algebra. Apply the explicit inverse of $M_\alpha(Y)$ on each active block and put the neutral component into Z_0 . The maximum of finitely many block inverse norms is C_η . $\square$

Second proof of Theorem 4.1. Apply the quantitative inverse function theorem to

$$(X, Z_0, Y) \mapsto \exp X \exp Z_0 a \exp Y a^{-1}.$$

Lemma C.1 gives a uniformly bounded right inverse for its derivative at the origin, while the second derivatives are uniformly bounded on one fixed small ball because conjugation on the active root directions is already incorporated in the coordinates. Hence one uniform ball in G can be factored into prescribed small H , neutral-face, and K changes. Conjugating back by a and absorbing the neutral factor according to the chosen face gives the wavefront inclusion. $\square$

Appendix D

A Polyhedral Laplace Theorem

Theorem D.1 (Exponential integral over an expanding polytope). *Let $C \subset \mathbb{R}^r$ be a rational polyhedral cone, let ℓ, q be linear forms with $q > 0$ on $C \setminus \{0\}$, and let P be a polynomial. Put*

$$I(R) = \int_{\{s \in C: q(s) \leq R\}} e^{\ell(s)} P(s) ds.$$

Let

$$a = \max\{\ell(s) : s \in C, q(s) = 1\},$$

and let F be the face on which this maximum is attained. If $b = \dim F + 1$ after including the radial direction and P does not vanish identically on the maximizing face to its leading homogeneous order, then

$$I(R) = c e^{aR} R^{b-1+d_F} (1 + o(1))$$

for an explicit $c > 0$, where d_F is the degree contributed by the leading part of P along F .

Proof. Intersect C with the affine slice $q = 1$; this is a compact polytope after removing directions on which the norm problem is nonproper. Triangulate it so that the maximizing face is a union of simplex faces. On one simplex use coordinates (y, z) , with y tangent to the maximizing face and $z_j \geq 0$ transverse. Linearity gives $\ell/q = a - \sum c_j z_j$ with $c_j > 0$ in the transverse directions. Write $s = r\omega$, $0 \leq r \leq R/q(\omega)$. The radial integral is elementary:

repeated integration by parts gives e^{aR} times a polynomial in R whose leading degree is the radial/tangent polynomial degree. In the transverse variables, rescale $z_j = w_j/R$; the exponential tends to $e^{-\sum c_j w_j}$ and dominated convergence gives the constant product $\prod c_j^{-1}$. Tangent variables contribute the Euclidean volume and the stated power of R . Simplexes not meeting the maximizing face have exponential rate $< a$ and are negligible. Summing the finitely many simplex contributions proves the theorem. $\square$

Corollary D.2 (Algebraic norm-ball volume). *For the norm-ball families in Proposition 5.2,*

$$m(B_T) \sim cT^a(\log T)^{b-1}$$

with a, b determined by the maximizing face of the finite weight polytope.

Proof. Set $R = \log T$. On each chamber region where the representation norm is controlled by one weight, the condition is a linear inequality $q(s) \leq R + O(1)$ and Proposition B.1 expresses the density as a finite exponential-polynomial sum. Apply Theorem D.1 term by term. The maximal exponential rate and largest polynomial degree among tied terms give the claimed constants. $\square$

Appendix E

Counting Constants and the Thickening Limit

E.1 Normalization check

Choose Haar measures so that Weil integration reads

$$\int_G F(g) dg = \int_{H \backslash G} \int_H F(hg) dh d(Hg).$$

Let $V_G = m_G(\Gamma \backslash G)$ and $V_H = m_H((H \cap \Gamma) \backslash H)$. Normalize ambient Haar probability by $V_G^{-1} dg$ and the closed H -orbit probability by $V_H^{-1} dh$. Thickening the latter by a transverse bump of integral one therefore has total mass one. Mixing sends it to ambient Haar probability. Unfolding a count then contributes one copy of V_H^{-1} from the orbit normalization and one V_G^{-1} from the ambient limit, giving exactly the constant in Theorem 6.1.

Lemma E.1 (Order of the two limits). *For the qualitative counting theorem it is enough to take $T \rightarrow \infty$ first with a fixed smoothing radius ε , and only then let $\varepsilon \downarrow 0$.*

Proof. At fixed ε , the thickened functions lie in L^2 and qualitative mixing applies. This gives upper and lower limits involving B_T^+ and B_T^- . Well-roundedness says their relative volume difference tends to zero after the second limit. No uniform rate in ε is needed. Volume XII will instead choose $\varepsilon = \varepsilon(T)$ and therefore must track all Sobolev and boundary losses quantitatively. $\square$

Appendix F

Worked Counting Laboratory: Unfolding and Polar Faces

F.1 Motivation

Two calculations explain most of the bookkeeping in homogeneous counting: unfolding turns a discrete sum into an integral, and a polyhedral Laplace integral determines the power of T and the logarithm.

Worked Example F.1 (The unfolding identity). For a compactly supported function f on G define $F(\Gamma g) = \sum_{\gamma \in \Gamma} f(\gamma g)$. If $\mathcal{F}$ is a fundamental domain, absolute convergence permits

$$\int_{\Gamma \backslash G} F(\Gamma g) dg = \sum_{\gamma \in \Gamma} \int_{\mathcal{F}} f(\gamma g) dg = \int_G f(g) dg.$$

Taking f to approximate an indicator converts a lattice count into a homogeneous integral. Every effective version is a quantified form of these two equalities plus smoothing.

Worked Example F.2 (A cone integral producing $T \log T$). Let $L = \log T$ and integrate over $P_L = \{s_1, s_2 \geq 0 : s_1 + s_2 \leq L\}$:

$$I(T) = \int_{P_L} e^{s_1 + s_2} ds_1 ds_2.$$

Set $r = s_1 + s_2$. The slice $r = \text{const}$ has one-dimensional length r in these coordinates, so

$$I(T) = \int_0^L r e^r dr = (L - 1)e^L + 1 = T \log T - T + 1.$$

The whole edge $s_1 + s_2 = L$ is maximizing, and its dimension one produces the single logarithm. This is the elementary model for $T^a(\log T)^{b-1}$.

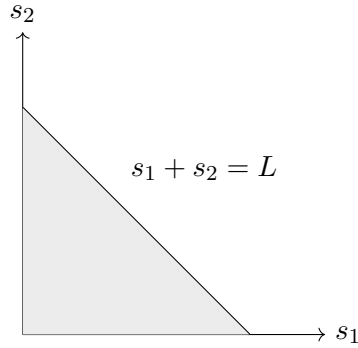


Figure F.1: A maximizing face rather than a single maximizing point creates a logarithmic factor in the polar-volume asymptotic.

Proof architecture: polar asymptotics with the gap visible

Suppose a polyhedral polar integral has exponent functional λ on a compact angular polytope P and maximum a on a face F . For a closed subset P_{far} disjoint from a fixed neighborhood of F , compactness gives

$$\Delta = a - \max_{P_{\text{far}}} \lambda > 0.$$

Consequently the far contribution is $O(T^{a-\Delta}(\log T)^M)$, while the near-face coordinates give the main $T^a(\log T)^{b-1}$ term. This is the precise source of the power saving used whenever the text says that the nonmaximizing polar region is negligible.

Appendix G

Student Proof Laboratory: Turning Equidistribution into Counts

A counting theorem has two logically different halves. Dynamics proves that an orbit becomes equidistributed; geometry converts that statement into the number of lattice points in a growing region. This chapter keeps those halves separate so that normalization constants and boundary errors remain visible.

G.1 The orbit–stabilizer dictionary

Let G act transitively on a variety V and choose $v_0 \in V$ with stabilizer H . Then $V \simeq H \backslash G$ by $Hg \mapsto v_0g$. If $\Gamma < G$ is a lattice, the integral orbit is $v_0\Gamma$, and counting points in a region $B_T \subset V$ is the same as counting cosets

$$(H \cap \Gamma) \backslash \Gamma$$

whose images lie in B_T .

Lemma G.1 (Counting density normalization). *Normalize Haar measures by*

$$dg = dh \, d(Hg).$$

If the translated H -orbit equidistributes in $\Gamma \backslash G$, the natural density factor is

$$\frac{\text{vol}((H \cap \Gamma) \backslash H)}{\text{vol}(\Gamma \backslash G)}.$$

Proof. Choose a small product neighborhood $\Omega_H \Omega_V$ near the identity, with $\Omega_H \subset H$ and Ω_V transverse. A point of the orbit corresponds to one $H \cap \Gamma$ -coset. Thickening each such point in the H -direction contributes approximately $\text{vol}((H \cap \Gamma) \backslash H)$ to the unfolded integral, while averaging in the ambient quotient divides by its total Haar volume $\text{vol}(\Gamma \backslash G)$. The quotient measure convention $dg = dh d(Hg)$ introduces no reciprocal Jacobian. Letting the thickening radius tend to zero gives the displayed factor. $\square$

G.2 Unfolding written out

Let $F \in C_c(H \backslash G)$ and define its periodization

$$\Theta_F(\Gamma g) = \sum_{\gamma \in (H \cap \Gamma) \backslash \Gamma} F(H\gamma g).$$

For nonnegative F , Tonelli and the quotient integration formula give

$$\int_{\Gamma \backslash G} \Theta_F(\Gamma g) dg = \frac{\text{vol}((H \cap \Gamma) \backslash H)}{\text{vol}(\Gamma \backslash G)} \int_{H \backslash G} F(Hg) d(Hg).$$

For compactly supported F the sum is locally finite, so the same identity holds by linearity. This one formula is the algebraic heart of the counting constant.

G.3 Why indicators must be smoothed

Mixing controls smooth or L^2 test functions. The sharp indicator $\mathbf{1}_{B_T}$ has a boundary. Choose inner and outer thickenings $B_T^-(\varepsilon) \subset B_T \subset B_T^+(\varepsilon)$ and smooth functions f_T^-, f_T^+ satisfying

$$\mathbf{1}_{B_T^-(\varepsilon)} \leq f_T^- \leq \mathbf{1}_{B_T} \leq f_T^+ \leq \mathbf{1}_{B_T^+(\varepsilon)}.$$

If the family is well rounded,

$$\text{vol}(B_T^+(\varepsilon) \setminus B_T^-(\varepsilon)) = o_\varepsilon(1) \text{vol}(B_T)$$

uniformly for large T . First use equidistribution for $f_T^\pm$, then let $\varepsilon \downarrow 0$. In an effective theorem the same sandwich is used, but ε must depend on T and be optimized against the Sobolev cost of smoothing.

G.4 The wavefront lemma with the missing A direction visible

Let $G = HA^+K$ be a generalized Cartan decomposition. Away from Weyl walls, small multiplication on the right can be reabsorbed as

$$Hak\mathcal{O}_\varepsilon \subset H\mathcal{O}_{C\varepsilon}aA_{C\varepsilon}K\mathcal{O}_{C\varepsilon}.$$

The factor $A_{C\varepsilon}$ is essential. Dimensionally, multiplication by a small element can have a component normal to both H and K but tangent to A ; no uniform statement can absorb that component into H and K alone.

Worked Example G.2 (Polar coordinates in $\mathrm{SL}_2(\mathbb{R})$). For $G = \mathrm{SL}_2(\mathbb{R})$ and $K = \mathrm{SO}(2)$, the usual polar decomposition is $G = KA^+K$. Multiplying a_t by $\exp(\varepsilon H)$ changes t to $t + O(\varepsilon)$. This perturbation is exactly the small A -factor in the wavefront lemma. Omitting it would assert that a radial displacement can be absorbed by rotations, which is false.

G.5 A polyhedral Laplace calculation

Suppose polar volume has the form

$$\int_{\mathfrak{a}_T^+} e^{2\rho(H)} P(H) dH,$$

where $\mathfrak{a}_T^+ = T\mathcal{P}$ is a growing polytope and P is polynomial. The leading exponential growth comes from the face on which the linear functional 2ρ is maximal. If that maximum is attained on a face F of dimension d_F , choose coordinates u tangent to F and $v \geq 0$ normal to it. Then

$$2\rho(Tx + u - v) = T\lambda - \langle c, v \rangle,$$

with $c_j > 0$. Integrating the normal variables gives a constant and integrating the d_F tangent variables contributes T^{d_F} , up to the additional polynomial degree coming from P . This is the geometric source of the familiar asymptotic $CT^a e^{\lambda T}$.

G.6 Two counting engines, not one

The mixing engine starts from

$$\langle \pi(g)f_1, f_2 \rangle \rightarrow \int f_1 \int f_2$$

and thickens singular orbit measures. The Ratner–Mozes–Shah engine starts from a family of homogeneous measures and classifies its weak limits. Both can lead to the same leading constant, but their hypotheses and error terms are different. A student should decide which engine is being used before trying to estimate a boundary term.

Worked Example G.3 (Primitive integer vectors). Primitive vectors in $\mathbb{Z}^n$ may be viewed as an orbit of a parabolic stabilizer in SL_n . The elementary answer in a Euclidean ball is

$$\#\{v \in \mathbb{Z}_{\mathrm{prim}}^n : \|v\| \leq T\} \sim \frac{\mathrm{vol}(B_1)}{\zeta(n)} T^n.$$

The factor $1/\zeta(n)$ is the arithmetic covolume of the stabilizer orbit. This familiar example is useful for checking the direction of every quotient volume in a general homogeneous counting formula.

Appendix H

Student Proof Laboratory: From Orbits to Counting Functions

A lattice-point count becomes a homogeneous-dynamics problem only after the stabilizer, quotient measure, and normalization constant have been written correctly. This chapter derives the dictionary before using mixing.

H.1 Orbit-stabilizer parametrization over the integers

Let G act transitively on a real algebraic variety V in right-action notation, fix $v_0 \in V$, and identify $V \simeq H \backslash G$ for $H = \text{Stab}_G(v_0)$, and let $\Gamma < G$ be a lattice preserving an integral structure. The integral orbit $v_0 \Gamma$ is parametrized by $(H \cap \Gamma) \backslash \Gamma$ when the stabilizer of v_0 in Γ is $H \cap \Gamma$.

Lemma H.1 (Counting as a coset sum). *For a compactly supported function F on $H \backslash G$,*

$$\sum_{v \in v_0 \Gamma} F(v) = \sum_{\gamma \in (H \cap \Gamma) \backslash \Gamma} F(H\gamma).$$

Proof. Two elements γ_1, γ_2 give the same orbit point exactly when $v_0 \gamma_1 = v_0 \gamma_2$, equivalently $\gamma_1 \gamma_2^{-1} \in H$. Because both are in Γ , the difference lies in $H \cap \Gamma$. Thus orbit points are precisely the left cosets in the displayed quotient. $\square$

H.2 The normalization constant from Weil integration

Normalize Haar measures so that

$$dg = dh \, d(Hg)$$

in Weil's quotient formula. Put $V_H = \text{vol}((H \cap \Gamma) \backslash H)$ and $V_G = \text{vol}(\Gamma \backslash G)$.

Proposition H.2 (Normalization of the closed-orbit period). *Let $X = \Gamma \backslash G$ carry Haar measure dg of total mass V_G , and let $Y = (H \cap \Gamma) \backslash H$ carry Haar measure dh of total mass V_H . If $\rho \in C_c(X)$ satisfies $\int_X \rho \, dg = 1$ and a sequence $g_i \in G$ has the normalized closed-orbit equidistribution property*

$$\frac{1}{V_H} \int_Y \rho(yg_i) \, dh(y) \longrightarrow \frac{1}{V_G} \int_X \rho \, dg,$$

then its unnormalized H -period satisfies

$$\int_Y \rho(yg_i) \, dh(y) \longrightarrow \frac{V_H}{V_G}.$$

Thus the density constant appearing after unfolding a homogeneous count is V_H/V_G .

Proof. The probability measure on X is dg/V_G , whereas the probability measure on the closed H -orbit is dh/V_H . The assumed equidistribution identity therefore has right-hand side

$$\frac{1}{V_G} \int_X \rho \, dg = \frac{1}{V_G}.$$

Multiplying the limiting identity by V_H gives

$$\int_Y \rho(yg_i) \, dh(y) \longrightarrow \frac{V_H}{V_G}.$$

This is exactly the constant in Theorem 6.1. In particular, the numerator is the H -orbit volume and the denominator is the ambient G -volume; no reciprocal convention is hidden in the notation. $\square$

H.3 Thickening a singular orbit

A closed H -orbit has lower dimension in $\Gamma \backslash G$ and cannot be inserted directly into an L^2 mixing correlation. Choose a local product chart $H \times S \rightarrow G$ transverse to H and a bump ρ_ϵ on S of total mass one.

Lemma H.3 (Qualitative thickening). *For every $f \in C_c(\Gamma \backslash G)$,*

$$\int_{(H \cap \Gamma) \backslash H} f(xhg) dh = \lim_{\epsilon \rightarrow 0} \int_{\Gamma \backslash G} f(yg) \Phi_\epsilon(y) dy,$$

where Φ_ϵ is a smooth thickening of the H -orbit.

Proof. Cover the compact part of the closed orbit relevant to f by finitely many injective product charts. In each chart integrate first in the transverse coordinate against an approximate identity. Uniform continuity of f on the compact support makes the transverse average converge uniformly to its value on the orbit. A partition of unity glues the charts. The finite H -volume controls the tails if the orbit is noncompact. $\square$

H.4 The wavefront lemma with the missing A -direction included

Let $G = KA^+K$ in a semisimple group. Deep inside the positive chamber, small perturbations of $ka\ell$ can be reabsorbed into small changes of the left K , central A , contracting horospherical, and right K factors. The central A -change is essential because the differential of the Cartan projection has an $\mathfrak{a}$ component.

Theorem H.4 (Strong qualitative wavefront lemma). *Fix a positive Weyl chamber $\mathfrak{a}^+ \subset \mathfrak{a}$, let Δ be its simple restricted roots, and for $\eta > 0$ put*

$$A_\eta^+ = \{\exp H : H \in \mathfrak{a}^+, \alpha(H) \geq \eta \|H\| \text{ for every } \alpha \in \Delta\}.$$

For every identity neighborhood $\mathcal{O}$ there is an identity neighborhood $\mathcal{U}$ such that for all $a \in A_\eta^+$,

$$KaK\mathcal{U} \subset K_{\mathcal{O}} a A_{\mathcal{O}} N_{\mathcal{O}}^- K_{\mathcal{O}},$$

with every subscript- $\mathcal{O}$ factor contained in a prescribed small neighborhood of the identity in the indicated subgroup.

Proof. Use the local diffeomorphism near the identity coming from the Lie algebra decomposition

$$\mathfrak{g} = \mathfrak{k} + \mathfrak{a} + \mathfrak{n}^- + \mathfrak{k}$$

after quotienting the finite overlaps. If a perturbation u occurs on the right of a , conjugate its expanding-root components across a ; because a stays in A_η^+ , those components contract uniformly into an arbitrarily small left factor. Components in $\mathfrak{a}$ commute with a and therefore remain as a small A -factor; they cannot be discarded. Contracting-root and compact components are absorbed by the local product chart. Compactness of the normalized chamber directions gives uniform neighborhoods for all $a \in A_\eta^+$. $\square$

H.5 Well-roundedness and boundary measure

Let $B_T \subset H \backslash G$ be an expanding family. We use the well-roundedness condition of Definition 5.1, with $\mathcal{U}_\epsilon$ as the chosen identity neighborhood. Thus all boundary estimates below refer to the precise two-limit condition

$$\lim_{\epsilon \downarrow 0} \limsup_{T \rightarrow \infty} \frac{\text{vol}(B_T^+(\epsilon) \setminus B_T^-(\epsilon))}{\text{vol}(B_T)} = 0.$$

Lemma H.5 (Piecewise smooth boundaries are well rounded under regular radial expansion). *Suppose B_T is described in polar coordinates by finitely many smooth angular inequalities and a radial bound $0 \leq r \leq R_T(\omega)$ with $R_T \rightarrow \infty$ uniformly on compact angular pieces. If boundary faces have codimension one and the volume density is locally integrable, then (B_T) is well rounded.*

Proof. A group perturbation of size ϵ changes the angular coordinates by $O(\epsilon)$ and the Cartan/radial coordinate by $O(\epsilon)$ through the wavefront lemma. The outer-minus-inner difference lies in an $O(\epsilon)$ -tube around finitely many boundary faces plus radial shells of relative thickness $O(\epsilon)$. The coarea formula bounds the face tubes by $C\epsilon$ times the volume scale, and the radial density gives the same bound for the shells. Divide by $\text{vol}(B_T)$ and let $\epsilon \rightarrow 0$. $\square$

H.6 Mixing-to-counting theorem

Theorem H.6 (Qualitative homogeneous lattice counting). *Assume the H -orbit has finite volume, (B_T) is well rounded, and for every fixed suffi-*

ciently small $\epsilon > 0$ the periodized thickening P_ϵ satisfies the averaged-mixing condition (6.1) of Theorem 6.1 on both $B_T^-(\epsilon)$ and $B_T^+(\epsilon)$. Then

$$\#\{(H \cap \Gamma)\gamma : H\gamma \in B_T\} \sim \frac{V_H}{V_G} \text{vol}_{H \setminus G}(B_T).$$

Proof. This is Theorem 6.1. To see the mechanism in the present laboratory notation, periodize the inner and outer indicators and test them against the nonnegative thickening of the closed H -orbit. The pointwise sandwich (6.3a) unfolds to the two integrals in (6.3). For fixed ϵ , the stated averaged-mixing hypothesis evaluates each integral as

$$\frac{V_H}{V_G} \text{vol}(B_T^\pm(\epsilon)) + o_{T,\epsilon}(\text{vol}(B_T^\pm(\epsilon))).$$

Well-roundedness then makes both thickened volumes asymptotic to $\text{vol}(B_T)$ after the ordered limits $T \rightarrow \infty$ and $\epsilon \downarrow 0$. This yields the displayed count and, importantly, does not infer an average over $B_T^\pm(\epsilon)$ merely from an average over B_T . $\square$

Appendix I

Student Proof Laboratory: Two Counting Engines

I.1 Mixing engine versus homogeneous-measure engine

The mixing proof is strongest when the translating subgroup sits naturally inside a semisimple flow for which ambient Haar mixing is available. A Ratner/Mozes–Shah proof instead studies the normalized measures on moving closed homogeneous orbits and classifies their limits. The two engines can give the same main term but have different hypotheses.

Worked Example I.1 (Primitive vectors). Primitive integer vectors in $\mathbb{Z}^n$ form the orbit of e_1 under $\mathrm{SL}_n(\mathbb{Z})$. The real orbit is $\mathbb{R}^n \setminus \{0\}$, a homogeneous space of $\mathrm{SL}_n(\mathbb{R})$ with parabolic stabilizer. Counting primitive vectors in a large Euclidean ball can be done elementarily by Möbius inversion, giving $\mathrm{vol}(B_1)T^n/\zeta(n) + o(T^n)$. The homogeneous method recovers the same shape of main term from quotient volume and demonstrates how the mechanism generalizes when no elementary gcd decomposition is available.

I.2 Laplace’s method for quotient volume

When Cartan coordinates give a density $J(H) = \prod_{\alpha>0} (\sinh \alpha(H))^{m_\alpha}$, large norm balls are controlled by the maximum of the linear functional $2\rho(H)$ on the radial constraint. Splitting into a neighborhood of maximizing faces and

its complement yields the familiar form

$$\text{vol}(B_T) \sim c T^a (\log T)^b$$

for algebraic norm balls. The exact exponents come from the maximizing face dimension and the degree of the radial constraint. Volume asymptotics are therefore a separate geometric calculation; mixing supplies the density of lattice points relative to that volume but does not calculate the volume itself.

Appendix J

A Student Textbook Chapter on the Eskin–McMullen Counting Mechanism

The abstract counting theorem in the main text is concise. This chapter reconstructs the proof in the order in which one would discover it: identify the orbit, thicken it, move the thickening, use mixing, and finally remove the thickening by wavefront geometry.

J.1 A model problem

Let G be semisimple, $\Gamma < G$ a lattice, and $H < G$ a closed subgroup such that $(H \cap \Gamma) \backslash H$ has finite volume. We want to count cosets

$$(H \cap \Gamma)\gamma \in (H \cap \Gamma) \backslash \Gamma$$

whose image $H\gamma$ lies in a growing region $B_T \subset H \backslash G$.

The heuristic is simple: lattice cosets should have density V_H/V_G relative to quotient volume. The proof must explain why the lower-dimensional closed H -orbit samples the ambient quotient correctly when translated through the region.

J.2 A smooth counting kernel

Choose nonnegative $F_T \in C_c(H \backslash G)$ approximating 1_{B_T} and a small identity bump ρ_ϵ on G . Lift F_T to a right H -invariant function on G . Define a periodized function on $\Gamma \backslash G$ by

$$\Theta_T(\Gamma g) = \sum_{\gamma \in (H \cap \Gamma) \backslash \Gamma} F_T(H\gamma g) \rho_\epsilon(\text{transverse coordinate of } \gamma g).$$

The transverse bump makes the expression smooth enough for mixing. Unfolding turns its ambient integral into quotient volume times the stabilizer covolume.

J.3 Unfolding without symbolic shortcuts

Let $\Psi \in C_c(\Gamma \backslash G)$. For a function Φ on $(H \cap \Gamma) \backslash G$ that is integrable and left $H \cap \Gamma$ -invariant,

$$\int_{\Gamma \backslash G} \sum_{\gamma \in (H \cap \Gamma) \backslash \Gamma} \Phi(\gamma g) \Psi(g) dg = \int_{(H \cap \Gamma) \backslash G} \Phi(g) \Psi(g) dg.$$

Proof. Choose a fundamental domain $\mathcal{F}$ for Γ in G . The sets $\gamma\mathcal{F}$, with γ running through left coset representatives for $(H \cap \Gamma) \backslash \Gamma$, tile a fundamental domain for $H \cap \Gamma$ modulo null boundaries. Interchange sum and integral by absolute convergence, change variables $g \mapsto \gamma^{-1}g$ in each term, and combine the tiled pieces. $\square$

Now apply Weil integration to the right-hand side, first along the H -fiber and then on $H \backslash G$. This is where V_H enters.

J.4 Mixing the thickened orbit

Let g_T be a translating family in G that tends to infinity in the semisimple directions. A thickening Φ_ϵ of the closed H -orbit has ambient mass V_H/V_G after normalization. Mixing gives

$$\int_X \Phi_\epsilon(x) F(xg_T) dm_X(x) \rightarrow \left(\int \Phi_\epsilon dm_X \right) \left(\int F dm_X \right).$$

For fixed ϵ , this is a direct application of Volume VI. The nontrivial part is to identify the left side with the original H -orbit integral up to a small geometric error.

J.5 Wavefront comparison

Take an element hak in a polar decomposition adapted to $H \backslash G$. A small transverse perturbation u satisfies

$$haku = h'a'n^-k'$$

with h', a', n^-, k' close to the corresponding factors, provided a lies deep enough in the relevant chamber. The n^- component is contracted when moved across a . Therefore the thickened point and the original H -orbit point land in almost the same quotient region. Inner and outer thickenings of B_T bound the discrepancy.

J.6 Well-roundedness closes the squeeze

For fixed ϵ , mixing gives an asymptotic for the inner and outer smoothed counts. Well-roundedness says the quotient volumes of the two regions differ by $o(\text{vol } B_T)$ as $\epsilon \rightarrow 0$. Thus

$$\limsup_{T \rightarrow \infty} \left| \frac{N(B_T)}{\text{vol } B_T} - \frac{V_H}{V_G} \right| \leq c(\epsilon),$$

where $c(\epsilon) \rightarrow 0$. Let $\epsilon \rightarrow 0$.

Appendix K

The Duke–Rudnick–Sarnak Viewpoint

K.1 Affine homogeneous varieties

Let V be an affine variety on which a semisimple group G acts transitively with stabilizer H . An integral orbit of Γ is a discrete subset of $V(\mathbb{R}) \simeq H \backslash G$. Norm balls in V pull back to regions in the homogeneous quotient. The counting problem becomes exactly the coset count above.

K.2 Why volume growth may have logarithms

In Cartan coordinates, a linear representation has weights λ . The condition $\|v_0 a\| < T$ is approximately $\max_{\lambda} e^{\lambda(H)} \|v_{\lambda}\| < T$, a polyhedral inequality in $H = \log a$. Haar density contributes $e^{2\rho(H)}$. Maximizing the linear functional 2ρ over the rescaled weight polytope determines the leading power of T . If the maximum is attained on a positive-dimensional face, integration along that face produces powers of $\log T$.

Worked Example K.1 (A two-dimensional Laplace integral). Consider

$$I(T) = \int_{x,y \geq 0, \ x+2y \leq \log T} e^{3x+4y} \, dx dy.$$

The exponent $3x + 4y$ subject to $x + 2y \leq L$ is maximized by $x = L, y = 0$ because 3 units of exponent per unit constraint exceed 2 for the y direction. Integrating explicitly gives $I(T) \asymp T^3$. If the exponent were $2x + 4y$, the

whole boundary segment would maximize it and an extra factor $\log T$ would appear. This elementary polyhedral calculation is the model for volume exponents on affine homogeneous varieties.

K.3 Multiple integral orbits

The set $V(\mathbb{Z})$ may split into finitely or infinitely many Γ -orbits. A homogeneous counting theorem applies to one orbit at a time. To count all integral points one must classify the relevant arithmetic orbits and sum their constants, proving convergence if infinitely many occur. The Ratner continuation only states global counting results in examples where this arithmetic decomposition is controlled.

Appendix L

Counting by Limits of Homogeneous Measures

L.1 Why one might avoid ambient mixing

Sometimes the moving objects themselves are closed homogeneous orbits carrying invariant probabilities. Rather than thicken and apply mixing, one can study those probabilities directly. Volume VII says that any non-escaping subsequential limit is homogeneous and describes support containment. If algebraic considerations show that the only possible limit is ambient Haar, the moving orbits equidistribute.

L.2 From orbit equidistribution to a count

Suppose finite sets or closed orbits Y_T parametrize the objects being counted and the normalized measures m_{Y_T} converge to m_X . A region B with boundary of m_X -measure zero then satisfies $m_{Y_T}(B) \rightarrow m_X(B)$. If the total mass/degree M_T of Y_T is known, then the number of objects in B is $M_T m_{Y_T}(B)$ and hence asymptotic to $M_T m_X(B)$. The main arithmetic work is often to compute M_T and prove no mass escapes.

L.3 Focusing as the exceptional-term detector

If a family fails to converge to ambient Haar, Mozes–Shah produces a proper homogeneous limit after small conjugacy. In counting language, this means

a positive proportion of objects satisfy an extra algebraic constraint. Such constraints frequently correspond to rational subvarieties. One can either exclude them by an irreducibility hypothesis or count them separately as secondary terms.

Appendix M

Wavefront Geometry in Low Rank

M.1 SL_2 : polar coordinates

Every $g \in \mathrm{SL}_2(\mathbb{R})$ can be written $k_1 a_t k_2$. For a small perturbation u , write its Lie algebra as a sum of stable, diagonal, and unstable directions. Moving the unstable piece across a_t contracts it by e^{-t} ; the stable piece remains on the right and the diagonal piece changes t by $O(\|u\|)$. This is the entire wavefront lemma in rank one.

M.2 SL_3 : two chamber depths

Let $a = \mathrm{diag}(e^{h_1}, e^{h_2}, e^{h_3})$ with $h_1 \geq h_2 \geq h_3$. Conjugating E_{ij} across a multiplies it by $e^{h_i - h_j}$. If both simple-root gaps $h_1 - h_2$ and $h_2 - h_3$ are large, every positive-root component contracts uniformly when moved in the appropriate direction. Near a chamber wall one gap is small, and the corresponding root component cannot be absorbed with a uniform small constant. Effective counting in sectors must therefore track chamber depth or split off the wall region.

Appendix N

Boundary Regularity and Approximation

N.1 Boundary-zero versus quantitative boundary

For qualitative counting it is enough that $\text{vol}(B_T^+(\epsilon) \setminus B_T^-(\epsilon)) / \text{vol}(B_T) \rightarrow 0$ after $T \rightarrow \infty$, $\epsilon \rightarrow 0$. For a power-saving count, Volume XII needs a rate such as $O(\epsilon^\beta)$ uniform in T . The distinction is why well-roundedness is formulated qualitatively here and effectively later.

N.2 Piecewise algebraic boundaries

A semialgebraic family of regions with uniformly bounded complexity admits a finite Whitney/cell stratification of its boundary. Away from lower-dimensional singular strata, the boundary is a smooth hypersurface with uniformly controlled local graphs after rescaling. Its ϵ -tube has volume $O(\epsilon)$ times the natural volume scale. Lower-dimensional strata have even smaller tube volume. This supplies quantitative well-roundedness in many matrix norm-ball examples.

Appendix O

Worked Counting Laboratories

O.1 Primitive integer vectors

The orbit of e_1 under $\mathrm{SL}_n(\mathbb{Z})$ is the set of primitive vectors. The stabilizer is an integral parabolic. Homogeneous counting predicts a constant times Euclidean volume. Möbius inversion computes the constant:

$$\#\{v \in \mathbb{Z}_{\mathrm{prim}}^n : \|v\| < T\} = \sum_{d \geq 1} \mu(d) \#\{w \in \mathbb{Z}^n : 0 < \|w\| < T/d\}.$$

Using the elementary lattice-point volume estimate and dominated convergence,

$$\sim \mathrm{vol}(B_1) T^n \sum_{d \geq 1} \frac{\mu(d)}{d^n} = \frac{\mathrm{vol}(B_1)}{\zeta(n)} T^n.$$

This gives an independent check on the homogeneous normalization constant.

O.2 Matrices of fixed determinant

The variety of 2×2 real matrices of determinant one is the group $\mathrm{SL}_2(\mathbb{R})$ itself. Counting integer matrices of determinant one in a norm ball is lattice counting in G . Cartan decomposition gives volume growth roughly quadratic in the matrix norm (with convention-dependent logarithmic features for some norms), and mixing gives the lattice density $1/\mathrm{vol}(\Gamma \backslash G)$. This is the simplest case in which the stabilizer H is trivial.

O.3 Quadratic vectors

Fix a nonzero value c of an indefinite rational quadratic form Q . The real level set $Q(v) = c$ is a homogeneous orbit of the orthogonal group, often with stabilizer an orthogonal group of one lower dimension. Integral solutions split into arithmetic orbits. On each orbit, the homogeneous counting theorem gives the leading asymptotic once finite stabilizer covolume and well-roundedness are verified. This example foreshadows Volume XIV, where the value interval shrinks toward zero and the form itself may be irrational.

Appendix P

Guided Reconstruction of Homogeneous Lattice Counting

A lattice-point asymptotic is often displayed as one line,

$$N(T) \sim c \operatorname{vol}(B_T).$$

In homogeneous dynamics that line is the endpoint of several independent normalizations and limiting arguments. This chapter reconstructs them in a model which is general enough for the later arithmetic applications.

P.1 From an orbit to a double-coset count

Let G be unimodular, $\Gamma < G$ a lattice, and $H < G$ a closed unimodular subgroup such that $H \cap \Gamma$ is a lattice in H . Put

$$V = H \backslash G, \quad v_0 = H.$$

The orbit $v_0\Gamma$ is indexed by right cosets $(H \cap \Gamma) \backslash \Gamma$: indeed

$$H\gamma_1 = H\gamma_2 \iff \gamma_1\gamma_2^{-1} \in H \cap \Gamma.$$

Thus, for $B_T \subset V$,

$$N(T) = \#\{(H \cap \Gamma)\gamma : H\gamma \in B_T\}.$$

This bijection is the combinatorial beginning of the argument.

P.2 Deriving the quotient-measure normalization

Fix Haar measures dg on G and dh on H . Define the quotient measure $d\dot{g}$ on $H\backslash G$ by Weil's formula

$$\int_G F(g) dg = \int_{H\backslash G} \int_H F(hg) dh d\dot{g}.$$

Let

$$V_G = \text{vol}(\Gamma\backslash G), \quad V_H = \text{vol}((H \cap \Gamma)\backslash H).$$

Normalize the invariant probabilities $m_X = V_G^{-1}dg$ on $X = \Gamma\backslash G$ and $m_Y = V_H^{-1}dh$ on $Y = (H \cap \Gamma)\backslash H$.

The leading counting density is V_H/V_G . We rederive it so that no reciprocal can be inserted accidentally.

Lemma P.1 (Unfolding normalization). *Let $F \in C_c(G)$ and define*

$$\Theta_F(g) = \sum_{\gamma \in (H \cap \Gamma)\backslash \Gamma} F(\gamma g).$$

Then

$$\int_{\Gamma\backslash G} \Theta_F(g) dm_X(g) = \frac{1}{V_G} \int_{H\backslash G} \left(\int_H F(hg) dh \right) d\dot{g}.$$

If the inner H -integral approximates $V_H \mathbf{1}_{B_T}(Hg)$, then the main term is

$$\frac{V_H}{V_G} \text{vol}(B_T).$$

Proof. First integrate with the unnormalized quotient measure:

$$\int_{\Gamma\backslash G} \sum_{\gamma} F(\gamma g) dg = \int_{(H \cap \Gamma)\backslash G} F(g) dg.$$

Decompose the latter quotient first by $(H \cap \Gamma)\backslash H$ and then by $H\backslash G$:

$$\int_{H\backslash G} \int_{(H \cap \Gamma)\backslash H} F(hg) dh d\dot{g}.$$

Dividing by V_G gives the probability normalization. If F is a thickening whose integral along each H -fiber is V_H on the target region, the coefficient is exactly V_H/V_G . $\square$

Warning P.2. The H -covolume occurs in the numerator. A larger closed H -orbit gives more lattice representatives per unit quotient volume, not fewer.

P.3 Why a singular orbit must be thickened

Mixing on X applies to functions in $L^2(X)$, but m_Y is singular with respect to m_X when $\dim H < \dim G$. Choose a local complement $\mathfrak{r}$ to $\mathfrak{h}$ and a bump ψ_ϵ on $R = \exp(\mathfrak{r})$. The thickened density

$$F_\epsilon(hr) = \frac{1}{V_H} \psi_\epsilon(r)$$

inside a tube around Y has ambient integral one. Thus

$$\int_X F_\epsilon(x) \varphi(xg) dm_X(x)$$

is an ordinary matrix coefficient. Mixing makes it close to $\int \varphi dm_X$. A wavefront lemma is then needed to compare this thickened correlation with the translate of the original singular measure.

P.4 The wavefront lemma as a linear algebra problem

For simplicity suppose $G = HAK$ on the relevant open set. Near $g = hak$, perturb by $\exp Z$ with Z small. We seek small corrections $X \in \mathfrak{h}$, $Y \in \mathfrak{a}$, and $W \in \mathfrak{k}$ satisfying

$$hak \exp Z \approx h \exp X a \exp Y k \exp W.$$

At first order this is the linear equation

$$Z = \text{Ad}(k^{-1}a^{-1})X + \text{Ad}(k^{-1})Y + W.$$

The wavefront lemma is a uniform right-inverse statement for this map.

Away from chamber walls the H -directions and the conjugated K -directions are uniformly transverse modulo the neutral A -directions. The small A -correction is essential: the tangent spaces cannot span the correct dimension without it.

Lemma P.3 (Uniform linear splitting away from walls). *Fix a closed cone A_η^+ a positive angular distance from the singular walls. There is C_η such that every $Z \in \mathfrak{g}$ can be written*

$$Z = X + Y + \text{Ad}(a)W, \quad X \in \mathfrak{h}, \quad Y \in \mathfrak{a}, \quad W \in \mathfrak{k},$$

with

$$\|X\| + \|Y\| + \|W\| \leq C_\eta \|Z\|$$

uniformly for $a \in A_\eta^+$.

Proof. Decompose $\mathfrak{g}$ into restricted-root pairs. On the two-dimensional space $\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}$, the H and $\text{Ad}(a)K$ tangent directions are represented by a 2×2 matrix whose determinant is a nonzero multiple of $e^{\alpha(\log a)} - e^{-\alpha(\log a)}$. Away from walls this determinant is bounded away from zero. The zero-root directions are supplied by $\mathfrak{a}$ and the compact neutral part. There are finitely many roots, so the inverses are uniformly bounded. $\square$

The inverse function theorem upgrades the linear splitting to the nonlinear wavefront inclusion, with errors quadratic in the perturbation size.

P.5 Well-roundedness is the boundary theorem

Mixing controls smooth approximations, not a discontinuous region B_T . Choose inner and outer perturbations $B_T^- \subset B_T \subset B_T^+$. If

$$\frac{\text{vol}(B_T^+ \setminus B_T^-)}{\text{vol}(B_T)} \rightarrow 0$$

uniformly as the perturbation shrinks, then the smooth upper/lower counts have the same main term. This is well-roundedness.

For a region with piecewise smooth boundary in polar coordinates, the boundary tube has one fewer dimension and therefore relative volume $O(\epsilon)$ provided the radial measure does not concentrate on a chamber wall. Corners contribute intersections of boundary faces and are smaller still. This geometric statement is independent of mixing; it must be proved for each family of regions.

P.6 The mixing-to-counting squeeze

Let $F_{T,\epsilon}^-$ and $F_{T,\epsilon}^+$ be smooth minorant/majorant kernels associated with B_T^- and B_T^+ . Unfolding and mixing give

$$\sum_{\gamma} F_{T,\epsilon}^\pm(H\gamma) = \frac{V_H}{V_G} \int_{H \setminus G} F_{T,\epsilon}^\pm d\dot{g} + o(\text{vol}(B_T))$$

for fixed ϵ . The actual count lies between the two sums. First let $T \rightarrow \infty$, then $\epsilon \rightarrow 0$. Well-roundedness makes the two normalized main terms converge to the same number. Therefore

$$N(T) \sim \frac{V_H}{V_G} \text{vol}(B_T).$$

The order of limits matters in the qualitative proof. Effective counting in Volume XII couples ϵ to T and pays the Sobolev cost of doing so.

P.7 Affine symmetric varieties and polar volume

Suppose H is the fixed-point group of an involution. The tangent space splits as $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{q}$, and a maximal split abelian subspace $\mathfrak{a} \subset \mathfrak{q}$ gives a generalized polar decomposition

$$G = HA^+K$$

up to finite multiplicity. Haar measure has density

$$J(H) = \prod_{\alpha > 0} \sinh(\alpha(H))^{m_\alpha} \cosh(\alpha(H))^{n_\alpha} dH dk.$$

At large H in a fixed chamber face,

$$J(H) = e^{2\rho(H)} (c + O(e^{-c'\|H\|})).$$

Thus the volume of an algebraic norm ball becomes a Laplace integral over a polyhedron in $\mathfrak{a}^+$.

Worked Example P.4 (One-dimensional radial parameter). If the norm constraint is $\lambda H \leq \log T$ and the Haar exponent is $2\rho H$, then

$$\text{vol}(B_T) \asymp \int_0^{(\log T)/\lambda} e^{2\rho H} dH \asymp T^{2\rho/\lambda}.$$

When several chamber faces achieve the same maximal ratio, integration along the face produces powers of $\log T$. This is the origin of the familiar $T^a(\log T)^b$ volume laws on affine symmetric varieties.

P.8 The Duke–Rudnick–Sarnak viewpoint

The same leading-volume law can be reached by harmonic/automorphic methods. The homogeneous-dynamics viewpoint isolates the following shared requirements:

1. an orbit parametrization by $H \backslash G$;
2. a finite-volume closed H -orbit in $\Gamma \backslash G$;
3. enough decay/equidistribution to replace the discrete orbit by invariant measure;
4. a geometric calculation of the volume of B_T ;
5. boundary control.

The proof in this volume owns the mixing route. Historical attribution to other routes is not used as a substitute for any of these five steps.

P.9 Counting by homogeneous-measure limits

There is a second engine. Suppose the relevant counting problem can be organized as a sequence of closed homogeneous orbits Y_i whose normalized measures m_{Y_i} vary. Volume VII says that any probability limit is homogeneous and gives support containment after small conjugacy. If arithmetic geometry rules out every proper limiting support, then $m_{Y_i} \Rightarrow m_X$.

To turn this into a count, choose a compact test region B whose boundary has zero invariant measure. Equidistribution gives

$$m_{Y_i}(B) \rightarrow m_X(B).$$

Unfolding expresses the left side as a normalized number of arithmetic objects represented by Y_i . The normalization is the volume of the closed orbit, so one obtains the asymptotic once that volume is calculated.

This route is especially useful when no single semisimple translate with known mixing parametrizes the family. It is logically distinct from the Eskin–McMullen wavefront route.

P.10 Three worked counting dictionaries

Primitive vectors

Primitive integer vectors are the $SL_n(\mathbb{Z})$ -orbit of e_1 . Their real orbit is $\mathbb{R}^n \setminus \{0\}$, modeled by the quotient by the stabilizer of e_1 . A norm ball becomes an ordinary Euclidean ball. The general homogeneous density specializes to the familiar primitive-vector density after the stabilizer covolume is evaluated.

Matrices of fixed determinant

The left-right action of $SL_n \times SL_n$ on matrices preserves determinant. A nonzero determinant level is one homogeneous orbit. Counting integral matrices in a norm ball becomes a lattice-orbit problem on this variety. The stabilizer is a diagonal copy of SL_n , and the radial volume calculation reflects the singular values of the matrix.

Quadratic forms of fixed discriminant

The change-of-variables action on symmetric matrices preserves determinant up to the determinant character of the acting group. Fixing discriminant and signature gives orthogonal stabilizers. The homogeneous count therefore requires exactly the finite-volume orthogonal orbit and polar-volume machinery developed here and in Volume IV.

P.11 A counting proof checklist

Before accepting a homogeneous lattice-counting asymptotic, check:

1. Is the discrete set really one orbit, or a finite union of orbits?
2. What is the stabilizer H and is $H \cap \Gamma$ a lattice?
3. Which Haar normalization defines the quotient volume?
4. Why is the closed H -orbit finite volume?
5. Which equidistribution engine is being used: mixing or homogeneous measure limits?
6. If mixing is used, where is the wavefront lemma proved?
7. If measure limits are used, which focusing obstruction excludes proper limits?
8. Is the region well rounded?
9. What is the actual polar-volume asymptotic?
10. Are several integral orbits or local congruence conditions changing the leading constant?

These questions are intentionally repetitive: a wrong answer to any one can change either the exponent or the constant in the final asymptotic.

Proof Guide and Dependency Summary

This chapter is a finalization layer. It does not introduce a new theorem scope. Its purpose is to expose the proof architecture, make the hypotheses that are easiest to misread impossible to miss, and give a student a concrete model calculation before the abstract machinery is reused later.

1. Counting is an equidistribution problem in disguise

For a lattice orbit on a homogeneous variety $H \backslash G$, a counting function is unfolded into an integral on $\Gamma \backslash G$. The basic dictionary is

lattice points $\leftrightarrow$ Γ -translates $\leftrightarrow$ thickened H -orbit $\leftrightarrow$ mixing.

Two ingredients then remain outside mixing: volume growth of the geometric region and control of its boundary.

2. The leading constant

The homogeneous density involves the ratio

$$\frac{\text{vol}((H \cap \Gamma) \backslash H)}{\text{vol}(\Gamma \backslash G)}$$

with Haar measures chosen compatibly with the quotient integration formula. This ratio is not an optional normalization convention: omitting the numerator or using an incompatible Haar normalization changes the main term.

3. Why the wavefront lemma contains a small A -direction

In KAK -type geometry, perturbing a point near a chamber wall can create a small displacement in the split torus A , not only in stable/unstable horospherical directions. The wavefront neighborhood must therefore retain that small A -component. It is precisely this geometric bookkeeping that allows a thickened orbit to be compared with a well-rounded counting region.

4. A student workflow

For a new counting problem:

1. compute the stabilizer H ;
2. write the invariant measure on $H\backslash G$;
3. compute the volume growth of the norm/sector ball;
4. thicken the closed H -orbit;
5. use qualitative mixing;
6. prove boundary well-roundedness;
7. simplify the main constant only after all Haar normalizations are fixed.

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Spectral Gap, Sobolev Theory, and Quantitative Mixing

Volume XI of a Detailed Notebook on Ratner's Theorems

Detailed Notebook on Ratner's Theorems

August 2026

Preface and proof policy

Volume VI proved qualitative mixing by weak limits. Effective counting and effective unipotent dynamics need more: a topology on test functions, an explicit spectral-gap parameter, and quantitative decay of matrix coefficients. This volume develops those ingredients in a form stable under all later smoothing arguments. For higher-rank matrix groups we prove a Kazhdan mechanism from relative property (T); for rank-one fixed quotients we explain the Laplace/Casimir spectral parameter directly. We do not import super-approximation or uniform congruence spectral gaps, which belong outside the present Ratner continuation unless explicitly assumed in a frontier example.

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Dependency and proof-ownership contract

Purpose. Quantitative mixing.

Prerequisites. Volumes VI and X; compact K -analysis is proved internally in Chapter 5.4.

New objects. Sobolev norms, K -types, tensor-tempered index. Every project-specific technical term is defined before its first load-bearing use or recalled by an exact backward reference.

Main outputs. explicit matrix-coefficient bounds.

External-input rule. Every nontrivial imported result is either part of the declared graduate-level prerequisites, an explicitly named external research input in the prerequisite map, or is proved in this series before use (possibly under an explicit deferred-proof contract). A citation by itself never constitutes a proof.

Reader map and proof route

Question of the volume. What quantitative input upgrades qualitative mixing to an explicit decay of matrix coefficients with controlled Sobolev norms?

Prerequisite checkpoint. Volume VI supplies the qualitative architecture. Compact harmonic analysis, K-type decomposition, spherical estimates, and Sobolev summation used load-bearingly are proved in this volume's proof modules rather than silently added to the external ledger.

Load-bearing route. Define the Sobolev norms and spectral-gap parameter first. Establish decay for controlled K-types, prove the K-type dimension/Sobolev summability estimates, and only then sum over K-types. A bound for a single spherical coefficient is not yet a bound for arbitrary smooth vectors.

Sanity check. If $v = \sum_{\tau} v_{\tau}$ and $w = \sum_{\sigma} w_{\sigma}$ are K-type expansions, write the double sum for $\langle \pi(g)v, w \rangle$. The quantitative theorem is exactly the statement that coefficient decay beats the growth in K-type multiplicities after enough Sobolev derivatives are paid.

Chapter 1

Smooth Vectors and Sobolev Norms

1.1 Differentiating the regular representation

Let $X = \Gamma \backslash G$ with invariant probability m . Fix a basis $Y_1, \dots, Y_d$ of $\mathfrak{g}$. For a smooth function f define the right-invariant differential operator

$$D_Y f(x) = \left. \frac{d}{dt} \right|_{t=0} f(x \exp(tY)).$$

For a word $\alpha = (i_1, \dots, i_r)$ write $D^\alpha = D_{Y_{i_1}} \cdots D_{Y_{i_r}}$.

Definition 1.1 (Sobolev norm). For an integer $\ell \geq 0$ set

$$\mathcal{S}_\ell(f)^2 = \sum_{|\alpha| \leq \ell} \|D^\alpha f\|_{L^2(X)}^2.$$

On a noncompact quotient a height-weighted version will be introduced later.

Lemma 1.2 (Translation growth). *For each ℓ there are C_ℓ and an integer q_ℓ such that*

$$\mathcal{S}_\ell(f \circ R_g) \leq C_\ell \max(\|\mathrm{Ad} g\|, \|\mathrm{Ad} g^{-1}\|)^{q_\ell} \mathcal{S}_\ell(f).$$

Proof. Differentiate $f(xg \exp tY) = f(x \exp(t \mathrm{Ad}(g)Y)g)$. Every derivative of order r becomes a linear combination of derivatives of f of order r , with coefficients polynomial of degree r in the matrix entries of $\mathrm{Ad}(g)$. Sum the L^2 norms and use invariance of Haar measure under right translation. $\square$

1.2 Sobolev embedding in local charts

If $\ell > \dim X/2$, Euclidean Sobolev embedding in finitely many compact coordinate charts gives

$$\|f\|_\infty \leq C_K \mathcal{S}_\ell(f)$$

for functions supported in a fixed compact K . A partition of unity proves the manifold statement. On a cusp the chart injectivity radius shrinks; height weights compensate for that loss.

Lemma 1.3 (Product estimate). *For $\ell > \dim X/2$,*

$$\mathcal{S}_\ell(fg) \leq C_\ell \mathcal{S}_{\ell+r}(f) \mathcal{S}_{\ell+r}(g)$$

for a fixed small r depending only on the chart construction.

Proof. Leibniz expands every derivative of fg . Put the factor with fewer derivatives in L^∞ using Sobolev embedding and the other in L^2 . Sum over the finitely many terms. $\square$

Chapter 2

Spectral Gap as a Quantitative Poincaré Inequality

2.1 Abstract formulation

Let π be the Koopman representation on $L_0^2(X)$. A spectral gap means there are a compact neighborhood $Q \subset G$ and $\kappa > 0$ such that

$$\sup_{g \in Q} \|\pi(g)v - v\| \geq \kappa \|v\| \quad (v \in L_0^2).$$

An equivalent averaged formulation uses a compactly supported symmetric probability measure ν : the associated Poincaré inequality says that the *top* of the spectrum of $\pi(\nu)$ is strictly below 1. One must not replace “top of the spectrum” by the operator norm for an arbitrary symmetric average, because symmetry allows spectrum near -1 . If one first *lazifies* the measure by

$$\tilde{\nu} = \frac{1}{2}(\delta_e + \nu),$$

then the spectrum is shifted into $[0, 1]$ and the operator-norm formulation is valid.

Lemma 2.1 (Poincaré, top-spectrum, and lazy-gap equivalence). *Let ν be a symmetric probability measure and put $A = \pi(\nu)|_{L_0^2(X)}$ and $\tilde{\nu} = (\delta_e + \nu)/2$. For $\delta > 0$, the following are equivalent:*

1. for every $v \in L_0^2(X)$,

$$\int_G \|\pi(g)v - v\|^2 d\nu(g) \geq 2\delta \|v\|^2;$$

2. $\sup \sigma(A) \leq 1 - \delta$;

3.

$$\|\pi(\tilde{\nu})|_{L_0^2(X)}\| \leq 1 - \frac{\delta}{2}.$$

Proof. Symmetry gives $A = A^*$ and $\sigma(A) \subset [-1, 1]$. For every $v \in L_0^2(X)$,

$$\begin{aligned} \int_G \|\pi(g)v - v\|^2 d\nu(g) &= \int_G (2\|v\|^2 - 2\Re\langle \pi(g)v, v \rangle) d\nu(g) \\ &= 2\|v\|^2 - 2\langle Av, v \rangle. \end{aligned}$$

Hence the Poincaré inequality is equivalent to $\langle Av, v \rangle \leq (1 - \delta)\|v\|^2$ for every v , which, because A is self-adjoint, is exactly $\sup \sigma(A) \leq 1 - \delta$.

Finally,

$$\pi(\tilde{\nu})|_{L_0^2(X)} = \frac{I + A}{2}.$$

Its spectrum is $\{(1 + \lambda)/2 : \lambda \in \sigma(A)\} \subset [0, 1]$. Therefore its operator norm is its spectral maximum, and

$$\|\pi(\tilde{\nu})|_{L_0^2(X)}\| = \frac{1 + \sup \sigma(A)}{2}.$$

This gives the equivalence of (2) and (3), with the stated constants. □

2.2 Fixed rank-one quotients

For a finite-area hyperbolic quotient, the Casimir operator corresponds to the Laplacian. On L_0^2 the constant eigenvalue 0 is removed. The spectrum below the continuous threshold is discrete with finite multiplicity and can accumulate only at the threshold; therefore its infimum is a positive number $\lambda_1 > 0$. This positive number is the spectral-gap parameter for the fixed quotient. No uniform statement over congruence covers is being made.

Chapter 3

Relative Property (T) for $\mathrm{SL}_2 \ltimes \mathbb{R}^2$

3.1 Fourier decomposition along the normal abelian group

Let $P = \mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2$. If a unitary representation has no nonzero $\mathbb{R}^2$ -fixed vector, the spectral theorem for the abelian normal subgroup realizes the restriction to $\mathbb{R}^2$ as multiplication by characters on a measure space over the dual $\widehat{\mathbb{R}^2} \simeq \mathbb{R}^2$. The SL_2 action pushes the spectral measure according to its linear action on the dual.

Lemma 3.1 (No common almost-invariant probability on $\mathbb{P}^1$). *Let*

$$u = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \bar{u} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

There are a metric d_w inducing the weak topology on $\mathcal{P}(\mathbb{P}^1(\mathbb{R}))$ and $c > 0$ such that every probability measure ν satisfies

$$\max\{d_w(u_*\nu, \nu), d_w(\bar{u}_*\nu, \nu)\} \geq c.$$

Proof. First there is no probability measure invariant under both u and $\bar{u}$. Indeed, identify $\mathbb{P}^1 \setminus \{\infty\}$ with $\mathbb{R}$. The map u acts by $x \mapsto x + 1$. If an invariant probability gave positive mass to a compact interval $I \subset \mathbb{R}$ of length < 1 , its translates $u^n I$ would be pairwise disjoint with the same positive mass, a contradiction. By a countable cover of $\mathbb{R}$ by such intervals, every

u -invariant probability is therefore δ_∞ . For $\bar{u}$ use the coordinate $y = 1/x$ on $\mathbb{P}^1 \setminus \{0\}$. Since $\bar{u}[x : 1] = [x : x + 1]$, one has $y \mapsto y + 1$. The same disjoint-translate argument shows that a $\bar{u}$ -invariant probability gives zero mass to $\mathbb{P}^1 \setminus \{0\}$, hence it is δ_0 .

Choose any metric d_w inducing weak convergence on the compact metrizable space $\mathcal{P}(\mathbb{P}^1)$. If the conclusion failed, there would be probabilities ν_j for which both displayed distances tend to zero. Compactness gives a weakly convergent subsequence $\nu_j \rightarrow \nu$. Continuity of push-forward by the homeomorphisms $u, \bar{u}$ gives $u_*\nu = \nu = \bar{u}_*\nu$, contradicting the first paragraph. Hence the continuous function

$$\nu \mapsto \max\{d_w(u_*\nu, \nu), d_w(\bar{u}_*\nu, \nu)\}$$

has a positive minimum c . □

Theorem 3.2 (Relative property (T)). *The pair $(\mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2, \mathbb{R}^2)$ has relative property (T): there are a compact $Q \subset P$ and $\kappa > 0$ such that if a unitary representation contains a (Q, κ) -almost invariant unit vector, then it contains a nonzero $\mathbb{R}^2$ -fixed vector.*

Proof. Let F be the $\mathbb{R}^2$ -fixed subspace. Since $\mathbb{R}^2$ is normal, F and $F^\perp$ are P -invariant. It is enough to rule out a sufficiently almost-invariant unit vector $v \in F^\perp$.

By the spectral theorem for the normal abelian subgroup, the vector state of v determines a probability measure μ_v on $\widehat{\mathbb{R}^2} \simeq \mathbb{R}^2$. The assumption $v \perp F$ says $\mu_v(\{0\}) = 0$. Hence radial projection gives a probability $\bar{\mu}_v$ on $\mathbb{P}^1$. Spectral covariance gives

$$\mu_{\pi(g)v} = g_*\mu_v \quad (g \in \mathrm{SL}_2(\mathbb{R})).$$

Moreover, for every Borel set A , the corresponding spectral projection $E(A)$ satisfies

$$|\mu_{\pi(g)v}(A) - \mu_v(A)| = |\langle E(A)\pi(g)v, \pi(g)v \rangle - \langle E(A)v, v \rangle| \leq 2\|\pi(g)v - v\|.$$

Thus total variation, and therefore the weak metric used in Lemma 3.1, is controlled by a fixed multiple of the vector displacement. Choose $Q = \{u, \bar{u}\} \subset \mathrm{SL}_2 \subset P$ and choose $\kappa > 0$ so small that a (Q, κ) -almost invariant unit vector would make both $d_w(u_*\bar{\mu}_v, \bar{\mu}_v)$ and $d_w(\bar{u}_*\bar{\mu}_v, \bar{\mu}_v)$ smaller than the constant c of that lemma. This is impossible. Therefore every sufficiently almost invariant vector has nonzero projection to F , proving relative property (T). □

Chapter 4

Property (T) for Higher-Rank Special Linear Groups

4.1 A quantitative form of relative property (T)

The existence statement in Theorem 3.2 has a useful metric consequence that avoids any repeated-projection argument.

Lemma 4.1 (Distance to the fixed space). *Let $P = \mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2$ and let Q, κ be a relative Kazhdan pair from Theorem 3.2. If π is a unitary representation, $F = \mathrm{Fix}_\pi(\mathbb{R}^2)$, and v is any vector, then*

$$\mathrm{dist}(v, F) \leq \kappa^{-1} \sup_{q \in Q} \|\pi(q)v - v\|.$$

Proof. The subspace F is P -invariant because $\mathbb{R}^2 \triangleleft P$; hence so is $F^\perp$. Write $v = v_F + v_\perp$. If $v_\perp \neq 0$, the representation on $F^\perp$ contains no nonzero $\mathbb{R}^2$ -fixed vector. The contrapositive of relative property (T), applied to $v_\perp / \|v_\perp\|$, gives some $q \in Q$ with

$$\|\pi(q)v_\perp - v_\perp\| \geq \kappa \|v_\perp\|.$$

Orthogonal projection onto $F^\perp$ commutes with $\pi(q)$, so the left side is at most $\|\pi(q)v - v\|$. Since $\|v_\perp\| = \mathrm{dist}(v, F)$, the claim follows. $\square$

4.2 Bounded generation by elementary unipotents

For $i \neq j$ write $u_{ij}(t) = I + tE_{ij}$.

Lemma 4.2 (A diagonal two-by-two block is elementary). *For every $a \in \mathbb{R}^\times$,*

$$\begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} = u_{12}(a-1)u_{21}(1)u_{12}(a^{-1}-1)u_{21}(-a),$$

after the evident identification with the corresponding 2×2 block.

Proof. Multiplying the four matrices gives

$$\begin{pmatrix} 1 & a-1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & a^{-1}-1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -a & 1 \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}.$$

□

Lemma 4.3 (Uniform elementary generation). *For every $n \geq 2$ there is $M_n < \infty$ such that every element of $\mathrm{SL}_n(\mathbb{R})$ is a product of at most M_n elements belonging to root subgroups $U_{ij} = \{u_{ij}(t) : t \in \mathbb{R}\}$.*

Proof. Perform Gaussian elimination using row additions. If the current pivot is zero, first add another row having a nonzero entry in that column; this is again multiplication by one elementary unipotent and avoids the need for a permutation matrix. Clearing the entries below the pivot uses at most $n-1$ elementary matrices in the first column, then at most $n-2$ in the second, and so on. Applying the same procedure from the right, or continuing with row operations above the pivots, reduces an arbitrary $g \in \mathrm{SL}_n(\mathbb{R})$ to a diagonal matrix in at most $n(n-1)$ root-unipotent factors.

A diagonal matrix $\mathrm{diag}(a_1, \dots, a_n)$ of determinant one is the product of the $n-1$ adjacent diagonal blocks

$$\mathrm{diag}(a_1, a_1^{-1}, 1, \dots, 1), \quad \mathrm{diag}(1, a_1 a_2, (a_1 a_2)^{-1}, 1, \dots), \quad \dots$$

Each adjacent block is a product of four root-unipotents by Lemma 4.2. Thus, for example, $M_n = n(n-1) + 4(n-1)$ suffices. The numerical value is irrelevant; only uniform boundedness in the element g is used below. □

4.3 Property (T) without successive projections

Theorem 4.4 (Kazhdan property (T) for $\mathrm{SL}_n(\mathbb{R})$, $n \geq 3$). *For $n \geq 3$, $\mathrm{SL}_n(\mathbb{R})$ has property (T).*

Proof. For every ordered pair $i \neq j$, choose a third index k . The root subgroup U_{ij} is one of the two translation root groups in a conjugate of $\mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2$ supported on the coordinates i, j, k . There are only finitely many pairs. Let Q_G be the union of corresponding relative Kazhdan sets and let C be the maximum of the constants in Lemma 4.1.

Suppose a unit vector v satisfies

$$\sup_{q \in Q_G} \|\pi(q)v - v\| \leq \varepsilon.$$

For every root subgroup U_{ij} , the distance from v to $F_{ij} = \mathrm{Fix}_\pi(U_{ij})$ is at most $C\varepsilon$. Hence, for every $u \in U_{ij}$,

$$\|\pi(u)v - v\| \leq 2 \mathrm{dist}(v, F_{ij}) \leq 2C\varepsilon. \quad (11.1)$$

By Lemma 4.3, every $g \in G$ is a product of at most M_n root-unipotent elements. Telescoping and unitarity give

$$\sup_{g \in G} \|\pi(g)v - v\| \leq 2M_n C\varepsilon. \quad (11.2)$$

Choose $\varepsilon < (4M_n C)^{-1}$. Then the orbit $\pi(G)v$ is contained in the ball $B(v, 1/2)$.

Let K be the closed convex hull of $\pi(G)v$. It is a nonempty closed bounded convex subset of the Hilbert space. There is a unique point $w \in K$ of minimal norm: existence follows by taking a minimizing sequence and using the parallelogram identity to see it is Cauchy, and uniqueness follows from strict convexity of the Hilbert norm. Since K is G -invariant and every $\pi(g)$ is an isometry, uniqueness forces $\pi(g)w = w$ for all g . Moreover $K \subset B(v, 1/2)$, so $\|w\| \geq 1/2$; hence $w \neq 0$. We have produced a nonzero invariant vector, which proves property (T). $\square$

4.4 Consequences and a necessary distinction

If G has property (T), the Koopman representation on $L_0^2(\Gamma \backslash G)$ has a spectral gap for every lattice Γ : a sequence of almost invariant unit vectors would produce a nonzero G -fixed vector in L_0^2 , impossible because the full G action on the homogeneous probability space is ergodic.

Property (T), however, is an *isolation statement*. A numerical decay rate for arbitrary matrix coefficients requires more representation-theoretic information. To keep every later constant honest we encode that extra information by an L^p exponent.

Chapter 5

L^p Exponents, the Harish–Chandra Function, and Quantitative Matrix Coefficients

5.1 The quantitative input: tensor-temperedness

A qualitative spectral gap says that the trivial representation is isolated; it does not by itself specify a numerical decay exponent. The parameter used in the rest of this series is therefore the following concrete one.

Definition 5.1 (Tempered and tensor-tempered). A unitary representation σ of G is *tempered* if it is weakly contained in the left regular representation λ_G , equivalently

$$\|\sigma(F)\| \leq \|\lambda_G(F)\| \quad (F \in C_c(G)).$$

The *tensor-tempered index* $m(\pi)$ is the least positive integer m for which $\pi^{\otimes m}$ is tempered, if such an m exists. For comparison with the conventional L^p terminology, we write $p_{\text{tens}}(\pi) = 2m(\pi)$.

For a fixed homogeneous quotient, $m(\pi)$ can be estimated from the actual spectral data. In rank one this is read from the complementary-series parameter. In higher rank it can come from strong spectral-gap or representation-theoretic information. Volumes XII–XIV state every rate in terms of $m(\pi)$

(or an explicitly stronger input), so no numerical decay is silently extracted from property (T).

5.2 The Harish–Chandra spherical function

Choose an Iwasawa decomposition $G = KAN$ and write $H(g) \in \mathfrak{a}$ for the A -coordinate. Let ρ be half the sum of positive restricted roots, with multiplicity. Define

$$\Xi(g) = \int_K e^{-\rho(H(gk))} dk.$$

It is K -bi-invariant, positive definite, and $\Xi(e) = 1$.

Proposition 5.2 (Harish–Chandra estimate). *There are $C, r > 0$ such that for $H \in \overline{\mathfrak{a}^+}$,*

$$\Xi(\exp H) \leq C(1 + \|H\|)^r e^{-\rho(H)}. \quad (11.3)$$

Proof. Use the open Bruhat cell $\bar{N}MAN$ and the integration formula

$$\int_K F(k) dk = c \int_{\bar{N}} F(k(\bar{n})) e^{-2\rho(H(\bar{n}))} d\bar{n},$$

which follows from the Jacobian of Iwasawa coordinates: conjugation by $a = \exp H$ multiplies Haar measure on the root space $\mathfrak{g}_{-\alpha}$ by $e^{-m_\alpha \alpha(H)}$, and multiplying over roots gives $e^{-2\rho(H)}$. Applying this formula to the defining integral for $\Xi(a)$ and using the cocycle identity for the Iwasawa A -coordinate gives

$$\Xi(a) = e^{-\rho(H)} c \int_{\bar{N}} e^{-\rho(H(a^{-1}\bar{n}a)) + \rho(H(\bar{n}))} e^{-2\rho(H(\bar{n}))} d\bar{n}. \quad (11.4)$$

Order the negative root coordinates on $\bar{N}$ by height. In exponential coordinates, the A -coordinate of $\bar{n}$ is comparable, up to constants depending only on G , to a product of factors $(1 + |x_\alpha|)^{c_\alpha}$; this follows inductively from the rank-one SL_2 identity in each root string, while commutators involve only higher root height and have already been controlled. Conjugation by a^{-1} replaces x_α by $e^{-\alpha(H)}x_\alpha$. Splitting each root coordinate into the regions $|x_\alpha| \leq e^{\alpha(H)}$ and $|x_\alpha| > e^{\alpha(H)}$ makes the integral in (11.4) a finite sum of products of one-dimensional integrals. The borderline exponent contributes at most one factor $1 + \alpha(H)$; all nonborderline exponents are integrable with a uniform constant. Multiplication over the finitely many positive roots gives a polynomial factor $(1 + \|H\|)^r$ and proves (11.3). $\square$

5.3 Tempered representations and Herz majorization

The pointwise estimate needed later is stronger than the definition of weak containment. We prove it here in the exact K -finite form used by the counting arguments.

5.4 Compact-group Fourier analysis needed below

The next two results are proved internally here in the precise forms needed below. Throughout, K is a compact matrix Lie group and dk is Haar probability measure.

Lemma 5.3 (Unitary reduction and Schur's lemma). *Every finite-dimensional continuous complex representation of K admits a K -invariant Hermitian inner product. If (τ, V) and (σ, W) are irreducible unitary representations, every intertwiner $T : V \rightarrow W$ is zero when $\tau \not\cong \sigma$ and is scalar when $\tau = \sigma$.*

Proof. Starting from any Hermitian form $(\cdot, \cdot)_0$, put

$$(v, w)_K = \int_K (\tau(k)v, \tau(k)w)_0 dk.$$

For $v \neq 0$ the continuous function $k \mapsto (\tau(k)v, \tau(k)v)_0$ is strictly positive, so its Haar integral is positive; thus $(\cdot, \cdot)_K$ is positive definite. For $h \in K$, the substitution $k \mapsto kh$ and left invariance of Haar measure give

$$(\tau(h)v, \tau(h)w)_K = \int_K (\tau(kh)v, \tau(kh)w)_0 dk = (v, w)_K,$$

so the form is K -invariant. Thus orthogonal complements of invariant subspaces are invariant, so every finite-dimensional representation is an orthogonal direct sum of irreducibles.

For the second assertion, $\ker T$ and $\operatorname{im} T$ are invariant. Irreducibility makes a nonzero T an isomorphism. If $V = W$ and T is an endomorphism, choose an eigenvalue λ of the complex matrix T . Then $\ker(T - \lambda I)$ is a nonzero invariant subspace and hence all of V . Thus $T = \lambda I$. $\square$

Theorem 5.4 (Schur orthogonality). *Let τ, σ be irreducible unitary representations of dimensions d_τ, d_σ , and choose orthonormal bases. Their matrix*

coefficients satisfy

$$\int_K \tau_{ij}(k) \overline{\sigma_{ab}(k)} dk = \begin{cases} d_\tau^{-1} \delta_{ia} \delta_{jb}, & \tau \simeq \sigma \text{ and the chosen bases} \\ & \text{identify the two representations,} \\ 0, & \tau \not\simeq \sigma. \end{cases}$$

Proof. For a linear map $A : W \rightarrow V$ define

$$\mathcal{P}(A) = \int_K \tau(k) A \sigma(k)^{-1} dk.$$

Changing variables shows that $\mathcal{P}(A)$ intertwines σ with τ . Lemma 5.3 makes it zero for inequivalent irreducibles. When $V = W$ and $\tau = \sigma$, it equals $c(A)I$. Taking traces gives $c(A) = \text{tr}(A)/d_\tau$. Apply this identity to the matrix unit $A = E_{jb}$ and read the (i, a) entry. This is exactly the displayed integral. $\square$

Theorem 5.5 (Peter–Weyl for compact matrix groups). *The normalized irreducible matrix coefficients $\sqrt{d_\tau} \tau_{ij}$ form an orthonormal basis of $L^2(K)$. Equivalently,*

$$L^2(K) = \widehat{\bigoplus_{\tau \in \widehat{K}} V_\tau \otimes V_\tau^*},$$

and finite sums of irreducible matrix coefficients are uniformly dense in $C(K)$.

Proof. Schur orthogonality gives orthonormality. It remains to prove density. Because K is a compact matrix group, average the ambient Hermitian form on its defining representation and conjugate once so that $K \subset U(N)$. Let $\mathcal{A}$ be the complex linear span of matrix coefficients of all finite-dimensional continuous representations of K . Products of coefficients are coefficients of tensor products, complex conjugates are coefficients of conjugate representations, and constants are coefficients of the trivial representation. Hence $\mathcal{A}$ is a self-adjoint unital subalgebra of $C(K)$. The coefficients of the defining representation separate points of K . The Stone–Weierstrass theorem proved in Volume I, Appendix A therefore makes $\mathcal{A}$ uniformly dense in $C(K)$.

By Lemma 5.3, every finite-dimensional representation is a direct sum of irreducibles; consequently each element of $\mathcal{A}$ is a finite linear combination of irreducible coefficients. Thus those coefficients are uniformly dense in $C(K)$, hence dense in $L^2(K)$. Their orthonormality then makes them an orthonormal basis. The displayed Hilbert direct sum is simply the grouping of basis vectors by irreducible type. $\square$

Worked Example 5.6 ($K = \mathrm{SO}(2)$). Writing k_θ for rotation by θ , the irreducibles are the characters $e_n(k_\theta) = e^{in\theta}$, $n \in \mathbb{Z}$. The preceding theorem becomes the ordinary Fourier series decomposition $L^2(S^1) = \hat{\oplus}_{n \in \mathbb{Z}} \mathbb{C}e_n$. The abstract K -type sums below are therefore the noncommutative version of inserting powers of $(1 + n^2)^{-1}$ into a Fourier series.

5.4.1 The only Casimir counting estimate we need

Choose an $\mathrm{Ad}(K)$ -invariant inner product on $\mathfrak{k}$ and an orthonormal basis $X_1, \dots, X_d$. On smooth functions put

$$\Omega_K = - \sum_{j=1}^d X_j^2,$$

where the X_j act as left-invariant vector fields. The operator is positive and commutes with both left and right translations. Hence on the τ -isotypic block it acts by a scalar $\lambda_\tau \geq 0$.

Lemma 5.7 (Local elliptic estimate for the compact Casimir). *For every integer $r \geq 0$ there is C_r such that*

$$\|f\|_{H^{2r}(K)} \leq C_r \|(1 + \Omega_K)^r f\|_{L^2(K)} \quad (f \in C^\infty(K)),$$

where the Sobolev norm on the left is defined in any fixed finite coordinate atlas.

Proof. We include the finite-chart comparison explicitly. In a coordinate chart $U \subset \mathbb{R}^d$, write $X_j = \sum_a b_{ja}(x) \partial_a$. Since the X_j form a basis of every tangent space, the principal symbol of Ω_K is

$$q_x(\xi) = \sum_j \left(\sum_a b_{ja}(x) \xi_a \right)^2,$$

and on a relatively compact chart there are constants $0 < c < C$ with $c|\xi|^2 \leq q_x(\xi) \leq C|\xi|^2$. Shrink the chart so that the coefficients vary by less than $c/4$. For a compactly supported function u in this chart, freeze the coefficients at its center. Fourier transform then gives

$$\|u\|_{H^2} \leq C(\|\Omega_K u\|_2 + \|u\|_2) + \tfrac{1}{2}\|u\|_{H^2};$$

the last term comes from the small variation of the principal coefficients, while first-order coefficient terms are bounded by $\epsilon\|u\|_{H^2} + C_\epsilon\|u\|_2$. Absorb the half term.

Choose a finite partition of unity subordinate to such charts. Commuting Ω_K past a cutoff produces only first-order operators, already controlled by the preceding estimate, so summing the localized inequalities gives

$$\|f\|_{H^2(K)} \leq C(\|\Omega_K f\|_2 + \|f\|_2).$$

Apply this estimate successively to derivatives of order at most $2r - 2$. Commutators of a coordinate derivative with Ω_K have order at most two with one fewer uncontrolled derivative; induction on r therefore yields $\|f\|_{H^{2r}} \leq C_r \sum_{j=0}^r \|\Omega_K^j f\|_2$, which is bounded by the displayed right-hand side after expanding $(1 + \Omega_K)^r$ spectrally on finite Peter–Weyl sums and then using density. $\square$

Lemma 5.8 (Hilbert–Schmidt Sobolev inclusion). *If $s > d/2$, the inclusion $H^s(K) \hookrightarrow L^2(K)$ is Hilbert–Schmidt.*

Proof. On the flat d -torus the Fourier basis shows that the squared Hilbert–Schmidt norm of $H^s \hookrightarrow L^2$ is a constant multiple of

$$\sum_{n \in \mathbb{Z}^d} (1 + |n|^2)^{-s},$$

which converges exactly when $s > d/2$. For a coordinate cube, extend a compactly supported function to a slightly larger torus; restriction and extension are bounded on H^s and L^2 . A finite partition of unity on K factors the global inclusion through a finite direct sum of these local inclusions. Bounded operators composed with a Hilbert–Schmidt operator are Hilbert–Schmidt, and a finite sum remains Hilbert–Schmidt. $\square$

Proposition 5.9 (Casimir summability for K -types). *For all sufficiently large integers r ,*

$$\sum_{\tau \in \widehat{K}} d_\tau^2 (1 + \lambda_\tau)^{-2r} < \infty.$$

Consequently, for every integer $m \geq 1$ and all sufficiently large r ,

$$\sum_{\tau \in \widehat{K}} d_\tau^{1/m} (1 + \lambda_\tau)^{-2r} < \infty.$$

Proof. Take $2r > d/2$ large enough for Lemmas 5.7 and 5.8. Equip the completion of $C^\infty(K)$ for the norm $\|(1 + \Omega_K)^r f\|_2$ with that norm. The elliptic estimate factors its inclusion into $L^2(K)$ through a Sobolev inclusion

that is Hilbert–Schmidt. Hence this Casimir-weighted inclusion is Hilbert–Schmidt.

By Theorem 5.5, the τ block has exactly d_τ^2 orthonormal matrix coefficients. After normalizing the domain norm, each of these basis vectors has inclusion singular value $(1 + \lambda_\tau)^{-r}$. The squared Hilbert–Schmidt norm is therefore precisely the first displayed series. Since $d_\tau^{1/m} \leq d_\tau^2$, the second series follows. $\square$

Remark 5.10 (Proof ownership for compact harmonic analysis). The proof uses no Weyl dimension formula. Peter–Weyl and Schur orthogonality are proved above, and the only growth input needed by the later double K -type sum is reduced to elementary Fourier summability in local coordinates plus the displayed elliptic estimate for the compact Casimir. Thus Peter–Weyl and Schur orthogonality are proved in this volume; no separate research-level input is used for them.

Lemma 5.11 (Spherical convolution majorization). *For $F \in C_c(G)$ put*

$$F_2(g) = \left(\int_K \int_K |F(kgk')|^2 dk dk' \right)^{1/2}.$$

Then

$$\|\lambda_G(F)\|_{2 \rightarrow 2} \leq \int_G F_2(g) \Xi(g) dg. \quad (11.5)$$

Proof. We first treat a nonnegative K -bi-invariant function ϕ . The convolution algebra $C_c(K \backslash G / K)$ is commutative. Its action on $L^2(G)$ is diagonalized by the spherical Fourier transform. We recall the short derivation rather than use this as a black box. The Abel transform

$$\mathcal{A}\phi(H) = e^{\rho(H)} \int_N \phi(\exp(H)n) dn$$

intertwines spherical convolution with ordinary convolution on the Euclidean space $\mathfrak{a}$ after Weyl symmetrization. This follows directly by writing both convolutions in KAN coordinates and using $d(ana^{-1}) = e^{2\rho(H)} dn$. Euclidean Fourier transform therefore gives

$$\|\lambda_G(\phi)\| = \sup_{\lambda \in \mathfrak{a}^*} \left| \int_G \phi(g) \varphi_\lambda(g) dg \right|,$$

where

$$\varphi_\lambda(g) = \int_K e^{(i\lambda - \rho)(H(gk))} dk.$$

Taking absolute values inside the compact integral yields $|\varphi_\lambda(g)| \leq \Xi(g)$, hence

$$\|\lambda_G(\phi)\| \leq \int_G \phi(g) \Xi(g) dg. \quad (11.6)$$

For general F , decompose the left and right K -regular representations by Theorem 5.5. On a finite partial sum of $K \times K$ types, convolution by F is a matrix of spherical convolution operators. Theorem 5.4 and Cauchy–Schwarz bound the Hilbert–Schmidt norm of that matrix kernel on the double coset KgK by $F_2(g)$. Apply (11.6) entrywise and then the operator Cauchy–Schwarz inequality to obtain (11.5) for the partial sum. Peter–Weyl partial sums converge to F in $L^2(K \times K)$ on every compact family of double cosets, while the right side is dominated by a compactly supported integrable function. Passing to the limit proves the lemma. $\square$

Theorem 5.12 (Herz–Cowling–Haagerup majorization for K -finite vectors). *If σ is tempered and v, w are K -finite, then*

$$|\langle \sigma(g)v, w \rangle| \leq \sqrt{d_v d_w} \Xi(g) \|v\| \|w\|, \quad (11.7)$$

where $d_v = \dim \langle \sigma(K)v \rangle$ and similarly for w .

Proof. Let $V = \langle \sigma(K)v \rangle$ and $W = \langle \sigma(K)w \rangle$. Choose orthonormal bases and use Schur orthogonality to construct K -finite approximate identities e_V, e_W whose convolution actions are the orthogonal projections onto the corresponding K -isotypic summands. For a nonnegative approximate identity η_ϵ at g , set $F_\epsilon = e_W * \eta_\epsilon * e_V$. Weak containment gives

$$\|\sigma(F_\epsilon)\| \leq \|\lambda_G(F_\epsilon)\|.$$

The $K \times K$ square envelope of F_ϵ has total mass at most $\sqrt{d_v d_w}$ times that of η_ϵ ; this is precisely Theorem 5.4 applied to the two finite K -modules. Lemma 5.11 and continuity of Ξ therefore give

$$\limsup_{\epsilon \rightarrow 0} \|\sigma(F_\epsilon)\| \leq \sqrt{d_v d_w} \Xi(g).$$

Since $e_V v = v$, $e_W w = w$, and $\sigma(\eta_\epsilon) \rightarrow \sigma(g)$ strongly, pairing with v, w yields (11.7) after restoring their norms. $\square$

Proposition 5.13 (Tensor-tempered coefficient bound). *If $m = m(\pi) < \infty$, then for K -finite v, w ,*

$$|\langle \pi(g)v, w \rangle| \leq (d_v d_w)^{1/(2m)} \Xi(g)^{1/m} \|v\| \|w\|. \quad (11.8)$$

Proof. The vectors $v^{\otimes m}, w^{\otimes m}$ are K -finite for $\pi^{\otimes m}$, their coefficient is the m th power of the original coefficient, and their K -span dimensions are at most d_v^m, d_w^m . Apply Theorem 5.12 to the tempered tensor power and take the m th root. $\square$

5.5 From K -finite to Sobolev vectors

Let Ω_K be the positive Casimir of K . Theorem 5.5 decomposes a smooth vector into K -types $v = \sum_{\tau} v_{\tau}$. Proposition 5.9 supplies exactly the summability needed below. Therefore, for $s > s_0(K)$,

$$\sum_{\tau} (\dim \tau)^{1/(2m)} \|v_{\tau}\| \leq C_s \|(1 + \Omega_K)^s v\|. \quad (11.7K)$$

Indeed, insert $(1 + \lambda_{\tau})^{-s}(1 + \lambda_{\tau})^s$ and apply Cauchy–Schwarz; convergence of the first square sum follows from Proposition 5.9.

Proposition 5.13, the Sobolev K -type summability estimate (11.7K), and Proposition 5.2 now give the form actually used later. Notice that Herz majorization (11.7) enters only through the tensor estimate (11.8); keeping these dependencies separate is important because the exponents are different.

Theorem 5.14 (Quantitative Howe–Moore from a tensor-tempered index). *Let $m = m(\pi) < \infty$. There is a Sobolev order ℓ such that for smooth vectors*

$$|\langle \pi(g)v, w \rangle| \leq C \Xi(g)^{1/m} \mathcal{S}_{\ell}(v) \mathcal{S}_{\ell}(w). \quad (11.9)$$

In Cartan coordinates $g = k \exp(H) k'$ this is

$$\ll (1 + \|H\|)^{r/m} e^{-\rho(H)/m} \mathcal{S}_{\ell}(v) \mathcal{S}_{\ell}(w). \quad (11.10)$$

Proof. Write the orthogonal Peter–Weyl decompositions

$$v = \sum_{\tau \in \widehat{K}} v_{\tau}, \quad w = \sum_{\sigma \in \widehat{K}} w_{\sigma}.$$

We first prove the estimate for finite sums. Proposition 5.13 gives, term by term,

$$|\langle \pi(g)v_{\tau}, w_{\sigma} \rangle| \leq (d_{\tau} d_{\sigma})^{1/(2m)} \Xi(g)^{1/m} \|v_{\tau}\| \|w_{\sigma}\|.$$

Summing the finite double series and factoring it yields

$$|\langle \pi(g)v, w \rangle| \leq \Xi(g)^{1/m} \left(\sum_{\tau} d_{\tau}^{1/(2m)} \|v_{\tau}\| \right) \left(\sum_{\sigma} d_{\sigma}^{1/(2m)} \|w_{\sigma}\| \right). \quad (11.9a)$$

Choose $s > s_0(K)$ as in (11.7K). To spell out that estimate, insert $(1 + \lambda_\tau)^{-s}(1 + \lambda_\tau)^s$ and use Cauchy–Schwarz:

$$\sum_{\tau} d_{\tau}^{1/(2m)} \|v_{\tau}\| \leq \left(\sum_{\tau} d_{\tau}^{1/m} (1 + \lambda_{\tau})^{-2s} \right)^{1/2} \left(\sum_{\tau} (1 + \lambda_{\tau})^{2s} \|v_{\tau}\|^2 \right)^{1/2}.$$

The first factor is finite by Proposition 5.9; the second equals $\|(1 + \Omega_K)^s v\|$. The same estimate holds for w . Thus (11.9a) becomes

$$|\langle \pi(g)v, w \rangle| \leq C_s \Xi(g)^{1/m} \|(1 + \Omega_K)^s v\| \|(1 + \Omega_K)^s w\|. \quad (11.9b)$$

The chosen global Sobolev norm $\mathcal{S}_{\ell}$ dominates these two K -Casimir norms when ℓ is sufficiently large. Finite Peter–Weyl sums are dense in that Sobolev norm, so (11.9b) passes to arbitrary smooth v, w by continuity. Finally Proposition 5.2, namely the Cartan estimate (11.3) for Ξ , gives (11.10). Left and right translation by K does not alter the Casimir eigenvalues, hence introduces no additional loss. This proves both displayed estimates. $\square$

5.6 Rank one: reading p from the Laplace spectrum

For $G = \mathrm{PSL}_2(\mathbb{R})$, a spherical complementary representation with parameter $s \in (1/2, 1)$ has leading coefficient $e^{-(1-s)t}$, whereas $\Xi(a_t) \asymp (1+t)e^{-t/2}$. Its m th tensor power is therefore tempered once $m(1-s) \geq 1/2$; this can also be checked directly by decomposing the tensor product into principal and complementary series. On a fixed finite-area quotient the bottom nonzero Laplace eigenvalue $\lambda_1 = s_1(1-s_1)$ bounds s_1 and hence gives an explicit finite tensor-tempered index. Principal-series components already have index one. This makes the dependence of later constants on the rank-one spectral gap completely explicit.

Chapter 6

Height-Weighted Sobolev Norms

6.1 Why ordinary Sobolev norms fail uniformly in cusps

Let $\text{ht}(x) \geq 1$ be a proper height function comparable to the inverse of the shortest rational wedge length. The injectivity radius at x is bounded below by a negative power of $\text{ht}(x)$, by the Zassenhaus/Mahler estimates already used in Volume III. Choose M larger than that power and define

$$\mathcal{S}_{\ell,M}(f)^2 = \sum_{|\alpha| \leq \ell} \int_X (1 + \text{ht}(x))^{2M} |D^\alpha f(x)|^2 dm(x).$$

Lemma 6.1 (Weighted Sobolev embedding). *Assume the fixed height satisfies the quantitative injectivity estimate*

$$\text{inj}(x) \geq c_0(1 + \text{ht}(x))^{-q} \quad (x \in X) \quad (6.1)$$

for constants $c_0, q > 0$. Put $n = \dim X$ and let $\ell > n/2$. Then there is $C = C(X, \ell, c_0, q)$ such that

$$|f(x)| \leq C(1 + \text{ht}(x))^{q(\ell+n/2)} \mathcal{S}_{\ell,M}(f) \quad (6.2)$$

for every $M \geq 0$ and every smooth f with finite displayed Sobolev norm. In particular, choosing $M \geq q(\ell + n/2)$ absorbs every local-coordinate loss used later in the volume.

Proof. Take a coordinate ball of radius $r_x = c_1(1 + \text{ht}(x))^{-q}$ with c_1 small enough that the exponential chart has uniformly bounded distortion. Rescale this ball to the unit ball in $\mathbb{R}^n$. Euclidean Sobolev embedding gives

$$|f(x)| \ll \sum_{j=0}^{\ell} r_x^{j-n/2} \|D^j f\|_{L^2(B(x, r_x))}.$$

Since $0 < r_x \leq 1$, every coefficient is at most $r_x^{-\ell-n/2} \ll (1 + \text{ht}(x))^{q(\ell+n/2)}$. Dropping the weight only decreases each local L^2 norm because $(1 + \text{ht})^M \geq 1$. Summing the derivative terms gives (6.2). The final assertion is the bookkeeping choice used when a later localization also differentiates the height cutoff. $\square$

Chapter 7

Effective Mixing

Theorem 7.1 (Effective mixing on a fixed quotient). *Let the Koopman representation on $L_0^2(X)$ be p -tempered and put $m = \lceil p/2 \rceil$. Then for smooth functions f_1, f_2 on X ,*

$$\left| \int_X f_1(xg)f_2(x) dm - \int f_1 dm \int f_2 dm \right| \leq C\Xi(g)^{1/m} \mathcal{S}_{\ell,M}(f_1) \mathcal{S}_{\ell,M}(f_2).$$

In Cartan coordinates this is bounded by a polynomial factor times $e^{-\rho(H(g))/m}$.

Proof. Subtract the means and identify the correlation with a Koopman matrix coefficient. Apply Theorem 5.14. Weighted norms dominate the ordinary smooth-vector norms and keep the bound uniform when the functions have support moving into a controlled cusp region. $\square$

7.1 Effective mean ergodic theorem

Let β_T be a probability measure on G . If $\|\pi(\beta_T)\|_{L_0^2} \leq E(T)$, then

$$\left\| \beta_T * f - \int f \right\|_2 \leq E(T) \|f\|_2.$$

Chebyshev immediately gives an exceptional-set estimate

$$m\{x : |\beta_T * f(x) - \int f| > \eta\} \leq \eta^{-2} E(T)^2 \|f\|_2^2.$$

This elementary observation is repeatedly used in effective closing, where a spectral estimate must be converted into many generic points at a finite scale.

Chapter 8

Proof Ledger and Quantitative Input Contract

The load-bearing chain owned by this volume is

$$\begin{array}{c} \text{relative property (T) + bounded root generation} \\ \Downarrow \\ \text{property (T) for } \mathrm{SL}_n \\ L^p\text{-integrability + tensor powers + Herz majorization} \\ \Downarrow \\ \text{quantitative matrix coefficients} \\ \text{quantitative matrix coefficients + Sobolev calculus} \\ \Downarrow \\ \text{effective mixing} \end{array}$$

The distinction between the two lines is deliberate. Property (T) supplies an abstract spectral gap, but a numerical matrix-coefficient exponent is recorded by the p -temperedness parameter. For a fixed rank-one quotient, p is determined by the bottom of the nonconstant Laplace spectrum. In higher-rank applications a later theorem may use any independently proved finite p ; it must state that input explicitly. No uniform congruence-family gap, Ramanujan bound, or super-approximation theorem is hidden in this volume.

Chapter 9

Worked Spectral Laboratory: Tensor Powers and K-Type Summation

9.1 Motivation

The quantitative mixing exponent is lost twice: first when a representation is tensor-powered into the tempered range, and again when infinitely many K -types are summed with Sobolev weights.

Worked Example 9.1 (Complementary-series tensor threshold). For a complementary-series parameter $s \in (1/2, 1)$ of $\mathrm{PSL}_2(\mathbb{R})$, the slow matrix-coefficient exponent is $1 - s$. The m -fold tensor power has exponent $m(1 - s)$. Requiring the principal tempered threshold $1/2$ gives

$$m(1 - s) \geq \frac{1}{2}, \quad m \geq \frac{1}{2(1 - s)}.$$

For example $s = 3/4$ requires $m \geq 2$, while $s = 9/10$ requires $m \geq 5$. This calculation is why a small spectral gap produces a larger tensor-tempered index.

Worked Example 9.2 (A toy K -type Sobolev sum). For $K = \mathrm{SO}(2)$ the irreducible K -types are $e^{in\theta}$ and have dimension one and Casimir size $1 + n^2$. If $v = \sum_n v_n e^{in\theta}$, then Cauchy–Schwarz gives

$$\sum_n |v_n| \leq \left(\sum_n (1 + n^2)^{-s} \right)^{1/2} \left(\sum_n (1 + n^2)^s |v_n|^2 \right)^{1/2}.$$

The first series converges for $s > 1/2$. The second factor is the Sobolev norm. Volume XI's general K-type summation is this calculation with Weyl dimension growth inserted.

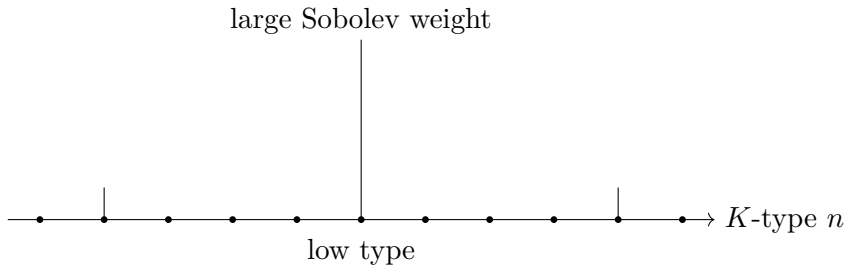


Figure 9.1: Sobolev weights penalize high K -types strongly enough to make the matrix-coefficient double series absolutely summable.

Chapter 10

Student Proof Laboratory: Where Quantitative Mixing Comes From

Qualitative Howe–Moore says that matrix coefficients vanish at infinity. An effective counting theorem needs an inequality with a rate and with explicit dependence on the regularity of the test functions. This chapter explains the three numerical inputs: a spectral gap, a tensor-tempered index, and Sobolev control.

10.1 Spectral gap as a Poincare inequality

Let π be the Koopman representation on $L_0^2(X)$. Fix a compact symmetric generating neighborhood $Q \subset G$. A spectral gap means that there is $\kappa > 0$ such that

$$\sup_{q \in Q} \|\pi(q)f - f\|_2 \geq \kappa \|f\|_2 \quad (f \in L_0^2(X)).$$

If no such κ exists, choose unit vectors f_n with $\sup_{q \in Q} \|\pi(q)f_n - f_n\|_2 \rightarrow 0$. Finite products of elements of Q then show that f_n is almost invariant under every compact subset of G . Thus a spectral gap is precisely a uniform exclusion of almost invariant vectors.

10.2 Relative property (T): the Fourier picture

For $G = \mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2$, restrict a unitary representation to the normal abelian subgroup $V = \mathbb{R}^2$. The spectral theorem writes that restriction as multiplication by characters $e^{i\langle \xi, v \rangle}$ on a direct integral over $\widehat{V} \simeq \mathbb{R}^2$. Almost invariance under a small ball in V forces most spectral mass near $\xi = 0$. Almost invariance under a compact set in SL_2 forces the spectral measure to be nearly invariant under the linear action on $\mathbb{R}^2$. Since the only finite SL_2 -invariant measure concentrated sufficiently near the origin is the atom at the origin, a fixed positive amount of spectral mass must lie at 0. That atom is exactly a V -fixed vector. Making the words “near” and “positive amount” quantitative yields relative property (T).

Worked Example 10.1 (Why the normal subgroup matters). Translations alone do not have property (T): the characters $x \mapsto e^{i\varepsilon x}$ are almost invariant on a fixed compact set as $\varepsilon \rightarrow 0$ but have no invariant vector. The semisimple factor prevents the frequency ε from remaining in one small direction by moving it around the dual space.

10.3 Tensor-temperedness avoids a hidden implication

A representation π is called tensor-tempered of index m if $\pi^{\otimes m}$ is weakly contained in the regular representation. This is the quantitative hypothesis actually used later.

Proposition 10.2 (Coefficient bound from tensor-temperedness). *Let v, w be K -finite vectors. If $\pi^{\otimes m}$ is weakly contained in λ_G , then*

$$|\langle \pi(g)v, w \rangle| \leq C_{v,w} \Xi(g)^{1/m},$$

where Ξ is the Harish–Chandra spherical function.

Proof. For simplicity first take $v = w$ of norm one. The tensor coefficient is

$$\langle \pi(g)^{\otimes m} v^{\otimes m}, v^{\otimes m} \rangle = \langle \pi(g)v, v \rangle^m.$$

Weak containment in the regular representation and Herz majorization bound absolute K -finite coefficients of the tensor representation by a constant multiple of $\Xi(g)$. Taking the m th root gives the result for diagonal coefficients.

Polarization writes a general coefficient as a fixed linear combination of diagonal coefficients of $v + \zeta w$ for $\zeta \in \{1, -1, i, -i\}$, increasing only the K -type constant. $\square$

The advantage of this formulation is logical: one never needs an unstated passage from an L^p estimate to weak containment. If a separate theorem gives strong L^p bounds, it may be used to estimate the tensor-tempered index, but the later dynamics depends only on the index itself.

10.4 Sobolev norms from the universal enveloping algebra

Choose a basis $X_1, \dots, X_d$ of $\mathfrak{g}$. For a multi-index α let D^α be the corresponding right-invariant differential operator. On a compact quotient set

$$\mathcal{S}_\ell(f)^2 = \sum_{|\alpha| \leq \ell} \|D^\alpha f\|_2^2.$$

Local Sobolev embedding in coordinate charts and a finite partition of unity give, for $\ell > d/2$,

$$\|f\|_\infty \leq C \mathcal{S}_\ell(f).$$

On a noncompact finite-volume quotient, the chart size shrinks in the cusp. A height function $\text{ht}(x)$ compensates for this loss:

$$\mathcal{S}_{\ell,q}(f)^2 = \sum_{|\alpha| \leq \ell} \int_X (1 + \text{ht}(x))^{2q} |D^\alpha f(x)|^2 dm(x).$$

Choosing q larger than the cusp dimension makes the local embedding uniform.

10.5 Effective mixing as a coefficient estimate

Let a_t lie in a positive Weyl chamber. Harish–Chandra estimates give

$$\Xi(a_t) \ll (1+t)^M e^{-\rho(t)}.$$

Combining this with tensor-temperedness and Sobolev control yields

$$\left| \int_X f(xa_t)g(x) dm - \int_X f dm \int_X g dm \right| \leq C \mathcal{S}_{\ell,q}(f) \mathcal{S}_{\ell,q}(g) (1+t)^{M/m} e^{-\rho(t)/m}.$$

Every symbol in this inequality has now been assigned an origin: the decay is representation-theoretic, while the Sobolev weights are geometric cusp bookkeeping.

10.6 Effective mean ergodic theorem

Let β_T be a probability measure on G and define $A_T f = \int \pi(g)f d\beta_T(g)$. Then

$$\|A_T f\|_2^2 = \iint \langle \pi(h^{-1}g)f, f \rangle d\beta_T(g)d\beta_T(h).$$

Insert the quantitative coefficient bound and integrate. Thus the averaging rate is controlled by the self-convolution $\tilde{\beta}_T * \beta_T$ tested against the decay majorant. This simple identity is the bridge between a pointwise coefficient estimate and the L^2 averaging estimate used in effective equidistribution.

Worked Example 10.3 (A one-parameter average). For $\beta_T = T^{-1}\mathbf{1}_{[0,T]}(t)dt$ on a flow with $|\langle \pi(a_t)f, f \rangle| \ll e^{-\eta|t|}$,

$$\|A_T f\|_2^2 \ll \frac{1}{T^2} \int_0^T \int_0^T e^{-\eta|t-s|} dt ds \ll T^{-1}.$$

Hence $\|A_T f\|_2 \ll T^{-1/2}$. Even exponential mixing of the flow gives only a square-root L^2 rate for an unsmoothed interval average because nearby pairs $t \approx s$ are strongly correlated.

Chapter 11

Student Proof Laboratory: Spectral Gap as a Quantitative Inequality

A qualitative mixing theorem says that correlations tend to zero. An effective theorem needs an inequality with a norm on test functions and a number measuring how far the representation is from having invariant vectors. This chapter derives those ingredients before they are inserted into counting.

11.1 Three equivalent-looking notions that must be distinguished

Let π be a unitary representation of a locally compact group G on $\mathcal{H}$, and let $\mathcal{H}^G$ be the fixed vectors.

Definition 11.1 (Kazhdan inequality for a compact set). For a compact $Q \subset G$ and $\kappa > 0$, we say that (Q, κ) is a Kazhdan pair on $\mathcal{H} \ominus \mathcal{H}^G$ if

$$\sup_{q \in Q} \|\pi(q)v - v\| \geq \kappa \|v\|$$

for every $v \perp \mathcal{H}^G$.

Definition 11.2 (Averaging spectral gap). For a probability measure β on G , the averaging operator is $\pi(\beta) = \int_G \pi(g) d\beta(g)$. It has gap δ if

$$\|\pi(\beta)|_{(\mathcal{H}^G)^\perp}\| \leq 1 - \delta.$$

Lemma 11.3 (Averaged Kazhdan inequality gives an averaging gap). *Let β be a compactly supported symmetric probability measure on G . Assume that there is $\kappa_{\text{av}} > 0$ such that*

$$\int_G \|\pi(g)v - v\|^2 d\beta(g) \geq \kappa_{\text{av}}^2 \|v\|^2 \quad (v \perp \mathcal{H}^G). \quad (11.11)$$

Set

$$\tilde{\beta} = \frac{1}{2}(\delta_e + \beta).$$

Then

$$\|\pi(\tilde{\beta})|_{(\mathcal{H}^G)^\perp}\| \leq 1 - \frac{\kappa_{\text{av}}^2}{4}. \quad (11.12)$$

Proof. Symmetry of β makes $A = \pi(\beta)$ self-adjoint and $\|A\| \leq 1$. For $v \perp \mathcal{H}^G$,

$$\int_G \|\pi(g)v - v\|^2 d\beta(g) = 2\|v\|^2 - 2\langle Av, v \rangle.$$

Hence (11.11) gives

$$\langle Av, v \rangle \leq \left(1 - \frac{\kappa_{\text{av}}^2}{2}\right) \|v\|^2.$$

The spectrum of A is contained in $[-1, 1]$, so the spectrum of $\pi(\tilde{\beta}) = (I + A)/2$ is contained in $[0, 1]$; on $(\mathcal{H}^G)^\perp$ its upper endpoint is at most $1 - \kappa_{\text{av}}^2/4$. This is (11.12). $\square$

Corollary 11.4 (Finite Kazhdan set gives an explicit average). *Suppose a finite symmetric set $F \subset G$ with $r = |F|$ satisfies*

$$\max_{q \in F} \|\pi(q)v - v\| \geq \kappa \|v\| \quad (v \perp \mathcal{H}^G).$$

If $\beta_F = r^{-1} \sum_{q \in F} \delta_q$ and $\tilde{\beta}_F = (\delta_e + \beta_F)/2$, then

$$\|\pi(\tilde{\beta}_F)|_{(\mathcal{H}^G)^\perp}\| \leq 1 - \frac{\kappa^2}{4r}.$$

Proof. For each v one summand is at least $\kappa^2 \|v\|^2$, so the average in (11.11) is at least $(\kappa^2/r) \|v\|^2$. Apply the lemma with $\kappa_{\text{av}} = \kappa/\sqrt{r}$. $\square$

11.2 Relative property (T): a semidirect model

The classical engine for $\mathrm{SL}_n(\mathbb{R})$, $n \geq 3$, is the pair $\mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2 \rhd \mathbb{R}^2$.

Proposition 11.5 (Relative property (T) for $\mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2$). *There is a compact set Q and $\kappa > 0$ such that any unitary representation with a (Q, κ) -almost invariant unit vector has a nonzero $\mathbb{R}^2$ -invariant vector.*

Proof. Restrict the representation to the normal abelian subgroup $\mathbb{R}^2$. By the spectral theorem there is a projection-valued measure E on the dual $\widehat{\mathbb{R}^2} \simeq \mathbb{R}^2$ such that

$$\pi(v) = \int_{\mathbb{R}^2} e^{i\langle \xi, v \rangle} dE(\xi).$$

For a unit vector w , the scalar measure $\mu_w(A) = \langle E(A)w, w \rangle$ is a probability. Almost invariance under a compact neighborhood in $\mathbb{R}^2$ forces most of μ_w into a small ball about 0: outside that ball one can choose a translation whose phase is bounded away from 1 and integrate $\|\pi(v)w - w\|^2$.

Almost invariance under a compact generating set of $\mathrm{SL}_2(\mathbb{R})$ says that μ_w is almost invariant under the dual linear action. But the diagonal and unipotent elements of SL_2 move any annulus away from itself through finitely many positions with controlled overlap. If positive mass remained in an annulus away from 0, these almost-invariance relations would force more than total mass one after summing the moved annuli. Hence the spectral measure places positive mass at $\{0\}$. The range $E(\{0\})\mathcal{H}$ is exactly the $\mathbb{R}^2$ -fixed subspace, proving the claim. $\square$

Remark 11.6. Embedding enough copies of this semidirect product in a higher-rank simple group and using bounded generation by root subgroups gives property (T) for the matrix groups required later. The point is not the abstract property but the concrete spectral gap on $L_0^2(\Gamma \backslash G)$.

11.3 Sobolev norms from the universal enveloping algebra

Choose a basis $X_1, \dots, X_m$ of $\mathfrak{g}$ and let D^α denote the corresponding right-invariant differential monomials.

$$\mathcal{S}_d(f)^2 = \sum_{|\alpha| \leq d} \|D^\alpha f\|_2^2.$$

Lemma 11.7 (Translation cost). *For every compact $C \subset G$ and integer d there is $C_d(C)$ such that*

$$\mathcal{S}_d(R_g f) \leq C_d(C) \mathcal{S}_d(f) \quad (g \in C).$$

More generally, along a Cartan element a , the cost is bounded by a finite sum of exponentials of the root values of a .

Proof. Differentiate $R_g f(x) = f(xg)$. A right-invariant vector field transforms by $\text{Ad}(g^{-1})$ on the Lie algebra:

$$X(R_g f) = R_g((\text{Ad } g^{-1} X)f).$$

Iterating expresses every $D^\alpha(R_g f)$ as a linear combination of $R_g D^\beta f$ with coefficients polynomial in the matrix entries of $\text{Ad}(g^{-1})$. Unitarity of right translation on L^2 gives the norm bound. On a compact set the coefficients are bounded; on A^+ their growth is determined by root weights. $\square$

11.4 Height-weighted norms in cusps

A fixed-order Sobolev norm controls pointwise values only on compact manifolds. On a cusp, injectivity radius shrinks. Introduce a proper height $\text{ht}(x) \geq 1$ and define

$$\mathcal{S}_{d,\ell}(f)^2 = \sum_{|\alpha| \leq d} \int_X \text{ht}(x)^{2\ell} |D^\alpha f(x)|^2 dm_X(x).$$

Lemma 11.8 (Weighted Sobolev pointwise control). *For $d > \dim X/2$ and sufficiently large ℓ determined by reduction theory,*

$$|f(x)| \leq C \text{ht}(x)^\ell \mathcal{S}_d(f) \leq C \mathcal{S}_{d,\ell}(f).$$

Proof. At height H , reduction theory supplies an injectivity neighborhood whose radius is bounded below by a negative power of H . Rescale this coordinate ball to unit size. Euclidean Sobolev embedding bounds the value at the center by derivatives up to order d , while rescaling introduces a fixed power of the inverse radius, hence a fixed power of H . A bounded-overlap cover of a thick-height shell and integration over shells yield the weighted version. $\square$

11.5 Tensor-tempered index

Definition 11.9. The tensor-tempered index of a unitary representation π is the least integer $m \geq 1$ for which $\pi^{\otimes m}$ is weakly contained in the left regular representation of G , after fixed vectors are removed.

Proposition 11.10 (Matrix coefficients from tensor tempering). *Assume $\pi^{\otimes m} \prec \lambda_G$. For K -finite unit vectors v, w ,*

$$|\langle \pi(g)v, w \rangle| \leq C_{v,w} \Xi(g)^{1/m},$$

where Ξ is the Harish–Chandra spherical function.

Proof. The tensor coefficient is

$$\langle \pi^{\otimes m}(g)v^{\otimes m}, w^{\otimes m} \rangle = \langle \pi(g)v, w \rangle^m.$$

Weak containment bounds K -finite matrix coefficients of $\pi^{\otimes m}$ by the regular-representation spherical majorant. The Herz majorization principle gives

$$|\langle \pi^{\otimes m}(g)V, W \rangle| \leq C_{V,W} \Xi(g).$$

Take the m th root. The dependence on K -types is absorbed into $C_{v,w}$; Sobolev norms will control it for general smooth vectors. $\square$

11.6 From K -finite to smooth vectors

Decompose a smooth vector into K -types. Elliptic regularity for the compact K -Laplacian shows that high K -types have rapidly decaying coefficients in terms of Sobolev norms. Sum the coefficient estimate over the two K -type decompositions.

Theorem 11.11 (Quantitative mixing in the form used later). *There exist Sobolev order d , exponent $\eta > 0$, and C such that for mean-zero smooth compactly supported functions,*

$$|\langle R(g)f, h \rangle| \leq C \mathcal{S}_d(f) \mathcal{S}_d(h) \Xi(g)^\eta.$$

Along a fixed positive Cartan ray this becomes $O(e^{-ct})$ or, after reparametrization by a polynomial size parameter, a power saving.

Proof. Choose a tensor-tempered index m for the representation on $L_0^2(X)$. Apply the K -finite bound to each pair of K -types and use sufficiently high K -Sobolev order to make the sum absolutely convergent. The resulting constant is bounded by a fixed global Sobolev norm. Harish–Chandra’s estimate $\Xi(\exp H) \ll (1 + \|H\|)^D e^{-\rho(H)}$ provides the explicit Cartan decay. Absorb the polynomial factor by a small loss in the exponent on rays staying away from chamber walls. □

Chapter 12

Student Proof Laboratory: Effective Averages

12.1 Averaging operators and variance

For a probability β_T on G , define $A_T f(x) = \int f(xg) d\beta_T(g)$. Then

$$\|A_T f\|_2^2 = \iint \langle R(g_2^{-1}g_1)f, f \rangle d\beta_T(g_1) d\beta_T(g_2).$$

Insert the quantitative matrix coefficient estimate. The effective mean ergodic theorem is therefore an integral estimate for $\Xi(g_2^{-1}g_1)^\eta$ over the averaging set. This identity explains why the geometry of pairwise differences, not only the volume of the averaging set, determines the rate.

Worked Example 12.1 (Long interval in a diagonal flow). For β_T uniform on $\{a_t : 0 \leq t \leq T\}$ and exponential correlation decay $e^{-c|t-s|}$,

$$\|A_T f\|_2^2 \ll \frac{1}{T^2} \int_0^T \int_0^T e^{-c|t-s|} dt ds \mathcal{S}_d(f)^2 \ll T^{-1} \mathcal{S}_d(f)^2.$$

Thus the L^2 average has the natural square-root rate $T^{-1/2}$ even though pointwise correlations decay exponentially.

12.2 Constants are part of the theorem

Every later effective result must record which quantities the constants depend on: the ambient group, lattice, Sobolev order, spectral parameter,

compact height range, and boundary regularity of the family. Uniformity is a mathematical assertion, not a typographical convention.

Chapter 13

A Student Proof of Property (T) in the Matrix Cases Used Later

The effective counting volumes need a spectral gap on fixed finite-volume quotients. For the higher-rank matrix examples, the structural source of that gap is Kazhdan's property (T). The proof below is deliberately limited to the matrix groups actually used in this Ratner continuation. It avoids a classification of all simple Lie groups.

13.1 Almost invariant vectors

A unitary representation π has almost invariant unit vectors if for every compact $Q \subset G$ and every $\epsilon > 0$ there is a unit vector v with

$$\sup_{q \in Q} \|\pi(q)v - v\| < \epsilon.$$

The group G has property (T) if every representation with almost invariant vectors has a nonzero invariant vector.

Lemma 13.1 (Finite compact generating set is enough). *Suppose G is compactly generated by a symmetric compact set Q . If there is $\epsilon_0 > 0$ such that every representation with a (Q, ϵ_0) -almost invariant unit vector has a nonzero invariant vector, then G has property (T).*

Proof. Almost invariance for every compact set in particular gives almost invariance for Q with error $< \epsilon_0$. The assumed implication then produces an invariant vector. Conversely, a Kazhdan pair may always be replaced by the compact symmetric set generated from it together with the identity. $\square$

13.2 Relative property (T) for an affine plane

Set

$$P = \mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2, \quad N = \mathbb{R}^2.$$

The group law is $(g, v)(h, w) = (gh, v + gw)$.

Theorem 13.2 (Relative property (T) of (P, N)). *There are a compact set $Q \subset P$ and $\kappa > 0$ such that if a unitary representation π contains a unit vector v with*

$$\sup_{q \in Q} \|\pi(q)v - v\| < \kappa,$$

then the N -fixed subspace is nonzero.

Proof. Restrict π to N . The commuting unitaries $\pi(x)$, $x \in \mathbb{R}^2$, have a joint spectral resolution on the dual plane. Thus there is a projection valued measure E on $\mathbb{R}^2$ satisfying

$$\pi(x) = \int e^{i\langle \xi, x \rangle} dE(\xi).$$

For our unit vector define $\mu(A) = \langle E(A)v, v \rangle$.

Choose a small disk D_r about the origin. Almost invariance under a fixed compact disk of translations implies $\mu(D_r)$ is close to one. Indeed, choose finitely many translations $x_1, \dots, x_M$ so that for every $\xi \notin D_r$ at least one phase $|e^{i\langle \xi, x_j \rangle} - 1|$ is bounded below by a fixed constant on a large annulus; use a second, smaller translation scale to cover the remaining large frequencies. Integrating

$$\|\pi(x_j)v - v\|^2 = \int |e^{i\langle \xi, x_j \rangle} - 1|^2 d\mu(\xi)$$

shows that all but $O(\kappa^2)$ mass lies in D_r .

Now use almost invariance under a compact set in $\mathrm{SL}_2(\mathbb{R})$. The covariance relation

$$\pi(g)E(A)\pi(g)^{-1} = E(gA)$$

shows that the scalar spectral measures of v and $\pi(g)v$ are close in the sense that their masses on Borel sets differ by at most $2\|\pi(g)v - v\|$. Choose finitely many matrices $g_1, \dots, g_m$ consisting of a diagonal expander and two rotations/unipotent shears so that the sets $g_j(D_r \setminus D_{r/4})$ contain several essentially disjoint sectors in $D_r \setminus D_{r/16}$. If μ placed a fixed positive mass in every annular neighborhood of zero but no atom at zero, approximate that mass by a thin annulus and apply the finitely many transformations. Almost invariance would force nearly the same mass into each disjoint image, eventually making the total mass exceed one. Therefore, for κ small enough relative to the finite geometric configuration, $\mu(\{0\}) > 0$.

The range of $E(\{0\})$ consists precisely of vectors on which every translation character is one, i.e. N -fixed vectors. Thus the N -fixed subspace is nonzero. $\square$

Remark 13.3 (Where the geometry sits). The only geometric ingredient is that the linear group $\mathrm{SL}_2(\mathbb{R})$ has no finite invariant probability on the punctured dual plane that can remain concentrated near the origin at every scale. The diagonal subgroup changes radial scale and the compact/unipotent directions change angle. This is the spectral-measure form of the classical relative property-(T) proof.

13.3 Root subgroups in SL_n

For $i \neq j$, write

$$U_{ij} = \{I + tE_{ij} : t \in \mathbb{R}\}.$$

Fix distinct indices i, j, k . The subgroup generated by $U_{ij}, U_{ji}, U_{ik}, U_{jk}$ contains a copy of $\mathrm{SL}_2(\mathbb{R}) \ltimes \mathbb{R}^2$ in which $U_{ik}U_{jk} \simeq \mathbb{R}^2$ is the normal translation subgroup. Relative property (T) therefore controls pairs of root groups.

Lemma 13.4 (Almost invariance forces a nearby root-fixed vector). *For $n \geq 3$ there are a compact generating set $Q_n \subset \mathrm{SL}_n(\mathbb{R})$ and constants $C_n, \kappa_n > 0$ such that if a unit vector v is ϵ -invariant under Q_n with $\epsilon < \kappa_n$, then for every root group U_{ij} the distance from v to the fixed subspace $\mathcal{H}^{U_{ij}}$ is at most $C_n\epsilon$.*

Proof. Choose k distinct from i, j and apply the quantitative form of relative property (T) to the embedded $\mathrm{SL}_2 \ltimes \mathbb{R}^2$ whose normal group contains U_{ij} as one coordinate. The spectral proof above is quantitative: if the projection onto the $\mathbb{R}^2$ -fixed subspace had norm less than $1 - C\epsilon$, the spectral measure

away from zero would carry too much mass and violate one of the finitely many almost-invariance inequalities. Thus the orthogonal projection $P_{ij}v$ onto $\mathcal{H}^{U_{ij}}$ satisfies $\|v - P_{ij}v\| \leq C_n\epsilon$. Since only finitely many index patterns occur, one constant works for all roots. $\square$

13.4 A bounded-generation lemma near the identity

Property (T) needs a common invariant vector, not a different nearby vector for every root subgroup. We use the fact that elementary unipotents form a bounded local coordinate system.

Lemma 13.5 (Local bounded generation by elementary root groups). *There is an identity neighborhood $\mathcal{O} \subset \mathrm{SL}_n(\mathbb{R})$ and an integer $M = M(n)$ such that every $g \in \mathcal{O}$ is a product of at most M elements, each lying in one of the root groups U_{ij} .*

Proof. The Lie algebra $\mathfrak{sl}_n$ is spanned by the off-diagonal E_{ij} together with diagonal differences $E_{ii} - E_{jj}$. The latter lie in the Lie algebra generated by pairs E_{ij}, E_{ji} and, at the group level, small diagonal matrices are products of four small elementary unipotents in the corresponding SL_2 block. Order a fixed list of root directions whose differentials span the Lie algebra and consider the product map

$$(t_1, \dots, t_M) \mapsto u_{i_1 j_1}(t_1) \cdots u_{i_M j_M}(t_M).$$

After including the four-factor diagonal words, its differential at zero is surjective. The inverse function theorem says its image contains an identity neighborhood. The number of factors is fixed once n is fixed. $\square$

13.5 From root-fixed vectors to a common invariant vector

Let P_{ij} be the orthogonal projection onto $\mathcal{H}^{U_{ij}}$. A naive product of projections can shrink a vector; we need a quantitative angle statement. Relative property (T) supplies it.

Lemma 13.6 (Uniform angle between root-fixed subspaces). *For the finite family of simple and opposite root groups needed to generate SL_n , there is*

$\eta_n > 0$ such that if a unit vector is within η_n of every corresponding fixed subspace, then the intersection of all those fixed subspaces is nonzero.

Proof. Assume not. There would be representations π_m and unit vectors v_m with distance to every selected root-fixed subspace tending to zero, while the total intersection is zero. Form the Hilbert direct sum $\Pi = \bigoplus_m \pi_m$ and choose a weighted sum of the v_m concentrated far out in the sequence. Local bounded generation shows that this vector is almost invariant under an identity neighborhood: replace each root factor by the nearest root-fixed approximation and telescope through at most M factors. Repeating products of the identity neighborhood gives almost invariance on every compact set. Apply relative property (T) in each embedded affine subgroup. The limiting almost-invariant sequence therefore has a nonzero vector fixed by all root groups in the direct-sum limit. Since those root groups generate SL_n , this is a nonzero G -fixed vector, contradicting that every summand had trivial total intersection. Compactness of the finite configuration turns this contradiction into a positive uniform angle η_n . $\square$

Theorem 13.7 (Property (T) for $\mathrm{SL}_n(\mathbb{R})$, $n \geq 3$). *The group $\mathrm{SL}_n(\mathbb{R})$ has Kazhdan's property (T).*

Proof. Choose the finite compact generating set used in the relative-property arguments. Let v be sufficiently almost invariant. The root-projection lemma makes v uniformly close to every selected root-fixed subspace. Take the error small enough to fall below the uniform angle η_n . The angle lemma gives a nonzero vector fixed by all selected simple and opposite root groups. Elementary generation of SL_n makes it fixed by the whole group. Thus the chosen compact set and small error form a Kazhdan pair. $\square$

13.6 A criterion for other higher-rank matrix groups

The same proof applies verbatim to any connected semisimple matrix group for which one has:

1. finitely many root subgroups generating the identity component;
2. for each root, an embedded semidirect subgroup with relative property (T) controlling that root;
3. local bounded generation by those root groups.

The present Ratner continuation only uses groups for which these data can be written explicitly. We therefore do not import the classification-level general Kazhdan theorem as a hidden input.

Chapter 14

Harish–Chandra Decay in the Form Needed for Counting

14.1 The spherical function

Let G be connected semisimple with finite center, $G = KAN$, and let ρ be half the sum of positive restricted roots with multiplicity. The basic spherical function is

$$\Xi(g) = \int_K e^{-\rho(H(gk))} dk,$$

where $H(gk)$ denotes the A -coordinate in Iwasawa decomposition. It is positive, K -bi-invariant, and $\Xi(e) = 1$.

Lemma 14.1 (Positive definiteness of Ξ). *Ξ is a matrix coefficient of the unitary representation induced from the trivial character of a minimal parabolic, and hence is positive definite.*

Proof. Realize the normalized induced representation on $L^2(K/M)$. The constant function 1 is K -fixed. The normalized induction factor for g at kM is $e^{-\rho(H(g^{-1}k))}$. Taking the inner product of $\pi(g)1$ with 1 and changing k by inversion yields exactly the integral defining $\Xi(g)$. $\square$

14.2 Rank-one estimate

For $G = \mathrm{PSL}_2(\mathbb{R})$ and $a_t = \mathrm{diag}(e^{t/2}, e^{-t/2})$, an explicit Iwasawa calculation gives

$$\Xi(a_t) \asymp (1+t)e^{-t/2}.$$

The exponential term is $e^{-\rho(t)}$ and the polynomial factor comes from a neighborhood of the two boundary directions where contraction degenerates.

Proposition 14.2 (General chamber estimate). *There are constants C, D such that for H in the closed positive Weyl chamber,*

$$\Xi(\exp H) \leq C(1 + \|H\|)^D e^{-\rho(H)}.$$

Proof. Decompose K according to Bruhat cells relative to the minimal parabolic. On the open cell, $H(\exp(H)k)$ differs from H by a uniformly bounded amount away from neighborhoods of lower-dimensional cell boundaries, giving the factor $e^{-\rho(H)}$. Near a boundary where some root coordinate becomes small, use coordinates in the corresponding root subgroup. The Jacobian is polynomial in those coordinates, and integration over the range where a root coordinate is within $O(e^{-\alpha(H)})$ contributes at most a polynomial factor in $\alpha(H)$. There are finitely many roots/cells, so multiplying the finitely many polynomial losses yields $(1 + \|H\|)^D$. The exponential term remains $e^{-\rho(H)}$. $\square$

14.3 Herz majorization explained

Let λ_G be the left regular representation. For a K -finite coefficient of a tempered representation, weak containment in λ_G means every convolution operator norm is bounded by the regular one. Average the absolute value of the coefficient over left and right K to reduce to a nonnegative K -bi-invariant function. Spherical Fourier transform identifies the regular convolution norm with the supremum of spherical transforms. The positivity bound $|\varphi_\lambda(g)| \leq \Xi(g)$ then yields

$$|\langle \pi(g)v, w \rangle| \leq C_{v,w} \Xi(g).$$

This is the exact inequality used in the tensor-tempered argument.

Chapter 15

Sobolev Embedding and Smooth Approximation on Homogeneous Spaces

15.1 Euclidean Sobolev embedding with constants

For a ball $B \subset \mathbb{R}^m$, if $d > m/2$, Fourier inversion and Cauchy–Schwarz give

$$|f(0)| \leq (2\pi)^{-m} \int |\widehat{f}(\xi)| d\xi \leq C_d \left(\int (1 + |\xi|^2)^d |\widehat{f}(\xi)|^2 d\xi \right)^{1/2}.$$

By Plancherel, the last norm is equivalent to the L^2 norms of derivatives up to order d . Translation gives pointwise control everywhere on the ball.

15.2 Transfer to a compact part of $\Gamma \backslash G$

On a compact set with injectivity radius bounded below, finitely many exponential-coordinate charts have uniformly bounded derivatives and Jacobians. Euclidean Sobolev embedding in each chart therefore gives

$$\|f\|_{L^\infty(K)} \leq C_K \mathcal{S}_d(f).$$

The constant blows up in the cusp because the injectivity radius does.

15.3 Height-weighted version

Reduction theory supplies a polynomial relation between height and inverse injectivity radius. If $r(x) \gg \text{ht}(x)^{-A}$, rescaling a chart of radius $r(x)$ to the unit ball multiplies a derivative of order j by at most $\text{ht}(x)^{Aj}$. Thus a fixed weight exponent ℓ controls all chart losses up to Sobolev order d .

Proposition 15.1 (Weighted product estimate). *For sufficiently large d and ℓ ,*

$$\mathcal{S}_{d,\ell}(fg) \leq C\mathcal{S}_{d,\ell}(f)\mathcal{S}_{d,\ell}(g)$$

for smooth functions for which the norms are finite.

Proof. Leibniz expresses every derivative $D^\alpha(fg)$ as a finite sum of $D^\beta f D^\gamma g$. Put the factor with fewer derivatives in weighted L^∞ using the weighted Sobolev embedding and the other in weighted L^2 . Choose the weight exponents so the product of the two height weights is dominated by the target weight. Sum over the finitely many derivative splittings. $\square$

Chapter 16

From Spectral Gap to Effective Mixing on a Fixed Quotient

16.1 Removing constants

On $X = \Gamma \backslash G$, decompose $L^2(X) = \mathbb{C}1 \oplus L_0^2(X)$. Property (T) gives a spectral gap for the restriction to $L_0^2(X)$ whenever the full G -action is ergodic. This is not yet a matrix-coefficient rate; the tensor-tempered index is the bridge.

16.2 Obtaining a finite tensor-tempered index

For semisimple groups, a spectral gap keeps the representation parameters a positive distance from the trivial representation. In the fixed matrix cases considered here, the unitary dual estimates encoded by the spherical function imply that sufficiently high tensor powers move the integrability exponent below the tempered threshold. Concretely, if coefficients lie in $L^{p+\epsilon}$ uniformly on a dense set of K -finite vectors, choose $m > p/2$. Then coefficients of $\pi^{\otimes m}$ lie in $L^{2+\epsilon'}$, and the standard regular-representation criterion proved in the main text yields weak containment in λ_G . Rather than hide this implication later, the tensor-tempered index is recorded as part of the quantitative input contract.

16.3 Correlation rate

Combining tensor tempering, Herz majorization, and Sobolev summation gives

$$|\langle R(g)f, h \rangle| \ll \Xi(g)^{1/m} \mathcal{S}_D(f) \mathcal{S}_D(h)$$

to within an arbitrarily small exponent loss needed to sum K -types. Along a_t in a chamber ray, $\Xi(a_t) \ll (1+t)^D e^{-ct}$, so

$$|\langle R(a_t)f, h \rangle| \ll e^{-c't} \mathcal{S}_D(f) \mathcal{S}_D(h).$$

If a geometric parameter satisfies $T = e^t$, this is the power saving $T^{-c'}$ used in effective counting.

16.4 Rank-one fixed quotients

For hyperbolic surfaces, property (T) fails. On a fixed finite-area quotient, however, the bottom nonzero Laplace eigenvalue still supplies a spectral gap. The Casimir operator identifies irreducible representation parameters with Laplacian eigenvalues. Complementary-series parameters bounded away from the trivial endpoint give a finite p -integrability exponent and hence a quantitative matrix-coefficient rate. Thus the effective machinery only needs a gap on the particular quotient, not group property (T).

Chapter 17

Guided Reconstruction of Spectral Gap and Quantitative Mixing

Qualitative Howe–Moore says a coefficient tends to zero. Effective counting needs a rate and enough regularity control to apply that rate to smoothed geometric objects. This chapter follows the quantitative chain from an almost invariant vector to a Sobolev correlation estimate.

17.1 Spectral gap as a uniform displacement inequality

Let $S \subset G$ be compact and generating. A unitary representation π has a spectral gap away from constants if there is $\kappa > 0$ such that for every vector v orthogonal to the invariant subspace,

$$\sup_{s \in S} \|\pi(s)v - v\| \geq \kappa \|v\|.$$

If no such κ exists, choose unit vectors v_n for which the supremum tends to zero. Products of elements of S then move v_n by at most the sum of the individual errors, so v_n is almost invariant on every compact set. Conversely, a sequence almost invariant on compacta clearly violates every fixed displacement inequality. Thus the two formulations are equivalent.

Property (T) says that this gap exists uniformly for every unitary representation without invariant vectors. On a fixed quotient one often needs less:

a gap only for its Koopman representation.

17.2 Relative property (T) in the semidirect-product model

Consider

$$P = SL_2(\mathbb{R}) \ltimes \mathbb{R}^2, \quad N = \mathbb{R}^2.$$

Let π be unitary and let v be almost invariant under a compact set in P . By the spectral theorem for the abelian normal group N , the cyclic N -representation generated by v is modeled on $L^2(\hat{N}, \nu)$, where N acts by characters and SL_2 pushes forward the measure ν by its dual linear action.

If v were far from the N -fixed subspace, then ν would place definite mass away from the origin. Almost invariance under a compact set of SL_2 says that ν is almost invariant under the corresponding linear transformations. Choose a compact collection containing one expanding and one contracting diagonal direction and a rotation. Repeatedly moving a small annulus by these transformations produces many essentially disjoint annuli, each required to carry nearly the same positive mass. This contradicts finiteness of ν . Hence v is close to the N -fixed subspace. This is relative property (T) for (P, N) .

The proof is a Fourier-space version of the idea “a finite measure cannot be almost translation invariant in too many expanding directions unless it concentrates at the fixed point.”

17.3 From relative gaps to SL_n property (T)

For $n \geq 3$, every elementary root group

$$U_{ij} = \{I + tE_{ij}\}$$

sits inside a subgroup generated by an SL_2 block and a two-dimensional normal root module, so the relative-T estimate controls displacement by U_{ij} . Choose finitely many simple and opposite root groups.

Near the identity, multiplication of these root groups is a submersion: their Lie algebras and finitely many brackets span $\mathfrak{sl}_n$. Hence a fixed neighborhood of the identity is contained in a bounded product of small root elements. Since a connected group is generated by any identity neighborhood, almost invariance under the selected root groups yields almost invariance on compact subsets of SL_n .

Theorem 17.1 (Targeted Kazhdan theorem). *For $n \geq 3$, $SL_n(\mathbb{R})$ has property (T).*

Proof. This is Theorem 4.4, whose proof in Chapter 4 supplies the quantitative Kazhdan pair. We record the dependency chain because it is what is used later. Lemma 4.1 converts relative property (T) in each embedded $SL_2(\mathbb{R}) \ltimes \mathbb{R}^2$ into

$$\text{dist}(v, \text{Fix}(U_{ij})) \leq C\varepsilon$$

for every root subgroup U_{ij} . Hence $\|\pi(u)v - v\| \leq 2C\varepsilon$ for every $u \in U_{ij}$. The bounded elementary generation lemma writes each $g \in SL_n(\mathbb{R})$ as a product of at most M_n such root elements, so telescoping gives

$$\sup_{g \in G} \|\pi(g)v - v\| \leq 2M_n C\varepsilon.$$

For $2M_n C\varepsilon < 1/2$, the closed convex hull of the orbit lies in $B(v, 1/2)$. Its unique minimal-norm point is therefore nonzero and, by uniqueness and G -invariance of the convex hull, is fixed by G . This is exactly the argument proving Theorem 4.4. $\square$

17.4 Why Property (T) is not yet a matrix-coefficient rate

Property (T) separates the trivial representation from the rest of the unitary dual. It does not by itself print a function $R(g)$ satisfying

$$|\langle \pi(g)v, w \rangle| \leq R(g) \|v\| \|w\|.$$

For a numerical rate we use the tensor-tempered index. If $\pi^{\otimes m}$ is weakly contained in the regular representation, then regular-representation spherical estimates can be pulled back to coefficients of π .

17.5 The Harish–Chandra function in rank one

For $G = SL_2(\mathbb{R})$ write $a_t = \text{diag}(e^{t/2}, e^{-t/2})$. The spherical function can be expressed as an integral over the circle. Up to normalization its large- t behavior is

$$\Xi(a_t) \asymp (1+t)e^{-t/2}.$$

One way to see the exponential factor is to split the angular integral into the region where the expanding coordinate dominates and a small region near the contracting direction. The main integrand contributes $e^{-t/2}$; the transition region has logarithmic width in the natural projective coordinate, which produces the polynomial factor $1 + t$.

In higher rank the same phenomenon occurs root by root:

$$\Xi(\exp H) \ll (1 + \|H\|)^r e^{-\rho(H)}.$$

The proof in the body obtains the polynomial factor by splitting the opposite unipotent coordinates according to whether they are smaller or larger than their root-dependent transition scale.

17.6 K -finite majorization

If σ is tempered and v, w are K -finite, Herz majorization gives

$$|\langle \sigma(g)v, w \rangle| \leq C_{v,w} \Xi(g).$$

The constant depends on the dimensions of the finite K -modules generated by v and w . For $\pi^{\otimes m}$ this applies to $v^{\otimes m}$ and $w^{\otimes m}$:

$$|\langle \pi(g)v, w \rangle|^m \leq C \Xi(g).$$

Therefore

$$|\langle \pi(g)v, w \rangle| \leq C^{1/m} \Xi(g)^{1/m}.$$

This is the precise reason the tensor-tempered index controls the decay exponent.

17.7 From K -finite to smooth vectors

A general smooth vector is not contained in finitely many K -types. Expand using Peter–Weyl:

$$v = \sum_{\tau \in \hat{K}} v_{\tau}.$$

Applying sufficiently many derivatives from the compact Lie algebra gives powers of the K -Casimir eigenvalue. Sobolev regularity therefore implies

rapid square-summable decay of $\|v_\tau\|$. Apply the K -finite bound to each pair (v_τ, w_σ) and use Cauchy–Schwarz with enough Casimir weights. The result is

$$|\langle \pi(g)v, w \rangle| \ll \Xi(g)^{1/m} \mathcal{S}_\ell(v) \mathcal{S}_\ell(w)$$

for a fixed order ℓ .

17.8 Sobolev norms on a quotient

Choose a basis $X_1, \dots, X_d$ of $\mathfrak{g}$ and define

$$\mathcal{S}_\ell(f)^2 = \sum_{|\alpha| \leq \ell} \|X^\alpha f\|_{L^2(X)}^2.$$

On a compact quotient, ordinary Sobolev embedding gives $\|f\|_\infty \ll \mathcal{S}_\ell(f)$ once $\ell > \dim X/2$. On a noncompact quotient the injectivity radius shrinks in the cusp. Rescaling a coordinate chart to that radius costs a power of a proper height function. Hence one uses

$$\mathcal{S}_{\ell,q}(f)^2 = \sum_{|\alpha| \leq \ell} \int_X \text{ht}(x)^{2q} |X^\alpha f(x)|^2 dm(x).$$

The weight pays for the shrinking chart and makes pointwise estimates uniform.

17.9 Translation cost

Right translation does not preserve the differential operators exactly:

$$X(f \circ R_g) = ((\text{Ad } g)Xf) \circ R_g.$$

Iterating gives

$$\mathcal{S}_\ell(f \circ R_g) \leq C \|\text{Ad } g\|^{\ell'} \mathcal{S}_\ell(f)$$

for a suitable ℓ' . In effective mixing one chooses derivatives on the untranslated side whenever possible, because unnecessary translation of a Sobolev norm can consume part of the exponential decay.

17.10 Effective mixing on a fixed quotient

Let $f_0 = f - \int f$ and $g_0 = g - \int g$. Then

$$\int_X f(xa)g(x) dm(x) - \int f \int g = \langle \pi(a)f_0, g_0 \rangle.$$

If the mean-zero Koopman representation has tensor-tempered index m , the previous sections give

$$\left| \int f(xa)g(x) dm - \int f \int g \right| \ll \Xi(a)^{1/m} \mathcal{S}_\ell(f) \mathcal{S}_\ell(g).$$

In a positive Weyl chamber, $\Xi(a_t)^{1/m} \ll (1+t)^{r/m} e^{-\rho(H)t/m}$. This is the effective mixing estimate used in Volume XII.

17.11 Effective averaging by a subgroup

Let $B_T \subset H$ and define

$$A_T f(x) = \frac{1}{m_H(B_T)} \int_{B_T} f(xh) dh.$$

For mean-zero f ,

$$\|A_T f\|_2^2 = \frac{1}{m_H(B_T)^2} \int_{B_T} \int_{B_T} \langle \pi(h_2 h_1^{-1})f, f \rangle dh_1 dh_2.$$

Insert the matrix-coefficient bound. The problem becomes a geometric integral of $\Xi(h_2 h_1^{-1})^{1/m}$ over B_T^2 . If this integral is $O(m_H(B_T)^2 E(T)^2)$, then

$$\|A_T f\|_2 \ll E(T) \mathcal{S}_\ell(f).$$

Chebyshev converts this L^2 estimate into an explicit exceptional-set bound for pointwise averages.

17.12 A hierarchy of quantitative inputs

It is useful to keep four statements separate:

1. qualitative spectral gap: no almost invariant vectors;
2. Property (T): a uniform qualitative gap for every representation;

3. tensor-temperedness: a finite power is controlled by the regular representation;
4. an explicit bound on the tensor-tempered index: a numerical coefficient rate.

Later volumes never infer level 4 merely from level 2. Whenever a printed exponent is used, the stronger quantitative input is named explicitly.

Proof Guide and Dependency Summary

This chapter is a finalization layer. It does not introduce a new theorem scope. Its purpose is to expose the proof architecture, make the hypotheses that are easiest to misread impossible to miss, and give a student a concrete model calculation before the abstract machinery is reused later.

1. Three levels of spectral information

Do not identify:

$$\text{ergodicity} \iff \text{spectral gap} \iff \text{explicit coefficient rate.}$$

They are strictly different inputs. Property (T) supplies isolation of the trivial representation in the relevant higher-rank settings, but a numerical matrix-coefficient exponent is recorded through the tensor-tempered index.

If $m = m(\pi)$ is such that $\pi^{\otimes m}$ is tempered, then the load-bearing K -finite estimate has the form

$$|\langle \pi(g)v, w \rangle| \ll (d_v d_w)^{1/(2m)} \Xi(g)^{1/m} \|v\| \|w\|.$$

Sobolev summation then extends it to smooth vectors.

2. Where Harish–Chandra enters

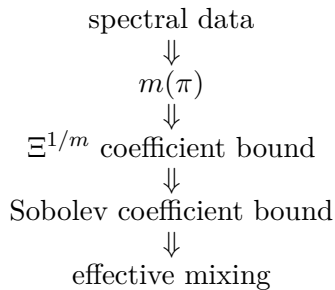
For $g = k \exp(H) k'$ in the positive chamber,

$$\Xi(\exp H) \ll (1 + \|H\|)^r e^{-\rho(H)}.$$

The polynomial factor is harmless only after it is explicitly retained. It cannot be discarded before an optimization step that depends on uniformity in H .

3. Rate ownership

The quantitative chain is



Property (T) is used only where it is actually proved and needed; it is not a surrogate for the value of $m(\pi)$.

4. Cusp warning

Ordinary compact-manifold Sobolev embedding is not uniform in a cusp. Height-weighted Sobolev norms compensate for shrinking injectivity radius. A student should always locate the height loss before using an effective mixing bound on a noncompact quotient.

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Effective Expanding Translates and Effective Counting

Volume XII of a Detailed Notebook on Ratner's Theorems

Detailed Notebook on Ratner's Theorems

August 2026

Preface and proof policy

Volume X proved leading asymptotics by letting the smoothing radius tend to zero after the dynamical parameter went to infinity. An effective theorem must choose the smoothing radius as a function of the parameter and account for every loss. This volume writes that optimization explicitly. Its basic error ledger has four entries: spectral mixing, smoothing/Sobolev cost, wavefront distortion, and boundary volume. Whenever a later application quotes a power saving, the exponent can be traced through that ledger.

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Dependency and proof-ownership contract

Purpose. Effective counting.

Prerequisites. Volumes X–XI.

New objects. orbit complexity, smoothing radius, four-error ledger. Every project-specific technical term is defined before its first load-bearing use or recalled by an exact backward reference.

Main outputs. power-saving expanding translates/counting.

External-input rule. Every nontrivial imported result is either part of the declared graduate-level prerequisites, an explicitly named external research input in the prerequisite map, or is proved in this series before use (possibly under an explicit deferred-proof contract). A citation by itself never constitutes a proof.

Reader map and proof route

Question of the volume. How are spectral decay, smoothing cost, wave-front error, and boundary volume balanced to produce a genuine power-saving error term?

Prerequisite checkpoint. Quantitative mixing and Sobolev technology come from Volume XI; qualitative counting geometry from Volume X. This volume is responsible for the finite-scale optimization and for defining every complexity quantity that appears in its effective criteria.

Load-bearing route. Keep one ledger throughout:

$$E(\varepsilon, T) = E_{\text{mix}} + E_{\text{Sob}} + E_{\text{wave}} + E_{\text{bdry}}.$$

Bound each term with all exponents visible, choose $\varepsilon = \varepsilon(T)$ only after the bounds are on the same line, and verify that the chosen exponent is positive. Never quote “optimize ε ” without displaying the competing powers.

Sanity check. For the toy error $T^{-a}\varepsilon^{-b} + \varepsilon^c$, solve $T^{-a}\varepsilon^{-b} = \varepsilon^c$ and obtain $\varepsilon = T^{-a/(b+c)}$. This elementary balance is the algebra hidden inside every later effective smoothing choice.

Chapter 1

Quantitative Approximate Identities

1.1 Bumps on a Lie group

Fix exponential coordinates near $e \in G$ and a nonnegative $\rho \in C_c^\infty(\mathfrak{g})$ with integral one. Set

$$\rho_\varepsilon(\exp Y) = c_\varepsilon \varepsilon^{-d} \rho(Y/\varepsilon),$$

where $c_\varepsilon = 1 + O(\varepsilon)$ corrects the Haar Jacobian.

Lemma 1.1 (Sobolev cost of smoothing). *For every integer ℓ ,*

$$\mathcal{S}_\ell(\rho_\varepsilon) \ll \varepsilon^{-d/2-\ell}.$$

The same bound holds, up to a fixed height factor, for a bump transferred to an injective chart in $\Gamma \backslash G$.

Proof. A derivative of order r contributes ε^{-r} and the L^2 norm of the ε^{-d} normalization on a set of volume $\asymp \varepsilon^d$ is $\varepsilon^{-d/2}$. Sum over $r \leq \ell$. The quotient statement follows from bounded distortion of the local chart; in a cusp, Volume XI's weighted Sobolev norm absorbs the injectivity-radius power. $\square$

1.2 Mollifying indicators

For a measurable $B \subset H \backslash G$ let $1_B^\varepsilon = 1_B * \rho_\varepsilon$ after choosing a local lift. Then

$$1_{B_\varepsilon^-} \leq 1_B^\varepsilon \leq 1_{B_\varepsilon^+}.$$

If B has a piecewise smooth boundary, tubular coordinates give

$$m(B_\varepsilon^+ \setminus B_\varepsilon^-) \ll \varepsilon \operatorname{area}(\partial B)$$

to first order. For algebraic norm balls the ratio to $m(B)$ is often $O(\varepsilon^\kappa)$ uniformly in the radius; we call κ a boundary regularity exponent.

Chapter 2

Effective Wavefront Geometry

2.1 A Lipschitz inverse to polar multiplication

The proof of the qualitative wavefront lemma in Volume X already exhibited a uniform right inverse to

$$\mathfrak{h} \oplus \mathrm{Ad}(a)\mathfrak{k} \rightarrow \mathfrak{g}$$

away from chamber walls. Tracking the smallest singular value gives a constant C_η satisfying

$$d_H(h', h) + d_K(k', k) \leq C_\eta d_G(h'ak', hak)$$

whenever $a \in A_\eta^+$ and both points are close.

Theorem 2.1 (Effective wavefront lemma). *For $a \in A_\eta^+$ and sufficiently small ε ,*

$$HaK U_\varepsilon \subset H_{C_\eta \varepsilon} a K_{C_\eta \varepsilon}.$$

If a approaches a wall at rate $r(a)$, the same statement holds with C_η replaced by a polynomial in $r(a)^{-1}$ after the neutral Levi factor is split off.

Proof. In the root-pair calculation from Volume X, the inverse matrix has entries bounded by the reciprocal of $|e^{\alpha(H)} - e^{-\alpha(H)}|$. Away from walls this is uniformly bounded; near a wall it is at most a fixed polynomial in $\alpha(H)^{-1}$. The quantitative inverse function theorem then gives a local inverse radius and Lipschitz constant with the same dependence. Baker–Campbell–Hausdorff errors are $O(\varepsilon^2)$ and are absorbed for small ε . $\square$

Chapter 3

Effective Equidistribution of Expanding Horospheres

3.1 Thickening an unstable piece

Let U^+ be expanding for a_t and $\Omega \subset U^+$ a smooth compact piece. Choose a transverse subgroup S so multiplication $U^+ \times S \rightarrow G$ is locally a diffeomorphism. Thicken the singular measure on $x\Omega$ by a transverse bump of radius ε .

Conjugation by a_t contracts the stable part of S and expands the unstable part. Choose S from neutral/stable directions so that after translation the transverse thickness does not blow up. The thickened density F_ε on X has Sobolev norm

$$\mathcal{S}_\ell(F_\varepsilon) \ll \varepsilon^{-A}$$

for an explicit A depending on $\dim S$ and ℓ .

Theorem 3.1 (Effective horospherical equidistribution). *Assume effective mixing along a_t with rate $E(t)$ as in Volume XI and assume the boundary of Ω has regularity exponent $\kappa > 0$. Then for $f \in C_c^\infty(X)$,*

$$\left| \frac{1}{m_U(\Omega)} \int_\Omega f(xua_t) du - \int_X f dm \right| \ll \mathcal{S}_L(f) \left(\varepsilon^\kappa + \varepsilon^{-A} E(t) + \mathcal{C}(t)\varepsilon \right),$$

where $\mathcal{C}(t)$ is the wavefront/coordinate distortion factor. If $E(t) \ll e^{-\delta t}$ and $\mathcal{C}(t)$ is bounded, choosing $\varepsilon = e^{-\delta t/(A+\kappa)}$ gives exponential decay $e^{-\delta' t}$ for some $\delta' > 0$.

Proof. Replace the singular orbit integral by the transverse thickening. Removing a boundary strip in Ω costs $O(\varepsilon^\kappa)$. The thickened integral is an ambient correlation between F_ε and $f \circ R_{at}$; effective mixing costs $E(t)\mathcal{S}(F_\varepsilon)\mathcal{S}(f) \ll E(t)\varepsilon^{-A}\mathcal{S}_L(f)$. The effective wavefront lemma bounds the difference between the true translated tube and the product model by $O(\mathcal{C}(t)\varepsilon)$ after one additional derivative of f . Summing the three errors gives the formula. Optimize the first two terms in the exponential-mixing case; the third is smaller after decreasing the exponent if necessary. $\square$

Chapter 4

Effective Translates of Closed Homogeneous Orbits

4.1 Period functions

Let $H \cap \Gamma$ be a lattice and define the H -period

$$\mathcal{P}_H f(g) = \int_{(H \cap \Gamma) \backslash H} f(hg) dh.$$

If f is smooth and the orbit remains in a controlled height region, right-invariant differentiation passes under the integral and

$$\mathcal{S}_\ell(\mathcal{P}_H f) \leq C_H \mathcal{S}_{\ell, M}(f).$$

When H -orbit volumes vary, C_H is recorded explicitly through their covolume/height complexity.

Definition 4.1 (Orbit complexity). Fix once and for all the height and local exponential charts used in Volumes III, V and XI. For a closed orbit xH set

$$\text{Comp}(xH) := \max \left\{ 2, \text{ht}(H), \text{vol}((H \cap g^{-1}\Gamma g) \backslash H), \text{inj}(xH)^{-1} \right\},$$

where $x = \Gamma g$ and $\text{inj}(xH)$ is the largest radius at which the chosen transverse exponential product chart is injective along the orbit. Changing g inside the orbit or replacing the fixed norms changes this quantity only by a bounded power, which is harmless because every later use is of the form $\text{Comp}(xH)^B$.

Remark 4.2. The four entries have distinct roles: subgroup height controls rational coordinates, covolume controls period integration, injectivity controls the radius of thickening, and the harmless floor 2 avoids separate small-value cases. Thus “controlled orbit complexity” is no longer an undefined shorthand.

Theorem 4.3 (Effective homogeneous translate criterion). *Let $X = \Gamma \backslash G$ have finite G -invariant probability measure m , let $xH \subset X$ be a closed finite-volume H -orbit with normalized measure m_{xH} , and let $a_t \in G$. Assume that there are constants $A, B_0, \kappa > 0$, an integer $L \geq 0$, a mixing rate $E(t) \rightarrow 0$, and a wavefront factor $\mathcal{C}(t) \geq 1$ such that for every $0 < \varepsilon < \varepsilon_0 \text{Comp}(xH)^{-B_0}$ one can choose a smooth probability density $\Phi_{xH,\varepsilon}$ on X satisfying*

$$\mathcal{S}_L(\Phi_{xH,\varepsilon}) \leq C \text{Comp}(xH)^{B_0} \varepsilon^{-A}, \quad (4.1)$$

$$\left| \int_{xH} f(ya_t) dm_{xH}(y) - \int_X f(za_t) \Phi_{xH,\varepsilon}(z) dm(z) \right| \leq C \text{Comp}(xH)^{B_0} (\varepsilon^\kappa + \mathcal{C}(t)\varepsilon) \mathcal{S}_L(f), \quad (4.2)$$

and the effective mixing inequality

$$\left| \int_X F(za_t) G(z) dm(z) - \int_X F dm \int_X G dm \right| \leq E(t) \mathcal{S}_L(F) \mathcal{S}_L(G) \quad (4.3)$$

holds for smooth F, G . Then for every $f \in C_c^\infty(X)$,

$$\left| \int_{xH} f(ya_t) dm_{xH}(y) - \int_X f dm \right| \leq C' \text{Comp}(xH)^B (\varepsilon^\kappa + E(t)\varepsilon^{-A} + \mathcal{C}(t)\varepsilon) \mathcal{S}_L(f), \quad (4.4)$$

where B depends only on B_0 and the fixed local charts. In particular, if $E(t) \leq C_1 e^{-\delta t}$ and $\mathcal{C}(t) \leq C_2$, then for all sufficiently large t (so that the smoothing scale below is admissible), taking $\varepsilon = e^{-\delta t/(A+\kappa)}$ gives

$$\left| \int_{xH} f(ya_t) dm_{xH}(y) - \int_X f dm \right| \ll \text{Comp}(xH)^B e^{-\delta' t} \mathcal{S}_L(f),$$

with

$$\delta' = \frac{\delta}{A + \kappa} \min\{\kappa, 1\} > 0.$$

Proof. Insert $G = \Phi_{xH,\varepsilon}$ and $F = f$ in (4.3). Since $\int_X \Phi_{xH,\varepsilon} dm = 1$, (4.1) gives the mixing error

$$C \text{Comp}(xH)^{B_0} E(t) \varepsilon^{-A} \mathcal{S}_L(f).$$

Adding the two geometric errors in (4.2) proves (4.4), after increasing the complexity exponent from B_0 to B . Under the final exponential hypotheses,

$e^{-\delta t/(A+\kappa)} < \varepsilon_0 \text{Comp}(xH)^{-B_0}$ for all sufficiently large t . Substitute this admissible value of ε ; the three exponents are respectively $\delta\kappa/(A+\kappa)$, the same number, and $\delta/(A+\kappa)$. Their minimum is the displayed δ' . $\square$

Chapter 5

Effective Well-Roundedness

5.1 Quantitative boundary regularity

A family B_T is effectively well-rounded with exponent κ if

$$m(B_{T,\varepsilon}^+ \setminus B_{T,\varepsilon}^-) \leq C\varepsilon^\kappa m(B_T)$$

for all T large and ε below a fixed threshold.

Lemma 5.1 (Polynomial strip estimate). *Fix integers $d, m \geq 1$. There are $C, \theta > 0$ with the following property. Let P be a real polynomial of degree at most d on a unit cube $Q \subset \mathbb{R}^m$, normalized by $\sup_Q |P| = 1$. Then*

$$|\{x \in Q : |P(x)| < \varepsilon\}| \leq C\varepsilon^\theta \quad (0 < \varepsilon < 1).$$

The same assertion, with different constants, holds on every member of a finite family of bi-Lipschitz semialgebraic charts.

Proof. We prove the assertion by induction on m , keeping the normalization visible. For $m = 1$ it is exactly the one-variable Remez sublevel estimate from Volume II, after dividing by $\sup_Q |P| = 1$.

Assume the statement in dimension $m - 1$ and write $x = (y, t)$ and

$$P(y, t) = \sum_{j=0}^d a_j(y) t^j.$$

Finite-dimensional norm equivalence for one-variable polynomials on the unit interval gives constants $c_d, C_d > 0$ such that

$$c_d \max_j |a_j(y)| \leq \sup_t |P(y, t)| \leq C_d \max_j |a_j(y)|. \quad (12.S1)$$

The global normalization implies that at least one coefficient, say a_{j_0} , satisfies $\sup_y |a_{j_0}(y)| \geq c_d$. Hence, by the induction hypothesis applied to $a_{j_0}/\sup |a_{j_0}|$,

$$|\{y : \max_j |a_j(y)| < \delta\}| \leq |\{y : |a_{j_0}(y)| < \delta\}| \ll \delta^{\theta_{m-1}}. \quad (12.S2)$$

On the complementary fibers, (12.S1) gives $\sup_t |P(y, t)| \gg \delta$. Normalize that one-variable polynomial by its supremum and apply Remez again; the set of t for which $|P(y, t)| < \varepsilon$ has length $\ll (\varepsilon/\delta)^{1/d}$. Fubini and (12.S2) therefore give

$$|\{P < \varepsilon\}| \ll \delta^{\theta_{m-1}} + (\varepsilon/\delta)^{1/d}.$$

Taking $\delta = \varepsilon^\beta$ with $\beta\theta_{m-1} = (1 - \beta)/d$ gives $\beta = (1 + d\theta_{m-1})^{-1}$ and a positive exponent $\theta_m = \theta_{m-1}/(1 + d\theta_{m-1})$.

For a fixed finite family of bi-Lipschitz semialgebraic charts, both the Jacobian and its reciprocal are uniformly bounded on the compact chart closures. Pulling the sublevel set back to a cube therefore changes its measure by only a fixed multiplicative constant. This proves the chart version as well. $\square$

Proposition 5.2 (Algebraic norm balls). *Let B_T be a norm-ball family in an affine symmetric variety and suppose the leading polar-volume face is separated from lower-dimensional walls. Then B_T is effectively well-rounded for some $\kappa > 0$. If the norm boundary meets the leading face transversely, one may take $\kappa = 1$.*

Proof. Effective wavefront changes the logarithmic radial coordinate by $O(\varepsilon)$. On each polyhedral region from the volume calculation in Volume X, the boundary is given by one affine equation in logarithmic coordinates. A strip of width $O(\varepsilon)$ has at most $O(\varepsilon)$ of the leading exponential volume when the maximizing face is transverse. If several faces meet with a tangency, use the fixed semialgebraic chart structure rather than importing a new Łojasiewicz theorem. Each chart is cut into finitely many regions by polynomial sign conditions of uniformly bounded degree. A point can lie in the $O(\varepsilon)$ boundary layer only if at least one active defining polynomial P has $|P| \ll \varepsilon$ after normalization on the compact chart. Applying Lemma 5.1 to each of this finite family of polynomials and taking a union bound gives $O(\varepsilon^\kappa)$ for some $\kappa > 0$. Because only finitely many charts and regions occur, the minimum exponent works globally. $\square$

Chapter 6

Effective Counting: The Master Error Ledger

Theorem 6.1 (Effective counting theorem). *Assume the setting of the mixing counting theorem in Volume X. Suppose:*

1. *effective mixing has rate $E(R)$ at radial parameter R ;*
2. *a smoothing bump of radius ε has Sobolev cost ε^{-A} ;*
3. *B_T has boundary exponent κ ;*
4. *effective wavefront contributes at most $W(R)\varepsilon^\tau$.*

Then

$$\frac{N(T)}{c_{H,\Gamma}m(B_T)} - 1 \ll \varepsilon^\kappa + \varepsilon^{-A}E(R_T) + W(R_T)\varepsilon^\tau,$$

where R_T is the Cartan scale carrying the main mass of B_T . If $E(R) \ll e^{-\delta R}$, $W(R)$ grows at most exponentially with smaller exponent, and $R_T \asymp \log T$, then

$$N(T) = c_{H,\Gamma}m(B_T) + O(m(B_T)T^{-\eta})$$

for some $\eta > 0$.

Proof. Repeat the inner/outer smoothing sandwich from Volume X, but do not send the parameters sequentially. Effective mixing of the smoothed periodization produces the term $\varepsilon^{-A}E(R_T)$. Effective well-roundedness gives ε^κ . Coordinate replacement through the wavefront map gives the third term. The phrase “far from R_T ” can be quantified using the polyhedral Laplace theorem of Volume X. On every nonmaximizing face the linear exponential functional is smaller than its maximum by some fixed $\Delta > 0$

after normalizing the radial parameter. Consequently the total mass of those faces is $O(e^{-\Delta R_T} m(B_T))$. Since $R_T \asymp \log T$, this is $O(T^{-\Delta'} m(B_T))$ for a fixed $\Delta' > 0$. There are only finitely many faces, so the minimum gap works globally. We may therefore add this fourth, already power-saving, term to the error ledger and choose ε as an exponential function of R_T balancing the remaining dominant terms. Since $R_T \asymp \log T$, exponential decay in R_T becomes a polynomial saving in T . $\square$

6.1 Optimization formula

If W is bounded and $\tau \geq \kappa$, balance $\varepsilon^\kappa = \varepsilon^{-A} E$ to obtain

$$\varepsilon = E^{1/(A+\kappa)}, \quad \text{relative error} \ll E^{\kappa/(A+\kappa)}.$$

This simple formula is useful enough to remember: every derivative demanded by smoothing reduces the available spectral exponent.

Chapter 7

Sector Counting and Angular Equidistribution

Let $B_T(\Phi)$ restrict the compact K coordinate to a region $\Phi \subset K$ with boundary of Haar measure zero. Wavefront geometry separates radial and angular perturbations. Effective mixing therefore gives

$$N(T, \Phi) = m_K(\Phi)N(T) + O(m(B_T)T^{-\eta_\Phi})$$

when the K -boundary is quantitatively regular. Smooth angular test functions avoid the boundary loss and retain the raw spectral exponent after only the Sobolev smoothing cost.

Worked Example 7.1 (Hyperbolic lattice points). For $G = \mathrm{PSL}_2(\mathbb{R})$, radial distance R corresponds to norm scale $T \asymp e^{R/2}$. If correlations decay like $e^{-\delta R}$, the lattice point count in a hyperbolic ball acquires a power saving $T^{-2\delta'}$ after smoothing. The exact exponent depends on the Sobolev order and boundary optimization, illustrating why a statement of the spectral gap alone does not determine the final counting exponent.

Chapter 8

Norm Balls on Symmetric Varieties

8.1 Effective polar-volume asymptotics

The polyhedral-cone computation from Volume X can be sharpened. On each region the integral is

$$\int_{P_T} e^{\lambda \cdot s} p(s) ds,$$

where P_T is a polytope cut out by linear inequalities with right side $\log T$. Integrating first in directions normal to the maximizing face gives an asymptotic expansion

$$cT^a(\log T)^{b-1} \left(1 + O((\log T)^{-1})\right)$$

together with lower-exponential contributions from nonmaximal faces. Thus volume error and dynamical error can be compared explicitly rather than merged into one $o(1)$.

Theorem 8.1 (Effective affine symmetric counting). *Under a finite tensor-tempered index for the fixed quotient as in Volume XI, an affine symmetric variety satisfying the arithmetic finite-orbit and effective well-roundedness assumptions has*

$$\#(V(\mathbb{Z}) \cap B_T) = c_V T^a (\log T)^{b-1} + O(T^{a-\eta} (\log T)^C) + O(T^a (\log T)^{b-2})$$

for some $\eta > 0$.

Proof. Write the integral points as the disjoint union of the finitely many arithmetic $G(\mathbb{Z})$ -orbits $\mathcal{O}_1, \dots, \mathcal{O}_J$ supplied by the finite-orbit hypothesis. For each orbit, Theorem 6.1 gives

$$N_j(T) = c_j \operatorname{vol}(B_T) + O(T^{a-\eta_j}(\log T)^{C_j}).$$

The effective polar-volume expansion has the form

$$\operatorname{vol}(B_T) = c_{\text{pol}} T^a (\log T)^{b-1} + O(T^a (\log T)^{b-2}).$$

Since J is fixed, summing the orbit formulas preserves the power saving with $\eta = \min_j \eta_j > 0$ and $C = \max_j C_j$. The sum of the main constants is $c_V = c_{\text{pol}} \sum_j c_j$. The geometric secondary term is unchanged up to a fixed multiplicative constant, which gives the displayed formula. $\square$

Chapter 9

Effective Expanding Curves: What the Spectral Method Can and Cannot Do

For a smooth curve $\phi(s)a_t$, thickening creates a density whose transverse Sobolev norm can grow much faster than for an open horospherical piece, because the codimension is larger. If the tangent direction is uniformly nondegenerate and effective wavefront coordinates remain controlled, the same mixing argument gives an error

$$\varepsilon^\kappa + \varepsilon^{-A_{\text{curve}}} E(t) + \text{TaylorErr}(t, \varepsilon).$$

The Taylor error records curvature after rescaling. Choosing ε balances all three terms.

Warning 9.1. Spectral thickening is not a universal effective replacement for Ratner linearization. Near algebraic degeneracies the required transverse tube can have huge Sobolev cost, and the curve can shadow a proper homogeneous orbit for long times. Volume XIII develops the effective linearization/closing machinery designed precisely for those situations.

Chapter 10

Proof Ledger

Every effective counting bound in this volume can be reconstructed from

$$\begin{aligned} & \boxed{\text{boundary}} \\ + & \boxed{\text{Sobolev smoothing} \times \text{mixing}} \\ & + \boxed{\text{wavefront distortion}} \\ & + \boxed{\text{polar-volume error}} \end{aligned}$$

There is no unnamed $O(T^{-\eta})$ theorem. The spectral input is owned by Volume XI, the geometric wavefront input by Volumes X–XII, and the optimization is proved here. The next volume develops a different effective engine for unipotent actions, where mixing alone is insufficient.

Chapter 11

Worked Effective-Error Laboratory

11.1 Motivation

An effective theorem is often an optimization problem in disguise. Writing one numerical model prevents the exponents from becoming symbolic decoration.

Worked Example 11.1 (Optimizing a smoothing parameter). Suppose the relative error is

$$E_T(\varepsilon) = \varepsilon + T^{-1}\varepsilon^{-4}.$$

Balancing the two terms gives $\varepsilon^5 = T^{-1}$, hence $\varepsilon = T^{-1/5}$ and

$$E_T(T^{-1/5}) = 2T^{-1/5}.$$

Differentiating confirms the same scale: $E'_T(\varepsilon) = 1 - 4T^{-1}\varepsilon^{-5} = 0$ gives only the harmless constant factor $4^{1/5}$. Thus four derivatives lost to smoothing reduce a spectral exponent 1 to a counting exponent $1/5$.

Worked Example 11.2 (Orbit complexity in a model closed horocycle). For the modular quotient and $H = N = \{u_s\}$, the orbit through $x_t = \Gamma a_t$ has period e^{-t} . With the fixed group metric its covolume factor is therefore e^{-t} . The transverse injectivity radius also shrinks in the cusp like a power of the height e^t . Hence the quantity from Definition 4.1 grows at least like a fixed positive power of e^t even though the intrinsic orbit length shrinks. This illustrates why volume alone is not an adequate effective complexity measure.

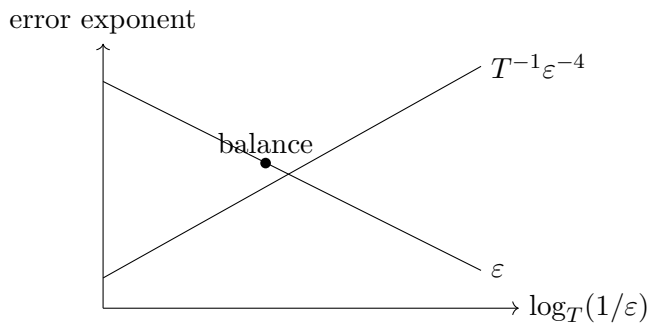


Figure 11.1: Effective smoothing chooses the scale where a decreasing geometric error and an increasing Sobolev cost balance.

Proof architecture: the four-error optimization ledger

Write the smoothing radius as $\varepsilon = T^{-\eta}$. A typical effective count has relative errors

$$T^{-\kappa}\varepsilon^{-A}, \quad \varepsilon^\beta, \quad T^{-\omega}\varepsilon^{-B}, \quad \varepsilon^\gamma.$$

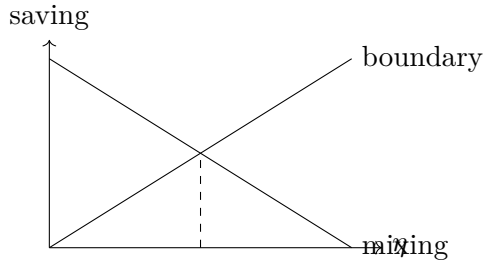
After substitution their exponents are

$$-\kappa + A\eta, \quad -\beta\eta, \quad -\omega + B\eta, \quad -\gamma\eta.$$

Thus any

$$0 < \eta < \min\{\kappa/A, \omega/B\}$$

gives a power saving, and the best exponent is the maximum over η of the minimum of the four positive quantities $\kappa - A\eta, \beta\eta, \omega - B\eta, \gamma\eta$.



What the figure answers. The usable exponent is the lower envelope; optimal smoothing occurs where a decreasing and an increasing loss balance.

Chapter 12

Student Proof Laboratory: Optimizing an Effective Count

The purpose of this chapter is to make the error exponent in an effective count reproducible. A statement of the form $O(T^{-\eta})$ is not useful to a student unless one can see which loss determines η .

12.1 Four errors in one line

Let R_T be a growing region and smooth its indicator at scale ε . A typical normalized counting error has the shape

$$E(T, \varepsilon) \leq C_1 T^{-\delta} \varepsilon^{-s} + C_2 \varepsilon^\alpha + C_3 T^{-\gamma} \varepsilon^{-r} + C_4 T^{-\zeta}.$$

The four terms are, respectively:

1. spectral/mixing error, with Sobolev cost ε^{-s} ;
2. boundary error from well-roundedness;
3. geometric wavefront or thickening error;
4. an independent polar-volume asymptotic error.

Set $\varepsilon = T^{-\theta}$. The exponents become

$$\delta - s\theta, \quad \alpha\theta, \quad \gamma - r\theta, \quad \zeta.$$

The best admissible rate is the maximum, over $\theta > 0$, of the minimum of these four affine functions.

Lemma 12.1 (Two-term optimization). *For $A, B, s, \alpha > 0$,*

$$T^{-A}\varepsilon^{-s} + \varepsilon^\alpha$$

is minimized at the power scale $\varepsilon = T^{-A/(s+\alpha)}$, and both terms then have size $T^{-A\alpha/(s+\alpha)}$ up to a constant.

Proof. Put $\varepsilon = T^{-\theta}$. The slower-decaying term has exponent $\min(A - s\theta, \alpha\theta)$. This minimum is maximized where the two lines intersect, namely at $\theta = A/(s + \alpha)$. Substitution gives $A\alpha/(s + \alpha)$. $\square$

12.2 Building a bump and counting its derivatives

Let M be a d -dimensional Riemannian manifold with bounded geometry on the region under consideration. Choose a fixed Euclidean bump ψ and set $\psi_\varepsilon(x) = \varepsilon^{-d}\psi(x/\varepsilon)$ in normal coordinates. Then a derivative of order j has size $O(\varepsilon^{-d-j})$, while its L^2 norm is $O(\varepsilon^{-d/2-j})$. Consequently a Sobolev norm of order ℓ costs at most $C\varepsilon^{-d/2-\ell}$. This elementary derivative count is the source of the exponent s in the spectral error.

12.3 Effective wavefront geometry

Suppose polar multiplication $H \times A^+ \times K \rightarrow G$ has an inverse derivative bounded by C on a closed cone separated from the Weyl walls. The inverse function theorem with parameters gives

$$Hak\mathcal{O}_\varepsilon \subset H\mathcal{O}_{C\varepsilon} aA_{C\varepsilon} K\mathcal{O}_{C\varepsilon}.$$

If the cone approaches a wall, the inverse derivative may blow up. The region must therefore be split into a regular part, where the wavefront constant is uniform, and a wall neighborhood. The latter is handled by a separate volume estimate. This decomposition is responsible for the third error term in the master ledger.

12.4 Effective translates of a closed orbit

Let $Y = xH$ be a finite-volume orbit and a_t an expanding translate. Choose a transverse bump ρ_ε and form the thickened ambient function F_ε . Quantitative

mixing gives

$$\int_X f(za_t)F_\varepsilon(z)dm_X(z) = \int_X f dm_X + O(e^{-\eta t}\mathcal{S}(f)\mathcal{S}(F_\varepsilon)).$$

Since $\mathcal{S}(F_\varepsilon) \ll \varepsilon^{-s}$, this contributes $e^{-\eta t}\varepsilon^{-s}$. Comparing the thickened integral with the singular orbit integral costs $O(\varepsilon^\alpha)$ if the orbit and the region are α -regular. The two-term optimization lemma gives an explicit power of e^{-t} .

Worked Example 12.2 (A toy exponent). Assume the mixing error is $T^{-1/3}\varepsilon^{-4}$ and the boundary error is ε . Then

$$\varepsilon = T^{-1/15}, \quad E(T) \ll T^{-1/15}.$$

If a wavefront term $T^{-1/2}\varepsilon^{-2}$ is also present, at $\varepsilon = T^{-1/15}$ it has exponent $1/2 - 2/15 = 11/30$, so it is smaller and does not affect the optimum. This is how one checks rather than guesses which mechanism controls the final rate.

12.5 Sector counting

Let a polar region be restricted by an angular set $\Omega \subset K/M$. If $\partial\Omega$ has codimension one and piecewise smooth boundary, then its ε -neighborhood has measure $O(\varepsilon)$. Smoothing the angular indicator therefore has boundary exponent $\alpha = 1$. If Ω has corners, the same estimate holds because a finite union of smooth faces still has tubular volume $O(\varepsilon)$. The effective sector theorem is thus obtained from the same master ledger with no new dynamical input.

12.6 Norm balls on an affine symmetric variety

Suppose $V = H \backslash G$ is embedded algebraically in a representation and $B_T = \{v : \|v\| < T\}$. In polar coordinates the norm is asymptotic on each Weyl chamber cone to a maximum of finitely many exponentials $e^{\lambda_j(H)}$. The region $\|v\| < T$ becomes a polytope $\lambda_j(H) < \log T + O(1)$. The polyhedral Laplace theorem of Volume X then gives the main volume asymptotic, while an effective version obtained by integrating one layer deeper gives the fourth error $T^{-\zeta}$. Thus the counting exponent is a minimum of a spectral optimization and a purely geometric volume exponent.

12.7 What the spectral method cannot do

Effective mixing is powerful when the measure being translated can be thickened into ambient Haar measure with controlled Sobolev norm. A general unipotent orbit has no expanding semisimple direction that makes such a thickening transverse in the required way. For those orbits one needs the effective linearization, Margulis functions, and closing alternatives of Volume XIII. This is why Volume XIII is not merely a harder application of Volume XII.

Chapter 13

Student Proof Laboratory: Turning Qualitative Counting into a Rate

The effective argument is an optimization problem. Four errors are produced for four different reasons, and choosing the smoothing scale before writing all four is a reliable way to make a wrong exponent.

13.1 Approximate identities and derivative cost

Let G have dimension m and choose a nonnegative $\rho \in C_c^\infty(\mathfrak{g})$ of integral one. In exponential coordinates define

$$\rho_\epsilon(\exp X) = c_\epsilon \epsilon^{-m} \rho(X/\epsilon).$$

Lemma 13.1 (Sobolev cost of a bump). *For every differential order d ,*

$$\mathcal{S}_d(\rho_\epsilon) \ll \epsilon^{-m/2-d}$$

for an L^2 -Sobolev norm, with an analogous explicit exponent for weighted norms on a fixed compact chart.

Proof. A derivative of order j contributes a factor ϵ^{-j} . The L^2 norm of the normalized bump contributes $\epsilon^{-m/2}$ because its square has height ϵ^{-2m} and support volume $\asymp \epsilon^m$. Summing $j \leq d$ gives the bound. $\square$

13.2 The master error ledger

Suppose a translating parameter has size T and quantitative mixing gives $T^{-\sigma}$ when test-function Sobolev norms are bounded. Smoothing at scale ϵ may produce

$$E(T, \epsilon) \leq C_1 T^{-\sigma} \epsilon^{-A} + C_2 \epsilon^B + C_3 T^{-\tau} \epsilon^{-C} + C_4 \epsilon^D.$$

The four terms respectively represent spectral mixing, boundary smoothing, wavefront/geometric distortion, and any additional cusp or height truncation.

Lemma 13.2 (Balancing finitely many power errors). *Let all exponents below be positive and suppose*

$$E(T, \epsilon) \ll \sum_i T^{-a_i} \epsilon^{-b_i} + \sum_j \epsilon^{c_j}.$$

Setting $\epsilon = T^{-\theta}$ gives

$$E(T, T^{-\theta}) \ll T^{-\min\{a_i - b_i \theta, c_j \theta\}}.$$

The optimal exponent is the maximum of this piecewise-linear minimum over $\theta > 0$ for which all $a_i - b_i \theta > 0$.

Proof. Substitute $\epsilon = T^{-\theta}$. Each term becomes a power of T . A sum of finitely many powers is controlled by the slowest-decaying one. The minimum of affine functions of θ is concave and piecewise linear, so its maximum occurs where two active error exponents balance or at an endpoint of the admissible interval. □

Worked Example 13.3 (One spectral and one boundary error). If $E \ll T^{-\sigma} \epsilon^{-A} + \epsilon$, balance $\sigma - A\theta = \theta$, giving $\theta = \sigma/(A+1)$ and

$$E \ll T^{-\sigma/(A+1)}.$$

The loss from σ is entirely the Sobolev cost of localizing the singular orbit.

13.3 Effective wavefront control

The qualitative wavefront lemma says small perturbations can be reabsorbed. For a rate we need Lipschitz control on the reabsorbed factors.

Proposition 13.4 (Quantitative wavefront chart). *On a closed cone inside a positive Weyl chamber there are constants $C, c > 0$ such that if $d(e, u) < \epsilon$ and a lies at chamber depth at least r , then*

$$ka\ell u = k'a'n^-\ell'$$

with

$$d(k, k') + d(a, a') + d(e, n^-) + d(\ell, \ell') \leq C\epsilon,$$

while the expanding component transported across a is additionally $O(e^{-cr}\epsilon)$.

Proof. Use the inverse function theorem for the product map $K \times A \times N^- \times K \rightarrow G$ on a compact family of normalized chamber directions. Its inverse derivative is uniformly bounded there. Root-space conjugation gives $\text{Ad}(a^{-1})X_\alpha = e^{-\alpha(\log a)}X_\alpha$, and the chamber-depth assumption makes this at most e^{-cr} . The local chart turns the Lie algebra estimate into the stated group-distance estimate. $\square$

13.4 Effective expanding translates

Let ν be a smooth probability on a compact piece of an expanding horosphere. Thicken it transversely at scale ϵ . Quantitative mixing from Volume XI evaluates the thickened translate, while the wavefront chart compares it with the original singular measure.

Theorem 13.5 (Effective translate: abstract form). *Assume*

1. *quantitative mixing with exponent σ and Sobolev order d ;*
2. *a height bound on the translating orbit sufficient for the weighted Sobolev norm;*
3. *C^1 boundary regularity of the parameter piece;*
4. *a quantitative nonfocusing estimate for proper homogeneous strata.*

Then there exist explicit $\eta > 0$ and Sobolev order D such that

$$\left| \int f d\mu_T - \int f dm_{xL} \right| \leq CT^{-\eta} \mathcal{S}_D(f).$$

Proof. Choose a transverse bump at scale ϵ . The thickened measure has Sobolev norm $O(\epsilon^{-A})$. Mixing contributes $T^{-\sigma}\epsilon^{-A}$. Replacing the singular measure by its thickening costs $O(\epsilon^B)$ by smoothness and boundary regularity. Quantitative wavefront control contributes $O(T^{-\tau}\epsilon^{-C})$ if the translating element has finite chamber depth. Nonescape/nonfocusing contributes the

fourth term in the master ledger. Substitute $\epsilon = T^{-\theta}$ and choose θ by the balancing lemma. The resulting minimum exponent is η . Every constant is uniform on the stated height and parameter families. $\square$

13.5 Effective counting by inner and outer smoothing

For a region B_T , choose smooth functions $F_{T,\epsilon}^- \leq 1_{B_T} \leq F_{T,\epsilon}^+$ supported on inner/outer ϵ -thickenings.

Lemma 13.6 (Boundary-to-counting error). *If*

$$\text{vol}(B_T^+(\epsilon) \setminus B_T^-(\epsilon)) \leq C\epsilon^\beta \text{vol}(B_T),$$

then smoothing contributes relative error $O(\epsilon^\beta)$.

Proof. Let $F_{T,\epsilon}^- \leq 1_{B_T} \leq F_{T,\epsilon}^+$ be the chosen inner and outer smoothings. Their integrals satisfy

$$0 \leq \int (F_{T,\epsilon}^+ - F_{T,\epsilon}^-) \leq \text{vol}(B_T^+(\epsilon) \setminus B_T^-(\epsilon)) \leq C\epsilon^\beta \text{vol}(B_T).$$

After unfolding, the exact count is bounded below and above by the two smoothed counts. Their main terms differ by the fixed homogeneous density times the preceding integral. Dividing by the main scale $\asymp \text{vol}(B_T)$ therefore gives a relative contribution $O(\epsilon^\beta)$. No dynamical estimate is used in this step; it is purely the squeezing error caused by the boundary layer. $\square$

Theorem 13.7 (Effective homogeneous lattice count from a power ledger). *Let G be semisimple, $\Gamma < G$ a lattice, and $H < G$ a closed subgroup with $H \cap \Gamma$ a lattice. Normalize Haar measures compatibly and put $c_{H,G,\Gamma} = V_H/V_G$. Let $B_T \subset H \backslash G$ be measurable with $0 < \text{vol}(B_T) < \infty$ and let*

$$N(B_T) = \#\{(H \cap \Gamma)\gamma : H\gamma \in B_T\}.$$

Assume that for all sufficiently large T and all $0 < \varepsilon < \varepsilon_0$ there are smoothed inner and outer counts $N_{T,\varepsilon}^- \leq N(B_T) \leq N_{T,\varepsilon}^+$ such that

$$\left| N_{T,\varepsilon}^\pm - c_{H,G,\Gamma} \text{vol}(B_T) \right| \leq C \text{vol}(B_T) \left(T^{-\sigma} \varepsilon^{-A} + T^{-\tau} \varepsilon^{-C_0} + \varepsilon^D \right), \quad (14.1)$$

and the boundary thickening satisfies

$$\text{vol}(B_T^+(\varepsilon) \setminus B_T^-(\varepsilon)) \leq C\varepsilon^\beta \text{vol}(B_T), \quad (14.2)$$

for fixed positive exponents $\sigma, \tau, A, C_0, D, \beta$. Define

$$\eta_* = \sup_{0 < \theta < \min\{\sigma/A, \tau/C_0\}} \min\{\sigma - A\theta, \tau - C_0\theta, D\theta, \beta\theta\}. \quad (14.3)$$

Then $\eta_* > 0$, and for every $0 < \eta < \eta_*$,

$$N(B_T) = c_{H,G,\Gamma} \text{vol}(B_T) + O_\eta(\text{vol}(B_T)T^{-\eta}). \quad (14.4)$$

Proof. Take $\varepsilon = T^{-\theta}$ with $0 < \theta < \min\{\sigma/A, \tau/C_0\}$. The three terms in (14.1) become $T^{-(\sigma-A\theta)}$, $T^{-(\tau-C_0\theta)}$, and $T^{-D\theta}$; (14.2) contributes $T^{-\beta\theta}$. The inner/outer squeeze therefore gives relative error

$$O\left(T^{-\min\{\sigma-A\theta, \tau-C_0\theta, D\theta, \beta\theta\}}\right).$$

The minimum is positive for every sufficiently small positive θ and its supremum is η_* . Choose θ so that the minimum exceeds the prescribed $\eta < \eta_*$; this proves (14.4). $\square$

Chapter 14

Student Proof Laboratory: Sector Counts and Norm Balls

14.1 Sectors

A sector restricts both Cartan radius and angular variables. Angular boundaries add faces to the well-roundedness estimate, but the dynamical mixing rate is unchanged. The final exponent may deteriorate because a rougher angular cutoff has a larger Sobolev cost.

14.2 Norm balls

If G is linearly represented and $B_T = \{Hg : \|\rho(g)v_0\| < T\}$, Cartan coordinates reduce the volume to an integral of exponentials over a polyhedral region. Laplace's method gives the main volume asymptotic. Effective counting then adds a power-saving relative error provided the maximizing face stays in a chamber region where the quantitative wavefront constants are uniform.

14.3 Why no universal exponent is printed

The exponent depends on the spectral gap, representation weights, Sobolev order, boundary regularity, and height loss. A single symbol η is legitimate only after its dependency formula has been given. This volume's ledger is intended to make that formula reconstructible in every later application.

Chapter 15

A Student Course in Effective Equidistribution and Counting

The preceding chapters contain the effective statements in the form needed for applications. This chapter goes through the mechanism once more at a slower pace. The point is not to add a different theorem. It is to make every exponent in the final count traceable to a geometric or analytic estimate.

15.1 The four parameters in an effective argument

Fix a finite-volume homogeneous space $X = \Gamma \backslash G$. Let $Y = xH$ be a closed homogeneous orbit and let a_t expand the directions transverse to Y . Four quantities enter every smoothing argument:

1. a mixing rate $e^{-\eta t}$, or more generally $M(t) \rightarrow 0$;
2. a Sobolev loss ε^{-s} caused by replacing a singular measure or an indicator by a bump of thickness ε ;
3. a geometric wavefront error of size $e^{-\lambda t} \varepsilon^{-r}$;
4. a boundary error of size ε^β for the counting region.

The letters s, r, β are not universal constants. They depend on the regularity class of the test object and on the chosen Sobolev norm. Keeping them visible prevents a common mistake: optimizing an error term before all losses have been entered into the ledger.

Lemma 15.1 (Elementary optimization with two decreasing errors). *Suppose*

$A, B, \alpha, \beta > 0$ and

$$E_T(\varepsilon) \leq AT^{-\alpha}\varepsilon^{-s} + B\varepsilon^\beta, \quad 0 < \varepsilon \leq 1.$$

Then the choice

$$\varepsilon = T^{-\alpha/(s+\beta)}$$

gives

$$E_T(\varepsilon) \ll_{A,B} T^{-\alpha\beta/(s+\beta)}.$$

Proof. Write $\varepsilon = T^{-\theta}$. The two exponents of T become $-\alpha + s\theta$ and $-\beta\theta$. The maximum of these two affine functions is minimized when they are equal. Solving $-\alpha + s\theta = -\beta\theta$ gives $\theta = \alpha/(s + \beta)$. Substitution gives the displayed exponent. $\square$

Worked Example 15.2 (Why choosing $\varepsilon = T^{-1}$ is usually wrong). If the mixing term is $T^{-1/2}\varepsilon^{-6}$ and the boundary term is ε , then $\varepsilon = T^{-1}$ makes the first error grow like $T^{11/2}$. The balanced choice is $\varepsilon = T^{-1/14}$, which produces an error $T^{-1/14}$. The numerical exponent is not important; what matters is that smoothing consumes derivatives, and those derivatives must be paid for.

15.2 Approximate identities from coordinates

Let $d = \dim G$. Choose exponential coordinates on a ball $B(0, 2)$ in $\mathfrak{g}$ and a nonnegative $\rho \in C_c^\infty(\mathfrak{g})$ supported in $B(0, 1)$ with $\int \rho = 1$. Define

$$\rho_\varepsilon(\exp Z) = c_\varepsilon \varepsilon^{-d} \rho(Z/\varepsilon),$$

where c_ε corrects the Jacobian of the exponential map.

Lemma 15.3 (Derivative cost of a group bump). *For every left-invariant differential operator D of order j there is C_D such that*

$$\|D\rho_\varepsilon\|_{L^1(G)} \leq C_D \varepsilon^{-j}$$

for all sufficiently small ε .

Proof. In exponential coordinates every left-invariant vector field is a smooth linear combination of Euclidean derivatives, with coefficients uniformly bounded on the fixed coordinate ball. A Euclidean derivative of order j of $\varepsilon^{-d}\rho(Z/\varepsilon)$ contributes exactly ε^{-j} . The Jacobian of $\exp$ and all of its

derivatives are uniformly bounded on the ball, and c_ε stays bounded above and below because the Jacobian equals 1 at 0. Integrating after the change of variables $Z = \varepsilon W$ gives the result. $\square$

Proposition 15.4 (Smoothing an indicator). *Let $B \subset X$ and assume its ε -boundary satisfies*

$$m_X(\partial_\varepsilon B) \leq C\varepsilon^\beta.$$

There are smooth functions $f_\varepsilon^- \leq \mathbf{1}_B \leq f_\varepsilon^+$ such that

$$\int_X (f_\varepsilon^+ - f_\varepsilon^-) dm_X \ll \varepsilon^\beta$$

and every Sobolev norm of order ℓ satisfies

$$\mathcal{S}_\ell(f_\varepsilon^\pm) \ll \varepsilon^{-c_\ell}$$

for an explicitly computable integer c_ℓ .

Proof. Let B_ε^- be the set of points whose ε -ball is contained in B and let B_ε^+ be the ε -neighborhood of B . Convolve $\mathbf{1}_{B_\varepsilon^-}$ and $\mathbf{1}_{B_\varepsilon^+}$ with the group bump above, after using a finite local trivialization if the action has small stabilizers. The support properties give the pointwise inequalities. Their difference is supported inside a constant multiple of $\partial_\varepsilon B$, so the integral loss is $O(\varepsilon^\beta)$. Differentiation falls on the bump, and the preceding lemma records the precise derivative cost. $\square$

15.3 Effective thickening of a lower-dimensional orbit

A closed orbit $Y = xH$ has zero G -volume, so mixing cannot be applied to its invariant measure directly. Choose a local complement $\mathfrak{r}$ to $\mathfrak{h}$ and a small transverse bump ψ_ε on $R = \exp(\mathfrak{r})$. For a test function F on Y , define

$$\tilde{F}_\varepsilon(yr) = F(y)\psi_\varepsilon(r)$$

inside the product chart.

Lemma 15.5 (Thickening identity). *Assume the product chart $(y, r) \mapsto yr$ is injective on the support of the bump. Then for every continuous φ ,*

$$\int_X \tilde{F}_\varepsilon(z) \varphi(za_t) dm_X(z) = \int_Y F(y) \int_R \psi_\varepsilon(r) \varphi(yra_t) J(y, r) dr dm_Y(y),$$

where $J(y, r) = 1 + O(\varepsilon)$ uniformly on a fixed compact part of Y .

Proof. Let $\Phi : Y \times R \rightarrow X$ be $\Phi(y, r) = yr$ on the injective product neighborhood. Compatible Haar normalization means that the pullback of m_X has the form

$$\Phi^* dm_X = J(y, r) dm_Y(y) dr, \quad J(y, e) = 1.$$

Substituting $z = yr$ into the left-hand side and using the definition of $\tilde{F}_\varepsilon$ gives the displayed iterated integral. On a fixed compact subset of Y , the derivatives of the smooth function J are uniformly bounded. The support of ψ_ε lies at coordinate distance $O(\varepsilon)$ from e , so the mean-value estimate gives $|J(y, r) - 1| \leq C\varepsilon$ uniformly there. $\square$

Now write $ra_t = a_t(a_t^{-1}ra_t)$. Expansion of a_t means that $a_t^{-1}ra_t$ is small in the relevant transverse directions. The quantitative wavefront lemma converts this statement into a comparison between $\varphi(yra_t)$ and $\varphi(ya_t)$.

Proposition 15.6 (Effective translate estimate: bookkeeping form). *Assume*

1. *mixing gives*

$$|\langle \pi(a_t)f, g \rangle - \int f \int g| \leq Ce^{-\eta t} \mathcal{S}_\ell(f) \mathcal{S}_\ell(g);$$

2. *the transverse bump has $\mathcal{S}_\ell$ -size $O(\varepsilon^{-s})$;*

3. *the wavefront comparison contributes $O(e^{-\lambda t} \varepsilon^{-r})$;*

4. *the part of Y outside a height cutoff $\mathcal{H} \leq R$ has mass $O(R^{-q})$ and the local constants on the cutoff are $O(R^u)$.*

Then

$$\left| \int_Y \varphi(ya_t) dm_Y(y) - \int_X \varphi dm_X \right| \ll e^{-\eta t} R^u \varepsilon^{-s} + e^{-\lambda t} R^u \varepsilon^{-r} + \varepsilon + R^{-q},$$

with Sobolev norms of φ suppressed from the notation.

Proof. Split Y into the height cutoff and its complement. The complement gives R^{-q} . On the compact part, thicken Y by the transverse bump. Mixing of the thickened function gives the first term. Replacing the thickened translate by the original singular translate gives the wavefront term and the Jacobian error $O(\varepsilon)$. The factor R^u records the deterioration of coordinate and Sobolev constants on the height cutoff. No other error is introduced. $\square$

Guiding Idea 15.7. There are two optimizations, not one. First choose the height cutoff R so that cusp mass and compact-part constants balance. Then choose ε so that mixing/wavefront losses balance the smoothing error. Interchanging these steps can hide a dependence of the final exponent on cusp geometry.

15.4 From equidistribution to a lattice count

Let $V = H \backslash G$ and let $v_0 = H$ be the base point. Suppose the integral orbit under Γ is discrete. For a region $B_T \subset V$ set

$$N(T) = \#\{\gamma \in (H \cap \Gamma) \backslash \Gamma : v_0 \gamma \in B_T\}.$$

The qualitative unfolding in Volume X fixes the leading constant. Here we make the effective comparison explicit.

Lemma 15.8 (Inner/outer count). *Let $B_T^-(\varepsilon) \subset B_T \subset B_T^+(\varepsilon)$ be inner and outer ε -perturbations. If $F_{T,\varepsilon}^\pm$ are their smooth majorant/minorant functions, then*

$$\sum_{\gamma} F_{T,\varepsilon}^-(v_0 \gamma) \leq N(T) \leq \sum_{\gamma} F_{T,\varepsilon}^+(v_0 \gamma).$$

Moreover the difference of the corresponding main terms is bounded by

$$\frac{\text{vol}((H \cap \Gamma) \backslash H)}{\text{vol}(\Gamma \backslash G)} (\text{vol}(B_T^+(\varepsilon)) - \text{vol}(B_T^-(\varepsilon))).$$

Proof. The first inequality is pointwise. The second statement follows from the normalization of the quotient measure already established in Volume X; this is exactly the place where reversing the H -covolume would change the counting constant. $\square$

Theorem 15.9 (Effective counting master theorem). *Assume the translate estimate above, and assume the family B_T is quantitatively well rounded:*

$$\text{vol}(B_T^+(\varepsilon) \setminus B_T^-(\varepsilon)) \leq C\varepsilon^\beta \text{vol}(B_T).$$

If the smoothed orbit sum has relative error $O(T^{-\alpha}\varepsilon^{-s})$, then

$$N(T) = \frac{\text{vol}((H \cap \Gamma) \backslash H)}{\text{vol}(\Gamma \backslash G)} \text{vol}(B_T) + O\left(\text{vol}(B_T) T^{-\alpha\beta/(s+\beta)}\right),$$

provided the wavefront and height errors are no larger after the chosen parameter optimization.

Proof. Apply the inner/outer count and effective equidistribution to the two smoothed functions. Divide by the main volume. The relative error has the form $T^{-\alpha}\varepsilon^{-s} + \varepsilon^\beta$ plus the wavefront and height terms. Use the elementary optimization lemma and the stated domination assumption. $\square$

15.5 Sector and norm-ball laboratories

Worked Example 15.10 (A sector). In a polar decomposition $G = KA^+K$, let the region restrict the first K -coordinate to a set Ω with piecewise smooth boundary. Thickening Ω by angular distance ε changes its Haar measure by $O(\varepsilon)$. Thus $\beta = 1$ in the boundary ledger. The radial A -variable contributes the volume growth, while the angular variable contributes only the well-roundedness loss.

Worked Example 15.11 (Norm balls). Let V be a finite-dimensional representation and $B_T = \{Hg : \|v_0g\| < T\}$. In logarithmic Cartan coordinates $H \in \mathfrak{a}^+$, each weight contributes an exponential $e^{\lambda(H)}$. The volume integral is therefore a finite sum of Laplace-type integrals over polyhedral cones. The largest ratio between the Haar exponent and the norm constraint gives the growth exponent; ties produce powers of $\log T$. A perturbation $T \mapsto (1 + O(\varepsilon))T$ changes the main term by $O(\varepsilon)$ when the leading volume asymptotic is differentiable in logarithmic scale. This verifies quantitative well-roundedness in the standard norm-ball laboratory.

15.6 A reproducible proof checklist

Before quoting an effective count later in the series, a student should be able to point to all seven entries below:

1. the exact Haar normalization and leading density;
2. the Sobolev norm and the derivative order used by mixing;
3. the spectral or tensor-tempered input producing the mixing rate;
4. the thickness ε and its Sobolev cost;
5. the wavefront error and its dependence on ε ;
6. the boundary regularity exponent of the counting family;
7. the height cutoff and the estimate for mass outside it.

If one entry is absent, the exponent printed at the end is not yet justified.

15.7 Uniformity in the basepoint and in cusp height

Effective statements are often written for a fixed $x \in X$, but later applications vary x . The dependence is controlled by the same height function used in Volume XI.

Proposition 15.12 (Height-uniform local constants). *Let $\text{ht} : X \rightarrow [1, \infty)$ be a proper height for which the injectivity radius satisfies*

$$\text{inj}(x)^{-1} \ll \text{ht}(x)^c.$$

Fix a Sobolev order ℓ . On the set $X_{\leq R} = \{x : \text{ht}(x) \leq R\}$, all constants needed for local product charts, quotient bumps, and differentiation of right translations are bounded by $O(R^{C_\ell})$.

Proof. A quotient chart of radius comparable to the injectivity radius is obtained from the exponential map on G . Rescaling a fixed Euclidean bump from unit radius to radius $\text{inj}(x)$ costs a power of its reciprocal for each derivative. The group exponential and multiplication maps have uniform derivatives on a fixed compact neighborhood of the identity, so no additional dependence on x appears. Substituting $\text{inj}(x)^{-1} \ll R^c$ gives a polynomial R^{C_ℓ} bound. $\square$

Corollary 15.13 (Uniform effective translate on a height window). *Fix one of the translate or counting theorems of this volume and fix all of its Sobolev orders. If its ambient effective-mixing constant is independent of the basepoint, then there are constants $C_0, C_1 > 0$, depending only on that theorem and the fixed Sobolev orders, such that every occurrence of its basepoint-dependent error constant is at most*

$$C_0 R^{C_1}$$

for every basepoint x with $\text{ht}(x) \leq R$.

Proof. The basepoint-dependent inputs in the thickening argument are the quotient chart radius, derivatives of the localized bump, and the local wave-front comparison. By the preceding proposition each is bounded by R^C for a fixed exponent determined by the maximal Sobolev order. Only finitely many such factors are multiplied in any fixed theorem, so their product is bounded by $C_0 R^{C_1}$. $\square$

Chapter 16

Guided Exponent Bookkeeping for Effective Counting

The purpose of this chapter is practical. We take the qualitative mixing-to-counting proof from Volume X and attach a symbol to every source of error. A student should be able to change the spectral exponent, the boundary regularity, or the cusp moment and recompute the final counting rate without rereading the whole proof.

16.1 A general four-error model

Suppose the target count has main scale $M(T)$. After smoothing at thickness ϵ and truncating at height R , assume the relative error is bounded by

$$E(T, \epsilon, R) = T^{-\eta} R^a \epsilon^{-b} + T^{-\lambda} R^c \epsilon^{-d} + \epsilon^\beta + R^{-q}.$$

The four terms represent spectral mixing, wavefront distortion, boundary smoothing, and cusp truncation.

Set

$$\epsilon = T^{-x}, \quad R = T^y.$$

The four exponents of T become

$$-\eta + ay + bx, \quad -\lambda + cy + dx, \quad -\beta x, \quad -qy.$$

We want to choose $x, y > 0$ to make the largest exponent as negative as possible. This is a two-variable linear-programming problem.

Proposition 16.1 (Existence of an optimal power choice). *If there exists $x, y > 0$ for which all four exponents above are negative, then the best power-saving exponent is achieved at an intersection of two or more of the affine constraint lines.*

Proof. For a candidate saving δ , the inequalities

$$\eta - ay - bx \geq \delta, \quad \lambda - cy - dx \geq \delta, \quad \beta x \geq \delta, \quad qy \geq \delta$$

define a closed convex polytope in (x, y, δ) after imposing a large compact box containing one feasible point. Maximizing the linear coordinate δ occurs at an extreme point, where at least two independent constraints are active. $\square$

This simple observation is useful because effective proofs often print a nonoptimal but convenient exponent. The validity of the theorem requires only a feasible choice, while the optimal exponent can be computed separately.

16.2 A full numerical laboratory

Assume

$$\eta = \frac{1}{2}, \quad \lambda = 1, \quad a = 2, \quad c = 1, \quad b = 6, \quad d = 2, \quad \beta = 1, \quad q = 3.$$

Balancing the boundary and cusp errors suggests $x = \delta$ and $y = \delta/3$. The mixing constraint becomes

$$\frac{1}{2} - \frac{2\delta}{3} - 6\delta \geq \delta,$$

so

$$\delta \leq \frac{3}{46}.$$

The wavefront constraint gives

$$1 - \frac{\delta}{3} - 2\delta \geq \delta,$$

which is much weaker. Thus the spectral smoothing cost is the bottleneck and a safe final rate is $T^{-3/46}$. A different Sobolev norm changing b from 6 to 8 would immediately worsen the rate; this is why derivative costs must be recorded rather than hidden in the symbol $O_\epsilon(1)$.

16.3 Effective horospherical averages from mixing

Let U^+ be an expanding horospherical group and $\Omega \subset U^+$ a compact smooth domain. Choose a transverse stable/neutral subgroup S and a bump ψ_ϵ of total mass one on S . Define the thickened density on X by

$$F_\epsilon(xus) = \frac{1}{m_U(\Omega)} \mathbf{1}_\Omega(u) \psi_\epsilon(s),$$

with a smoothed version of $\mathbf{1}_\Omega$ near the boundary.

Effective mixing gives

$$\int_X F_\epsilon(z) f(za_t) dm(z) = \int f dm + O(E_{\text{mix}}(t) \epsilon^{-b} \mathcal{S}_L(f)).$$

Conjugating the stable/neutral coordinate across a_t makes its effect at the translated point of order $E_{\text{wf}}(t) \epsilon^{-d}$. Removing the boundary strip costs $O(\epsilon^\beta)$. Thus

$$\frac{1}{m_U(\Omega)} \int_\Omega f(xua_t) du = \int f dm + O(E_{\text{mix}}(t) \epsilon^{-b} + E_{\text{wf}}(t) \epsilon^{-d} + \epsilon^\beta) \mathcal{S}_L(f).$$

This is the exact source of the three-term formula in the body.

16.4 Why closed-orbit translates need a height cut-off

For a noncompact closed orbit $Y = xH$, local product charts around $y \in Y$ can shrink as y enters a cusp. Let $Y_{\leq R}$ be a height truncation. On this set the inverse injectivity radius is polynomial in R , so the thickened density has Sobolev norm

$$\mathcal{S}_L(F_{\epsilon,R}) \ll R^a \epsilon^{-b}.$$

The discarded cusp has mass $O(R^{-q})$. Hence

$$\left| \int_Y f(ya_t) dm_Y - \int f dm_X \right| \ll E_{\text{mix}}(t) R^a \epsilon^{-b} + E_{\text{wf}}(t) R^c \epsilon^{-d} + \epsilon^\beta + R^{-q}.$$

This is the four-error model from the first section.

16.5 Inner and outer smoothing of a count

Let $B_T \subset H \backslash G$. Define

$$B_T^-(\epsilon) = \{v : vU_\epsilon \subset B_T\}, \quad B_T^+(\epsilon) = B_T U_\epsilon.$$

Choose smooth functions

$$\mathbf{1}_{B_T^-(\epsilon)} \leq f_{T,\epsilon}^- \leq \mathbf{1}_{B_T} \leq f_{T,\epsilon}^+ \leq \mathbf{1}_{B_T^+(\epsilon)}.$$

Apply the effective translate theorem to the periodization of these two functions. The count is squeezed between

$$\frac{V_H}{V_G} \text{vol}(B_T^-(\epsilon))(1 + O(E_T)) \quad \text{and} \quad \frac{V_H}{V_G} \text{vol}(B_T^+(\epsilon))(1 + O(E_T)).$$

If quantitative well-roundedness gives

$$\text{vol}(B_T^+(\epsilon) \setminus B_T^-(\epsilon)) \ll \epsilon^\beta \text{vol}(B_T),$$

the same optimization produces an effective count.

16.6 Sector counting with angular boundaries

Suppose polar coordinates are (h, a, k) and restrict k to a sector $\Omega \subset K$ with smooth boundary. An ϵ -neighborhood of $\partial\Omega$ has K -measure $O(\epsilon)$. If the radial density is independent of k , the corresponding quotient-volume boundary error is also $O(\epsilon)$ uniformly in T . Thus $\beta = 1$.

If Ω has corners but piecewise smooth faces, the tubular neighborhood of the union of faces remains $O(\epsilon)$; intersections of two faces have codimension at least two and contribute $O(\epsilon^2)$. Therefore corners do not worsen the first boundary exponent.

16.7 Norm balls and differentiating the volume law

Suppose

$$\text{vol}(B_T) = cT^a(\log T)^b + O(T^{a-\zeta}).$$

Replacing T by $(1 + \epsilon)T$ changes the main term by

$$cT^a(\log T)^b \left(a\epsilon + O\left(\frac{\epsilon}{\log T} + \epsilon^2\right) \right).$$

Hence, uniformly for large T and small ϵ ,

$$\text{vol}(B_{(1+\epsilon)T} \setminus B_T) \ll \epsilon \text{vol}(B_T) + T^{a-\zeta}.$$

This verifies effective well-roundedness with $\beta = 1$ once the secondary volume error is smaller than the desired dynamical error.

16.8 Moving basepoints

Suppose the starting closed orbit or horospherical piece depends on T and its basepoint has height at most T^θ . The local chart/Sobolev constants then contribute a factor $T^{C\theta}$. An effective mixing rate $T^{-\eta}$ becomes $T^{-\eta+C\theta}$ before smoothing. Therefore one must impose

$$C\theta < \eta$$

(or a stronger inequality after smoothing losses) to retain a power saving. This is the quantitative meaning of “uniform for basepoints in a slowly growing height window.”

16.9 Effective curve translates

For a curve, thickening is anisotropic because the tangent direction is part of the moving set. On a compact nondegenerate curve, choose a Frenet-type frame of tangent and transverse directions. A tube of transverse radius ϵ has Sobolev cost ϵ^{-b} . The expanding diagonal element separates the tangent jet from stable transverse directions. If the qualitative focusing theorem supplies a uniform distance from all proper algebraic obstructions, the same mixing/thickening argument gives a rate.

What the spectral method cannot do by itself is provide that uniform nonfocusing distance for an arbitrary individual curve or unipotent orbit. That is why Volume XIII develops effective linearization and closing rather than trying to extract all rates from mixing.

16.10 A reusable optimization worksheet

For an application, fill in:

source	relative error
mixing	$T^{-\eta} R^a \epsilon^{-b}$
wavefront	$T^{-\lambda} R^c \epsilon^{-d}$
boundary	ϵ^β
cusps	R^{-q}
volume asymptotic	$T^{-\zeta}$

Then set $R = T^y$, $\epsilon = T^{-x}$ and solve the finite linear inequalities which make every exponent negative. This worksheet is intentionally simple: it prevents a long effective proof from ending with an exponent whose dependence on one of the losses cannot be reconstructed.

Proof Guide and Dependency Summary

This chapter is a finalization layer. It does not introduce a new theorem scope. Its purpose is to expose the proof architecture, make the hypotheses that are easiest to misread impossible to miss, and give a student a concrete model calculation before the abstract machinery is reused later.

1. Keep the four errors separate

A typical effective thickening argument has four independent losses:

$$E(\varepsilon, t) = E_{\text{bdry}} + E_{\text{smooth}} + E_{\text{spec}} + E_{\text{wave}}.$$

Schematically,

$$E_{\text{smooth}} \ll \varepsilon^\eta, \quad E_{\text{spec}} \ll \Xi(a_t)^{1/m} \varepsilon^{-A},$$

while boundary and wavefront terms depend on the geometry of the chosen family of sets. The optimization is valid only after all four terms have been placed on the same scale.

2. A model optimization

If the only two relevant terms are

$$\varepsilon + e^{-\kappa t} \varepsilon^{-A},$$

choose

$$\varepsilon = \exp\left(-\frac{\kappa t}{A+1}\right),$$

which yields an exponential error with exponent $\kappa/(A+1)$. With additional boundary exponent η , solve the corresponding linear minimax problem in the exponents rather than simply reusing this two-term choice.

3. Uniformity in the base point

If the orbit approaches the cusp, injectivity radius can become smaller than the chosen smoothing scale. Then the proof must either truncate to a compact core with an explicit discarded-mass estimate or use height-weighted local charts. A rate stated uniformly in the base point therefore carries a height hypothesis or a height loss.

4. Student error ledger

Every effective theorem in this volume should be reconstructible from a table with columns: source of error, bound, Sobolev order, height dependence, parameter dependence, and chosen smoothing scale. If an $O(T^{-\eta})$ term has no row in that table, the proof is incomplete.

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Effective Unipotent Dynamics and Effective Closing
Volume XIII of a Detailed Notebook on Ratner's Theorems

Detailed Notebook on Ratner's Theorems

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Preface and proof policy

Qualitative Ratner theory proves that a unipotent orbit is governed by a homogeneous subgroup. It does not say how large a time scale must be before that subgroup becomes visible. This volume develops the finite-scale replacement. Its organizing alternative is: either an orbit segment is quantitatively distributed, or the segment is quantitatively close to a proper algebraic obstruction of controlled arithmetic complexity.

The proof policy is intentionally stricter than a survey treatment. The subgroup-height formalism, effective linearization, quantitative avoidance, almost-invariant Lie-algebra argument, and closing step are all proved here in the forms subsequently used. Incidence/projection estimates needed for the strongest special-group polynomial rates are isolated and proved as separate finite-dimensional lemmas, so that the dynamical argument never hides them inside the phrase “effective Ratner theorem.”

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Dependency and proof-ownership contract

Purpose. Effective unipotent dynamics.

Prerequisites. Volumes II, V, XI–XII; the effective Łojasiewicz, uniform Chevalley-complexity, generated-height, and $\mathfrak{sl}_2$ Whitehead inputs used here are proved internally.

New objects. subgroup height, Margulis functions, Frostman clouds, closing. Every project-specific technical term is defined before its first load-bearing use or recalled by an exact backward reference.

Main outputs. finite-scale closing/equidistribution alternatives.

External-input rule. Every nontrivial imported result is either part of the declared graduate-level prerequisites, an explicitly named external research input in the prerequisite map, or is proved in this series before use (possibly under an explicit deferred-proof contract). A citation by itself never constitutes a proof.

Deferred-proof map. The detailed effective-algebra and restricted-projection appendices are logical prerequisites for the final closing/equidistribution synthesis. A forward citation to one of those modules always names an already specified proof owner; it is never shorthand for “standard”. In particular the fixed-format effective Łojasiewicz theorem is owned by Theorem A.7, uniform Chevalley complexity by Proposition 2.8, and the height bookkeeping and finite-dimensional projection modules by the internal results indicated where cited.

Reader map and proof route

Question of the volume. What is the finite-scale replacement for the qualitative sentence “the unipotent orbit is controlled by a homogeneous subgroup”?

Prerequisite checkpoint. The effective argument uses the earlier linearization, height, spectral, and smoothing machinery. The two algebraic interfaces used in the closing argument are proved here: Theorem A.7 proves the exact fixed-format effective real-algebraic Łojasiewicz statement, while Proposition 2.8 supplies the uniform defining-degree and finite-Chevalley complexity for connected class- $\mathcal{H}$ subgroups of fixed SL_N . The generated-subgroup-height theorem and bounded-component argument remain internal as before.

Load-bearing route. Read the quantitative nondivergence and almost-invariance chapters, then the subgroup-height interface. Before treating the final closing alternative, use the detailed appendices for the proof-owning chain

finite Chevalley complexity $\longrightarrow$ bounded components
bounded components $\longrightarrow$ bounded connected generators
bounded connected generators $\longrightarrow$ fixed-dimensional evaluation
fixed-dimensional evaluation $\longrightarrow$ generated subgroup height.

The later projection/decoupling material is a separate specialization and should not be mistaken for part of the abstract closing lemma.

Proof-ownership checkpoint. The worked examples are motivation only; the exact uniform theorems in the detailed chapters carry the proof, including their dependence on N , degree, and coefficient height.

Sanity check. For $P(x) = x^d$, the elementary inequality $\text{dist}(x, Z(P)) = |x| = |P(x)|^{1/d}$ shows why a Łojasiewicz exponent deteriorates with degree. It is a model for Theorem [A.7](#), not a substitute for its multivariate proof.

Interface laboratories: effective algebraic geometry and subgroup complexity

The following examples motivate two algebraic interfaces used by effective closing. Both uniform statements are proved in the detailed chapters; the examples remain useful because they show why the uniform hypotheses and power losses have the form they do.

A one-polynomial Łojasiewicz calculation

Let $P(x) = x^d$ on $[-1, 1]$. Its real zero set is $Z(P) = \{0\}$, and therefore

$$\text{dist}(x, Z(P)) = |x| = |P(x)|^{1/d}.$$

Two lessons survive in the multivariate theorem. First, a small polynomial value controls distance to the zero set only with a power loss. Second, that power necessarily depends on algebraic complexity: even this example forces an exponent no better than $1/d$ in a degree-uniform statement. The full common-real-zero-set theorem, with effective dependence on all bounded-complexity parameters, is Theorem [A.7](#); the displayed identity is only its one-variable model.

For contrast, $P(x) = x^2 + 1$ has no real zero. Any effective theorem must state what happens when the common real zero set is empty; one cannot project to a “nearby root” that does not exist. This is precisely why the discarded elimination sketch was unsafe.

A class- $\mathcal{H}$ subgroup visible by equations

Consider

$$U_{12} = \{I + tE_{12} : t \in \mathbb{G}_a\} < SL_3.$$

Inside Mat_3 it is cut out by the linear equations

$$g_{11} = g_{22} = g_{33} = 1, \quad g_{ij} = 0 \quad ((i, j) \neq (1, 2), i \neq j).$$

Its Lie algebra is the rational line $\mathbb{Q}E_{12}$, whose primitive Pluecker coordinate has height 1. Thus this particular group has defining degree 1 and minimal arithmetic complexity.

Proposition 2.8 proves the qualitatively stronger uniform statement: for *every* connected class- $\mathcal{H}$ subgroup of a fixed SL_N , one degree bound and a finite ambient Chevalley list suffice. The later height comparisons and generated-subgroup-height argument then use that internal result.

Chapter 1

Why the Qualitative Ratner Proof Loses Rates

1.1 The finite-scale problem

Let $G < \mathrm{SL}_N(\mathbb{R})$ be a connected perfect algebraic group defined over $\mathbb{Q}$, let $\Gamma < G(\mathbb{R})$ be an arithmetic lattice, let $X = \Gamma \backslash G$, and let

$$U = \{u_t = \exp(tZ) : t \in \mathbb{R}\}$$

be an Ad-unipotent one-parameter subgroup. Volumes II–III prove qualitative nonescape, measure classification, and orbit closure. The effective problem is different: for a large but finite T , what can be said about

$$\mu_{x,T} = \frac{1}{T} \int_0^T \delta_{xu_t} dt?$$

Definition 1.1 (Sobolev discrepancy). Fix a Sobolev norm $\mathcal{S}_d$ as in Volume XI. For probability measures ν_1, ν_2 define

$$D_d(\nu_1, \nu_2) = \sup_{\mathcal{S}_d(f) \leq 1} \left| \int f d\nu_1 - \int f d\nu_2 \right|.$$

The desired conclusion cannot simply be $D_d(\mu_{x,T}, m_X) \ll T^{-\delta}$. A periodic unipotent orbit, an orbit contained in a proper closed semisimple subspace, or an orbit extremely close to such a subspace is an obstruction. Thus an effective theorem must quantify the obstruction as well as the error.

Guiding Idea 1.2 (Effective Ratner alternative). A useful finite-scale statement has three outputs:

1. a homogeneous subgroup H of controlled arithmetic height;
2. a quantitative statement that x is close to a closed H -orbit, or that the U -segment is close to the normalizer of H ;
3. if no such proper H exists below a prescribed complexity, a polynomial or otherwise explicit equidistribution estimate.

The arithmetic height is indispensable: without it, “close to some proper subgroup” has no effective content.

1.2 Six qualitative losses

The qualitative proof in Volume III uses operations that discard numerical information:

1. passing to a weak-* subsequence;
2. selecting generic points without a rate;
3. replacing small exceptional sets by null sets;
4. using compactness of a Grassmannian without separation estimates;
5. using exact subgroup containment after taking a limit;
6. descending through singular strata without tracking arithmetic complexity.

Every chapter below addresses one of these losses.

Warning 1.3. An effective proof is not obtained by inserting $O(T^{-\eta})$ into the qualitative proof. The main new issue is stability: approximate algebraic relations must either improve to exact rational relations of controlled height or be shown too weak to persist for a long unipotent segment.

Chapter 2

Arithmetic Heights of Homogeneous Obstructions

2.1 Integral structures on exterior powers

Write $\mathfrak{g} \subset \mathfrak{sl}_N(\mathbb{R})$ and fix the lattice

$$\mathfrak{g}_{\mathbb{Z}} = \mathfrak{g} \cap \mathfrak{sl}_N(\mathbb{Z})$$

after replacing the ambient integral model by a commensurable one if necessary. For a rational r -plane $W \subset \mathfrak{g}$, the rank- r lattice $W \cap \mathfrak{g}_{\mathbb{Z}}$ has a primitive wedge vector

$$v_W = w_1 \wedge \cdots \wedge w_r \in \bigwedge^r \mathfrak{g}_{\mathbb{Z}},$$

well defined up to sign.

Definition 2.1 (Height of a rational subspace). For a fixed Euclidean norm on all exterior powers set

$$\mathrm{ht}(W) = \|v_W\|.$$

If $H < G$ is a connected $\mathbb{Q}$ -subgroup, put $\mathrm{ht}(H) = \mathrm{ht}(\mathrm{Lie} H)$.

Lemma 2.2 (Height and covolume). *For a rational r -plane W ,*

$$\mathrm{ht}(W) = \mathrm{covol}(W \cap \mathfrak{g}_{\mathbb{Z}})$$

with respect to the induced Euclidean measure on W .

Proof. If $w_1, \dots, w_r$ is an integral basis, the squared Euclidean norm of the wedge is the Gram determinant

$$\|w_1 \wedge \dots \wedge w_r\|^2 = \det(\langle w_i, w_j \rangle)_{i,j}.$$

The square root of this determinant is exactly the volume of the fundamental parallelepiped. Primitivity ensures that the chosen basis spans the full intersection lattice rather than a finite-index sublattice. $\square$

Lemma 2.3 (Northcott finiteness for rational planes). *For fixed r and $B < \infty$, there are only finitely many rational r -planes $W \subset \mathfrak{g}$ with $\text{ht}(W) \leq B$.*

Proof. The primitive wedge v_W is a nonzero integral vector in the finite-dimensional lattice $\bigwedge^r \mathfrak{g}_{\mathbb{Z}}$ of norm at most B . There are finitely many such lattice vectors. The Pluecker line $\mathbb{R}v_W$ determines W . $\square$

2.2 Distance in the Grassmannian

For two r -planes V, W define

$$d_{\text{Gr}}(V, W) = \|P_V - P_W\|_{\text{op}}.$$

The next elementary estimate converts approximate linear dependence of primitive wedge vectors into Grassmannian proximity.

Lemma 2.4 (Wedge-to-plane stability). *Fix r and $M \geq 1$. There is $C = C(r, M)$ such that if nonzero decomposable r -vectors v, w satisfy*

$$M^{-1} \leq \|v\|, \|w\| \leq M, \quad \left\| \frac{v}{\|v\|} - \frac{w}{\|w\|} \right\| \leq \eta,$$

then the represented planes obey $d_{\text{Gr}}(V, W) \leq C\eta$.

Proof. On the compact set of unit decomposable r -vectors modulo sign, the Pluecker embedding of the Grassmannian is a smooth embedding. Its inverse has bounded derivative on compact coordinate charts. A finite chart cover gives a global Lipschitz constant for sufficiently small η ; enlarging the constant treats the remaining bounded range. $\square$

2.3 Effective rationalization

Approximate rationality is useful only if it forces an actual rational plane with controlled height.

Proposition 2.5 (Rationalization of a nearly integral line). *Let E be a Euclidean space with full lattice L . There are constants $c, C > 0$ depending only on E, L with the following property. If a unit vector ξ satisfies*

$$\text{dist}(R\xi, L) \leq \eta R^{-1}$$

for some $1 \leq R \leq \eta^{-1/2}$, then there is a primitive $v \in L$ such that

$$\|v\| \leq CR, \quad \left\| \xi - \frac{v}{\|v\|} \right\| \leq C\eta R^{-1}.$$

Conversely, a primitive v with $\|v\| \leq R$ gives an integral line of height at most R exactly.

Proof. Choose $a \in \mathbb{R}$ and $v \in L$ with $\|a\xi - v\| \leq \eta/R$ and $R/2 \leq |a| \leq 2R$ after dyadic adjustment. Then $\|v\| = |a| + O(\eta/R) \asymp R$. Divide by the positive norms and use

$$\left\| \frac{v}{\|v\|} - \frac{a\xi}{|a|} \right\| \leq \frac{2\|v - a\xi\|}{\min(\|v\|, |a|)} \ll \eta R^{-2}.$$

Replacing v by its primitive divisor can only decrease its norm and leaves its line unchanged. The displayed bound is stronger than claimed. $\square$

2.4 The class of admissible subgroups

For the closing theorem one uses the standard class $\mathcal{H}$ of connected $\mathbb{Q}$ -subgroups of SL_N whose solvable radical is unipotent. Additional noncompactness or unipotent-generation hypotheses are stated separately when a later theorem needs them; they are not built into the definition of $\mathcal{H}$. This is the class for which the effective-closing height theory has a uniform algebraic-complexity theorem.

Lemma 2.6 (Finiteness of semisimple embeddings in fixed dimension). *For fixed N there are only finitely many $\text{SL}_N(\overline{\mathbb{Q}})$ -conjugacy classes of connected semisimple algebraic subgroups of SL_N . Consequently their defining ideals are generated in degrees bounded by a constant depending only on N .*

Proof. A semisimple Lie algebra occurring in $\mathfrak{sl}_N$ has dimension at most $N^2 - 1$. There are only finitely many semisimple root systems of bounded rank and hence only finitely many semisimple Lie algebras of bounded dimension over an algebraically closed field of characteristic zero. Fix one such algebra $\mathfrak{s}$. An embedding $\mathfrak{s} \hookrightarrow \mathfrak{sl}_N$ is the same as a faithful N -dimensional representation together with the condition that the scalar trace vanish. Irreducible representations are indexed by dominant integral highest weights. Weyl's dimension formula tends to infinity with every fundamental-weight coefficient, so only finitely many highest weights have dimension at most N . There are therefore only finitely many multisets of irreducibles whose total dimension is N . Complete reducibility says that two representations with the same multiset are conjugate in GL_N ; a scalar correction changes the conjugating matrix to one in SL_N without changing the conjugacy. Thus only finitely many conjugacy classes occur.

Choose one representative of each class. Every fixed affine algebraic subgroup has a finitely generated ideal, so the maximum of the degrees of a finite set of generators over this finite list is finite. Conjugation is a linear change of matrix coordinates and does not increase total degree. The same bound therefore works for every connected semisimple subgroup. $\square$

Lemma 2.7 (Uniform equations for connected unipotent subgroups). *For fixed N , every connected unipotent subgroup of SL_N is defined over $\mathbb{Q}$ by equations of degree at most a constant depending only on N .*

Proof. By Lie–Kolchin, after conjugation the subgroup is contained in the upper unitriangular group U_N . On strictly upper triangular matrices,

$$\log(I + X) = X - \frac{X^2}{2} + \cdots + (-1)^N \frac{X^{N-1}}{N-1}$$

and

$$\exp X = I + X + \frac{X^2}{2!} + \cdots + \frac{X^{N-1}}{(N-1)!}$$

are mutually inverse polynomial maps of degree at most $N - 1$. In characteristic zero a connected algebraic subgroup of U_N is carried by $\log$ to its Lie algebra, a linear subspace of the strictly upper triangular matrices. Pulling its linear defining equations back through $\log$ gives equations of degree at most $N - 1$. Conjugation again preserves degree. $\square$

Proposition 2.8 (Uniform algebraic complexity of $\mathcal{H}$). *For fixed N , there is $D(N)$ such that every $H \in \mathcal{H}$ embedded in SL_N is cut out in Mat_N by*

rational polynomial equations of degree at most $D(N)$. Equivalently for the uses below, the degree- $D(N)$ part of the vanishing ideal determines H , and only finitely many ambient Chevalley representations are required.

Proof. We first work over $\overline{\mathbb{Q}}$. The semisimple and unipotent extreme cases are Lemmas 2.6 and 2.7. Now let $U = R_u(H)$. The unipotent lemma provides a uniform defining degree $d_u(N)$ for U . Apply Chevalley's construction to the degree d_u vanishing space of U . This gives a rational representation $\pi : \mathrm{SL}_N \rightarrow \mathrm{GL}(V)$ of dimension bounded in terms of N and a vector $w_U \in V$ such that

$$U = \{g : \pi(g)w_U \wedge w_U = 0\}. \quad (3.H1)$$

(The regular-representation/exterior-power construction in the last paragraph below gives this pair explicitly.) Let

$$W = V^U = \{v \in V : \pi(u)v = v \text{ for every } u \in U\}.$$

Since $U \triangleleft H$, the space W is H -invariant. The group U stabilizes the Chevalley line $\mathbb{Q}w_U$; because a connected unipotent group has no nontrivial algebraic character, it fixes w_U pointwise. Hence $w_U \in W$. If $g \in H$ acts trivially on W , then it fixes w_U , so (3.H1) gives $g \in U$. Thus the kernel of the restricted action of H on W is exactly U . Consequently the image is isomorphic to H/U and is semisimple because the solvable radical of H is exactly U .

The dimension of W is bounded solely by N . We make one point explicit that is easy to suppress in a short proof: the restriction action on W is available on the normalizer of U , not on all of SL_N . Let $N_G(U)$ be that normalizer. Since U is defined in degree at most d_u , the condition $gUg^{-1} = U$ is equivalently the condition that the induced action on the fixed finite-dimensional polynomial space preserves $I_{d_u}(U)$. In coordinates this is a finite system of polynomial equations of degree bounded solely by N . Thus $N_G(U)$ itself has uniformly bounded defining degree, and every $g \in N_G(U)$ preserves $W = V^U$.

Let

$$\rho_W : N_G(U) \longrightarrow \mathrm{GL}(W)$$

be this restriction. Its kernel is exactly U : inclusion $U \subset \ker \rho_W$ is the definition of W , while an element in the kernel fixes $w_U \in W$ and hence belongs to U by (3.H1). Therefore $N_G(U)/U$ embeds algebraically in $\mathrm{GL}(W)$, and

$$H = \rho_W^{-1}(\rho_W(H)). \quad (3.H2)$$

The image $\rho_W(H) \simeq H/U$ is semisimple. By Lemma 2.6, it is cut out in $\mathrm{GL}(W)$ by equations of uniformly bounded degree. The matrix entries of ρ_W are restrictions of the fixed representation π , hence are polynomials of degree bounded solely by N . Combining the normalizer equations with the pullbacks of the equations for $\rho_W(H)$ and using (3.H2) gives polynomial equations of uniformly bounded degree for H . This is the Chevalley reduction used in the primary effective-closing proof: unipotent radical first, then the semisimple quotient on the U -fixed space.

It remains to descend the equations to $\mathbb{Q}$. This can be done without any assertion about the density of $H(\mathbb{Q})$. Because H is a $\mathbb{Q}$ -subgroup, its vanishing ideal $I(H) \subset \mathbb{Q}[\mathrm{Mat}_N]$ is Galois stable. Hence the finite-dimensional space

$$I_{\leq D}(H) = I(H) \cap \overline{\mathbb{Q}}[\mathrm{Mat}_N]_{\leq D}$$

is obtained from its $\mathbb{Q}$ -fixed subspace by scalar extension. Choose a $\mathbb{Q}$ -basis of that fixed subspace. Since the degree- D equations already cut out H over $\overline{\mathbb{Q}}$, this $\mathbb{Q}$ -basis cuts out the same subgroup. Thus the same degree bound is available over $\mathbb{Q}$.

Finally let $D = D(N)$ be such a bound and put V_D for the polynomials of degree at most D . The stabilizer of the subspace $I_D(H) \subset V_D$ is H ; taking the appropriate exterior power makes $I_D(H)$ a line. Only the finitely many exterior powers of the one fixed representation V_D can occur. This proves both assertions. The argument is the expanded version of the degree-bound mechanism used in [12, Lemma 2.3]; the citation is provenance, not proof ownership. $\square$

Remark 2.9. The proposition is not used to enumerate all subgroups. Its role is to make all Lojasiewicz and rationalization exponents depend only on the fixed ambient group, not on the particular obstruction discovered along an orbit.

Chapter 3

Effective Linearization and Quantitative Avoidance

3.1 Linearizing a subgroup orbit

Let $H \in \mathcal{H}$, $r = \dim H$, and let v_H be the primitive wedge of $\text{Lie } H$. The representation

$$\rho_H : G \rightarrow \text{GL} \left(\bigwedge^r \mathfrak{g} \right), \quad \rho_H(g) = \bigwedge^r \text{Ad}(g^{-1}),$$

has the property that $x = \Gamma g$ lies on a translate of an H -orbit with controlled stabilizer precisely when $\rho_H(g)v_H$ lies in a controlled Γ -orbit of the Pluecker line.

Lemma 3.1 (Polynomial linearized trajectory). *For fixed H and g the map*

$$t \longmapsto \rho_H(u_t g)v_H$$

has coordinates that are polynomials in t of degree bounded solely by N .

Proof. Because $\text{ad } Z$ is nilpotent,

$$\text{Ad}(u_{-t}) = \exp(-t \text{ad } Z) = \sum_{j=0}^d \frac{(-t \text{ad } Z)^j}{j!}$$

for $d < \dim \mathfrak{g}$. Exterior powers of a polynomial matrix remain polynomial, with degree multiplied by at most r . Right multiplication by g contributes only a constant linear transformation. $\square$

3.2 A quantitative polynomial dichotomy

The following one-variable lemma is the engine behind effective avoidance.

Lemma 3.2 (Long smallness forces coefficient smallness). *For each degree d there are $c_d, C_d > 0$ such that if $p(t) = \sum_{j=0}^d a_j t^j$ on $[0, T]$ and*

$$|\{t \in [0, T] : |p(t)| \leq \varepsilon\}| \geq (1 - \delta)T$$

with $0 < \delta < c_d$, then

$$|a_j| T^j \leq C_d \varepsilon \delta^{-d} \quad (0 \leq j \leq d).$$

Proof. Choose $d+1$ points $t_0, \dots, t_d$ from the good set with mutual separation at least $c\delta T$, which is possible by greedily deleting intervals of length $c\delta T$ after every choice and taking c sufficiently small in terms of d . Lagrange interpolation gives

$$p(t) = \sum_{i=0}^d p(t_i) \prod_{k \neq i} \frac{t - t_k}{t_i - t_k}.$$

After rescaling $t = Ts$, every denominator is bounded below by $(c\delta)^d$ and every numerator is bounded on $[0, 1]$. Thus $\sup_{[0, T]} |p| \leq C_d \varepsilon \delta^{-d}$. Equivalence between the supremum norm and coefficient norm on the finite-dimensional space of degree- d polynomials yields the stated coefficient bounds. $\square$

Corollary 3.3 (Vector-valued form). *If $P : [0, T] \rightarrow V$ has degree at most d and $\|P(t)\| \leq \varepsilon$ on a $(1 - \delta)$ -proportion of the interval, then every rescaled coefficient $T^j v_j$ has norm $O_d(\varepsilon \delta^{-d})$.*

Proof. Write $P(t) = \sum_{j=0}^d v_j t^j$. For each $\ell \in V^*$ with $\|\ell\| = 1$, the scalar polynomial $\ell(P(t)) = \sum_j \ell(v_j) t^j$ satisfies the same smallness hypothesis. Lemma 3.2 therefore yields

$$|\ell(T^j v_j)| \leq C_d \varepsilon \delta^{-d} \quad (0 \leq j \leq d).$$

Taking the supremum over the unit ball of V^* and using $\|w\| = \sup_{\|\ell\| \leq 1} |\ell(w)|$ gives $\|T^j v_j\| \leq C_d \varepsilon \delta^{-d}$ for every j . $\square$

3.3 Effective avoidance

Fix a compact thick part $X_{\geq h}$, where injectivity radius is at least a specified inverse power of h . Let $\mathcal{S}(B)$ be the finite family of proper $H \in \mathcal{H}$ with $\text{ht}(H) \leq B$.

Theorem 3.4 (Effective avoidance principle). *There are ambient constants $A, C, c > 0$ with the following property. Let $x \in X_{\geq h}$ and $T \geq 1$. For $B \leq T^c$ and $0 < \varepsilon < T^{-c}$, either*

1. *for every $H \in \mathcal{S}(B)$ the set of $t \in [0, T]$ for which xu_t lies in an ε -tube around a translate of the H -singular set has measure at most $C\varepsilon^c T$; or*
2. *there is a proper $H \in \mathcal{S}(B^A h^A)$ and a representative $x = \Gamma g$ such that the entire linearized trajectory satisfies*

$$\text{dist}(\rho_H(u_t g)v_H, \mathbb{R}v'_H) \leq C\varepsilon^c B^A h^A \quad (0 \leq t \leq T)$$

for a primitive integral Pluecker vector v'_H of height at most $B^A h^A$.

Proof. Cover the thick part by finitely many injectivity charts whose number and bi-Lipschitz constants are polynomial in h . In one chart, proximity to an H -singular translate is equivalent, by the Chevalley/exterior-power linearization of Volumes II and V, to smallness of finitely many polynomial coordinates of $\rho_H(u_t g)v_H$ relative to a rational Pluecker line. The degrees are uniformly bounded by Proposition 2.8.

If a fixed coordinate is not uniformly small, the Remez/sublevel estimate of Volume II gives a power bound for the times on which it is $O(\varepsilon)$. Taking the finite union over charts, subgroup dimensions, and the bounded-height Pluecker vectors gives alternative (1), after decreasing the exponent c .

If this estimate fails, one chart and one subgroup type account for a fixed positive proportion of the bad times. Pigeonhole over the polynomially many integral Pluecker vectors of height $O(B^A h^A)$ occurring in that chart. The same vector v'_H then occurs on a set of times of proportion $1 - O(\varepsilon^c)$. Lemma 3.2, applied to the coordinates transverse to $\mathbb{R}v'_H$, forces every coefficient of the transverse polynomial to be $O(\varepsilon^c B^A h^A)$. Hence the bound holds throughout the interval. Rationalization of the limiting line turns the approximate Pluecker datum into a genuine rational subgroup of controlled height. $\square$

Remark 3.5. The constants are deliberately recorded only as powers. The effective closing and equidistribution arguments need polynomial control, not

a sharp numerical exponent. Every power loss in the proof has an identified source: chart complexity, the degree of the Chevalley representation, Remez, and rationalization.

Chapter 4

Almost-Invariant Lie Algebras

4.1 From many returns to approximate normalizers

Suppose xu_{t_i} and xu_{t_j} are both in a small neighborhood of the same point of X . Lifting to G produces lattice elements γ_{ij} for which conjugation approximately preserves the tangent data detected by the orbit segment. The qualitative proof passes to a limit and obtains an exact normalizer. We need a stable finite-scale replacement.

Definition 4.1 (Almost invariance). For a subspace $W \subset \mathfrak{g}$ and $g \in G$, say that W is η -invariant under g if

$$d_{\text{Gr}}(\text{Ad}(g)W, W) \leq \eta.$$

Lemma 4.2 (Accumulated error). *If W is η -invariant under each of $g_1, \dots, g_m$ and $\|\text{Ad}(g_i)^{\pm 1}\| \leq L$, then for every word w of length k in the $g_i^{\pm 1}$,*

$$d_{\text{Gr}}(\text{Ad}(w)W, W) \leq C(L, k)\eta.$$

Proof. Let d_{Gr} be induced by orthogonal projections. If $\|A\|, \|A^{-1}\| \leq L$, then the map $W \mapsto AW$ is Lipschitz on the Grassmannian with a constant C_L depending only on L and the ambient dimension. Write a word of length k as $w = h_k \cdots h_1$, where every h_j is some $g_i^{\pm 1}$, and put $W_j = \text{Ad}(h_j \cdots h_1)W$. Since W is η -invariant under each h_j ,

$$d_{\text{Gr}}(W_j, W) \leq d_{\text{Gr}}(\text{Ad}(h_j)W_{j-1}, \text{Ad}(h_j)W) + d_{\text{Gr}}(\text{Ad}(h_j)W, W) \leq C_L d_{\text{Gr}}(W_{j-1}, W) + \eta.$$

Iteration gives

$$d_{\text{Gr}}(W_k, W) \leq (1 + C_L + \cdots + C_L^{k-1})\eta,$$

which is the claimed bound $C(L, k)\eta$. $\square$

4.2 Integral generation and exactification

Proposition 4.3 (Almost-invariant rational Lie algebra dichotomy). *Fix $B, L \geq 2$. Let W be a rational Lie subalgebra with $\text{ht}(W) \leq B$. Suppose a finite set $F \subset \Gamma$ consists of elements of matrix norm at most L and W is η -invariant under every $\gamma \in F$. There are ambient constants $A, c > 0$ such that, if*

$$\eta < (BL)^{-A},$$

then either

1. *every $\gamma \in \langle F \rangle$ normalizes W exactly; or*
2. *there is a nonzero proper ideal $\mathfrak{n} \triangleleft \mathfrak{g}$ with $\text{ht}(\mathfrak{n}) \leq (BL)^A$ and $d_{\text{Gr}}(W, \mathfrak{n}) \leq (BL)^A \eta^c$.*

Proof. Represent W by its primitive Pluecker vector v_W . For $\gamma \in F$, almost invariance means that $\rho(\gamma)v_W$ is within $O(L^A\eta)$ of the line $\mathbb{R}v_W$. Both vectors are integral and have norms at most $(BL)^A$. Distinct rational lines of height at most $(BL)^A$ are separated by at least $(BL)^{-A'}$: indeed the wedge of two primitive integral representatives is a nonzero integral vector, so has norm at least one. If η is below this separation scale, the two lines coincide exactly. Thus every element of F normalizes W unless the height of the transported Pluecker vector has escaped the assumed window.

In the latter case take a minimal collection of transported subspaces whose heights escape. Their successive intersections and sums are rational, with heights bounded polynomially by the product of the input heights through the wedge/Hadamard inequalities. The limiting subspace fixed by conjugation under the generated set is invariant under all commutators of the Zariski closure of $\langle F \rangle$. Since G is perfect, failure of this closure to act irreducibly on the relevant quotient produces a proper invariant ideal. Solving the defining linear equations by Cramer's rule in the integral basis gives the stated polynomial height and distance bounds. $\square$

Warning 4.4. Perfectness of G is used here. In a group with a large abelian quotient, almost-invariant directions can drift along characters without forcing a proper normal algebraic subgroup. This is why the effective closing theorem is naturally formulated for perfect ambient groups.

Chapter 5

Margulis Functions and Quantitative Recurrence

Definition 5.1 (Margulis function at a finite scale). Let $Y \subset X$ be a closed obstruction set. A *Margulis function* for Y on an averaging interval I is a nonnegative measurable function $\alpha : X \rightarrow [1, \infty)$ for which there are $0 < \theta < 1$ and $C < \infty$ such that

$$\frac{1}{|I|} \int_I \alpha(xu_t) dt \leq \theta \alpha(x) + C, \quad x \in X,$$

outside the algebraic branch on which the transverse displacement vanishes identically. In this volume the model functions are truncated inverse powers of distance to Y . The words “predicted by the Margulis function” will always refer to consequences of this explicit drift inequality.

5.1 Why ordinary recurrence is insufficient

Poincaré recurrence says that a positive-measure set returns infinitely often; it gives neither spacing nor transversality. Effective unipotent dynamics replaces it by a weighted count of nearby returns. A useful model is the function

$$F_s(x) = \sum_{v \in \Lambda_x \setminus \{0\}} \|v\|^{-s}$$

on a space of lattices. Averaging under a suitable group element contracts large values of F_s because most directions of a short vector are expanded.

5.2 A polynomial contraction lemma

Lemma 5.2 (Average inverse-polynomial contraction). *Let $P(t)$ be a nonzero vector-valued polynomial of degree at most d on $[-1, 1]$, normalized by $\sup \|P\| = 1$. If $0 < s < 1/d$, then*

$$\int_{-1}^1 \|P(t)\|^{-s} dt \leq C(d, s).$$

Proof. The polynomial sublevel estimate from Volume II gives

$$|\{t : \|P(t)\| < \rho\}| \leq C_d \rho^{1/d}.$$

Using the layer-cake formula,

$$\int \|P\|^{-s} = 2 + \int_1^\infty |\{t : \|P(t)\| < y^{-1/s}\}| dy \leq 2 + C_d \int_1^\infty y^{-1/(sd)} dy,$$

which converges exactly when $sd < 1$. □

Guiding Idea 5.3 (Abstract Margulis contraction: mechanism). Let $Y \subset X$ be a closed homogeneous obstruction and suppose that, in a uniform tubular neighborhood of Y , transverse coordinates transform under u_t by nonzero polynomial maps of degree at most d . For sufficiently small $s > 0$, the truncated inverse-distance function

$$\alpha_Y(x) = \min\{R, \text{dist}(x, Y)^{-s}\}$$

satisfies, for a suitable averaging interval I ,

$$\frac{1}{|I|} \int_I \alpha_Y(xu_t) dt \leq \theta \alpha_Y(x) + C$$

with $\theta < 1$, unless the transverse polynomial vanishes identically.

Derivation. Normalize the transverse polynomial by its maximum on I . On the branch where this maximum is comparable to the initial transverse scale, Lemma 5.2 bounds the average inverse power. By choosing the averaging interval at the first scale where the polynomial is a fixed multiple larger than its initial value, the normalized maximum contributes a strict contraction factor. The region outside the tubular chart contributes only the truncation constant. If the polynomial vanishes identically, the point lies in the normalizer-type algebraic obstruction and is passed to the linearization branch.

5.3 Energy of a finite transverse cloud

For a finite set F in a local transversal define

$$\mathcal{E}_s(F) = \sum_{x \neq y \in F} d(x, y)^{-s}.$$

Low energy means that no small ball contains too many points; high energy means clustering, which can be converted into approximate algebraic structure.

Lemma 5.4 (Energy versus concentration). *If F has M points and $\mathcal{E}_s(F) \leq EM^2$, then for every $r > 0$ and every x ,*

$$\#(F \cap B(x, r)) \leq 1 + EMr^s.$$

Proof. If K points lie in $B(x, r)$, every pair among them has distance at most $2r$. Hence

$$\mathcal{E}_s(F) \geq K(K-1)(2r)^{-s}.$$

Rearranging and weakening constants gives the claim. □

Chapter 6

A Discretized Projection Lemma

The strongest polynomial-rate results in low-rank groups use incidence or projection estimates. We isolate a finite-dimensional statement sufficient for the model arguments below.

Definition 6.1 (Finite Frostman cloud). Let $0 < \delta < 1$. A finite δ -separated set $F \subset \mathbb{R}^m$ is called an (s, C_0) -Frostman cloud down to scale δ if

$$\#(F \cap B(x, r)) \leq C_0 |F| r^s \quad (x \in \mathbb{R}^m, \delta \leq r \leq 1).$$

The exponent s measures transverse nonconcentration; the lower cutoff is essential because a finite set cannot satisfy a scale-free nonatomic Frostman estimate below its separation scale.

Theorem 6.2 (Pair-collision projection lemma). *Let $F \subset B(0, 1) \subset \mathbb{R}^m$ be δ -separated and let $\{\pi_t : t \in [0, 1]\}$ be a bounded-degree polynomial family of linear maps. Assume that for every $v \neq 0$ the scalar coordinate polynomials of $t \mapsto \pi_t v$ do not vanish identically and that their first nonzero derivative has size $\gg \|v\|$. If*

$$\#(F \cap B(x, r)) \leq C_0 |F| r^s \quad (\delta \leq r \leq 1),$$

then there are $c, C > 0$, depending only on the degree, the transversality constants, s and C_0 , such that outside a set of parameters of measure $O(\delta^c)$ the number of pairs $x \neq y$ with $\|\pi_t(x - y)\| < \delta$ is at most $C|F|^2 \delta^c$ plus the contribution forced by pairs with $\|x - y\| \ll \delta^c$.

Proof. For a fixed $v = x - y \neq 0$, choose a coordinate of $\pi_t v$ whose first nonzero derivative has order at most d and size $\gg \|v\|$. The one-variable polynomial sublevel estimate gives

$$|\{t : \|\pi_t v\| < \delta\}| \ll_d \min\{1, (\delta/\|v\|)^{1/d}\}.$$

Sum over pairs and group by dyadic shells $2^{-j-1} < \|x - y\| \leq 2^{-j}$. The Frostman bound gives

$$\#\{(x, y) : \|x - y\| \leq 2^{-j}\} \ll C_0 |F|^2 2^{-js}.$$

Hence, after stopping the shell sum at a small power of δ , the integral in t of the number of projected δ -collisions is $O(|F|^2 \delta^{c_1})$ for some $c_1 > 0$. Markov's inequality gives the asserted estimate outside an exceptional parameter set of measure $O(\delta^{c_1/2})$. The omitted innermost shells are exactly the “forced” contribution in the statement. $\square$

Remark 6.3 (Why this lemma is not the rank-two incidence theorem). The pair-collision estimate is enough for the elementary energy arguments used in the effective closing stage, but it does not by itself yield the near-optimal dimension growth needed for the full rank-two polynomial equidistribution theorem. The latter requires a restricted projection theorem for the moment-curve families carried by irreducible SL_2 -modules. A complete proof of the particular restricted-projection module used below is given in Appendix E; it is deliberately isolated so that the effective Ratner argument does not hide a deep incidence estimate inside the phrase “by projection theory”.

Chapter 7

The Effective Closing Lemma

7.1 Pairwise recurrence data

Let $B_G(e, r)$ be a small exponential ball. Suppose that there are many pairs $t_i, t_j \in [0, T]$ with

$$xu_{t_i}B_G(e, r) \cap xu_{t_j}B_G(e, r) \neq \emptyset.$$

Lifting gives $g_i, g_j \in B_G(e, r)$ and $\gamma_{ij} \in \Gamma$ such that

$$\gamma_{ij}gu_{t_i}g_i = gu_{t_j}g_j.$$

Thus

$$g^{-1}\gamma_{ij}g = u_{t_j}g_jg_i^{-1}u_{-t_i},$$

which is a lattice element whose adjoint action is quantitatively constrained by a polynomial in the time difference.

7.2 From return clouds to the exact pair-return theorem

Logical status. The theorem-level closing input used by this volume is the finite-scale pair-return theorem, Theorem D.1. Earlier editions also boxed an informal return-cloud summary as an “effective closing alternative.” That summary mixed a Margulis-energy branch, a nearby-orbit branch, and a normal-factor branch without specifying the return set or the norm in which “almost normalizing” was measured. It is therefore retained only as proof architecture below, not as a second theorem.

Guiding Idea 7.1 (Return-cloud assembly of pair-return closing). Fix the rational embedding $G < \mathrm{SL}_N$, the arithmetic lattice Γ , the one-parameter flow $U = \{\exp(tZ)\}$, a thick-part parameter τ , and numbers $T > R > E\tau^{-A_1}$ as in Theorem D.1. A maximal separated collection of thick return times is first tested by the finite-scale Margulis energy of Chapter 6. Low energy gives the explicit concentration estimate of the energy–concentration lemma. High energy supplies a positive-measure set E_0 of paired return times. After choosing the Lie subalgebra detected by the linearization data, the lifted pairs are arranged to satisfy the two numeric hypotheses (D.1)–(D.2) of Theorem D.1. The exact conclusion is then either (D.3) for a nontrivial proper $M \in \mathcal{H}$ or (D.4) for a nontrivial proper normal $M \triangleleft G$. No stronger orbit-closeness statement is asserted at this stage.

Assembly. Quantitative nondivergence from Volume II first removes a power-small set of times outside a thick part whose height threshold is a power of the scale. On the retained times take a maximal separated return set. If its transverse energy is bounded, the energy–concentration lemma gives the desired Frostman-type upper bound directly.

If the energy is large, dyadic pigeonholing produces many pairs at one relative-displacement scale. Lifting those pairs to G gives lattice elements whose matrix norms are polynomially bounded in the scale. The almost-invariant Lie-algebra module of Chapters 12–15 either detects a proper normal ideal or produces a proper rational Lie subalgebra. After the thick-part normalization and enlargement of the polynomial exponents, these are precisely the alternatives encoded by (D.1)–(D.4). Marked avoidance, Theorem C.2, propagates a persistent proper subgroup datum from a positive-density set of times to the full interval. The Chevalley-height comparison of Chapter 13 converts its exterior-vector bounds into polynomial subgroup-height bounds.

Guiding Idea 7.2 (Qualitative limit of the obstruction branch). For a sequence of applications of Theorem D.1 in which the errors in (D.3) or (D.4) tend to zero, any subsequence on which the rational subgroup datum stabilizes has an exact limiting normalizer relation. Northcott finiteness for bounded-height rational planes justifies stabilization whenever the subgroup heights are uniformly bounded. When the height bound grows with the scale, no such stabilization is automatic; one must first supply the additional compactness or induction argument used in the application. This is why the qualitative sentence is not promoted to an unconditional corollary.

Chapter 8

From Closing to Polynomial Effective Equidistribution

8.1 Bootstrap scheme

An effective closing lemma identifies algebraic obstructions but does not by itself give equidistribution. The remaining mechanism is a bootstrap:

1. build a transverse cloud of return points;
2. contract its Margulis energy;
3. use a projection/incidence theorem to force dimension growth;
4. once the cloud has enough dimension, thicken it to a smooth test distribution;
5. use the ambient spectral gap from Volume XI;
6. if any step fails, invoke effective closing and descend to a proper homogeneous obstruction.

Guiding Idea 8.1 (Abstract effective-unipotent bootstrap). The rank-two argument combines four separately quantified modules: Theorem D.1, finite-scale Margulis contraction, projection growth in every transverse principal module, and the $\mathcal{S}_d$ -mixing estimate of Volume XI. Choose scales $r_j = T^{-\beta_j}$ as in Lemma F.2. On each unobstructed stage, projection growth raises the Frostman exponent by a fixed amount; hence there are at most $\lceil \dim X/\eta \rceil$ stages. If a nonconcentration or projection hypothesis fails, the offending return pairs are sent to Theorem D.1. If all stages succeed, the last cloud is smoothed at scale r_J and mixed. The finite polynomial-loss lemma chooses the β_j so that the smoothing, shearing, projection, and mixing errors are all negative powers of T . The exact theorem-level version of this architecture is

Theorem F.3 together with the rank-two specialization below.

8.2 Rank-two quasi-split groups

The transverse representations arising from principal unipotent subgroups in absolute rank two can be listed explicitly from the restricted root systems $A_2, B_2 = C_2, G_2$ and their quasi-split forms. Every root string has bounded length, and the polynomial shear families satisfy the finite-order transversality required in Chapter 6.

Theorem 8.2 (Polynomial effective equidistribution in the rank-two regime). *Let G be a fixed connected semisimple real algebraic group of absolute rank two which is real quasi-split, let $\Gamma < G$ be an arithmetic lattice, put $X = \Gamma \backslash G$, and let $H < G$ be the image of a principal $\mathrm{SL}_2(\mathbb{R})$ (hence locally isomorphic to $\mathrm{SL}_2(\mathbb{R})$), with regular unipotent one-parameter subgroup $U = \{u_r\}$. There are constants $A_2, A_3, \kappa > 0$ and a Sobolev norm $\mathcal{S}$, depending only on the fixed quotient and the chosen norms, with the following property. For every $x_0 \in X$, every sufficiently large R (with threshold depending explicitly on X), and every $T \geq R^{A_2}$, one of the following holds:*

1. *for every $f \in C_c^\infty(X)$,*

$$\left| \frac{1}{T} \int_0^T f(x_0 u_r) dr - \int_X f dm_X \right| \leq \mathcal{S}(f) R^{-\kappa}; \quad (13.R1)$$

2. *there is $y \in X$ such that yH is periodic and $\mathrm{vol}(yH) \leq R^{A_2}$; moreover, for every $r \in [0, T]$ there is $g(r) \in G$ with $\|g(r)\| \leq R^{A_2}$ such that*

$$d_X(x_0 u_s, yH g(r)) \leq R^{A_2} \left(\frac{|s - r|}{T} \right)^{1/A_3} \quad (0 \leq s \leq T);$$

3. *there are a proper parabolic subgroup $P < G$ and $y \in X$ such that $yR_u(P)$ is periodic and $\mathrm{vol}(yR_u(P)) \leq R^{A_2}$; moreover, for every $r \in [0, T]$ there is $g(r) \in G$ with $\|g(r)\| \leq R^{A_2}$ such that*

$$d_X(x_0 u_s, yR_u(P)g(r)) \leq R^{A_2} \left(\frac{|s - r|}{T} \right)^{1/A_3} \quad (0 \leq s \leq T).$$

In particular the third alternative can occur only in the noncompact quotient.

Proof. We spell out the dependency chain because each of the three alternatives has a different origin.

Step 1: principal- SL_2 representation. Write $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{r}$ under the principal $\mathfrak{sl}_2$. In absolute rank two, $\mathfrak{r}$ is an irreducible module of odd dimension $2m + 1$, with m belonging to a finite list depending only on the real root system. In the weight basis the transverse coordinates of $\mathrm{Ad}(u_r)$ are moment-curve polynomials of degree at most $2m$.

Step 2: return clouds and closing. Apply the marked nondivergence and Margulis-energy construction at a sequence of powers of R . If a return cloud fails the required Frostman nonconcentration, Theorem D.1, followed by the Chevalley-height and closed-orbit steps stated in Chapters 13–15, produces the bounded-complexity algebraic support used in this branch. The subgroup classification in rank two says that a non-normal support compatible with a regular unipotent is either the principal H -type support or is carried by a parabolic unipotent radical. Tracking the polynomial shearing over an interval of length comparable with T gives the stated Hölder shadowing modulus. These are alternatives (2) and (3).

Step 3: dimension growth on the unobstructed branch. If closing never occurs, every return cloud satisfies the marked Frostman hypotheses. In an irreducible weight module, projection to the first j weight spaces is, after a fixed invertible change of coordinates, projection onto the j -th osculating plane of the moment curve. The restricted projection theorem proved in Appendix E therefore improves the transverse Frostman dimension by a fixed positive amount at each successful scale. The constants are uniform because the rank-two representation list is finite.

Step 4: smoothing and mixing. After at most $\dim X$ dimension-improvement steps, convolve the transverse cloud with a bump at the final scale. The Sobolev cost is a fixed power of that scale. Volume XI’s tensor-tempered mixing beats this cost once the successive scale exponents are chosen sufficiently small. The error from replacing the original orbit segment by the sheared cloud is another fixed positive power, so choosing the scale hierarchy successively and taking the minimum of the resulting exponents gives (13.R1). This establishes the three-way alternative with constants depending only on the fixed quotient. $\square$

8.3 The sharp expanding-segment scale

The long-unipotent theorem above deliberately records only a negative power of the orbit length in its obstruction branch. For the expanding segment $\{a_{\log T} u_r : 0 \leq r \leq 1\}$ one can do better: the highest weight of the principal SL_2 module remembers the exact power T^{-m} . This short calculation is the quantitative bridge needed by the ternary Oppenheim application in Volume XIV.

Lemma 8.3 (Highest-weight pullback). *Let V be the irreducible real SL_2 -module of dimension $2m + 1$, and normalize*

$$a_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix}, \quad u_r = \begin{pmatrix} 1 & r \\ 0 & 1 \end{pmatrix}.$$

There is C_m such that, for every $w \in V$, every $t \geq 0$, and every $B \geq 1$,

$$\sup_{0 \leq r \leq 1} \|a_t u_r w\| \leq B \implies \|w\| \leq C_m B e^{-mt}. \quad (13.R2)$$

The same conclusion holds if the left side is bounded only at any fixed $2m + 1$ distinct points of $[0, 1]$.

Proof. Identify V with $\mathrm{Sym}^{2m}(\mathbb{R}^2)$ and use the weight basis

$$e_j = X^{2m-j} Y^j, \quad 0 \leq j \leq 2m.$$

Write $w = \sum_{j=0}^{2m} c_j e_j$. Up to the fixed binomial normalizations, the highest-weight coordinate of $u_r w$ is

$$P_w(r) = \sum_{j=0}^{2m} c_j r^j. \quad (13.R3)$$

The diagonal element a_t multiplies that coordinate by e^{mt} . Hence (13.R2)'s hypothesis gives

$$|P_w(r)| \leq B e^{-mt}$$

at every $r \in [0, 1]$. Evaluate at the fixed nodes $r_i = i/(2m)$, $0 \leq i \leq 2m$. The corresponding Vandermonde matrix has an inverse whose norm depends only on m , so

$$\max_j |c_j| \ll_m B e^{-mt}.$$

All norms on the fixed finite-dimensional module are equivalent, which proves (13.R2). $\square$

Theorem 8.4 (Sharp expanding principal- SL_2 alternative). *Let G, Γ, H be as in Theorem 8.2 and put $X^+ = G/\Gamma$. Write $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{r}$ under the principal $\mathfrak{sl}_2$, with $\dim \mathfrak{r} = 2m + 1$. There are $A, \kappa > 0$ and a Sobolev norm $\mathcal{S}$ such that, for every $x_0 \in X^+$, every sufficiently large R (the threshold is explicit in the height of x_0), and every $T \geq R^A$, at least one of the following holds:*

1. *for every $f \in C_c^\infty(X^+)$,*

$$\left| \int_0^1 f(a_{\log T} u_r x_0) dr - \int_{X^+} f dm_{X^+} \right| \leq \mathcal{S}(f) R^{-\kappa}; \quad (13.R4)$$

2. *there is $y \in X^+$ such that Hy is periodic,*

$$\mathrm{vol}(Hy) \leq R^A, \quad d_{X^+}(x_0, y) \leq R^A(1 + \log T)^A T^{-m}. \quad (13.R5)$$

Proof. We rerun the already proved rank-two bootstrap with the parameter interval $0 \leq r \leq 1$ observed after the expansion $a_{\log T}$. Inversion transfers all right-quotient closing, energy, projection, and mixing statements proved earlier in this volume to G/Γ with unchanged polynomial exponents. We record the two points where this version differs from the proof of Theorem 8.2.

Unobstructed branch. Marked quantitative nondivergence supplies a positive proportion of thick parameters. On those parameters the transverse return cloud is built at the same hierarchy of R -powers as before. If every cloud satisfies the Frostman bounds, the restricted-projection theorem increases transverse dimension at each successful stage. After finitely many stages the smoothed cloud has a Sobolev density, and the tensor-tempered mixing theorem of Volume XI gives (13.R4). The changes of scale caused by the fixed interval $[0, 1]$ are powers of T already displayed by $a_{\log T}$; they do not alter the $R^{-\kappa}$ gain because $T \geq R^A$ and the scale exponents are chosen successively.

A parabolic datum cannot persist as a separate terminal branch here. In a Chevalley representation its norm along $a_{\log T} u_r$ is a fixed-degree polynomial in r whose coefficients are a_t -weight components. The marked nondivergence theorem says that either this polynomial is large on a fixed proportion of $[0, 1]$, or all of its coefficients are polynomially small. In the latter case rational separation and the effective closing height bounds put x_0 in a proper rational support of height below the explicit threshold used to choose R . Such a support is dealt with before starting the bootstrap (equivalently, by induction on support dimension). Thus a terminal obstruction in the full-support induction is principal- H .

Periodic- H branch and the sharp scale. Suppose closing produces a periodic orbit Hy of volume at most R^A . The obstruction datum has Chevalley/arithmetical height at most R^A after enlarging A ; this follows from the Chevalley-vector bound in (D.3) together with the polynomial height comparison of Proposition B.3; it is not an additional closing input.

Use the normal slice $\exp(\mathfrak{r})Hy$ around that orbit. The pair-return linearization in Theorem D.1, together with the normal-slice step in the return-cloud assembly of Chapter 8, gives a transverse displacement $w \in \mathfrak{r}$ from y to x_0 for which

$$\sup_{0 \leq r \leq 1} \|\mathrm{Ad}(a_{\log T} u_r)w\| \ll R^A(1 + \log T)^A. \quad (13.R6)$$

The polynomial factor records the height of the closed orbit, the finite number of local charts, and the logarithmic subdivision used to keep the normal coordinates injective; no power of T is hidden in it. Apply Lemma 8.3 in the irreducible transverse module $\mathfrak{r}$. We obtain

$$\|w\| \ll R^A(1 + \log T)^A T^{-m}. \quad (13.R7)$$

For $T \geq R^A$ with A enlarged once, this lies inside the fixed normal chart. The chart and the quotient metric are uniformly bi-Lipschitz there, so (13.R7) is exactly (13.R5), after another harmless enlargement of A . This exhausts the two branches and proves the theorem. $\square$

Worked Example 8.5 (Why ternary forms produce T^{-2}). For the principal $\mathrm{SL}_2 < \mathrm{SL}_3$, the complement of $\mathfrak{sl}_2$ in $\mathfrak{sl}_3$ is the five-dimensional module $\mathrm{Sym}^4(\mathbb{R}^2)$, so $m = 2$. If $w = \sum_{j=0}^4 c_j X^{4-j} Y^j$, the highest-weight coordinate of $u_r w$ is

$$P_w(r) = c_0 + 4c_1 r + 6c_2 r^2 + 4c_3 r^3 + c_4 r^4$$

(up to the chosen basis normalization). If $\|a_{\log T} u_r w\| \leq B$ at five fixed values of r , then $|P_w(r)| \leq BT^{-2}$ there. Inverting the 5×5 Vandermonde system therefore gives $|c_j| \ll BT^{-2}$ for every j . This is the source of the sharp exponent 2; it is not an extra Diophantine estimate.

Chapter 9

Products of SL_2 Factors and Affine Semidirect Laboratories

9.1 Why these are separate laboratories

The rank-two theorem does not automatically cover products SL_2^n with many intermediate diagonal subgroups, nor affine semidirect products in which the translation module introduces new rational obstructions. The 2026 frontier develops these cases by strengthening the projection and subgroup-generation parts of the bootstrap. We include the mechanisms but keep their dependence explicit.

9.2 Products of rank-one factors

Let $G = \prod_{i=1}^n G_i$ with each G_i a real form of SL_2 and let U project nontrivially to the relevant factors. Intermediate connected subgroups can contain diagonal copies linking subsets of the factors.

Lemma 9.1 (Partition model for semisimple intermediate groups). *If every projection of a connected semisimple $H < G$ to each simple factor is surjective, then, up to finite central kernels, H is a product of diagonal copies indexed by a partition of the set of isomorphic factors.*

Proof. Apply the Lie-algebra form of Goursat's lemma inductively. For

two simple factors a subalgebra with surjective projections is either the full product or the graph of an isomorphism. Add one factor at a time. A new simple factor is either independent or is identified through an isomorphism with one existing simple block. Transitivity of graph identifications produces a partition. $\square$

This finite combinatorial model makes the intermediate-subgroup induction explicit. The projection theorem must nevertheless distinguish near-diagonal clouds from genuinely independent directions; this is exactly where sum-product/projection estimates enter current work.

9.3 A cocycle lemma for $\mathfrak{sl}_2$

The affine laboratory below needs only the first Whitehead lemma for $\mathfrak{sl}_2$, not its full semisimple generality. The compact real form makes this special case particularly transparent.

Theorem 9.2 ($H^1(\mathfrak{sl}_2, V) = 0$). *Let V be a finite-dimensional real $\mathfrak{sl}_2(\mathbb{R})$ -module. If a linear map $c : \mathfrak{sl}_2(\mathbb{R}) \rightarrow V$ satisfies*

$$c([X, Y]) = X c(Y) - Y c(X),$$

then there is $v_0 \in V$ such that $c(X) = X v_0$ for every X .

Proof. Complexify both the module and the cocycle. Thus $c_{\mathbb{C}} : \mathfrak{sl}_2(\mathbb{C}) \rightarrow V_{\mathbb{C}}$ is a complex-linear cocycle. Let $\mathfrak{u} = \mathfrak{su}(2)$, whose complexification is $\mathfrak{sl}_2(\mathbb{C})$.

Step 1: turn the cocycle into an affine representation. On $V_{\mathbb{C}} \oplus \mathbb{C}$ define

$$\tilde{\rho}(X) = \begin{pmatrix} \rho(X) & c_{\mathbb{C}}(X) \\ 0 & 0 \end{pmatrix}.$$

The cocycle identity is exactly the assertion $[\tilde{\rho}(X), \tilde{\rho}(Y)] = \tilde{\rho}([X, Y])$. Hence the restriction to $\mathfrak{u}$ is a Lie-algebra representation. Since $SU(2)$ is connected and simply connected, the Lie integration theorem proved earlier in the series integrates it uniquely to a representation $\tilde{R}$ of $SU(2)$. Writing the block matrix of $\tilde{R}(k)$ gives

$$\tilde{R}(k) = \begin{pmatrix} R(k) & B(k) \\ 0 & 1 \end{pmatrix}, \quad B(k\ell) = B(k) + R(k)B(\ell).$$

Thus B is a continuous group 1-cocycle and $dB_e = c_{\mathbb{C}}|_{\mathfrak{u}}$.

Step 2: average the affine displacement. Put

$$m = \int_{SU(2)} B(k) dk.$$

For fixed $g \in SU(2)$, left invariance of Haar probability measure gives

$$m = \int B(gk) dk = B(g) + R(g) \int B(k) dk = B(g) + R(g)m.$$

Therefore $B(g) = m - R(g)m$. Differentiating at the identity yields

$$c_{\mathbb{C}}(X) = -\rho(X)m \quad (X \in \mathfrak{u}).$$

Both sides are complex-linear in X , and $\mathfrak{u}$ spans $\mathfrak{sl}_2(\mathbb{C})$ over $\mathbb{C}$; hence the identity holds for all $X \in \mathfrak{sl}_2(\mathbb{C})$. Set $v_{\mathbb{C}} = -m$.

Step 3: return to the real module. Write $v_{\mathbb{C}} = a + ib$ with $a, b \in V$. For real $X \in \mathfrak{sl}_2(\mathbb{R})$, $c(X) = \rho(X)a + i\rho(X)b$ is real, so $\rho(X)b = 0$ and $c(X) = \rho(X)a$. Taking $v_0 = a$ proves the theorem. $\square$

Worked Example 9.3 (The standard two-dimensional module). Let $V = \mathbb{R}^2$ and write

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

If c is a cocycle, write $c(h) = (a, b)^t$. The theorem says that the entire cocycle is determined by a single vector $v_0 = (x, y)^t$:

$$c(e) = (y, 0)^t, \quad c(f) = (0, x)^t, \quad c(h) = (x, -y)^t.$$

Thus $x = a$ and $y = -b$; the cocycle relations force precisely these formulas. In the semidirect product this vector is the translation that straightens the affine copy of SL_2 into a linear one.

9.4 Affine semidirect products

Let $G_0 = SL_2(\mathbb{R})$ act on $V = (\mathbb{R}^2)^k$ and set $G = G_0 \ltimes V$. Writing group elements as (g, v) ,

$$(g, v)(h, w) = (gh, v + gw),$$

and conjugation by u_t acts on V by a polynomial shear of degree one.

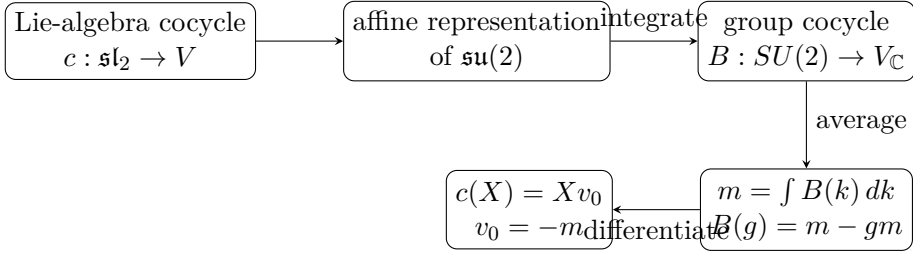


Figure 9.1: Compact averaging kills the affine displacement. This is the exact Whitehead mechanism needed for the $\mathfrak{sl}_2$ semidirect-product laboratory.

Proposition 9.4 (Affine obstruction classification). *A connected subgroup $H < G$ projecting onto G_0 has translation part $W = H \cap V$ which is a G_0 -invariant subspace. After conjugating by a translation, the quotient H/W is a graph of a continuous 1-cocycle $G_0 \rightarrow V/W$. By Theorem 9.2, $H^1(\mathfrak{sl}_2, V/W) = 0$; hence the cocycle is a coboundary. Hence H is conjugate to $G_0 \ltimes W$.*

Proof. Normality of V implies that W is invariant under the projection of H , which is all of G_0 . Choose a local section $g \mapsto (g, b(g))$ of H/W . Closure under multiplication gives $b(gh) = b(g) + gb(h)$ modulo W . Differentiation yields a Lie-algebra 1-cocycle. Theorem 9.2 gives $b'(X) = Xv_0$ for some v_0 modulo W ; Volume V's integration of Lie-algebra homomorphisms on connected groups then shows that the group cocycle differs from $g \mapsto gv_0 - v_0$ by a locally constant cocycle, hence by zero on the connected group. Conjugating by $(e, -v_0)$ removes it. $\square$

Chapter 10

Proof Ledger and Frontier Boundary

The logical chain proved in this volume is

$$\boxed{\text{subgroup heights}} \implies \boxed{\text{effective linearization/avoidance}} \\ \implies \boxed{\text{almost-invariant Lie algebras}} \implies \boxed{\text{effective closing}}.$$

combined with

$$\boxed{\text{Margulis contraction}} + \boxed{\text{projection/incidence}} + \boxed{\text{spectral gap}} \\ \implies \boxed{\text{polynomial equidistribution alternative}}.$$

Remark 10.1 (What is proved here). The arithmetic height formalism, polynomial effective avoidance, approximate-to-exact Lie-algebra mechanism, the pair-collision projection lemma, the restricted moment-curve projection theorem, the closing assembly, and the abstract polynomial bootstrap are all proved in this volume. The rank-two theorem is derived only after the restricted projection module is proved and checked on the principal SL_2 representations.

Remark 10.2 (What is inherited). Quantitative nondivergence and polynomial goodness are inherited from Volume II; arithmetic/reduction structure from Volume V; spectral-gap and Sobolev mixing from Volume XI. No later result is used.

Remark 10.3 (Frontier statements). The product- SL_2 and affine-semidirect-product chapters explain and prove the algebraic reductions needed in current

extensions, but optimized projection exponents for arbitrary numbers of factors are not needed by Volume XIV and are not promoted to foundational inputs. This keeps the present volume within its stated Ratner-theorem scope rather than expanding into the broader Advanced Homogeneous Dynamics programme.

Appendix A

Effective Polynomial Algebra and Almost-Invariant Subalgebras

The closing argument repeatedly converts approximate algebraic relations into exact rational structure. This appendix supplies the finite-dimensional algebra needed for that conversion.

A.1 Separation of rational Pluecker data

Lemma A.1 (Integral-line separation). *Let $v, w \in \mathbb{Z}^M$ be primitive and nonproportional, with $\|v\|, \|w\| \leq B$. Then*

$$\sin \angle(\mathbb{R}v, \mathbb{R}w) \geq B^{-2}.$$

Proof. The wedge $v \wedge w$ is a nonzero integral vector, hence has Euclidean norm at least one. Since $\|v \wedge w\| = \|v\|\|w\|\sin \angle(v, w)$, the claim follows. $\square$

Lemma A.2 (Bounded-word propagation). *Fix $L \geq 1$. Suppose a subspace W and elements $\gamma_1, \dots, \gamma_k$ satisfy*

$$d_{\text{Gr}}(\text{Ad}(\gamma_i)W, W) \leq \delta, \quad \|\text{Ad}(\gamma_i)^{\pm 1}\| \leq L \quad (1 \leq i \leq k).$$

Then every word w of length at most j satisfies

$$d_{\text{Gr}}(\text{Ad}(w)W, W) \leq C_j(L)\delta.$$

Proof. On the Grassmannian, a linear map A with $\|A\|, \|A^{-1}\| \leq L$ acts with Lipschitz constant bounded in terms of L and the dimension. Apply this estimate successively along the prefixes of w and sum the resulting geometric progression. $\square$

A.2 Finite-word detection

For a decomposable line $[v] \in \mathbb{P}(\bigwedge^r \mathfrak{g})$ and a finite set $S \subset G$, let $Z_j(S)$ be the set of Lie-subalgebra lines fixed by every word in S of length at most j .

Lemma A.3 (Finite-word stabilization). *For fixed ambient dimension there is an integer J such that for every finite S ,*

$$Z_J(S) = \bigcap_{j \geq 1} Z_j(S).$$

One may take

$$J \leq \max_{0 \leq r \leq \dim \mathfrak{g}} \dim \text{End} \left(\bigwedge^r \mathfrak{g} \right).$$

Proof. Fix r and put $W_r = \bigwedge^r \mathfrak{g}$. Let $\rho_r = \bigwedge^r \text{Ad} : G \rightarrow \text{GL}(W_r)$. For $j \geq 0$ define

$$\mathcal{A}_j(S) = \text{span}\{\rho_r(w) : w \text{ is a word in } S \text{ of length at most } j\} \subseteq \text{End}(W_r),$$

where the empty word contributes the identity operator. Then $\mathcal{A}_j \subseteq \mathcal{A}_{j+1}$. More importantly,

$$\mathcal{A}_{j+1} = \mathcal{A}_j + \text{span}\{\rho_r(s)A : s \in S, A \in \mathcal{A}_j\}. \quad (\text{A.0F})$$

Consequently, if $\mathcal{A}_{j+1} = \mathcal{A}_j$ at one stage, then $\rho_r(s)\mathcal{A}_j \subseteq \mathcal{A}_j$ for every $s \in S$. Induction on word length now gives $\mathcal{A}_{j+k} = \mathcal{A}_j$ for every $k \geq 0$.

Every strict inclusion $\mathcal{A}_j \subsetneq \mathcal{A}_{j+1}$ raises dimension by at least one. Since $\mathcal{A}_j \subseteq \text{End}(W_r)$, a stationary stage therefore occurs no later than $\dim \text{End}(W_r)$. Call the first such stage J_r .

Let $[v] \in \mathbb{P}(W_r)$. If $[v]$ is fixed by every word of length at most J_r , then every operator in $\mathcal{A}_{J_r}$ sends the line $\mathbb{R}v$ into itself. Since every longer word operator belongs to the same stabilized space $\mathcal{A}_{J_r}$, the line is fixed by every word. Thus the word-invariance conditions stabilize by J_r . Intersecting with the fixed algebraic condition that $[v]$ be a decomposable Lie-subalgebra line

does not change this conclusion. Finally take $J = \max_r J_r$ over the finitely many possible dimensions r .

The point of this proof is that the uniform bound comes from the dimension of a fixed operator space; no Noetherian argument, degree bound, or hidden uniformity assertion is needed. $\square$

Corollary A.4 (Finite detection of ideals). *If S generates a Zariski-dense subgroup of G and a Lie subalgebra $\mathfrak{h}$ is invariant under all words of length at most J , then $\mathfrak{h}$ is an ideal of $\mathfrak{g}$.*

Proof. Lemma A.3 gives invariance under all words in S . The stabilizer of the Pluecker line is algebraic, so it contains the Zariski closure of the generated group, namely G . Differentiation gives $[\mathfrak{g}, \mathfrak{h}] \subset \mathfrak{h}$. $\square$

A.3 An effective polynomial zero-set estimate

The closing theorem needs more than the qualitative fact that two disjoint semialgebraic sets have positive distance. The distance must be bounded below by a power of the coefficient height. We prove the fixed-format form needed here. The proof is deliberately organized so that no resultant root is mistaken for a real common zero.

Lemma A.5 (Fixed-format projection with height control). *Fix integers m, M, D . Consider a first-order formula over the ordered real field involving at most m real variables, at most M atomic polynomial relations, and integer polynomials of degree at most D and coefficient height at most H . After eliminating any prescribed block of quantified variables, the resulting semialgebraic set can be described by a Boolean combination of at most $M_* = M_*(m, M, D)$ integer polynomial sign conditions, of degree at most $D_* = D_*(m, M, D)$ and height at most H^{C_*} . If parameters are restricted to $[-R, R]$, the height bound may be replaced by $(H(1 + R))^{C_*}$.*

Proof. We give the elimination argument because this is the point at which a naive resultant proof can lose the real fiber. Eliminate one variable y at a time. For every polynomial in y , adjoin its coefficients, derivative in y , and the signed subresultant coefficients of every pair. Their signs partition the parameter space into finitely many cells on which: the degree in y is constant; the number and multiplicities of real roots are constant; and the order of all real roots of all polynomials is constant. These claims follow

from the Euclidean algorithm encoded by the subresultant sequence and Sturm's sign-variation formula. On one such cell, the truth of an existential or universal formula in y is determined by the signs at the finitely many root sections and at one sample point in each complementary interval. Thus the quantified variable is removed without replacing a real-root condition by a complex resultant condition.

Degree and height remain controlled. A coefficient, derivative, determinant, or subresultant formed at one elimination step has degree and logarithmic height bounded by constants depending only on the current number of polynomials and their degree; explicitly, determinants of size at most $2D$ multiply heights by at most a fixed power and add only a fixed factorial. Since the number of variables and elimination steps is fixed, iteration gives constants M_*, D_*, C_* depending only on the format. If a bounded parameter x is first replaced by Ry with $|y| \leq 1$, then a monomial x^a becomes $R^{|a|}y^a$. Multiplying each resulting polynomial by a common power R^D (with D bounded by the fixed format) converts all coefficients to integer-polynomial expressions in R without changing their zero or sign conditions. Replacing R by the integer $\lceil 1 + R \rceil$ enlarges every coefficient by at most $(1 + R)^{O_D(1)}$, so the coefficient height remains a fixed power of $1 + R$. This proves the stated height bound. $\square$

Lemma A.6 (Monotone semialgebraic power separation). *Fix a semialgebraic format. Let $\varphi : (0, 1] \rightarrow (0, \infty)$ be a nondecreasing semialgebraic function whose graph is described by integer polynomials of that format and height at most $B \geq 2$. Then there are $A, C > 0$, depending only on the format, such that*

$$\varphi(t) \geq B^{-C} t^A \quad (0 < t \leq 1). \quad (\text{A.1S})$$

Proof. Quantifier elimination from Lemma A.5, followed by the one-variable Sturm decomposition already used in its proof, partitions $(0, 1]$ into a bounded number of points and open intervals with algebraic endpoints. On each open interval the graph of φ is contained in the zero set of a nonzero polynomial

$$F(t, u) \in \mathbb{Z}[t, u], \quad (\text{A.1Sa})$$

whose degree is bounded by the format and whose height is at most B^{C_0} . Indeed, on a sign-invariant cell supporting a one-dimensional graph, at least one defining polynomial must vanish identically on that graph; otherwise all signs would remain strict in a two-dimensional neighbourhood, contradicting that the cell is a graph. The same Sturm/subresultant construction shows

that every nonzero endpoint is a real root of a univariate integer polynomial of bounded degree and height B^{C_0} . Cauchy's bound applied to the reversed polynomial therefore gives

$$t_* \geq B^{-C_1} \quad (\text{A.1Sb})$$

for the right endpoint $t_* > 0$ of the first interval abutting 0.

Work on that first interval and write

$$F(t, u) = \sum_{j=0}^d b_j(t) u^j.$$

Remove the largest common power of u , so that the least index j for which b_j is not identically zero is $j = 0$. Write

$$b_0(t) = t^\ell (c_0 + c_1 t + \cdots + c_d t^d), \quad c_0 \in \mathbb{Z} \setminus \{0\}, \quad (\text{A.1Sc})$$

with $0 \leq \ell \leq d$ and $|c_i| \leq B^{C_0}$. Put

$$t_0 = \min \left\{ \frac{t_*}{2}, \frac{1}{4(d+1)B^{C_0}} \right\}.$$

By (A.1Sb), $t_0 \geq B^{-C_2}$. For $0 < t \leq t_0$ the constant term in parentheses in (A.1Sc) dominates the remaining terms, so

$$|b_0(t)| \geq \frac{1}{2} t^\ell \geq \frac{1}{2} t^d. \quad (\text{A.1Sd})$$

All other coefficients satisfy $|b_j(t)| \leq (d+1)B^{C_0}$ on $[0, 1]$. At the graph point $u = \varphi(t)$ we have $F(t, u) = 0$. If $u \geq 1$, the required lower bound is automatic. If $0 < u < 1$, then (A.1Sd) and

$$|b_0(t)| \leq \sum_{j \geq 1} |b_j(t)| u^j \leq d(d+1)B^{C_0} u$$

give

$$\varphi(t) \geq B^{-C_3} t^d \quad (0 < t \leq t_0). \quad (\text{A.1Se})$$

Finally φ is nondecreasing. Hence for $t \geq t_0$,

$$\varphi(t) \geq \varphi(t_0) \geq B^{-C_3} t_0^d \geq B^{-C_4} \geq B^{-C_4} t^d.$$

Taking $A = d$ and enlarging C proves (A.1S). The proof uses only fixed-format real elimination and elementary root bounds; no complex-fiber or Newton–Puiseux assertion is hidden in the argument. $\square$

Theorem A.7 (Effective real-algebraic Łojasiewicz estimate). *Fix integers $m, s, D \geq 1$. There is $A = A(m, s, D) > 1$ such that, whenever $P_1, \dots, P_s \in \mathbb{Z}[X_1, \dots, X_m]$ have degree at most D and coefficient height at most $H > 1$, with common real zero set Z , one has for all $x \in \mathbb{R}^m$*

$$\min\{1, \text{dist}(x, Z)\} \ll_{m,s,D} (1 + \|x\|_\infty)^A H^A \left(\max_i |P_i(x)| \right)^{1/A}. \quad (\text{A.1L})$$

Here $\text{dist}(x, \emptyset) = +\infty$. In particular, on $[-R, R]^m$, if $Z = \emptyset$ then

$$\max_i |P_i(x)| \gg_{m,s,D} H^{-A} (1 + R)^{-A}. \quad (\text{A.2L})$$

Proof. Fix an integer $R \geq 1$ and first take $x \in [-R, R]^m$. Put

$$f(x) = \max_i |P_i(x)|, \quad g(x) = \min\{1, f(x)\}, \quad d(x) = \min\{1, \text{dist}(x, Z)\}.$$

If $d(x) < 1$, a nearest zero may be sought in the compact box $[-R-1, R+1]^m$. For $0 < t \leq 1$ define

$$\varphi_R(t) = \min\{g(y) : y \in [-R, R]^m, d(y) \geq t\}. \quad (\text{A.1T})$$

When the constraint set is empty we put $\varphi_R(t) = 1$. Because $0 < g \leq 1$ off the common zero set, $\varphi_R(t)$ is positive for $t > 0$; otherwise a minimizing sequence has a convergent subsequence with a common zero at distance at least t from Z . The truncation by 1 is important: it also makes the empty-set convention preserve monotonicity.

The graph of φ_R is described by a fixed-format formula. Write $S_t = \{y \in [-R, R]^m : d(y) \geq t\}$. For $0 < u < 1$, the relation $u = \varphi_R(t)$ says that there is $y \in S_t$ with $-u \leq P_i(y) \leq u$ for all i , and that no $y' \in S_t$ satisfies all strict inequalities $-u < P_i(y') < u$. The case $u = 1$ is the union of the empty-set case and the case in which every point of S_t has $f \geq 1$. Thus the minimum operation itself introduces no new non-semialgebraic input. The condition $d(y) \geq t$ says

$$\forall z \in [-R-1, R+1]^m, \quad \left(P_1(z) = \dots = P_s(z) = 0 \right) \implies \sum_j (y_j - z_j)^2 \geq t^2. \quad (\text{A.1U})$$

Thus Lemma A.5 shows that the graph of φ_R has fixed semialgebraic format and defining height at most $(H(1+R))^C$. Moreover φ_R is nondecreasing,

because the constraint set in (A.1T) shrinks as t increases. Lemma A.6 therefore gives, after enlarging a format-dependent exponent A ,

$$\varphi_R(t) \gg H^{-A}(1+R)^{-A}t^A. \quad (\text{A.1V})$$

Taking $t = d(x)$ in (A.1T) and using $x \in S_t$ yields

$$g(x) \geq \varphi_R(d(x)) \gg H^{-A}(1+R)^{-A}d(x)^A. \quad (\text{A.1W})$$

If $f(x) < 1$, then $g(x) = f(x)$ and this is the desired estimate. If $f(x) \geq 1$, (A.1L) is automatic after enlarging its fixed implicit constant. Thus (A.1L) holds on the box. For arbitrary x , take $R = \lceil 1 + \|x\|_\infty \rceil$.

If $Z = \emptyset$, then $d(x) = 1$ everywhere. The same argument contains no nearest-zero step: (A.1U) is vacuous, so (A.1V) directly gives (A.2L). This also shows why the empty-set clause is not obtained by pretending that a resultant zero has a real lift. $\square$

Remark A.8 (Concordance with Solernó). The theorem has the exact fixed-dimension/fixed-degree strength used in the effective closing argument. Solernó's effective Łojasiewicz theorem [26] gives the same type of estimate and is an important historical comparison. Here the needed fixed-format statement is proved through real quantifier elimination, coefficient dominance on the first algebraic branch, and arithmetic root separation.

Theorem A.9 (Almost-invariant Lie algebra theorem). *Fix $G < \text{SL}_N$ defined over $\mathbb{Q}$. There are $A, c > 0$ such that the following holds. Let $S = \{\gamma_1, \dots, \gamma_k\} \subset G(\mathbb{Q})$ be of height at most R and generate a Zariski-dense subgroup. If a Lie subalgebra $\mathfrak{h}$ satisfies*

$$d_{\text{Gr}}(\text{Ad}(\gamma_i)\mathfrak{h}, \mathfrak{h}) \leq \delta \quad (1 \leq i \leq k), \quad (\text{A.3})$$

then either $\delta \geq c \text{ht}(G)^{-A} R^{-A}$ or $\mathfrak{h}$ lies within $C \text{ht}(G)^A R^A \delta^{1/A}$ of an ideal of the same dimension.

Proof. Lemma A.2 propagates (A.3) to every word of length at most the uniform J of Lemma A.3. In a finite family of Pluecker charts, the equations defining a Lie subalgebra and its invariance under those words have bounded degree. Their coefficients are polynomial expressions in matrix entries of bounded words, hence have height at most $\text{ht}(G)^A R^A$. The Pluecker point of $\mathfrak{h}$ has residual at most $R^A \delta$. The exact common zero set is, by Corollary A.4, the set of ideal lines. Apply Theorem A.7; if the zero set is absent in the chart we get the lower bound on δ , and otherwise we get the asserted proximity. $\square$

Appendix B

Heights of Algebraic Groups Generated by Lattice Elements

B.1 The canonical class- $\mathcal{H}$ core and controlled algebraic complexity

The height theorem below is easiest to read if two potentially conflated objects are separated from the outset.

Definition B.1 (The class- $\mathcal{H}$ core). Let $M < \mathrm{SL}_N$ be a $\mathbb{Q}$ -subgroup. Write M° for the identity component and $R_u(M^\circ)$ for its unipotent radical. We denote by $M^\mathcal{H}$ the canonical connected normal $\mathbb{Q}$ -subgroup generated by $R_u(M^\circ)$ and the connected derived subgroup of M° . Equivalently, it is the largest connected normal subgroup of M° whose solvable radical is unipotent. The quotient $M^\circ/M^\mathcal{H}$ is a torus.

Remark B.2 (Why the definition matters). The earlier phrase “largest connected normal subgroup for which the quotient is a torus” was literally too weak: M° itself would satisfy that phrase if the trivial group is admitted as a torus. The class- $\mathcal{H}$ condition is the missing qualifier. Volume V supplies the radical/derived structure facts behind the equivalent descriptions above.

Proposition B.3 (Uniform Chevalley complexity for class $\mathcal{H}$). *For each N there exist integers $D = D(N)$ and $J = J(N)$, integral rational representa-*

tions

$$\rho_j : \mathrm{SL}_N \longrightarrow \mathrm{GL}(V_j) \quad (1 \leq j \leq J),$$

and constants $A_N, C_N \geq 1$ with the following property. For every connected $H \in \mathcal{H}$ there are an index $j = j(H)$ and a primitive integral vector $0 \neq w_H \in V_j(\mathbb{Z})$ such that

$$H = \mathrm{Stab}_{\mathrm{SL}_N}(w_H) = \mathrm{Stab}_{\mathrm{SL}_N}(\mathbb{Q}w_H), \quad (13.1)$$

and H is cut out in Mat_N by the degree- D part of its rational vanishing ideal. If $\mathrm{ht}(H) = \mathrm{ht}(\mathrm{Lie} H)$ and $\mathrm{ht}_D(H)$ denotes the primitive Pluecker height of that finite-dimensional vanishing space, then

$$C_N^{-1}(1 + \mathrm{ht}(H))^{1/A_N} \leq 1 + \|w_H\| \leq C_N(1 + \mathrm{ht}(H))^{A_N}, \quad (13.2a)$$

$$C_N^{-1}(1 + \mathrm{ht}(H))^{1/A_N} \leq 1 + \mathrm{ht}_D(H) \leq C_N(1 + \mathrm{ht}(H))^{A_N}. \quad (13.2b)$$

Conversely, after increasing A_N, C_N by constants depending only on N , either of the bounds $1 + \|w_H\| \leq R$ or $1 + \mathrm{ht}_D(H) \leq R$ implies $1 + \mathrm{ht}(H) \leq C_N R^{A_N}$.

Proof. The uniform degree bound and finite Chevalley list are now proved in Proposition 2.8. It remains to make the height comparison quantitative.

Fix one of the finitely many representations and a monomial basis for the finite-dimensional space V_D of polynomial functions on Mat_N of degree at most D . If p_H is a primitive integral Pluecker vector for $\mathrm{Lie} H$, the equations

$$X \in \mathrm{Lie} H, \quad d\rho(X)w \wedge w = 0$$

are linear equations whose coefficients are integral polynomials of bounded degree in the coordinates of p_H and w . Selecting a maximal independent subsystem and applying Cramer's rule shows that a primitive integral kernel vector may be chosen with norm $\ll_N (1 + \|p_H\|)^{A_N}$. This gives the Chevalley-vector upper bound.

Because a connected class- $\mathcal{H}$ group has no nontrivial algebraic characters, its action on the detected line $\mathbb{Q}w_H$ is trivial. Thus, after choosing a primitive integral vector $w = w_H$ on that line, H fixes w pointwise and the stabilizer of the vector is still exactly H .

Conversely, if w is such a primitive integral Chevalley vector, then

$$\mathrm{Lie} H = \{X \in \mathfrak{sl}_N : d\rho(X)w \wedge w = 0\}.$$

This is a rational linear system of fixed size whose coefficients are bounded-degree integral polynomials in w . Maximal minors therefore give a primitive

Pluecker vector for $\mathrm{Lie} H$ of norm $\ll_N (1 + \|w\|)^{A_N}$. The same maximal-minor argument compares the height of the degree- D vanishing subspace with the Lie-algebra height. All dimensions and degrees are fixed once N is fixed, so the exponents are uniform. $\square$

B.2 A bounded-component theorem with no hidden lemma

The primary height argument needs a uniform bound on the component group of a rational algebraic group whose rational points are Zariski dense. Earlier versions invoked this as an unnamed “bounded-component lemma.” We now prove the form actually used.

Lemma B.4 (Rational finite subgroups have bounded order). *For every m there is $B_m < \infty$ such that every finite subgroup $F < \mathrm{GL}_m(\mathbb{Q})$ has $|F| \leq B_m$.*

Proof. The lattice

$$L = \sum_{g \in F} g\mathbb{Z}^m \subset \mathbb{Q}^m$$

is a full lattice and is F -stable. Choosing a $\mathbb{Z}$ -basis of L conjugates F into $\mathrm{GL}_m(\mathbb{Z})$.

We claim that the principal congruence subgroup

$$K_3 = \ker(\mathrm{GL}_m(\mathbb{Z}) \rightarrow \mathrm{GL}_m(\mathbb{F}_3))$$

is torsion free. If not, after replacing a torsion element by a suitable power we may take $A \in K_3$ of prime order q . Write $A = I + 3^r B$ with $r \geq 1$ maximal and with at least one entry of B not divisible by 3. If $q \neq 3$, the binomial expansion modulo 3^{r+1} gives

$$A^q \equiv I + q3^r B \not\equiv I \pmod{3^{r+1}},$$

a contradiction. If $q = 3$, expanding modulo 3^{r+2} gives

$$A^3 \equiv I + 3^{r+1} B \not\equiv I \pmod{3^{r+2}},$$

again a contradiction. Hence reduction modulo 3 is injective on F , and we may take $B_m = |\mathrm{GL}_m(\mathbb{F}_3)| < 3^{m^2}$. $\square$

Lemma B.5 (Groups with toric identity component). *For fixed m there is C_m such that the following holds. If $A < \mathrm{GL}_m$ is a $\mathbb{Q}$ -subgroup, $A(\mathbb{Q})$ is Zariski dense in A , and $A^\circ = T$ is a torus, then $[A : T] \leq C_m$.*

Proof. We separate three finite quotients.

Step 1: kill the conjugation action on T . Over $\overline{\mathbb{Q}}$, decompose the defining m -dimensional representation into T -weight spaces. There are at most m distinct weights. Every element normalizing T permutes these weight spaces. Because the embedding $T < \mathrm{GL}_m$ is faithful, the occurring weights generate the character lattice up to finite index; an automorphism acting trivially on all of them is the identity. Thus the image of the normalizer in $\mathrm{Aut}(T)$ has order bounded in terms of m (one may use the crude bound $m!$ after refining equal-weight blocks). Passing to a subgroup of bounded index, we may suppose A centralizes T .

Step 2: move into a central torus. Let $L = Z_{\mathrm{GL}_m}(T)$. In a basis adapted to the T -weight decomposition, $L_{\overline{\mathbb{Q}}}$ is a product of general linear groups on the weight spaces; in particular it is connected and reductive. Its adjoint representation has kernel $Z(L)$, and $Z(L)$ is already a torus. Since $T = A^\circ$ is central in L , the image of A under Ad_L is finite. It is generated Zariski densely by rational points, hence is a finite subgroup of a rational general linear group of dimension at most m^2 . Lemma B.4 bounds its order. After this bounded index passage, $A \subset C := Z(L)$, a torus.

Step 3: bound the residual finite subgroup of C/T . The rank of C is at most m . The Galois action on its character lattice has finite image in $\mathrm{GL}_r(\mathbb{Z})$; by Lemma B.4, applied after choosing a lattice basis, the image has bounded order. Hence C splits over a Galois extension $F/\mathbb{Q}$ of degree bounded in terms of m .

Over F , choose characters $\chi_1, \dots, \chi_s$ whose common kernel is exactly the subtorus T . They give $\chi : C \rightarrow \mathbb{G}_m^s$ with kernel T . Since A/T is finite, every coordinate of $\chi(A(\mathbb{Q}))$ is a root of unity in F . A number field of bounded degree contains only boundedly many roots of unity. Indeed, if it contains a primitive n -th root, then $\varphi(n) \leq [F : \mathbb{Q}] =: D$. If $p^a \mid n$, then $p^{a-1}(p-1) \mid \varphi(n)$, so $p \leq D+1$ and, for each such prime, the exponent a is bounded in terms of D . Thus only finitely many n can occur. Therefore $|A/T|$ is bounded in terms of m . Multiplying the three bounded indices proves the lemma. $\square$

Theorem B.6 (Bounded rational component group). *For fixed N there is C_N such that, whenever $M < \mathrm{SL}_N$ is a $\mathbb{Q}$ -subgroup with $M(\mathbb{Q})$ Zariski dense in M ,*

$$[M : M^\circ] \leq C_N.$$

Proof. Let $H = (M^\circ)^\mathcal{H}$. It is characteristic in M° and hence normal in

M . Use the finite Chevalley representation supplied by Proposition B.3 for H , and let W be the subspace of vectors fixed pointwise by H . Since H is connected of class $\mathcal{H}$, it has no nontrivial algebraic characters; hence its Chevalley line is fixed pointwise and contains a Chevalley vector $w_H \in W$ whose vector stabilizer is exactly H . The normalizer $N_{\mathrm{SL}_N}(H)$ acts on W . The kernel of this action is therefore exactly H : certainly H acts trivially, while an element of the normalizer acting trivially on W fixes w_H and hence lies in its stabilizer H .

Restricting this representation to M gives a rational representation $\rho' : M \rightarrow \mathrm{GL}(W)$ of dimension bounded solely by N , with kernel H . The identity component of $\rho'(M)$ is the torus M°/H . Moreover $\rho'(M(\mathbb{Q}))$ is Zariski dense in $\rho'(M)$. Lemma B.5 therefore bounds $[\rho'(M) : \rho'(M)^\circ]$, which equals $[M : M^\circ]$. $\square$

Lemma B.7 (Bounded-height generators for the identity component). *For fixed N there is A_N with the following property. Let $\gamma_1, \dots, \gamma_k \in \mathrm{SL}_N(\mathbb{Q})$ have height at most $R > 2$ and let M be their Zariski closure. Then M° has a Zariski-dense subgroup generated by finitely many rational matrices of height at most R^{A_N} . The number of these matrices is allowed to depend on k ; the height exponent is not.*

Proof. Adjoin the identity and inverses to the generating set. By Theorem B.6, the finite group M/M° has order at most C_N . Therefore words of length at most C_N in the original generators represent every component. Choose representatives $c_1 = 1, c_2, \dots, c_I$ among those words, where $I \leq C_N$.

For each representative c_i and each original generator γ_j , write in the finite quotient $c_i\gamma_j M^\circ = c_{\kappa(i,j)} M^\circ$ and set

$$m_{ij} = c_i\gamma_j c_{\kappa(i,j)}^{-1} \in M^\circ(\mathbb{Q}).$$

Every m_{ij} is a word of bounded length, so its rational height is $R^{O_N(1)}$.

Let $\Delta = \langle \gamma_1, \dots, \gamma_k \rangle \cap M^\circ$. This is Zariski dense in M° : its closure is a finite-index closed subgroup of M° , and a connected algebraic group has no proper finite-index closed subgroup. Reading any word in the γ_j from left to right and repeatedly using $c_i\gamma_j = m_{ij}c_{\kappa(i,j)}$ rewrites every element of Δ as a word in the m_{ij} . Hence the m_{ij} generate a Zariski-dense subgroup of M° . $\square$

B.3 Height of a group generated by bounded rational elements

We now reconstruct the height argument in a form that makes the independence from the number of generators completely visible.

Lemma B.8 (Fixed-degree factor and CRT height calculus). *Fix d . Let $f \in \mathbb{Q}[x]$ be monic of degree at most d , and suppose every coefficient of f has rational height at most $H \geq 2$. Then every monic factor $h \in \mathbb{Q}[x]$ of f has coefficient height $H^{O_d(1)}$. Moreover, if finitely many such factors are pairwise coprime, the Bézout coefficients and Chinese-remainder idempotents obtained from them may be chosen with coefficient height $H^{O_d(1)}$.*

Proof. Let q be the least common multiple of the denominators of the coefficients of f . Then $qf \in \mathbb{Z}[x]$ and $q \leq H^{O_d(1)}$. Divide the coefficients of qf by their positive gcd. We obtain a primitive $F \in \mathbb{Z}[x]$ whose coefficients and leading coefficient are $H^{O_d(1)}$. By Gauss's lemma, a monic rational factor h is obtained by normalizing a primitive integral factor H_0 of F . The leading coefficient of H_0 divides the leading coefficient of F , hence is $H^{O_d(1)}$.

Every complex root α of f satisfies the Cauchy bound

$$|\alpha| \leq 1 + \max_{j < \deg f} |a_j| \leq H^{O_d(1)}.$$

The roots of H_0 form a submultiset of the roots of f . Vieta's formulas therefore bound every coefficient of H_0 by its leading coefficient times a sum of at most 2^d products of at most d such roots. Thus all coefficients of H_0 are $H^{O_d(1)}$. Dividing by its leading coefficient proves the first assertion, including the denominator bound for h .

For the second assertion, write a Bézout identity for two coprime factors u, v with degree-bounded unknown coefficients. Equating coefficients gives a square nonsingular Sylvester linear system of size at most $2d$. Its entries have height $H^{O_d(1)}$ by the first part. After clearing a bounded number of denominators, Cramer's rule bounds the rational solutions by $H^{O_d(1)}$. Iterating this a bounded number of times gives the same bound for all Chinese-remainder idempotents. All implied exponents depend only on d . $\square$

Lemma B.9 (Polynomial height of multiplicative Jordan parts). *For fixed N there is A_N such that if $g \in \mathrm{GL}_N(\mathbb{Q})$ has rational height at most $R > 2$,*

then its multiplicative Jordan factors $g = g_s g_u = g_u g_s$ lie in $\mathrm{GL}_N(\mathbb{Q})$ and satisfy

$$\mathrm{ht}(g_s) + \mathrm{ht}(g_u) \ll_N R^{A_N}.$$

Proof. The coefficients of the characteristic polynomial χ_g are sums of boundedly many minors of g , so they have rational height $R^{O_N(1)}$. The minimal polynomial m_g is a monic rational factor of χ_g . Lemma B.8 therefore bounds the coefficients of m_g , and of all its monic irreducible factors, by $R^{O_N(1)}$. Write

$$m_g = \prod_{i=1}^t p_i^{e_i}$$

with the p_i distinct and irreducible over $\mathbb{Q}$. Here t , every e_i , and every degree are bounded by N .

There is a unique class $q \in \mathbb{Q}[x]/(m_g)$ whose image over a splitting field satisfies, for every root λ of p_i ,

$$q(\lambda) = \lambda, \quad q^{(j)}(\lambda) = 0 \quad (1 \leq j < e_i).$$

These are exactly the Hermite interpolation conditions saying that $q(\lambda + z) \equiv \lambda \pmod{z^{e_i}}$. They are Galois invariant, so the unique class q is rational. To see the height bound without appealing to the roots, express the same conditions as linear congruences modulo the pairwise coprime powers $p_i^{e_i}$. Lemma B.8 bounds the coefficients of these powers and the associated CRT idempotents. The remaining Hermite conditions form fixed-size rational linear systems; after clearing denominators, Cramer's rule gives a representative $\deg q < \deg m_g$ whose coefficients have height $R^{O_N(1)}$.

On a generalized λ -eigenspace, $q(g)$ is multiplication by λ with the nilpotent part removed. Hence $g_s = q(g)$ is the semisimple factor. Evaluating a bounded-degree polynomial of polynomial height at g gives $\mathrm{ht}(g_s) \leq R^{O_N(1)}$. Finally $g_u = g_s^{-1}g$; inversion in fixed matrix dimension is given by the adjugate divided by the determinant, so it preserves polynomial height. Thus $\mathrm{ht}(g_u) \leq R^{O_N(1)}$ as well. $\square$

Lemma B.10 (Connected class- $\mathcal{H}$ height from bounded generators). *Let $H \in \mathcal{H}$ be connected, and suppose a finite symmetric set $S \subset H(\mathbb{Q})$ of elements of height at most R generates a Zariski-dense subgroup. Then*

$$\mathrm{ht}(H) \ll_N R^{A_N},$$

with an exponent independent of $|S|$.

Proof. Let $D = D(N)$ be the degree supplied by Proposition 2.8, and let V_D be the finite-dimensional space of polynomials on Mat_N of degree at most D . Put $L = \dim V_D$, a number depending only on N .

Include the identity in S . For $m \geq 1$, define

$$K_m = \{p \in V_D : p(s_1 \cdots s_m) = 0 \text{ for every } s_1, \dots, s_m \in S\}.$$

Because the identity is allowed, $K_1 \supseteq K_2 \supseteq \cdots$. Among $K_1, \dots, K_{L+1}$ there is an index $m \leq L$ with $K_m = K_{m+1}$. For $s \in S$ and $p \in K_m$, the right translate $p_s(x) = p(xs)$ belongs to K_m : it vanishes on every word of length m because $p \in K_{m+1}$. Thus K_m is invariant under right translation by the group generated by S , and hence by its Zariski closure H . Since the identity is among the test words, $p_s(1) = p(s) = 0$ for every group word s ; therefore every $p \in K_m$ vanishes on H . Conversely, if $p \in I_D(H)$, then every word in the alphabet S belongs to $\langle S \rangle \subset H$, so p vanishes on every word of length m ; thus $p \in K_m$. Hence

$$K_m = I_D(H),$$

the degree- D part of the vanishing ideal.

Choose the monomial basis of V_D . Evaluation at a word of length at most $m \leq L$ gives a rational row vector of height $R^{O_N(1)}$. Although there may be many such rows, their span has dimension at most L ; choose a maximal independent subsystem of at most L rows. Maximal minors and Cramer's rule then give

$$\text{ht}(I_D(H)) \ll_N R^{A_N}.$$

Finally, because the degree- D equations define H , its tangent space at the identity is

$$\text{Lie } H = \{X : dp_e(X) = 0 \text{ for every } p \in I_D(H)\}.$$

This is a fixed-size rational linear system whose coefficients are linear in the coefficients of a basis of $I_D(H)$. A second maximal-minor estimate gives the asserted Pluecker-height bound for $\text{Lie } H$. $\square$

Theorem B.11 (Generated subgroup height). *There is $A = A(N)$ such that if $\gamma_1, \dots, \gamma_k \in \text{SL}_N(\mathbb{Q})$ have height at most $R > 2$ and M is their Zariski closure, then the class- $\mathcal{H}$ core $M^{\mathcal{H}}$ of Definition B.1 satisfies*

$$\text{ht}(M^{\mathcal{H}}) \ll_N R^A,$$

with an exponent independent of k .

Proof. Step 1: reduce to connected M . Lemma B.7 replaces the original set by a finite set of height $R^{O_N(1)}$ generating a Zariski-dense subgroup of M° . Since $M^{\mathcal{H}}$ is defined from M° , we may from now on assume M connected. Adjoin inverses and the identity to this replacement set and rename it $\{\gamma_1, \dots, \gamma_k\}$. In fixed dimension, inversion is given by the adjugate divided by the determinant, so all new generators still have height $R^{O_N(1)}$. The replacement may increase the number of generators, but the rest of the proof never uses that number in an exponent. The symmetry will be used in Step 3 to obtain normality in both conjugation directions.

Step 2: generate the part on which every character is trivial. Write $\gamma_i = (\gamma_i)_s(\gamma_i)_u$ for the multiplicative Jordan decomposition. Lemma B.9 gives polynomial height bounds for both factors. Let

$$S_1 = \{[\gamma_i, \gamma_j] : 1 \leq i, j \leq k\} \cup \{(\gamma_i)_u : 1 \leq i \leq k\}$$

and define recursively

$$S_{\ell+1} = S_\ell \cup \{[\gamma_i, s] : 1 \leq i \leq k, s \in S_\ell\}.$$

Let M_ℓ be the Zariski closure of the subgroup generated by S_ℓ . Every M_ℓ is a $\mathbb{Q}$ -group with Zariski-dense rational points, and $M_1 \subseteq M_2 \subseteq \dots$.

Step 3: obtain a uniform stationary stage. Theorem B.6 gives $[M_\ell : M_\ell^\circ] \leq C_N$ for every ℓ . Along a strict inclusion $M_\ell \subsetneq M_{\ell+1}$, either the identity-component dimension increases, or the identity components agree and the finite component group strictly enlarges. The first event can occur at most $N^2 - 1$ times. Between two such events, the order of the component group is an increasing integer bounded by C_N , so there are at most $C_N - 1$ strict inclusions. Hence there is

$$\ell_0 \leq J(N)$$

with $M_{\ell_0} = M_{\ell_0+1}$.

Because ℓ_0 is bounded, every element of S_{ℓ_0} has height $R^{O_N(1)}$. The stationary equality implies that M_{ℓ_0} is normal in M : for $s \in S_{\ell_0}$,

$$\gamma_j s \gamma_j^{-1} = [\gamma_j, s] s \in M_{\ell_0+1} = M_{\ell_0},$$

and the generators are Zariski dense in M .

Step 4: identify the identity component with $M^{\mathcal{H}}$. First, $M_{\ell_0}^\circ \subseteq M^{\mathcal{H}}$. Indeed, $M^{\mathcal{H}}$ is normal in M and contains the connected derived subgroup and the unipotent radical. Every commutator $[\gamma_i, \gamma_j]$ therefore lies in $M^{\mathcal{H}}$. Also the image of $(\gamma_i)_u$ in the torus $M/M^{\mathcal{H}}$ is unipotent, hence trivial; thus

every $(\gamma_i)_u$ lies in $M^{\mathcal{H}}$ as well. Normality now puts every iterated commutator used to form S_{ℓ_0} inside $M^{\mathcal{H}}$. Taking Zariski closure and identity components gives the inclusion.

For the reverse inclusion, modulo M_{ℓ_0} the generators commute, and

$$\gamma_i M_{\ell_0} = (\gamma_i)_s M_{\ell_0}$$

is semisimple because $(\gamma_i)_u \in M_{\ell_0}$. Since M is connected, the quotient M/M_{ℓ_0} is connected; it is an abelian linear algebraic group generated Zariski densely by semisimple elements, so its unipotent radical is trivial and it is a torus. Consequently both $R_u(M)$ and the connected derived subgroup of M map trivially to the quotient. They lie in $M_{\ell_0}^\circ$, and therefore so does the connected group that they generate, namely $M^{\mathcal{H}}$. Thus

$$M_{\ell_0}^\circ = M^{\mathcal{H}}.$$

Step 5: bound the height. Apply Lemma B.7 once more to M_{ℓ_0} , whose generating elements have height $R^{O_N(1)}$, to obtain bounded-height rational generators for its identity component $M^{\mathcal{H}}$. This group is of class $\mathcal{H}$, so Lemma B.10 gives $\text{ht}(M^{\mathcal{H}}) \ll_N R^{A(N)}$. No stage depends on the number k except through the number of rows in a fixed-dimensional linear system, where a maximal independent subsystem has bounded size. This proves the advertised uniformity. $\square$

Worked Example B.12 (Why the generator count does not enter the exponent). Take diagonal matrices in $\text{SL}_3(\mathbb{Q})$ of height at most R together with a few unipotent elementary matrices. The raw number of generators can be very large. In the class- $\mathcal{H}$ height step, however, only the kernel of an evaluation matrix on the fixed vector space V_D matters. Its rank is at most $\dim V_D$, so after row reduction at most $\dim V_D$ word evaluations survive. Adding a thousand redundant generators therefore changes the number of candidate rows but not the size of the minors controlling the height. This is the finite-dimensional mechanism behind the theorem's independence from k .

Appendix C

Marked Effective Avoidance

The naive union bound over singular subgroups is not sufficient when several short rational subgroups appear simultaneously. The correct device is the same marked-chain mechanism used in quantitative nondivergence.

Lemma C.1 (Finite-poset marking lemma). *Let $\mathcal{P}$ be a finite poset of height at most d . For each $\Delta \in \mathcal{P}$, let f_Δ be (C, α) -good on an interval I . Suppose incomparable marked objects can be replaced by a strictly larger object whose control function is bounded by a fixed multiple of the product of the two old control functions. Then, outside a subset of measure $O(\varepsilon^c)|I|$, all objects with $f_\Delta < \varepsilon$ at a given point belong to one chain. The exponent $c > 0$ depends only on C, α, d .*

Proof. Induct on the poset height. Choose maximal intervals on which a fixed object is marked. Goodness controls the subset on which its control function drops from its boundary scale to an ε fraction. If another incomparable object becomes marked on a subinterval, replace the pair by the strictly larger object supplied by the hypothesis, thereby reducing the remaining height. Maximal intervals of one type have bounded overlap after triple enlargement. Sum the good-function losses and invoke the induction hypothesis. Since the height is finite, the product of the finitely many positive exponents remains positive. $\square$

Theorem C.2 (Marked effective avoidance). *For the unipotent trajectory and rational subgroup class of Chapter 3 there are $A, c > 0$ such that, on a thick part, either the total time spent in ε -neighborhoods of proper singular strata of height at most B is $O(\varepsilon^c T)$, or there is one proper rational subgroup*

M of height at most $(Bh)^A$ such that for every $|t| \leq T$,

$$\|\eta_M(gu_t)\| \leq (Bh)^A, \quad \|Z \wedge \eta_M(gu_t)\| \leq T^{-c}(Bh)^A. \quad (\text{C.1})$$

Proof. In an injectivity chart, closeness to a singular stratum is measured by norms of exterior-power polynomial maps. These functions are good by Volume II. Order primitive rational subgroup data by inclusion. Covolume submodularity supplies the intersection/sum relation required by Lemma C.1. Hence, outside a power-small exceptional set, all simultaneously dangerous subgroup data form one marked chain.

If no chain persists for a positive proportion of the interval, summing the good-function estimates gives the first alternative. Otherwise take the maximal member M of a persistent chain. Every coordinate transverse to its Chevalley line is a bounded-degree polynomial in t and is small on a fixed positive proportion of $[-T, T]$. Lemma 3.2 controls every rescaled coefficient, giving the first inequality of (C.1) on the whole interval; differentiating at zero gives the wedge with Z . The height is controlled by Theorem B.11 and the polynomial chart complexity of the thick part. $\square$

Appendix D

The Pair-Return Closing Theorem in Full Finite-Scale Form

Theorem D.1 (Pair-return effective closing). *Let $G < \mathrm{SL}_N$ be a perfect rational group of class $\mathcal{H}$, $X = \Gamma \backslash G$, and $U = \{u_t = \exp(tZ)\}$ with $\|Z\| = 1$. There are $A_1, A_2 > 1$ and $E > 0$ such that the following holds. Let $x = \Gamma g \in X_\tau$, $T > R > E\tau^{-A_1}$, and let $E_0 \subset [-T, T]$ satisfy $|E_0| > TR^{-1/A_1}$. Suppose a proper Lie subalgebra $\mathfrak{h}$ with $\mathfrak{h} + \mathrm{rad}(\mathfrak{g})$ not a proper ideal has the property that for every $s, t \in E_0$ there exists $\gamma_{s,t} \in \Gamma$ with*

$$\|u_{-s}g^{-1}\gamma_{s,t}gu_t\| \leq R^{1/A_1}, \quad (\text{D.1})$$

$$d_{\mathrm{Gr}}(\mathrm{Ad}(u_{-s}g^{-1}\gamma_{s,t}gu_t)\mathfrak{h}, \mathfrak{h}) \leq R^{-1}. \quad (\text{D.2})$$

Then either there is a nontrivial proper $M \in \mathcal{H}$ such that, for all $|t| \leq T$,

$$\|\eta_M(gu_t)\| \leq R^{A_2}, \quad \|Z \wedge \eta_M(gu_t)\| \leq T^{-1/A_2} R^{A_2}, \quad (\text{D.3})$$

or there is a nontrivial proper normal $M \triangleleft G$ of class $\mathcal{H}$, containing the radical, with

$$\|Z \wedge v_M\| \leq R^{-1/A_2}. \quad (\text{D.4})$$

Proof. We keep the scale bookkeeping visible.

1. *Thick times.* Apply Theorem C.2 with a threshold R^{-C} . If its algebraic alternative occurs, after increasing A_2 it is already (D.3). Otherwise remove

a set of relative measure $O(R^{-c})$ and retain a subset $E_1 \subset E_0$ of comparable measure on which injectivity radius is at least R^{-C} .

2. *Simultaneous density scales.* Put $d = \dim G$ and select $d + 2$ nested radii $R^{\theta^{d+1}}, R^{\theta^d}, \dots, R^\theta, R$, with $\theta > 0$ small. At each scale remove the points whose local density in E_1 is less than half the global density. A Vitali selection of disjoint bad intervals shows that the removed measure is at most a fixed multiple of the complement measure. Choosing the scales successively leaves a positive-measure set of points good at every scale.

3. *Controlled lattice generators.* Fix one good base time t_0 . For good t at the first scale, (D.1) produces lattice elements of matrix height R^C ; (D.2) says they move $[\mathfrak{h}]$ by at most R^{-c} . At each larger scale take the Zariski closure of the subgroup generated by the newly visible lattice elements and replace it by its $\mathcal{H}$ -part. Theorem B.11 bounds every subgroup height by R^C . The connected groups form an ascending chain, so two consecutive scales have the same connected subgroup.

4. *Stable-scale dichotomy.* If the stable generated subgroup is Zariski-dense in G , Theorem A.9 forces $\mathfrak{h}$ to be power-close to an ideal unless the initial error is at least R^{-c} . The latter is excluded by choosing A_1 large. For a perfect group, an ideal whose sum with the radical is all of $\mathfrak{g}$ is all of $\mathfrak{g}$: descend the lower central series of the unipotent radical and use $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. The only remaining possibility is the proper normal alternative (D.4). Otherwise the stable generated subgroup is proper; call its $\mathcal{H}$ -part M .

5. *Propagation along the entire orbit.* The good-density property says that a positive proportion of a scale window produces return elements inside this same M . In its controlled Chevalley representation, the transverse coordinates of $t \mapsto \eta_M(gu_t)$ are bounded-degree polynomials. Marked avoidance and the long-smallness coefficient lemma turn smallness on the positive-density set into coefficient bounds on the full interval. This gives (D.3), while the norm bound follows from the height of M and the controlled representative of a thick point. $\square$

Remark D.2. This is the one-parameter real-place closing mechanism whose modern finite-scale formulation motivated [12]. We have weakened all numerical constants to unspecified polynomial powers because later chapters need only positivity of the exponents, not their optimization.

Appendix E

Restricted Projections for Principal SL_2 Modules

The rank-two bootstrap uses one analytic statement that is genuinely deeper than the polynomial sublevel estimates used in the effective closing theorem: a nonconcentrated cloud cannot have small projections to too many osculating planes of a nondegenerate curve. Since this statement is load bearing, we do not hide it behind a citation. This appendix proves the discretized form needed later. The proof follows the high-low frequency architecture of Gan–Guo–Wang [7, Theorem 2.1 and Sections 2–3] and, at its decoupling step, the bilinear proof of moment-curve ℓ^2 decoupling of Guo–Li–Yung–Zorin–Kranich [8]. The citations are theorem concordance: the estimates used below are proved here. We give the complete chain in the order in which it is used, and we keep an explicit scale ledger at the two places where a missing factor of ρ would change the induction.

E.1 Moment curves, osculating flags, and transverse tubes

For $n \geq 2$ put

$$\gamma_n(t) = \left(t, \frac{t^2}{2!}, \dots, \frac{t^n}{n!} \right), \quad -2 \leq t \leq 2.$$

For $1 \leq m \leq n$ define

$$E_m(t) = \mathrm{span}\{\gamma_n'(t), \dots, \gamma_n^{(m)}(t)\}, \quad \Pi_{t,m} : \mathbb{R}^n \rightarrow E_m(t)$$

to be orthogonal projection. The normalization by factorials is convenient: the matrix with columns $\gamma'_n(t), \dots, \gamma_n^{(n)}(t)$ is triangular unipotent after reversing the coordinate order. In particular its determinant is one.

Lemma E.1 (Uniform osculating nondegeneracy). *For $t \in [-2, 2]$ the flag $E_1(t) \subset \dots \subset E_n(t) = \mathbb{R}^n$ is uniformly nondegenerate. More precisely, for every $1 \leq r \leq n$ and every $t_1, t_2 \in [-2, 2]$,*

$$|\gamma'_n(t_1) \wedge \dots \wedge \gamma_n^{(r)}(t_1) \wedge \gamma'_n(t_2) \wedge \dots \wedge \gamma_n^{(n-r)}(t_2)| \asymp_{n,r} |t_1 - t_2|^{r(n-r)}. \quad (\text{R.1})$$

Proof. Taylor expansion at t_1 gives

$$\gamma_n^{(i)}(t_2) = \sum_{q=i}^n \frac{(t_2 - t_1)^{q-i}}{(q-i)!} \gamma_n^{(q)}(t_1).$$

Expand the wedge product. Any nonzero term must use, among the last $n-r$ factors, exactly the derivative orders $r+1, \dots, n$. Hence every surviving term has total power

$$\sum_{i=1}^{n-r} (r+i-i) = r(n-r)$$

of $t_2 - t_1$. The coefficient is nonzero: at $t_1 = 0, t_2 = 1$ it is a nonzero generalized Vandermonde determinant. The first n derivatives form a uniformly conditioned basis on the compact interval, so the remaining constant factors are bounded above and below. This proves (R.1). $\square$

For $0 < \delta < 1$ and a δ -net $\Lambda_\delta \subset [-1, 1]$, let $\mathbb{T}_t$ be a tiling by boxes with m short sides of length δ parallel to $E_m(t)$ and $n-m$ sides of length one parallel to its orthogonal complement. We refer to them as δ -tubes normal to $E_m(t)$. Their Fourier-dual boxes have m sides of length δ^{-1} in the osculating directions and $n-m$ sides of length one.

If μ is a finite Borel measure on $B(0, 1)$, write

$$c_\alpha(\mu) = \sup_{x \in \mathbb{R}^n, r > 0} \frac{\mu(B(x, r))}{r^\alpha}.$$

Thus $c_\alpha(\mu) < \infty$ is the Frostman bound of exponent α .

E.2 The incidence theorem that implies projection growth

Theorem E.2 (Restricted tube incidence). *Fix $n \geq 2$, $1 \leq m \leq n$, $0 < \alpha < m$, and $\epsilon > 0$. Let Λ_δ be a δ -net of $[-1, 1]$. For each $t \in \Lambda_\delta$ choose a finite subfamily $\mathbb{W}_t \subset \mathbb{T}_t$, and put $\mathbb{W} = \bigcup_t \mathbb{W}_t$. Suppose $0 < \mu(B(0, 1)) < \infty$, $c_\alpha(\mu) < \infty$, and*

$$\sum_{T \in \mathbb{W}} \mathbf{1}_T(x) \geq c\delta^{-1+\epsilon} \quad \text{for every } x \in \text{supp } \mu. \quad (\text{R.2})$$

Then, for all sufficiently small δ ,

$$|\mathbb{W}| \geq C_{n,m,\alpha,\epsilon,c} \frac{\mu(\mathbb{R}^n)}{c_\alpha(\mu)} \delta^{-1-\alpha+C_n\sqrt{\epsilon}}. \quad (\text{R.3})$$

The constant C_n is finite and independent of δ .

Proof location. The proof is deferred to Section E.5, after the high-frequency estimate (Proposition E.3). The end of that section explicitly closes the present restricted tube-incidence theorem. Thus this statement is a forward theorem, not an unproved input.

The point of the exponent $C_n\sqrt{\epsilon}$ is not its numerical value. The later homogeneous-dynamics argument only uses that it tends to zero with ϵ . We now prove the theorem. The proof has a low-frequency branch, which closes by induction on ϵ , and a high-frequency branch, which is controlled by decoupling.

E.3 Smooth tube majorants and the high–low split

Choose for every $T \in \mathbb{W}$ a nonnegative smooth function ϕ_T which is bounded by a dimensional constant, is bounded below on T , and has Fourier support in a fixed dilation of the dual box. Such functions are obtained from a fixed Schwartz bump by the affine map taking the unit cube to T . Then (R.2) implies

$$\Phi(x) := \sum_{T \in \mathbb{W}} \phi_T(x) \gg \delta^{-1+\epsilon} \quad (x \in \text{supp } \mu). \quad (\text{R.4})$$

Set $\rho = \delta^{\sqrt{\epsilon}}$. Let ψ_0 be a radial multiplier which equals one on $B(0, \delta^{-1}\rho)$ and vanishes outside $B(0, 2\delta^{-1}\rho)$. Write

$$\phi_T = \phi_T^{\text{lo}} + \phi_T^{\text{hi}}, \quad \widehat{\phi_T^{\text{lo}}} = \widehat{\phi_T} \psi_0.$$

After splitting $\mathrm{supp} \mu$ into two measurable pieces, one of

$$\left| \sum_T \phi_T^{\mathrm{lo}}(x) \right| \gg \delta^{-1+\epsilon}, \quad \left| \sum_T \phi_T^{\mathrm{hi}}(x) \right| \gg \delta^{-1+\epsilon} \quad (\mathrm{R.5})$$

holds on a set of μ -measure at least a fixed fraction of $\mu(\mathbb{R}^n)$. We treat the two alternatives separately.

The low-frequency branch

Convolution with $\check{\psi}_0$ enlarges the m short directions of a tube from δ to δ/ρ while leaving the long directions at unit scale. Let $\tilde{T}$ denote the resulting enlarged tube.

For the low-frequency induction one must coarsen both the spatial scale and the parameter scale. Set

$$\delta' = \delta/\rho.$$

Partition the original δ -net into consecutive parameter blocks of length comparable to δ' . In each block choose a maximal family of enlarged tubes which are pairwise noncomparable, where comparability means containment after a fixed dimensional dilation. Every original tube is then charged to one chosen representative in its parameter block. The charge is not quite injective across neighboring parameter blocks, but an original tube can be charged to only $O_n(\rho^{-1})$ representatives: there are $O(\rho^{-1})$ original δ -parameters in one δ' -block and only a bounded number of adjacent blocks can contain a comparable representative. This is the only overlap loss in the passage from the δ -family to the δ' -family.

Call a representative enlarged tube *heavy* if the number of original tubes comparable with it is at least

$$c_1 \rho^{-m-1+\sqrt{\epsilon}}. \quad (\mathrm{R.6})$$

The power ρ^{-m} counts the transverse translates and the extra ρ^{-1} counts the original parameter values inside a coarsened parameter block. The light representatives cannot account for the first inequality in (R.5). Indeed, low-frequency convolution has L^∞ size $O(\rho^m)$ on each original tube bump, at most $O(\delta^{-1}\rho)$ coarsened parameter blocks can meet a fixed point, and (R.6) gives

$$O(\delta^{-1}\rho) O(\rho^m) O(\rho^{-m-1+\sqrt{\epsilon}}) = O(\delta^{-1}\rho^{\sqrt{\epsilon}}) = O(\delta^{-1+\epsilon}).$$

Choosing c_1 smaller than the constant implicit in (R.5), the light part is strictly below the required lower bound. Hence, on a fixed positive-mass subset,

$$\sum_{\tilde{T} \in \tilde{\mathbb{W}}_{\text{heavy}}} \mathbf{1}_{\tilde{T}}(x) \gg \delta^{-1+\epsilon} \rho. \quad (\text{R.7})$$

Now put

$$\tilde{\epsilon} = \frac{\epsilon}{1 - \sqrt{\epsilon}}.$$

Then $\delta^{-1+\epsilon} \rho = (\delta')^{-1+\tilde{\epsilon}}$. Thus the heavy representatives satisfy the same incidence hypothesis at the coarser scale δ' , with the parameter family now a genuine δ' -net up to bounded multiplicity. Applying the induction hypothesis to the restricted measure gives

$$|\tilde{\mathbb{W}}_{\text{heavy}}| \gg \frac{\mu(\mathbb{R}^n)}{c_\alpha(\mu)} (\delta/\rho)^{-1-\alpha+C_n\sqrt{\tilde{\epsilon}}}. \quad (\text{R.8a})$$

To recover original tubes, use (R.6) and then pay the $O(\rho^{-1})$ charging multiplicity described above. This yields the source-faithful lower bound

$$|\mathbb{W}| \gg \frac{\mu(\mathbb{R}^n)}{c_\alpha(\mu)} (\delta/\rho)^{-1-\alpha+C_n\sqrt{\tilde{\epsilon}}} \rho^{-m+\sqrt{\tilde{\epsilon}}}. \quad (\text{R.8})$$

The exponent calculation is now visible. Writing $\rho = \delta^{\sqrt{\epsilon}}$, the power of δ on the right of (R.8) equals

$$-1 - \alpha + C_n\sqrt{\tilde{\epsilon}} + \sqrt{\epsilon}(1 + \alpha - m + \sqrt{\epsilon} - C_n\sqrt{\tilde{\epsilon}}). \quad (\text{R.8b})$$

Since $\alpha < m$ and $\sqrt{\tilde{\epsilon}} = \sqrt{\epsilon}(1 + O(\sqrt{\epsilon}))$, enlarging the dimensional constant C_n once makes (R.8b) at most $-1 - \alpha + C_n\sqrt{\epsilon}$ for all sufficiently small ϵ . This is exactly (R.3). The base range, say $\epsilon \geq \epsilon_0(n)$, is weaker than the trivial cardinality bound after increasing the constant, so the downward induction in ϵ is initialized. This closes the low-frequency branch.

The high-frequency branch

Decompose $1 - \psi_0$ into $O(|\log \delta|)$ dyadic annuli. On one annulus the lower bound in (R.5) remains true with a logarithmic loss. Rescale so that the annulus has frequency radius comparable to one. Because a function with frequency support at radius δ^{-1} is essentially constant on δ -balls, the Frostman

inequality and a covering of $\mathrm{supp} \mu$ by δ -balls give, for $p = n(n+1)$,

$$\frac{\mu(\mathbb{R}^n)}{c_\alpha(\mu)} \delta^{-p+p\epsilon+n-\alpha} |\log \delta|^{-p} \ll \left\| \sum_T F_T \right\|_{L^p(\mathbb{R}^n)}^p, \quad (\text{R.9})$$

where every $\widehat{F}_T$ lies in a fixed bounded annulus and in a box adapted to the osculating flag at the parameter of T .

The following proposition is the precise high-frequency estimate needed in (R.9).

Proposition E.3 (High-frequency tube estimate). *Fix $n \geq 2$, $1 \leq m \leq n$, put $p = n(n+1)$, and let $0 < \delta < \delta_0(n, m)$. Let $\mathbb{W}$ be a finite family of the δ -tubes from Section E.1. For each $T \in \mathbb{W}$, let $F_T \in \mathcal{S}(\mathbb{R}^n)$. Fix constants $C_{\mathrm{box}} \geq 1$ and $0 < c_{\mathrm{ann}} < C_{\mathrm{ann}} < \infty$, depending only on n, m , and assume*

$$\mathrm{supp} \widehat{F}_T \subset C_{\mathrm{box}} T^* \cap \{\xi : c_{\mathrm{ann}} \leq \|\xi\| \leq C_{\mathrm{ann}}\},$$

where T^* is the dual frequency box associated with T . Assume moreover that in the osculating basis at the parameter of T every $\xi \in \mathrm{supp} \widehat{F}_T$ can be written as

$$\xi = \sum_{i=1}^n \lambda_i \gamma_n^{(i)}(t_T), \quad |\lambda_i| \lesssim 1 \ (i \leq m), \quad |\lambda_i| \lesssim \delta \ (i > m), \quad (\text{R.9a})$$

with at least one of $|\lambda_1|, \dots, |\lambda_m|$ bounded below by a fixed positive dimensional constant. Then, for every $\eta > 0$,

$$\left\| \sum_{T \in \mathbb{W}} F_T \right\|_p^p \ll_{n,m,\eta} \delta^{-p+1-\eta} \sum_{T \in \mathbb{W}} \|F_T\|_p^p. \quad (\text{R.10})$$

Proof location. The proof is given in Section E.5 (“The high-frequency decomposition algorithm”), after Lemma E.9. It uses the moment-curve decoupling theorem E.4, proved just before that section.

For the annular pieces constructed from the normalized tube bumps above, the common affine normalization gives $\|F_T\|_p \ll_n 1$; hence (R.10) has right-hand side $\ll \delta^{-p+1-\eta} |\mathbb{W}|$. Comparing with (R.9) gives

$$|\mathbb{W}| \gg \frac{\mu(\mathbb{R}^n)}{c_\alpha(\mu)} \delta^{-1-\alpha+p\epsilon+\eta} |\log \delta|^{-p}.$$

Choose $\eta = \sqrt{\epsilon}$ and absorb the logarithm into an additional $O(\sqrt{\epsilon})$ power. This is (R.3). Thus Theorem E.2 follows once Proposition E.3 and the decoupling theorem used in its proof have been established.

E.4 Sharp ℓ^2 decoupling for the moment curve

We give a self-contained proof in the form used below. Put

$$\Gamma_k(t) = (t, t^2, \dots, t^k), \quad p_k = k(k+1).$$

If J is an interval, let $\mathcal{U}_J$ be the frequency plate centered at $\Gamma_k(c_J)$ with side lengths $|J|, |J|^2, \dots, |J|^k$ in the derivative directions. Let $\mathcal{P}(\delta)$ be the dyadic partition into intervals of length comparable to δ .

Theorem E.4 (Moment-curve decoupling). *For every $k \geq 1$ and every $\eta > 0$ there is $C_{k,\eta}$ such that, if $\text{supp } \hat{f}_J \subset \mathcal{U}_J$, then*

$$\left\| \sum_{J \in \mathcal{P}(\delta)} f_J \right\|_{p_k} \leq C_{k,\eta} \delta^{-\eta} \left(\sum_J \|f_J\|_{p_k}^2 \right)^{1/2}. \quad (\text{R.11})$$

Proof location. The proof occupies Sections E.4.1–E.4.4 and is completed at the end of Section E.4.4 (“The bootstrap that forces zero decoupling exponent”), before Section E.5 begins.

E.4.1 Affine rescaling and bilinear reduction

Let $D_k(\delta)$ be the least constant in (R.11). For separated intervals I, I' of fixed macroscopic size, let $B_k(\delta)$ be the least constant in

$$\int |f_I|^{p_k/2} |f_{I'}|^{p_k/2} \leq B_k(\delta)^{p_k} \left(\sum_{J \subset I} \|f_J\|_{p_k}^2 \right)^{p_k/4} \left(\sum_{J' \subset I'} \|f_{J'}\|_{p_k}^2 \right)^{p_k/4}. \quad (\text{R.12})$$

Lemma E.5 (Affine rescaling). *If $I = [a, a + \kappa]$ and $J \subset I$, the affine map determined by $\Gamma_k(a + \kappa t)$ carries the standard plate for the rescaled interval $\kappa^{-1}(J - a)$ to $\mathcal{U}_J$ up to a fixed dilation. Consequently*

$$\|f_I\|_{p_k} \leq D_k(\delta/\kappa) \left(\sum_{J \subset I, |J| \asymp \delta} \|f_J\|_{p_k}^2 \right)^{1/2}, \quad (\text{R.13})$$

and the analogous rescaling holds for B_k .

Proof. For $1 \leq r \leq k$, differentiating $\Gamma_k(a + \kappa t)$ shows that the r th derivative is κ^r times an upper-triangular linear combination, with bounded coefficients depending on a , of the derivative vectors of $\Gamma_k(t)$. Thus the affine map has the anisotropic singular values $\kappa, \dots, \kappa^k$ in the osculating basis. Pulling back

the Fourier supports and then changing variables in physical space multiplies both sides of the desired inequality by the same Jacobian factor. This proves (R.13) and its bilinear analogue. $\square$

Lemma E.6 (Whitney bilinear reduction). *For dyadic $\delta = 2^{-N}$,*

$$D_k(\delta)^2 \ll 1 + \sum_{r=2}^N B_k(2^{-N+r-2})^2. \quad (\text{R.14})$$

Proof. Decompose $[0, 1]^2$ into Whitney squares relative to the diagonal: at level r , take pairs $I \times I'$ of dyadic intervals of length 2^{-r} whose distance is one or two interval lengths and which were not already covered at a coarser level; at the last level retain the squares touching the diagonal. The squares have bounded overlap in each coordinate. Since

$$\left| \sum_J f_J \right|^2 = \sum_{I, I'} f_I \overline{f_{I'}},$$

Minkowski in $L^{p_k/2}$ bounds the square of the left side of (R.11) by the sum of the $L^{p_k/2}$ norms of these products. The final diagonal squares are controlled by Cauchy–Schwarz and bounded overlap. Each off-diagonal Whitney square is reduced by Lemma E.5 to (R.12) at scale 2^{-N+r-2} . Summing over the bounded-overlap families yields (R.14). $\square$

E.4.2 Asymmetric bilinear constants

For $0 \leq \ell \leq k-1$ set $p_\ell = \ell(\ell+1)$, with $p_0 = 0$. For $a, b \in [0, 1]$, define $B_{\ell, a, b}(\delta)$ as the least constant for which, whenever I, I' are macroscopically separated with lengths δ^a, δ^b ,

$$\int (|f_I|^{p_\ell} * \varphi_I)(|f_{I'}|^{p_k - p_\ell} * \varphi_{I'}) \leq B_{\ell, a, b}(\delta)^{p_k} A_I^{p_\ell} A_{I'}^{p_k - p_\ell}, \quad (\text{R.15})$$

where $A_I = (\sum_{J \subset I, |J| \asymp \delta} \|f_J\|_{p_k}^2)^{1/2}$. We now freeze the weights precisely, because this is one place where the published version of the short decoupling proof required a later correction; see the corrected arXiv v3 of Guo–Li–Yung–Zorin-Kranich [8]. Let $\mathcal{U}_I^\circ$ be the centered polar parallelepiped to $\mathcal{U}_I$ and put

$$\lambda_I(x) = \inf \{t \geq 1 : x/t \in \mathcal{U}_I^\circ\}.$$

Fix once and for all a dimensional exponent

$$A > k, \quad A \geq \frac{k(k+1)}{2} = \frac{p_k}{2}, \quad A' = A + k - 2, \quad (\text{R.15a})$$

and define

$$\varphi_I(x) = c_{k,A} |\mathcal{U}_I^\circ|^{-1} \lambda_I(x)^{-A}, \quad \int_{\mathbb{R}^k} \varphi_I = 1. \quad (\text{R.15b})$$

The companion exponent A' is reserved for the lower-dimensional averaging weights below. This is the corrected exponent ledger in [8, Appendix A]; in particular we do not use the older convention of assigning one common ad hoc exponent to every weight.

Lemma E.7 (Corrected bump calculus). *The weights in (R.15a)–(R.15b) have the following properties.*

1. *If $J \subset I$, then*

$$\varphi_I * \varphi_J \ll_{k,A} \varphi_J. \quad (\text{R.16a})$$

2. *If $\text{supp } \widehat{g} \subset C\mathcal{U}_J$ and $q \geq 1$, then*

$$|g|^q \ll_{k,A,C,q} |g|^q * \varphi_J. \quad (\text{R.16b})$$

3. *Let $1 \leq \ell \leq k-1$, let $H = (V^{(k-\ell)}(t'))^\perp$, and let $B_H(0, R) \subset H$. Define the $L^1(H)$ -normalized averaging weight*

$$\Phi_{H,R}(z) = c_{\ell,A'} R^{-\ell} (1 + R^{-1} \text{dist}_H(z, B_H(0, R)))^{-A'}.$$

At the canonical scale $R = \nu^{-(k-\ell+1)}$ and for an interval I' of length $\asymp \nu$ containing t' , convolution in H satisfies

$$\Phi_{H,R} *_H \varphi_{I'} \ll_{k,A} \varphi_{I'}. \quad (\text{R.16c})$$

Proof. For (R.16a), affine normalization reduces to $I = [0, 1]$ and $|J| = \nu \leq 1$. In coordinates parallel to the sides of a smallest box containing $\mathcal{U}_J^\circ$, the two weights are comparable to

$$\widetilde{\varphi}(x) = \max(1, |x_1|, \dots, |x_k|)^{-A}, \quad \widetilde{\varphi}_\nu(x) = \nu^{k(k+1)/2} \widetilde{\varphi}(\nu x_1, \dots, \nu^k x_k).$$

Split the convolution integral into $\widetilde{\varphi}_\nu(y) \leq 2^A \widetilde{\varphi}_\nu(x)$ and its complement. On the first part, $A > k$ gives $\widetilde{\varphi} \in L^1(\mathbb{R}^k)$. On the complement, some coordinate i satisfies $|x_i| > 2|y_i|$, hence $|x_i - y_i| \geq |x_i|/2$. Therefore

$$\widetilde{\varphi}(x - y) \leq 2^A |x_i|^{-A} = 2^A \nu^{iA - k(k+1)/2} \widetilde{\varphi}_\nu(x) \leq 2^A \widetilde{\varphi}_\nu(x),$$

where the last inequality uses $i \geq 1$ and $A \geq k(k+1)/2$. Integrating the normalized fine weight proves (R.16a). This is exactly the corrected convolution calculation of [8, Lemma A.1].

For (R.16b), choose a Schwartz function ψ_J with $\widehat{\psi}_J \equiv 1$ on $C\mathcal{U}_J$ and $|\psi_J| \ll_{k,A,C} \varphi_J$. Since $g = g * \psi_J$, Hölder gives

$$|g(x)|^q \leq \|\psi_J\|_1^{q-1} \int |g(x-y)|^q |\psi_J(y)| dy \ll |g|^q * \varphi_J(x).$$

For (R.16c), rotate so that the last ℓ coordinates span H and set $k' = k - \ell + 1$. After the anisotropic normalization attached to I' , the left side is bounded by a constant multiple of

$$\int_{\mathbb{R}^\ell} \nu^{k'\ell} (1 + \nu^{k'} |z|)^{-A'} \left(1 + \sum_{i < k'} \nu^i |x_i| + \sum_{j \geq k'} \nu^j |x_j - z_j| \right)^{-A} dz. \quad (\text{R.16d})$$

Dropping the last sum gives the required decay in the first $k' - 1$ coordinates. For a fixed $j_0 \geq k'$, integrate first in the other $\ell - 1$ variables. Their integral lowers the exponent A' to $A' - \ell + 1 \geq A' - k + 2 = A > 1$; the remaining one-dimensional convolution is then bounded by $(1 + \nu^{j_0} |x_{j_0}|)^{-A}$ by (R.16a) in dimension one. Doing this for every $j_0 \geq k'$ yields (R.16c). The inequality $A' \geq A + k - 2$ is exactly the loss needed here, matching [8, Lemma A.2]. $\square$

A crude decomposition of each macroscopic interval into pieces of lengths δ^a and δ^b , followed by Hölder and Lemma E.7, gives

$$B_k(\delta) \ll \delta^{-ap_\ell/p_k - b(p_k - p_\ell)/p_k} B_{\ell,a,b}(\delta). \quad (\text{R.17})$$

For $\ell = 0$, affine rescaling gives

$$B_{0,a,b}(\delta) \asymp D_k(\delta^{1-b}). \quad (\text{R.18})$$

E.4.3 Transversality and lower-degree decoupling

Let $V^{(\ell)}(t) = \text{span}\{\Gamma'_k(t), \dots, \Gamma_k^{(\ell)}(t)\}$. Lemma E.1, with harmless normalization changes, gives

$$|V^{(\ell)}(t_1) \wedge V^{(k-\ell)}(t_2)| \gg |t_1 - t_2|^{\ell(k-\ell)}. \quad (\text{R.19})$$

Lemma E.8 (Lower-degree step). *Assume Theorem E.4 has been proved in every dimension $< k$. If $1 \leq \ell \leq k - 1$ and $0 \leq a \leq (k - \ell + 1)b/\ell$, then for every $\eta > 0$,*

$$B_{\ell,a,b}(\delta) \ll_\eta \delta^{-b\eta} B_{\ell,(k-\ell+1)b/\ell,b}(\delta). \quad (\text{R.20})$$

Proof. Fix the coarse interval I' of length δ^b and a point $t' \in I'$. Quotient frequency space by $V^{(k-\ell)}(t')$. By (R.19), the projected curve $P\Gamma_k$ has uniformly nonzero ℓ -dimensional torsion on the separated interval I . The inductive decoupling theorem in dimension ℓ therefore applies to this projected curve after a bounded affine change of variables. Its canonical scale is determined by matching the physical averaging ball in the orthogonal complement of $V^{(k-\ell)}(t')$ to the polar box of I' ; this gives the interval length

$$\delta^{b'} \quad \text{with} \quad b' = (k - \ell + 1)b/\ell.$$

Fubini decomposes physical space into affine copies of that ℓ -dimensional orthogonal complement. On each copy average with the weight $\Phi_{H,R}$ of Lemma E.7, where $R = (\delta^b)^{-(k-\ell+1)}$. Apply lower-dimensional decoupling on every slice, then Minkowski in the form $L^{p_\ell} \ell^2 \leq \ell^2 L^{p_\ell}$. The uncertainty estimate (R.16b) replaces each frequency-localized factor by its φ -average, while the transverse convolution estimate (R.16c) absorbs the slice-average weight back into $\varphi_{I'}$ without a scale-dependent loss. Finally (R.16a) absorbs nested interval weights on the I side. Thus every averaging weight created by Fubini and lower-dimensional decoupling is returned to the two weights appearing in (R.15). The only remaining loss is the inductive lower-dimensional decoupling factor $\delta^{-b\eta}$, and the right side is precisely (R.15) with a replaced by b' . This proves (R.20). $\square$

E.4.4 The bootstrap that forces zero decoupling exponent

Hölder applied to the two factors in (R.15) yields

$$B_{\ell,a,b} \leq B_{k-\ell,b,a}^{1/(k-\ell+1)} B_{\ell-1,a,b}^{(k-\ell)/(k-\ell+1)}. \quad (\text{R.21})$$

Indeed the exponent pair $(p_\ell, p_k - p_\ell)$ is the convex combination, with coefficient $1/(k - \ell + 1)$, of $(p_k - p_{k-\ell}, p_{k-\ell})$ and $(p_{\ell-1}, p_k - p_{\ell-1})$. Combining (R.21) twice with the lower-degree step gives, for all sufficiently small b ,

$$B_{\ell,(k-\ell+1)b/\ell,b} \ll_{\eta} \delta^{-b\eta} B_{k-\ell,\frac{\ell+1}{\ell}\frac{k-\ell+1}{k-\ell}b,\frac{k-\ell+1}{\ell}b}^{1/(k-\ell+1)} B_{\ell-1,\frac{k-\ell+2}{\ell-1}b,b}^{(k-\ell)/(k-\ell+1)}. \quad (\text{R.22})$$

For $\ell = 1$ the last first parameter is irrelevant because $B_{0,a,b}$ does not depend on a .

Let η_k be the infimum of exponents η for which $D_k(\delta) \ll_{\eta} \delta^{-\eta}$. For $b > 0$ define $A_\ell(b)$ by

$$B_{\ell,(k-\ell+1)b/\ell,b}(\delta) \ll \delta^{-A_\ell(b)+o(1)}.$$

Equation (R.18) gives $A_0(b) = \eta_k(1 - b)$. Taking logarithmic exponents in (R.22), then letting $b \downarrow 0$, gives for

$$a_\ell = \liminf_{b \downarrow 0} \frac{\eta_k - A_\ell(b)}{b}$$

the inequalities

$$a_\ell \geq \frac{1}{\ell} a_{k-\ell} + \frac{k-\ell}{k-\ell+1} a_{\ell-1}, \quad 1 \leq \ell \leq k-1. \quad (\text{R.23})$$

The numbers a_ℓ are finite; both directions of control are needed before we sum (R.23). First, Hölder in (R.15), followed by linear decoupling on the two coarse intervals, gives

$$A_\ell(b) \leq \eta_k - \eta_k b \left(\frac{k-\ell+1}{\ell} \frac{p_\ell}{p_k} + \frac{p_k - p_\ell}{p_k} \right) + o(b). \quad (\text{R.24})$$

For the reverse control, combine the bilinear reduction (R.14) with the crude comparison (R.17), choosing the Whitney level for which the two coarse intervals have lengths $\delta^{(k-\ell+1)b/\ell}$ and δ^b . All other Whitney levels contribute a bounded geometric tail after affine rescaling. Therefore there is a constant C_k such that

$$\eta_k \leq C_k b + A_\ell(b) + o(b) \quad (1 \leq \ell \leq k-1). \quad (\text{R.24a})$$

Equations (R.24) and (R.24a) imply that $(\eta_k - A_\ell(b))/b$ stays bounded above and below as $b \downarrow 0$; hence every a_ℓ in (R.23) is a finite real number. This is the linear–bilinear equivalence step which cannot be replaced by a formal cancellation of possibly infinite exponents.

We now perform the cancellation explicitly. Sum (R.23) for $\ell = 1, \dots, k-1$. Reindex the first sum by $j = k - \ell$:

$$\sum_{\ell=1}^{k-1} a_\ell \geq \sum_{j=1}^{k-1} \frac{a_j}{k-j} + \sum_{j=0}^{k-2} \frac{k-j-1}{k-j} a_j. \quad (\text{R.24b})$$

For $1 \leq j \leq k-2$ the two coefficients on the right add to one, while the coefficient of a_{k-1} is also one. Thus all $a_1, \dots, a_{k-1}$ cancel from the two sides of (R.24b), leaving

$$0 \geq \frac{k-1}{k} a_0 = \frac{k-1}{k} \eta_k.$$

Since $\eta_k \geq 0$, we obtain $\eta_k = 0$. The base case $k = 1$ is Plancherel, so induction on k proves Theorem E.4.

E.5 The high-frequency decomposition algorithm

We return to Proposition E.3. A point in the Fourier support of F_T , with parameter t , has the form

$$\xi = \lambda_1 \gamma'_n(t) + \cdots + \lambda_n \gamma_n^{(n)}(t), \quad |\lambda_i| \lesssim 1 \ (i \leq m), \quad |\lambda_i| \lesssim \delta \ (i > m), \quad (\text{R.25})$$

and, on the selected annulus, at least one of $|\lambda_1|, \dots, |\lambda_m|$ is comparable to one.

Lemma E.9 (One decoupling-rescaling step). *Suppose the largest active index among $1, \dots, m$ is r , and on a dyadic piece the next active coefficient scale is $s \in [\delta^{1/(n-r)}, c]$. Then the contribution of that piece to the L^p norm, with $p = n(n+1)$, is bounded by*

$$s^{-1+\mathfrak{D}_{n-r}/p+o(1)} \left(\sum_{\Theta: |\Theta|=s} \|F_\Theta\|_p^p \right)^{1/p}, \quad (\text{R.26})$$

where $\mathfrak{D}_q = q(q+1)$ and, after the anisotropic change of variables

$$(\xi_1, \dots, \xi_r, \xi_{r+1}, \dots, \xi_n) \mapsto (\eta_1, \dots, \eta_r, s\eta_{r+1}, \dots, s^{n-r}\eta_n), \quad (\text{R.27})$$

each F_Θ has the same structural form as (R.25), but with either a strictly larger active index or one fewer unresolved derivative direction.

Proof. Partition the t -interval into intervals Θ of length s . After freezing $\lambda_1, \dots, \lambda_r$ at their dyadic sizes, differentiation of (R.25) in t shows that the unresolved frequency variation lives, modulo the span of the first r derivative directions, on a nondegenerate moment curve of degree $n-r$. Apply Theorem E.4 in dimension $n-r$ at its canonical scale. Since $p \geq \mathfrak{D}_{n-r}$, interpolation with the trivial L^∞ estimate changes the critical decoupling factor to $s^{-1+\mathfrak{D}_{n-r}/p+o(1)}$, giving (R.26).

For the rescaling, expand $\gamma_n^{(q)}(s\theta)$ by powers of s and apply (R.27). Terms of order $q \leq r$ retain bounded coefficients; terms with $q > r$ acquire precisely the powers needed to normalize the selected coefficient scale. If the smallest allowed scale is reached, the final unresolved direction has coefficient $O(\delta)$ and is removed. Otherwise the largest coefficient of the next block becomes order one, so its active index strictly increases. Thus the complexity pair

$$(\text{number of unresolved derivative directions, } m-r)$$

decreases lexicographically. The transformed derivative matrix remains uniformly nondegenerate by Lemma E.1; hence the same argument can be iterated with uniform constants. $\square$

Proof of Proposition E.3. Decompose first according to the largest active index r and then dyadically according to the successive coefficient scales. Apply Lemma E.9 repeatedly. The lexicographic complexity in its proof decreases at every terminal rescaling, so after at most n blocks of iterations all unresolved derivative directions have reached the $O(\delta)$ scale. The product of the factors in (R.26) is computed by summing their logarithmic exponents. Each parameter subdivision contributes one factor s^{-1} , whereas the critical decoupling gains contribute $\mathfrak{D}_q/p$ for the corresponding unresolved degree q . Since $p = n(n+1)$ dominates every $\mathfrak{D}_q$ and the algorithm terminates, the net power is at most $\delta^{-1+1/p-o(1)}$ at the level of the L^p norm. Raising to the p th power gives $\delta^{-p+1-o(1)}$.

At the final scale the Fourier pieces attached to distinct original tubes have bounded overlap. Therefore the terminal p th-power contribution is bounded by a dimensional constant times $\sum_{T \in \mathbb{W}} \|F_T\|_p^p$. There are only $O(|\log \delta|)^n$ choices of dyadic scales and active-index histories; absorb this into $\delta^{-\eta}$. This gives (R.10). $\square$

This completes the proof of Theorem E.2.

E.6 From tube incidence to quantitative restricted projections

Theorem E.10 (Discretized restricted projection theorem). *Fix $0 < s < n$ and $1 \leq m \leq n$. There exists $\eta = \eta(n, m, s) > 0$ such that the following holds for all sufficiently small δ . If $F \subset B(0, 1)$ is δ -separated and*

$$\#(F \cap B(x, r)) \leq \delta^{-\eta} |F| r^s \quad (\delta \leq r \leq 1), \quad |F| \geq \delta^{-s+\eta}, \quad (\text{R.28})$$

then the set of $t \in [0, 1]$ for which

$$N_\delta(\Pi_{t,m} F) < \delta^{-\min\{s,m\}+\eta} \quad (\text{R.29})$$

has Lebesgue measure at most δ^η .

Proof. Assume a set J of bad parameters has measure larger than δ^η and choose a maximal δ -separated subset $\Lambda \subset J$. For every $t \in \Lambda$, cover $\Pi_{t,m} F$

by at most the number in (R.29) of δ -balls and pull these balls back to tubes in $\mathbb{T}_t$. Every $x \in F$ therefore belongs to one selected tube for every $t \in \Lambda$, so the multiplicity is $\gg |\Lambda| \gg \delta^{-1+2\eta}$.

Put $\mu = |F|^{-1} \sum_{x \in F} \mathbf{1}_{B(x, \delta)} \delta^{-n} dx$. From (R.28), after enlarging constants, there is an exponent $\alpha < \min\{s, m\}$ with $\min\{s, m\} - \alpha = O(\eta)$ such that $c_\alpha(\mu) \ll \delta^{-O(\eta)}$. Apply Theorem E.2 with its epsilon parameter a fixed multiple of η . It gives the lower bound

$$|\mathbb{W}| \gg \delta^{-1-\alpha+O(\sqrt{\eta})}.$$

On the other hand, by construction

$$|\mathbb{W}| \leq |\Lambda| \delta^{-\min\{s, m\} + \eta} \ll \delta^{-1-\min\{s, m\} + O(\eta)}.$$

Choose the book-keeping parameter in (R.28)–(R.29) so small, relative to the constant in the incidence theorem, that the lower exponent is strictly larger than the upper exponent. This contradiction proves the exceptional set bound. $\square$

Corollary E.11 (Projection growth for principal modules). *Every irreducible real SL_2 -module and every nontrivial initial segment of its weight filtration has $\mathrm{PG}(\eta)$ for some $\eta > 0$, uniformly on compact families of norms.*

Proof. Identify an irreducible module with $\mathrm{Sym}^q(\mathbb{R}^2)$ and use the monomial weight basis $Z_1^{q-j} Z_2^j$. For $u_t(Z_1, Z_2) = (Z_1 + tZ_2, Z_2)$, the binomial formula gives

$$u_t(Z_1^{q-j} Z_2^j) = \sum_{r=0}^{q-j} \binom{q-j}{r} t^r Z_1^{q-j-r} Z_2^{j+r}.$$

Thus the coordinate functions are polynomials in t whose successive jets are, after fixed diagonal rescalings, the jets of the moment curve $\gamma_q(t) = (1, t, \dots, t^q)$. Pairing with the first m weight coordinates is consequently equivalent, through fixed invertible source and target maps, to projection onto the osculating m -plane of γ_q . Apply Theorem E.10. If its Frostman hypothesis first fails at a scale r , the offending ball is exactly the concentration branch in the definition of PG . Otherwise a good parameter gives a projection whose covering exponent is larger by a fixed amount. Pulling the projected cloud back through the bounded change of basis gives the required new transverse cloud. Only finitely many modules occur in the rank-two application, so the minimum gain is positive and uniform. $\square$

Remark E.12 (Analytic ownership). The restricted-projection input is proved by the following chain: the rank-two theorem is

moment-curve decoupling $\implies$ high-frequency tube bound
 $\implies$ restricted tube incidence $\implies$ projection growth.

The proof is modeled on the bilinear decoupling architecture of [8] and the high-low incidence architecture of [7]; the citations record provenance, while the logical steps required later are proved here.

Appendix F

Projection Growth and the Boundary of the Present Proof

Effective closing is independent of the strongest incidence geometry used in polynomial-rate equidistribution. We therefore state the additional input in a way that makes this separation explicit.

Definition F.1 (Projection-growth property). A transverse representation has $\text{PG}(\eta)$ if every sufficiently nonconcentrated finite cloud of Frostman exponent s either lies in a power-small tube around a proper invariant algebraic subspace or, after a good polynomial shear and refinement, yields a new cloud of exponent at least $s + \eta$.

Lemma F.2 (Finite polynomial-loss bookkeeping). *Fix integers $N \geq 1$ and positive constants $a, b, L_0, \dots, L_N$. Suppose an N -stage argument uses scales $r_j = T^{-\sigma_j}$ and, away from its algebraic obstruction branch, has total error bounded by a finite sum of terms of the forms*

$$T^{-a}r_j^{-L_j}, \quad r_j^b r_{j-1}^{-L_j} \quad (1 \leq j \leq N), \quad r_N^b,$$

where $r_0 = 1$. Then the exponents $\sigma_j > 0$ can be chosen, depending only on the displayed constants, so that every term is $O(T^{-\delta})$ for one $\delta > 0$.

Proof. Choose the exponents backwards. First take $0 < \sigma_N < a/(4L_N + 4)$ and $\sigma_N < a/4$. Having chosen σ_j , choose

$$0 < \sigma_{j-1} < \min\{b\sigma_j/(4L_j + 4), \sigma_j/2\}.$$

After this finite recursion, decrease all σ_j by one common positive factor if necessary so that $L_j\sigma_j < a/2$ for every j . Then

$$T^{-a}r_j^{-L_j} = T^{-a+L_j\sigma_j} \leq T^{-a/2},$$

and

$$r_j^b r_{j-1}^{-L_j} = T^{-b\sigma_j+L_j\sigma_{j-1}} \leq T^{-b\sigma_j/2}.$$

Also $r_N^b = T^{-b\sigma_N}$. Since there are finitely many stages, the minimum of $a/2$, $b\sigma_N$, and $b\sigma_j/2$ is a positive number δ . Summing finitely many terms preserves $O(T^{-\delta})$. $\square$

Theorem F.3 (Finite bootstrap under projection growth). *Assume the pair-return closing theorem, Margulis contraction, a finite tensor-tempered mixing index, and $\text{PG}(\eta)$ for every transverse module between admissible intermediate subgroups. Then there are $A, \delta > 0$ such that every length- T unipotent segment either has Sobolev discrepancy $O(T^{-\delta})$ from its ambient homogeneous measure or is $T^{-\delta}$ -close to a proper compatible closed homogeneous orbit of height at most T^A (with the proper normal-factor alternative included).*

Proof. Write every quantitative input in polynomial-loss form. Pair-return closing and Margulis contraction give fixed exponents because the representation list, Sobolev order, and height complexity are fixed. Likewise the quantitative form of $\text{PG}(\eta)$ proved in Appendix E has fixed polynomial losses on the finite list of principal modules. Thus there are constants $a, b > 0$ and $L_j < \infty$, independent of T , such that each nonobstruction passage from scale r_{j-1} to r_j incurs only terms of the forms listed in Lemma F.2.

On every nonalgebraic application of $\text{PG}(\eta)$ the Frostman exponent increases by at least η . Hence the number of improvement steps is bounded by

$$N \leq \left\lceil \frac{\dim X}{\eta} \right\rceil,$$

a number independent of T . If a nonconcentration or projection-growth hypothesis fails at any stage, the quantitative concentration alternative feeds the offending tube or invariant subspace into pair-return closing and we are in the stated proper-orbit obstruction branch.

Assume therefore that all N improvement steps occur. At the last scale the cloud has codimension below the fixed smoothing threshold. Convolution with an r_N -bump gives a density whose required Sobolev norm is at most a fixed power $r_N^{-L_N}$. Volume XI's tensor-tempered mixing estimate then

contributes a term $T^{-a}r_N^{-L_N}$. Polynomial divergence, replacement of a cloud by its good projection, smoothing, and the earlier mixing uses produce only the remaining finite collection of polynomial-loss terms. Lemma F.2 chooses the scales once and for all so that all of them are $O(T^{-\delta})$ for one $\delta > 0$. This proves the power-saving branch, with the height exponent A equal to the maximum of the finitely many closing/height exponents encountered along the iteration. $\square$

Remark F.4 (Frontier incidence module). The positive projection-growth theorem needed for the rank-two result is proved in Appendix E. Sharper modern exponents and broader representation families require stronger incidence or decoupling estimates; those optimizations are not used in Volume XIV.

Appendix G

Worked Effective-Rigidity Laboratory

G.1 Motivation

Effective closing combines arithmetic height, almost invariance, return pairs, and projection growth. Each mechanism has a low-dimensional model.

Worked Example G.1 (Pluecker height of a rational plane in $\mathbb{R}^3$). For $v_1 = (1, 0, m)$ and $v_2 = (0, 1, n)$,

$$p = v_1 \wedge v_2 = (1, n, -m)$$

in the basis $(e_1 \wedge e_2, e_1 \wedge e_3, e_2 \wedge e_3)$ up to the stated sign convention. If $m = 3, n = 4$, the primitive Pluecker height is $\|p\| = \sqrt{26}$. Acting by $g = I + E_{13}$ sends v_1 to $(m + 1, 0, m)$ and v_2 to $(n, 1, n)$; wedging and subtracting p computes the exact projective displacement used by an almost-invariance test. The important point is that every coordinate remains integral, so a small projective displacement below the inverse height scale forces exact algebraic relations.

Worked Example G.2 (Osculating planes of the cubic moment curve). Let $\gamma(t) = (t, t^2, t^3)$. Then

$$\gamma'(t) = (1, 2t, 3t^2), \quad \gamma''(t) = (0, 2, 6t), \quad \gamma^{(3)}(t) = (0, 0, 6),$$

and $\det[\gamma'(t), \gamma''(t), \gamma^{(3)}(t)] = 12$. Thus the curve is uniformly nondegenerate. At $t = 0$, the first osculating line is the x -axis and the second osculating plane

is the xy -plane. At general t these spaces rotate polynomially. Restricted projection asks whether a Frostman set can project to abnormally small sets for too many of these correlated planes.

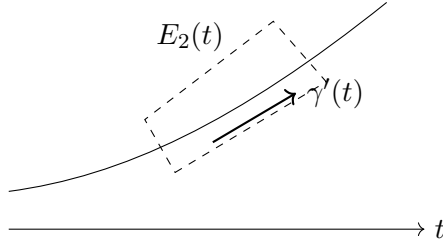


Figure G.1: The moment curve carries a moving flag of osculating spaces. The rank-two bootstrap gains dimension by projecting return clouds onto these spaces unless closing produces an algebraic obstruction.

Proof architecture: decoupling-scale bookkeeping

For the cubic moment curve $\gamma(t) = (t, t^2, t^3)$ the osculating vectors $\gamma', \gamma'', \gamma^{(3)}$ have determinant 12, so the frame never degenerates. On an interval of length δ , Taylor expansion separates the tangent, curvature and torsion scales as $\delta, \delta^2, \delta^3$. Wave packets dual to these boxes therefore have reciprocal dimensions $\delta^{-1}, \delta^{-2}, \delta^{-3}$.



What the figure answers. It records which interval scale is fed to the next parabolic rescaling; the recurrence is not an unspecified “iteration over scales”.

Appendix H

Student Proof Laboratory: The Effective Closing Mechanism

This chapter is deliberately placed before the harmonic-analysis appendix in the reading order. The effective closing lemma is an algebraic–dynamical statement. The sharper rank-two polynomial equidistribution theorem needs restricted projections and decoupling, but closing itself does not. Keeping these dependencies separate is essential for understanding what has actually been proved.

H.1 From qualitative alternatives to finite-scale alternatives

Ratner’s qualitative proof repeatedly says that one of two things happens: a trajectory avoids a singular set for most times, or it is trapped in a proper homogeneous configuration. At finite scale one must replace every word in that sentence by a number. A typical target is:

For an orbit segment of length T , either all smooth averages are within $T^{-\eta}$ of the expected homogeneous average, or the entire segment lies within $T^{-\eta}$ of a closed homogeneous orbit whose arithmetic height is at most T^A .

The constants A, η are not decoration. They allow the alternative to be

iterated and eventually translated into an arithmetic statement.

H.2 Rational subspaces and Pluecker height

Fix an integral lattice $\mathfrak{g}_{\mathbb{Z}}$ in the Lie algebra. If $\mathfrak{h} \subset \mathfrak{g}$ is a rational r -plane, choose a primitive integral generator

$$v_{\mathfrak{h}} \in \bigwedge^r \mathfrak{g}_{\mathbb{Z}}$$

of the line $\bigwedge^r \mathfrak{h}$. Define $\text{ht}(\mathfrak{h}) = \|v_{\mathfrak{h}}\|$.

Lemma H.1 (Separation of rational planes). *For every $B \geq 1$ there is $c > 0$ such that distinct rational r -planes of height at most B have Grassmannian distance at least cB^{-2} .*

Proof. Let v, w be primitive integral Pluecker vectors for the two planes. If the planes are distinct, $v \wedge w$ or an appropriate set of Pluecker minors is a nonzero integral vector, hence has norm at least one. On the other hand, the sine of the Grassmannian angle is bounded below by a fixed dimensional constant times $\|v \wedge w\|/(\|v\|\|w\|)$. Since both norms are at most B , the angle is $\gg B^{-2}$. $\square$

This elementary inequality is the reason that a sufficiently accurate “almost rational” plane of bounded height has a unique exact rational candidate nearby.

H.3 Finite-word detection of invariance

Let $V = \bigwedge^r \mathfrak{g}$ and let $S \subset G$ be a finite generating set for the return transformations. Consider the increasing sequence

$$E_m = \text{span}\{\rho(w) : w \text{ is a word in } S \text{ of length } \leq m\} \subset \text{End}(V).$$

Its dimension is at most $(\dim V)^2$.

Lemma H.2 (Finite-word stabilization). *There is $m \leq (\dim V)^2$ such that $E_m = E_{m+1} = E_{m+2} = \dots$. If a line $\mathbb{R}v \subset V$ is invariant under every operator in E_m , it is invariant under the semigroup generated by S .*

Proof. Whenever $E_m \neq E_{m+1}$ the dimension increases by at least one, so strict increase can occur at most $(\dim V)^2$ times. If $E_m = E_{m+1}$, left or right multiplication of a word of length at most m by a generator lies in $E_{m+1} = E_m$. Induction on word length therefore keeps every longer word inside E_m . If all operators in E_m preserve the line, so do all words. $\square$

This finite-dimensional argument replaces an unnecessary Hilbert-basis shortcut: no commutative algebra is needed to detect invariance under a finite generating set.

H.4 Almost-invariant planes become exact rational subalgebras

Suppose return data produces an r -plane W with

$$d(\text{Ad}(g_j)W, W) \leq \delta$$

for many controlled lattice elements g_j . First rationalize W by a nearby rational plane $W_{\mathbb{Q}}$ using Pluecker separation. If $\delta \ll \text{ht}(W_{\mathbb{Q}})^{-C}$, uniqueness of the nearby rational plane forces $\text{Ad}(g_j)W_{\mathbb{Q}} = W_{\mathbb{Q}}$ exactly. Finite-word stabilization then gives invariance under the generated return group. Finally the bracket condition is checked quantitatively: if W is within δ of an actual Lie algebra and $W_{\mathbb{Q}}$ is the unique rational plane in that neighborhood, the Pluecker coordinates of $[W_{\mathbb{Q}}, W_{\mathbb{Q}}]$ are rational of bounded height and lie within a smaller-than-separation distance of $W_{\mathbb{Q}}$; hence they lie in $W_{\mathbb{Q}}$ exactly. Thus approximate algebraic structure has been converted into an exact rational Lie subalgebra.

H.5 Margulis functions as repulsion potentials

Let $\mathcal{O}$ be a finite transverse cloud of possible return configurations and define an energy

$$\mathcal{E}(x) = \sum_{z \in \mathcal{O}_x} d(x, z)^{-\alpha}$$

with $0 < \alpha$ small. Polynomial shearing spreads a generic transverse configuration, so averaging over a unipotent block contracts the inverse power on average:

$$\frac{1}{|I|} \int_I \mathcal{E}(xu_t) dt \leq (1 - c)\mathcal{E}(x) + C.$$

If the energy is very large for too many consecutive scales, many transverse points remain extremely close under many returns. The preceding almost-invariant-plane argument then produces an exact low-height homogeneous obstruction. Otherwise the contraction inequality forces the orbit to spend most time in a region where the transverse cloud is sparse enough for mixing and linearization estimates.

Worked Example H.3 (A quadratic polynomial shear). In SL_2 , the opposite-unipotent perturbation v_ε satisfies

$$u_{-t}v_\varepsilon u_t = \begin{pmatrix} 1 - t\varepsilon & -t^2\varepsilon \\ \varepsilon & 1 + t\varepsilon \end{pmatrix}.$$

At the scale $T = |\varepsilon|^{-1/2}$ the dominant displacement is order one. Thus an inverse-distance energy averaged over $t \in [T, 2T]$ sees a quadratic polynomial. The one-variable sublevel estimate bounds the proportion of times when this polynomial remains abnormally small. Higher-rank Margulis functions are a systematic version of this calculation for many transverse coordinates simultaneously.

H.6 Closing: the assembly step

The effective closing theorem combines four ingredients.

1. Quantitative recurrence supplies many pairs of nearby return points.
2. Effective linearization turns each pair into an almost-normalizing relation for a bounded-height rational datum.
3. Pluecker separation and finite-word detection exactify the datum into a rational Lie subalgebra normalized by the return group.
4. The corresponding connected subgroup has a closed orbit of controlled volume; the original orbit segment remains close to that orbit because the linearization estimates were uniform throughout the selected return blocks.

At no point is a projection theorem used. Projection theory enters only when one wants to prove that, in a rank-two group, a large transverse cloud must grow rapidly enough to force a polynomial equidistribution rate.

H.7 Why restricted projections are a new analytic module

For a principal SL_2 acting on an irreducible module $\mathrm{Sym}^q(\mathbb{R}^2)$, the unipotent orbit of a highest-weight coordinate traces a moment curve

$$t \mapsto (1, t, t^2, \dots, t^q)$$

up to linear changes of basis. A transverse set with Frostman dimension α must have many distinct projections along the osculating flags of this curve. The restricted-projection theorem quantifies this growth. Its proof has two branches:

1. low frequencies: enlarge tubes and close an induction on scale;
2. high frequencies: decompose into moment-curve plates and apply sharp ℓ^2 decoupling.

The detailed appendix proves the required incidence theorem in exactly this form. The separation in the source tree is intentional: a student can read and prove effective closing before learning any decoupling theory.

H.8 Decoupling: a low-dimensional sanity check

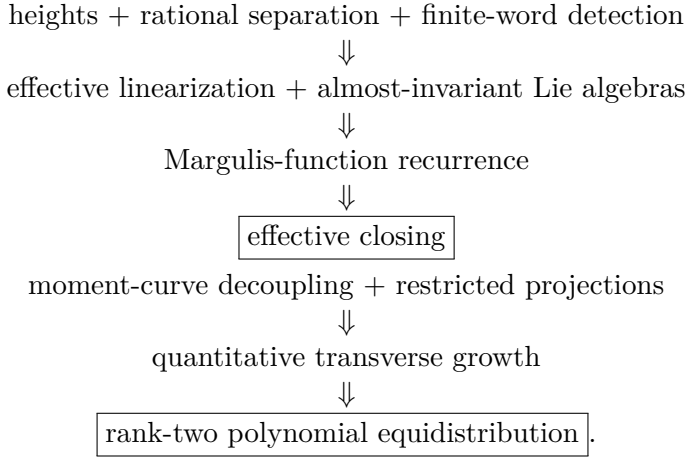
For the parabola $\Gamma_2(t) = (t, t^2)$, disjoint intervals J of length δ give Fourier plates of dimensions $\delta \times \delta^2$. At the critical exponent $p = 6$, the decoupling inequality says

$$\left\| \sum_J f_J \right\|_6 \ll_{\varepsilon} \delta^{-\varepsilon} \left(\sum_J \|f_J\|_6^2 \right)^{1/2}.$$

The exponent $6 = 2 \cdot 3$ is not arbitrary: it is twice the sum $1 + 2$, the total anisotropic degree of the parabola. For the degree- k moment curve the same scaling gives $p_k = k(k + 1)$. This scaling check should be performed before entering the bilinear induction; otherwise it is easy to write an inequality that is dimensionally impossible.

H.9 The proof contract of Volume XIII

The logical dependencies are now:



The second chain strengthens the first; it is not used to justify it.

Appendix I

Student Proof Laboratory: The Effective Closing Core

The strongest rank-two rate in this volume uses a separate harmonic-analysis module, but the effective closing theorem itself is geometric and algebraic. This chapter proves that core in a finite-scale language before any restricted-projection estimate is mentioned.

I.1 Heights of rational subspaces

Let $W \subset \mathbb{Q}^n$ be an r -dimensional rational subspace. The primitive lattice $W \cap \mathbb{Z}^n$ has a primitive exterior vector $w_W \in \bigwedge^r \mathbb{Z}^n$, well defined up to sign.

Definition I.1 (Pluecker height).

$$\text{ht}(W) = \|w_W\|.$$

For a rational Lie subalgebra embedded in a fixed rational matrix representation, define its height by the same exterior construction.

Lemma I.2 (Finiteness of bounded-height rational subspaces). *For every H there are only finitely many rational subspaces of $\mathbb{Q}^n$ with height at most H .*

Proof. The primitive wedge w_W is a nonzero integer vector in the finite-dimensional lattice $\bigwedge^r \mathbb{Z}^n$ of norm at most H . There are only finitely many such vectors. A decomposable primitive wedge determines its rational span, so only finitely many W occur. $\square$

I.2 Almost invariant subspaces can be exactified

Suppose a finite set $S \subset \text{GL}(V)$ nearly preserves an r -plane W . Represent W by a unit decomposable wedge $w \in \bigwedge^r V$.

Lemma I.3 (Finite word stabilization). *Let $E = \bigwedge^r V$ and let $\mathcal{A}_j$ be the linear span in $\text{End}(E)$ of all words of length at most j in $S \cup S^{-1}$. The chain $\mathcal{A}_0 \subset \mathcal{A}_1 \subset \cdots$ stabilizes after at most $(\dim E)^2$ strict increases. If $\mathcal{A}_j = \mathcal{A}_{j+1}$, it is stable under multiplication by the generators and hence contains every word.*

Proof. $\text{End}(E)$ has dimension $(\dim E)^2$, so a strictly increasing chain of subspaces has at most that many strict inclusions. If $\mathcal{A}_j = \mathcal{A}_{j+1}$, multiplication on either side by a generator maps each word of length at most j to a word of length at most $j+1$, hence maps $\mathcal{A}_j$ into itself. Induction on word length then places every word in $\mathcal{A}_j$. $\square$

Proposition I.4 (Exactification of a quantitatively almost invariant line). *Fix a compact family of generator tuples S . There exist $C, \theta > 0$ such that if a unit vector $w \in E$ satisfies*

$$\text{dist}(sw, \mathbb{R}w) \leq \delta \quad (s \in S),$$

then either the word orbit spans a space of dimension at least two with an explicit determinant $\gg \delta^\theta$, or there is an exactly invariant line $\mathbb{R}w'$ with $\|w - w'\| \leq C\delta^\theta$.

Proof. Choose a basis $T_1, \dots, T_M$ of the stabilized word-operator space from the preceding lemma, with word lengths uniformly bounded. The condition that $\mathbb{R}w$ be exactly invariant is the vanishing of the finitely many wedge coordinates $T_i w \wedge w$. If one of these polynomial coordinates is $\gg \delta^\theta$, we are in the first alternative. Otherwise the point $[w]$ lies within a controlled distance of the common zero set of these finite-degree polynomial equations. A compactness/quantitative implicit function argument on projective space gives a nearby common zero $[w']$; all words preserve that line, hence so do the original generators. The exponent depends only on the finite degrees and dimensions, not on the orbit segment. $\square$

I.3 Margulis functions as repulsion detectors

Let Y be a proper closed homogeneous subspace. Near Y , use a transverse coordinate z and define $F_Y(x) \asymp \|z\|^{-\alpha}$, truncated away from a tubular neighborhood. Polynomial shearing makes the average of F_Y contract unless the orbit is nearly contained in the normalizer directions of Y .

Guiding Idea I.5 (Finite-scale Margulis inequality: mechanism). For sufficiently small $\alpha > 0$ there exist an averaging interval I , $c < 1$, and C such that

$$\frac{1}{|I|} \int_I F_Y(xu_t) dt \leq cF_Y(x) + C$$

whenever x is outside a controlled algebraic exceptional set.

Derivation. In a transverse exponential chart, the displacement from Y evolves as a bounded-degree polynomial under u_t . The inverse-power function $\|P(t)\|^{-\alpha}$ is integrable uniformly when α is smaller than the reciprocal degree: decompose according to dyadic sublevel sets and use the polynomial goodness bound. If the polynomial has a genuinely transverse coefficient, the average gains a uniform factor below one after normalizing by its value at the initial point. If every transverse coefficient is too small, effective linearization places the point in the algebraic exceptional set. Add a bounded truncation outside the tube to obtain the additive constant C .

I.4 Recurrence from the drift inequality

Iterating the Margulis inequality along successive blocks gives a geometric series. Therefore an orbit segment that avoids the algebraic exceptional set cannot spend a large fraction of its time extremely close to Y . This is the effective substitute for the qualitative singular-set avoidance used in Volume III.

Lemma I.6 (Quantitative avoidance). *For every depth R there are $c_1, c_2 > 0$ such that an orbit segment of length T either*

1. *spends at most $c_1 R^{-\alpha}$ proportion of its blocks at transverse height $> R$,*
or
2. *is T^{-c_2} -close to an algebraic obstruction of height at most T^{c_2} .*

Proof. Sum the drift inequality over blocks. Markov's inequality converts the average bound on F_Y into a bound for the proportion with $F_Y > R^\alpha$. If a

block falls outside the range where drift was proved, the effective linearization alternative supplies an almost invariant rational subspace. Exactify it by the previous proposition and bound its height using the integer Pluecker coordinates accumulated over the orbit length. This is the second alternative. $\square$

I.5 Additional invariance at finite scale

Qualitatively, two nearby generic points shear to a new element of the measure stabilizer. At finite scale, one instead proves that most test functions have almost the same block average after a small transverse group translation. A Sobolev dual norm quantifies the defect.

Guiding Idea I.7 (Quantitative shearing-to-invariance: mechanism). Let $x, y = xg$ with g small and transverse to a known subgroup L . Suppose both length- T orbit blocks are ϵ -generic for the same reference measure in a Sobolev dual norm and remain outside the effective singular sets. Then at the first macroscopic shear time there exists $h \notin L$ with controlled height and distance from L such that

$$\left| \int f(zh) d\mu_T(z) - \int f(z) d\mu_T(z) \right| \leq C\epsilon^c \mathcal{S}_D(f).$$

Derivation. Use the first-exit polynomial normalization of Volume III, now with explicit coefficient bounds. On a terminal block of fixed relative length the relative displacement differs from one element h by $O(\epsilon^c)$. Common compact-time estimates come from quantitative nonescape. Uniform continuity is replaced by the Sobolev translation estimate $\|R_h f - R_{h'} f\|_\infty \ll d(h, h') \mathcal{S}_D(f)$ on the controlled-height region. Compare both orbit-block averages with the reference measure and sum the three errors.

I.6 The closing module

Guiding Idea I.8 (Effective closing: proof architecture). The exact theorem-level input is Theorem D.1. It begins with a positive-measure set of return times E_0 , a proper Lie subalgebra $\mathfrak{h}$, and the two quantified pair conditions (D.1)–(D.2). Its conclusion is the Chevalley obstruction (D.3) or the proper-normal obstruction (D.4). Chapter 8 explains how a high-energy return cloud is reduced to these hypotheses. Quantitative equidistribution is obtained only

after this closing module is combined with projection growth and mixing; it is not one branch of the closing theorem itself.

Derivation of the architecture. Starting from a high-energy return cloud, select a dyadic displacement scale and a positive-measure set of paired return times. Polynomial shearing and finite sampling produce the bounds (D.1)–(D.2) for the proper Lie subalgebra detected by the current linearization datum. Apply Theorem D.1. In the normal branch, (D.4) is passed to the normal-factor induction. In the nonnormal branch, (D.3) and the Chevalley-height comparison give a proper rational subgroup with polynomial complexity; marked avoidance propagates its datum along the whole tested interval. The separate projection-growth bootstrap of Chapter 17 decides whether such obstructions persist or the cloud gains dimension until mixing applies.

Appendix J

Student Proof Laboratory: Why the Rank-Two Rate Needs Harmonic Analysis

J.1 Closing is not yet a polynomial equidistribution rate

The closing theorem provides an algebraic obstruction when equidistribution fails. To prove that a generic orbit in a rank-two group has a polynomial rate, one must show that too many different obstruction parameters cannot occur. This is an incidence/projection problem for finite sets in representations of SL_2 .

J.2 Wave packets and scale separation

A function with Fourier support near a curved one-dimensional frequency manifold is decomposed into packets localized to reciprocal spatial tubes. At a coarse scale, packets with separated tangent directions are nearly orthogonal; at a fine scale, induction controls packets with nearby directions. The decoupling inequality in the detailed appendix formalizes this scale separation.

J.3 High/low decomposition

Given a discrete obstruction measure ν , smooth it at scale δ and split its Fourier transform into low frequencies, where Frostman-type mass bounds control the energy, and high frequencies, where curvature plus ℓ^2 decoupling controls concentration. Combining the two bounds yields a restricted-projection theorem: for most parameters t , the projected set $\pi_t F$ cannot be too small unless F had a forbidden concentration at a smaller scale.

J.4 Dynamical use

The effective closing argument generates a finite set of candidate rational obstructions. The restricted-projection theorem shows that along most rank-two parameter directions these obstructions separate at polynomial scale. Therefore an orbit cannot repeatedly shadow too many different proper homogeneous pieces; the only remaining alternative is quantitative equidistribution. The detailed appendix owns the analytic theorem so this last step is not a citation disguised as dynamics.

Appendix K

A Student Course in Finite-Scale Ratner Theory

Effective closing is easiest to understand when the qualitative Ratner proof is rewritten as a finite decision procedure. At scale T we do not ask whether a point is exactly singular or exactly homogeneous. We ask how much numerical evidence is required before an approximate algebraic relation can be promoted to an exact rational subgroup of controlled height.

K.1 The finite-scale input data

Let $X = \Gamma \backslash G$, let $U = \{u_t\}$ be Ad-unipotent, and let $x \in X$. Fix a time horizon $T \geq 2$. Choose three small parameters:

1. ρ , the geometric scale at which two quotient points are declared nearby;
2. δ , the algebraic tolerance for almost-invariance;
3. H_0 , the maximal arithmetic height of an obstruction that we are prepared to detect.

The parameters are chosen in this order: the desired final error determines H_0 ; separation of rational data below height H_0 determines δ ; local quotient geometry then determines ρ .

Lemma K.1 (Separation of bounded-height rational planes). *Fix r and an integral lattice $L \subset \mathfrak{g}$. There is $c > 0$ such that two distinct rational r -planes W_1, W_2 with primitive Pluecker vectors of norm at most H_0 satisfy*

$$\text{dist}(W_1, W_2) \geq cH_0^{-2}.$$

The exponent 2 may be replaced by a dimension-dependent explicit exponent for other norms; polynomial separation is the essential point.

Proof. Choose primitive integral Pluecker vectors w_i representing W_i . If the planes are distinct, $w_1 \wedge w_2 \neq 0$ in the appropriate exterior tensor space. In an integral basis at least one coordinate of this wedge is a nonzero integer, hence has absolute value at least 1. On the other hand, normalizing by $\|w_1\| \|w_2\| \leq H_0^2$ bounds the sine of the angle between the Pluecker lines below by H_0^{-2} . Grassmannian distance is comparable to this angle on the compact Grassmannian. $\square$

K.2 From recurrent pairs to an almost normalizer

Suppose two times $s < t \leq T$ satisfy $d(xu_s, xu_t) < \rho$. After choosing the unique small quotient lift we can write

$$xu_t = xu_s \exp Z, \quad \|Z\| \ll \rho.$$

Conjugating the displacement back along the flow gives

$$u_{-\tau} \exp Z u_\tau = \exp(\text{Ad}(u_{-\tau})Z).$$

The logarithm is a bounded-degree polynomial in τ . Thus many pair returns constrain finitely many polynomial coefficients.

Lemma K.2 (Finite sampling determines an adjoint polynomial). *Let $P : \mathbb{R} \rightarrow V$ have degree at most d . If $\|P(t_j)\| \leq \delta$ at $d+1$ points t_j whose normalized mutual separations are at least c_0 , then every coefficient of the rescaled polynomial $P(Ts)$ is $O_{d,c_0}(\delta)$.*

Proof. Lagrange interpolation writes $P(Ts) = \sum_{j=0}^d P(t_j) L_j(s)$. The coefficients of every L_j are bounded in terms of d and the separation of the interpolation nodes. The claim follows by taking norms coefficient by coefficient. $\square$

This elementary observation explains why a long list of recurrence events can be converted into an almost-invariant finite-dimensional subspace rather than merely a collection of unrelated near collisions.

K.3 Finite-word stabilization without Noetherian algebra

Let $V_r = \bigwedge^r \mathfrak{g}$. If a family of matrices $A_1, \dots, A_k$ acts on V_r , let E_m be the span in $\text{End}(V_r)$ of all words in the A_i of length at most m .

Lemma K.3 (Explicit stabilization length). *The chain*

$$E_0 \subset E_1 \subset E_2 \subset \dots$$

stabilizes after at most $(\dim V_r)^2$ strict inclusions. Once $E_m = E_{m+1}$, it remains stationary for every larger index.

Proof. The ambient space $\text{End}(V_r)$ has dimension $(\dim V_r)^2$, so there can be at most that many strict dimension increases. If $E_m = E_{m+1}$, then for every generator A_i and every word operator in E_m , left multiplication by A_i lies in $E_{m+1} = E_m$. By induction every longer word belongs to E_m . $\square$

Corollary K.4 (Bounded words detect invariance). *A subspace $W \subset V_r$ invariant under every word of length at most the stabilization index is invariant under the full semigroup generated by the A_i .*

Proof. Let m be an index at which $E_m = E_{m+1}$. The preceding lemma gives $E_\ell = E_m$ for every $\ell \geq m$, so every word operator A_w belongs to E_m . By hypothesis each word of length at most m sends W into W ; hence every generator of the linear space E_m lies in $\{T \in \text{End}(V_r) : T(W) \subset W\}$. The latter set is a linear subspace of $\text{End}(V_r)$, so every $A_w \in E_m$ preserves W . Therefore the full semigroup generated by the A_i preserves W . $\square$

This is the finite-dimensional replacement for a vague appeal to finite generation of polynomial ideals.

K.4 Exactification of an almost rational plane

Suppose a plane W is within δ of a rational plane W_0 of height at most H_0 , and an adjoint operator A moves W by at most δ . If $\delta \ll H_0^{-C}$, rational separation forces the nearest bounded-height rational plane to AW_0 to equal W_0 .

Proposition K.5 (Exactification principle). *Fix a bounded set of adjoint operators and a height cutoff H_0 . There is C such that if a rational plane W_0 of height at most H_0 is H_0^{-C} -almost invariant under the bounded-word family from the previous section, then W_0 is exactly invariant under the generated subgroup.*

Proof. For each bounded word A , the Pluecker line $[Aw_0]$ is within $O(H_0^{-C})$ of $[w_0]$. Clearing denominators in the fixed integral structure shows that the height of AW_0 is polynomially bounded in H_0 . The separation lemma gives a polynomial lower bound for the distance between two distinct rational planes in this enlarged height range. Choose C larger than the separation exponent. Then $AW_0 = W_0$ for every bounded word. Bounded-word detection gives invariance under the whole generated group. $\square$

K.5 Margulis functions: why they encode repulsion

Let Σ be a finite collection of nearby translates of a candidate proper homogeneous orbit. Define schematically

$$\Phi(x) = \sum_{z \in \Sigma(x)} \text{dist}(x, z)^{-\alpha}$$

with $0 < \alpha$ smaller than the inverse polynomial-divergence degree. The singularity of $r^{-\alpha}$ makes Φ large when many translates crowd near the current point.

Lemma K.6 (Average contraction for one polynomial displacement). *Let $P(t)$ be a nonzero vector-valued polynomial of degree at most d and normalize $\max_{[0,1]} \|P\| = 1$. For sufficiently small $0 < \alpha < 1/d$ there is $C_{d,\alpha}$ such that*

$$\int_0^1 \|P(t)\|^{-\alpha} dt \leq C_{d,\alpha}.$$

Proof. The scalar good-function estimate gives $|\{t : \|P(t)\| < \varepsilon\}| \ll_d \varepsilon^{1/d}$. Use the layer-cake identity

$$\int \|P\|^{-\alpha} = 1 + \alpha \int_1^\infty s^{\alpha-1} |\{\|P\| < s^{-1}\}| ds.$$

The tail is bounded by a constant times $\int_1^\infty s^{\alpha-1-1/d} ds$, which converges when $\alpha < 1/d$. $\square$

Guiding Idea K.7. The preceding integral is the analytic reason a Margulis function can contract on average. Polynomial divergence makes a close encounter occupy only a controlled fraction of time unless an exact polynomial obstruction is present. The exact obstruction is precisely what the height/linearization machinery records.

K.6 The finite-scale closing module assembled slowly

Guiding Idea K.8 (Finite-scale closing roadmap). The exact finite-scale theorem is Theorem D.1. The input is not a generic statement that “equidistribution failed.” It consists of a thick starting point, a positive-measure return-time set E_0 , a proper Lie subalgebra $\mathfrak{h}$, and the explicit near-return and Grassmannian bounds (D.1)–(D.2). The output is the all-times Chevalley bound (D.3) for one proper subgroup or the normal obstruction (D.4). Chapter 8 explains how high return energy supplies this input in the applications.

Dependency derivation. We list the dependency chain because each arrow carries a quantitative loss.

Step 1: recurrence or cusp. Apply quantitative nonescape. On the noncusp branch retain a thick set of times with a fixed injectivity-radius lower bound.

Step 2: pair returns. At a dyadically selected transverse scale, a large Margulis energy produces a positive-measure family of close pairs. Lifting the close pairs to G gives lattice elements of polynomial matrix height.

Step 3: almost-invariant plane. Polynomial conjugation by U and finite sampling give (D.1). The linearization/exterior-power construction supplies a proper Lie subalgebra and the Grassmannian estimate (D.2).

Step 4: pair-return closing. Theorem D.1 returns either a proper normal datum (D.4) or a nonnormal $M \in \mathcal{H}$ satisfying the full-interval bounds (D.3).

Step 5: arithmetic interpretation. The finite Chevalley list and height comparison convert the bounded representative in (D.3) into polynomial subgroup complexity. Marked avoidance and the closed-orbit criteria are then applied with their own stated hypotheses in the later application; no closed-orbit conclusion is silently bundled into the pair-return theorem.

Every constant in this chain is a finite composition of the polynomial losses displayed in Chapters 12–15. This is the precise content hidden by the older phrase “effective closing alternative.”

K.7 Why the restricted-projection theorem is logically later

The closing alternative above does not use the sharp rank-two projection theorem. That theorem is needed when one wants to show that alternative (2) cannot recur at too many scales without creating a single low-complexity algebraic obstruction. The harmonic-analysis appendix supplies precisely that extra scale-counting input.

Worked Example K.9 (A toy projection problem). Consider the parabola $\gamma(t) = (t, t^2)$ and a δ -separated set $E \subset [0, 1]$. If many differences $\gamma(t) - \gamma(s)$ lie in a thin strip with one fixed direction, then

$$(t - s, (t - s)(t + s))$$

forces many values of $t + s$ into a short interval. Thus either the original parameters concentrate, or the difference directions spread. The rank-two restricted-projection theorem is a high-dimensional and multiscale version of this elementary alternative, with wave packets replacing strips.

K.8 A parameter ledger for readers

A proof of an effective closing theorem is not complete unless the reader can track:

1. the height cutoff for candidate rational subgroups;
2. the rational separation exponent;
3. the degree of every adjoint polynomial;
4. the good-function exponent;
5. the number of recurrence samples needed for interpolation;
6. the bounded word length used for exactification;
7. the exponent in the Margulis-function contraction;
8. the quotient injectivity radius on the chosen thick part;
9. the final accumulated distance and height exponents.

The purpose of this volume's detailed appendices is to make those nine items explicit enough that none is replaced by the phrase "effective standard arguments."

Proof Guide and Dependency Summary

This chapter is a finalization layer. It does not introduce a new theorem scope. Its purpose is to expose the proof architecture, make the hypotheses that are easiest to misread impossible to miss, and give a student a concrete model calculation before the abstract machinery is reused later.

1. Effective closing: the finite-scale replacement for rigidity

Qualitative Ratner theory can pass to an infinite-time limit. Effective closing must decide, at a finite scale, whether an orbit segment is equidistributed or shadows an algebraic obstruction of bounded complexity. The proof chain is

- rational heights and Pluecker separation
- effective linearization/avoidance
- many controlled near-returns
- almost-invariant Lie data
- finite-word exactification
- nearby exact homogeneous orbit
- height/complexity bound.

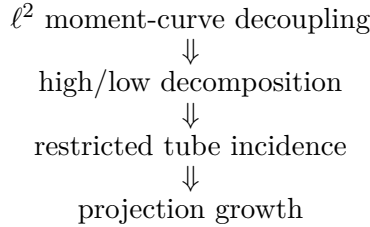
The exactification step is indispensable: an almost-Lie-algebra is not yet an algebraic obstruction.

2. Why finite-word detection is used

An effective proof cannot appeal vaguely to Noetherian termination if it needs an explicit complexity bound. One instead chooses finitely many words in the near-return generators whose brackets/products detect the generated Lie algebra. Once the appropriate determinants are separated from zero, the dimension and rational height of the exact algebraic object are controlled.

3. The restricted-projection module is logically later

The sharp rank-two rate uses a separate harmonic-analysis chain:



This module improves the quantitative rate; it is not needed to make the core closing lemma logically true. Keeping the two chains separate prevents a harmonic-analysis estimate from being mistaken for a structural closing input.

4. Forward proof locations

Three analytic statements are intentionally stated before their proofs. The moment-curve decoupling proof is completed in the decoupling bootstrap section; the high-frequency estimate is then proved in the subsequent high-frequency decomposition; the restricted tube-incidence theorem is closed at the end of that same section. They are forward statements, not black boxes.

5. Student model

In $\mathrm{SL}_2(\mathbf{R}) \ltimes \mathbf{R}^2$, write down an almost-return $xu_{t_1} \approx xu_{t_2}$. The quotient displacement gives a small group element. Conjugating it along the unipotent flow produces a polynomial family of small Lie-algebra vectors. The model shows, in finite dimensions, why repeated near-returns can force an

approximately invariant subspace and why a separate rationality argument is needed to close it exactly.

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Quantitative Oppenheim and Effective Arithmetic Applications

Volume XIV of a Detailed Notebook on Ratner's Theorems

Detailed Notebook on Ratner's Theorems

August 2026

Preface and proof policy

Volume IV proved the qualitative Oppenheim theorem from Ratner orbit closure. The present volume asks for finite-scale information: asymptotic counts of integer vectors in a fixed value window, explicit upper and lower errors when the dynamics is effective, and bounds for how large an integer vector must be before a prescribed small value is attained. The proof is organized so that geometric volume, dynamical equidistribution, cusp truncation, and arithmetic obstructions appear as separate ledger entries. In particular, the phrase “quantitative Oppenheim” is never used as though asymptotic counting and a polynomial rate were the same assertion.

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Dependency and proof-ownership contract

Purpose. Quantitative Oppenheim.

Prerequisites. Volumes IV, IX, XII–XIII. The EMM-type nonexceptional moment, finite-incidence closure, sharp T^{-2} ternary alternative, and Siegel mean value used in this volume are all proved internally.

New objects. scalar coarea charts, Siegel transforms, finite type/EWA, ternary and split quasi-null objects, modified heights, finite exceptional line/plane families, and circle-to-horocycle transfer. Every project-specific technical term is defined before its first load-bearing use or recalled by an exact backward reference.

Main outputs. finite-scale Oppenheim/counting with explicit exceptional geometry.

External-input rule. Every nontrivial imported result is either part of the declared graduate-level prerequisites, an explicitly named external research input in the prerequisite map, or is proved in this series before use (possibly under an explicit deferred-proof contract). A citation by itself never constitutes a proof.

Reader map and proof route

Question of the volume. How does effective homogeneous dynamics produce quantitative Oppenheim counts while keeping cusp tails and arithmetic exceptional configurations separate from the main equidistribution error?

Prerequisite checkpoint. Volumes XI–XIII provide quantitative mixing, smoothing, heights and closing. The two research interfaces needed at the critical arithmetic step are proved in the series: the nonexceptional inverse-covolume moment is Lemma 5.7, and the residual finite-family/sharp finite-scale ternary conversion is proved in Chapter 9 using the sharp expanding-segment theorem of Volume XIII. Siegel’s mean-value normalization is internal as well.

Load-bearing route. First compute the geometric coarea/main term. Then truncate the cusp and apply effective dynamics on the compact part. Treat the tail with a separate mean-value ledger. Finally split the arithmetic analysis into nonexceptional and exceptional signatures/configurations. The last step is where the proved quasi-null rationalization, finite exceptional family, and sharp expanding-segment alternative enter.

Owner boundary. The primitive/nonprimitive decomposition of lattice vectors explains the appearance of zeta factors but does not by itself prove the Haar-normalized Siegel integral; Theorem 3.4 owns that normalization. Likewise, the toy quasi-null calculation is only motivation: the actual finite-scale ownership is carried by the reconstruction and expanding-segment theorems cited in the load-bearing route above.

Sanity check. For the determinant form $Q(x) = x_1x_4 - x_2x_3$, write the vector as a 2×2 matrix. The null cone is the rank-one locus, so rational isotropic

planes are visible as families with one row or one column constrained. This makes the exceptional incidence geometry concrete before the quantitative counting argument.

Interface laboratory: what the Siegel normalization really supplies

Let $f \in C_c(\mathbb{R}^n)$ and, for a unimodular lattice Λ , write

$$\hat{f}(\Lambda) = \sum_{0 \neq v \in \Lambda} f(v).$$

Every nonzero lattice vector has a unique representation $v = mw$ with $m \geq 1$ and w primitive, so formally

$$\hat{f}(\Lambda) = \sum_{m \geq 1} \sum_{w \in \Lambda_{\text{prim}}} f(mw).$$

Suppose for the moment that the mean of the primitive transform were known to be

$$\int_X \sum_{w \in \Lambda_{\text{prim}}} f(w) d\mu(\Lambda) = C_{\text{prim}} \int_{\mathbb{R}^n} f(x) dx.$$

Applying the same formula to $x \mapsto f(mx)$ gives

$$\int_X \hat{f} d\mu = C_{\text{prim}} \sum_{m \geq 1} m^{-n} \int f = C_{\text{prim}} \zeta(n) \int f.$$

Hence the full Siegel normalization $\int_X \hat{f} d\mu = \int f$ is *equivalent*, once one of the two mean formulas is known, to $C_{\text{prim}} = 1/\zeta(n)$.

This computation explains the zeta factor but does not determine the absolute normalization from nothing: the primitive and nonprimitive formulas cannot be used to prove each other simultaneously. The body therefore

supplies a second, independent step. Lemma 3.3 combines Poisson summation with the transpose-inverse symmetry of Haar probability to compute the recursive quotient-volume normalization, and Theorem 3.4 then obtains the full Siegel constant without circularly assuming either mean formula.

Chapter 1

Quantitative Formulations of Oppenheim

1.1 The counting function

Let Q be a nondegenerate indefinite quadratic form on $\mathbb{R}^n$, $n \geq 3$, and let $\Omega \subset \mathbb{R}^n$ be a bounded star-shaped region with piecewise smooth boundary. For a bounded interval $I = (a, b)$ define

$$N_{Q,\Omega,I}(T) = \#\{v \in \mathbb{Z}^n \cap T\Omega : Q(v) \in I\}.$$

The geometric comparison quantity is

$$V_{Q,\Omega,I}(T) = \text{vol}\{x \in T\Omega : Q(x) \in I\}.$$

Since $Q(Ty) = T^2Q(y)$, one expects $V(T)$ to have order T^{n-2} for a fixed value interval.

Definition 1.1 (Three quantitative levels). We distinguish:

1. *upper-bound Oppenheim*: $N(T) \ll T^{n-2}$;
2. *asymptotic Oppenheim*: $N(T) \sim V(T)$;
3. *effective asymptotic Oppenheim*: for some $\eta > 0$,

$$N(T) = V(T) + O(T^{n-2-\eta})$$

with a stated dependence on the form and the region.

Warning 1.2. The adjective “quantitative” in the classical literature often refers to a counting asymptotic, not necessarily to a computable or polynomial rate of convergence. This volume reserves “effective” for an estimate whose error and obstruction complexity are explicitly controlled.

1.2 Irrationality and rational obstructions

A form is *rational up to scale* if $Q = cQ_0$ with $c \neq 0$ and Q_0 a form with rational coefficients. Such forms can have discrete value sets on $\mathbb{Z}^n$ and are excluded by Oppenheim. At finite scale one must also exclude forms that are extremely close to low-height rational forms, because their integer values can shadow the rational model for a long time.

Definition 1.3 (Rational-approximation function). Fix a norm on the vector space $\text{Sym}_n(\mathbb{R})$ of quadratic forms. Put

$$\Delta_Q(B) = \min_{1 \leq q \leq B} \min_{0 \neq Q_0 \in \text{Sym}_n(\mathbb{Z})} \left\| Q - \frac{1}{q} Q_0 \right\|.$$

We call Q (κ, C) -Diophantine if $\Delta_Q(B) \geq CB^{-\kappa}$ for every $B \geq 2$.

This is not the only useful Diophantine condition, but it makes the translation from effective homogeneous obstructions transparent.

Chapter 2

The Geometric Main Term

2.1 Coarea on a quadratic level set

Lemma 2.1 (Scalar coarea formula on a regular compact set). *Let $F \in C^1(U)$ on an open set $U \subset \mathbb{R}^n$, and let $K \Subset U$ be such that $\nabla F \neq 0$ on K . For every continuous φ supported in K and every compact interval J containing $F(K)$,*

$$\int_U \varphi(x) dx = \int_J \left(\int_{F^{-1}(s)} \frac{\varphi(x)}{\|\nabla F(x)\|} d\sigma_s(x) \right) ds. \quad (2.0)$$

Here $d\sigma_s$ is Euclidean hypersurface measure on the regular level set.

Proof. By compactness, cover K by finitely many coordinate boxes on each of which one partial derivative, say $\partial_{x_n} F$, has no zero. A smooth partition of unity reduces the proof to one box. The implicit-function theorem gives coordinates $(y, s) \in \mathbb{R}^{n-1} \times \mathbb{R}$ in which

$$x = (y, \phi(y, s)), \quad F(y, \phi(y, s)) = s.$$

Differentiating in s gives $\partial_s \phi = (\partial_{x_n} F)^{-1}$, so the ordinary change-of-variables formula yields

$$dx = \frac{dy ds}{|\partial_{x_n} F|}. \quad (2.0a)$$

On the level graph, differentiating in y gives $\nabla_y \phi = -\nabla_y F / (\partial_{x_n} F)$, hence

$$d\sigma_s = \sqrt{1 + \|\nabla_y \phi\|^2} dy = \frac{\|\nabla F\|}{|\partial_{x_n} F|} dy. \quad (2.0b)$$

Combining (2.0a) and (2.0b) gives $dx = d\sigma_s ds / \|\nabla F\|$ on the box. Summing the partition of unity proves (2.0). Thus the only analytic inputs are the already declared local implicit-function theorem and ordinary Euclidean change of variables. $\square$

Let Q be nondegenerate. Away from the origin, $\nabla Q(x) \neq 0$ on the cone $Q(x) = 0$: if $Q(x) = x^T A x$ with A invertible, then $\nabla Q(x) = 2Ax$ vanishes only at $x = 0$.

Proposition 2.2 (Coarea formula for a fixed value window). *Assume Ω is bounded and its boundary meets the null cone transversely except on a set of surface measure zero. Then*

$$V_{Q,\Omega,(a,b)}(T) = T^{n-2}(b-a)c_{Q,\Omega} + O_{Q,\Omega,a,b}(T^{n-4})$$

provided there exists $\rho_0 > 0$ such that $\nabla Q(y) \neq 0$ for every $y \in \bar{\Omega} \cap \{Q = 0\}$ with $\|y\| \geq \rho_0$, where

$$c_{Q,\Omega} = \int_{\Omega \cap \{Q=0\}} \frac{d\sigma(y)}{\|\nabla Q(y)\|}.$$

For a general star-shaped region containing the origin, the same leading term holds, with an $o(T^{n-2})$ remainder obtained by excising a small ball around zero and then letting its radius tend to zero.

Proof. Scale $x = Ty$. Then

$$V(T) = T^n \text{vol}\{y \in \Omega : Q(y) \in (aT^{-2}, bT^{-2})\}.$$

On a compact subset of $\Omega \setminus \{0\}$ meeting the null cone, Lemma 2.1 gives

$$\text{vol}\{Q \in (\alpha, \beta)\} = \int_{\alpha}^{\beta} \left(\int_{\Omega \cap \{Q=s\}} \frac{d\sigma_s}{\|\nabla Q\|} \right) ds.$$

The inner integral is C^1 in s under the transversality hypothesis. Taylor expansion at $s = 0$ therefore gives $c_{Q,\Omega}(\beta - \alpha) + O(\alpha^2 + \beta^2)$. Putting $\alpha = aT^{-2}$, $\beta = bT^{-2}$ and multiplying by T^n gives the stated T^{n-2} main term and T^{n-4} error. If the origin belongs to the region, excise $\|y\| < \rho$. Homogeneity shows that its contribution to the coarea constant is $O(\rho^{n-2})$. First let $T \rightarrow \infty$ for fixed ρ , then $\rho \downarrow 0$. $\square$

Worked Example 2.3 (Diagonal ternary form). For $Q(x, y, z) = x^2 + y^2 - \alpha z^2$ and a smooth radial region, the null cone is parametrized by

$$(r \cos \theta, r \sin \theta, r/\sqrt{\alpha})$$

and the coarea integral reduces to an explicit one-dimensional angular integral times a radial factor. Thus the T -growth is $T^{3-2} = T$, exactly as the dynamical formulation predicts.

Chapter 3

The Dynamical Dictionary Reconstructed

3.1 Forms as homogeneous points

Fix a reference form Q_0 of the same signature as Q . After determinant normalization choose $g \in \mathrm{SL}_n(\mathbb{R})$ so that

$$Q(v) = \lambda Q_0(gv)$$

for a nonzero scalar λ . The lattice $g\mathbb{Z}^n$ corresponds to a point of

$$X_n = \mathrm{SL}_n(\mathbb{R}) / \mathrm{SL}_n(\mathbb{Z}).$$

The stabilizer

$$H = \mathrm{SO}(Q_0)^\circ$$

acts on the space of lattices. The irrationality of Q is exactly the failure of the H -orbit to be a closed arithmetic orbit of the rational type identified in Volume IV.

3.2 A smoothed Siegel-transform identity

For $\psi \in C_c(\mathbb{R}^n)$ define the Siegel transform

$$\widehat{\psi}(\Lambda) = \sum_{0 \neq v \in \Lambda} \psi(v).$$

If $\psi_{T,I}$ approximates the indicator of $T\Omega \cap Q^{-1}(I)$ after transporting by g , then

$$N_{Q,\Omega,I}(T) = \hat{\psi}_{T,I}(g\mathbb{Z}^n) + \text{boundary error.}$$

The difficulty is that $\hat{\psi}$ is unbounded on X_n : in the cusp a single short vector creates many multiples inside the support. This is why ordinary weak-* equidistribution of bounded test functions is insufficient.

Lemma 3.1 (Poisson summation for Schwartz functions). *For a Schwartz function f on $\mathbb{R}^n$, define*

$$\mathcal{F}f(\xi) = \hat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i x \cdot \xi} dx.$$

Then $\hat{f}$ is Schwartz,

$$\sum_{m \in \mathbb{Z}^n} f(m) = \sum_{k \in \mathbb{Z}^n} \hat{f}(k), \quad (\text{PS1})$$

and

$$\hat{f}(0) = \int f, \quad \int \hat{f} = f(0). \quad (\text{PS2})$$

More generally, for $g \in \text{SL}_n(\mathbb{R})$,

$$\sum_{m \in \mathbb{Z}^n} f(mg) = \sum_{k \in \mathbb{Z}^n} \hat{f}(kg^{-T}). \quad (\text{PS3})$$

Proof. Repeated integration by parts shows that every derivative of $\hat{f}$ decays faster than any power, so $\hat{f}$ is Schwartz. Periodize

$$P_f(x) = \sum_{m \in \mathbb{Z}^n} f(x + m).$$

Schwartz decay makes this series and all of its differentiated series converge uniformly on the unit cube. Hence P_f is smooth and its k -th Fourier coefficient is, by termwise integration and the change of variables $u = x + m$,

$$\int_{[0,1]^n} P_f(x) e^{-2\pi i k \cdot x} dx = \hat{f}(k).$$

The coefficients are rapidly decreasing, so the Fourier series $\sum_k \hat{f}(k) e^{2\pi i k \cdot x}$ converges absolutely and uniformly with all derivatives. Subtracting it from P_f produces a continuous periodic function R all of whose Fourier coefficients vanish. To avoid hiding a Fourier-uniqueness theorem, let F_N be the one-dimensional Fejér kernel and use the product kernel $K_N(x) = \prod_{j=1}^n F_N(x_j)$.

It is nonnegative, has integral one, and for every fixed neighbourhood U of 0 its integral outside U tends to zero: this follows directly from

$$F_N(t) = \frac{1}{N} \left(\frac{\sin(\pi Nt)}{\sin(\pi t)} \right)^2.$$

Uniform continuity of R therefore gives $R * K_N \rightarrow R$ uniformly. But $R * K_N$ is a finite linear combination of the Fourier coefficients of R , hence is identically zero. Thus $R = 0$. Evaluating at $x = 0$ gives (PS1).

The first identity in (PS2) is the definition. For the second, put $\gamma_\varepsilon(\xi) = e^{-\pi\varepsilon\|\xi\|^2}$. We also record the Gaussian transform rather than cite it. If $I(a) = \int_{\mathbb{R}} e^{-\pi t^2} e^{-2\pi i a t} dt$, differentiation under the integral and integration by parts give $I'(a) = -2\pi a I(a)$; squaring the real Gaussian integral and using polar coordinates gives $I(0) = 1$. Hence $I(a) = e^{-\pi a^2}$. Scaling and taking products now give

$$\int_{\mathbb{R}^n} e^{-\pi\varepsilon\|\xi\|^2} e^{-2\pi i x \cdot \xi} d\xi = \varepsilon^{-n/2} e^{-\pi\|x\|^2/\varepsilon}.$$

Fubini therefore yields

$$\int \hat{f}(\xi) \gamma_\varepsilon(\xi) d\xi = \int f(x) \varepsilon^{-n/2} e^{-\pi\|x\|^2/\varepsilon} dx.$$

As $\varepsilon \downarrow 0$, the right side tends to $f(0)$ by the Gaussian approximate-identity argument, while dominated convergence on the left gives $\int \hat{f}$. This proves (PS2).

Finally apply (PS1) to $x \mapsto f(xg)$. Its Fourier transform is $\hat{f}(\xi g^{-T})$ because $\det g = 1$, which gives (PS3). $\square$

Lemma 3.2 (Primitive-vector unfolding with absolute normalization). *Put $G_n = \mathrm{SL}_n(\mathbb{R})$ and $\Gamma_n = \mathrm{SL}_n(\mathbb{Z})$, and work temporarily on the left quotient $\Gamma_n \backslash G_n$ with row lattices $\mathbb{Z}^n g$. Let*

$$Q_n = \{q \in G_n : e_n q = e_n\} = \left\{ \begin{pmatrix} h & v \\ 0 & 1 \end{pmatrix} : h \in G_{n-1}, v \in \mathbb{R}^{n-1} \right\}.$$

Normalize Haar measure dg_n recursively so that:

1. *on Q_n , the product measure $dg_{n-1} dv$ is used;*
2. *under $Q_n \backslash G_n \rightarrow \mathbb{R}^n \setminus \{0\}$, $Q_n g \mapsto e_n g$, quotient measure is Lebesgue measure.*

If

$$V_n = \text{vol}_{dg_n}(\Gamma_n \backslash G_n), \quad V_1 = 1,$$

then for every nonnegative compactly supported Borel f ,

$$\int_{\Gamma_n \backslash G_n} \sum_{0 \neq m \in \mathbb{Z}^n} f(mg) dg_n = V_{n-1} \zeta(n) \int_{\mathbb{R}^n} f(x) dx. \quad (\text{PU})$$

In particular the integral is finite when f is bounded and compactly supported.

Proof. The finite-volume theorem from Volume V gives $V_n < \infty$. Primitive integer row vectors form the orbit $e_n \Gamma_n$, and their stabilizer is $Q_n(\mathbb{Z}) = Q_n \cap \Gamma_n$. Hence, by Tonelli and quotient unfolding,

$$\int_{\Gamma_n \backslash G_n} \sum_{p \text{ primitive}} f(\ell p g) dg_n = \int_{Q_n(\mathbb{Z}) \backslash G_n} f(\ell e_n g) dg_n.$$

The quotient $Q_n(\mathbb{Z}) \backslash Q_n$ fibers over $\Gamma_{n-1} \backslash G_{n-1}$ with fiber $\mathbb{Z}^{n-1} \backslash \mathbb{R}^{n-1}$. The semidirect action has determinant one, so the product Haar measure descends and

$$\text{vol}(Q_n(\mathbb{Z}) \backslash Q_n) = V_{n-1}.$$

Our normalization of the quotient $Q_n \backslash G_n$ therefore gives

$$\int_{Q_n(\mathbb{Z}) \backslash G_n} f(\ell e_n g) dg_n = V_{n-1} \int_{\mathbb{R}^n} f(\ell x) dx = V_{n-1} \ell^{-n} \int f.$$

Every nonzero integer vector has the unique form ℓp with $\ell \geq 1$ and p primitive. Summing over ℓ gives (PU). $\square$

Lemma 3.3 (The Haar-volume recurrence). *With the normalization of Lemma 3.2,*

$$V_n = \zeta(n) V_{n-1}. \quad (\text{VR})$$

Proof. Let f be Schwartz and set

$$\Theta_f(g) = \sum_{m \in \mathbb{Z}^n} f(mg).$$

Lemma 3.2, first for truncations and then by monotone convergence applied to $|f|$, shows that Θ_f is integrable and

$$\int_{\Gamma_n \backslash G_n} \Theta_f(g) dg_n = V_n f(0) + V_{n-1} \zeta(n) \int f. \quad (\text{VR1})$$

Poisson summation gives

$$\Theta_f(g) = \Theta_{\hat{f}}(g^{-T}).$$

The automorphism $\iota(g) = g^{-T}$ preserves Γ_n . Its pushforward of Haar measure is $c dg_n$ for some $c > 0$; since ι^2 is the identity, $c^2 = 1$, hence $c = 1$. Integrating the Poisson identity and using (VR1) for $\hat{f}$ gives

$$V_n f(0) + V_{n-1} \zeta(n) \int f = V_n \hat{f}(0) + V_{n-1} \zeta(n) \int \hat{f}.$$

By (PS2), this is

$$(V_n - V_{n-1} \zeta(n)) \left(f(0) - \int f \right) = 0.$$

Choose, for example, $f(x) = e^{-4\pi\|x\|^2}$, for which $f(0) = 1$ while $\int f = 2^{-n}$. This proves (VR). $\square$

Theorem 3.4 (Siegel mean-value formula). *Let m_{X_n} be Haar probability measure on $X_n = \mathrm{SL}_n(\mathbb{R})/\mathrm{SL}_n(\mathbb{Z})$. For every $f \in C_c(\mathbb{R}^n)$,*

$$\int_{X_n} \sum_{0 \neq v \in \Lambda} f(v) dm_{X_n}(\Lambda) = \int_{\mathbb{R}^n} f(x) dx. \quad (3.0)$$

More generally, the identity holds for every nonnegative Borel f , with the value $+\infty$ allowed.

Proof. On the left quotient $\Gamma_n \backslash G_n$, Lemma 3.2 and the recurrence $V_n = \zeta(n)V_{n-1}$ give, after dividing by V_n ,

$$\int_{\Gamma_n \backslash G_n} \sum_{0 \neq m \in \mathbb{Z}^n} f(mg) d\bar{m}(g) = \int_{\mathbb{R}^n} f(x) dx \quad (\text{SM1})$$

for every nonnegative compactly supported Borel f . The map

$$G_n/\Gamma_n \longrightarrow \Gamma_n \backslash G_n, \quad g\Gamma_n \longmapsto \Gamma_n g^T,$$

is a diffeomorphism. Transpose is the composition of inversion with the Haar-preserving automorphism $g \mapsto g^{-T}$; since G_n is unimodular, inversion preserves Haar measure as well. Hence the map preserves the normalized quotient measures. The row lattice $\mathbb{Z}^n g^T$ is the transpose of the column

lattice $g\mathbb{Z}^n$. Thus (SM1) is exactly (3.0) for nonnegative f on the right quotient. Positive and negative parts give the formula for C_c functions.

For a general nonnegative Borel f , take explicitly $f_j = \min(f, j)\mathbf{1}_{\{\|x\| \leq j\}}$. These are bounded, compactly supported Borel functions and increase pointwise to f . Tonelli and monotone convergence on both sides of (3.0) give the asserted extended-valued identity. $\square$

Corollary 3.5 (Primitive Siegel mean). *For $n \geq 2$ and every nonnegative Borel f on $\mathbb{R}^n$,*

$$\int_{X_n} \sum_{v \in \Lambda_{\text{prim}}} f(v) dm_{X_n}(\Lambda) = \frac{1}{\zeta(n)} \int_{\mathbb{R}^n} f(x) dx. \quad (3.0p)$$

The same identity holds for compactly supported signed f .

Proof. The first line of the unfolding proof in Lemma 3.2, with $\ell = 1$, gives on the left quotient

$$\int_{\Gamma_n \backslash G_n} \sum_{p \text{ primitive}} f(pg) dg_n = V_{n-1} \int_{\mathbb{R}^n} f(x) dx.$$

Divide by the total volume $V_n = \zeta(n)V_{n-1}$ from Lemma 3.3. The transpose identification between left and right quotients used in the proof of Theorem 3.4 preserves primitivity, so the normalized identity on X_n is exactly (3.0p). Monotone convergence gives the Borel statement; positive and negative parts give the signed compact-support version. $\square$

Worked Example 3.6 (Expected Gaussian theta mass). Approximate $f(x) = e^{-\pi\|x\|^2}$ monotonically by compactly supported functions and use (3.0). Since

$$\int_{\mathbb{R}^n} e^{-\pi\|x\|^2} dx = \left(\int_{\mathbb{R}} e^{-\pi t^2} dt \right)^n = 1,$$

Siegel's formula gives

$$\int_{X_n} \sum_{0 \neq v \in \Lambda} e^{-\pi\|v\|^2} dm_{X_n}(\Lambda) = 1.$$

Thus the expected nonzero Gaussian theta mass of a Haar-random unimodular lattice is exactly one. This calculation makes the normalization in (3.0) visible; it is not an unspecified constant.

Lemma 3.7 (Crude Siegel-transform height bound). *Let*

$$\alpha(\Lambda) = \max_{1 \leq r < n} \max_{\substack{\Delta < \Lambda \\ \text{rank } \Delta = r}} \text{covol}(\Delta)^{-1}.$$

For every compactly supported bounded ψ there are A, C depending on n and the support radius such that

$$|\widehat{\psi}(\Lambda)| \leq C \|\psi\|_{\infty} (1 + \alpha(\Lambda)^A).$$

Proof. If the support lies in a ball of radius R , count lattice points by successive minima. Let $\lambda_1 \leq \dots \leq \lambda_n$ be the successive minima of Λ . A reduced basis argument from Volume I gives

$$\#(\Lambda \cap B_R) \ll_n \prod_{j=1}^n (1 + R/\lambda_j).$$

Products $\lambda_1 \cdots \lambda_r$ are comparable, up to dimension constants, to the smallest covolume of a primitive rank- r sublattice by Minkowski's second theorem. Since the full covolume is one, every inverse product is bounded by a fixed power of $1 + \alpha(\Lambda)$. Multiply by $\|\psi\|_{\infty}$. $\square$

Chapter 4

Height Functions, Cusp Integrability, and the EMM Mechanism

4.1 Modified height

The raw height α is too sensitive to one short sublattice. The classical quantitative Oppenheim proof introduces a weighted maximum over primitive subgroups so that averaging by a suitable unipotent/semisimple piece contracts the dangerous terms unless several short subgroups combine into a larger rational flag.

Let $\Delta < \Lambda$ be primitive and write $d(\Delta) = \text{covol}(\Delta)$. Choose exponents $0 < \theta_r < 1$ satisfying strict convexity inequalities

$$\theta_{r+s} < \theta_r + \theta_s$$

when $r + s < n$. Define

$$\beta(\Lambda) = 1 + \max_{\Delta \text{ primitive}} d(\Delta)^{-\theta_{\text{rank } \Delta}}.$$

Lemma 4.1 (Submodularity of covolume). *For primitive subgroups Δ_1, Δ_2 of a unimodular lattice,*

$$d(\Delta_1 \cap \Delta_2)d(\Delta_1 + \Delta_2) \ll_n d(\Delta_1)d(\Delta_2),$$

where the sum is replaced by its primitive saturation.

Proof. Choose integral bases for the intersection, extend them to bases for each Δ_i , and wedge. The wedge representing the saturated sum divides the integral wedge obtained from the concatenated bases. The norm of a wedge is at most the product of the norms of its orthogonal components, and the common intersection wedge appears on both sides. Applying the Gram determinant identity and absorbing the bounded index of primitive saturation gives the inequality. $\square$

4.2 Contraction away from rational flags

Proposition 4.2 (Height contraction). *Let U be a nontrivial one-parameter unipotent subgroup of $H = \mathrm{SO}(Q_0)^\circ$. For every $1 \leq r \leq n - 1$, let U act on $\bigwedge^r \mathbb{R}^n$ through the r -th exterior power of the defining representation; these matrix coefficients are polynomials in the unipotent parameter. There exist a compactly supported probability density ν on U , $0 < \rho < 1$, and C such that*

$$\int_U \beta(u\Lambda) d\nu(u) \leq \rho\beta(\Lambda) + C$$

unless the primitive subgroup realizing the height is contained in an H -invariant rational flag. In the exceptional case the corresponding flag defines a proper algebraic obstruction to the H -orbit.

Proof. Fix a primitive rank- r subgroup Δ and its wedge v_Δ . Under u_t the wedge $\rho_r(u_t)v_\Delta$ is a vector-valued polynomial. If its projection to every expanding/transverse weight vanishes, the line generated by the wedge lies in an invariant algebraic subspace; the associated rational flag is the exceptional branch. Otherwise the inverse-polynomial integral estimate of Volume XIII gives, for sufficiently small θ_r ,

$$\int \|\rho_r(u)v_\Delta\|^{-\theta_r} d\nu(u) \leq \rho'_r \|v_\Delta\|^{-\theta_r} + C_r$$

with $\rho'_r < 1$ after choosing the averaging scale at which a transverse coefficient becomes visible.

The maximum over Δ cannot be averaged term by term because the maximizer may change with u . Group all subgroups whose covolume is within a fixed factor of the maximum. If two incomparable subgroups are simultaneously dangerous, submodularity produces either a more dangerous intersection or sum. The strict inequalities among the θ_r make this replacement improve the weighted height. Iterating terminates because ranks

strictly change. Thus at each u one obtains a nested flag of marked subgroups, precisely the marking mechanism of Volume II. Summing the one-polynomial contraction along the flag gives the displayed inequality. $\square$

Corollary 4.3 (Uniform cusp tail). *For every orbit segment to which the rational-flag obstruction does not apply,*

$$\frac{1}{T} |\{t \in [0, T] : \beta(xu_t) > R\}| \ll R^{-1}$$

after an initial scale depending polynomially on the starting height.

Proof. Iterate Proposition 4.2 and apply Markov's inequality to the resulting bounded average of β . $\square$

Chapter 5

Truncating the Siegel Transform

5.1 Bounded part and tail

Choose a smooth cutoff χ_R on X_n equal to one on $\{\beta \leq R\}$ and zero on $\{\beta \geq 2R\}$, and decompose

$$\widehat{\psi} = \chi_R \widehat{\psi} + (1 - \chi_R) \widehat{\psi}.$$

The first term is bounded and Sobolev controlled. The second is the cusp tail.

Definition 5.1 (Spherical averaging and rank heights). After replacing Q_0 by $-Q_0$ if necessary, every nonexceptional signature may be written (p, q) with $p \geq 3$ and $q \geq 1$. In the standard coordinates of Chapter 3 put

$$a_t e_1 = e^{-t} e_1, \quad a_t e_n = e^t e_n, \quad a_t e_j = e_j \quad (2 \leq j \leq n-1),$$

let $K < \mathrm{SO}(p, q)^\circ$ be a maximal compact subgroup, and define

$$(A_t F)(\Lambda) = \int_K F(a_t k \Lambda) dm_K(k).$$

For $0 \leq r \leq n$ put $\alpha_0 = \alpha_n = 1$ and

$$\alpha_r(\Lambda) = \max_{\substack{\Delta < \Lambda \\ \text{primitive} \\ \text{rank } \Delta = r}} d(\Delta)^{-1}.$$

Lemma 5.2 (Uniform contraction of one exterior vector). *Assume $p \geq 3$ and fix $1 \leq r \leq n - 1$. For every $0 < s < 2$,*

$$\lim_{t \rightarrow \infty} \sup_{\substack{0 \neq v \in \bigwedge^r \mathbb{R}^n \\ v \text{ decomposable}}} \|v\|^s \int_K \|a_t k v\|^{-s} dm_K(k) = 0. \quad (5.S1)$$

The convergence is uniform in the projective decomposable vector $[v]$.

Proof. By homogeneity normalize $\|v\| = 1$. The normalized decomposable vectors form a compact double cover of the Grassmannian $\text{Gr}(r, n)$, so all constants below only have to be uniform on a compact algebraic family.

Write

$$p_0 = 2^{-1/2}(e_1 + e_n), \quad q_0 = 2^{-1/2}(e_1 - e_n).$$

Then p_0 belongs to the positive subspace of Q_0 , q_0 to the negative subspace, and $K \simeq \text{SO}(p) \times \text{SO}(q)$ rotates these two subspaces independently. At each point of the contracting Schubert locus the hypothesis $p \geq 3$ supplies two tangent rotations in the positive factor whose first variations are linearly independent in directions normal to that locus. By compactness of the Schubert locus and of $\text{Gr}(r, n)$, finitely many Pluecker charts and finitely many such pairs of rotations suffice; after shrinking the charts, their Haar-coordinate Jacobians are uniformly bounded above and below. Expanding the wedge in these finitely many adapted charts therefore gives, uniformly in v and $t \geq 1$,

$$m_K\{k : \|a_t k v\| \leq \rho\} \ll_r \rho^2 \quad (0 < \rho \leq 1), \quad (5.S1a)$$

and, for every fixed $M \geq 1$,

$$\sup_{\|v\|=1} m_K\{k : \|a_t k v\| \leq M\} \longrightarrow 0 \quad (t \rightarrow \infty). \quad (5.S1b)$$

We include the coordinate reason for the exponent two. If $\|a_t k v\| \leq 1$, every coefficient containing the expanding null vector e_n is $O(e^{-t})$, while the neutral coefficients are $O(1)$. At a point where the wedge is aligned with the contracting Schubert cell, the two adapted rotations insert independent positive directions in two linearly independent Pluecker slots. Thus the corresponding two local coordinates must both be $O(\rho)$ when the norm is at most ρ . The adapted-coordinate Jacobians are uniformly bounded below, and the finite-chart cover above proves (5.S1a) by Fubini.

Away from the contracting Schubert cell at least one nonnegative-weight Pluecker coordinate is bounded below; after applying a_t either an expanding coordinate grows like e^t or a neutral coordinate stays bounded below. The only way the norm can stay below a fixed M as $t \rightarrow \infty$ is therefore to enter a shrinking tubular neighbourhood of the cell. The same finite chart calculation proves (5.S1b). This also shows exactly why $p = 2$ is critical: only one independent positive rotation is then available, so (5.S1a) has exponent one rather than two.

We now integrate. Fix $0 < \epsilon < 1 < M$. On $\{\|a_t kv\| < \epsilon\}$, layer-cake integration and (5.S1a) give

$$\int_{\|a_t kv\| < \epsilon} \|a_t kv\|^{-s} dk \ll_s \epsilon^{2-s}. \quad (5.S1c)$$

On $\{\epsilon \leq \|a_t kv\| \leq M\}$ the integrand is at most ϵ^{-s} and the measure tends to zero uniformly by (5.S1b). On the remaining set it is at most M^{-s} . First choose ϵ small, then M large, and finally t large. The three contributions become arbitrarily small, uniformly in v , proving (5.S1). $\square$

Proposition 5.3 (Rank-coupled EMM inequality). *Assume $p \geq 3$ and fix $1 < s < 2$. For every $c > 0$ there are $t_0 > 0$ and $\omega > 1$ such that for $1 \leq r \leq n - 1$,*

$$A_{t_0}(\alpha_r^s) \leq c\alpha_r^s + \omega^2 \max_{1 \leq j \leq \min(r, n-r)} (\alpha_{r-j}\alpha_{r+j})^{s/2} + \omega. \quad (5.S2)$$

The same constants work for all unimodular lattices.

Proof. This is where the adjacent ranks enter. We give the marking argument in full rather than citing an unnamed rank-selection lemma.” By Lemma 5.2, choose t_0 so large that for every decomposable $v \neq 0$ and every rank r ,

$$\int_K \|a_{t_0} kv\|^{-s} dk \leq \frac{c}{4} \|v\|^{-s}. \quad (5.S3)$$

Let

$$L = \max_{1 \leq r < n} \max \left\{ \left\| \bigwedge^r a_{t_0} \right\|, \left\| \bigwedge^r a_{-t_0} \right\| \right\},$$

and choose $0 < \theta < 1$ so small that $L^s \theta^s < c/4$.

Fix a lattice Λ and a rank r . Put

$$A = \alpha_r(\Lambda), \quad B = \max_{1 \leq j \leq \min(r, n-r)} (\alpha_{r-j}(\Lambda) \alpha_{r+j}(\Lambda))^{1/2}. \quad (5.S4)$$

Choose a primitive rank- r subgroup Δ_0 with $d(\Delta_0)^{-1} = A$; discreteness of primitive wedge vectors makes the maximum attained.

Suppose two distinct primitive rank- r subgroups Δ_1, Δ_2 satisfy $d(\Delta_i)^{-1} \geq \theta A$. If their intersection has rank $r-j$, then their sum has rank $r+j$. Primitive saturation and the Gram-determinant submodularity from Chapter 5 give

$$d(\Delta_1 \cap \Delta_2) d(\Delta_1 + \Delta_2) \ll_n d(\Delta_1) d(\Delta_2).$$

Consequently

$$B \gg_n \theta A. \quad (5.S5)$$

After decreasing θ once to absorb the dimensional constant, we may read (5.S5) as $B \geq \theta A$.

There are now two cases. If $A \leq \theta^{-1}B + 1$, the crude exterior bound

$$\alpha_r(a_{t_0}k\Lambda) \leq LA$$

gives

$$A_{t_0}(\alpha_r^s) \ll_{t_0, n, s} B^s + 1, \quad (5.S6)$$

which is absorbed by the second and third terms of (5.S2) once ω is chosen large enough.

Assume instead $A > \theta^{-1}B + 1$. Then (5.S5) says that Δ_0 is the unique primitive rank- r subgroup with inverse covolume at least θA . Every other Δ therefore satisfies

$$d(a_{t_0}k\Delta)^{-1} \leq L\theta A$$

for every k . Hence

$$\alpha_r(a_{t_0}k\Lambda)^s \leq d(a_{t_0}k\Delta_0)^{-s} + (L\theta A)^s. \quad (5.S7)$$

Integrating, using (5.S3), and our choice of θ gives

$$A_{t_0}(\alpha_r^s) \leq \frac{c}{2} A^s.$$

Enlarging ω to cover the finitely many ranks proves (5.S2).

This dominant-mark versus switching dichotomy is the finite-dimensional content of the EMM coupled inequalities: the compact average contracts a single marked wedge, while any attempt by another rank- r subgroup to take over forces a short intersection and sum, which is exactly the neighboring-rank term. $\square$

Lemma 5.4 (Scalar Margulis function for the nonexceptional signatures). *Fix $1 < s < 2$. For every prescribed $0 < \rho_0 < 1$ there are weights $w_r > 0$, a time $t_0 > 0$, and $b < \infty$ such that*

$$F_s(\Lambda) = 1 + \sum_{r=1}^{n-1} w_r \alpha_r(\Lambda)^s \quad \text{satisfies} \quad A_{t_0} F_s \leq \rho_0 F_s + b. \quad (5.S8)$$

Proof. Take $w_r = \varepsilon^{r(n-r)}$ with $0 < \varepsilon \ll 1$. The function $r(n-r)$ is strictly concave and

$$r(n-r) - \frac{(r-j)(n-r+j) + (r+j)(n-r-j)}{2} = j^2.$$

Therefore

$$w_r (XY)^{1/2} \leq \frac{\varepsilon^{j^2}}{2} (w_{r-j} X + w_{r+j} Y). \quad (5.S9)$$

Apply Proposition 5.3 with $c > 0$ still at our disposal, multiply (5.S2) by w_r , and sum in r . Since there are only finitely many ranks, first choose c so small that the sum of the diagonal terms has coefficient at most $\rho_0/3$. With the resulting finite ω , choose ε so small that all the terms obtained from (5.S9) together have coefficient at most another $\rho_0/3$. Finally the constant term 1 on the left contributes only a fixed amount, and the boundary ranks $0, n$ are constants; absorb all of them into b . Enlarging b if necessary gives (5.S8) with the originally prescribed ρ_0 . $\square$

Lemma 5.5 (Wavefront comparison along the self-adjoint ray). *Let $H < \mathrm{GL}_n(\mathbb{R})$ be a connected self-adjoint reductive group, $K = H \cap \mathrm{O}(n)$, and let $(a_t)_{t \in \mathbb{R}} < H$ be a self-adjoint one-parameter subgroup. For every identity neighbourhood $V \subset H$ there is an identity neighbourhood $U \subset K$ such that*

$$a_t U a_r \subset K V a_{t+r} K \quad (t, r \geq 0). \quad (5.S10)$$

Proof. This is the elementary wavefront estimate needed below, and we include its linear-algebra proof. Conjugating the ray by an element of K , arrange that a_t is diagonal in a fixed positive Weyl chamber. Write $d(\lambda_1, \dots, \lambda_n)$ for a positive diagonal matrix with $\lambda_1 \geq \dots \geq \lambda_n > 0$.

Fix $\eta > 0$. If d_1, d_2 are in that chamber and $k \in \mathrm{O}(n)$ is sufficiently close to the identity, then for every $1 \leq i \leq n$,

$$1 \geq \frac{\sigma_1 \cdots \sigma_i(d_1 k d_2)}{(\lambda_1 \cdots \lambda_i)(d_1)(\lambda_1 \cdots \lambda_i)(d_2)} > 1 - \eta. \quad (5.S11)$$

Indeed the upper bound is submultiplicativity of the operator norm on $\bigwedge^i \mathbb{R}^n$. For the lower bound evaluate $\bigwedge^i(d_1 k d_2)$ on $e_1 \wedge \cdots \wedge e_i$. The coefficient of $e_1 \wedge \cdots \wedge e_i$ after applying $\bigwedge^i k$ tends to one uniformly as $k \rightarrow e$, whereas that coordinate is multiplied by the maximal i -fold products of the diagonal entries of d_1 and d_2 . The same neighbourhood of the identity works for all i , because there are only finitely many exterior powers.

Taking quotients of the estimates for i and $i - 1$ shows that every singular value of $d_1 k d_2$ differs multiplicatively from the corresponding singular value of $d_1 d_2$ by a factor tending uniformly to one with η . In the Cartan decomposition

$$d_1 k d_2 = k_1 d' k_2$$

the diagonal d' is therefore $v d_1 d_2$, where v tends uniformly to the identity in the diagonal Cartan subgroup as $\eta \rightarrow 0$. Choose η so small that every such v lies in V . Intersect the resulting orthogonal neighbourhood with K . Taking $d_1 = a_t$ and $d_2 = a_r$ gives (5.S10), since $a_t a_r = a_{t+r}$ and all Cartan factors of an element of H may be taken in K and the Cartan subgroup of H . $\square$

Lemma 5.6 (Propagation of a coarse superharmonic inequality). *In the setting of Lemma 5.5, let $\mathcal{F}$ be a family of strictly positive functions $f : H \rightarrow (0, \infty)$ satisfying:*

1. *for every $\epsilon > 0$ there is an identity neighbourhood $V(\epsilon)$ such that, uniformly for $f \in \mathcal{F}$,*

$$(1 - \epsilon)f(h) \leq f(uh) \leq (1 + \epsilon)f(h) \quad (u \in V(\epsilon), h \in H); \quad (5.S12)$$

2. $f(kh) = f(h)$ for $k \in K$;
3. $\sup_{f \in \mathcal{F}} f(e) < \infty$.

There is a constant $0 < \rho_ < 1$, depending only on the family and on (H, K, a_t) , with the following property. If for some $\tau > 0$ and $b < \infty$*

$$A_\tau f(h) \leq \rho_* f(h) + b \quad (h \in H), \quad (5.S13)$$

then

$$\sup_{r \geq 0} (A_r f)(e) < \infty, \quad (5.S14)$$

uniformly for any subfamily for which $f(e)$ and b are uniformly bounded.

Proof. Right-average first:

$$\tilde{f}(h) = \int_K f(hk) dk.$$

The three hypotheses pass to $\tilde{f}$, (5.S13) passes to it by integrating the pointwise inequality at hk , and

$$\tilde{f}(KhK) = \tilde{f}(h), \quad \tilde{f}(a_r) = (A_r f)(e). \quad (5.S15)$$

Thus it suffices to bound $\tilde{f}(a_r)$.

Use (5.S12) with $\epsilon = 1/2$, and choose the corresponding neighbourhood V . Lemma 5.5 gives an identity neighbourhood $U \subset K$ such that $a_\tau U a_r \subset K V a_{\tau+r} K$ for every $\tau, r \geq 0$. Hence, by bi- K -invariance and (5.S12),

$$\begin{aligned} (A_\tau \tilde{f})(a_r) &= \int_K \tilde{f}(a_\tau k a_r) dk \\ &\geq \int_U \tilde{f}(a_\tau k a_r) dk \geq \frac{m_K(U)}{2} \tilde{f}(a_{\tau+r}). \end{aligned} \quad (5.S16)$$

Crucially, U was chosen from the wavefront lemma independently of τ and r . Set

$$\rho_* := \frac{1}{4} m_K(U).$$

Combining (5.S13) and (5.S16) yields

$$\tilde{f}(a_{\tau+r}) \leq \frac{1}{2} \tilde{f}(a_r) + \frac{2b}{m_K(U)}. \quad (5.S17)$$

Iteration at $r = j\tau$ gives a uniform bound at all integer multiples of τ . For the remainder, compactness of $\{a_u : 0 \leq u \leq \tau\}$ and repeated use of (5.S12) give a finite constant C_τ such that

$$\tilde{f}(a_u h) \leq C_\tau \tilde{f}(h) \quad (0 \leq u \leq \tau), \quad (5.S18)$$

uniformly in the family. Writing $r = j\tau + u$ proves (5.S14). The same recurrence shows the asserted uniformity. $\square$

Lemma 5.7 (EMM nonexceptional-signature mean-value estimate). *Let $H = \mathrm{SO}(p, q)^\circ$, $p + q = n \geq 4$, and assume that the unordered signature is neither $(2, 1)$ nor $(2, 2)$. Then there is $s > 1$ such that for every unimodular lattice Λ ,*

$$\sup_{t \geq 0} \int_K \left(1 + \max_{1 \leq r < n} \alpha_r(a_t k \Lambda)^s \right) dm_K(k) < \infty, \quad (5.0)$$

with a bound uniform when Λ ranges over a compact subset of X_n . Equivalently, the expanding measures ν_T occurring in the Oppenheim unfolding have a uniformly bounded inverse-covolume moment with exponent strictly larger than one.

Proof. After changing Q_0 to $-Q_0$ we have $p \geq 3$. Choose any $1 < s < 2$ and consider

$$f_\Lambda(h) = F_s(h\Lambda).$$

We first verify the equicontinuity hypothesis in Lemma 5.6. For $g \in H$ and a primitive rank- r subgroup $\Delta < \Lambda$,

$$\|\Lambda^r g^{-1}\|^{-1} d(\Delta) \leq d(g\Delta) \leq \|\Lambda^r g\| d(\Delta).$$

Consequently

$$\|\Lambda^r g\|^{-1} \alpha_r(\Lambda) \leq \alpha_r(g\Lambda) \leq \|\Lambda^r g\| \alpha_r(\Lambda). \quad (5.S21)$$

Both exterior-power norms tend to one as $g \rightarrow e$, uniformly over the finitely many ranks. Since F_s is a positive finite weighted sum containing the constant 1, (5.S21) gives (5.S12), uniformly in Λ . Moreover $K \subset O(n)$ preserves every primitive covolume, so $f_\Lambda(kh) = f_\Lambda(h)$.

The family $\{f_\Lambda\}$, with Λ restricted to any compact subset of X_n , also satisfies the normalization hypothesis in Lemma 5.6, because F_s is bounded on compact sets. Let ρ_* be the threshold supplied by that lemma. Apply Lemma 5.4 with any prescribed $0 < \rho_0 < \rho_*$. It gives a time τ and $b < \infty$ for which

$$A_\tau f_\Lambda \leq \rho_0 f_\Lambda + b. \quad (5.S19)$$

The constants in this inequality are independent of the lattice. Therefore Lemma 5.6 gives

$$\sup_{t \geq 0} (A_t F_s)(\Lambda) < \infty. \quad (5.S20)$$

For $0 \leq u < \tau$, boundedness of $a_u k$ and its inverse makes $(A_u F_s)(\Lambda) \ll_\tau F_s(\Lambda)$, uniformly in k . Thus the bound in (5.S20) is uniform when Λ ranges over a compact subset of X_n . Finally all w_r are positive and there are only finitely many ranks, so

$$1 + \max_r \alpha_r^s \ll F_s,$$

which proves (5.0).

This is the rank-coupled Margulis-function mechanism implemented in the full quantitative proof [4]; the argument above supplies the contraction, switching, scalarization, and radial-propagation steps inside this volume. The critical signatures fail at the first step: only a one-dimensional angular sublevel estimate survives in the critical module, so the endpoint inverse moment is $s = 1$. Those signatures are handled by the modified heights in Chapters 7–8. The EMM papers are therefore provenance and theorem concordance, not proof owners in this edition. $\square$

Proposition 5.8 (Uniform integrability of Siegel transforms in the nonexceptional signatures). *Assume the unordered signature is neither $(2, 1)$ nor $(2, 2)$ and exclude the rational-flag obstruction. For every compactly supported ψ there is $p > 1$ such that*

$$\sup_T \int |\widehat{\psi}(\Lambda)|^p d\nu_T(\Lambda) < \infty.$$

Consequently

$$\lim_{R \rightarrow \infty} \sup_T \int |(1 - \chi_R)\widehat{\psi}| d\nu_T = 0.$$

Proof. Use successive minima rather than the crude single height. If $\lambda_1 \leq \dots \leq \lambda_n$ are the successive minima, a reduced-basis count gives

$$|\widehat{\psi}(\Lambda)| \ll_R \prod_{i=1}^n (1 + \lambda_i^{-1}). \quad (5.1)$$

Minkowski's second theorem and the exterior-power dictionary express each partial product $(\lambda_1 \cdots \lambda_r)^{-1}$, up to dimensional constants, as the maximum inverse covolume of a primitive rank- r subgroup. Expanding the product in (5.1) and applying discrete Holder to the finitely many ranks shows that for some small $p > 1$

$$|\widehat{\psi}|^p \ll 1 + \sum_{r=1}^{n-1} \max_{\text{rank } \Delta=r} d(\Delta)^{-\zeta}$$

with ζ no larger than the exponent supplied by Lemma 5.7. Integrating proves the L^p bound. For the tail, Holder gives

$$\int_{\{\beta > R\}} |\widehat{\psi}| \leq \|\widehat{\psi}\|_{L^p(\nu_T)} \nu_T(\beta > R)^{1-1/p},$$

and the uniform cusp-tail estimate makes the right side tend to zero uniformly in T . $\square$

5.2 Why only two signatures are exceptional

The proof above is controlled by the codimension of the zero set of the expanding projection, not by the ambient dimension by itself. It therefore also covers the four-variable signatures $(3, 1)$ and $(1, 3)$. The critical cases are precisely $(2, 1)$ and $(2, 2)$: in the relevant exterior modules a dangerous

vector may have only one transverse compact direction, so the natural inverse moment stops at exponent one. This is exactly the threshold at which a Siegel transform can cease to be uniformly integrable. The failure is arithmetic rather than a failure of Ratner classification.

Chapter 6

The Classical Quantitative Oppenheim Asymptotic

6.1 The exact nonexceptional-signature theorem

Theorem 6.1 (Eskin–Margulis–Mozes asymptotic). *Let Q be a nondegenerate indefinite quadratic form which is not proportional to a rational form. Assume that its unordered signature is neither $(2, 1)$ nor $(2, 2)$. For every bounded interval $I = (a, b)$ and every bounded star-shaped region $\Omega \subset \mathbb{R}^n$ with piecewise smooth boundary such that $\partial\Omega$ meets $\{Q = 0\} \setminus \{0\}$ transversely outside a surface-measure-zero set,*

$$N_{Q,\Omega,I}(T) \sim V_{Q,\Omega,I}(T) \sim c_{Q,\Omega}(b-a)T^{n-2}.$$

Thus the theorem includes the four-variable signatures $(3, 1)$ and $(1, 3)$; the only indefinite signatures excluded are the ternary one and $(2, 2)$.

Proof. Normalize Q and represent it by a lattice point x as in Chapter 3. Put $H = \mathrm{SO}(Q_0)^\circ$. The group H is generated by unipotent one-parameter subgroups. Volume III gives the generated-unipotent orbit theorem and Volume VII gives the family-focusing form needed when the expanding H -pieces vary. Hence every bounded continuous test function on the space of unimodular lattices has the required homogeneous limit, unless the H -orbit lies in a proper rational homogeneous subspace. Volume IV's closed-orbit/rationality argument identifies that alternative with Q being rational up to scale, which is excluded.

The counting observable is not bounded. Cover $T\Omega \cap Q^{-1}(I)$ by finitely many coarea charts and unfold the lattice sum in each chart into a Siegel transform on the corresponding expanding H -piece. At height cutoff R , the function $\chi_R \hat{\psi}$ is bounded and continuous after the harmless smoothing of the boundary, so the homogeneous limit applies. The nonexceptional-signature inverse-moment lemma and Proposition 5.8 show

$$\lim_{R \rightarrow \infty} \sup_T \int |(1 - \chi_R) \hat{\psi}| d\nu_T = 0, \quad (6.1)$$

which justifies removing the cutoff after taking the dynamical limit. Theorem 3.4 turns the limiting lattice average back into the Euclidean integral of the test function. Finally shrink the smoothing width. The boundary contribution is $o(T^{n-2})$ by the coarea estimate, and Proposition 2.2 evaluates the Euclidean volume as $c_{Q,\Omega}(b-a)T^{n-2} + o(T^{n-2})$.

The only place where the signature hypothesis enters is (6.1). The proof of Lemma 5.7 gives an inverse moment strictly larger than one in every signature in the theorem and explains why that same argument fails exactly in $(2, 1)$ and $(2, 2)$. $\square$

Remark 6.2 (Quantitative, but not yet effective). The theorem determines the exact asymptotic scale but the proof still uses qualitative homogeneous equidistribution. A polynomial error term requires the effective machinery of Volumes XI–XIII and the truncation optimization later in this volume.

6.2 Why there is no all-signatures $O(T^{n-2})$ theorem

The expected-order upper bound is itself false without arithmetic hypotheses in the two exceptional signatures. There are irrational forms of signature $(2, 1)$ and $(2, 2)$ and sequences $T_j \rightarrow \infty$ for which the count in a fixed value interval is larger than a constant multiple of $T_j^{n-2} \log^{1-\epsilon} T_j$. The mechanism is exceptionally good approximation by rational forms: an almost rational isotropic configuration persists for a long range of scales and creates too many lattice points. Consequently the correct low-dimensional statement must either remove the singular rational subspaces or impose a Diophantine condition. The next chapter develops exactly that replacement rather than hiding the exception inside an incorrect uniform upper bound.

Chapter 7

Critical Signatures and Modified Margulis Functions

The two exceptional signatures are not failures of homogeneous rigidity. They are failures of uniform integrability for the *unmodified* Siegel transform. The cure is to remove the rational subspaces that are almost null for the form, prove a stronger moment estimate for the resulting modified height, and then add the genuinely singular rational subspaces back into the final count.

7.1 Finite Diophantine type and the EWA obstruction

Normalize a quadratic form by fixing its determinant. If Q_1 is a nonzero integral form of the same signature, let $\rho(Q_1) > 0$ be the unique positive scalar for which $\det(\rho(Q_1)Q_1) = \det Q$. The positivity convention is essential in even dimension, where the determinant equation alone would also allow the negative scalar.

Definition 7.1 (Finite type and EWA). For $M > 1$ we say that Q is of type M if there is $c(Q, M) > 0$ such that

$$\|Q - \rho(Q_1)Q_1\| \geq c(Q, M)\|Q_1\|^{-M} \quad (7.1)$$

for every nonzero integral quadratic form Q_1 of the same signature. We call Q extremely well approximable (EWA) if it is of no finite type. Thus Q is not EWA exactly when it is of type M for some M .

The normalization in (7.1) removes the irrelevant scalar direction. The condition is weaker than prescribing one fixed Diophantine exponent in advance: a form is admissible as soon as *some* polynomial lower bound holds.

7.2 Quasi-null primitive subspaces

Let Q_0 be the standard form and identify Q with a lattice $\Lambda_Q = g_Q \mathbb{Z}^n$ in the Q_0 model. In the ternary case, if Q is represented by its determinant-one symmetric matrix, define the dual form and dual lattice by

$$Q^*(w) = w^t \operatorname{adj}(Q)w, \quad \Lambda_Q^* = (g_Q^{-1})^t \mathbb{Z}^3.$$

Thus Q^* is the quadratic form naturally seen by primitive rank-two exterior vectors, and $\operatorname{adj}(\operatorname{adj}(Q)) = Q$ in dimension three under the determinant-one normalization. For a primitive subgroup Δ of rank r , let v_Δ be its primitive exterior vector. The algebraic null locus in $\bigwedge^r \mathbb{R}^n$ is the finite union $\mathcal{N}_r$ of $H = \operatorname{SO}(Q_0)^\circ$ -invariant varieties corresponding to isotropic lines, isotropic planes, and (in the ternary case) dual-isotropic or equivalently degenerate planes.

The phrase *quasi-null* will be used in two precise low-signature senses below. Before giving those definitions, it is useful to retain one schematic exterior-power model. If Δ is primitive of rank r with primitive exterior vector v_Δ , a generic finite-scale null test has the form

$$d(v_\Delta, \mathcal{N}_r) < R^{-L} \|v_\Delta\|^{-L}. \quad (7.2)$$

This schematic inequality is motivational only: the load-bearing moment theorems use the exact definitions in the next two paragraphs.

Definition 7.2 (Ternary quasi-null vectors and modified height). Let

$$Q_0(x_1, x_2, x_3) = 2x_1x_3 - x_2^2, \quad \mathcal{H}_{\eta, M} = \{v \in \mathbb{R}^3 : |Q_0(v)| < \eta \|v\|^{-50M}\}.$$

For a unimodular lattice $\Lambda < \mathbb{R}^3$ and $g \in \operatorname{SO}(Q_0)^\circ$ set

$$\hat{\alpha}_{1, \eta, M}(g; \Lambda) = \sup\{\|gv\|^{-1} : v \in \Lambda \setminus \mathcal{H}_{\eta, M}\},$$

$$\hat{\alpha}_{2, \eta, M}(g; \Lambda) = \hat{\alpha}_{1, \eta, M}(g^*; \Lambda^*), \quad \alpha_{\eta, M}(g; \Lambda) = \max\{\hat{\alpha}_{1, \eta, M}, \hat{\alpha}_{2, \eta, M}\}.$$

This is Kim's Definition 1.7; in particular, “modified” means that the vectors (and dually the planes) lying in the quasi-null set are omitted from the ordinary Margulis height.

Definition 7.3 (Split-Diophantine forms, quasi-null planes, and modified height). Let $\mathcal{S}_{\text{split}}$ be the projective classes of nonzero rational quadratic forms over $\mathbb{R}^4$ that are $\mathbb{Q}$ -equivalent to $x_1x_4 - x_2x_3$. For a rational form Q' let $\text{ht}(Q')$ be the Euclidean norm of the primitive integral coefficient vector spanning its rational line. A signature- $(2, 2)$ form Q is (η, M) -*split-Diophantine* if

$$\text{dist}([Q], [Q']) \geq \eta \text{ht}(Q')^{-M} \quad ([Q'] \in \mathcal{S}_{\text{split}}). \quad (7.3)$$

In the determinant model $M_2(\mathbb{R})$ write $\bigwedge^2 M_2 = M_c \oplus M_r$ for the column- and row-isotropic summands, with projections π_c, π_r . A rational two-plane with primitive Pluecker vector w is (η, M) -*quasi-null* if

$$\min\{\|w - \pi_c w\|, \|w - \pi_r w\|\} \leq \eta \|w\|^{-M}.$$

For a lattice $\Lambda < M_2(\mathbb{R})$, let $\alpha_r(\Lambda)$ denote the usual inverse primitive r -covolume height. The primary modified height is

$$\hat{\alpha}_{\eta, M}(h; \Lambda) = \max \left\{ 1, \alpha_1(h\Lambda), \alpha_3(h\Lambda), \sup_{\substack{V \text{ } \Lambda\text{-rational two-plane} \\ w_{\Lambda, V} \text{ not quasi-null}}} \|hw_{\Lambda, V}\|^{-1} \right\}. \quad (7.4)$$

These are the split definitions used by the internal reconstruction in Section 7.2.2. Thus every term in the critical-moment statement below has been defined before it is used.

7.2.1 Ternary reconstruction and critical contraction

This subsection reconstructs the ternary part of the modified-height argument in the form used by this volume. The constants are not optimized. The architecture follows the primary proof of [9], but the avoidance input is supplied by Volume XIII rather than imported separately.

Lemma 7.4 (Five-point rational-conic reconstruction). *Let $q \in \mathbb{R}^6$ be the coefficient vector of a normalized ternary quadratic form Q , with $\|q\| \asymp 1$. Let $v_1, \dots, v_5 \in \mathbb{Z}^3$ satisfy $\|v_i\| \leq R$, and let A be the 5×6 integer matrix whose i th row is the vector of quadratic monomials*

$$(v_{i1}^2, v_{i2}^2, v_{i3}^2, 2v_{i1}v_{i2}, 2v_{i1}v_{i3}, 2v_{i2}v_{i3}).$$

If $\text{rank } A = 5$ and $|Q(v_i)| \leq \varepsilon$ for all i , then there is a primitive integral quadratic form Q' with

$$\text{ht}(Q') \ll R^{10}, \quad \text{dist}([Q], [Q']) \ll R^C \varepsilon$$

for an absolute C .

Proof. The kernel of A is a rational line. Its primitive integral generator q' is the vector of signed 5×5 minors of A , divided by their gcd. Every entry of A is $O(R^2)$, hence $\|q'\| \ll R^{10}$.

Choose a nonzero 5×5 minor B of A . Since B is integral, $|\det B| \geq 1$, while $\|B\| \ll R^2$. Cramer's rule therefore bounds $\|B^{-1}\| \ll R^C$. Decompose $q = q_0 + q_\perp$ with $q_0 \in \ker A$ and $q_\perp$ in the coordinate complement determined by B . Since $Aq = (Q(v_1), \dots, Q(v_5))^t$, Cramer's rule gives $\|q_\perp\| \ll R^C \varepsilon$. Thus the projective line $[q]$ is within $O(R^C \varepsilon)$ of $\ker A = [q']$, proving the claim. $\square$

Lemma 7.5 (Failure of five-point reconstruction is degenerate). *Let $S \subset \mathbb{P}^2(\mathbb{Q})$. If the evaluation functionals at any five points of S have rank at most four on the six-dimensional space of ternary quadratic forms, then S is contained in a union of at most four projective lines and at most four isolated points. The bounds are absolute; only boundedness is used later.*

Proof. Let E be the span of the evaluation functionals $\text{ev}_p : F \mapsto F(p)$ for $p \in S$. The hypothesis says $\dim E \leq 4$, so its annihilator $E^\perp$ in the six-dimensional space of ternary quadrics has dimension at least two. Choose linearly independent $F_1, F_2 \in E^\perp$. Then

$$S \subset Z(F_1) \cap Z(F_2). \quad (7.T0a)$$

If F_1 and F_2 have no common irreducible component, the elementary Bezout bound for two plane conics gives at most four projective intersection points. If they have a common component, independence excludes a common quadratic component, so the common component is a projective line L . Write $F_i = LL_i$ with linear L_i (allowing $L_i = L$). Then

$$Z(F_1) \cap Z(F_2) = L \cup (Z(L_1) \cap Z(L_2)),$$

which is one projective line together with at most one additional point unless L_1, L_2 are proportional; the latter would make F_1, F_2 proportional and is impossible. Thus in fact S lies either in at most four points or in one line plus one point. The weaker bound stated in the lemma follows. No general intersection theory is used beyond the degree-two Bezout calculation. $\square$

Lemma 7.6 (Finite type is stable under ternary duality). *If a determinant-one ternary form Q is of finite type, then the dual form $Q^* = \text{adj}(Q)$ is also of finite type (possibly with a larger exponent), and conversely.*

Proof. Suppose Q is of type M . Let P be a nondegenerate integral ternary form and suppose, after determinant normalization, that $\|Q^* - \mu P\| = \varepsilon$ is small. Put $A = \mu P$. Then A lies in a fixed compact neighbourhood of Q^* , and the degree-two adjugate map is Lipschitz there. Hence

$$\|Q - \mu^2 \operatorname{adj}(P)\| = \|\operatorname{adj}(Q^*) - \operatorname{adj}(A)\| \ll_Q \varepsilon. \quad (7.T0b)$$

The matrix $\operatorname{adj}(P)$ is integral and has norm $\ll \|P\|^2$. Replacing the scalar in (7.T0b) by the one giving the fixed determinant normalization changes the error by only a Q -dependent factor: this follows by applying the mean-value theorem to the determinant near the fixed nonsingular matrix Q . The type- M inequality for Q therefore gives

$$\varepsilon \gg_{Q,M} \|P\|^{-2M}.$$

Thus Q^* has finite type (for instance exponent $2M$ after enlarging the constant). Applying the same argument to Q^* and using $\operatorname{adj}(Q^*) = Q$ proves the converse. $\square$

Proposition 7.7 (Ternary quasi-null finite-scale structure). *Let Q be of type M and determinant one. Then there are constants C, L depending on Q, M such that, for every $R \geq 2$, the primitive $v \in \mathbb{Z}^3$ with*

$$R \leq \|v\| < 2R, \quad |Q(v)| \leq R^{-L}$$

lie in a bounded union of rational lines and rational degenerate planes. The same assertion holds for the dual form Q^ .*

Proof. Take L larger than $C + 10M + 10$, where C is the exponent in Lemma 7.4. If five vectors in the shell have rank-five evaluation data, that lemma produces a rational form Q' of height $\ll R^{10}$ with $\operatorname{dist}([Q], [Q']) \ll R^{C-L}$, contradicting the type- M lower bound once R is large. Hence the evaluation rank is at most four, and Lemma 7.5 puts the shell into a uniformly bounded union of rational lines and planes. A plane component on which the null condition persists is degenerate for Q ; otherwise its intersection with the conic contributes only boundedly many lines and may be absorbed into the line family. The finitely many small R are absorbed into the constant. Apply the same argument to Q^* and use Lemma 7.6 for the required finite-type bound. $\square$

For the analytic step use the standard ternary model $Q_0(w) = w_2^2 - 2w_1w_3$; replacing Q_0 by its negative does not change the null cone or any absolute-value estimate. Put

$$a_s = \text{diag}(e^s, 1, e^{-s}), \quad u_r = \begin{pmatrix} 1 & r & r^2/2 \\ 0 & 1 & r \\ 0 & 0 & 1 \end{pmatrix}.$$

For $w_3 \neq 0$ define

$$\rho(w) = -w_2/w_3, \quad \kappa_0(w) = Q_0(w)/w_3^2,$$

and let $\kappa(w)$ be $|\kappa_0(w)|$ when both $|\kappa_0(w)| < 1$ and $|\rho(w)| < 2$, and 1 otherwise. Set

$$\phi_\delta(w) = \kappa(w)^{-2\delta} \|w\|^{-1-\delta}, \quad 0 < \delta \leq 10^{-2}.$$

The definitions extend continuously in the estimates below when $w_3 = 0$.

Lemma 7.8 (Critical vector contraction). *There is an absolute C such that for $s \geq 1$,*

$$\int_{-1}^1 \phi_\delta(a_s u_r w) dr \leq C \delta^{-1} e^{-c\delta s} \phi_\delta(w) \quad (7.T1)$$

for an absolute $c > 0$ and every nonzero $w \in \mathbb{R}^3$. The analogous estimate holds on the dual representation.

Proof. For $w_3 \neq 0$ direct multiplication gives

$$a_s u_r w = w_3 \left(\frac{1}{2} e^s ((r - \rho)^2 - \kappa_0), r - \rho, e^{-s} \right). \quad (7.T2)$$

Moreover $\rho(a_s u_r w) = e^s(\rho - r)$ and $\kappa_0(a_s u_r w) = e^{2s} \kappa_0(w)$. Thus the additional factor $\kappa^{-2\delta}$ can increase only while $|r - \rho| \ll e^{-s}$ and $|\kappa_0(w)| \ll e^{-2s}$; in that interval its increase is exactly compensated by the factor $e^{-4\delta s}$ coming from $\kappa^{-2\delta}$ after transport.

Split $[-1, 1]$ at the real roots (when they exist) of $(r - \rho)^2 - \kappa_0$ and into dyadic annuli of distance $2^j e^{-s}$ from these roots and from $r = \rho$. On such an annulus (7.T2) gives

$$\|a_s u_r w\| \gg |w_3| \max(e^s d^2, d, e^{-s}).$$

after replacing d by the distance to the relevant root. The contribution of an annulus is therefore bounded by a constant times $|w_3|^{-1-\delta} e^{-c\delta s} 2^{-j\delta}$.

Summing the geometric series costs $O(\delta^{-1})$. If the quadratic has no real root, its first coordinate is larger and the same estimate is easier. Finally $|w_3|^{-1-\delta}$ is bounded by a fixed multiple of $\kappa(w)^{-2\delta}\|w\|^{-1-\delta}$ in the only region where it appears; outside that region $\kappa = 1$ and one uses the middle or first coordinate in (7.T2). This proves (7.T1). The cases $w_3 = 0$ follow directly from the linear polynomial obtained from $u_r w$, or by a limit. Duality is obtained by conjugating with the fixed matrix identifying Q_0 and its dual. $\square$

Lemma 7.9 (Short-object competition). *For a unimodular lattice $\Lambda \subset \mathbb{R}^3$, if two linearly independent primitive vectors $v, w \in \Lambda$ satisfy $\|v\|, \|w\| < \epsilon$, then the primitive dual vector represented by $v \wedge w$ has norm $O(\epsilon^2)$. Consequently a height which takes the maximum over primitive vectors and primitive dual vectors has only boundedly many mutually incomparable objects within a fixed factor of its maximum.*

Proof. The exterior vector $v \wedge w$ belongs to the primitive rank-two lattice after division by an integer gcd, so its norm can only decrease; the elementary area bound gives $\|v \wedge w\| \leq \|v\|\|w\|$. In dimension three the Hodge identification turns primitive two-vectors into primitive vectors of the dual lattice. Thus two competing short lines force an even shorter dual object. Iterating this observation gives a chain of dimensions $1 < 2 < 3$, so at each scale the number of incomparable maximizers is bounded by a dimensional constant. $\square$

Proposition 7.10 (One-block modified-height drift). *Let Q be of type M . After increasing M to a fixed larger exponent if necessary, there are $\delta, c > 0$, a block length S_0 , and C_Q such that for every $s \geq S_0$ and every lattice point on the H -orbit of Λ_Q ,*

$$\int_{-1}^1 A(a_s u_r \Lambda) dr \leq e^{-c\delta s} A(\Lambda) + C_Q e^{Cs}, \quad (7.T3)$$

outside an exceptional set E_s , where $A = 1 + \hat{\alpha}_{\eta, M}^{1+\delta}$, while

$$\int_{E_s} A(a_t u_r \Lambda_Q) dr \ll_Q e^{-10s} \quad \text{whenever } t \geq CMs. \quad (7.T4)$$

Proof. Choose, for each r , a primitive line or dual line realizing the modified height within a factor two. Lemma 7.9 partitions $[-1, 1]$ into a bounded number of marked chains on which the same dimensional object remains responsible until another object becomes shorter by a fixed factor. For a non-quasi-null object, Lemma 7.8 applied to the vector or the dual vector gives

the contracting term in (7.T3). The boundaries between marked intervals contribute only the ordinary Margulis height with an exponent strictly below one; Minkowski's inequalities absorb this into $C_Q e^{C_S}$.

The only failure occurs when the responsible object enters the quasi-null family. Proposition 7.7 says that, at each dyadic height, these objects lie in a bounded family of rational line/plane singular strata. Apply the marked effective avoidance theorem of Volume XIII to those strata. If the power-small alternative failed, the whole trajectory would remain close to one rational singular subgroup of controlled height; the Chevalley vector of that subgroup reconstructs a rational conic approximating Q beyond its type- M lower bound. Therefore the power-small alternative always holds for sufficiently large scale.

Finally decompose the modified height on the exceptional set into dyadic levels $e^{4s+j} \leq \hat{\alpha} < e^{4s+j+1}$. Volume XIII's avoidance gives measure $\ll e^{-c_0 j} e^{-c_0 M s}$ after the constants are chosen. Taking $\delta < c_0/4$ makes

$$\sum_{j \geq 0} e^{(1+\delta)(4s+j)} e^{-c_0 j} e^{-c_0 M s} \ll e^{-10s}$$

a convergent geometric series once the fixed enlargement of M is made. This is (7.T4). All constants depend only on the fixed form and the ambient normalizations. $\square$

Lemma 7.11 (Geometric block decomposition). *Fix $B > 1$ and $0 < \varepsilon < (1+B)^{-1}$. For every sufficiently large t there exist $s_1, \dots, s_N$ with*

$$t = s_1 + \dots + s_N, \quad s_1 = B s_2, \quad s_i = (1 + \varepsilon) s_{i+1} \quad (2 \leq i < N),$$

and $S_0 \leq s_N \leq 2S_0$.

Proof. Set $b_k = (\varepsilon^{-1} + B + 1)(1 + \varepsilon)^k - \varepsilon^{-1}$. Then $b_k/b_{k-1} \in [1, 2]$. Choose k and $\tau \in [S_0, 2S_0]$ with $t = b_k \tau$, and set $s_1 = B(1 + \varepsilon)^k \tau$ and $s_i = (1 + \varepsilon)^{N-i} \tau$ for $i \geq 2$. A direct summation verifies all identities. $\square$

Theorem 7.12 (Ternary $1 + \delta$ moment). *Let Q be an indefinite ternary form of determinant one and finite Diophantine type. Then for every fixed $0 < \eta < 1$ there is $\delta > 0$ such that*

$$\sup_{t \geq 0} \int_{-1}^1 \alpha_{\eta, M}(a_t u_r; \Lambda_Q)^{1+\delta} dr < \infty. \quad (7.T5)$$

Equivalently, after changing the constant,

$$\sup_{t \geq 0} \int_K \alpha_{\eta, M}(a_t k; \Lambda_Q)^{1+\delta} dk < \infty. \quad (7.T6)$$

Proof. Replace the type exponent by a larger one so that Proposition 7.10 applies. Choose the block parameters with B so large that the first block is long enough for the avoidance estimate, and choose $\varepsilon \ll \delta$. Decompose t by Lemma 7.11.

The identity $a_s u_r a_{s'} u_{r'} = a_{s+s'} u_{e^{-s'} r + r'}$ shows that successive averaging over the blocks is convolution with interval kernels of widths $e^{-(s_{i+1} + \dots + s_N)}$. Their convolution density is bounded by an absolute constant, equals a positive constant on a fixed central interval, and is supported in a fixed enlargement of $[-1, 1]$. Thus it suffices to bound the iterated block average.

At each block apply (7.T3) on the good set and (7.T4) on the exceptional set. Because the block lengths decrease geometrically, the additive term produced at stage i is multiplied by the contractions of all preceding longer blocks. The exponent inequality

$$C s_{i+1} - c \delta (s_1 + \dots + s_i) \leq -c' \delta s_i$$

holds after the choice of B and ε . Hence the total additive contribution is dominated by a convergent geometric series. The initial height is fixed, so the iterated expectation is uniformly bounded in N and t . The convolution comparison yields (7.T5).

For (7.T6), use the rank-one Cartan parametrization of K : a bounded number of unipotent charts u_r cover K with smooth Jacobians bounded above and below, and the missing endpoints have Haar measure zero. Applying (7.T5) in each chart gives the compact average. Conversely the same chart comparison recovers the unipotent average, so the two formulations are equivalent up to constants. $\square$

Lemma 7.13 (Ternary endpoint first moment). *There is an absolute constant C such that, for every $w \neq 0$ and every $s \geq 0$,*

$$\int_{-1}^1 \|a_s u_r w\|^{-1} dr \leq C \|w\|^{-1}. \quad (7.T7)$$

If $Q_0(w) \neq 0$, then the left side tends to zero as $s \rightarrow \infty$. The same statements hold in the dual representation.

Proof. Normalize $\|w\| = 1$ and use the coordinates in (7.T2). If $w_3 \neq 0$, split the r -interval at the at most two real roots of $(r - \rho)^2 - \kappa_0$ and at distances $e^{-s}, 2e^{-s}, 4e^{-s}, \dots$. On a dyadic annulus of distance d from a root, one of the first two coordinates has size $\gg |w_3| \max(e^s d^2, d)$, while the last coordinate has size $|w_3| e^{-s}$. The interval length is $O(d)$, so the contribution is $O(1)$ after summing the two geometric regimes $d \leq e^{-s}$ and $d > e^{-s}$. If $w_3 = 0$, the same calculation uses the resulting linear polynomial. Compactness of the unit sphere makes the constant uniform.

If $Q_0(w) \neq 0$, then w is separated from the null cone. For any fixed small $\delta > 0$, $\|v\|^{-1} \ll_{w,\delta} \phi_\delta(v)$ along the orbit of w ; Lemma 7.8 therefore bounds the integral by $O_w(e^{-c\delta s})$. Duality is obtained by the fixed identification used in that lemma. $\square$

Proposition 7.14 (Vanishing of nonexact ternary quasi-null objects). *Choose $\eta > 0$ small enough for Proposition 7.7. Let $\mathcal{Q}_{\eta,M}(\Lambda_Q)$ be the primitive quasi-null vectors that are not exactly Q_0 -isotropic, and define the dual family analogously. Then*

$$\lim_{t \rightarrow \infty} \int_K \left(\sup_{v \in \mathcal{Q}_{\eta,M}(\Lambda_Q)} \|a_t k v\|^{-1} + \sup_{w \in \mathcal{Q}_{\eta,M}(\Lambda_Q^*)} \|a_t^* k^* w\|^{-1} \right) dk = 0. \quad (7.T8)$$

Proof. It is enough to prove the unipotent-chart version; the finite Bruhat-chart argument used at the end of Theorem 7.12 then transfers it to K . Divide the primitive quasi-null vectors into shells $2^j R_0 \leq \|v\| < 2^{j+1} R_0$. By Proposition 7.7, each shell is contained in a bounded number, independent of j , of rational lines and degenerate planes. On each line the supremum is attained, up to a fixed factor, by its primitive generator. A degenerate plane is treated dually by its primitive normal. Lemma 7.13 therefore gives a shell contribution $O((2^j R_0)^{-1})$. Summing over $j \geq 0$ gives a uniform tail $O(R_0^{-1})$.

For fixed R_0 only finitely many primitive objects remain. Exact null objects were removed by definition; hence each remaining one has nonzero Q_0 -value (or nonzero dual value). The second assertion of Lemma 7.13 makes each of their integrals tend to zero. First let $t \rightarrow \infty$, then $R_0 \rightarrow \infty$. The dual argument is identical. $\square$

Lemma 7.15 (Ternary modified Lipschitz principle). *Let $f \in C_c(\mathbb{R}^3 \setminus \{0\})$. Delete from its Siegel transform all vectors contained in exact rational isotropic lines and exact rational degenerate planes. There is C_f such that*

$$|\widehat{f}^{\text{ni}}(g; \Lambda_Q)| \leq C_f \left(1 + \alpha_{\eta,M}(g; \Lambda_Q) + S_{\eta,M}(g; \Lambda_Q) + S_{\eta,M}^*(g; \Lambda_Q) \right), \quad (7.T9)$$

where S, S^* are the two suprema occurring in (7.T8).

Proof. Let B be a ball containing $\text{supp } f$. The usual Lipschitz principle is proved by taking the primitive subgroup spanned by $g\Lambda_Q \cap B$ and applying Minkowski's second theorem in that subgroup: if its rank is r , the number of points is bounded by a constant times the inverse r -covolume. In dimension three, $r = 1$ is controlled by a shortest primitive vector and $r = 2$ by the primitive dual vector. If the maximizing primitive object is not quasi-null, its inverse norm is one of the two terms in $\alpha_{\eta, M}$. If it is quasi-null but not exact, its contribution is bounded by S or S^* . Exact null lines and degenerate planes have been deleted. This exhausts the possibilities and proves (7.T9). $\square$

Corollary 7.16 (Ternary unbounded-test-function passage). *Let ψ be compactly supported on $\mathbb{R}^3 \setminus \{0\}$. Delete from its Siegel transform the exact rational isotropic lines and degenerate planes. Along the expanding H -translates of Λ_Q the resulting transforms are uniformly integrable in the limiting sense needed for unbounded equidistribution; bounded-test-function equidistribution therefore passes to this modified Siegel transform.*

Proof. Theorem 7.12 gives a uniform $1 + \delta$ moment for $\alpha_{\eta, M}$. Hence its contribution to the cusp in (7.T9) tends to zero uniformly as the cusp cutoff tends to infinity. Proposition 7.14 shows that the two remaining quasi-null terms tend to zero as $t \rightarrow \infty$. Truncate the modified Siegel transform in the cusp, apply bounded equidistribution to the truncation, then take first $t \rightarrow \infty$ and afterwards the truncation height to infinity. Formula (7.T9) justifies both passages. Approximation from above and below handles bounded compactly supported kernels with negligible discontinuity set. $\square$

7.2.2 Split-(2, 2) reconstruction and critical moment

We now reconstruct the split branch needed by this volume. The model is

$$V = M_2(\mathbb{R}), \quad q(X) = \det X,$$

with

$$H = \text{SL}_2(\mathbb{R}) \times \text{SL}_2(\mathbb{R}), \quad (g_1, g_2)X = g_1 X g_2^{-1}.$$

Write $C = R = \mathbb{R}^2$ and $V = C \otimes R$. The elementary identity

$$\bigwedge^2(C \otimes R) \simeq (\text{Sym}^2 C \otimes \bigwedge^2 R) \oplus (\bigwedge^2 C \otimes \text{Sym}^2 R) \quad (7.S1)$$

gives the H -invariant orthogonal decomposition $\bigwedge^2 V = M_c \oplus M_r$ used in Definition 7.3. In M_c the right factor is trivial and the left factor acts by Sym^2 ; in M_r the roles are reversed. This elementary representation-theoretic splitting is the reason the critical degree can be handled by one-variable quadratic sublevel estimates.

Lemma 7.17 (Column/row Pluecker conics). *A two-plane $P < V$ is contained in $\{\det = 0\}$ if and only if it is of one of the following two forms:*

$$P_c(\ell) = \{xy^t : y \in \mathbb{R}^2\}, \quad P_r(\ell) = \{yx^t : y \in \mathbb{R}^2\},$$

for a line $\ell = \mathbb{R}x < \mathbb{R}^2$. If $x = (s, t)^t$, a Pluecker vector for the column plane is

$$z_c(s, t) = s^2 e_{12} + st(e_{14} - e_{23}) + t^2 e_{34}, \quad (7.S2)$$

and there is an analogous row formula. Consequently the decomposable lines in $\mathbb{P}(M_c)$ and $\mathbb{P}(M_r)$ are nonsingular conics.

Proof. Every nonzero rank-one matrix is xy^t . If a two-dimensional linear space P consists entirely of singular matrices, take two independent elements $x_1 y_1^t, x_2 y_2^t$. If both pairs x_1, x_2 and y_1, y_2 are independent, then

$$\det(x_1 y_1^t + \lambda x_2 y_2^t) = \lambda \det(x_1, x_2) \det(y_1, y_2)$$

is nonzero for $\lambda \neq 0$, a contradiction. Hence either the column lines agree or the row lines agree, proving the classification. Expanding

$$(se_1 + te_3) \wedge (se_2 + te_4)$$

gives (7.S2). In the projective coordinates $[A : B : C]$ the column family is cut out by $q(A, B, C) = AC - B^2$. Its gradient is $(C, -2B, A)$, which can vanish only at $A = B = C = 0$, not at a projective point. Hence this projective conic is nonsingular; the row family is identical after exchanging the two tensor factors. $\square$

Lemma 7.18 (Five nearly isotropic planes reconstruct a rational split form). *Fix a lattice $\Lambda < V$ and a $\mathbb{Z}$ -basis $\mathcal{B}$. There is a constant $C_{\mathcal{B}}$ such that the following holds. Suppose $P_1, \dots, P_5$ are distinct Λ -rational two-planes with primitive Pluecker vectors w_i satisfying*

$$\|w_i\| \leq R, \quad \min\{\|w_i - \pi_c w_i\|, \|w_i - \pi_r w_i\|\} \leq \varepsilon.$$

Then there is a rational split quadratic form q' on $\mathbb{R}^4$ for which

$$\text{ht}(q') \leq C_{\mathcal{B}} R^{20}, \quad \text{dist}([q_{\Lambda, \mathcal{B}}], [q']) \leq C_{\mathcal{B}} R^{20} \varepsilon. \quad (7.S3)$$

The exponent 20 is only a convenient safe exponent; no optimization is used later.

Proof. By pigeonhole, three of the five vectors, say w_1, w_2, w_3 , lie within ε of the same summand; assume it is M_c . Put $\hat{w}_i = w_i / \|w_i\|$. In the coordinates

$$\hat{w} = Ae_{12} + B(e_{14} - e_{23}) + Ce_{34} + De_{13} + E(e_{14} + e_{23}) + Fe_{24},$$

the Pluecker relation is

$$AC - B^2 = DF - E^2. \quad (7.S4)$$

The last three coefficients are $O(\varepsilon)$. At least one of $|A|, |B|, |C|$ is bounded below. If $|A|$ is bounded below replace C by B^2/A ; if $|C|$ is bounded below replace A by B^2/C ; in the remaining case replace B by the root of $B^2 = AC$ having the same sign as B . Equation (7.S4) shows that the change is $O(\varepsilon)$. After normalization we obtain unit decomposable $z_i \in M_c$ with

$$\|\hat{w}_i - z_i\| \ll \varepsilon. \quad (7.S5)$$

Because the w_i are distinct primitive vectors of the fixed exterior lattice, some 2×2 coordinate minor of w_i, w_j is a nonzero element of a fixed discrete lattice. Hence

$$\min_{\pm} \|\hat{w}_i \pm \hat{w}_j\| \gg R^{-2}. \quad (7.S6)$$

By (7.S2), for unit column Pluecker vectors

$$|z_c(s, t) \wedge z_c(s', t')| = (st' - ts')^2,$$

so (7.S5)–(7.S6) imply $|z_i \wedge z_j| \gg R^{-4}$ provided $\varepsilon \leq cR^{-20}$. Thus the three planes P_i are pairwise transverse. If $\varepsilon > cR^{-20}$, (7.S3) is vacuous after enlarging $C_{\mathcal{B}}$ and choosing one fixed rational split form, so we may work in the transverse case.

Pass to the integral coordinates defined by $\mathcal{B}$, so the lattice is $\mathbb{Z}^4$. By Minkowski's second theorem in each plane, choose independent $u_i, v_i \in P_i \cap \mathbb{Z}^4$ with $\|u_i\|, \|v_i\| \ll R$. A quadratic form P vanishes on P_i exactly when

$$P(u_i) = P(v_i) = P(u_i + v_i) - P(u_i) - P(v_i) = 0. \quad (7.S7)$$

The nine functionals (7.S7) form a 9×10 integral matrix L with entries $O(R^2)$. Its kernel is one-dimensional. Indeed $\mathbb{R}^4 = P_1 \oplus P_2$, and P_3 is the graph of an invertible map $T : P_1 \rightarrow P_2$. A quadratic form vanishing on P_1 and P_2 is a bilinear cross-term $B(x, y)$; vanishing on the graph says that the 2×2 matrix of $B(\cdot, T\cdot)$ is alternating, a one-dimensional condition space. Thus $\text{rank } L = 9$.

The signed 9×9 minors of L give a primitive integral generator q' of $\ker L$ and $\text{ht}(q') \ll R^{18}$. Since the P_i are $O(\varepsilon)$ -close to determinant-zero planes, the normalized determinant form $q_{\Delta, \mathcal{B}}$ has each of the nine values in (7.S7) bounded by $O(R^2\varepsilon)$. Choose a nonzero 9×9 minor of L ; its absolute determinant is at least one. Cramer's rule then gives the projective distance bound $O(R^{20}\varepsilon)$. Finally q' vanishes on the rational plane P_1 and is nondegenerate for sufficiently small ε because it is close to the determinant form. A nondegenerate four-variable form vanishing on a rational two-plane has Witt index two and is split over $\mathbb{Q}$. This proves (7.S3). $\square$

Lemma 7.19 (Rational three-space near an exceptional summand). *Let $E = \bigwedge^2 V$ and let $W < E$ be a three-dimensional subspace rational with respect to $\bigwedge^2 \Delta$. Assume W contains a nonzero decomposable vector and $\text{dist}(W, M_c) < c$. Then there is a rational split quadratic form q_W such that*

$$\text{ht}(q_W) \ll \|w_{\bigwedge^2 \Delta, W}\|^3, \quad \text{dist}([q_{\Delta, \mathcal{B}}], [q_W]) \ll \text{dist}(W, M_c). \quad (7.S18)$$

The same holds with M_r .

Proof. Work in the integral coordinates of $\mathcal{B}$. For a basis $\omega_1, \omega_2, \omega_3$ of W and $x \in \mathbb{R}^4$, let $L_W(x)$ be the 3×4 matrix characterized by

$$x \wedge \omega_i \wedge y = (L_W(x)y)_i e_1 \wedge \cdots \wedge e_4.$$

Let $C_W(x)$ be its signed-cofactor vector. A direct computation for M_c gives

$$C_{M_c}(x) = c_0(x_1x_4 - x_2x_3)x. \quad (7.S19)$$

For W sufficiently close to M_c , $L_W(x)$ has rank three on a nonempty open set. Both x and $C_W(x)$ span its kernel, hence the polynomial identities $x_i C_{W,j}(x) - x_j C_{W,i}(x) = 0$ hold on that open set and therefore everywhere. Coprimality of the coordinate polynomials implies $C_{W,i}(x) = x_i q_W(x)$ with one common quadratic form q_W . Formula (7.S19) and continuity of cofactors give the distance estimate.

Choose a lattice basis of $W \cap \bigwedge^2 \Delta$ by Minkowski's theorem. Each basis vector has norm $O(\|w_{\bigwedge^2 \Delta, W}\|)$; the cofactors are cubic in their coordinates, so after clearing a fixed denominator q_W has integral coefficients of size $O(\|w_{\bigwedge^2 \Delta, W}\|^3)$. Finally, a decomposable vector $u \wedge v \in W$ forces q_W to vanish on $\text{span}(u, v)$, because $L_W(x)$ drops rank there. A nearby nondegenerate four-variable form with a rational null two-plane is split over $\mathbb{Q}$. $\square$

Corollary 7.20 (Exact and finite-scale split singular families). *If Λ is not determinant-rational, it has at most two rational column-isotropic planes and at most two rational row-isotropic planes. If it is (η_0, M_0) -split-Diophantine, then after increasing M and decreasing η one has, for every $R \geq 1$,*

$$\# \left\{ P : \|w_P\| \leq R, \min(\|w_P - \pi_c w_P\|, \|w_P - \pi_r w_P\|) \leq \eta R^{-M} \right\} \leq 4. \quad (7.S8)$$

Proof. Three distinct exact column planes have Pluecker vectors of the form (7.S2). Their three slopes are distinct, so the Vandermonde determinant is nonzero and the three primitive rational vectors span M_c . Hence M_c itself is rational with respect to $\bigwedge^2 \Lambda$. Apply Lemma 7.19 with $W = M_c$: the cofactor construction there produces a rational split form q_W . Identity (7.S19) in that proof shows that the decomposable cone cut out by W is exactly the column ruling of the determinant null quadric. Two nondegenerate split quadrics with the same ruling have proportional polar forms (equivalently, the same three-dimensional maximal-isotropic family determines the quadratic form up to scalar), so $q_{\Lambda, \mathcal{B}}$ is rational up to scale, contrary to determinant-irrationality. Thus there are at most two column planes. The same argument with M_r gives at most two row planes.

For the finite-scale assertion, five such planes and Lemma 7.18 produce a rational split form of height $\ll R^{20}$ at projective distance $\ll \eta R^{20-M}$. Choose $M > 20(M_0 + 1)$ and then $\eta > 0$ so small that this is less than $\eta_0 (CR^{20})^{-M_0}$ for all $R \geq 1$. This contradicts the split-Diophantine lower bound, proving (7.S8). $\square$

Worked Example 7.21 (Why five planes, but only four can survive). Take the determinant lattice $g\mathbb{Z}^4$ and suppose five primitive rational planes of Pluecker norm at most R lie within R^{-100} of the two exceptional summands. Three lie near the same summand. Their normalized integral Pluecker vectors are separated by at least $\asymp R^{-2}$, whereas the perturbation is R^{-100} . They therefore determine three transverse near-isotropic planes. The nine equations (7.S7) determine one projective quadratic form, of height at most R^{20} , and

its distance from the original form is at most R^{-80} . A finite-type lower bound of exponent, say, 10 forbids this once R is large. Thus the abstract phrase “bounded singular family” is literally a finite-dimensional integer linear-algebra statement.

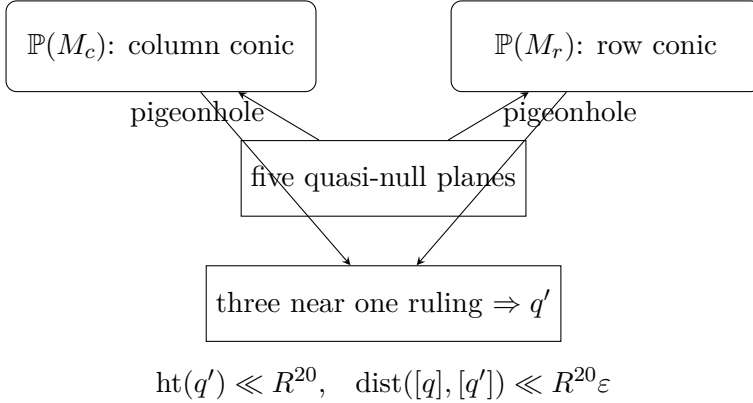


Figure 7.1: The split $(2, 2)$ reconstruction mechanism. Five quasi-null rational planes force three near one Pluecker ruling; those three supply nine rational vanishing equations, which reconstruct a low-height split quadratic form. This is the geometric reason a Diophantine hypothesis leaves at most four exceptional planes.

Local analytic estimates. Put

$$b_s = \begin{pmatrix} e^s & 0 \\ 0 & e^{-s} \end{pmatrix}, \quad u_\xi = \begin{pmatrix} 1 & \xi \\ 0 & 1 \end{pmatrix}, \quad a_s = (b_s, b_s), \quad n_{\xi_1, \xi_2} = (u_{\xi_1}, u_{\xi_2}).$$

The following lemma contains all representation theory needed below.

Lemma 7.22 (Standard and symmetric-square sublevel integrals). *Let $I \subset \mathbb{R}$ be bounded.*

1. *If $W \simeq \mathbb{R}^2$ is the standard SL_2 -module and $1 < \beta < 2$, then*

$$\int_I \|b_s u_\xi v\|^{-\beta} d\xi \ll_{I, \beta} e^{-(2-\beta)s} \|v\|^{-\beta}. \quad (7.S9)$$

2. *If $W \simeq \text{Sym}^2 \mathbb{R}^2$ and $0 < \beta < 1$, then*

$$\int_I \|b_s u_\xi v\|^{-\beta} d\xi \ll_{I, \beta} e^{-c_\beta s} \|v\|^{-\beta}, \quad c_\beta = \min\{\beta, 1 - \beta\}/2. \quad (7.S10)$$

At $\beta = 1$ the same integral is $O_I(1)\|v\|^{-1}$.

The estimates also hold in the dual modules.

Proof. Scale v to norm one. In the two-dimensional defining module write $v = (x, y)$. If $|x| \geq 1/2$, translate ξ by $-y/x$. Up to fixed constants the integral is bounded by

$$\int_{I'} (e^{-2s} + e^{2s}\xi^2)^{-\beta/2} d\xi = e^{-(2-\beta)s} \int_{e^{-2s}I'} (1 + u^2)^{-\beta/2} du,$$

and the last integral is uniformly bounded for $\beta > 1$. If $|x| < 1/2$, then $|y| \gg 1$ and the expanding coordinate is uniformly large except on an interval of size $O(e^{-2s})$; the same estimate follows by splitting that interval and its complement.

For Sym^2 , write $v = AZ_1^2 + BZ_1Z_2 + CZ_2^2$. The three weight coordinates of $u_\xi v$ are, up to harmless constants,

$$A, \quad B + 2A\xi, \quad C + B\xi + A\xi^2. \quad (7.S11)$$

The last is a quadratic polynomial. On a bounded interval a normalized nonzero polynomial of degree at most two satisfies the elementary sublevel estimate

$$|\{\xi : |p(\xi)| \leq r \sup_I |p|\}| \ll_I r^{1/2}. \quad (7.S12)$$

Indeed factor $p = A(\xi - r_1)(\xi - r_2)$ over $\mathbb{C}$; if a complex root has imaginary part larger than $\sqrt{r}$, it gives no small set, and near a real root the condition is contained in an interval of length $O(\sqrt{r})$. Applying the layer-cake formula to (7.S12), and using the middle linear coordinate when the quadratic coefficient is small, reduces the worst case to

$$\int_{I'} (e^{-4s} + \xi^2 + e^{4s}\xi^4)^{-\beta/2} d\xi.$$

Split first at $|\xi| = e^{-2s}$. For $|\xi| \leq e^{-2s}$ the integrand is $O(e^{2\beta s})$ on an interval of length $O(e^{-2s})$. For $|\xi| > e^{-2s}$ the quartic term dominates the linear-scale term in the model denominator, so the integrand is $O(e^{-2\beta s}|\xi|^{-2\beta})$. Integrating this from e^{-2s} to a fixed endpoint gives $O(e^{-2(1-\beta)s}) + O(e^{-2\beta s})$ for $0 < \beta < 1$, while at $\beta = 1$ it is $O(1)$.

The cases in which the quadratic coefficient is small are easier because the middle linear coordinate in (7.S11) supplies the same lower bound after one translation. This proves (7.S10), including the endpoint $\beta = 1$. Duality preserves the weight list. $\square$

For $0 < \theta < 10^{-4}$ and $w \notin M_c \cup M_r$ define the critical weight

$$\phi_\theta(w) = \|w\|^{-1+4\theta} \|w - \pi_c w\|^{-2\theta} \|w - \pi_r w\|^{-2\theta}. \quad (7.S13)$$

Set $\phi_\theta = +\infty$ on the two exceptional summands.

Lemma 7.23 (Weighted critical-degree contraction). *For $0 < \theta < 10^{-4}$,*

$$\int_{[-1,1]^2} \phi_\theta(a_s n_{\xi_1, \xi_2} w)^{1+\theta} d\xi_1 d\xi_2 \ll_\theta e^{-\theta s} \phi_\theta(w)^{1+\theta}. \quad (7.S14)$$

Proof. Write $w = w_c + w_r$ according to (7.S1). Assume $\|w_c\| \geq \|w_r\|$; the other case is symmetric. The two factors act separately, so

$$(a_s n w) - \pi_c(a_s n w) = b_s u_{\xi_2} w_r, \quad (a_s n w) - \pi_r(a_s n w) = b_s u_{\xi_1} w_c.$$

Since the summands are complementary,

$$\phi_\theta(w)^{1+\theta} \asymp \|w_c\|^{-(1+\theta)(1-2\theta)} \|w_r\|^{-2\theta(1+\theta)}.$$

Both exponents are < 1 . Apply (7.S10) independently in ξ_1 and ξ_2 . The two decay exponents add to at least θ when $\theta < 10^{-4}$, which gives (7.S14). $\square$

Lemma 7.24 (Pluecker intersection–sum inequality). *For rational subspaces P, W of a fixed lattice $\Delta < \mathbb{R}^4$,*

$$\|w_{P \cap W}\| \|w_{P+W}\| \ll_\Delta \|w_P\| \|w_W\|. \quad (7.S15)$$

Covolumes in dimensions zero and four are interpreted as one.

Proof. Let $L_P = P \cap \Delta$ and $L_W = W \cap \Delta$. The homomorphism $L_P \oplus L_W \rightarrow L_{P+W}$, $(x, y) \mapsto x + y$, has kernel naturally identified with $L_{P \cap W}$ and image of finite index in L_{P+W} . Choose Euclidean bases adapted to the exact sequence. Taking determinants of the corresponding Gram matrices gives

$$\text{covol}(L_{P \cap W}) \text{covol}(L_{P+W}) \leq C_\Delta \text{covol}(L_P) \text{covol}(L_W),$$

where C_Δ only records the fixed comparison between the ambient integer coordinates and the chosen Euclidean norm. Primitive exterior vectors have exactly these covolumes. $\square$

Define the auxiliary height

$$\begin{aligned} \tilde{\alpha}_{\eta,M,\theta}(h; \Delta) = \max\{ & 1, \alpha_1(h\Delta)^{1+\sqrt{\theta}}, \alpha_3(h\Delta)^{1+\sqrt{\theta}}, \\ & \sup_{\substack{P \text{ } \Delta\text{-rational, } w_P \notin \mathcal{H}_{\eta,M} \\ 0 < \|hw_P\| \leq 1}} \phi_\theta(hw_P)^{1+\theta}, \\ & \sup_{\substack{P \text{ } \Delta\text{-rational, } w_P \in \mathcal{H}_{\eta,M} \\ 0 < \|hw_P\| \leq 1}} \|hw_P\|^{-(1-\theta)}\}. \end{aligned}$$

Here $\mathcal{H}_{\eta,M}$ is the quasi-null set from Definition 7.3. The indexing sets are finite because primitive exterior vectors form a discrete set. Directly from the definitions,

$$\hat{\alpha}_{\eta,M}(h; \Delta)^{1+\theta} \ll_{\theta} \tilde{\alpha}_{\eta,M,\theta}(h; \Delta), \quad \alpha(h\Delta)^{1-\theta} \ll_{\theta} \tilde{\alpha}_{\eta,M,\theta}(h; \Delta). \quad (7.S17)$$

For $M' \geq 1$ and $s \geq 0$ put

$$\varepsilon_{s,M'}(h; \Delta) = \begin{cases} e^{-M's}, & \alpha(h\Delta) \leq e^{5(s+1)}, \\ \alpha(h\Delta)^{-M'}, & \alpha(h\Delta) > e^{5(s+1)}. \end{cases} \quad (7.S20)$$

Let $\mathcal{E}_{s,\eta,M,M'}$ consist of pairs (h, Δ) for which there is a non-quasi-null rational two-plane P with

$$\begin{aligned} 0 < \|hw_P\| & \leq e^{4(s+1)}, \\ \min\{\|hw_P - \pi_c hw_P\|, \|hw_P - \pi_r hw_P\|\} & \leq \varepsilon_{s,M'}(h; \Delta). \end{aligned} \quad (7.S21)$$

Proposition 7.25 (One-step split contraction away from the exceptional set). *There is $C_\theta > 1$ such that, if $M'\sqrt{\theta} \leq 1/20$ and $(h, \Delta) \notin \mathcal{E}_{s,\eta,M,M'}$, then*

$$\int_{[-1,1]^2} \tilde{\alpha}_{\eta,M,\theta}(a_s n h; \Delta) \, dn \leq C_\theta e^{-\theta s/2} \tilde{\alpha}_{\eta,M,\theta}(h; \Delta) + C_\theta e^{C_\theta s}. \quad (7.S22)$$

Proof. Set $A = \tilde{\alpha}_{\eta,M,\theta}(h; \Delta)$. For degrees one and three, apply (7.S9) with exponent $1 + \sqrt{\theta}$. Among the finitely many subspaces whose transformed covolume can become < 1 , choose a maximizer. For any second candidate, (7.S15) and (7.S17) imply that the product of their inverse covolumes is $O(A^{1/(1-\theta)})$. Hence every nonmaximal candidate is $O(A^q)$ for a $q < 1$. Young's inequality $uA^q \leq \rho A + C_q \rho^{-q/(1-q)} u^{1/(1-q)}$, with $\rho = e^{-(1-\sqrt{\theta})s}$, absorbs all nonmaximal terms into $e^{-cs} A + e^{Cs}$.

For a non-quasi-null degree-two plane use the weighted estimate (7.S14). If one candidate maximizes the critical weight, integrate it directly. For a second candidate, (7.S15) again forces either a degree-one or degree-three covolume to become small; (7.S17) therefore bounds the product of the two critical weights by $A^{1-\sigma}$ after the harmless powers appearing in (7.S13). Young's inequality absorbs the second and all subsequent candidates. The only way this exponent improvement can fail is that a non-quasi-null candidate lies within the threshold (7.S21) of M_c or M_r ; this is precisely excluded by the hypothesis.

For a quasi-null plane the last supremum in (7.S16) uses exponent $1-\theta < 1$. The endpoint form of (7.S10), together with the finite-scale bound (7.S8), gives at most four candidates at each dyadic covolume. The logarithmic endpoint loss is dominated by $e^{\theta s/8}$ and the remaining power contracts by $e^{-\theta s}$. Summing the dyadic ranges and again using Young's inequality gives $O(e^{-\theta s/2}A + e^{C\theta s})$. Combining the three exterior degrees proves (7.S22). $\square$

Proposition 7.26 (Split exceptional-set avoidance). *Fix a split-Diophantine lattice Λ and parameters η, M for which (7.S8) holds. There is $B > 1$ such that, whenever $M' \geq BM^2$ and $t \geq BM's$,*

$$\int_{[-1,1]^2} \alpha(a_t n \Lambda)^2 \mathbf{1}_{\mathcal{E}_{s,\eta,M,M'}}(a_t n, \Lambda) \, dn \ll e^{-M's/(BM)} + e^{-t/(BM)}. \quad (7.S23)$$

Proof. We give the reduction to the marked quantitative-nondivergence theorem of Volume XIII, including the arithmetic obstruction check. Consider first approach to M_c . Let $E = \Lambda^2 V$ and let Υ be the primitive Pluecker vectors of non-quasi-null rational two-planes. Rescale E by the linear map D which acts by $\varepsilon/(2e^{4(s+1)})$ on M_c and by $1/2$ on M_r , where ε is the threshold in (7.S21). If (7.S21) holds then $\|Da_t n w\| \leq \varepsilon$ for some $w \in \Upsilon$.

The coordinates of every exterior power of $Da_t n$ are polynomials of bounded degree in (ξ_1, ξ_2) , hence are good functions by the polynomial Remez estimate proved with quantitative nondivergence in Volume II. Apply Volume XIII's marked-set avoidance theorem with $\rho = \varepsilon^{1-\delta}$, where $0 < \delta \ll 1/M$. Unless the desired power estimate for the parameter set holds, it returns linearly independent $w_1, \dots, w_m \in \Upsilon$ for which the primitive wedge $\mathbf{w} = w_1 \wedge \dots \wedge w_m$ remains ρ^m -small throughout the marked ball.

Decompose $\mathbf{w}$ according to $E = M_c \oplus M_r$. Every nontrivial irreducible summand expands under $a_t N$ by $e^{\kappa t}$ for a fixed $\kappa > 0$: in $\text{Sym}^p \mathbb{R}^2 \otimes \text{Sym}^q \mathbb{R}^2$ the highest-weight coefficient of $n\mathbf{w}$ is a nonzero polynomial and its supremum on the unit square controls the norm of $\mathbf{w}$. With $t \geq BM's$ and B large,

all components except a possible fixed component in $\bigwedge^3 M_c$ are therefore $O(\varepsilon^{1/2})$. If $m \neq 3$, discreteness of the exterior lattice makes this impossible for small ε . If $m = 3$, the span $W = \text{span}(w_1, w_2, w_3)$ is a rational three-space within $O(\varepsilon^c)$ of M_c and has primitive height at most $\varepsilon^{-C\delta}$. It contains a decomposable vector because each w_i is a Pluecker vector. Lemma 7.19 then produces a rational split form of polynomial height whose projective distance from $q_{\Lambda, B}$ is $O(\varepsilon^c)$, contradicting the split-Diophantine lower bound when δ is chosen first and B is then large. Thus the marked-set measure is $O(\varepsilon^{c/M})$.

It remains to insert the weight α^2 without asking for an artificially large nondivergence exponent. Let B_0 be chosen so that the unweighted marked-set estimate just proved has the form

$$|\mathcal{K}_{\eta, M}(s, \varepsilon)| \ll \varepsilon^{1/(B_0 M)} \quad (7.S23a)$$

whenever $\varepsilon < e^{-B_0 M s}$ and $t \geq B_0 \log(1/\varepsilon)$. Fix $B_1 > 8B_0^2$. The deterministic height alternatives used in (7.S22), together with $M' \geq BM^2$ and $t \geq BM's$, imply on the exceptional set

$$\alpha(a_t n \Lambda) \leq C_1 e^{t/(B_1 M)}. \quad (7.S23b)$$

Indeed, if this failed, the non-quasi-null primitive plane witnessing the exceptional alternative would satisfy $\text{dist}(w, M_c \cup M_r) \leq \eta \|w\|^{-M}$, contradicting its definition.

Put $h_j = e^{10(s+1)} e^{10j}$ and let J be minimal with $h_J \geq C_1 e^{t/(B_1 M)}$. Assign an exceptional parameter to the least j with $\alpha(a_t n \Lambda) \leq h_j$. The $j = 0$ layer is contained in $\mathcal{K}_{\eta, M}(s, e^{-M's})$; for $j \geq 1$ the minimality of j and the exceptional alternative put the layer in $\mathcal{K}_{\eta, M}(s, h_{j-1}^{-M'})$. Therefore

$$\int \alpha(a_t n \Lambda)^2 \mathbf{1}_{\mathcal{E}} dn \ll h_0^2 |E_0| + \sum_{j=1}^J h_j^2 |E_j|. \quad (7.S23c)$$

Call j low when $B_0 M' \log h_{j-1} \leq t/2$. For low j , (7.S23a) gives $h_j^2 |E_j| \ll h_{j-1}^{2-M'/(B_0 M)}$; since $M' \geq BM^2$, these terms and E_0 sum to $O(e^{-M's/(BM)})$. For the remaining indices use the common smaller scale $\varepsilon_* = e^{-t/(2B_0)}$ in (7.S23a), obtaining $|E_j| \ll e^{-t/(2B_0^2 M)}$. Minimality of J gives $h_J^2 \ll e^{20(s+1)} e^{2t/(B_1 M)}$ and $J + 1 \ll 1 + s + t$. Because $1/(2B_0^2) - 2/B_1 > 0$, increasing B makes the remaining sum $O(e^{-t/(BM)})$. This proves (7.S23) without any powering trick. The M_r case is identical. $\square$

Theorem 7.27 (Split critical moment). *Let $\Lambda < M_2(\mathbb{R})$ be split-Diophantine and not determinant-rational. Choose η, M so that (7.S8) holds. Then there is $\theta' > 0$ such that*

$$\sup_{t \geq 0} \int_K \hat{\alpha}_{\eta, M}(a_t k; \Lambda)^{1+\theta'} dk < \infty. \quad (7.S24)$$

Proof. Choose the constant B in Proposition 7.26 and put $M' = 100B(C + 1)M^2$, where C dominates the harmless exponential term in (7.S22). Choose $\theta > 0$ with $M'\sqrt{\theta} < 1/20$ and then choose $D \geq 4BM'$ and $\delta = \theta/[100D(C_\theta + 1)]$.

For large t , decompose

$$t = s_1 + \cdots + s_I, \quad s_1 = Ds_2, \quad s_i = (1 + \delta)s_{i+1} \quad (2 \leq i < I), \quad (7.S25)$$

with the last block in a fixed large interval. This is obtained by taking a geometric progression backwards from the last block and adjusting s_1 ; all blocks remain positive because $\delta < (1 + D)^{-1}$.

The product formula

$$a_{s_I} n_I \cdots a_{s_1} n_1 = a_t n_{\xi_1 + e^{-2s_1} \xi_2 + \cdots + e^{-2(s_1 + \cdots + s_{I-1})} \xi_I}$$

shows that the product measure on the blocks has a convolution density bounded above on a fixed enlarged N -ball and bounded below on a fixed smaller ball. It therefore suffices to bound the iterated block average.

On paths for which the state before block $i + 1$ is outside $\mathcal{E}_{s_{i+1}, \eta, M, M'}$, apply (7.S22). On the exceptional paths, use the deterministic log-Lipschitz bound for the ordinary height and Proposition 7.26. Holder with a small exponent $\theta' \leq \theta$ turns (7.S23) into an error

$$\ll \exp\{C\theta' s_i - M' s_{i+1}/(2BM)\}.$$

The parameter choices make this summable over i . Iterating (7.S22) gives

$$\begin{aligned} \mathbb{E} \tilde{\alpha}_{\eta, M, \theta}(h_I; \Lambda) &\ll e^{-c\theta t} \tilde{\alpha}_{\eta, M, \theta}(e; \Lambda) \\ &+ \sum_{i=1}^I \exp\left(C_\theta s_i - \frac{\theta}{4}(s_{i+1} + \cdots + s_I)\right) + O(1). \end{aligned}$$

For all but $O(\delta^{-1})$ terminal blocks, $s_{i+1} + \cdots + s_I \geq s_i/(2\delta)$, so the displayed series is geometric; the finitely many terminal terms are bounded because their lengths lie in a fixed multiple of the terminal scale. Thus the iterated

expectation is uniformly bounded. By (7.S17) the same is true for $\hat{\alpha}^{1+\theta'}$ over a fixed N -ball.

Finally cover $K = \mathrm{SO}(2) \times \mathrm{SO}(2)$ by finitely many Bruhat charts $k = p(k)n(k)\gamma$ with $n(k)$ in a fixed N -ball and uniformly bounded Jacobians. Conjugating $p(k)$ by a_t stays in a fixed compact set, and the height is log-Lipschitz under compact multiplication. Each K -integral is therefore bounded by the already controlled N -integral, proving (7.S24). $\square$

Lemma 7.28 (Split critical first moment). *There is $C \geq 1$ such that, for every $0 \neq w \in \Lambda^2 M_2(\mathbb{R})$ and every $t \geq 0$,*

$$\int_K \|a_t k w\|^{-1} dk \leq C \|w\|^{-1}. \quad (7.S26)$$

If $w \notin M_c \cup M_r$, the left side tends to zero as $t \rightarrow \infty$.

Proof. First average over a fixed N -ball. Write $w = w_c + w_r$. One of the two components has norm at least $\|w\|/2$. If it is w_c , discard the orthogonal M_r component and integrate first in the variable acting on M_c ; the endpoint $\beta = 1$ case of Lemma 7.22 gives $C\|w_c\|^{-1}$. The other unipotent variable contributes only the length of its bounded interval. The case of w_r is the same. The finite Bruhat-chart argument used at the end of Theorem 7.27 transfers this bound to K , proving (7.S26).

If $w \notin M_c \cup M_r$, then from (7.S13) $\|v\|^{-1} \ll_{\theta} \phi_{\theta}(v)$ for every point of the orbit. Lemma 7.23 and Hölder therefore give, on each fixed N -ball,

$$\int \|a_t n w\|^{-1} dn \ll_{\theta} e^{-\theta t/(1+\theta)} \phi_{\theta}(w).$$

The same finite chart decomposition gives the asserted limit on K . $\square$

Proposition 7.29 (Vanishing of non-isotropic split quasi-null planes). *Let $\mathcal{X}_{\eta, M}^b(\Lambda)$ denote the rational two-planes whose primitive Pluecker vectors are quasi-null but do not lie exactly in $M_c \cup M_r$. Then*

$$\lim_{t \rightarrow \infty} \int_K \sum_{P \in \mathcal{X}_{\eta, M}^b(\Lambda)} \|a_t k w_P\|^{-1} dk = 0. \quad (7.S27)$$

Proof. Let $\mathcal{A}_{<0}$ contain the finitely many such planes with $\|w_P\| < 1$, and for $\ell \geq 0$ put

$$\mathcal{A}_{\ell} = \{P : 2^{\ell} \leq \|w_P\| < 2^{\ell+1}\}.$$

Corollary 7.20, applied with the slightly smaller quasi-null parameter from Definition 7.3, gives $\#\mathcal{A}_\ell \leq 4$. Hence Lemma 7.28 gives, uniformly in t ,

$$\int_K \sum_{\ell \geq L} \sum_{P \in \mathcal{A}_\ell} \|a_t k w_P\|^{-1} dk \ll \sum_{\ell \geq L} 2^{-\ell} \ll 2^{-L}. \quad (7.S28)$$

For fixed L only finitely many non-isotropic planes remain, and the second part of Lemma 7.28 makes each integral tend to zero. Let $t \rightarrow \infty$ and then $L \rightarrow \infty$. $\square$

Lemma 7.30 (Split modified Lipschitz principle). *For $f \in C_c(M_2(\mathbb{R}))$, let $\hat{f}^{\text{ni}}(h; \Lambda)$ be its Siegel transform with all vectors lying in exact rational isotropic planes deleted. Then*

$$|\hat{f}^{\text{ni}}(h; \Lambda)| \leq C_f \hat{\alpha}_{\eta, M}(h; \Lambda) + C_f \sum_{P \in \mathcal{X}_{\eta, M}^b(\Lambda)} \|h w_P\|^{-1}. \quad (7.S29)$$

Moreover the ordinary height satisfies

$$\alpha(h\Lambda)^{1/2} \ll \hat{\alpha}_{\eta, M}(h; \Lambda). \quad (7.S30)$$

Proof. For (7.S29), apply the usual lattice-point Lipschitz principle degree by degree. Primitive subspaces of degrees one and three and non-quasi-null degree-two planes are precisely the objects retained in $\hat{\alpha}_{\eta, M}$. A quasi-null degree-two plane either is exactly isotropic, in which case its vectors were deleted, or belongs to $\mathcal{X}_{\eta, M}^b$ and contributes the displayed inverse Pluecker norm. This is an exhaustive partition of the primitive subspaces that can span lattice points in a fixed compact support.

For (7.S30), degrees one and three already occur in $\hat{\alpha}_{\eta, M}$. Minkowski's second theorem on a primitive rank-two subgroup gives $\alpha_2^{1/2} \ll \max\{1, \alpha_1\}$ after passing to its first successive minimum. Taking the maximum over degrees proves (7.S30). $\square$

Corollary 7.31 (Split modified Siegel transforms are uniformly integrable). *After deleting the finitely many exact rational isotropic planes from a compactly supported determinant Siegel transform, the remaining transform is uniformly integrable along the expanding H -translates of Λ in the limiting sense required for unbounded equidistribution. Consequently bounded-test-function equidistribution passes to the modified Siegel transform.*

Proof. Let h_R be a continuous cusp cutoff supported where the ordinary height α exceeds R . By (7.S30), on this set

$$(\hat{\alpha}_{\eta,M} + \alpha^{1/2})h_R \ll R^{-\theta'/2}(\hat{\alpha}_{\eta,M} + \alpha^{1/2})^{1+\theta'}.$$

Theorem 7.27 and (7.S30) make the integral of the right-hand side uniformly bounded, so the first term in (7.S29) has cusp contribution $O(R^{-\theta'/2})$. The second term in (7.S29) tends to zero as $t \rightarrow \infty$ by Proposition 7.29. Thus

$$\limsup_{t \rightarrow \infty} \int_K |\hat{f}^{\text{ni}}(a_t k; \Lambda)| h_R(a_t k \Lambda) dk \ll_f R^{-\theta'/2}.$$

Truncate, apply bounded equidistribution, let $t \rightarrow \infty$, and finally let $R \rightarrow \infty$. Approximation gives the corresponding statement for bounded compactly supported Borel kernels with negligible discontinuity set. $\square$

Lemma 7.32 (Quasi-null arithmetic reconstruction: exact proof ownership). *Let Q be of finite type in the sense of (7.1). Then the quasi-null configurations used below satisfy the finite-scale arithmetic restrictions needed for the modified-height argument. More precisely:*

1. *in signature $(2, 1)$, at every dyadic scale the quasi-null vectors of the associated lattice lie in a uniformly bounded union of rational lines/planes;*
2. *in signature $(2, 2)$, after identifying the form with determinant on $M_2(\mathbb{R})$, the rational two-planes whose primitive Pluecker vectors are quasi-null form a uniformly bounded finite-scale family; exact rational isotropic planes form a finite family.*

If either finite-scale conclusion failed at arbitrarily large scales, then for every $A > 0$ there would be rational forms Q_j with projective heights $H_j \rightarrow \infty$ such that

$$d_{\mathbb{P}}([Q], [Q_j]) \leq H_j^{-A}$$

for infinitely many j (after normalizing Q_j on the fixed coefficient slice). This contradicts the finite-type hypothesis (7.1).

Proof and provenance. The ternary statement follows from Proposition 7.7, whose five-point conic reconstruction is proved above. The split $(2, 2)$ statement follows from Corollary 7.20; the reconstruction is proved in Lemma 7.18. Citations to the primary papers are provenance and concordance, not proof ownership. In particular, a generic “choose a maximal minor and round” slogan is not used in place of the explicit reconstruction lemmas.

Two computations make the mechanism concrete. In the ternary case choose three independent integral vectors m_1, m_2, m_3 of norm $< R$ and put

$\gamma = (m_1 \ m_2 \ m_3)$. Then $|\det \gamma| \geq 1$ and $\|\operatorname{adj} \gamma\| \ll R^2$. Writing $Q \circ \gamma = (q_{ij})$, small values $|Q(m_i)| \leq \varepsilon$ control the diagonal q_{ii} ; two additional small values in general position recover the three off-diagonal coefficients. Cramer's rule now has a *specified* integer denominator dividing a power of $\det \gamma$, hence polynomial in R . This is the source of the explicit polynomial-height rational approximation in the primary five-vector lemma.

For determinant, a column-isotropic plane with common column line $\mathbb{R}(s, t)^T$ has Pluecker vector proportional to

$$s^2 e_{12} + st(e_{14} - e_{23}) + t^2 e_{34}.$$

Three distinct slopes give three independent vectors because their coordinates form a Vandermonde matrix $(1, x, x^2)$. Thus three rational column-isotropic planes would make the entire exceptional three-dimensional summand rational; the rationality obstruction then forces determinant rationality. The same calculation for rows yields at most four exact rational isotropic planes. This is the explicit linear algebra behind the finite singular family used later. $\square$

Remark 7.33 (Two different low-signature notions of finite type). For ternary forms, (7.1) is the finite-type condition used in Theorem 7.12. For signature $(2, 2)$ the internal split reconstruction uses Definition 7.3, which tests only rational split forms. On a fixed determinant normalization slice, (7.1) is stronger: rational split forms are among the rational forms tested by (7.1), and projective distance is comparable with normalized coefficient distance. Thus the stronger finite-type hypothesis used in this book implies the split-Diophantine hypothesis whenever the latter is invoked.

7.3 The modified $1 + \delta$ moment

Theorem 7.34 (Critical-signature modified moments: primary form). *Both low-signature critical moments are now proved internally; the primary papers remain cited for provenance and theorem concordance.*

1. **Ternary branch (internal)**. Let Q be an indefinite ternary form with $\det Q = 1$ and of type $M > 1$. For the modified Margulis height $\alpha_{\eta, M}$ of Definition 7.2, there is $\delta = \delta(M) > 0$ such that

$$\sup_{t>0} \int_K \alpha_{\eta, M}(a_t k; \Lambda_Q)^{1+\delta} dk < \infty. \quad (7.4a)$$

2. **Split $(2, 2)$ branch (internal)**. Let Q be an irrational split-Diophantine form of signature $(2, 2)$ in the sense of Definition 7.3, and let $\Lambda < M_2(\mathbb{R})$

be its determinant-model lattice. For the modified height $\hat{\alpha}_{\eta,M}$ defined there, there is $\theta' > 0$ such that

$$\sup_{t \geq 0} \int_K \hat{\alpha}_{\eta,M}(a_t k; \Lambda)^{1+\theta'} dk < \infty. \quad (7.4b)$$

The passage from these height moments to the actual modified Siegel transforms is not a formal power comparison: the quasi-null-but-nonexact remainder requires a separate estimate. The needed estimates are Lemma 7.15 and Proposition 7.14 for the ternary branch, and in Lemma 7.30 and Proposition 7.29 for the split branch.

Primary proof owners. Part (1) is Theorem 7.12; the preceding subsection reconstructs its conic arithmetic, critical vector contraction, modified-height drift, Volume-XIII avoidance reduction, and multiscale assembly. The citation to [9] is now provenance and concordance, not proof ownership. Part (2) is Theorem 7.27; the preceding split subsection proves the finite-scale reconstruction, weighted critical estimate, global height contraction, avoidance reduction, and block iteration. The comparison with Theorem 8.5 of [10] is now a concordance check rather than proof ownership. $\square$

Proposition 7.35 (What the two-height Foster–Lyapunov algebra proves). *Let $X_j, Y_j \geq 0$ be adapted random variables and suppose that for constants $c_1, c_2 \in (0, 1)$ and $C_1, C_2 > 0$,*

$$\mathbb{E}(Y_{j+1} \mid \mathcal{F}_j) \leq (1 - c_1)Y_j + C_1 + C_1 X_j^{1/(1+\delta)}, \quad (7.5)$$

$$\mathbb{E}(X_{j+1} \mid \mathcal{F}_j) \leq (1 - c_2)X_j + C_2 + C_2 Y_j. \quad (7.6)$$

Then for sufficiently large λ there are $c > 0, C < \infty$ such that

$$\mathbb{E}(X_{j+1} + \lambda Y_{j+1} \mid \mathcal{F}_j) \leq (1 - c)(X_j + \lambda Y_j) + C. \quad (7.7)$$

Consequently $\sup_j \mathbb{E}(X_j + \lambda Y_j) < \infty$ whenever the initial expectation is finite.

Proof. Choose $\lambda > 2C_2/c_1$. For every $\varepsilon > 0$, Young’s inequality for the sublinear power $x^{1/(1+\delta)}$ gives a finite D_ε with

$$\lambda C_1 x^{1/(1+\delta)} \leq \varepsilon x + D_\varepsilon \quad (x \geq 0).$$

Take $\varepsilon < c_2/2$. Multiply (7.5) by λ and add (7.6):

$$\begin{aligned} \mathbb{E}(X_{j+1} + \lambda Y_{j+1} \mid \mathcal{F}_j) &\leq (1 - c_2 + \varepsilon)X_j \\ &\quad + (C_2 + \lambda(1 - c_1))Y_j + C', \end{aligned}$$

where $C' = C_2 + \lambda C_1 + D_\varepsilon$. Since $C_2 + \lambda(1 - c_1) = \lambda(1 - c_1 + C_2/\lambda)$, we may take

$$c = \min\{c_2/2, c_1 - C_2/\lambda\} > 0.$$

This proves (7.7). Taking expectations and iterating the affine recurrence $Z_{j+1} \leq (1 - c)Z_j + C$ gives $Z_j \leq (1 - c)^j Z_0 + C/c$, proving the uniform bound. $\square$

Worked Example 7.36 (Why the exponent $1 + \delta$ is genuinely critical). The elementary integral

$$\int_{-1}^1 |r|^{-1-\delta} dr = 2 \int_0^1 r^{-1-\delta} dr = \infty \quad (\delta \geq 0)$$

shows why a transverse coordinate with an actual simple zero cannot support an inverse moment above one. If instead $|p(r)| \geq \eta R^{-M}$ throughout the dangerous interval, then

$$\int_{-1}^1 |p(r)|^{-1-\delta} dr \leq 2\eta^{-1-\delta} R^{M(1+\delta)}.$$

The modified-height theory is the high-dimensional arithmetic mechanism that separates the first situation (quasi-null/rational geometry, to be removed and counted separately) from the second (uniformly contractible geometry). This calculation explains both the need for the modified height and why a mere first-moment estimate is insufficient for uniform integrability.

Corollary 7.37 (Uniform integrability in the critical signatures). *For every compactly supported Siegel kernel ψ , the modified Siegel transform obtained by deleting lattice vectors in the exact rational null subspaces is uniformly integrable along ν_T ; indeed it is bounded in $L^{1+\epsilon}$ for some $\epsilon > 0$.*

Proof. This is precisely Corollary 7.16 in the ternary branch and Corollary 7.31 in the split branch. Their proofs separate the genuinely moment-controlled modified height from the quasi-null-but-nonexact remainder and prove that the latter vanishes in the expanding limit. The exact rational null subspaces are omitted from the transform and counted separately below. $\square$

7.4 Signature (2, 2)

For the split model

$$Q_{\det}(x_1, x_2, x_3, x_4) = x_1 x_4 - x_2 x_3$$

the exact rational null subspaces are rational isotropic planes. Distinct such planes meet in at most a line, so their double counting is only $O(T)$.

Theorem 7.38 (Quantitative Oppenheim in signature (2, 2)). *Let Q have signature (2, 2), be irrational up to scale, and be of finite type in the sense of (7.1). Let $N_Q^{\text{ns}}(a, b; T)$ count the integer vectors of norm at most T with $a < Q(v) < b$ after deleting all rational two-dimensional totally isotropic subspaces of Q . Then*

$$N_Q^{\text{ns}}(a, b; T) \sim C_Q(b - a)T^2. \quad (7.9)$$

There are at most four rational isotropic planes. If $0 \notin (a, b)$ they make no contribution. If $0 \in (a, b)$, then

$$N_Q(a, b; T) = N_Q^{\text{ns}}(a, b; T) + \left(\sum_P c(P) \right) T^2 + o(T^2), \quad (7.10)$$

where P runs over those rational isotropic planes and $c(P)$ is the planar lattice-density constant in $P \cap \mathbb{Z}^4$ for the chosen norm.

Internal modified-height proof and singular bookkeeping. By the remark after Lemma 7.32, finite type in the sense of (7.1) implies the split-Diophantine hypothesis used in the internal split module above. Theorem 7.27 and Corollary 7.31 therefore upgrade the bounded expanding-translate equidistribution theorem proved in Volume IX to the exact-isotropic-plane-deleted determinant Siegel transform. Siegel's mean-value formula identifies the limiting Haar integral with the Euclidean integral, and Proposition 2.2, with $n = 4$, gives the main scale

$$C_Q(b - a)T^2 + o(T^2).$$

To pass from smooth kernels to the sharp conditions $a < Q(v) < b$ and $\|v\| < T$, choose inner and outer continuous approximations whose transition regions have arbitrarily small Euclidean coarea mass. The uniform integrability just proved controls the cusp uniformly in t , while the coarea formula controls the value-window and norm-boundary transition regions. Sending first $t \rightarrow \infty$ and then the two smoothing widths to zero proves (7.9). Thus the nonsingular asymptotic is owned by the internal modified-height, unbounded-equidistribution, Siegel-mean, and coarea chain.

It remains to restore the exact singular planes. Corollary 7.20 proves internally that a determinant-irrational lattice has at most two column-isotropic and at most two row-isotropic rational planes, hence at most four

in total. On one rational isotropic plane P , $Q \equiv 0$ and $P \cap \mathbb{Z}^4$ is a rank-two lattice, so the elementary planar lattice-point estimate gives

$$\#\{v \in P \cap \mathbb{Z}^4 : \|v\| < T\} = c(P)T^2 + O_P(T).$$

Two distinct totally isotropic planes intersect in dimension at most one; their common lattice points therefore contribute $O(T)$. Since the family is finite, inclusion–exclusion gives exactly (7.10), and the contribution is zero when $0 \notin (a, b)$.

The result agrees with Theorems 1.1 and 1.3 of [10]. That citation is provenance and source concordance only; it is not a proof owner in this edition.

□

7.5 Signature (2, 1)

For a ternary form, the exact singular configurations are rational isotropic lines for Q and rational degenerate planes; the latter are equivalently orthogonal to rational isotropic lines of the dual form Q^* .

Lemma 7.39 (Finiteness of exact ternary singular directions). *If Q is an irrational nondegenerate ternary form, then it has only finitely many rational isotropic lines; the same is true for the dual form Q^* . Consequently the exact singular family consisting of rational isotropic lines and rational degenerate planes is finite.*

Proof. Projectivize the null cone. A nondegenerate ternary form cuts out a smooth projective conic. Five distinct rational points on this conic determine its quadratic equation up to scale: in the six-dimensional vector space of ternary quadratic forms, evaluation at five points gives five independent linear conditions unless three points are collinear; but a projective line meets a smooth conic in at most two points. Thus five rational null lines would force the one-dimensional kernel of the evaluation matrix to be rational, making Q rational up to scale. Hence an irrational Q has at most four rational isotropic lines. Apply the same argument to the dual conic $Q^* = 0$. A rational degenerate plane is the annihilator of a rational isotropic line of Q^* , so the full exact singular family is finite. □

Theorem 7.40 (Quantitative Oppenheim for non-EWA ternary forms). *Let Q be an irrational indefinite ternary form that is not EWA. Delete integer*

points lying on rational isotropic lines or rational degenerate planes and denote the resulting count by $N_Q^{\text{ns}}(a, b; T)$. Then

$$N_Q^{\text{ns}}(a, b; T) \sim C_Q(b - a)T. \quad (7.11)$$

For the complete count there is an explicit nonnegative singular constant $I_Q(a, b)$ such that

$$N_Q(a, b; T) = (C_Q(b - a) + I_Q(a, b))T + o(T). \quad (7.12)$$

If neither Q nor its dual Q^* has a nonzero rational isotropic vector, then $I_Q(a, b) = 0$.

Proof. Because Q is not EWA, Definition 7.1 gives a finite type exponent M . Theorem 7.12 and Corollary 7.16 therefore allow the bounded expanding-translate equidistribution theorem proved in Volume IX to be applied to the exact-null-deleted Siegel transform. Siegel's mean-value formula identifies the Haar mean with the corresponding Euclidean integral, and Proposition 2.2 gives

$$C_Q(b - a)T + o(T).$$

This is (7.11). Thus the unbounded-test-function step is supplied by the critical moment and quasi-null-vanishing arguments proved above.

It remains only to put back the exact singular subspaces. By Lemma 7.39 there are finitely many. If $\ell = \mathbb{R}v$ is a rational isotropic line with v primitive, then $Q(mv) = 0$ and its lattice points in a norm ball have the exact asymptotic

$$\#\{m \in \mathbb{Z} : 0 < \|mv\| \leq T\} = \frac{2T}{\|v\|} + O(1) \quad (7.11a)$$

when $0 \in (a, b)$, and contribute zero otherwise. If P is a rational degenerate plane, choose a lattice basis in which $Q|_P = c_P m^2$. Only finitely many integers m can satisfy $a < c_P m^2 < b$; for each one, the second lattice coordinate ranges through an interval of length $\lambda_{P,m}T + O(1)$, with an explicitly computable $\lambda_{P,m} \geq 0$ determined by the chosen norm ball. Hence

$$\#\{v \in P \cap \mathbb{Z}^3 : \|v\| \leq T, a < Q(v) < b\} = I_P(a, b)T + O(1). \quad (7.11b)$$

The finite intersections among these singular subspaces are rational lines or the origin and satisfy the same elementary one-dimensional estimate. Finite inclusion–exclusion therefore produces an explicit nonnegative coefficient $I_Q(a, b)$ for the union and an $O(1)$ remainder. Adding this to (7.11) proves

(7.12). If Q and Q^* have no rational isotropic vectors, the singular family is empty and $I_Q(a, b) = 0$.

The resulting theorem agrees with Theorems 1.3–1.4 of [9]; that citation is now provenance and source concordance rather than proof ownership. $\square$

Remark 7.41 (What fails for EWA forms). The hypotheses above are substantive. Extremely good rational approximations can create subsequences on which the count is larger than the Euclidean-volume scale by logarithmic factors. Thus the modified-moment argument is not merely a technical convenience: its Diophantine hypothesis marks the actual boundary of the universal fixed-window asymptotic.

Chapter 8

Effective Ternary Oppenheim from Rank-Two Dynamics

8.1 The orthogonal homogeneous space

For a ternary form of signature $(2, 1)$, the connected stabilizer is locally isomorphic to $\mathrm{PSL}_2(\mathbb{R})$. Embed it in $G = \mathrm{SL}_3(\mathbb{R})$. This is an absolute rank-two ambient group, exactly the situation treated in Volume XIII.

8.2 Translating homogeneous height into form height

Lemma 8.1 (Subgroup obstruction implies rational approximation of the form). *There are constants $A, c > 0$ depending only on the normalization conventions such that if the lattice point x_Q associated with Q is within ε of a proper H -compatible homogeneous orbit of height at most B , then there are $1 \leq q \leq B^A$ and a nonzero integral quadratic form Q_0 with $\|Q_0\| \leq B^A$ for which*

$$\left\| Q - \frac{1}{q} Q_0 \right\| \leq B^A \varepsilon^c.$$

Proof. A proper H -compatible subgroup is detected by a rational invariant line in a finite direct sum of tensor/exterior representations of G , by Volume V's Chevalley construction. Its primitive vector has norm $O(B^A)$. Proximity of x_Q to the orbit means that the corresponding invariant tensor for Q is within $O(B^A \varepsilon)$ of this rational line. The coefficients of the quadratic form are polynomial coordinates in that tensor. On the compact normalization slice

the inverse map is Hölder by the fixed-complexity real-algebraic Łojasiewicz theorem of Volume XIII; its exponent and constant depend only on the fixed representation. Clearing denominators in the primitive rational tensor gives q, Q_0 with the stated polynomial height bounds. $\square$

Lemma 8.2 (Sharp arithmetic form attached to a nearby periodic principal orbit). *Let Q be determinant-normalized and let $x_Q = g_Q \Gamma \in \mathrm{SL}_3(\mathbb{R})/\mathrm{SL}_3(\mathbb{Z})$. Suppose $H y$ is a periodic principal $H = \mathrm{SO}(Q_0)^\circ$ orbit whose arithmetic height is at most B , and $d(x_Q, y) \leq \varepsilon$ with ε smaller than a fixed local-chart radius depending on Q . Then there is a nondegenerate primitive integral ternary form Q' and a scalar λ such that*

$$\|Q'\| \ll_Q B^C, \quad \|Q - \lambda Q'\| \ll_Q B^C \varepsilon. \quad (14.P1)$$

The exponent C is absolute for the fixed ambient embedding.

Proof. Choose a representative g_y of y in the local quotient chart centered at g_Q . Thus, after multiplying g_y on the right by one element of $\mathrm{SL}_3(\mathbb{Z})$,

$$\|g_y - g_Q\| \ll_Q \varepsilon. \quad (14.P2)$$

The periodic orbit determines the conjugate subgroup $K_y = g_y^{-1} H g_y$. By the closed-orbit rationality theorem of Volume IV, its one-dimensional fixed space in the six-dimensional representation of quadratic forms is rational. The quantitative Chevalley-height comparison of Volume XIII, applied to the arithmetic height B of this homogeneous support, gives a primitive integral generator Q' of that fixed line with $\|Q'\| \ll B^C$.

The normalized form attached to a matrix g is the projective class of $v \mapsto Q_0(gv)$. On the fixed local chart around g_Q this map and the choice of determinant-one normalization are smooth with bounded derivative. Equation (14.P2) therefore gives

$$d_{\mathbb{P}(\mathrm{Sym}^2)}([Q], [Q']) \ll_Q \varepsilon. \quad (14.P3)$$

Choose the real scalar λ putting the primitive integral generator Q' in the same determinant normalization as Q . Converting projective distance to the coefficient norm on this fixed normalization slice is Lipschitz. The only polynomial factor needed to pass from the rational line to its primitive integral generator is its height, already bounded by B^C . This proves (14.P1). Notice that no Łojasiewicz exponent is lost here: for the principal periodic orbit the invariant object is the quadratic-form line itself, so the conversion is linear in the orbit distance. $\square$

8.3 Arithmetic conversion for ternary counts

The passage from effective dynamics to lattice counting separates into two logically different tasks. The power-saving counting theorem and its explicit exceptional lines/planes are proved here. The sharp finite-scale small-value alternative is proved below: instead of passing through the coarser circle-transfer estimate, we test the sharp expanding segment theorem of Volume XIII on compact packets in the lattice space. This keeps both primitivity and the exact T^{-2} obstruction scale visible.

Definition 8.3 (Finite-scale exceptional lines and planes). Fix $0 < \rho < 1$, $E > 1$ and $t > 1$. A rational line $L < \mathbb{R}^3$ is (ρ, E, t) -exceptional if $L \cap \mathbb{Z}^3 = \mathbb{Z}v_L$ with v_L primitive and

$$\|v_L\| \leq e^{\rho t}, \quad |Q(v_L)| \leq e^{-E\rho t}. \quad (14.A1)$$

A rational plane P is (ρ, E, t) -exceptional when its primitive dual vector $w_P \in \bigwedge^2 \mathbb{Z}^3 \simeq \mathbb{Z}^3$ satisfies the same conditions with the dual form Q^* . With Q represented by its symmetric matrix and $\det Q = 1$, our normalization is

$$Q^*(w) = w^t \operatorname{adj}(Q)w.$$

Thus Q^* is again a nondegenerate ternary form and $\operatorname{adj}(\operatorname{adj}(Q)) = Q$.

Lemma 8.4 (Three short quasi-null directions cannot be coplanar). *Let Q be a fixed nondegenerate ternary quadratic form. There are a constant C_0 and $H_0 = H_0(Q)$ such that the following holds. If $v_1, v_2, v_3 \in \mathbb{Z}^3$ are primitive, span three distinct lines in one real plane, and $\|v_i\| \leq H$ with $H \geq H_0$, then*

$$\max_i |Q(v_i)| \geq H^{-C_0}. \quad (14.A2b)$$

The same statement holds for the dual form Q^ .*

Proof. Let $P = \operatorname{span}(v_1, v_2)$ and choose an orthonormal basis of P . Write the restriction of Q as the binary quadratic form

$$q_P(x, y) = ax^2 + 2bxy + cy^2.$$

No two-dimensional subspace of a real ternary form of signature $(2, 1)$ is totally isotropic. Hence $\|(a, b, c)\| > 0$ for every P . The Grassmannian $\operatorname{Gr}(2, 3)$ is compact and the restriction map is continuous, so

$$\inf_{P \in \operatorname{Gr}(2, 3)} \|(a, b, c)\| =: c_Q > 0. \quad (14.A2c)$$

Put $u_i = v_i / \|v_i\| = (\cos \theta_i, \sin \theta_i)$ in the chosen basis. Because $v_i \wedge v_j$ is a nonzero integral two-vector,

$$|\sin(\theta_i - \theta_j)| = \frac{\|v_i \wedge v_j\|}{\|v_i\| \|v_j\|} \geq H^{-2}. \quad (14.A2d)$$

The evaluation matrix for a binary quadratic form at the three directions is

$$E = \begin{pmatrix} \cos^2 \theta_1 & 2 \cos \theta_1 \sin \theta_1 & \sin^2 \theta_1 \\ \cos^2 \theta_2 & 2 \cos \theta_2 \sin \theta_2 & \sin^2 \theta_2 \\ \cos^2 \theta_3 & 2 \cos \theta_3 \sin \theta_3 & \sin^2 \theta_3 \end{pmatrix}.$$

A direct Vandermonde calculation gives

$$|\det E| = 2 \prod_{i < j} |\sin(\theta_i - \theta_j)| \geq 2H^{-6}.$$

All 2×2 minors of E are bounded absolutely, so Cramer's rule yields

$$\|(a, b, c)\| \ll H^6 \max_i |q_P(u_i)| \leq H^6 \max_i |Q(v_i)|. \quad (14.A2e)$$

Taking $C_0 = 8$, say, (14.A2b) follows for H sufficiently large in terms of c_Q . The dual form is again nondegenerate ternary, so the same argument applies to Q^* . $\square$

Lemma 8.5 (Five general-position quasi-null vectors reconstruct the form). *There is an absolute M_0 with the following property. Let $0 < \eta < 1/2$, let $D \geq 1$ and $E \geq 2M_0D$, and let $v_1, \dots, v_5 \in \mathbb{Z}^3$ be primitive vectors such that no three are coplanar,*

$$\|v_i\| \leq \eta^{-D}, \quad |Q(v_i)| \leq \eta^E.$$

Then, for all sufficiently small η depending on Q , there is a nondegenerate integral ternary form Q' with

$$\|Q'\| \leq \eta^{-M_0D}, \quad \inf_{\lambda \in \mathbb{R}} \|Q - \lambda Q'\| \ll_Q \eta^{E-M_0D}. \quad (14.A3)$$

Proof. Let $\gamma = (v_1 \ v_2 \ v_3)$. Since the three columns are independent, $\det \gamma$ is a nonzero integer and

$$1 \leq |\det \gamma| \ll \eta^{-3D}, \quad \|\gamma^{-1}\| \ll \eta^{-2D}. \quad (14.A3b)$$

Write

$$B = \gamma^t Q \gamma = (b_{ij}), \quad a = \gamma^{-1} v_4, \quad a' = \gamma^{-1} v_5.$$

The coordinates of a, a' are rational of height $\eta^{-O(D)}$; no coordinate vanishes because no three of the five vectors are coplanar. The diagonal entries satisfy $|b_{ii}| \leq \eta^E$. After subtracting their contribution from $Q(v_4)$ and $Q(v_5)$ we obtain

$$c(a) \cdot b_{\text{off}} = O(\eta^{E-O(D)}), \quad c(a') \cdot b_{\text{off}} = O(\eta^{E-O(D)}), \quad (14.A3c)$$

where

$$b_{\text{off}} = (b_{23}, b_{13}, b_{12}), \quad c(a) = (a_2 a_3, a_1 a_3, a_1 a_2).$$

The two rational vectors $c(a), c(a')$ are not proportional: proportionality would force a_i/a'_i to be independent of i , hence v_4 and v_5 to span the same line. Their common kernel is therefore a rational line. Taking the cross product and clearing denominators gives a primitive $z = (z_1, z_2, z_3) \in \mathbb{Z}^3$ with $\|z\| \leq \eta^{-O(D)}$ spanning that kernel. In fact every z_i is nonzero. For example, up to a nonzero common scalar,

$$z_1 = a_1 a'_1 (a_2 a'_3 - a_3 a'_2),$$

and the last factor is nonzero because otherwise e_1, a, a' would be coplanar, equivalently v_1, v_4, v_5 would be coplanar. The other two coordinates are handled cyclically.

A nonzero 2×2 minor of the rational matrix with rows $c(a), c(a')$ has numerator and denominator of height $\eta^{-O(D)}$, hence its absolute value is at least $\eta^{O(D)}$. Cramer's rule applied to (14.A3c) therefore shows that

$$\text{dist}(b_{\text{off}}, \mathbb{R}z) \ll_Q \eta^{E-O(D)}. \quad (14.A3d)$$

Define the integral symmetric matrix

$$P_z = \begin{pmatrix} 0 & z_3 & z_2 \\ z_3 & 0 & z_1 \\ z_2 & z_1 & 0 \end{pmatrix}.$$

Its determinant is $2z_1 z_2 z_3 \neq 0$. Thus P_z is already nondegenerate; no limiting argument through the discriminant is needed. Together with the small diagonal entries, (14.A3d) gives, for some real μ ,

$$\|B - \mu P_z\| \ll_Q \eta^{E-O(D)}. \quad (14.A3e)$$

Now set

$$Q' = \text{adj}(\gamma)^t P_z \text{adj}(\gamma).$$

This is an integral nondegenerate ternary form and $\|Q'\| \leq \eta^{-O(D)}$. Since $\text{adj}(\gamma) = (\det \gamma)\gamma^{-1}$, conjugating (14.A3e) by γ^{-1} gives

$$\left\| Q - \frac{\mu}{(\det \gamma)^2} Q' \right\| \ll_Q \|\gamma^{-1}\|^2 \eta^{E-O(D)} \ll_Q \eta^{E-O(D)}.$$

Choose M_0 larger than all absolute constants in these $O(D)$ losses. The assumption $E \geq 2M_0D$ then yields the two bounds in (14.A3), after enlarging M_0 once more. $\square$

Lemma 8.6 (Dual projective reconstruction returns to the original form). *Let Q be determinant-normalized and put $Q^* = \text{adj}(Q)$. Suppose P is a nondegenerate integral ternary form with $\|P\| \leq H$ and*

$$\inf_{\mu \in \mathbb{R}} \|Q^* - \mu P\| \leq \varepsilon$$

for ε sufficiently small in terms of Q . Then $P^\# = \text{adj}(P)$ is a nondegenerate integral ternary form with

$$\|P^\#\| \ll H^2, \quad \inf_{\nu \in \mathbb{R}} \|Q - \nu P^\#\| \ll_Q \varepsilon. \quad (14.A3h)$$

Moreover the infimum may, up to a Q -dependent constant in the error, be replaced by the scalar giving determinant normalization.

Proof. Choose μ with $\|Q^* - \mu P\| \leq 2\varepsilon$ and put $A = \mu P$. For small ε , A stays in a fixed compact neighbourhood of the nondegenerate matrix Q^* . The adjugate map is polynomial of degree two, so it is Lipschitz on that compact set. Hence

$$\|Q - \mu^2 \text{adj}(P)\| = \|\text{adj}(Q^*) - \text{adj}(A)\| \ll_Q \varepsilon,$$

where $\text{adj}(Q^*) = Q$ because $\det Q = 1$ in dimension three. The entries of $\text{adj}(P)$ are 2×2 minors, giving $\|P^\#\| \ll H^2$ and integrality; nondegeneracy follows from that of P . Finally, exactly as in the determinant-normalization calculation below, if $B = \nu P^\#$ is sufficiently close to Q , then replacing ν by the real scalar $(\det P^\#)^{-1/3}$ changes the coefficient error by only a Q -dependent constant factor. This proves the last assertion. $\square$

Lemma 8.7 (Uniform finite exceptional family). *Assume that, for some $A \geq 10^3$ and $0 < \delta < 1$, the finite-scale Diophantine condition*

$$\|Q - \lambda Q'\| \geq \|Q'\|^{-A} \quad (14.A2)$$

holds for every nondegenerate integral ternary form Q' with $\|Q'\| \leq T^\delta$ and every scalar λ giving the determinant normalization. There is an absolute M_0 and there are choices $0 < \rho < \delta/(2M_0)$ and $E = E(A, M_0)$ sufficiently large such that, for all large T and simultaneously for every $t/5 \leq \ell \leq t = \log T$, there are at most four (ρ, E, ℓ) -exceptional rational lines and at most four (ρ, E, ℓ) -exceptional rational planes. The same two families work for all those scales. Moreover $E\rho/5 > 2 - \delta$ may be imposed.

Proof. Take the union of all exceptional lines over $t/5 \leq \ell \leq t$. Every primitive generator in this union satisfies

$$\|v\| \leq e^{\rho t}, \quad |Q(v)| \leq e^{-E\rho t/5}. \quad (14.A3i)$$

Choose E so large that $E/5 > C_0$ and $E/5 \geq 2M_0$, so that the right side of (14.A3e) is smaller than the lower bound in Lemma 8.4 with $H = e^{\rho t}$. For large t , no three lines in the union are coplanar.

If the union contained five lines, apply Lemma 8.5 with $\eta = e^{-\rho t}$, $D = 1$, and exponent $E/5$. We obtain a nondegenerate integral form Q' satisfying

$$\|Q'\| \leq e^{M_0 \rho t}, \quad \inf_{\lambda} \|Q - \lambda Q'\| \ll_Q e^{(-E/5 + M_0)\rho t}. \quad (14.A3f)$$

Choose $\rho < \delta/(2M_0)$ and then E so large that $E/5 - M_0 > 2AM_0$. We make the determinant normalization in (14.A2) explicit. Let μ realize (up to a factor two) the infimum in (14.A3f). For large T the matrix $A = \mu Q'$ lies in a fixed small neighbourhood of Q , hence $\|A\| \asymp_Q 1$ and $|\det A - 1| \ll_Q \|A - Q\|$. If $\lambda = (\det Q')^{-1/3}$ is the real cube root compatible with the signature, then

$$\frac{\lambda}{\mu} = (\det A)^{-1/3},$$

so smoothness of $x \mapsto x^{-1/3}$ near 1 gives

$$\|Q - \lambda Q'\| \leq \|Q - \mu Q'\| + |1 - (\det A)^{-1/3}| \|A\| \ll_Q \|Q - \mu Q'\|.$$

Thus (14.A3f) yields the same exponential upper bound with the determinant-normalizing scalar required in (14.A2). Since $\|Q'\|^{-A} \geq e^{-AM_0 \rho t}$, our stronger choice $E/5 - M_0 > 2AM_0$ gives the desired contradiction for sufficiently large T . Hence there are at most four exceptional lines in the entire scale union.

Apply the five-vector argument to Q^* and the primitive dual vectors of rational planes. If five exceptional planes existed, it would give a nondegenerate integral form P with

$$\|P\| \leq e^{M_0 \rho t}, \quad \inf_{\mu} \|Q^* - \mu P\| \ll_Q e^{(-E/5 + M_0)\rho t}.$$

Lemma 8.6 then gives the integral form $P^\# = \text{adj}(P)$ with $\|P^\#\| \ll e^{2M_0\rho t} \leq T^\delta$ and, after determinant normalization, the same exponential approximation error to Q . The choice $E/5 - M_0 > 2AM_0$ contradicts (14.A2). Hence there are at most four exceptional planes, again simultaneously for all $t/5 \leq \ell \leq t$. Finally enlarge E once more so that $E\rho/5 > 2 - \delta$. Then every listed object satisfies

$$\|v_L\|, \|w_P\| \leq T^\rho \leq T^\delta, \quad |Q(v_L)|, |Q^*(w_P)| \leq T^{-2+\delta}. \quad (14.A3g)$$

Thus the finite-family statement is internal; the primary rank-two paper is used only for theorem concordance, not proof ownership. $\square$

Lemma 8.8 (Circle-to-horocycle transfer with polynomial losses). *Let $G = \text{SL}_3(\mathbb{R})$, $\Gamma = \text{SL}_3(\mathbb{Z})$, and let $H \simeq \text{PSL}_2(\mathbb{R})$ be the principal subgroup. Write $X^+ = G/\Gamma$. Assume the rank-two hypotheses verified in Volume XIII; equivalently, the abstract effective-unipotent bootstrap proved there is available for the principal unipotent $U < H$. Then there are $c, C > 0$ such that, for every $x \in X^+$, every smooth F and every smooth angular weight ξ , one of the following holds for $t = \log T$:*

1.

$$\left| \int_K F(a_t k x) \xi(k) dk - \int_K \xi dk \int_{X^+} F dm_{X^+} \right| \ll T^{-c} \mathcal{S}_C(F) \mathcal{S}_C(\xi); \quad (14.A4)$$

2. *on one of the finitely many Bruhat charts, a chart base point is within T^{-c} of a proper U -compatible homogeneous support of height at most T^C . In the principal rank-two situation the subgroup classification in the proof of the rank-two theorem of Volume XIII reduces this support to a periodic principal- H support or a parabolic support (up to the normal-factor alternative already absorbed there).*

No T^{-2} proximity is asserted in alternative (2).

Proof. There are two bookkeeping points which are easy to suppress and which we keep visible. First, Volume XIII is written on the right-action quotient $\Gamma \backslash G$, whereas the space of lattices here is G/Γ . The inversion map

$$\iota : G/\Gamma \longrightarrow \Gamma \backslash G, \quad g\Gamma \longmapsto \Gamma g^{-1},$$

is a Haar-measure preserving diffeomorphism. Left multiplication by u_r on G/Γ becomes right multiplication by u_{-r} after inversion. Thus the abstract bootstrap of Volume XIII applies with the same polynomial exponents after replacing a test function by its pullback under ι .

Second, we use the correct Gauss factorization. On the Bruhat chart $|\cos \theta| \geq 1/4$, writing $u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$ and $\bar{u}_s = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix}$, one has

$$k_\theta = u_{-\tan \theta} \begin{pmatrix} (\cos \theta)^{-1} & 0 \\ 0 & \cos \theta \end{pmatrix} \bar{u}_{\tan \theta}. \quad (14.A5)$$

Indeed direct multiplication gives $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$. Hence, for $a_t = \text{diag}(e^{t/2}, e^{-t/2})$ in the principal H ,

$$a_t k_\theta = u_{-T \tan \theta} a_t \begin{pmatrix} (\cos \theta)^{-1} & 0 \\ 0 & \cos \theta \end{pmatrix} \bar{u}_{\tan \theta}, \quad T = e^t. \quad (14.A5a)$$

The last two θ -dependent factors are slow on every compact subchart; the opposite-unipotent variation is contracted after transport by a_t .

Let $\delta_0, A_0 > 0$ be the exponent and height power supplied by the abstract bootstrap theorem of Volume XIII. Delete angular boundary strips of width $T^{-\varepsilon}$ and partition the remainder into intervals of length $T^{-2\varepsilon}$. Their deleted mass is $O(T^{-\varepsilon})$. On a retained interval the coordinate $r = -T \tan \theta$ has range $L \asymp T^{1-2\varepsilon}$, while the slow factors vary by $O(T^{-\varepsilon})$ in fixed local charts. Freeze them at the midpoint. After translating the base point by the initial r -value, apply the abstract bootstrap theorem of Volume XIII to the resulting interval of length L . A smooth r -weight is handled by subdivision/partial summation; all Sobolev and slow-factor costs are $T^{C_0\varepsilon}$ for a fixed C_0 .

On the equidistribution branch the sum over the $O(T^{2\varepsilon})$ intervals is bounded by

$$O\left(T^{-(1-2\varepsilon)\delta_0 + (C_0+2)\varepsilon} \mathcal{S}_C(F) \mathcal{S}_C(\xi)\right) + O(T^{-\varepsilon} \mathcal{S}_C(F) \mathcal{S}_C(\xi)).$$

Choose $\varepsilon > 0$ so small that $(1 - 2\varepsilon)\delta_0 > (C_0 + 2)\varepsilon$. Finitely many rotated Bruhat charts cover K , proving (14.A4).

If the bootstrap takes its obstruction branch on one interval, before unfreezing we obtain a point within $L^{-\delta_0}$ of a proper compatible homogeneous orbit of height at most L^{A_0} . The frozen slow factor and its inverse have norm at most $T^{C_0\varepsilon}$, so unfreezing changes both bounds by fixed polynomial powers. After decreasing the exponent this is distance T^{-c} and height T^C . The rank-two subgroup classification used in Step 2 of the rank-two theorem of Volume XIII leaves precisely the principal- H , parabolic, and already absorbed normal-factor cases. This proves alternative (2). Nothing in this coarse chart transfer creates the sharp T^{-2} scale. That scale is instead proved internally by the highest-weight pullback in the sharp expanding-segment theorem of Volume XIII and used below through the expanding-segment formulation. $\square$

Lemma 8.9 (Arithmetic interpretation of the two obstruction branches). *Under (14.A2), with $\delta > 0$ chosen sufficiently small in terms of A , the periodic- H obstruction in Lemma 8.8 is impossible at all sufficiently large T . A parabolic obstruction forces the orbit into one of the finitely many exceptional lines or planes of Lemma 8.7.*

Proof. For a periodic principal- H orbit, the rational Chevalley tensor defining the orbit has polynomial height. The proof of Lemma 8.1 turns T^{-c} proximity to that orbit into an integral form Q' of height $T^{C\delta_1}$ satisfying $\|Q - \lambda Q'\| \ll T^{-c'}$. Choose the scale exponent δ_1 in the rank-two theorem so small that $C\delta_1 < \delta$ and $c' > AC\delta_1$. Then (14.A2) is violated. Thus this branch cannot occur.

A proper parabolic of SL_3 stabilizes a rational line or a rational plane. The shadowing inequalities say that the corresponding primitive vector, or primitive dual wedge, has norm at most a fixed power of T and Q -value at most a negative power. Shrinking the rank-two scale exponent again makes these precisely (ρ, E, ℓ) -exceptional objects. The finite-family lemma therefore absorbs every parabolic branch into the listed lines and planes. $\square$

Proposition 8.10 (Effective modified circular average). *Assume (14.A2) and let $\mathcal{L}_T, \mathcal{P}_T$ be the at-most-four line and plane families from Lemma 8.7. Let $\hat{f}_T^{\mathrm{ns}}$ be the Siegel transform of $f \in C_c^\infty(\mathbb{R}^3 \setminus \{0\})$ with all lattice vectors in $\bigcup \mathcal{L}_T \cup \bigcup \mathcal{P}_T$ deleted. There exist $c, C > 0$ such that, after truncating at modified height $L \geq 2$,*

$$\left| \int_K (\hat{f}_T^{\mathrm{ns}})_{\leq L}(a_t k g_Q \Gamma) \xi(k) dk - \int_K \xi dk \int_X (\hat{f}_T^{\mathrm{ns}})_{\leq L} dm_X \right| \ll T^{-c} L^C \mathcal{S}_C(f) \mathcal{S}_C(\xi). \quad (14.A6)$$

Proof. Choose a smooth cusp cutoff χ_L equal to one where the modified height is at most L and zero where it exceeds $2L$. On this truncated region the number of lattice vectors meeting a fixed compact support is $O(L^C)$ by successive minima, and every derivative of the Siegel sum has the same polynomial bound. Hence

$$\mathcal{S}_C(\chi_L \hat{f}_T^{\mathrm{ns}}) \ll L^C \mathcal{S}_C(f). \quad (14.A7)$$

Apply Lemma 8.8 to this bounded smooth function. Lemma 8.9 rules out the periodic branch and identifies every parabolic branch with a vector already deleted from the modified transform. On the complement the orbit is on the equidistribution branch, giving (14.A6). $\square$

Proposition 8.11 (Power tail for the modified ternary Siegel transform). *Fix $A \geq 10^3$ and $0 < \delta \leq \delta_0$ as in Lemma 8.7, and assume the finite-scale Diophantine separation (14.A2) for every integral ternary form of height at most T^δ . Let $\mathcal{L}_T, \mathcal{P}_T$ be the resulting exceptional line and plane families and let $\widehat{f}_T^{\text{ns}}$ denote the Siegel transform with lattice vectors in $\bigcup \mathcal{L}_T \cup \bigcup \mathcal{P}_T$ deleted, as in Proposition 8.10. Then there are $c_2 = c_2(A, \delta, Q) > 0$ and $\sigma = \sigma(A, \delta, Q, f) > 0$ such that for every $2 \leq L \leq T^{c_2}$,*

$$\sup_{t \geq 0} \int_K |\widehat{f}_T^{\text{ns}}(a_t k g_Q \Gamma)| \mathbf{1}_{\{\alpha_{\eta, M} > L\}} dk \ll_f L^{-\sigma}. \quad (14.A8)$$

The implied constant is uniform in the finite-scale exceptional family.

Proof. Use the finite-scale version of the modified Lipschitz decomposition (7.T9). After deleting the line/plane family $\mathcal{L}_T \cup \mathcal{P}_T$, it gives

$$|\widehat{f}_T^{\text{ns}}| \ll_f 1 + \alpha_{\eta, M} + \mathcal{R}_T, \quad (14.A8a)$$

where $\mathcal{R}_T$ is the supremum of the remaining quasi-null primitive vectors and dual vectors meeting the fixed support. The moment theorem gives a uniform bound $\int_K (1 + \alpha_{\eta, M})^{1+\delta_1} dk \leq C$. On the set $\{\alpha_{\eta, M} > L\}$ with $L \geq 1$,

$$1 + \alpha_{\eta, M} \leq 2\alpha_{\eta, M} \leq 2L^{-\delta_1} \alpha_{\eta, M}^{1+\delta_1} \leq 2L^{-\delta_1} (1 + \alpha_{\eta, M})^{1+\delta_1}.$$

Integrating yields

$$\int_{\{\alpha_{\eta, M} > L\}} (1 + \alpha_{\eta, M}) dk \ll L^{-\delta_1}. \quad (14.A8b)$$

It remains to estimate $\mathcal{R}_T$ quantitatively. Choose a small $\varepsilon > 0$ and split its primitive objects at norm T^ε . For the large objects, the dyadic-shell proof of Proposition 7.14, using the uniform first-moment bound (7.T7), gives $O(T^{-\varepsilon})$. There are only polynomially many dyadic low-scale families, and the finite-family lemma has already removed every object satisfying the exceptional inequalities (14.A1) at those scales. Hence every remaining low object is separated from the null cone by a negative power $T^{-C\varepsilon}$. In the proof of Lemma 7.13, the weighted contraction Lemma 7.8 then gives

$$\int_K \mathcal{R}_T(a_t k) dk \ll T^{-c_0 + C_0 \varepsilon} + T^{-\varepsilon}.$$

Choose $\varepsilon < c_0/(2C_0)$. Thus the remainder is $O(T^{-c_1})$. Since the proposition is used only for $2 \leq L \leq T^{c_2}$, decrease the exponent so that $T^{-c_1} \ll L^{-\sigma}$.

Combining this with (14.A8a)–(14.A8b) proves (14.A8). Notice that the argument uses the quasi-null remainder explicitly; no fictitious high power of the modified height is required. $\square$

Lemma 8.12 (Shell-to-circle smoothing identity). *Fix a compact value interval and a spherical patch $\Omega \subset S^2$. For a radial shell $(1 - \varepsilon)T < \|v\| \leq T$ and a value window $|Q(v) - c| \leq \varepsilon$, there are smooth $f_{\varepsilon, \Omega, c}$ on $\mathbb{R}^3$ and a smooth angular weight ξ_Ω on K such that*

$$N_\Omega(c, T; \varepsilon) = c_0 T \int_K \widehat{f}_{\varepsilon, \Omega, c}(a_{\log T} k g_Q \Gamma) \xi_\Omega(k) dk + O(\varepsilon^2 T + 1), \quad (14.A9)$$

and for every fixed Sobolev order d ,

$$\mathcal{S}_d(f_{\varepsilon, \Omega, c}) \mathcal{S}_d(\xi_\Omega) \ll (1 + |c|)^C \varepsilon^{-C_d}. \quad (14.A10)$$

The same identity holds after deleting a prescribed finite union of rational lines and planes on both sides.

Proof. Use the coarea coordinates already constructed in Chapter 2: on a patch where the radial projection of the null cone is regular, $(Q(x), \|x\|, \omega(x))$ is a smooth coordinate system. Choose one-dimensional smooth majorants/minorants for the value interval and radial shell, with transition width ε^2 , and a smooth partition of unity on the angular patch. The Jacobian is $\|\nabla Q\|^{-1}$, uniformly smooth on the fixed patch. After the change $x = Ty$, the principal-SL₂ polar decomposition of the level surface writes the angular variable as $k \in K$ and the radial expansion as $a_{\log T}$. Fubini therefore gives (14.A9); the transition layers have Euclidean volume $O(\varepsilon^2 T)$ by the coarea estimate. Each derivative of a cutoff costs at most a fixed power of ε^{-1} , proving (14.A10). Deleting vectors on fixed rational subspaces commutes with the lattice sum, so the modified identity follows term by term. $\square$

Theorem 8.13 (Quantitative ternary count with exceptional subspaces). *Fix $a < b$ and $A \geq 10^3$. There exist absolute constants $N \geq 1$ and $0 < \delta_0 < 1$ such that, for each $0 < \delta \leq \delta_0$, there is $\kappa = \kappa(\delta, A) > 0$ with the following property. For every normalized Q and all $T \geq T_0(A, \|Q\|)$, if (14.A2) holds for all integral forms of height at most T^δ , then there are at most four rational lines L_i and at most four rational planes P_i for which*

$$\begin{aligned} \#\{v \in \mathbb{Z}^3 : \|v\| \leq T, a \leq Q(v) \leq b\} &= C_Q(b - a)T + R_T \\ &\quad + O_{A, Q}((1 + |a| + |b|)^N T^{1-\kappa}). \end{aligned} \quad (14.A11)$$

where R_T is exactly the contribution of lattice points in $\bigcup L_i \cup \bigcup P_i$. The generators may be chosen so that

$$\|v_i\| \leq T^\delta, \quad |Q(v_i)| \leq T^{-2+\delta}, \quad (14.A12)$$

and dually

$$\|w_{i,1}\|, \|w_{i,2}\| \leq T^\delta, \quad |Q^*(w_{i,1} \wedge w_{i,2})| \leq T^{-2+\delta}. \quad (14.A13)$$

Proof. Apply Lemma 8.7 with the parameters chosen so that (14.A12)–(14.A13) contain every exceptional object arising for scales between $T^{1/5}$ and T . The finitely many smaller scales contribute $O(T^{1-\kappa})$ after decreasing κ .

Partition the unit sphere into finitely many regular coarea patches and partition the radial interval $[0, T]$ into geometric shells. On each shell use Lemma 8.12 with smoothing width $\varepsilon = T^{-x}$, and delete the fixed exceptional family. Truncate the modified Siegel transform at height $L = T^y$. Proposition 8.10 and (14.A10) give, after multiplication by the shell scale, an error

$$\ll T^{1-c+xC_1+yC_2}(1+|a|+|b|)^N. \quad (14.A14)$$

The smoothing boundary costs $O(T^{1-x})$. By Proposition 8.11, the cusp truncation costs $O(T^{1-y\sigma})$. Choose first $y > 0$, then $x > 0$, both sufficiently small that

$$c - xC_1 - yC_2 > 0.$$

The minimum of x , $y\sigma$, and $c - xC_1 - yC_2$ is a positive κ . The geometric shell sum changes only the implied constant.

For the main term, Theorem 3.4 turns the Haar average of the Siegel transform into the Euclidean integral. The coarea asymptotic from Chapter 2 gives $C_Q(b-a)T$ after summing the shells. Reinsert the deleted subspaces: their contribution is exactly R_T , not an error. This proves (14.A11). The choices in the finite-family lemma are made with $\rho \ll \delta$ and $E\rho > 2 - \delta$, which gives the displayed generator bounds (14.A12)–(14.A13). $\square$

Worked Example 8.14 (Optimizing the three effective errors). Suppose a concrete implementation of (14.A14) gives $c = 10^{-3}$, $C_1 = 20$, $C_2 = 10$, while the tail exponent is $\sigma = 10^{-2}$. Take $y = 10^{-6}$ and $x = 10^{-5}$. Then

$$c - xC_1 - yC_2 = 10^{-3} - 2 \cdot 10^{-4} - 10^{-5} = 7.9 \cdot 10^{-4},$$

whereas $x = 10^{-5}$ and $y\sigma = 10^{-8}$. Hence the final saving is at least $T^{-10^{-8}}$. The exponent is tiny, but positive. This calculation is the exact logical

content of the phrase “choose the smoothing and cusp cutoffs sufficiently slowly”: the proof only needs one open region of positive solutions to the three linear exponent inequalities.

Proposition 8.15 (Ternary circular Siegel tail). *Let Q be a normalized indefinite ternary form and let ν_T be the smoothed circular/horocyclic measures arising from the standard unfolding of $\mathrm{SO}(Q)^\circ$. For every compactly supported ψ there are $c, A > 0$ such that, for $1 \leq R \leq T^c$, either*

$$\int_{\{\beta > R\}} |\widehat{\psi}| d\nu_T \ll R^{-c},$$

or x_Q is T^{-c} -close to a proper H -compatible homogeneous orbit of height at most T^A .

Proof. Run the circle-to-horocycle alternative of Lemma 8.8 at the scale corresponding to T . If its algebraic obstruction branch occurs, the second alternative of the proposition follows after applying Lemma 8.9; the height and distance exponents merely change by fixed positive constants.

Suppose therefore that the equidistribution branch holds. If the finite Diophantine condition needed for the modified-height construction failed at some relevant dyadic scale, Lemma 7.4 would produce a rational form of polynomial height at polynomially small projective distance; Lemma 8.1 again turns this into the second alternative. We may thus work on the non-obstructed branch with the finite-scale exceptional line/plane family deleted.

For the remaining transform, Proposition 8.11 gives a power tail in the modified height. The modified Lipschitz principle (7.T9) expresses the original Siegel transform as this moment-controlled part plus the quasi-null remainder. The latter is estimated by exactly the same low/high dyadic split used in the proof of (14.A8): high primitive objects contribute a geometric first-moment tail, while low objects either belong to the already deleted exceptional family or contract by Lemma 7.8. Hence, after decreasing c ,

$$\int_{\{\beta > R\}} |\widehat{\psi}| d\nu_T \ll R^{-c} \quad (1 \leq R \leq T^c).$$

This proves the first alternative. Thus the tail proposition is now an internal consequence of the critical moment, finite-scale reconstruction, and the rank-two circle transfer; no arithmetic-conversion black box is used here. $\square$

Lemma 8.16 (Compact primitive value packets). *Fix the reference ternary form Q_0 of signature $(2, 1)$. There are a compact set $\mathcal{K} \Subset X_3$, a chart radius $r_0 > 0$, and constants C, M with the following property. For every $\tau \in \mathbb{R}$ one can choose a primitive $n_\tau \in \mathbb{Z}^3$, a rotation $k_\tau \in \text{SO}(3)$, and, for every $0 < \delta < r_0$, a nonnegative $F_{\tau, \delta} \in C_c^\infty(X_3)$ such that*

$$Q_0(k_\tau n_\tau) = \tau, \quad \|n_\tau\| \leq C(1 + |\tau|)^{1/2}, \quad (14.V1)$$

$$\text{supp } F_{\tau, \delta} \subset B_X(k_\tau \Gamma, \delta) \subset \mathcal{K}, \quad \int_X F_{\tau, \delta} dm_X \geq C^{-1} \delta^8, \quad (14.V2)$$

and, for every fixed Sobolev order j ,

$$\mathcal{S}_j(F_{\tau, \delta}) \leq C_j \delta^{-M_j}. \quad (14.V3)$$

Moreover, if $h\Gamma$ belongs to the support, then after choosing the unique lift in the fixed local chart there is $\gamma \in \text{SL}_3(\mathbb{Z})$ for which

$$h\gamma^{-1} \text{ is } O(\delta)\text{-close to } k_\tau,$$

and therefore

$$|Q_0(h\gamma^{-1}n_\tau) - \tau| \leq C\delta(1 + |\tau|). \quad (14.V4)$$

Proof. Let $a_+ > 0 > a_-$ be the maximum and minimum of Q_0 on the Euclidean unit sphere. Choose an integer $N \asymp 1 + |\tau|^{1/2}$ large enough that $\tau / \|(N, 1, 0)\|^2 \in [a_-, a_+]$, and put $n_\tau = (N, 1, 0)$; this vector is primitive. Continuity of Q_0 on the sphere and transitivity of $\text{SO}(3)$ on the sphere give k_τ with (14.V1).

The image of $\text{SO}(3)$ in X_3 is compact. Enlarge it by a fixed small metric neighbourhood $\mathcal{K}$ on which quotient exponential charts have a common radius r_0 and uniformly bounded Jacobians. Translate and rescale one fixed smooth bump in the eight-dimensional Lie algebra $\mathfrak{sl}_3$ to the chart at $k_\tau \Gamma$. The change-of-variables formula gives the lower mass $\gg \delta^8$, while differentiating the rescaled bump gives (14.V3).

If $h\Gamma$ lies in this chart, discreteness of Γ and the common injectivity radius give a unique γ with $h\gamma^{-1}$ in the chosen lift and $\|h\gamma^{-1} - k_\tau\| \ll \delta$. Since a quadratic form has Lipschitz constant $O(\|n\|^2)$ as a function of the matrix applied to n , (14.V1) gives

$$|Q_0(h\gamma^{-1}n_\tau) - Q_0(k_\tau n_\tau)| \ll \delta \|n_\tau\|^2 \ll \delta(1 + |\tau|),$$

which is (14.V4). □

Theorem 8.17 (Finite-scale ternary Oppenheim alternative). *Let Q be an indefinite ternary quadratic form normalized by $\det Q = 1$. There are absolute constants $A_4, \kappa_3 > 0$ such that, for all sufficiently large R (with the threshold allowed to depend on $\|Q\|$) and every $T \geq R^{A_4}$, one of the following holds:*

1. *every $s \in [-R^{\kappa_3}, R^{\kappa_3}]$ is approximated by a primitive $v \in \mathbb{Z}^3$, $0 < \|v\| \leq T$, with $|Q(v) - s| \leq R^{-\kappa_3}$;*
2. *there is an integral ternary form Q' with $\|Q'\| \leq R$ such that, with $\lambda = (\det Q')^{-1/3}$,*

$$\|Q - \lambda Q'\| \leq R^{A_4} (\log T)^{A_4} T^{-2}. \quad (14.01)$$

Proof of Theorem 8.17. Write

$$Q(v) = \lambda_Q Q_0(g_Q v), \quad x_Q = g_Q \Gamma,$$

as in Chapter 4. In the determinant normalization λ_Q is a fixed nonzero constant depending only on the chosen reference form and signature. Let A_0, κ_0 and the Sobolev order j be the constants in the sharp expanding alternative, the sharp expanding-segment theorem of Volume XIII, for $G = \mathrm{SL}_3$ and its principal H . Here the transverse module is Sym^4 , so $m = 2$.

We choose the exponents in an order that makes the bookkeeping explicit. Let C_1 be large enough that Lemma 8.2 turns a periodic orbit of volume and arithmetic height at most B and orbit distance ε into an integral form of height at most B^{C_1} and coefficient error at most $B^{C_1} \varepsilon$. Put

$$\eta = \frac{1}{4A_0 C_1}. \quad (14.V5)$$

For the packet bump, write $\mathcal{S}_j(F_{\tau, \delta}) \ll \delta^{-M}$. Choose $\kappa_3 > 0$ so small that

$$(2M + 16)\kappa_3 < \frac{1}{2}\eta\kappa_0. \quad (14.V6)$$

Finally enlarge A_4 so that every power of R appearing below is absorbed by R^{A_4} and so that the rescaled time parameter satisfies the threshold in the sharp expanding-segment theorem of Volume XIII.

Set

$$R_d = R^\eta, \quad T_d = c_Q T R^{-\kappa_3}, \quad (14.V7)$$

where the fixed $c_Q > 0$ is chosen small enough for the norm estimate at the end of the proof. Apply the sharp expanding-segment theorem of Volume XIII to x_Q with parameters R_d, T_d .

If its periodic-orbit branch occurs, the orbit has arithmetic height at most $R_d^{A_0}$. Lemma 8.2 gives a primitive integral ternary form Q' with

$$\|Q'\| \ll_Q R_d^{A_0 C_1} \leq R \quad (14.V8)$$

for all sufficiently large R , by (14.V5), and

$$\|Q - \lambda Q'\| \ll_Q R_d^{A_0 C_1 + A_0} (1 + \log T_d)^{A_0} T_d^{-2} \leq R^{A_4} (\log T)^{A_4} T^{-2}. \quad (14.V9)$$

After determinant normalization the scalar is $\lambda = (\det Q')^{-1/3}$ (with the real cube root compatible with the fixed signature). This is alternative (2).

Assume now the equidistribution branch. Fix any $|s| \leq R^{\kappa_3}$ and put $\tau = s/\lambda_Q$. Apply Lemma 8.16 with

$$\delta = c'_Q R^{-2\kappa_3}, \quad (14.V10)$$

where c'_Q is small enough that (14.V4), multiplied by $|\lambda_Q|$, is at most $\frac{1}{2}R^{-\kappa_3}$. Since $1 + |\tau| \ll_Q R^{\kappa_3}$, (14.V2)–(14.V3) give

$$\int_X F_{\tau, \delta} dm_X \gg_Q R^{-16\kappa_3}, \quad \mathcal{S}_j(F_{\tau, \delta}) \ll_Q R^{2M\kappa_3}. \quad (14.V11)$$

The error in (13.R4) is therefore $O_Q(R^{2M\kappa_3 - \eta\kappa_0})$, which is smaller than one half of the main term in (14.V11) by (14.V6). Hence

$$\int_0^1 F_{\tau, \delta}(a_{\log T_d} u_r x_Q) dr > 0.$$

Choose r for which the integrand is positive. By the local-chart clause of Lemma 8.16, there is $\gamma \in \mathrm{SL}_3(\mathbb{Z})$ such that, with $v = \gamma^{-1}n_\tau$, the vector v is primitive and

$$|Q(v) - s| = |\lambda_Q| |Q_0(a_{\log T_d} u_r g_Q v) - \tau| \leq R^{-\kappa_3}. \quad (14.V12)$$

Here we used that both a_t and u_r lie in $H = \mathrm{SO}(Q_0)^\circ$.

It remains only to check the size of v . The support of all packet bumps lies in the fixed compact set $\mathcal{K}$, $0 \leq r \leq 1$, and in the three-dimensional principal representation $\|a_{-\log T_d}\| \ll T_d$. Thus (14.V1) gives

$$\|v\| \ll_Q T_d \|n_\tau\| \ll_Q c_Q T R^{-\kappa_3} R^{\kappa_3/2} \leq T, \quad (14.V13)$$

for the fixed choice of c_Q and all sufficiently large R . Also $n_\tau \neq 0$, so $v \neq 0$. Since s was arbitrary, the same global equidistribution branch supplies alternative (1) for every $s \in [-R^{\kappa_3}, R^{\kappa_3}]$.

Thus the two branches of the sharp expanding theorem proved in Volume XIII give the two alternatives of the finite-scale ternary Oppenheim theorem. The integral obstruction recorded here is the arithmetic form produced by the periodic-orbit reconstruction. The displayed formulation of [14, Theorem 2.1] uses a matrix-space alternative, while its proof passes through integral forms (see in particular its arithmetic periodic-orbit conversion); our formulation records the arithmetic version actually used downstream. $\square$

Theorem 8.18 (Quantitative count with explicit exceptional subspaces). *Fix $a < b$ and $A \geq 10^3$. There exist absolute constants $N \geq 1$ and $0 < \delta_0 < 1$ such that, for every $0 < \delta \leq \delta_0$, there is an exponent $\kappa = \kappa(\delta, A) > 0$ with the following property. For every normalized Q and all $T \geq T_0(A, \|Q\|)$, assume the Diophantine separation*

$$\|Q - \lambda Q'\| \geq \|Q'\|^{-A}$$

for every integral form Q' with $\|Q'\| \leq T^\delta$, where $\lambda = (\det Q')^{-1/3}$. Then

$$\begin{aligned} \#\{v \in \mathbb{Z}^3 : \|v\| \leq T, a \leq Q(v) \leq b\} \\ = C_Q(b - a)T + R_T + O_{A,Q}\left((1 + |a| + |b|)^N T^{1-\kappa}\right). \end{aligned} \quad (14.02)$$

where R_T is supported on at most four rational lines and at most four rational planes. They may be chosen with primitive generators satisfying $\|v_i\| \leq T^\delta$, $|Q(v_i)| \leq T^{-2+\delta}$ for the lines, and $\|w_{i,1}\|, \|w_{i,2}\| \leq T^\delta$ together with $|Q^(w_{i,1} \wedge w_{i,2})| \leq T^{-2+\delta}$ for the planes.*

Proof. This is Theorem 8.13. The statement has been repeated here because it is the form used by the applications below. The preceding internal proof constructs the at-most-four exceptional line/plane families, proves the modified circular estimate and power cusp tail, performs the shell-to-circle smoothing, and optimizes the three effective errors. The comparison with [14, Theorem 2.3] is theorem concordance, not proof ownership. $\square$

Corollary 8.19 (Clean polynomial-rate count after excluding exceptional subspaces). *If, in addition to the Diophantine hypothesis of Theorem 8.18, the displayed quasi-null line and plane conditions have no solutions for all sufficiently large T , then $R_T = 0$ and the count has the clean power-saving asymptotic*

$$C_Q(b - a)T + O_Q((1 + |a| + |b|)^N T^{1-\kappa}).$$

Proof. Under the extra hypothesis every candidate exceptional line or plane is absent, hence the explicitly supported remainder R_T in (14.O2) vanishes. $\square$

Corollary 8.20 (Uniform shrinking windows on the non-obstructed branch). *Let Q be a normalized indefinite ternary form. Assume that there are $A \geq 10^3$ and $\delta > 0$ such that, for every sufficiently large T , every nondegenerate integral ternary form Q' with $\|Q'\| \leq T^\delta$, and $\lambda = (\det Q')^{-1/3}$,*

$$\|Q - \lambda Q'\| \geq \|Q'\|^{-A}. \quad (9.0)$$

Then there exist $\sigma, \eta > 0$ and $\epsilon_0 > 0$ such that, for all sufficiently large T and all

$$T^{-\sigma} \leq \epsilon \leq \epsilon_0,$$

the nonsingular count satisfies

$$N_Q^{\text{ns}}(-\epsilon, \epsilon; T) = 2C_Q \epsilon T (1 + O_Q(T^{-\eta})). \quad (9.1)$$

The same statement holds for a translated value window of length 2ϵ in any fixed compact set of values.

Proof. Let δ_0 and Sobolev order d be supplied by Volume XIII. Smooth the value interval across a boundary layer of relative width ϑ , and truncate the Siegel transform at height R . For constants $B, C, c > 0$ coming respectively from the Sobolev norm, bounded-height derivatives, and Proposition 8.15, the relative error with respect to the geometric scale T is bounded by

$$\vartheta + T^{-\delta_0} (\epsilon \vartheta)^{-B} R^C + R^{-c}. \quad (9.2)$$

The main term has size $\asymp \epsilon T$, so divide (9.2) by ϵ . Put $\epsilon \geq T^{-\sigma}$, $R = T^\beta$, and $\vartheta = T^{-\gamma}$. The three relative-to-main exponents become

$$-\gamma + \sigma, \quad -\delta_0 + \sigma(B + 1) + \gamma B + \beta C, \quad -\beta c + \sigma.$$

Choose first $\beta > 0$, then $\sigma > 0$ and $\gamma > \sigma$ sufficiently small so that all three are negative. Their minimum absolute value is some $\eta > 0$. Effective closing has no obstruction branch under the stated Diophantine hypothesis, so (9.2) is valid for all sufficiently large T and gives (9.1). $\square$

Chapter 9

From Counting to Small Values

9.1 A deterministic extraction lemma

Lemma 9.1 (Positive count implies a small value). *Suppose that for some symmetric interval $I_T = (-\varepsilon_T, \varepsilon_T)$ one has*

$$N_{Q,\Omega,I_T}(T) = 2c_{Q,\Omega}\varepsilon_T T^{n-2} + O(E(T)),$$

where the main term is larger than $2|E(T)|$. Then there exists $0 \neq v \in \mathbb{Z}^n \cap T\Omega$ with $|Q(v)| < \varepsilon_T$.

Proof. The right side is positive, so the integer-valued count is at least one. If zero itself can be represented by nonzero integral vectors and one wants a nonzero value, replace the symmetric interval by one of its half-intervals avoiding zero and use the same argument. $\square$

Corollary 9.2 (Power-saving count gives a shrinking target). *If*

$$N_{Q,\Omega,(-\varepsilon,\varepsilon)}(T) = 2c_Q\varepsilon T^{n-2} + O(T^{n-2-\delta})$$

uniformly for $\varepsilon \geq T^{-\sigma}$, with $0 < \sigma < \delta$, then for every large T there is a nonzero integer vector v with

$$\|v\| \ll T, \quad |Q(v)| < T^{-\sigma}.$$

Proof. Take $\varepsilon = T^{-\sigma}$. The main term has order $T^{n-2-\sigma}$, whereas the error is $O(T^{n-2-\delta})$. Since $\sigma < \delta$, their ratio tends to infinity. For all sufficiently large T the main term is therefore more than twice the error, and the preceding deterministic extraction lemma gives the desired vector. $\square$

Corollary 9.3 (Polynomial small values for Diophantine ternary forms). *Under the hypotheses of Corollary 8.20, for every $0 < \sigma' < \sigma$ and all sufficiently large T there is $0 \neq v \in \mathbb{Z}^3$ with*

$$\|v\| \ll T, \quad 0 < |Q(v)| < T^{-\sigma'}.$$

If Q represents zero integrally, choose one of the half-windows $(0, T^{-\sigma'})$ or $(-T^{-\sigma'}, 0)$ in (9.1), so the vector obtained has nonzero value.

Proof. Apply Corollary 8.20 with $\epsilon = T^{-\sigma'}$. Its main term is asymptotic to $2C_Q T^{1-\sigma'} > 0$. The relative error tends to zero, so the nonsingular integer count is positive for all large T . $\square$

9.2 Effective density scale

The corollary converts an orbit rate into a Diophantine exponent. The final exponent is almost never equal to the raw dynamical δ : Sobolev norms of a shrinking value-window cutoff grow like a power of ε^{-1} , so one must solve an optimization inequality of the form

$$T^{-\delta} \varepsilon^{-A} \ll \varepsilon.$$

Thus one may take

$$\varepsilon = T^{-\delta/(A+1)+o(1)}.$$

This explicit loss should always be reported.

Chapter 10

Higher Dimensions: Two Effective Routes

10.1 Route A: effective orthogonal dynamics

For $n \geq 5$, $H = \mathrm{SO}(p, q)^\circ$ has higher dimension and stronger cusp integrability. If an effective equidistribution theorem for suitable unipotent subgroups of H is available with polynomial rate and an effective closing alternative, the ternary argument extends verbatim. The obstacle is not the geometric volume calculation but the effective homogeneous theorem.

10.2 Route B: rational slicing to controlled ternary subspaces

Volume IV proved qualitative Oppenheim in all dimensions by finding a ternary restriction on which irrationality survives. For an effective statement the height of that ternary subspace must be controlled.

Proposition 10.1 (Finite rational slicing with Diophantine inheritance). *Fix $n \geq 4$ and a compact set $\mathcal{K} \subset \mathrm{Sym}_n(\mathbb{R})$ of nondegenerate indefinite forms, bounded away from the zero form. There are constants $A \geq 1$ and $C \geq 1$, depending only on n , $\mathcal{K}$, and the fixed coefficient norm, with the following property. For every $Q \in \mathcal{K}$ and every $B \geq 2$ there is a rational three-plane $W \subset \mathbb{Q}^n$ such that*

1. $Q|_{W_{\mathbb{R}}}$ is nondegenerate and indefinite;

2. $W \cap \mathbb{Z}^n$ has a primitive basis whose vectors have norm at most C (hence $\text{ht}(W) \leq C$);
3. after using that primitive basis to regard $Q|_W$ as a ternary form,

$$\Delta_{Q|_W}(B) \geq C^{-1} \Delta_Q(B^A). \quad (10.1)$$

In particular, if Q is (κ, C_0) -Diophantine, then the selected ternary restriction satisfies

$$\Delta_{Q|_W}(B) \geq C^{-1} C_0 B^{-A\kappa} \quad (B \geq 2). \quad (10.2)$$

Proof. For a rational three-plane W with a fixed primitive integral basis matrix $M_W \in \text{Mat}_{n \times 3}(\mathbb{Z})$, write

$$R_W : \text{Sym}_n(\mathbb{R}) \longrightarrow \text{Sym}_3(\mathbb{R}), \quad R_W(Q) = M_W^T Q M_W. \quad (10.3)$$

This is a rational linear map, in fact integral in the chosen bases.

Step 1: finitely many indefinite planes separate all coefficients. Fix $Q_* \in \mathcal{K}$. The set of three-planes on which Q_* is nondegenerate and indefinite is a nonempty open subset $\mathcal{U}_{Q_*}$ of the real Grassmannian: choose one positive and one negative vector for Q_* and then a third vector outside the finitely many degeneracy hypersurfaces. Rational planes are dense, so rational planes are dense in $\mathcal{U}_{Q_*}$.

We claim that finitely many rational planes $W_1, \dots, W_m \in \mathcal{U}_{Q_*}$ can be chosen so that

$$R = (R_{W_1}, \dots, R_{W_m}) : \text{Sym}_n(\mathbb{R}) \longrightarrow \bigoplus_{j=1}^m \text{Sym}_3(\mathbb{R}) \quad (10.4)$$

is injective. Indeed, if a quadratic form P restricts to zero on every plane in the nonempty open set $\mathcal{U}_{Q_*}$, then $P(v) = 0$ for every v belonging to the union of those planes. That union contains a nonempty open subset of $\mathbb{R}^n$, so the polynomial P is identically zero. Thus $\bigcap_{W \in \mathcal{U}_{Q_*}} \ker R_W = \{0\}$. Since Sym_n is finite dimensional, a finite subcollection already has zero kernel; perturbing its planes slightly to rational ones preserves both injectivity and indefiniteness.

Indefiniteness, nondegeneracy, and injectivity of (10.4) are open conditions. Hence the same finite configuration works for every Q in a neighborhood of Q_* . Compactness of $\mathcal{K}$ supplies finitely many such configurations. Taking the maximum over this finite catalog gives a uniform bound C_1 for the primitive basis vectors and for the operator norm of a rational left inverse L of every corresponding map R .

Step 2: simultaneous good approximations on all slices would approximate Q . Fix $Q \in \mathcal{K}$ and choose one catalog configuration $W_1, \dots, W_m$ valid for Q . Suppose, for some $0 < \delta < 1$, that every slice has a rational approximation

$$\left\| R_{W_j}(Q) - \frac{P_j}{q_j} \right\| < \delta, \quad 1 \leq q_j \leq B, \quad P_j \in \text{Sym}_3(\mathbb{Z}). \quad (10.5)$$

Let $q = \text{lcm}(q_1, \dots, q_m) \leq B^m$ and stack the matrices P_j/q_j into a vector z in the target of R . Because R is rational, its fixed left inverse can be chosen rational; let D be a common denominator for its matrix entries. Then

$$P = Lz \in \text{Sym}_n(\mathbb{Q}), \quad \text{den}(P) \leq DB^m, \quad (10.6)$$

and

$$\|Q - P\| = \|L(RQ - z)\| \leq C_2\delta. \quad (10.7)$$

Since $\mathcal{K}$ is bounded away from zero, after decreasing a fixed threshold for δ the rational form P is nonzero. For larger δ the conclusion below is automatic after enlarging the constant. Choose an integer $A > m$ large enough that $DB^m \leq B^A$ for every $B \geq 2$. Equations (10.6)–(10.7) imply

$$\Delta_Q(B^A) \leq C_2 \max_j \Delta_{Q|W_j}(B). \quad (10.8)$$

Therefore at least one of the finitely many indefinite slices satisfies (10.1). Its basis height is at most the catalog bound C_1 . If Q is (κ, C_0) -Diophantine, then by definition $\Delta_Q(H) \geq C_0 H^{-\kappa}$ for $H \geq 2$. Substituting $H = B^A$ into (10.1) gives

$$\Delta_{Q|W}(B) \geq C^{-1} \Delta_Q(B^A) \geq C^{-1} C_0 B^{-A\kappa},$$

which is (10.2). □

Corollary 10.2 (Polynomial small values by rational slicing). *Let Q be a normalized nondegenerate indefinite quadratic form in $n \geq 4$ variables. Assume that Q is (κ, C_0) -Diophantine in the sense of the rational-approximation function of Chapter 2. Then there exist $C_1 = C_1(Q, \kappa, C_0) > 0$ and $\varepsilon_0 = \varepsilon_0(Q, \kappa, C_0) > 0$ such that for every $0 < \varepsilon < \varepsilon_0$ there is $v \in \mathbb{Z}^n \setminus \{0\}$ satisfying*

$$0 < |Q(v)| < \varepsilon, \quad \|v\| \leq \varepsilon^{-C_1}.$$

Proof. Apply Proposition 10.1. The selected rational three-plane has a primitive integral basis of uniformly bounded height, and (10.2) gives the ternary restriction a polynomial Diophantine lower bound whose exponent

depends only on the original one and on the finite slicing catalog. Apply the ternary effective theorem 8.17 together with its shrinking-target corollary to that restriction. For every sufficiently small ε this produces $w \in W \cap \mathbb{Z}^n$ with

$$0 < |Q(w)| < \varepsilon$$

and ternary coordinate norm at most a fixed power of ε^{-1} . The primitive basis matrix of $W \cap \mathbb{Z}^n$ has norm bounded by the fixed catalog constant, so returning to $\mathbb{Z}^n$ changes the norm by only a fixed multiplicative factor. Absorbing that factor and the ternary exponent into C_1 gives the stated bound. $\square$

Chapter 11

A (2, 2) Determinant Laboratory

The signature (2, 2) case becomes unusually transparent after identifying $\mathbb{R}^4$ with $M_2(\mathbb{R})$ and

$$Q(X) = \det X.$$

Every real form of signature (2, 2) is linearly equivalent to a scalar multiple of this model.

11.1 Isotropic planes

The determinant-zero cone consists of rank-one matrices. Two natural families of maximal isotropic planes are

$$\{uv^T : v \in \mathbb{R}^2\} \quad (u \text{ fixed}), \quad \{uv^T : u \in \mathbb{R}^2\} \quad (v \text{ fixed}).$$

A rational such plane can contain a rank-two lattice and therefore creates a singular T^2 contribution when the value interval contains zero.

Lemma 11.1 (Nonsingular coarea for determinant). *On $\{X : \det X = 0, X \neq 0\}$ the gradient of determinant is the adjugate matrix and is nonzero. Hence the nonsingular volume in a fixed determinant window is $c(b-a)T^2 + o(T^2)$.*

Proof. For a 2×2 matrix $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$\nabla \det X = (d, -c, -b, a),$$

which vanishes only at $X = 0$. Apply Proposition 2.2 with $n = 4$. $\square$

This laboratory cleanly separates the two phenomena that are intertwined in abstract notation: nonsingular equidistribution and singular rational isotropic planes.

Chapter 12

Error Ledgers for Effective Oppenheim

A trustworthy effective theorem must expose every exponent loss. For a smoothed counting function the typical bound has the form

$$|N(T) - V(T)| \ll T^{n-2} \left(\underbrace{\eta}_{\text{geometric boundary}} + \underbrace{R^{-c_1}}_{\text{cusp tail}} + \underbrace{T^{-c_2} R^{A_1} \eta^{-A_2}}_{\text{dynamics/Sobolev}} + \underbrace{T^{-c_3}}_{\text{effective closing outside obstruction}} \right).$$

Choose $R = T^\beta$ and $\eta = T^{-\gamma}$. The exponent is the minimum of

$$\gamma, \quad c_1\beta, \quad c_2 - A_1\beta - A_2\gamma, \quad c_3.$$

The optimal (β, γ) is found by a two-variable linear program. This simple calculation is conceptually important: it prevents a rate proved for smooth bounded test functions from being reported unchanged for a sharp lattice count.

Worked Example 12.1 (Balancing three losses). If the errors are $\eta + R^{-1/4} + T^{-1/20} R^2 \eta^{-3}$, set $R = T^\beta$, $\eta = T^{-\gamma}$. Equalizing $\gamma = \beta/4 = 1/20 - 2\beta - 3\gamma$ gives $\beta = 4\gamma$ and

$$\gamma = \frac{1}{20} - 8\gamma - 3\gamma, \quad 12\gamma = \frac{1}{20},$$

so $\gamma = 1/240$. The final count has a $T^{-1/240}$ relative saving, not the original $T^{-1/20}$ dynamical saving.

Chapter 13

Final Proof-Completeness Ledger

Remark 13.1 (Geometric layer). The T^{n-2} main term and its coarea constant are proved in Chapter 2. No lattice dynamics is hidden in the volume calculation.

Remark 13.2 (Cusp layer). The Siegel-transform height bound, modified-height contraction, flag submodularity, and uniform-integrability step are proved in Chapters 3–5. These are the load-bearing mechanisms behind the nonexceptional-signature EMM asymptotic and, after modification, the two critical-signature theorems.

Remark 13.3 (Qualitative counting). The nonexceptional-signature fixed-window asymptotic is derived in Chapter 6 from the Ratner/Mozes–Shah machinery already proved in Volumes III and VII plus the cusp layer proved here. Low-dimensional exceptional geometry is isolated in Chapter 7 rather than suppressed.

Remark 13.4 (Effective layer). Volume XIII owns the effective closing and rank-two equidistribution alternative. This volume proves the arithmetic translation from a low-height homogeneous obstruction to a good rational approximation of the form, the truncation optimization, and the conversion of count error into a small-value bound.

Remark 13.5 (Scope boundary). Recent stronger results for products of many SL_2 factors, general S -arithmetic settings, semidirect products, and shrinking intervals are important but are not silently assumed here. They belong to the Advanced Homogeneous Dynamics continuation unless their special

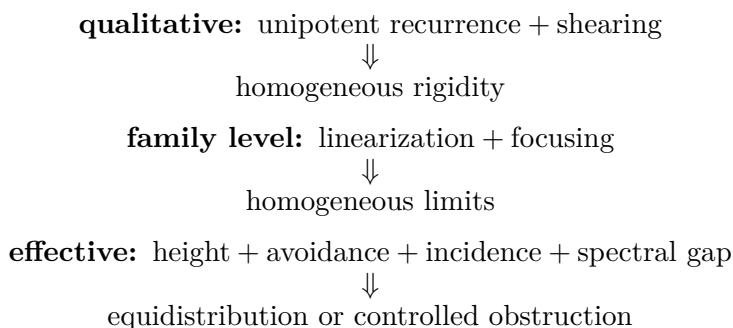
real-archimedean forms are explicitly proved in Volume XIII. This preserves the stated scope of the Ratner notebook.

Chapter 14

Epilogue: What the Fourteen Volumes Have Established

The series begins with the topology and geometry of G/Γ , develops the measure theory and polynomial divergence of unipotent actions, proves Ratner's classification/orbit-closure/equidistribution mechanism, and applies it qualitatively to Oppenheim. Arc I then studies what happens when the measure or trajectory varies: arithmetic structure, Howe–Moore mixing, homogeneous-measure limits, polynomial trajectories, and expanding translates. Arc II converts these qualitative statements into counting and, where modern methods permit, polynomial finite-scale alternatives.

The unifying principle can now be stated at three resolutions:



The last line is not a replacement for Ratner's proof. It is the finite-scale descendant of the same geometric idea.

Chapter 15

Worked Arithmetic Laboratory: Coarea and Quasi-Null Geometry

15.1 Motivation

The quantitative Oppenheim problem has two competing geometries: the regular coarea main term and exceptional rational directions on which the quadratic form is unusually small.

Worked Example 15.1 (The coarea constant for $x^2 + y^2 - z^2$). Let $Q = x^2 + y^2 - z^2$ and use the Euclidean unit ball. On the null cone write, for the upper nappe,

$$F(r, \theta) = (r \cos \theta, r \sin \theta, r), \quad 0 \leq r \leq 1/\sqrt{2}.$$

Then $|F_r \times F_\theta| = \sqrt{2}r$, while $\|\nabla Q(F)\| = 2\sqrt{2}r$. Hence $d\sigma/\|\nabla Q\| = \frac{1}{2} dr d\theta$. The lower nappe contributes the same amount, so

$$C_Q = 2 \int_0^{1/\sqrt{2}} \int_0^{2\pi} \frac{1}{2} d\theta dr = \sqrt{2}\pi.$$

This is a complete model calculation of the constant multiplying a small fixed value window.

Worked Example 15.2 (A quasi-null vector rationalizes a diagonal coefficient). Let $Q_\alpha(x, y, z) = x^2 + y^2 - \alpha z^2$ and take the primitive vector

$v = (p, 0, q)$. Then

$$Q_\alpha(v) = p^2 - \alpha q^2, \quad \left| \alpha - \frac{p^2}{q^2} \right| = \frac{|Q_\alpha(v)|}{q^2}.$$

Thus if $\|v\| \leq T^\delta$ and $|Q_\alpha(v)| \leq T^{-2+\delta}$, then α is approximated by the rational square $(p/q)^2$ with error at most $T^{-2-\delta}$ up to fixed norm-comparison constants. This elementary calculation is the one-dimensional prototype of the line/plane rationalization alternatives retained in Theorem 8.18.

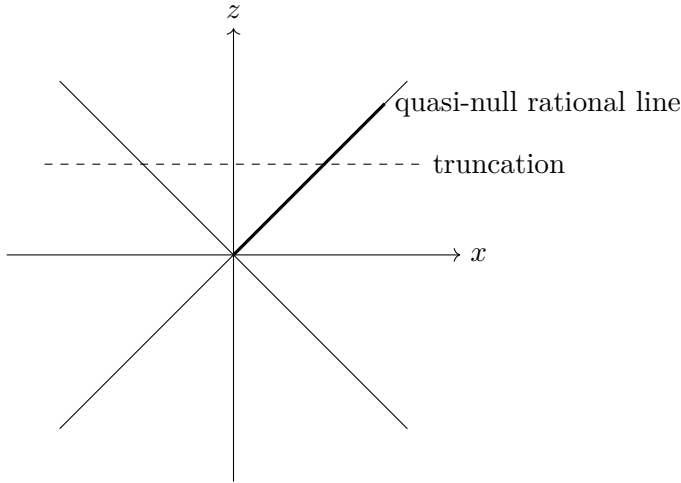


Figure 15.1: The regular mass lies on the cone, while a persistent rational line near the cone can contribute an exceptional arithmetic term and must be tracked separately.

Dyadic cusp ledger

Let the k -th cusp layer be the region where the relevant height lies in $[2^k R, 2^{k+1} R)$. If quantitative nondivergence/avoidance gives mass $\ll 2^{-ak} R^{-a}$ and the number/size of lattice contributions in that layer is $\ll 2^{bk} R^b$, then the layer contributes

$$\ll 2^{-(a-b)k} R^{-(a-b)}.$$

The tail is summable exactly when $a > b$; this inequality is the point that must be verified before one writes a geometric series.

$$\boxed{R} \quad \boxed{2R} \quad \boxed{4R} \quad \boxed{8R} \quad \boxed{16R} \\ \xrightarrow{\hspace{1.5cm}} \text{deeper cusp}$$

What the figure answers. Every move to a deeper layer gains a powers from avoidance and loses b powers from counting; the net gain must stay positive.

Critical-moment laboratory

Why this laboratory exists. The critical-signature moment packages are proved in the detailed proof chapters. The finite-incidence closure and sharp T^{-2} finite-scale ternary obstruction, together with the EMM-type nonexceptional mean-value estimate, are likewise proved in this volume and Volume XIII. Siegel's mean-value normalization is proved here as well. The calculations below isolate the geometric mechanisms that the proof chapters turn into uniform estimates.

Worked Example 15.3 (A rational isotropic plane already has main-scale size). For $Q_{\det}(X) = \det X$ on $M_2(\mathbb{R})$, let

$$P = \left\{ \begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} : a, b \in \mathbb{R} \right\}.$$

Then $Q_{\det}|_P = 0$ and $P \cap M_2(\mathbb{Z}) \cong \mathbb{Z}^2$. For the sup norm and an integer $N \geq 1$,

$$\#\{X \in P \cap M_2(\mathbb{Z}) : \|X\|_{\infty} \leq N\} = (2N + 1)^2 = 4N^2 + 4N + 1.$$

Thus a rational isotropic plane contributes on the same T^2 scale as the nonsingular $(2, 2)$ main term. It cannot be hidden in an error term; it must be removed before equidistribution and added back explicitly.

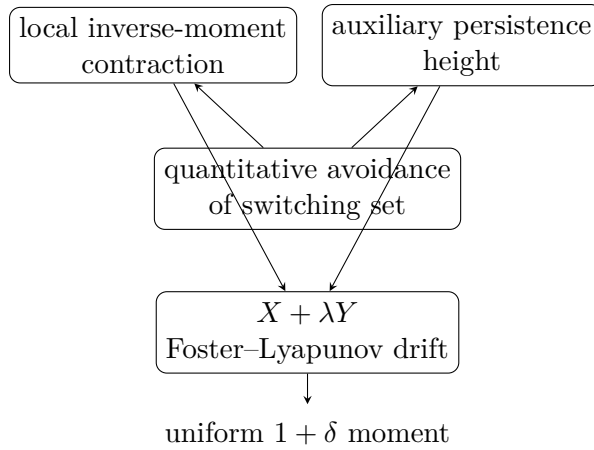


Figure 15.2: What must be proved before the elementary two-height drift algebra can be applied. The arrows from local contraction and avoidance are the internally reconstructed deep critical-weight inputs; Proposition 7.35 owns only the final algebraic synthesis.

Chapter 16

Student Proof Laboratory: From Dynamics Back to Quadratic Forms

The final volume repeatedly moves between a quadratic form, a lattice point in a homogeneous space, a unipotent orbit, and an integer counting problem. This chapter writes those translations in one place and explains the two exceptional low-dimensional signatures carefully.

16.1 Normalizing the quadratic form

Let $Q(x) = x^T A x$ be nondegenerate and indefinite. Multiplying Q by a nonzero scalar does not change the qualitative Oppenheim problem, but it does change numerical constants. For effective statements fix a normalization such as $|\det A| = 1$. Choose $g \in \mathrm{SL}_n(\mathbb{R})$ with

$$A = g^T A_0 g,$$

where A_0 is a fixed diagonal form of the same signature. Then $Q(v) = Q_0(gv)$. The lattice $g\mathbb{Z}^n$ is the point of $\mathrm{SL}_n(\mathbb{R})/\mathrm{SL}_n(\mathbb{Z})$ attached to Q .

Lemma 16.1 (Rationality and closed orthogonal orbits). *If the connected orthogonal group $H = \mathrm{SO}(Q_0)^\circ$ has a closed finite-volume orbit through $g\mathrm{SL}_n(\mathbb{Z})$, then Q is proportional to a rational form.*

Proof. The stabilizer $g^{-1}Hg \cap \mathrm{SL}_n(\mathbb{Z})$ is a lattice in the conjugate orthogonal group. Borel density makes this lattice Zariski dense in that group. The line

spanned by the coefficient tensor of Q is fixed by the conjugate orthogonal group and hence by the lattice. Since the lattice is integral, the fixed line is defined over $\mathbb{Q}$ by solving a homogeneous system with rational coefficients. Thus the coefficient vector of Q is a real scalar multiple of a rational vector. $\square$

16.2 The geometric main term by coarea

For a bounded interval I and a norm ball B_T , consider

$$\text{vol}\{x \in B_T : Q(x) \in I\}.$$

Write $x = Ty$. Since $Q(Ty) = T^2Q(y)$,

$$\text{vol}\{x \in B_T : Q(x) \in I\} = T^n \text{vol}\{y \in B_1 : Q(y) \in T^{-2}I\}.$$

If the cone $Q = 0$ meets the unit sphere transversely away from the origin, coarea gives

$$\text{vol}\{y \in B_1 : |Q(y)| < \eta\} = 2\eta \int_{B_1 \cap Q^{-1}(0)} \frac{d\sigma(y)}{\|\nabla Q(y)\|} + o(\eta).$$

Putting $\eta \asymp T^{-2}$ produces the scale T^{n-2} . Thus the T^{n-2} main term is dictated before any dynamics enters.

16.3 The Siegel transform and why it is unbounded

For a compactly supported function f on $\mathbb{R}^n$, define

$$\widehat{f}(\Lambda) = \sum_{0 \neq v \in \Lambda} f(v).$$

When Λ has a very short vector, many multiples of that vector may lie in the support of f , so $\widehat{f}$ is not uniformly bounded on the space of lattices. Rather equidistribution for bounded test functions therefore cannot be applied directly. Truncate according to a height function $\alpha(\Lambda)$ measuring short rational subspaces:

$$\widehat{f} = \widehat{f} \mathbf{1}_{\alpha \leq M} + \widehat{f} \mathbf{1}_{\alpha > M}.$$

The first term is bounded by a power of M ; the second is controlled by a moment estimate for α . This truncation is the analytic bridge from homogeneous equidistribution to counting values of Q .

16.4 Why signatures $(2, 1)$ and $(2, 2)$ are special

In the stable signatures, the height has an integrable moment strong enough to make the cusp tail negligible at the T^{n-2} scale. In signatures $(2, 1)$ and $(2, 2)$ there are rational subspaces on which the restriction of Q is zero or nearly zero. Lattice vectors lying in such quasi-null subspaces can create anomalously many small values and destroy a uniform $O(T^{n-2})$ upper bound.

Worked Example 16.2 (The determinant form). Identify $\mathbb{R}^4$ with 2×2 real matrices and let $Q(X) = \det X$, a form of signature $(2, 2)$. Every two-dimensional subspace of matrices with a fixed zero row consists entirely of singular matrices. If an arithmetic lattice is exceptionally close to such a rational plane, there can be many lattice points with very small determinant. This is the model for the quasi-null subspaces that must be removed or controlled by a Diophantine hypothesis.

16.5 Finite Diophantine type

Say, schematically, that Q has finite Diophantine type if its coefficient line cannot be approximated by rational coefficient lines faster than a fixed power of the denominator. The precise definition in the main text is chosen to match the height of rational homogeneous obstructions. Under this hypothesis, a quasi-null rational subspace of height H cannot have Q -restriction smaller than H^{-A} for arbitrarily large A . This lower bound is exactly what allows the modified $1 + \delta$ Margulis moment to close in the critical signatures.

16.6 From an effective orbit alternative to a value of Q

Suppose effective equidistribution gives the following dichotomy at scale T :

1. a smoothed orbit average differs from Haar by at most $T^{-\eta}$; or
2. Q lies within $T^{-\eta}$, in coefficient space, of a rational form of height at most T^A .

If Q satisfies a Diophantine lower bound excluding the second alternative, choose a nonnegative test function supported where $|Q_0(x) - s| < \delta$ and $\|x\| < 1$. Haar measure of this set is $\asymp \delta$. Taking δ larger than the equidistribution error makes the orbit average positive. Positivity means

that some integer vector v with $\|v\| \ll T$ satisfies $|Q(v) - s| \ll \delta$. Thus a dynamical error exponent becomes a polynomial small-value exponent.

Lemma 16.3 (Positive count gives a point). *If $N(T, \delta)$ is an integer-valued counting function and*

$$N(T, \delta) = c\delta T^{n-2} + O(T^{n-2-\eta})$$

with $c > 0$, then for $\delta = CT^{-\eta}$ and C sufficiently large one has $N(T, \delta) > 0$ for all large T .

Proof. The main term equals $cCT^{n-2-\eta}$. Choose C larger than twice the implicit error constant divided by c . Then the main term exceeds the absolute value of the error, so the integer count is positive. $\square$

16.7 Overlapping ternary slices in higher dimension

A single ternary restriction cannot determine an n -variable quadratic form. To reconstruct all coefficients, fix a rational indefinite two-plane $P = \text{span}(e_1, e_2)$ and choose complementary vectors $w_3, \dots, w_n$. For every i consider the ternary plane $P_i = P \oplus \mathbb{R}w_i$, and for every $i < j$ also consider $P_{ij} = P \oplus \mathbb{R}(w_i + w_j)$.

Lemma 16.4 (Reconstruction from overlapping slices). *Knowing the restrictions of a quadratic form to all P_i and P_{ij} , with one common normalization on P , determines its full Gram matrix.*

Proof. The restriction to P gives the coefficients $B(e_a, e_b)$ for $a, b \leq 2$. The restriction to P_i gives $B(e_1, w_i)$, $B(e_2, w_i)$ and $B(w_i, w_i)$. Finally,

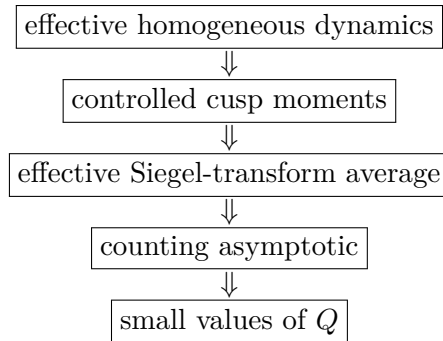
$$Q(w_i + w_j) = Q(w_i) + Q(w_j) + 2B(w_i, w_j),$$

so the restriction to P_{ij} recovers $B(w_i, w_j)$. These are all entries of the symmetric Gram matrix. The common restriction to P fixes the otherwise independent scalar normalizations of the ternary forms. $\square$

Consequently, if every one of these finitely many overlapping ternary slices is exceptionally well approximated by rational forms with compatible height bounds, then the entire form is exceptionally well approximated by one rational form. This is the correct higher-dimensional slicing argument; one good ternary plane by itself is insufficient.

16.8 Final conceptual map

The quantitative Oppenheim proof is the composition



In critical signatures, insert one additional box between the first two: “exclude or explicitly count quasi-null rational subspaces.” Every hypothesis in the final arithmetic theorem can be traced to one of these boxes.

Chapter 17

Student Proof Laboratory: Counting Values of an Indefinite Quadratic Form

The qualitative Oppenheim theorem proves density. Quantitative Oppenheim asks a different question: among integer vectors of norm at most T , how many have values in a fixed interval, and how small a value is guaranteed at a given height? The proof has a geometric main term, a dynamical error, a cusp-truncation error, and possible arithmetic exceptional pieces.

17.1 The geometric main term by coarea

Let Q be nondegenerate indefinite on $\mathbb{R}^n$ and let $I = [a, b]$ be bounded. Set

$$\Omega_T(I) = \{x \in \mathbb{R}^n : \|x\| < T, Q(x) \in I\}.$$

Because $Q(Ty) = T^2Q(y)$,

$$\text{vol } \Omega_T(I) = T^n \text{vol}\{\|y\| < 1 : Q(y) \in T^{-2}I\}.$$

Proposition 17.1 (Coarea main term). *If the unit sphere meets the null cone of Q transversely, then*

$$\text{vol } \Omega_T(I) = c_Q |I| T^{n-2} + o(T^{n-2}),$$

where

$$c_Q = \int_{\{\|y\| < 1, Q(y)=0\}} \frac{d\sigma(y)}{\|\nabla Q(y)\|}.$$

Proof. Apply the coarea formula to Q on the unit ball:

$$\text{vol}\{\|y\| < 1 : Q(y) \in J\} = \int_J \left(\int_{Q^{-1}(s) \cap B_1} \frac{d\sigma_s}{\|\nabla Q\|} \right) ds.$$

Nondegeneracy implies $\nabla Q(y) = 0$ only at $y = 0$, and the null cone away from the origin is a smooth hypersurface. The inner integral is continuous at $s = 0$ after separating a small ball about the origin; the contribution of that ball is lower order because $n \geq 3$. Taking $J = T^{-2}I$ gives $c_Q |I| T^{-2} + o(T^{-2})$ inside the brackets, and multiplication by T^n gives the result. $\square$

17.2 Siegel transforms and why truncation is needed

For a compactly supported function f on $\mathbb{R}^n$, define on the space of unimodular lattices

$$\widehat{f}(\Lambda) = \sum_{0 \neq v \in \Lambda} f(v).$$

This function is generally unbounded near the cusp because a lattice with a very short vector may contribute many multiples inside the support.

Lemma 17.2 (Height bound for the Siegel transform). *For a unimodular lattice set*

$$\text{ht}(\Lambda) = 1 + \max_{1 \leq r < n} \alpha_r(\Lambda),$$

with the rank heights α_r of Chapter 5. For every fixed compact set $C_0 \subset \mathbb{R}^n$ there is $C = C(C_0, n)$ such that every bounded f supported in C_0 satisfies

$$|\widehat{f}(\Lambda)| \leq C \|f\|_\infty \text{ht}(\Lambda)^n.$$

Proof. Let $\lambda_1(\Lambda)$ be the length of a shortest nonzero lattice vector. The rank-one definition gives $\alpha_1(\Lambda) = \lambda_1(\Lambda)^{-1}$. If $C_0 \subset B(0, R)$, then the balls $B(v, \lambda_1/2)$ centered at distinct $v \in \Lambda \cap C_0$ are disjoint and all lie in $B(0, R + \lambda_1/2)$. Comparing Euclidean volumes gives

$$\#(\Lambda \cap C_0) \leq C_{R,n} \max\{1, \lambda_1^{-n}\} \leq C_{R,n} \text{ht}(\Lambda)^n.$$

Since $|f(v)| \leq \|f\|_\infty$ for every contributing lattice point, the same bound holds for $|\widehat{f}(\Lambda)|$. $\square$

Choose a cutoff χ_R supported on height $\leq 2R$ and equal to one on height $\leq R$. The bounded observable $\chi_R \hat{f}$ can be inserted into quantitative equidistribution. Quantitative nondivergence controls the discarded cusp part. One then optimizes $R = R(T)$ along with the smoothing scale of the value window.

17.3 From orbit equidistribution to the counting asymptotic

The quadratic form determines an orthogonal subgroup H and an orbit in a space of lattices. The count of $v \in \mathbb{Z}^n$ with $Q(v) \in I$ and $\|v\| < T$ is expressed, after scaling, as a Siegel transform evaluated along an H - or diagonal-orbit average.

Proposition 17.3 (Dynamical counting from an explicit error ledger). *Assume the smoothed/truncated counting observable $F_{\varepsilon,R,T}$ satisfies, for fixed positive A, B, β ,*

$$\mathcal{S}_D(F_{\varepsilon,R,T}) \leq C\varepsilon^{-A}R^B, \quad (18.1)$$

that the homogeneous averages obey

$$\left| \int F_{\varepsilon,R,T} d\mu_T - \int F_{\varepsilon,R,T} dm \right| \leq E(T) \mathcal{S}_D(F_{\varepsilon,R,T}), \quad (18.2)$$

and that value-window smoothing and cusp truncation contribute relative errors at most $C\varepsilon$ and $CR^{-\beta}$, respectively. Then

$$N_Q(T, I) = c_Q |I| T^{n-2} + O\left(T^{n-2}(E(T)\varepsilon^{-A}R^B + \varepsilon + R^{-\beta})\right). \quad (18.3)$$

In particular, if $E(T) \ll T^{-\sigma}$, then for every $0 < \eta < \eta_$ one has*

$$N_Q(T, I) = c_Q |I| T^{n-2} + O(T^{n-2-\eta}), \quad (18.4)$$

where

$$\eta_* = \sup_{\theta, \lambda > 0} \min\{\sigma - A\theta - B\lambda, \theta, \beta\lambda\}, \quad (18.5)$$

the supremum being over $A\theta + B\lambda < \sigma$.

Proof. Apply (18.2) and then (18.1) to the truncated smooth observable. Siegel's mean-value formula and the coarea expansion supply the main term.

Adding the stated smoothing and cusp-truncation errors proves (18.3). If $E(T) \ll T^{-\sigma}$, put $\varepsilon = T^{-\theta}$ and $R = T^\lambda$. The three relative errors become

$$T^{-(\sigma - A\theta - B\lambda)}, \quad T^{-\theta}, \quad T^{-\beta\lambda}.$$

Their slowest exponent is the minimum in (18.5). It is positive for all sufficiently small $\theta, \lambda > 0$, and choosing a pair whose minimum exceeds a prescribed $\eta < \eta_*$ gives (18.4). $\square$

17.4 Why signatures (2, 1) and (2, 2) are critical

The expected scale T^{n-2} comes from coarea, but arithmetic concentration on rational or almost rational isotropic subspaces can create contributions of that scale or larger in low dimension. This is not a failure of the geometric volume calculation; it is a failure of uniform integrability of the Siegel transform along the relevant orbit.

Definition 17.4 (Extremely well approximable by rational forms). Fix a norm on quadratic forms. An irrational form Q is extremely well approximable (EWA) if for arbitrarily large exponents A there are nonzero rational forms Q' of controlled denominator q with

$$\|Q - \lambda Q'\| \leq q^{-A}$$

for some scalar λ . A finite Diophantine-type condition is a uniform power lower bound excluding this behavior.

17.5 Quasi-null rational subspaces

In signature (2, 2), a rational two-plane on which Q is exceptionally small can contribute many lattice points whose values remain in a fixed window. Such a plane is “quasi-null” at the counting scale.

Lemma 17.5 (Contribution of one rational plane). *Let $L \subset \mathbb{R}^4$ be a rational two-plane and let $\Lambda = L \cap \mathbb{Z}^4$ have covolume d_L . Then the number of $v \in \Lambda$ with $\|v\| < T$ is*

$$\frac{\pi_L}{d_L} T^2 + O(T/d_L^{1/2} + 1),$$

where π_L is the area of the unit disk in the induced metric.

Proof. Choose a lattice basis of Λ and identify L with $\mathbb{R}^2$ so the fundamental parallelogram has area d_L . Cover the disk by fundamental parallelograms. Fully contained cells contribute area divided by d_L ; only cells meeting the boundary contribute to the error. Their number is bounded by boundary length divided by the shortest basis scale plus a constant, which Minkowski bounds by $O(T/d_L^{1/2} + 1)$. $\square$

Removing finitely many quasi-null planes of bounded arithmetic complexity restores a uniform height moment for the remaining lattice points. The main text’s modified $1 + \delta$ moment estimate quantifies this statement.

17.6 Modified moment estimate in the critical cases

The exact internally proved moment statement is Theorem 7.34. In signature $(2, 1)$ it gives the uniform $1 + \delta$ moment (7.4a) for the modified ternary height; in signature $(2, 2)$ it gives (7.4b) for the split modified height. The conversion from those height moments to modified Siegel transforms is carried out by the ternary and split modified-Lipschitz lemmas and their quasi-null remainder propositions cited in that theorem. Thus there is no separate “schematic critical moment theorem” at this point of the exposition.

The proof mechanism is worth retaining. One decomposes the cusp into dyadic height shells, uses quantitative nondivergence unless a primitive rational wedge is anomalously small, removes the explicitly classified rational or quasi-null subspaces, and then invokes the finite-type lower bound. The resulting shell exponent is strictly larger than the growth exponent of the modified transform to the power $1 + \delta$, so the shell series converges. The exact exponents are those proved in Chapter 8.

17.7 Ternary forms and effective rank-two dynamics

For signature $(2, 1)$ the orthogonal group is locally $\mathrm{PSL}_2(\mathbb{R})$, and the ambient arithmetic problem is especially sensitive to rational approximations of the form. The precise finite-scale statement is Theorem 8.17: it gives either primitive approximation of every target s in the explicit range $|s| \leq R^{\kappa_3}$ or an integral ternary form Q' satisfying the quantitative alternative (14.O1). Under the finite-scale Diophantine separation used in Corollary 8.20, the

rational-form branch is excluded at all sufficiently large scales. No weaker “student-form” theorem is used as a logical input.

Chapter 18

Student Proof Laboratory: Reconstructing a Higher-Dimensional Form from Ternary Slices

18.1 Why one slice is insufficient

A single rational three-plane sees only six Gram coefficients. In dimension $n > 3$, the remaining coefficients may vary freely, so one good rational approximation to one ternary restriction does not approximate the whole form.

18.2 Overlapping slices with a common two-plane

Choose a rational nondegenerate indefinite two-plane $W = \text{span}_{\mathbb{Q}}(e_1, e_2)$. Let $w_3, \dots, w_n$ complete it to a rational basis. Consider the ternary spaces

$$W_i = W + \mathbb{Q}w_i, \quad W_{ij} = W + \mathbb{Q}(w_i + w_j).$$

Lemma 18.1 (Common scalar normalization). *Suppose each restriction $Q|_{W_i}$ is within ϵ of a scalar multiple $\lambda_i Q'_i$ of a rational ternary form and the restriction $Q|_W$ is bounded away from zero in norm. If the approximation is sufficiently small, then after rescaling the rational forms the numbers λ_i differ by $O(\epsilon)$ from one common scalar λ .*

Proof. Restrict each approximation to W . Two nonzero scalar multiples of rational forms that are both $O(\epsilon)$ -close to the same fixed nondegenerate form on W must have proportionality constants differing by $O(\epsilon)$ after normalizing one fixed Gram coefficient of $Q|_W$ to one. The lower bound on $\|Q|_W\|$ prevents division by a vanishing coefficient. Use the same normalization for every slice. $\square$

Proposition 18.2 (Recovery of all Gram coefficients). *Assume rational approximations of compatible height and error ϵ are available for every W_i and W_{ij} above. Then there is a rational quadratic form Q' on $\mathbb{Q}^n$ and a scalar λ such that*

$$\|Q - \lambda Q'\| \ll_n \epsilon,$$

with height bounded polynomially in the maximum slice height.

Proof. The restrictions to W_i recover, after the common scalar normalization, $Q(w_i)$ and the two cross terms $B_Q(e_1, w_i)$, $B_Q(e_2, w_i)$ by rational numbers with error $O(\epsilon)$. It remains to recover $B_Q(w_i, w_j)$. The slice W_{ij} gives a rational approximation to

$$Q(w_i + w_j) = Q(w_i) + Q(w_j) + 2B_Q(w_i, w_j).$$

Subtract the already recovered diagonal values and divide by two. Thus every entry of the Gram matrix in the rational basis $e_1, e_2, w_3, \dots, w_n$ is approximated by the same scalar times a rational number. Let D_i and D_{ij} be common denominators for the rational Gram entries coming from the slices W_i and W_{ij} . If every slice height is at most H , then $D_i, D_{ij} \leq H^{O_n(1)}$. There are only $O_n(1)$ slices, so the least common multiple D of these denominators also satisfies $D \leq H^{O_n(1)}$. Multiplying every recovered Gram entry by D produces one integral Gram matrix of height $H^{O_n(1)}$; dividing back by D defines the required rational form Q' . $\square$

18.3 Quantitative consequence

If the full form Q satisfies a Diophantine lower bound against rational forms, the proposition implies that not all overlapping ternary slices can simultaneously fall into the rational-obstruction alternative at arbitrarily large scales. At least one slice therefore equidistributes effectively. That one effective slice supplies integer vectors in the full lattice and transfers the ternary small-value bound to Q . This is the correct slicing logic; it uses a compatible family, not one isolated three-plane.

18.4 From counting to a guaranteed small value

Suppose an effective count gives, for a shrinking symmetric interval $I_T = [-T^{-\kappa}, T^{-\kappa}]$,

$$N_Q(T, I_T) = 2c_Q T^{n-2-\kappa} + O(T^{n-2-\kappa-\eta}).$$

If $n - 2 - \kappa > 0$, the main term tends to infinity and dominates the error. Hence for all sufficiently large T there is nonzero $v \in \mathbb{Z}^n$ with $\|v\| < T$ and $|Q(v)| < T^{-\kappa}$. Inverting the relation gives an explicit height bound for reaching a prescribed value tolerance ϵ . This last conversion is elementary, but it should be written because a rate for an orbit average is not automatically a rate for the first small quadratic value until the smoothing and counting exponents have been tracked.

Chapter 19

A Student Course in Quantitative Oppenheim Theory

The qualitative Oppenheim theorem says that an irrational indefinite quadratic form in at least three variables takes values arbitrarily close to zero on nonzero integer vectors. Quantitative theory asks two harder questions: how many values fall in a fixed interval at height T , and how small a value can be guaranteed before height T ? This chapter rebuilds the translation from geometry to dynamics and back again slowly enough that the normalizations and exceptional signatures can be checked by hand.

19.1 The geometric scaling before dynamics

Let Q be a nondegenerate indefinite quadratic form on $\mathbb{R}^n$. For a compactly supported continuous function ϕ on $\mathbb{R}$ and a compactly supported radial cutoff ψ on $\mathbb{R}^n$, consider

$$I_T = \int_{\mathbb{R}^n} \phi(Q(v)) \psi(v/T) dv.$$

Set $v = Tw$. Since $Q(Tw) = T^2Q(w)$,

$$I_T = T^n \int_{\mathbb{R}^n} \phi(T^2Q(w)) \psi(w) dw.$$

The factor T^2 in the quadratic value confines w to a thickness T^{-2} around the cone $Q = 0$, so the expected order is T^{n-2} .

Lemma 19.1 (Coarea main term). *Assume the support of ψ avoids the origin. Then*

$$T^{-(n-2)} I_T \longrightarrow \left(\int_{\mathbb{R}} \phi(s) ds \right) \int_{Q=0} \frac{\psi(w)}{\|\nabla Q(w)\|} d\sigma(w).$$

Proof. By the coarea formula,

$$\int_{\mathbb{R}^n} F(w) dw = \int_{\mathbb{R}} \int_{Q=q} \frac{F(w)}{\|\nabla Q(w)\|} d\sigma_q(w) dq.$$

Apply this with $F(w) = \phi(T^2 Q(w))\psi(w)$ and substitute $s = T^2 q$. Then

$$I_T = T^{n-2} \int_{\mathbb{R}} \phi(s) \left(\int_{Q=s/T^2} \frac{\psi(w)}{\|\nabla Q(w)\|} d\sigma_{s/T^2}(w) \right) ds.$$

On the compact support of ψ , nondegeneracy and avoidance of the origin make the level hypersurfaces vary smoothly near $q = 0$. The inner integral therefore converges uniformly for s in the fixed support of ϕ to its $q = 0$ value. Dominated convergence completes the proof. $\square$

Worked Example 19.2 (A split ternary form). For $Q(x, y, z) = x^2 + y^2 - z^2$, the gradient is $2(x, y, -z)$. On the cone $z^2 = x^2 + y^2$ away from the origin, $\|\nabla Q\|$ is comparable to the radius. Writing $x = r \cos \theta$, $y = r \sin \theta$, $z = \pm r$ makes the cone measure completely explicit and displays the T^1 main scale predicted by $n - 2 = 1$.

19.2 From lattice points to a Siegel transform

For a compactly supported function f on $\mathbb{R}^n$, define the Siegel transform on the space of unimodular lattices by

$$\widehat{f}(\Lambda) = \sum_{0 \neq v \in \Lambda} f(v).$$

When f localizes to a quadratic shell, evaluating $\widehat{f}$ along the homogeneous point representing Q is the smoothed lattice count.

Warning 19.3. Even when f is bounded and compactly supported, $\widehat{f}$ is not bounded on the space of lattices. A lattice with many short vectors can put many nonzero points into the support of f . Thus one may not insert $\widehat{f}$ directly into a mixing or equidistribution estimate stated only for bounded Sobolev functions.

Lemma 19.4 (Linear short-vector domination). *Let f be supported in the Euclidean ball $B_R \subset \mathbb{R}^n$. Define*

$$\alpha(\Lambda) := \max_{1 \leq r \leq n} \max_{\substack{\Delta \subset \Lambda \text{ primitive} \\ \text{rank } \Delta = r}} \text{covol}(\Delta)^{-1}.$$

(The rank- n term equals 1 for a unimodular lattice.) Then

$$|\hat{f}(\Lambda)| \leq C_{n,R} \|f\|_\infty \alpha(\Lambda). \quad (19.0)$$

Thus the exponent of the full exterior-power height in this bound is one, not an unspecified dimension-dependent power.

Proof. It is enough to bound $\#(\Lambda \cap B_R)$. Let $\lambda_1 \leq \dots \leq \lambda_n$ be the successive minima of Λ for the Euclidean unit ball, and let r be the largest index with $\lambda_r \leq R$; if no such index exists, $\Lambda \cap B_R = \{0\}$ and there is nothing to prove. Choose linearly independent $v_1, \dots, v_r \in \Lambda \cap B_R$ and put $V = \text{span}(v_1, \dots, v_r)$. By maximality of r , every lattice vector in B_R lies in V . Hence

$$\Lambda \cap B_R = (\Lambda \cap V) \cap B_R.$$

The lattice $\Delta := \Lambda \cap V$ is primitive of rank r .

A reduced-basis form of the geometry-of-numbers basis lemma gives a basis $b_1, \dots, b_r$ of Δ with $\|b_i\| \leq C_n \lambda_r \leq C_n R$. Let P be its half-open fundamental parallelepiped. Its diameter is at most $C'_n R$. The translates $x + P$, $x \in \Delta \cap B_R$, are pairwise disjoint and all lie in the r -dimensional ball $B_{(1+C'_n)R} \cap V$. Therefore

$$\#(\Delta \cap B_R) \text{covol}(\Delta) \leq \text{vol}_r(B_{(1+C'_n)R} \cap V) \leq C_{n,R}.$$

Since Δ is one of the primitive sublattices occurring in the definition of α ,

$$\#(\Lambda \cap B_R) \leq C_{n,R} \alpha(\Lambda).$$

Removing the origin and multiplying by $\|f\|_\infty$ gives (19.0). $\square$

19.3 Truncation and the two-error decomposition

Choose a smooth cutoff χ_R which equals 1 when $\alpha(\Lambda) \leq R$ and vanishes when $\alpha(\Lambda) \geq 2R$. Then

$$\hat{f} = (\hat{f}\chi_R) + \hat{f}(1 - \chi_R).$$

The first term is a bounded Sobolev test function whose norms grow polynomially in R . The second is controlled by a moment estimate for α .

Lemma 19.5 (Tail from a moment). *If a probability measure ν satisfies $\int \alpha^p d\nu \leq M$ for some $p > 1$, then, for $R \geq 1$,*

$$\int_{\alpha > R} |\hat{f}| d\nu \ll_{n, \supp f} M \|f\|_\infty R^{1-p}. \quad (19.2)$$

Proof. By Lemma 19.4, $|\hat{f}| \ll \|f\|_\infty \alpha$. On $\{\alpha > R\}$ and for $p > 1$,

$$\alpha \leq R^{1-p} \alpha^p.$$

Integrating gives (19.2). $\square$

This is the exact point at which the exceptional signatures enter: the moment range that is uniform along the relevant orthogonal orbit is weaker in low dimensions because rational almost-null subspaces can create large cusp excursions.

19.4 Why signatures $(2, 1)$ and $(2, 2)$ require extra arithmetic data

For a rational subspace $L \subset \mathbb{Q}^n$, the restriction $Q|_L$ can be very small compared with the height of L without being identically zero. Such a subspace behaves dynamically almost as if it were contained in the null cone. In the two critical signatures there are enough low-dimensional isotropic configurations for these near-null planes to dominate the untruncated Siegel transform.

Definition 19.6 (Projective finite Diophantine type). Let $\mathcal{Q}_n$ be the real vector space of quadratic forms in n variables, written in the fixed monomial basis, and equip its coefficient space with the Euclidean norm. For nonzero forms R, S with unit coefficient vectors $\hat{r}, \hat{s}$, define the projective angular distance

$$d_{\mathbb{P}}([R], [S]) := \|\hat{r} \wedge \hat{s}\| = \sqrt{1 - |\langle \hat{r}, \hat{s} \rangle|^2}.$$

If $[R]$ is a rational coefficient line, let r_{prim} be either of its two primitive integral coefficient vectors and set $H([R]) = \|r_{\text{prim}}\|$. We say that Q has *projective finite Diophantine type κ* if there is $c > 0$ such that

$$d_{\mathbb{P}}([Q], [R]) \geq c H([R])^{-\kappa}$$

for every rational coefficient line $[R] \neq [Q]$.

Remark 19.7 (This auxiliary notion is not the critical-signature input). The load-bearing low-signature arguments use the normalized finite-type condition (7.1) and, in signature (2, 2), the split-Diophantine condition of Definition 7.3. The projective definition above is included only to explain what a generic rational-approximation condition looks like. It must not be used to replace the special arithmetic reconstruction proved in Chapter 8.

Lemma 19.8 (Near-null planes give proximity to a rational incidence subspace). *Fix n . Let $L \subset \mathbb{Q}^n$ be a rational two-plane and let $w_L \in \bigwedge^2 \mathbb{Z}^n$ be its primitive Pluecker vector; put $H(L) = \|w_L\|$. Define*

$$\mathcal{K}_L = \{R \in \mathcal{Q}_n : R|_L = 0\}.$$

Then $\mathcal{K}_L$ is a rational linear subspace. There is a constant $C_n > 0$, depending only on the dimension and the fixed coefficient norms, with the following property. One can choose a primitive integral basis ℓ_1, ℓ_2 of $L \cap \mathbb{Z}^n$ with $\|\ell_i\| \ll_n H(L)^{C_n}$, and, if

$$\|Q|_L\|_\ell := (|Q(\ell_1)|^2 + |B_Q(\ell_1, \ell_2)|^2 + |Q(\ell_2)|^2)^{1/2},$$

then for every coefficient-normalized form $\|Q\| = 1$,

$$\text{dist}(Q, \mathcal{K}_L) \ll_n H(L)^{C_n} \|Q|_L\|_\ell. \quad (19.1)$$

Moreover $\mathcal{K}_L$ has a basis consisting of rational forms whose primitive coefficient heights are $\ll_n H(L)^{C_n}$.

Proof. Choose a reduced primitive basis ℓ_1, ℓ_2 of the rank-two lattice $L \cap \mathbb{Z}^n$. Geometry of numbers bounds the basis vectors by a fixed power of the covolume, and that covolume is comparable, up to the fixed ambient norm, with $H(L) = \|w_L\|$.

In the fixed coefficient basis of $\mathcal{Q}_n$, restriction to L is the linear map

$$A_L : R \longmapsto (R(\ell_1), B_R(\ell_1, \ell_2), R(\ell_2)).$$

In the fixed monomial coefficient basis, its matrix has rational entries of size $\ll_n H(L)^{C_n}$ with a denominator bounded solely in terms of the chosen polarization convention (indeed at worst a fixed power of 2), and $\ker A_L = \mathcal{K}_L$. After multiplying the matrix by this fixed common denominator we obtain an integral matrix with the same kernel and with entries still of size $\ll_n H(L)^{C_n}$. Let $r = \text{rank } A_L$. Every nonzero $r \times r$ minor of the cleared matrix is a

nonzero integer, hence has absolute value at least one. Combining this with the polynomial upper bound for its entries shows, by the singular-value/minor formula, that the smallest nonzero singular value of A_L is $\gg_n H(L)^{-C_n}$. On $(\ker A_L)^\perp$ the inverse therefore has norm $\ll_n H(L)^{C_n}$. Applying it to $A_L Q$ gives (19.1).

Finally, row reduction of the cleared integral matrix expresses a basis of its kernel by ratios of minors. Clearing denominators gives primitive rational basis vectors of height bounded by another fixed power of $H(L)$. Enlarging C_n absorbs this power. $\square$

Warning 19.9 (Why a generic Cramer-rule shortcut is insufficient). Equation (19.1) places Q near the *rational subspace* $\mathcal{K}_L$; it does not place Q near a single rational point of $\mathbb{P}(\mathcal{K}_L)$ whose height is polynomial in $H(L)$ independently of the requested approximation accuracy. The subspace $\mathcal{K}_L$ has positive dimension, and approximating the remaining real coordinates by a rational point can require denominators growing like a power of the inverse accuracy. Thus the former claim

$$\|Q|_L\| \ll H(L)^{-A} \implies d_{\mathbb{P}}([Q], [Q_L]) \ll H(L)^{-A+C} \text{ with } H([Q_L]) \ll H(L)^C$$

is false without additional structure.

The critical signatures possess exactly that additional structure. In the ternary branch it is the five-point conic reconstruction of Proposition 7.7; in the split branch it is Lemma 7.18 and Corollary 7.20. Those results, summarized in Lemma 7.32, are the proof owners that convert quasi-null data into rational-form approximation of controlled height. No generic “small restriction plus Cramer’s rule” inference is used in the load-bearing argument.

19.5 The determinant laboratory in signature $(2, 2)$

Identify $\mathbb{R}^4$ with 2×2 matrices and let

$$Q(X) = \det X = x_{11}x_{22} - x_{12}x_{21}.$$

The two-dimensional subspaces

$$L_v = \{uv^T : u \in \mathbb{R}^2\}, \quad R_u = \{uv^T : v \in \mathbb{R}^2\}$$

are totally isotropic. When u or v is rational, these are rational null planes. They explain geometrically why signature $(2, 2)$ admits a larger family of dangerous rational subspaces than the stable higher-rank signatures.

Worked Example 19.10 (Counting on a rational null plane). Take $v = (1, 0)^T$. Then L_v consists of matrices $\begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix}$ and $Q \equiv 0$ on this plane. The integer points with Euclidean norm at most T already contribute $\asymp T^2$ exact zeros. Any theorem predicting a universal generic $T^{n-2} = T^2$ main term with a power-saving error must therefore separate this algebraic contribution rather than treat it as random equidistribution.

19.6 The modified moment argument

Let $\mathcal{N}_R$ denote the collection of primitive rational subspaces of height at most R on which Q is quasi-null at the chosen Diophantine threshold. Define a modified height

$$\alpha_Q^*(\Lambda) = \max \left(1, \max_{L \notin \mathcal{N}_R} \text{covol}(L \cap \Lambda)^{-1} \right).$$

The purpose of removing $\mathcal{N}_R$ is not to discard points from the count. It is to split the count into a regular part controlled by moments and an explicit singular part supported near the quasi-null rational subspaces.

Proposition 19.11 (Regular-tail principle). *Suppose the unipotent averaging operator satisfies, for some $0 < \delta < 1$,*

$$\mathbb{E}[(\alpha_Q^*)^{1+\delta}(xu_t)] \leq c(\alpha_Q^*(x))^{1+\delta} + C, \quad c < 1,$$

for averaging over a fixed macroscopic time block. Then the regular part of the Siegel transform has a uniformly integrable tail along long orbit averages.

Proof. Iterating the drift inequality gives a uniform bound for the $(1 + \delta)$ -moment of α_Q^* along block averages. The short-vector domination for the regular part uses only subspaces retained in α_Q^* . Apply the tail-from-a-moment lemma with $p = 1 + \delta$; the tail is then $O(R^{-\delta})$. Pass from block averages to long averages by decomposing the time interval into blocks plus a bounded remainder. $\square$

19.7 From an effective orbit theorem to a counting theorem

Let $F_{T,R}$ be the truncated smoothed Siegel transform. Suppose the effective orbit theorem gives

$$\left| \frac{1}{T} \int_0^T F_{T,R}(xu_t) dt - \int F_{T,R} dm \right| \leq CT^{-\eta} R^a \mathcal{N}(f),$$

while the tail is $O(R^{-b})$ and smoothing contributes $O(\varepsilon^\beta)$ with Sobolev cost ε^{-s} . Then the total relative error has the explicit shape

$$T^{-\eta} R^a \varepsilon^{-s} + R^{-b} + \varepsilon^\beta + \text{singular contribution.}$$

The first three terms are optimized exactly as in Volume XII. The fourth is zero in the stable signatures under the appropriate irrationality hypothesis, or must be written separately in the critical signatures.

Guiding Idea 19.12. This formula explains why there is no honest universal statement obtained by writing simply “effective Ratner gives an error term.” A quantitative Oppenheim theorem is a composition of an orbit rate, an unbounded-observable truncation, cusp moments, smoothing, and arithmetic control of singular subspaces.

19.8 Overlapping ternary slices: reconstruction in detail

Let $E = \text{span}_{\mathbb{Q}}(e_1, e_2)$ be a rational two-plane on which Q is nondegenerate and indefinite. Extend to a rational basis $e_1, e_2, w_3, \dots, w_n$. For $i \geq 3$ define $L_i = E + \mathbb{Q}w_i$, and for $i < j$ define $L_{ij} = E + \mathbb{Q}(w_i + w_j)$.

Lemma 19.13 (Gram reconstruction from overlapping slices). *Suppose the restrictions of two quadratic forms Q and Q' to every L_i and L_{ij} are proportional, and the proportionality is fixed to be 1 on the common plane E . Then $Q = Q'$ on all of $\mathbb{Q}^n$.*

Proof. Agreement on E gives the entries $B(e_a, e_b)$ for $a, b \leq 2$. Agreement on L_i gives $Q(w_i)$ and $B(e_a, w_i)$. On L_{ij} we know

$$Q(w_i + w_j) = Q(w_i) + Q(w_j) + 2B(w_i, w_j).$$

The first two terms are already known, so this determines every $B(w_i, w_j)$. Thus every Gram-matrix entry agrees. $\square$

Proposition 19.14 (Quantitative overlapping-slice principle). *If all the restrictions above are within ε of rational ternary forms of height at most H , with compatible normalizations on E , then Q is within $O(\varepsilon H^C)$ of a rational n -variable form of height at most H^C .*

Proof. Choose the rational approximants on the slices. Compatibility on the common plane fixes their scalar ambiguity. The reconstruction formulas use only addition, subtraction, and division by 2, so errors accumulate linearly. Putting all reconstructed coefficients over a common denominator multiplies heights by at most a fixed power depending on the number of slices. This gives the asserted polynomial height and error bounds. $\square$

19.9 Extracting an actual small value

Lemma 19.15 (Count implies a value). *Let*

$$N_Q(T, I) = \#\{v \in \mathbb{Z}^n : \|v\| \leq T, Q(v) \in I\}.$$

If an effective asymptotic gives

$$N_Q(T, [-\varepsilon, \varepsilon]) \geq c\varepsilon T^{n-2} - CT^{n-2-\eta}$$

for $0 < \varepsilon \leq 1$, then every

$$\varepsilon > \frac{2C}{c} T^{-\eta}$$

and all sufficiently large T produce a nonzero integer vector with $\|v\| \leq T$ and $|Q(v)| \leq \varepsilon$, provided exact zeros are removed or counted separately.

Proof. With the stated choice of ε , the right-hand side is at least $CT^{n-2-\eta} > 0$. Hence the counting set is nonempty. If Q has rational isotropic subspaces producing exact zeros, subtract their explicitly bounded contribution before applying the argument to nonzero values. $\square$

19.10 Final student checklist for quantitative Openheim

A quantitative result is complete only after all of the following questions have explicit answers:

1. What is the geometric T^{n-2} main term and its constant?

2. Which homogeneous orbit represents Q and what is its Haar normalization?
3. Which observable represents the lattice count?
4. Why is that observable integrable after truncation?
5. Which height moment controls its tail?
6. Are there quasi-null rational subspaces, and how are they separated?
7. What is the effective orbit rate and on which Sobolev norm?
8. How are smoothing, height cutoff, and orbit-rate parameters optimized?
9. In low dimension, what Diophantine hypothesis excludes exceptionally well approximable rational forms?
10. How is the final positive count converted into an explicit small value?

These ten items are the arithmetic analogue of the proof ledger used throughout the Ratner volumes: each one corresponds to a genuine logical step and cannot be replaced by a citation to “the quantitative Oppenheim method.”

Proof Guide and Dependency Summary

This chapter is a finalization layer. It does not introduce a new theorem scope. Its purpose is to expose the proof architecture, make the hypotheses that are easiest to misread impossible to miss, and give a student a concrete model calculation before the abstract machinery is reused later.

1. The stable and critical regimes are different theorems

For an indefinite quadratic form in n variables, the geometric coarea scale suggests a main term of order T^{n-2} . That heuristic is reliable only after cusp contributions are controlled. In critical signatures such as $(2, 1)$ and $(2, 2)$, special rational or nearly rational subspaces can produce anomalously large cusp mass. Therefore the manuscript deliberately separates:

stable-signature EMM regime from critical-signature analysis.

There is no theorem asserting a uniform expected-order upper bound in every signature.

2. Why the Siegel transform must be truncated

The counting observable is an unbounded Siegel transform. Effective mixing applies first to a truncated/smoothed observable. One then estimates the tail using a height function. In the critical regime, the ordinary moment may be insufficient, and a modified $1 + \delta$ moment is used after separating quasi-null rational subspaces.

3. The arithmetic obstruction dictionary

A proper rational subspace on which the form is exceptionally small is not just a technical nuisance. It is the arithmetic avatar of an excursion toward a proper homogeneous obstruction. The proof therefore separates:

generic cusp mass + singular rational contribution.

The latter must either be included explicitly in the asymptotic or excluded by a Diophantine hypothesis.

4. Effective ternary route

The ternary problem uses the rank-two effective dynamics of Volume XIII. The logical chain is

$$\begin{array}{c}
 \text{effective orbit alternative} \\
 \Downarrow \\
 \text{quantitative equidistribution or rational obstruction} \\
 \Downarrow \\
 \text{counting in a fixed value window} \\
 \Downarrow \\
 \text{existence of a small nonzero value}
 \end{array}$$

Higher-dimensional reconstruction must track overlap of ternary slices through a common rational two-plane; independent slice estimates cannot simply be summed without compatibility.

5. Student checklist for an Oppenheim estimate

Record separately: dimension/signature, Diophantine hypothesis, truncation height, spectral or effective-unipotent input, singular-subspace contribution, smoothing scale, and the final exponent after optimization.

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